Core Algebra

— *EYNTKA* Series —

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Contents

Part I

Groups

The concept of a group is the culmination of multiple branches of mathematics converging to a single unifying object that works in each domain. Arguably the first branch of mathematics where the notion of a group was used first in the modern sense was in *algebraic equations*, that is the study of polynomials. A young mathematician by the name of Évariste Galois found the answer around 1832[1] to an over 2000 year old problem: does there exist a "quadratic equation" for all polynomials, what mathematicians would call the *solvability of degree n polynomials* problem. The degree 3 (cubic) polynomials and degree 4 (quartic) polynomials case has been solved in the affirmative[2] around 300 years before Galois, but the degree 5 (quintic) polynomial remained a mystery. Galois proved that there exists quintic polynomials with *no* formula, in particular, there exist quintic polynomial $ax^5+bx^4+cx^3+dx^2+ex+f$ where a root x_0 cannot be written as a sum, difference, product, quotient, or root $(+, -, \times, \text{div}, \sqrt[n]{-})$ of the coefficients! We shall prove this in chapter 20, section 20.2.4.

Around the same time, mathematicians started to discover that there are "more geometries" than just the usual flat geometry of euclidean space (now known as euclidean geometry). It was shown that one of the foundational axiom of geometry ever since the time of the ancient Greeks (the 5th postulate, or parallel postulate) is in fact an assumption that can be removed without causing a contradiction! Near the beginning of these developments, a mathematician by the name of Felix Klein saw that the notion of a group seems to be intimately connected with understanding these new geometries, and a Norwegian mathematician named Sophus Lie (pronounced "Lee") came up with *Lie Groups* which solidified the importance of groups within modern geometry[3].

While all of this was happening, a notion very similar to groups was already being used in *number theory* with mathematicians such as Gauss and Dirichlet noticing inherent symmetries in primes that showed up while solving Diophantine equations. We shall cover these results (in modern algebraic language) in chapter 50.

In these early days, these three disciplines were defining groups in slightly different ways. It was the effort of mathematicians such as Camille Jordan and Ernst Kummer to coalesce these concepts, resulting in the modern definition of a group as we know it today. The formalization of the definition of a group became a huge stepping stone for an enormous amount of disciplines to notice its practicality within their domain. Today, groups have become a major, if not central, object in almost every branch of mathematics, but also have been extensively used in physics, chemistry, cryptography, computer science, and many more fields. The reason groups are a fundamental object in each field is that it can interpreted as representing the *symmetry* of some system.

A modern mathematicians would now consider groups to be a rather universal and "natural" object. However, groups came late in the development of mathematics, partly due to the mathematical build-up that needed to happen and partly due to their abstract nature that does not make them seem natural to a first-time viewer. A consequence of this is that it makes teaching about groups introduce a new level of abstraction that many students have not experienced. The difficulties are further compounded by the fact that visualizations of many groups are scarce or impractical[4]. This is however not a bad thing: a whole new way of learning and "experiencing" mathematical objects is the reward for internalizing groups and all the further objects that got based off of them.

In this Part, we shall cover in detail all the fundamental concepts of groups. Most of these concepts shall be instrumental in the rest of the textbook, and the reader shall find that this part of the book

[1] Évariste Galois accepted a challenge to a dual. The night before the duel he wrote down what is now considered *Galois Theory* before passing away in the duel the next day

[2] see this link for the cubic formula and this link for the quartic

[3] This material belongs in a class in geometry, but we shall cover some important algebraic aspects of it in chapter 54

[4] We shall see that there are natural ways of visualizing groups, but it sometimes worse to think about the visual picture, for example some groups can only be visualized by the symmetries of an over 100'000 dimensional shape

is the nexus for most algebraic tools that we shall cover in the rest of the book.

0.0.1 For Graduate Student The interpretation of a group representing symmetry is in fact very natural: all groups can be represented as a category with one objects whose morphisms are isomorphisms. From this perspective, it is not that it is *an* interpretation, but *the* interpretation.

There is another way that I (currently) personally see groups. They feel like naturally "2nd order" objects. If we are trying to study some object (numbers, shapes, graphs, etc), we will want to understand some property of the object. This property can be local (like all loops based at a certain point, or the functions that exist on a certain subset), or it can be global (ex. all loops that go everywhere up to homotopy, symmetries of the object, etc.). This can usually be identified by defining a certain action on the object of study. Groups arise naturally when considering actions. Thus, groups are a way of capturing a type of algebraic property of an object. There can be other types of properties of an object (ex. curvature, twisting, etc.) which are not represented algebraically, and hence groups are just a part of the puzzle of looking for information about an object.

1

Introduction to Groups

In this chapter, you will cover the basic properties of groups. In the process, we will prove statements that you perhaps thought were axiomatic. For example, if $ab = ac$, does that mean $b = c$? We will classify two very important sub-categories: groups with the axiom of *commutativity* ($ab = ba$), and those without it. These two categories of groups will set the narrative for how we study groups, meaning this one axiom plays an incredibly central role in the study of groups! We will cover many examples of groups to get a taste on what groups are, and define the groups which will be referenced indefinitely throughout the rest of the book. We will then cover how to compare two groups, and define the notion of equality between two groups. We'll then see how we can use groups to "represent" a system of objects. For example, we'll use a group to represent all possible ways of rotating an object. In this case, the system will consist of all possible rotations of the object.

1.1 Basic Properties and Definitions

As mentioned in the preface, algebraic structures are sets along with a function (or functions) which must follow certain rules. The most common type of function used for an algebraic structure is called a *binary operation*:

Definition 1.1.1: Binary Operation

1. A *binary operation* $*$ on a set G is a function $* : G \times G \to G$. For any $a, b \in G$, we shall write $a * b$ or $*(a, b)$. Often the notation is abbreviated to ab.
2. A binary operation $*$ on a set G is *associative* if for every $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$
3. If $*$ is a binary operation on a set G we say elements a and b *commute* if $a * b = b * a$. We say $*$ (or G) is *commutative* if for every $a, b \in G$, $a * b = b * a$

The star of the show in group theory is the binary operation. This function is the key mathematical structure added to a set which we study when talking about groups. When talking about different group structures, we're often talking about different binary operations.

Example 1.1.2: Binary Operations

1. Arithmetic operations are a good source of binary operations. As we've discussed in the preliminary chapter, $+$ and $\times$ are functions that take in two numbers as input, and output one number. If you want to practice proving something is a function, you can try showing that $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is a function (and since the domain is the cartesian product of two copies of the codomain, it's a binary operation). Similarly, you can do the same with $- : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.
2. Is $-$ a binary operation with $\mathbb{N}$ (i.e. is $- : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ a binary operation)? What about $\times$ and div? Which of these binary operations are associative? Are they commutative? What if you switch $\mathbb{N}$ with $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{R}$?
3. When working with sets, it is common to take the union or intersection of sets. These are binary operations. If A is a set and $P(A)$ is its powerset, then $\cup : P(A) \times P(A) \to P(A)$ is a binary operation. Similarly for $\cap$
4. Take $\mathbb{Z}$, and recall the *gcd* and *lcm* from the preliminary chapter (ref:HERE). Then $gcd : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is a binary operation. Similarly for *lcm*. Can you replace $\mathbb{Z}$ with $\mathbb{N}$?
5. If you have taken some multi-variable calculus, then the cross product $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ is an example of a binary operation.
6. Take $E : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ to be $E(x, y) = x^y$. Check that $E(x, E(y, z))$ is not necessarily equal to $E(E(x, y), z)$ (that is $x^{(y^z)}$ is not necessarily equal to $(x^y)^z$). Hence, exponentiation is a non-assotiative operation.
7. Given two functions f and g over the same domain and codomain ($f : X \to X$ and $g : X \to X$), their composition. Thus composition of functions with the same domain and codomain forms a binary operation.

With these intuitions, we can move on to defining a group:

Definition 1.1.3: Group

A *group* is an ordered pair $(G, *)$, where G is a set and $*$ is a binary operation on G satisfying the following axioms:

1. associativity: $(a * b) * c = a * (b * c)$
2. there exists an element e in G, called the *identity* of G, such that for every $a \in G$, we have $e * a = a$
3. For each $a \in G$ there is an element $b \in G$ called an *inverse* of a, such that $ab = ba = e$. We denote this element b by a^{-1}, so that we can equivalently write $a^{-1} * a = e$

The group $(G, *)$ is called *abelian* (or commutative) if $a * b = b * a$ for every $a, b \in G$.

If $(G, *)$ is a group, we say that G is a group under $*$.

1.1.4 Note Some authors like to make it explicit that it's closed under the operation, meaning if $a, b \in G$, then $a * b \in G$. However, that is implied in the definition of binary operation (i.e. a function), since the codomain of $*$ is G.

Example 1.1.5: First Encounter With Groups

In the following examples, when we're working with numbers, think of them only as a set, and define the operations $+$, $\times$ or others in the way they were intuitively defined when learning arithmetic's.

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are groups. We would say, for example, that $\mathbb{Z}$ is a group under $+$. For each of these examples, we would have the identity to be $e = 0$, and for each a, $a^{-1} = -a$. Note that technically, the $+$ in each group is not the same $+$, because the domains and co-domains are different. Note that $\mathbb{N}$ is *not* a group under $+$, since not every element has an inverse: There does not exist an $a \in \mathbb{N}$ such that $2 + a = 0$ (i.e. $-2 \notin \mathbb{N}$). [a]
2. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ under $\times$ are *not* groups. $\mathbb{Q} - \{0\}$, $\mathbb{R} - \{0\}$ $\mathbb{C} - \{0\}$, $\mathbb{Q}_{>0}$, $\mathbb{R}_{>0}$ are groups under $\times$, with $e = 1$, and $a^{-1} = \frac{1}{a}$. Why is that? Take a moment to think about why removing the zero will make those sets groups. Is $\mathbb{Z} - \{0\}$ a group under $\times$? Is $(\mathbb{Z} - \{0\})$ a group under $+$?
3. For any $n \in \mathbb{N}$, $\mathbb{Z}/n\mathbb{Z}$ is a group under modulo $+$ $(\mathbb{Z}/n\mathbb{Z}, +)$. Can you show that 0 is the identity? Can you show that every number has an inverse? Note that $((\mathbb{Z}/n\mathbb{Z}), \times)$ under $\times$ *not* a group in general! It *is* a group when p is prime. You can actually pick some non-prime n and try it (ex. $n = 4$).The general proof that $(\mathbb{Z}/n\mathbb{Z}, \times)$ is a group when n is prime will come later (after we introduce theorem 2.3.11)
4. If $(A, \bullet_a)$ and $(B, \bullet_b)$ are groups, we can form a group on their cartesian product $A \times B$ called the *direct product group*, whose elements are those in the Cartesian product

   $$A \times B = \{(a, b) | a \in A, b \in B\}$$

   and the binary operations is defined point wise:

   $$(a, b) * (c, d) = (a \bullet_a c, b \bullet_b d)$$

We can denote this group and its binary operation $(A \times B, (\bullet_a, \bullet_b))$. A common example of a direct product of group is $(\mathbb{R} \times \mathbb{R}, (+, +))$, with the usual notation of $\mathbb{R}^2$. We will dedicate a lot of energy understanding the details of the Cartesian product of groups in chapter 5.

5. Define $+_2 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ $x +_2 y = 2 + x + y$. The operation $+_2$ is associative, since

$$(x +_2 y) +_2 z = (2 + x + y) +_2 z = (2 + x + y) + 2 + z = 2 + x + (2 + y + z) = x +_2 (y +_2 z)$$

Furthermore, $e = -2$, since
$$x +_2 -2 = x + 2 - 2 = x$$
and $x^{-1} = -x - 4$ since
$$x^{-1} +_2 x = -x - 4 + 2 + x = -2$$
showing that $(\mathbb{Z}, +_2)$ is indeed a group. Notice that $(\mathbb{Z}, +_2)$ and $(\mathbb{Z}, +)$ are different (for example, their identity is different)[b]. When introducing multiplication to integers, we shall want $+$ and $\cdot$ to behave well together; the usual $+(x, y) = x + y$ will insure this[c].

6. $G = (\mathrm{Mat}_n(\mathbb{R}), \cdot)$ is almost forms a group. A binary operation exists, identity exists, and associativity holds. However, there are non-invertible matrices, meaning that $\forall M \in G$, it's not necessarily the case that $\exists N \in G$ such that $NM = \mathrm{I}$. We will explore this in section 1.2.5⋆

7. Take $\{e\}$ to be the set with a single element, an a binary operation $\cdot$ such that $e \cdot e = e$ (that is, the binary operation that treats e like an identity). Then $(\{e\}, \cdot)$ is a group! This group is called the *trivial group*.

8. Take a set A and its powerset $P(A)$. Is $(P(A), \cup)$ a group? What about $(P(A), \cap)$? answer: [d]

9. Take $(\mathbb{Z}, -)$, where $-$ is the operation of subtraction. Then $(\mathbb{Z}, -)$ is not a group! Can you see which axiom fails?

10. Here is a more complex example which fails for the same reason as the previous one. Take $(\mathbb{R}^3, \times)$ where $\times$ represents the cross product:

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_1b_3 - a_3b_1 \\ a_1b_2 - a_2b_1 \end{pmatrix}$$

or, if you've taken a physics or linear algebra class, you might have seen it defined as

$$a \times b = \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

where we represent the three components of the vector as $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, and we use the determinant to shorten notation. Then $(\mathbb{R}^3, \times)$ is *not* a group! The cross product is not associative! Take

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

Then

$$\begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}2\\3\\4\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\end{pmatrix} \qquad \begin{pmatrix}-1\\2\\-1\end{pmatrix} \times \begin{pmatrix}3\\4\\5\end{pmatrix} = \begin{pmatrix}14\\2\\-10\end{pmatrix}$$

and

$$\begin{pmatrix}2\\3\\4\end{pmatrix} \times \begin{pmatrix}3\\4\\5\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\end{pmatrix} \qquad \begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}-1\\2\\-1\end{pmatrix} = \begin{pmatrix}-8\\-2\\4\end{pmatrix}$$

meaning $(\mathbb{R}^3, \times)$ is *not* associative, meaning it's not a group. The cross product is a different binary operation called the *lie bracket*[e].

11. If you know some complex analysis, then $G = \{z \in \mathbb{C} | z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$ is a group! Visually, this group looks like points on the unit circle in the complex plane. The main idea here is since $z^n = 1$, then elements in G will be of the form $e^{\frac{2\pi i}{n} \cdot k}$ for $1 \leq k \leq n$. If you're comfortable with multiplying complex numbers and trust me that all the elements of G are of this form, then you can check that this is a group.

    More generally, $G = \{z \in \mathbb{C} \mid \|z\| = 1\}$ is a group with multiplication (i.e. the unit circle in the complex plane). You can verify that $1 \in G$ and for any $z, w \in G$, $\|zw\| = \|z\| \, \|w\| = 1 \cdot 1 = 1$. Using these two fact, showing there exists an inverse and that G is associative follows from properties of $\mathbb{C}$ (exercise ref:HERE)

12. If you know some complex arithmetic's, show that $G = \{z \in \mathbb{C} \mid 0.1 < |z| < 10\}$ with the usual multiplication structure of $\mathbb{C}$ satisfies associativity, having an identity, and having an inverse, but is *not* a group.

13. Let A be any set. Consider the collection of every bijective function from A to itself:

    $$G = \{f : A \to A \mid f \text{ is bijective}\}$$

    Then with the binary operation of composition, $(G, \circ)$ is a group. What is the identity, and why do inverses always exist? The collection of bijective functions is a very common group that shows up in many different ways in different areas of mathematics and so has lots of notations:

    $$\mathrm{Bij}(A) \qquad \mathrm{Sym}(A) \qquad \mathrm{Aut}_{\mathbf{Set}}(A)$$

Part of the motivation behind learning group theory is in the abundance of areas groups show up in mathematics. The more domains of mathematics you learn, the more types of groups you will encounter. For example, field theory (which we explore in part IV) have *Galois groups* at its heart, Differential Geometry have *lie groups* that allow for a large classification of and construction of "manifolds" (a generalization of the notion of Euclidean space), Algebraic Geometry (which we explore in part V) use *algebraic groups* to represent more complicated objects in more familiar terms, Analysis uses *topological groups* (for example, Amenable and Locally Compact groups) in the study of Fourier Analysis and Functional Analysis, Algebraic topology has the *fundamental group* and *homology group* at its heart, and combinatorics uses *geometric group theory* (for example, the *Kleinian group* or *hyperbolic groups* shows up frequently).

Group can also be thought of as one of the "basic" structures that many other structures will build on or act on. Of the algebraic structures that will be explored in this book, rings, fields, vector-spaces, modules, and (lie and associative) algebras are all also groups.

Overall, it is worth remembering that groups are a very common mathematical object that show up a little bit everywhere.

[a] $\mathbb{N}$ is a weaker structure called a monoid which doesn't require inverses. More on them in section 7.1

[b] I'm careful in saying they are different instead of saying they have *different group structures*. There is a subtly here on how we define two group structures being the same or different. This is addressed in section 1.4

[c] we shall desire distributivity, or for the expert that $\mathbb{Z}$ is representable in its endomorphism ring

[d] the answer is no! Like $\mathbb{N}$, these are example of monoids

[e] Explored in section **??**

In general, when we write a group, the group operation will be clear from context, so instead of writing $(G, *)$, we will write G. We will take on this convention form here on out unless it's ambiguous. We will also assume the standard binary operation of addition for

$$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/n\mathbb{Z}$$

unless stated otherwise (ex, if you see $\mathbb{Z}$, assume it is $(\mathbb{Z}, +)$). If we want the multiplicative structure of these numbers, we'll add a $\times$ in the superscript and implicitly omit the zero:

$$\mathbb{Z}^\times, \mathbb{Q}^\times, \mathbb{R}^\times, \mathbb{C}^\times (\mathbb{Z}/n\mathbb{Z})^\times$$

These sets are $(\mathbb{Z} - \{0\}, \times)$ [1], $(\mathbb{Q} - \{0\}, \times)$, $(\mathbb{R} - \{0\}, \times)$, and so forth. The sets $\mathbb{R}^+$ and $\mathbb{Q}^+$ shall represent the collection of positive real numbers and positive rational number with the usual multiplication. Furthermore, since we are always working with binary operations, the word "binary" is often omitted for convenience.

1.1.6 Note In many of the previous examples, there is some more structure usually associated with those symbols. For example, in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$, there is a notion of some numbers being bigger than others, that is, $1 < 2$, or $n < n + 1$. These are *not* part of the definition of a group, and so cannot be used when proving general statements about groups! It will actually be very common that you work with groups with *more* structure than their group structure, so be quite careful that the examples you hold when doing exercises *only* represent the structure you want to work with at that moment

You might have noticed that some of the underlying sets of the above groups are subsets of each other: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$. Does this imply that there is a relation between their group structure? For example does the usual addition in $\mathbb{Z}$ works as in $\mathbb{Q}$ with usual addition? If you think about this particular example, you'll see that the answer is yes. But being a subset doesn't always means you'll have the same (if any) group structure: $\mathbb{N} \subseteq \mathbb{Z}$, but $\mathbb{N}$ is not even a group! On the other hand, if we take $2\mathbb{Z} \subseteq \mathbb{Z}$, and give $2\mathbb{Z}$ the same operation as $\mathbb{Z}$ (that is $(2\mathbb{Z}, +)$ is our group), then $(2\mathbb{Z}, +)$ acts exactly the same way as $\mathbb{Z}$, with just fewer elements. So, we'll give a name to subsets which inherit group properties of their parent group:

Definition 1.1.7: Subgroup

Let G be a group and $H \subseteq G$. Then H is a subgroup of G if it is still a group under the same operation as G. We denote the subgroup by $H \leqslant G$ [a]

[a] In some textbooks, the symbol for subgroup is $H \preceq G$

[1] Note that $\mathbb{Z}^\times$ is not a group under $\times$ as we've pointed out earlier. However, the binary operations $\times$ is still defined on this set

You might have flinched to it saying "the same operation", since it was explicitly mentioned in the definition that for a group G, the operation is $G \times G \to G$, so H cannot have the same operation since the domain and co-domain are different. This is a common convention in algebra for structures which are similar: if we restrict the operation to only the elements of H, then $(H, *|_H)$ should be a group. In particular, if $* : G \times G \to G$ is the binary operation on G, then we say we *restrict* the binary operation to H if we can define $*|_H : H \times H \to H$, where for any two elements $h_1, h_2 \in H$ $*|_H (h_1, h_2) = *(h_1, h_2)$. If $(H, *|_H)$ is a group, we say it's a subgroup of $(G, *)$. In this way, we say that H has "the same" binary operation as G.

Example 1.1.8: Subgroups

1. Take $2\mathbb{Z} \subseteq \mathbb{Z}$ – the set of all even integers. The identity is there: $0 \in 2\mathbb{Z}$. If you add two even integers, you get an even integer $(2x + 2y = 2(x + y))$ [a]. Furthermore, if $2x \in 2\mathbb{Z}$, then there exists $-2x \in 2\mathbb{Z}$ such that $2x - 2x = 0$, meaning every element has an inverse. Finally, associativity is implied by associativity in $\mathbb{Z}$: for any $2x, 2y, 2z \in 2\mathbb{Z}$, we have that $2x, 2y, 2z \in \mathbb{Z}$, so $2x + (2y + 2z) = (2x + 2y) + 2z$. Thus $2\mathbb{Z} \leqslant \mathbb{Z}$

2. Right before the previous definition, it was mentioned that $\mathbb{Z} \subseteq \mathbb{Q}$ is an example of a subgroup (i.e. $\mathbb{Z} \leqslant \mathbb{Q}$). This is an exercise worth doing to get comfortable with the concept. Mix an match with other types of numbers to get more practice.

3. Let $\mathbb{Z}^\times \subseteq \mathbb{Q}^\times$ (i.e. $Z - \{0\} \subseteq \mathbb{Q} - \{0\}$). Is $\mathbb{Z}^\times$ a subgroup?

4. Let $H \leqslant G$ and $K \leqslant H$. Is $K \leqslant G$?

5. Here is an exercise in understanding the nuance of binary operations: Is $\mathbb{Z}/5\mathbb{Z} \leqslant \mathbb{Z}$? answer: [b]

[a] Note here how I *did* use extra properties that are not necessarily part of a group (I factored the two out). This is fine, since we're working with a *particular* example, this would not be a proof we can do on general groups

[b] It is no, since the binary operation does not "act" the same way $(3 + 3 = 1)$ in $\mathbb{Z}/5\mathbb{Z}$ but $3 + 3 = 6$ in $\mathbb{Z}$

Groups are missing one key property that numbers usually have: commutativity. Though we already talked about them in the definition of a group, I will add it here to emphasize their importance:

Definition 1.1.9: Abelian Group

Given a group G, if $ab = ba$ for all $a, b \in G$, we call G a *commutative* or *abelian* group

Example 1.1.10: Abelian Groups

$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/n\mathbb{Z}$ are all Abelian Groups [a]. So are $\mathbb{Q}^\times, \mathbb{R}^\times, \mathbb{C}^\times$. In fact, all examples in the example 1.1.5 were abelian groups. We haven't yet introduced groups which are not abelian. The first such group will be the Dihedral group (definition 1.2.25)

[a] This is an example where the binary operation is implied. The binary operations shown in example 1.1.5 are the standard binary operations for these sets

As mentioned earlier, being abelian/commutative is insanely powerful. Of all the properties the reader will cover in basic group theory, a lot of them will be properties a group being "almost" or "kind of" abelian. In this text, we will come across many definitions and sections that are centered around how close they are to being abelian [2]

The reason we relate a lot of conditions to commutativity is because Abelian Groups behave in much more predictable ways as compared to general groups, as shall be explored. The power of commutativity actually permeates a lot of algebraic structures. Because of this, we often split up domains in mathematics in "commutative" and "not-commutative". As an example of this split, some mathematicians like to reference the set of Abelian groups as **Ab** and the set of all groups as **Group** to show how different a general group versus a general abelian group is [3]. There are entire areas of mathematics where the problem has been solved in the "abelian case", while the "nonabelian case" is an active area of research[4].

For the visual reader, the following is a helpful tool to understand the group operation:

Definition 1.1.11: Cayley Table (Multiplication Table)

Let G be a finite group (that is, a group on a finite set). Label the elements of G as $\{g_1, g_2, ..., g_n\}$, where g_1 represents the identity.
Then *Cayley table* or *multiplication table* of G is the $n \times n$ matrix whose i, j entry is the group element $g_i g_j$.

I personally prefer the term Cayley table since not all binary operations are called "multiplication"; it can be confusing to do a multiplication table for $(G, +)$

Example 1.1.12: Cayley Table

Take $G = \mathbb{Z}/4\mathbb{Z}$. Then the Cayley tables for addition and multiplication are:

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	4
2	2	3	4	0
3	3	4	0	1

$\times$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

This can be done for any finite group, and sometimes gives valuable insights instantly. For example, notice that in the Cayley table for multiplication, for row 0, 1, and 3, they contain every element of $\mathbb{Z}/4\mathbb{Z}$, yet the row for 2 doesn't. Notice further that both tables are symmetrical, that is, position ij is equal to position ji.

In general, it can be quite cumbersome to write out an entire Cayley table, especially when the order the group will be higher than 4, if not infinite. So throughout Part I, we develop algebraic tools to deduce information about groups.

Exercise 1.1.1

[2] for reference, see definition 3.1.9, definition 2.2.4, and section 5.3, 6.1, and 6.1

[3] There are some technicalities here in saying it's the *set* of all groups, due to Russel's paradox. We should instead say the *class* of all sets. This is addressed in Chapter **??**

[4] There are a huge number of such examples, but to point out one relevant to algebra class field theory completely classifies abelian extension of fields, while the nonabelian counterpart is a very deeply researched problem and is called *Langland's Program*. Another example that creates a split between the commutative and non-commutative case is within the study of algebraic geometry where the structures are built out of commutative rings, while the non commutative case would require more machinery from analysis such as Fourier Transforms

1. Let $G = \mathbb{Q}$, and the binary operation be average : $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$, where $\text{average}(x, y) = \frac{x+y}{2}$. Is $(G, \text{average})$ a group? What if $G = \mathbb{Z}/4\mathbb{Z}$?
2. Can a group structure be put on $\emptyset$ (i.e., can there exist a function $* : \emptyset \times \emptyset \to \emptyset$ such that $(\emptyset, *)$ is a group?) (In prelim chapter, make an exercise saying that $f : \emptyset \to \emptyset$ is a function, while $f : A \to \emptyset$, $|A| > 0$ is *not*, and also that $\emptyset \times \emptyset$ is vacuously well-defined)
3. (If you know some complex analysis) In example 1.1.5[10], it was stated that
$$G = \{z \in \mathbb{C} \mid |z| = 1\}$$
is a group. Prove that G is indeed a group.
4. Let $\text{Bij}(X)$ be the set of all bijections from X to X. Prove that $\text{Bij}(X)$ is a group. In section 1.2.2, we will relabel this as $\text{Sym}(X)$.
5. look at this link to get some ideas
6. (Perhaps this a superficial question, but the identity and inverse condition's can be replaced by functions, and so G would be a set with 1 binary operation, 1 "1-ary" operation, and a 0-ary operation. Introduce this to make groupoids and universal algebras more intuitive way later?
7. perhaps a question on how the identity can be replaced by a monary operation? (i.e. a perspective that aims to be a prelude to Universal Algebras?)

1.1.1 Consequences of Group Axioms

With the definition of groups down, we can deduce a lot of direct consequences form the axioms. One thing common with mathematical constructs is an abundance of propositions. These are vital to be able to prove by oneself, and once comfortable should be treated as knowledge as important as the definition of a group – I would even go so far as to say that not knowing the properties of groups means you don't really know what a group is. For example, do you really know what a frying pan is/can do if you haven't cooked with it before? Do you know the in's an out of what utensils can you use to not damage it, or what type of food will stick to it or not? You will only know once you've practiced using it.

I would strongly suggest that the reader attempts proving *all* of these propositions before looking at the proof, especially if this is the first time the reader has seen algebraic structures! It is crucial moving forward that these properties are second nature, since everything we build form here on out will constantly be relying on these properties.

The first thing to point out is that if we have $a * e = a$, we can deduce $e * a = e$.

Proposition 1.1.13: one-sided identity implies two-sided identity

Let G be a group and e an identity. If $a * e = a$ then $e * a = a$ (or vice-versa)

***Proof*:**
Let's say $e * a = a$. We want to show that $a * e = a$ (1). If we multiply equation (1) on the right

by a^{-1}, we get

$$\begin{aligned} &= (a * e) * a^{-1} \\ &= a * (e * a^{-1}) && \text{associativity} \\ &= a * a^{-1} && \text{Identity definition} \\ &= e && \text{inverse definition} \end{aligned}$$

Thus, $(a * e) * a^{-1} = e$. Multiplying both sides by a on the right, we get

$$\begin{aligned} ((a * e) * a^{-1}) * a &= e * a \\ \Leftrightarrow\ (a * e) * (a^{-1} * a) &= e * a \\ \Leftrightarrow\ (a * e) * e &= e * a \\ \Leftrightarrow\ a * e &= e * a \end{aligned}$$

as we sought to show.

Some author's put $a * e = e * a = a$ as a definition. This is a matter of preference: since it's a direct consequence of the axioms, it can be put as one of the fundamental rules a group has to follow without worry of constraining the definition. Some authors disagree with this, and would rather keep the definition of group in fuller generality. This preference for generality is a balancing act: the more general you go, the more "complicated" the concept gets. However, if done right, you in general gain a more "nuanced" perspective with generality. In this book, the more general idea was taken, since some early proofs fail for more subtle reasons. For example, $(\mathbb{Z}, -)$ fails to be a group because of associativity, however if we put $a * e = e * a = a$ in the definition, we can also say it fails because $-a = 0 - a \neq a - 0 = a$ (for $a \neq 0$). However, as we've seen in the above proof, this fact *relies on associativity*, and therefore is hiding the fact that $(\mathbb{Z}, -)$ is still failing to be a group because of associativity.

Corollary 1.1.14: one-sided Inverse Implies two-sided Inverse

Let a, a^{-1} be elements of the group G and let e be the identity of G. Then $a * a^{-1} = e$ if and only if $a^{-1} * a = e$

***Proof*:**
Let's say $b = a^{-1}$ so that $a * b = e$. Then $b = b * e = b * a * b$. Multiplying both sies by b^{-1}, we get

$$\begin{aligned} e &= b * b^{-1} \\ &= (b * a * b) * b^{-1} \\ &= b * a * (b * b^{-1}) \\ &= b * a \\ &= a^{-1} * a && b = a^{-1} \end{aligned}$$

as we sought to show.

Proposition 1.1.15: Basic properties of Group elements

1. The identity of G is unique. Furthermore, if for some $x \in G$, $f * x = x$, then is equal to the identity
2. For each $a, \in G$, a^{-1} is uniquely determined
3. $(a^{-1})^{-1} = a$, for all $a \in G$
4. $(a * b)^{-1} = (b^{-1}) * (a^{-1})$.
5. generalized associative law (induction on associativity).

***Proof*:**
Again, I highly encourage you to attempt all of these before looking at these answers!!

1. Let $e \in G$ be the identity, that is, for all $x \in g$ $e * x = x$. Let's say there is another element $f \in G$ that is also the identity. Then

$$\begin{aligned} e &= e * f && \text{definition of identity} \\ &= f && \text{definition of identity} \end{aligned}$$

therefore, $e = f$, showing that e is unique.

Furthermore, if there was an f such that for only *some* $x \in G$, $f * x = x$, then

$$\begin{aligned} (f * x) * x^{-1} &= (x * x^{-1}) = e \\ \Leftrightarrow f * (x * x^{-1}) &= e \\ \Leftrightarrow f &= e \end{aligned}$$

thus, there cannot be a "partial" identity.

2. exercise

With these in mind, here's some standard notation to simplify writing: One usually write the operation $*$ as $\cdot$ to represent an abstract binary operation, unless otherwise specified; we write $a \cdot b$ as ab. We usually also denote the identity e as 1. We'll also implement the following short hand: [5]

$$xxxxxxx \cdots x = x^n, \qquad x^{-1}x^{-1} \cdots x^{-1} = x^{-n}, \qquad x^0 = 1$$

Don't forget that last one is the identity of G. If we're using a group where specific notation is already defined (like with the $+$ operation), we can use it's specific convention. So $a + a + \cdots + a = na$, $-a + -a + \cdots + -a = -na$, and $0a = 0$ [6]. Furthermore, $a + (-a)$ is sometimes shorthanded to $a - a$.

The next result we'll cover let's us generalize our intuition of pre-abstract algebra. When learning

[5] $x^0 = 1$ since $x^0 = x^{1-1} = x^1 x^{-1} = xx^{-1} = 1$

[6] If you have covered rings, this might look familiar to the first property in proposition 9.1.9. It is quite similar in that this shorthand secretly is a ring-action on the group (but not a $\mathbb{Z}$ module since G is not necessarily abelian). This means that the intuition behind the notation is similar to the proof in 9.1.9: $0a = (1-1)a = a - a = 0$

algebra for the first time, you've probably encountered equations of the form:

$$3x = 5$$

and where asked to solve for x. The way you would do this is by dividing both sides by 3 to get the equation:

$$x = \frac{5}{3}$$

This might seem like the obvious thing to do in any situation. However, you can't always cancel out a number on one side and get a unique result! Take a moment to review your proof of uniqueness of identity and uniqueness of the inverse, and consider this thought experiment:

1.1.16 Thought Experiment Let's take a set for which every element does *not* have an inverse. Recall (or prove) that $\mathbb{Z}/6\mathbb{Z}$ is not a group under $\cdot$. In particular, $[2]$ has no multiplicative inverse. Since $[2][2] = [0]$, then:

$$[4] = [2][2] = [2][5] = [10] = [4]$$

However, $[2] \neq [5]$. In other words, I couldn't "cancel" out the $[2]$ on the left side, i.e. for the equation $[2]x = [4]$ we get $x = [2]$ or $x = [5]$.

This cancelling out of a number on the left [or right] side is something you probably took for granted, but it's actually the consequence of the fact that every number in a group must have a unique inverse! This fact is proven in the next proposition:

Proposition 1.1.17: Cancellation Property: Unique solution to monomials

Let G be a group and let $a, b \in G$. The equation $ax = b$ and $ya = b$ have unique solutions for $x, y \in G$. In particular, the left and right cancellation properties hold in G:

1. If $au = av$ then $u = v$
2. if $ub = vb$ and $u = v$

Sometimes, this proposition is called the *Cancellation Law*

***Proof*:**
Simply use uniqueness of inverse. The point of this theorem is that if $ab = e$, for any element $a \in G$ and some element $b \in G$, then $b = a^{-1}$.

This proposition is very powerful, it means we can bring in our elementary algebra knowledge. In this sense, groups are the structure on which we generalize the idea of doing elementary algebra on. They are limited in that they only allow one operation: you probably have worked with equation with both addition and multiplication (ex. $3x^2 - 5 = 0$). We will work with such equations with 2 binary operations once we study *rings* (see part II)

With these properties, I'll take a moment to explain the power of associativity and commutativity. In all type of numbers systems we saw $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$, for any two numbers a and b:

$$aaabbbbaaabaaaabbbbbaa$$

is unambiguous. and so, we can write

$$a^3b^3a^3ba^4b^5a^2$$

Which is really powerful. If you didn't have associativity, then $aaabbbbaaabaaaabbbbbaa$ would be ambiguous, since it's possible that

$$(aaabbbbaaabaaaabbbbba)a \neq a(aabbbbaaabaaaabbbbbaa)$$

So every pair of elements would have to be bracketed since

$$abc$$

would be ambiguous [7] since it's possible that

$$(ab)c \neq a(bc)$$

and so you would be forced to have an expression like

$$((((aa)b)((bb)(ba)))(((((aa)(ba))a))((a(a(bb)))((b(bb))a))))a$$

with no simpler way of representing the expression – quite a monstrosity. There is more than $2^{22} = 4'194'304$ ways of bracketing this!! The idea is that associativity means that given a sequence of operations that must be performed in a certain order:

$$a_1 * a_2 * a_3 * \cdots * a_n$$

We are allowed to "break up" the problem into smaller problems that can be grouped and operated and then recombined in the right order. For example if we must perform 4 actions in order:

$$a_1 * a_2 * a_3 * a_4$$

Then associativity means that we could have considered $a_3 * a_4 = b_2$ as a new combined actions and $a_1 * a_2 = b_1$ also as a combined action:

$$b_1 * b_2$$

Or we could have taken $a_1 * a_2 = b_1$ and then combine it with the action of a_3, $b_1 * a_3 = b_3$ and then with a_4 and get the same result:

$$a_1 * a_2 * a_3 * a_4 = b_1 * b_2 = b_3 * a_4$$

This can also be thought of as "pre-combining" the steps before performing the result in the same order, which if the operation is associative would give you the same result.

Example 1.1.18: Associativity and The Lack of It

1. An example of real world phenomena where this is not applicable rock-paper-scissors. Let's says the (commutative) binary operation takes in two choices and gives the winner (and if the two input are the same return the input). Then:

   $$(\text{rock} * \text{paper}) * \text{scissors} = \text{paper} * \text{rock} = \text{paper} \neq \text{rock} * \text{scissors} = \text{rock} * (\text{paper} * \text{scissors})$$

2. *Divide and Conquer (Parallel Computing)*: Associativity is key in parallel computing. If a

[7] like $\mathbb{R}^3$ with the cross product

sequence of operations is associative, we don't have to wait for a single processor to calculate from left to right:

$$((((a_1 * a_2) * a_3) * a_4) \dots)$$

Instead, we can dispatch the problem locally to different processors:

(a) Processor 1 solves $a_1 * a_2$ locally.

(b) Processor 2 solves $a_3 * a_4$ locally.

They then combine their local results[a]. Such "reduction" operations must be associative precisely so they can be solved locally on separate servers and combined later without changing the final outcome.

3. Exponentiation is *not* associative: $(a^b)^c$ is not in general equal to $(a)^{b^c}$.

4. The averaging operation is not associative

5. Convince your self that cooking instruction are usually not associative.

[a]Frameworks like Google's MapReduce rely entirely on this

Since functions are associative, most structures in mathematics will be associative, however there are entire domain of study which concentrates on a non-associative structure [8] or drop the condition of associativity[9]. We will briefly see an alternative to associativity for an algebraic structure we will cover in part X, chapter 54, but structures without associativity are outside the scope of this book[10]. As a consequence of associativity, we can unambiguously write:

$$x^{a+b} = x^{b+a}$$

since the order of multiplication doesn't effect the final result. We can also visualize associativity through the use of directed graphs. If we take abc, then we can visualize which tuple of elements we choose to multiply first:

$$\begin{tikzcd}
 & ((ab)c) = (a(bc)) & \\
(ab)c \arrow[ur] & & a(bc) \arrow[ul] \\
 & abc \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

this is more clear if we have 4 elements, any way we choose to bracket our elements leads to the same

[8]the most famous of is called Lie Algebra, with something called the Jacobi Identity replacing associativity. Another is the logical implications: $(a \to b) \to c$ does not always give the same truth value as $a \to (b \to c)$

[9]Three such examples are *loops*, *Quasigroups*, and *magmas*

[10]See ref:HERE for books covering such topics

result:

$$\begin{tikzcd}
 & & & (abcd) & & & \\
((ab)c)d \arrow[urrr] & & (a(bc))d \arrow[ur] & (ab)(cd) \arrow[u] & a((bc)d) \arrow[ul] & & a(b(cd)) \arrow[ulll] \\
 & (ab)cd \arrow[ul] \arrow[urr] & & a(bc)d \arrow[ul] \arrow[ur] & & ab(cd) \arrow[ull] \arrow[ur] & \\
 & & & abcd \arrow[ull] \arrow[u] \arrow[urr] & & &
\end{tikzcd}$$

Note that adding commutativity makes this even more powerful, since

$$aba = aab = a^2b$$

and so we can group all like terms together, and write

$$aaabbbbaaabaaaabbbbbaa \qquad \text{as} \qquad aaaaaaaaaaaabbbbbbbbbb$$

and collapse it to:

$$a^{12}b^{10}$$

That is, the only information we need is the number of times we multiply by a and b, with no regard to the order we multiply them! This added axiom is so powerful that it will turn out that it will constrain the number of possible (finite and finitely generated) groups, which are relatively few[11] (see section 5.1.1)

or:exercises

(Assuming some basic analysis knowledge) Because of general associativity we have that for any finite number of terms, we can bracket them however we want and will get the same result. Some curious among you might wonder if we can expand this to be able to take an infinite number of terms and have a similar result. However, this immediately becomes tricky. For example, if we take $\mathbb{Z}$, then what is $1+1+1+1+\cdots$? It should be verified that no number in $\mathbb{Z}$ will satisfy this conditions (you can always pick a larger number). Even worse, what if we take $1+1-1+1-1+\cdots$? Due to the infinite nature of this series, it does not have a resolution. We can try to limit ourselves to all infinite series that *converge*. However, there is even a limitation in this case. (move the terms around using commutativity of a non-absolutely convergent sequence). Because of this, the study of figuring out how to "add" infinitely many elements is relegated to another field of mathematics called analysis.

As an exercise to show the power of commutativity, show that if G is abelian, then :$(ab)^{-1} = a^{-1}b^{-1}$. (Once homomorphisms are introduced, show $x \mapsto x^{-1}$ and $x \to x^a$ are also homomorphisms

1.2 Examples of Groups

Now that we covered the basics, it is important to build up a family of common groups to get familiar with the how to recognize groups, and to feel very comfortable manipulating groups. We already

[11]The notion of finitely generated will be covered in section 2.3. This result shall be proved in theorem 5.1.8. As of writing this book, the project of classifying all abelian groups is still a very unsolved problem

showed how many previous mathematical structures were already groups (the canonical examples being $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$). We will now define new groups to show further what is possible with the definition of the group, and to build-up a repertoire of more elaborate examples.

1.2.1 Integers Modulo n and Cyclic Groups

Recall in section 0.2.2 that $\mathbb{Z}/n\mathbb{Z}$ is a collection of equivalences classes satisfying $a \sim_n b$ if and only if there exists a $k \in \mathbb{Z}$ st $a - b = kn$. Given that $\mathbb{Z}$ is a group and the results proposition 0.2.15, it is reasonable to think that it may form a group:

Proposition 1.2.1: Modular Arithmetic's Groups

Let $\mathbb{Z}/n\mathbb{Z}$ be the integesr modulo n. Then $\mathbb{Z}/n\mathbb{Z}$ forms a group under modular addition:

$$(a \bmod n) +_n (b \bmod n) = (a + b) \bmod n$$

with the binary operation denoted as $+_n$, or if the context permits simply as $+$

In almost all cases, the notion $+$ shall be used, for it will be clear from context that the elements will be part of a certain set of modular integers.

***Proof*:**

(Well-defined) By proposition 0.2.15(1), the operation is well-defined

(Idenitty) For any n, there is always $\overline{0}_n \in \mathbb{Z}/n\mathbb{Z}$ for $n - n = 0$. Then for any $\overline{a}_n \in \mathbb{Z}/n\mathbb{Z}$:

$$\overline{0}_n + \overline{a})n = \overline{0 + a}_n$$

Then notice that

$$a - (0 + a) = 0 = kn \qquad (\text{with } k = 0)$$

Thus, $\overline{0 + a}_n = \overline{a}_n$, and so $\overline{0}_n$ is the identity for any $n \in \mathbb{N}_{>0}$

(Inverse) For any $\overline{a}_n \in \mathbb{Z}/n\mathbb{Z}$, pick $\overline{-a}_n$ (i.e., the equivalence class that can be represented by $-a$). Then:

$$(a + -a) - 0 = 0$$

implying that $\overline{a}_n = \overline{-a}_n = \overline{a + (-a)}_n = \overline{0}_n$, showing that $\overline{a}_n$ is indeed the inverse. Usually, a representative between $0 \leq t \leq n - 1$ so that $\overline{-a}_n = \overline{t}_n$.

(Associaivity) let $\overline{a}_n, \overline{b}_n, \overline{b}_n \in \mathbb{Z}/n\mathbb{Z}$. We want to show that

$$(\overline{a}_n + \overline{b}_n) + \overline{c}_n = \overline{a}_n + (\overline{b}_n + \overline{c}_n)$$

expanding the definition of the binary operation, we get:

$$\overline{(a + b) + c}_n = \overline{a + (b + c)}_n$$

In particular, we want to verify that:

$$[(a + b) + c] - [a + (b + c)] = kn$$

which by the associativity of the integers, we get that their difference is 0, which implies both sides are equal, completing the proof.

Definition 1.2.2: Integers Modulo n

The set $\mathbb{Z}/n\mathbb{Z}$ forms a group under modulo addition called *integers modulo n*.

Hence, for each $n \in \mathbb{N}_{>0}$, there is a group with with n elements. It has yet to be explored whether the modulo multiplication is a viable binary operation, namely if:

$$(a \bmod n) \cdot (b \bmod n) = (ab) \bmod n \tag{1.1}$$

can form a binary operation. Some more care must be put in:

Example 1.2.3: Not Always Group

Take $n = 4$. The reader should verify that $\overline{1}_4$ is the identity given the binary operation defined in equation (1.1), and by proposition 1.1.15(1) it is unique. However, notice that $\overline{2}_4$ has *no* multiplicative inverse:

$$\overline{0 \cdot 2}_4 = \overline{0}_4 \qquad \overline{1 \cdot 2}_4 = \overline{2}_4 \qquad \overline{2 \cdot 2}_4 = \overline{0}_4 \qquad \overline{3 \cdot 2}_4 = \overline{2}_4$$

Hence, $(\mathbb{Z}/n\mathbb{Z}, \cdot_4)$ is *not* a group. However, if we pick $n = 3$, notice that this forms a group!

The reader may have observed that 3 is a prime and may wonder if is the key difference. This is in fact close: the important part is to choose all elements that are *coprime* with n:

Definition 1.2.4: Euler Totient

Define $\varphi : \mathbb{N} \to \mathbb{N}$ to be

$$\varphi(n) := |\{k \in \mathbb{N} \mid \gcd(n, k) = 1\}| = \#\{k \in \mathbb{N} \mid \gcd(n, k) = 1\}$$

that is, $\varphi(n)$ counts the number of coprime

The notation φ is often re-used and so shouldn't be considered a "universal" symbol for the Euler Totient.

Definition 1.2.5: Integer Modulo n With Multiplication

Let $\mathbb{Z}/n\mathbb{Z}$ be the integers modulo n and choose representatives for each element so that the n equivalence classes may be represented with $0, ..., n-1$. Take $(\mathbb{Z}/n\mathbb{Z})^\times \subseteq \mathbb{Z}/n\mathbb{Z}$ defined to be

$$(\mathbb{Z}/n\mathbb{Z})^\times = \left\{\overline{t}_n \in \mathbb{Z}/n\mathbb{Z} \;\middle|\; \varphi(t) = 1\right\}$$

Define the binary operation to be:

$$(a \bmod n) \cdot (b \bmod n) = (ab) \bmod n \tag{1.2}$$

Then $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group modulo n with multiplication.

The reader should verify as an exercise that this is indeed a group.

Cyclic Subgroups

Each group G will have something similar to the integers modulo n, which is worth exploring:

Proposition 1.2.6: Creating a Cyclic Subgroup

Let G be a group and take any $x \in G$. Let $\langle x \rangle = \left\{ x^a \middle| a \in \mathbb{Z} \right\}$. Then $\langle x \rangle$ is a subgroup of G

***Proof*:**
I would recommend you do this as an exercise!

Let $\langle x \rangle = \{x^a | a \in \mathbb{Z}\}$. We'll run through the group-axioms:

1. (associativity) Associativity come's from associativity in G. Let $r, s, t \in \langle x \rangle$, and consider

$$(r + s) + t$$

then since $r, s, t \in G$, and G is associative, we get that

$$(r + s) + t = r + (s + t)$$

and since $r, s, t \in \langle x \rangle$ were arbitrary 3 elements, this holds for any elements of $\langle x \rangle$, completing the proof

2. (identity) We need to show that in $\langle x \rangle$, there exists an element e such that $ex^a = x^a$. Since $0 \in \mathbb{Z}$, $x^0 = e \in \langle x \rangle$. And since for any element $y \in x$, $ey = y$, and every x^a is an element of G, $ex^a = x^a e = x^a$

3. (inverse). Let $y \in \langle x \rangle$ be an element of $\langle x \rangle$. Then by definition, for some $a \in \mathbb{Z}$, $y = x^a$. Then since $-a \in \mathbb{Z}$, $x^{-a} \in \langle x \rangle$, and as we've seen earlier:

$$x^a x^{-a} = x^{a-a} = x^0 = e$$

showing that x^a has an inverse for every a.

Thus, $\langle x \rangle$ is a group. Notice that the binary operation is identical to that of G restricted to $\langle x \rangle$, making $\langle x \rangle$ a subgroup of G, completing the proof.

Given any element $x \in G$, we can form a subgroup $\langle x \rangle \leqslant G$. The reason for it being called a cyclic subgroup comes when we consider a finite group G [12]

Example 1.2.7: Why is $\langle x \rangle$ called Cyclic?

Let G be a finite group (i.e. $|G| = n$ for some $n \in \mathbb{N}$), take $x \in G$ and consider $\langle x \rangle \leqslant G$.

1. Show that $\langle x \rangle$ is finite, that is, there is some $k \in \mathbb{N}$ such that $|\langle x \rangle| = k$

2. Show that for any x^{-a}, where $a \in \mathbb{N}$, there exists some number $b \in \mathbb{N}$ such that

$$x^{-a} = x^b$$

[12]More generally, subgroup $\langle x \rangle \leqslant G$ where $|\langle x \rangle|$ is finite

Conclude that $\langle x\rangle$ can be generated by raising number *only* to the power of positive number, i.e.

$$\{x^a \mid a \in \mathbb{Z}\} = \{x^a \mid a \in \mathbb{N}\}$$

3. show that if $|\langle x\rangle| = k$, then $x^k = e$. Use this fact to show that for any $x^a \in \langle x\rangle$, $a \leq k$.

With these facts, we can visualize a finite cyclic subgroup $\langle x\rangle$ using a graph. Let's say 1 is the identity. If we multiply 1 by x $(1 \cdot x)$, we get x. We can represent this visually with

$$1 \xrightarrow{\cdot x} x$$

if we multiply x by x, we get

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2$$

continuing in this fashion, we will make long "chain":

$$1 \xrightarrow{\cdot x} x \xrightarrow{\cdot x} x^2 \xrightarrow{\cdot x} x^3 \xrightarrow{\cdot x} \cdots$$

until we reach 1, at which point we can circle back to it in the graph:

$$\begin{tikzcd}
1 \arrow[r, "\cdot x"] & x \arrow[r, "\cdot x"] & x^2 \arrow[r, "\cdot x"] & x^3 \arrow[r, "\cdot x"] & \cdots \arrow[r] & x^{k-1} \arrow[lllll, bend right, "\cdot x"']
\end{tikzcd}$$

which more conveniently can be written as

$$\begin{tikzcd}
 & \dots \arrow[dl, "\cdot x"] & x^3 \arrow[l, "\cdot x"] & \\
x^{k-1} \arrow[dr, "\cdot x"] & & & x^2 \arrow[ul, "\cdot x"] \\
 & 1 \arrow[r, "\cdot x"] & x \arrow[ur, "\cdot x"] &
\end{tikzcd}$$

producing us a cycle!

Since every element can form a cyclic group, and their structure is characterized by the cardinality of $\langle x\rangle$, we will give a special name to this cardinality:

Definition 1.2.8: Order of an element

For a group G and $x \in G$, define the *order of* x to be the smallest positive integer n such that $x^n = 1$, and denote this integer by $|x|$. In this use, x is said to be of order n. If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order
Equivalently, we can also say $|x| = |\langle x\rangle|$, that is, the order of x is the order of the group generated by x

Another word for element with finite order is *torsion elements*. Elements that are not torsion elements (i.e. with infinite order) are called *torsion-free elements*.

Example 1.2.9: Order of Elements

1. Every element of $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under $+$ has infinite except 0. Is the same true for $\mathbb{R}^\times$ (i.e. $\mathbb{R} - \{0\}$ (answer: [a]) Note the identity always has order 1.
2. Can you find the order of the elements in $\mathbb{Z}/p\mathbb{Z}$ for prime p? What if p is not prime? What are the order of the elements of $\mathbb{Z}/60\mathbb{Z}$?

[a] no, take $-1 \in \mathbb{R}^\times$

1.2.10 Note The order of the element depends on *both* the set and the binary operator. See exercise ref:HERE
Also, the word *order* is sometimes used to mean the cardinality of the group.

I highly suggest you now do a couple of exercises, since these groups will be at the foundation of finding all possible abelian group [13]

Exercise 1.2.1

1. for $n > 1$, $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication (hint: think of identity)
2. prove that every element in a finite group has finite order
3. if $x \in G$, G a group, and $|x| = n$, then $x^{-1} = x^{n-1}$.
4. $x \in G$, G a group. Then $|x| = |x^{-1}|$. (hint: $x \cdot x \cdots x = 1$, so keep multiplying by x^{-1} till you get $1 = x^{-1} \cdot x^{-1} \cdots x^{-1}$ If x has infinite order). Though in certain infinite order groups, like $\mathbb{R} - \{0\}$ under multiplication, -1 has order 2, but 1 has order 1. Is this only true when the operation is $+$?
5. 21. is there a catch? why should n be odd?
6. $x, g \in G$, G a group. Then $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$. (hint: simply assign an order to x, and expand $g^{-1}xg$. things will cancel out nicely. For the second part, start with ab, and multiply like so $babb^{-1}$. By what you've shown, this has the same order).
7. $x \in G$, $|x| = n < \infty$. If $n = st$ for some positive integers s and t, prove that $|x^s| = t$. (hint: how many times do you need to multiply x^s to get to 1)
8. if $x^2 = 1$ for all $x \in G$ then G is abelian.
9. if x is an element of the group G, then $H = \{x^n | n \in \mathbb{Z}\}$ is a subgroup (cf. show for all $h, k \in H$, $hk \in H$, $h^{-1} \in H$) of G (called the *cyclic subgroup* of G generated by x)
10. $A \times B$ is abelian if and only if A and B is abelian
11. Let $A \times B$ be the *direct product* group formed by the cartesian product of the group A, B as shown in example 1.1.5. Show that $(a, 1)$ and $(1, b)$ commute. Deduce that the order of (a, b)

[13] In particular, finitely generated abelian group, see section ref:HERE

is the least common multiple of $|a|$ and $|b|$.

12. any finite group G of even order contains an element of order 2. (hint: let $t(G)$ be the set $\left\{g \in G | g \neq g^{-1}\right\}$. Show that $t(G)$ has even number of elements and every non-identity element of $G - t(G)$ has order 2).

13. Other exercises you can do I'd recommend: 32, do 33, 34, 35, 36

14. Challenging exercise. Let A be a set containing numbers. Define the following

$$\mathcal{A} = \bigoplus_{i=1}^{\infty} A$$

to be the set of infinitely long tuples were *all but finitely many* elements are non zero. For example, if $A = \{1, 2, 3\}$, then

$$(0, 0, 1, 0, 2, 0, 0, 0, 0, 1, 3, 3, 2, 0, ...)$$

where the ... means there are zeros going on forever. Then this tuples is in $\mathcal{A}$. On the other hand

$$(1, 1, 1, 1, 1, ...)$$

where the ... means the ones are going on forever is not in $\mathcal{A}$.

If G is a group, let

$$\mathcal{G} = \bigoplus_{i=1}^{\infty} G$$

be the set of tuples where all but finitely many elements are non-zero, and define the operation on $\mathcal{G}$ to be pointwise:

$$(a_1, a_2, a_3, ...) \cdot (b_1, b_2, b_3, ...) = (a_1 b_1, a_2 b_2, a_3 b_3, ...)$$

Show that $\mathcal{G}$ along with this operation is a group.

15. Let G be a finite group. Show that G can be generated by elements to the power of $\mathbb{N}$ (move into the generators and relations section maybe?)

16. In some textbooks, groups are introduced more through the use of *graphs*. We will present these ideas here. (Google Frucht's Theorem)

17. Let p be a prime number. Show that $(\mathbb{Z}/p\mathbb{Z})$ is a cyclic group.

18. Let $n = 8$. What is $(\mathbb{Z}/8\mathbb{Z})^\times$? What if $n = 9$. Conclude that in general, $(\mathbb{Z}/n\mathbb{Z})^\times$ is not necessarily cyclic, but it can be.

19. (perhaps a question on the structure of $(\mathbb{Z}/n\mathbb{Z})^\times$ in general?)

1.2.2 Symmetric Groups

Let X be any set. As covered in the preliminary chapter, a bijective function $f : X \to X$ always has a (two-sided) inverse function[14], say f^{-1}. Furthermore, function composition is associative, and since

[14] This in fact characterizes bijective functions

the domain and codomain match, the composition of two bijective functions $f, g : X \to X$ give's another bijective function $g \circ f : X \to X$ (as the reader ought to check if they're not convinced). The final property to check for the collection of bijective functions on a set X to form a group is whether there exists a bijective function $\mathrm{id}_X : X \to X$ such that for all bijective functions $f : X \to X$, $\mathrm{id}_X \circ f = f \circ \mathrm{id}_X = f$. The reader should take a moment to think what this function

1.2.11 Moment Taking a moment

The reader may have noticed that this function arises naturally by taking $f \circ f^{-1}$, namely it's the function defined to be $s \mapsto s$. Thus, given any set X, there is a natural group structure on the collection of bijections on X. This set is given a name:

Definition 1.2.12: Symmetric Group

Let X be any set. Then then:

$$S_X := \{f : X \to X \mid f \text{ is a bijection}\}$$

is called the *symmetric group* or *premutation group* of X [with respect to composition $\circ$]. If the cardinality of X is finite, say $|X| = n$, then we denote S_X by S_n.

Other common notations for the symmetric group are:

$$\mathrm{Bij}(A) \qquad \mathrm{Sym}(A) \qquad \mathrm{Aut}_{\mathbf{Set}}(A)$$

If X is finite, then the bijections S_n are determined only by the place to which elements map to and not by any property of the elements, making the bijections only dependent on the number of elements and hence the S_n notation being unambiguous. For this reason, it is often assumed that the set over which the bijections are being taken is given by[15]:

$$\{1, 2, ..., n\}$$

A mention on notation is in order: if we take $f, g \in S_X$, then the current short-hand notation for groups would suggest that the product of f and g ought to be fg and represents the short-hand $f \circ g$. However, the prevailing notation for compositions reads the functions *right-to-left*:

$$f * g = g \circ f \qquad \Rightarrow \qquad (g \circ f)(x) = g(f(x))$$

This notational quirk comes from the fact that the conventions for function composition comes much earlier than the convention for group operations. Mathematically, this shall not effect the underlying theory[16].

Symmetric groups are rather large; the number of bijections between finite sets grows:

$$|S_n| = n!$$

And hence even for $|X| = 100$, $|S_{100}| \gg 10^{100}$. For $n \in \{1, 2\}$, the bijections are rather simple, namely:

$$f_1 : \{1\} \to \{1\},\ f_1(1) = 1 \qquad g_1 : \{1, 2\} \to \{1, 2\},\ g_1 = \mathrm{id}_{\{1,2\}} \quad g_2(1) = 2\ g_2(2) = 1$$

[15] This set is also sometimes called the *indexing set* and represents labeling each element of some set of size n
[16] It shall complicate notation a little bit when defining left-group actions, which shall be addressed in due time

For larger sets, it may get difficult to track the information a bijection holds. For example, if σ is a bijection on a set with 9 elements:

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

then we after the σ permutes, we might get

$$(\sigma(1), \sigma(2), \sigma(3)\sigma(4), \sigma(5), \sigma(6), \sigma(7), \sigma(8), \sigma(9)) = (2, 4, 1, 3, 7, 9, 8, 5, 6)$$

It is cumbersome to write-out the permutation by saying what each value of $\sigma \in S_n$ maps to. Instead, there is the following *matrix notation* as a shorthand:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 1 & 3 & 7 & 9 & 8 & 5 & 6 \end{pmatrix} \tag{1.3}$$

Representing f_1, g_1, g_2 in matrix notation, we would get:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

This is better notation, however we can do even better taking advantage of the fact that a bijection is one-to-one and onto. This implies that every element is only associated with one element and each element must be represented. This allows for the following “cycle” notation. Take equation 1.3, and notice that 1 maps to 2, which maps to 4, which maps to 3, which maps to 1:

$$1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$$

Furthermore, 5 maps to 7, which maps to 8, which maps to 5:

$$5 \mapsto 8 \mapsto 7 \mapsto 5$$

Finally, 6 maps to 9, and 9 maps to 6:

$$6 \mapsto 9 \mapsto 6$$

This behavior can be represented via the following *cycle notation*:

$$(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)$$

Each “chain” in the above is called a *cycle*, and the combination of each is called the *cycle representaiton* If a cycle contains n elements, it is called an n-cycle. In the above example, there were not any elements that map to itself. If the element 10 is added to the set and we extend σ to $\overline{\sigma}$so that $\overline{\sigma}(10) = 10$, then we would write (1 2 4 3)(5 8 7)(6 9)(10). By convention, any element that maps to itself is dropped in cycle notation, and hence $\overline{\sigma}$ would in cycle notation also be represented as:

$$(1\ 2\ 4\ 3)(5\ 8\ 7)(6\ 9)$$

Hence, there is an inherent ambiguity in cycle notation on the size of the domain/codomain, since any elements that map to themselves is not taken into accost, however it is usually the element that map to a non-trivial element that are of interest and hence this ambiguity does not effect the theory. Note that:

$$(1\,2\,3) = (3\,1\,2) = (2\,3\,1)$$

Since they all actually do the same permutation, you’re just starting at different positions when converting to the cycle notation.

Example 1.2.13: Cycle Representation

1. Let τ be a bijective function on a set $\{1, 2, 3, 4, 5\}$ such that:
$$(\sigma(1), \sigma(2), \sigma(3), \sigma(4), \sigma(5)) = (5, 4, 3, 2, 1)$$
What is the cycle notation for τ
2. Cycles can be used to define permutations. If $X = \{1, 2, 3, 4\}$. Find the bijection represented by the cycle representation:
$$(1\ 2\ 3)(2\ 3\ 4)$$
Notice that the two cycles overlap. To find τ, first find where 1 maps to in the right-cycle, and the where the resulting element maps to the left cycle; in function notation if f is the left cycle and g is the right cycle then the reader is looking for $f(g(1))$. Proceed with this for all 4 element. Remember that if an element is not in a cycle, then it is considered to map to itself.
3. Is it true that the functions represented by $(1\,2)(3\,4)$ and $(3\,4)(1\,2)$ are equal, that is:
$$(1\ 2)(3\ 4) = (3\ 4)(1\ 2)$$
More generally, do disjoint cycles commute?

To insure the reader has the right calculation, here is an example of the product of two cycles as reference:
$$(1\ 4\ 5\ 2)(4\ 1\ 3) = (1\ 3\ 5\ 2)$$

With those extra properties of cycles, we will name one more concept for ease of labeling cycles:

Definition 1.2.14: Cycle Type

Given a cycle-representation of a permutation, you can make the cycles disjoint. Then, order them from smallest to largest. Then the cycle representation is the "length" of each cycle in that order.

Example 1.2.15: Cycle-Type

$(1\,2\,4\,3)(5\,7\,8)(6\,9)$ is equivalent to $(6\,9)(5\,7\,8)(1\,2\,4\,3)$ which we can label as a $(2, 3, 4)$-cycle type. If you have a $(1\,2)(3\,4)$ premutation, then it's a $(2, 2)$-cycle type.

A cycle of particular importance are 2-cycles.

Definition 1.2.16: Transposition

2-cycles are usually referred to as ***transpositions***

Using transpositions, any n-cycle can be decomposed into the product of transpositions. For example, take:
$$(1\,2\,4\,3)$$

Then the reader should verify that:

$$(1\,3)(1\,4)(1\,2) = (1\,2\,4\,3)$$

So, the cycle broke up into 3 transpositions. In fact, any n-cycles will break up into $n-1$ transpositions. It will break up to exactly (at most and at least) $n-1$ transpositions.

Theorem 1.2.17: n-Cycle Breakdown To Exactly $n-1$ Transpositions

Let $\sigma \in S_n$ be an n-cycle. Then σ can be broken down to the composition of exactly $n-1$ transpositions

***Proof*:**
exercise

Because of this, given any σ, which can then be written as disjoint cycles, and then into transpositions will always have a fixed number of transpositions in the product, in other words this is an invariant of a cycle σ, which deserves then a name:

Definition 1.2.18: Parity Of Cycle

Given $\sigma \in S_n$, then the parity of the cycle is the parity of the number of transpositions so if we have an n-cycle, where n is even, then it's parity is odd, since it will break down to $n-1$ transpositions. A cycle with even parity is called an *even permutation*, similarly called *odd-permutation* when the parity is odd.

Parity might seem like a weird thing to care about, but it will in fact come in very useful. So useful in fact, that there's a special name for groups based around parity:

Definition 1.2.19: Alternating Group

Let S_n be the permutation group permuting n Elements. Then $A_n \subseteq S_n$ is all cycles with *even* permutations. Note that the compositions of two cycles from A_n is even, and that the inverse of an even permutation is even, thus

$$A_n \leqslant S_n$$

Why this group is interesting requires more algebraic tools, so we'll leave at defining it for now and get back to it to show it's use.

Before moving on to properties of symmetric groups, some reader may find it interesting that at the origins of group theory, groups were defined to be subgroups of symmetric groups! Later, ________and ________have abstracted these properties and formed the modern axiomatic representation. Later in this book, it shall be shown that all groups are in fact subgroups of the appropriate symmetric group (see theorem 4.5.3).

Example 1.2.20: Interesting Exercises on Commutativity

We earlier mentioned how being commutative is a nice property, and that groups which are commutative are quite nice. It was also mentioned that S_n is in a sense the most general group that possible. So, in a sense, if we can establish properties of S_n for particular orders, that can be quite insightful.

Here's one such property worth proving. If we take S_n, $n \geq 3$, then S_n is *not* commutative To prove this, try composing a n-cycles and a 2-cycle which are not disjoint form the left and right and see what you get

Properties of Symmetric Groups

Proposition 1.2.21: Order of element is p iff composition of disjoint p-cycles

Let p be a prime. Show that an element of S_n has order p if and only if its cycle decomposition is a product of (commuting) p-cycles.

***Proof*:**
exercise

Note that if p is not prime, the "if and only if" is not necessarily true.

Example 1.2.22: Counter-Examples

1. Take
$$s = (1\,2)(3\,4\,5\,6) = s_1 s_2$$
Then $(s_1 s_2)^4 = 1$, but $|s_1| = 2$.
2. Take $(1\,2)(3\,4\,5) \in S_5$. Then
$$\begin{aligned}(1\,2)(3\,4\,5)^2 &= (3\,5\,4)\\(1\,2)(3\,4\,5)^3 &= (1\,2)\\(1\,2)(3\,4\,5)^4 &= (3\,4\,5)\\(1\,2)(3\,4\,5)^5 &= (1\,2)(3\,5\,4)\\(1\,2)(3\,4\,5)^6 &= id\end{aligned}$$
but $6 > 5$. Thus, we found a cycle of order 6 in S_5. The trick was to find two disjoint cycles such that the $\text{lcm}(\sigma, \tau) > n$.

Proposition 1.2.23: σ is p-cycle for prime p then σ^k is p-cycle

If $\sigma \in S_n$ has prime order, then σ^k for every $1 \leq k \leq p$ is a p-cycle

***Proof*:**
This is a simple application of proposition 1.2.21, and since we're taking $\langle\sigma\rangle$, we'll use theorem 2.3.11 (the theorem on cyclic groups). Since $|\langle\sigma\rangle| = p$, then $\langle\sigma^k\rangle = p$ for every $k < p$ ($\sigma^p = e$). This means that σ^k has prime order. Thus, by proposition 1.2.21, it is composed of p-cycles. Since $\langle\sigma\rangle = p$, and σ is a bijection (meaning we never go to another element while composing σ), then every σ^k is a p-cycle

Note that if σ is not a p-cycle, this is not necessarily true. For example:

$$(1\,2\,3\,4)\,,(1\,2\,3\,4)^2 = (1\,3)(2\,4)\,,(1\,2\,3\,4) = (1\,4\,3\,2)$$

Notice how σ^2 broke down to two 2-cycles. The way I remember this to recall that if σ is a 4-cycle, then $|\sigma^2| = 2$, so $\sigma^2\sigma^2 = id$, which by what we've covered in the previous proposition (1.2.21), σ must be disjoint 2-cycles!! This holds true for any n-cycle, giving you an easy way of thinking about σ^k for any n-cycle with $1 \leq k \leq n$.

Proposition 1.2.24: Conjugation preserves permutation-type

If $\sigma, \tau \in S_n$, then

$$\sigma\tau\sigma^{-1} = (\sigma(1)\,\sigma(2)\,\cdots\sigma(t_k))$$

i.e., it's like applying the function σ to every element of τ

***Proof*:**
This proof relies on this key observation:

$$(a\,b)(a\,c)(a\,b)^{-1} = (a\,b)(a\,c)(a\,b) = (b\,c)$$

that is, we shifted a to b (so in a sense if $\sigma = (a\,b)$ and $\tau = (a\,c)$, $\sigma\tau\sigma^{-1} = (\sigma(a), \sigma(b))$).

The rest of this proof is now a matter of keeping track of notation [a]

[a] This proof is easier prove once your introduced to group action and conjugacy classes (see definition 4.6.1)

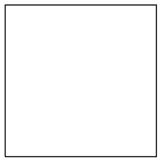
1.2.3 Dihedral Groups

In classical geometry, an n-gon is a 2D shape with all sides of the same length. For example, a triangle is a 3-gon, a square is a 4-gon, a pentagon is a 5-gon, and so forth, while a rectangle and a circle are not n-gons. Each of these shapes can be rotated or reflected in such a way that we get back the same shape: these are called the *rigid-motion symmetries* or just *symmetries* of the n-gon. The term *rigid motion* comes from analysis/linear-algebra where a function $f : X \to X$ is called a *rigid motion* has the property that the distance between each point remains the same after the application of the function. Notationally, this would be represented as:

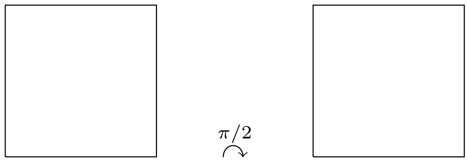
$$\|f(x) - f(y)\| = \|x - y\|$$

where the $\| - \| : X \to \mathbb{R}_{>0}$ is a special kind of function called the *norm* which defines size and distance information on a space[17]. We shall not use rigid motions explicitly, however we shall insure that when transforming n-gons, their they shall never stretch (ex. a square shall not become a rectangle). We shall further place the restriction that the rigid motions of the n-gons have to map to themselves, this is another way of saying that it is a *symmetry*. For example, if we choose a square (a 4-gon):

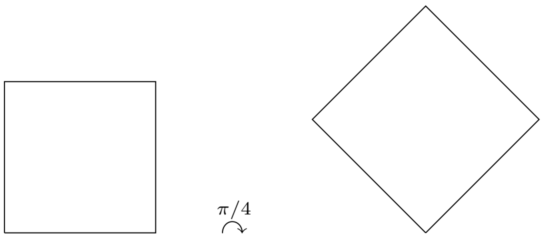
[17] Norms are introduced formally in *Linear Algebra* [12].



and rotated it a quarter turn to the right (90 degrees to the right or $\pi/2$ radians to the right), then we'd get back a square



if we were to rotate it an eighth of a turn, we would not get the exact shape back:



Let's label quarter turns to the right r (for **r**otate). Compounding rotations, the resulting square get's rotated by π, $3\pi/2$, and 2π radians; each of these can be labeled r^2, r^3, and r^4 respectively. Another symmetry of the square would be reflections, namely there is a vertical, horizontal, and two diagonal reflections: $\sigma_h, \sigma_v, \sigma_{d1}, \sigma_{d_2}$. These are not all the symmetries of a square (for example, we transpose the left-edge with the right edge), however it is the only symmetries that are also *rigid motions* (the reader who is already comfortable with norms form linear algebra is free to prove this fact).

More formally, we may represent any n-gon with n-points, with the restrictions that any symmetry (i.e. bijection) f applied to these n-points must further follow the restrictions that the n-gon these points represent must transform as a rigid-motion (all points must remain equi-distant). This special subset of S_n with this added restriciton is called the *dihedral group*:

Definition 1.2.25: Dihedral Groups

Let S_n be the symmetric group on n elements and take the subset $D_n \subseteq S_n$ for which the premutations further follow the restriction of rigid-motions (so that these points represent an n-gon).Then D_n is called the *dihedral group*.

In order for the set D_n to be a group, one ought to prove this is indeed the csae

***Proof*:**

1. Identity comes from the identity function $\mathrm{id}_n \in D_n$ since it is certainly a rigid motion.
2. The inverse come from the fact that if σ is a rigid motion, it's inverse is as well

3. The set is associative since its a collection of functions

4. The composition of two rigid motions is a rigid motion, and hence the binary operation is well-defined

Proposition 1.2.26: Properties of D_n

Let $G = D_n$. Let f represent rotation by by $2\pi/n$, and s represent a reflection about some fixed axis. Then:

1. $1, r, \ldots, r^{n-1}$ are all distinct and $r^n = 1$, so $|r| = n$

2. $|s| = 2$

3. $s \neq r^i$ for any i

4. $sr^i \neq sr^j$ for all $0 \leq i, j \leq n-1$ with $i \neq j$, so

$$D_n = \left\{1, r, r^2, \cdots, r^{n-1}, , sr, sr^2, \cdots, sr^{n-1}\right\}$$

   i.e. each element can be written uniquely in the form $s^k r^i$ for some $s = 0$ or 1, and $r = 0, \cdots, n-1$

5. $rs = sr^{-1}$. In particular this shows D_n is not abelian!

6. $r^i s = sr^{-i}$ (use induction)

***Proof*:**
exercise

In particular, we know now any element in D_n can be written as $s^k r^i$, with $k = 0 \vee 1$ and $0 \leq i \leq n-1$. For example, if $n = 12$:

$$(sr^9)(sr^6) = sr^9 sr^6 = sr^9 r^{-6} s = sr^3 s = s^2 r^{-3} = r^{-3} = r^9$$

This group will come up very often since it's a particularly simple.

Proposition 1.2.27: Order of Dihedral Groupq

Let D_n be a dihedral group. Then:

$$|D_n| = 2n$$

***Proof*:**
exercise.

Some authors like to put the order of the Dihedral group in the subscript, that is they write D_{2n}. This convention is rather confusing given the original intuition behind Dihedral groups, and so the D_n notation is used instead for the clarity it brings.

1.2.4 Quaternion Group

The Quaternion (or Hamiltonian) group is a group with 2 fundamental interpretation:

1. The Quaternions are a generalization of the complex numbers! We can think of the fact that $i^2 = -1$ to be the key property which distinguishes the complex numbers from other number systems. If we take $C = \{-1, 1, i, -i\} \subseteq \mathbb{C}$, then

$$\langle C | i^2 = -1 \rangle$$

   will form the group which represents the rotation of $\pi/2$ in the cartesian plane. William Hamilton around the 1840s started a project to try an make 3D numbers $(1, i, j)$, the same way $\mathbb{C}$ can be thought of as 2D $(1, i)$. He couldn't manage to make them 3D, but realized the key generalization is making them "4D" $(1, i, j, k)$. He had this flash of inspiration near a bridge in 1843 and wrote down the idea under that very bridge. Today, there is a plaque commemorating the moment of inspiration.

2. It got discovered later that Quaternions efficiently capture the idea of rotating a 3D object in 3D space. At the beginning of vector calculus, some mathematician considered using quaternion instead of vectors as the pedagogical tool used to introduce the concept of 3D rotation, but history went with vector notation. With leaps in quantum mechanics in the late 20th century, quaternions were also used to represent *spin*, a fundamental property of sub-atomic particles.

Let:

$$Q = \{1, -1, i, -i, j, -j, k, -k\}$$

Such that the following relations hold:

$$i^2 = j^2 = k^2 = -1 \tag{1.4}$$
$$1 \cdot a = 1 \cdot a \qquad \forall a \in Q_8 \tag{1.5}$$
$$ij = k \qquad ji = -k \tag{1.6}$$
$$ik = -j \qquad ki = j \tag{1.7}$$
$$jk = i \qquad kj = -i \tag{1.8}$$

Or, in terms of group presentation:

$$Q_8 = \langle Q | i^2 = j^2 = k^2 = ijk = -1 \rangle$$

This group is interesting, for it's the 2nd non-abelian group you've seen with order 8, D_8 being the other group. Furthermore, both of them share a similar intuitive background, being the based on rotation. However, Q_8 rotates 3D objects, while D_8 rotates 2d objects, which actually produce different group structures, a these Cayley tables show:

Example 1.2.28: Comparing D_8 and Q_8

Element	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	$-j$	j
$-i$	$-i$	i	1	-1	$-k$	k	j	$-j$
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Figure 1.1: Q_8 Multiplication Table

While D_8 has

Element	e	a	a^2	a^3	x	ax	a^2x	a^3x
e	e	a	a^2	a^3	x	ax	a^2x	a^3x
a	a	a^2	a^3	e	ax	a^2x	a^3x	x
a^2	a^2	a^3	e	a	a^2x	a^3x	x	ax
a^3	a^3	e	a	a^2	a^3x	x	ax	a^2x
x	x	a^3x	a^2x	ax	e	a^3	a^2	a
ax	ax	x	a^3x	a^2x	a	e	a^3	a^2
a^2x	a^2x	ax	x	a^3x	a^2	a	e	a^3
a^3x	a^3x	a^2x	ax	x	a^3	a^2	a	e

Figure 1.2: Q_8 Multiplication Table

which the reader will notice is different

We will actually be able to show that two groups are not the same once we introduce the concept of isomorphisms (section 1.4).

Exercise 1.2.2

1. Show that there exists 4 homomorphisms $\varphi : Q_8 \to \mathbb{R}^\times$

1.2.5⋆ Matrix Groups

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

This section assumes either some linear algebra knowledge or that you have internalized some of section 0.5 (however, some of these collection of objects goes beyond that section). All prerequisite knowledge assumed in this section shall be covered in great detail in chapter 13 and section 13.2; for the inner-product background behind the orthogonal and unitary groups, see *Linear Algebra* [12]. Let's first recall the notation we use to describe the collection of matrices:

Definition 1.2.29: Matrix Group Under Addition

Let G be an abelian group. Then The set $M_{n\times m}(\mathbb{F})$ forms a group under element-wise operation of G, namely:

$$A * B \equiv (a_{ij} * b_{ij})_{ij}$$

If the reader is familiar with linear algebra, they may notice this group is not actually that interesting: it is equivalent to the group G^{nm}[18]. Sticking to $G = k$ is a field (notice that all fields are Abelian groups under addition), the more interesting binary operation that can be performed on matrices is matrix multiplication, since it combines elements in much more interesting ways. As you might recall, matrix multiplication requires dimensions to line up properly, so for associative to hold, we'll have the dimensions be the same. Furthermore, the identity must be I, the matrix with 1's on the diagonal, and zero's everywhere else. If there are matrices that multiply and *don't* get to I, then they cannot be in the group. Hence, the subset of invertible matrices from $M_n(k)$ must be selected in order to make a group, which the reader may recall from linear algebra implies we are taking the set for which all matrices have a non-zero determinant

Definition 1.2.30: General Linear Group

$$GL_n(F) = (\{A|A \text{ is an } n\times n \text{ matrix with entries from } F \text{ and } \det(A) \neq 0)\}$$

We will also name 3 special subgroups of GL_n since it comes up often enough. Check that these are indeed groups:

Definition 1.2.31: Special Linear Group

$$SL_n(k) = (\{A|A \text{ is an } n\times n \text{ matrix with entries from } F \text{ and } \det(A) = 1)\}$$

If $k \in \{\mathbb{R}, \mathbb{C}\}$, then geometrically these can be thought of as the volume and orientation preserving functions.

Definition 1.2.32: Orthogonal Group

$$O_n(\mathbb{F}) = \big(\{A|A^tA = AA^t = I\big)\}$$

In this group, the transformations are *distance-preserving*, which as an exercise the reader can check that for $O_2(\mathbb{R})$, all transformations are rotations and reflections across lines passing through the

[18]We can actually show this once we get to isomorphisms (definition 1.4.12)

origin[19]. Limiting to orthogonal matrices with determinant 1 gives us the set of all rotations:

Definition 1.2.33: Special Orthogonal Group

$$SO_n(\mathbb{F}) = \{A | A^t A = AA^t = I \,, \det(A) = 1\}$$

If the underlying field is clear, we sometimes write $SO(n)$.

Exercise 1.2.3

1. 6, 7. 11 If you feel like getting comfortable with Heisenberg groups
2. (If you know the dot product) Recall that if v, w are vectors then the dot product can be represented as $v \cdot w = v^T w$ where v^T is the transpose. Show that matrices in $O(\mathbb{R})$ preserve the dot product (namely $Av \cdot Aw = v \cdot w$).

Proposition 1.2.34: Order of $GL_n(\mathbb{F}_p)$

Let $G = GL_n(\mathbb{F}_p)$ where $\mathbb{F}_p$ is a finite field (and thus, has prime order). Then

$$|GL_n(\mathbb{F}_p)| = \prod_{i=0}^{n-1} (p^n - p^i) = (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1})$$

***Proof*:**
The proof for this comes down to remembering that $GL_n(\mathbb{F}_p)$ has linearly independent columns since for any $A \in GL_n(\mathbb{F}_p)$ $\det(A) \neq 0$. We will first show it for $n = 2$. We can separate it in cases where matrices are definitely invertible, and matrices which might not might be invertible. Matrices which only have 1 entry are always non-invertible. These type of matrices are always invertible (since their determinant is always non-zero):

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \quad \begin{pmatrix} a & b \\ c & 0 \end{pmatrix} \quad \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ b & c \end{pmatrix} \quad \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$$

for $a, b, c > 0$ However

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Is not invertible if the first column is a scalar multiple of the second.

There are a total of $(p-1)^2$ types of unique columns (corresponding to the $p-1$ possibilities per element in the first column). Since the scalars are in $\mathbb{Z}/p\mathbb{Z}$, then there is $p-1$ unique scalar multiples of any row. Since there are $(p-1)^4$ total non-zero matrices, then we have

$$(p-1)^4 - (p-1)^3$$

invertible matrices with all non-zero entries.

[19] Higher dimensional analogues of "rotation" and "reflection" come from taking this interpretation in lower dimensions and claiming it generalizes to higher dimensions

What's left is to calculate the other types of matrices which are

$$2(p-1)^2$$
$$4(p-1)^3$$

So we have a total of

$$(p-1)^4 - (p-1)^3 + 2(p-1)^2 + 4(p-1)^3$$

possible invertible matrices.

TBD

Exercise 1.2.4

1. Show that there is a bijective correspondence between the platonic solids and the finite subgroups of $SO(3)$ that are not cyclic or dihedral, hence the platonic solids is a special case of the classification "non-trivial" finite subgroups of $SO(3)$, where non-trivial here is a bit more than just the trivial subgroup.

1.3 Generators and Relations

(When dealing with numbers, they can represent anything, for example 5 apples, 5 cars, 5 textbooks, and so forth. However, when we defined the dihedral group. it was specifically linked to the rotation and reflections of n-gons. In this section, we will show a way to abstract away this dependency on a concrete object to define a group).

Notice that D_n can be written with r and s (reflections and rotations). Similarly, $\langle x \rangle \leqslant G$ is an element from which we can find every other element of $\langle x \rangle$. In a sense, they *generated* the group. This notion will be treated rigorously with free groups later (section 2.3.3), but intuitively this idea makes sense. For now, here how we'll define them

Definition 1.3.1: Generators

A subset S of elements of a group G with the property that every element of G can be written as a (finite) product of elements of S and their inverses is called the set of *generators* of G. If $x \subseteq G$ generates x, then we write $\langle x \rangle$. Similarly, if $x, y \in G$ are needed to generate G, we write $\langle x, y \rangle$. More generally, if $A \subseteq G$ generates G we write $\langle A \rangle$.

Example 1.3.2: Generators

1. An easy example would be 1 for $\mathbb{Z}$, since every $x \in \mathbb{Z}$ could be written as adding 1 or it's inverse. So $\mathbb{Z} = \langle 1 \rangle$.

2. By property (4) for D_n, $S = \{r, s\}$ is a set of generators of D_n, so $D_n = \langle r, s \rangle$.

Later, we'll see for finite groups that we don't need the inverse operation to generate finite groups.

For any set of generators, we have a set of relationships that they satisfy[20]. For D_n, they are

$$r^n = 1, \qquad s^2 = 1, \qquad rs = sr^{-1}$$

Why are these the relations? Because any additional relation between elements of groups may be derived from these three (p.40). Note that there can be more than one set of relations for a group (we'll see later that D_n has another set of relations that can be used).

Definition 1.3.3: Presentation of a group

if some group G is generated by a subset S and there is some collection of relations, say $R_1, \cdots, R_m$, which are all equations with elements from $S \cup \{1\}$, such that any relation among the elements of S an be deduced from these, we shall call these generators and relations a *presentation* of G, and write

$$G = \langle S | R_1, \cdots, R_m \rangle$$

Example 1.3.4: Presentation

1. Show that the presentation:

$$\langle r, s | r^n = 1, s^2 = 1, rs = sr^{-1} \rangle$$

represents the Dihedral group D_n

2. Let A be any set. Show that

$$\langle A | ab = 1, \forall a, b \in A \rangle$$

is group. In particular, show that from any set, we can define the trivial group. In section 1.2.2 and 2.3.3, we'll show two other (those times non-trivial) examples of creating groups from arbitrary sets.

1.3.5 Prelude to Word Problem the right-hand side is, in the general case, a collection of arbitrary equations. It may be hard to determine when two elements in the group are equal! it may also be hard to determine the order of the group. For example

$$\langle x, y | x^2 = y^2 = (xy)^2 = 1 \rangle$$

is a presentation of a group of order 4, whereas

$$\langle x, y | x^3 = y^3 = (xy)^3 = 1 \rangle$$

is a presentation of an infinite group (exercise)

It might be tempting to only work with presentations, or to try and come up with every possible group working with generators. A considerable amount of time has been poured into this problem, and absolutely incredibly, it turns out that you can come up with presentations which are *insoluble*:

[20]With only one exception. When a set of generators has no relationships, we call the group generated by that set a *free group*

Example 1.3.6: Insoluble Presentation
The group representation by this monstrous presentation:

$$\left\langle \begin{array}{lll} a,b,c,d,e,p,q,r,t,k & | & \\ p^{10}a = ap, & pacqr = rpcaq, & ra = ar, \\ p^{10}b = bp, & p^2adq^2r = rp^2daq^2, & rb = br, \\ p^{10}c = cp, & p^3bcq^3r = rp^3cbq^3, & rc = cr, \\ p^{10}d = dp, & p^4bdq^4r = rp^4dbq^4, & rd = dr, \\ p^{10}e = ep, & p^5ceq^5r = rp^5ecaq^5, & re = er, \\ aq^{10} = qa, & p^6deq^6r = rp^6edbq^6, & pt = tp, \\ bq^{10} = qb, & p^7cdcq^7r = rp^7cdceq^7, & qt = tq, \\ cq^{10} = qc, & p^8ca^3q^8r = rp^8a^3q^8, & \\ dq^{10} = qd, & p^9da^3q^9r = rp^9a^3q^9, & \\ eq^{10} = qe, & a^{-3}ta^3k = ka^{-3}ta^3 & \end{array} \right\rangle$$

Figure 1.3: Group for which we cannot write out what the elements are

has elements which we *cannot* write out!

The details of this is are well beyond this course, but it's existence is still interesting to know about

https://en.wikipedia.org/wiki/Word_problem_for_groups

You can get familiar with presentations by doing exercises in the book (p.41,42).

Proposition 1.3.7: Generators For Symmetric Group

$$S_n = \langle (i, i+1) \mid 1 \leq i \leq n \rangle$$

***Proof*:**
This proof relies on this key observation:

$$(a\,b)(a\,c)(a\,b)^{-1} = (a\,b)(a\,c)(a\,b) = (b\,c)$$

that is, we shifted a to b (so in a sense if $\sigma = (a\,b)$ and $\tau = (a\,c)$, $\sigma\tau\sigma^{-1} = (\sigma(a), \sigma(b))$).
With that in mind, we'll proceed by *induction*. If $n = 2$, then S_2 is generated by this set, so assume it generate S_{n-1} and consider S_n. Take some cycle with n in it:

$$(a_1\,a_2\,...\,a_k\,n)$$

which we've shown earlier can be decomposed as

$$(a_1\,a_2)(a_1\,a_3)\cdots(a_1\,a_k)$$

All the $(a_1\,a_j)$ are in G by the induction hypothesis (they are not in because of the generating set because we don't know if $(a_1\,a_j) = (i\,i+1)$). What's left to show is whether $(a_1\,n)$ is in G. To do this, we use the conjugation fact we brought up earlier:

$$(a_1\,n) = (a_1\,n-1)(n-1\,n)(a_1\,n-1)$$

which since $(n-1\,n) \in S$, and the other two terms are in G by induction hypothesis, it means that $(a_1\,n) \in G$. And Since any cycle can be broken down into the aforementioned form (where if we have a cycle, we can move k to the back, since its the same cycle that represents it), any cycle can be broken down in this fashion, and thus, we completed the proof

As a bonus, if you want to represent such a group:

$$\langle \sigma_i | \sigma_i^2 = 1\,, \sigma_i \sigma j = \sigma_j \sigma i \text{ for } |i-j| > 1\,, (\sigma_i, \sigma_{i+1})^3 = 1 \rangle$$

Though this is not necessary to know, but it can be good to know if you want to show something to be isomorphic to the symmetric group (since if something is isomorphic, it has the same relations).

https://en.wikipedia.org/wiki/Symmetric_group#Generators_and_relations.

There is a way to prove these once we introduce either quotient groups or conjugacy classes (in the group-action section). Thus, we won't prove these here for the nicest way to prove them is once we have those extra tools.

Exercise 1.3.1

1. exercise 7, 16, simply to know how to solve a concrete problem.
2. exercise ,17,18.
3. Do 10 to see that you can find all symmetries
4. Take $(1, i)$ for $2 \leq i \leq n$, that is, the set where you fix one element, then the 2nd position is all other elements.
5. This one is probably the most useful theoretically. You can actually generate a symmetric group given only 2 cycles!! Take any n-cycles from S_n, and a 2-cycle of adjacent elements in the n-cycle!

1.4 Homomorphisms and Isomorphisms

When comparing D_8 and Q_8 in example 1.2.28, we resolved at looking at the Cayley tables to distinguish the two. However, this becomes very tedious to do for larger groups, and downright impossible for infinite groups. In this section, we'll make precise the notion of comparing groups.

What does it mean to compare groups? If we were comparing tuples, $(a, b) \neq (b, a)$, since the order matters. On the other hand, if we're comparing sets, then $\{a, b\} = \{b, a\}$ since the order doesn't matter, as long as the elements are the same. For groups, the star of the show is the binary operation. If two groups G and H shared *exactly* the same binary operation, they'd be identical in terms of the binary operation. We often encounter groups with different elements, but the binary operation acts identically in both groups.

Example 1.4.1: "Identical" groups

Let $G = \{0, 1, 2\}$ and $H = \{a, b, c\}$, and define $(G, +_G)$ and $(H, +_H)$ with binary operation's which have Cayley tables:

$+_G$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

and

$+_H$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

Then notice we can replace 0 with a, 1 with b, and 2 with c, and the two tables would be identical. In other words, the operation $+_H$ acts exactly the same as $+_G$.

It is sometimes restrictive that the two binary operations act *exactly* the same. sometimes that are similar, but the two groups share a different structure. The take-away is that when working with homomorphisms and isomorphissm, what we care about is the binary operation between two groups behave similarly.

1.4.1 Homomorphisms

Definition 1.4.2: Homomorphism

Let $(G, \cdot)$, and $(H, \bullet)$ be groups. A map $\psi : G \to H$ such that for all $x, y \in G$,

$$\psi(x \cdot y) = \psi(x) \bullet \psi(y)$$

is called a *homomorphism*

Remember that the left side is the domains operation, and the right side is the co-domains operation. A good way of remembering the rule of a homomorphism is that the resulting image $\varphi(G) = \{\varphi(g) \mid g \in G\}$ is also a group (this is proved in proposition 1.4.4). Another way remembering the rule of a homomorphism is that both if there is a homomorphism, then in some way the binary operation must be *preserved*. It is not necessarily perfectly preserved – it's possible that only some of it is preserved. The following examples of homomorphisms may help in gaining an intuition

Example 1.4.3: Homomorphisms

1. Take the integers with the usual addition. Let e represent even integers; o represent odd integers. Then we know that adding even with even is even, even with odd is odd, odd with even is odd, and odd with odd is even:

$$e + e = e,\ o + o = e,\ o + e = e + o = o$$

On the other hand, if we take 1 and -1 and find all possible outcomes of multiplication, we get

$$1 \cdot 1 = 1, \quad (-1) \cdot (-1) = 1, \quad 1 \cdot (-1) = (-1) \cdot 1 = -1$$

Notice how 1 can represent even integers, and -1 represent odd. In this way, multiplication on -1 and 1 "captures" the idea of adding even and odd numbers.

Let's formalize this idea by making an explicit homomorphism. First, $\{-1, 1\}$ is a group under usual multiplication $\cdot$. The element 1 is the identity since $1 \cdot 1 = 1$, $-1 \cdot 1 = -1$. Furthermore -1 it's own inverse ($-1 \cdot -1 = 1$). For associativity, most cases are essentially immediate to check, with the following case being the non-trivial case:

$$(-1 \cdot -1) \cdot -1 = 1 \cdot -1 = -1 \cdot 1 = -1 \cdot (-1 \cdot -1)$$

Any case with 1 in the question be more easily checked since $a \cdot 1 = 1 \cdot a = 1$, making the cases of $1 \cdot -1 \cdot -1$, $-1 \cdot 1 \cdot -1$, $1 \cdot 1 \cdot -1$, and so forth easy to check. Thus, $\{-1, 1\}$ is a group.

Next, let $\psi : \mathbb{Z} \to \{-1, 1\}$ be a function such that

$$\psi(x) = \begin{cases} 1 & x \text{ is even} \\ -1 & x \text{ is odd} \end{cases}$$

Then if e represents an arbitrary even number and o an arbitrary odd number:

$$\begin{gathered} \psi(e + e) = \psi(e) = 1 = 1 \cdot 1 = \psi(e) \cdot \psi(e) \\ \psi(o + o) = \psi(o) = 1 = (-1) \cdot (-1) = \psi(o) \cdot \psi(o) \\ \psi(o + e) = \psi(e + o) = \psi(o) = -1 = -1 \cdot 1 = 1 \cdot -1 = \psi(e) \cdot \psi(o) = \psi(o) \cdot \psi(e) \end{gathered}$$

Showing that ψ is indeed a homomorphism.

Generalizing the previous result, notice that our group $\{-1, 1\}$ is homomorphic to $\mathbb{Z}/2\mathbb{Z}$ (remember that by default, if we say $\mathbb{Z}/2\mathbb{Z}$ is a group, we mean $(\mathbb{Z}/2\mathbb{Z}, +)$ with modulo 2 addition). In particular

$$f : \{-1, 1\} \to \mathbb{Z}/2\mathbb{Z} \qquad f(1) = 0, \ f(-1) = 1$$

Then one should verify that that f is indeed a homomorphism. It is further a bijection, meaning that every element of the group $\mathbb{Z}/2\mathbb{Z}$ represents exactly one element from $\{-1, 1\}$, showing we can inter-change the two.

Now, take $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $\varphi(x) = [x]$. Then φ is a homomorphism. Indeed:

$$\begin{aligned} \varphi(x + y) &= [x + y] \\ &= [x] + [y] && \text{property of } \mathbb{Z}/n\mathbb{Z} \\ &= \varphi(x) + \varphi(y) \end{aligned}$$

If $n = 2$, then this will be exactly the homomorphism we defined earlier.

2. If you have done any calculus or analysis, you might have seen before that:

$$\frac{d}{dx}(f + g) = \frac{d}{dx}(f) + \frac{d}{dx}(g) \qquad \int f + g\, dx = \int f, dx + \int g\, dx$$

so $\frac{d}{dx}$ and $\int$ both satisfy this condition! Are they homomorphisms? In fact, they are, with the group being all differentiable functions for the 1st one, and all integrable functions for the second one. These groups require more build-up within Calculus or Analysis, and so we will not cover these operation in more detail. (More words here! Feels incomplete)[a].

3. For those with knowledge of the exponential function e^z (which we'll write as exp), then there is a natural way of relating the addition and multiplication operation of $\mathbb{C}$ and $\mathbb{C}^\times$ (as well as $\mathbb{R}$ and $\mathbb{R}^\times$). The map $\exp : \mathbb{C} \to \mathbb{C}^\times$ is surjective group homomorphism. Restricting the domain and co-domain, we also get an injective (but not surjective) map $\exp : \mathbb{R} \to \mathbb{R}^\times$.

4. Verify that there is no bijective homomorphism from D_8 to Q_8. At the moment, this may be a tedious task, it will become easier to show later with some more properties of bijective homomorphisms.

5. Let $G_1 \times G_2$ be the direct product of two groups G_1 and G_2 as defined in example 1.1.5(4). The map $\pi_1 : G_1 \times G_2 \to G_1$ and $\pi_2 : G_1 \times G_2 \to G_2$ defined by

$$\pi_1((g_1, g_2)) = g_1 \qquad \pi_2((g_1, g_2)) = g_2$$

are homomorphisms.

6. Let G, H be any group. Then $\varphi : G \to H$ defined by $\varphi(g) = e_H$ where e_H is the identity of H is a homomorphism. This is called the *trivial homomorphism*.

7. (If you know about $\mathbb{C}$) Show that the subset $\{1, i, -1, -i\} \subseteq \mathbb{C}$ is a subgroup of $\mathbb{C}$ and that there is a bijective group homomorphism $\varphi : \mathbb{Z}/4\mathbb{Z} \to \{1, i, -1, -i\}$.

[a]However, $\frac{d}{dx}(fg)$ is not necessarily equal to $\left(\frac{d}{dx}(f)\right)\left(\frac{d}{dx}(g)\right)$ and $\int fg\,dx$ is not necessarily equal to $\left(\int f\,dx\right)\left(\int g\,dx\right)$. The first is studied in a course on Lie Groups and Lie Algebras, while the other is studied in a course in Measure Theory, in particular L^p spaces and Hölder's inequality

We will often simplify notation since lots of information is usually deduced from context. We will drop the binary operation and simply write $\varphi(ab) = \varphi(a)\varphi(b)$, given that it's clear that when we write ab the binary operation happens in the domain, and that when we werite $\varphi(a)\varphi(b)$ the binary operation happens in the codomain.

One more interpretation of a homomorphism that is a little more abstract but is enlightening in it's own way is that a homomorphism f is a function that "commutes" with the binary operation: If $*$ represent any binary operation (of any group, not just one group), then f is a homomorphism if and only if

$$f \circ * = * \circ f$$

Concretely, if we let $*$ for a group G be represented as $*_G$ and $*$ for a group H be represented as $*_H$, then

$$f \circ *_G = *_H \circ f$$

or, if written in terms of elements and with the standard notation for binary operation:

$$f(x *_G y) = f(x) *_H f(y)$$

This again show's how we can think of commutativity as a powerful property. This idea can be captured in a diagram like so:

$$\begin{tikzcd}
G \times G \arrow[r, "\varphi \times \varphi"] \arrow[d, "*_G(a,b)"'] & H \times H \arrow[d, "*_H(\varphi(a),\varphi(b))"] \\
G \arrow[r, "\varphi(ab)"'] & H
\end{tikzcd}$$

which can be read as "start at $G \times G$, then either go down then right, or go right then down, and both paths should give the same result", i.e. $\varphi(ab) = \varphi(a)\varphi(b)$.

Now that homomorphisms are defined, you might ask yourself "given two groups G, and H, how do I know there is a homomorphism between them?" There is always a trivial answer to this, the homomorphism sending everything to the identity always exist [21]. This is called the *trivial homomorphism* (you should check that it's indeed a homomorphism):

$$\varphi : G \to \{e\} \qquad \varphi(g) = e$$

So the better question would be "How can I find the non-trivial homeomorphisms between these two groups?" This in some cases can be a very difficult question. If we just look at them as set-functions from G to H, then if there are n objects in one category and m in the other, there are minimum n^m possible functions! That's a ton to check. We can deduce properties of homomorphisms which will let us diminish that number:

Proposition 1.4.4: Properties of Homomorphisms

Let G, H be groups and φ be a homomorphism. Then:

1. $\varphi(1_G) = 1_H$
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$
3. $\varphi(g^n) = \varphi(g)^n$
4. $\ker\varphi$ is a subgroup of G
5. $\mathrm{Im}(\varphi)$, the image of G under φ is a subgroup of H.
6. if $\psi : H \to K$ is also a homomorphism, then $\psi \circ \varphi : G \to K$ is a homomorphism, i.e., the composition of homomorphisms is a homomorphism.

***Proof*:**

1. $\varphi(1_G) = \varphi(1_G 1_G) = \varphi(1_G)\varphi(1_G)$, so $\varphi(1_G) = \varphi(1_G)\varphi(1_G)$. Then by proposition 1.1.17
$$\varphi(1_G)\varphi(1_G)^{-1} = \varphi(1_G)\varphi(1_G)\varphi(1_G)^{-1}$$
the left hand side becomes 1_H, and the right hand side becomes $\varphi(1_G)1_H = \varphi(1_G)$, giving us
$$\varphi(1_G) = 1_H$$
2. Hint: take $\varphi(1_G) = \varphi(gg^{-1})$, and apply linearity and proposition 1.1.17
3. Hint: $\varphi(g^n) = \varphi(g^{n-1}g)$ and use the fact that φ is a homomorphism
4. Hint: Take the set $\ker(\varphi) := \{g \in G \mid \varphi(g) = 1_H\}$, and check the 3 axioms of a group
5. Hint: Take $\varphi(G) := \{h \in H \mid \exists g \in G \text{ such that } \varphi(g) = h\}$ and check the 3 axioms of a group
6. Hint: Apply the homomorphism condition twice.

[21] Elaborated in proposition 1.4.5

These propositions can even help us deduce whether [non-trivial] homomorphisms exist. For example, is there a non-trivial homomorphism from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$? Let's say $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ were such a map. By the above proposition we have $\varphi(0) = 0$. Let's say $\varphi(1) = n$. Then

$$0 = \varphi(0) = \varphi(1 + 1) = \varphi(1) + \varphi(1) = n + n$$

giving us $0 = n + n$, which re-arranging is $-n = n$. The only integer equal to its negative is 0, hence $n = 0$. This shows that there is no non-trivial map from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}$.

The particular structure of the group that is being preserved depends on the operation and the cardinality of the domain and co-domain. In general, to get an idea of what part of the group is being preserved, you'll have to check by doing a couple of example.

Exercise 1.4.1

1. Find all non-trivial homomorphisms between D_8 and Q_8
2. (If you know some complex analysis) Let S^1 be the circle in the complex plane. show that $f : \mathbb{R} \to S^1$, $f(x) = e^{2\pi ix}$ is a homomorphism.
3. Perhaps another exercise like the one above? Perhaps a $G \to G$ example? I think getting some ideas between which groups are homomorphic to each other to be useful for long-term mathematical career.
4. is there a non-trivial homomorphism between $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}$? answer:[22]
5. Let $\varphi : G \to H$ be a homomorphism. Show that $\varphi(a + b + c) = \varphi(a) + \varphi(b) + \varphi(c)$. More generally, if $g_1, g_2, ..., g_n \in G$, then $\varphi(g_1 + g_2 + \cdots + g_n) = \varphi(g_1) + \varphi(g_1) + \cdots + \varphi(g_1)$. [hint: recall assocaitivity and induction]
6. Does there exist a surjective homomorphism $\varphi : G \to H$ where G is abelian and H is not?
7. Let S where $|S| = 3$ be a set and let $\mathrm{Sym}(S)$ all bijections on S (i.e. the symmetric group). Show that there exists a homomorphism $\sigma : D_6 \to \mathrm{Sym}(S)$. (it is actually an isomorphism, move this exercise?) (This is a particular example of a *group action*, which we will explore in detail in chapter 4)
8. Define the map $\varphi_g : G \to G$, $\varphi(x) = gx$. Is this map in general a homomorphism? If not in general, in a particular case?
9. Define the map $\psi_g : G \to G$, $\psi(x) = gxg^{-1}$. Is this map in general a homomorphism? If not in general, in a particular case?
10. Let $\varphi : G \to H$ be a homomorphism and $\ker(\varphi) = \{e_G\}$. Prove that φ is injective.
11. Show that there exist an injective homomorphism $\varphi_n : D_n \hookrightarrow O_2(\mathbb{R})$.

Important examples

I'll go over some homomorphisms to build a deeper understanding of the concept

[22] There does not exist a non-trivial homomorphism $\varphi : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}$, since every non-identity element of $\mathbb{Z}/n\mathbb{Z}$ ha finite order, but all elements of $\mathbb{Z}$ has infinite order.

The first one is deceptively boring: take $\varphi : G \to \{e\}$ such that $\varphi(g) = e$. The function φ is a homomorphism, since for any $g, h \in G$

$$\varphi(g \cdot h) = e = e \cdot e = \varphi(g) \cdot \varphi(h)$$

This seemingly boring homomorphism has actually a very important property: for *any* group G, there *always* exist a *unique* isomorphism from G to $\{e\}$. This is also true for homomorphisms with $\{e\}$ as the domain! This fact is important enough to make a proposition:

Proposition 1.4.5: Homomorphism To/From Trivial Group

Let G be any group and I represent the trivial group. Then there always exists a unique homomorphism

$$\varphi : G \to I \qquad \psi : I \to G$$

to and from I

***Proof*:**
Let e_G represent the identity of G and e_I represent the element of I. We've already shown φ is a homomorphism, so let's now show uniqueness. Let's say φ' is another homomorphim with the same domain/codomain. Note that

$$\varphi'(e_G) = \varphi'(e_G \cdot e_G) = \varphi'(e_G)\varphi'(e_G) \;\Rightarrow\; \varphi'(e_G) = \varphi'(e_G)\varphi'(e_G)$$

and by proposition 1.1.17, we can multiply each side by$(\varphi')^{-1}$ and get

$$e_I = \varphi'(e_G)$$

meaning the identity *must* map to the identity. But that would then make φ' identical to φ, thus

$$\varphi = \varphi'$$

showing that φ must be unique.

A similar proof works for ψ.

So why is this property important? What can we do with it? In this case with groups, it is not used often [23]. what's more important is the concept that there *always* exists a homomorphism from any group, and that it is a *unique* homomorphism. These properties for the right homomorphisms will come back big-time when we talk about *quotient groups* (chapter 3): For *every* homomorphism there *always* exists a *unique* group (called the quotient group) which will "represent" the homomorphism.

Thus, we'll give a name to groups with this property:

[23]It is used in the context of Category theory, where $\{e\}$ is both the initial and final group in the category of **Group**, making it an intuitive first step towards introducing universal properties. More on this in section 8

Definition 1.4.6: Initial And Final Groups

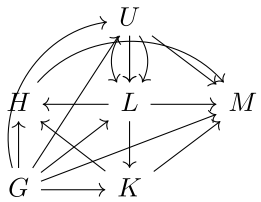
Let G be a group. If for any other group H, then

1. if there exists a unique group $\varphi : G \to H$, then we say that G is an *initial group* (or *initial object*)

2. if there exists a unique homomorphism $\varphi : H \to G$, then we say that G is an *final group* (or *final object*)

3. if G is either initial or final, we say that G is a *terminal group* (or *terminal object*)

The reason for this vocabulary is clearer is you think of a graph where the edges are homomorphisms and the nodes are groups:



(1.9)

You can see how there is only one homomorphism from G to any other group, and M has only one homomorphism from every other group. Thus, G is initial, H is final, and both G and H are terminal.

1.4.7 Note An object *can* be both initial and final. The trivial group is one such example

For now, we do not have the tools to properly analyze this result; we will gain those in chapter 2, and revisit this idea in chapter 3.

Redefining Subgroup

Moving back towards things we can prove, I'll present an important way of using homomorphisms to *define* subgroups, using the fact that subgroups as a similar structure to the original groups, and homomorphisms preserve some structure of a group. It might already be somewhat intuitive that if $H \leqslant G$, then H is pretty similar to G, especially that it must have the same group operation. With homomorphisms, we can formalize this idea:

Proposition 1.4.8: Defining Sub Group With Homomorphism

Let $(G, \cdot)$ and $(H, \bullet)$ be two groups, and $H \subseteq G$. Then H is a subgroup if and only if there exists a inclusion map $i : H \to G$ which is also a homomorphism

***Proof*:**
We'll show that this definition is equivalent to definition 1.1.7, that is, that $(H, \cdot)$ is a group. Let $i : H \to G$ be an homomorphism which is also an inclusion map. All of these will come from proposition 1.4.4.

1. ($e_G = e_H$) We need to how that H has the same identity. This comes for free with proposition 1.4.4

2. (inverse) We need to show that for any $a \in (H, \cdot)$, then there is a $b \in (H, \cdot)$ such that $ab = ba = e_G$. Since $i(a^{-1}) = a^{-1}$, then a^{-1} is a candidate. Indeed, since

$$i(e_H) = i(aa^{-1}) = i(a)i(a^{-1}) = i(a)i(a)^{-1} = aa^{-1} = e_G$$

then a^{-1} will indeed be our desired inverse

3. (associativity) Similarly to how we've down the other two, but the proof is a little more tedious to carry out. One can do this to try proving associativity at least once in their career.

This is probably one of my favourite proposition using homomorphisms: really showing how homomorphism captures the idea that an algebraic structure is preserved by defining subgroups with an inclusion map that is also a homomorphism.

Collection of Homomorphisms

Next, we will address a little more complicated object. We might ask ourselves in what other way can we use homomorphisms to study groups? If a homomorphism preserves group structure, what possible homomorphisms can we have between a group and itself? What would that give us? Let's first label the set of all such homomorphisms:

Definition 1.4.9: Set of Endomorphisms

1. Given a group G, an endomorphism is a homomorphism from an object to itself: $\varphi : G \to G$.

2. Let $\text{End}(G) := \{\varphi : G \to G \mid \forall a, b \in G,\ \varphi(ab) = \varphi(a)\varphi(b)\}$ represent the set of all endomorphisms on G

What $\text{End}(G)$ represent is the set of all possible group structure's that can be derived form G. In the example with $\mathbb{Z}$ being mapped to $\{-1, 1\}$ (example 1.4.3), notice that we can make the codomain all of $\mathbb{Z}$, and the function remains identical. In this way, we would make that function an element of $\text{End}(\mathbb{Z})$, and so this shows that the function in this example captures one of the possible sub structures of $\mathbb{Z}$ (in particular, a special kind of subgroup, we ill explore this in more detail in chapter ref:HERE)

If we can figure out some structure on $\text{End}(G)$, we can start getting a good idea on what are all sub-structures in G. More excitingly, if $\text{End}(G)$ can be proven to be a group (with composition, $\circ$, as the operation), then we can use all the technics of group theory to study it. However, this turns out not to be the case. The problem comes when trying to prove the existence of an inverse, which doesn't necessarily exist (try proving it's a group!).

So you might ask if there is still a way to make $\text{End}(G)$ into a group; perhaps with a nice operation, we can define a [non-trivial] group structure? The one that is usually first attempted is defining addition *pointwise*, that is

$$f, g \in \text{End}(G) \qquad f + g \equiv (f + g)(x) = f(x) + g(x)$$

Where $\equiv$ means we are showing how the ne function "$f+g$" is defined by how it processes every element. With the operation defined, to check that $f+g$ is in $\text{End}(G)$, we need to check if it's a homomorphism, that is

$$(f+g)(x+y) = (f+g)(x) + (f+g)(y)$$

and going through the computation:

$$\begin{aligned}(f+g)(x+y) &= f(x+y) + g(x+y)\\ &= f(x) + f(y) + g(x) + g(y)\end{aligned}$$

and here is were we get stuck, since G is not necessarily commutative.

However, what if G was commutative? Well, the result will actually follow much more nicely!

$$\begin{aligned}&= f(x) + f(y) + g(x) + g(y)\\ &= f(x) + g(x) + f(y) + g(y)\\ &= (f+g)(x) + (f+g)(y)\end{aligned}$$

showing that $(f+g)$ is indeed a homomorphism!! Thus, $f+g$ is a well-defined operation. From here, you can check the axioms of a group to check that $\text{End}(G)$ is indeed a group. It's even an abelian group! This is an example of how being commutativity changes how a group effects.

1.4.10 Remark In fact, more is true. If the domain of the functions are arbitrary sets, then $\text{Hom}(A, G)$ is *still* an abliean group! This fact will be very useful to us much further don the line, in part II, III, and X.

Generalization of 2D and 3D rotations; intuition on 4D

Point Groups? (`https://en.wikipedia.org/wiki/Point_groups_in_two_dimensions`) (and the 3D one too?)

Finally, we will take a moment to detour into trying to use groups to build an intuition for 4+ dimensional objects! As we've seen, D_n represents rotations of 2D polygons, and Q_8 represents rotations of 3D objects. What about rotations and symmetries of higher dimensional objects? To find such a group, we will show that there is a homomorphism $\varphi_D : D_n \Rightarrow SO_3(\mathbb{R})$ and $\varphi_Q : Q_8 \to SO_3(\mathbb{R})$

First, we'll define φ_D:

$$r \to \begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix} \qquad d \to \begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix}$$

TBD (do same with quaternion. Perhaps take a look at representation theory to figure out Q_8, Generalize to 4D with $SO(4)$)

Example 1.4.11: exercicss
gaining intuition on matrix multiplication

1. matrix multiplication could've been defined component-wise, so why wasn't it? Build up to (using the appropriate vocabulary, not the following one) that given $\dim M = n$, $\dim N = n$, $\text{Hom}_{k-\mathbf{Vect}}(M, N) \cong_{k-\mathbf{Vect}} M_n(k)$, and we'll get $\Psi(f \circ g) = A_f \cdot A_g$ (i.e. composition

becomes matrix multiplication)

2. elaborate if dimensions change?

Exercise 1.4.2

1. Let G be an arbitrary group.
   1. Prove that there always exists a homomorphism from $\mathbb{Z}$ to G
   2. Why does this not contradict the fact that I is the only initial group?

1.4.2 Isomorphisms

While homomorphisms preserve some algebraic structure, isomorphisms preserve the *entire* algebraic structure. An isomorphism can be thought of as the generalization of the concept of equality: If two groups are isomorphic, they are identical (as groups). They will only differ in the symbols which are used in representing them. The idea of an isomorphism extends far beyond group theory. Here's a fun linguistic example: if I say

one, two, three, four, five

or

un, deux, trois, quatre, cinq

Then both are capturing the idea of $1, 2, 3, 4, 5$, except in one I'm writing it in English, and in the other in French. Those two sentences ostensibly capture the same [numerical] ideas, but with different symbols

When trying to capture this idea in groups, we have to consider what it means for two group to be identical. First, the two groups would need to be of the same size, if not we cannot say they are the same. Therefore, the homomorphism must also be bijective. Next, all the relation between the two groups need to be the same. In example ref:HERE, we only preserved part of the relations.

Definition 1.4.12: Isomorphism

The map $\psi : G \to H$ is called an *isomorphism* and G and H are said to be *isomorphic* or of the same *isomorphism* type, written $G \cong H$ if

1. ψ is a homomorphism
2. ψ is a bijection

1.4.13 Note Dummit showed this as the definition of isomorphism. But in a more categorical framework, this is not in general how you would introduce this. You would instead say that there exists an inverse function: $g : H \to G$ which is also homomorphic. It can be shown that this (for groups) implies a bijection. [a]

[a] If you've already studied topology, there's an example where a homomorphic bijection is not an isomorphism

When two groups are isomorphic, it means they only differ by the symbols we choose to use, they are in practice identical. The trivial isomorphism is the identity map $\mathrm{id} : G \to G$. We can also form an equivalence class of isomorphic groups: take $\mathcal{G}$ to the set of groups and let the equivalence relation be isomorphisms.

Example 1.4.14: Isomorphism Examples

Let V_4 (*Vierergruppe*, meaning 4-group, sometimes called Klein 4-group), is the group that represents the symmetries of a [non-square] rectangle. You can define it as

$$V_4 = \langle a, b | a^2 = b^2 = (ab)^2 = e \rangle$$

Show that:

$$V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

Where $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has the same operation defined in the cartesian product example in ref:HERE.

A less obvious example of a group isomorphism is $\psi : \mathbb{R} \to \mathbb{R}^+$, $\exp(x) = e^x$. e^x is invertible, and thus bijective, and $e^{x+y} = e^x e^y$, preserving group structure.

Proposition 1.4.15: Properties Of Isomorphisms

(Motivated by properties of diffeomorphism in John Lee. In particular, I want to say that if $g \circ f$ and f is an isomorphism, [do we need to force g to be either injective or surjective?]so must g be)

***Proof*:**

1. The composition of isomorphism is an isomorphism
2. If $g \circ f$ and g are isomorphisms, then since g is an isomorphism, so if $g^{-1} \circ g \circ f = f$, and similarly if f is an isomorphism: $g \circ f \circ f^{-1} = g$

An interesting exercise to show is if A and B are sets of the same cardinality, and S_A and S_B are two symmetric groups, then $S_A \cong S_B$. The book explains it on p.37, but personally I feel it's more intuitive if you think of it as relabeling since A and B are arbitrary sets.

Going back for a moment to hen e tried to define a group $(\mathrm{End}(G), \circ)$ the fatal flaw as that not all homomorphisms have inverses. However, e have seen that all isomorphisms can be reversed, that is, i $\varphi : G \to H$ is an isomorphism, then there exists a ψ, usually denoted φ^{-1} such that $\varphi^{-1} : H \to G$ is an isomorphism and $\varphi^{-1} \circ \varphi = \varphi \circ \varphi^{-1} = \mathrm{id}$ (here id is either id_G or id_H depending on whether φ is on the right or left in the composition respectively). If e take $\varphi : G \to G$ to e an isomorphism, then φ^{-1} ill *exactly* have an inverse. Excitingly, this in fact define as group!

Definition 1.4.16: Automorphism

1. if $\varphi : G \to G$ is an isomorphism from a group to itself, then φ is called an ***automorphism***
2. Let $\mathrm{Aut}(G) := \{\varphi : G \to G \mid \varphi \text{ is an automorphism}\}$ be the collection of all automorphisms of G

Proposition 1.4.17: $\mathrm{Aut}(G)$ is a Group

Let G be a group and $\mathrm{Aut}(G)$ be the collection of automorphisms. Then $(\mathrm{Aut}(G), \circ)$ forms a group

***Proof*:**
Proving it's a group should be doable at this point

Example 1.4.18: Automorphisms
Simplest example are identity maps. Other examples come form Linear Algebra with bijective linear functions (since $\varphi(a)\varphi(b) = \varphi(ab)$ for a linear function, it *is* an example)

1.4.19 Foreshadowing The concept of automorphisms will not be limited to groups. Many algebraic and non-algebraic structures will have the concept of automorphism defined. Most excitingly, the set of automorphisms for *any* of these objects *always* forms a group! Because of this, we will take a hole chapter to study (TBD chap 4? better transition?)

:exercises

(make sub questions) example 1.1.5[5], we showed that $(\mathbb{Z}, +)$ and $(\mathbb{Z}, +_2)$ had a "different structure" by showing the identity and inverses are different. Show that they are in fact isomorphic. This means that they are not a good example of two different groups on the same set since algebraically, they are identical.

Show that $D_8 \ncong \mathbb{Z}/8\mathbb{Z}$. With this knowledge, conclude that we can define two different groups on the set of 8 elements, giving a proper example of two different groups structure on the same underlying set.

(Challenging) Try to come up with two different structure on the underlying set of $\mathbb{Z}$ (Hint: recall the structure of $\bigoplus_i^\infty(\mathbb{Z}/2\mathbb{Z})$ and show there is a bijection between $\mathbb{Z}$ and this set, so that you can define a group structure on $\mathbb{Z}$ relative to this bijection. Is this group isomorphic to $\mathbb{Z}$?)

In the following section, we will explore classifying all possible groups on n elements where n has small order (larger orders of n will come when we have more tools)

First Use of Isomorphisms

In mathematics, there's a whole set of theorems that we use to show that classes of mathematical objects, like groups, are isomorphic to each other or not. Such theorems are called *classification theorems*. As a simple example, we can classify all non-abelian groups of order 6, in particular:

1.4.20

$$\text{any non-abelian group of order 6 is isomorphic to } S_3$$

Holding the theorem to be true for now, we can conclude that $D_6 \cong S_3$, and $GL_2(\mathbb{F}_2) \cong S_3$ without needing to find an explicit map. Not all groups of order 6 are isomorphic to S_3 because they're not all abelian. However, all abelian groups of order 6 are isomorphic to $\mathbb{Z}/6\mathbb{Z}$! Thus, any group of order 6 is isomorphic to either S_3 or $\mathbb{Z}/6\mathbb{Z}$. This theorem will be proved in exercise ref:HERE.

In general, it is difficult, sometime unfeasible, to find an explicit isomorphism mapping. Here are some necessary, but not sufficient, conditions for isomorphisms:

Proposition 1.4.21: Isomorphism Conditions

Let $\psi : G \to H$ be an isomorphism. Then:

1. $|G| = |H|$
2. G is abelian if and only if H is abelian
3. for all $x \in G$, $|x| = |\psi(x)|$

***Proof*:**
Fairly simple - left as an exercise.

Example 1.4.22
This proposition shows that S_3 and $\mathbb{Z}/6\mathbb{Z}$ are not isomorphic, because one is abelian and the other is not. Additionally, $(\mathbb{R} - \{0\}, \times)$ and $(\mathbb{R}, +)$ are not isomorphic, because $-1 \in \mathbb{R}^\times$ has order 2, and no element in $(\mathbb{R}, +)$ has order 2

There's a cool introduction to how two groups could be homomorphic given the presentation of a group, which will require free groups to prove formally. I'll omit this for now (p.52, bottom).

Example 1.4.23

1. Take D_n and it's presentation $\langle r, s | r^n = s^2 = 1, sr = r^{-1}s\rangle$. Now take $n = mk$, $k \geq 3$ and consider d_{2k}. Then

$$\psi : d_n \to d_{2k} \qquad \text{by} \qquad \psi(r) = r_1, \text{ and } \psi(s) = s_1$$

is a homomorphism. In other words, the operation between Dihedral groups which are "multiples" are the same (think of a triangle and hexagon. two hexagon rotations is 2 triangle rotation, angle wise).

2. an example showing D_6 and S_3 are isomorphic. p.53

Proposition 1.4.24: Isomorphism properties

Let $\psi : G \to H$ be an isomorphism, and $\Psi : G \to H$ an isomorphism (if the context permits it)

1. Let $\psi : G \to H$ is a homomorphism. Prove that the identity maps to the identity
2. $\psi(x^n) = \psi(x)^n$, for all $n \in \mathbb{Z}$.
3. $|\Psi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true for ψ? (hint: think of the "trivial homomorphism" which sends every element from G to $\{1\}$.)
4. If Ψ exists, then G is isomorphic if and only if H is isomorphic (easy)
5. If you know where the generators map, you know where all the entries map

***Proof*:**
I'll do one of them:

1. $$1_H\psi(x) = \psi(x) = \psi(1_G \cdot x) = \psi(1_G) \cdot \psi(x) \Rightarrow 1_H = \psi(1_G)$$

The rest is good to do yourself to really understand the proposition.

The last point in proposition 1.4.24 is one to keep in mind. I found it particularly useful when trying to construct isomorphisms between groups and S_n (which you will extensively once you cover group-actions).

Example 1.4.25

Prove that $SL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is isomorphic to the quaternion group of order 8

Proof. Label the first matrix A and the second matrix B. To start, bring up the presentation of Q_8 to get an idea of it's structure. See if you can bring it down to two generators. What are the properties of these generators (i.e. representation). Does A and B match it? If so, what's then the isomorphism. $\square$

Prove that a homomorphism on S_n means it will preserve the parity of the cycle

Proof. Check cases $\square$

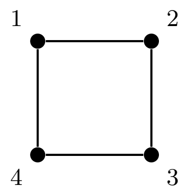
1.5 Symmetry and Transformations: A Prelude to Group Actions

We have so far studied groups as abstract algebraic objects, defined by axioms and explored through subgroups, homomorphisms . But groups did not arise historically from axioms—they arose from *symmetry.* Évariste Galois studied permutations of roots of polynomials. Arthur Cayley studied transformations of space. In every early appearance of what we now call a “group,” the central idea was the same: a collection of *reversible transformations* of some object, where transformations can be composed and undone.

In this section, we pause the axiomatic development to revisit this original intuition. The goal is simple: to see that groups *act on things*, and that this perspective is the natural way to think about them. We will make all of this precise in Chapter 4; for now, we build intuition.

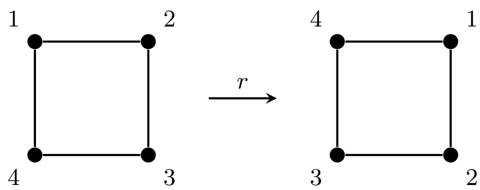
The Symmetries of a Square

Recall from Section 1.2.3 that the dihedral group D_8 consists of the 8 symmetries of a square: four rotations and four reflections. Let us label the corners $1, 2, 3, 4$:



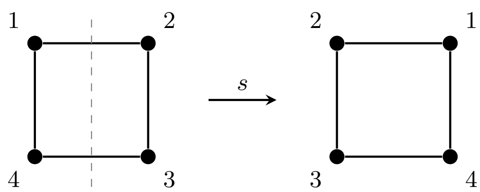
Each symmetry of the square *moves the corners around.* A 90° clockwise rotation r sends each corner to the next one clockwise:

$$1 \longmapsto 2, \quad 2 \longmapsto 3, \quad 3 \longmapsto 4, \quad 4 \longmapsto 1.$$



Similarly, the reflection s across the vertical axis swaps the left and right columns:

$$1 \longmapsto 2, \quad 2 \longmapsto 1, \quad 3 \longmapsto 4, \quad 4 \longmapsto 3.$$

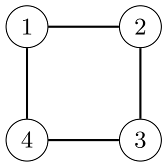


Each element of D_8 determines a specific way of rearranging the four corners—that is, a *bijection* from $\{1, 2, 3, 4\}$ to itself. We can compose these rearrangements: rotating by r twice gives the same result as rotating by r^2. In other words, the group D_8 is *acting* on the set $\{1, 2, 3, 4\}$.

Two observations are worth highlighting. First, every symmetry is *reversible*: if r rotates the square clockwise, then r^{-1} rotates it back. This is why we need a *group* (with inverses) rather than just a monoid. Second, the identity element $e \in D_8$ does nothing to the corners—it is the "trivial symmetry" that leaves everything in place.

Beyond the Square: Symmetry is Everywhere

The group D_8 does not only describe the symmetries of a geometric square. Consider the *cycle graph* C_4 on four vertices:



A *graph automorphism* is a bijection on the vertex set that preserves adjacency. One can verify that the automorphism group $\mathrm{Aut}(C_4)$ is precisely D_8: the same rotations and reflections that preserve the square also preserve which vertices are joined by an edge. The group D_8 captures an abstract *pattern of symmetry* that manifests in both geometry (the square) and combinatorics (the cycle graph). This is the key insight: a single group can describe the symmetries of *many different objects*.

Here are several more instances of this phenomenon, drawn from across mathematics. In each case, a group "acts" on a set by rearranging its elements in a structured, reversible way.

1. **Linear transformations.** The general linear group $GL_n(\mathbb{R})$ acts on $\mathbb{R}^n$: each invertible matrix A transforms a vector $\vec{x}$ to $A\vec{x}$. Composing two transformations corresponds to multiplying the matrices, and the identity matrix leaves every vector fixed. Rotations, reflections, scalings, and shears of n-dimensional space are *all* captured by this single action.

2. **Permutations.** The symmetric group S_n acts on $\{1, 2, \ldots, n\}$ by evaluation: each permutation σ sends i to $\sigma(i)$. This is, in a sense, the "most general" action—every rearrangement of n objects corresponds to some element of S_n. As we will see in Chapter 4, *every* group action can be viewed through the lens of permutations.

3. **The Rubik's Cube.** A standard 3×3 Rubik's Cube has six faces, each consisting of nine colored squares. The center square of each face is fixed (it defines the face's color), leaving $6 \times 8 = 48$ movable *facelets*. Each legal move—a quarter-turn or half-turn of a single face—permutes these 48 facelets, and the set of all finite sequences of legal moves forms a group $\mathcal{R} \leq S_{48}$. This group is enormous:

$$|\mathcal{R}| \;=\; 43{,}252{,}003{,}274{,}489{,}856{,}000 \;\approx\; 4.3 \times 10^{19},$$

yet it is a *tiny* fraction of $|S_{48}| = 48! \approx 1.2 \times 10^{61}$. Not every conceivable rearrangement of the facelets can be reached by legal moves. The structure of the group $\mathcal{R}$—its subgroups, generators, and quotients—is precisely what determines which configurations are reachable from a solved cube and which are not.

4. **Translations of polynomials.** Consider the set of all real polynomials. The additive group $(\mathbb{R}, +)$ acts on this set by *translation*: given $t \in \mathbb{R}$ and a polynomial f, define

$$(t \cdot f)(x) \;=\; f(x+t).$$

Translating by t and then by s is the same as translating by $t+s$, and translating by 0 changes nothing. This simple action connects algebra to analysis: Taylor's theorem, for instance, describes exactly how a polynomial changes under translation.

What All These Examples Share

In every example above, three things are true:

1. Each element of the group "does something" to every element of the set—it rearranges, transforms, or moves things around.
2. The **identity element does nothing**: it leaves every element of the set exactly where it was.
3. The group operation is **compatible with composition**: performing one element's transformation followed by another's is the same as performing the transformation of their product.

These three observations are the soul of what we will call a *group action*. In Chapter 4, we will distill them into a precise definition—a single homomorphism from the group into a symmetric group—and develop a rich theory around it. But the underlying philosophy is already clear:

A group is not just an abstract set with a binary operation.
It is a collection of symmetries, waiting to act.

Groups Acting on Themselves: A First Glimpse

The examples above involve a group acting on an "external" object: a square, a vector space, a set of facelets. But some of the best applications of group actions come from letting a group act *on its own elements*. There are two natural ways to do this. Given a group G and elements $g, a \in G$:

- Left multiplication: define $g \cdot a = ga$. Each element g "shuffles" the elements of G by multiplying on the left. This will lead us to *Cayley's Theorem*, which asserts that every group is isomorphic to a group of permutations.
- Conjugation: define $g \cdot a = gag^{-1}$. This action detects the internal structure of G: which elements "behave similarly" under the group operation, and how the center $Z(G)$ relates to the rest of the group. It will lead us to the *Class Equation* and, ultimately, to the celebrated *Sylow Theorems*, which describe how a finite group decomposes relative to its prime factors.

Both of these self-actions will be central to Chapter 4. The remarkable fact is that by studying how G transforms *itself*, we can deduce deep structural information about G that is invisible from the axioms alone.

From Actions to Cayley Graphs

We have seen that each element g of a group can be thought of as a transformation: it "moves" every element a to ga. If we focus on a set of generators $\{s_1, s_2, \ldots, s_k\}$, then each generator defines a specific *step* that can be taken from any group element. Starting from the identity e and repeatedly taking these steps, we trace out the entire group.

This is precisely the idea behind a *Cayley graph* (Definition 1.6.1): we place a vertex at each element of G and draw a directed edge from a to $s_i a$ for each generator s_i. The resulting picture is a vivid record of the group's structure, and it is nothing other than a *visual record of the left-multiplication action* restricted to the generators. As we explore Cayley graphs in the next section, keep this action-based perspective in mind—it is the thread that connects the algebra of groups to the geometry of their visual representations, and it is a thread we will pick up again, with full force, in Chapter 4.

1.6 Cayley Graphs

Another way we try to find group-structure is by creating a graph using the generators of a group. This graph gives a visual on how the group elements interact with each other:

Definition 1.6.1: Cayley Graph

Let G be a group and let S be a generating set of G. The *Cayley Graph* $\Gamma = \Gamma(G, S)$ is a colored directed graph constructed as follows:

1. Each element $g \in G$ is assigned a *vertex*: the vertex set $V(\Gamma)$ of Γ bijectivity corresponds with G
2. Each generator $s \in S$ is assigned a colour c_s
3. For any $g \in G$ and $s \in S$, the vertices corresponding to the elements g and gs are joined by a direct edges of colour c_s. Thus, the edge set $E(\Gamma)$ consists of pairs of the form (g, gs), with $s \in S$ providing the colour

some popular, but not necessary, conventions are: [24]

1. The set S is finite
2. $S = S^{-1} = \left\{s^{-1} \middle| s \in S\right\}$
3. $e \notin S$ (i.e., the identity is not in S)

Example 1.6.2: Cayley Graphs

1. Let $G = \mathbb{Z} = \langle 1 \rangle$. Then we can say that $S \cup S = \{1, -1\}$. This gives us the graph:

$$\cdots \qquad -3 \longleftarrow -2 \longleftarrow -1 \longleftarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \qquad \cdots$$

Take a moment to try and see what the Cayley graph of $\mathbb{Z} \times \mathbb{Z}$ would be (the answer is given

[24] These conventions are usually followed in *geometric group theory*

in a later example)

2. Let $G = S_3 = \langle a, b | a^2 = b^3 = 1, ab = b^2 a \rangle$. Then if b is represented by red edges and a is represented by green vertices, we get:

$$\begin{tikzcd}
1 \arrow[r] \arrow[d, bend right] & b \arrow[r] \arrow[d, bend right] & b^2 \arrow[ll, bend right] \arrow[d, bend right] \\
a \arrow[r] \arrow[u, bend right] & ab \arrow[r] \arrow[u, bend right] & ab^2 \arrow[ll, bend left] \arrow[u, bend right]
\end{tikzcd}$$

3. Let $G = \mathbb{Z}/4\mathbb{Z}$. Then the Cayley graph is:

$$\begin{tikzcd}
0 \arrow[r] & 1 \arrow[r] & 2 \arrow[r] & 3 \arrow[lll, bend right]
\end{tikzcd}$$

4. Let $G = \mathbb{Z} \times \mathbb{Z}$. Then the Cayley graph is:

$$\begin{tikzcd}
 & & & \vdots & & & \\
 & (-2,2) & (-1,2) \arrow[l] & (0,2) \arrow[l] \arrow[r] & (1,2) \arrow[r] & (2,2) & \\
 & (-2,1) \arrow[u] & (-1,1) \arrow[l] \arrow[u] & (0,1) \arrow[l] \arrow[r] \arrow[u] & (1,1) \arrow[r] \arrow[u] & (2,1) \arrow[u] & \\
\cdots & (-2,0) \arrow[u] \arrow[d] & (-1,0) \arrow[l] \arrow[u] \arrow[d] & (0,0) \arrow[l] \arrow[r] \arrow[u] \arrow[d] & (1,0) \arrow[r] \arrow[u] \arrow[d] & (2,0) \arrow[u] \arrow[d] & \cdots \\
 & (-2,-1) \arrow[d] & (-1,-1) \arrow[l] \arrow[d] & (0,-1) \arrow[l] \arrow[r] \arrow[d] & (1,-1) \arrow[r] \arrow[d] & (2,-1) \arrow[d] & \\
 & (-2,-2) & (-1,-2) \arrow[l] & (0,-2) \arrow[l] \arrow[r] & (1,-2) \arrow[r] & (2,-2) & \\
 & & & \vdots & & &
\end{tikzcd}$$

if we let $S = \{\text{here}\}$, then the Cayley graph is

show that there is no twist

5. Let $G = D_\infty = \langle r, s \mid rs = sr^{-1} \rangle$ (i.e this is D_n but with no $r^n = 1$ relation) and let $S = \{r, s, r^{-1}, s^{-1}\}$. Then the Cayley graph is:

show the twist

6. $GL_2(\mathbb{Z}_2)$

Just by looking at the graphs, you might be able to see some interesting patterns. For example, by taking the direct product, we found it produces parallel graphs. We see in S_3 and $\mathbb{Z}/n\mathbb{Z}$ that arrow point *to* the identity represent the *order* of the element. Different Cayley graphs can also give different insights: notice that the Cayley graph for $\mathbb{Z} \times \mathbb{Z}$ with $S = \{\text{here}\}$ was simply two parallel lines with every node being connected with the one above it. On the other hand, the Caley graph for D_∞ looks like the one for $\mathbb{Z} \times \mathbb{Z}$, except it has a "twist" introduced to it[25].

In most of this textbook, Cayley graphs will not play a prominent role. They come back here and there to give a visual way of looking at new concepts introduced[26]. However, they are the foundation for an area of group theory called *geometric group theory.* For the study of geometric group theory, some knowledge of analysis is needed (ex. metrics). For those interested in such explorations, cite:HERE

[25]In chapter 5, we will learn about a way to construct D_∞ from $\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$, and will denote it as $\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$. Some mathematicians will call "$\rtimes$" the *crossed product* because it introduced a "cross" or "twist" in the Cayley graph

[26]For example, they will come back when working with free groups and semi-direct products

2

Sub Groups and Generating Groups

Now that we've studied some fundamental aspects of group theory and important examples, let's dive deeper in the mechanics used in analyzing and finding groups. The first way we'll be studying group is by trying to take smaller groups from the larger group and seeing if we can find information through the pieces of the group.

We will start by trying to find subgroups that exist for *any* group G. These subgroups are sometimes called *natural subgroups*, since they always exist for a group, as though the axioms of the group "naturally" let these subgroups exist. We will then move on to constructing groups given some subset of a group, and show how the basic operations of $\cup$ and $\cap$ interact with groups. Finishing off our exploration of generating groups, we will define the concept of "most general group"; it will turn out that the free groups can be thought of as the groups that "generate" all other possible groups. We will end this chapter by exploring a new way of visualizing the subgroups of finite groups through the use of a graph (in particular a lattice) that will become useful when trying to find how to construct larger groups.

2.1 Basic Definition and Properties

First, a reminder of what it means to take a smaller group from the same group:

Definition 2.1.1: Subgroup

Let G be a group with operation $*$. The subset H of G is a *subgroup* of G if H is a nonempty, closed under the restricted operation $*|_H$, that is, it is closed under products and inverse. We shall write $H \leqslant G$.

Recall that the binary operation $*|_H$ on H is a restriction of the binary operation $*$ on G. In particular, if $* : G \times G \to G$ is the binary operation on G, then we say we *restrict* the binary operation to H if we can define $*|_H : H \times H \to H$, where for any two elements $h_1, h_2 \in H$ $*|_H (h_1, h_2) = *(h_1, h_2)$. If $(H, *|_H)$ is a group, we say it's a subgroup of $(G, *)$. In this way, we say that H has "the same" binary operation as H. We will abuse notation and denote the binary operation of the subgroup $H \leqslant G$ as the same operation as the group G, that is, we will write $(G, \cdot)$ and $(H, \cdot)$ instead of $(G, \cdot)$ and $(H, \cdot|_H)$. If we have $H \leqslant G$ and $H \subsetneq G$, then we will denote this as $H < G$

Example 2.1.2

1. If $H = \{1\}$, then for any G, $H \leqslant G$. H is called the trivial subgroup. Similarly, by definition $G \leqslant G$ always.
2. $\mathbb{Z} \leqslant \mathbb{Q} \leqslant \mathbb{R}$ with the operation of addition.
3. $\mathbb{Z}^+ \nleqslant \mathbb{Z}$ under addition. Can you prove it?
4. Take the set of all symmetries ($\{1, \sigma\} \subseteq D_8$) and all rotations ($\{1, r, r^2, r^3\} \subseteq D_8$). These are both subgroups.

The subgroup H inherits a lot of properties we have proven in section 1.1.1. For example, for any equation of elements in H, it is also an equation of elements in G, and so will have the same solution(s) or lack thereof. Furthermore, the identity will remain the same, since every group must have a unique identity. Similarly for inverses.

For computational purposes, here is a useful proposition that warps up the conditions of a subgroup to be able to more quickly identify whether a subset is or is not a subgroup:

Proposition 2.1.3: The Subgroup Criterion

A subset H of G is a subgroup if and only if

1. $H \neq \emptyset$
2. for all $x, y \in H$, $xy^{-1} \in H$

Furthermore, if H is finite, then it suffices to check that H is non-empty and closed under its operation.

***Proof*:**
Let $H \subseteq G$

($\Rightarrow$) Let $H \leqslant G$. Then $1 \in H$, so $H \neq \emptyset$. Take $x, y \in H$. If $y \in H$, then $y^{-1} \in H$. Since H is closed under its operation, then $xy^{-1} \in H$

($\Leftarrow$) Let $H \subseteq G$ and consider $(H, \cdot|_H)$.

(a) **(associativity)** Associativity comes for free since the binary operation on H is the same, so if $x, y, z \in H$, then $x(yz) = (xy)z$

(b) **(identity)** Since $H \neq \emptyset$, there must exist some $x \in H$. Then by the 2nd condition, $xx^{-1} = 1 \in H$, so $1 \in H$.

(c) **(inverse)** Let $x \in H$. Since $1 \in H$ then $1x^{-1} = x^{-1} \in H$.

(d) **(well-defined)** Let's make sure that if $x, y \in H$ then $xy \in H$ (making sure that the binary operation is still a function). By what we've just shown, $y^{-1} \in H$. Thus $x(y^{-1})^{-1} = xy \in H$ (where $(x^{-1})^{-1} = x$ was proved in proposition 1.1.15). Thus, $\cdot|_H$ is indeed a function

Thus, $H \leqslant G$

Now, let H be a finite group. Then for any $x \in H$, there can only be finitely many distinct elements $x_1, x_2, ..., x_n, ...$, meaning for some $a, b \in \mathbb{N}$, $x^a = x^b$ with $a < b$. This means that $x^{b-a} = 1$, and so for any $x \in H$, x has finite order. Thus, we get that $x^{-1} = x^{n-1}$, where $x^n = 1$. Thus, we automatically get H is a subgroup if H is closed under its binary operation

There are a lot of good exercises if you don't feel comfortable showing a particular subset is a subgroup or not on p.48

Exercise 2.1.1

1. Prove or disprove that the following are groups: (make sub question thing later): $\mathbb{Q}^\times \leqslant \mathbb{R}$, $D_3 \leqslant D_4$
2. Let $H, K \leqslant G$. Is $H \cup K$ always a group? Prove or give a counter-example. If not, then identify what condition(s) is needed for $H \cup K$ to be a group.
3. Prove that G cannot have a subgroup H with $|H| = n - 1$, where $n = |G| > 2$.
4. Let $K \leqslant H$ and $H \leqslant G$. Prove that $K \leqslant G$

2.2 Examples of Subgroups

In this section, we will explore "natural" subgroups as well as subgroups that arise given any homomorphism

2.2.1 Kernel and Image

Given any homomorphism $\varphi : G \to H$, are there subsets that can be created using φ which will be subgroups of either G or H? Naturally, the image of a homomorphism is a subgroup of H ($\varphi(G) \subseteq H$), essentially by definition. We might ask ourselves if there exists some subgroup of G that we can construct using φ?

commented old text here:

For the first subgroup, consider the following set

$$\ker(\varphi) \subseteq G \qquad \ker(\varphi) = \{g \in G \mid \varphi(g) = 1\}$$

then $\ker(\varphi) \leqslant G$

Proof. We will use the subgroup criterion. Take $a, b \in \ker(\varphi)$. Then using properties of homomorphisms:

$$1 = \varphi(ab^{-1}) = \varphi(a)\varphi(b^{-1}) = \varphi(a)\varphi(b)^{-1}$$

and since $\varphi(a) = \varphi(b) = 1$, we get

$$\varphi(a)\varphi(b)^{-1} = 11^{-1} = 11 = 1$$

so $ab^{-1} \in \ker\varphi$

$\square$

the set $\ker(\varphi)$ will thus be given a name:

Definition 2.2.1: Kernel

Let $\varphi : G \to H$ be a homomorphism. Then the set

$$\ker(\varphi) = \{g \in G \mid \varphi(g) = 1\}$$

is the *kernel* of φ

2.2.2 Question to Ponder Let's say $\varphi : G \to H$ and $\psi : G \to H$ are two homomorphisms with the same domain and codomain. If $\ker(\varphi) = \ker(\psi)$, does that imply $\varphi = \psi$?

As we have discussed earlier, $\varphi(G)$ is a group:

$$\varphi(G) \subseteq H, \quad \varphi(G) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

For completeness of this section, we put the proof here:

Definition 2.2.3: Image

Let $\varphi : G \to H$ be a homomorphism. Then the set

$$\varphi(G) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

or

$$\operatorname{im}(\varphi) = \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}$$

is the *image* of G with φ.

Proof. We will use the subgroup criterion as well as proposition 1.4.4. Since $\varphi(e_G) = e_H$, we have that $\varphi(G)$ is non-empty.

Next, take $a, b \in \varphi(G)$, meaning there exists an $x, y \in G$ such that $\varphi(x) = a$ and $\varphi(y) = b$. Since $\varphi(y^{-1}) = \varphi(y)^{-1}$, we have

$$ab^{-1} = \varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(G)$$

thus, $ab^{-1} \in \varphi(G)$, meaning $\varphi(G)$ passes the subgroup criterion, as we sought to show □

Given that $\varphi(G) \subseteq H$, $\varphi(G)$ has structure of H. However, since $\varphi(G)$ also has some of it's structural information based on G, it would not be surprising if we can somehow characterize $\varphi(G)$ through G. In fact, this will the first link done in the (probably now illusive) chapter 3.

Exercise 2.2.1

1. Do exercises with group actions showing that the kernel of the conjugation action is the centralizer

2.2.2 Center, Centralizers, and Normalizers

Given any group G, does there exist [non-trivial][1] subsets $H \subseteq G$ which will *always* be subgroups (i.e. $H \leqslant G$), regardless of the operation? Not all subsets will form subgroups. For example, take the set

$$T(G) = \{x \in G \mid x^n = 1, \text{ for some } n \in \mathbb{N}_{>0}\}$$

this set is called the *torsion subset* (i.e. the set of all element of finite order). If $T(G) \leqslant G$, then it must pass the subgroup criterion for all possible groups G. However, take

$$G = \langle x, y \mid x^2 = y^2 = 1 \rangle$$

both $x, y \in T(G)$. However, $xy^{-1} = xy \notin T(G)$ (note that $y^2 = 1 \Rightarrow y = y^{-1}$) – the element $(xy)^n = xyxyxyxy\cdots$ will never equal to 1, there are no relation that will permit the appropriate cancellation to occur. In particular, note that there is no relation for G which makes $xy = yx$ which would allow the values to commute and cancel. This is rather difficult to prove at this stage in the book; we will come back to this in section 2.3.3. Overall, $T(G) \nleqslant G$ for all groups G. Notice that if G was abelian then $T(G)$ *will* always be a subgroup (see exercise ref:HERE).

The question arises then if there are properties for which a subset will *always* be a subgroup for any group G regardless of the operation on G. These types of natural subgroups are in general quite hard to find, however they do exist – the rest of this section will be exploring examples of such subsets.

The first one we will explore is a great example of the power of commutativity: Let

$$Z(G) \subseteq G, \; Z(G) = \{g \in G \mid gx = xg \text{ for all } x \in G\}$$

then $Z(G) \leqslant G$

Proof. We will use the subgroup criterion. For this whole proof, let x be an arbitrary element of G. First note that $1x = x1 = x$, and so $1 \in Z(G)$. Next, take any $a, b \in Z(G)$. Since $bx = xb$, multiplying on the left and right by b^{-1} on both sides of the equation:

$$b^{-1}bxb^{-1} = b^{-1}xbb^{-1} \quad \Rightarrow \quad xb^{-1} = b^{-1}x$$

[1] the trivial subset being just the identity $\{e_G\}$, which is always the simplest subgroup which has essentially no "interesting" structure, and hence trivial

and since equality is symmetric, $b^{-1}x = xb^{-1}$, meaning $b^{-1} \in Z(G)$. Finally, we have

$$ab^{-1}x = axb^{-1} = xab^{-1}$$

meaning $ab^{-1} \in Z(G)$, proving it's a subgroup by the subgroup criterion □

This subgroup is special enough to be given a name:

Definition 2.2.4: Center

Let G be an arbitrary group. Define

$$Z(G) := \{g \in G | gx = xg \text{ for all } x \in G\}$$

to be the set of elements commuting with all the elements of G. This subset is called the *center* of G.

2.2.5 trivia The letter Z is because the German word for center is *Zentrum*.

Example 2.2.6: Center of Groups

The provided examples should be verified by the reader as practice

1. If G is abelian, then $Z(G) = G$. Thus $Z(\mathbb{Z}) = \mathbb{Z}$, $Z(\mathbb{R}) = \mathbb{R}$, and so forth
2. $Z(D_4) = \left\{1, r^2\right\}$
3. $Z(Q_8) = \{1, -1\}$

The Center of other subgroups we've encountered in section 1.2 will be left as exercises.

This should start giving you an idea of the power of commutativity. In fact, it is even more powerful than we need it to be. The notion of the center can be weakened a bit and still remain a group. Take $A \subseteq G$ (notice that A is just a subset). Then define

$$C_G(A) \subseteq G,\ C_G(A) = \left\{g \in G \mid gag^{-1} = a \text{ for all } a \in A\right\}$$

Note that $gag^{-1} = a$ is equivalent to $ga = ag$. The change in equation is due to a further generalization we will shortly make. Before we get to that, we must first show that $C_G(A)$ is indeed a group

Proof. We will use the subgroup criterion. First, since for any $a \in A$, $1a = a1 = a$, $1 \in C_G(A)$
Next, take $x, y \in C_G(A)$. Since $y \in C_G(A)$, $yay^{-1} = a$ for all $a \in A$. Multiplying on the left by y^{-1} and right by y, we get $a = y^{-1}ay = (y^{-1})a(y^{-1})^{-1}$, showing that $y^{-1} \in C_G(A)$. Finally, to show $xy^{-1}a(xy^{-1})^{-1} = xy^{-1}ayx^{-1} = a$, we use associativity and the fact that $x, y, y^{-1} \in C_G(A)$:

$$\begin{aligned} xy^{-1}axy^{-1} &= xay^{-1}yyx^{-1} \\ &= xax^{-1} \\ &= a \end{aligned}$$

and so $xy^{-1} \in C_G(A)$, showing that $C_G(A) \leqslant G$ by the subgroup criterion □

Since it's a subgroup that will always exist, it is special enough to be given a name:

Definition 2.2.7: Centralizer

Let G be an arbitrary group, and $A \subseteq G$[a]. Define

$$C_G(A) := \{g \in G | gag^{-1} = a \text{ for all } a \in A\}$$

This subset of G is called the *centralizer* of A in G. Since $gag^{-1} = a$ if and only if $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A.
If A is a subgroup of G and $C_G(A) = G$, then A is called a *central subgroup*

[a]Notice A is a *subset*, not necessarily a subgroup

If we take, $A = G$ then $C_G(G) = Z(G)$. Equivalently, notice that:

$$\bigcap_{g \in G} C_G(g) = Z(G)$$

that is, it is the elements which are commutative with *all* elements (see exercise ref:HERE)

If $A \leqslant G$ such that $C_G(A) = G$, that means that every element of G commutes with A. It is left as an exercise (ref:HERE) to show that all central subgroups are contained in the center (hence the name central)

Example 2.2.8: Centralizer of Groups

Let $A = \{1, r, r^2, r^3\} \subseteq D_8$ and $B = \{1, i\} \subseteq Q_8$

1. $C_A(D_8) = A$. Notice how this set is larger than $Z(D_8)$, since $r^3 \in C_A(D_8)$ but $r^3 \notin Z(D_8)$
2. $C_B(Q_8) = \{1, -1, i, -i\}$. Similar to the previous example, notice that $C_B(Q_8)$ is larger than $Z(Q_8)$.

We can generalize this result even further: so far, we have written $ga = ag$ for all $g \in G$. we can re-write this as $g\{a\} = \{a\}g$, where we define $g\{a\} = \{gx \mid x \in \{a\}\}$ and $\{a\}g = \{xg \mid x \in \{a\}\}$ This might seem to be making the situation overly-complex, however the great advantage of this approach is we can now general to taking a general set A

$$gA = Ag \qquad gA = \{ga \mid a \in A\} \;\; Ag = \{ag \mid a \in G\}$$

What this difference does is that it weakens the condition for an element to "directly" commute ($ga = ag$) to "indirectly" commuting, or more precisely commuting with respect to the set A: for every $a \in A$, there exists an $a' \in A$ such that $ga = a'g$. It will not always be the case that if $A \subseteq G$, then $gA = Ag$

Example 2.2.9: Counter-Example and Example

Let $A = \{r, \sigma\} \subseteq D_4$. Take $g = r$. Then:

$$rA = \{r^2, r\sigma\} \qquad Ar = \{r^2, \sigma r\} = \{r^2, r^3\sigma\}$$

notice that $rA \neq Ar$

However, if $A = \{1, r, r^2, r^3\}$ then for any $g \in D_4$, $gA = Ag$. As an example, let $g = \sigma$. Then

$$\sigma A = \{\sigma, \sigma r, \sigma r^2, \sigma r^3\} \qquad A\sigma = \{\sigma, r\sigma, r^2\sigma, r^3\sigma\} = \{\sigma, \sigma r, \sigma r^2, \sigma r^3\}$$

verifying other values of D_4 by hand shows that $gA = Ag$ for all $g \in D_4$

Therefore, we can define the following set of elements that satisfy $gA = Ag$:

$$N_G(A) \subseteq G, \ N_G(A) = \left\{g \in G \;\middle|\; gAg^{-1} = A \text{ for all } g \in G\right\}$$

Again, $N_G(A)$ is a subgroup of G. The proof is almost identical to the proof for $C_G(A)$ (change a for A in the proof). We therefore give a name to this special subgroup:

Definition 2.2.10: Normalizer

Take $gAg^{-1} = \left\{gag^{-1} | a \in A\right\}$. Define the *normalizer* of A in G to be the set

$$N_G(A) := \left\{g \in G | gAg^{-1} = A \text{ for all } g \in G\right\}$$

If A is a group and $N_G(A) = G$ (that is, $gAg^{-1} = A$ for all $g \in G$) we call A a *normal subgroup* of G

Notice that if we have $g \in C_G(A)$, then $gag^{-1} = a$ with $a \in A$, which means that $gAg^{-1} = A$, and so $g \in N_G(A)$. It follows quickly that $C_G(A) \leqslant N_G(A)$ (see exercise ref:HERE)

Example 2.2.11: Normalizer of Groups

Let $A = \{1, r, r^2, r^3\} \subseteq D_8$ and $B = \{1, i\} \subseteq Q_8$

1. $N_A(D_8) = D_8$. Notice how $Z(G) \subseteq C_A(D_8) \subseteq N_A(D_8)$, that is

$$\{1, r^2\} \subseteq \{1, r, r^2, r^3\} \subseteq D_8$$

Since A is also a subgroup of G, A is also an example of a normal subgroup of D_8

2. $N_B(Q_8) = \{1, -1, i, -i\}$ First, $1 \in N_B(Q_8)$. To show that $i \in N_B(Q_8)$, notice that

$$iB = \{i, -1\} = Bi$$

so $i \in N_B(Q_8)$. Similarly, for -1 and $-i$. For $j, k \in Q_8$, we know that $ij = k$, $ji = -k$ and $ik = -j$, $ki = j$ (see section 1.2.4).

$$jB = \{j, -k\} \neq \{j, k\} = Bj \qquad kB = \{k, j\} \neq \{k, -j\} = Bk$$

meaning $j, k \notin N_B(Q_8)$. Similarly for $-j, -k$. Thus, we are left with $N_B(Q_8) = \{1, -1, i, -i\}$. Notice that $C_B(Q_8) = N_B(Q_8)$, that is, the normalizer doesn't necessarily have to be bigger than the centralizer

3. If $D = \{1, (1\ 2)\} \subseteq S_3$, what is $N_D(S_3)$? What about $C_D(S_3)$?

The notion of the normalizer is not simply generalization for the sake of generalization, it in fact comes up as a very important property: the kernel of a homomorphism has is always normal!

Proposition 2.2.12: Kernels Are Normal Subgroups

Let $\varphi : G \to H$ be a homomorphism, and take $\ker(\varphi) \leqslant G$. Then $\ker(\varphi)$ is a normal subgroup, that is

$$N_G(\ker(\varphi)) = G$$

In other words, for any $g \in G$, $g(\ker(\varphi)) = (\ker(\varphi))g$

***Proof*:**
Let $\varphi : G \to H$, and for notational convenience, let $K = \ker(\varphi)$.

Take any $g \in G$, and $k \in K$, that is, $k \in G$ such that $\varphi(k) = e_H$. We want to show that $gK = Kg$, or equivalently $gKg^{-1} = K$, for all $g \in G$. Take $gkg^{-1} \in gKg^{-1}$. Then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)1\varphi(g^{-1}) = \varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$$

thus, $gkg^{-1} \in K$, so for some k', $k' = gkg^{-1}$. Since this is true for arbitrary $k \in K$, we get $gKg^{-1} = K$, as we sought to show

The fact that kernels are normal subgroups will become very important in chapter 3 This brings us to the end of our generalizations. This section will be closed with a quick remark about normal subgroups. We did not talk about them much here. However, they are at the heart of chapter 3. For now, you can keep them in the back of your mind.

Exercise 2.2.2

1. $C_G(Z(G)) = G$ and deduce that $N_G(Z(G)) = G$
2. If $A, B \subseteq G$, and $A \subseteq B$, then $C_G(B) \leqslant C_G(A)$
3. For each S_3, D_8, Q_8, compute the centralizers of each element and find the center of each group. Does Lagrange's Theorem (ex 19 section 1.7) simplify the work?

   These next 3 subgroups from the next 3 questions will come back when we talk about group actions (and the conjugacy action, section 4.6)
4. Find the centre of $GL_3(\mathbb{C})$
5. Find the centralizer of $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ in $GL(3, \mathbb{C})$
6. Find the centralizer of $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}$ in $SL(5, \mathbb{C})$, the 5×5 complex matrices having determinant equal to 1
7. there are a lot of good exercises in the book
8. . (re-work exercise since $T(G)$ given as an example of why $T(G)$ not considered *natural* torsion

group) Let G be an abelian group. Take $T(G) = \{x \in G \mid \exists n \in \mathbb{N} \text{ such that } x^n = 1\}$. This group is called the *torsion subgroup* of G. Show that $T(G)$ is indeed a group. is $T(G)$ a group is G is not abelian?

9. prove that $\bigcap_{g \in G} C_G(g) = Z(G)$

10. Let $A \leqslant G$ such that $C_G(A) = G$. Prove that $A \leqslant Z(G)$

11. (advanced, maybe move to another section?) Show that $C_G(S) \leqslant N_G(S)$ by showing $C_G(S)$ is the kernel of the homomorphsims $N_G(S) \to \mathrm{Sym}(S)$ by $g \mapsto (x \mapsto gxg^{-1})$ (i.e. $T(g)(x) = T_g(x) = gxg^{-1}$)

12. Maybe do a Galois correspondence question? this wiki-entry made me think of it: here

13. For any group G, and any subset $A \subseteq G$, is it true that $Z(G) \leqslant C_G(A)$?

2.3 Subgroups Generated by Subsets

All groups have an underlying set. We can thus ask ourselves many questions about groups related to it's underlying set. For example, If we have two subgroups $H, K \leqslant G$, is $H \cap K$ or $H \cup K$ also a subgroup? What if we extend to finite or infinite intersection or union? On another hand, we can ask ourselves if there exists subsets $A \subseteq G$ that can in some way generate G using the binary operation. If generating all of G is too strong of a condition, can A always generate a subgroup of G? These are the types of questions that will be asked in this section.

Note that the technics in this section are very common throughout algebra (and many other areas of mathematics). Asking whether a subset of the underlying of an algebraic object "generates" the entire object (or part of the object) will occupy our time in many of the following chapters of this book (especially chapter 13).

2.3.1 Cyclic Groups and Cyclic subgroups

We will start our exploration subsets $A \subseteq G$ which are singletons: $|A| = 1$. The first question we will ask ourselves is if there is a subgroup $H \leqslant G$ which contains A. The answer to this is always yes: $G \leqslant G$ is a subgroup and by definition $A \subseteq G$. Though this is a valid answer, this is a very boring answer (in mathematical lingo, we'd say it is the trivial case). Thus, we look for a more interesting question: what is the *smallest* subgroup of G that contains A[2]. If $H \leqslant G$ is the smallest subgroup containing A, H is called a *cyclic subgroup*. If the entire group G is the smallest subgroup containing A, then G is called a *cyclic group*. To find the cyclic subgroup or groups given a subset $\{a\} = A$, a similar strategy will be used from when the order of an element (definition 0.1.14) was discussed. Cyclic groups they pop-up quite often due to their simplicity.

[2] the notion of "smallest" is well-defined if $|G|$ is finite, but get's a bit tricker if $|G|$ is infinite (ex. $2\mathbb{Z} \trianglelefteq \mathbb{Z}$ but $|2\mathbb{Z}| = |\mathbb{Z}|$. We will show in the section 2.3.2 how to be more precise with the notion of smallest

Definition 2.3.1: Cyclic groups

Let G be a group. The group G is *cyclic* if G can be generated by a single element, i.e., there is some element $x \in G$ such that $G = \{x^n | n \in \mathbb{Z}\}$ (where the operation is usually denoted as multiplication)

Sometimes, the notation C_n used to represent arbitrary cyclic groups of order n (with multiplication notations)

We say G is generated by x (or x generated G), and write $G = \langle x \rangle$. It was proven in proposition 1.2.6 that $\langle x \rangle$ is indeed a group. The existence of a C_n for every n is because there exists the cyclic group $\mathbb{Z}/n\mathbb{Z}$ for every n (or $\langle r_n \rangle$ for $r \in D_n$). Later in this section, we will show that $\mathbb{Z}/n\mathbb{Z}$ is the *only* cyclic group of order n up to isomorphism, meaning we will be able to conclude that $C_n \cong \mathbb{Z}/n\mathbb{Z} \cong \langle r_n \rangle$. Since C_n usually has multiplication notation, for this section we will use 1 to represent the identity.

2.3.2 Note The fact that $\langle x \rangle$ is the smallest subgroup containing x forced by construction: if there was a subgroup $H \leqslant G$ such that $H \subsetneq \langle x \rangle$, then there would be number n such that $x^n \in \langle x \rangle$ but $x^n \notin H$. Do you see how this contradicts that H is a subgroup?

Example 2.3.3

1. As mentioned earlier, the usual candidate for this example if $\mathbb{Z}/n\mathbb{Z}$, that is, addition modular n, which can be generated by $\overline{1}$. So $\langle \overline{1} \rangle = \mathbb{Z}/n\mathbb{Z}$. Notice there is some ambiguity here: How can you know if $\overline{1}$ represents the of order $5, 10, 324$? In general, you can't. It is up using the context to determine the order. If the context requires clarity, a subscript will be added: $\langle \overline{1}_n \rangle = \mathbb{Z}/n\mathbb{Z}$ or $\langle [1]_n \rangle = \mathbb{Z}/n\mathbb{Z}$

2. D_n is *not* cyclic. Try finding a single element in D_n which generates all of D_n

3. Is Q_8 cyclic? What about $\mathbb{Z}$? What about $\mathbb{Z} \times \mathbb{Z}$?

Proposition 2.3.4: $|C_n| = |x|$

If $C_n = \langle x \rangle$, then $|C_n| = |x|$ (where if one side of this equality is infinite so is the other).

***Proof*:**
The idea is to show that that if $|x| < \infty$, then if two elements differ in exponent and are smaller than the group, $a \neq b < n$, then $x^a \neq x^b$, which can be shown by contradiction. For the case where the exponent is greater than or equal to n, $t \geq n$, x^t, such an elements can always equal to an element of order less than n. In particular, if $t \geq n$, then $x^t = x^{t-kn}$ for appropriate k for which $0 \leq t - kn < n$. Thus, in the $t \geq n$ case, the proof comes down to the first case that was proved.

For the $|x| = \infty$ case, assume $x^a = x^b$, $a \neq b$, and proceed by contradiction (i.e. $x^{b-a} = 1$).

2.3.5 Note This does not imply that for *any* $x \in C_n$, $|x| = |C_n|$. For example, $\mathbb{Z}/6\mathbb{Z}$ is cyclic since $\mathbb{Z}/6\mathbb{Z} = \langle \overline{1} \rangle$. However, $|\overline{2}| = 3$, meaning $\overline{2}$ cannot generate $\mathbb{Z}/6\mathbb{Z}$. On the other hand, $\overline{1}$ is not the only element of order 6 in $\mathbb{Z}/6\mathbb{Z}$, can you think of a second? Is there more than two?

Corollary 2.3.6: Cyclic $\Rightarrow$ Abelian

If G is cyclic, then G is abelian

***Proof*:**
Let G be a cyclic group. Take some $g \in G$ that generates it. Take some $a, b \in G$ so $g^n = a$ and $g^m = b$ then

$$ab = g^n g^m = g^{n+m} = g^{m+n} = g^m g^n = ba$$

thus, G is also abelian.

You might not be satisfied with this proof: why in the exponent can we commute n and m? The answer comes the associativity of groups. Since

$$a(aa) = (aa)a$$

it makes no difference to write

$$a^{1+2} \qquad a^{2+1}$$

or more generally $a^{n+m} = a^{m+n}$ due to proposition 1.1.15[5].

Since every element in a cyclic is determined by the exponent, having some information about what power of the generator leads to the identity can give us some information on smaller powers of the generator that also lead to the identity.

Proposition 2.3.7: Powers of an Element

Let G be a group, $x \in G$ and let $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $\gcd(m, n) = d$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x|$ divides m, $|x| \big| m$

***Proof*:**
This proof relies on Euclid Algorithm (see section 0.2. Let $d = \gcd(a, b)$, so that $m = dm'$ and $n = dn'$. Since $x^m = x^n = 1$, then we have $x^{dm'} = x^{dn'} = 1$. Thus, we have

$$1 = x^{dm'} = \underbrace{x^d x^d \cdots x^d}_{m' \text{ times}}$$

$x^d = 1$ or $x^d = k$ ($k \neq 1$) so $k^{m'} = 1$. If $x^d = 1$, then we're done, so let's say $x^d = k$. Then, similarly for $n = dn'$:

$$1 = x^{dn'} = \underbrace{x^d x^d \cdots x^d}_{n' \text{ times}}$$

where $x^d = 1$ or $x^d = k$ (same k) so $k^{n'} = 1$. Thus, we get $k^{n'} = k^{m'} = 1$. However, here lies the contradiction. Without loss of generality, let's say $n' < m'$ (if $n' > m'$, simply switch the order of the elements in the proceeding proof) so that $m' = n' + \ell$. Then since

$\gcd(n', m') = 1$, $k^{m'} = k^{n'+\ell} = k^{n'}k^{\ell} = k^{\ell} \neq 1$. If it did, then m' would be a multiple of n', but then $\gcd(m', n') \neq 1$

Thus, $x^{\gcd(n,m)} = 1$, as we sought to show

Corollary 2.3.8

Let G be a group (not necessarily cyclic), let $x \in G$, and let $a \in \mathbb{Z} - \{0\}$.

1. if $|x| = \infty$, then $|x^a| = \infty$
2. if $|x| = n < \infty$, then $|x^a| = \frac{n}{\gcd(n,a)}$. If a divides n, then $|x^a| = \frac{n}{a}$.

***Proof*:**

1. If $|x^a| < \infty$, then there would exist an n such that $(x^a)^n = x^{an} = 1$. (if an is negative, take $-an$). But then x would have finite order, a contradiction
2. Since $x^n = 1$, $1 = 1^a = (x^n)^a = x^{na}$. By proposition 2.3.7, $x^{gcd(a,an)} = 1$

With this idea in mind, let's do what we did before and try to see if we can classify cyclic groups.

For cyclic groups, there's in fact an incredibly simple isomorphism which will be introduced showing that if two cyclic groups have the cardinality (same order), they are in fact isomorphic, and thus can be treated as interchangeable! This means that when dealing with future theorems about cyclic groups, once we found it applies to a cyclic group of a certain order, we know it applies to any cyclic group of the same order! This also means there are only countably many finite cyclic groups, each with a predictable structure.

Theorem 2.3.9: Isomorphism between same-order Cyclic Groups

Any two cyclic groups of the same order are isomorphic. More specifically,

1. if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map

$$\psi : \langle x \rangle \to \langle y \rangle \qquad x^k \mapsto y^k$$

   is well defined and is an isomorphism
2. if $\langle x \rangle$ is an infinite cyclic group, the map

$$\psi : \mathbb{Z} \to \langle x \rangle \qquad k \mapsto x^k$$

   is well defined and is an isomorphism

This proof is not too difficult and so a guide has being laid out for the reader to deduce the proof

***Proof*:**
First, show the function is well defined, that is, if $x^a = x^b$ (for possibly different a and b), then $y^a = y^b$. This uses proposition 2.3.7. Then show the map is a homomorphism. To prove it's an

isomorphism, you can either

1. Show there exists a homomorphism $\varphi : \langle y \rangle \to \langle x \rangle$ such that $\varphi \circ \psi = \psi \circ \varphi = \text{id}$ (technically, $\text{id}_{\langle x \rangle}$ or $\text{id}_{\langle y \rangle}$ respectively)
2. Show ψ is injective. At which point subjectivity comes from the fact that both groups have the same cardinality and are finite.
3. Show ψ is surjective. At which point injectivity comes from the fact that both groups have the same cardinality and are finite.

For the infinite case, the map is "clearly" well-defined: if $a = b$, then $x^a = x^b[= x^a]$. By the exponent laws, this map is homomorphic. Then to prove this map is an isomorphism, you can either:

1. Show there exists a homomorphism $\varphi : \langle x \rangle \to \mathbb{Z}$ such that $\varphi \circ \psi = \psi \circ \varphi = \text{id}$ (technically, $\text{id}_{\mathbb{Z}}$ or $\text{id}_{\langle x \rangle}$ respectively)
2. Show the map is injective and surjective (not that injectivity (resp. surjectivity) no longer implies surjectivity (resp injectivity) in the infinite case, even if the cardinalities are the same. Can you think of a surjective but not injective map from $\mathbb{Z}$ to $\mathbb{Z}$? An injective but not surjective map from $\mathbb{Z}$ to $\mathbb{Z}$? see exercise ref:HERE (in Cardinality section))

This let's us properly conclude what was stated at the beginning of this section: if G is s cyclic group of order n (ex. $\mathbb{Z}/n\mathbb{Z}$ or $\langle r \rangle$, $r \in D_n$) then

$$\mathbb{Z}/n\mathbb{Z} \cong \langle r \rangle$$

Hence, instead of referencing a specific cyclic group, it is sometimes useful to just write C_n (some authors like Z_n) to represent some arbitrary cyclic group of order n, and use $\mathbb{Z}$ for an infinite cyclic group (instead of C_∞ or Z_∞)

Now that we classified all cyclic groups, we can ask ourselves the question of which elements generate one of these groups. this next proposition leads us to the answer

Now that we know how many cyclic sub-groups there are, and how these groups can be generated, we can explore what kind of sub-groups we get from cyclic groups. Quite naturally, every subgroup of a cyclic sub group is cyclic. It will turn out that if you have a cyclic group of infinite order, it has an infinite number of cyclic subgroups, and if its of finite order, lets say n, then every sub groups is of an order that divides n. That is , if $H \leqslant G$, G is cyclic, $|H| = a$, then $a|n$.

Proposition 2.3.10: x relation to cyclic group H

Let $H = \langle x \rangle$.

1. Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.
2. Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of H is $\psi(n)$, where ψ is Euler's totient function. (which gives the number of relatively prime numbers to it).

Proof. Dummit p.57-58 □

Finally, we'll put together all our knowledge into one theorem for easy reference:

Theorem 2.3.11: elements that generate cyclic subgroup

Let $H = \langle x \rangle$ be a cyclic group

1. Every subgroup of H is cyclic. More precisely, if $K \leqslant H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$
2. If $|H| = \infty$, then for any distinct non negative integers a, b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of h correspond bijectivity with the integers $1, 2, 3, ...$
3. If $|H| = n < \infty$, then for each positive integer a dividing n, there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{gcd(n,m)} \rangle$, so that the subgroups of H correspond bijectivity with the positive divisors of n.

intuition

***Proof*:**
This proof involves a lot of counting.

1. exercise
2. exercise

Also, once you have Lagrange's Theorem (3.3.1), this proof will become much easier.

Note that the finite group examples of theorem 2.3.11 will eventually easily generalize to perhaps the most use theorem of the course: Lagrange Theorem (theorem 3.3.1), which sates that the order of any subgroup divides the order of a group ($H \leqslant G \Rightarrow |H|||G|$).

The next theorem is in a sense a generalization of part (3) of 2.3.11, since it uses this fact. Once I proof Lagrange, I'll reference this theorem to make sure this makes sense. I do want to introduce this fact here because it feels quite appropriate as a generalization of the previous theorem:

Corollary 2.3.12: if G has order p, then it is cyclic

If G is a group of prime order p, then G is cyclic, hence $G \cong Z_p$.

***Proof*:**
First show that G is cyclic. Let $x \in G$. since $|G| = p$, then $|G| > 1$ so let $x \neq e$. We can always take $\langle x \rangle \subseteq G$. Then by Lagrange theorem (generalization of part 3 of theorem 2.3.11), we know that $\langle x \rangle ||G|$, However, $|G|$ is prime, so only 1 and G divide it. Since $x \neq e$, then $|\langle x \rangle| \neq 1$, so $|\langle x \rangle| = p$, meaning that G was generated by a single element, and so is cyclic. By corollary 2.3.6, it is abelian.

To show its isomorphic to $\mathbb{Z}_p$ should now be easy.

Note that this doesn't mean that if $H \leqslant G$ where $|H| = p$ for some prime that $H \in Z(G)$!

Example 2.3.13: : $H \notin Z(G)$
Take D_8. Note that $\langle \sigma \rangle \leqslant D_8$ and $|\langle \sigma \rangle| = 2$, and so it's cyclic and abelian. However. $r\sigma = \sigma r^3$ meaning that $\sigma \notin Z(D_8)$. However, it is in the centralizer $C_{D_8}(\sigma) = \{g \in G | g\sigma = \sigma g\}$ for $\sigma\sigma = e = \sigma\sigma$, which can also be "obvious" since $\langle \sigma \rangle$ is abelian, and so it would be in the centralizer.

The intuition behind this is that for something to be in the center, it must commute with *all* of G. It is possible to take a subgroup that only commutes (as we've seen with the idea of the centralizer). This is why the intersection under definition 2.2.4 is true: it must commute with everything.

2.3.14 Note You cannot take $\langle x \rangle \leqslant G$ and do $G - \langle x \rangle$. G is not necessarily a group. Take D_8, and take $K = \langle \sigma \rangle$. Then consider $H = D_8 - K$. Then $r^3 \cdot r\sigma = \sigma$, showing that H is not closed under multiplication. Thus, subtraction of groups is not well-defined. However, it will turn out that under the right circumstances, *dividing* and *multiplying* groups will be well-defined! This will be the subject of the next chapter (3)

interesting exercise

1. number 9
2. number 13
3. number 16
4. Knowing Cyclic groups, we can look at another great example of a homomorphism: $\psi : \mathbb{Z} \to Z_n$, where $Z_n = \langle x \rangle$,
$$\psi(a) = x^a$$
Prove that it's a homomorphism, then figure out what elements of $\mathbb{Z}$ map to an element of Z_n.
5. the product of two cyclic subgroups is not necessarily cyclic!

2.3.2 Generating Subgroups

We now turn our attention to generating groups or subgroups using more than 1 element. When studying cyclic groups, we asked whether given an element $x \in G$, does there exists a smallest group $H \leqslant G$ such that $x \in H$. It was shown that taking the closure of x by taking all powers of x resulted in such a group, namely $H = \langle x \rangle$. Phrased differently, the subgroup $\langle x \rangle$ is the smallest subgroup of G that contains the set $\{x\}$. What it means to be the *smallest* is if $L \leqslant G$ is another subgroup where and $x \in L$, then $\langle x \rangle \subseteq L$. We now replace the singleton $\{x\}$ with some arbitrary subset $A \subseteq G$.

As mentioned in the beginning of this chapter, generating subgroups from a subset of a group is not unique to Group Theory. In the following chapters studying other algebraic structures (ex. Ideals, rings, modules, vector-spaces, algebras, fields, etc.), the theme of trying to find minimal sub-object $S \leqslant X$ (subgroup, subspace, subfield, subring, etc.) that contain a subset $A \subseteq X$ from the original object (so that $A \subseteq S \leqslant X$) will repeat itself, and so will the general method for finding the minimal subobject. To find a minimal subobject there are two possible general approach:

1. (Generation Method) If $A \subseteq X$ is a subset of the object, take all possible subobjects containing A and intersect them. In the case of groups, if $A \subseteq G$, then take all possible subgroup $H_A \leqslant G$ containing A (so $A \subseteq H_A \leqslant G$ and intersect them all.

2. (Closure Method) The above approach can be thought as the "global approach". Another way of finding the minimal subobject is instead by the "element approach" (or "local approach"): If A is a subset of the object, then let $\overline{A}$ be the set of all possible combinations of elements of A using the binary operation of G. In the case of groups, of $A \subseteq G$, then $\overline{A}$ will be all the possible combination of elements of A with the binary operation of G.

The main result of this section is that these two methods will coincide and form the same subgroup. In proceeding chapters covering subobjects (subrings, ideals, subspaces, etc.), some details will be omitted due to them being covered in this section, so it is advised that this section be well understood.

We start with the first method. We need show that the intersection of subgroups still forms a group:

Proposition 2.3.15: intersection of Groups

Let G be a group and let $\mathcal{H}$ be any collection of subgroups of G. Then

$$\bigcap_{H \in \mathcal{H}} H$$

Is a subgroup of G.

***Proof*:**
Let $K = \bigcap_{H\in\mathcal{H}} H$. First, $e \in H$ for all $H \in \mathcal{H}$, so $e \in K$. Next, take $a, b \in K$. Since $b \in K$, b is in every subgroup H in $\mathcal{H}$. Thus, b^{-1} is in every subgroup H in $\mathcal{H}$, and so $b^{-1} \in K$. Similarly, ab^{-1} would be in every subgroup H in $\mathcal{H}$ (since each subgroup passes the subgroup test), so $ab^{-1} \in K$. Thus, by the subgroup test, K is a subgroup of G

Therefore, the following is well-defined:

Definition 2.3.16: Minimal Subgroup via Generation

let G be a group and $A \subseteq G$. Then

$$\langle A \rangle = \bigcap_{\substack{H \leqslant G \\ A \subseteq H}} H$$

is called the *closure of* A, or the subgroup *generated by* A. If A is a finite set $\{a_1, a_2, ..., a_n\}$, we can write $\langle a_1, a_2, ..., a_n \rangle$. Furthermore, $\langle A, B \rangle := \langle A \cup B \rangle$ for the union of two sets.

We have already been using this notation when $A = \{x\}$ is a singleton, in particular cyclic subgroups are the closure of singletons. Note that $\langle A \rangle$ always exists, since at minimum $A \subseteq G$ (and naturally $G \leqslant G$), so G is part of the sets in be intersection. This is important to point out as there exists sets where $\langle A \rangle = G$:

Definition 2.3.17: Generating Set and Generators

Let $A \subseteq G$. Then if $\langle A \rangle = G$, then A is called a *generating set* for G, and the elements of A are called *generators*.

If $A = \emptyset$, then A is containing in every group, including the trivial group $\{e_G\}$. As the intersection $\{e_G\}$ with any group is $\{e_G\}$, we see that $\langle \emptyset \rangle = \{e_G\}$.

Proposition 2.3.18: Minimal Subgroup using Generation

Let $A \subseteq G$. Then $\langle A \rangle \leqslant G$ is the unique smallest subgroup containing A.

***Proof*:**
If $H' \leqslant G$ and $A \subseteq H'$, then by definition H' is one of the subgroup that were intersected in the aforementioned definition, since $\langle A \rangle$ is equal to this intersection, $\langle A \rangle \subseteq H'$, so $\langle A \rangle$ is minimal.

Next, $\langle A \rangle$ is the unique smallest subgroup containing A (similarly to how $\langle x \rangle$ is unique in the sense that all cyclic subgroups of the same order are isomorphic to it). If there exists another group K' that was also the smallest subgroup containing A, then K' would be in the intersection above, so $\langle A \rangle \subseteq K'$. But since K' is supposed to be the smallest subgroup containing A, it is also the case that $K' \subseteq \langle A \rangle$. Therefore $\langle A \rangle = K'$, showing uniqueness as well, completing the proof.

The limit with this approach is that all the possible subgroups containing A must be found, and that is not always feasible. There isn't a general method to find elements of $\langle A \rangle$ (except, naturally, that $e_G \in A$). The 2nd approach will give a better insight into this problem by changing perspective away from the high-level and starting with the elements of A and trying to "generate" a subgroup containing A. To that end, define the set

$$\overline{A} := \{a_1^{\sigma_1} a_2^{\sigma_2} \cdots a_n^{\sigma_n} \mid n \in \mathbb{Z} \text{ and } a_i \in A,\ \sigma_i = \pm 1, 1 \leq i \leq n\}$$

where if $A = \emptyset$ then $\overline{A} = \{e_G\}$. The set $\overline{A}$ contains all possible finite products (using the binary operation of G) of the elements from A. Note that the a_i's need not be distinct, for example a^2 for $a \in A$ would be written $a^1 a^1$. It is also easy to verify that this set is indeed a subgroup of G using the subgroup criterion (consider that n can vary in any possible length, see exercise 2.3.1(2.3.1 (1))). The interesting question is whether this set is also the smallest minimal subgroup containing A, and indeed it is:

Proposition 2.3.19

Let G be a group and $A \subseteq G$ be a subset of G. Then

$$\langle A \rangle = \overline{A}$$

***Proof*:**
exercise; here is an outline:

1. Let $x \in \overline{A}$ so that they both can be written as:

$$x = a_1^{\sigma_1} \cdots a_n^{\sigma_n}$$

   By using closure properties, show that $x \in \langle A \rangle$.

2. Conversely, assume $x \in \langle A \rangle$ so that $x \in H_A$ for each H containing A. For the sake of contradiction, assume that x is not of the above form. Show then that $\overline{A} \subsetneq \langle A \rangle$ contradicting minimality of $\langle A \rangle$.

From now on, the notation $\langle A \rangle$ will be used to represent the subgroup generated by A.

We can ask some basic question about $\langle A \rangle$ given A and G. If G is abelian, then the order the elements are written in doesn't matter, and so if $A = \{a_1, ..., a_k\}$, every element can be written as:

$$a_1^{n_1} a_2^{n_2} \cdots a_k^{n_k}$$

This is not the case if G is not abelian. Consider for example:

$$G = \langle a, b \mid a^2 = b^2 = 1 \rangle \tag{2.1}$$

Notably, $ab \neq ba$. Next, if each $a_i \in A$ has finite order, then if G is abelian we have that G is finite. This is because:

$$(ab)^n = a^n = b^n$$

See exercise 2.3.1(2.3.1 (2)). This is not necessarily the case if G is non-abelian. Consider again the group in equation (2.1) where $|a| = 2$ and $|b| = 2$. Then we see that:

$$abababa....$$

If G is not abelian, then as mentioned in remark 1.3.5 and example 1.3.6, it is exceedingly difficult to determine the structure of G.

We may also ask if there is any "optimal" set of generators for a group, that is a "best" subset $A \subseteq G$ st $\langle A \rangle = G$. Generally, there can be many sets that generate a group: for example $\langle r, \sigma \rangle = \langle r\sigma, \sigma \rangle = D_n$, or $\langle 2, 3 \rangle = \langle 7, 5 \rangle = \mathbb{Z}/10\mathbb{Z}$. Notice that no proper subset of $\{2, 3\}$ generates $\mathbb{Z}/10\mathbb{Z}$, however there do exist smaller subsets of $\mathbb{Z}/10\mathbb{Z}$ that generate $\mathbb{Z}/10\mathbb{Z}$ (namely, $\{1\}$). Thus, the cardinality of the minimal generating subset cannot be determined by simply finding a minimal generating subset: it would have to be $\min \{|A| \mid A \subseteq G,\ \langle A \rangle = G\}$[3].

Notice also that in the case of D_4 with generators $a = \sigma$ and $b = r\sigma$ have order 2, while D_4 has order 8. Using this show how it is not always the case that all elements of D_4 will be written of the form $a^\alpha b^\beta$ $(\alpha, \beta \in \mathbb{Z})$ like in the abelian case, in particular show that aba cannot be written as $a^\alpha b^\beta$ for any $\alpha, \beta \in \mathbb{Z}$.

[3] If G is infinite, it might be that we require the stronger notion of an Infimum: $\inf \{|A| \mid A \subseteq G,\ \langle A \rangle = G\}$

2.3.20 Food for Thought If G is finitely generated, is every subgroup of G finitely generated? In other words, if

$$G = \langle S|R\rangle$$

where S is a set of elements and R is the relation on those elements, then can you find a subgroup $H \leqslant G$ such that you need an infinite number of elements to generate H? There is an answer to this question which will be given as an exercise 2.3.2(2.3.2 (1)), but it can be useful to contemplate this question now as an exercise in the types of questions a mathematicians might ask themselves. You can further ask if $\varphi : G \to H$ is a homomorphsims, and G is finitely generated, is $\varphi(G)$ finitely generated?

Exercise 2.3.1

1. Show that $\overline{A}$ is indeed a subgroup
2. Let G be abelian and $A \subseteq G$ a finite set containing only elements of finite order. Show that $\langle A\rangle$ is finite.
3. Show that every finitely generated subgroup of $\mathbb{Q}$ is cyclic
4. Let (Presentation for V_4. This group is called the Klein 4-group, or *Viergruppe*. Prove that $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, showing that a group has a non-unique representation.
5. If $H, K \leqslant G$, is $H \cup K$ a group? If $H \cup K$ is a subgroup in some case, what relation would there be between H and K (does on include the other?)
6. Let $G = k[x, 1/x]$ be the group of polynomials under multiplication.

2.3.3 Free Groups

(somewhere define "formal combination". Will need it for defining polynomial rings and tensors, and other free objects more generally)

So far, when we were generating groups from sets, we either took a subset of G (and so took the relation between elements of G), or we defined the set given a presentation. For example, We can make a group from $\{a, b\}$ by defining [4]

$$\langle a, b|a^n = 1, b^2 = 1, ab = b^{-1}a\rangle$$

what happens if we get rid of one or two of these relations? What about if we get rid of all three – we make a group "free" of relations besides necessary the ones needed to form a group? We explore this question in this section.

Since this group has no restrictions, it can be thought of as the "most general" group on that number of elements. In fact, something more subtle is going on here: since $\{a, b\}$ (or any set A to that matter) is essentially an arbitrary set, we're asking if from *any* arbitrary set, is there a way of defining a group using as little "extra group structure" as possible, or put another way, as free from relations as possible. If we can define such a group given $\{a, b\}$, is there a way of "defining" another group generated by two elements using this more "free group" by adding relations to the "free group"?

[4] This is D_n

The answers to this question lies in exploring how to make a *function* (or to be pedantic a *set-function*) from any set A to a group G

$$f : A \to G$$

be "lifted" to a *group-homomorphism*

$$\varphi : F(A) \to G$$

where $F(A)$ will be our "free group" on A (the notation $F(A)$ is suggestive that this group will indeed be called a free group). The word "lifted" here means that if we restrict $F(A)$ to A, then $\varphi|_A = f$. This can be visualized with the following diagram (called a commutative diagram):

$$\begin{tikzcd} F(A) \arrow[r] & G \\ A \arrow[u] \arrow[ur] & \end{tikzcd}$$

Thus, we turn a set function into a group-homomorphism while still keeping the structure of the set-function. More generally, if we have $f : A \to G_2$, then if there exists a function $f_2 : A \to G_1$ and a homomorphism $\varphi : G_1 \to G_2$ such that $f_2 = \varphi \circ f_1$, then φ is called a lift of f_1. Equivalently, φ is a lift of f_1 is the following diagram commutes:

$$\begin{tikzcd} G_1 \arrow[r, "\varphi"] & G_2 \\ A \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

Why are lifts the key concept? It is because not all set-function can be lifted to group homomorphism:

Example 2.3.21: Lifts and non-Lifts

1. Let $A = \{a\}$, $G_1 = \mathbb{Z}/5\mathbb{Z}$, and $G_2 = \mathbb{R}$. Define
   (a) $f_1 : A \to G_1,\ f_1(a) = 0$
   (b) $f_2 : A \to G_2,\ f_2(a) = 4$

   Does A lift to G_1, that is:

$$\begin{tikzcd} \mathbb{Z}/5\mathbb{Z} \arrow[r] & \mathbb{R} \\ \{a\} \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

   that is, does there exist a homorphism $\varphi : \mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$ such that $f_2 = \varphi \circ f_1$ If we can find a homomorphism φ such that

$$\begin{tikzcd} \mathbb{Z}/5\mathbb{Z} \arrow[r, "\varphi"] & \mathbb{R} \\ \{a\} \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

   then a lift exists. Let's say there was such a φ. Then we have that

$$4 = f_2(a) = (\varphi \circ f_1)(a) = \varphi(0) = 0$$

that is $4 = 0$ in $\mathbb{R}$. Note that since φ is a homomorphism, we are forced to have $\varphi(0) = 0$ as we showed in proposition 1.4.4. However, we already know that $4 \neq 0$. Thus, no such homomorphism can exist. Therefore, these is lift from $\{a\} \xrightarrow{f_1} \mathbb{R}$ to $\mathbb{Z}/5\mathbb{Z} \to \mathbb{R}$

2. What if we swap 0 and 4? Then:

$$0 = f_2(a) = (\varphi \circ f_1(a)) = \varphi(4)$$

Before, we were forced to say $\varphi(0) = 0$. However, this time we are only restricted to saying $\varphi(4) = 0$. There does indeed exist a homomorphism from $\mathbb{Z}/5\mathbb{Z}$ to $\mathbb{R}$ which satisfies this property: $\varphi(x) = 0$ (i.e., the trivial homomorphism). No other homomorphism can exist, since if $\varphi(4) \neq 0$, then $0 = \varphi(0) = \varphi(4+4+4+4+4) = 5 \cdot \varphi(4) \neq 0$ – a contradiction.

3. Let's take the same example as before, but let $f_1'(a) = 1$. Like last time, let's assume a homomorphism φ existed such that $f_2 = \varphi \circ f_1'$. Thus

$$1 = f_2(a) = (\varphi \circ f_1')(a) = \varphi(1)$$

Thus, $\varphi(1) = 1$. As a consequence

$$\varphi(1+1) = \varphi(1) + \varphi(1) = 1 + 1 = 2 \cdots$$

meaning $\varphi(n) = n$. However, this implies that

$$0 = \varphi([0]) = \varphi([5]) = 5$$

but $0 \neq 5$. Thus, no such homomorphism exists, and so there is no morphism from (f_1', G_1) to (f_2, G_2).

4. Let $G_1 = G_2 = \mathbb{Z}/5\mathbb{Z}$ and let $A = \{a\}$. Let

   (a) $f_1 : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$, $f_1(a) = 1$

   (b) $f_2 : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$, $f_2(a) = 2$

   (TBD: want to show how there can be more than 1 (non-trivial) morphism between these two objects of $\mathcal{F}^A$)

So what group can take the place of $F(A)$ (if any) that will always satisfy a lift? Perhaps it is a group we have already encountered. We have shown in example 1.3.4 and section 1.2.2 how to define the trivial and symmetric group from arbitrary sets. Let's label the trivial group on A as $T(A)$, and the symmetric group on A as S_A. Though this shows that there is more than one way of defining a group operation on arbitrary sets, this does not help us in finding the most "general" group that can be defined on a set. Let's take the earlier presentation of D_n. The trivial group has no information about this group. The group S_2 has 2 element: the trivial element and (1 2). However, $|D_n| = 2n$, meaning it has many more elements than S_2. Thus, such a group cannot be used as the "most general group on the set $\{a, b\}$"

In terms of intuition in relation to lifts, we see that:

$$A \to G \implies T(A) \to G$$

will not in general be a lift (choose some set $|A| > 1$ and a nontrivial group G). Similarly,

$$A \to G \implies S_A \to G$$

will not be a lift in general.

The problem with both these examples is that they in fact define a lot of structure on their underlying set. Instead, we will try to make a structure which puts the minimal amount of relations between objects to make it a group so that hopefully $F(A) \to G$ can always be a lift! Before I show the answer, I encourage you to stop reading for a moment try to think of another way you can define a group on some arbitrary set A, and perhaps think about how this group can be thought of as the "most general".

The High-level Idea

The first time you'll see this construction, you might think we added a lot of fluff in the way of the answer. However, I assure the reader that the longer build-up will have larger rewards! The construction we're doing here will repeat itself multiple times in the book in subtly different ways. This longer construction allows us to link all these constructions as examples of *free objects* [5], and more generally *universal properties* [6] later down the line. This will be the first example in the book of a universal property, it's the *universal property of free groups.*

Let's say A is an arbitrary set. Our goal is to construct a group, which we will label $F(A)$, from which every other group which contains A can be derived. First, we will be "relating" A to any group G. Let's take all set-functions:

$$f : A \to G$$

Where A is a set that maps into the group G [7]. Let's label this function along with its co-domain G as (f, G). Choosing all possible f and G will make all possible functions from A to every possible group, essentially telling us all ways A relates to any group G. The fact that there is no restriction on the functions from A to G is a way of saying there is *no* group structure on A which limits the types of maps we can form. Remember that when we lift the map $f : A \to G$ to $\varphi : F(A) \to G$, we need that $\varphi|_A = f$, so the fact that the map can be arbitrary means that $F(A)$ must accommodate for any function from A to G. As we have shown in example 2.3.21, not all groups can accommodate this. Let's label the set of all possible (f, G) as $\mathcal{F}^A$:

$$\mathcal{F}^A = \{(f, G) | f : A \to G\}$$

Next, we want to be able to relate the groups in the co-domain. We want to define $F(A)$ to be able to lift from a set-function to a group-homomorphsims. What we will do is define the following relation between any two set's (f_1, G_1), (f_2, G_2): we will say there is a *morphism* between (f_1, G_1), (f_2, G_2)

$$(f_1, G_1) \xrightarrow{\varphi} (f_2, G_2)$$

if the following diagram commutes:

[5] If you're curious to know where else we will encounter free objects, we will see *free monoids* in section 7.1 (which will turn out to be isomorphic to the set of all polynomials!), and *free-modules* in section 12.3.2

[6] You'll see these everywhere, to name a few: theorem 3.1.14, theorem ref:HERE (product), theorem ref:HERE (coproduct), proposition 9.2.4, theorem 9.5.4, theorem 15.3.4, theorem 15.1.12, and more!

[7] It might be more intuition for every f to be *injective*, since this way we'd get that every G contains the set A. We can indeed do this, and we will actually get the same result. For reasons that will be elaborated later in this section, the more general approach is desirable to be able to repeat this trick later in the textbook

$$\begin{tikzcd} G_1 \arrow[r, "\varphi"] & G_2 \\ A \arrow[u, "f_1"] \arrow[ur, "f_2"'] & \end{tikzcd}$$

that is, where $f_2 = f_1 \circ \varphi$. You can think of the morphism φ as a homomorphism that is restricted by f_1 and f_2. In fact, every example in examples 2.3.21 can be re-written to be morphisms from $\mathcal{F}^A$. Take a moment to go back and translate the previous notation to the new one, and think about what $\mathcal{F}^A$ with its objects and morphisms represent. We've essentially taken any group homomorphism between any two groups $G_1 \xrightarrow{\varphi} G_2$ and also made a relation between A and these groups. In fact, as reader's who have read section 0.4 and 8 might have guessed, we have defined a category

Proposition 2.3.22: $\mathcal{F}^A$ is a category

The set $\mathcal{F}^A$ along with the morphisms form a category (we can write $\mathfrak{F}^A S$ as a shorthand to represent $\mathcal{F}^A$ and it's morphisms). In particular:

1. For every $(f, G) \in \mathfrak{F}^A$, there exists an identity morphism $1_{(f,G)} : (f, G) \to (f, G)$
2. If $\varphi, \psi \in \mathfrak{F}^A$ are morphisms

$$\varphi : (f_1, G_1) \to (f_2, G_2) \qquad \psi : (f_2, G_2) \to (f_3, G_3)$$

then $\psi \circ \varphi$ is a morphism.

***Proof*:**
First, composition of morphisms will be a morphism in $\mathcal{F}$. If $\varphi, \psi \in \mathcal{F}$ are composable morphisms (i.e. the codomain of φ matches the domain of ψ) then

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[rr, bend left, "\psi\circ\varphi"] & H \arrow[r, "\psi"] & K \\ & A \arrow[ul, "f"] \arrow[u, "f'"] \arrow[ur, "f''"'] & \end{tikzcd} \quad \Rightarrow \quad \begin{tikzcd} G \arrow[rr, "\psi\circ\varphi"] & & K \\ & A \arrow[ul, "f"] \arrow[ur, "f''"'] & \end{tikzcd}$$

that is, the morphism $\psi \circ \varphi \in \mathcal{F}^A$ with domain (f, G) and codomain (f'', K) exists

Next, we will check each object $(f, G) \in \mathcal{F}^A$ has an identity morphism. Take an arbitrary $(f, G) \in \mathcal{F}^A$, so that $A \xrightarrow{f} G$. Then the homomorphism $\mathrm{id}_G : G \to G$, $\mathrm{id}_G(x) = x$ satisfies

$$\begin{tikzcd} G \arrow[r, "\mathrm{id}_G"] & G \\ A \arrow[u, "f"] \arrow[ur, "f"'] & \end{tikzcd}$$

If we take any $\varphi \in \mathcal{F}^A$ where $\varphi : (f, G) \to (f', H)$, then $\mathrm{id}_H \varphi = \varphi \mathrm{id}_G = \varphi$, showing that identities exist.

The fact that any compossible triple is associative comes from function composition.

With the category $\mathfrak{F}^A$ finally defined, we can ask the following question

Does $\mathfrak{F}^A$ have an initial object[8]?

That is, does there exists (f^*, G^*) such that for any object $(f, G) \in \mathfrak{F}^A$, there exists a unique morphism φ^* such that

$$\begin{tikzcd} G^* \arrow[r, "\varphi^*"] & G \\ A \arrow[u, "f^*"] \arrow[ur, "f"'] & \end{tikzcd}$$

Notice that this is the same lifting question we have asked about a theoretical "free group" $F(A)$ near the beginning of this section, but re-phrased! You might ask if such a group even exists? If such a group does exist, then have found a group which let's us relate it to any other group. We will call such a group (if it exists) a *free group*. Essentially, $F(A)$ must be able to relate to any other group, making it the "most general" group. But what is $F(A)$? In the following section, we will construct such a group that will indeed be the initial object of $\mathfrak{F}^A$.

Constructions of $F(A)$

Let's first give the answer to this question when $A = \{a\}$. Then the group $F(\{a\})$ is $\mathbb{Z}$. and set map $j : \{a\} \to \mathbb{Z}$ is $j(a) = 1$. Let (f, G) be any element of $\mathcal{F}^A$. Let's say $f(a) = g$. Then $g = f(a) = (\varphi \circ j)(a) = \varphi(j(a)) = \varphi(1)$. That is, the generator of $\mathbb{Z}$ maps to g. By proposition ref:HERE,

$$\varphi(n) = g^n$$

and since this was forced by proposition ref:HERE, φ is unique.

In the case of $\emptyset$, it would be $\{0\}$, which checks out since we saw that $\{0\}$ is the initial objects of groups (proposition 1.4.5). In the case were $A = \{a\}$ (that is, an arbitrary singleton). Then HERE ($\mathbb{Z}$).

For the more general case, we will define a new group.

Let A be an arbitrary set. For every element $a \in A$, let it correspond to some element $b \in B$. For the sake of what's to come, we will instead of label these element a^{-1}, and the set they belong to A'. Then, define a *word* to be any ordered tuple

$$(a_1, a_2, ..., a_n) \qquad a_i \in A \cup A'$$

We usually write a word as the literal putting together of letters side by side:

$$a_1 a_2 \cdots a_n$$

Definition 2.3.23: Word

Let A be some set. Then let $W(A)$ denote all possible combinations of *words*. TBD

[8] see definition 1.4.6

Example 2.3.24: Example

1. here
2. here

Define reduction, define reduced word,

Lemma 2.3.25: reduced word lemma

reduced word lemma

***Proof*:**
here

Proposition 2.3.26: Word Group

$(W(A), R)$ is a group

***Proof*:**

1. (associativity)
2. ($\exists$ identity)
3. ($\exists$ inverse)

Theorem 2.3.27: Universal Property Of Free Groups

The following are each the same statement, just different ways of stating it. Let $f : A \to G$ be a function from a set to a group, $i : A \to F(A)$ to be $i(a) = a$ (i.e. $i : A \hookrightarrow F(A)$), and $F(A) = (W(A), R)$.

1. $(i, F(A))$ is the initial object of $\mathfrak{F}$.
2. The set $(i, F(A))$ satisfies the universal property for free groups on A.
3. Then there exist a unique homomorphism $\varphi : F(A) \to G$ such that f factors through i, that is, $f = \varphi \circ i$, i.e., the following diagram commutes

$$\begin{tikzcd} F(A) \arrow[r, "\varphi"] & G \\ A \arrow[u, "i"] \arrow[ur, "f"'] & \end{tikzcd}$$

we call the group component of $(i, F(A))$ the *free group* on A.

***Proof*:**
We're essentially showing that given map $f : A \to G$, we can extend it to a map $\varphi : F(A) \to G$. So, let f be any map from A to some group G, and i be defined as in the theorem. TO define φ, we can use the information provided by f essentially completely. For any a and $a^{-1} \in A \cup A'$,

$$\varphi(a) = f(a) \qquad \varphi(a^{-1}) = f(a)^{-1}$$

notice that since A doesn't contain A, so the function f doesn't map a^{-1} anywhere. However, by construction, $F(A)$ does have a^{-1}, and so we define a place for it to map. To define φ on words with multiple letters by concatenating the result of the image, that is

$$\varphi(a_1 a_2 \cdots a_n) := \varphi(a_1)\varphi(a_2)\cdots\varphi(a_n)$$

This function is well-defined, since if $w = w'$, then

$$\varphi(w) = \varphi(a_1)\varphi(a_2)\cdots\varphi(a_n) = \varphi(w')$$

furthermore, this functions splits f since for any $a \in A$

$$f(a) = \varphi(i(a)) = \varphi(a)$$

which is true by definition

Now, notice that for any $w \in W(A)$, $\varphi(w) = \varphi(R(w))$. For example, if $w = abb^{-1}a^{-1}a$, then

$$\begin{aligned}
\varphi(w) &= \varphi(abb^{-1}a^{-1}a) \\
&= \varphi(a)\varphi(b)\varphi(b^{-1})\varphi(a^{-1})\varphi(a) \\
&= f(a)f(b)f(b)^{-1}f(a)^{-1}f(a) \\
&= f(a) \\
&= \varphi(a) \\
&= \varphi(R(w))
\end{aligned}$$

With this fact, we have that φ will be a homomorphism If $w, w' \in W(A)$, then

$$\varphi(w \cdot w') = \varphi(R(ww')) = \varphi(ww') = \varphi(w)\varphi(w')$$

establishing that $\varphi : F(A) \to G$ is a homomorphism!

Finally, we need to show that it's unique. This comes down to seeing that by the properties of homomorphisms, the way we've defined is the only way we can define a homomorphism from $F(A)$ to G. Let's say φ' was another homomorphsims from $F(A)$ to G. Then by the commutative diagram, it must be that

$$f(a) = \varphi'(i(a)) = \varphi'(a)$$

but that is exactly the definition of φ. The fact that φ' must split with f forces $\varphi = \varphi'$.

Since f an G was arbitrary, we showed that there always exists a unique homomorphism φ from $F(A)$ to G such that $f = \varphi \circ f$ – as we sought to show

Corollary 2.3.28: Free Group Is Unique

let A and B be sets such that $|A| = |B|$. Then

$$F(A) \cong F(B)$$

***Proof*:**
This comes down to properties of terminal objects in categories (initial objects are *a fortiori* terminal objects).

Since free groups on sets of the same cardinality are isomorphic, we usually write F_n to denote the free group with n generators.

Definition 2.3.29: Freely Generated

Let S be a set. Then we say that S *freely generates* a group G if $G = F(S)$.

Example 2.3.30: Free Groups

1. Earlier we said that $F(A) \cong \mathbb{Z}$. If we take $f : \{a\} \to \mathbb{Z}$, $f(a) = 1$, then by the universal property of free groups, there exists a group-homomorphism: $\varphi : F(A) \to \mathbb{Z}$, where $\varphi(1) = 1$. Then it's true that φ defines an isomorphism between $F(A)$ and $\mathbb{Z}$ (which you should check if it's unclear), and so
$$F(A) \cong \mathbb{Z}$$
2. The free set on 2 elements, say $F(\{x, y\})$ is a group we have encountered. The easiest way to show this group is by showing part of its Cayley graph:

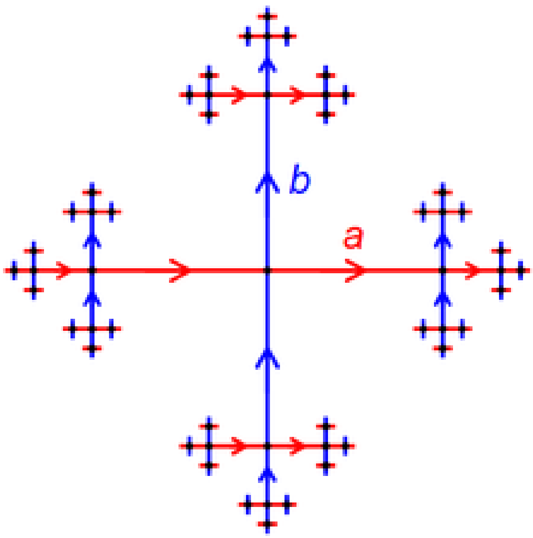


Figure 2.1: Cayley Graph of F_2

this graph in fact goes on forever. The length of the edges where halved every time to make the diagram nicer. [a]

[a] If you have done any algebraic topology, this is the fundamental group of the figure 8!

Some other properties from Clara Löh's book? Geometric group theory p. 17

2.3.4 Free Abelian Group

We can repeat this same construction, except limit our attention to abelian groups. In particular, given a set A an any map $f : A \to G$ to an abelian group, we want to lift it to an abelian group $F^{ab}(A)$ in the domain of the homomorphism $\varphi : F^{ab}(A) \to G$.

The construction is actually basically identical if we impose a commutativity condition on $F(A)$, that is, for any $a, b \in F^{ab}(A)$, $ab = ba$. If you were uncomfortable with the proof of the universal property of free group, it is worth going over it again, then go over it a 2nd time but with only abliean groups and $F^{ab}(A)$ in mind.

What we will be focusing on this section is finding a nicer group that is isomorphic to $F^{ab}(A)$.

As a closing remark on the construction of free and free abelian groups, (How it is plausible that $F(A)$ is empty or a singleton or something smaller than A so this idea of "lift" is not true. However, our construction at the end will show that it is indeed a lift. In many similar constructions in the book (ref:HERE), this construction will still be lifts, however we will encounter some (the first one in chapter 15 in which there is an initial object which is not a lift)

2.3.5 *Free-Forgetful Adjoint

Is this the right place to ask this question? It's asking whether we have really defined the "right" idea of general group. The way we make this idea rigorous is by asking if we have a functor $F : \textbf{Group} \to \textbf{Set}$, that forgets the structure, is there a functor $G : \textbf{Set} \to \textbf{Group}$ that "regains" some structure? It cannot regain all of it because sets don't have enough information for that, but is there a way in which we can make sure that our free construction is the "mirror-image" of forgetting the structure?

Build up to this: If $L(\_) : \textbf{Group} \to \textbf{Set}$ takes a group G and returns the underlying set: $\mathcal{L}((G, \cdot)) = G$, and $\mathcal{F}(\_) : \textbf{Set} \to \textbf{Group}$ takes a set and returns a group: $\mathcal{F}(S) = ((F(S), \cdot_{F(S)})$, then

$$\mathrm{Hom}_{\textbf{Set}}(S, \mathcal{L}(G)) \cong_{\textbf{Set}} \mathrm{Hom}_{\textbf{Group}}(\mathcal{F}(S), G)$$

where we there is $\cong_{\textbf{Set}}$ in the middle because Hom_{Set} doesn't have a group structure. If it did, we would ask it to preserve that too.

Exercise 2.3.2

1. Let $G = \langle a, b \rangle$ be the free (non-abelian) group generated on two elements. Show that $H = \langle a^n b a^{-n} \mid n \in \mathbb{Z} \rangle$ is an infinitely generated subgroup
2. (harder) Let $G + \mathrm{SL}_2(\mathbb{Z})$. Show it contains a subgroup isomorphic to the free subgroup with two elements.

2.4 Subgroup Lattices

In this section, we will show how to use graphs in order to visually see the subgroup structure of a group.

Before introducing a rigorous definition, let's first do an example: Let's do the subgroup lattice of Q_8. We must first know all the subgroups of Q_8. With some computation, the subgroups that are found are

$$1,\ \langle -1\rangle,\ \langle i\rangle,\ \langle j\rangle,\ \langle k\rangle,\ Q_8$$

where $1 = \{e\}$. By the definition of Q_8, any other set of generators will fall into these groups. For example, notice that $\langle i, j\rangle = \langle k\rangle$ (similarly for i, k and j, k), and $\langle i, j, k\rangle = Q_8$. To construct the lattice, start by imagining Q_8 on the "top" and the identity group on the "bottom"‘

$$\begin{array}{c} Q_8 \\ \\ \\ \\ 1 \end{array}$$

then, apply the first rule of doing the lattice:

1. draw a line if one group is a subgroup of another group.

Since $1 \leqslant Q_8$, then

$$\begin{tikzcd} Q_8 \arrow[d, dash] \\ 1 \end{tikzcd}$$

Next, pick one of the groups, let's say it's $\langle -1\rangle$. Now apply the second rule of drawing a lattice:

2. if $H \leqslant K \leqslant G$, then remove the line from H to G (if one is already drawn) and add a line from H to K, and K to G. In other words, only draw a line between subgroups A and B if and only if there is no group H such that $A \leqslant H \leqslant B$.

Since $1 \leqslant \langle -1\rangle \leqslant Q_8$, our diagram will now look like this:

$$\begin{tikzcd} Q_8 \arrow[d, dash] \\ \langle -1\rangle \arrow[d, dash] \\ 1 \end{tikzcd}$$

similarly, we can add $\langle i \rangle$ like so:

$$\begin{tikzcd}
Q_8 \arrow[d, dash] \\
\langle i \rangle \arrow[d, dash] \\
\langle -1 \rangle \arrow[d, dash] \\
1
\end{tikzcd}$$

Next, notice that $\langle i \rangle \nleqslant \langle j \rangle$, and $\langle j \rangle \nleqslant \langle i \rangle$, however both $\langle i \rangle$ and $\langle j \rangle$ have $\langle -1 \rangle$ as a subgroup and are both subgroups of Q_8. The same situation occurs for $\langle k \rangle$. Thus, the updated lattice will be:

$$\begin{tikzcd}
 & Q_8 \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle i \rangle \arrow[dr, dash] & \langle j \rangle \arrow[d, dash] & \langle k \rangle \arrow[dl, dash] \\
 & \langle -1 \rangle \arrow[d, dash] & \\
 & 1 &
\end{tikzcd}$$

since we have exhausted our list of possible subgroups, we have finished the subgroup lattice of Q_8.

More generally, the orientation doesn't matter (we could've had 1 on top and Q_8 on bottom, or going from left to right). What matter's is we know how to translate a line into an appropriate subgroup (i.e. If 1 is on top and Q_8 is on bottom, then a line now becomes a supergroup inclusion $H \geqslant G$). Note also that the particular shape the lattice took doesn't really matter, a pretty diamond-like shape was chosen to make the lattice more readable.

If G is an infinite group, then a lattice doesn't always exist. For example, if $G = \mathbb{Q}$, then for any group $1 \leqslant H \leqslant G$, there exists a group K such that $1 \leqslant K \leqslant H$, meaning we cannot follows condition (2). Thus, we usually work with finite groups. Certain infinite groups permit subgroup lattices (see exercise ref:HERE)

With this build-up, we can formally define a subgroup lattice:

Definition 2.4.1: Group Lattice

Let G be a finite group and $\mathcal{G}$ be the collection of subgroups of G. Then $\Gamma = (\mathcal{G}, E_{\leqslant})$ is the graph where there is an edge between a group $A, B \in \mathcal{G}$ if and only if there does not exist a group K such that $A \leqslant K \leqslant B$. When Γ is visualized, we put G on top and the identity group 1 on bottom.

Example 2.4.2: Group Lattices

Many examples, and the problem's that might arise with infinite lattices, and how some infinite groups permit lattices

1. Let $G = \mathbb{Z}/2\mathbb{Z} = C_n$. Then by theorem ref:HERE, the subgroup lattice is easily determined by the divisors of n. Thus ,we get:

$$\begin{tikzcd}
 & & \mathbb{Z}/8\mathbb{Z} \arrow[d, dash] \\
 & \mathbb{Z}/4\mathbb{Z} \arrow[d, dash] & \langle 2\rangle \arrow[d, dash] \\
\mathbb{Z}/2\mathbb{Z} \arrow[d] & \langle 2\rangle \arrow[d, dash] & \langle 4\rangle \arrow[d, dash] \\
0 & 0 & 0
\end{tikzcd}$$

   Note that $\mathbb{Z}/2\mathbb{Z} = \langle 1\rangle$, or $\mathbb{Z}/4\mathbb{Z} = \langle 1\rangle = \langle 3\rangle$, or that $\mathbb{Z}/8\mathbb{Z} = \langle 1\rangle = \langle 3\rangle = \langle 5\rangle = \langle 7\rangle$

2. More generally, if $G = \mathbb{Z}/p^n\mathbb{Z}$, then

$$\begin{tikzcd}
\mathbb{Z}/p^n\mathbb{Z} = \langle 1\rangle \arrow[d, dash] \\
\langle p\rangle \arrow[d, dash] \\
\langle p^2\rangle \arrow[d, dash] \\
\vdots \arrow[d, dash] \\
\langle p^n\rangle = 0
\end{tikzcd}$$

3. If $G = \mathbb{Z}/n\mathbb{Z}$ where n is a non-prime power, we can get some interesting diagram. For example:

$$\begin{tikzcd}
 & \mathbb{Z}/20\mathbb{Z} \arrow[dl, dash] \arrow[d, dash] \\
2 \arrow[d, dash] \arrow[dr, dash] & 5 \arrow[d, dash] \\
4 \arrow[dr, dash] & 10 \arrow[d, dash] \\
 & 0
\end{tikzcd}$$

4. Let $G = V_4$ (the Klein 4 group from exercise ref:HERE). Then

$$\begin{tikzcd} & V_4 \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\ \langle a \rangle \arrow[dr, dash] & \langle b \rangle \arrow[d] & \langle c \rangle \arrow[dl, dash] \\ & 0 & \end{tikzcd}$$

   Notice that this is the same lattice as the Quaternions (i.e. Q_8) that we have shown at the beginning. However, $V_4 \ncong Q_8$ (worth doing as an exercise). This shows that having the same group lattice is not a sufficient condition for two groups to be isomorphic (though it is a necessary condition)

Theorem 2.4.3: Subgroup And Sublattices

Let $H \leqslant G$. Then the lattice of H is the sublattice of G of groups contained in H

***Proof*:**
This fact comes straight from the definition of a lattice. Since $H \leqslant G$, then if the subgroup lattice of H where to differ from the subgroup lattice of G whose subgroups are contained in H, then the lattice of G would be wrong – a contradiction.

Example 2.4.4: Example
Give D_4, Q_8, $\mathbb{Z}/n\mathbb{Z}$.

(TBD on phrasing) We can also ask ourselves if homomorphisms preserve some group lattice. In particular, if $\varphi : G \to H$ is a homomorphism, and $\varphi(G)$ is the image of φ, can we say something about the group lattice of $\varphi(G)$ given the group lattice of G? We know that $\varphi(G)$ will be sublattice of H by theorem 2.4.3. However, finding the lattice of $\varphi(G)$ is a little more tricky. We will recieve our answer next chapter – the name of the theorem giving the answer will be the 4th isomorphism theorem.

2.4.5 Foreshadowing $\varphi(G)$ will be have "dual" to the subgroup lattice condition. That is, while subgroup's will have a sublattice of groups contained in H, $\varphi(G)$ will have the subgroup lattice of groups *contained* within a special group (which we will discover in chapter 3

Exercise 2.4.1

1. Let $G = \mathbb{Q}$. If $1 \leqslant H \leqslant G$, show that there always exists a group H' such that $1 \leqslant 1 \leqslant H' \leqslant H$.

2.5 *Universal Property of kernel and coKernels

This section can be comfortably skipped, as its contents will not come back until chapter ref:HERE. This section covers how to think about the kernel as a universal object. As you can probably guess by the fact that there is a universal property, the concept of the kernel of a homomorphism is not limited to groups. They will appear often in other sections. Most importantly, in chapter ref:HERE, (why do we need kernels when working with derived categories)

In the following seciton, $0 : G \to H$ will represent the homomorphism $\varphi(G) = e_H[= 0]$. This is indeed a homomorphism (as we've seen in proposition 1.4.5).

Theorem 2.5.1: Universal Property Of The Kernel

Let $\varphi : G \to H$ be a homomorphism. Then the inclusion map $i : \ker(\varphi) \hookrightarrow H$ is the final object in the category of group-homomorphsism $\alpha : G \to H$ such that $\varphi \circ \alpha$ is the trivial map. In other words, every group homomorphism $\alpha : K \to G$ such that $\varphi \circ \alpha$ is trivial factor through

$$\begin{tikzcd} K \arrow[r, "\alpha"'] \arrow[rr, bend left, "0"] \arrow[dr, "\exists!\alpha'"'] & G \arrow[r, "\varphi"'] & H \\ & \ker(\varphi) \arrow[u, hook, "i"'] & \end{tikzcd}$$

(put this somewhere as a reference to double-check if they got the category right?) Let $C_{\varphi,K}$ be the category who's objects be homomorphsism $\alpha : K \to G$ such that

$$\begin{tikzcd} K \arrow[r, "\alpha"'] \arrow[rr, bend left, "0"] & G \arrow[r, "\varphi"'] & H \end{tikzcd}$$

If (K, α) and (J, β) are two objects in this category, define the morphism $\gamma : (K, \alpha) \to (J, \beta)$ to be a homomorphism from K to J such that

$$\begin{tikzcd} K \arrow[r, "\alpha"'] \arrow[rr, bend left, "0"] \arrow[dr, "\gamma"'] & G \arrow[r, "\varphi"'] & H \\ & J \arrow[u, "\beta"'] & \end{tikzcd}$$

commutes.

***Proof*:**
Let $\alpha : K \to G$ such that $\varphi \circ \alpha = 0$, that is

$$\varphi(\alpha(K)) = 0$$

then $\alpha(K) \subseteq \ker(\varphi)$. If there was a homomorphism $\overline{\alpha}K \to \ker(\varphi)$, it would have to factor through:

$$\alpha = i \circ \overline{\alpha}$$

but this forces $\overline{\alpha}$ to be α restricted to K, forcing uniqueness and existnece.

(Also explain the idea of the dual question, or that it exists), namely

Theorem 2.5.2: Universal Property Of The coKernel

Let $\varphi : G \to H$ be a homomorphism. Then the inclusion map $i : \ker(\varphi) \hookrightarrow H$ is the final object in the category of group-homomorphsism $\alpha : G \to H$ such that $\varphi \circ \alpha$ is the trivial map. In other words, every group homomorphism $\alpha : K \to G$ such that $\varphi \circ \alpha$ is trivial factor through

$$\begin{tikzcd}
G \arrow[r, "\varphi"'] \arrow[rr, bend left, "0"] & H \arrow[r, "\beta"'] \arrow[d, "\pi"] & Q \\
& G/H \arrow[ur, "\exists!\beta'"'] &
\end{tikzcd}$$

which we will anwser in chapter 3?

Also, for cokernel, this useful exact sequence might be helpful:

$$0 \to \ker(f) \to V \xrightarrow{f} W \to \operatorname{coker}(f) \to 0$$

3

Homomorphisms and Quotient Groups

In this chapter, we take a closer look at homomorphisms and how they relate different groups. We start our exploration by getting a better understanding of the image of a homomorphsim. For example, if $\varphi : G \to H$ is a homomorphism, then we saw that $\varphi(G)$ is a group (re-prove this if it is not clear). Since φ is a function, then as we saw in φ making $(G/\sim_\varphi)$ bijective to $\varphi(G)$ (in particular, $x \sim_\varphi y \;\Leftrightarrow\; \varphi(x) = \varphi(y)$), which we can see by the φ making $(G/\sim_\varphi)$ bijective to H (in particular, $x \sim_\varphi y \;\Leftrightarrow\; \varphi(x) = \varphi(y)$), which we can see by the function:

$$[g] \mapsto \varphi(g)$$

The collection $(G/\sim_\varphi)$ is called the *fibers* of φ[1]. Since there is a bijection between $\varphi(G)$ and $(G/\sim_\varphi)$, and $\varphi(G)$ is a group, can we define a group structure on $(G/\sim_\varphi)$? Furthermore, is there some "minimal" and "unique" group structure that we can put on $(G/\sim_\varphi)$ so that there is a cannonical identification between $\varphi(G)$ and $(G/\sim_\varphi)$; those comfortable with the category theory presented so far will recognize this as asking for the universal properties of quotient objects in the category **Grp**. Due to this universal property, the name of the group structure on $(G/\sim_\varphi)$ satisfying the universal property will be the *quotient group*. The exploration of this question is at the center of the first section of this chapter. Due to the axioms of groups, it will turn out that there is a lot of "regularity" in the collection of fibers which allow for an easy representation of $(G/\sim_\varphi)$.

After finishing the first part, we will have developed the appropriate tools to get back to what we started in section 1.3 and define these notions more formally, i.e., how do we impose relations on groups. about algebraic structures more generally. about algebraic structures mor egenerally.

For the rest of the chapter, we will use our new theory to give us some very powerful results. We will see that when working with finite groups, the order of the group will give us a lot of information

[1]Given $\varphi : G \to H$ and an element $h \in H$, the fiber of h is all elements that map to h by φ

about it's subgroup structure, we will use our understanding of quotient groups to learn how to build some new types of groups, and we will learn 4 isomorphism theorems which give us lots of information on group structure.

We will end this chapter by introducing one of the most fundamental classification theorem in finite group theory: *Jordan Hölder's Program.* In the last few section, we will show how there are groups that can be considered the "building blocks" of any other finite groups, and show how we can put these groups together to form other groups.

3.1 Quotient Groups

Let $\varphi : G \to H$ be a homomorphism. We start by exploring the structure of $\varphi(G)$. Since $\varphi(G)$ is a set, we know by proposition 0.1.28 in the preliminary chapter that there is a bijection between $G/\sim_\varphi$ and $\varphi(G)$ such that for any function $\varphi : G \to H$ where $a \sim b \Leftrightarrow \varphi(a) = \varphi(b)$, we have

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[d, "\pi"'] & H \\ G/\sim \arrow[ur, "\bar{\varphi}"'] & \end{tikzcd}$$

We now try to see if there is an equivalent universal property for quotients in **Grp**. To that end, we would then want $G \xrightarrow{\pi} G/\sim$ to be a homomorphism, that is

$$\pi(a \cdot b) = \pi(a) \bullet \pi(b)$$

or, written in terms of equivalence relations,

$$[a \cdot b] = [a] \bullet [b]$$

We would need to have that $\bullet$ is independent of representation. That is, if we have $x, y \in [a]$, then we want

$$x \sim y \Leftrightarrow xb \sim xb \tag{3.1}$$

since we would also want that

$$[b \cdot a] = [b] \bullet [a]$$

we must have that

$$x \sim y \Leftrightarrow bx \sim by \tag{3.2}$$

These are necessary condition in order for $(G/\sim, \bullet)$ to be a group, since if once of these condition's fail, then we won't have a group. For examle, if $x \sim y \not\Rightarrow xb \sim yb$, then $[x][b] \neq [xb]$.

These are necessary conditions, since if $G/\sim$ was a group, then these must hold. We can ask if they are sufficient conditions, that is, condition's such that if they are followed, then $(G/\sim, \bullet)$ must be a group. These two condition's are also sufficient, meaning defining a group structure on $G/\sim$ is equivalent to the equivalence relation respecting these two properties:

Theorem 3.1.1: $(G/\sim, \bullet)$ is the quotient object of Grp

Let G be a group and $G/\sim$ be the set of equivalence relations given $\sim$. Then $G/\sim$ is a group under $\bullet$ if and only if

$$\forall g \in G,\ (x \sim y \Leftrightarrow gx \sim gy \text{ and } xg \sim yg)$$

Furthermore, If $\varphi : G \to H$ is another homomorphism such that $g \sim g' \Rightarrow \varphi(g) = \varphi(g')$, then there exists a homomorphism φ such that the following diagram commutes:

$$\begin{tikzcd} G/\sim \arrow[r, "\varphi"] & H \\ G \arrow[u, "\pi"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

We say that $\sim$ is *compatible* with the group-structure on G

***Proof*:**
The proof of the $\Rightarrow$ direction we have just showed above the theorem, so we'll show that these two equivalence relations are sufficient conditions for to define $(G/\sim, \bullet)$.

First, note that the operation $\bullet : G/\sim \times G/\sim \to G/\sim$ $[a] \bullet [b] = [ab]$ is well-defined, since if we take any other two representative $a' \in [a]$ and $b' \in [b]$, then since

$$b \sim b' \Leftrightarrow ab \sim ab' \text{ and } a'b \sim a'b'$$

and

$$a \sim a' \Leftrightarrow ab \sim a'b \text{ and } ab' \sim a'b'$$

Thus, putting pieces together using transitivity, we get

$$ab \sim ab' \sim a'b' \text{ or } ab \sim a'b \sim a'b'$$

both showing that $ab \sim a'b'$, meaning $\bullet$ is well-defined. Next, we need to show that $(G/\sim, \bullet)$ is indeed a group

1. Let $[a], [b], [c] \in G/\sim$. Then

$$\begin{aligned} ([a][b])[c] &= ([ab])[c] \\ &= [(ab)][c] \\ &= [(ab)c] \\ &= [a(bc)] \\ &= \vdots \\ &= [a]([b][c]) \end{aligned}$$

2. I claim that $[e_G] \in G/\sim$ is the identity. For any $[g] \in G/\sim$,

$$[g][e_G] = [ge_G] = [g] = [e_G g] = [e_G][g]$$

as we sought to show

3. Finally, for any $[g] \in G/\sim$, I claim that $[g]^{-1} = [g^{-1}]$. Indeed:

$$[g][g^{-1}] = [gg^{-1}] = [e_G]$$

showing that $[g^{-1}]$ is indeed the inverse of G

Thus, $(G/\sim, \bullet)$ is indeed a group.

Finally, we must show that G is universal with respect to homomorphisms $\varphi : G \to G'$ such that $g \sim g' \Rightarrow \varphi(g) = \varphi(g')$. Note that we already have that such a function exists, since φ is also a set function, so we already have that there exists a unique set-funciton $\varphi : G/\sim \to G'$ such that $\varphi([g]) = \varphi(g)$. Thus, we only need to check that φ is a homomorphism. We get this pretty much be definition:

$$\varphi([a][b]) = \varphi([ab]) = \varphi(ab) = \varphi(a)\varphi(b) = \varphi([a])\varphi([b])$$

showing that φ is indeed a homomorphism, and must be unique since there is only one possible set function (and hence homomorphism) that satisfies this property (there are no more functions to check).

Since $\bullet$ is the unique structure on $G/\sim$ that respects the universal property, we will drop it without worry Since $\bullet$ is the unique structure on $G/\sim$ that respects the universal propety, we will drop it without worry that there be any confusion, that is well write

$$[a][b] = [ab]$$

.

Definition 3.1.2: Quotient Group Via Congruence Relation

Let G be a group and $\varphi : G \to H$ a group homomorphism giving the induced equivalence relation $\sim_\varphi$. Then the group $G/\sim_\varphi$ from theorem 3.1.1 is called a *quotient group* and the equivalence relation $\sim_\varphi$ is called a *congrunce relation.*

The fact that equivalence relations that respect 3.1 and 3.2 are the only ones that are The fact that equivalence relations that respect 3.2 and 3.2 are the only ones that are congruence relations is very powerful, since this greatly limits what type of equivalence relations we have. For the 3.1, 3.2, or both, that is we shall find all possible congruence relations, as well as all possible one-sided congruence relations. We start by characterizing the equivalences classes of $G/\sim$ satisfying 3.1. We start by characterizing the equivalences classes of $G/\sim$ satisfying 3.1. there is any information we can figure out about the equivalence classes:

Proposition 3.1.3: Equivalence Classes Using property 3.1

Let $G/\sim$ be a quotient set given $\sim$ with proprety 3.1. Then

1. The equivalence class $K \in G/\sim$ where $e_G \in K$ is a subgroup of G ($K \leqslant G$)
2. Given any other equivalence calss $[g] \in G/\sim$, for any $a, b \in [g]$

$$a \sim b \;\Leftrightarrow\; a^{-1}b \in K \;\Leftrightarrow\; aK = bK$$

where the $aK = \{ak \mid k \in K\}$ and $bK = \{bk \mid k \in K\}$

Proof:

1. Since $K \subseteq G$, we need to show that for any $a, b \in K$, $ab^{-1} \in K$. Since $e_G \in K$, then

$$e_G \sim a$$

since $\sim$ is a congruence relation:

$$a^{-1} \sim e_G$$

using property 3.1. Using property 3.1 again, we get

$$a^{-1}b \sim b \sim e_G$$

where the last part comes from the transitivity of equivalence relations, and so

$$a^{-1}b \sim e_G \Rightarrow a^{-1}b \in K$$

meaning $K \leqslant G$. Note that if we worked with property 3.2, we would instead be showing that $ab^{-1} \in K$.

2. Let $[g]$ be an equivalence class and $a, b \in [g]$ so that $a \sim b$.

Since $\sim$ respects 3.1 $a^{-1}b \sim e_G$, implying $a^{-1}b \in K$. Next, take any element $x \in a^{-1}bK$. Then $x = a^{-1}bk$. Next, since $a^{-1}b \in K$ and $k \in K$, $x \in K$. Thus, $a^{-1}bK \subseteq K$. This implies that $bK \subseteq aK$, since if $x \in bK$, then $x = bk$, and so $x = a(a^{-1}bk)$, and since $a^{-1}b \in K$ and $k \in k$, we get $x = ak'$ for some $k' = a^{-1}bk \in K$, an so $x \in aK$. Using symmetry of $\sim$, we'll get that $aK \subseteq bK$, implying $aK = bK$. Thus, we showed that

$$a \sim b \Rightarrow a^{-1}b \in K \Rightarrow aK = bK$$

Note that if we were using property 3.2, then we'd get $Ka = Kb$ since we'd be multiplying on the right.

To make this chain of implication go full circle, assume $aK = bK$. By setting $k = e_G$, we get $ae_G = a \in bK$, and so $ab^{-1} \in K$. Since K is the equivalence class containing e_G, we have

$$ab^{-1} \sim e_G \Leftrightarrow a \sim b$$

by 3.1, thus showing that $aK = bK \Rightarrow a \sim b$, completing the loop of implication, thus showing that

$$a \sim b \Leftrightarrow a^{-1}b \in K \Leftrightarrow aK = bK$$

as we sought to show

This shows that for any equivalence relation satisfying 3.1 (and similarly 3.2), it's equivalence classes will be of the form aK resp. Ka aK Ka where a is the representative of an equivalence class and K is the equivalence class containing e_G. Thus, an equivalence class will determine a subgroup K which then determines a representation for the equivalence classes of $G/\sim$. equivalence class will determine a subgroup K which then determines the equivalence classes of $G/\sim$.

Corollary 3.1.4: Quotient Set Size

let G be a a finite group and $G/\sim$ be the quotient sets with respect to an equivalence relation $\sim$ satisfying equation 3.1 or 3.2 (or both). Then every quotient set is of the same size, that is

$$|aK| = |bK| \qquad \forall a, b \in G$$

***Proof*:**
exercise (see ref:HERE)

Since these cosets represent our equivalence relations, we will give a special name to these quotient sets:

Definition 3.1.5: Cosets

Let $H \leqslant G$. Then for every $g \in G$, we can define

$$gH = \{gh | h \in H\}$$

we call this a [left]-***coset***. We can also define the ***right***-coset:

$$Hg = \{hg | h \in H\}$$

If additive notation is used for G ($(G, +)$), it is common to denote the coset as

$$g + H \qquad H + g$$

The set of left-cosets is represented as G/H, an the set of right-cosets is represented as $G \setminus H$.

What equivalence relations that satisfies poposition 3.1.3? In other words, we're asking for a converse to proposition 3.1.3[1], which stated that if $\sim$ respected property 3.1 (or 3.2), then the equivalence class containing e_G is a subgroup. The answer is yes:

Proposition 3.1.6: Subgroups And Equivalence Classes

Let $H \leqslant G$ be a subgroup of G. Define the equivalence relation

$$a \sim_L b \Leftrightarrow a^{-1}b \in H$$

then $\sim_L$ respects property 3.1

***Proof*:**
Let's first show that $\sim_L$ is indeed an equivalence relation

1. If $a \sim a$, then $a^{-1}a = e \in H$, which is the case since H is a group
2. Let $a \sim b$. Then $a^{-1}b \in H$, implying $(a^{-1}b)^{-1} = b^{-1}a \in H$, implying $b \sim a$
3. Let $a \sim b$ and $b \sim c$. Then $a^{-1}b, b^{-1}c \in H$. Then $a^{-1}bb^{-1}c = a^{-1}c \in H$, implying $a \sim c$.

Thus, $\sim_L$ is indeed an equivalence relation.

To show it satisfies property 3.1, notice that for any $g \in G$

$$a \sim b \to a^{-1}b \in Ha^{-1}(g^{-1}g)b \in H \Rightarrow (ga)^{-1}gb \in H \Rightarrow ga \sim_L gb$$

showing that $\sim_L$ satisfies property 3.1

This proposition shows that any subgroup of H will define an equivalence relation $\sim_L$ respecting property 3.1, and thus by proposition 3.1.3[2] partition G (even better, equally partition G due to corollary 3.1.4). Similarly, we can show that the relatoin $a \sim_R b \Leftrightarrow ab^{-1} \in H$ gives an equivalence relation which satisfies property 3.2. Note that since H partitioned G, only $eH = H$ is a subgroup of G since only it contains the identity.

Combining proposition 3.1.3 and 3.1.6 we get the following one to one correspondence:

$$\{H | H \leqslant G\} \leftrightarrow \left\{ \begin{array}{c} \text{equivalence relations } \sim_L \\ \text{on } G \text{ satisfying} \\ \text{property 3.1} \\ \\ \\ \end{array} \right\}$$

Similarly:

$$\{H | H \leqslant G\} \leftrightarrow \left\{ \begin{array}{c} \text{equivalence relations } \sim_R \\ \text{on } G \text{ satisfying} \\ \text{property 3.2} \\ \\ \\ \end{array} \right\}$$

3.2, showing that the subgroups of G give us all possible such equivalence relations, that is all one-sided congruence relations. Because of this, we can without ambiguity write G/H to represent $G/\sim$ with an equivalence relation $\sim$ satisfying property 3.1, and similarly for $G \setminus H$. 3.2, showing that the subgroups of G give us all possible such equivalence relations. Because of this, we can without ambiguity write G/H to represent $G/\sim$ with an equivalence relation $\sim$ satisfying property 3.1, and similarly for $G \setminus H$.

$G \setminus H$ satisfying $\sim_R$? The answer is actually no: G/H satisfying $\sim_R$? The answer is actually no:

3.1.7 counter-example Take $G = D_6$. Then $H = \langle \sigma \rangle$ is a subgroup, and has left cosets $eH(= H), rH, r^2H$. giving us

$$\{e, \sigma\}, \{r, r\sigma\}, \{r^2, r^2\sigma\}$$

and the right-cosets are

$$\{e, \sigma\}, \{r, \sigma r\}, \{r^2, \sigma r^2\}$$

using the fact that $r\sigma = \sigma r^2$ and $r^2\sigma = \sigma r^2$, we can re-write the right cosets as

$$\{e, \sigma\}, \{r, r^2\sigma\}, \{r^2, r\sigma\}$$

showing that they partition G differently than the left cosets!
This means that $\sim_L$ defined by H does not respect property 3.2. If it did, then

$$r \sim_L r\sigma \Rightarrow r(r\sigma)^{-1} = r\sigma r^2 = rr\sigma \in H$$

however, $r^2\sigma \notin H$, showing that it does not satify the propety

Remember that we showed in proposition 3.1.3 that *both* the properties need to be satisfied in order for G/H to be a group. Since every $\sim$ with these properties corresponds to a subgroup H, it will come down to a special subgroup H which has both properties

(build-up normal subgroups somehow!) We have in fact already encountered the property which will always make H induce both $\sim_L$ and $\sim_R$

(I don't like this "we've already encountered". Would be better to build up the same way we did before, with the "neccessary and sufficient" narrative from before)

Proposition 3.1.8: When H Induces Both $\sim_L$ And $\sim_R$

The relatoins $\sim_L$ and $\sim_R$ induce the same cosets corresponding to H if and only if H is normal

***Proof*:**
If H is normal, then $gH = Hg$, meaning the left and right cosets are the same and so H induces both $\sim_L$ and $\sim_R$. Conversly, if $\sim_L$ and $\sim_R$ are both respected, and take $x \in aH$, so $xa^{-1} \in H$, meaning $xa^{-1} = h$. Manipulating this, we get that $ax^{-1} = h^{-1}$ so $x^{-1} = a^{-1}h^{-1}$, and so $x^{-1} \in a^{-1}H$. Then since $\sim_R$ is respected, $x \sim a$. Since $\sim_L$ is repsected then $x^{-1}a \in H$, meaining $x^{-1}a = h$ or $x^{-1} \in Ha^{-1}$. Then by similar reasoning we get $x \in Ha$, showing that $aH \subseteq Ha$. doing the same argument backwards we get that $Ha \subseteq aH$ giving us $aH = Ha$, which is exactly the condition of normality

(TBD, the converse of my proof is very messy)

Definition 3.1.9: Normal Subgroup

Let $H \leqslant G$ such that $aH = Ha$. Then H is a normal subgroup, and we denoted it $H \trianglelefteq G$

We've already encountered normal subgroups before

Example 3.1.10: Normal Subgroups

1. for any group G, $Z(G) \leqslant G$. Take any $y \in G$ and consider $yx \in yZ(G)$. Then by definition of $Z(G)$, $yx = xy$, and so $[xy] = yx \in Z(G)y$, showing that $Z(G)$ is normal
2. (another such example?)

(after seeing how important normal subgroups are, here are a couple of properties about them:)

Proposition 3.1.11: properties of normal sub-groups

Let N be a normal sub-group of a group G. Then the following are equivalent

1. $N \trianglelefteq G$
2. $N_G(N) = G$
3. $gN = Ng$
4. the operation on left cosets of N in G as described in the previous proposition turns the set of cosets into a group.
5. $gNg^{-1} \subseteq N$ for all $g \in G$
6. if G is abelian and $H \leqslant G$ then $H \trianglelefteq G$

Proposition 3.1.8 is very important. The equivalence relatoin $\sim$ corresponding to H has both property 3.1 and 3.2 by proposition 3.1.6 and so by proposition 3.1.3 G/H is a group! Since normal subgroups are so important, we'll use the notation $\trianglelefteq$ instead of $\leqslant$ to say that H is a normal subgroup of G ($H \trianglelefteq G$) Given our new found knowledge, we will give a new definition of quotient group

Definition 3.1.12: Quotient Group

Let $H \trianglelefteq G$ be a normal subgroup of G. The *quotient group* G/H, sometimes written G mod H (and read G modulo H or G mod H for short) is the group $(G/\sim, \bullet)$ where $\sim$ is defined in proposition 3.1.6, and $\bullet$ is defined by

$$aHbH := (ab)H$$

An easy way to remember that this operation is defined when is H is normal is by remembering $a(Hb)H = abHH = abH$. For notation, we can write $aH = \bar{a} = [a]$ for some representative u and $G/H = \bar{G}$. Thus, going all the way back to $\pi : G \to G/\sim$, and use our knowledge to re-write this as Thus, going all the way back to $\pi : G \to G/\sim$, and use our knowledge to re-write this as

$$\pi : G \to H \qquad \pi(g) = gH$$

where π is a homomorphism if $H \trianglelefteq G$. If $H \ntrianglelefteq G$, then π is *not* a homomorphism.

Since π is a homomorphism, it's kernel is a subgroup:

$$\ker(\pi) = \{g \in G \mid \pi(g) = eH\}$$

and since $eH = H$, this means that $\ker(\pi) = H$. Since H can be any normal subgroup, this means that H is always the kernel of some map

$$\text{Normal} \Rightarrow \text{Kernel}$$

Is the converse true, that is, the kernel is always normal? If so, that would be really powerful, since the kernel would then always be a normal subgroup of G, and so any homomorphism $\varphi : G \to H$ would induce a quotient group G/K, where $K = \ker(\varphi)$. Kernels are always normal subgroups!

Proposition 3.1.13: Kernel Then Normal

Let $\varphi : G \to H$ be a homomorphism and let $K = \ker(\varphi)$. Then $K \trianglelefteq G$

***Proof*:**
Let K be the kernel of $\varphi : G \to H$, and take any $g \in g$. then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g)\varphi(g)^{-1} = e_H$$

and so $gkg^{-1} \in K$, showing that $gK = Kg$, meaning $K \trianglelefteq G$

This means that given any homomorphism $\varphi : G \to H$, we can *always* define an equivalence relation on G given the kernel of φ since $\ker(\varphi)$ is normal!

Since any normal subgroup will be the kernel of some homomorphism, and the kernel of any homomorphism will be normal, we get that the two are essentially synonymous in group theory:

$$\text{Normal} \Leftrightarrow \text{Kernel}$$

(A word how with this fact, we can find the structure of G/H by relating it to the image of $\varphi(G)$?)

With our new knowledge on quotient groups, we re-visit the universal property of quotients (theorem 3.1.1) with the intend on being able

(words here on bringing back intuition from uni. prop. of set, but now in Grp it's more prevalent, since we're linking group structures. Notice that since $H \subseteq \ker(\varphi)$, that means H will make a "finer" as compared to $\ker(\varphi)$, and since both will respect the corresponding $\sim_H$ induced by H, we know by theorem 3.1.1 that it must factor through G/H. THis result is important enough to be given a name:)

Theorem 3.1.14: The First Isomorphism Theorem

Let $\varphi : G \to H$ be a homomorphism and let $K = \ker(\varphi)$. If $N \trianglelefteq G$ where $N \subseteq \ker(\varphi)$, then there exists a unique *homomorphism* $\Phi : G/N \to H$ such that the following diagram commute:

$$\begin{tikzcd} G/N \arrow[r, "\Phi"] & H \\ G \arrow[u, "\pi"] \arrow[ur, "\varphi"] & \end{tikzcd}$$

where $\varphi = \Phi \circ \pi$, and π is the natural projection

***Proof*:**
We already know this universal property in terms of congruence relations. We simply must translate the information of this theorem to the one required for theorem 3.1.1

So far, the question is saying that if

$$N \subseteq \ker(\varphi) \Leftrightarrow (\forall n \in N,\ \varphi(n) = e_H) \Rightarrow \varphi(ab^{-1}) = e_H$$

for $a, b \in N$ moving things around we get

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_G \ \Leftrightarrow\ \varphi(a) = \varphi(b)$$

thus giving us

$$(\forall a, b \in N),\ ab^{-1} \in N \Rightarrow \varphi(a) = \varphi(b)$$

and, since $ab^{-1} \in N$ is the condition defining the induced equivalence relation by N (proposition 3.1.6), we get

$$a \sim b \Rightarrow \varphi(a) = \varphi(b)$$

which put's us exactly at the inition condition for theorem 3.1.1, which gives us that the diagram commutes, as we sought to show.

What theorem 3.1.14 (the universal property of quotient groups) shows is that all the information a homomorphism φ gives us can be captured via the quotient group G/K (TBD?)

Corollary 3.1.15: Canonical Decomposition For Groups

Let $\varphi : G \to H$ be any group homomorphism. Then it may be decomposed into

$$\begin{tikzcd} G \arrow[r, two heads, "\pi"'] \arrow[rrr, bend left, "\varphi"] & G/\ker(\varphi) \arrow[r, "\cong", "\Phi"'] & \mathrm{im}(G) \arrow[r, hook, "\iota"'] & H \end{tikzcd}$$

where Φ is an isomorphism

***Proof*:**
This proof is simply piecing together all the results with figured out so far. Since $\mathrm{im}(G) \leqslant H$, then there exists an inclusion map ι which is a homomorphism. We've established that $\ker(\varphi)$ is normal, and that any normal subgroup will induce a quotient group $G/\ker(\varphi)$, meaning π (the natural projection) is a well-defined homomorphism.

To show $\Phi : G/\ker(\varphi) \to \mathrm{im}(G)$, is an isomorphism, note that we already have that Φ is given by the canonical decomposition of quotient sets. By the universal property of quotient groups, we have that it's a homomorphism. Since bijective homomorphisms are isomorphisms, we have that it's an isomorphism, completing the proof.

Corollary 3.1.16: Surjective Homomorphisms

Let $\varphi : G \to H$ be a surjective homomorphism, and let $K = \ker(\varphi)$. Then

$$G/K \cong H$$

***Proof*:**
For $G/K \cong H$, notice that $H = \text{im}(G)$ in corollary 3.1.15.

This last corollary is very useful and one you will use constantly in both theoretical and practical context. It is saying that if we want to know the structure of G/K, if we can find a surjective homomorphism from G to H with kernel K, then $G/K \cong H$, i.e., H will represent the structure!

(Integarte this somehow?? Maybe later)

$$1 \xrightarrow{\iota_1} K \xhookrightarrow{\iota_K} G \xrightarrow{\varphi} H \xrightarrow{\pi} 1$$

where 1 is the trivial group, ι_1 and ι_K are the inclusion homomorphisms, and π is the projection of H onto the trivial group.

The 2nd form presented in corollary 3.1.16 lets us nicely split up the pieces of any given homomorphism. This representation is called a *short exact sequence*. We will return to short exact sequences soon to flesh out many intersting results when precieving which groups can be put in the 3 non-trivial positions, that is

$$1 \xrightarrow{\iota_1} A \hookrightarrow B \twoheadrightarrow C \xrightarrow{\pi} 1$$

(bunch of examples: perhaps a reminder that if $H \leqslant G$ the $i : H \to G$ shows that G contains H??)

Exercise 3.1.1

Throughout these exercises, let $\varphi : G \to H$ be a homomorphism between the two groups G and H, and $K = \ker(\varphi)$

1. Show that $\varphi^{-1}(Q)$ (the pre-image of Q) is a group. If Q was a normal subgroup, $Q \trianglelefteq H$, show that $\varphi^{-1}(Q)$ is also normal. Is it true that if $H \trianglelefteq G$ then $\varphi(H) \trianglelefteq \varphi(G)$?
2. Let $h \in H$. We call $\varphi^{-1}(h)$ a *fiber* of φ over h. Let $F_h = \varphi^{-1}(H)$. Show that
   1. For any $u \in F_h$, $F_h = uK$
   2. For any $u \in F_h$, $F_h = Ku$
3. Let $aKbK = \{akbk' \mid a, b \in G, k, k' \in K\}$ be a set, and $K \trianglelefteq G$. Show that

   $$aKbK = abK$$

   where $abK = \{abk \mid a, b \in G, k \in K\}$ [hint: for $aKbK \subseteq abK$, recall that $aK = Ka$ since K is normal. For $aKbK \supseteq abK$, note that $1 \in K$ so $1 \cdot a \in Ka$. Take advantage of normality to complete the proof]
4. Let $R \leqslant G$. Show that the cosets of G/R are all of equal size.
5. Show that if $aB = 1B$, then $a \in B$

6. Let $H, K \trianglelefteq G$. Show that $H \cap K \trianglelefteq G$
7. Generalizing the previous result, If $\{H_i\}_{i \in I}$ is some collection of normal subroups of G show that $\cap_{i \in I} H_i \trianglelefteq G$
8. Let G be finitely generated and let $N \trianglelefteq G$ be finitely generated. Prove that G/N is finitely generated.
9. For any group G, show that $Z(G) \trianglelefteq G$
10. Show that $n\mathbb{Q} = \mathbb{Q}$, where we're abusin' notation in writing $n\mathbb{Q}$ to mean $\{nq \mid q \in \mathbb{Q}\}$ even though multiplication is not defined on $\mathbb{Q}$ (Groups that have this property are called *divisible* or *injective* groups. More on them in section 17.3) [maybe any divisible group has a natural 2nd group structure? cause in a sense, every element can be "divided", and so it respects the property of inverse, which is also unique by the fact that the element is a unique group element, and the other axioms follow naturally]
11. Let $G \cong H$, $N \trianglelefteq G$ and $K \trianglelefteq H$. If $N \cong K$ is it true that $G/N \cong H/K$? Prove or show a counter example. hint: [2]
12. (Useful in Lie Groups) Let $GL_n(\mathbb{R})$ be the set of invertible $n \times n$ matrices over $\mathbb{R}$ (more generally, if you know linear algebra, over a field $\mathbb{F}$ with characteristic $\neq 2$ or 3 if $n = 2$). What are the normal subgroups of $GL_n(\mathbb{R})$?
13. We've shown that if we have a map $f : G \to H$ and $N \trianglelefteq H$, we can define a map $\overline{f} : G \to H/N$. Is the converse true? That is, if we have a map $f : G \to H/N$, do we have a map $F : G \to H$ such that $f = \pi \circ F$? [Hint: Consider maps with domains and codomains of $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$, $n > 1$]
14. Show that every subgroup of Q_8 is normal. Hence, every subgroup being normal does not imply the group is abelian[3]

3.2 Presentations

We have seen in section 2.3.3 that if we have any set map from a set S to a group G, $f : S \to G$, then it can be extended it to a homomorphism from the group $F(S)$ to G: $\varphi : F(S) \to G$, $\varphi|_S = f$. We've also seen that if we have a surjective homomorphism $G \to H$, then $G/N \cong H$ (where N is the kenrel of the homomorphism). We can combine these results: Converting to $(G, *)$ notation for groups for a moment, given an identity map $\mathrm{id} : G \to (G, *)$, which is a prefectly valid surjective set function from the set G to the underlying $\varphi : (F(G), *_F) \to (G, *)$. By the first isomorphism theorem, if N is the kernel, then $(F(G)/N, (*_F)_\varphi) \cong (G, *)$. Thus, $\varphi : F(G) \to (G, *)$. By the first isomorphism theorem, if N is the kernel, then $F(G)/N \cong (G, *)$. Thus, every group is the quotient of a free group! We now go back to our usual convention of groups being written as G where we drop the binary operation, where we will consider it clear from context when a map is a set map and when it is a homomorphism.

[2] reverse text? This might be giving the answer? Consider $G = H = N = \mathbb{Z},\ K = 2\mathbb{Z}$

[3] On the other hand, Q_8 is special: Every non-abelian group for which all subgroups are normal is of the form $G \cong Q_8 \times B \times D$ where B is something called an elementary abelian 2-group and D is called a torsion abelian group whose elements are of odd order. A group where all subgroups are normal is called a Dedekind group, and a non-abelian Dedekind group is called a Hamiltonian group.

Let's generalize this result a bit: Let's say $S \subseteq G$ such that $\langle S \rangle = G$, and consider $f : S \to G$. Then by the universal property of free groups, we have that $\varphi : F(S) \to G$ is a group homomorphism. Since $\langle S \rangle = G$, and all possible words are in $F(S)$, then by construction of $F(S)$ and φ, the map φ is a surjective group homomorphism. Thus, for the appropriate kernel N, $F(S)/N \cong G$.

Since $F(S)$ is free from relations, but G has relations, φ somehow imposed relations onto $F(S)$. This means that N captured the idea of the relations imposed on $F(S)$ to make $F(S)/N \cong G$. As observed in section 2.3, any relations of a group can be written as

$$r_1 = r_2 = \cdots = r_k = 1$$

If we let $r_i = \prod_{i \in I_{r_i}} s_i$ to be some arbitrary finite[4] product of elements of S and it's inverses. Then if If we let $r_i = \prod_{i \in I_{r_i}} s_i$ to be some arbitrary product of elements of S and it's inverses. Then if we let $R = \{r_1, r_2, ..., r_k\}$, we can write $G = \langle S | r_1 = r_2 = \cdots = r_k = 1 \rangle$ as

$$G = \langle S | R \rangle$$

for short hand. Since $F(S)/N \cong G$, it must be that $R \subseteq N$. We might be tempted to say $\langle R \rangle \subseteq N$, however the closure *need not be normal*, while N must be a normal subset. Instead, we'll define the noraml closure:

Definition 3.2.1: Normal Closure

Let $S \subseteq G$ be a subset of a group G. Then define the intersection of all normal subgroups of G containing S

$$\langle S \rangle^{\trianglelefteq} = \bigcap_{\substack{N_i \trianglelefteq G \\ S \subseteq N_i}} N_i$$

to be *normal closure of* S.

$\langle S \rangle^{\trianglelefteq}$ is indeed a normal subgroup by proposition ref:HER. Since $r_1 = r_1 = \cdots = r_k = 1$ in G, then $[r_1] = [r_2] = \cdots = [r_k] = [1] \in F(S)/N$, and so $r_1, r_2 ... r_k \in N$. We thus have that $\langle R \rangle^{\trianglelefteq} \subseteq N$.

Next, let's say we have some $x \in N$. Then that means then $[x] = [1]$, so for some $n' \in N$, $xn' = 1$. But then, xn' must be some combinations of elements $r_1, r_2, \cdots, r_k$, since by the presentation of G, those are the elements that can equal to 1. Thus $N \subseteq \langle R \rangle^{\trianglelefteq}$. Therefore $N = \langle R \rangle^{\trianglelefteq}$

From this, we can formally define a presentation

Definition 3.2.2: Presentation

Let G be any group. Then a *presentation* of G is a pair of sets (S, R) where $\langle R \rangle^{\trianglelefteq}$ is the kernel of the homomorphism $F(S) \to G$. The elements of S are called the *generators* of G, and the elements of R are called the *relations* of G:

$$\langle S | R \rangle := F(S)/\langle R \rangle^{\trianglelefteq}$$

We say G is finitely generated if given a presentation (S, R), $|S|$ is finite. We say G is *finitely presented* if both $|S|$ and $|R|$ are finite.

[4] Infinite products would require more care to be well-defined, see EYNTKA real analysis I

Notice how this statement is like the dual equivalent of Cayley's theorem (4.5.3): In Cayley's theorem, every group is the subgroup of a symmetric group. Here, every group is the quotient of a free group.

Example 3.2.3

1. $\langle A|\emptyset\rangle$ is the presentation of the free group $F(A)$
2. dihedral group

3.2.4 Note Let $N \trianglelefteq G$. Clearly, $\langle N\rangle^{\trianglelefteq} = N$. However, it is rather difficult to determine whether $S \subseteq N$ is a smallest set that satisfies $\langle S\rangle^{\trianglelefteq} = N$. (More words about the word problem?)

Corollary 3.2.5: Finitely Generated Groups

Let G be a group. Then G is finitely generated if and only if G is isomorphic to the quotient group of a finitely generated free group.

***Proof*:**
Let $F(S)/N \cong G$. If $F(S)$ is finitely generated, then $F(S)/N$ is finitely generated (see exercise ref:HERE) (Though in that exercise, $N \trianglelefteq G$ is finitely generated given the regular closure, not the normal closure... must think about that)

On the other hand, let G is finitely generated, say $S \subseteq G$ such that $\langle S\rangle = G$ and $|S| = n$. Then there exists an inclusion map $S \hookrightarrow G$. By the universal property of free groups, this extends to a map $\varphi : F(S) \twoheadrightarrow G$. This map is surjective since S is a generating set for G. Then by the 1st isomorphism theorem, $F(S)/\ker(\varphi) \cong G$, fulfilling our requirement.

(ref:HERE if you do geometric group theory or if this is referenced in other places)

In this book, this will be the end our theoretical journey at looking at groups as $F(S)/N \cong G$. A whole field of study has emerged because of this universal property called *geometric group theory*, which we do some exposition for in the appendix (see ref:HERE). For those interested in geometric group theory, I recommend (ref:HERE, perhaps Clara Löh's book and that othe guy whose book is harder).

3.3 Application to Finite Groups

As mentioned earlier, thinking about cosets in themselves is also very useful. In fact, a theorem which you will you to death in this course relies on cosets. With the fact that cosets partition your group, you can deduce this strikingly important result:

Theorem 3.3.1: Lagrange Theorem

If G is a finite group and H is a subgroup of G, then the order of H divides the order of G (i.e., $|H|\,\big|\,|G|$), and the number of left cosets of H in G equals $\frac{|G|}{|H|}$

***Proof*:**
Let $H \leqslant G$. Then you know that H partitions G into equal-sized cosets by proposition ref:HERE (it is an exercise), and so if $|G| = n$, and $|H| = k$, then you know $k|n$, implying $|H|\big|G$. Furthermore, since if $H \trianglelefteq G$, then

$$|G/H| = |G|/|H|$$

Example 3.3.2

1. As a quick reminder, we used this theorem in (*corollary* 2.3.12)to get a useful result a bit early. You can now go back and perhaps check your intuition to make sure it makes sense.
2. Try going back to (*theorem* 2.3.11) and proving it using Lagrange's Theorem. You will find it much easier, almost trivial even.

As the examples show, Lagrange is incredibly powerful as it let's us relate the order of a group and it's subgroup, which is a fact used extensively.

Definition 3.3.3: Index

If G is a group (possibly infinite) and $H \leq G$, the number of left cosets of H in G is called the index of H in G and is denoted by $|G : H|$.

With this notation, we can re-write Lagrange's Theorem as

$$|G| = |H||G : H|$$

Corollary 3.3.4: x to the order of G

If G is a finite group and $x \in G$, then the order of x divides the order of G. In particular $x^{|G|} = 1$ for all $x \in G$.

Corollary 3.3.5: index of H is 2 $\Rightarrow$ H is normal

If The $H \leqslant G$ and the index of H is 2, then H is normal

***Proof*:**
Since $[G : H] = 2$, we know that there are only 2 cosets: eH and xH for the correct $x \in G$. We want to show that H is normal, that is, for every element $g \in G$, $gH = Hg$ (which in our case there is only e and x which form distinct cosets) . Since $eH = He$, by commutativity of identity element from last homework, it sufices to show that

$$xH = Hx$$

for this, we'll take advantage of the index of H. Since the index of H is 2, we know that

$$H \cup xH = G \qquad \text{and} \qquad H \cup Hx = G$$

since the cosets of H partition G, as we showed in part 1 of this homework. Since $xH \subset G$, we know that

$$xH \subset H \cup Hx$$

therefore

$$\forall h \in H, \text{ either } xh = h'x \qquad \text{or} \qquad xh = h' \qquad \text{for some appropriate } h' \in H$$

let's say that for the sake of contradiction $xH \neq Hx$, so we can rule out the first option for a moment, leaving us with $xh = h'$. However, that would imply that $xh \in H$, which is means xh is in the first coset. However, we've shown in the first question of this homework that cosets are distinct, making this a contradiction. Therefore, we're forced to say $xh = h'x$ for appropriate h', however that implies that $xh \in Hx$. Since this holds for any $h \in H$, and we can repeat the same process for Hx, we have that

$$xH = Hx$$

meaning that all cosets "commute", and so G is normal, as we sought to show.

3.3.6 Why this doen't necessarily work when index greater than 2

Counter-example: Take $G = D_3$, i.e. the triangle group, and $H = e, \sigma$. Then

$$\{e, \sigma\} \cup \{r, r\sigma\} \cup \left\{r^2, r^2\sigma\right\} = G$$

and

$$\{e, \sigma\} \cup \{r, \sigma r\} \cup \left\{r^2, \sigma r^2\right\} = G$$

i.e. the union of the left and right cosets both must equal G. This idea is very important as it forces a relation between left and right cosets. now consider rH. Since the unions of both sets must match, we know that

$$rH \subseteq G \Rightarrow rH \subseteq H \cup Hr \cup Hr^2$$

we knwo that $rH \not\subseteq H$ since that would make the two cosets *not* distinct, and so we have

$$rH \subseteq Hr \cup Hr^2$$

which is true! test it out (while recalling that $r\sigma = \sigma r^2$). In the case we since the index being 2 situation, that would make $xH \subseteq Hx$, and vice versa for Hx too, and so it forces normality.

Because of the power of Lagrange, one might ask themselves if the converse is true: if a number divides the order of a group $n \big| |G|$, does that mean there's a subgroup $H \leqslant G$ such that $|H| = n$? That turns out to be not true:

Example 3.3.7: : Counter-Example

We'll show that $6 \big| |A_4| \not\Rightarrow H \leqslant A_4, |H| = 6$. First, let's figure out the elements of A_4:

1. the identity
2. all even permutation with max order of 4: $(\cdot\,\cdot)(\cdot\,\cdot)$
3. Either the 2 transpositions are disjoint, or they overlap with 1 element. They cannot overlap

with 2 elements because then $(\cdot\,\cdot)(\cdot\,\cdot) = (\cdot\,\cdot)$, making it an odd-permutation.

4. If they overlap with 1 element, then $(\cdot\,\cdot)(\cdot\,\cdot) = (\cdot\;\cdot\;\cdot)$, thus we have all 3-cycles, and all permutation represented by 2 disjoint 2-cycles

5. By counting, we can find 8 3-cycles, and 3 $(2,2)$-cycle types, for a total of 12 cycles (when including the identity), meaning $6\big||A_4|$

Let's now suppose there exists a subgroup $H \leqslant A_4$ of order 6. To analyze it we'll take advantage of the fact that the intersection of two subgroups is a subgroup (2.3.15). If you take all $(2,2)$ cycle type elements, plus the identity, you will get a group $(K \leqslant A_4)$ [a]. Take

$$H \cap K = H' \qquad \Rightarrow \qquad H' \leqslant H\,, H' \leqslant K$$

By Lagranges Theorem, $|H'|\big||H|$ and $|H'|\big||K|$ which implies the order is either 1 or 2. If $|H'| = 1$, then it only has the identity. That would mean that H and K are practically disjiont, meaning that if we take 2 non-identity elements $h \in H, k \in K$, then $h \cdot k = g$ we cannot produce an element $g \in H$ or $g \in K$ since then by closure, we'd get $k \in H$ and $h \in K$. Doing this for the case of H:

$$\begin{aligned} hk &\in H \\ \Rightarrow h^{-1}hk &\in H \\ \Rightarrow k &\in H \\ \Rightarrow H \cap K &\supseteq \{e\} \end{aligned}$$

Thus, we'd get $|H| \cdot |K|$ total elements. But $|H| \cdot |K| = 6 \cdot 4 = 24$, which is greater than the number of elements we have: $|A_4| = 12$. Thus, $|H'| \neq 1$, so we must have that $|H'| = 2$. This must also imply that H has an element $h \in H$ that is both a product of two disjoint cycles, while the other 4 elements are 3-cycles.

If $|H'| = 2$, then we'll show that it will also lead to a contradiction. Note that since $|H| = 6$, then $|A_4 : H| = 2$, so by corollary (3.3.5), $H \trianglelefteq A_4$, meaning $\forall \sigma \in A_4$, $\sigma H = H\sigma$, or equivalently $\sigma H \sigma^{-1} = H$

Now, take take the cycle of type $(2,2)$ in H from earlier: $h = (i\,j)(k\,l)$ with $\{i,j,k,l\} = \{1,2,3,4\}$ and the 3-cycle $\sigma = (i\,j\,k) \in A_4$. Then we have that

$$\sigma h \sigma^{-1} = (j\,k)(i\,l) \neq (i\,j)(k\,l)$$

And since H is normal, $\sigma k \sigma^{-1} \in H$. However, that would imply that there is a *second* cycle of type $(2,2)$, meaning that $|H \cap V| \geq 3$, again a contradiction.

Therefore, there cannot be a subgroup of order 6 in A_4, as we sought to show

[a] Trivia: This group is isomorphic to the Klein 4-group V_4

We showed how the converse is not true, but are there partial converses that are true? There are actually two famous theorems which talk about partial converses: *Cauchy's Theorem* and *Sylow's Theorem*. Both these theorems rely on prime-number properties to give us a partial converse. Sylow's theorem goes a step further in telling us how many groups of a particular order we can find (so more than there exists one). However, by the way we'll present Sylow's theorem, it will require group-actions, the topic of the next section. However, Cauchy's theorem technically doesn't. There is an exercises in the book for doing this theorem, so I'll link it here for now for when I have time to do it:

Theorem 3.3.8: Cauchy's Theorem

If G is a finite group and p is prime diving $|G|$, then G has an element of order p

***Proof*:**
proof outlined in exercise 9.

There is a simpler proof which is related to another theorem called Sylow's Theorem (4.8.5) in a similar way that the Inverse Function Theorem and Implicit function Theorem are related: proving one of them is relatively hard, but then proving the other from the first is a super easy consequence. Since Sylow's theorem *is* proved and covered extensively, I proved this theorem using Sylow's theorem once it is introduced (see 4.8.6)

Cauchy's theorem is super important. It pop's a lot when trying to find groups of smaller order:

Example 3.3.9
If G is a finite abelian group with no normal subroups (besides 0 and G)[a], then $G \cong Z_p$.

Proof. Since G is abelian, if $H \leqslant G$, then $H \trianglelefteq G$ (since $aH = Ha$ by use of commutative property). However, G is simple, so either $H = \{e\}$ or $H = G$. By Cauchy's theorem, we also know that if $p \big| |G|$ (which once must as $|G|$ is composed of prime-factors), then there exists an element of order p (meaning a subgroup of order p). However, that's a contradiction as we've just shown. Therefore, either $G = \{e\}$ or $G = \langle x \rangle$ where $|G| = p$. Either way, G, is cyclic and abelian, which by 2.3.9 means that

$$G \cong Z_p$$

□

[a] maybe introduce simple groups earlier?

Note that G was a general finite group. If we have that G is abelian, then the converse is actually *completely* true! This turns out to simply be by the fact that if $\langle x \rangle = p$ and $\langle y \rangle = q$ for prime p, q then $\langle xy \rangle = pq$

Corollary 3.3.10: Cauchy's Theorem For Abelian Groups

If G is a finite abelian group, and n divides G, then there is a group of order n

First, we prove it for if $n = p$ is prime

***Proof*:**
p.102 in book

Now let's prove it in general

***Proof*:**
We'll for the first time proceed by induction! If $|G| = 1$, we're done, so let's say $|G| = n$, and if there is a an element $m < n$ such that $m \big| |G|$ then there exists an $H \leqslant G$ such that $|H| = m$. The edge cases are the same as Cauchy's theorem, so they'll be omitted.

By Cauchy's theorem, we know that if $p \big| |G|$ then $P \leqslant G$ such that $|P| = p$. Since $|G| > 1$, we can assume such a p exists. Then since G is abelian, $P \trianglelefteq G$, so G/P is defined. In particular

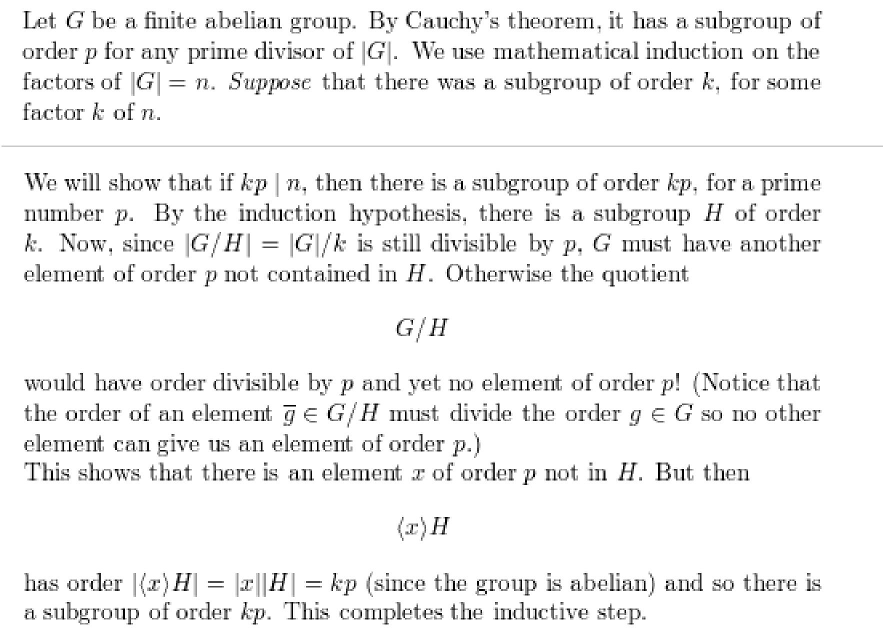
$$|G/P| = n/p$$

Which means we found our smaller group G/P. Now for any $m \big| |G/P|$, there exists a $\overline{H}$ such that $\overline{H} = m$. By the 4th Isomorphism theorem, we know there exists a $H \leqslant G$ such that

$$H = H/N$$

meaning that $|H| = \frac{n}{p} \cdot p = n$, showing that there is group of order H, as we sought to show.

Here's also another proof I found



Let G be a finite abelian group. By Cauchy's theorem, it has a subgroup of order p for any prime divisor of $|G|$. We use mathematical induction on the factors of $|G| = n$. *Suppose* that there was a subgroup of order k, for some factor k of n.

We will show that if $kp \mid n$, then there is a subgroup of order kp, for a prime number p. By the induction hypothesis, there is a subgroup H of order k. Now, since $|G/H| = |G|/k$ is still divisible by p, G must have another element of order p not contained in H. Otherwise the quotient

$$G/H$$

would have order divisible by p and yet no element of order p! (Notice that the order of an element $\overline{g} \in G/H$ must divide the order $g \in G$ so no other element can give us an element of order p.)
This shows that there is an element x of order p not in H. But then

$$\langle x \rangle H$$

has order $|\langle x \rangle H| = |x||H| = kp$ (since the group is abelian) and so there is a subgroup of order kp. This completes the inductive step.

Figure 3.1: Alternative proof

This theorem should show you the power of commutativity! If a group is commutative, you have an an incredibly simple group structure! If you know the order of the group, you know the entire substrucutres of the group, giving you a lot of information! This very fact will be key when categorizing every abelian group in chapter 5! This means that a good chunk of the groups studied in group theory are non-commutative groups, since the structure of commutative groups is pretty simple and well-understood.

Be mindful that Lagrange's theorem gives you information about *finite* groups. In general, arithmetics on the infinite won't be defined in a meaningful way for us. In fact, we can get quite some weird results, like subgroups being isomorphic to their parent group!

Example 3.3.11: Subgroups Of $\mathbb{Z}$
Take the group $\mathbb{Z}$, and any subgroup $n\mathbb{Z} \leqslant \mathbb{Z}$ for $n \in \mathbb{N}_+$. Show that

$$n\mathbb{Z} \cong \mathbb{Z}$$

that is, every (non-trivial) subgroup of $\mathbb{Z}$ is isomorphic to $\mathbb{Z}$!

3.4 Internal Direct Product

Though it might seem inconspicuous at first, the introduction of a normal subgroup will let us be able to define the multiplication of subgroups that from subgroups in themselves!! The idea behind this idea is to decompose a group into smaller subgroups, giving an insight on its structure. In chapter 5, we will explore in greater detail the results of this decomposition. For now, we will explore a couple of properties of internal direct product.

We'll first define some notation:

Definition 3.4.1: HK (Internal Direct Product)

Let H, K be subgroups of a group and define

$$HK = \{hk | h \in H, k \in K\}$$

3.4.2 Note This is *not* necessarily a group! Take $\langle(123)\rangle = H \leqslant S_4$ and $\langle(3\,4)\rangle = K \leqslant S_4$. Then

$$HK = \{\mathrm{id}, (123), (132), (123)(34), (132)(34), (34)\}$$

and $(132)(34) = (3421)$, but $(3421)^2 \notin HK$, meaning HK is not closed!

Proposition 3.4.3: cardinality of HK

Let H, K be finite subgroups of a group. Then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

***Proof*:**
Notice that HK is the union of left-cosets of K:

$$HK = \bigcup_{h \in H} hK$$

Thus, it suffices to find the number of distinct left-cosets hK.

If $h_1K = h_2K$ then $h_2^{-1}h_1K = K \Rightarrow h_2^{-1} \in K$. Since $h_2^{-1}h_1 \in H$, this means that

$$h_2^{-1}h_1 \in H \cap K \Rightarrow h_2^{-1}h_1H \cap K = H \cap K \Rightarrow h_1H \cap K = h_2H \cap K$$

And so we reduced the problem to counting the number of distinct $H \cap K$ cosets. This is useful, since $|H \cap K|\big|H$, and so by Lagrange's theorem

$$|H : H \cap K| = \frac{|H|}{|H \cap K|}$$

Thus, there are that many left cosets of K. Since K partitioned HK, it means that every element of the cosets are unique, so to find $|HK|$ we now get

$$|HK| = \frac{|H|}{|H \cap K|}|K| = \frac{|H||K|}{|H \cap K|}$$

as we sought to show

Proposition 3.4.4: When is HK a group

If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$

***Proof*:**
Let $H, K \leqslant G$.

($\Rightarrow$) Let's say $HK \leqslant G$. We want to show that $HK = KH$, that is, for any $hk \in HK$, there exists an $h' \in H$, $k' \in K$ such that $hk = k'h'$. Since HK is a group, it is closed under its operation. Thus, for any $h_1k_1, h_2k_2 \in HK$, $h_1k_1h_2k_2 \in HK$. Since $HK = \langle hk | h \in H, k \in K\rangle$, there is a $h_3 \in H$ and $k_3 \in K$ such that

$$h_1k_1h_2k_2 = h_3k_3$$

re-arranging, we get that

$$h_3h_1k_1 = k_3k_2^{-1}h_2$$

By letting $h = h_3h_1$, $k = k_1$, $h' = h_2$, and $k' = k_3k_2^{-1}$, we get that

$$hk = k'h'$$

as we sought to show (more words on how we can choose these values and still let h, h', k, k' be "arbitrary")

($\Leftarrow$) Let's say $HK = KH$. We want to conclude that HK is a subgroup of G. We'll show that HK is closed under its binary operation – the other axioms follows straightforwardly.

Corollary 3.4.5: Condition on K when Internal Direct Product is Present

If H and K are subgroups of G and $H \leq N_G(K)$, then HK is a subgroup of G. In particular, if $K \leqslant G$ then $HK \leqslant G$ for any $H \trianglelefteq G$. Furthermore, if H and K are normal subgroups, then $HK \trianglelefteq G$.

***Proof*:**
The first part of the corrollary was proven in the last proposition. What is left to prove if both H and K are normal subgroups, then $HK \trianglelefteq G$. (TBD – is this true?)

Defining when HK is a group is intersting, but it would be even more useful if we can figure out the structure of HK. For example, Can we say that $HK \cong H \times K$? The answer is not always! In fact, we will find a way of exactly figuring out the structure of HK in chapter 5 once we have learned the appropriate tools in chapter 4

3.4.6 Foreshadowing Thinking about HK like in corollary 3.4.5 will become the defining feature when we discuss *semi-direct* products in chapter 5.

Since we have now defined how we can take the "product" of two groups, we can ask the following intersting question: If $\mathcal{N}$ is some subset of normal subgroup G that is "closed" under multiplication, does it form a group? The reader should fiddle around with this before the answer is given. The same question will be asked in the next part of the book where we will have the equivalent of normal subgroup for rings (an algebraic structure with 2 binary operations). It with the "normal" objects of rings. The answer for groups shall be presented in chapter 7. with the "normal" objects of rings. In the case for groups, such a collection will not form a group.

Thus, we will set internal direct products aside, with only an exception made for the next section, until we have accumulated more tools to discuss them again.

3.5 Isomorphism Theorems

We know turn back to how quotient groups are inseparably related to homomorphisms. Mathematicians who were studying homomorphisms wanted to understand the relation between groups given certain properties, for then we can gain new perspective into problems being solved. For example, if we have a group $HK \leqslant G$, and $K \trianglelefteq G$, can we always determine what HK/K is? Is there *always* an answer to this question, no matter what group HK we pick? Another question is what if we "recusively" take quotients: If $H \trianglelefteq K \trianglelefteq G$, what is $(G/H)/(K/H)$? Is there a simplification that will make this more understandable? Yet another such question is how much can we deduce of the subgroups of G/N (i.e. the lattice of G/N) given what we know about G?

In this section, we'll show these results:

1. We'll sum up the results of of theorem 3.1.14, but give it the name that is more commonly used
2. We will look at equivalent ways of looking at internal direct products, since they pop up a lot (especially chapter 5)

3. An easier way of thinking of quotient groups whose quotient is a quotient group.
4. the subgroup lattice of G and the subgroup lattice of G/N

We first recall theorem 3.1.14, but given it its classical name:

Theorem 3.5.1: First Isomorphism Theorem

If $\Phi : G \to H$ is a homomorphism of groups, then $\ker(\Phi) \trianglelefteq G$ and $G/\ker\Phi \cong \Phi(G)$. Similarly, if $N \trianglelefteq G$ and $N \subseteq \ker(\Phi)$, there exists a unique *homomorphism* $\varphi : G/N \to H$ such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[dr, "\Phi"] & G/N \arrow[d, "\varphi"] \\ & H \end{tikzcd}$$

where $\Phi = \varphi \circ \pi$, and π is the natural projection

Recall that the powerful part of this theorem is that it is the universal property of quotient groups: for *any* homomorphism $\varphi : G \to H$, there exists a unique group G/N (which represents the equivalence class, and thus the homomorphism, within its group structure) along with the unique homomoprhism π [5]which let's us link G to G/N such that $\Phi = \pi \circ \varphi$, meaning we can study the results of the homomorphism Φ by looking at group information of G/N.

Also, remember that if we take $\Phi(G) \leqslant H$, and $N = \ker(\Phi)$ so that

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[dr, "\Phi"] & G/\ker(\Phi) \arrow[d, "\varphi"] \\ & \Phi(G) \end{tikzcd}$$

then

$$G/\ker(\Phi) \cong \Phi(G)$$

Example 3.5.2

1. we can get some information about S_n. We can define the function $\epsilon : S_n \to \{-1, 1\}$ whether if σ is even then $\epsilon(\sigma) = 1$, and if σ is odd then $\epsilon(\sigma) = -1$. Note that we can treat $\{-1, 1\}$ as a group if $-1 \cdot -1 = 1, 1 \cdot -1 = -1, -1 \cdot 1 = 1, 1 \cdot 1 = 1$. Then ϵ would actually be a homomorphism. Then $\ker(\epsilon) = A_n$, that is, the cycles of even permutation, also known as the alternating groups. By the first isomorphism theorem, we then know that

$$S_n/A_n \cong \epsilon(S_n) = \{\pm 1\}$$

[5] In particular, it's a *natural homomorphism*. The word natural here has a technical meaning which we explore in section **??**. We will explore the concept once we explore the non-natural choice for basis for vector-spaces, which will let us contrast what it means to have a "natural" versus a "non-natural" function

and so, using Lagrange, we get that

$$2 = |S_n/A_n| = \frac{|S_n|}{|A_n|} \qquad \Rightarrow \qquad |A_n| = \frac{1}{2}|S_n|$$

2. Let's say we have two groups G, H with $\varphi : G \to H$ being a surjective homomorphism. Then we know that

$$G/\ker(\varphi) = H$$

in other words, if we have a surjective homomorphism, we can always view the structure of H through the perspective of the quotient group $G/\ker(\varphi)$!

Corollary 3.5.3: properties as consequence

Let $\varphi : G \to H$ be a group homomorphism

1. φ is injective if and only if $\ker\varphi = 1$
2. $|G : \ker\varphi| = |\varphi(G)|$ [a]

[a]If you have taken linear algebra, this is the Rank-Nullity theorem equivalent for Group Theory

The second Isomorphism theorem let's us view the quotients of internal direct products (definition 3.4.1) form a different angle.

Theorem 3.5.4: The Second or Diamond Isomorphism Theorem

Let G be a group, let A and B be subgroups of G and assume $A \leq N_G(B)$. Then AB is a subgroup of G, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.

The name diamond comes form this diagram where the arrow represent subgroups, and a red arrow represents norma subgroup:

$$\begin{tikzcd}
 & G \arrow[d] & & & & G \arrow[d] & \\
 & AB \arrow[dl] \arrow[dr, "\trianglelefteq"] & & & & AB \arrow[dl] \arrow[dr] & \\
A \arrow[dr] & & B \arrow[dl] & \Rightarrow & A \arrow[dr, "\trianglelefteq"] & & B \arrow[dl] \\
 & A \cap B \arrow[d] & & & & A \cap B \arrow[d] & \\
 & \{e\} & & & & \{e\} &
\end{tikzcd}$$

and

$$\frac{AB}{B} \cong \frac{A}{A \cap B}$$

The reason this theorem is intersting is because we often enough want to take the internal direct prodcut of groups, and deduce some structural information form it with quotients, and this theorems gives us a a tool do do so (see exercise ref:HERE)

***Proof*:**
Since $B \trianglelefteq G$, by proposition 3.4.4, $AB \leqslant G$. Another way of seeing this is that AB is the collection of cosets Ab. Then if $\pi : G \to G/A$ is the cannonical projection, $AB = \pi^{-1}(\pi(B))$. The image of a subgroup is a group (ref:HERE) and the pre-iamge of a subgroup is a group (see exercise ref:HERE). Thus AB is a subgroup.

To show $B \trianglelefteq AB$, we need to show that for any $ab \in AB$, $abB(ab)^{-1} = abBb^{-1}a^{-1} = B$. To do so, recall that B is normal, so that $kBk^{-1} = B$, and so

$$abBb^{-1}a^{-1} \overset{!}{=} aBa^{-1} = B$$

where $\overset{!}{=}$ comes from the fact that B is closed. Since ab was a general element of AB, it follows that $B \trianglelefteq AB$.

Finally, to show that $\frac{B}{A\cap B} \cong \frac{AB}{B}$, define the map $\varphi : A \to AB/B$, where $\varphi(a) = aB$. We'll show this is a surjective homomorphism. This function is indeed a homomorphism, since

$$\varphi(ab) = abB \overset{!}{=} aBbB = \varphi(a)\varphi(b)$$

where $\overset{!}{=}$ comes from exercise ref:HERE. Alternativly, Since $B \trianglelefteq AB$, $\pi : AB \to AB/B$ is a well-defined group-homomorphism. Then one can show that restricting to $\pi|_A\, A \to AB/B$ is also a group-homomorphism.

To show that φ is surjective, notice that every element of AB/B can be written as abB. Since $b \in B$, then

$$abB = aB$$

it follows that an arbitrary coset abB has a map to it (i.e. $\varphi(a) = aB = abB$), showing that φ is surjective. Thus, by corollary 3.1.16, $A/\ker(\varphi) \cong AB/B$. To find $\ker(\varphi)$, we simply put together the information given to us:

$$\ker(\varphi) = \{a \in A \mid \varphi(a) = 1B = B\} = \{a \in A \mid aB = B\} \overset{!}{=} \{a \in A \mid a \in B\} = A \cap B$$

where $\overset{!}{=}$ comes from exercise ref:HERE. Thus, substituing $\ker(\varphi)$ by $A \cap B$, we get

$$\frac{A}{A \cap B} \cong \frac{AB}{B}$$

as we sought to show

One way to remember this theorme is through the following mnemonic:

> If we have $\frac{AB}{B}$, then add a $A \cap \_$ to the top and bottom, giving us
>
> $$\frac{A \cap AB}{A \cap B} = \frac{A}{A \cap B}$$

Another way to think about the 2nd isomorphism theorem is to consider the case where $A \cap B = \{e\}$.

Then this theorem shows that

$$\frac{AB}{B} \cong A \qquad \frac{AB}{A} \cong B$$

Heuristically, we can think in this situation that we are "cancelling out" the B's. However, be mindful to take this as a heuristic – this is *not* what is actually happening!

The third Isomorphism theorem lets quickly simplify quotient groups quotiented *by* a quotient group:

Theorem 3.5.5: The Third Isomorphism Theorem

Let G be a group and let H and K be normal subgroups of G with $H \leq K$. Then K/H, G/H and

$$(G/H)/(K/H) \cong G/K$$

If we denote the quotient by H with a bar, this can be written

$$\bar{G}/\bar{K} \cong G/K$$

***Proof*:**
Since $H \leqslant K$, then $K/H \trianglelefteq G/H$. To show this, first note that K/H is indeed a subgroup of G/H. For any $[x], [y] \in K/H$, $[x][y]^{-1} = [xy^{-1}] \in K/H$ since $xy^{-1} \in K$. Furthermore, for any $[g] \in G/H$, and any coset $[k] \in K/H$, $[g][k] = [k'][g]$, $k' \in K$ (i.e. $[g](K/H) = (K/H)[g]$), since both K and H are normal. Thus $K/H \trianglelefteq G/H$.

Now, define the map

$$\varphi : G/H \to G/K \qquad gH \mapsto gK$$

This map is well-defined: If $g_1H = g_2H$, then $g_1h_1 = g_2h_2$. Since $H \leqslant K$, then $h_1, h_2 \in K$, so $g_1h_1 = g_2h_2$ still holds in G/K for any $h_i \in H$. Thus $\varphi(g_1H) = \varphi(g_2H)$, showing the function is well-defined.

Now, notice this map is a homomorphism and surjective. It is a homomorphism since

$$\varphi(g_1Hg_2H) \stackrel{!}{=} \varphi(g_1g_2H) = g_1g_2K \stackrel{!}{=} g_1Kg_2K = \varphi(g_1H)\varphi(g_2H)$$

where the $\stackrel{!}{=}$ equalities come from exercise ref:HERE. For surjectivity, since $H \leqslant K$, then for any gK, we know there is at least the coset gH that maps to it (there might be more, since $H \leqslant K$). Hence, by corrollary 3.1.16

$$(G/H)/\ker(\varphi) \cong G/K$$

to find $\ker(\varphi)$, we put together the information we have so far:

$$\begin{aligned}
\ker(\varphi) &= \{gH \in G/H \mid \varphi(gH) = K\} \\
&= \{gH \in G/H \mid gK = K\} \\
&= \{gH \in G/H \mid g \in K\} \\
&= K/H
\end{aligned}$$

Thus, replacing $\ker(\varphi)$ with K/H we get

$$(G/H)/(K/H) \cong G/K$$

as we sought to show

An easy heuristic to remember $(G/H)/(K/H) \cong G/K$ is to think about how to simply fraction's of fractions, that is, the "flip and cancel" trick

$$\frac{\frac{1}{2}}{\frac{4}{2}} = \frac{1}{2}\frac{2}{4} = \frac{1}{4}$$

Note however that this is *not* waht's happening here – it is just a useful heuristic to remember the 3rd Isomorphism Theorem. Note too that for the simplification to occur, you need that H is in the denominator of both quotients, you cannot have a different value.

Example 3.5.6: Example

Using the 3rd isomorphism theorem?

Finally, we will relate the subgroup structure (the lattice) of G/N with G. It will actually be precisely a sublattice of G that contains N! This means that going to G/N will preserve subgroup relations:

Theorem 3.5.7: The Fourth or Lattice Isomorphism Theorem (Correspondence Theorem)

Let G be a group and let N be a normal subgroup of G. Then there is a bijection from the set of subgroups H of G which contain N onto the set of subgroups $\bar{H} = H/N$ of G/N.

$$\{H | H \leqslant G,\ H \text{ contains } N\} \leftrightarrow \left\{\overline{H} \leqslant \overline{G}\right\}$$

In particular, every subgroup of $\bar{G}$ is of the form H/N for some subgroup H of G containing N (namely, its pre-image in G under the natural projection homomorphism from G to G/N). This bijection has the following properties between the two lattices

***Proof*:**

Let $\overline{G} = G/N$. Let $\overline{H} \leqslant \overline{G}$. Then since the pre-image of a subgroup is a group, take $H = \pi^{-1}(\overline{H})$. Then by definition, $\pi(H) = \overline{H}$, meaning for every subgroup of $\overline{G}$, there is a subgroup of G that maps to it.

Furthermore, $N \leqslant H$. Since $\pi : H \to H/N$, $\pi^{-1}(1N) \subseteq H$ (note that it's a subset and not element since it's possible that the pre-image contains multiple elements). By definition

$$\pi^{-1}(1N) = \{h \in H \mid \pi(h) = N\} = \{h \in H \mid hN = N\} = \{h \in H \mid h \in N\}$$

and since $\pi^{-1}(1N) \subseteq H$, it must be that $N \subseteq H$, as we sought to show

This fact has many very nice consequences:

Corollary 3.5.8: Equivalent way of looking at 4th Isomorphism Theorem

1. $A \leqslant B$ if and only if $\overline{A} \leqslant \overline{B}$
2. if $A \leqslant B$ then $|B : A| = |\overline{B} : \overline{A}|$
3. $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$
5. $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$

***Proof*:**
Each of these can be proved using a simialr technic as the proof for the 4th Isomorphism Theorem

Another way of looking at the first part of the Correspondence Theorem is in the following slight generalization: Let G be a group and N a normal subgroup so that G/N is well-defined. If $\mathcal{G}$ contains all subgroups of G, for any $H \in \mathcal{G}$, the map $\pi : H \to G/N$, $\pi(h) = hN$ is well-defined. So, H maps to a subgroup of G/N. If $\mathcal{N}$ represents all subgroups of G/N, then there is a surjective map

$$\mathcal{G} \twoheadrightarrow \mathcal{N}$$

and if we take $\mathcal{G}|_N$ to be all subgroups that contain N, then there is a bijection between $\mathcal{G}|_N$ and $\mathcal{N}$:

$$\mathcal{G}|_N \leftrightarrow \mathcal{N}$$

Example 3.5.9: Using Correspondence Theorem

1. Let $G = Q_8$, and take $\langle -1\rangle \trianglelefteq Q_8$. Then in the following lattice, the double lines represent the sublattice of Q_8 which is an isomorphic copy of the lattice of $Q_8/\langle -1\rangle$

$$\begin{tikzcd}
& Q_8 \arrow[dl, Rightarrow] \arrow[d, Rightarrow] \arrow[dr, Rightarrow] & \\
\langle i\rangle \arrow[dr, Rightarrow] & \langle i\rangle \arrow[d, Rightarrow] & \langle i\rangle \arrow[dl, Rightarrow] \\
& \langle i\rangle \arrow[d, dash] & \\
& \{1\} &
\end{tikzcd}$$

2. Let $G = D_4$ and let $\langle r^2 \rangle \trianglelefteq D_4$. Then with the same definitions as the last example, we get:

$$\begin{tikzcd}
& & D_4 \arrow[dl, Rightarrow] \arrow[d, Rightarrow] \arrow[dr, Rightarrow] & & \\
& \langle s, r^2 \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, Rightarrow] & \langle r \rangle \arrow[d, Rightarrow] & \langle rs, r^2 \rangle \arrow[dl, Rightarrow] \arrow[d, dash] \arrow[dr, dash] & \\
\langle s \rangle \arrow[drr, dash] & \langle r^2 s \rangle \arrow[dr, dash] & \langle r^2 \rangle \arrow[d, dash] & \langle rs \rangle \arrow[dl, dash] & \langle r^3 s \rangle \arrow[dll, dash] \\
& & 1 & &
\end{tikzcd}$$

Notice that several subgroup of G (for example $\langle s \rangle$) become elements in a larger subgroup (in this case $\langle s, r^2 \rangle$). Thus, every subgroup of $\overline{G}$ might have multiple subgroups of G that map to it, but a unique subgroup which contains N.

A good example how the map of subgroup of G to G/N is surjective is by the following example: Notice that $Q_8/\langle -1 \rangle \cong D_8/\langle r^2 \rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and $\langle -1 \rangle \cong \langle r^2 \rangle$, but $Q_8 \ncong D_8$. We will explore this idea in more depth in the next section.

Exercise 3.5.1

1. In the following questions, we're going to prove what's sometimes called the 5th Isomorphism Theorem (also known as the *Butterfly Lemma*), which for our purposes will offer an alternative proof to the main result of section 3.7. HERE

3.6 Extensions

Recall (or revisit) that if $H \leqslant G$ (and G is finite), then the subgroup lattice of H is the sublattice of G of all groups *contained* in H (theorem 2.4.3), so *a fortiori* if $K \trianglelefteq G$ is a normal subgroup then the subgroup lattice of K is the sublattice of G of all subgroups contained in K. Even hen G is infinite, the set of subgroup of K is the set o subgroup of G which are contained in K. The Correspondence Theorem has shown us that the "dual" is true for quotient groups (except it orks for *all* groups, not just finite): If $K \trianglelefteq G$, then the subgroup lattice of G/K is the sublattice of G of all groups *containing* K. In this way, K and G/K together capture the subgroup structure of a group! In this section, we will explore how we can find a group structure given *only* the information from K and G/K.

More succintely, Given $K \trianglelefteq G$, the group K and G/K contain the subgroup structure of G. However, K and G/K don't capture the full structure of a group: in the process of quotienting, G/K losses some information:

Example 3.6.1: Subgroup And Quotient Group Limits

Take $G = D_8$ and $K = \{1, r, r^2, r^3\} \cong C_4$. Then $D_8/K \cong C_2$ – let's label this K'. Then given just K and K', e can't guarantee that $K' = D_8/K$, other groups can take the place of D_8. For example:

$$H = C_4 \times C_2 \qquad \varphi : C_4 \times C_2 \to C_2,\ \varphi((a,b)) = ((0,b))$$

Here, H is another group here the kernel is C_4 and the quotient is C_2.

It must be noted though that there is still information within cosets: for any $x \in G$, then $[x] = xK \in G/K$, so for any $y \in xK$, $y = xk$ for appropriate $k \in K$ (proposition 3.1.3). This means for every coset, there is some information about how the elements relate. An easy example of this can be seen if $K = Z(G)$.

Example 3.6.2: Relation Within Cosets

Take $G/Z(G)$ (as an asside, try proving that $Z(G) \trianglelefteq G$ alway, exercise ref:HERE). Take any $[x] \in G/Z(G)$, and any $a, b \in [x]$. Then $a = xz_a$ and $b = xz_b$, so $ab = xz_axz_b = xz_bxz_a = ba$ (since $z_a, z_b \in Z(G)$). Thus, within any coset of $G/Z(G)$, the elements commute! This is not nesesarily the case if the elements are *not* within the same coset (ex. $G = D_7$).

However, as we saw in example 3.6.1, there can be more than one group (say G') we can construct from K and K' (where $G'/K = K'$). We will call groups which can be built in this fashion *extension*:

Definition 3.6.3: Extension

Let A and B be groups. If $G/A \cong B$, then we say *G is an extension of A by B.*

The idea behind this name is that we want to be able to restrict G to A (i.e. A must be able to map injectively and homomorphically into G; $A \hookrightarrow G$), similarly to how we can restrict functions (the dual of restricting functions is also called *extending* functions, where the extended function must be exactly the same as the orginal funciton when restricting).

As we saw at the beginning of this sectino, quotient groups are the prefect candidate for extension of groups since every group can be decomposed into a [normal] subgroup and quotient group (TBD – perhaps put this earlier)

Since all normal subgroups are also kernels, we can re-write what it means to be an extension in terms of homomorphisms. In the following definition, let I represent the trivial group.

Definition 3.6.4: Short Exact Sequence

Let A, B, C be group. If there exists an injective homomorphism f and a surjective homomorphism g such that

$$I \xrightarrow{\iota} A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{\pi} I$$

where $\mathrm{im}(f) = \ker(g)$, ι is the trivial injective $(0 \mapsto 0)$ map and π is the total projective map $(c \mapsto 0)$, then we say that *B is an extension of A by C.*

Notice that $\mathrm{im}(\iota) = \ker(f)$ since f is injective, $\mathrm{im}(g) = \ker(\pi)$ since g is surjective, and $\mathrm{im}(f) = \ker(g)$ by definition. The fact that the image of the previous function is the kernel of the next function is no coincidence – this property is called *exactness* [6]. If we make a longer chain of homomorphisms with this property (ex. $X_1 \xrightarrow{f_1} X_2 \xrightarrow{f_2} X_3 \xrightarrow{f_3} X_4 \xrightarrow{f_4} \cdots$), then such a chain is called an *exact sequence* (we will return to these in chapter 17⋆). The exact sequence in definition 3.6.4 is the

[6] You might have seen the term *exact vector-field*. This concept is in fact the same, however it would take us a long time before we can link the two (see ref:HERE)

shortest non-trivial sequence with the trivial group at the beginning and end, and so it is called a short exact sequence.

For our purposes, the fact that $\mathrm{im}(f) = \ker(g)$ means that $B = C/A$, meaning that B is an extension of A by C. (exercise: does $G \to H$ imply a short exact sequence? If no, can it)

Example 3.6.5: Short Exact Sequences

In the following examples, let 1 represent the trivial group

1. If $\varphi : G \to H$ is a homomorphism and $K = \ker(\varphi)$, we always have
$$1 \to K \hookrightarrow G \xrightarrow{\varphi} \varphi(G) \to 1$$
is a short exact sequence. If φ is surjective, then
$$1 \to K \hookrightarrow G \xrightarrow{\varphi} H \to 1$$
In this way, every surjective homomorphism represents an extension G of K by H.

2. Let A and B be two arbitrary groups. Then
$$1 \to A \hookrightarrow A \times B \xrightarrow{(a,b)\mapsto(0,b)} B \to 1$$
is a short exact sequence. In this way, any group can be extended by any other group to the extensions $A \times B$. This is sometimes called the *trivial extension* [a]

3. Take $A = B = 2$. Then
$$1 \to C_2 \xrightarrow{f} C_4 \xrightarrow{g} C_2 \to 1$$
where $f(x) = 2x$ and $g(0)$ and $g(2)$ both map to 0, while $g(1)$ and $g(3)$ map to 1. It is easy to verify that f and g are indeed homomorphism and that the above sequence is exact.

4. Similarly to the last example, Let $A = C_2$ and $B = C_8$ and $C = C_4$. Can you find the appropriate maps such that
$$1 \to C_2 \to C_8 \to C_4 \to 1$$
is a short exact sequence? Is the following
$$1 \to C_2 \to D_4 \to C_4 \to 1$$
also a short exact sequence with the appropriate maps?

[a] In fact, once we cover homological algebra in chapter 17⋆, the idea of this being the trivial extension will become rigorous

An extension of an X group by an X group is X. When is this statement true when X is:

1. cyclic
2. If M and A are abelian, G need not be abelian
3. finite
4. finitely generated

5. Noetherian

It is in general difficult to tell when when given a short exact sequence if the sequence splits. In section 17.2, we will classify which *abelian groups* in a short exact sequence of abelian groups split[7] (abelian groups which always split in a short exact sequence of abelian groups be called *projective groups*). For general groups, The best tool we have is trying to identify when C acts on B, (mabye there are some identification of when this is possible??)

(maybe make HNN extensions an exericise? Or more it to an Algebraic topology section if one ever exists?)

(

You may also think of the short exact sequence:

$$1 \to H \to G \to G/H \to 1$$

Then since H is the kernel, we can break down G into zH for some representative $z \in G/H$. It is natural to ask if we can think of as G as "$(G/H) \cdot H$". However, this is not so simple. It sometimes works, it is called a split extension, and in general you require the fundamental theorme of extension of groups (which shows that G is a subgroup of the wearth product of H and G/H. We alos use homology to see how much G "fails" to be a direct product of H and G/H)

)

Exercise 3.6.1

1. Let $0 \to \mathbb{Z} \to B \to A \to 0$ be an extension, where B is an abelian group and A is free. Show that B splits. [Interestingly, The converse of this statement (known as Whitehead's Problem) is undecidable in ZFC, that is, it is a statement that could neither be true nor false! This idea would be explored with Gōdel's Incompleteness Theorems]

2. Is it save to say that if $0 \to A \to G \to B \to 0$ where $|B|$ is finite, then G is A-by-finite (and so virtually A)? That would be another cool way of thinking about the B term in a short exact sequence!

3.7 Classifying all Finite Groups: Hölder Program

With the isomorphism theorems under our belt, we are ready to tackle one of the central questions posed by group-theorists: How can we classify all groups? This sections start by building up the mental models used in order to comprehend the way classification efforts.

We can approach the problem of classification in many different ways. We briefly mentioned the idea trying to construct arbitrary presentation's of groups to try and create groups, but we've mentioned that this can create "insoluble presentation", meaning presentaion in which we cannot represent every element. In general, it is very hard to find groups in that manner. Perhaps instead we can start with the groups we knwo and see if there is a way to "decompose" them. The first thing you might try is decomposing the group into cyclic sub-groups. This is certainly possible (take an element,

[7]It will turn out that all abelian groups are $\mathbb{Z}$-modules, and so this will be a consequence of classifying when modules split with the space case of when the module is over $\mathbb{Z}$

generate a group from it, remove those elements from your origianl group, continue doing this till you've exhauted the group [8]). But how would you then put them back together? For example, we know that $\langle r \rangle, \langle s \rangle \leqslant D_4$. With what we've learnt, is there a way to represent D_4 using these? In fact, we have just seen that we can but them back together using short exact sequences! The group D_4 is an extension of C_2 by C_4.

However, we must be more careful using extensions when we are trying to classify groups; are cyclic subgroups really the way to go? In fact, this is not the case: there exists cyclic groups which are an extension of smaller groups. We would want to be able to have some fixed subgroups we can use as our "building puzzles" which are not extensions of any other groups. In this section we will define these "smalles puzzle pieces" or "atoms of groups". Try taking a moment to see if you can come up with a more suitable definition of "smallest puzzle piece" for groups and have an idea of an example of a group following your definition.

We will end this section by being able to declare a theorem that took group theorists the better part of the 20th century to prove – the classification of all finite groups! As a quick note before proceeding, this classification effort is in fact much simpler for *abelian groups*. In chapter 6.1, section ref:HERE, we will continue this exploration in the finite abelian case.

Defining our Terms

First, we will have to define what would be our "puzzle piece". Since we are using extensions, we know there is a non-trivial homomorphism from a group G to another group H if G has a non-trivial normal subgroup. If G has *only* trivial subgroups, the structure of G is not "composed" of other possible structure but is as "simple" as possible. This motivates the folloing definition.

Definition 3.7.1: Simple Groups

A (finite or infinite) group G is called *simple* if $|G| > 1$ and the only normal subgroups of G are 1 and G.

These groups are called "simple" because there are no more group-structures within them [9]. To undersatnd what "no more group-structures within them" means, recall the 4th isomorphism theorem. If N is a noraml subgroup, then ther is a direct correspondence between the subgroups of G/N and the subgroups of G containing N. If there is no $N \trianglelefteq G$ besides $\{1\}$ and G, then the lattice of G can't be split into G/N and G. So in a sense, that lattice is as small as it gets. (comment on infinite case?)

Example 3.7.2: Simple Groups

We know that by Lagrange's theorem that if G is prime, then it's only subgroups must be 1 and G, and so groups of order p are examples of simple groups. If G is abelian, then it can in fact be shown that every simple subgroup is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ for appropriate prime p, since those are the only subgroups which no non-trivial subgroups (and so no normal subgroups)

There are also examples of non-abelian simple Groups, both infinite and finite (the smallest finite

[8] This naturally works well in groups of *countable* size, one has to be more careful for groups of *uncountable* size – the axiom of choice is involved (see chapter A for details on the axiom of choice)

[9] The term simple is one that comes up often enough in matheamtics. It has a sligthly more nuanced meaning than just the simplest object (ex. the trivial group is not a simple group, it's "too simple to be simple". See this nlab article on simple objects

one (of order 60) will be explored in detail in chapter 4, section ref:HERE).

Looking at simple groups a little more abstractly for a moment, they play an analogous role to primes in arithmetic's of $\mathbb{N}$ as in they cannot be "factored" into smaller [meaningful] pieces[10] (i.e. N and G/N). There is even a "unique factorization theorem" based on this idea. First, we'll define what is meant by "factorize" a group

(link to extensions)

Definition 3.7.3: Subnormal Series

Let G be a group. Then

$$G = G_1 \geqslant G_2 \geqslant G_3 \geqslant \cdots \geqslant G_n = \{1\}$$

is called a *series* (or *tower*) for the group G. If every $G_i \trianglelefteq G$, then G the series is called a *normal series*. If $G_i \trianglelefteq G_{i-1}$ for every $1 \leq i \leq n$, then we call the series a *subnormal series*:

$$G = G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

Sometimes, [normal/subnomral] series are written in the ascending, rather than descending order:

$$\{1\} = G_0 \leqslant G_1 \leqslant \cdots \leqslant G_n = G$$

I chose to write them in descending order since down the line we will be comparing some types of series to other types, and if everything is written in descending order, then we'd have a scendario were we start writing $G_{n-i} \trianglelefteq B_i$ ineated of $G_i \trianglelefteq B_i$, which I feel will make the simple relations a little more notationaly cumbersome than it needs to be.

You might ask whether this a subnormal series would imply a noraml series, that is, that $N_i \trianglelefteq G$? However, that doesn't need to be the case:

Example 3.7.4: Normality *not* transitive

Here's an example of how normality is *not* transitive. Let $G = D_8$. Let $H \trianglelefteq G$ such that $H = \langle r\sigma, \sigma r\rangle$. Next, take $K \trianglelefteq H$ such that $K = \langle r\sigma\rangle$. Note that

$$K \trianglelefteq H \trianglelefteq G$$

but $K \ntrianglelefteq G$

thus a subnormal series need not be a noraml series. However, a normal series is a subnormal series: if $N_{i+1} \trianglelefteq G$, then $gN_{i+1}g^{-1} = N_{i+1}$ for every $g \in G$. If we restrict to every $n \in N_i$, then since $N_i \subseteq G$, it is already the cse that $nN_{i+1}n^{-1} = N_{i+1}$. Thus $N_{i+1} \trianglelefteq N_i$. Thus:

$$\text{normal series} \Rightarrow \text{subnormal series} \qquad \text{subnormal series} \not\Rightarrow \text{normal series}$$

Thus, subnormal series can be thought of as the decomposition of a group.

[10] meaningful here meaning there is a notion of combining groups using short exact sequence, unlike general groups can't be

Example 3.7.5: Example

here. In particualr, give a two subnromal series for a group, one more refined than the other

In (reference previous example), the subnormal serise cannot be refined further, since every N_i/N_{i+1} is simple. We give a name to such a serise:

Definition 3.7.6: Composition Series

In a group G, a sequence of groups

$$1 = N_0 \leqslant N_1 \leqslant N_2 \leqslant \cdots \leqslant N_{k-1} \leqslant N_k = G$$

is called a *composition series* if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group, for all $0 \le i \le k-1$. If the above sequence is a composition series, the quotient group N_{i+1}/N_i are called the *composition factors* of G.

Every finite group has a composition series. Since the group is finite, there exists a maximal subgroup [11], so that $N \trianglelefteq G$ so that G/N is simple. If $N = \{1\}$, then G is already simple, since the only proper normal subgroup of G is $\{1\}$, meaning the only other normal subgroup is G. If the group G/N was not simple, then there would exist a group $N'/N \trianglelefteq G/N$. Then by the 4th isomorphism theorem, $N' \trianglelefteq G$, where N' is a group containing N. However, N is a maximal subgroup, so it must be that $N = N'$, so we'd get $N'/N = N/N \cong \{1\} \trianglelefteq G/N$, and so G/N is simple. We can then continue the same process with N, taking $N_1 \trianglelefteq N$ to be the maximal normal subgroup. Since G was finite this process will be end.

Example 3.7.7: Some Composition Series

Take $\mathbb{Z}_p \times \mathbb{Z}_q$. Then

$$\{e\} \trianglelefteq \mathbb{Z}_p \times \{e\} \trianglelefteq \mathbb{Z}_p \times \mathbb{Z}_q$$
$$\{e\} \trianglelefteq \{e\} \times \mathbb{Z}_q \trianglelefteq \mathbb{Z}_p \times \mathbb{Z}_q$$

The quotient of any two will produce simple groups

Take Q_8 Then

$$\{e\} \trianglelefteq \langle -1 \rangle \trianglelefteq \langle i \rangle \trianglelefteq Q_8$$
$$\{e\} \trianglelefteq \langle -1 \rangle \trianglelefteq \langle j \rangle \trianglelefteq Q_8$$
$$\{e\} \trianglelefteq \langle -1 \rangle \trianglelefteq \langle k \rangle \trianglelefteq Q_8$$

Where I figured out the composition series with the help of the lattice of subgroups to trace possible paths and figuring out whether they're normal or not. The quotient so any of these factors will be isomorphic to $\mathbb{Z}_2$, which is simple, and thus there are composition series.

another example is:

$$1 \trianglelefteq \langle s \rangle \trianglelefteq \langle s, r^2 \rangle \trianglelefteq D_8 \qquad \text{and} \qquad 1 \trianglelefteq \langle r^2 \rangle \trianglelefteq \langle r \rangle \trianglelefteq G$$

[11] note that maximal subgroups by definition are proper subgroups

Note that not all infinite groups admit a compositions series

Example 3.7.8: Example

I think $M_2(\mathbb{Q})$ or $GL_2(\mathbb{Q})$ is an example.

However, some groups do have compositions series. Take the group

$$D_\infty = \langle x, y | y^2 = 1,\ xy = y^{-1}x \rangle$$

this is called the *infinite dihedral group*. Notice that $\langle x \rangle \trianglelefteq D_8$: if we take some $g \in \langle x \rangle$, then since $\langle x \rangle$ is cyclic, then $g = x^a$ for some a. Conjgating x^a, we get $yx^a y^{-1} = yyx^a = x^a$, meaning $yxy^{-1} \in \langle x \rangle$. Using this, we can show that $\langle x \rangle \trianglelefteq D_\infty$. Thus, $D_1 = D_\infty / \langle x \rangle$ is well-defined. We need to show that D_1 is Let's take some $g \notin \langle x \rangle$ so that $[g] \neq [e]$. (buidlling up to show that)

$$C_2 \cong D_\infty / \langle x \rangle$$

Which is simple. Further more, C_2 is itself a simple abelian group. Thus, our composition seris is

$$D_\infty \trianglerighteq \langle x \rangle \trianglerighteq \{1\}$$

Some of infinite groups are close to being composition series, in the sense that $G = \cup_{i=1}^{\infty} G_i$ where G_i / G_{i+1}. Such groups will be addressed further [book ref:HERE]

(A word on how composition series are just simple extensions:

We saw that a finite group G always has a composition series, and hence a subnormal series:

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

Then notice that every subnormal series can be broken down into short exact sequences:

$$1 \longrightarrow G_{n-1} \longrightarrow G_{n-2} \longrightarrow G_{n-2}/G_{n-1} \longrightarrow 1$$

$$1 \longrightarrow G_i \longrightarrow G_{i-1} \longrightarrow G_{i-1}/G_i \longrightarrow 1 \quad 1 \longrightarrow G_1 \longrightarrow G_0 \longrightarrow G_0/G_1 \longrightarrow 1$$

thus, every subnormal series can be thought of as a series of series of extensions G_0/G_1, G_1/G_2, ... G_{n-1}.)

One might ask about if a composition series is supposed to be unique factorization, what about there being 3 composition series for Q_8, and what about the re-arranging we've done of $\mathbb{Z}_p \times Z_q$? The *quotients* of each N_{i+1}/N_i will be unique *up to rearrangement*! That is the content of the next theorem:

Theorem 3.7.9: Jordan-Hölder's Theorem

Let G be a finite group with $G \neq 1$. Then

1. G has a composition series
2. The composition factors in a composition series are unique, namely, if $1 = N_0 \leqslant N_1 \leqslant \cdots \leqslant N_r = G$, and $1 = M_0 \leqslant M_1 \leqslant \cdots \leqslant M_s = G$, are two composition series of G, then $r = s$ and there is some permutation π of $\{1, 2, ..., r\}$ such that

$$M_{\pi(i)}/M_{\pi(i)-1} \cong N_{\pi(i)}/N_{\pi(i)-1} \qquad 1 \leq i \leq r$$

***Proof*:**
apparently doable. This material wasn't covered in class and there's a test coming up so I'll get back to it another time

With that definition of factorization, we can now ask one of the biggest questions in group theory:

Theorem 3.7.10: The Hölder Program

1. Classify *all* finite simple groups
2. Find all ways of "putting simple groups together" to from other groups

***Proof*:**
The proof for the 1st point took around 100 years of work, with the effort of around 100 mathematicians, writing between 300-500 papers, totalling 5000-10000 pages. It is left as some light reading for the reader ☺

The second point is yet to be complete.

The result of all that work means the following single incompassing theorem can confidentily be write down:

Theorem 3.7.11: Types of Finite Groups

There is a list consisting of 18 (infinite) families of simple groups and 26 (or 27 [a]) simple groups not belonging to these families (the *sporadic* simple groups) such that every finite simple group is isomorphic to one of the groups in this list

[a] some include Tits group

Example 3.7.12
Here are a couple such family of groups:

1. $\{Z_p \big| p \text{ is prime }\}$
2. $\{SL_n(\mathbb{F})/Z(SL_n(\mathbb{F})) \big| n \in \mathbb{Z}^+, n \geq 2, \text{ and } \mathbb{F} \text{ is a finite field}\}$
3. A_n, $n \geq 5$.

> Examples after this start to get more complicated. I found this wikiepdia article outlining more of the history of this process.
>
> here

(somehow mention "generalization of abelian group section", where I can expan on how nilpotent and solvable groups give us some nice properties to make some groups easier to reprsent)

:exercise Let $H \trianglelefteq G$ such that there does not exist a $K \trianglerighteq H$ such that $H \trianglelefteq K \trianglelefteq G$. In other words if $K \trianglerighteq H$, then either $K = H$ or $K = G$. Such a normal subgroup is called the maximal normal subgroup. Show that H is a maximal normal subgroup if and only if G/H is a simple group

3.8 Quotient Objects

Link to Epimorphisms, link the idea of the "dual" is sub-object. Some intuition then perhaps on how the sub and quotient object together capture the idea of the entire group G. Leads to the idea of extensions.

In the previous chapter, we introduced the equivalent definition of subgroup in proposition 1.4.8. In it, we had an injective map $i : H \to G$. We will now ask what mathematicians call the natural *dual* question: What happens if we instead consider a surjective function $j : G \to H$ which is also a homomorphism?

(Link to Ker Object?)

4

Group Actions

group action is functor $* \to X$

In Section 1.5, we saw that groups arise naturally as *symmetries*: the dihedral group captures the symmetries of a polygon, the symmetric group captures every possible rearrangement of a finite set, and the general linear group captures every invertible linear transformation of a vector space. In each case, the group *acted* on something.

Historically, groups *were* symmetry groups before they were axiomatized: Galois studied permutations of roots. Cayley studied transformations of coordinates. Klein, in his 1872 *Erlangen Programme*, went so far as to propose that *geometry itself* should be defined by specifying a group of transformations and studying the properties they leave invariant (see the aside below). The abstract axioms we introduced in Section 1.1 are a *distillation* of what all these concrete symmetry groups have in common.

In this chapter, we reverse the process. Having built the abstract theory we now reconnect it to its roots by developing the theory of *group actions*: the study of how a group can act on a set (or on itself). We will show:

1. The Orbit-Stabilizer Theorem (§4.3). If a group G acts on a set X, then X decomposes into orbits, and the size of each orbit is controlled by a subgroup of G (the stabilizer). This is, in spirit, a generalization of Lagrange's Theorem (theorem 3.3.1). We shall see that if the action is *transitive*, then X can be represented as cosets of G, leading the way to motivate why it is in fact natural (perhaps better to say more universal) to study actions of a group on itself

2. Cayley's Theorem (§4.5). Every group is isomorphic to a subgroup of a symmetric group. In other words, every abstract group *is* a group of permutations—the symmetry perspective is not just a heuristic, it is a theorem.

3. The Class Equation (§4.6). By letting a group act on itself by conjugation, we obtain an

arithmetic identity that splits $|G|$ into contributions from the center $Z(G)$ (definition 2.2.4) and the non-central conjugacy classes. This will immediately imply that every p-group has a nontrivial center.

4. The Sylow Theorems (§4.8). For any prime p dividing $|G|$, Sylow's three theorems guarantee the *existence* of subgroups of maximal p-power order, prove they are all *conjugate* (hence isomorphic), and tightly constrain their *number*. These theorems are among the most powerful tools in finite group theory.

Along the way, we will see several important *types* of actions—faithful, transitive, free, and regular—and see how each captures a different facet of the relationship between a group and the object it acts on.

A recurring theme will be that the most important applications of group actions come not from letting a group act on some external set, but from letting a group act *on itself* or on its own substructure (elements, subsets, cosets). When a group acts on itself by left multiplication, we get Cayley's Theorem. When it acts on its cosets, we get structural information about normal subgroups. When it acts on itself by conjugation, we get the Class Equation. When it acts on the set of its own Sylow subgroups, we get the Sylow counting theorem. Each of these is an instance of the same machine; what changes is the *choice of action*, and each choice reveals different information.

If there is one philosophical takeaway from this chapter, it is this:

To understand a group, watch how it acts.

4.0.1 Applied Aside: Klein's Erlangen Programme In 1872, Felix Klein presented a new vision of geometry at the University of Erlangen. Rather than defining geometries axiomatically (as Euclid had done), Klein proposed that a *geometry* is a set X together with a group G acting on it, and that the "geometric properties" of X are precisely those invariant under the action of G.

Under this lens, different choices of group yield different geometries:

Group G	**Set** X	**Geometry**
Euclidean isometries	$\mathbb{R}^2$	Euclidean geometry
Affine transformations	$\mathbb{R}^2$	Affine geometry
Projective transformations	$\mathbb{RP}^2$	Projective geometry

Euclidean geometry studies properties like distance and angle (invariant under rigid motions); affine geometry studies parallelism (invariant under the larger group of affine maps); projective geometry studies incidence (invariant under the still larger projective group). More symmetry means fewer invariants, hence a "coarser" geometry.

Klein's Programme was one of the first explicit formulations of the idea that *group actions define structure*. It remains a guiding principle: in differential geometry, in topology, and throughout modern mathematics, the question "what is the symmetry group, and how does it act?" is often the first one asked.

4.1 Definitions and First Examples

In Section 1.5, we observed informally that a group can "act" on a set by rearranging its elements. Our goal now is to make this precise. We will give two equivalent formulations of the definition—one emphasizing the homomorphism, the other emphasizing axioms on the action map—and prove that they are interchangeable.

The Homomorphism Definition

Let us return to the motivating example from the prelude. The dihedral group D_{2n} (definition 1.2.25) acts on the set of vertices $\{1, 2, \ldots, n\}$ of a regular n-gon. Each element $g \in D_{2n}$ determines a bijection of the vertex set: a rotation cyclically permutes the labels, a reflection swaps them. Different elements yield different (or sometimes the same) bijections, and—crucially—performing one symmetry followed by another corresponds to composing the associated bijections.

In other words, there is a map ρ that assigns to each group element $g \in D_{2n}$ a permutation $\rho(g) \in \mathrm{Sym}(\{1, \ldots, n\})$, and this map *respects the group structure*: $\rho(gh) = \rho(g) \circ \rho(h)$. That is, ρ is a homomorphism (definition 1.4.2). This observation is the definition.

Definition 4.1.1: Group Action

Let G be a group and let X be a set. A *group action* of G on X is a group homomorphism

$$\rho\colon G \to \mathrm{Sym}(X),$$

where $\mathrm{Sym}(X)$ is the symmetric group on X (definition 1.2.12), i.e. the group of all bijections $X \to X$ under composition.
We write $G \curvearrowright X$ (read "*G acts on X*") to indicate that such a homomorphism has been specified. The set X, together with the action ρ, is called a ***G-set***.

Let us unpack what this definition gives us concretely.

- For each $g \in G$, the image $\rho(g)$ is an element of $\mathrm{Sym}(X)$—that is, a *bijection* from X to X. We can *evaluate* this bijection at any point $x \in X$ to obtain an element $\rho(g)(x) \in X$. One should read "$\rho(g)(x)$" as: *"the result of applying the permutation $\rho(g)$ to the element x."*
- Since ρ is a homomorphism, it respects composition: $\rho(gh) = \rho(g) \circ \rho(h)$ for all $g, h \in G$.
- Since ρ is a homomorphism, it sends the identity to the identity (proposition 1.4.4): $\rho(e) = \mathrm{id}_X$, the identity permutation.

It is convenient to introduce a shorthand that suppresses the homomorphism ρ:

$$g \cdot x \;:=\; \rho(g)(x) \qquad \text{for all } g \in G,\ x \in X.$$

In this notation, the two properties above translate into:

1. Compatibility: $\quad (gh) \cdot x \;=\; g \cdot (h \cdot x) \qquad$ for all $g, h \in G$ and $x \in X$.
2. Identity: $\quad e \cdot x \;=\; x \qquad$ for all $x \in X$.

These are not new axioms—they are automatic consequences of ρ being a homomorphism. The natural question is: are they *sufficient*? That is, if we have a rule "$g \cdot x$" satisfying just these two properties, does it necessarily come from a homomorphism to $\mathrm{Sym}(X)$? The answer is yes.

The Axiomatic Characterization

Many textbooks take the following as the *definition* of a group action, and then derive the homomorphism as a consequence. Since we have gone the other direction, we state it as a theorem establishing the equivalence.

Theorem 4.1.2: Equivalence of Definitions

Let G be a group and X a nonempty set. There is a bijective correspondence between:

(a) group homomorphisms $\rho\colon G \to \mathrm{Sym}(X)$, and

(b) functions $\cdot\colon G \times X \to X$, written $(g, x) \mapsto g \cdot x$, satisfying

- *(A1)* $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$, and
- *(A2)* $e \cdot x = x$ for all $x \in X$.

The correspondence is given by $g \cdot x = \rho(g)(x)$.

***Proof*:**
(a)⇒(b). Given a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, define $g \cdot x := \rho(g)(x)$. We check the two axioms.

For (A1): since ρ is a homomorphism, $\rho(gh) = \rho(g) \circ \rho(h)$, so

$$(gh) \cdot x \;=\; \rho(gh)(x) \;=\; \big(\rho(g) \circ \rho(h)\big)(x) \;=\; \rho(g)\big(\rho(h)(x)\big) \;=\; g \cdot (h \cdot x).$$

For (A2): since ρ sends identities to identities (proposition 1.4.4), $\rho(e) = \mathrm{id}_X$, so

$$e \cdot x \;=\; \rho(e)(x) \;=\; \mathrm{id}_X(x) \;=\; x.$$

(b)⇒(a). Given a function $\cdot\colon G \times X \to X$ satisfying (A1) and (A2), we must construct a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$. For each $g \in G$, define the function

$$\sigma_g\colon X \to X, \qquad \sigma_g(x) = g \cdot x.$$

We must show two things: that each σ_g is a bijection (hence an element of $\mathrm{Sym}(X)$), and that the map $g \mapsto \sigma_g$ is a group homomorphism.

Step 1: Each σ_g is a bijection. We claim that $\sigma_{g^{-1}}$ is a two-sided inverse for σ_g. For any $x \in X$:

$$(\sigma_g \circ \sigma_{g^{-1}})(x) = \sigma_g(g^{-1} \cdot x) = g \cdot (g^{-1} \cdot x) \overset{\text{(A1)}}{=} (gg^{-1}) \cdot x = e \cdot x \overset{\text{(A2)}}{=} x,$$

so $\sigma_g \circ \sigma_{g^{-1}} = \mathrm{id}_X$. An identical computation gives $\sigma_{g^{-1}} \circ \sigma_g = \mathrm{id}_X$. Thus σ_g is a bijection, and $\sigma_g \in \mathrm{Sym}(X)$.

Step 2: $g \mapsto \sigma_g$ is a homomorphism. Define $\rho\colon G \to \mathrm{Sym}(X)$ by $\rho(g) = \sigma_g$. For any $g, h \in G$ and $x \in X$:

$$\rho(gh)(x) = \sigma_{gh}(x) = (gh) \cdot x \overset{(A1)}{=} g \cdot (h \cdot x) = \sigma_g(\sigma_h(x)) = (\sigma_g \circ \sigma_h)(x) = \big(\rho(g) \circ \rho(h)\big)(x).$$

Since this holds for every $x \in X$, we have $\rho(gh) = \rho(g) \circ \rho(h)$, so ρ is a homomorphism.

Finally, these two constructions are inverse to each other: starting from ρ, forming $g \cdot x = \rho(g)(x)$, and then recovering the homomorphism $g \mapsto \sigma_g$ gives back $\sigma_g = \rho(g)$; conversely, starting from $\cdot$, forming $\rho(g) = \sigma_g$, and then recovering the action gives $\rho(g)(x) = \sigma_g(x) = g \cdot x$. Thus the correspondence is bijective.

4.1.3 Remark: Currying The equivalence above is an instance of a general phenomenon in mathematics (and computer science) called *currying*. A function of two variables $\cdot\colon G \times X \to X$ can be equivalently viewed as a function of one variable $\rho\colon G \to X^X$ that returns a *function* $X \to X$. The group action axioms are exactly the conditions ensuring that ρ lands in $\mathrm{Sym}(X) \subseteq X^X$ and is a homomorphism. This point of view—recasting a "two-variable" structure as a "one-variable" homomorphism—will reappear when we study modules and representations.

Thanks to theorem 4.1.2, we may define group actions using whichever formulation is most convenient. In practice:

- To define an action on a concrete set, it is usually easiest to write down a rule $g \cdot x =$ (some formula) and verify axioms (A1) and (A2).
- To reason abstractly about actions (e.g. to apply the isomorphism theorems), it is usually more powerful to work with the homomorphism $\rho\colon G \to \mathrm{Sym}(X)$.

We give a name to this homomorphism when it arises from an action:

Definition 4.1.4: Permutation Representation

If $G \curvearrowright X$ is a group action, the associated homomorphism $\rho\colon G \to \mathrm{Sym}(X)$ is called the *permutation representation* of the action. We say the action *affords* or *induces* the permutation representation ρ.

Since ρ is a homomorphism, all the tools we developed for homomorphisms apply: its kernel $\ker(\rho)$ is a normal subgroup of G (proposition 2.2.12), its image $\rho(G)$ is a subgroup of $\mathrm{Sym}(X)$ (proposition 1.4.8), and the First Isomorphism Theorem (theorem 3.1.14) gives $G/\ker(\rho) \cong \rho(G) \leq \mathrm{Sym}(X)$. We will exploit these facts heavily starting in Section 4.2.

We close this subsection by recording why we chose the homomorphism formulation as our leading definition. This is a matter of taste, but the reasoning is forward-looking:

- A group action is a homomorphism $G \to \mathrm{Sym}(X)$.
- A linear representation (Chapter X, if applicable) will be a homomorphism $G \to GL(V)$.
- A module (Part III) will be defined via a ring homomorphism $R \to \mathrm{End}(M)$.

In each case, the pattern is the same: the algebraic object "acts" by mapping homomorphically into the automorphisms of the thing it acts on. Remembering this single pattern—*action = homomorphism into an automorphism group*—will serve you well across all of algebra.

A Menagerie of Examples

We now build fluency with the definition by working through a collection of examples. In each case, we specify the group G, the set X, and the action rule $g \cdot x$, and we either verify the axioms or indicate that the verification is a straightforward exercise.

Example 4.1.5: Fundamental Group Actions

1. *The tautological action of S_n.* Let $G = S_n$ and $X = \{1, 2, \ldots, n\}$. Define

$$\sigma \cdot i \;=\; \sigma(i).$$

   The permutation representation is the identity map $\rho = \mathrm{id}\colon S_n \to \mathrm{Sym}(\{1, \ldots, n\}) = S_n$, which is trivially a homomorphism. This is the most elementary group action: each permutation simply *does what it says*.

2. *Dihedral symmetries of a regular n-gon.* Let $G = D_{2n}$ (definition 1.2.25) and let $X = \{1, 2, \ldots, n\}$ represent the vertices of a regular n-gon, labeled clockwise. The rotation r acts by $r \cdot i = i + 1 \pmod n$, and the reflection s acts by $s \cdot i = n + 1 - i \pmod n$ (with an appropriate labeling convention). The permutation representation $\rho\colon D_{2n} \to S_n$ is an injective homomorphism, embedding D_{2n} as a subgroup of S_n.

3. *Matrix groups acting on vectors.* Let $G = GL_n(\mathbb{R})$ (definition 1.2.30) and $X = \mathbb{R}^n$. Define

$$A \cdot \vec{x} \;=\; A\vec{x} \qquad \text{(matrix–vector multiplication)}.$$

   Axiom (A1) is the associativity of matrix multiplication: $(AB)\vec{x} = A(B\vec{x})$. Axiom (A2) is the identity matrix property: $I_n\vec{x} = \vec{x}$. Thus this defines a group action.

   This action is the foundation of linear algebra: every invertible linear transformation $\mathbb{R}^n \to \mathbb{R}^n$ is encoded by an element of $GL_n(\mathbb{R})$, and composition of transformations corresponds to matrix multiplication.

4. *Two actions of $\mathbb{R}^\times$ on $\mathbb{R}^2$.* Let $G = (\mathbb{R}^\times, \cdot)$, the multiplicative group of nonzero reals, and $X = \mathbb{R}^2$. Here are two *different* actions of the same group on the same set:

$$\textbf{Scaling:}\quad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ ry \end{pmatrix}, \qquad\qquad \textbf{Hyperbolic:}\quad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ r^{-1}y \end{pmatrix}.$$

   Both satisfy (A1) and (A2) (exercise—note that commutativity of $\mathbb{R}^\times$ is used in the hyperbolic case). The scaling action dilates every vector uniformly, while the hyperbolic action stretches the x-axis and compresses the y-axis (or vice versa), preserving the hyperbolas $\{xy = c\}$.

   This example makes an important point: *a group action is not determined by the group and the set alone.* One must specify the rule.

5. *Left multiplication.* Let G be any group, and let $X = G$ (the underlying set of the group itself). Define
$$g \cdot a \;=\; ga \qquad \text{(group multiplication)}.$$
Axiom (A1) is exactly the associativity of the group operation: $(gh) \cdot a = (gh)a = g(ha) = g \cdot (h \cdot a)$. Axiom (A2) is the identity property: $e \cdot a = ea = a$.

   Note that for a given g, the map $a \mapsto ga$ is *not* a group homomorphism (unless $g = e$)—it is only a bijection $G \to G$. This action is our first example of a group acting on *itself*, and it will lead to Cayley's Theorem in Section 4.5.

6. *Conjugation.* Let G be any group, and again let $X = G$. Define
$$g \cdot a \;=\; gag^{-1}.$$
Let us verify the axioms. For (A1):
$$(gh) \cdot a = (gh)\, a\, (gh)^{-1} = g\, h\, a\, h^{-1} g^{-1} = g \cdot (hah^{-1}) = g \cdot (h \cdot a).$$
For (A2): $e \cdot a = eae^{-1} = a$.

   Unlike left multiplication, conjugation "preserves" much of the group's internal structure: it fixes the identity, sends subgroups to (isomorphic) subgroups, and preserves the order of elements. This action will lead to the Class Equation in Section 4.6.

7. *Rotations of a tetrahedron.* Let $G = A_4$ (definition 1.2.19), the alternating group on four symbols, and let $X = \{1, 2, 3, 4\}$ represent the vertices of a regular tetrahedron. The 12 elements of A_4 correspond to the 12 rotational symmetries of the tetrahedron:

   - e: the identity (no rotation).
   - 8 rotations by 120° and 240° about axes through a vertex and the center of the opposite face, e.g. $(1\,2\,3)$ and $(1\,3\,2)$ fix vertex 4.
   - 3 rotations by 180° about axes connecting the midpoints of opposite edges, e.g. $(1\,2)(3\,4)$ swaps the two pairs.

   The action is $\sigma \cdot i = \sigma(i)$, i.e. the restriction of the tautological action of S_4 to A_4.

8. *Translation of polynomials.* Let $G = (\mathbb{R}, +)$ and let $X = \mathbb{R}[x]$, the set of all polynomials with real coefficients. Define
$$t \cdot f \;=\; f_t, \qquad \text{where } f_t(x) = f(x + t).$$
For (A1): $\big((s + t) \cdot f\big)(x) = f(x + s + t)$ and $\big(s \cdot (t \cdot f)\big)(x) = (t \cdot f)(x + s) = f(x + s + t)$. For (A2): $(0 \cdot f)(x) = f(x + 0) = f(x)$.

   This action connects algebra to analysis: Taylor's theorem describes how the coefficients of f_t depend on the coefficients of f and the shift t.

4.1.6 Applied Aside: Symmetry in Chemistry and Crystallography The symmetry group of a molecule—the collection of all rotations and reflections that map the molecule to itself—is called its *point group*. These point groups are finite subgroups of the orthogonal group $O(3)$ (definition 1.2.32). Remarkably, there are only 32 distinct crystallographic point groups compatible with the periodic structure of a crystal lattice.
Group actions enter directly: a point group acts on the set of atoms in a molecule, and the *orbits* of this action (a concept we formalize in Section 4.2) determine which atoms are "chemically equivalent." In spectroscopy, *selection rules*—which govern which transitions between energy levels are allowed—are derived from the representation theory of these point groups. In crystallography, the 32 point groups combine with translational symmetries to yield the *230 space groups*, which classify all possible crystal structures in three dimensions.

A Word on Left and Right Actions

What we have defined above is, more precisely, a *left* group action: the group element appears on the *left* in the expression "$g \cdot x$." There is an equally valid notion of a right action.

Definition 4.1.7: Right Group Action

A *right action* of a group G on a set X is a function $X \times G \to X$, written $(x, g) \mapsto x \cdot g$, satisfying:

(R1) $(x \cdot g) \cdot h = x \cdot (gh)$ for all $g, h \in G$ and $x \in X$, and

(R2) $x \cdot e = x$ for all $x \in X$.

Equivalently, a right action is a group homomorphism $\rho \colon G^{\mathrm{op}} \to \mathrm{Sym}(X)$, where G^{op} is the *opposite group*[a] of G.

[a] The opposite group G^{op} has the same underlying set as G but with the multiplication reversed: $a \cdot_{\mathrm{op}} b = ba$.

Observe the key difference in (R1): the parentheses associate to the *left* ($(x \cdot g) \cdot h$), and the group product is gh in its original order. Compare with the left-action axiom (A1), where the parentheses associate to the *right* ($g \cdot (h \cdot x)$).

Left and right actions are completely interchangeable in the following sense:

Proposition 4.1.8: Converting Between Left and Right Actions

If $\cdot_L \colon G \times X \to X$ is a left action, then

$$x \cdot_R g \;:=\; g^{-1} \cdot_L x$$

defines a right action. Conversely, if $\cdot_R$ is a right action, then $g \cdot_L x := x \cdot_R g^{-1}$ defines a left action.

***Proof*:**
We verify (R1). For any $g, h \in G$ and $x \in X$:

$$\begin{aligned}(x \cdot_R g) \cdot_R h = h^{-1} \cdot_L (x \cdot_R g) &= h^{-1} \cdot_L (g^{-1} \cdot_L x) \\ &= (h^{-1}g^{-1}) \cdot_L x = (gh)^{-1} \cdot_L x = x \cdot_R (gh).\end{aligned}$$

For (R2): $x \cdot_R e = e^{-1} \cdot_L x = e \cdot_L x = x$. The converse is analogous.

Because of this correspondence, there is no essential loss of generality in working with one type exclusively.

4.1.9 Convention For the remainder of this book, "group action" means *left* group action unless explicitly stated otherwise.

4.2 Orbits, Stabilizers, and the Kernel

Given a group action $G \curvearrowright X$, two fundamental questions arise:

1. From the perspective of X: Given a point $x \in X$, where can x be "moved to" by the elements of G?
2. From the perspective of G: Given a point $x \in X$, which elements of G "leave x in place"?

The answers to these questions—*orbits* and *stabilizers*—are two of the most important concepts in the theory of group actions. As we will see, they are intimately related, and much of the power of group actions comes from playing them off against each other.

Orbits

Consider the special orthogonal group $SO(3)$ (definition 1.2.33) acting on the unit sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ by matrix multiplication. If we start at the north pole $N = (0, 0, 1)$ and apply every rotation in $SO(3)$, we reach *every* point on the sphere: $SO(3) \cdot N = S^2$. Now restrict to the subgroup $SO(2) \leq SO(3)$ of rotations about the z-axis. From the north pole, $SO(2)$ goes nowhere: $SO(2) \cdot N = \{N\}$. But from a point P at latitude θ, $SO(2)$ traces out the entire latitude circle through P. The latitude circles—together with the two poles—form a family of non-overlapping subsets that cover the sphere.

This is the geometric prototype of a general phenomenon: under any group action, the set X decomposes into non-overlapping "layers" of points that can be reached from one another.

Definition 4.2.1: Orbit

Let $G \curvearrowright X$. For any $x \in X$, the ***orbit*** of x under the action of G is the set

$$G \cdot x \;=\; \{g \cdot x \big| g \in G\}.$$

In words, $G \cdot x$ is the set of all points in X that x can be "sent to" by some element of G.

The notation $\mathrm{Orb}_G(x)$ or $\mathcal{O}_x$ is also common in the literature; we will use $G \cdot x$ throughout.

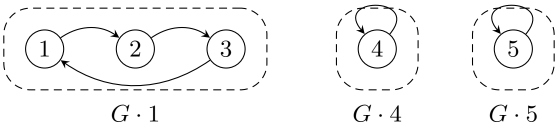
Example 4.2.2: Orbits Under a Cyclic Group

Let $G = \langle(1\ 2\ 3)\rangle = \{e, (1\ 2\ 3), (1\ 3\ 2)\} \cong \mathbb{Z}/3\mathbb{Z}$, viewed as a subgroup of S_5, and let $X = \{1, 2, 3, 4, 5\}$ with the tautological action $\sigma \cdot i = \sigma(i)$. Then:

$$\begin{aligned}
G \cdot 1 &= \{e(1), (1\,2\,3)(1), (1\,3\,2)(1)\} = \{1, 2, 3\},\\
G \cdot 4 &= \{e(4), (1\,2\,3)(4), (1\,3\,2)(4)\} = \{4\},\\
G \cdot 5 &= \{5\}.
\end{aligned}$$

Note that $G \cdot 1 = G \cdot 2 = G \cdot 3$ (the three points can all reach each other), and the set X decomposes as the disjoint union

$$X \;=\; \{1, 2, 3\} \;\sqcup\; \{4\} \;\sqcup\; \{5\}.$$



The decomposition into orbits holds for any group action.

Proposition 4.2.3: Orbits Partition the Set

Let $G \curvearrowright X$. Define a relation $\sim$ on X by

$$x \sim y \quad \Longleftrightarrow \quad y = g \cdot x \text{ for some } g \in G.$$

Then $\sim$ is an equivalence relation on X, and the equivalence class of x is precisely the orbit $G \cdot x$. Consequently, the orbits of the action partition X:

$$X \;=\; \bigsqcup_{i \in I} G \cdot x_i,$$

where $\{x_i\}_{i \in I}$ is any set of orbit representatives (one element chosen from each orbit).

***Proof*:**

We verify the three properties of an equivalence relation.

- *Reflexive:* $x = e \cdot x$, so $x \sim x$. (Uses axiom (A2).)
- *Symmetric:* If $x \sim y$, then $y = g \cdot x$ for some $g \in G$. Applying g^{-1}:

  $$g^{-1} \cdot y = g^{-1} \cdot (g \cdot x) \overset{(A1)}{=} (g^{-1}g) \cdot x \overset{(A2)}{=} e \cdot x = x,$$

  so $x = g^{-1} \cdot y$, which gives $y \sim x$. (Uses the existence of inverses in G.)
- *Transitive:* If $x \sim y$ and $y \sim z$, then $y = g \cdot x$ and $z = h \cdot y$ for some $g, h \in G$. Then

  $$z = h \cdot y = h \cdot (g \cdot x) \overset{(A1)}{=} (hg) \cdot x,$$

so $x \sim z$. (Uses closure of the group operation.)

The equivalence class of x is $[x] = \{y \in X : y \sim x\} = \{g \cdot x : g \in G\} = G \cdot x$. Since equivalence classes partition a set, the orbits partition X.

The margin annotations in the proof are worth reflecting on: *reflexivity* uses the identity axiom, *symmetry* uses the existence of inverses, and *transitivity* uses closure. In other words, the group axioms are precisely what is needed to make the orbit relation an equivalence relation. A monoid action (where inverses may not exist) would give reflexivity and transitivity but *not* symmetry, and the orbits would generally fail to partition the set.

We introduce notation for the *set of orbits*:

Definition 4.2.4: Set of Orbits

If $G \curvearrowright X$, the set of all orbits is denoted

$$X/G \;=\; \{G \cdot x : x \in X\}.$$

This is called the ***orbit space*** (or *set of orbits*) of the action.

Counting orbits will become a central theme once we prove Burnside's Lemma (§4.3).

Stabilizers and Fixed Points

We now turn to the second question: which group elements "do nothing" to a given point?

Definition 4.2.5: Stabilizer

Let $G \curvearrowright X$ and let $x \in X$. The ***stabilizer*** (or *isotropy subgroup*) of x in G is

$$G_x \;=\; \{g \in G \big| g \cdot x = x\}.$$

The notation $\mathrm{Stab}_G(x)$ is also standard; we use the subscript notation G_x for brevity.

Proposition 4.2.6: The Stabilizer is a Subgroup

For any $x \in X$, the stabilizer G_x is a subgroup of G.

***Proof*:**
We apply the subgroup criterion (proposition 2.1.3).

- *Non-empty:* $e \cdot x = x$ by axiom (A2), so $e \in G_x$.
- *Closed under products and inverses:* Let $g, h \in G_x$. Then

  $$(gh^{-1}) \cdot x \overset{\text{(A1)}}{=} g \cdot (h^{-1} \cdot x).$$

  Since $h \in G_x$, we have $h \cdot x = x$, and so $h^{-1} \cdot x = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = e \cdot x = x$. Therefore $(gh^{-1}) \cdot x = g \cdot x = x$, giving $gh^{-1} \in G_x$.

The fact that G_x is a *subgroup* (not just a subset) is crucial: it means we can bring the full weight of our subgroup theory to bear. In particular, G_x has an index in G given by Lagrange's Theorem (theorem 3.3.1), and—as we will see in the next section—this index is precisely the size of the orbit $G \cdot x$.

Example 4.2.7: Computing Stabilizers

Consider $G = D_{2\cdot 4} = D_8$ acting on the vertices $X = \{1, 2, 3, 4\}$ of a square, with the labeling from Section 1.5. We compute the stabilizer of vertex 1. An element $g \in D_8$ stabilizes 1 if and only if g maps vertex 1 to itself. Among the 8 elements of D_8:

- The identity e fixes 1.
- The rotations r, r^2, r^3 all move vertex 1.
- Among the four reflections, exactly one fixes vertex 1: the reflection sr^3 across the diagonal through vertex 1 (depending on the labeling convention).

Thus $G_1 = \{e, sr^3\} \cong \mathbb{Z}/2\mathbb{Z}$, a subgroup of order 2. Note that $|G_1| = 2$ and $|G \cdot 1| = |\{1, 2, 3, 4\}| = 4$, so $|G_1| \cdot |G \cdot 1| = 2 \cdot 4 = 8 = |D_8|$. This is the orbit-stabilizer theorem, proved below.

There is a dual notion to the stabilizer that will be useful later:

Definition 4.2.8: Fixed-Point Sets

Let $G \curvearrowright X$.

1. For a single element $g \in G$, the *fixed-point set* of g is
$$X^g = \{x \in X \big| g \cdot x = x\}.$$
2. The *fixed-point set* of the entire group G is
$$X^G = \{x \in X \big| g \cdot x = x \text{ for all } g \in G\} = \bigcap_{g \in G} X^g.$$

Observe the duality: the stabilizer G_x is a subset of G (the group elements that fix a *given point*), while the fixed-point set X^g is a subset of X (the points fixed by a *given group element*). Both notions have the same underlying condition, "$g \cdot x = x$," but viewed from opposite sides:

	Lives in. . .	**Defined by fixing. . .**
Stabilizer G_x	G	the point $x \in X$
Fixed-point set X^g	X	the element $g \in G$

4.2.9 A Mnemonic for Orbits and Stabilizers Here is a useful way to keep the two concepts straight. Both G_x (stabilizer) and $G \cdot x$ (orbit) have an element of X in their notation:

- If G_x is *large* (many elements of G fix x), then $G \cdot x$ is *small* (the point x is not moved very far).
- If G_x is *small* (few elements of G fix x), then $G \cdot x$ is *large* (the point x is moved to many different places).

In the extreme: if $G_x = G$ (everything fixes x), then $G \cdot x = \{x\}$ (a singleton orbit). Conversely, if $G_x = \{e\}$ (only the identity fixes x), then the orbit is as large as possible. This inverse relationship will be made precise by the Orbit-Stabilizer Theorem in Section 4.3.

We close this subsection with two remarks connecting stabilizers to concepts from Chapter 2 that will be developed fully later in this chapter.

1. Left multiplication (item 5). When G acts on itself by $g \cdot a = ga$, the stabilizer of any $a \in G$ is
$$G_a = \{g \in G : ga = a\} = \{e\}.$$
Every stabilizer is trivial: no element other than the identity can "fix" a point under left multiplication.
2. Conjugation (item 6). When G acts on itself by $g \cdot a = gag^{-1}$, the stabilizer of a is
$$G_a = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$
the centralizer of a (definition 2.2.7). Thus the orbit-stabilizer framework gives us a direct link between orbits (conjugacy classes) and centralizers. We will exploit this connection in Section 4.6.

The Kernel of an Action

The stabilizer G_x captures which group elements fix a *single* point. We now ask: which group elements fix *every* point?

Definition 4.2.10: Kernel of a Group Action

Let $G \curvearrowright X$ with permutation representation $\rho\colon G \to \mathrm{Sym}(X)$. The ***kernel*** of the action is
$$\ker(G \curvearrowright X) \;=\; \{g \in G \big| g \cdot x = x \text{ for all } x \in X\} \;=\; \bigcap_{x \in X} G_x.$$

An element in the kernel "acts trivially"—it is a group element that induces the identity permutation on X. This means $\rho(g) = \mathrm{id}_X$, and so the kernel of the *action* is exactly the kernel of the *homomorphism* ρ:
$$\ker(G \curvearrowright X) \;=\; \ker(\rho).$$

Proposition 4.2.11: The Kernel is a Normal Subgroup

$$\ker(G \curvearrowright X) \trianglelefteq G$$

***Proof*:**
Since $\rho\colon G \to \mathrm{Sym}(X)$ is a group homomorphism, its kernel $\ker(\rho)$ is a normal subgroup of G by proposition 2.2.12.

The kernel measures how much "information about G" is lost by the action. If the kernel is large, many group elements are "wasted"—they all induce the same (trivial) permutation. If the kernel is trivial, then distinct group elements always induce distinct permutations. This motivates:

Definition 4.2.12: Faithful Action

A group action $G \curvearrowright X$ is called *faithful* (or *effective*) if its kernel is trivial:

$$\ker(G \curvearrowright X) = \{e\}.$$

Equivalently, the action is faithful if and only if the permutation representation $\rho\colon G \to \mathrm{Sym}(X)$ is *injective*.

In a faithful action, distinct group elements always produce distinct permutations of X: if $g \neq h$, then there exists some $x \in X$ with $g \cdot x \neq h \cdot x$. The group G is thus "faithfully represented" as a group of permutations of X.

Example 4.2.13: A Non-Faithful Action
Let $G = \mathbb{Z}/4\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$ and $X = \{1, 2\}$. Define an action by $\bar{n} \cdot i = \tau^n(i)$, where $\tau = (1\ 2) \in S_2$. Then:

$$\bar{0} \cdot i = i, \quad \bar{1} \cdot i = \tau(i), \quad \bar{2} \cdot i = \tau^2(i) = i, \quad \bar{3} \cdot i = \tau^3(i) = \tau(i).$$

The elements $\bar{0}$ and $\bar{2}$ both act as the identity on X, so

$$\ker(\rho) \;=\; \{\bar{0}, \bar{2}\} \;\cong\; \mathbb{Z}/2\mathbb{Z}.$$

The action is *not* faithful. By the First Isomorphism Theorem (theorem 3.1.14),

$$G/\ker(\rho) \;\cong\; \rho(G) \;\leq\; \mathrm{Sym}(X) \;=\; S_2,$$

so $(\mathbb{Z}/4\mathbb{Z})/(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ embeds into S_2—consistent with the fact that the *effective* part of the action is just "swap or don't swap."

The example above illustrates a general principle: even if an action is not faithful, we can always *make* it faithful by passing to the quotient.

Proposition 4.2.14: Quotient by the Kernel Acts Faithfully

Let $G \curvearrowright X$ with kernel $K = \ker(\rho)$. Then the quotient group G/K acts faithfully on X via

$$(gK) \cdot x \;=\; g \cdot x.$$

***Proof*:**
We first check well-definedness: if $gK = hK$, then $h^{-1}g \in K$, so $(h^{-1}g) \cdot x = x$ for all x, whence $g \cdot x = h \cdot x$ for all x. Thus the action of gK does not depend on the choice of representative.

The axioms (A1) and (A2) are inherited directly from the action of G. Finally, if gK is in the kernel of this new action, then $g \cdot x = x$ for all $x \in X$, so $g \in K$, meaning $gK = eK$. The kernel of the G/K-action is trivial, so the action is faithful.

In the language of the permutation representation: ρ factors through the quotient as

$$G \;\twoheadrightarrow\; G/K \;\overset{\bar{\rho}}{\hookrightarrow}\; \mathrm{Sym}(X),$$

where $\bar{\rho}$ is injective. This is simply the First Isomorphism Theorem (theorem 3.1.14) applied to ρ.

We summarize the key objects introduced in this section for quick reference.

Object	**Notation**	**Lives in**	**Definition**
Orbit of x	$G \cdot x$	subsets of X	$\{g \cdot x : g \in G\}$
Stabilizer of x	G_x	subgroups of G	$\{g \in G : g \cdot x = x\}$
Fixed-point set of g	X^g	subsets of X	$\{x \in X : g \cdot x = x\}$
Kernel of the action	$\ker(\rho)$	normal subgroups of G	$\bigcap_{x \in X} G_x$

4.3 The Orbit-Stabilizer Theorem

In the previous section, we observed an inverse relationship between orbits and stabilizers: the larger the stabilizer, the smaller the orbit, and vice versa. We also noticed, in the D_8 example (example 4.2.7), that $|G_1| \cdot |G \cdot 1| = |D_8|$. The Orbit-Stabilizer Theorem makes this relationship precise, and reveals it to be a direct consequence of Lagrange's Theorem.

Statement and Proof

Theorem 4.3.1: The Orbit-Stabilizer Theorem

Let G be a group acting on a set X, and let $x \in X$. There is a bijection

$$G/G_x \longleftrightarrow G \cdot x, \qquad gG_x \longmapsto g \cdot x,$$

between the set of left cosets of the stabilizer G_x in G and the orbit of x. In particular,

$$|G \cdot x| \;=\; [G : G_x].$$

If G is finite, this gives $|G \cdot x| = |G| \,/\, |G_x|$, or equivalently,

$$|G| \;=\; |G \cdot x| \cdot |G_x|.$$

***Proof*:**
Define the map

$$\varphi \colon G/G_x \to G \cdot x, \qquad \varphi(gG_x) = g \cdot x.$$

We verify that φ is a well-defined bijection.

Well-defined. Suppose $gG_x = hG_x$. Then $h^{-1}g \in G_x$, so $(h^{-1}g) \cdot x = x$. By axiom (A1):

$$g \cdot x = h \cdot \big((h^{-1}g) \cdot x\big) = h \cdot x.$$

Thus $\varphi(gG_x) = \varphi(hG_x)$, and the map does not depend on the choice of coset representative.

Injective. Suppose $\varphi(gG_x) = \varphi(hG_x)$, i.e. $g \cdot x = h \cdot x$. Then

$$(h^{-1}g) \cdot x = h^{-1} \cdot (g \cdot x) = h^{-1} \cdot (h \cdot x) = (h^{-1}h) \cdot x = e \cdot x = x,$$

so $h^{-1}g \in G_x$, hence $gG_x = hG_x$.

Surjective. Every element of $G \cdot x$ has the form $g \cdot x$ for some $g \in G$, and $g \cdot x = \varphi(gG_x)$.

Since φ is a bijection, $|G \cdot x| = |G/G_x| = [G : G_x]$. When G is finite, Lagrange's Theorem (theorem 3.3.1) gives $[G : G_x] = |G|/|G_x|$.

4.3.2 Remark The formula $|G| = |G \cdot x| \cdot |G_x|$ has exactly the same shape as Lagrange's Theorem: $|G| = [G : H] \cdot |H|$. It *is* Lagrange's Theorem, applied to the subgroup $H = G_x$. The content of the Orbit-Stabilizer Theorem is the bijection φ that identifies the abstract index $[G : G_x]$ with the concrete, countable object $G \cdot x$.

When X is finite, we can combine the Orbit-Stabilizer Theorem with the fact that orbits partition X (proposition 4.2.3) to obtain a useful counting formula.

Corollary 4.3.3: The Orbit-Counting Formula

Let G be a finite group acting on a finite set X, and let $x_1, \ldots, x_k$ be representatives from the distinct orbits. Then

$$|X| \;=\; \sum_{i=1}^{k} |G \cdot x_i| \;=\; \sum_{i=1}^{k} [G : G_{x_i}].$$

***Proof*:**

Since the orbits partition X (proposition 4.2.3), we have $|X| = \sum_{i=1}^{k} |G \cdot x_i|$. Replacing each $|G \cdot x_i|$ by $[G : G_{x_i}]$ via theorem 4.3.1 yields the result.

This formula will be the key ingredient in the Class Equation (Section 4.6), where we apply it to the conjugation action and split the sum according to whether orbit representatives lie in the center.

Examples

Example 4.3.4: Orbits of Polynomials Under S_4

Let $R = \mathbb{Z}[x_1, x_2, x_3, x_4]$, the ring of polynomials in four variables with integer coefficients, and let S_4 act on R by permuting the indices:

$$\sigma \cdot p(x_1, x_2, x_3, x_4) \;=\; p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)}).$$

We compute the orbit of $f = x_1 + x_2$ under this action.

Rather than applying all 24 elements of S_4 to f, we first identify the stabilizer. A permutation σ fixes $x_1 + x_2$ if and only if the set $\{\sigma(1), \sigma(2)\} = \{1, 2\}$ (since addition is commutative, the order within the pair does not matter). This means σ must permute $\{1, 2\}$ among themselves and $\{3, 4\}$ among themselves. The stabilizer is therefore

$$(S_4)_f \;=\; \langle (1\ 2),\, (3\ 4) \rangle \;\cong\; \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

a copy of the Klein four-group, with $|(S_4)_f| = 4$. By the Orbit-Stabilizer Theorem,

$$|S_4 \cdot f| = \frac{|S_4|}{|(S_4)_f|} = \frac{24}{4} = 6.$$

We can now enumerate the orbit with confidence that we are looking for exactly 6 polynomials. Choosing one representative from each coset of $(S_4)_f$ yields:

$$S_4 \cdot (x_1 + x_2) = \{x_1 + x_2,\; x_1 + x_3,\; x_1 + x_4,\; x_2 + x_3,\; x_2 + x_4,\; x_3 + x_4\}.$$

The Orbit-Stabilizer Theorem saved us from checking all 24 permutations: knowing the stabilizer immediately told us the orbit size, and we only needed to find 6 distinct images.

Example 4.3.5: S_n Acting on $\{1, \ldots, n\}$

Consider S_{10} acting on $\{1, \ldots, 10\}$ by $\sigma \cdot i = \sigma(i)$. For any $i, j \in \{1, \ldots, 10\}$, there exists a permutation sending i to j (for instance, the transposition $(i\ j)$), so the action has a single orbit: $S_{10} \cdot i = \{1, \ldots, 10\}$ for every i.

The stabilizer of, say, 1 consists of all permutations that fix 1:

$$(S_{10})_1 = \{\sigma \in S_{10} : \sigma(1) = 1\} \cong S_9.$$

The Orbit-Stabilizer Theorem confirms: $|S_{10} \cdot 1| = |S_{10}|/|(S_{10})_1| = 10!/9! = 10$.

Example 4.3.6: D_8 Acting on Vertices, Revisited

The group D_8 acts on $X = \{1, 2, 3, 4\}$ (the vertices of a square). Since any vertex can be sent to any other by a suitable rotation, the action has a single orbit of size 4. From example 4.2.7, the stabilizer of vertex 1 has order 2. The Orbit-Stabilizer Theorem gives $|G \cdot 1| = 8/2 = 4$, as expected.

Now consider D_8 acting on the set of two diagonals, $\mathcal{D} = \{d_1, d_2\}$, where $d_1 = \{1, 3\}$ and $d_2 = \{2, 4\}$. A rotation by $90°$ swaps the two diagonals, while a rotation by $180°$ preserves both. Here $|G \cdot d_1| = 2$ and $|G_{d_1}| = 8/2 = 4$. The stabilizer G_{d_1} consists of the symmetries that preserve the diagonal d_1 as a set: $G_{d_1} = \{e,\ r^2,\ sr,\ sr^3\}$, a copy of the Klein four-group.

Burnside's Lemma

The Orbit-Stabilizer Theorem gives the size of each individual orbit. But in many applications—particularly in combinatorics—what we really want to know is the *number* of orbits. The following classical result answers this question.

Theorem 4.3.7: Burnside's Lemma

Let G be a finite group acting on a finite set X. Then the number of orbits is

$$|X/G| \;=\; \frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where $X^g = \{x \in X : g \cdot x = x\}$ is the fixed-point set of g (definition 4.2.8).

In words: the number of distinct orbits equals the average number of fixed points, averaged over all group elements.

4.3.8 Historical Note Despite its name, this result was known to Cauchy (1845) and proved in full generality by Frobenius (1887). Burnside included it in his influential 1897 textbook, citing Frobenius, and subsequent authors attributed it to Burnside. For this reason, some authors call it the *Cauchy–Frobenius Lemma*. We follow the (ahistorical but widespread) convention of calling it Burnside's Lemma.

***Proof*:**
The proof is a double-counting argument. Consider the set of pairs

$$\mathcal{S} \;=\; \{(g,x) \in G \times X \big| g \cdot x = x\}.$$

We count $|\mathcal{S}|$ in two ways.

Counting by group elements (summing over $g \in G$). For each g, the number of $x \in X$ with $g \cdot x = x$ is $|X^g|$. Therefore

$$|\mathcal{S}| = \sum_{g \in G} |X^g|. \tag{4.1}$$

Counting by set elements (summing over $x \in X$). For each x, the number of $g \in G$ with $g \cdot x = x$ is $|G_x|$. Therefore

$$|\mathcal{S}| = \sum_{x \in X} |G_x|. \tag{4.2}$$

We now simplify (4.2) by grouping the sum according to orbits. Let $\mathcal{O}_1, \ldots, \mathcal{O}_k$ be the distinct orbits. For any x in an orbit $\mathcal{O}_i$, the Orbit-Stabilizer Theorem (theorem 4.3.1) gives $|G_x| = |G|/|\mathcal{O}_i|$. Summing over the $|\mathcal{O}_i|$ elements of the orbit:

$$\sum_{x \in \mathcal{O}_i} |G_x| \;=\; |\mathcal{O}_i| \cdot \frac{|G|}{|\mathcal{O}_i|} \;=\; |G|.$$

Each orbit contributes exactly $|G|$ to the sum, and there are $k = |X/G|$ orbits, so

$$\sum_{x \in X} |G_x| \;=\; |X/G| \cdot |G|.$$

Equating (4.1) and (4.2):

$$\sum_{g \in G} |X^g| \;=\; |X/G| \cdot |G|,$$

and dividing by $|G|$ yields the result.

The double-counting technique at the heart of this proof is worth internalizing: whenever a condition involves both a group element and a set element (here, "g fixes x"), counting the pairs from both sides often yields a clean identity.

Let us see Burnside's Lemma in action.

Example 4.3.9: Counting Colorings of a Square
How many distinct ways can we color the vertices of a square using 2 colors (say, black and white), if two colorings are considered "the same" when one can be obtained from the other by a symmetry of the square?

Let X be the set of all 2-colorings of 4 labeled vertices, so $|X| = 2^4 = 16$. The symmetry group $G = D_8$ acts on X by permuting the vertices (and hence the colors assigned to them). The distinct colorings "up to symmetry" are precisely the orbits, so we seek $|X/G|$.

We compute $|X^g|$ for each $g \in D_8$. Using the vertex labeling 1 (top-left), 2 (top-right), 3 (bottom-right), 4 (bottom-left), the eight elements act as the following permutations:

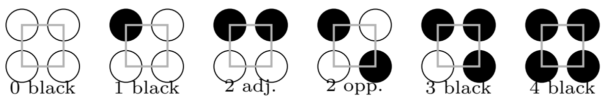
g	Permutation	Cycle structure	$\lvert X^g \rvert$
e	$()$	four fixed points	$2^4 = 16$
r	$(1\,2\,3\,4)$	one 4-cycle	$2^1 = 2$
r^2	$(1\,3)(2\,4)$	two 2-cycles	$2^2 = 4$
r^3	$(1\,4\,3\,2)$	one 4-cycle	$2^1 = 2$
s	$(1\,2)(3\,4)$	two 2-cycles	$2^2 = 4$
sr	$(2\,4)$	one 2-cycle, two fixed	$2^3 = 8$
sr^2	$(1\,4)(2\,3)$	two 2-cycles	$2^2 = 4$
sr^3	$(1\,3)$	one 2-cycle, two fixed	$2^3 = 8$

The rule for each row: a coloring is fixed by g if and only if all vertices within each cycle of g receive the same color. So $|X^g| = 2^{(\text{number of cycles of } g)}$, counting fixed points as 1-cycles.

Burnside's Lemma gives:

$$|X/G| \;=\; \frac{1}{8}(16+2+4+2+4+8+4+8) \;=\; \frac{48}{8} \;=\; 6.$$

The six distinct colorings are:



4.3.10 Applied Aside: Counting Molecular Structures The same technique applies in chemistry. Consider a benzene ring C_6H_6, where each hydrogen atom can be replaced by a substituent (say, chlorine). How many distinct chlorinated benzene derivatives exist? The symmetry group of the hexagonal ring is D_{12}, acting on the six positions. Burnside's Lemma (or its refinement, Pólya's Enumeration Theorem) answers this question systematically. For instance, with two possible substituents (H or Cl) at each of 6 positions, there are

$$\frac{1}{12}\left(2^6 + \cdots\right) = 13$$

distinct molecules, a count that matches experimental chemistry. This interplay between symmetry and enumeration is foundational in chemical graph theory.

4.4 Types of Actions

Not all group actions are created equal. Some actions allow different group elements to "look the same" on X; others ensure that every group element acts distinctly. Some actions can move any point to any other; others split X into multiple unreachable regions. In this section, we introduce a taxonomy of four properties—faithful, transitive, free, and regular—that classify the most important qualitative features of an action.

Faithful Actions

We have already defined this notion in definition 4.2.12: an action $G \curvearrowright X$ is *faithful* if $\ker(\rho) = \{e\}$, i.e. if the only element that fixes every point of X is the identity. Equivalently, the permutation representation $\rho\colon G \to \mathrm{Sym}(X)$ is injective, so G embeds into $\mathrm{Sym}(X)$.

Faithfulness is a *global* condition: it says nothing about any individual stabilizer G_x, only about their intersection $\bigcap_{x \in X} G_x$. Individual stabilizers may well be nontrivial.

Example 4.4.1: Faithful Actions

1. The tautological action of S_n on $\{1, \dots, n\}$ is faithful: any nontrivial permutation must move at least one element.
2. The action of D_{2n} on the vertices of a regular n-gon is faithful for $n \geq 3$: distinct symmetries of the polygon induce distinct permutations of the vertices. Note, however, that stabilizers are nontrivial—the stabilizer of a vertex is the order-2 subgroup generated by the reflection fixing that vertex.
3. The conjugation action of G on itself (item 6) has kernel
$$\ker(\rho) = \{g \in G : gag^{-1} = a \text{ for all } a \in G\} = Z(G),$$
the center of G (definition 2.2.4). This action is faithful if and only if $Z(G) = \{e\}$, i.e. if and only if G has trivial center.

As we saw in proposition 4.2.14, any action can be made faithful by passing to the quotient $G/\ker(\rho)$. In this sense, the "faithful part" of an action is always available; the kernel measures the redundancy.

Transitive Actions

Definition 4.4.2: Transitive Action

A group action $G \curvearrowright X$ is called *transitive* if it has exactly one orbit, i.e. $G \cdot x = X$ for every $x \in X$. Equivalently, for any $x, y \in X$, there exists some $g \in G$ such that $g \cdot x = y$.

In a transitive action, every point can be reached from every other point. No point is "isolated" from the rest.

Example 4.4.3: Transitive and Non-Transitive Actions

1. The action of S_n on $\{1, \ldots, n\}$ is transitive: for any i, j, the transposition $(i\ j)$ sends i to j.
2. The action of D_8 on the four vertices of a square is transitive: r cyclically permutes all four vertices.
3. The action of $\langle(1\ 2\ 3)\rangle$ on $\{1, 2, 3, 4, 5\}$ from example 4.2.2 is *not* transitive: it has three orbits.
4. The conjugation action of G on itself is transitive if and only if every pair of elements is conjugate. For finite groups, this can only happen when $|G| = 1$ (since the identity is conjugate only to itself, so if $|G| > 1$, the singleton $\{e\}$ is always an orbit separate from any non-identity element).

Transitive actions enjoy a useful structural property: the stabilizers of any two points are conjugate subgroups.

Proposition 4.4.4: Stabilizers of Points in the Same Orbit

Let $G \curvearrowright X$. If $y = g \cdot x$ for some $g \in G$, then

$$G_y = g\, G_x\, g^{-1}.$$

In particular, if the action is transitive, then the stabilizers of any two points in X are conjugate subgroups of G.

***Proof*:**
Let $h \in G_y$, so that $h \cdot y = y$. Since $y = g \cdot x$:

$$g \cdot x = y = h \cdot y = h \cdot (g \cdot x) = (hg) \cdot x,$$

so $(g^{-1}hg) \cdot x = g^{-1} \cdot (g \cdot x) = x$, giving $g^{-1}hg \in G_x$, i.e. $h \in gG_xg^{-1}$. This shows $G_y \subseteq gG_xg^{-1}$.
Conversely, if $h \in gG_xg^{-1}$, write $h = gkg^{-1}$ for some $k \in G_x$. Then

$$h \cdot y = (gkg^{-1}) \cdot (g \cdot x) = g \cdot (k \cdot x) = g \cdot x = y,$$

so $h \in G_y$. Thus $gG_xg^{-1} \subseteq G_y$.

This proposition has an important consequence: in a transitive action, knowing the stabilizer of *any single point* determines (up to conjugacy) the stabilizer of every point.

Even more is true. A transitive action is completely determined, up to a natural notion of equivalence, by the stabilizer of any chosen point. To make this precise, we need a brief definition.

Definition 4.4.5: Isomorphism of G-sets

Let $G \curvearrowright X$ and $G \curvearrowright Y$ be two actions of the same group. A bijection $\varphi\colon X \to Y$ is an *isomorphism of G-sets* (or a *G-equivariant* bijection) if

$$\varphi(g \cdot x) = g \cdot \varphi(x) \qquad \text{for all } g \in G,\ x \in X.$$

If such a bijection exists, we write $X \cong_G Y$ and say the two G-sets are *isomorphic*.

In other words, φ is a relabeling of X as Y that is compatible with the group action: it does not matter whether we act first and then relabel, or relabel first and then act.

Theorem 4.4.6: Structure Theorem for Transitive Actions

Let $G \curvearrowright X$ be a transitive action, and fix any point $x_0 \in X$. Then the bijection

$$\varphi\colon G/G_{x_0} \to X, \qquad gG_{x_0} \mapsto g \cdot x_0,$$

from the Orbit-Stabilizer Theorem (theorem 4.3.1) is an isomorphism of G-sets, where G acts on G/G_{x_0} by left multiplication: $h \cdot (gG_{x_0}) = (hg)G_{x_0}$.
In particular, every transitive G-set is isomorphic to G/H for some subgroup $H \leq G$.

***Proof*:**
We already know from theorem 4.3.1 that φ is a well-defined bijection (using transitivity so that $G \cdot x_0 = X$). It remains to verify equivariance. For any $h \in G$ and any coset gG_{x_0}:

$$\varphi(h \cdot gG_{x_0}) = \varphi((hg)G_{x_0}) = (hg) \cdot x_0 = h \cdot (g \cdot x_0) = h \cdot \varphi(gG_{x_0}).$$

Thus φ respects the action, and the two G-sets are isomorphic.

This theorem is one of the most clarifying results in the theory: it says that, up to relabeling, *every* transitive action is a coset action. We will study coset actions in detail in Section 4.5.

Free Actions

Faithful actions require the *intersection* of all stabilizers to be trivial. Free actions require something much stronger: every *individual* stabilizer must be trivial.

Definition 4.4.7: Free Action

A group action $G \curvearrowright X$ is called *free* if

$$G_x = \{e\} \qquad \text{for every } x \in X.$$

Equivalently, if $g \cdot x = x$ for some $x \in X$, then $g = e$.

There is a useful reformulation: an action is free if and only if distinct group elements never produce the same result at any point.

Proposition 4.4.8: Characterization of Free Actions

An action $G \curvearrowright X$ is free if and only if for all $g, h \in G$ and all $x \in X$,

$$g \cdot x = h \cdot x \quad \Longrightarrow \quad g = h.$$

***Proof*:**
If $g \cdot x = h \cdot x$, then $(h^{-1}g) \cdot x = x$, so $h^{-1}g \in G_x$. If the action is free, $G_x = \{e\}$, so $h^{-1}g = e$, giving $g = h$. Conversely, if the implication holds for all g, h, x, then taking $h = e$: $g \cdot x = x$ implies $g = e$, so every stabilizer is trivial.

Example 4.4.9: Free Actions

1. The left-multiplication action $G \curvearrowright G$ (item 5) is free: if $ga = a$, then $g = e$ by cancellation (proposition 1.1.17). This holds for any group G.
2. The action of $(\mathbb{Z}, +)$ on $\mathbb{R}$ by $n \cdot x = x + n$ is free: if $x + n = x$, then $n = 0$.
3. The action of D_8 on the vertices of a square is *not* free: the stabilizer of each vertex has order 2.
4. The action of $\mathbb{Z}/2\mathbb{Z} = \{e, \sigma\}$ on $\{1, 2, 3, 4\}$ via $\sigma = (1\ 2)(3\ 4)$ is free: σ has no fixed points. It is not transitive, however (the orbits are $\{1, 2\}$ and $\{3, 4\}$).

Free actions interact cleanly with the Orbit-Stabilizer Theorem.

Corollary 4.4.10: Orbit Sizes in Free Actions

If G is a finite group acting freely on X, then every orbit has size $|G|$. In particular, $|G|$ divides $|X|$.

***Proof*:**
For every $x \in X$, $|G_x| = 1$, so $|G \cdot x| = |G|/|G_x| = |G|$ by theorem 4.3.1. Since the orbits partition X (proposition 4.2.3), $|X|$ is a sum of copies of $|G|$.

Note that every free action is faithful: if $G_x = \{e\}$ for all x, then $\ker(\rho) = \bigcap_x G_x = \{e\}$. The converse is false, as the example of D_8 on the vertices of a square shows (faithful, but not free).

Regular Actions

The strongest condition arises when we ask for an action that is simultaneously free and transitive.

Definition 4.4.11: Regular Action

A group action $G \curvearrowright X$ is called *regular* if it is both free and transitive.

Regular actions have a particularly elegant characterization.

Proposition 4.4.12: Characterization of Regular Actions

An action $G \curvearrowright X$ is regular if and only if for every pair $x, y \in X$, there exists a *unique* $g \in G$ such that $g \cdot x = y$.

***Proof*:**
Transitivity says "at least one such g exists." Freeness says "at most one such g exists" (by proposition 4.4.8: if $g \cdot x = y = h \cdot x$, then $g = h$). Together, they give exactly one.

Conversely, if for every x, y there exists a unique g with $g \cdot x = y$, then existence gives transitivity, and uniqueness gives freeness.

Example 4.4.13: Regular Actions

1. The left-multiplication action $G \curvearrowright G$ is regular: it is free (example 4.4.9) and transitive (given $a, b \in G$, the element $g = ba^{-1}$ satisfies $g \cdot a = ba^{-1}a = b$). This is the *canonical* regular action.
2. The action of $(\mathbb{Z}, +)$ on $\mathbb{R}$ by translation ($n \cdot x = x + n$) is free but not transitive ($\frac{1}{2}$ is not in the orbit of 0), hence not regular.
3. The action of D_8 on the four vertices of a square is transitive but not free, hence not regular.

By the Orbit-Stabilizer Theorem, a regular action of a finite group G on a set X satisfies $|X| = |G \cdot x| = |G|$ for every x: the set and the group have the same cardinality. In fact, the Structure Theorem for Transitive Actions (theorem 4.4.6) shows precisely what every regular action looks like:

Corollary 4.4.14: Regular Actions are Left Multiplication

If G acts regularly on X, then $X \cong_G G$ as G-sets. That is, up to isomorphism, the left-multiplication action of G on itself is the *only* regular G-action.

***Proof*:**
By theorem 4.4.6, any transitive G-set is isomorphic to G/G_{x_0} for some $x_0 \in X$. Since the action is also free, $G_{x_0} = \{e\}$, so $X \cong_G G/\{e\} = G$.

Summary: A Taxonomy of Group Actions

We collect the four types and their interrelationships in a single reference table.

Type	*Condition*	*Via Orbit-Stabilizer*	*Canonical example*
Faithful	$\bigcap_{x \in X} G_x = \{e\}$	$\rho\colon G \hookrightarrow \mathrm{Sym}(X)$	$D_{2n} \curvearrowright$ vertices
Transitive	$G \cdot x = X$ for all x	$\lvert X\rvert = [G : G_x]$	$G \curvearrowright G/H$
Free	$G_x = \{e\}$ for all x	$\lvert G \cdot x\rvert = \lvert G\rvert$ for all x	$G \curvearrowright G$ by left mult.
Regular	Free + Transitive	$\lvert X\rvert = \lvert G\rvert$ and $G_x = \{e\}$	$G \curvearrowright G$ by left mult.

The logical implications among the four types are:

$$\text{Regular} \implies \text{Free} \implies \text{Faithful}, \qquad \text{Regular} \implies \text{Transitive}.$$

None of the converse implications hold in general, as the following examples illustrate:

- Faithful $\not\Rightarrow$ Free: $D_8 \curvearrowright \{1, 2, 3, 4\}$ (faithful, but $\lvert G_1\rvert = 2$).
- Free $\not\Rightarrow$ Transitive: $\mathbb{Z}/2\mathbb{Z} \curvearrowright \{1, 2, 3, 4\}$ via $(1\ 2)(3\ 4)$ (free, but two orbits).
- Transitive $\not\Rightarrow$ Faithful: The conjugation action of S_3 on $\{1, 2, 3\}$ by relabeling is transitive, but if we instead consider $\mathbb{Z}/6\mathbb{Z}$ acting on $\{1, 2, 3\}$ via $\bar{1} \mapsto (1\ 2\ 3)$, the action is transitive with kernel $\{\bar{0}, \bar{3}\} \cong \mathbb{Z}/2\mathbb{Z}$.
- Transitive $\not\Rightarrow$ Free: $D_8 \curvearrowright \{1, 2, 3, 4\}$ (transitive, but not free).

4.5 A Group Acting on Itself

So far, our group actions have involved a group acting on an external set: vertices of a polygon, vectors in $\mathbb{R}^n$, polynomial rings. But as we saw, this often leads to G-isomorphisms between the G-set X and some quotient of G. Thus, we turn our attention to a group acting on *itself* (or on structures derived from itself, such as cosets). These self-actions will produce some of the key results in the chapter.

Action by Left Multiplication

The simplest self-action is left multiplication: each group element shuffles the entire group by multiplying on the left.

Let G be a group and let $X = G$ (the underlying set). The action is

$$g \cdot a = ga \qquad \text{for all } g, a \in G.$$

We verified in item 5 that this is a valid group action. We can now classify it using the taxonomy of Section 4.4:

Proposition 4.5.1: Left Multiplication is a Regular Action

The left-multiplication action $G \curvearrowright G$ is regular (free and transitive).

***Proof*:**
The action is free: if $g \cdot a = a$, then $ga = a$, so $g = e$ by cancellation (proposition 1.1.17). The action is transitive: given any $a, b \in G$, taking $g = ba^{-1}$ gives $g \cdot a = ba^{-1}a = b$.

Since the action is faithful (every free action is), the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ is injective. Let us see what ρ looks like concretely.

Example 4.5.2: Permutation Representation of the Klein Four-Group
Let $G = \{e, a, b, c\}$ be the Klein four-group, with multiplication defined by $a^2 = b^2 = c^2 = e$ and $ab = c$, $ac = b$, $bc = a$. Label $e \leftrightarrow 1$, $a \leftrightarrow 2$, $b \leftrightarrow 3$, $c \leftrightarrow 4$. For each $g \in G$, we compute the permutation $\sigma_g(i) = j$ where $g \cdot g_i = g_j$:

g	$g \cdot e$	$g \cdot a$	$g \cdot b$	$g \cdot c$	σ_g
e	e	a	b	c	$()$
a	a	e	c	b	$(1\ 2)(3\ 4)$
b	b	c	e	a	$(1\ 3)(2\ 4)$
c	c	b	a	e	$(1\ 4)(2\ 3)$

The image $\rho(G) = \{(), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ is a subgroup of S_4 isomorphic to the Klein four-group. One can verify directly that $g \mapsto \sigma_g$ is a homomorphism, and injectivity is visible from the table: distinct rows give distinct permutations.

Cayley's Theorem

The example above embeds a group of order 4 into S_4. The same construction works for any group.

Theorem 4.5.3: Cayley's Theorem

Every group is isomorphic to a subgroup of a symmetric group. More precisely, if G is a group, then the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ of the left-multiplication action is an injective homomorphism. If $|G| = n$, then G is isomorphic to a subgroup of S_n.

***Proof*:**
The left-multiplication action $G \curvearrowright G$ is free (proposition 4.5.1), hence faithful (definition 4.2.12). Thus the permutation representation $\rho\colon G \to \mathrm{Sym}(G)$ has trivial kernel, so ρ is injective. By the First Isomorphism Theorem (theorem 3.1.14), $G \cong \rho(G) \leq \mathrm{Sym}(G)$.

If $|G| = n$, we may label the elements of G as $\{g_1, \ldots, g_n\}$ and identify $\mathrm{Sym}(G) \cong S_n$, so G embeds into S_n.

Cayley's Theorem is one of the most philosophically significant results in group theory. It shows that the abstract axioms (closure, associativity, identity, inverses) do not create anything genuinely "new" beyond permutation groups. Every group, no matter how abstractly defined, *is* a group of symmetries—a subgroup of a symmetric group. The Erlangen Programme's vision (see the aside in box 4.0.1) is vindicated: groups and symmetries are one and the same.

4.5.4 Historical Note When group theory began in the early 19th century, a "group" simply *meant* a group of permutations. Cayley was among the first to propose the abstract axiomatic definition. His 1854 theorem then showed that the abstraction lost nothing: the original permutation-based intuition remains fully applicable to every abstract group.

That said, the embedding $G \hookrightarrow S_n$ is rarely practical. The symmetric group S_n has order $n!$, which is vastly larger than n for even modest values: if $|G| = 100$, then $|S_{100}| = 100! \approx 9.3 \times 10^{157}$. Working inside S_n is typically no easier than working directly with G. The theorem's value is *conceptual*: it guarantees that any general statement proved about subgroups of symmetric groups automatically applies to all groups.

The following example illustrates a subtler limitation: Cayley's bound $n = |G|$ is sometimes optimal—you genuinely cannot embed the group into any smaller symmetric group.

Example 4.5.5: The Minimum Faithful Degree of Q_8
The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ has order 8, so Cayley's Theorem gives $Q_8 \hookrightarrow S_8$. We claim that Q_8 cannot act faithfully on any set X with $|X| \leq 7$.

Suppose $Q_8 \curvearrowright X$ with $|X| \leq 7$. For any $x \in X$, the Orbit-Stabilizer Theorem (theorem 4.3.1) gives

$$|G_x| = \frac{|Q_8|}{|Q_8 \cdot x|} = \frac{8}{|Q_8 \cdot x|} \geq \frac{8}{7} > 1,$$

so $G_x \neq \{e\}$ for every x. Now, a particular property of Q_8 is that every nontrivial subgroup contains -1 (one can verify this by inspecting the subgroup lattice of Q_8, cf. example 1.2.28). Since G_x is a nontrivial subgroup for every x, we have $-1 \in G_x$ for every x. Therefore $-1 \in \bigcap_{x \in X} G_x = \ker(\rho)$, and the action is not faithful.

Thus, the Cayley embedding into S_8 is the smallest symmetric group that can faithfully contain Q_8.

Action on the Coset Space

Left multiplication acts on the full group G. We now generalize by letting G act on the set of left cosets of a subgroup.

Let $H \leq G$, and let $G/H = \{gH : g \in G\}$ denote the set of left cosets (this is a set, not necessarily a group, since H need not be normal). Define

$$g \cdot (aH) = (ga)H \qquad \text{for all } g \in G,\ aH \in G/H.$$

This is well-defined: if $aH = bH$, then $b^{-1}a \in H$, so $(gb)^{-1}(ga) = b^{-1}a \in H$, giving $(ga)H = (gb)H$. The two axioms are immediate: $(gh) \cdot aH = (gh)aH = g \cdot (haH)$ and $e \cdot aH = aH$.

Note that when $H = \{e\}$, the cosets of H are simply the elements of G, and this action reduces to left multiplication. In this sense, the coset action *generalizes* the construction of the previous subsection.

The following theorem extracts the structural information that the coset action provides.

Theorem 4.5.6: Properties of the Coset Action

Let $H \leq G$, and let G act on G/H by left multiplication. Let $\pi_H : G \to \mathrm{Sym}(G/H)$ be the associated permutation representation. Then:

1. The action is transitive.
2. The stabilizer of the coset eH is H:
$$G_{eH} = H.$$
3. The kernel of the action is the largest normal subgroup of G contained in H:
$$\ker(\pi_H) = \bigcap_{g \in G} gHg^{-1}.$$

Proof:

1. Given any two cosets $aH, bH \in G/H$, take $g = ba^{-1}$. Then $g \cdot aH = (ba^{-1})aH = bH$. Since a and b were arbitrary, the action is transitive.
2. The stabilizer of eH is $\{g \in G : g \cdot eH = eH\} = \{g \in G : gH = H\}$. Now $gH = H$ if and only if $g \in H$ (recall that $gH = H \Leftrightarrow g \in H$). Thus $G_{eH} = H$.
3. The kernel is $\ker(\pi_H) = \bigcap_{aH \in G/H} G_{aH}$. The stabilizer of an arbitrary coset aH is:
$$\begin{aligned} G_{aH} &= \{g \in G : g \cdot aH = aH\} = \{g \in G : gaH = aH\} \\ &= \{g \in G : a^{-1}ga \in H\} = \{g \in G : g \in aHa^{-1}\} = aHa^{-1}. \end{aligned}$$
Therefore
$$\ker(\pi_H) = \bigcap_{a \in G} aHa^{-1}.$$
It remains to show this is the *largest* normal subgroup of G contained in H. Since $\ker(\pi_H) \leq G_{eH} = H$ (part 2), the kernel is contained in H. It is normal in G since it is the kernel of a homomorphism (proposition 2.2.12).
Conversely, let $N \trianglelefteq G$ with $N \leq H$. For any $a \in G$: $N = aNa^{-1} \leq aHa^{-1}$ (since $N \leq H$ implies $aNa^{-1} \leq aHa^{-1}$). Thus $N \leq \bigcap_{a \in G} aHa^{-1} = \ker(\pi_H)$, confirming maximality.

Example 4.5.7: D_8 Acting on Cosets of $\langle s \rangle$

Let $G = D_8$ and $H = \langle s \rangle = \{e, s\}$. The index $[D_8 : H] = 8/2 = 4$, so there are four left cosets:
$$C_1 = \{e,\, s\}, \quad C_2 = \{r,\, sr^3\}, \quad C_3 = \{r^2,\, sr^2\}, \quad C_4 = \{r^3,\, sr\}.$$
The coset action gives a homomorphism $\pi_H : D_8 \to S_4$. To find the image, we compute where r and s send each coset:
$$\begin{aligned} r&\colon\ C_1 \mapsto C_2,\ C_2 \mapsto C_3,\ C_3 \mapsto C_4,\ C_4 \mapsto C_1 && \Longrightarrow && \sigma_r = (1\,2\,3\,4), \\ s&\colon\ C_1 \mapsto C_1,\ C_2 \mapsto C_4,\ C_3 \mapsto C_3,\ C_4 \mapsto C_2 && \Longrightarrow && \sigma_s = (2\,4). \end{aligned}$$

The image $\pi_H(D_8) = \langle(1\ 2\ 3\ 4), (2\ 4)\rangle$ is a subgroup of S_4 isomorphic to D_8. In particular, $\ker(\pi_H) = \{e\}$, so the action is faithful: D_8 embeds into S_4. This is smaller than the Cayley embedding into S_8, and it recovers the natural representation of D_8 as the symmetry group of a square.

theorem 4.5.6 predicted this: the kernel is $\bigcap_{g\in G} g\langle s\rangle g^{-1}$. Since the only conjugates of $\langle s\rangle$ are $\{e, s\}$ and $\{e, sr^2\}$, their intersection is $\{e\}$.

Example 4.5.8: Recovering the Index-2 Theorem

If $[G : H] = 2$, the coset action gives a homomorphism $\pi_H : G \to S_2 \cong \mathbb{Z}/2\mathbb{Z}$. Since $|S_2| = 2$, the kernel has index at most 2 in G. But $\ker(\pi_H) \leq H$ and $[G : H] = 2$, so $\ker(\pi_H) = H$. Since kernels are normal, $H \trianglelefteq G$.

We have thus re-derived corollary 3.3.5 as a special case of the coset action.

Subgroups of Smallest Prime Index

The observation in the previous example generalizes far beyond index 2. The key insight is that if $[G : H] = p$ for a small prime p, then the coset action maps G into S_p, and S_p is "small enough" to force the kernel to be large.

Theorem 4.5.9: Subgroups of Smallest Prime Index are Normal

Let G be a finite group and let p be the smallest prime dividing $|G|$. If $H \leq G$ with $[G : H] = p$, then $H \trianglelefteq G$.

***Proof*:**
The coset action gives a homomorphism $\pi_H : G \to \mathrm{Sym}(G/H) \cong S_p$. Let $K = \ker(\pi_H)$. By theorem 4.5.6, $K \trianglelefteq G$ and $K \leq H$. We will show $K = H$.

By the First Isomorphism Theorem (theorem 3.1.14), $G/K \cong \pi_H(G) \leq S_p$, so $|G/K|$ divides $|S_p| = p!$. Since $K \leq H$, the index decomposes as

$$[G : K] = [G : H] \cdot [H : K] = p \cdot [H : K].$$

Therefore $p \cdot [H : K]$ divides $p!$, which gives $[H : K]$ divides $(p - 1)!$.

On the other hand, $[H : K]$ divides $|H|$ by Lagrange's Theorem (theorem 3.3.1), and $|H|$ divides $|G|$, so every prime factor of $[H : K]$ is also a prime factor of $|G|$. Since p is the *smallest* prime dividing $|G|$, every prime factor of $[H : K]$ is at least p.

But every prime factor of $(p - 1)!$ is strictly less than p. The only positive integer all of whose prime factors are simultaneously $\geq p$ and $< p$ is 1. Therefore $[H : K] = 1$, so $K = H$, and $H = \ker(\pi_H) \trianglelefteq G$.

Example 4.5.10: Applications of the Index-p Theorem

1. If $|G| = 2m$ with m odd, the smallest prime dividing $|G|$ is 2. Any subgroup of index 2 is normal. This recovers corollary 3.3.5.

2. Let $|G| = 2 \cdot 3 \cdot 5 = 30$. The smallest prime dividing $|G|$ is 2. If G has a subgroup of index 2 (i.e. order 15), it is automatically normal. Any subgroup of index 3 need not be normal, since 3 is not the smallest prime dividing 30.

3. Let $|G| = 3 \cdot 7 = 21$. The smallest prime is 3. If G has a subgroup of index 3 (i.e. order 7), it is normal.

4. Let $|G| = 5 \cdot 11 \cdot 13 = 715$. The smallest prime is 5. A subgroup of index 5 (order 143), should one exist, is normal.

4.5.11 Remark A group of order n need not have a subgroup of every possible index. For instance, A_4 has order 12 and the smallest prime dividing 12 is 2, but A_4 has no subgroup of order 6 (cf. the counterexample to the converse of Lagrange's Theorem, example 3.3.7). The theorem says that *if* a subgroup of index p exists, it must be normal; it does not guarantee existence. The question of existence will be addressed by the Sylow Theorems in Section 4.8.

We pause to take stock. Starting from the simple idea of a group multiplying its own elements, we have obtained:

- Cayley's Theorem: every group embeds into a symmetric group.
- The coset action: a versatile tool that converts subgroup information ($H \leq G$) into homomorphism information ($G \to S_p$).
- The index-p theorem: a powerful normality criterion that requires nothing more than knowing $[G : H]$ and the prime factorization of $|G|$.

In the next section, we will see that a different self-action—conjugation—yields an entirely different class of structural results.

4.6 The Conjugation Action and the Class Equation

In Section 4.5, a group acted on itself by left multiplication, and the payoff was Cayley's Theorem: every group is a group of permutations. We now let a group act on itself by *conjugation*, and the payoff will be of a completely different character: arithmetic information about the internal structure of the group, encoded in an identity called the *Class Equation*.

Conjugation on Elements

Let G be a group and let $X = G$. The *conjugation action* is defined by

$$g \cdot a \;=\; gag^{-1} \qquad \text{for all } g, a \in G.$$

We verified in item 6 that this is a valid group action. Let us now identify its orbits and stabilizers.

Definition 4.6.1: Conjugacy Class

Two elements $a, b \in G$ are *conjugate* in G if $b = gag^{-1}$ for some $g \in G$, i.e. if they lie in the same orbit of the conjugation action. The orbit

$$\mathrm{cl}(a) \;=\; G \cdot a \;=\; \{gag^{-1} : g \in G\}$$

is called the *conjugacy class* of a.

Since orbits partition the set (proposition 4.2.3), conjugacy classes partition G: every element belongs to exactly one conjugacy class.

The stabilizer of the conjugation action has a familiar name:

$$G_a = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\} = C_G(a),$$

the centralizer of a in G (definition 2.2.7). The Orbit-Stabilizer Theorem (theorem 4.3.1) therefore gives:

$$|\,\mathrm{cl}(a)| \;=\; [G : C_G(a)]. \tag{4.3}$$

An element a has a singleton conjugacy class, $\mathrm{cl}(a) = \{a\}$, if and only if $gag^{-1} = a$ for all $g \in G$, i.e. if and only if a commutes with every element of G. This is precisely the condition $a \in Z(G)$ (definition 2.2.4). Thus:

The singleton conjugacy classes are exactly the elements of the center.

Example 4.6.2: Conjugacy Classes of Small Groups

1. If G is abelian, then $gag^{-1} = a$ for all g, a, so every conjugacy class is a singleton: $\mathrm{cl}(a) = \{a\}$ for all a. The conjugation action is trivial.

2. Let $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. The center is $Z(Q_8) = \{1, -1\}$ (example 1.2.28). The non-central elements form three conjugacy classes:

$$\mathrm{cl}(i) = \{i, -i\}, \qquad \mathrm{cl}(j) = \{j, -j\}, \qquad \mathrm{cl}(k) = \{k, -k\}.$$

To verify: $jij^{-1} = ji(-j) = ji \cdot j^{-1}$. Using $ji = -k$ and $(-k)(-j) = kj = -i$, so $jij^{-1} = -i$. Each conjugacy class has size $[Q_8 : C_{Q_8}(i)] = 8/4 = 2$, since $C_{Q_8}(i) = \langle i \rangle$ has order 4.

3. Let $G = D_8$. The center is $Z(D_8) = \{e, r^2\}$. The non-central conjugacy classes are:

$$\mathrm{cl}(r) = \{r, r^3\}, \qquad \mathrm{cl}(s) = \{s, sr^2\}, \qquad \mathrm{cl}(sr) = \{sr, sr^3\}.$$

Each has size 2, consistent with $[D_8 : C_{D_8}(r)] = 8/4 = 2$, etc.

The Class Equation

We now translate the orbit decomposition of the conjugation action into an arithmetic identity.

Theorem 4.6.3: The Class Equation

Let G be a finite group, and let $g_1, \ldots, g_r$ be representatives of the distinct conjugacy classes of G that are *not* contained in $Z(G)$. Then

$$|G| \ = \ |Z(G)| \ + \ \sum_{i=1}^{r} [G : C_G(g_i)].$$

Each term $[G : C_G(g_i)]$ in the sum is strictly greater than 1 and divides $|G|$.

***Proof*:**
The conjugacy classes partition G, so $|G| = \sum |\operatorname{cl}(a)|$ as a ranges over a set of orbit representatives. We split this sum into two parts:

- Elements $a \in Z(G)$ have $|\operatorname{cl}(a)| = 1$. There are $|Z(G)|$ such elements.
- Elements $a \notin Z(G)$ have $|\operatorname{cl}(a)| = [G : C_G(a)] > 1$ (since $C_G(a) \neq G$ when $a \notin Z(G)$). Let $g_1, \ldots, g_r$ be representatives of these non-central classes.

Combining:

$$|G| = \sum_{a \in Z(G)} 1 \ + \ \sum_{i=1}^{r} |\operatorname{cl}(g_i)| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

Each $[G : C_G(g_i)]$ divides $|G|$ by Lagrange's Theorem (theorem 3.3.1), and is > 1 since $g_i \notin Z(G)$.

Example 4.6.4: The Class Equation in Practice

1. For $G = Q_8$: $|Z(Q_8)| = 2$ and the three non-central classes each have size 2, giving $8 = 2 + 2 + 2 + 2$.
2. For $G = D_8$: $|Z(D_8)| = 2$ and the three non-central classes each have size 2, giving $8 = 2 + 2 + 2 + 2$.
3. For $G = S_3$: $Z(S_3) = \{e\}$, and the conjugacy classes are $\{e\}$, $\{(1\ 2), (1\ 3), (2\ 3)\}$, $\{(1\,2\,3), (1\,3\,2)\}$, giving $6 = 1 + 3 + 2$.
4. If G is abelian, the Class Equation reads $|G| = |G|$, which is vacuously true but uninformative. The Class Equation is a tool for *non-abelian* groups.

p-Groups Have Nontrivial Center

The Class Equation becomes especially powerful when $|G|$ is a prime power. The following theorem is the first result that students typically cannot prove *without* group actions.

Theorem 4.6.5: p-Groups Have Nontrivial Center

Let p be a prime and let G be a group of order p^α for some $\alpha \geq 1$. Then $Z(G) \neq \{e\}$; more precisely, p divides $|Z(G)|$.

***Proof*:**
The Class Equation gives

$$p^\alpha \;=\; |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

For each i, the centralizer $C_G(g_i)$ is a proper subgroup of G (since $g_i \notin Z(G)$), so $[G : C_G(g_i)] > 1$. Since $C_G(g_i) \leq G$ and $|G| = p^\alpha$, Lagrange's Theorem gives $|C_G(g_i)| = p^{\beta_i}$ for some $\beta_i < \alpha$. Therefore $[G : C_G(g_i)] = p^{\alpha-\beta_i}$, which is divisible by p.

The left-hand side p^α is divisible by p, and every term in the sum is divisible by p. Therefore $|Z(G)|$ is divisible by p, so $|Z(G)| \geq p > 1$.

The argument is clean and short, but notice how much machinery is involved beneath the surface: the conjugation action, the Orbit-Stabilizer Theorem (to get $|\operatorname{cl}(g_i)| = [G : C_G(g_i)]$), Lagrange's Theorem (to deduce $p \big| [G : C_G(g_i)]$), and the orbit partition (to write the Class Equation). This is a vivid demonstration of the power of group actions as a *proof technique*.

Corollary 4.6.6: Groups of Order p^2 Are Abelian

If $|G| = p^2$ for some prime p, then G is abelian. More precisely, $G \cong \mathbb{Z}/p^2\mathbb{Z}$ or $G \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

***Proof*:**
By theorem 4.6.5, $p \big| |Z(G)|$. Since $Z(G) \leq G$ and $|G| = p^2$, Lagrange gives $|Z(G)| \in \{p, p^2\}$.

If $|Z(G)| = p^2$, then $Z(G) = G$ and G is abelian.

If $|Z(G)| = p$, then $|G/Z(G)| = p^2/p = p$, so $G/Z(G)$ has prime order and is therefore cyclic (corollary 2.3.12). We claim this forces G to be abelian, contradicting $Z(G) \neq G$.

Indeed, let $G/Z(G) = \langle gZ(G) \rangle$. Then every element of G can be written as $g^k z$ for some $k \in \mathbb{Z}$ and $z \in Z(G)$. For any two elements $g^{k_1} z_1$ and $g^{k_2} z_2$:

$$(g^{k_1} z_1)(g^{k_2} z_2) = g^{k_1} g^{k_2} z_1 z_2 = g^{k_2} g^{k_1} z_2 z_1 = (g^{k_2} z_2)(g^{k_1} z_1),$$

where we used that powers of g commute with each other, that $z_1, z_2 \in Z(G)$ commute with everything, and that $Z(G)$ is abelian. Thus G is abelian, a contradiction.

Therefore $|Z(G)| = p^2$, and G is abelian of order p^2. By the classification of finite abelian groups (or by direct argument using corollary 2.3.12), the only abelian groups of order p^2 are $\mathbb{Z}/p^2\mathbb{Z}$ and $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

4.6.7 Remark The key step—"if $G/Z(G)$ is cyclic, then G is abelian"—is worth remembering in its own right. More generally, if $G/Z(G)$ is cyclic, the same argument shows every pair of elements commutes, so G is abelian and $Z(G) = G$. This means $G/Z(G)$ is trivial, a stronger conclusion than cyclic. In other words: $G/Z(G)$ *can never be a nontrivial cyclic group.*

Conjugation on Subsets and Subgroups

So far, conjugation has acted on *elements* of G. There is a natural extension to *subsets*: let $\mathcal{P}(G)$ denote the power set of G, and define

$$g \cdot S \;=\; gSg^{-1} \;=\; \{gsg^{-1} : s \in S\} \qquad \text{for } g \in G,\ S \subseteq G.$$

This defines a group action $G \curvearrowright \mathcal{P}(G)$. (Verification of the axioms is identical to the element case.)

Definition 4.6.8: Conjugate Subsets

Two subsets $S, T \subseteq G$ are *conjugate* in G if $T = gSg^{-1}$ for some $g \in G$.

When we restrict attention to the set of all *subgroups* of G, conjugation sends subgroups to (isomorphic) subgroups, so this restriction is also a valid action.

The stabilizer and orbit of a subset S under this action connect to familiar concepts:

- The *stabilizer* of S is
$$G_S = \{g \in G : gSg^{-1} = S\} = N_G(S),$$
the normalizer of S in G (definition 2.2.10).
- The *orbit* $G \cdot S = \{gSg^{-1} : g \in G\}$ is the set of all conjugates of S.

The Orbit-Stabilizer Theorem immediately yields:

Corollary 4.6.9: Number of Conjugates

The number of distinct conjugates of a subset $S \subseteq G$ is $[G : N_G(S)]$. In particular, S is the only member of its conjugacy class if and only if $N_G(S) = G$, i.e. if and only if $S \trianglelefteq G$ (when S is a subgroup) or S is a union of conjugacy classes (when S is an arbitrary subset).

This corollary will be indispensable in the proof of Sylow's Theorems (Section 4.8), where we count the number of conjugates of a Sylow p-subgroup.

Example 4.6.10: Conjugates of a Subgroup

Let $G = S_4$ and $H = \langle (1\ 2) \rangle = \{e, (1\ 2)\}$. The normalizer $N_{S_4}(H)$ consists of all $\sigma \in S_4$ with $\sigma H \sigma^{-1} = H$, i.e. $\sigma(1\ 2)\sigma^{-1} \in \{e, (1\ 2)\}$. Since conjugation preserves cycle type (proposition 1.2.24), $\sigma(1\ 2)\sigma^{-1}$ is always a transposition. It equals $(1\ 2)$ precisely when σ fixes the set $\{1, 2\}$, so

$$N_{S_4}(H) = \{\sigma \in S_4 : \{\sigma(1), \sigma(2)\} = \{1, 2\}\} = \{e,\ (1\ 2),\ (3\ 4),\ (1\ 2)(3\ 4)\}.$$

Therefore H has $[S_4 : N_{S_4}(H)] = 24/4 = 6$ conjugates. These are the six subgroups $\langle (i\ j) \rangle$ for $1 \le i < j \le 4$, one for each transposition.

Conjugacy in S_n and Cycle Types

Since S_n plays a distinguished role throughout this chapter, we take a moment to characterize its conjugacy classes completely. The answer is as clean as one could hope: conjugacy in S_n is determined entirely by cycle type.

Theorem 4.6.11: Conjugacy Classes in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type (definition 1.2.14).

***Proof*:**
One direction was established in proposition 1.2.24: if $\tau = \sigma\pi\sigma^{-1}$, then τ has the same cycle type as π, because conjugation by σ sends the cycle $(a_1\ a_2\ \cdots\ a_k)$ to $(\sigma(a_1)\ \sigma(a_2)\ \cdots\ \sigma(a_k))$, preserving the cycle length.

For the converse, suppose π and τ have the same cycle type. Write them in disjoint cycle notation (including 1-cycles for fixed points):

$$\pi = (a_1\ \cdots\ a_{k_1})(b_1\ \cdots\ b_{k_2})\cdots(z_1\ \cdots\ z_{k_m}),$$
$$\tau = (c_1\ \cdots\ c_{k_1})(d_1\ \cdots\ d_{k_2})\cdots(w_1\ \cdots\ w_{k_m}),$$

where the cycle lengths $k_1, k_2, \ldots, k_m$ are the same in both (this is what "same cycle type" means). Define $\sigma \in S_n$ by

$$\sigma(a_i) = c_i, \quad \sigma(b_j) = d_j, \quad \ldots, \quad \sigma(z_l) = w_l$$

for all indices. Since both decompositions account for every element of $\{1, \ldots, n\}$ exactly once, σ is a well-defined bijection. Then

$$\sigma\pi\sigma^{-1} = (\sigma(a_1)\ \cdots\ \sigma(a_{k_1}))(\sigma(b_1)\ \cdots\ \sigma(b_{k_2}))\cdots = (c_1\ \cdots\ c_{k_1})(d_1\ \cdots\ d_{k_2})\cdots = \tau.$$

So π and τ are conjugate.

Example 4.6.12: Conjugacy Classes of S_4
The partitions of 4 are $4 = 4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$, giving five cycle types and hence five conjugacy classes:

Cycle type	*Example*	*Description*	*Class size*	$\lvert C_{S_4}(\cdot)\rvert$
(1^4)	e	identity	1	24
$(2, 1^2)$	$(1\ 2)$	transpositions	6	4
(2^2)	$(1\ 2)(3\ 4)$	double transpositions	3	8
$(3, 1)$	$(1\ 2\ 3)$	3-cycles	8	3
(4)	$(1\ 2\ 3\ 4)$	4-cycles	6	4

The class sizes sum to $1+6+3+8+6 = 24 = |S_4|$. The centralizer sizes in the last column satisfy

$|\operatorname{cl}(\pi)| \cdot |C_{S_4}(\pi)| = 24$ by (4.3).

Since $Z(S_4) = \{e\}$ (for $n \geq 3$, the center of S_n is trivial), the Class Equation reads:

$$24 = 1 + 6 + 3 + 8 + 6.$$

More generally, there is a formula for the size of each conjugacy class in S_n. If a permutation has cycle type $(1^{a_1}\, 2^{a_2}\, \cdots\, n^{a_n})$ (meaning a_k cycles of length k, with $\sum k\, a_k = n$), then the size of its centralizer is

$$|C_{S_n}(\pi)| \;=\; \prod_{k=1}^{n} k^{a_k} \cdot a_k! \tag{4.4}$$

and the conjugacy class has size $n!/|C_{S_n}(\pi)|$. (The factor k^{a_k} accounts for rotations within each k-cycle, and $a_k!$ accounts for permuting cycles of the same length.) We leave the verification as an exercise.

4.7 Automorphisms and Inner Automorphisms

In Section 4.6, the conjugation action $g \cdot a = gag^{-1}$ gave us the Class Equation by viewing each g as a *permutation* of the underlying set of G. But conjugation does more than permute elements: it preserves the group operation. That is, the map $a \mapsto gag^{-1}$ is not just a bijection $G \to G$ but an *automorphism*. Upgrading the permutation representation from $\operatorname{Sym}(G)$ to $\operatorname{Aut}(G)$ will yield one of the most important structural results in the chapter.

The Conjugation Homomorphism

Fix an element $g \in G$ and consider the map

$$\varphi_g \colon G \to G, \qquad \varphi_g(x) = gxg^{-1}.$$

Proposition 4.7.1: Conjugation by g is an Automorphism

For each $g \in G$, the map φ_g is an automorphism of G.

***Proof*:**
We verify that φ_g is a bijective homomorphism.

Homomorphism. For any $x, y \in G$:

$$\varphi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\,\varphi_g(y).$$

Bijection. The map $\varphi_{g^{-1}}$ is a two-sided inverse:

$$(\varphi_g \circ \varphi_{g^{-1}})(x) = g(g^{-1}xg)g^{-1} = x = g^{-1}(gxg^{-1})g = (\varphi_{g^{-1}} \circ \varphi_g)(x).$$

Thus $\varphi_g \in \operatorname{Aut}(G)$ (definition 1.4.16).

Since each φ_g is an automorphism, we can ask whether the assignment $g \mapsto \varphi_g$ is compatible with the group structure.

Proposition 4.7.2: The Conjugation Map is a Homomorphism

The map
$$\Phi\colon G \to \mathrm{Aut}(G), \qquad \Phi(g) = \varphi_g,$$
is a group homomorphism, with $\ker(\Phi) = Z(G)$.

***Proof*:**
For any $g, h \in G$ and $x \in G$:
$$\Phi(gh)(x) = \varphi_{gh}(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = \varphi_g(\varphi_h(x)) = (\varphi_g \circ \varphi_h)(x).$$

Since this holds for all x, we have $\Phi(gh) = \Phi(g) \circ \Phi(h)$, so Φ is a homomorphism.

The kernel consists of those g for which φ_g is the identity automorphism:
$$\ker(\Phi) = \{g \in G : gxg^{-1} = x \text{ for all } x \in G\} = \{g \in G : gx = xg \text{ for all } x \in G\} = Z(G).$$

Let us pause to compare this with the conjugation action from Section 4.6. There, we had a homomorphism $\rho\colon G \to \mathrm{Sym}(G)$ (the permutation representation of the conjugation action). Now we have a homomorphism $\Phi\colon G \to \mathrm{Aut}(G)$. Since $\mathrm{Aut}(G) \le \mathrm{Sym}(G)$ (proposition 1.4.17), these are compatible: Φ is simply ρ with its codomain refined from $\mathrm{Sym}(G)$ to $\mathrm{Aut}(G)$. The kernel is the same in both cases—$Z(G)$—but the refined codomain gives us access to the richer structure of $\mathrm{Aut}(G)$.

Inner Automorphisms and the Isomorphism $G/Z(G) \cong \mathrm{Inn}(G)$

Definition 4.7.3: Inner Automorphism

The image of the conjugation homomorphism $\Phi\colon G \to \mathrm{Aut}(G)$ is called the group of *inner automorphisms* of G:
$$\mathrm{Inn}(G) \;=\; \Phi(G) \;=\; \{\varphi_g : g \in G\} \;\le\; \mathrm{Aut}(G).$$

An automorphism that is *not* inner is called an *outer automorphism*.

Since Φ is a homomorphism with kernel $Z(G)$ and image $\mathrm{Inn}(G)$, the First Isomorphism Theorem (theorem 3.1.14) immediately yields:

Theorem 4.7.4: $G/Z(G) \cong \mathrm{Inn}(G)$

For any group G,
$$G/Z(G) \;\cong\; \mathrm{Inn}(G).$$

***Proof*:**
Apply the First Isomorphism Theorem to $\Phi\colon G \to \mathrm{Aut}(G)$:

$$G/\ker(\Phi) \;\cong\; \Phi(G), \qquad \text{i.e.,} \qquad G/Z(G) \;\cong\; \mathrm{Inn}(G).$$

Short as it is, this theorem encodes a great deal of structural information. It says that the "non-abelian part" of G—measured by the quotient $G/Z(G)$—is faithfully captured by the inner automorphisms. The larger the center, the fewer inner automorphisms there are, and vice versa.

Example 4.7.5: Inner Automorphisms of Familiar Groups

1. If G is abelian, then $Z(G) = G$, so $\mathrm{Inn}(G) \cong G/G = \{e\}$. Every automorphism of G is outer.
2. For $G = S_3$: $Z(S_3) = \{e\}$, so $\mathrm{Inn}(S_3) \cong S_3/\{e\} \cong S_3$. Since $|\mathrm{Aut}(S_3)| = 6 = |S_3|$ (one can verify that every automorphism of S_3 is inner), we have $\mathrm{Inn}(S_3) = \mathrm{Aut}(S_3)$.
3. For $G = D_8$: $Z(D_8) = \{e, r^2\}$, so $\mathrm{Inn}(D_8) \cong D_8/\{e, r^2\}$, a group of order 4. Since $D_8/\{e, r^2\}$ is non-cyclic (it is generated by two elements of order 2), $\mathrm{Inn}(D_8)$ is isomorphic to the Klein four-group. In fact, $|\mathrm{Aut}(D_8)| = 8$, so $[\mathrm{Aut}(D_8) : \mathrm{Inn}(D_8)] = 2$: there are automorphisms of D_8 that cannot be realized by conjugation.

The construction above has a natural generalization. Instead of conjugating G by itself, we can conjugate any subgroup H by elements that normalize it.

Theorem 4.7.6: The N/C Theorem

Let $H \leq G$. The map

$$\Psi\colon N_G(H) \to \mathrm{Aut}(H), \qquad \Psi(g)(h) = ghg^{-1},$$

is a well-defined homomorphism with $\ker(\Psi) = C_G(H)$. Consequently,

$$N_G(H)\,/\,C_G(H) \;\hookrightarrow\; \mathrm{Aut}(H).$$

In particular, $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$.

***Proof*:**
The map $\Psi(g) = \varphi_g|_H$ (conjugation by g, restricted to H) is well-defined because $g \in N_G(H)$ means $gHg^{-1} = H$ (definition 2.2.10), so φ_g sends H to H. That $\varphi_g|_H$ is an automorphism of H follows from proposition 4.7.1 by restriction.

The verification that Ψ is a homomorphism is identical to the proof of proposition 4.7.2. The kernel is

$$\ker(\Psi) = \{g \in N_G(H) : ghg^{-1} = h \text{ for all } h \in H\} = C_G(H) \cap N_G(H) = C_G(H),$$

where the last equality uses $C_G(H) \leq N_G(H)$ (if g commutes with every element of H, then certainly $gHg^{-1} = H$). The embedding follows from the First Isomorphism Theorem (theorem 3.1.14).

When $H \trianglelefteq G$, we have $N_G(H) = G$, so the theorem gives $G/C_G(H) \hookrightarrow \mathrm{Aut}(H)$. When $H = G$ (which is always normal in itself), we recover $G/Z(G) \cong \mathrm{Inn}(G)$ as a special case.

4.7.7 Foreshadowing: Semi-Direct Products The N/C Theorem has an important special case. Suppose $H \trianglelefteq G$ and $K \leq G$. Then $K \leq N_G(H) = G$, so restricting Ψ to K gives a homomorphism

$$\psi\colon K \to \mathrm{Aut}(H), \qquad k \mapsto (h \mapsto khk^{-1}).$$

This homomorphism ψ encodes how K "twists" H by conjugation, and it will be the key ingredient in the definition of the *semi-direct product* (chapter 5). Knowing H, K, and ψ is enough to reconstruct the group G (under suitable conditions).

4.7.1 Outer Automorphisms

To define the quotient $\mathrm{Aut}(G)/\mathrm{Inn}(G)$, we first verify that $\mathrm{Inn}(G)$ is normal in $\mathrm{Aut}(G)$.

Proposition 4.7.8: $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$

For any group G, $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$.

***Proof*:**
Let $\sigma \in \mathrm{Aut}(G)$ and $\varphi_g \in \mathrm{Inn}(G)$. For any $x \in G$:

$$(\sigma \circ \varphi_g \circ \sigma^{-1})(x) = \sigma\big(g \cdot \sigma^{-1}(x) \cdot g^{-1}\big) = \sigma(g) \cdot x \cdot \sigma(g)^{-1} = \varphi_{\sigma(g)}(x).$$

Thus $\sigma \circ \varphi_g \circ \sigma^{-1} = \varphi_{\sigma(g)} \in \mathrm{Inn}(G)$, and $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$.

Definition 4.7.9: Outer Automorphism Group

The *outer automorphism group* of G is the quotient

$$\mathrm{Out}(G) \;=\; \mathrm{Aut}(G)\,/\,\mathrm{Inn}(G).$$

The outer automorphism group measures how much of $\mathrm{Aut}(G)$ is *not* explained by conjugation. A group with $\mathrm{Out}(G) = \{e\}$ is called *complete* (when it also has trivial center); in such a group, every automorphism is inner.

Example 4.7.10: Outer Automorphisms

1. If G is abelian, then $\mathrm{Inn}(G) = \{e\}$, so $\mathrm{Out}(G) \cong \mathrm{Aut}(G)$. All automorphisms are outer.
2. $\mathrm{Out}(S_n) = \{e\}$ for $n \neq 6$: every automorphism of S_n is inner. The case $n = 6$ is exceptional: $|\mathrm{Out}(S_6)| = 2$, giving S_6 a unique outer automorphism (up to composition with inner ones).
3. $\mathrm{Out}(D_8) \cong \mathbb{Z}/2\mathbb{Z}$: there is one nontrivial coset of $\mathrm{Inn}(D_8)$ in $\mathrm{Aut}(D_8)$.

We record for later use the short exact sequence that encodes the relationship between the center, the group, and its inner automorphisms:

$$1 \longrightarrow Z(G) \hookrightarrow G \xrightarrow{\Phi} \operatorname{Inn}(G) \longrightarrow 1 \tag{4.5}$$

This sequence says: $Z(G)$ measures how far G is from being faithfully represented by its inner automorphisms. $\operatorname{Inn}(G)$ captures the "non-central" part of G; the center is precisely the part that acts trivially by conjugation.

4.7.2 Characteristic Subgroups

Normal subgroups are defined by invariance under *inner* automorphisms: H is normal in G if $\varphi_g(H) = H$ for every $\varphi_g \in \operatorname{Inn}(G)$. A natural strengthening is to require invariance under *all* automorphisms.

Definition 4.7.11: Characteristic Subgroup

A subgroup $H \leq G$ is *characteristic* in G, written $H \operatorname{char} G$, if

$$\sigma(H) = H \qquad \text{for every } \sigma \in \operatorname{Aut}(G).$$

Proposition 4.7.12: Properties of Characteristic Subgroups

1. If $H \operatorname{char} G$, then $H \trianglelefteq G$.
2. If H is the unique subgroup of G of a given order, then $H \operatorname{char} G$.
3. If $H \operatorname{char} K$ and $K \trianglelefteq G$, then $H \trianglelefteq G$.

***Proof*:**

1. Since $\operatorname{Inn}(G) \leq \operatorname{Aut}(G)$, invariance under all automorphisms implies invariance under inner automorphisms: $gHg^{-1} = \varphi_g(H) = H$ for all $g \in G$.
2. Let $\sigma \in \operatorname{Aut}(G)$. Since σ is an isomorphism, $\sigma(H)$ is a subgroup of G with $|\sigma(H)| = |H|$. By uniqueness, $\sigma(H) = H$.
3. Let $g \in G$. Since $K \trianglelefteq G$, the inner automorphism φ_g restricts to an automorphism $\varphi_g|_K \in \operatorname{Aut}(K)$ (because $gKg^{-1} = K$). Since $H \operatorname{char} K$, this automorphism preserves H:

$$gHg^{-1} = \varphi_g(H) = (\varphi_g|_K)(H) = H.$$

Since this holds for every $g \in G$, $H \trianglelefteq G$.

Item 3 is the principal reason characteristic subgroups are useful. Recall that normality is *not* transitive: $H \trianglelefteq K$ and $K \trianglelefteq G$ does *not* imply $H \trianglelefteq G$ in general (example 3.7.4). Characteristic subgroups provide a partial remedy: if the inner link is characteristic (rather than only normal), transitivity is restored.

Example 4.7.13: Characteristic Subgroups

1. The center $Z(G)$ is characteristic in G: if $\sigma \in \mathrm{Aut}(G)$ and $z \in Z(G)$, then for any $x \in G$,
$$\sigma(z)\,x = \sigma(z)\,\sigma(\sigma^{-1}(x)) = \sigma(z \cdot \sigma^{-1}(x)) = \sigma(\sigma^{-1}(x) \cdot z) = x\,\sigma(z),$$
so $\sigma(z) \in Z(G)$. Thus $\sigma(Z(G)) \subseteq Z(G)$, and applying the same argument to σ^{-1} gives equality.

2. The commutator subgroup $G' = \langle xyx^{-1}y^{-1} : x, y \in G\rangle$ is characteristic in G: any automorphism σ sends commutators to commutators ($\sigma(xyx^{-1}y^{-1}) = \sigma(x)\sigma(y)\sigma(x)^{-1}\sigma(y)^{-1}$), so $\sigma(G') = G'$.

3. Every Sylow p-subgroup of a group G is normal if and only if it is the *unique* Sylow p-subgroup. By part 2 of proposition 4.7.12, such a subgroup is automatically characteristic. We will use this in Section 4.8.

4.7.14 Remark: Normality vs. Characteristic The logical chain is:
$$\text{characteristic} \implies \text{normal} \implies \text{subgroup},$$
and neither implication reverses. A normal subgroup that is not characteristic: consider $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $H = \{(0,0),(1,0)\}$. Then $H \trianglelefteq G$ (since G is abelian), but the automorphism swapping the two coordinates sends H to $\{(0,0),(0,1)\} \neq H$. So H is normal but not characteristic.

We summarize the automorphism-related groups and their relationships in a single diagram:

$$\begin{tikzcd}
1 \arrow[r] & Z(G) \arrow[r, hook] & G \arrow[r, two heads, "\Phi"] & \mathrm{Inn}(G) \arrow[r, hook] \arrow[d, phantom, "\rotatebox{90}{$\trianglelefteq$}"] & \mathrm{Aut}(G) \arrow[r, two heads] & \mathrm{Out}(G) \arrow[r] & 1 \\
 & & & \mathrm{Aut}(G) & & &
\end{tikzcd}$$

The top row is exact: $Z(G)$ is the kernel of Φ, $\mathrm{Inn}(G)$ is its image (hence the kernel of the projection to $\mathrm{Out}(G)$), and $\mathrm{Out}(G)$ is the cokernel. This picture captures the hierarchy at a glance: $Z(G)$ measures commutativity, $\mathrm{Inn}(G)$ measures conjugation, and $\mathrm{Out}(G)$ measures the "exotic" automorphisms that conjugation cannot produce.

4.8 Sylow's Theorems

Lagrange's Theorem (theorem 3.3.1) shows that if $H \leq G$, then $|H|$ divides $|G|$. The converse is false in general: A_4 has order 12 but no subgroup of order 6 (example 3.3.7). Sylow's Theorems provide a remarkable *partial converse*: for every prime power p^α dividing $|G|$, a subgroup of order p^α not only exists but is essentially unique up to conjugation, and the number of such subgroups is tightly constrained. These are among the most powerful and frequently used results in finite group theory, and their proofs are an applications of everything we have built in this chapter: the Orbit-Stabilizer Theorem, the Class Equation, and the coset action.

p-Groups and Sylow Subgroups

Definition 4.8.1: p-Group

Let p be a prime. A group P is called a *p-group* if $|P| = p^k$ for some $k \geq 0$. A subgroup $H \leq G$ that is a p-group is called a *p-subgroup* of G.

Definition 4.8.2: Sylow p-Subgroup

Let G be a finite group and write $|G| = p^\alpha m$ where $\alpha \geq 0$ and $p \nmid m$ (so p^α is the largest power of p dividing $|G|$). A subgroup $P \leq G$ of order p^α is called a *Sylow p-subgroup* of G. We write $\mathrm{Syl}_p(G)$ for the set of all Sylow p-subgroups of G, and $n_p = n_p(G) = |\mathrm{Syl}_p(G)|$ for their number.

Example 4.8.3: Sylow Subgroups of Small Groups

1. $|S_3| = 6 = 2 \cdot 3$. Sylow 2-subgroups have order 2: these are $\langle(1\ 2)\rangle$, $\langle(1\ 3)\rangle$, $\langle(2\ 3)\rangle$. Sylow 3-subgroups have order 3: only $\langle(1\ 2\ 3)\rangle$. So $n_2 = 3$ and $n_3 = 1$.
2. $|A_4| = 12 = 2^2 \cdot 3$. Sylow 2-subgroups have order 4; Sylow 3-subgroups have order 3.
3. If $|G| = p^\alpha$ (a p-group), then G itself is its only Sylow p-subgroup, so $n_p = 1$.
4. $|\mathbb{Z}/n\mathbb{Z}| = n$. For each prime p dividing n, the unique subgroup of order p^α (where $p^\alpha \| n$) is the unique Sylow p-subgroup. (Here uniqueness follows from the fact that $\mathbb{Z}/n\mathbb{Z}$ has exactly one subgroup of each order dividing n.)

Sylow's Theorems

Before proving Sylow's Theorems, we isolate a lemma that will be used repeatedly. It says that when a p-group acts on a finite set, the number of fixed points is congruent to the total size of the set modulo p.

Lemma 4.8.4: Fixed-Point Congruence for p-Group Actions

Let P be a finite p-group acting on a finite set X. Then

$$|X| \equiv |X^P| \pmod{p},$$

where $X^P = \{x \in X : g \cdot x = x \text{ for all } g \in P\}$ is the set of fixed points (definition 4.2.8).

***Proof*:**
The orbits of P on X partition X. By the Orbit-Stabilizer Theorem (theorem 4.3.1), each orbit has size $[P : P_x] = |P|/|P_x|$, which divides $|P| = p^k$, so every orbit size is a power of p. An orbit has size 1 (i.e. p^0) if and only if it is a fixed point. Every other orbit has size p^j with $j \geq 1$,

hence divisible by p. Therefore

$$|X| = |X^P| + \sum_{\substack{\text{non-fixed}\\\text{orbits}}} p^{j_i} \equiv |X^P| \pmod{p}.$$

This lemma is the mechanism behind many counting arguments in p-group theory. For instance, theorem 4.6.5 (p-groups have nontrivial center) is a special case: applying the lemma to the conjugation action $G \curvearrowright G$ with $P = G$ gives $|G| \equiv |Z(G)| \pmod{p}$, so $p \big| |Z(G)|$.

Theorem 4.8.5: Sylow's Theorems

Let G be a finite group with $|G| = p^\alpha m$, where p is prime and $p \nmid m$.

1. *(Existence.)* G has a Sylow p-subgroup, i.e. a subgroup of order p^α.
2. *(Conjugacy.)* Any two Sylow p-subgroups of G are conjugate. More generally, if P is a Sylow p-subgroup and Q is any p-subgroup of G, then $Q \leq gPg^{-1}$ for some $g \in G$.
3. *(Counting.)* The number n_p of Sylow p-subgroups satisfies:
   (a) $n_p = [G : N_G(P)]$ for any $P \in \mathrm{Syl}_p(G)$,
   (b) n_p divides m,
   (c) $n_p \equiv 1 \pmod{p}$.

We prove each part in turn.

***Proof of Sylow I*:**
We proceed by strong induction on $|G|$. If $|G| = 1$ (so $\alpha = 0$), the trivial subgroup has order $p^0 = 1$ and there is nothing to prove. Suppose $|G| > 1$ and that the result holds for all groups of order less than $|G|$. Write $|G| = p^\alpha m$ with $\alpha \geq 1$ and $p \nmid m$.

Consider the Class Equation (theorem 4.6.3):

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

Case 1: $p \big| |Z(G)|$. Since $Z(G)$ is abelian, Cauchy's Theorem for abelian groups (corollary 3.3.10) gives an element $z \in Z(G)$ of order p. Let $N = \langle z \rangle$. Since $N \leq Z(G)$, we have $N \trianglelefteq G$, and $|G/N| = p^{\alpha-1}m < |G|$.

By the induction hypothesis, G/N has a subgroup $\overline{P}$ of order $p^{\alpha-1}$. By the Correspondence Theorem (theorem 3.5.7), there exists $P \leq G$ with $N \leq P$ and $P/N = \overline{P}$. Then

$$|P| = |P/N| \cdot |N| = p^{\alpha-1} \cdot p = p^\alpha.$$

Case 2: $p \nmid |Z(G)|$. Since $p \big| |G|$ and the Class Equation gives $|Z(G)| = |G| - \sum [G : C_G(g_i)]$, if p divided every term $[G : C_G(g_i)]$, then p would divide $|Z(G)|$, a contradiction. So there exists some g_j with

$$p \nmid [G : C_G(g_j)].$$

Since $|G| = |C_G(g_j)| \cdot [G : C_G(g_j)]$ and $p^\alpha \big| |G|$ while $p \nmid [G : C_G(g_j)]$, we conclude $p^\alpha \big| |C_G(g_j)|$. But $g_j \notin Z(G)$, so $C_G(g_j) \neq G$ and $|C_G(g_j)| < |G|$.

By the induction hypothesis, $C_G(g_j)$ has a subgroup P of order p^α. Since $P \leq C_G(g_j) \leq G$ and $|P| = p^\alpha$, P is a Sylow p-subgroup of G. □

***Proof of Sylow II*:**
Let $P \in \mathrm{Syl}_p(G)$ and let Q be any p-subgroup of G. Consider the set of left cosets G/P, and let Q act on G/P by left multiplication: $q \cdot gP = (qg)P$.

Since $|G/P| = [G : P] = m$ and $p \nmid m$, the Fixed-Point Congruence (lemma 4.8.4) gives

$$m = |G/P| \equiv |(G/P)^Q| \pmod{p}.$$

Since $p \nmid m$, we have $(G/P)^Q \neq \varnothing$: there is at least one fixed point.

A coset gP is fixed by all $q \in Q$ if and only if $qgP = gP$ for all $q \in Q$, i.e. $g^{-1}qg \in P$ for all $q \in Q$, i.e. $g^{-1}Qg \leq P$, i.e. $Q \leq gPg^{-1}$.

So there exists $g \in G$ with $Q \leq gPg^{-1}$. If Q is itself a Sylow p-subgroup, then $|Q| = p^\alpha = |gPg^{-1}|$, and the inclusion forces $Q = gPg^{-1}$: all Sylow p-subgroups are conjugate. □

***Proof of Sylow III*:**
Let $\mathcal{S} = \mathrm{Syl}_p(G)$ and $n_p = |\mathcal{S}|$.

Part (a): $n_p = [G : N_G(P)]$. The group G acts on $\mathcal{S}$ by conjugation: $g \cdot P_i = gP_ig^{-1}$. By Sylow II, this action is transitive (any two Sylow p-subgroups lie in the same orbit). The stabilizer of P under this action is

$$G_P = \{g \in G : gPg^{-1} = P\} = N_G(P).$$

By the Orbit-Stabilizer Theorem (theorem 4.3.1), $n_p = |\mathcal{S}| = [G : N_G(P)]$.

Part (b): $n_p \big| m$. Since $P \leq N_G(P) \leq G$, the index decomposes as

$$[G : P] = [G : N_G(P)] \cdot [N_G(P) : P].$$

Thus $n_p = [G : N_G(P)]$ divides $[G : P] = m$.

Part (c): $n_p \equiv 1 \pmod{p}$. Fix $P_1 \in \mathcal{S}$ and let P_1 act on $\mathcal{S}$ by conjugation. By the Fixed-Point Congruence (lemma 4.8.4),

$$n_p \equiv |\mathcal{S}^{P_1}| \pmod{p},$$

where $\mathcal{S}^{P_1}$ is the set of Sylow p-subgroups fixed by every element of P_1.

We claim $\mathcal{S}^{P_1} = \{P_1\}$. Suppose $P_j \in \mathcal{S}^{P_1}$, so $P_1 \leq N_G(P_j)$. Since $P_j \trianglelefteq N_G(P_j)$ and $P_1 \leq N_G(P_j)$, the product P_1P_j is a subgroup of $N_G(P_j)$ (proposition 3.4.4). Its order is

$$|P_1P_j| = \frac{|P_1| \cdot |P_j|}{|P_1 \cap P_j|} = \frac{p^\alpha \cdot p^\alpha}{|P_1 \cap P_j|} = \frac{p^{2\alpha}}{|P_1 \cap P_j|},$$

which is a power of p (since $P_1 \cap P_j$ is a p-group). So P_1P_j is a p-subgroup of G containing P_1. Since P_1 is a *maximal* p-subgroup (it has order p^α), we must have $P_1P_j = P_1$, which gives $P_j \leq P_1$. Since $|P_j| = |P_1| = p^\alpha$, it follows that $P_j = P_1$.

Thus $|\mathcal{S}^{P_1}| = 1$, and $n_p \equiv 1 \pmod{p}$. □

Cauchy's Theorem

With Sylow's Theorems in hand, we can immediately prove the full version of Cauchy's Theorem for arbitrary finite groups, completing the proof that was deferred in theorem 3.3.8.

Theorem 4.8.6: Cauchy's Theorem

If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.

***Proof*:**
Write $|G| = p^\alpha m$ with $\alpha \geq 1$ and $p \nmid m$. By Sylow I, G has a Sylow p-subgroup P of order p^α. Let $x \in P$ with $x \neq e$. Then $|\langle x \rangle|$ divides $|P| = p^\alpha$ (theorem 3.3.1), so $|x| = p^\beta$ for some $1 \leq \beta \leq \alpha$. The element $y = x^{p^{\beta-1}}$ satisfies

$$|y| = \frac{|x|}{(|x|,\, p^{\beta-1})} = \frac{p^\beta}{p^{\beta-1}} = p,$$

where we used theorem 2.3.11. Since $y \in P \leq G$, G has an element of order p.

Note the logical structure: the abelian case of Cauchy's Theorem (corollary 3.3.10) was used *in the proof of Sylow I*, and the full Cauchy's Theorem is now derived *from* Sylow I. There is no circular reasoning.

Consequences

The following corollary collects the most frequently used structural consequences.

Corollary 4.8.7: Characterizations of a Unique Sylow Subgroup

Let $P \in \mathrm{Syl}_p(G)$. The following are equivalent:

1. $n_p = 1$ (i.e. P is the unique Sylow p-subgroup of G).
2. $P \trianglelefteq G$.
3. $P \operatorname{char} G$.

***Proof*:**
1 $\Rightarrow$ 3: If P is the unique subgroup of G of order p^α, then P is characteristic by proposition 4.7.122.

3 $\Rightarrow$ 2: Characteristic implies normal (proposition 4.7.121).

2 $\Rightarrow$ 1: If $P \trianglelefteq G$ and Q is any other Sylow p-subgroup, then by Sylow II, $Q = gPg^{-1}$ for some $g \in G$. Since P is normal, $gPg^{-1} = P$, so $Q = P$. Thus $n_p = 1$.

In particular, to show that a Sylow subgroup is normal, it suffices to show $n_p = 1$. The congruence and divisibility conditions from Sylow III often force this conclusion.

Example 4.8.8: Using Sylow's Theorems

1. Let $|G| = 15 = 3 \cdot 5$. Then $n_3 \mid 5$ and $n_3 \equiv 1 \pmod 3$, giving $n_3 \in \{1\}$. Similarly, $n_5 \mid 3$ and $n_5 \equiv 1 \pmod 5$, giving $n_5 \in \{1\}$. Both Sylow subgroups are normal and unique, each cyclic of prime order.

2. Let $|G| = 12 = 2^2 \cdot 3$. Then $n_3 \mid 4$ and $n_3 \equiv 1 \pmod 3$, giving $n_3 \in \{1, 4\}$. Also, $n_2 \mid 3$ and $n_2 \equiv 1 \pmod 2$, giving $n_2 \in \{1, 3\}$. Both possibilities genuinely occur: for A_4, $n_3 = 4$ and $n_2 = 1$; for $\mathbb{Z}/12\mathbb{Z}$, $n_3 = 1$ and $n_2 = 1$.

3. Let $|G| = 99 = 9 \cdot 11$. Then $n_{11} \mid 9$ and $n_{11} \equiv 1 \pmod{11}$. The divisors of 9 are $1, 3, 9$, and only $1 \equiv 1 \pmod{11}$. So $n_{11} = 1$ and the Sylow 11-subgroup is normal.

4.8.9 A Strategy for Sylow Problems Many problems in finite group theory reduce to the following systematic procedure:

1. Write $|G| = p^\alpha m$ with $p \nmid m$ for each prime p dividing $|G|$.

2. For each p, list the candidates for n_p: they must simultaneously divide m and be congruent to $1 \pmod p$.

3. Use the candidates to deduce structural information: if $n_p = 1$, the Sylow p-subgroup is normal; if $n_p > 1$, count the elements of order p^k across all Sylow subgroups and check for contradictions.

We will apply this strategy systematically in Section 4.9.

4.8.10 Applied Aside: Noether's Theorem in Physics While Sylow's Theorems concern finite groups, the underlying philosophy—that symmetries constrain structure—extends far beyond algebra. In 1918, Emmy Noether proved that every continuous symmetry of a physical system corresponds to a conserved quantity:

Symmetry (group action)	*Conservation law*
Time translation	Energy
Spatial translation	Momentum
Rotational symmetry	Angular momentum

Noether's Theorem uses *Lie groups* (continuous groups acting smoothly on manifolds), but the conceptual root is identical to everything in this chapter: a group acts on a space, and the structure of the action—its orbits, its stabilizers, its kernel—reveals deep invariants. The conservation laws of physics are, in a precise sense, consequences of group actions.

4.9 Applications of Sylow's Theorems

We now put the Sylow machinery to work. The goal is to demonstrate, through a sequence of increasingly sophisticated examples, how the existence, conjugacy, and counting theorems combine to constrain—and sometimes completely determine—the structure of a finite group from its order alone.

Groups of Order pq

Let $p < q$ be distinct primes and let $|G| = pq$. This is the simplest case beyond prime and prime-squared orders, and Sylow's Theorems give a nearly complete classification.

Theorem 4.9.1: Classification of Groups of Order pq

Let $p < q$ be primes and $|G| = pq$.

1. G has a unique (hence normal) Sylow q-subgroup.
2. If $p \nmid (q-1)$, then G is cyclic: $G \cong \mathbb{Z}/pq\mathbb{Z}$.
3. If $p \big| (q-1)$, then there are exactly two isomorphism classes of groups of order pq: the cyclic group $\mathbb{Z}/pq\mathbb{Z}$, and a non-abelian group (which can be constructed as a semi-direct product, see Section 5).

***Proof*:**
Part I. By Sylow III, $n_q \big| p$ and $n_q \equiv 1 \pmod{q}$. The divisors of p are 1 and p. Since $p < q$, we have $p \not\equiv 1 \pmod{q}$ (as $1 < p < q$). Therefore $n_q = 1$, and the unique Sylow q-subgroup Q is normal in G (corollary 4.8.7).

Part II. By Sylow III, $n_p \big| q$ and $n_p \equiv 1 \pmod{p}$. Since q is prime, $n_p \in \{1, q\}$. If $n_p = q$, then $q \equiv 1 \pmod{p}$, i.e. $p \big| (q-1)$. Contrapositively, if $p \nmid (q-1)$, then $n_p = 1$, and the Sylow p-subgroup P is also normal.

Now $P \cong \mathbb{Z}/p\mathbb{Z}$ and $Q \cong \mathbb{Z}/q\mathbb{Z}$ (both have prime order, hence are cyclic by corollary 2.3.12). Since $P \cap Q$ is a subgroup of both P and Q, its order divides $\gcd(p, q) = 1$, so $P \cap Q = \{e\}$. By proposition 3.4.3, $|PQ| = |P||Q|/|P \cap Q| = pq = |G|$, so $PQ = G$.

We claim G is abelian. For any $x \in P$ and $y \in Q$, the commutator $xyx^{-1}y^{-1}$ lies in Q (since $Q \trianglelefteq G$ gives $xyx^{-1} \in Q$, so $xyx^{-1}y^{-1} \in Q$) and also in P (since $P \trianglelefteq G$ gives $yx^{-1}y^{-1} \in P$, so $xyx^{-1}y^{-1} \in P$). Thus $xyx^{-1}y^{-1} \in P \cap Q = \{e\}$, giving $xy = yx$.

Since every element of $G = PQ$ is a product of an element of P and an element of Q, and these all commute, G is abelian. Let x generate P and y generate Q. Then $|xy| = \operatorname{lcm}(p, q) = pq$ (since $\gcd(p, q) = 1$), so $G = \langle xy \rangle \cong \mathbb{Z}/pq\mathbb{Z}$.

Part III. If $p \big| (q-1)$, then $n_p = q$ is possible, and a non-abelian group of order pq exists. Its construction requires the semi-direct product, which we defer to Section 5. One can show (using the semi-direct product theory) that the non-abelian group is unique up to isomorphism.

Example 4.9.2: Concrete Cases

1. $|G| = 15 = 3 \cdot 5$. Here $p = 3, q = 5$, and $3 \nmid 4$. So $G \cong \mathbb{Z}/15\mathbb{Z}$: there is only one group of order 15.
2. $|G| = 35 = 5 \cdot 7$. Here $p = 5, q = 7$, and $5 \nmid 6$. So $G \cong \mathbb{Z}/35\mathbb{Z}$.
3. $|G| = 6 = 2 \cdot 3$. Here $p = 2, q = 3$, and $2 \big| 2$. There are two groups: $\mathbb{Z}/6\mathbb{Z}$ and the non-abelian group S_3.
4. $|G| = 21 = 3 \cdot 7$. Here $p = 3, q = 7$, and $3 \big| 6$. There are two groups: $\mathbb{Z}/21\mathbb{Z}$ and a non-abelian group of order 21.

Groups of Order p^2q: A Worked Example

We illustrate the general strategy for $|G| = p^2q$ ($p \neq q$ primes) through the concrete case $|G| = 12 = 2^2 \cdot 3$.

Example 4.9.3: Sylow Analysis of Groups of Order 12
Let $|G| = 12$. The Sylow constraints give:

$$\begin{aligned} n_3 \big| 4 \text{ and } n_3 &\equiv 1 \pmod 3 && \implies && n_3 \in \{1, 4\}, \\ n_2 \big| 3 \text{ and } n_2 &\equiv 1 \pmod 2 && \implies && n_2 \in \{1, 3\}. \end{aligned}$$

Case $n_3 = 4$. There are 4 Sylow 3-subgroups, each cyclic of order 3 with 2 non-identity elements. Since distinct Sylow 3-subgroups intersect trivially (their intersection has order dividing 3 and is a proper subgroup of each, so it is $\{e\}$), there are $4 \times 2 = 8$ elements of order 3. The remaining $12 - 8 = 4$ elements (including e) must form the unique Sylow 2-subgroup. Thus $n_2 = 1$ and the Sylow 2-subgroup P is normal.

Since P has order 4, $P \cong \mathbb{Z}/4\mathbb{Z}$ or $P \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (corollary 4.6.6). The group A_4 is an example of this case, with $P \cong V_4$ and $n_3 = 4$.

Case $n_3 = 1$. The unique Sylow 3-subgroup $Q \cong \mathbb{Z}/3\mathbb{Z}$ is normal. The Sylow 2-subgroup P (order 4) acts on Q by conjugation, giving a homomorphism $P \to \mathrm{Aut}(Q) \cong \mathbb{Z}/2\mathbb{Z}$ (by theorem 4.7.6). The structure of G depends on this homomorphism: different choices yield the cyclic group $\mathbb{Z}/12\mathbb{Z}$, the dihedral group D_{12}, and the dicyclic group Dic_3 (among others, depending on whether $P \cong V_4$ or $P \cong \mathbb{Z}/4\mathbb{Z}$).

The key technique on display is *element counting*: when n_p is large, the Sylow p-subgroups account for many elements, leaving fewer elements for Sylow subgroups at other primes. This often forces $n_q = 1$ at a different prime, yielding a normal subgroup.

The same strategy applies to any order p^2q. In general:

1. Compute the Sylow constraints for n_p and n_q.
2. If both $n_p = 1$ and $n_q = 1$ are forced, both Sylow subgroups are normal, and G is a semi-direct product (or direct product) of a p-group and a q-group.

3. If $n_q > 1$ is possible, use element counting to check whether $n_p = 1$ is forced. If so, the normal Sylow p-subgroup provides structural leverage.
4. The full classification typically requires the semi-direct product machinery of Section 5.

4.9.1 Simplicity of A_5

We now prove one of the most celebrated results in group theory: the alternating group A_5 is simple. This fact is significant far beyond group theory—it is the algebraic reason that the general quintic equation cannot be solved by radicals, as Galois discovered.

We begin by determining the conjugacy classes of A_5.

Lemma 4.9.4: Conjugacy Classes of A_5

The group A_5 has five conjugacy classes, with sizes 1, 12, 12, 15, 20.

***Proof*:**
The even permutations in S_5 have four possible cycle types:

Cycle type	*Example*	*Count in* A_5
(1^5)	e	1
$(3, 1^2)$	$(1\ 2\ 3)$	$\binom{5}{3} \cdot 2 = 20$
$(2^2, 1)$	$(1\ 2)(3\ 4)$	$\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15$
(5)	$(1\ 2\ 3\ 4\ 5)$	$5!/5 = 24$

Total: $1 + 20 + 15 + 24 = 60 = |A_5|$.

In S_5, permutations of the same cycle type form a single conjugacy class (theorem 4.6.11). In A_5, an S_5-conjugacy class of even permutations either remains a single A_5-class or splits into two classes of equal size. The criterion is: a class *splits* in A_5 if and only if the centralizer in S_5 consists entirely of even permutations.

- 3-*cycles:* $C_{S_5}((1\ 2\ 3))$ contains the odd permutation $(4\ 5)$. The class does not split: one A_5-class of size 20.
- $(2^2, 1)$-*type:* $C_{S_5}((1\ 2)(3\ 4))$ contains the odd permutation $(1\ 2)$. The class does not split: one A_5-class of size 15.
- 5-*cycles:* $C_{S_5}((1\ 2\ 3\ 4\ 5)) = \langle(1\ 2\ 3\ 4\ 5)\rangle \cong \mathbb{Z}/5\mathbb{Z}$, all even. The class *splits* into two A_5-classes, each of size $24/2 = 12$.

The representatives of the two 5-cycle classes are $(1\ 2\ 3\ 4\ 5)$ and $(1\ 3\ 5\ 2\ 4)$ (or equivalently $(1\ 2\ 3\ 4\ 5)$ and $(1\ 2\ 3\ 5\ 4)$—any 5-cycle not conjugate to the first within A_5). The five class sizes are therefore $1, 12, 12, 15, 20$.

Theorem 4.9.5: A_5 is Simple

The alternating group A_5 has no nontrivial proper normal subgroups.

***Proof*:**
Suppose $N \trianglelefteq A_5$ with $N \neq \{e\}$. Since N is normal, it is a union of conjugacy classes and must include $\{e\}$. The class sizes are $1, 12, 12, 15, 20$. Since $|N|$ divides $|A_5| = 60$ (theorem 3.3.1), we check all possible unions:

Sum including 1	*Divides* 60*?*
$1 + 12 = 13$	No
$1 + 15 = 16$	No
$1 + 20 = 21$	No
$1 + 12 + 12 = 25$	No
$1 + 12 + 15 = 28$	No
$1 + 12 + 20 = 33$	No
$1 + 15 + 20 = 36$	No
$1 + 12 + 12 + 15 = 40$	No
$1 + 12 + 12 + 20 = 45$	No
$1 + 12 + 15 + 20 = 48$	No
$1 + 12 + 12 + 15 + 20 = 60$	Yes (all of A_5)

No proper nontrivial union of conjugacy classes (including $\{e\}$) has order dividing 60. Therefore $N = \{e\}$ or $N = A_5$, and A_5 is simple.

4.9.6 Remark: The Quintic and Simplicity The simplicity of A_5 is the key obstruction in Galois's proof that there is no formula (using radicals) for the roots of a general polynomial of degree ≥ 5. The rough idea: the "solvability" of a polynomial equation by radicals corresponds to a property of its Galois group called *solvability* (which we will define in Section 6.1.1), and A_5 is the smallest non-solvable group. The chain of normal subgroups required by solvability cannot exist in a simple group with more than one element. Since A_5 arises as the Galois group of certain quintic polynomials, those polynomials are not solvable by radicals.

Exercise 4.9.1

1. Let $|G| = pqr$ where $p < q < r$ are distinct primes. Show that G has a normal Sylow subgroup for at least one of the primes p, q, r.

   Hint: Apply Sylow III to all three primes. If $n_r > 1$ and $n_q > 1$, count the elements of orders r and q (they contribute $(n_r)(r-1) + (n_q)(q-1)$ non-identity elements from distinct cyclic subgroups). Show that if all three $n_p, n_q, n_r > 1$, the total exceeds pqr.

2. Let $|G| = 30 = 2 \cdot 3 \cdot 5$.

1. Show that $n_5 \in \{1, 6\}$ and $n_3 \in \{1, 10\}$.
2. Using 4.9.1 (1), conclude that G has a normal Sylow 5-subgroup or a normal Sylow 3-subgroup (or both).
3. Deduce that G has a normal subgroup of order 15.

   Hint: If $Q \trianglelefteq G$ is a Sylow 5-subgroup, consider G/Q and use the result on groups of order 6. If the Sylow 3-subgroup R is also normal, argue using $|QR| = 15$.

3. Let $|G| = 36 = 2^2 \cdot 3^2$.

   1. Show that $n_3 \in \{1, 4\}$.
   2. If $n_3 = 4$, show that the action of G on $\mathrm{Syl}_3(G)$ by conjugation gives a homomorphism $\pi\colon G \to S_4$. Conclude that $\ker(\pi)$ is a nontrivial normal subgroup of G.

      Hint: $|G| = 36 > 24 = |S_4|$, so π cannot be injective.

   Deduce that every group of order 36 has a nontrivial proper normal subgroup (and hence is not simple).

4. Show that the only simple group of order 60 is A_5 (up to isomorphism).

   Hint: Let G be simple of order 60. Show $n_5 = 6$, giving $[G : N_G(P)] = 6$ for $P \in \mathrm{Syl}_5(G)$. The action of G on the 6 cosets of $N_G(P)$ gives an injective homomorphism $G \hookrightarrow S_6$. Argue that the image lies in A_6, and identify it with A_5.

5. Let p be an odd prime. Show that every Sylow p-subgroup of D_{2n} is cyclic and normal.

   Hint: The cyclic subgroup $\langle r\rangle \trianglelefteq D_{2n}$ has index 2.

4.10 Summary

This chapter developed the theory of group actions from first principles and applied it to obtain structural results about finite groups. We collect the main threads here for reference.

The Core Framework

A *group action* of G on a set X is a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$ (definition 4.1.1), equivalently a map $G \times X \to X$ satisfying the compatibility and identity axioms (theorem 4.1.2). Every action decomposes X into *orbits* (definition 4.2.1), and the size of each orbit is governed by the *stabilizer* via the Orbit-Stabilizer Theorem (theorem 4.3.1):

$$|G \cdot x| = [G : G_x].$$

The number of orbits can be computed by Burnside's Lemma (theorem 4.3.7): $|X/G| = \frac{1}{|G|}\sum_{g\in G} |X^g|$.

Types of Actions

Type	*Defining condition*	*Consequence*
Faithful	$\ker(\rho) = \{e\}$	$G \hookrightarrow \mathrm{Sym}(X)$
Transitive	One orbit: $G \cdot x = X$	$X \cong_G G/G_{x_0}$ for any x_0
Free	All stabilizers trivial: $G_x = \{e\}$	Every orbit has size $\lvert G\rvert$
Regular	Free + Transitive	$X \cong_G G$ (left multiplication)

The Key Self-Actions

The most powerful applications came from letting G act on itself or on structures derived from itself. The following table summarizes the three principal self-actions and their consequences.

Action	*Set X*	*Orbit of x*	*Stabilizer G_x*	*Kernel*	*Key result*
Left mult. $g \cdot a = ga$	G	G (one orbit)	$\{e\}$	$\{e\}$	Cayley's Thm
Conjugation $g \cdot a = gag^{-1}$	G	$\mathrm{cl}(a)$	$C_G(a)$	$Z(G)$	Class Equation
Coset action $g \cdot aH = gaH$	G/H	G/H (one orbit)	H (at eH)	$\bigcap_g gHg^{-1}$	Index-p Thm

Two further actions played essential roles:

- *Conjugation on subgroups* ($g \cdot S = gSg^{-1}$): the stabilizer is the normalizer $N_G(S)$, and the number of conjugates of S is $[G : N_G(S)]$ (corollary 4.6.9).
- *Conjugation on Sylow subgroups*: transitivity (Sylow II) gives $n_p = [G : N_G(P)]$, and the Fixed-Point Congruence (lemma 4.8.4) gives $n_p \equiv 1 \pmod p$.

The Major Theorems

1. *Orbit-Stabilizer Theorem* (theorem 4.3.1): $|G \cdot x| = [G : G_x]$.
2. *Burnside's Lemma* (theorem 4.3.7): $|X/G| = \frac{1}{|G|}\sum_g |X^g|$.
3. *Cayley's Theorem* (theorem 4.5.3): Every group embeds into a symmetric group.
4. *The Class Equation* (theorem 4.6.3): $|G| = |Z(G)| + \sum [G : C_G(g_i)]$.
5. *p-Groups have nontrivial center* (theorem 4.6.5).
6. $G/Z(G) \cong \mathrm{Inn}(G)$ (theorem 4.7.4).
7. *The N/C Theorem* (theorem 4.7.6): $N_G(H)/C_G(H) \hookrightarrow \mathrm{Aut}(H)$.

8. *Sylow's Theorems* (theorem 4.8.5): Existence, conjugacy, and counting of Sylow p-subgroups.

9. *Cauchy's Theorem* (theorem 4.8.6): If $p \big| |G|$, then G has an element of order p.

Looking Ahead

The idea that a group *acts* on something—and that the structure of the action reveals the structure of the group—is not a technique that ends with this chapter. In the chapters ahead, group actions will reappear in new forms:

- In the study of *semi-direct products* (Section 5), the N/C Theorem and the conjugation action will be the key ingredients for constructing new groups from old ones.
- In *Galois theory*, a Galois group acts on the roots of a polynomial and on intermediate field extensions; the Fundamental Theorem of Galois Theory is, at its core, a correspondence between subgroups (stabilizers) and orbits (field extensions).
- In *module theory* (Part III), the group action framework generalizes to rings acting on abelian groups. The single condition "ρ is a homomorphism" will become the defining property of a module.

The unifying philosophy of this chapter bears repeating one last time:

> *Groups are not static objects. They are agents of symmetry, and the way to understand them is to watch how they act—on sets, on themselves, and on each other.*

4.11⋆ Group Actions in Advanced Mathematics

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

This section is addressed to the mathematically curious reader who wishes to see where the ideas of this chapter lead. No proofs are given, and nothing here is prerequisite for the rest of the book.

The theory of group actions developed in this chapter—homomorphisms to symmetric groups, orbits, stabilizers, counting—is the finite, discrete case of a theme that pervades nearly every branch of modern mathematics. In each new setting, the pattern is the same: a group acts on a structured object, and the interplay between the action and the structure produces deep theorems. We sketch three such settings.

4.11.1 Representation Theory

In this chapter, a group action was a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, where X was only a *set*. What happens if we upgrade X to a *vector space*?

A *(linear) representation* of a group G is a homomorphism

$$\rho\colon G \to GL(V),$$

where V is a vector space over a field $\mathbb{F}$ and $GL(V)$ is the group of invertible linear maps $V \to V$. Each group element acts on V not as a bijection alone, but as an *invertible linear transformation*: it respects addition and scalar multiplication.

Every group action $G \curvearrowright X$ on a finite set gives rise to a representation in a canonical way. Let $V = \mathbb{F}^X$ be the vector space with basis $\{e_x : x \in X\}$, one basis vector per element of X. Then G acts on V by permuting the basis: $g \cdot e_x = e_{g\cdot x}$. This is the *permutation representation* in the linear sense, and it converts our set-theoretic theory into linear algebra.

The power of representation theory lies in the tools of linear algebra that become available: eigenvalues, traces, direct-sum decompositions. The *character* of a representation is the function $\chi\colon G \to \mathbb{F}$ defined by $\chi(g) = \mathrm{tr}(\rho(g))$. Characters satisfy remarkable orthogonality relations, and for finite groups over $\mathbb{C}$, the irreducible characters form a basis for the space of class functions (functions constant on conjugacy classes). The number of irreducible representations equals the number of conjugacy classes—a striking connection to the Class Equation of Section 4.6.

Two landmark results illustrate the depth of the theory:

- *Maschke's Theorem (theorem 53.1.9):* If $\mathrm{char}(\mathbb{F}) \nmid |G|$, then every representation decomposes as a direct sum of irreducible representations. This is the analogue of diagonalizability for group actions.
- *Burnside's p^aq^b Theorem (theorem 53.2.24):* Every group of order p^aq^b (two prime factors) is solvable. Remarkably, the only known proof for over a century used *representation theory*, not pure group theory—a vivid demonstration that "linearizing" group actions can unlock results inaccessible by other means. (A purely group-theoretic proof was found by Thompson and others only much later.)

4.11.2 Algebraic Topology

Group actions enter topology through the theory of *covering spaces*.

Let X be a topological space and $\widetilde{X} \to X$ a covering map. The *deck transformations* (or covering transformations) are the homeomorphisms $\varphi\colon \widetilde{X} \to \widetilde{X}$ that commute with the projection to X. These form a group $\mathrm{Deck}(\widetilde{X}/X)$ that acts on $\widetilde{X}$ by homeomorphisms. When the covering is *normal* (or *regular*—note the terminology!), the deck group acts *freely and transitively* on each fiber. This is precisely a regular action in the sense of definition 4.4.11, and the base space is recovered as the orbit space:

$$X \;\cong\; \widetilde{X} \,/\, \mathrm{Deck}(\widetilde{X}/X).$$

The fundamental group $\pi_1(X, x_0)$ acts on the fiber $p^{-1}(x_0)$ by *monodromy*: lifting a loop in X to a path in $\widetilde{X}$ and tracking where the endpoint lands. This action encodes the topology of the covering:

- The *orbits* of the monodromy action correspond to the connected components of $\widetilde{X}$ (if the covering is connected, there is one orbit—the action is transitive).
- The *stabilizer* of a point in the fiber corresponds to the image of $\pi_1(\widetilde{X})$ in $\pi_1(X)$—the "surviving" loops.
- The *orbit-stabilizer correspondence* becomes the classical bijection between connected coverings of X (up to isomorphism) and conjugacy classes of subgroups of $\pi_1(X)$—a topological analogue of the Structure Theorem for Transitive Actions (theorem 4.4.6).

More broadly, whenever a group G acts on a topological space X, one can study *equivariant topology*: homology and cohomology theories that respect the action. The *equivariant cohomology* $H^*_G(X)$ combines the topology of X with the algebra of G, and its computation often reduces to understanding the fixed points and stabilizers of the action—exactly the tools developed in this chapter.

4.11.3 Category Theory

Category theory provides the most general and conceptually clean framework for understanding group actions.

A group G can be viewed as a category $\mathbf{B}G$ with a single object $*$ and one morphism $* \to *$ for each element of G, with composition given by the group operation. (The invertibility of morphisms makes $\mathbf{B}G$ a *groupoid*—a category in which every morphism is an isomorphism.)

From this viewpoint:

Algebraic concept	*Categorical translation*	*Target category*
Group action on a set	Functor $\mathbf{B}G \to \mathbf{Set}$	Sets and functions
Linear representation	Functor $\mathbf{B}G \to \mathbf{Vect}_{\mathbb{F}}$	Vector spaces
Module over a group ring	Functor $\mathbf{B}G \to \mathbf{Ab}$	Abelian groups
Topological G-space	Functor $\mathbf{B}G \to \mathbf{Top}$	Topological spaces

In each row, the pattern from definition 4.1.1 is visible: a group action is a homomorphism into the automorphism group of some object. The categorical perspective replaces "automorphism group" with "endomorphism monoid" and "homomorphism" with "functor," but the essential idea is identical.

The collection of all G-sets itself forms a category, denoted G-**Set**, whose morphisms are the G-equivariant maps (definition 4.4.5). The Structure Theorem for Transitive Actions (theorem 4.4.6) becomes the statement that every transitive G-set is isomorphic to a coset space G/H in this category. Burnside's Lemma can be reinterpreted as a statement about the *colimit* of the functor $\mathbf{B}G \to \mathbf{Set}$.

Perhaps most strikingly, the passage from group actions to module theory—which will occupy much of Part III—is simply the passage from functors into **Set** to functors into **Ab**. The group ring $\mathbb{Z}[G]$ arises as the *free abelian group* on the morphisms of $\mathbf{B}G$, and a $\mathbb{Z}[G]$-module is exactly a functor $\mathbf{B}G \to \mathbf{Ab}$. In this way, the entire theory of modules over group rings—and by extension, much of representation theory—is a natural continuation of the ideas introduced in this chapter.

A group action is a functor from a one-object groupoid.
Change the target category, and the whole of algebra unfolds.

5

Composing Groups: Direct and Semi-Direct Products

(Current intuition: We've learned that any finite group is an extension of simple groups. How now do we concretely know the structure of G given H and $G/H = K$? This can in general be a very hard question, however there are cases in which this question is much more managable: the *direct produc* and *semi-direct product*. As it turns out, if G is an extention of H by K, then the relations of G can purely be described by H and K by something called the *wreath product* which a sort of action on H given by K including the semi-direct product; this is the *Krasner–Kaloujnine universal embedding theorem*. This theorem is more difficult and of little use to the general project of introducing algebra in this book, but the spirit of undersatnding the structure of G given H and K is going to be a theme that shall crop up a lot, and deeply explored in part XI)

I got a couple of intuition on direct products

1. Adding group structure to products. There is a notion of products with sets: X, Y can give $X \times Y$. It is very easy to show that if X and Y have group structures $((X, *_X), (Y, *_Y))$, then the product of these form a very natural group. This product also doesn't break under infinitely many products too [1]

2. Sort of the dual to the previous intuition: instead of building up groups, we're going to *decompsoe* groups into direct product of groups. We've learnt how to make subgroups out of groups by looking for elements with special properties (ex. commutativity) or generating subgroups with a subset of elements. We've found smaller groups by learning about how to properly define the idea of quotienting two groups with the help of homomorphisms. Now, we'll sort of combine the two. We are going to try and decompoes groups (ex. G) to into smaller

[1] If you've seen topology, you know that this is not always true in every Category. In the **Top** Category, you will/have seen that the product topology on infinitely many spaces need some special conditions

groups whose direct product would be G (ex. $G = H \times K$), or when we *almost* have a direct product ($G = H \rtimes K$)

3. This intuition overlaps with the last point. If we have $H, K \trianglelefteq G$, so that we don't have to think about $H \cap K$, then we can form $HK = G$ (or if $H \cap K = \{e\}$, $H \times K \cong G$). We can generalize this, and have $H \trianglelefteq G$ and $K \leqslant G$, an we will from here define a new product $H \rtimes K = G$ to be able to construct some differnet groups. Remebmer how I said a lot of stuff in math defines how you're *almost* abelian? Here is another example of the idea of being commutative as being the point from which we are trying to generalize. This one will heavily rely on intuition presented about the automorphisms! TBD

5.1 Classical Definitions And Properties

You might recall example 1.1.5[4], we showed we can create a group-structure given two groups. We revisit this definition here in generality:

Definition 5.1.1: Direct Product

The ***direct product*** of $G_1 \times G_2 \times \cdots$ of groups $G_1, ..., G_n, ...$ with the operations $*_1, ..., *_n, ...$ respectively is the set of sequences $(g_1, ..., g_n, ...)$ (or n-tuples if the number of groups is finite) where $g_9 \in G_i$ with the operation defined componentwise:

$$(g_1, ..., g_n, ...) * (h_1, ..., h_n, ...) = (g_1 *_1 h_1, ..., g_n *_n h_n, ...)$$

Sometimes, the direct product is called the *external direct product* to differentiate it from the *internal direct product* (see definition 3.4.1)

Example 5.1.2

1. Think of $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ with component-wise addition: $(\mathbb{R} \times \mathbb{R}, (+, +))$ or $(\mathbb{R}^2, (+, +))$.
2. The groups can also be arbitrary. Let $G_1 = \mathbb{Z}$, $G_2 = S_3$ and $G_3 = GL_2(\mathbb{R})$. Then $G_1 \times G_2 \times G_3$ with the operation defined component wise is indeed a group

For the direct product, there are a few general properties you can prove as an exercise:

Proposition 5.1.3: properties of Direct Products

Let every G_i be a group.

1. $|\prod_i G_i| = \prod_i |G_i|$
2. For each fixed i, the set of elements of G which have the identity of G_j in the j^{th} position for all $j \neq i$ and arbitrary elements of G_i in position i is a subgrou pof G isomorphic to G_i:
$$G_i \cong (1_{g_1}, ..., g_i, 1_{g_{i+1}}, ...)$$
3. For each fixed i, define $\pi_i : G \to G_i$ by
$$\pi_i((g_1, ..., g_n)) = g_i$$
then π_i is a surjective homomorphsim with
$$\begin{aligned}\ker(\pi_i) &= \{(g_1, ..., g_{i-1}, 1, g_{i+1}, ..., g_n) | g_j \in G_j \text{ for all } j \neq i\} \\ &\cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times G_n\end{aligned}$$
4. Under the identificatiosn in part (1), if $x \in G_i$ and $y \in G_j$ for some $i \neq j$, then $xy = yx$

***Proof*:**
exercise

(Comment on that last part of the proposition with $xy = yx$. Since $(X, 0)$ and $(0, Y)$ are "independent" from each other (i.e., they both kind of stay in their own lanes, when multiplying we just do component wise), it comes at the "cost", or a necessary consequence of it, is that we must treat those element having an *abelian* relation. This is another example of how something being abelian makes thing very powerful – if we want those two to be "independent", we impose an abeilan condition, we impose an abeilan condition)

Exercise 5.1.1

1. Does there exist a group such that $G \times G \cong G$? hint: [2]

   Let G be a finitely generated abelian group. Show that every subgroup is also a finitely generated abelian group.

In definition 5.1.1, the components of the direct product need not be the same. If they are the same, we can introduce the following notation:

$$G^4 = G \times G \times G \times G$$

which repesents the direct product of the group G 4 times.

We can equivalently write it this way

is equivalent to $\{f : \{0, 1, 2, 3\} \to G \mid f \text{ is a function}\}$ where f is any function! (Poof of this here).

[2] Think about $G = \prod_{n=1}^{\infty} \mathbb{Z}$

Example 5.1.4: Example
here

Show how this is useful in generalizing to (for example):

$$(\mathbb{Z}/2\mathbb{Z})^{\mathbb{R}} = \{f : \mathbb{R} \to \mathbb{Z}/2\mathbb{Z}\}$$

more generally, we can write A^B!

Maybe as a bonus (so stared section) show the link between A^B and the universal property of the product.

Example 5.1.5: Exercies

1. If $H \leqslant G_1 \times G_2$, is it true that there always exists subgroups $H_1 \leqslant G_1$ and $H_2 \leqslant G_2$ such that
$$H \cong H_1 \times H_2$$
if yes prove it, if not give a counter-example.

5.1.1 Fundamental theorem of Finitely Generated Abelian Groups

This theorem is stated, but not proved till till much later (chapter 14, theorem 14.1.9). The main idea that is trying to be presented here is the power of of being abelian: If we have an abelian, it is *such* a nice group that we can classify them all with isormophism to $\mathbb{Z}$ and $\mathbb{Z}_{p^\alpha}$ for appropriate power-prime. It will also help in our indeavour to classify groups. With this representation, it becomes much much easier. Here are two terms we introduce now to present the theorem:

Definition 5.1.6: Finitely Generated Group

A group is finitely generated if $\exists A \subseteq G$ such that $\langle A \rangle = G$.

Definition 5.1.7: Free Abelian Group

$\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$. $\mathbb{Z}^r$ is called the *free abelian group of rank r*.

Theorem 5.1.8: Fundamental Theorem of Finitely Generated Abelean Groups

Let G be a finitely generated group G. Then:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_s}$$

for some integers $r, n_1, ..., n_s$ with the following conditions:

1. $r \geq 0$ and $n_j \geq 2$ for all j, and
2. $n_{i+1} | n_i$ for $1 \leq i \leq s - 1$

Furthermore, the expression is unique (p.173 for clarification).

For those interested in a proof of this theorem for finite group see section ref:HERE [3]

> ***Proof***:
> We prove this in theorem 14.1.9. We will give a purely group-theoretical proof of this fact in ref:HERE.

What we'll do for the rest of this section is give the vocabulary to talk about abelian groups.

Notice also that it's for *finitely generated abelian groups*. What about infinitely generated abelian groups? We can simplify for a moment and ask about countably generated abelian groups. We have solved this for p-groups, but we have yet to solve it in the general case:

https://math.stackexchange.com/questions/3927834/classification-of-countably-infinite-abelian-groups

Definition 5.1.9: free rank and invariant factors

r is called the *free rank* or *betti number*, while $n_1, ..., n_s$ are called the *invariant factors*. The whole expression is called the *invariant factor decomposition of G*.

will finish later

Exercise 5.1.2

1. Let $A \subseteq B$ two bwo finitely generated free abelian groups each of rank n. Show that B/A has rank 0.
2. Let $G = (\mathbb{Z}/n\mathbb{Z})^\times$. Decompose G into a product of cyclic groups with additive notation.

5.2 Recognizing Direct Products

Let's say we have a (not necessarily abelian) group G. Can G be decomposed into the direct product of two smaller groups? This question is what we'll be exploring in this section.

The 4th identity of proposition 5.1.3 gives us a very valuable insight on how to think of direct products. If we have two groups G_1 and G_2, we showed that creating a set of 2-tulpes with component-wise operation creates a new group. With the fact that components commute ($xy = yx$, if $x \in G_1$, $y \in G_2$), we can also define the direct product in terms of the following:

$$G_1 \times G_2 = \langle x, y | xy = yx, x \in G_1, y \in G_2 \rangle$$

where the operation is done when everything is "moved to the appropriate place"

> **Example 5.2.1: presentation example**
> Let $G_1 = S_3$ and $G_2 = \mathbb{Z}_5$. Take $x = (1\ 2)4$ and $y = 4(2\ 3)2$. Then
>
> $$xy = (1\ 2)44(2\ 3)2 = (1\ 2)(2\ 3)442 = (1\ 2\ 3)2$$

[3] For those who want to see how the difficulties of generalizing this theorem to infinitely (particularly countably) generated abelian groups and want to see a particular generalization, google Prüfer Theorems

You might ask yourself what's the point of this 2nd perspective? The idea is that we can now ask ourselves when can a group be decomposed into smaller groups. In other words, if we had a group G, can we split it into two parts G_1, G_2 such that $G \cong G_1 \times G_2$?

Thus, we will explore exactly how far a two subgroups (or more generally subsets) of a group are from being commutative, and then use that information to find out when we can decompose groups.

We start with defining a way to capture *how commutative* a groups/subgroups/elements are:

Definition 5.2.2: Commutator and Commutator Group

Let G be a group, let $x, y \in G$, and let A, B be nonemtpy subset of G

1. Define $[x,y] = x^{-1}y^{-1}xy$. $[x,y]$ is called the *commutator* of x and y
2. Define $[A,B] = \langle [a,b] \big| a \in A,\ b \in B \rangle$, the group geenrated by the commutators of elements from A and B
3. Define $G' = \langle [x,y] | x, y \in G \rangle$ to be the subgroup of G generated by commutators of all element sfrom G. G' is called the *commutator subgroup* of G

Notice that if x and y commute, then $[x,y] = 1$. If all the element of G commute, then $G' = \{1\}$. Thus, the commutator captures an idea of how much the elements of G commute

Proposition 5.2.3: Commutator Subgroup Properties

Let G be a group, $x, y \in G$, and $H \leqslant G$. Then

1. $xy = yx[x,y]$
2. $H \trianglelefteq G$ if and only if $[H,G] \leqslant H$
3. for any automorphism σ of G, $\sigma([x,y]) = [\sigma(x), \sigma(y)]$, that is, $G' \mathrm{char} G$. As a consequence, G/G' is abelian
4. G/G' is the largest abelian quotient of G. That is, if $H \trianglelefteq G$ and G/H is abelian, then $G' \leqslant H$. Conversely, if $G' \leqslant H$, then $H \trianglelefteq G$ and G/H is abelian
5. (Universal Abelian Quotient) if $\varphi : G \to A$ is any homomorphism of G into an abelian group A, then φ factors through G', i.e., $G' \leqslant \ker(\varphi)$ and the following diagram commutes:

$$\begin{tikzcd} G \arrow[r] \arrow[dr, "\varphi"] & G/G' \arrow[d] \\ & A \end{tikzcd}$$

(A word on how:) These proofs are a good review of many previous concepts covered in this textbook, and so I highly recommend you try to do all of these yourself before doing the proof – you'll gain a much stronger intuition behind commutator subgroups.

***Proof*:**

1. $xy = yxx^{-1}y^{-1}xy = xy$
2. Let $H \trianglelefteq G$. Then $gH = Hg$, meaning every element of H, $gh = h'g$ for some $h, h' \in H$. Thus
$$[H, G] = \langle h^{-1}g^{-1}hg | h \in H, g \in G\rangle = \langle hh' | h, h' \in H\rangle$$
thus, $[H, G]$ is generated by elements of H, meaning $[H, G] \leqslant H$. Conversely, if $[H, G] \leqslant G$, then for every $h^{-1}g^{-1}hg \in [H, G]$, there is an h' such that
$$h^{-1}g^{-1}hg = h' \Rightarrow g^{-1}hg = hh'$$
and since this is true for any $h \in H$, $g^{-1}Hg = H$, or if we replace g with g^{-1}, we get the more conventional $gHg^{-1} = H$, meaning $H \trianglelefteq G$.
3. Let $[x, y] \in G'$. If $\sigma \in \mathrm{Aut}(G)$, then
$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) = [\sigma(x), \sigma(y)]$$
and since $\sigma(x), \sigma(y) \in G$, that means' σ maps commutators to commutators. Since σ is bijective (i.e., has a two-sided inverse):
$$\sigma(G') = G$$
for any σ, and so $G'\mathrm{char}G$, and so $G' \trianglelefteq G$ (proposition 4.7.12).
Since G/G' is well-defined, notice that for $[x], [y] \in G/G'$,
$$[xy] = [xyy^{-1}x^{-1}yx] = [yx]$$
and so, G/G' is abelian.
4. Let $H \trianglelefteq G$ such that G/H is abelian. Then for any $[x], [y] \in G/H$, $[xy] = [yx]$, implying
$$[xyx^{-1}y^{-1}] = [1] \Rightarrow xyx^{-1}y^{-1} \in H$$
showing that $G' \leqslant H$. Conversely, if $G' \leqslant H$, then for any $g \in G$ and any $h \in H$
$$ghg^{-1}h^{-1} \in H \Rightarrow ghg^{-1}h^{-1} = h'$$
for some $h' \in H$, but then $ghg^{-1} = h'h$. Since h, h', g were all arbitrary, that means that $gHg^{-1} = H$ for any g, and so H is normal, and so G/H is well-defined. Furthermore, for any $[x], [y] \in G/H$,
$$[xy] = [xyy^{-1}x^{-1}yx] = [yx]$$
since $y^{-1}x^{-1}yx \in H$. Thus, G/H is also abelian.
Another way of proving this is by taking advantage of isomorphism theorems (take a moment to think about it if you didn't see it this way as good practice!).
If G/G' is abelian, then every subgroup is normal, and since $G' \leqslant H$, $H/G' \trianglelefteq G/G'$ is well-defined. By the 4th isomorphism theorem, $H \trianglelefteq G$, and so G/H is well-defined. Then, by the 3rd isomorphism theorem:
$$G/H \cong (G/G')(H/G')$$
and since G/G' is abelian, so is it's quotient, and so G/H is abelian.

5. This is another way of writing (4), but in terms of homomorphisms.

   Let $\varphi : G \to A$ be a homomorphism from G to any abelian group G, and let $\ker(\varphi)$ be the kernel. We need to find a map from G to G/G' and a map G/G' to A such that the diagram commutes, that is, if π is the map from G ot G/G' and ψ is the map from G/G' to A, then $\varphi = \psi \circ \pi$.

   Since G is abelian, $G/\ker(\varphi)$ is abelian. Thus, by part (4), $G' \leqslant \ker(\varphi)$. Since G' is also a normal subgroup of G, G/G' is well-defined, the natrual projection map exists and is well-defined:

   $$\pi : G \to G/G' \qquad \pi(g) = gG'$$

   Now, for the map from G/G' to A, we need to make sure that it splits with φ, meaning it is dependent on where φ sends it's elements, and that it is independent of the representatives of G/G', that is, if

   $$[g] = [gx^{-1}y^{-1}xy] \Rightarrow \psi(g) = \psi(gx^{-1}y^{-1}xy)$$

   In fact, φ with the co-domain of G/G' will do just that. Let $\psi = \overline{\varphi}$ represent the function:

   $$\overline{\varphi} : G/G' \to A \qquad \overline{\varphi}([g]) = \varphi(g)$$

   this function is well-defined, since if $[g] = [gx^{-1}y^{-1}xy]$ for any $x, y \in G$

   $$\begin{aligned}
   \overline{\varphi}(gx^{-1}y^{-1}xy) &= \varphi(gx^{-1}y^{-1}xy) && \text{by definition} \\
   &= \varphi(g)\varphi(x^{-1})\varphi(y^{-1})\varphi(x)\varphi(y) && \varphi \text{ is a homomorphism} \\
   &= \varphi(g)\varphi(x)\varphi(x^{-1})\varphi(y)\varphi(y^{-1}) && A \text{ is abelian} \\
   &= \varphi(g)\varphi(xx^{-1})\varphi(yy^{-1}) && \varphi \text{ is a homomorphism} \\
   &= \varphi(g)\varphi(1)\varphi(1) && \\
   &= \varphi(g) && \varphi(1) = 1 \\
   &= \overline{\varphi}([g]) && \text{by definition}
   \end{aligned}$$

   Thus, this function is indeed well-defined! It came down to φ already being defined for us, and A being abelian. One should also check that $\overline{\varphi}$ is indeed a homomorphism (exercise).

   Finally, we need to show that $\varphi = \overline{\varphi} \circ \pi$. If $g \in g$. Then

   $$(\overline{\varphi} \circ \pi)(g) = \overline{\varphi}(\pi(g)) = \overline{\varphi}([g]) = \varphi(g)$$

   and since this is true for all $g \in G$, then $\varphi = \overline{\varphi} \circ \pi$, meaning φ does indeed split! Thus, the diagram does indeed commute:

   $$\begin{tikzcd}
   G \arrow[r] \arrow[dr, "\varphi"] & G/G' \arrow[d] \\
   & A
   \end{tikzcd}$$

(develop universal property and commutative groups)

Example 5.2.4

1. If G is abelian, then for any $x, y \in G$, $[x, y] = xyx^{-1}y^{-1} = xx^{-1}yy^{-1} = 1$, and so $G' = 1$, and $G/G' \cong G$.
2. Example with D_8
3. Example with D_n
4. Example with S_5
5. Example with F_2

With this, we can answer now give criterion to recognize direct products. First, a lemma:

Lemma 5.2.5: Representation's of HK

Let H and K be subgrousp of a group G. The number of distinct ways of writing each element of the set HK in the form hk, for some $h \in H$ and $k \in K$ is $|H \cap K|$. In particualr, if $H \cap K = 1$, then each element of HK can be written uniquely as a product of hk, for some $h \in H$ and $k \in K$.

***Proof*:**
see proposition 3.4.3

Theorem 5.2.6: Recognition Theorem For Direct Products

Let G. If H, K are subgroups such that

1. H and K are normal in G, and
2. $H \cap K = \{1\}$

Then

$$HK \cong H \times K$$

Conersly, if $HK \cong H \times K$, then $H \cap K = \{1\}$ and both H and K are normal subgroups of G.

If $HK = G$, then $G \cong H \times K$. Similarly, If G is isomorphic to the diret product of M and N (where M and N need not be subgroups of G) if $H \cong M$, $K \cong N$, and $G = HK$.

The propety that $H \cap K = 1$ has a name:

Definition 5.2.7: Complement

Let H be a subgroup of G. A subgroup K of G is called a *complement* for H in G if $G = hK$ and $H \cap K = 1$.

Thus, we can rephrase the previous theroem as a group is a derict product if there exists a normal complement for some proper normal subgroup of G.

***Proof*:**
We'll first prove a lemma that we'll use for both direction: If HK is a group and $|H \cap K| = 1$, then $hk = kh$.

Proof. Since $H \trianglelefteq G$, $khk^{-1} \in H$, and so $(khk^{-1})h^{-1} \in H$. Similarly, $k(hk^{-1}h^{-1}) \in H$. Since $H \cap K = 1$, we have

$$khk^{-1}h^{-1} = 1 \Rightarrow kh = hk$$

showing that every element of H commutes with every element of K. Note that does does not show that HK is abelian. For example, we did not show that for any $k, k' \in HK$, $kk' = k'k$. □

If $HK \cong H \times K$, we know that $|HK| = |H \times K| = |H||K|$, implying that $|H \cap K| = 1$, fulfilling the first requirement (ONLY IN THE FINITE CASE!!). Next, since HK is indeed a group (being isomorphis to $H \times K$), then by corolalry 3.4.5, at least one of H or K are normal. We need to show that both H and K are normal. We know by proposition 5.1.3 that $(H, 1)$ and $(1, K)$ are normal. We can't just push H or K through to $(H, 1)$ or $(1, K)$ since we don't have control over what isomorphism we have, and we can't force H to map to $(H, 1)$ [a]. Since $HK \cong H \times K$ their pre-image are also normal groups. Let's say φ is the isomorphism. Then since both of these preimage are isomorphic to H and K repsectively, then we can create an isomorphism $\psi : HK \to HK$ where $\psi(hk) = \varphi^{-1}(h, k)$. Since φ^{-1} is an isomorphism, this map is well-defined and bijective. THis map is also a homomorphism since:

$$\begin{aligned}
\psi(h_1k_1h_2k_2) &= \psi(h_1h_2k_1k_2) && \text{lemma at the beginning} \\
&= \varphi^{-1}((h_1h_2, k_1k_2)) \\
&= \varphi^{-1}((h_1, k_1) \cdot (h_2, k_2)) \\
&= \varphi^{-1}(h_1, k_1)\varphi^{-1}(h_2, k_2) \\
&= \psi(h_1k_1)\psi(h_2'k_2)
\end{aligned}$$

Thus, H and K can be asociated with $(H, 1)$ and $(1, K)$, and must also be normal – completing this direciton.

For the other way, we need to find an isomorphism from HK to $H \times K$. We'll first establish some properties of HK to show such a function exists.

Since H and K are normal, then *a fortiori*, we met the condition for HK to be a subgroup by proposition 3.4.4, and so by the lemma done earlier, every element of H commutes with elements of K.

Now, by lemma 5.2.5, the elements of HK can be uniqueley written as hk, and so we can define the function

$$\varphi : HK \to H \times K \qquad hk \mapsto (h, k)$$

This function is a homomorphism, since $h_1k_1h_2k_2 = h_1h_2k_1k_2$. Since the right hand side is of the form $h'k'$ for some $h' \in H$, $k' \in K$, this is the unique way of representing it, and so:

$$\begin{aligned}
\varphi(h_1k_1h_2k_2) &= \varphi(h_1h_2k_1k_2) \\
&= (h_1h_2, k_1k_2) \\
&= (h_1, k_1)(h_2k_2) \\
&= \varphi(h_1k_1)\varphi(h_1k_2)
\end{aligned}$$

From here, φ is a bijection, since every element of $H \times K$ must uniquely be mapped to by an element hk in HK, and so φ is an isomrophism.

[a]Ex. Take $(\mathbb{Z}/6\mathbb{Z})(\mathbb{Z}/6\mathbb{Z}) \to \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ and map the element order 2 and 3 into different places in the image

Another way of thinking about this theorems is by saying that the *interal direct product* (definition 3.4.1) and *external direct product* (definition 5.1.1) of two subgroups H and K of G are equivalent if the two conditions in theorem 5.2.6 match. If $H \cap K \neq \{1\}$ or one of them is not normal, then $HK \ncong H \times K$. In the next section, we'll explore a generalization of the direct product, the *semi-direct product*, and show that HK can *always* be written as $HK \cong H \rtimes K$, where $\rtimes$ represents the semi-direct product

Note that if we have an abelian group G, finitely generated or not, (theorem 5.1.8 can't be used), and we have $H, K \leqslant G$, (really $\trianglelefteq G$ since G is abelian), then it's always the case that $HK \cong H \times K$. This is another example of the niceties of being commutative.

5.2.1 A word on Perfect Groups

We saw that if a group G is abelian, then $G' = \{1\}$. What if we had the complete opposite: if $G = G'$? Such a group has a name:

Definition 5.2.8: Perfect Group

Let G be a group and G' be the commutator subgroup. If $G = G'$, then G is called a *prefect group*

Example 5.2.9: Perfect Groups

1. A_5 is an example of a perfect group.

I'm not sure yet how yet they are important. They seemed to be linked to solvable groups.

5.3 Semi-Direct Products

As we mentioned earlier, not all groups can be broken down into [non-trivial] direct products:

Example 5.3.1: $G \ncong H \times K$

Let $G = D_6 = \langle \sigma, r | \sigma^2 = r^3 = 1,\ \sigma r = r^{-1}\sigma \rangle$. By Lagrange, we know non-trivial proper subgroups of D_6 would have order 2 or 3. We can computaitonly determine that $\langle \sigma \rangle$ and $\langle r \rangle$ are the only subgroups of order 2 and 3 respectively. Let $\langle \sigma \rangle$ represent the subgroup of order 2, and $\langle r \rangle$ represent the subgroup of order 3. Since $|D_6| = 6$, then $\langle r \rangle \trianglelefteq D_6$ (corollary 3.3.5). Since one of the subgroups are normal, by proposition 3.4.4 $\langle r \rangle \langle \sigma \rangle$ is a group, and by proposition 3.4.3, the cardinality of the group is 6. Since $|D_6| = 6$, it must be that $\langle r \rangle \langle \sigma \rangle \cong D_6$. However, we saw under the definition of 3.1.12 that $\langle 2 \rangle \ntrianglelefteq D_6$, and so by the recognition theorem for direct products:

$$D_6 \ncong \langle \sigma \rangle \times \langle r \rangle$$

If there were an isomorphism between these two, then no matter what isomorphism we chose, we will encounter a problem from the fact that $rs = sr^{-1}$. let's say φ was some isomorphism from $\langle s\rangle\langle r\rangle$ to D_6. Then $\varphi(ab) = (s, r)$ for appropriate a, b where $a \in \{e, \sigma\}$ and $b = \left\{e, r, r^2\right\}$ Then:

$$(e, r^2) = (s, r) \cdot (s, r) = \varphi(abab) = \varphi(aab^{-1}b) = \varphi(e) = (e, e)$$

but $(e, r^2) \neq (e, e)$, and therefore this isomrophism *must* fail, no matter how it's defined.

Since, $\langle\sigma\rangle \cong \mathbb{Z}/2\mathbb{Z}$ and $\langle r\rangle \cong \mathbb{Z}/3\mathbb{Z}$, we can swtich over to that notation. Then we can say that

$$D_6 \ncong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$$

The question might arise on whether there is a suitable generalization of the direct product to classify the above group. More generally, is there a way to capture the structure of HK, where H and K are *arbitrary* groups (i.e. not subgroups of a larger group G). That question is explored in this section. Note that if G was abelian then automatically, $K \trianglelefteq G$, since every subgroup of an abelian group is normal. Thus, this distinction becomes interesting only when G is non-abelian.

The way we'll do this is by trying to fix the fact that elements of the group H and K will no longer always commute with each other. Note that we always have that $HK = KH$ (proposition 3.4.4), but that didn't imply $hk = kh$. In the dihedral example, we had $rs = sr^{-1}$. If we were to try to write this as a product, we'd get

$$(r, 1)(1, s) = (1, s)(r^{-1}, 1)$$

Another example would be S_3. The group S_3 is another group of order 6 with subgroups of order 2 and 3, for example: $(1\ 2)$ and $(1\ 2\ 3)$). Since $|S_3| = 6$, we again have $|(1\ 2\ 3)| \trianglelefteq S_3$ and $\langle(1\ 2\ 3)\rangle\langle(1\ 2)\rangle \cong S_3$. However, one can easily check that $(1\ 2) \ntrianglelefteq S_3$ (try taking $g = (1\ 2\ 3)$ and see if $g\langle(1\ 2)\rangle g^{-1} = \langle(1\ 2)\rangle$). We again get that $ab \neq ba$, and instead, they commute in a different manner. let $a = (1\ 2)$ and $b = (1\ 2\ 3)$. Then you can verify that:

$$ab = b^2 a$$

meaning when a and b comute, b becomes b^2. Thus, we can write

$$S_3 = \langle a, b | a^2 = b^3 = 1,\ ab = b^2 a\rangle$$

If were to try to write this as a product, we'd get that

$$(a, 1)(1, b) = (1, b^2)(a, 1)$$

So the question is "Is there a more general way to capture how elements in a group commute"?

As a way to motivate the coming definition, Let's start by having a larger group G, and recall that to define HK as a group, we need $H \trianglelefteq G$ and $K \leqslant G$.

Since HK is a group, we already have that $(h_1k_1)(h_2k_2) = h_3k_3$ for some $h_3k_3 \in HK$ (similarly to how in a direct product, $(a_1, b_1)(a_2b_2) = (a_1a_2, b_1b_2) = (a_3, b_3)$). More precicely, we take $h_1k_1, h_2k_2 \in HK$, we can re-write $h_1k_1h_2k_2$ as $h_3k_3 \in HK$. We showed this in proposition 3.4.4. Doing it again here, notice exactly how we get to write $h_1k_1h_2k_2$ as h_3k_3:

$$\begin{aligned}
(h_1k_1)(h_2k_2) &= h_1k_1h_2(k_1^{-1}k_1)k_2 \\
&= h_1(k_1h_2k_1^{-1})k_1k_2 \\
&= (h_1(k_1h_2k_1^{-1}))k_1k_2 \\
&= h_3k_3
\end{aligned}$$

where $h_3 = h_1(k_1h_2k_1^{-1})$ (since $H \trianglelefteq G$, $gh_1g^{-1} \in H$) and $k_3 = k_1k_3$.

To get to this construction, we heavily relied on *starting* with the group G and all the properties of the subgroups H, and K (ex. $H \cap K = \{e\}$, $H \trianglelefteq G$). Relying so heavily on a bigger group G does not help in our generalization of the direct product: when defining an arbitrary direct product $G_1 \times G_2$, we did not rely on some bigger group G from which G_1 and G_2 were normal subgroups. Thus, we have to figure out a way to do this construction by only relying on information on H and K.

To do this, we must be able to describe the operation between $(h_1k_1) \cdot (h_2k_2)$ without using the normality of $H \trianglelefteq G$. It is easy for the k elements since $k_3 = k_1k_2$, so the question about about defining $k_1h_2k_1^{-1}$.

We have in fact already encountered a solution to this problem. Since H is normal in G, the group K *acts* on H by conjugation:

$$k \cdot h = khk^{-1} \qquad \text{for } h \in H, k \in K$$

Since H is normal, $khk^{-1} = h' \in H$, meaning if we know how every k acts on every h, we can drop the necessity for a larger group G. In fact, all this information *can* be captured with a group-action:

$$\varphi : K \to \mathrm{Aut}(H)$$

meaning we now only rely on the product of K, H, and the homomorphism φ which captures how H and K interact. Importantly for defining a group with this, $\mathrm{Aut}(H) = \mathrm{Aut}_{\mathbf{Grp}}(H)$, not $\mathrm{Aut}_{\mathbf{Set}}(H)$, and so

$$\varphi(x)(ab) = \varphi(x)(a)\varphi(x)(b)$$

Thus, we've eliminated the need for a larger group G!

With this idea, we can define a our new product:

Definition 5.3.2: Semi-Direct Product

Let H and K be groups and let φ be a homomorphism form K into $\mathrm{Aut}(H)$. Let $\cdot$ denote the (left) action of K on H determined by φ. Let G be the set of ordered pairs (h, k) with $h \in H$ and $k \in K$ and define the following multiplication on G:

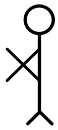
$$(h_1, k_1)(h_2, k_2) = (h_1k_1 \cdot h_2, k_1k_2)$$

we usually denote $G = H \rtimes_\varphi K$, or $H \rtimes K$ if the context permits it

Notice how we are not using a bigger group, but a lot of intuition can be derived from thinking about H and K as subgroups of a larger group, showing how subgroups still play an important role in the analysis and conception of semi-direct products, but it is certainly not necessary as we will point out in the examples.

The $\rtimes$ notation is to remember that H is the normal subgruop of $H \rtimes K$ (as we'll prove soon). It might be evident that in most cases $H \rtimes K \ncong K \rtimes H$, since we'd be defining different actions. More on this when we show some examples.

Another way of remembering which group is acting on which is with a pop-culture reference (curtosy of one of my professors): if you are aware of what a "Kamehameha" is from dragon ball, then the symbol "$\rtimes$" can be thought of as Goku's hand preforming the Kamehameha.



If you're unaware of the cultural reference, here's a youtube video I found (or you can google it):

https://www.youtube.com/watch?v=_ct3ccrU4AI

As silly as it may be, this turned out to be the most efficient way for me to remember which groups acts on which.

Before we get too hasty, we should prove that this is indeed a group and that all the properties we were talking about earlier hold:

Proposition 5.3.3: Properties Of Semi-Direct Product

1. This multiplication makes G into a group of order $|G| = |H||K|$
2. The sets $\{(h,1)|h \in H\}$ and $\{(1,k)|k \in K\}$ are subgroups of G and the maps $h \mapsto (h,1)$ and $k \mapsto (1,k)$ for $k \in K$ are isomorphisms of these subgroups with the groups H and K respectively:
$$H \cong \{(h,1)|h \in H\} \qquad K \cong \{(1,k)|k \in K\}$$
3. $H \trianglelefteq G$
4. $H \cap K = \{e\}$
5. for all $h \in H$ and $k \in K$, $khk^{-1} = k \cdot h = \varphi(k)(h)$

***Proof*:**

1. To show $H \rtimes K$ is a group, we need to show it obeys the axioms of a group
   (a) **associativity**. Let $(a,x),(b,y),(c,z) \in H \rtimes Z$. Then
$$\begin{aligned}((a,x)(b,y))(c,z) &= (a\ x \cdot b, xy)(c,z)\\ &= (a\ x \cdot b\ (xy) \cdot c, xyz)\\ &= (a\ x \cdot b\ x \cdot (y \cdot c), xyz)\\ &= (a\ \varphi(x)(b)\ \varphi(x)(y \cdot c), xyz)\\ &= (a\ \varphi(x)(b\ y \cdot c), xyz)\\ &= (a\ x \cdot (b\ y \cdot c), xyz)\\ &= (a,x)(by \cdot c, yz)\\ &= (a,x)((b,y),(c,z))\end{aligned}$$
   where I switched over to homomorphism notation to show that since $\varphi(x)$ is a homomorphism (note that $\varphi(x) : K \to K$, not $\varphi : K \to K$), then $\varphi(x)(st) = \varphi(x)(s)\varphi(x)(t)$.

(b) **Identity**: I claim that $(1, 1)$ is an identity. In particular, for any $(x, y) \in H \rtimes K$

$$(1, 1)(x, y) = (1\ 1 \cdot x, 1y) = (x, y) = (x, y)(1, 1)$$

(c) **Inverse**: I claim that for any $(h, k) \in H \rtimes K$, $(h, k)^{-1} = (k^{-1} \cdot h^{-1}, k^{-1})$ is an inverse. Indeed:

$$\begin{aligned}(h, k)(k^{-1} \cdot h^{-1}, k^{-1}) &= (hk \cdot (k^{-1} \cdot h^{-1}), kk^{-1}) \\ &= (h(kk^{-1}) \cdot h^{-1}, 1) \\ &= (h\ 1 \cdot h^{-1}, 1) = (hh^{-1}, 1) \\ &= (1, 1)\end{aligned}$$

and similarly for $(h, k)^{-1}(h, k)$.

Thus, $H \rtimes K$ is a group.

2. Let $\hat{H} = \{(h, 1)|h \in H\}$ and $\hat{K} = \{(1, k)|k \in K\}$. Notice that for any $a, b \in H$,

$$(a, 1)(b, 1) = (ab \cdot 1, 1) = (ab, 1)$$

and for any $a, b \in K$

$$(1, a)(1, b) = (1a \cdot 1, ab) = (1, ab)$$

where $a \cdot 1 = 1$ was shown in ref:HERE (in the Aut**Grp**(G) section should be said we can now do)

$$\varphi(x)(a + b) = \varphi(x)(a) + \varphi(x)(b)$$

leading us to this fact.

Thus, we can see that the inculsion map is a homomorphism, and that the map from H to $\hat{H}$ and K to $\hat{K}$ is isomorophic. By this identification,

3. Since $K \cong \hat{K}$ and $H \cong \hat{H}$, it's clear then that $H \cap K = \{1\}$.

4. If we identify elements k with $(1, k)$ and h with $(h, 1)$ Then notice that

$$\begin{aligned}khk^{-1} &= (1, k)(h, 1)(1, k)^{-1} \\ &= ((1, k)(h, 1))(1, k^{-1}) \\ &= (k \cdot h, k)(1, k^{-1}) \\ &= (k \cdot h\ k \cdot 1, kk^{-1}) \\ &= (k \cdot h, 1) \\ &= k \cdot h\end{aligned}$$

i.e., we have that $khk^{-1} = k \cdot h$. In other words $kh = \varphi(k)(h)k$, so commuting kh made k act on h.

5. Apparnelty, $\hat{K} \leqslant N_G(\hat{H})$. Since $G = HK$ (I guess from earliers intuition?) then "certainly" $H \leqslant N_G(H)$, so we have $N_G(H) = G<$ i.e, $H \trianglelefteq G$, completing (3)

Point (4) in particular is worth pointing out since this is were we generalized our intuition. In the coming example, play close attention to exactly how the action commute elements.

Remember that the semi-direct products are a generalization of direct products. Exactly how is what this next theorem identifies:

Proposition 5.3.4: When The Semi-Direct Product is a Direct Product

Let H and K be groups and let $\varphi : K - \mathrm{Aut}(H)$ be a homomorphism. Then the following are equivalent:

1. The identity (set) map between $H \rtimes K$ and $H \times K$ is a group homomorphism hence an isomrophism
2. φ is the trivial homomorphism form K into $\mathrm{Aut}(H)$
3. $K \trianglelefteq H \rtimes K$

***Proof*:**
p.178

Example 5.3.5: Semi-Direct Products

1. Let H be any abelian group (even of infinite order) and let $K = \langle x \rangle \cong Z_2$. be the group of order 2. Define $\varphi : K \to \mathrm{Aut}(H)$ by mapping x to the automorphism of inversion on H, so that the associated action is $x \cdot h = h^{-1}$, or, written in homomorphism form:

$$\varphi(x)(h) = h^{-1} \qquad \varphi(e)(h) = h$$

Note that this is an automorphism, for

$$f_x : H \to H, \qquad f_x(h) = h^{-1} \qquad f_x(xy) = (xy)^{-1} = x^{-1}y^{-1} = f_x(x)f_x(y)$$

since H is abelian, the map is it's own inverse (making it bijective).

Thus, we can define $H \rtimes_\varphi K$. To distinguish between the group-action and the group operation, I'll ues the + symbol to represent the operation for both groups. To get comfortable with this, let's do a computation:

$$(a_1, x)(a_2, x) = (a_1 + x \cdot a_2, x + x) = (a_1 a_2^{-1}, 0)$$

$$\begin{aligned}
xhx^{-1} &= (0, x)(h, 0)(0, x^{-1}) \\
&= (0 + x \cdot h, 0 + x)(0, x^{-1}) = (-h, x)(0, x) \\
&= (-h + x \cdot 0, x + x) \\
&= (-h - h, 0) \\
&= (-h, 0) \\
&= h^{-1}
\end{aligned}$$

Thus, we have:

$$xhx^{-1} = h^{-1} \Rightarrow xh = h^{-1}x$$

in the special case where $H = Z_n$, then $\mathbb{Z}/n\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z} \cong D_n$!!! if $H = \mathbb{Z}$, then we'd say $G = D_\infty$.

2. Let $H = \mathbb{R}^2$ and $K = SO(2)$.

3. Since simple groups have no normal subgroups, simple groups cannot be decomposed into semi-direct products, showing further how they are like "prime numbers" for groups.

More examples in the book p.178

With the idea of the semi-direct product in mind and a couple of examples, we end this section by given sufficient and necessary conditions to recognize semi-direct products:

Theorem 5.3.6: Recognition Theorem For Semi-Direct Products

Suppose G is a group with subgrosup H and K such that

1. $H \trianglelefteq G$
2. $H \cap K = 1$

Let $\varphi : K \to \mathrm{Aut}(H)$ be the homomorphism define dby mapping $k \in K$ to the automrophis mof left conjugation by k on H. Then

$$HK \cong H \rtimes K$$

In particualr, if $G = HK$, with H and K satisfying the aforementioend conditions, then

$$G \cong H \rtimes K$$

Put another way, a group is a semi-derict product if there exists a complement for some proper normal subgroup of G.

***Proof*:**
p. 180

(A word on how not all groups are semi-direct products. Show example. So Semi-direct product is actually a special kind of extension (will show in exercises). This type of extensions (called a split extension) will come back again later with modules (part III, chapter 17⋆).

There is a product that does represent any extension by the two groups on the left and right (sort of); it is called teh wreath product, and it comes along with the universal embedding theorem.

https://en.wikipedia.org/wiki/Wreath_product

Definition 5.3.7: Split Sequecne

Given a short exact sequence

$$1 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 1$$

we say that it *splits* if there eexists a homomorphism $\psi : H \to G$ such that $\psi \circ f = \mathrm{id}_G$ (so in a sense, also embedd C in B). We can write this as the commutative diagram:

$$\begin{tikzcd}
1 \arrow[r] & A \arrow[r] & B \arrow[r, "f"] & C \arrow[r] & 1 \\
& & & C \arrow[u, "\mathrm{id}"] \arrow[ul, "\psi"] &
\end{tikzcd}$$

Point out how with this we can make $B \cong C \rtimes_\alpha \ker(\varphi) = C \rtimes A$, where the action $\alpha : C \to \mathrm{Aut}(A)$, $\varphi(c)(a) = c \cdot a = \psi(c)^{-1}a\psi(c)[\in A]$.

If the action is trivial, then the extension can be represented as a direct product

$$1 \longrightarrow A \longrightarrow A \times C \longrightarrow C \longrightarrow 1$$

So is every short exact sequence split? The answer is unfortunately no

Example 5.3.8: Counter-Example

Take

$$1 \longrightarrow C_2 \longrightarrow C_4 \longrightarrow C_2 \longrightarrow 1$$

With the first component mapping $0 \mapsto 0$ and $1 \mapsto 2$. while in the 2nd map, $0, 2$ map to 0 and $1, 3$ map to 1.

5.4 A Categorical approach of the Product

The idea of being able to take the "product" of two objects is not unique to sets and groups. As we continue to explore more and more algebraic structures, we will keep asking ourselves a similar question: Does algebraic structure X have the flexibility/restriction to define products? [4] What we'll be doing then is finding a property which we can use to define what a product is in different algebraic settings.

Let's pick two arbitrary groups A and B. We defined product in the classical definition as the new group $(A \times B, (\cdot_A, \cdot_B))$. Let's denote this new group C. The key idea is that C is a group that contains groups A and B, and you can "recover" A and B from C. The way we can recover these two groups from C is simply by projcting to them:

$$\pi_A : C \to A,\ (a, b) \mapsto a \qquad \pi_B : C \to B,\ (a, b) \mapsto b$$

[4]As usual with category theory, the question of whether the product exists goes beyond algebra: Topological spaces, uniform spaces, graphs, ordered sets, and other domains of mathematics will also have products!

It should be checked that π_A and π_B are both surjective homomorphisms. Notice that $\pi_A(C) \cong A$ and $\pi_A(C) \cong B$. In this way, we have "recovered" A and B from C. Furthermore, C contains no more information than information about A and B. So essentially, if there exists a group C that most efficienly stores A and B, in particular if there are two functions (in our case π_A, π_B) which let's us recover A and B, then we can say C is the product of A and B. let's write this group and these two functions (C, π_A, π_B)

But is C the most efficient group for storing information about the group A and B? We can answ3re this by relating C to any other group, as is common in category theory. Let's say we have a group $G \in \text{Obj}(Grp)$ were we have a homomorphism $G \xrightarrow{f_A} A$ and $G \xrightarrow{f_B} B$ (so some structe of G is in A and B). Then does there exist a homomorphism $f : G \to C$ which can compose with π_A and π_B and produce f_1 and f_2? In other words: (Rushed)

$$\begin{tikzcd}
& G \arrow[d, "f"] \arrow[ddl, bend right, "f_A"'] \arrow[ddr, bend left, "f_B"] & \\
& C \arrow[dl, "\pi_A"] \arrow[dr, "\pi_B"'] & \\
A & & B
\end{tikzcd}$$

In a sense, it's saying that whatever structure G has of A and B, C must have it too, since we can pass through C wihtout loosing that structure.

We can now foramlize this idea with categories: Let $C_{A,B}$ be a category. The objects are

$$\begin{tikzcd}
& X \arrow[dl, "f_A"'] \arrow[dr, "f_B"] & \\
A & & B
\end{tikzcd}$$

and it can be represented as (X, f_A, f_B). If (X, f_A, f_B) and (Y, f_A, f_B) are objects, then a morphism $f : (X, f_A, f_B) \to (Y, f_A, f_B)$ exists if the following diagram commutes:

$$\begin{tikzcd}
& X \arrow[d, "f"] \arrow[ddl, bend right, "f_A"'] \arrow[ddr, bend left, "f_B"] & \\
& Y \arrow[dl, "\pi_A"] \arrow[dr, "\pi_B"'] & \\
A & & B
\end{tikzcd}$$

Example 5.4.1: Morphissm In $C_{A,B}$

here. Examples and counter-examples

Proposition 5.4.2: $C_{A,B}$ Froms A Category

Let $C_{A,B}$ be defined as above. Then $C_{A,B}$ forms a category

***Proof*:**
Left as an exercise that everything works out.

We can ask whether $C_{A,B}$ has *final* objects, That is, an object $(G, f_A, f_B) \in C_{A,B}$ where for any $(H, g_A, g_B) \in C_{A,b}$, $\text{Hom}((H, g_A, g_B), (G, f_A, F_B))$ is a singleton. In fact, (C, π_A, π_B) is such an object. In other words, we'll show that (C, π_A, π_B) must be the only group with this property since the axioms of groups and homomorphisms forces it to be.

Theorem 5.4.3: Universal Property Of The Product For Groups

Let A and B be groups in G. Then $A \times B$ is a group, $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ are homomorphisms. For any group H, if $f_A : H \to A$ and $f_B : H \to B$, then there exists a unique f such that $f_A = \pi_A \circ f$ and $f_B = \pi_B \circ f$. The morphism f is denoted $f_A \times f_B$.

***Proof*:**
We showed that $(A \times B, \cdot_A, \cdot_B)$ in proposition ref:HERE. Next, let $(A \times B, \pi_A, \pi_B)$ be as above and let $(H, f_A, f_B) \in C_{A,B}$. Define

$$f : H \to A \times B \qquad f(h) = (f_A(h), f_B(h))$$

Then we can see that $f_A(h) = (\pi_A \circ f)(h) = \pi_A(f_A(h), f_B(h)) = f_A(h)$, and similarly for f_B, showing that the diagram indeed comutes, and so f is indeed a morphism in $C_{A,B}$.

To show that f is unique, let's assume there is another morphism f'. Then

$$f_A(h) = (\pi_A \circ f')(h) = \pi_A(f'(h)) = \pi_A(f'(h)_A, f'(h)_B) = f'(h)_A$$

where $f'(h)_A$ and $f'(h)_B$ represent the components of $f'(h)$. Similarly, $f'(h)_B = f_B(h)$. Therefore, $f'(h) = (f_A(h), f_B(h))$. However, that is exactly what f does:

$$f(h) = (f_A(h), f_B(h)) = f'(h)$$

thus $f = f'$, showing uniqueness. Thus (C, π_A, π_B) is the final object of $C_{A,B}$. In other words, $A \times B$ respects the universal property of the product in Group.

We can further generalize this for an arbitrary direct product, that is if $(G_i)_{i \in I}$ is a family of groups, then their direct product $G = \prod_{i \in I} G_i$ along with $\pi_i : G \to G_i$ for every $i \in I$ will be a final object in $C_{(G_i)_{i \in I}}$.

Theorem 5.4.4: Universal Property For Arbitrary Direct Product Of Groups

For every group H, and every I-indexed family of morphisms $f_i : H \to G_i$, there exists a unique morphism $f : Y \to X$ such that the following diagrams commute for every $i \in I$:

$$\begin{tikzcd}
H \arrow[d, "f"] \arrow[dr, "f_i"] & \\
G \arrow[r, "\pi_i"] & G_i
\end{tikzcd}$$

***Proof*:**
Similar to 5.4.4. All the steps generalize without a problem.

Exercise 5.4.1

1. Given that you've seen the universal property of products for groups, what do you think it would be for sets? What would be the generalization for an aribtary category? (For a hint, see the preliminary chapter or section 5.4)
2. let $1 \to A \to B \to C \to 1$ be an exact sequence. Show there exists an action $\rho : C \to \mathrm{Out}(A)$

5.4.1 The dual of the Product

What happens if you flip the arrows? Instead of looking at $C \xrightarrow{\pi_A} A$ and $C \xrightarrow{\pi_B} B$, we look at $A \xrightarrow{i'_A} C$ and $B \xrightarrow{i'_A} C$, and for all groups G, if we have $A \to G$ and $B \to G$, then there exists a homomorphism $f : C \to G$ such that the following diagram commutes

$$\begin{tikzcd}
A \arrow[dr, "i'_A"] \arrow[ddr, bend right, "f'_A"'] & & B \arrow[dl, "i'_B"'] \arrow[ddl, bend left, "f'_B"] \\
& C \arrow[d, "F"] & \\
& G &
\end{tikzcd}$$

Does $(A \times B, i_A, i_B)$ where $i_A(a) = (a, 1)$ and $i_B(b) = (1, b)$ work? Let's break down the restrictions of this commutative diagram to check. By the above commutative diagram, we need that $f'_A(a) = F(i_A(a)) = F((a, 1))$ and that $f'_B(b) = F(i_B(b)) = F((1, b))$. So we have

$$F((a, 1))F((1, b)) = f'_A(a)f'_B(b)$$

Since F must be a homomorphism, then:

$$F((a, 1))F((1, b)) = F((a, 1)(1, b)) = F((a, b)) = F((1, b)(a, 1)) = F((1, b))F((a, 1))$$

that is $f'_A(a)f'_B(b) = f'_B(b)f'_A(a)$, that is, the image of A and B must always commute. However, we can see that there will exist a group G and homomorphisms f'_A f'_B where $f'_A(a)f'_B(b) \neq f'_B(b)f'_A(a)$:

Example 5.4.5: Counter-example
Let $A = \mathbb{Z}_2$ and $B = D_6$ so that we get

$$\begin{tikzcd}
\mathbb{Z}_2 \arrow[dr, "i_A"] & & D_6 \arrow[dl, "i_B"] \\
& \mathbb{Z}_2 \times D_6 &
\end{tikzcd}$$

Let $G = D_6$, $f'_A : \mathbb{Z}_2 \to D_6$, $f'_A(x) = \sigma^x$, $f'_B : D_6 \to D_6$, $f'_B(x) = x$ (i.e. f'_B is the identity map).

Then is it true that there exists a homomorphism F such that:

$$\begin{tikzcd} \mathbb{Z}_2 \arrow[dr, "i_A"] \arrow[ddr, bend right, "x\mapsto\sigma^x"'] & & D_6 \arrow[dl, "i_B"'] \arrow[ddl, bend left, "\mathrm{id}"] \\ & \mathbb{Z}_2 \times D_6 \arrow[d, "F"] & \\ & D_6 & \end{tikzcd}$$

By definition $F((a,1)) = F(i_A(a)) = f'_A(a) = \sigma^a$ and $F((1,b)) = F(i_B(b)) = f'_B(b) = b$, so

$$F((a,b)) = \sigma^a b$$

and by what we just saw, we must also have that $F((a,b)) = b\sigma^a$ for every $a \in A$ and $b \in B$. However

$$F((1,2)) = \sigma r^2 \neq r^2\sigma$$

showing that F is *not* a well-defined homomorphism!

It should be noted that if we *only* stuck to abelian group, then this wouldn't be a problem, since we always have that $ab = ba$, so $F((a,b))$ is well-defined. In fact, it is worth checking that if A and B are abelian groups, $((A \times B), i_A, i_B)$ *does* respect the commutative diagram, and is the initial object in the category who's objects are triples (G, f_A, f_B) (where G *has* to be abelian) and morphsism are the aformentioned commutaive diagram.

The problem here came that $A \times B$ actually added structure on it. As we saw in proposition 5.1.3[5], for any $x, y \in A \times B$, $xy = yx$. (Talk about how then the product is a way to store information by somehow adding conditions to let the storage happen??)

So does there exist a group and maps will be the initial object in this category? The answer to this question is actually yes! It is a group we have not yet encountered. The main idea here is that this group is supposted to be the most generic way of having A and B (so as little relations btween A and B) so that the homomorphism from this group to any G does not preserve these properties. We have encountered a similar situation like this before when defining the *free group* (see section 2.3.3). Something analogous is defined called the *free product*.

Definition 5.4.6: Free Product

Let H and K be groups, S_H be a generating set for H and S_K be a generating set for K, R_H and R_K be the relations given the generating sets respectively (at this point, I already showed presentations, so I should be able to introduce R_H and R_K rigorously). Then

$$H * K = \langle S_H \cup S_k | R_H \cup R_K \rangle$$

Example 5.4.7: Free Products

Take $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$. Mention that this is the group mentioned in section re:HERE (chap 2 near beginning) that as a group for which the torsion subset as not a group

Exercise 5.4.2

1. Show that $(H * K, i_H, i_K)$ respects the universal property of the co-product in **Grp**.

Infinite abelian case

It should be mentioend again that if we restrict to **Ab**, then the universal property of the coproduct is fulfilled by $H \times K$ (notice that $H * K = H \times K$). Is this also true in the infinite case: given some groups $G_1, G_2, \cdots$, does $G_1, G_2 \times ...$ respect the universal property of the coproduct? The answer is no!

Example 5.4.8: Example

Let's say $G = \prod_{i=1}^{\infty} \mathbb{Z}$. Let's say that every component group is represented by G_i. For every G_i, define the inclusion map $G_i \xrightarrow{\iota_i} G$. Let's say $G_i \xrightarrow{f_i} \mathbb{Z}$, $f_i(x) = x$ Then if G respects the universal property of the coproduct, then there exists a function F such that

$$f_i(g_i) = F(0, 0, ..., g_i, ..., 0, ...)$$

This is true for every component of F. Now here is the question: for some elements $1 \in G_1$, $1 \in G_2 ... 1 \in G_i, ...$, what is:

$$F(1, 1, ..., 1, ...)$$

By linearity, we can see that

$$F(1, 1, ..., 1, ...) = 1 + F(0, 1, 1, ..., 1, ...)$$

and we can keep going this way. However, the problem is that we will continue indefenetily. More precicely, there is no number $y \in \mathbb{Z}$ such that $y = F(1, 1, 1, ..., 1, ...)$. Thus, F is *not* a well-defined function!

The problem came from the fact that with the flexibility of having an infinite sum, we run into the possibility of never terminating our sum, that is, we can create a scenario where we have series which do not converge (TBD?).

A group still exists which can fix our scenario and respct the universal property for coproducts in the infinite case:

Definition 5.4.9: Direct Sum Of Groups

Let $\{A_i\}_{i \in I}$ be a family of groups. Then the *direct product*

$$A = \bigoplus_{i \in I} A_i$$

is a subset of $\prod_{i \in I} A_i$ consisting of all elements $(x_i)_{i \in I}$ with $x_i \in A_i$ such that $x_i = 0$ for all but finitely many indices i (cofinitely many indices).

The proof that this is a group was an exercise ref:HERE (in section 1.1). Note that in the finite case, $\bigoplus_{i=1}^{n} G_i = \prod_{i=1}^{n} G_i$. It only differs once we reach the infinite case

Theorem 5.4.10: Universal Property Of Coproducts In Ab

Let I be an index for the family of groups $G_{i \in I}$. Then $G = \bigoplus_{i \in I} G_i$ respects the universal property of the corpoduct in **Ab**.

***Proof*:**
This proof is very similar to the infinte case one and the one for the usual universal property of the direct product.

5.4.2 Universal Property of Semi-Direct Products?

I want to bring in the intuition of a product that when elements commute it introduces a twist. Show this through Cayley graphs

here

Then generalize the how the universal property of products with coslice categories to fibered categories.

However, it also seems like there isn't a universal property for semi-direct products, since it's not unique (saw that wiki entry where many semi-direct products shared the same structure, so I'm not sure where this yet goes in the "global landscape of mathematics"

https://en.wikipedia.org/wiki/Semidirect_product#Generalizations

Also, to show lack of uniqueness:

$$(C_3 \rtimes D_8) \cong (Q_{12} \rtimes C_2) \cong (D_{12} \rtimes C_2) \cong (V \rtimes D_6)$$

There is also no necessary and sufficient conditions that are known for semi-direct products, although they can be characterzied by being a right-split exact sequence.

Exercise 5.4.3

1. Show that $\mathrm{GL}(F) \cong \mathrm{SL}(F) \rtimes F^{\times}$.

6

Further Topic in Group Theory

Miscelaneous things I don't know how to classify yet

More on J.H. program? More on solvable groups? introduce metabelian groups? nilpotent goups? (Have to mention Krasner–Kaloujnine universal embedding theorem and make exercises defining the wreath product)

6.1 Generalization of Abelian Groups

So far we saw that

$$\text{cyclic} \Rightarrow \text{Abelian}$$

We found a way to find how close agroup is to being abelian using commutator groups. If $G' = \{1\}$, then G is abelian. Otherwise, G is not abelian. We'll develop further generalization's from this concept.

Definition 6.1.1: Metabelian Group

Let G be a group. If $G^{(1)} \neq \{1\}$, but $G^{(2)} = \{1\}$, we sy that G is a *metablean* group

Proposition 6.1.2: Properties Of Metabelian Groups

Let G is metabelian, then

1. any subgroup is metabelian
2. If $\varphi : G \to H$ is a homomorphism, then $\varphi(G)$ is metabelian

***Proof*:**

1. Let $H \leqslant G$ be a subgroup of G, and consider H''. If, $H'' \subseteq G'' = \{1\}$, and so H is metabelian as well. This fact can easily be shown direclty. Notice that $H' \subseteq G'$, since if $x = aba^{-1}b^{-1} \in G$, then this is also a commutator for the group G, and so $x \in G'$. By the same logic, the group H'' will be a subset of G'', meaning H'' is trivial as well.
2. Let $\varphi : G \to H$ be a homomorphism and G a metabelian group, and consider $\varphi(G) \leqslant H$. We'll show that homomorphisms preserve commutator subgroups to show that $\varphi(G)$ is metabelian. In paticualr, if $N \subseteq G$, then $\varphi(N') = \varphi(N)'$.
   (a) ($\subseteq$) Let $x \in \varphi(N')$. Then $x = \varphi(xyx^{-1}y^{-1}) = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1}$. And so by definition, $x \in \varphi(N)'$.
   (b) ($\supseteq$) Let $x \in \varphi(N)'$. Then $x = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1} = \varphi(xyx^{-1}y^{-1})$. And so by definition $x \in \varphi(N')$.

   Thus
   $$\varphi(G)'' = \varphi(G')' = \varphi(G'') = \varphi(\{1\}) = \{1\}$$
   and so $\varphi(G)$ is also metabelian.

Example 6.1.3: Metabelian Groups

1. Any abelian group is always metabelian
2. if A and B are abelian groups, and $A \rtimes B$ is well-defined, then $A \rtimes B$ is metabelian.

Proposition 6.1.4: Extension of Abelian Groups

Let G be a group. If $H \trianglelefteq G$ is abelian and G/H is abelian, then G is metabelian. More generally, if G extends an abelian group N by an abelian group M, then G is metabliean, i.e, the following is a short exact sequence

$$1 \longrightarrow A \longrightarrow G \longrightarrow M \longrightarrow 1$$

Furthermore, the converse is true too, namely if $H \trianglelefteq G$ and G/H are abelian, then $G'' = \{1\}$, showing this is an equivalent definition

***Proof*:**
Let $A \trianglelefteq G$ and G/A be abelian. Then by proposition ref:HERE, $G' \subseteq A$, meaning $G' = \{1\}$. Evidentally then, $G'' = \{1\}$. Conversly, if $G'' = \{1\}$, then G' the elements of G' commute. Note that this does not imply $G' \trianglelefteq G$ immediately, since we did not esatblish those elements commute with every element of G, and so we must show that $G' \trianglelefteq G$. Let $g \in G$ and $h \in G'$. By definition, $ghg^{-1}h^{-1} \in G'$. Since $h \in G'$, $ghg^{-1}h^{-1}h = ghg^{-1} \in G'$, and thus $G' \trianglelefteq G$. Thus, by proposition ref:HERE, G/G' is also abelian completing the other direction.

Thus this is an equivalent definition of metabelian.

6.1.1 Solvable Groups

Metabelian groups are a simple example of a group called a *solvable* group. Instead of looking at the 2nd commutator subgroup being trivial, we can ask if the nth commutator subgroup will be trivial:

Definition 6.1.5: nth Commutator Group

Let G be a group and $G' = G_{(1)}$.

$$G_{(n)} = [G_{(n-1)}, G_{(n-1)}] = \left\{aba^{-1}b^{-1} \middle| a, b \in G_{(n-1)}\right\} = (G_{(n-1)})'$$

represents the nth commutator group.

Definition 6.1.6: Solvable Groups

A group G is *solvable* if there is a subnormal series with commutator subgroup:

$$G = G_0 \trianglerighteq G_{(1)} \trianglerighteq G_{(2)} \trianglerighteq \cdots \trianglerighteq G_{(n)} = \{1\}$$

such a subnormal series is called the *derived series*. I $G_{(n)} = 1$, we say G has *solvable length* n.

Example 6.1.7: Solvable Groups

1. Every abelian group is solvable, since $1 \trianglelefteq G$ is a normal series.
2. Any metabelian group is solvable, with the series is called the *derived series of* G.

$$1 = G'' \trianglelefteq G' \trianglelefteq G$$

   Since G/G' is abelian, and $G'/G'' \cong G'$ which is abelian.

There are equivalent definitions of solvability that are worth keeping in mind to have a variety of intuitions.

Lemma 6.1.8: characteristic of nth commutator

Let G be a group. Then $G_{(i+1)} \text{char} G_{(i)}$ for any $i \in \mathbb{N}$. Consequently, $G_{(i+1)} \trianglelefteq G_{(i)}$.

***Proof*:**
We have already proved this in proposition 5.2.3. Then normality comes from proposition 4.7.12.

We can also prove normality directly. We want to show that for any $g \in G_{(i)}$, $gG_{(i+1)}g^{-1} = G_{(i+1)}$. Take $h \in G_{(i+1)}$. Then by definition of $G_{(i+1)}$, $ghg^{-1}h^{-1} \in G_{(i+1)}$. Since $h \in G_{(i+1)}$,

$$ghg^{-1}h^{-1}h = ghg^{-1} \in G_{(i+1)}$$

implying $gG_{(i+1)}g^{-1} = G_{(i+1)}$, and so $G_{(i+1)} \trianglelefteq G_i$. since i was arbitrary, we see that this holds true for any nth commutator subgroup, as we sought to show.

Theorem 6.1.9: Equivalent Definitions Of Solvable group

The following are equivlent

1. G is solvable
2. There exists a subnormal series

$$G = A_0 \trianglerighteq A_1 \trianglerighteq A_2 \trianglerighteq \cdots A_n = \{1\}$$

such that A_i/A_{i+1}. Such a subnormal series is called a *subnormal series with abelian factors*.

***Proof*:**
$(1 \Rightarrow 2)$ We want to construct a derived series. However, we already have one: since $G_{(i+1)} \trianglelefteq G_{(i)}$, let $A_i = G_i$.

$(2 \Rightarrow 1)$ Since $G/A_{(1)}$ is abelian, then by proposition 5.2.3 $G_{(1)} \leqslant A_1$. Then inductively, let $G_{(i)} \leqslant A_i$. then since A_i/A_{i+1} is abelian, there exists a group $G^* \leqslant A_{i+1}$ such that A_i/G^* is abelian. Then, take $f : G_{(i)} \to A_i/G^*$, $x \mapsto [x]$. Consider $[x], [y] \in f(G_{(i)})$. Then since $f(G_{(i)})/G^*$ is a subgroup of A_i/A_{i+1}, it is also abelian, so

$$[xy] = [yx] \Rightarrow xyg = yx \qquad g \in G*$$

moving variables around we get

$$xyx^{-1}y^{-1} = [x, y] = g$$

since x^{-1}, y^{-1} was arbitrary, we see that $G_{(i+1)} \leqslant G* \leqslant A_{i+1}$, that is

$$G_{(i+1)} \trianglelefteq A_i$$

Thus, once we reach A_n, we see that

$$G_{(n)} \trianglelefteq A_n = \{1\}$$

thus $G_{(n)} = 1$, as we sought to show.

The use of the 2nd definition is to show that finding any solvable series for a group is enough to show a group is solvable. The advantage of the 1st definition is that it is the most refined solvable series of a group, since the commutator subgroup is smallest subgroup which makes the quotients A_i/A_{i+1} abelian.

Proposition 6.1.10: Properties Of Solvable Groups

Let G be solvable. Then

1. subgroups of solvable groups are solvable
2. if $\varphi : G \to H$ is a homomorphism, then $\varphi(G)$ is solvable
3. if M is an extension of a solvable group by a solvable group, then M is solvable.

***Proof*:**
For (1) and (2), the commutator subgroup definition makes these proofs much easier. To get used to both definitions, we'll prove the statement using both:

1. Let $G_{(n)} = \{1\}$, and consider $H \leqslant G$. Then as we've seen, $H_{(n)} \leqslant G_{(n)}$ meaning H is solvable.

   Using the other definition, we'll construct a solvable series for H. Let $G \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n$ be a solvalbe series for G. Take the homomorphism

   $$\varphi_i : H \cap G_i \to G_i/G_{i+1} \qquad \varphi(x) = xG_{i+1}$$

   which has kernel $(H \cap G_i) \cap G_{i+1} = H \cap G_{i+1}$. Since it's a kernel, it is a normal subgruop of $H \cap G_i$, and the quotient $H \cap G_i/H \cap G_{i+1}$ maps injectivly into G_i/G_{i+1}. Since it is commutative, then so is $H \cap G_{i+1}$. Thus

   $$H \trianglerighteq H \cap G_1 \trianglerighteq H \cap G_2 \trianglerighteq \cdots \trianglerighteq H \cap G_n$$

   is a solvable series for H.

2. We've shown that $\varphi(G'') = \varphi(G')' = \varphi(G)''$ (proposition 6.1.2). Continuing inductively, we'd get $\varphi(G_{(n)}) = \varphi(G)_{(n)}$. We can see that the derived series will be preserved.

   Using the other definition, let $\overline{G}$ be the quotient gorup of G, and let $\overline{G}_i$ be the image of G_i into $\overline{G}$. Then

   $$\overline{G} \trianglerighteq \overline{G}_1 \trianglerighteq \cdots \trianglerighteq \overline{G}_n = \{1\}$$

   is a solvable series for $\overline{G}$

3. Let $N \trianglelefteq G$ and G/N be solvable. We'll show then that G has a series. Since N and $\overline{G}$ is solvable, then they both have a solvable series:

   $$\overline{G}\overline{G}_1 \trianglerighteq \overline{G}_2 \trianglerighteq \cdots \trianglerighteq \overline{G}_n = \{1\}$$

   $$N \trianglerighteq N_1 \trianglerighteq N_2 \trianglerighteq \cdots \trianglerighteq N_m = \{1\}$$

   Thus, by the 3rd isomrophism theorem $\overline{G}_i$ in G we get $G_i/G_{i+1} \cong \overline{G}_i/\overline{G}_{i+1}$. Combining these series, we get

   $$G \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = N \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_m = \{1\}$$

   which is a solvable series for G, completing the proof.

Corollary 6.1.11: Finite P-Groups

Let G be a finite p-group. Then G is solvable.

***Proof*:**
Let's use induction on the order of G. The base case is trivial, so let's consider an order $|G| > 1$. Then by hw exexerice (or corollary 4.16 in Milne Group Theory book) $Z(G)$ has non-trivial center. Since $Z(G)$ is normal in G, $G/Z(G)$ is well-defined. Since $Z(G)$ is non-trivial $G/Z(G)$ has smaller order. Thus, by the induction hypothesis, $G/Z(G)$ is solvable. Since $Z(G)$ as a group is commutative, by proposition 6.1.10(2), G is solvable.

There is actually one more incredibly powerful theorem for which groups are solvable:

Theorem 6.1.12: Feit-Thompson's Theorem

Let G be a finite group of odd order. Then G is solvable.

***Proof*:**
Huge proof requiring 255 pages.

From this statement, we can state that either a finite group is solvable, or it has an element of order 2 (see exercice 1.2.1)!

Theorem 6.1.13: Solvability Condition

The finite group is solvable if and only if for every divisor n of $|G|$ such that $\left(n, \frac{|G|}{n}\right) = 1$, G has a subgroup of order n .

***Proof*:**
very difficult proof not in textbook.

6.1.2 Nilpotent Groups

(put somehere ho to classify finite abelian groups) We've defined solvable groups using nth commutator groups to give a notion of generalization from being commutative. What if we used the center instead? SDince $Z(G) \trianglelefteq G$, $G/Z(G)$ is well-defined. To generalize, we can define:

$$Z^2(G) = Z(G/Z(G))$$

that is, the 2nd center is the center of the quotient $G/Z(G)$. If we can continue this chain with $Z^n(G)$, and eventually get a commutative group $z^n(G) = G$, we call such a group a *nilpotent group*:

Definition 6.1.14: Nilpotent Groups

Let G be a group. If $Z^m(G) = G$. for some $m \in \mathbb{N}$, then G is said to be nilpotent. The smallest such m is called the *(nilpotency) class of G*.

Example 6.1.15: Nilpotent Groups

1. If G is abelian, then $Z(G) = G$, meaning all abelian groups are nilpotent
2. If G is meta-abelian, Then $G/Z(G)$ is abelian (since $G' \subseteq Z(G)$), meaning $Z(G/Z(G)) = G/Z(G)$, and so is nilpotent with nilpotency class 2.
3. Every p-groups are an example of nilpotent groups. Let G be a finite p-group, so that $|G| = p^a$ for some $a \in \mathbb{N}$. If $Z(G) = G$, we're done. If not, then since $Z(G)$ is non-trivial (proposition ref:HERE), $|Z(G)|||G|$, so $|Z(G)| = p^{a_2}$ for some $a_2 < a$. But then $Z(G)$ is again a p-group, so we can continue this process. Thus, we get an ever shrinking sequence of nubmers $a > a_2 > a_3 \cdots a_n$, till eventually it must be that either $a_n = 1$ or $a_n = 0$. If $a_n = p$, then the associated group is cyclic, implying it's abelian, and we're done. If $a_n = 0$, then we have the trival subgroup, which is trivially abelian.
4. This is not so much an example as future motivation: Nilpotent groups make an appearance in the classification of Lie Groups.

Notice that for any $[g] \in Z^2(G)$, $[g, x] \in Z(G)$ for any $x \in G$. To see this, pick $[x] \in G/Z(G)$ (i.e., the coset which can have x as it's reprsentative). Then since $[g] \in Z^2(G)$, $[gx] = [xg]$. Then $gxz = xgz'$ for some $z, z' \in Z(G)$. Then$g^{-1}x^{-1}gx = z'z^{-1} \in Z(G)$. Thus, $Z^2(G) \subseteq Z(G)$. This holds true for any $Z^n(G)$ (defined inductively). Thus, can also get a subnormal series, called the *upper central series*:

$$\{1\} \subseteq Z(G) \trianglelefteq Z^2(G) \cdots Z^n(G) \cdots$$

If this series terminates with $Z^n(G) = G$, then we say that G is a *nilpotent group*.

Proposition 6.1.16: Equivalent Definition

The following are equivalent:

1. G is nilpotent (i.e. there exists an *upper central series*)
2. There exists a *lower central series*
$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$
where $G_i = [G, G_{i-1}] = \left\{ghg^{-1}h^{-1} \middle| g \in G, h \in G_{i-1}\right\}$.
3. There exists a *central series*
$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$
where $G_i/G_{i+1} \leqslant Z(G/G_i)$, or equivalently, $[G, G_{i+1}] \leqslant G_i$

***Proof*:**
(1) $\Rightarrow$ (2) Let G be nilpotent. Then we saw that if $g \in Z^{i+1}(G)$, then $[g, x] \in Z^i(G)$, meaning $[G, Z^{i+1}(G)] \leqslant Z^i(G)$. Then decomposing recusively, we see that $G_{i+1} = [G, G_i] \trianglelefteq Z^i(G)$. Showing that every G_i exists, and will have to terminate at G_n. Furthermore, by the same justification as $G_{(i)}$, $G_{i+1} \trianglelefteq G_i$. Thus, they form a subnormal series

(1) $\Rightarrow$ (3) easy to do

(2) $\Rightarrow$ (1)

With the lower central series, we can gain an intuition on why the group is called nilpotent. Let G be any group and $g \in g$ be any element. Define the *adjoint action*

$$\mathrm{ad}_g : G \to G \qquad \mathrm{ad}_g(x) = [g, x]$$

Then if G has nilpotency class n, then $(\mathrm{ad}_g)^n(x) = e$, for all $x \in G$. In this sense, the adjoint action is nilpotent. Note that if this is true for each element, this does not imply that G is nilpotent

Example 6.1.17: Example
here

Such groups are called *Engel groups*, and come up in Lie Algebra.

Proposition 6.1.18: Nilpotent Then Solvable

Let G be a nilpotent group. Then G is solvable

***Proof*:**
Notice that $G_{(n)} \trianglelefteq G_n$, meaning the there must be a solvable length of G, since for some n, $G_{(n)} \subseteq G_n = \{1\}$, implying $G_{(n)} = 1$, showing that G is solvable.

Note that the converse is not true

Example 6.1.19: Solvable, but not Nilpotent
let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \middle| a, b, c \in \mathbb{R}, ab \neq 0 \right\} \leqslant GL_2(\mathbb{R})$$

Then I encourage you to check that $Z(G) = \{aI | a \neq 0\}$ where I is the identity matrix. Furthermore $Z^2(G) = Z(G/Z(G)) \cong \{1\}$, i.e., we don't resolve to the an abelian group but a trivial group, and so G is not nilpotent.

However, G is solvable. We can form a solvable series using $H = \left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} | a \in \mathbb{R} \right\}$. It should be verified that $H \trianglelefteq G$, and that $G/H \cong \mathbb{R}^\times \times \mathbb{R}^\times$ (meaning it's abelian), and that $H \cong (\mathbb{R}, +)$, meaning H is abelian, so $G'' = H' = \{1\}$, showing it is indeed solvable.

Another example would be S_3. Since $S_3 \cong \mathbb{Z}_3 \rtimes \mathbb{Z}_2$, and both $\mathbb{Z}_3$ and $\mathbb{Z}_2$ are nilpotent (since their abelian), then the semi-direct product of nilpotent groups is not necessarily nilpotent. More generally, the extension of a nilpotent group by a nilpotent group is not necessarily nilpotent.

Thus, we have the folowing chain of implications

$$\text{cyclic} \Rightarrow \text{abelian} \Rightarrow \text{nilpotent} \Rightarrow \text{solvable} \Rightarrow \text{finitely generated (show } A_5 \text{ not solvable)}$$

So nilpotent groups don't interact well with extensions. However, the fac that their slightly stronger than extensions, but more general then abelian groups, brings us some useful properties:

Proposition 6.1.20: Properties Of Nilpotent Groups

Let G be a nipotent group. Then

1. If H is a proper subgruop of G, then H is a proper normal subgruop of $N_G(H)$. We call this property the *normalizer property*, and we sometimes say that the "normalizer grows"
2. Every Sylow subgroup of G is normal
3. G is the direct product of its Sylow subgroups
4. if d divides the oder of G, then G has a noraml subgroup of order d

In fact, (4) implies that G is nilpotent, giving us another definition of nilpotency.

***Proof*:**

1. Let G be a nilpotent group and $H \leqslant G$. We want to show that $H \trianglelefteq N_G(H)$. We'll do this by induction on the order of G. This is clearly true when $|G| = 1$, so let $|G| = n$. IF G is abelian, then for any $H \leqslant$, $N_G(H) = G$, so we get our result trivially. So let's say G was not abelian.

   if $Z(G) \nleqslant H$, then
   $$here$$
   If $Z(G) \leqslant H$, then $H/Z(G)$ is contained in $G/Z(G)$. By proposition ref:HERE, $G/Z(G)$ is nilpoent. Thus, there es a subgroup of $G/Z(G)$ which normalizer $H/Z(G)$ and $H/Z(G)$ is a proper subgroup. Thus, by the correspondence theorem, $H \trianglelefteq N_G(H)$. TBD

2. Let P be a Sylow p-subgroup, and consider $N = N_G(P)$. Since $P \trianglelefteq N$, by proposition 4.8.7, $P\text{char}N$. Since
   $$P\text{char}N \trianglelefteq N_G(N)$$
   Then by transitivity of characteristic subgruops, we have that $P \trianglelefteq N_G(N)$. Then by Sylow's Theorem, $N_G(N) \trianglelefteq N$. By part (1), we get that $N = G$, showing that $P \trianglelefteq G$, that is, Sylow p-subgroups are normal subgroups of G

3. Let $p_1, p_2, ..., p_n$ represent distinct pimres dividing the order of G

https://en.wikipedia.org/wiki/Nilpotent_group#Properties

Exercise 6.1.1

1. Prove that any nilpotent group of class 3 or lower is metabelian
2. Show that if $\text{Aut}(G)$ is abelian, then G is nilpotent of class 2

6.1.3 Polycyclic Groups

We've extended a abelian group using *central extensions* and *abelian extensions* (TBD). In this section, we'll once again be extending an abelian group, but this time we'll ask what happens if we extend a cyclic group in such a way that G_i/G_{i+1} is cyclic?

Definition 6.1.21: Polycyclic Group

Let G be a group. Then if

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

st G_i/G_{i+1} is cyclic, Then G is said to be polycyclic. The series G admits is called a *subnormal series with cyclic factors*. If $G_n = \{1\}$, then we say that G is polycyclic with length n.
If $n = 2$, then we say that G is a *metacyclic group*.

Example 6.1.22: polycyclic groups
By the fundamental theroem of finitely generated group, you can prove that all finitely generated abelian group are in fact polycyclic (exercice).

word on saying *the* subnormal series with cyclic factors? (up to isomorphism?)

Proposition 6.1.23: Polycyclic Then Finitely Generated

Let G be a polycyclic group. Then G is finitely generated.

***Proof*:**
Let G be polycyclic, so that

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{1\}$$

is the subnormal series with cyclic factors. Since G_0/G_1 is cyclic, let' say $[g_0] \in G_0/G_1$ generates G_0/G_1. Similarly, let's say $[g_i] \in G_i/G_{i+1}$ such that $\langle g_i \rangle = G_i/G_{i+1}$. Eventually we'll get that $\langle g_{n-1} \rangle = G_{n-1}/G_n \cong G_{n-1}$, meaning G_{n-1} is cyclic. Then we get that

$$1 \to G_{n-1} \to G_{n-2} \to G_{n-2}/G_{n-1} \to 1$$

i.e., we extend a cyclic group by a cyclic group. Notice if we take any $x \in G_{n-2}$ then we get $[x] \in G_{n-2}/G_{n-1}$, so $[x] = [g_{n-2}^\alpha] = [g_{n-2}]^\alpha$ so

$$x = (g_{n-2}g_{n-1})^\alpha$$

and since x was arbitrary, this mean that $\langle g_{n-2}, g_{n-1} \rangle = G_{n-2}$. continuing inductively, we'll get that

$$\langle g_i | 1 \leq i \leq n \rangle = G$$

showing that G is finitely generated.

Note that the converse is false:

Example 6.1.24: Finitely Generated, But Not Polycyclic
Let $G = A_5$, which is a simple non-cyclic group. Since it simple, it's subnormal series is only

$$A_5 \trianglerighteq \{1\}$$

and so, it cannot be polycyclic.

Then it can be checked that if we restrict to finitely generated groups, we get:

cyclic $\Rightarrow$ abelian $\Rightarrow$ nilpotent $\Rightarrow$ Polycyclic $\Rightarrow$ solvable $\Rightarrow$ finitely generated (show A_5 not solvable)

6.2 Non-finitely generated abelian groups

(Prüfer's Theroems!)

6.3 Linear Groups

Perhaps add a section here on linear groups and some more complicated properties because of how important they are to Lie Algebra and Invariant Theory?

7

Generalization of Groups

In this chapter, I will cover common algebraic objects which are generalizations of groups.

7.1 Monoid

Having a unique inverse is one of the more powerful axioms you can add to an algebraic structure. Recall how the uniqueness of a linear solution is bassed off of the fact that the invere exists and is unique (proposition 1.1.17). This property is also called the *cancellation property.* There are many mathematical structures which do not have this ($\mathbb{N}$ would be a group if inverses existed). For this reason, the study of this structure is also of interest to many

Definition 7.1.1: Monoid

A *Monoid* is an ordered pair $(M, *)$ where M is a set and $*$ is a binary operation on M satisfying the following axioms

1. associativity: $(a * b) * c = a * (b * c)$
2. there exists an element e in G, called the *identity* of G such that for every $a \in G$, we have $a * e = e * a = a$ [a]

The Monoid $(M, *)$ is called *abelian* (or commutative) if $a * b = b * a$ for every $a, b \in M$

[a] it can in fact be proven that if $a * e = a$ then $e * a = a$ (or vice-versa); an indispensable exercise worth doing

Example 7.1.2
Naturally, all groups are monoids. I'll give non-group example of monoids:

1. The natural numbers $\mathbb{N}$
2. Given a set A and it's powerset $P(A)$ (set of all substs), then $(P(A, \cup))$ is a monoid with $\emptyset$ as the identity.
3. Given a set A and it's powerset $P(A)$, then $(P(A), \cap)$ is a monoid with A as the identity.
4. The set $\{0, 1\}$ with any boolean operation ($\Rightarrow$, NOR, AND, etc.) is a monoid. More commonly, the set is relabled as $\{\text{False}, \text{True}\}$
5. The set of strings, given an alphabet is a free monoid.
6. Given a set A, the set of all functions $f : A \to A$ is a monoid! This is incredibly powerful, since we can now put an algebraic structure on sets themselves!! Any set has a default algebraic structure! Further wil be true soon: When we study rings, rings will be an abelian group, along with a second operation which will be a monoid!!

   There are far reaching consequences of this when relating algebra to other fields. If you can put a ablien group structure on something, then taking the set of all functions will now produce you a ring! In algebraic topology, this is exact trick is used to be able to compare rings with topologies, which all comes down to
7. Given the collection of all normal subgroups of a group, their multiplications form a monoid structure! functions having a natural monoid structure!

Recall that $\text{Aut}(G)$ is a group, but $\text{End}(G)$ was not since not every element has an inverse (section 1.4)? This then makes $\text{End}(G)$ a monoid

Proposition 7.1.3: G Abelian Implies End(G) Monoid

here

Proof. Let G be an ablien group and consider $\text{End}(G)$.

Associativity holds by

The identity naturally exists, since $1 : G \to G$, $1(g) = g$ is definable and $\forall r \in \text{End}(G)$, $r1 = 1r = r$. $\square$

Example 7.1.4
exercice: Show that if $G = D_6$, then $\text{End}(G)$ is not a group.

There is a lot of intereting theory for monoids which I will simply sum up for now to get to the main reason I wanted to bring them up

1. Recall that I said that groups have the cancellation property because of the unique inverse of every element. It is possible that there are monoids with this property that are not groups. Such monoids are called monoids *with the cancellation property.* The natural numbers is an

example of such a monoid. Monoids with the cancellation property can be extended to form groups (called the Grothendieck construction). If the monoid is finite, then it implies that it's a group

2. Recall that the set of all homomorphisms between a group and itself forms a group. With a monoid, we can weaken the condition of the functions being homomorphisms. Given a set S and every set-function $f : S \to S$ with the operation defined point-wise ($(f+g)(x) = f(x) + g(x)$) is also a monoid.

3. The same way we defined group-actions with groups, we can define monoid-actions with the same axioms. These pop-up in computer-science in the study of automata.

4. In functional programming, monads is a central tool in programming. Monads are monoids! The objects on which they are monoids is a more complicated subject, and will be left for the appendix on category theory [1]

5. free-monoids play a pivitol role in modeling the behavior of machines (ex. Turing machines). (DEVELOP ON THIS)

Though all these reaon are good reasons to look into monoids, I bring them up for their use in Category Theory. More particularly, we can define a monoid to be a category with a single element!

The Grothendieck construction named in (1) above is worth carrying out in full, because it is the mechanism by which a number of later invariants are built. Whenever a class of objects carries a notion of direct sum, its isomorphism classes form an abelian monoid, and the invariant one actually wants is the group this construction produces: the Grothendieck group of vector bundles, of finitely generated projective modules, and of a C^*-algebra are each defined this way. The construction is stated for an arbitrary abelian monoid, since cancellation is exactly what it does *not* need.

Definition 7.1.5: Grothendieck Completion

Let $(M, +)$ be an abelian monoid. Define a relation on $M \times M$ by

$$(m, n) \sim (m', n') \quad \text{if and only if} \quad m + n' + p = m' + n + p \;\text{ for some } p \in M$$

The *Grothendieck completion* of M is the quotient $M^{-1}M = (M \times M)/\sim$ with addition induced coordinatewise, together with the monoid homomorphism

$$\iota \colon M \to M^{-1}M, \qquad m \mapsto [(m, 0)]$$

The class of (m, n) is written $[m] - [n]$, which the group structure below justifies.

The auxiliary p in the relation is what makes $\sim$ transitive without assuming cancellation, and it is exactly the obstruction measured in part (3) below.

[1] For those who are impatient or know some Category theory, a monoid on a category of functors which are derived given by a particular adjoint

Proposition 7.1.6: Properties of the Grothendieck Completion

Let M be an abelian monoid. Then:

1. $M^{-1}M$ is an abelian group, and every element of it has the form $[m] - [n]$ for some $m, n \in M$.
2. $\iota(m) = \iota(n)$ if and only if $m + p = n + p$ for some $p \in M$.
3. ι is injective if and only if M has the cancellation property.
4. For every abelian group A, composition with ι is a bijection

$$\mathrm{Hom}_{\mathbf{Grp}}(M^{-1}M, A) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Mon}}(M, A)$$

natural in M and A; that is, $M \mapsto M^{-1}M$ is left adjoint to the forgetful functor from abelian groups to abelian monoids.

***Proof*:**
Step 1: $\sim$ is an equivalence relation. Reflexivity and symmetry are immediate. For transitivity, suppose $m + n' + p = m' + n + p$ and $m' + n'' + q = m'' + n' + q$. Adding the two equations gives

$$m + n'' + (n' + m' + p + q) = m'' + n + (n' + m' + p + q)$$

so $(m, n) \sim (m'', n'')$ with the witness $r = n' + m' + p + q$. Note that no cancellation was used, which is the point of carrying p at all.

Step 2: the group structure. Coordinatewise addition respects $\sim$, since adding a fixed pair to both sides of the defining equation preserves it. Associativity and commutativity are inherited from M, the class $[(0, 0)]$ is an identity, and $[(n, m)]$ is inverse to $[(m, n)]$ because $(m + n,\ n + m) \sim (0, 0)$ with $p = 0$. So $M^{-1}M$ is an abelian group, and $[(m, n)] = \iota(m) - \iota(n)$, which is (1).

Step 3: parts (2) and (3). By definition $\iota(m) = \iota(n)$ means $(m, 0) \sim (n, 0)$, that is $m + p = n + p$ for some p, which is (2). Then ι is injective exactly when $m + p = n + p$ forces $m = n$, which is the cancellation property, giving (3).

Step 4: the universal property. Let $f\colon M \to A$ be a monoid homomorphism into an abelian group. If $(m, n) \sim (m', n')$, say $m + n' + p = m' + n + p$, then applying f and cancelling in A, which is legitimate because A is a group, gives $f(m) - f(n) = f(m') - f(n')$. So

$$\tilde{f}([m] - [n]) := f(m) - f(n)$$

is well defined, and it is a homomorphism because addition in $M^{-1}M$ is coordinatewise. It satisfies $\tilde{f} \circ \iota = f$. For uniqueness, any homomorphism g with $g \circ \iota = f$ is determined on the elements $\iota(m)$, and these generate $M^{-1}M$ by (1), so $g = \tilde{f}$. Naturality in M and A is immediate from the formula, completing the proof.

Example 7.1.7: Grothendieck Completions

1. $\mathbb{N}$ under addition is cancellative, so ι is injective, and $\mathbb{N}^{-1}\mathbb{N} = \mathbb{Z}$. This is the motivating case named at the start of the section.

2. Let $M = \{0, 1\}$ with $1 + 1 = 1$. Cancellation fails, since $1 + 1 = 0 + 1$ while $1 \neq 0$, and indeed $M^{-1}M = 0$: taking $p = 1$ shows $(1, 0) \sim (0, 0)$, so $\iota(1) = 0$.

3. More generally, if M has an absorbing element ω, meaning $\omega + m = \omega$ for every m, then $M^{-1}M = 0$: for any m, taking $p = \omega$ gives $m + \omega = 0 + \omega$. This is why the construction is applied to *finitely generated* objects rather than arbitrary ones, an infinite direct sum being absorbing.

Exercise 7.1.1

1. standard monoid questions
2. In the following exercices, we'll test your understanding of the idea of defining quotient objects by building up to defining a quotient monoid
   a) Recall the difference between an equivalence relation and a congruence relation. What condition do we need in order that $[x][y] = [xy]$ in a monoid? (I saw that the regular one for groups fails, and it works if there's an inverse – investigate this further for it seems to be an interesting question to do!!)

7.2 Groupoid

Recall how at the very begining under definition 1.1.3, I remarked how some authors prefer to say groups are not closed under multiplication? Some authors introduce this convention to be able to later drop this axiom in order to define the more general notion of *groupoid*. A groupoid is a group which is not closed under the binary operation! [2]

Definition 7.2.1: Groupoid

A groupoid is a set G with a unary operation $^{-1} : G \to G$ and a patial function $* : G \times G \rightharpoonup G$ where the following axioms hold:

1. Where $a * b$ and $b * c$ are defined, $(a * b) * c = a * (b * c)$. Conversaly, if $(a * b) * c$ and $a * (b * c)$ is defined, then so are both $a * b$ and $b * c$ and $(a * b) * c = a * (b * c)$
2. $a^{-1} * a$ and $a * a^{-1}$ are always well-defined
3. if $a * b$ is defined, then $a * b * b^{-1} = a$ and $a^{-1} * a * b - b$

Lot's of cool stuff I'll fill in later

Semi-groups? If you introduce wreath product, can be intersting to talk about in conjuction with automata?

[2]Technically, this means it's not a binary operation, because binary operations are functions, which need to be closed, but the spirit of the sentence is correct

8

Category of Group

Throughout the book, we'll be introducing more and more algebraic structures of varying levels of similarity and differences[1]. It would be very convenient to be able to have a way to compare these algebraic structures. In this section, we elaborate more on the concept of *Categories* which were introduced in section 0.4 . When we introduce new algebraic structures, we will show how the algebraic structure also forms a *category*, and by doing so we will open up the ability to compare and contrast all sorts of properties. For example, we saw in example 1.1.5 that if G and H are groups we can define $G \times H$ to be a group under point-wise operations. Will the same type of construction be possible for algebraic structures in coming chapters? For rings? Group-actions? Fields? Vector-space? Will all of these objects have a notion similar to subgroup? Will they all have a notion of homomorphism/isomorphism, or a similar notion to group action? The answer will be yes to some and no to others! In order for these structure to have or not have a property, say for example the ability to do a "direct product", there must be a way of saying "the direct product of groups follows the same principle as the direct product of rings/group-actions/fields/vector spaces/etc." This will be accomplished with the language of category theory.

As mentioned in section 0.4, the core concept of categories is that we want to be able to have a structure that holds *objects* (ex. groups), and a particular way to *relate* these objects (ex. homomorphisms) – see the analogy in the aforementioned section for an analogous point of reference. Different categories can have the same objects but different relations, or the same relations but different objects, or different objects and relations. Commonly, a relation between two objects will be a way to change from one object to another while preserving some of the original structure of the object. These types of relations are called *structure preserving* relations. As an example of relations between objects that preserve some structure, take homomorphism, which relates two groups if there is some common group-structure. The fact that homomorphisms preserve (some or all of) the structure can also be

[1] In this book, there will be 15 algebraic objects that can be considered "key objects" to algebra, while we also work with 6 others non-algebraic objects throughout the book (ex. sets, graphs, ordered sets, etc.)

found in the etymology of the word: the word *homomorphism* come from *homo-* meaning "same" [2] and *morphism* meaning shape. Since homomorphsims preserve (some or all) group structure, they in a sense change a shape from one group to another while keeping some of the "sameness" between the two. Many different mathematical objects have a relation between them that preserve some of it's structure (topologies with topology-preserving functions, graphs and graph-preserving functions, or ordered sets with order-preserving functions to name few), which is why when categories were first introduced, the word *morphism* came to be the name used to describe relations between objects (i.e. if A and B are related in a category, there is a morphism from A to B). The word morphism is a generalization of the word homomorphism to capture the idea of relation's between objects.

I was purposefully being vague in the start by saying that a category has a "particular way to relate objects" instead of just jumping to saying that morphisms preserve some of the structure of the objects. At the beginning of category theory in the 1940s, the types of relations morphisms represented were relations that preserved (some or all of) the structure of the objects. As time went on, more mathematical objects were shown to form categories using morphisms that didn't preserve structure in a way we saw with homomorphisms, and the idea that some of the structure is preserved in some context can be a little of a stretch (see exercise ref:HERE for an example (order on sets)). This is good to know early on to not get the wrong idea of categories, however in the undergraduate level, 99% of the time morphism in categories will be about preserving some/all the structure of a particular object.

With this very high-level idea, let's give some concrete examples before introducing the definition of a category. If we take all the groups and all the possible homomorphism between groups; that will in fact from the category of groups, sometimes denoted **Grp** or **Group**. Another example would be the collection of all sets and all function between sets; that will form the category of sets, sometimes denoted **Set**. We'll soon bring in the definition of a category to prove that it will follow the axioms of a category, for now we will build-up some terminology in order to talk about these collections.

Definition 8.0.1: Category Of Grp and Set

1. Let **Grp** represent the collection of all groups and homomorphism between Groups and **Set** the collection of all sets and functions between sets.
2. Let $\text{Obj}(\mathbf{Grp})$ represent all the objects in the category of **Grp**. Similarly for $\text{Obj}(\mathbf{Set})$.
3. For any $G, H \in \text{Obj}(\mathbf{Grp})$, Let $\text{Hom}_{\mathbf{Grp}}(G, H)$ represent all homomorphisms from G to H. Similarly, if $A, B \in \text{Obj}(\mathbf{Set})$, let $\text{Hom}_{\mathbf{Set}}(A, B)$ represent all function form A to B.

In many ways, properties of these collections was the starting point for the first category theorists definition of a category, so it is worth taking a closer look at the properties. To not clutter the analysis, we will stick with **Grp** to build-up to the definition of a category.

When we relate two group using homomorphsims, we always have a domain and a co-domain:

$$\varphi : G \to H$$

This means that relations are *directional*: a homomorphism from G to H is different from a homomorphism from H to G. In some structure, direction of the relation doesn't matter (for example, in equivalence relations, $x \sim y$ if and only if $y \sim x$). In categories, we pay careful attention at the

[2] Other words with homo- would be homeostasis (same state), homochromatic (same colour), or homogeneous (same kind/nature, or uniform structure). It's antonym would be hetero- (heterostatis, heterochromatic, heterogenous)

direction of the arrow, and so when generalizing, a morphism from an object A to B is not the same from a morphism from an object B to A.

When we work with homomorphism, we showed in proposition 1.4.4 that the composition of two homomorphisms is a homomorphism. Particularly, if have a homomorphism from G to H, and a homomorphism from H to K, then there will always exist a homomorphism from G to K:

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & H \arrow[r, "g"] & K \end{tikzcd}$$

So, if there is a way to get from G to K through homomorphism, there must be a way to "get there directly" too. Morphisms will have the same property.

Finally, notice that for every group there always exists a "trivial" homomorphism:

$$1_G : G \to G \qquad 1_G(g) = g$$

that is, the homomorphism takes G to itself. In a sense, what is being said here is that every object trivially relates to itself. Another way of thinking that 1_G is trivial is that when composed with other homomorphisms (with appropriate domain or co-domain), it does not change them. If:

$$\alpha : G \to H,\ \beta : F \to G$$

were homomorphsims, then

$$\alpha \circ 1_G = \alpha \qquad 1_G \circ \beta = \beta$$

We want this same property to exist for categories: That every object in a category has a *trivial morphism* which can compose with morphisms (with appropriate domain or co-domain) and not affect it

You might have noticed that all of these properties are about the morphisms of a category. This is a common philosophy in category theory: What's important isn't the objects, but the how an objects relates to other object! It is arguable that the more mathematics you will encounter, the more it will be clear that it is the way objects can relate to object objects that i the defining feature of many objects. In fact, many algebraic objects that we will see will be defined by how they relate to other objects.

With this, we will finally define a category:

Definition 8.0.2: Category

A ***category*** C consists of[a]

1. A collection of *objects*, denoted $\text{Obj}(C)$
2. A collection of *morphisms*, denoted $\text{Mor}C$

where

- Each morphism has a specified ***domain*** and ***codomain*** object. Sometimes, this is written as $f : X \to Y$, where f is the morphism with domain X and codomain Y [b]. For any $A, B \in \text{Obj}(C)$, $\text{Hom}_C(A, B)$ denotes all morphisms form A to B.
- Each object has an ***identity morphism*** $1_X : X \to X$
- For each pair of morphisms f, g with the codomain of f equal to the domain of g (ex. $f \in \text{Hom}C(A, B)$, $g \in \text{Hom}_C(B, C)$), there exists a specified ***composite morphism*** gf (or $g \cdot f$)[c] whose domain is equal to the domain of f and whose codomain is equal to the codomain of g, i.e.

$$f : X \to Y \qquad g : Y \to Z \qquad \Rightarrow \qquad gf : X \to Z$$

such that

1. For any $f : X \to Y$, the composites $1_Y f$ and $f 1_X$ are both equal to f (i.e. unital with the identity homomorphism)
2. For any composable triple of morphisms f, g, h, the composites $h(gf)$ and $(hg)f$ are equal (i.e. associative), and so we can write hgf unambiguously.

[a]The word *collection* being used instead of *set* was purposeful; we will get to this
[b]Note that f **need not be a function**! Namely, the morphism need not have a unique codomain
[c]the notation is purposefully not $\circ$ so as to not draw too much of an analogy of gf with the process of composing functions to demonstrate that in some cases gf is *not* function composition

Example 8.0.3: Examples of Categories

1. It was mentioned that groups along with homomorphisms from a category. Indeed, by proposition 1.4.4 the composition of two homomorphsims is a homomorphism, so $gf = g \circ f$. Homomorphsims are associative since function composition is associative (proposition 0.1.22), and for any $G \in \text{Obj}(Grp)$, the map $\text{id} :\to G$, $\text{id}(g) = g$ is the identity morphisms since for any $f : F \to G$ and $h : G \to H$, $f \circ \text{id} = f$ and $\text{id} \circ h = h$.
2. If we take all groups, and restrict to taking only the isomorphisms, we will again have a category; check that all the axioms are still satisfied. Such a category is called a *groupoid* (the reason for this name will be made evident in exercise ref:HERE (show with 1 element is group, then define more general structure groupoid with no identity, which we expand more upon in section 7.2))
3. If we take all abelian groups and use homomorphsims, this will again be a category. This category is significant enough to deserve it's own notation: this category will denoted **Ab**.

4. If we take all sets and all functions between sets, we will get the category of **Set**.

   Notice that every morphism in **Grp** can be thought of as a morphism in **Set** if we "forget" the group structure of the domain and co-domain, and the homomorphism condition of the morphism of **Grp**. In This way, the category of **Grp** and the category of **Set** are related if we "forget" some of the additional structure of **Grp**. Is it possible to have the reverse? That is, starting with **Set**, is it possible to somehow "upgrade" it to **Grp**? Is there more than one way of doing this? Is there a "natural" way of doing it, that is, is there a way of performing this upgrade in such a way that it follows from the properties of groups and sets without needing to add any more conditions? These questions will be further explored in section 2.3.3. generally.

5. If we take the collection partially ordered sets, that the collection of $(S, \leq)$ where $S \in \mathrm{Obj}(Set)$ and $\leq$ is a possible partial order on S, then we can define a notion similar to homomorphisms that preserve (some or all of) the structure of the order, i.e. we can define a $f : (S, \leq_S) \to (R, \leq_R)$ where f is the appropriate morphism. Can you think of what this morphism should be?

6. In section 0.3, we introduced graphs and graph isomorphisms. These also form a category! There is also a notion of "graph homomorphisms" mapping vertices to adjacent vertices and a generalization of the notion of *graph colouring*. For those interested in this topic, see ref:HERE.

Before continuing with the language of category, we will add one more example of a category which is a little more abstract than the previous categories but one we will constantly use. Namely, we can take categories themselves to be objects and define a notion of morphism between categories! This category will be denoted **Cat**. Since morphisms relate objects, the morphisms in the category of **Cat** allows us to find commonalities between different categories! We have in fact already described a category morphism in example 8.0.3[4]: If we take **Grp**, we can forget the group structure of every group and the homomorphism property of every homomorphsims and we will end up with **Set**. In effect, we did $(G, *) \to G$ and mapped the homomorphism f where $f : (G, *_G) \to (H, *_H)$ to the function f $f : G \to H$ where the function f is identical to the homomorphsims f except it no longer requires the homomorphism property (in fact, it is impossible for it to have the homomorphism property since sets don't have a binary operation). We will call category morphisms *Functors* (as a generalization of the word function)

Definition 8.0.4: [Covariant] Functor

Let C, D be two categories. Then a *Functor* F will consist of:

1. a function between $\text{Obj}(C)$ and $\text{Obj}(D)$. If $a \in C$ is an object, then $F(a) \in D$ denotes the object in D.
2. a function between $\text{Mor}C$ and $\text{Mor}D$. If $f \in C$ is a morphism in C, then $F(f)$ denotes the morphism in D.

such that

1. F must preserves the domain and codomain of each morphism: $f : A \to B$ must map to $F(f) : F(A) \to F(B)$
2. F must map identities to identities: $F(\text{id}_f) = \text{id}_{F(f)}$
3. F must preserve composition: $F(g \cdot_C f) = F(g) \cdot_D F(f)$.

Example 8.0.5: Functor

1. For any category C, there is always a trivial functor sending objects to themselves and functions to themselves
2. The relation above between **Group** and **Set** is indeed a functor: $F : \textbf{Group} \to \textbf{Set}$ Such a functor is called a *forgetful functor* since it forgets some of the structure of the category. In the coming chapters, many of the objects are augmentation of sets. All of these categories will have a forgetful functor $F : \mathbf{C} \to \textbf{Set}$.

Some authors prefer to be more specific about the second condition and define it instead as the following: for any two object $a, b \in \text{Obj}(C)$ there exists a function $\text{Hom}_C(a, b) \to \text{Hom}_D F(a), F(b)$. We will see many more examples of functors throughout this book. For now, it is left as an optional exercise to prove that categories and functors together form a category[3].

So far, all the categories we have dealt were all in some fundamental sense based on sets: each object was either a set with an augmentation (ex. $(G, *)$ or $(S, \leq)$) or a collection of sets following certain restrictions (ex. graphs). Because of this, every object in every category so far has the notion of having "elements". However, there is nothing in the definition of an object in a category that restricts objects to needing elements:

Example 8.0.6: Non-Set Category

Let $\mathbb{N}$ be the set of all natural numbers and $\leq$ be the usual order. Then we define a category C whose objects are the numbers of $\mathbb{N}$ and whose morphism is $\leq$, that is for every $x \in \mathbb{N}$, $\leq$ is the morphism that relates x to numbers greater than or equal to number y, which we will label $x \leq y$. Notice that $\leq$ is not a function, since x is not "mapped" to a single element but to every element greater than it. Furthermore, $\leq$ is the only morphism in in this category. Composition will return $\leq$ (since it's the only morphism) and we see that if $x \leq y$ and $y \leq z$ then their composition results

[3]There is a technicality about the fact we are taking "the category of all categories" in the sense that this is self-referential creating a contradiction usually called Russel's Paradox. There is a definition of *small category* which circumnavigates this problem which was not introduced so as to no overload new terminology

in $x \leq z$.

This is indeed a category: $\leq$ has a domain and co-domain (if you like you can picture $\leq: x \to y$ as long as it is kept in mind that $\leq$ is not a function). For any $x \in \mathbb{N}$, x has an identity morphism, namely the only morphism $\leq$, (you can picture $\leq \cdot \leq = \leq$).

More generally, any set S with some partial order forms a category. For example, if X is a set, then $(P(X), \subseteq)$ is a category whose objects are all subsets of X and the morphism is the partial order induced by $\subseteq$ (i.e. $U \subseteq V$ then $U \to V$).

(Another would be to simply draw diagrams!!)

The set $\mathbb{N}$ was an arbitrary choice in the previous construction: we could have chosen $\mathbb{R}$ or $\{0\}$ or an arbitrary set with an arbitrary ordering. We can visualize the category in the example by imagining a directed graph where every edge represents the morphism between the domain and codomain:

$$\begin{tikzcd}
0 \arrow[r] \arrow[rr, bend right] \arrow[rrr, bend right] & 1 \arrow[r] \arrow[rr, bend left] & 2 \arrow[r] & 3 & \cdots
\end{tikzcd}$$

Such a directed graph is called a *commutative diagram*. The full reason behind the name is a little abstract without the proper build-up[4], however we can still understand the etymology given the following illustrative example: imagine we start at the number 1, and every horizontal arrow represents multiplying on the right by x, and every vertical arrow represents multiplying on the right by y, and we form a square like so:

$$\begin{tikzcd}
1 \arrow[r, "\cdot x"] \arrow[d, "\cdot y"] & x \arrow[d, "\cdot y"] \\
y \arrow[r, "\cdot x"] & yx = xy
\end{tikzcd}$$

Similarly, we could have started with $G \times G$ in the top left, let the right arrow be the swap operation $(x, y) \mapsto (y, x)$, the bottom arrow being the identity $(x, y) \mapsto (x, y)$, and then both arrow to the bottom right being the group multiplication arrows.

This square shape is so common that the any directed graph represented morphisms came to be called a commutative diagram. In section 8.2, we shall further explore the use of commutative diagram to understand properties of categories.

8.1 Basic properties of Morphisms

I clearly don't have enough knowledge to write this yet:

https://en.wikipedia.org/wiki/Monomorphism.

Read this and come back.

When we studied function, we started by classifying injective and surjective functions due to their simplicity and universality. It would be nice if there are some nice equivalent, or at least similar, notions of injectivity and subjectivity for morphisms in general categories. Since objects need not

[4]We will often care about morphisms between categories commuting in some "natural" way. This will lead to the study of natural transformation from which the name commutative diagram will takes it's name

have elements, the definition has to take this generality into account. Fortunately, there is already a property of injective and surjective functions that was characterized in terms of functions in proposition 0.1.26 and 0.1.28: it was shown that all injective functions have a left inverse and all surjective functions have a right-inverse. These notions can be generalized in the following way:

Definition 8.1.1: Monomorphism

Let $f : A \to B$ be a morphism. Then if for any morphism $\alpha : Z \to A$, $\beta : Z \to A$, if

$$f \circ \alpha = f \circ \beta \Rightarrow \alpha = \beta$$

then we call f a monomorphism

Intuitively, monomorphisms should take you to a "unique place" similarly to how injective functions have a single element to represent them.

Proposition 8.1.2: In Set, monomorphism if and only if Injective

Let $f : A \to B$ be a set function. Then f is injective if and only if it is a monomorphism

***Proof*:**

($\Rightarrow$) Let's say f is injective, and that $f \circ \alpha = f \circ \beta$. We want to show that $\alpha = \beta$. By proposition 0.1.26, there exists a left-inverse f^{-1}. Now, notice that

$$\begin{aligned}(f^{-1} \circ f) \circ \alpha &= f^{-1} \circ (f \circ \alpha) && \text{associativity of composition}\\ &= f^{-1} \circ (f \circ \beta) && \text{by assumption}\\ &= (f^{-1} \circ f) \circ \beta && \text{associativity of composition}\end{aligned}$$

meaning we get $\mathrm{id}_A \circ \alpha = \mathrm{id}_A \circ \beta$, and so

$$\alpha = \beta$$

as we sought to show

($\Leftarrow$) Let's say f is a monomorphism so that $f \circ \alpha = f \circ \beta$ implies $\alpha = \beta$. Let's say f wasn't injective, that is, for some $a, b \in A$, $f(a) = f(b)$ but $a \neq b$. Choose $B' = f(B)$ to be the domain of α and β where α and β both agree where to send elements with the exception of α sending $f(a)$ to a and β sending $f(a)$ to b. Then

$$f \circ \alpha = f \circ \beta$$

but $\alpha \neq \beta$ a contradiction. Since those elements were arbitrary, it must be the case that,

$$f(a) = f(b) \Rightarrow a = b$$

showing injectivity.

One can also prove this directly by letting $Z = \{p\}$ (i.e., a singleton) and using the monomorphism property to conclude that

$$f(a) = f(b) \Rightarrow a = b$$

Similarly, we can define an epimorphism:

Definition 8.1.3: Epimorphism

Let $f : A \to B$ be a morphism.

If for any function $\alpha : Z \to B$, $\beta : Z \to B$, if

$$\alpha \circ f = \beta \circ f \Rightarrow \alpha = \beta$$

then we call f a epimorphism

Proposition 8.1.4: In Set, epimorphism if and only if Surjective

Let $f : A \to B$ be a set function. Then f is surjective if and only if it is a epimorphism

Proof:
Exercise

We have mentioned earlier that in category theory, we define objects based on how they relate to other objects. We have in fact done this before: When we defined a subgroup based on a group. We mentioned how this more complicated definition generalizes we

Definition 8.1.5: Subobject

Let C be a category and $X \in C$ be an object in this category. We say $Y \in C$ is a *suboject* of X is there exists a monomorphism

$$Y \to X$$

(maybe details with the "up to isomorphism" thing)

To demonstrate what I mean, Let's take an arbitrary set X, and an arbitrary set Y. When can I say that Y is a subset of X, or vice versa? You probably guessed it: if one can be part of included in another, one is a subset of the other. Written in terms of functions: if there exists an inclusion map $i : Y \to X$, then Y is a subset of X ($Y \subseteq X$). We then can define a set being a subset as follows:

If there exists an inclusion map $i : X \to Y$, then X is a subset of Y ($Y \subseteq X$).

If both inclusion maps exist, then there is two injections, and so by Schröder–Bernstein theorem (see appendix A), there exists a bijection between X and Y ($X = Y$). In this way, we can actually *define* a set to be a *subset* of another set if there exists an inclusion map from Y *to* the bigger set. If we weaken our need for i to be an inclusion map, to i being an injective map, we can start comparing sets with different elements [5]

We can apply this now to groups! First, recall (or revisit) what we did in proposition 1.4.8. Since we have proven the proposition is equivalent to the definition of subgroup, we will bring it here, but label it as a second definition of subgroup:

[5] More philosophically, it doesn't actually matter what are the elements of a set; what matters is that there *are* arbitrary distinct elements in a set. So being an inclusion or injective is actually "equivalent" in a broader sense. How so? This question will be answered in appendix **??** with more build-up to the greater philosophy of mathematics

Since the same technic is used in defining both concepts, we can say each are similar concepts or types of “objects”. Thus, we can say what we have here is applied the more general idea of *sub-object* to both situations.

Notice how for sets, if there are is an injection going both way, we say that the two sets are bijective and $X = Y$? This is actually *exactly* the same with groups!

Example 8.1.6: sub-objects

1. **subobjects in Grp**: Recall in proposition 1.4.8 where we showed that if $(H, \bullet)$ is a group and $H \subseteq G$, then H is a subgroup if and only if the inclusion map $i : H \to G$ is a homomorphism. This proposition was the proof that **Grp** had subojects!
2. **subobjects in Set**:
3. sub-order?

(maybe mention the question of switching the monomorphism for an epimorphism and how that let's you define something we'll call *quotient objects*, which we'll get too soon?)

(build-up to the idea of isomorphism in category theory)

Definition 8.1.7: Isomorphism In C

Let C be a category, and f be a morphism. Then f is said to be an *isomorphism* if HERE.

Example 8.1.8: isomorphism in different Categories

1. In **Grp**, HERE
2. In **Set**, HERE
3. In the category of ordered sets, HERE

Proposition 8.1.9: Isomorphism Generalized

Let G, H be groups. Then if there exists an injective homomorphism $G \to H$ *and* and injective homomorphism $H \to G$, then we say that G and H are *isomorphic* and we write $G \cong H$.

Notice how I didn't say that $G = H$. That is because I said we have an *injective* homomorphism. I did not say that it's necessarily an *inclusion* map. If it were, we can justify saying $G = H$ completely. However, group theoretically, there is no difference between G and H, making them “equivalent”. In this way $G \sim H$ in terms of equality $=$. So, combining the symbols $\sim$ and $=$ gets us $G \cong H$.

The great advantage of defining subgroup/subset in this way is that we can now define what it means to be a sub-object in many other areas of mathematics! We can even ask whether it is *possible* to define the notion of sub-object; there are some areas in mathematics where the concept of an injective map cannot be defined!![6]

[6] The category of posets $(P, \leq)$ is an example. See appendix **??**

Also, perhaps define Hom, Aut, and End here? Perhaps Also give the category of **Set**? To be able to give an idea of what a category is and to bet able to define $\mathrm{Aut}_{\mathbf{Set}}(A)$ for group actions. Introduce the notation

$$\mathrm{Sym}(X) := \mathrm{Aut}_{\mathbf{Set}}(X)$$

to use it later?

Show how for any category C, and any object $X \in C$, then $\mathrm{Aut}_C(X)$ forms a group! In this way, groups appear everywhere. The fact that $\mathrm{Aut}_C(G)$ is a group will come back big-time when studying *group actions*.

Next talk about initial and final objects. This leads to universal properties. Bring up how $\{e\}$ was an initial and final object in **Group**. Bring up how initial and final objects are unique, so once we proved on exists, we proved it's the only one up to isomorphism.

8.2 Commutative diagrams

As mentioned countless times, categories try to capture the idea of relations between objects. Very often, this is done by forcing some relation between morphisms. We've already seen an example of this in the category of **Set** and **Grp** by saying the composition of two functions (or homomorphisms) is again a function (or homomorphism). We represented this by the diagram:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[rr, bend right, "h"'] & Y \arrow[r, "g"] & Z \end{tikzcd}$$

where we usually denote h as $g \circ f$. We can form all sorts of relations between objects and enforce the fact that compositions of functions must lead to other arrow (TBD). We call this a commutative diagram

(category can be projected down to a graph. I you only project (PROPER PROPERTY HERE), then you get a special graph called a *quiver*, which try to be a generalization of algebraic structures (TBD).)

Definition 8.2.1: Commutative Diagram

Give a nice definition here

Example 8.2.2: Example

Must re-phrase

1. Recall that a group homomorphsims is a function $\varphi : G \to H$ which preserves the group action. If we take G, and H represent the underlying set, and $(G, \cdot_G)$, $(H, \cdot_H)$ represent the set with the underlying operation, then we can what's the most *natural* requirement for the operation to be preserved? Note that if we drop the operation preserving part of φ, we can think of the inputs of the binary operation as [a]

$$(\varphi \times \varphi) : G \times G \to H \times H \qquad (\varphi \times \varphi)(a, b) = (\varphi(a), \varphi(b))$$

Then, we can combine this with the operations on the group to get the following diagram:

$$\begin{tikzcd}
G \times G \arrow[r, "\varphi\times\varphi"] \arrow[d, "\cdot_G"] & H \times H \arrow[d, "\cdot_H"] \\
G \arrow[r, "\varphi"] & H
\end{tikzcd}$$

Thus, from a categorical perspective, the best way to define an operation to be preserving is if this diagram *commutes*, meaning:

$$\begin{tikzcd}
(a,b) \arrow[r, dotted, no head] \arrow[d, mapsto] & {} \arrow[d, dotted, no head] \\
a \cdot b \arrow[r, mapsto] & \varphi(a \cdot b)
\end{tikzcd}
\qquad
\begin{tikzcd}
(a,b) \arrow[r, mapsto] \arrow[d, dotted, no head] & (\varphi(a), \varphi(b)) \arrow[d, mapsto] \\
{} \arrow[r, dotted, no head] & \varphi(a) \cdot \varphi(b)
\end{tikzcd}$$

which in it's familiar form would be $\varphi(a \cdot b) = \varphi(a)\varphi(b)$

2. maybe bring up how 1.9 is not necessarily a commutative diagram since the arrow's aren't necessarily restricted through composition.

[a]The reason for this construction, as we shall soon see, relies on the *universal property of products*. For now, since it's something all readers would know at this point, we leave it as intuitive

8.3 More abstract examples

(Give an example of a slice category or comma category, and a fibered category. Explain how the point of these categories and many like them is to capture certain relations between objects.

8.4 Summary

A category (definition 8.0.2) gives objects, morphisms between them, and a composition law; a functor (definition 8.0.4) is the corresponding map between categories; and a commutative diagram (definition 8.2.1) is the notation in which an equality between composites is asserted. Against that backdrop, each notion below replaces a statement about *elements* by a statement about *morphisms*. What licenses the replacement in each case is that the two agree in **Set**, where the element-level notion is the familiar one.

*Notion in **Set***	*Categorical form*	*Agreement in **Set***
Injective function	Monomorphism (definition 8.1.1)	proposition 8.1.2
Surjective function	Epimorphism (definition 8.1.3)	proposition 8.1.4
Subset $Y \subseteq X$	Subobject (definition 8.1.5)	inherited, the definition being via a monomorphism Y -

The pattern is the one every later categorical definition in this book follows: a property phrased by quantifying over elements is re-phrased by quantifying over morphisms into or out of the object, and the re-phrasing is checked against **Set** before being adopted elsewhere.

Part II

Rings

Now that we've the essentials covered groups, we can move on to structures which have 2 binary operations on the set. These structures are called *rings* [7] As with commutative groups, commutative rings will be very different from rings, so much so that one can think of ring theory as really being 2 seperate domains put under one label: commutative rings and non-commutative rings. However, similarly to how all groups share some common structure (a notion of homomorphism, isomorphism theorem, direct product, free group, group-action, etc.), so do rings, and so we will be building on the foundation of ring theory.

We may wonder how to interpret adding a second binary operation to an abelian group? Does this restrict our structure? Does this limit what we can prove? Does it give more tools to work with when proving? The way I like to perceive adding this second operation is that there are more properties that we have at our disposal to work with. The "type" of new property we add also determines what properties we work with (for example, as is briefly addressed in appendix C, section C.4, we can add a topological structure to groups which gives us different tools we can use to study groups). In the case of adding a new binary operations, we gain more "algebraic" tools to work with. For example, we have an understanding of how the prime numbers of $\mathbb{N}$ interact when we add them[8] or when we multiply them, but we have little knowledge of the interaction of prime numbers between these two binary operations. Furthermore, for any abelian group, we can always add at least one new binary operation that satisfies the axioms of a ring[9] (which we will present near the end of section 9.1.1). In that way, all groups can be made into rings, and we can study them as rings.

For the 1st operation, we will always use additive notation. For example, if we're taking the internal direct product of two groups in this notation, we will write $H + K$, since we will reserve the multiplicative notation for the 2nd binary operation.

[7] The term *ring* was probably a contraction of David Hilbert's term *Zahlring*, literally translated as *number ring*. The word ring was probably coined because of the cyclic nature of the numbers Hilbert was working with. More details on this discussion can be found here: https://math.stackexchange.com/questions/61497/why-are-rings-called-rings

[8] All be it to a very limited degree

[9] Similarly to how all abelian groups are topological groups with the discrete topology

9

Introduction To Rings

With Groups extensively covered, what's "basic" has now changed. We need not so extensively elaborate intuitions for homomorphisms, quotients, products, or actions. Instead, we'll show how these concept grow when considering *rings*. Fundamentally, a ring is a set with two operators (hence, is a triple $(R, +, \cdot)$ where $+$ and $\cdot$ satisfy certain axioms[1]). We will show some really interesting examples which we gain when we introduce a second operator, and take advantage of some of the categorical language we have built-up to shed some more light on the nature of rings (which will be noticably different than that of groups).

As with groups homomorphisms, "ring homomorphisms" will play a central role in the study of rings. There will be an equivalent to "normal groups" which we will extensively study: the equivalent for rings is called *ideals*. Manipulating and combining ideals is at the core of ring theory. After exploring ideals, we will take a moment to look at a way to embedd any ring R into a larger ring that permits the notion of "division" (as we'll see, unlike groups, not all elements of a ring will have an inverse with the second operation). Through this construction, we will show how to take $\mathbb{Z}$ and create $\mathbb{Q}$.

[1] These axioms can be said to be "natural" in some way. These exploration's will be done later in the chapter, see ref:HERE

9.1 Basic Properties and Definitions

Definition 9.1.1: Ring

A *ring* R is a set together with two binary operations $+$ and $\cdot$ (called addition and multiplication), $(R, +, \cdot)$ satisfying the following axioms:

1. $(R, +)$ is an *abelian* group, where the identity is denoted by 0
2. **(Mult. Associativity)** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$
3. **(Mult. Identity)** There exists an $e. \in R$ such that for every $r \in R$, $e. \cdot r = r \cdot e. = r$. The element $e.$ is usually labeled 1.
   - Sometimes, we use *rng* to denote a ring without multiplicative identity.
4. **(Distributivity)** For all $a, b, c \in R$

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \qquad \text{and} \qquad (a + b) \cdot c = (a \cdot c) + (b \cdot c)$$

   The two equations separately are called left and right distributively respectively.

If $a \cdot b = b \cdot a$ for all $a, b \in R$, we call the ring a *commutative ring*.

We'll usually write $a \cdot b$ as ab for convenience. The convention that the multiplicative identity is 1 and additive identity is 0 is essentially universal, and so it is safe enough to have it as part of the definition of a ring.

9.1.2 Ring always has identity? Some author's do not include 1 in the definition of a ring (we called rings without a (multiplicative) identity *rng*); in this case, a ring with a multiplicative identity is usually called a unital ring. Any *rng* R can be extended to a ring by taking $\mathbb{Z} \oplus R$ with identity $(1, 0)$ (What is the multiplication here that makes $\mathbb{Z} \oplus \mathbb{R}$ into a ring?)[a]. The most common place in mathematics where mathematicians usually don't want rings to have identities is in analysis; for those inclined towards examples see [19]. In Algebra, it is now pretty rare to find a textbook that define a ring without a multiplicative identity, however some classical textbooks have done so (for example [14] and [16])

[a] More generally, taking the direct sum with any [unital] ring will do, however different choices might lead to better result in different fields, for those interested see exercise ref:HERE

Just like associativity was a powerful axiom, so is distributivity for linking the $+$ operation and the $\times$ operation. If you have any arbitrary expression:

$$(a + b)(c + d + e(f + g(i + j)kl(m + n)op))q$$

It can be simplified to the sum of monomials:

$$\begin{gathered} ac + bc + ad + bd + aefq + befq + aegiklmopq + begiklmopq + aegjklmopq \\ + begjklmopq + aegiklnopq + begiklnopq + aegjklnopq + begjklnopqq \end{gathered}$$

There are non-distributive algebraic structures. However, the study of non-distributive structures seems to be much scarser than that of non-associative structures or structures without inverse. [2]. Distribuitivity seems to be how most structure link the two operations; at the minimum it usually would be distributive on only one side [3].

9.1.3 Note R being an abelian group is unavoidable given the distributive property

$$a+b+a+b=(1+1)(a+b)=a+a+b+b \Rightarrow a+b=b+a$$

Another way to think about G being abelian is that we want to concentrate on the *ring* operation now, so if we can make the group-operation very-well behaved (as we've shown commutativity does in chapter 5), it will not obtrude our study of the ring operation. This is a common technic.

Example 9.1.4: Ring Examples

1. Taking any abelian group G, and define $a \cdot b = 0$. This is the *trivial ring.* If If $G = \{0\}$, then we call this the *zero ring.* In fact, all $ab = 0$ for all $a, b \in R$ if and only if R is the zero ring:

   Proof. If R is the zero ring, then we already have that $0 \cdot 0 = 0$. Thus, we must prove the converse

   Let's say $\forall a, b \in R$, $ab = 0$. Then $1 \cdot 1 = 0$, meaning $1 = 0$

   Then, notice that

   $$\begin{aligned} &\Leftrightarrow\ 0\cdot a=(0+0)\cdot a=0\cdot a+0\cdot a \\ &\Leftrightarrow\ 0\cdot a=0\cdot a+0\cdot a \\ &\Leftrightarrow\ 0\cdot a-0\cdot a=0\cdot a+0\cdot a-0\cdot a && \text{additive inverse} \\ &\Leftrightarrow 0 = 0\cdot a \end{aligned}$$

   Thus, we have $0 = 0 \cdot a = 1 \cdot a = a$ for all a, and so all element equal 0, showing it's a zero ring. □

   In fact, this proof has a very powerful corollary: if $1 = 0$, then we have the zero ring! This property usually breaks many parts of many proofs, so we usually say that $1 \neq 0$ in most theorems and propositions. In some textbooks, this is even part of the axioms of a ring

2. The integers $\mathbb{Z}$ is the canonical example of a ring; when ring theory started, a lot of study was with trying to characterize properties of integers. Note that $ab = ba$, making $\mathbb{Z}$ a *commutative ring.*

3. $\mathbb{Z}/n\mathbb{Z}$ is a [commutative] ring with identity

4. $\mathbb{R}, \mathbb{Q}, \mathbb{C}$ are all [commutative] rings with identity. Naturally, $\mathbb{N}$ is not [a]

5. If R and S are rings, then $R \times S$ is also a ring with operation component wise. [b]. What is the identity of $R \times S$?

[2] I found this paper: generalized De Morgan algebra

[3] Something called a *Quasifields* is an example

6. If you have taken Linear Algebra, then all fields are an example of [commutative] rings

7. Recall the group Q_8 (see 1.2.4). Using it, we will define the set $\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$ where the elements bi, cj, an dk are distinct elements of $\mathbb{H}$, and are not another element in $\mathbb{R}$ or Q_8 (think of a, b, c, d as coefficients). Then we can make $\mathbb{H}$ into a ring by defining addition to be ponitwise:

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k$$

and multiplication through repeated use of the distributive property, which distributes to:

$$\begin{aligned}(a + bi + cj + dk)(a' + b'i + c'j + d'k) = (&aa' - bb' - cc' - dd') \\ &+ (ab' + ba' + cd' - cd')i \\ &+ (ac' - bd' + ca' + db')j \\ &+ (ad' + bc' - cb' + da')k\end{aligned}$$

Then $\mathbb{H}$ is called the *(real) Hamiltonians* (real since we could have defined them over $\mathbb{C}$). They form a non-commutative ring, and can be considered an extension of the complex numbers $\mathbb{C}$ (there are in fact two copies of $\mathbb{C}$ inside $\mathbb{H}$, more in that in exercise ref:HERE)

8. If X is a set and A is a ring, then the collection of all function $f : X \to A$ are a ring with function addition and multiplication being done point-wise: $(f + g)(x) = f(x) + g(x)$, $(fg)(x) = f(x)g(x)$. All ring structure on this function space comes from A. A concrete example of this is if $X = [0, 1]$, $A = \mathbb{R}$, and f is continuous $f : [0, 1] \to \mathbb{R}$. These types of rings show up a lot in analysis with the collection of function from a space X (a topological space, or measurable space, a manifold, etc.) to $\mathbb{R}$ or $\mathbb{C}$ giving many properties of the domain of the functions.

9. The set $2\mathbb{Z}$ is not a ring since it does not have an identity. It is thus a *rng*.

10. From section 1.2.5⋆, we saw that $(M_n(G), \cdot)$ is not necessarily a group since inverses were not not always defined. However, this is not a requirements for ring multiplication! It is very tedious and a bit hard at this stage to check that associativity and distributivity hold (there will be a simple proof of this in corrollary 13.2.7), but they are verifiable and show that $(M_n(R), +, \cdot)$ is a ring. Note that $M_n(R)$ is not commutative. For example

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

Matrix rings will be explored in more detail soon, and will have a whole chapter (in particular chapter 53) dedicated to the study of when matrix rings "act" on finite groups (in a similar way groups acted on sets in group-actions).

11. Recall when I asked if $P(X)$ is a group under $\cup$? Here is another similar question with rings. Take $P(X)$ again, and define addition and multiplication as

$$A + B = (A - B) \cup (B - A) \qquad A \cdot B = A \cap B$$

prove that it's a [commutative ring]. Another thing you can show is that for any $Y \in P(X)$, $Y^2 = Y$. If a ring has this property, it is said to be a *boolean ring*. [c]

12. Let G be any abelian group. Then $(\mathrm{End}_{\mathbf{Grp}}(G), +, \circ)$ is a ring. In this way, any ring is formed from an abelian group. [d]

[a] $\mathbb{N}$ is a semi-ring with $+$ and $\cdot$

[b] Though this might feel "trivially true", the fact that taking the direct product of two rings will produce the same algebraic structure, in this case a ring, is not always true! See 9.1.27 for a counter-example

[c] This ring comes up in *measure theory*, where the set of measurable sets can be made into a boolean ring

[d] The general term for taking [abelian] objects (ex. abelian groups) from an category C and working with $\mathrm{End}(X)$ as the new object and morphisms that also preserve $\circ$ is **Ab**-enriched category given the abelian category C

When working with rings, we will assume the usual ring structure for $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{Z}/n\mathbb{Z}$, and use the same notation if it's clear in context. The notation $\mathbb{Z}^\times, \mathbb{Q}^\times, \ldots$ will soon be repurposed for another use later in this section.

Note that every ring is naturally an abelian group as well as a monoid (see section 7.1). When working with homomorphisms for rings, both structure need to be preserved in some way for the image of a funciton to be a ring. There is in fact one more sneaky criterion that needs to be added in order for a the image of a function be be a ring, and hence a *ring homomorphism*:

Definition 9.1.5: Ring Homomorphism

Let R and S be rings. Then:

1. *ring homomorphism* is a map $\psi : R \to S$ satisfying
   (a) $\varphi(a + b) = \varphi(a) + \varphi(b)$
   (b) $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
   (c) $\varphi(1_R) = 1_S$
2. If $\varphi : R \to S$ is a ring homomorphism, and there exists a ring homomorphism $\psi : S \to R$, then φ is a ring isomorphism

Note that just like for group homomorphisms, it can be proven that definition 9.1.5[2] is equivalent to a ring homomorphism being bijective.

Note that the reason we add the condition that $\varphi(1_R) = \varphi(1_S)$ is so that the image $\varphi(R)$ is also a ring. If a ring is not defined to have a multiplicative identity (as is done for *rngs*), then the condition of $\varphi(1_R) = \varphi(1_S)$, is not necessary. Furthermore, where $\psi(e_G) = \psi(e_H)$ is a provable property given that ψ is a group-homomorphism and e_G, e_H are the respective identities of the domain and codomain, the same is not true for rings, that is, the first 2 conditions of a ring homomorphism can be satisfied, however the resulting image will not be a ring (take $f : \mathbb{Z} \to \mathbb{Z}$, $x \mapsto 2x$, see exercise ref:HERE).

Many times, mathematical objects will have multiples different structures defined on them, but we sometimes want to only work with some of them[4]. Thus, a group homomorphism between rings will only preseve group structure, a while ring homomorphism preserves both. Ring homomorphisms will be further explored in section 9.3, where we will show how to relate the structure of $\varphi(R) \subseteq S$ to R similarly to how it was done with quotient groups.

From definition 9.1.5, we can define a sub-ring:

[4] An example with full of different structures is $\mathbb{R}$ which can come with a group, ring, and order and structure, but also structures that we haven't covered, some of which are non-algebraic (ex. completeness (see uniformly continuous functions and uniform spaces), or topological)

Definition 9.1.6: Sub-Ring

Let R be a ring and $S \subseteq R$ a subset. Then S is a *subring* of R if it is still a ring under the same operations as R. In particular, a subring of a ring R is a subset $S \subseteq R$ with a ring operation such that the inclusion map $i : S \to R$ is a ring homomorphism [a].

[a]See proposition 1.4.8

Corollary 9.1.7: sub-ring condition

a *subring* of the ring R is a subgroup of R that is closed under multiplication

***Proof*:**
see exercise ref:HERE

It is simpler to prove a subset is a subring as compared to subgroups: you need to show it' an abelian sub-group that is still closed under multiplication. Notice that there is no special symbol for a subring like there as for subgroup. This will be a recurring pattern as we continue to explore algebra: there is no special symbol for "subobjects". We will always say "Let S be a subring of R" or "Let $S \subseteq R$ be a subring of R". Another consideration to keep in mind is that the identity of a subring need not be the same (since an inclusion homomorphism $S \hookrightarrow R$ need not map the identity to itslef): If $R \times S$ is a ring, $R \times \{0\}$ is a subring. What is it's [multiplicative] identity?

One subring is available in every ring and measures how far the multiplication is from being commutative. It is the ring-theoretic analogue of definition 2.2.4, and unlike the group case it is a subring rather than merely a subgroup.

Definition 9.1.8: [Ring] Center

Let R be a ring. The *center* of R is

$$Z(R) := \{c \in R \mid cx = xc \text{ for all } x \in R\}$$

the set of elements commuting with every element of R. A ring is commutative exactly when $Z(R) = R$.

That $Z(R)$ is a subring is immediate from corollary 9.1.7: it contains 0 and 1, it is closed under subtraction since $(c - c')x = cx - c'x = xc - xc' = x(c - c')$, and it is closed under multiplication since $(cc')x = c(c'x) = c(xc') = (cx)c' = x(cc')$. It is commutative, being a set of pairwise commuting elements, so the center of any ring is a commutative ring.

Proposition 9.1.9: Properties of Rings

Let R be a ring. Then:

1. $0a = a0 = 0$ for all $a \in R$
2. $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$ (recall $-a$ is the additive inverse)
3. $(-a)(-b) = ab$ for all $a, b \in R$
4. The multiplicative identity is unique and $-a = (-1)a$.
5. Generalized Distribuitivity: Distributivity can be extended inductively.

***Proof*:**
All of these proofs take advantage of the abelian structure of R and distributivity. For example:

1. Using the fact that 0 is an additive identity and distribuitivity, we get:

$$0 \cdot a = (0 + 0)a = 0 \cdot a + 0 \cdot a$$

Since $0 \cdot a \in R$, there exists an additive inverse, thus

$$0 \cdot a - 0 \cdot a = 0 \cdot a + 0 \cdot a - 0 \cdot a$$

implying $0 = 0 \cdot a$. A similar argument can be made for $0 = a \cdot 0$

the other 3 proofs continue in a similar fashion.

:exercise Show that $(-1)^2 = 1$.

Consequences of no Multiplicative Inverse

Now that we know what is needed to be a ring, let's put names to common rings with properties added on. For each of these properties, I would recommend to go back and see which rings we covered have these properties. If you want to verify your intuition, you can check glossary (see part XIV).

The first thing is that the multipication operation is missing one of the axioms of a group: elements need not have a multiplicative inverse. If every element has a multiplicative inverse, we will give a name to such a ring:

Definition 9.1.10: Division Ring

A ring R with identity 1, where $1 \neq 0$ is called a *division ring* (or a *skew field*) if every non-zero element $a \in R$ has a multiplicative inverse, $b \in R$ such that $ab = 1$. It can be proven this implies that $ba = 1$

If furthermore, multipication is commutative, the resulting ring is a commutative division ring, whcih is called *field*:

Definition 9.1.11: Field

Let F be a divison ring. If F is also *commutative*, then F is called a *field*.

Sometimes, the symbol k (from German Körperor[5]) or $\mathbb{F}$ will be used for a field The field is the "smallest" mathematical structure in which we can perform all the standard arithmetic operations $+, -, \times, \div$ in the usual way. If R is a field, then it is as though there is two compatible abelian group structures (both are associatve, have an identity, have an inverese, both commute), the two operations being linked through distributivity. In particular, it's like $(R, +)$ and $(R - \{0\}, \cdot)$ are combined through distributivity. . If R is a field, we can call the sub-rings of R a sub-field. For example, $\mathbb{Q}$ is a subfield of $\mathbb{R}$.

(TBD revise this in light of the following fact: If F and K are field, $(F, +) \cong_{\mathbf{Grp}} (K, +)$ and $F^{\times} \cong_{\mathbf{Grp}} K^{*}$, then is it true that $F \cong_{\text{Ring}} K$? I saw that no, and in principle there can be uncountably many combinations of possible field structures)

Example 9.1.12: Exercise

Prove that any subfield of $\mathbb{R}$ must contain $\mathbb{Q}$

Proof. Let $K \subseteq \mathbb{F}$ be a subifeld. First show the identity of $\mathbb{R}$ is the identity of K. Then $\mathbb{Z}$ is there (why?). Then $\mathbb{Q}$ is there (why?) $\square$

define a quandle (an exampleis matrices over non-commutative rings and $(a^b)^c = (a^c)^{(b^c)}$. Show that conjugation of a group defines a quandle. This is more useful in set theory

Zero divisors and Units

If a ring is not a division ring, then the cancellation property does not necessarily hold (see proposition 1.1.17). In groups, we constantly take advantage of the fact that $ab = ac \Rightarrow b = c$. This proof relied on the [unique] inverse of elements. However, there are rings in which this does not happen

Example 9.1.13

Let $R = \mathbb{Z}/6\mathbb{Z}$ with the usual addition and multiplication. First, show that it's a ring.

Then, notice that [2] times any number in $\mathbb{Z}/6\mathbb{Z}$ is not [1] (i.e. there does not exist $[a]$ such that $[2][a] = [1]$):

$$[2][1] = [2]\,, [2][2] = [4]\,, [2][3] = [0]\,, [2][4] = [2]\,, [2][5] = [4]$$

Furthermore, there exists an element ([3]) such that [2] times it gives the additive identity: $[2][3] = [0]$

So it breaks down! You might wonder if for any $a \in R$ if there exists an element b such that $ab = 0$ or $ab = 1$, but that need not be the case:

[5]some words about whatever German Mathematician pioneered fields that lead to this notation

Example 9.1.14: No Multiplicative Inverses

In $\mathbb{Z}$ with the usual addition and multiplication, take any $a \in \mathbb{Z}$, $a \notin \{0,1\}$. Then there does not exist an element b such that $ab = 1$, which is shown by $\mathbb{Z}$ not having multiplicative inverse, but also $ab = 0$ cannot happen in $\mathbb{Z}$ for any $b \in \mathbb{Z}$

Thus, one cannot imply the other in general

You might further wonder if for any $a \in R$ that only $ab = 0$ happens or only $av = 1$ happens, that is, if there exists an element $b \in R$ such that $ab = 0$, then there does *not* exist an element $v \in R$ such that $av = 1$ (or vice versa). However, this again is not the case! What if $ab = 0$, or $ab = 1$, does that means $ba = 0$ or $ba = 1$? Also no (in the non-commutative case)!

Example 9.1.15

1. Take S to be the set of all infinite sequences. This set, under component-wise addition, is a group. Then take $\text{End}(S)$, making function composition multiplication. Then this is a ring with identity.

2. Prove that

$$P(a_1, a_2, ..., a_n, ...) = (a_1, 0, ..., 0, ...) \tag{9.1}$$
$$L(a_1, a_2, ..., a_n, ...) = (a_2, a_3, ..., a_{n-1}, ...) \tag{9.2}$$
$$R(a_1, a_2, ..., a_n, ...) = (0, a_1, ..., a_{n+1}, ...) \tag{9.3}$$
$$0(a_1, a_2, ..., a_n, ...) = (0, 0, ..., 0, ...) \tag{9.4}$$
$$1(a_1, a_2, ..., a_n, ...) = (a_1, a_2, ..., a_n, ...) \tag{9.5}$$
$$\tag{9.6}$$

   are all elements in $\text{End}(S)$ ($L, R, P, 0, 1 \in \text{End}(S)$)

3. Show that $LP = 0$, but $LR = 1$. Then, show that $PL \neq 0$ (so $LP = 0$, but $PL \neq 0$)

4. You just showed that $LR = 1$. Show that $RL \neq 1$.

Since it's not commutative, it's not imediately clear that since we found an example where $ab = 0$ but $ac = 1$, then we can find the same but multiplying from the left side. However, we can construct exactly such an example from the previous example:

1. Define $\text{End}(S)^{op}$ to be the object where function composition is reversed:

$$f \cdot g = g \circ f$$

   show that $\text{End}(S)^{op}$ is still a ring. Convince yourself that all the previous elements $(1) - (5)$ are in $\text{End}(S)^{op}$.

2. Show that $PL = 0$, but $RL = 1$. Then, show that $PL \neq 0$ (so $LP = 0$, but $PL \neq 0$)

So, we cannot carry over our intuition of the cancellation property into general rings! However, thanks to associatvity, there is *one* relation we can prove: if $ab = 0$, then there does not exist a c such that $ca = 1$. That is, if there exists an element b such that $ab = 0$, then there does not exist a c

such that $ca = 1$:

Proposition 9.1.16: Partial Result for ab and ca

Let $a \in R$ such that there exists an element $b \in R$ such that $ab = 0$ (or $ba = 0$). Then there does not exist an element c such that $ca = 1$ (or $ac = 1$), that is:

$$\begin{matrix} ab = 0 \\ ca \neq 1 \end{matrix} \quad \text{and} \quad \begin{matrix} ab = 1 \\ ca \neq 0 \end{matrix}$$

***Proof*:**
Take $a \in R$ as in the proposition. Let's say there exists a $b \in R$, $b \neq 0$, such that $ab = 0$. Let's say there is a c such that $ca = 1$, $c \neq 0$. Then:

$$b = 1b = (ca)b = c(ab) = c(0) = 0$$

a contradiction since $b \neq 0$

Let's say there exists a $b \in R$, $b \neq 0$, such that $ba = 1$. Let's say there is a c such that $ac = 0$, $c \neq 0$.

$$c = 1c = (ba)c = b(ac) = b \cdot 0 = 0$$

a contradiction since $c \neq 0$

Note that as a consequence of this theorem, if an element a has a left-sided inverse l and ar right-sided inverse r, then $l = r$, since:

$$l = l1 = l(ar) = (la)r = 1r = r$$

Since multiplicative inverses can either multiply to 1, to 0 or to neither, we will take a moment to name elements that can multiply into 1, 0, or neither:

Definition 9.1.17: Zero Divisors and Units

Let R be a ring.

1. Let $a \in R$ be a nonzero element. Then a is called a *left* (resp. *right*) *zero divisor* if there is a nonzero element $b \in R$ such that ab (resp. ba) equals to 0. If $ab = ba = 0$, then a is called a *two-sided zero divisor*, or simply a *zero-divisor*

2. Assume $1 \neq 0$. An element u of R is called a left (resp. right) *unit* in R if there is some v in R such that vu (resp. uv) equals to 1. If $uv = vu = 1$. , then u is called a *two-sided unit*, or simply *unit* for short

If R is commutative, then there is no distinction between left and right zero-divisors/units.

The term zero-divisors comes from the following observation: if $ab = n$ in a commutative ring for simplicity, then we may say that $a|n$ and $b|n$ (i.e. a, b divide n). Then if $ab = 0$, then "$a\big|0$" and "$b\big|0$", so a, b divide 0 (they are *zero-divisors*). This notion of divisibility shall further be explored in chapter 10. Note that if u is a unit of R so that there is some v such that $uv = vu = 1$, then v is unique (think of the same proof for groups). The same is not the case for zero-divisors (try playing around in $\mathbb{Z}/10\mathbb{Z}$). Since any unit must be a two-sided inverse, we can define the following:

Proposition 9.1.18: Group of Units

Let R be a commutative ring, and let $R^\times$ denote the set of all units of R. Then $R^\times$ is a group under $\cdot$, and is called the *group of units of R*

Proof:
Associativity comes from associativity of multiplication, multiplicative identity comes from the fact that 1 is a unit, and inverses come from the definition of units.

It only remains to show that $\cdot$ is indeed a well-defined operation; does it map units of R to units of R. Take $a, b \in R^\times$. Since both are units, then there exists a u and v st $au = ua = bv = vb = 1$. Then I claim that vu is the element such that $(vu)(ab) = 1$. Indeed:

$$(vu)(ab) = v(ua)b = v \cdot 1 \cdot b = v \cdot b = 1$$

Thus $ab \in R^\times$, showing it is indeed well-defined as we sought to show

Sometimes, the group of units of R is denoted $U(R)$, R^*, or $E(R)$ (from German *Einheit*, meaning "oneness") Notice how this let's us characterize division rings and fields in a new way: if every non-zero element of a ring (resp. commutative ring) is a unit, the ring is then a division ring (resp. field). Note too that it is not always obvious when a ring can be characterized as a field in this way:

Example 9.1.19: Every Element is a Unit

Let D be a not-a-perfect square rational number [a] Then take $\mathbb{Q}(\sqrt{(D)}) \subseteq \mathbb{C}$,

$$\mathbb{Q}(\sqrt{(D)}) := \left\{ a + b\sqrt{(D)} \;\middle|\; a, b \in \mathbb{Q} \right\}$$

Note that if $D > 0$, then $\mathbb{Q}(\sqrt{D}) \subseteq \mathbb{R}$. This set is indeed a ring since

$$\begin{aligned} &\forall x, y \in \mathbb{Q}(\sqrt{5}) \qquad a + b\sqrt{D} - c - d\sqrt{D} = (a - c) + (b - d)\sqrt{D} \\ &\forall x, y \in \mathbb{Q}(\sqrt{5}) \qquad (a + b\sqrt{D})(c + d\sqrt{D}) = (ac + bdD) + (ad + bc)\sqrt{D} \end{aligned}$$

showing it's indeed a [commutative] ring.Note that D being not-a-square implies every element can be written uniquely. If it were a square, say 9, then

$$2\sqrt{9} = 6$$

meaning two elements were written the same way! Because of this, if a, b are non-zero, then $a^2 - b^2 D$ is non-zero. For example, if $D = 4$, then

$$4^2 - 2^2 4 = 16 - 16 = 0$$

but Since D is not-a-square,

$$a^2 - b^2 D = 0 \Rightarrow a^2 = b^2 D \Rightarrow \frac{a^2}{b^2} = D \Rightarrow \frac{a}{b} = \sqrt{D}$$

which since D is not-a-square, it's square-root is not a rational number [b], hence a contradiction.

Thus, $a^2 - b^2 D$ is non-zero if a, b are non-zero. The point of this is to notice that

$$(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - b^2 D$$

then $a + b\sqrt{D}$, $a - b\sqrt{D}$ must both be non-zero! Thus, we can define

$$\frac{a - b\sqrt{D}}{a^2 - b^2 D}$$

to be the inverse of $a + b\sqrt{D}$! This shows that every non-zero element of $\mathbb{Q}(\sqrt{D})$ is a unit, and hence it's a field!

[a]perfect square number: $x \in X$ such that $\sqrt{x} \in X$

[b]A similar proof to that of $\sqrt{2}$ is not rational!

This also means that since every element in a field has a multiplicative inverse, a field has *no* zero-divisors.

As we saw, if x is *not* a zero divisor, this does not imply that x *is* a unit, and vice-versa, as we have seen with the integers. However, in the finite case, this *does* hold

Proposition 9.1.20: In finite ring, not zero divisor $\Rightarrow$ unit

If R is a finite ring, and $x \in R$ is *not* a zero-divisor, then x is a unit

***Proof*:**
The key idea of finiteness is that we can take advantage of forcing injective maps being surjective. Withought loss of generality, when there are zero divisors, we take the left zero divisors. The proof for right zero-divisors works identically by switching the order of the elements.

Let's say $x \in R$ is not a left zero divisor so that for no element $y \in R$ there is $xy = 0$. If $x = 1$, we're done, so let's say $x \neq 1$. Define the map $y \mapsto xy$. [a]. Since for no y there is $xy = 0$, then this map is injective. Since R is finite and the map is from R to R, then this map is also surjective. Thus, there exists a y' such that $xy' = 1$. It remains to show that $y'x = 1$.

The element x can either be a right zero-divisor or it can not be. If it is not a right-zero divisor, then the map the map $y \mapsto yx$ would be injective, and hence surjective, and we can find an element y'' such that $y''x = 1$. So, let's say x was a right zero-divisor. If it was a right zero-divisor, then there would exist a $z \in R$ such that $zx = 0$, and x would be a right, but not a left zero-divisor. However, there exists a y' such that $xy' = 1$, which is a contradiction by proposition 9.1.16. Therefore, in the finite case, if x is a right zero-divisor, it would also have to be a left zero-divisor

Therefore, x is not a zero-divisor, so we can again define the map $y \mapsto yx$ which is also injective, and so there exists a y'' such that $y''x = 1$. Finally, notice that:

$$y' = 1 \cdot y' = (y''x) \cdot y' = y'' \cdot 1 = y''$$

showing that $y' = y''$ and so $xy' = y'x = 1$, fulfilling the condition for x being a unit, as we sought to show.

[a]Tangentially, note that this function is not in general a ring homomorphism – can you see why?

No Zero-Divisors: Integral Domains

Numbers as they are generally known do not have zero divisors (ex. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$). A large part of the study of rings is the study of number theory, and so defining rings which do not have zero divisors is important for ease of communication:

Definition 9.1.21: Integral Domain

A commutative ring with identity $1 \neq 0$ is called *integral domain* if it has a no zero divisors

Example 9.1.22: Integral Domains

1. $\mathbb{Z}$ is the canonical example, even the name-sake
2. $\mathbb{Q}$ and $\mathbb{R}$ are integral domains. Note that every (nonzero) elements are also units
3. $(\mathrm{GL}_n(\mathbb{F}), +, \cdot))$ is an integral domain
4. $M_n(F), +, \cdot)$ is not an integral domain, since
$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$
5. Let $F = \mathrm{Hom}_{\mathbf{Set}}(X, R)$ be a collection of functions where $|X| > 2$ $R \neq \{0\}$. As noted in example 9.1.4 this is a ring. This ring is not an integral domain. Let $a, b \in X$, $a \neq b$, and $x \in R$, $x \neq 0$. Then define a function where $f(a) = 0, f(b) = x$, and the rest of the elements of f map to zero, and define $g(a) = x, g(b) = 0$, and the rest of the elements of g map to zero. Note that f and g are not the zero function since they are not identically zero, but $fg = 0$. Importantly
$$fg(a) = f(a)g(a) = 0x = 0 \qquad fg(b) = f(b)g(b) = x0 = 0$$
The set of functions from a set X to a ring R is rarely an integral domain when $|X| > 0$ and $R \neq \{0\}$, since it would have to be the case that $fg = 0$ implies that either f is zero everywhere or g is zero everywhere. However, they do exist (an important example in complex analysis is the ring of power-series, or more generally the ring of analytic functions).

Sometimes the term *domain* used for the same type of ring without the commutative property:

Definition 9.1.23: Domain

A ring in which there are no zero-divisors is called a domain

The nicest property of integral domains is that the cancellation law holds!

Proposition 9.1.24: Cancellation Law for Integral Domains

Assume a, b, c are elements of any ring with a not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$.

***Proof*:**

$$\begin{aligned} ab &= ac \\ \Leftrightarrow a(b-c) &= 0 \\ \Leftrightarrow a = 0 \text{ or } b - c &= 0 \\ \Leftrightarrow a = 0 \text{ or } b &= c \end{aligned}$$

With this property of integral domains, proposition 9.1.20 can be expanded upon:

Corollary 9.1.25: Finite Integral Domain is a Field

If R is a finite integral domain, then R is a field

Just like in proposition 9.1.20, the trick in this proof is a neat use of function to capture the idea of how an element acts on every other element in an object, not unlike how we can think when working with group actions:

***Proof*:**
Let R be a finite integral domain. and let a be a nonzero element of R. By the cancellation law the map $x \mapsto ax$ is an injective function. To see this, let's say it was not injective so $ax = ay$ but $x \neq y$. Then

$$ax - ay = 0 \quad \Rightarrow \quad a(x-y) = 0$$

Since a is non-zero and there are no zero divisors, it must be that $x - y = 0$, so $x = y$.

Now, since the map is injctive, then $|\varphi(R)| = |R|$, meaning it must be surjcetive [a]. In particular, there is some $b \in R$ such that $ab = 1$. A similar procedure can be done to show that there exists a c such that $ca = 1$. Since units are unique (see exercise ref:HERE) $c = b$, so a is a unit in R. Since a was arbitrary nonzero element, R is a field.

[a] Note that for finite sets, it's impsosible to have a set with "less" elements, but be smaller cardinality, while it is possible for infinite sets, like $2\mathbb{Z} \subseteq \mathbb{Z}$, but $|2\mathbb{Z}| = |\mathbb{Z}$

9.1.26 Note If we drop commutativity and add that every element is a unit (getting a divsion ring), then this theorem still holds (partly by proving that every finite division ring is necessarily commutative)! The proof of this is relegated to Rerpresentation theory in part X since it requires a lot more build-up, but it's interesting to note.

Though integral domains are nice, they loose a every important property: If R is an integral domain, then $R \times R$ is *not* an ingeral domain

Example 9.1.27: product of integrals domains is not an integral domain
Let R be an arbitrary integral domain. Then take $R \times R$, with addition and multiplication done point-wise. Then

$$(1,0) \cdot (0,1) = (0,0)$$

meaning there are zero-divisors, and so $R \times R$ is not a integral domain A particular consequence of this is that the direct product of fields is not a field (since fields are integral domains) More generally, integral domains do not satisfy the universal property of products.

A summary of all these properties is in section 9.1.2.

Exercise 9.1.1

1. An element $e \in R$ is called *idempotent* if $e^2 = e$. Does there exist a ring R that has exactly 3 idempotent elements? Find one or show it's not possible.
2. We'll prove that every finite integral domain is a ring using the pigeonhole principle

   a) Show... click here for this proof
3. We will define the complex numbers. Take the real numbers $\mathbb{R}$, and let i be some variable. Define
$$\mathbb{C} := \langle \{\mathbb{R}, i\} \mid i^2 = -1 \rangle$$
   That is, we will take all possible combinations of the elements from $\mathbb{R}$ and the element i, and impose the relation $i^2 = -1$ on i. To get an idea of what's in $\mathbb{C}$:
$$i \in \mathbb{C},\ ci \in \mathbb{C},\ a + bi \in \mathbb{C} \cdots$$
   Prove that $\mathbb{C}$ is a ring. We will later give a slightly more nuanced definition of of $\mathbb{C}$, see ref:HERE.
4. (why not extend to 3 binary operations) (this exercise shows that if you want even more Yoneda compatibilities then you crash out and get trivial structures)
5. Let' say $s \neq 1$ and $s^2 = s$. This propety is called idempotence. Show that the only idempotent element in an integral domain is 0 and 1
6. Let R be a ring with $1 \neq 0$. Is $\varphi : R \to R$, $\varphi(r) = 0$ a ring-homomorphism? If so, prove it, and if not, would it be a rng-homomorphism (i.e. would the image be a rng)?
7. Show that the only idempotent elemnt in a field' is 0 or 1 (there exists a proof using properties of a field but not properties of integral domain)
8. A field is said to be characteristic n if there exists a number n st $\underbrace{1 + 1 + 1 \cdots + 1}_{n \text{ times}} = 0$. If no such number exists, we say that the field has characteristic 0. Give an
9. Let $u \in R$ be a unit and $R \to S$ a ring homomorphism. Then $\varphi(u)$ is a unit. Is the same true for zero-divisors? What about non-zero-divisors? example (or examples) of fields with characteristic 0, 2, and 5.
10. Show that if $\mathbb{F}$ is a field of characteristic 0, then $\mathbb{Q} \subseteq \mathbb{F}$.
11. (For those who know some Analysis and want examples of *rngs*) In the process of defining the Fourier series, it is necessary to define the convolution product between two continuous functions (on compact support, or more generally L^2)
12. (ibid last comment) Let X be a locally compact space (i.e. for every $x \in X$, there exists a compact set $K \subseteq X$ such that $x \in K$, that is every point has a compact neighbourhood). Let $C_0(X)$ be the set of continuous complex functions $f : X \to \mathbb{C}$ that vanish at infinity (for every $\epsilon > 0$, there exists a K such that $|f(x)| < \epsilon$ for all $x \in X - K$). Then $(C_0(X), +, \cdot)$ with function addition and multiplicatoin is a rng, but not a ring (namely, the function $1(x)$ does not vanish at infinity). This rng is used quite often in analysis: it comes up in Fourier

analysis, in defining Radon Measures, and in the study of C^*-algebra to name a few examples (see Folland or ref:HERE).

13. (do the example in Commutative algebra 2 (Rings, ideals, modules) at timestamp 13:33)

    (somehow make this a comment??) Loosely speaking, rngs correspond to locally compact spaces, while rings correspond to compact spaces. For analysts, working over compact spaces is much to restrictive ($\mathbb{R}^n$ is not compact), while locally compact spaces are much more natural to work with ($\mathbb{R}^n$ is locally compact). Furthermore, recall we have mentioned we can turn a rng into a ring by doing $\mathbb{Z} \oplus R$ to some rng R. If we were interested in keeping nice analytical properties, we would instead take $\mathbb{R} \oplus R$. (ex. $\mathbb{R} \oplus C_0(X)$). This would correspond to the 1-point compatification of X: $C_0(X \cup \{\infty\}) \equiv \mathbb{R} \oplus C_0(X)$!

14. Show that $(F, +, \cdot)$ be a ring. Show that $(F, +, \cdot)$ is a field if and only if $(F, +)$ is an abelian group and the set $F^\times$ is equal to $F - \{0\}$.

15. Let R be a ring and R^{op} denote the ring where the underlying set and addition operation is the same, and the multiplication is defined as $x * y = yx$ where yx happens in R. Show that $\text{id} : R \to R^{\text{op}}$ is not necessarily a ring isomorphism (it is sometimes called an *anti-isomorphism*). When is it a ring isomorphism? Show that if R is a division ring, then the map $x \mapsto x^{-1}$ *is* a ring isomorphism.

16. show that if $x^n = 0$, then $(x - 1)$ is an invertible element. An element where $x^n = 0$ is called a *nilpotent element*.

17. recall that $(x + y)^n = \sum_{k=1}^n \binom{n}{k} x^n y^{n-k}$. Show that the binomial formula holds if and only if $xy = yx$.

18. I the set of all left (resp. right) invertible elements a group? How that $Z(R)$ is a subring of R.

19. let G be a cyclic group of order n. For each $k \in \mathbb{Z}$, define $f : G \to G$, $f_k(g) = kg$. The n the map $k \mapsto f_k$ defines a ring isomorphism $\mathbb{Z}/n\mathbb{Z} \cong_{\mathbf{Ring}} \text{End}_{\mathbf{Ring}}(G)$ and a group isomorphism $(\mathbb{Z}/n\mathbb{Z})^* \cong_{\mathbf{Grp}} \text{Aut}_{\mathbf{Grp}}(G)$

9.1.1 Category of Ring

We now approach rings a 2nd time, but this time, we will take a more birds-eye view of the category or rings. We will see if this category has some properties different than the category of groups, and whether there is a general way of extending groups to form rings.

Proposition 9.1.28: Category of Ring

The collection of Rings along with ring homomorphisms form a category. This category will be denoted **Ring**.

***Proof*:**
See exercise ref:HERE

Since **Ring** is a category, it will be verified what are the initial/final objects, and whether there are product, sub-, and quotient objects, and whether quotient objects can be classified through kernels. We can further ask about the relation between **Group** and **Ring**, and ask whether there is an analogous concept of ring-actions. In this section we will drop the subscript and write $\mathrm{Hom}(R, S)$ or $\mathrm{End}(R)$ instead of $\mathrm{Hom}_{\mathbf{Ring}}(R, S)$ or $\mathrm{End}_{\mathbf{Ring}}(R)$

The fact that sub-objects and product objects exist in rings (that is, subrings and product rings) has already been explored in definition 9.1.6 and example 9.1.4[5], though not in categorical rigour (which the reader can practice in exercise ref:HERE if they are so inclined). We will mention some more about product of rings and the consequences of integral domains not having products as an interesting example of category with no products. Quotient objects will be revisited in more detail in section 9.3.

Rings and $\mathbb{Z}$

The first thing to check is whether **Ring** has initial and final objects. In definition 1.4.6, we introduced the concept of initial and final group. We found there to not really be an intersting initial and final group: only the identity group. The same is true for the final object of rings. However, the zero ring cannot be the initial: if R is the zero ring, and we have a homomorphism $R \to S$, then S must be the zero ring (see exercise ref:HERE). There is still an initial object in the category or rings: we have our first example of a non-trivial initial object!

Proposition 9.1.29: $\mathbb{Z}$ is an Initial Ring

The ring $\mathbb{Z}$ is an intial ring

***Proof*:**
Let R be any ring (with identity) and let $\varphi : \mathbb{Z} \to R$ such that

$$\varphi(z) = 1 + 1 + 1 \text{ (z times) } = z \cdot 1$$

this is a ring homomorphism. For

$$\varphi(a + b) = (a + b) \cdot 1 = a + b = \varphi(a)\varphi(b) \qquad \varphi(ab) = (ab) \cdot 1 = a(b \cdot 1) = (a \cdot 1)(b \cdot 1) = \varphi(a)\varphi(b)$$

Furthermore, if $\varphi' : \mathbb{Z} \to R$ was another ring homomorphism, then by proposition 9.1.9

$$\varphi'(1) = 1\,, \varphi'(2) = 1 + 1\,, \cdots$$

that is, by the fact that the identity *must* be distributed in this fashion, we are forced that

$$\varphi = \varphi'$$

making it unique

Note that if rings are not defined to have a multiplicative identity, then there is no unique ring wihtout a multiplicative identity, see exercise ref:HERE (exercise 2.16 in Aluffi part III)

So how can we use this result? The way I think about it is something "integer-like" in every ring [with identity]. It might not be commutative, it might not be injective (the codomain might even be finite), or it might be uncountably big (like $\mathbb{R}$ or $M_n(\mathbb{C})$) but there is still something integer like.

For this reason, asking questions like "what properties of the integers do rings have" is actually a very sensible question! In fact, that will be essentially what we do in chapter 10!

For example, if we take $x \in \mathbb{Z}$, then $x = p_1 p_2 \cdots p_n$, that is x has a *unique prime factorization*. In the next chapter, we will study rings which have this similar property (they will be called *unique factorization domains*). Similarly, to find the $gcd(a, b)$, we can use the euclidean algorithm which algorithmically shows how much the set of divisors of the two numbers intersect. This is very handy since we can in very few steps find the common divisors of two numbers of extremely large numbers. Rings in which the notion of the euclidean algorithm is well-dfined will be called *euclidean domains*. Finally, a little more difficult but important concept, any number can be decomposed into *unique* prime numbers in $\mathbb{Z}$. We will find the equivalent of "numbers" in rings to be "ideals" (the original name was "ideal numbers" of rings). If every ideal can be generated by a single element $x \in R$, the ideal is called a *principal ideal*. Thus, a ring which is has "numbers" (ideals) which are generated by one element (just like a number just needs itself to describe, well, itself) will be called a *principal ideal domain*.

Thus, in the folloing chapters, some basic theory will extensively be used. Take a moment to review section 0.2 for a refresher on some basic tools of number theory.

Rings and $\mathrm{End}(G)$

You might ask how should we choose which of the axioms are "nice" axioms to add to our structure. One way of doing so is by thinking about the multiplication action being permutations of the elements, not unlike ho the group operation can be thought of as the permutation of elements (i.e. any group $(G, +)$ is a subgroup of $\mathrm{Sym}(G)$) [7]

We saw in definition 1.4.9 that if G is an abelian group that $\mathrm{End}(G)$ is an agelian group under pointwise addition. Notice that there is actually already a 2nd operation, since function can compose:

$$(\mathrm{End}(G), +, \circ)$$

Then, it can be checked that composition here will make this into a ring (as shown in example 9.1.4)

You might then ask converse: given ring $(R, +, \cdot)$, can we find a ring isomorphism to $\mathrm{End}((R, +))$? The answer is yes! This will actually turn out to be the Cayley's theorem for rings (theorem 4.5.3)! We will not cover the ring version of Cayley's theorem, for a more general theorem called Yonneda's Lemma cover's the idea of using objects to "represent" other objects (ref:HERE)

If you want to check, the natural ring embedding from $(R, +, \cdot)$ to $\mathrm{End}(R, +)$ is

$$r \mapsto (x \mapsto xr) \qquad \text{or} \qquad r \mapsto (x \mapsto rx)$$

depending on the convention on composition.

:exercse

Let $\mathrm{Aut}_{\mathbf{Ab}}(R)$ be the set of group-automorphisms on R. Show that $U(R) \cong_{\mathbf{Ab}} \mathrm{Aut}_{\mathbf{Ab}}(R)$

let R be a commutative ring. Show that $\lambda : R \to \mathrm{End}_{\mathbf{Ab}}(R)$ by $r \mapsto (y \mapsto ry)$ is indeed a ring homomorphism. If R is not commutative, show that it's almost abelian.

[7]Research **Ab**-enriched categories for information about this process

Products of Rings

Rings have products: Given any to rings A and B, there exists a ring $A \times B$ along ring homomorphisms $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ such that for any C here $f_A : C \to A$ and $f_B : C \to B$, there exists a unique $f! : C \to A \times B$ such that the folloing diagram commutes:

$$\begin{tikzcd}
 & C \arrow[dl, bend right, "f_A"'] \arrow[d, "f!"] \arrow[dr, bend left, "f_B"] & \\
 & A \times B \arrow[dl, "\pi_A"] \arrow[dr, "\pi_B"'] & \\
A & & B
\end{tikzcd}$$

In this way, products in **Ring** are like products in **Grp**. A similar situation occurs for coproducts.

However, as mentioned earlier, if we retirct to *integral domains* and take the category whose objects are integral domains (this is indeed still a category as can be verified), they are no longer products, as demonstrated in exmaple 9.1.27!

Since since all fields are integral domains, this means that products are not defined for fields! As a consequence of this, we cannot repeat the same trick as in chapter 5 in creating larger groups using products. Thus, extensions of fields (similar to how they were defined for groups in section 3.6) will have to be used to study the creation of larger fields. To study field extensions, we must first study ring homomorphisms more extensively; the particular way in which ring homomorphissm will be used in constructing larger fields requires more build-up, and so the study of fields is relegated to part IV.

9.1.2 Summary

Keeping track of these properties and their relation to everything can be a bit to take in. It might help to have a visual too:

$$\begin{tikzcd}
 & \text{field} & \\
\text{integral domain} \arrow[ur, "\forall a \exists b \text{ } st \text{ } ab{=}1 \text{ } (+1)"] & & \text{division ring} \arrow[ul, "ab{=}ba"'] \\
\text{commutative ring} \arrow[u, "\forall a \nexists b \text{ } st \text{ } ab{=}0"] & & \text{domain} \arrow[u, "\forall a \exists b \text{ } st \text{ } ab{=}1 \text{ } (+1)"'] \arrow[ull, "ab{=}ba"] \\
 & \text{ring} \arrow[ul, "ab{=}ba"] \arrow[ur, "\forall a \nexists b \text{ } st \text{ } ab{=}0"'] & \\
 & \text{rng} \arrow[u, dashed, "\text{add identity } 1"] &
\end{tikzcd}$$

The *rng* can actually have more arrow's extending from it to commutative ring, domain, and integral domain, but the diagram would be cumbersome, so I opted to make it a dashed line from *rng* to *ring*, since, it is not needed to "upgrade" those rings until we get to division ring and fields. I added a (+1) where we really need the multiplicative identity to define the next concept.

Notice that if a commutative ring only has units, it is then a commutative division ring, and hence a field. The line as not drawn so as to not clutter the diagram.

look at comment:

The following categorical results are also noted:

1. $\mathbb{Z}$ is the initial object of rings
2. All rings are subrings of $\text{End}((R, +))$
3. Products exist for rings, but not for integral domains (and so not for fields).

Exercise 9.1.2

1. Since $\mathbb{Z}$ is initial, for any ring R and natural homomorphism $\mathbb{Z} \to \mathbb{R}$, $\text{im}(\mathbb{Z})$ exists is a subring. Show that $\text{im}(\mathbb{Z})$ is the *smallest* subring of any ring. Show further that $\text{im}(\mathbb{Z})$ is either $\mathbb{Z}_n$ for any $n > 0$ or $\mathbb{Z}$. If $|\text{im}(\mathbb{Z})| < \infty$, then we say the ring R has *characteristic* n, and if $\text{im}(\mathbb{Z}) \cong \mathbb{Z}$ then R has *characteristic 0*.
2. Let R be a ring and define R^{op} to be the ring with the same underlying set and addition operation, and multiplication defined by $x * y = yx$, where yx happens in R. Show that R^{op} is a ring, and that $(\_)^{\text{op}} : \mathbf{Ring} \to \mathbf{Ring}$ is a functor (where do ring homomorphisms map to?). Is it covariant or contravariant?
3. Let $a, b \in R$ such that $ab = ba$. Show that ab is invertible if and only if a and b are invertible. Generlize your result to show that if $ab \neq ba$, then if ab invertible, then a is left-invertible, and b is right-invertible.
4. Show that $\text{End}_{\mathbf{Ab}}(\mathbb{Z}) \cong_{\mathbf{Ring}} \mathbb{Z}$.

9.2 Examples of Rings

The following are rings that will show up time and time again, and being aquainted with them is very useful. Many of these rings will be the starting point to a new field of study in mathematics. For example, Polynomial rings are pivotal in Commutative Algebra (see part V), and when the polynomial ring takes on coefficients from a field (i.e. a polynomial ring over a field), then they form the foundation for field theory (see part IV). Matrix rings are a common the "representation objects", the same way $\text{Sym}(X)$ was an object that can be used to "represent" groups (see part X chapter 51). Group rings can be thought of as the dual of the group of unity of rings, and also appear frequently in representation theory (see part X chapter 53) and group cohomology (see part XI chapter 57).

There is also an optional section on formal power series covering the elementary operations that can be performed on formal power series. These results are important for a class in complex analysis, however the formal power series ring will come up as examples for certain concepts introduced later in the book as well. These examples of formal power series will not be critical in understanding any material, and so they can be comfortably skipped.

9.2.1 Polynomial Rings

Polynomial Rings are one of the first examples we'll give of a [commutative] extension of a ring. Let R be a commutative ring [8]. What we'll be doing is extending R by an arbitrary unkown element x. In a sense, what we're doing is saying x is now an element of R, an then we figure out how to define this new structure (which we label $R[x]$) so that it's also a ring. Since $x \in R[x]$, then $x \cdot x \in R[x]$. Since we know nothing of x, we can only say that $x \cdot x = x^2$ is another element. We can continue this process inductively getting us every x^k for all $k \in \mathbb{N}$. For any other element r: $r \cdot x = rx$ where rx is a new element of $R[x]$. A similar reasoning to creating new elements applies to adding: $r + x$ is a new element. However, it need not be the case that an element has an inverse, so we do not define an x^{-1} such that $xx^{-1} = 1$[9]. All of these properties together give us the following definition:

Definition 9.2.1: Polynomial Ring

Let R be a ring and x be an indeterminante. Then the sum

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

with $n \geq 0$ and each $a_i \in R$ is called a *polynomial* in x with coefficients a_i in R. The set of all such polynomials is called the ring of *polynomial in the variable x with coefficients in R* (or *polynomial ring* if context permits) and will be denoted $R[x]$. Addition is defined as

$$(a_n x^n + \cdots + a_1 x + a_0) + (b_n x^n + \cdots + b_1 x + b_0) = (a_n + b_n)x^n + \cdots (a_1 + b_1)x + (a_0 + b_0)$$

where a_n or b_n may be zero in order for addition of polynomials of different degrees to be defined. Multiplication is perfomed by defining $(a_i x^i)(b_j x^j) = a_i b_i x^{i+j}$ and then extending to all polynomials by the distributive law; this is called the *convolution product*
The following are pertinent definitions with regards to polynomials.

1. If $a_n \neq 0$, then the polynomial is said to be of *degree n*, and $a_n x^n$ is called the *leading coefficient* (where the leading coefficeint of the zero polynomial is taken to be 0).
2. The polynomial is *monic* if $a_n = 1$.

For reference, here is the general formula for polynomial multiplication (i.e. the convolution product)

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \sum_{l=0}^{k} p_l q_{k-l} x^k$$

9.2.2 Remark If you have done Fourier Analysis (or intend to in the future), then the convolution of two continuous (more generally measurable) functions follow's the exact same properties as polynomial multiplication. In fact, polynomial multiplication would be a discrete convolution.

[8] In the exercises, there are questions concerning the study of non-commutative R forming a polynomial ring and why it is often preferred for R to be commutative

[9] polynomial rings where the indeterminate x has an inverse are called the *ring of fractions of R[x]*. see section 9.5

Note that R is always a subring of $R[x]$, which we can see through the natural inclusion map [10] $R \hookrightarrow R[x]$ of $r \mapsto r$.

Usually, when we manipulate polynomials we treat them as functions. It must be stressed that we are *not* treating the elements of $R[x]$ as functions. The study of the elements of $R[x]$ as functions will be done in part V, which will establish the largest connection between algebra and geometry in the 20th century.

If $R = F$ is a field, then every coefficient is a unit, meaning we can divide each coefficient a polynomial to make it *monic*, that is,

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \overset{1/a_n}{\mapsto} x^n + a'_{n-1}x^{n-1} + \cdots + a'_1 x + a'_0$$

where $a'_i = \frac{a_i}{a_n}$. We shall see that often-times many results *up to multiplication by unit* (meaning if the result holds on p, it will hold on cp, where c is a unit). For this reason, it is worth keeping the term *monic* in mind.

We can extend the definition of a polynomial ring to a polynomial ring over several variables by simply making $R[x]$ a polynomial ring over y, that is

$$R[x, y] = (R[x])[y] = R([y])[x] = R[y, x]$$

Continuing inductively, we can define a ring over a finite number of variable this way:

$$R[x_1, x_2, ..., x_{n+1}] = (R[x_1, x_2, ..., x_n])[x_{n+1}]$$

This process can be extended "indefinitely"[11] to define $R[x_1, x_2, ..., x_n, ...]$, in which every polynomial (i.e. elements of $R[x_1, x_2, ...,])$ has finitely many terms, and each term can have multiples of any of variables of $R[x_1, x_2, ..., x_n, ...]$, for example:

$$x_{30}^2 x_7^{23523} x_2 - r x_{100} x^{39} \in R[x_1, x_2, ...]$$

Multiplication in $R[x_1, x_2, ...]$ is the same as the multiplication in polynomial rings over finitely many variables, where we know the value for any term be expanding out the multiplication far enough to group like-terms. If $x_1^{d_1} x_2^{d_2} \cdots x_k^{d_k}$ is a term, then the *total degree* of the term is

$$d = d_1 + d_2 + \cdots + d_k$$

and the tuple $(d_1, d_2, ..., d_k)$ is the multi-degree of the term.

Properties of Polynomial Rings

To motivate the next theorem, let's say we have two polynomials of degree 1, $ax + b, cx + d \in R[x]$. We can ask what is the degree of $(ax + b)(cx + d) = acx^2 = (ad + bc)x + bd$. We might expect the degree to be 2. However, it is possible that a and c are zero-divisors: take $4x + 1, 4x + 1 \in (\mathbb{Z}/8\mathbb{Z})[x]$. In fact, in the example just stated, $4x + 1$ is in fact a unit in $(\mathbb{Z}/8\mathbb{Z})[x]$! If R is an integral domain, then some of its structure will be simpler:

[10] the map is natural since there is no other information needed to define this map besides R and $R[x]$. For a more rigorous definition of natural, appendix **??** needs to be read to reach section **??**

[11] For those uncomfortable with how the fact that the above process of defining a polynomial ring will always only produce polynomial rings with finitely many variables, the construction of $R[x_1, x_2, ...]$ is done using a *limit*, see ref:HERE

Proposition 9.2.3: Degree Of Polynomials

Let R be an integral domain and let $p(x), q(x)$ be nonzero elements of $R[x]$. Then

1. degree $p(x)q(x) =$ degree $p(x)$ + degree $q(x)$
2. the units of $R[x]$ are the units of R
3. $R[x]$ is an integral domain

***Proof*:**

1. This comes from the definition of multiplication of polynomials and that R is an integral domain, as the highests order term will be $x^{\deg(p(x)+\deg(q(x)))}$ and there is no zero-divisors to cancelt out the term.
2. Let $u \in R[x]$ be a unit so that there exists a $v \in R[x]$ such that $uv = 1$. Then by the degree formula, $\deg(uv) = \deg(u) + \deg(v)$ [a]. Since $\deg(uv) = 0$, then it must be that $\deg(u) = \deg(v) = 0$, showing that $v \in R$
3. Let $p(x), q(x) \in R[x]$ such that $p(x)q(x) = 0$. Then $0 = \deg(0) = \deg(p(x)q(x)) = \deg(p(x) + \deg(q(x))$, and so both $p(x), q(x)$ must be of degree zero and so in R. Since R is an integral domain, then it must be that $p(x) = q(x) = 0$, showing that $R[x]$ is an integral domain.

[a] notice again that R must be an integral domain for this to work

The defining property of commutative rings is that they are the free objects of commutative rings.

Theorem 9.2.4: Universal Property of Polynomial Rings

Let S be a commutative ring and let

$$\varphi : R \to S$$

be any ring homomorphism and let $s \in S$ be any element of S. Then there is a unique ring homomorphism

$$\psi : R[x] \to S$$

such that $\psi(x) = s$ and which makes the following diagram commute

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, "f"] & S \\ R[x] \arrow[ur, "\psi"'] & \end{tikzcd}$$

A nuance that must be mentioned is that polynomial rings may be over different rings. We thus may thus ask how a ring S is generated with respect to a ring R. As demonstrated earlier, $\mathbb{Z}$ is the initial object of **Ring**, and so in the arbitrary case $f : X \to S$ for some set X, we shall have a polynomial ring on the elements X over $\mathbb{Z}$. For those who are categorically inclined, try figuring out what is the category involved behind the scenes (this is similar to the construction for free groups in section 2.3.3)

***Proof*:**
The proof becomes easy to see once you realize a polynomial:

$$a_n x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0$$

becomes

$$\varphi(a_n)s^n + \varphi(a_{n-1})s^{n-1} + \cdots \varphi(a_1)s + \varphi(a_0)$$

A particular case of this is when $\varphi : R \to R$ is an identity map. In this case, we may link the notion of a [formal] polynomial to a polynomial function:

Definition 9.2.5: Evalutation Map

Let R be a ring and let $\alpha \in R$. The natural ring homomorphism

$$\varphi : R[x] \to R$$

which acts as the identity on R and which sends x to α is called the *evaluation at* α and is often denoted ev_α.
We say that α is a *zero* (or root) of $f(x)$ if $f(x)$ is in the kernel of ev_α.

The evaluation map can come in quite handy in many proofs that will be seen later.

Example 9.2.6
Move this to somewhere else

1. Take $\mathbb{Q}[x,y]$ ($= (\mathbb{Q}[x])[y]$, i.e. a polynomial ring over $\mathbb{Q}[x]$ in y). Is (x) a prime ideal? Take $\mathbb{Q}[x,y] \to \mathbb{Q}[x]$ by $p(x,y) \mapsto p(x,0)$. This is the evaluation map at 0.

   This maps is clearly surjective, since for every $p(x)$, $p(x,0)$ will map to it. This map is also clearly linear since

   $$\varphi(p(x,y) + q(x,y)) = \varphi\left(\sum_{i=1}^{n}(a_i)x^i + \text{y terms}\right) = \sum_{a_i}^{n} a_i = \varphi(p(x,y)) + \varphi(q(x,y))$$

   and since it's linear on every term, to check the multiplicative operation, it suffices to check on monomials, which is easy:

   $$\varphi(ax^n bx^m) = \varphi(abx^{n+m}) = abx^{n+m} = ax^n bx^m = \varphi(ax^n)\varphi(bx^m)$$

   and if y is aterm, y geos to zero so mulitplicativity for those terms is automatic. Therefore, φ is a ring homomorphism. It's kernel is clealry (y), meaning

   $$\mathbb{Q}[x,y]/(y) \cong \mathbb{Q}[x]$$

   and since $\mathbb{Q}[x]$ is an integral domain (y) is a prime ideal.

When working with functions, they are determined by the value at each point of evaluation. It must be stressed that in general, a polynomial is *not* determined by it's evaluation

Example 9.2.7: Polynomial Not Determined By Evaluation
[Polynomial Not Determined By EvaluationPolynomial Not Determined By Evaluation] Take $x^2 - x \in (\mathbb{Z}/2\mathbb{Z})[x]$. Does the evaluation of this polynomial uniquely determine $x^2 - x$?

9.2.2⋆ Formal Power Series

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

(A realisation i had that maybe should be integrated?: I just realised. So we know from complex that [complex] polynomial behaves like waves (since they are harmonic). We know that multiplication of polynomials introduces the operator of convolution. We know Fourier somehow gives you a notion finding the fourier-coefficients, i.e. finding the waves. Well, if F is the fourier transform, the fact that $F(f * g) = F(f) \cdot F(g)$ feels like it makes more intuitive sense!)

Any polynomial is a finite linear combination of the terms x^n. If we allow for infinitely many terms, we call such an object a *formal power series*, and write it like so:

$$f(x) = \sum_{i=1}^{\infty} a_i x^i \qquad a_i \in R$$

The reader may ask what does it mean to take an infinite linear combination, as we have defined the + notation to only take in two inputs, and if we recurse over the inputs we may have a finite linear combination. These are valid concerns which will be addressed in *Commutative Algebra* [8] (completion). For now, we may consider $f(x)$ in the above equation as "actually" being an infinite tuple[12]

$$(a_0, a_1, ..., a_n) \in \prod_{i=0}^{\infty} R$$

however we should take this identification as simply insuring that $f(x)$ is well-defined; we shall not write $f(x)$ in the notation of tuples and from now on shall assume we can take infinite sums. The collection of all formal power series are certainly closed under termwise addition. We must make sure multiplication is still well-defined. For multipilcation to be well-defined, each coefficient must be well-defined. Fortunately, this is the case. If $p(x)$ and $q(x)$ where

$$\begin{aligned}
p(x)q(x) &= \left(\sum_{i=0}^{\infty} a_i x^i\right)\left(\sum_{j=0}^{\infty} b_j x^j\right) \\
&= a_0b_0 + (a_0b_1 + a_1b_0)z + (a_0b_2 + 2a_1b_1 + a_2b_0)z^2 + \cdots \\
&= \sum_{n=0}^{\infty}\left(\sum_{i=0}^{n} a_i b_{n-i}\right) x^n \\
&= \sum_{n=0}^{\infty} (A_n * B_n)\, x^n
\end{aligned}$$

[12] this may push the concern in asking how $\prod_{i=0}^{\infty}$ is defined. For those with this concern, they may skip to chapter 25 or they may take for granted that we may always make a countable number of choices and remember the order

where $A_n = \sum_{i=0}^{n} a_i$ and $B_n = \sum_{i=0}^{n} b_i$ and $A_n * B_n$ is equal to the the summand above and is called the *convolution* of the two finite sequences. Since addition and multiplication is well-defined, we have a ring:

Definition 9.2.8: Formal Power Series

The collection of all infinite R-linear combinations of $\{1, x, x^2, ...\}$ forms a ring known as the ring of *formal power series* over R, and is denoted $R[[x]]$

The term "formal" comes from how we often associate to an object $f \in R[[x]]$ the evaluation function. If we had a function with a domain and codomain for which $r \mapsto \sum_{i=0}^{\infty} a_i r^i$, we would call this function a *power series*. However, for such a function to be defined, we require the ability to unambiguously define summing infinitely many elements of a ring. Such considerations require the introduction of a topology on our ring, something we shall not cover. We shall remark that it is very commmon in compex analysis to work with power series and choose $R \in \{\mathbb{R}, \mathbb{C}\}$ with the usual euclidean topology, leading to many very important results about complex-differentiable functions. There is one value we may unambiguously define without the need for any convergence considerations:

$$f(0) := a_0$$

Proposition 9.2.9: Formal Power Series Integral Domain

Let $R[[x]]$ be the formal power series over R. Then $R[[x]]$ is an integral domain

***Proof*:**
proof here

Given that $R[[x]]$ is the extension of $R[x]$, it may see intriguing to think that these two rings share very similar properties. However, the "infinitude" of $R[[x]]$ will make the ring act rather differently. The following may be the most striking change when extending to infinite polynomials:

Lemma 9.2.10: Existence of More Units

Let $R[[x]]$ be the formal power series over a ring R. Then $(1 - x)$ has an inverse in $R[[x]]$

***Proof*:**
Notice that

$$(1 - x)(1 + x + x^2 + \cdots + x^n + \cdots) = 1$$

and hence by definition that means that $1 - x$ has an inverse, namely $\sum_{n=0}^{\infty} x^n$.

By convention, we denote inverses using quotient notation, and hence we would write:

$$\frac{1}{1-x} := \sum_{n=0}^{\infty} x^n$$

For those who have taken calculus, they may be familiar with an "analysis" version of this proof relying on absolute convergence. The above shows that this result is purely algebraic. In fact, we can fully classify all units in a formal power series

Proposition 9.2.11: Units In Formal Power Series

Let $f \in k[[x]]$. Then f is a unit (i.e. f has an inverse) if and only if $a_0 \neq 0$

***Proof*:**
The condition is certainly necessary since if

$$g(x) = \sum_n b_n x^n \qquad f(x)g(x) = 1$$

then we must have $a_0 b_0 = 1$, so $a_0 \neq 0$. If conversely we have $a_0 \neq 0$, we shall show that the formal power series $(a_0)^{-1} f(x) = f_1(x)$ has an inverse $g_1(x)$, and hence $(a_0)^{-1} g_1(x)$ is the inverse of $f(x)$. Decompose $f_1(x)$ into:

$$f_1(x) = 1 - u(x)$$

then substitute $u(x)$ for y in the proof of lemma 9.2.10 to get that $1 - u(x)$ has an inverse. But then we may conclude that $f(x)$ has an inverse, thus completing the proof.

Another question we may ask is the following: Given two polynomials $f(x), g(x)$, we may take $f(g(x))$, (replace x with $g(x)$). Can the same be done for power series? The following shows when this is possible:

Definition 9.2.12: Formal Derivative

Let $f \in k[[x]]$ be a formal power series. Then if $f(x) = \sum_{k=0}^{\infty} a_k x^k$, the formal derivative of f is the value

$$\frac{d}{dx}(f) = \sum_{k=0}^{\infty} k a_k x^{k-1}$$

we usually denote the derivative of $f(x)$ as $f'(x)$, and for higher derivatives we write $f^{(n)}(x)$ to represent taking recursively the derivative of f n times.

We may use the formal derivative recursively to find the coefficients of f. Namely, $f(0) = a_0$, $f'(0) = a_1$, ..., $f^{(n)}(0) = a_n$, and so forth.

Let f, g be power series. We may wonder when $g(f(z))$ is well-defined:

Proposition 9.2.13: Composing Power Series

Let f, g be formal power series. Then $g(f(z))$ is well-defined if $b_0 = 0$.

***Proof*:**
When composing power series, we get:

$$a_0 + a_1(b_1 z + b_2 z^2 + \cdots) + a_2(b_1 z = b_2 z^2 + \cdots)^2 + \cdots$$

where each coefficient is an infinite sum, and so we must ask for their convergence conditions. One thing we can do to guarantee this is by setting $b_0 = 0$, which will guarantee that all the coefficients are finite sums and hence well-defined.

We often denote the composition as $g \circ f$.

Proposition 9.2.14: Associativity Of Composition

Let $f, g, h \in k[[x]]$ be power series. Then

$$(h \circ g) \circ f = h \circ (g \circ f)$$

***Proof*:**
Henri Cartan Elementary Theory of Analytic functions p. 13

Define $I(x) = x$. Notice that for any $f \in k[[x]]$, $I \circ f = f \circ I = f$.

Proposition 9.2.15: Composition Has Inverse

Let $f \in k[[x]]$ and $I(x) = x$. Then a right inverse exists, $f \circ g$ if and only if $f(0) = 0$ and $f'(0) \neq 0$ (i.e. $a_1 \neq 0$). For g to also be a left-inverse, we require $g(0) = 0$, in which case

$$f \circ g = g \circ f = I$$

9.2.16 remark (If you know some calculus) Notice how the condition $f'(0) \neq 0$ bears resemblence to the inverse function theorem

***Proof*:**
We want to a g such that $f(g(z)) = z$. We want:

$$a_0 + a_1(b_1 z + b_2 z^2 + \cdots) + a_2(b_1 z = b_2 z^2 + \cdots)^2 = z$$

expanding and using the method of undetermined coefficients expanding the first two terms we get:

$$a_0 = 0 \qquad a_1 b_1 = 1$$

This shows us that $a_0 = 0$ and $a_1 = f'(0) \neq 0$ are necessary conditions. To show they are sufficient conditions, we can see that we may deduce all other coefficeints given these conditions.For example, we may deduce the coefficient of z^n as $a_0 + a_1 g(z) + \cdots + a_n g(z)^n$, so

$$a_1 b_n + P_n(a_2, ..., a_n, b_1, ..., b_{n-1}) = 0$$

Thus, we would start by doing $b_1 = 1/a_1$, and $b_2, b_3, ...$ are defined recursively (this resulting polynomials is called *Bells polynomial*). With our construction, we have that g satisfies $b_0 = 0$ and $b_1 \neq 0$. Thus, we can do the same thing to g, that is $g(f_1(w)) = w$. These inverses are equal:

$$f_1 = \mathrm{id} \circ f_1 = (f \circ g) \circ f_1 = f \circ (g \circ f_1) = f \circ \mathrm{id} = f$$

where we assumed the associativity of composition, something that can be checked. Finally, for the radius of convergence, we shall delay such considerations for now, noting that when we show that holomorphic functions are analytic we can use the inverse function theorem to estimate the values.

Example 9.2.17: Formal Power Series

Some classical examples of power series include:

1. $\exp(x) = e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$
2. $\cos(x) =$
3. $\sin(x) =$
4. $\log(x) =$

9.2.3 Matrix Rings

As was briefly mentioned in the preliminary chapter (section 0.5), many structures can be "represented as matrices", that is, instead of studying all these structure as separate algebraic objects, they all follow the same rules as matrix rings.

Definition 9.2.18: Matrix Ring

Let R be a ring. Then $M_n(R)$ is the collection of all $n \times n$ matrices with coefficients in R. Addition is defined term-wise and multiplication is defined via matrix-multiplication as given in section 0.5. A matrix may be represented by:

$$(a_{ij})_{ij} \in M_n(R)$$

where a_{ij} represents the value at position ij of the matrix. The identity of $M_n(R)$ is denoted I_n.

In the above notation, matrix multiplication AB of two matrices can be represented as

$$(c)_{ij} = \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right)_{ij}$$

Note that $M_1(R) \cong R$, and $M_2(R)$ is *not* commutative, and always has zero divisors, for:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

R can be embedded into R by taking by taking mapping it to the diagonal:

$$a \mapsto (a)_{ii}$$

where the matrix $(a)_{ii}$ has zero in all other entries. It should be check that the subring of diagonal entries with all the same value is a subring, sometimes denoted $\iota(R) \subseteq M_n(R)$ to emphasize that $\iota(R)$ is an embedding, or RI_n. It should be check that $I_n = \iota(1)$. If R is commutative, then:

$$Z(M_n(R)) = RI_n$$

that is, the center of the matrix ring is the diagonals with the same coefficients. Slightly more generally, the subset of diagonal matrices also form a subring., espec Diagonal matrices are rather common, and so they are often denoted

$$\mathrm{diag}(a_1, a_2, ..., a_n) = (a_i)_{ii}$$

Using the diagonal embedding, we can define a functor $M_n(-) : \mathbf{Ring} \to \mathbf{Ring}$ mapping R to the diagonal and φ to $M_n(\varphi)$ that maps $(a_{ij})_{ij} \mapsto (\varphi(a_{ij}))_{ij}$ (the reader should feel free to check this is indeed a functor). Thus, there is the following diagram:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, hook] & S \arrow[d, hook] \\ M_n(R) \arrow[r, "M_n(\varphi)"'] & M_n(S) \end{tikzcd}$$

Thus, if $S \subseteq R$, then $M_n(S) \subseteq M_n(R)$.

Matrix rings shall appear in many context, notably in the case when $R = k$ is a field in the theory of vector-spaces and when R is a division ring in representation theory.

9.2.4 Group Rings

We have seen how to take a set and turn it into a group in many ways (ex. trivial group $T(X)$, permutation group S_X, free group $F(X)$). Of these methods the free-group construction gave us a way to lift a set map to a group map. If you have read section 2.3.5, **Set** and **Grp** form a free-forgetful adjoint, which tells us that than the "lifting" of set functions to group homomorphisms happened in a "natural way". We can repeat this process with groups and rings, forming a "free ring" with respect to groups. This "free ring" will not be free with respect to *sets*, and so will differ from free groups[13]. However, after going through free rings with respect to groups, the reader might start seeing why both constructions are called *free*.

Definition 9.2.19: Group Ring

Let R be a commutative ring and let G be any group, where $g_0 = e$ is the identity, with the operation written multiplicity. Then define the *group ring* $R[G]$ (or RG) of G with respect to R to be the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where only finitely many a_i are non-zero where addition is done point wise and the product is defined per monomial as $(ag_i)(bg_j) = (ab)g_k$ where ab is performed in R and $g_i g_j = g_k$, and then extended this is extended to multiplication of the formal sums using convolution.

Since $g_0 = e$, we often write $a_0 g_0 = a_0$. In this way, we can identify $R \subseteq R[G]$, and R in as a subring of $R[G]$ has it's original ring operations. Similarly, taking $a_{ij} = \delta_{ij}$[14] where

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

[13] We will cover free rings later

[14] this is called the Kronecker delta

we can identify $G \subseteq R[G]$, and G as a subgroup of $R[G]$ has it's original group operation. Even better, $G \leqslant R[G]^{\times}$. It is an easy exercise to check that $R[G]$ is indeed a ring, and that $R[G]$ is commutative if and only if G is commutative. Another simple exercise to check is that R (identified as a subring of $R[G]$) is in the center of $R[G]$ ($R \subseteq Z(G)$), and that $1 \in R$ (identified as a subring of $R[G]$) is the identity of $R[G]$

The reader may see that group rings are indeed the free objects with respect to the forgetful functor **Ring** $\to$ **Grp** by repeating the construction given for free groups: Take the collection of set maps $f : G \to R$ where only finitely many $f(g)$ are nonzero (such maps are said to have *finite support*[15]). Then two elements are added pointwise, $h \mapsto f(h) + g(h)$, and multiplication has the following clever interpretation:

$$h \mapsto \sum_{ab=h} f(a)g(b) = \sum_{a \in G} f(a)g(a^{-1}h)$$

Notice this sum is indeed finite since f, g have finite support.

Example 9.2.20: Group Rings

1. Let $R = \mathbb{Z}$ and $G = D_8$. Then, for example $3r^2 - s, rs + 9 \in \mathbb{Z}[D_8]$.
2. If G is any finite subgroup, and $S \subseteq R$ is a subring, then $S[G] \subseteq R[G]$ is a subring.
3. (If you know about the Hamiltonian's $\mathbb{H}$), then $\mathbb{R}[Q_8] \neq \mathbb{H}$, even though they both conains Q_8. Can you see why?
4. Let $R = \mathbb{Q}$ and $G = S_3$. Let $a = 3(2\ 3) - 4(1\ 2\ 3)$ and $b = 1/2(1\ 2) + 5(3\ 2\ 1)$. Compute:

$$a + b \qquad ab$$

Note that if G has an element of finite order, then the group ring always has zero divisors: let $g \in G$ such that $m = |g| > 1$. Then

$$(1 - g)(1 + ... + g^{m-1}) = 1 - g^m = 1 - 1 = 0$$

Exercise 9.2.1

1. Is $R[x]$ isomorphic to $\oplus_{i=1}^{\infty} R$? Is $R[[x]]$ isomorphic to $\prod_{i=1}^{\infty} R$?)
2. Let $f \big| g$. Does this imply $\deg(f) \big| \deg(g)$?
3. Let R be a **non**-commmutative ring. Define the appropriate ring so that for any set function $f : A \to S$, there is a unique map $\iota : A \to R$ to the ring $F : R \to S$ such that the diagram commutes. That is, define the free object in the general case so that it is left-adjoint to the forgetful functor **Ring** $\to$ **Set**.
4. Take the full sub-category **Poly-Ring** $\subseteq$ **Ring**. What is the inital object of **Poly-Ring**?
5. Let $p(x) \in R[x]$ be a polynomial such that for all $r \in R$,

$$p(r) = a_n r^n + \cdots + a_1 r + a_0 \neq 0$$

[15]For those who know some calculus/analysis, this is similar to how many smooth functions are defined on compact support

Show that there is an extension $S \supseteq R$ such that there exists $s \in S$ where $p(s) = 0$. In particular, there exists an S such that there exists a natural inclusion $\iota : R \hookrightarrow S$, $p(x)$ naturally includes in S and some element $s \in S$ is a root to $p(x)$:

$$p(s) = a_n s^n + \cdots + a_1 s + a_0 = 0$$

6. Show that $R[\mathbb{N}] \cong R[x]$. Show that $R[\mathbb{Z}] \cong R[x, x^{-1}]$
7. Is $\mathbb{Z}[x, x^{-1}] \cong \mathbb{Z}[[x]]$

9.3 Ring Homomorphisms and Quotient Rings

In this section, we explore the structure of the quotient objects of rings, that is, if $\varphi : R \to S$ is a ring-homomorphism, we aim to find a way to talk about the structure of $\varphi(R) \subseteq S$ given structure for R. We first recall the definition of ring homomorphisms from the first section:

Definition 9.3.1: Ring Homomorphism Remainder

Let R and S be rings. Then:

1. *ring homomorphism* is a map $\psi : R \to S$ satisfying
   (a) $\varphi(a + b) = \varphi(a) + \varphi(b)$
   (b) $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
   (c) $\varphi(1_R) = 1_S$
2. The *kernel* of the ring homomorphism φ, denoted $\ker(\varphi)$ is the set of elements of R that map to 0 in S
3. A ring homomorphism with an inverse that is a ring homomorphism is a *ring isomorphism.*

If the context is clear, “homomorphism” will be used instead of ”ring homomorphism“. Also, if A, B are rings, then $A \cong B$ will always be interpreted as an isomorphism between the two rings.

Example 9.3.2

1. $\pi : Z \to \mathbb{Z}/2\mathbb{Z}$, here $\pi(x) = [x]$ is the natural projection map, is a ring homomorphism
2. $\varphi_n : \mathbb{Z} \to \mathbb{Z}$, $\varphi_n(x) = nx$ is *not* a ring-homomorphism.
3. $\varphi : \mathbb{Q}[x] \to \mathbb{Q}$, $\varphi(p(x)) = p(0)$ is a ring homomorphism, here $\ker \varphi = \{a_n x^n + \cdots + a_1 x\}$. Notice that every elements in the kernel has an x. We can also think of this as $p(x) \cdot x$ for any $p(x) \in \mathbb{Q}[x]$. This way of looking at it will become the main way we look at the kernel of a ring homomorphism shortly.
4. As shown earlier, if $S \subseteq R$ has a ring-structure, then if the inclusion map $i : S \to \mathbb{R}$ is a [ring] homomoprhism, then S is a subring.

5. The inclusion map $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is a ring homomorphism. Is there a non-trivial homomorphim from $\mathbb{Q}$ to $\mathbb{Z}$?

6. Embedd $k[x] \hookrightarrow k[[x]]$ by mapping $k[x]$ to be all formal power series with only finitely many nonzero elements. Then by proposition 9.2.11, any polynomial $p(x) \in k[x]$ such that $p(0) \neq 0$ under the evaluation map is invertible in $k[[x]]$. Hence, there is a power series that can be identified by $1/p(x)$. Thus, we may define polynomial quotients in $k[[x]]$

$$\frac{p(x)}{q(x)} \qquad q(0) \neq 0$$

7. Is the derivative map $\frac{d}{dx} : k[[x]] \to k[[x]]$ defined by

$$\frac{d}{dx}\left(\sum_{k=0}^{\infty} a_k x^k\right) = \sum_{k=0}^{\infty} k a_k x^{k-1}$$

   a ring homomorphism? what is $\frac{d}{dx}(fg)$?

8. Let $k[[x]]$ and $k[[y]]$ be two formal power series over a field k. Let $T \in k[[y]]$ such that $b_0 = 0$. If we define $f \circ g$ to be (HERE). Then $f \mapsto f \circ T$ is a ring homomorphism from $k[[x]]$ to $k[[y]]$. What happens if $b_0 \neq 0$?

Just as the study of group-homomorphisms is equivalent to finding a normal subgroup H and studying G/H, so too will there be a link between ring-homomorphisms and a special subset I of R for which R/I represents the homomorphism. Just like for group-homomorphisms, the set I is based on the kernel of the ring-homomorphism, and R/I will be isomorphic to the image of φ:

Proposition 9.3.3: Image and Kernel of Ring Homomorphism

Let R, S be rings and $\psi : R \to S$ be a homomorphism.

1. The image of ψ is a sub-ring of S (as should be clear as it is one of the "properties" that are sought after for defining a homomorphism)

2. the kernel of ψ is, in general, *not* a subring of R. Rather, the kernel of ψ retains the abelian structure of R (i.e. is closed under adition an $a - b \in I$), and is closder under multiplication (left/right multiplication)[a] from elements of R (for all $r \in R$ and $i \in I$, $ri \in \ker(\psi)$ and $ir \in \ker(\psi)$).

[a]note that it was specified that it was both left and right multiplication because the multiplication is not necessarily commutative, while addition is

As an example of a kernel, take $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[x]$, $p(x) \mapsto p(0)$. Then $\ker(\varphi) = x\mathbb{Z}[x]$, which is not a ring as $1 \notin x\mathbb{Z}[x]$. Kernels are the more general object *rng*. Note that if we define a ring to not have an identity, then the kernel would indeed be a sub-ring of R. Some authors define rings to not have an identity for this very reason. However, as a consequence, almost every theorem will have to be prefaced with "let R be a ring with identity", meaning the study of ring theory will ultimately be the study of a special case of rings with identity, which begs the question whether broadening the definition to not have an identity is worth the trade-off. However one sees it, it is simply a mathematical reality that kernels are different objects than rings (as the proposition suggests), and this author belives it is better to treat them as such[16]. Kernels will soon be defined as *ideals* (definition 9.3.6) since they have different properties from rings.

***Proof*:**

1. This should be an easy exercise to make sure homomorphisms can be comfortably worked with
2. Let $I = \ker\varphi$ and take $a, b \in I$. Then

$$\varphi(a-b) = \varphi(a) - \varphi(b) = 0 \Rightarrow a - b \in I$$

showing that I is closed addition, and in particular is an abelian group.

Now take $r \in R$. We want to show that for any $i \in I$, $ri, ir \in I$. The key to this proof is the 1st property of property 9.1.9. Let $I = \ker\varphi$. Then

$$\varphi(ri) = \varphi(r)\varphi(i) = \varphi(r)\cdot 0 = 0 \Rightarrow ri \in I$$

and similarly

$$\varphi(ir) = \varphi(i)\varphi(r) = 0\cdot\varphi(r) = 0 \Rightarrow ir \in I$$

Thus, I is closed under left-right multiplication of R.

9.3.4 Remark As a consequence of kernels not being subrings, the concept of extensions as discussed in section 3.6 don't apply so simply to rings! A group extension relied on the fact that both N and G/N were groups. Since ideals are not rings, the same concept does not apply for rings! There is an analogous concept of a "ring extension"[a], which generalize the notion of an extension by just requiring a ring extension S of R to be that there exists an injective ring homomorphism $\varphi : R \to S$ (similarly to how if $N \trianglelefteq G$ then we always have that $N \hookrightarrow G$). This notion of ring extension will be central to us in Part V [b], but the "block-building" story that happened for groups does not play out in the same way way for rings. However, in part III, we will see that any ring R is a natural R-modules, and modules will have extensions defined on them [c]. Thus, we may study the extensions of rings in a group-theoretic way in the context of module theory.

[a]not to be confused with ring extensions that extend a ring by an abelian group as described in Nagata Masayoshi's book *Local Rings* (see Wikipedia article)

[b]It will be the fundamental way we look at varieties and number fields

[c]The category of R-module is closed under kernels

This does not mean that we cannot define $R/\ker(\varphi)$. In fact, $R/\ker(\varphi)$ will be a well-defined ring, as we explore now. Note that since R is an abelian group, and φ is also always a group-homomorphism, $\ker\varphi$ is always be a normal group. If we label $I = \ker(\varphi)$ then R/I already has a group structure, that is:

$$(r + I) + (s + I) = (r + s) + I$$

is well defined. The question becomes if we can also add a multiplicative structure to the quotient

[16]The fact that the kernel is not a ring means that the category of **Ring** does not have kernel objects

group, that is:

$$(r+I)\cdot(s+I) = (rs)+I \qquad \text{(in equivalence class notation: } [x\cdot y] = [x]\cdot[y])$$

With I being the kernel of a ring homomorphism, this result is immediate (i.e. it is well-defined). For $i, k \in I$ and $r, s \in R$:

$$\varphi((r+i)(s+k)) = \varphi(r+i)\varphi(s+k) = (\varphi(r)+\varphi(i))(\varphi(s)+\varphi(k)) = \varphi(r)\varphi(s)+0+0+0 = \varphi(r)\varphi(s)$$

where we used both the additive and multiplicative linearity condition. Labelling $[x] = x+I$, we call $(R/I, +_\sim, \cdot_\sim)$ with $+_\sim$ and $\cdot_\sim$ being defined through φ the *quotient ring* [17]. Thus, I being the kernel of a ring is a sufficient condition to make R/I into a ring. Is it a necessary condition? That is, can there be weaker structures that still preserves the multiplicative structure as shown above? For example, can I simply be a normal subgroup and always preserve the multiplicative structure? The answer is no:

Example 9.3.5: : Quotient group without preserving multiplicative property

Take $\mathbb{Q}/\mathbb{Z}$ as a quotient group. Since $\mathbb{Q}$ is abelian, $\mathbb{Q}/\mathbb{Z}$ is a well-defined quotient group. Let's say it was a well-defined quotient ring. Consider

$$\left(\frac{a}{n}+\mathbb{Z}\right) = (a+\mathbb{Z})\left(\frac{1}{n}=\mathbb{Z}\right) = (0+\mathbb{Z})\left(\frac{1}{n}+\mathbb{Z}\right) = [0]$$

so, every element greater than 1 is zero. Through this, we can make the rest of the elements of $\mathbb{Q}/\mathbb{Z}$ zero. Take $\frac{1}{n}$. Then

$$[0] = \left[0\cdot\frac{1}{n}\right][0]\left[\frac{1}{n}\right] = [1]\left[\frac{1}{n}\right] = \left[1\cdot\frac{1}{n}\right] = \left[\frac{1}{n}\right]$$

Which means that multiplication forced $(\frac{a}{b}+\mathbb{Z}) = (0+\mathbb{Z})$, meaning that

$$(r+\mathbb{Z})(s+\mathbb{Z}) \neq (rs+\mathbb{Z})$$

unless $(r+I) = [0]$ or $(s+I) = 0$, meaning there is no salvageable sub-ring we can which would make this work.

So we must figure out what properties I requires for closure of multiplication. Let's say that I is a normal subgroup, and let's work with the cosets R/I. Let's say that multiplication of cosets is well defined. That would imply that

$$[r+i][s+j] = [rs] \qquad i, j \in I$$

that is, multiplication is independent of representation. If $r = s = 0$, then that means that

$$(i+I)(j+I) \in I \Rightarrow ij \in I$$

meaning that I is closed under multiplication. Next, what if $s = 0$ and r is arbitrary. Then that

[17] as before we'll delay defining the quotient object till we find the sufficient conditions on I independent of a ring homomorphism

would lead to $[r][0] = [0]$ by proposition 9.1.9, which, written in terms of cosets, will be:

$$\begin{aligned}
&((r+i)+I)\cdot(j+I) = I \\
\Rightarrow &(rj+ij)+I = I \\
\Rightarrow &(ri)+I = I \\
\Rightarrow &rb \in I
\end{aligned}$$

and since r is any arbitrary element $r \in R$, then I must be closed under left multiplication of any element of R. Similarly if $r = 0$, meaning $is \in I$.

Conversely, let's say I is closed under multiplication from the left and right-side: $\forall i \in I$, $ri \in I$, $ir \in I$ $\forall r \in R$. Then

$$\begin{aligned}
(r+i)+I\cdot(s+j)+I &= (r+i)(s+j)+I \\
&= (rs+rj+is+ij)+I \\
&= (rs+I)+(rj+I)+(is+I)+(ij+I)
\end{aligned}$$

and since I is closed under multiplication, $ij \in I$, and since I is closed under left and right multiplication by any element of R, $rj, is \in I$, meaning we are left with:

$$= (rs)+I$$

meaning we are once again representative independent. Thus, we showed that left and right R-closure of I is a necessary *and* sufficient condition for the cosets defined by I to have a 2nd operation.

As with cosets, the equivalence classes induced by ring homomorphisms are given a special name:

Definition 9.3.6: Ideals

Let R be a ring, and let I be a subset of R, and let $r \in R$. Then:

1. $rI = \{ra|a \in I\}$, and $Ir = \{ar|a \in I\}$
2. a subset I of R is a *left ideal* if I is closed under *left* multiplication by elements from R ($rI \subseteq I$ for all $r \in R$)
3. Similarly, I is a *right* ideal if I is closed under *right* multiplication by elements from R ($Ir \subseteq I$ for all $r \in R$)
4. A subset I that is *both* a left and right ideal is called an *ideal* (or, for added emphasis, a *two-sided* ideal) of R

Example 9.3.7

1. Take $m \in \mathbb{Z}$. Then $m\mathbb{Z} \subseteq \mathbb{Z}$ is an ideal. Similarly for $\mathbb{Z}m$.
2. Take $m\mathbb{Z} \subseteq \mathbb{R}$. Then $m\mathbb{Z}$ is not not an ideal (since $\frac{1}{2m}mk \notin m\mathbb{Z}$).
3. In $\mathbb{Z}_4 \times \mathbb{Z}_4$, $\langle(1,1)\rangle$ is a sub-ring, but not an ideal
4. Take any product of two rings $R\times S$. Then $R\times 0 \subseteq \times S$ is an ideal, since for any $(a,0) \in R\times 0$, $(R\times S)\cdot(a,0) \subseteq R\times 0$. Notice that $R\times 0$ is also a subring of $R\times S$ meaning it contains an

identity. This is the *only* exception in which a subring an ideal can contain an identity.

(A word somewhere on how ideals define the relatinships in rings, similarly to how normal groups did for rings. We do not use the same angle-braket notation $\langle\ \rangle$ as we've done for groups since this notation will be used for a different purpose (note to self: in constructing free-algebras. This genelizes better the angle-bracket notation since all ring's are subsumed as an algebra by being $\mathbb{Z}$-algebras))

9.3.8 Trivia Why is it called an ideal? In number theory, people tried finding more information about polynomials like

$$x^2 - d = 0 \Rightarrow x = \sqrt{d}$$

And we would construct the field $\mathbb{Q}(\sqrt{d}) = \left\{a + b\sqrt{d} | a, b \in \mathbb{Q}\right\}$ which was called $\mathbb{Q}$ with $\sqrt{d}$ adjoint. This ring (in fact field) has a sub-ring $\mathbb{Z}[\sqrt{d}]$ and Q, making the diagram:

$$\begin{tikzcd}
\mathbb{Q}[\sqrt{d}] \arrow[d] \arrow[dr] & \\
\mathbb{Z}[\sqrt{d}] \arrow[dr] & \mathbb{Q} \arrow[d] \\
& \mathbb{Z}
\end{tikzcd}$$

So, with this construction, we have the question of developping the concept of prime number in $\mathbb{Z}$ to it in $\mathbb{Z}[\sqrt{d}]$. If you try to take the same idea (a element that has no divisors in the ring except the element and ± 1). But it turns that if you try to define an element of that, elements don't necessarily have a unique factorization!
Example
This is so fundamental that this was dropped as a good definition as prime element in $\mathbb{Z}[\sqrt{d}]$. Insead, the proper generalization of this concept will turn out to be ideals in $\mathbb{Z}$ which will be called *prime ideals*, and then look at their natural extension into $\mathbb{Z}[\sqrt{d}]$. The word "ideal" here comes from how these elements would be the *ideal numbers*. The word number will eventaully get dropped, leaving us with ideal.

9.3.9 Computational Shortcut To prove that I is an ideal, we must show that I is non-empty, that it's closed under subtraction ($\forall r \in R$), closed under multiplication ($\forall r \in R$), and that it's a normal subgroup. If R has a 1, then you can replace "closed under subtraction" to "closed under multiplication" since $(-1)a = -a$ will cover the negative case

Another reason to treat ideals differently than rings is the following:

> Another way of how ideals are different form rings is that being a ideal is "ring-dependent". For example $(x) = \mathbb{R}[x]x$ is an ideal of $\mathbb{R}[x]$, but not an ideal of $\mathbb{C}[x]$, since $ix + 1$ notin $(x) = \mathbb{R}[x]x$. This points again to how ideals are different.

This discussion is summarized in the following proposition:

Definition 9.3.10: Quotient Ring

When I is an ideal of R. The ring R/I with the natural multiplicative operation is called the *quotient ring* of R by I.

Proposition 9.3.11: Quotient groups and Ring Homomorphism

Let R be a ring and let I be an ideal of I. Then the (additive) quotient group R/I is a ring under the binary operation:

$$(r+I)+(s+I)=(r+s)+I \qquad \text{and} \qquad (r+I)\cdot(s+I)=(r\cdot s)+I$$

for all $r, s \in \mathbb{R}$. Conversely, if I is any subgroup such that the above operations are well defined, then I is an ideal of R.

Just like for groups, we can ask if every ideal is the kernel of a ring homomorphism, essentially asking for the 1st isomorphism theorem for rings (i.e. the universal property of quotient rings). This turns out to be the case:

Theorem 9.3.12: The first isomorphism theorem for Rings

1. If $\varphi : R \to S$ is a homomorphism of rings, then the kernel of φ is an ideal of R, the image of φ is a subring of S and $R/\ker(phi)$ is isomorphic as a ring of $\varphi(R)$.
2. If I is any ideal of R, then the map

$$R \to R/I \qquad \text{defined by} \qquad r \mapsto r+I$$

   is a surjective ring homomorphism with kernel I (this homomorphism is called the *natural projection* of R onto R/I). Thus every ideal is the kernel of a ring homomorphism and vice versa.

***Proof*:**
This proof is similar to the fisrt isomorphism theorem (theorem 3.1.14), and we already did a good chunk of it in the preceeding paragarphs. The proof of the natural projection follows very similar lines to theorem 3.1.14.

As with groups, the bar notation $\overline{r}$ or $r \bmod I$ notation is used to represent the equivalence class $[r]$.

Example 9.3.13

1. R and $\{0\}$ are ideals. An ideal is proper if $I \neq R$. The set with only zero is called the *trivial* ideal.
2. $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.
3. Take $R = \mathbb{Z}[x]$, and let I be the subset whose polynomials are of degree at least 2. Note that I is closed under addition and multiplication, and thus an ideal. The cosets consist of

the elements with the same 0th and 1st term, so

$$\overline{1 + 2x + x^2} = \overline{1 + 2x - 5x^3} = \overline{1 + 2x + x^6 + 87x^{1231}}$$

Note that $\overline{x}\overline{x} = \overline{x^2} = \overline{0}$, thus R/I has a zero divisor, even though $\mathbb{Z}[x]$ does not .

4. We saw $\mathbb{Q}[x]/(x)$. Let's try another polynomial: $\mathbb{Q}[x]/(x^2 - 2)$. Since we can divide polynomials, we can divide any polynomial by $x^2 - 2$. We can in fact keep dividing it till we get a polynomial of lesser degree. We will actually then get that

$$\mathbb{Q}[x]/(x^2 - 2) = \{ax + b | a, b \in \mathbb{Q}\}$$

Note that $x^2 - 2$, so $x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}$. So, since $[x^2 - 2] = [0]$, this will make

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}) = \left\{x + y\sqrt{2} | x, y \in \mathbb{Q}\right\}$$

Or, more specifically, if we take the ring-homomorphism $\varphi : \mathbb{Q}[x] \to \mathbb{Q}(\sqrt{2})$, $p(x) \mapsto p(\sqrt{2})$, then This is a isomorphism.

5. If we take $k[x, y]$, then $k[x, y]/(x) \cong k[y]$

6. Let $G = \langle g \rangle$ be a cyclic group and k be any field. Define:

$$\varphi : k[G] \to k[x] \qquad \varphi(ag^k) = ax^k \qquad \forall a \in k$$

Show that:

$$k[G] \cong \frac{k[x]}{x^n - 1}$$

Example 9.3.14
Let's say you want to figure out all integer solutions to

$$x^2 + y^2 = 3z^2$$

Suppose such integers exist. Observe first that we may assume x, y and z have no factors in common, since otherwise we could divide through this equation by the square of this common factor and obtain another set of integer solutions smaller than the initial ones. This equation simply states a relation between these elements in the ring $\mathbb{Z}$. As such, the same relation must also hold in any quotient ring as well. In particular, this relation must hold in $\mathbb{Z}/n\mathbb{Z}$ for any integer n. The choice $n = 4$ is particularly efficacious, for the following reason: the squares mod 4 are just 0^2, 1^2, 2^2, 3^2, i.e., 0, 1 (mod 4). Reading the above equation mod 4 (that is, considering this equation in the quotient ring $\mathbb{Z}/4\mathbb{Z}$), we must have

$$\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = + \begin{Bmatrix} 0 \\ 3 \end{Bmatrix} \qquad (\text{mod} 4)$$

where the $\begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$ indicate that either a 0 or a 1 may be taken. The only possibility is if they're all even, but we assumed they weren't, thus there is no solution.

Note that this technique has limitation. The following equation has solution integers modulo any n, but no integer solutions (which is hard to prove).

$$3x^3 + 4x^3 + 5z^3 = 0$$

Eventually, we'll be using a similar technique to factor polynomial.

There is also the 2nd, 3rd, and 4th isomorphisms theorems. Each of these are quite easy to show: We already know they're true because R is an *additive group* (or *correspondence of groups*, in the case of the Fourth Isomorphism theorem). It only needs to be show that the group isomorphism is also a multiplicative map, and thus a ring-isomorphism.

Beforehand, we had $H, K \leqslant G$ and defined HK. Since we are now writing the group operation additively, we will convert our notation to saying $H + K$. If I, J are subrings of R, then since they're both normal (even abelian) subgroups, then $I + J$ is well-defined.

Theorem 9.3.15: 2nd, 3rd, 4th Isomorphism Theorem for Rings

(2) Let A be a subring and B be an ideal of R. Then $A + B$ is a subring of R, $A \cap B$ is an idea of A, and $\frac{A+B}{B} \cong \frac{A}{A \cap B}$

(3) Let I, J be ideals of R iwth $I \subseteq J$. Then J/I is an ideal of R/I and $(R/I)/(J/I) = R/I$.

(4) Let I be an ideal of F. The correspondence $A \leftrightarrow A/I$ is an inclusion preserving bijetion between the set of subrings A of R that contian I and the set of subrings of R/I. Furthermore, A (a subring containin I) is an ideal of R if and only if A/I is an ideal of R/I

***Proof*:**
The proof of each of these is an upgrade from the proof for the group version. Proving the group-isomorphism is immediate, then proving multiplicative linearity, it's an immediate consequence of the definition of the quotient ring.

(A word on simple ring, i.e. rings with no *two-sided* right-sided ideals? A word on $M_n(k)$ is simple, since no two-sided ideals (since two ided of ideals of $M_n(k)$ are of the form $M_n(I)$), but there exist left-sided ideals (ex. set of matrices with some fixed zero column) and right-sided ideals (set of with fixed zero row))

9.3.16 Remark simpleCommRingIffFieldRmk
We can define a *simple ring* to be a ring whose only two-sided ideals are R and itself[a], which mirrors the definition of simple groups. Then fields *are* all simple commutative rings[b]. For the non-commutative case, it is harder to find simple rings. (see exercise ref:HERE for an example, or proposition 51.2.5).

[a] the reason for sticking to two-sided ideals is because we can quotient R by two-sided ideals to get a quotient ring, but we cannot quotient by by 1-sided ideals and get a quotient ring

[b] The story is a bit different for *rngs*; see comment in document

Exercise 9.3.1

1. Let R, S be rings and consider the product ring $R \times S$. What are the ideals of $R \times S$?
2. Let I be an ideal of the ring R and let $(I) = I[x]$ denote the ideal of $R[x]$ generated by I (the set of polynomials with coefficient I). Then

$$R[x]/(I) \cong (R/I)[x]$$

   In particular, if I is a prime ideal of R then (I) is a prime ideal of $R[x]$. Conclude that $\mathbb{Z}[x]/(n\mathbb{Z}[x]) \cong (\mathbb{Z}/n\mathbb{Z})[x]$
3. Show that the set of upper-triangular matrices is a subring.

9.4 Properties and Further Structures of Ideals

Since the kernel of ring homomorphisms are ideals, they form the core of much of ring theory. Recall how when working with groups, they can be thought of as $F(A)/N$ for an appropriate normal set N which represents the *relations* on the generators A. Ideals will play a role of similar importance: Ideals will represent the relations. There will be an analogous notion of free ring, however the study of generation of free rings shall be delayed to part III since the two constructions are identical.

This is not to say that thinking of ideals as adding relations is un-useful! For example, since $\mathbb{Z}$ is initial in rings, it is of great interest to see how what relations ideals of $\mathbb{Z}$ can give us, what properties do they have, and how they translate with ring homomorphisms. (We will focus a lot on this in number theory).

Furthermore, closure is not well-defined on subrings, that is the "closure" of a set $A \subseteq R$, that is the smallest largest set that makes $+$ and $\cdot$ well-defined, need not be a ring. Take for example $\{2\} \subseteq \mathbb{Z}$, then if $\overline{\{2\}}$ is the "ring closure" of $\{2\}$, then $\overline{\{2\}} = (2)$, which is not a ring. Another example is $\{1+x\} \subseteq \mathbb{Z}[x]$. Then $\overline{\{1+x\}}$ is not a ring, since $1 \notin \overline{\{1+x\}}$[18], and it is not an ideal since $(x+2)(x+1) \notin \overline{\{x+1\}}$[19]. It is a rng, and so it makes senes to define finitely generated rngs, however due to the reason stated in remark ref:HERE, we continue to focus on rings.

9.4.1 Operations and Generation on Ideals

The first thing we might think of is way of extending how to manipulates ideals through binary operations (in essense, we find different ways of combining what relations we impose on a ring). We give a list of common operations of ideals

[18]This is actually a bit difficult to prove, that no polynomial $p(y)$ exists such that $p(1+x) = 0$, and so no polynomial $q = p + 1$for which $q(x+1) = 1$. The full answer will be given in part IV when discussing transcendental extensions.

[19]To see this, we require the fact that every polynomials factors uniquely into irreducibles, and so it's impossible that $p(x+1) = (x+2)(x+1)$ for any $p(y)$. We will show this in chapter 11

Definition 9.4.1: Operations on Ideals

Let R be a ring and $I, J \subseteq R$ be (left, right, or two-sided) ideals. Then:

1. $I \cap J$ is the intersection of two ideals. More generally,
$$\bigcap_{i \in \mathcal{I}} I_i$$
is the intersection of ideals I_i indexed by $\mathcal{I}$.

2. Define the *sum* of I and J by $I + J = \{a + b | a \in I, b \in J\}$. More generally,
$$\sum_{i \in \mathcal{I}} I_i = \left\{ \sum_{i \in N} a_i \;\middle|\; N \subseteq \mathcal{I},\ |N| \in \mathbb{N},\ a_i \in I_i \right\}$$
is the sums of ideals I_i indexed by $\mathcal{I}$.

3. Define the *product* of I and J, denoted IJ, to be the set of all finite sums of elements of he form ab wit $a \in I$ and $b \in J$.
$$\left\{ \sum_{i=1}^{n} a_i b_i \middle| a_i \in I,\ b_i \in J \right\}$$
Note that we must take finite sums since the ideal must be closed under addition. More generally:
$$\prod_{i \in \mathcal{I}} I_i = \left\{ \sum_{i=1}^{m} a_{i_1} \cdots a_{i_n} \;\middle|\; a_{i_1}, ..., a_{i_n} \in \mathcal{I}, n, m \in \mathbb{N} \right\}$$

4. For any $n \geq 1$, define the n^{th} power of I, denoted I^n to be the set consisting of all finite sums of elements of the form $a_1 a_2 \cdots a_n$ with $a_i \in I$ for all $i >$ Equivalently, I^n is defined inductively by defining $I^1 = I$, and $I^n = II^{n-1}$ for $n = 2, 3, ...$ By convention $I^0 = R$.

5. Let R be commutative. Define the *ideal quotient* $(I : J)$ to be:
$$(I : J) := \{x \in R \mid xI \subseteq J\}$$

6. Let R be commutative. Define the *radical* of I to be
$$\sqrt{I} := \{r \in R \mid r^n \in I \text{ for some } n \geq 1\}$$
An ideal satisfying $I = \sqrt{I}$ is called a *radical ideal*.

Note that if I, J are *both* ideals, then $I + J$ is an ideal (this set is closed under sums and $r(a + b) = ra + rb \in I + J$). It should be emphasized that the set IJ is the *finite sum* of elements of the form ab. This condition is required so that IJ is closed under addition, which is necessary for IJ to be an ideal. Just like groups and rings, the union of two ideals is not an ideal (for example, $2\mathbb{Z} \cup 3\mathbb{Z}$ does not contain 5, however $2 \in 2\mathbb{Z}$, $3 \in 3\mathbb{Z}$). Can you show that the quotient ideal $(I : J)$ is indeed an ideal? In some very lose way, if $a \in (x)$, then that means that x is a "factor" of a in the sense that $a = rx$ for some $r \in R$)

Example 9.4.2

1. Let $I = 6\mathbb{Z}$ and $J = 10\mathbb{Z}$. Then $I + J$ is the ideal of all integers of the form $6x + 10y$ for $x, y \in \mathbb{Z}$. Since 6 an 10 are divisible by 2, meaning that $6x + 10y = 2(3x + 5y)$, so that $2\mathbb{Z} \supseteq 6\mathbb{Z} + 10\mathbb{Z}$. Conversely, we see that $2 = 6(2) + 10(-1)$, so $2x = 6(2x) + 10(-x)$, meaning that every element of $2\mathbb{Z}$ can be written of the form $6\mathbb{Z} + 10\mathbb{Z}$, so $2\mathbb{Z} \subseteq 6\mathbb{Z} + 10\mathbb{Z}$. Thus $2\mathbb{Z} = 6\mathbb{Z} = 10\mathbb{Z}$. We can actually generalize this to $n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$ where $d = (m, n)$, in the next section. Similarly, it is worth checking that $6\mathbb{Z}10\mathbb{Z} = 60\mathbb{Z}$.
2. Take $I \subseteq \mathbb{Z}[x]$ to be the ideal whose constant term is even. Then (2) and (x) are in subsets of I. Thus, $4 = 2 \cdot 2$ and $x^2 = x \cdot x$ are in I^2, as is their sum $x^2 + 4$. However, $x^2 + 4$ cannot be written as a product $p(x)q(x)$ of two elements of I.
3. the ideal $((0) : I)$ known as
4. Let $(2), (6) \subseteq \mathbb{Z}$. Then $((2) : (6)) = (3)$, showing how the quotient ideal generlizes some notion of division. *annihilator of* I and consists of all elements of R such that $rx = 0$ for all $x \in I$. Usually, for any ideal $I = (r)$ generated by a single element, we write $(r : I)$ instead of $((r) : I)$. Using this, can you deduce what $(0 : R)$ is equal to?

The radical needs a word of its own, since unlike the other five operations it is not built from sums and products of elements already in the ideals, and that it is an ideal at all is a small argument rather than an observation.

Proposition 9.4.3: The Radical Is an Ideal

Let R be a commutative ring and $I \subseteq R$ an ideal. Then $\sqrt{I}$ is an ideal of R containing I, and $\sqrt{\sqrt{I}} = \sqrt{I}$.

***Proof*:**
Every $r \in I$ satisfies $r^1 \in I$, so $I \subseteq \sqrt{I}$; in particular $0 \in \sqrt{I}$.

For closure under the ring action, let $r \in \sqrt{I}$ with $r^n \in I$ and let $a \in R$. Since R is commutative, $(ar)^n = a^n r^n$, which lies in I because I absorbs multiplication, so $ar \in \sqrt{I}$.

For closure under addition, let $r^n \in I$ and $s^m \in I$. By the binomial theorem, valid in any commutative ring,

$$(r + s)^{n+m-1} = \sum_{i=0}^{n+m-1} \binom{n + m - 1}{i} r^i s^{n+m-1-i}$$

Consider a single term. If $i \geq n$ then $r^i = r^n r^{i-n} \in I$; and if $i \leq n - 1$ then $n + m - 1 - i \geq m$, so $s^{n+m-1-i} \in I$ by the same argument. Either way the term lies in I, and I is closed under addition, so $(r + s)^{n+m-1} \in I$ and $r + s \in \sqrt{I}$. Together with the previous paragraph this makes $\sqrt{I}$ an ideal.

Finally, if $r \in \sqrt{\sqrt{I}}$ then $r^n \in \sqrt{I}$ for some n, so $(r^n)^m = r^{nm} \in I$ for some m, giving $r \in \sqrt{I}$; the reverse inclusion is the first paragraph applied to $\sqrt{I}$, completing the proof.

We can also ask oursevles about the generation of ideals:

Definition 9.4.4: Generating Ideals

1. (A) is the smallest ideal containing A. Then
$$(A) = \cap_{A \subseteq I} I$$
where I is an ideal containing A.
2. RA is he set of all finite sums of elements of the form ra between R and A.
$$\{r_1 a_1 + \cdots + r_k a_k | r_i \in R, a_i \in A\}$$
Similarly for AR and RAR.
3. An ideal generated by a single element is a *principal ideal.*
4. an ideal generated by a finite set is called a *finitely generated ideal.*

If R is commutative, then $RA = AR = RAR = (A)$.

9.4.5 Moment To Ponder You might ask yourself, with addition and multiplication defined on the ideals, does the set of ideals form a ring? Try it out! Is this a way to study ideals? More on studying the ideal structure of a ring when studying Modules in part III (actually, wouldn't referencing Picard's group Pic(R) be a better reference here)

In the following chapter's, ideals generate by a singleton (a) will be of keen interest. As a prelude, notice that if $b \in (a)$, then $b = ra$ for some $r \in R$. Phrased differently, a divides b. Also, $b \in (a)$ if and only if $(b) \subseteq (a)$. In other words, being an element inside the ideal gives some idea of "divisibility" or "factorization".

Example 9.4.6

1. (0) and R are both principal ideals, since (0) is generated by 0, and $R = (1)$ (since we assumed R has an identity).
2. Every ideal of $\mathbb{Z}$ is a principal ideal

   Proof. We'll show that $(m) + (n) = gcd(m, n)$

   $(\Rightarrow)$ Let $x \in (m) + (n)$. Let's show that $x \in (m, n)$. Since $x \in (m) + (n)$, then:
$$x = ma + nb \qquad a, b \in \mathbb{Z}$$
   letting $p = (m, n)$, we see that
$$ma + nb = (m'a + n'b)p = cp \in (m, n)$$
   where m', n' are the remainders, and $c = m'a + n'b$, showing that $cp \in (m, n)$

($\Leftarrow$) Let $p = gcd(m, n)$. Without loss of generality, let's say $m > n$. We will take advantage of the euclidean algorithm. First, start with:

$$m = a_1 n - r_1$$

where r_1 is the remaidner by dividng m by n and a is the number of times n can be divided into m. This is the extended euclidean algorithm covered in MAT240. We can work our way down till we get

$$r_k = a_{k+1} p - 0$$

where $p = gcd(m, n)$. Then, we can use Bézouts identity from the results to get that:

$$p = k_1 m - k_2 n$$

with appropriate k_1, k_2. This then implies that

$$cp = ck_1 m - ck_2 n$$

for any $c \in \mathbb{Z}$, and so cp can be written of the form $am + bn$, meaning $cp \in (m) + (n)$, as we sought to show.

We can actually use this to show that *any* (A) is a principle ideal in $\mathbb{Z}$. Since $\mathbb{Z}$ is commutative, we can think of $(A) = RA$. Then we can do the same greatest common denominator trick to get down to our desied number such that $d\mathbb{Z}$ is equal to $\mathbb{Z}A$. We can use this to show that every ideal of $\mathbb{Z}$ are principal ideals. Since $\mathbb{Z}$ is also an integral domain, meaning it's also a domain, we will end up calling $\mathbb{Z}$ a *principal ideal domain*.

However, it's not always true that if (a, b) can be reduced to (c) that a ring is a principal ideal domain. There is a sub-class of rings called *Bézout Domains* for which every pair of number can be reduced to a principal ideal, but the ring itself is *not* a principal ideal domain. More on that in chapter 10. $\square$

3. Let $R = \mathbb{Z}[x]$. Then $(2, x)$ is *not* a principal ideal. Let

$$(2, x) = \{2p(x) + xq(x) | p(x), q(x) \in \mathbb{Z}[x]\}$$

that is, all polynomials with *even* constants (and hence, a proper ideals, since it doesn't containt 1). Let's say for some $a(x) \in \mathbb{Z}[x].(2, x) = (a(x))$. Since $2 \in (a(x))$, there must be some $p(x)$ such that $2 = p(x)a(x)$. By proposition 9.2.3, we know that degree $p(x)a(x) =$ degree $p(x)+$ degree$a(x)$, and since degree(2) is 0, then $a(x)$ and $p(x)$ must *both* be constant polynomials. Since 2 is a prime number, the only possibilities are $p(x), a(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $1 \in (a(x))$, and so by proposition 9.4.10, $(a(x)) = \mathbb{Z}[x]$. But that's a contradiction to $(2, x)$ being proper. Thus, it must be that $a(x) = \pm 2$. But then, $x \in (\pm 2)$, and so $x = p(x)(\pm 2)$. But this is impossible; we'd need $\frac{1}{2}x$ to be $p(x)$, but $\frac{1}{2} \notin \mathbb{Z}$. Thus, $(2, x)$ is *not* a principal ideal. Thus $\mathbb{Z}[x]$ is *not* a principal ideal domain.

I like to thinke of elements of (p, q) for some $R[x]$ as elements in which every term has p or q as a factor. This is because when taking $\alpha p + \beta q$ for $\alpha, \beta \in R[x]$, you can have α or β be 0, meaning you just need elements which have p or q as a factor.

4. If we take $\mathbb{Q}[x]$ instead of $\mathbb{Z}[x]$, then this would be solved since $(2) = \mathbb{Q}[x]$ since $\frac{1}{2}2 = 1$, maning (2) has a unit, meaning $(2, x)$ has a unit. In fact, $\mathbb{Q}[x]$ will become a principal ideal domain because of this! Try playing around with the following ideals to see they are indeed principal

- $(2, x)$
- $(x, x^2 + 1)$
- $(p(x), q(x))$ for $p, q \in \mathbb{Q}[x]$.

5. Take $R = k[x]$ and take $I = \{p(x) \in k[x] | p(q) = 0\}$ for some arbitrarry $q \in \mathbb{F}$. One can check that it is an ideal. More interestingly though, we know from high-school math that if $p(x) = 0$, then we can factor out $p(x) = (x - q)r(x)$ for any polynomial $r(x) \in \mathbb{F}[x]$. This will thus make $I = ((x - q))$, making these ideals principal ideals! However, if $I = \{p(x) \in \mathbb{F}[x] | p(a) = p(b) = 0, a \neq b\}$ for two arbitrary $a, b \in F$, $a \neq b$, then I is *not* a principal ideal.

As the intuition for ideals suggest, they are a sort of generalization of properties of $\mathbb{Z}$. For example, if $a \in R$ and (a) is an ideal, then $R/(a)$ is well-defined. Now, take $\bar{b} \in R/(a)$ and generate $(\bar{b})$. Then it should be verified that

$$(R/(a)/(\bar{b}) = R/(a, b)$$

Furthermore, notice that in $\mathbb{Z}$, the ideal (a, b) is equal to the ideal $(\gcd(a, b))$. For this reason, ideals capture some more general sense of divisibility for rings! Exploring the notion of divisibility further will be the goal of chapter 10.

Since ideals are so important, it is often the case that studying rings with simpler ideal structure are of much interest. Two particular cases are especially important due to the abundance of finiteness:

Definition 9.4.7: Noetherian rings

Let R be a commutative ring. Then if every ideal[a] $I \subseteq R$ is finitely generated, then R is called a *Noetherian ring*.

[a] Be sure not to mistake ideal for ring, for finitely generated rings have not been defined

Noetherian rings are particularly nice in that they are finite locally and globally. If we just require that the ring is finitely generated, it may be that a subring is non-finitely generated. Conversely. We shall have lots of experience with Noetherian rings later, for now it is useful to know these rings since since being Noetherian is stable under taking the quotient, and is also stable under the "converse" of taking the quotient (if you've covered section 3.6, the Noetherian condition is stable under extensions). For the next proposition, note that a finitely generated ideal is defined identically to a finitely generated ring (that it is generated by finitely many elements):

Proposition 9.4.8

let R be a Noetherian ring and I be any ideal of R. Then R/I is finitely generated. Conversely, if I and R/I is finitely generated, then R is finitely generated.

***Proof*:**
Make it a guided exercise, since it is easy to follow

One of the simplest Noetherian rings are those for which every ideal is not only finitely generated, but also principal. If the ideal is furthermore an integral domain (to avoid pathologies from zero-divisors) then it is a principal ideal domain:

Definition 9.4.9: Principal Ideal Domain

Let R be an integral domain. Then if every ideal $I \subseteq R$ is a principal ideal, then R is called a *Principal Ideal Domain.*

As a simple example, $\mathbb{Z}$ is a PID (can you see why?). It should be noted that *every* ideal is principal, even those that may seem infinitely-generated. In fact, there are domain in which only finitely-generated ideals are principle (such rings are called *Bézout* rings).

Another way of looking at principal ideal domains is that they are "locally free". Here, locality is taken to mean that any proper submodule is free. Since R is an integral domain, it is torsion free, and since R is principle, if $I \subseteq R$ is a submodule (i.e. ideal), then

$$I = (a) \qquad a \in R$$

This fact shall make PID's very powerful, and in fact the idea that PID's have "local freeness" shall be further explore in chapter 17⋆[20].

(TBD – somehow incorporate this?) *Principal Ideal Domains* (PIDs) will have the property of having "prime elements" well-defined and lend itself useful in many classification theorems (for example, study of Jordan Canonical Forms in chapter 14 will heavily rely on this type of ring). (The point is to introduce them later in the chapter on factorization)

Exercise 9.4.1

1. Michael Atiyah has some excellent elementary exercises!! Some of his propositions are also great: the sequence of propositions on radical ideals can totally be exercises here! p. 6-9.
2. $(I \cap J)(I + J) = IJ$

9.4.2 Ideals and Units

As briefly mentioned, some ideals seem to be related to a notion of divisibility. This intuition more clear with the following theorem:

Proposition 9.4.10: units in Ideals and ideals in fields

1. $I = R$ if and only if I contains a unit
2. R is a division ring if and only if it's only left- and right-sided ideals are 0 and itself. If R is commutative, then if the only ideals of R are 0 and itself, then R is a field.

***Proof*:**

1. one inclusion is easy. The key part to the other inclusion is that I must be closed under multiplication from all elements of R. If $I = R$ then I contains the unit 1. If I contains a unit u, then given the inverse $v \in I$ and any element $r \in R$

$$r = r \cdot 1 = r(uv) = (rv)u \in I$$

[20]A projective module over a PID is in fact a free module

Since I is an ideal.

2. If R is a division ring, then every element is a unit, and thus for any set A $RA, AR \subseteq R$ must contain units, and so $I = R$. Conversly, if the only left and right-sided ideals are R and 0, then for any $u \in R$, $Ru = uR = R$ (since it's the only ideal). Then $1 \in (u)$ meaning $1 = uv = vu$ for some $v \in R$. Since this holds for every element, then R is also a division ring. If R is also commutative, it is a commutative division ring, and thus a field.

9.4.11 Note As a consequence, if $1 \in I$, then $I = R$, since 1 is a unit!! This means that in a proper ideal, $1 \notin I$, and so, they cannot be sub-rings with identity $1 \neq 0$.

It should be pointed out it was necessary to have the condition that both let and right-sided ieals are trivial, instead of just picking one or saying that only two-sided ideals are trivial.

Example 9.4.12: Two-Sided Ideals Trivial, but non-trivial left/right-sided ideals

Let k be a field and $M_2(k)$ the matrix ring over k. Then it should be checked that $M_2(k)$ has non-trivial left and right-sided ideals (think of columns and rows of $M_2(k)$). However, $M_2(k)$ has *no* two-sided non-trivial ideals.

Now, $M_2(k)$ is not a division ring: the element

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

does not have an inverse (as can be shown using some lienar algebra).

(make an exercise somewhere finding the ideals of $M_n(R)$, namely $M_n(I)$)

Since any two-sided ideal is a left and right ideal, then division ring's naturally have no non-trivial two-sided ideals.

Corollary 9.4.13: Division Ring and Ring-Homomorphism

If R is a division ring, then any non-zero ring-homomorphism $\varphi : R \to S$ to any other ring must be injective.

***Proof*:**
Let φ be a ring-homomrophism. The kernel of φ is a two-sided ideal of R, but R is a divison ring, meaning the only ideals of R are 0 and R. Since φ is non-trivial, $\ker \varphi \neq R$, and so $\ker \varphi = 0$, making φ injective.

This might make one think that studying division rings, or more particularly fields, in terms of homomorphisms is less useful. However, we will *still* use ring-homomorphisms on fields, since mapping a field injectively into another field will turn out to be quite useful: we can study how fields *embedd* into one another (this will be called a *field extensions*, which will be defined in part IV, and an *integral extension* for more general rings like division rings, which will be defined in part V).

If a ring does only have 0 and R as two-sided ideals, we have the analogous concept to simple groups

Definition 9.4.14: Simple Rings

Let R be a ring. Then R is said to be simple of it's only [left/right/two-sided] ideals are the 0 ideal and itself

One might think that from here, there will be a similar story as in group theory: A lot of time and effort spent on classifying all rings (and finite rings) given this idea of a simple ring. It turns out though that the classification of finite rings are in fact much much easier: Every finite simple ring is isomorphic to the ring $M_n(\mathbb{F}_q)$ over a finite field of order q (i.e., intimately related to matrix rings)! In fact, the study of simple rings is the setting of representation theory! We'll cover simple ring in part X

9.4.3 Maximal Ideals

Many times, we might want if an ideals is *as large as possible* in a ring. That is, if I is an ideal, then there does not exist another J such that $I \subseteq J$ unless $J = R$

Definition 9.4.15: Maximal ideal

An ideal M in an arbitrary ring S is called a *maximal ideal* if $M \neq S$ and the only ideals containing M are M and S.

Example 9.4.16: Ring with maximal ideal

Take $\mathbb{Z}$ and $6\mathbb{Z}$. Since $6 = 2(x)$, $6\mathbb{Z} \subseteq 2\mathbb{Z}$. However, since 2 is a prime, we cannot repeat this same trick twice. Thus, $2\mathbb{Z}$ is a maximal ideal. In general, $p\mathbb{Z}$ for prime p is a maximal ideal

All rings have a maximal ring with the exception of a ring with the trivial ring structure:

Example 9.4.17: Ring without maximal ideal

Take any group with a non-maximal group (ex $\mathbb{Q}$) and take the trivial ring structure ($ab = 0$).

The only restriction on a ring that requires it to have a maximal ideal is that $1 \neq 0$. The way we will prove this will require a form of "super-induction". The use of this proof technique requires the axiom of choice. For those who want to have an idea of Zorn's lemma, see section A.3, and for those who want to see how it links to the axiom of choice, see appendix A

Proposition 9.4.18: Ring with $1 \neq 0$ have Maximal ideals

Let R be a non-trivial ring and $I \subsetneq R$ be a proper ideal. Then I is contained in a maximal ideal

***Proof*:**

Since $I \subsetneq R$, $1 \neq 0$), and $1 \notin I$ (or else $I = R$), a maximal ideal can exist. To show that it is the case, we will us Zorn's Lemma (axiom A.2.3). Let $\mathcal{I}$ be the collection of all proper ideals of R containing I, and let it be ordered by inclusion. Then $(\mathcal{I}, \subseteq)$ is a partial order. If e take any chain $\mathcal{C} \subseteq \mathcal{S}$, then

$$J = \bigcup_{A \in \mathcal{C}} A$$

is the ideal which is the upper-bound for $\mathcal{C}$. For this to be the case, J must be an ideal.

1. J is non-empty, since any ideal must contain 0, thus $0 \in J$
2. Take $a, b \in J$ and consider $a - b$. Then for some $A, B \in \mathcal{C}$, $a \in A$ and $b \in B$. Since $\mathcal{C}$ is a chain, $A \subseteq B$ or $B \subseteq A$. Witout loss of generality, let's assume $A \subseteq B$ and say $a, b \in B$ (if e pick $B \subseteq A$ do $a, b \in A$). Then since B is an ideal, $a - b \in B$, so $a - b \in J$, showing that J is closed under subtraction
3. Take $a \in J$ and consider $r \cdot a$ and $a \cdot r$. Then since $a \in J$, a is in some ideal, let's say $a \in A$. Then $r \cdot a \in A$ and $a \cdot r \in A$, so $r \cdot a, a \cdot r \in J$, showing it's closed uner left and right multiplication

Thus, J is an ideal. It is further a proper ideal. If it ere not, then $1 \in J$, meaning for some ideal A, $1 \in A$. But then $A = R$, showing it is *not* a proper ideal, a contradiction to the definition of J. Thus, J must also be a proper ideal.

Therefore, by Zorn's Lemma, $\mathcal{I}$ must have a maximal element M, that is, if there is an element M' such that $M \subseteq M'$, then it must be that $M' \subseteq M$, so $M = M'$, which fulfills the definition of a maximal ideal, as e sought to show.

Note that we took the union of ideals within the proof. In general, if $I, J \subseteq R$, then $I \cup J$ is not an ideal (consider $2\mathbb{Z} \cup 3\mathbb{Z}$). The reason it worked for this proof is because the ideals are in a chain:

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

It should be pointed out that it is possible to have an infinitely long chain of the form $(x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$ of which the union is not a maximal ideal, hence being maximal is a different condition that such a chain terminating in finite itterations, or even countably-infinite many iterations. This is part of the importance of using Zorn's lemma in the proof.

Since Zorn's Lemma was used, there is no way to computationally figure out what the maximal ideal is by just using this proposition. If a ring is also commutative, we can more easily characterize it with the structure of their quotient ring

Proposition 9.4.19: Maximal Ideals in Commutative Rings

Let R be a commutative ring with $1 \neq 0$. The ideal M is a maximal ideal if and only if the quotient R/M is a field

***Proof*:**
This follow's from the Correspondence Theorem and proposition 9.4.10. Let M be a maximal ideal so that there is no proper I such that $M \subsetneq I \subsetneq R$. Then by the Correspondence Theorem, the ideals I containing M correspond bijectively to the ideals of R/M, and since no proper ideal contains M, the only ideals of R/M are (0) and R/M, thus M is maximal if and only if the only ideals of R/M are (0) and R/M. By proposition 9.4.10, a ring is a field if and only if it's only

ideals or (0) and itself. Thus, M is an ideal if and only if R/M is a field, as we sought to sho

The use of the Correspondence theorem in proposition 9.4.19 can be visualized with such a diagram:

$$\begin{tikzcd} R \arrow[d, dash] & R/M \arrow[d, dash] \\ M & 0 \end{tikzcd}$$

Notice that if R was not commutative, then left ideals are not necessarily right ideals, and so the theorem does not apply: if we pick any one-sided ideal (say a left-ideal), and perform proposition 9.4.18 to show the existence of a maximal left-ideal (and hence, we have a non-empty collection of maximal left-ideals), it is not guarnateed that the set of maximal left-ideals will contain any proper right-ideals[21]. A different approach must be taken to study R/I when R is not commutative and I is a maximal left/right ideal, which will lead to many fruitful results in part X. This case will require some more module theory, and so will be left alone for now.

In the commutative case, this let's us study properties of R by studying the field R/M for all maximal ideal M. If R has a single maximal ideal (say $\mathfrak{m}$) then $R/\mathfrak{m}$ is called the *residue field*. An example of a local ring would be $\mathbb{Z}/4\mathbb{Z}$, with single maximal ideal $\{\overline{0}, \overline{2}\}$. Such are called local rings for a reason; they show up intensively in commutative algebra, and will be central in [13, Smooth Points and Regular Local Rings] in part V.

This last characterization of maximal ideals through quotienting for commutative rings will come to be very useful in the construction of some fields. For example this will be used to construct all finite fields by taking quotient of the ring $\mathbb{Z}[x]$ by maximal ideals. In a more specific case, we will be construction the "extension" of a field k by first defining a polynomial ring $k[x]$, and quotienting it by a maximal ideal $M \subseteq k[x]$. This will be the foundation of *field Theory*, and will be covered in part IV

Example 9.4.20

1. We've hand-waved a bit before how $p\mathbb{Z}$ is a maximal field. Since $\mathbb{Z}$ is commutative, we can now rigorously state this by noticing that $\mathbb{Z}/p\mathbb{Z}$ is a field
2. The ideal $(2, x)$ is a *maximal* in $\mathbb{Z}[x]$, since $\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z}$ is a field
3. Let $R = \mathcal{F}([0,1], \mathbb{R})$ be the ring of all functions from $[0,1]$ to $\mathbb{R}$, and let M_a be the kernel of evaluation at a ($M_a = \{x \in [0,1] | f(x) = a\}$). Show that M_a is maximal.

There are two more cool examples I'll get to when I see their importance

Note that a ring can have multiple maximal ideals. For example, $\mathbb{Z} \times \mathbb{Z}$ is a ring, and $I = ((1,0), (0,2))$ is an ideal generated by $(1,0)$ and $(0,2)$. Then $(\mathbb{Z} \times \mathbb{Z})/I \cong \mathbb{Z}/2\mathbb{Z}$, meaning I is a maximal ideal. Similarly, $J = ((2,0), (0,1))$ is an ideal and $(\mathbb{Z} \times \mathbb{Z})/J \cong \mathbb{Z}/2\mathbb{Z}$. Thus, both I and J are maximal. Note too that the two quotients need not be isomorphic (see exercise ref:HERE, thinking of $\mathbb{Z}$, with the fact that $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$ are not isomorphic).

[21] Indeed, we can modify example 9.4.12 to see that $M_2(k)/(0)$ is not a field, and hence a counter-example

9.4.4 Prime Ideals

The first definition given of prime numbers is that a number is prime if and only if it's only divisors are 1 and itself, that is the only numbers such that $a|p$ are 1 and p. This definition works well for integers, but does it generalize well when we try to talk about divisibility over more general rings. This is due to a sneaky "global" assumption when working with primes, namely that an element is prime if *any* factorization that may include the prime element *has* to include the element as a factor. The following example should illustrate how this isn't necessarily the case:

Example 9.4.21: The Need For Globalness

[The Need For Globalness] Take $R = \mathbb{Z}[2i]$. Then $4 = (-2i)(-2i)$, and $-2i$ can no longer be factor[a]. However, we also have $4 = 2 \cdot 2$, and 2 can no longer factor (which in this case can be verified by taking a few cases). Thus, even though 2 was a factor of 4, we may split 4 differently, showing that there are "local" factorizations.

[a] up to multiplying by a unit, but factorizaiton is only defined up to unt, think $2 \cdot 3 = (-2) \cdot (-3)$

This subtly becomes very important, and shall be explored in depth in chapter 10.

To capture this idea, consider a prime number p. Then p is a prime number and if p divides ab, $a, b \in \mathbb{Z}$, then p divides a or p divides b. For example, if $3|15$, then either $3|3$ or $3|15$. If p was not a prime number, this is not the case: $8|16[= 4 \cdot 4]$, but $8 \nmid 4$ and $8 \nmid 4$. The prime ideals for the integers are the sets that contain all the multiples of a given prime number

Definition 9.4.22: Prime Ideal

Assume R is commutative. An ideal P is called a ***prime ideal*** if $P \neq R$ and whenever the product ab of two element $a, b \in R$ is an element of P, then at least one of the a an b is an element of P.

Since R is commutative, another way of saying the condition is that if A, B where ideals and $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$. The non-commutative case does not generalize well, however we discuss a remedy for this case in chapter 50[22].

Example 9.4.23: Prime Ideals

The origins of the idea come from generalizing the notion of primes in $\mathbb{Z}$, so we'll do the same. The prime numbers p generate the prime-ideals (p).

Let's say $ab \in (p)$, meaning $ab = rp$ for some $r \in \mathbb{Z}$ (note that $\mathbb{R}$ is commutative so $(p) = Rp$). Since $ab = rp$ that impleis that $p|ab$ (p divides ab): If you think about what it means for a number x to divide y ($x|y$), then that means there exists another number a such that $ax = y$. For example $3|6$, so $2\dot{3} = 6$.

Now, since $p|ab$, then $p|a$ or $p|b$. If $p \nmid a$ and $p \nmid b$, then ab wouldn't have it as a prime number under their factorization, but then $p \nmid ab$ – a contradiction. So, $p|a$ or $p|b$. But that would mean $r_a p = a$ or $r_b p = b$ for appropriate r_a or r_b. But then, $a \in (p)$ or $b \in (p)$.

Here's a concrete exmpale. Take $3\mathbb{Z}$, an ideal of $\mathbb{Z}$. (can think of $\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$, $\varphi(x) = [x]$ to convince yourself it's an ideal). Now let's say $ab \in 3\mathbb{Z}$. That means that $3|ab$, meaning either $3|a$ or $3|b$. In either case, that would imply $a \in 3\mathbb{Z}$ or $b \in 3\mathbb{Z}$, showing that $3\mathbb{Z}$ is a prime ideal. On

[22] namely, the ideal group is abelian for certain non-abelian

the other hand, if we have $6\mathbb{Z}$, Then $2 \cdot 3 \in 6\mathbb{Z}$, but $2 \notin 6\mathbb{Z}$ and $3 \notin 6\mathbb{Z}$, meaning $6\mathbb{Z}$ is *not* a prime ideal.

Note that this is an *inclusvie* or, not exlusive. If $p = 2$, and $a = b = 2$, then $2|4$, and both $2|2$ and $2|2$.

Another way I like to think about is is that we already know that if $a \in I$ and $b \in I$, then $ab \in I$. However, it's not clear the converse is true $ab \in I$, $a \in I$ and $b \in I$. This condition is a little too strong: Since $1 \cdot x \in I$ then $1 \in I$, meaning $I = R$. So what if we loosen the condition a bit , and say $ab \in I$ then $a \in I$ *or* $b \in I$. In other words, we can to a limited degree work backwards from *any* combination of elements $ab \in I$ to find one of them keeps needing to be in I. For example, if $abc \in I$ (say we braketed $(ab)c$) then either $ab \in I$ or $c \in I$. Since $ab \in I$, then either $a \in I$ or $b \in I$. This idea will come back in section 10.1, when we talk about factorizing elements in an integral domain, and will be the key idea to a couple of proofs[23].

(Is all I said earlier a fancy way to say that prime ideals are multiplicatively closed?)

This interpretation has some validity if we consdider the following scenario: Take a ring R, and a prime element $p \in R$. Then let's say there was an element $r \in R$ such that $r|p$. This would imply that $p = rr'$ for some $r' \in R$. Since the two values are the same $(rr') = (p)$, implying $rr' \in (p)$. Here, we get to use the property of p being prime: either $r' \in (p)$ or $r \in (p)$, meaning $p|r$ or $p|r'$. If $p|r$, then we have $r|p$ and $p|r$, meaning $p = r$. Be careful here, this is not always true. If r, r' where unit, then we always have $r|r'$ and $r'|r$, but not awlays $r = r'$. For example, in $\mathbb{R}$, $1|2$, $2|1$ but $1 \neq 2$. However, it works here because HERE. So, we are forced to say $r' = u$, where u is a unit. For the othe way around, HERE.

Take $\mathbb{Z}$ again and let's say I was an ideal with this condition. then for some $x \in I$, $px \in I$ for every prime p. By our condition, $p \in I$. This will hold true for every prime-number, meaning I now has all but the identity. I'm not sure if this will always happen, but it's worth pondering.

Another thought: Take (x) and consider $ab \in (x)$. Then $ab = cx$ for appropriate $c \in R$. Then we no get the question $cx \in (x)$ too. This though is "obvious" since $x \in (x)$. But ab is not obvious. So the "representation" of the value that both ab and cx represents is important: it's a question about hether the constituent components of a represetnation is in there (a or b or c or x)

Here are some more example of prime ideals:

Example 9.4.24: Further Examples of Prime Ideals

1. Take $C([0, 1])$, that is, all continuous function to/from $[0, 1]$. Let $I \subseteq C([0, 1])$ be all polynomials with roots at $\frac{1}{2}$ and $\frac{1}{3}$ ($f(\frac{1}{2}) = f(\frac{1}{3}) = 0$). Is I an ideal? Is it a prime ideal? Hint [a].
2. As an exercise to show that you can no longer "go backards", prove that if R is a commutative ring, and P is a prime ideal with no zero-divisors, then R is an integral domain. (Hint [b]).

[a] Think about two polynomials with single roots such that when multiplied, you get an element of I
[b] Assume there is a zero-divisor

This last example is important enough to be referenced many times, and so we make it a proposition:

[23] for example, proposition 10.1.32 and ref:HERE

Proposition 9.4.25: Prime Ideals And Integral Domains

Assume R is commmutative. Then the ideal P is a prime ideal in R if and only if the quotient ring R/P is an integral domain

Corollary 9.4.26: Maximal Ideal Then Prime Ideal

Assume R is commutative. Then every maximal ideal of R is a prime ideal

***Proof*:**
If M is a maximal idea, then by proposition 9.4.19 R/M is a field. A field is an integral domain, so the corollary folllows from proposition 9.4.25

Example 9.4.27

1. The principal ideals generated by (p) where p is prime are both prime and maximal. Not all prime ideals are maximal: (0) is a prime ideal, but not maximal.
2. The ideal (x) is a prime ideal in $\mathbb{Z}[x]$ since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, but not maximal since $\mathbb{Z}$ is not a field. Or, you can diretly see that $(x) \subseteq (2, x)$, meaning that (x) is not maximal.

Exercise 9.4.2

1. Let R, S be rings and $R \times S$ be the product ring. What are the prime ideals of $R \times S$(answer: $R \times P$ and $Q \times S$ for primes P and Q. In general, only one component can be the ideal)
2. Let $a, b \subseteq R$ be ideals. Show that $(a + b)(a \cap b) \subseteq ab$ and $ab \subseteq a \cap b$. Show that if $(a, b) = (1)$, then $a \cap b = ab$.
3. exercise on quotient ideals $(a : b)$
4. Let $f : R \to S$ be a ring homomorphism and $p \subseteq S$ a prime ideal. Show that $f^{-1}(p)$ is a prime ideal. If $m \subseteq S$ is maximal, is $f^{-1}(m)$ maximal? (Consider $R = \mathbb{Z}$ and $S = \mathbb{Q}$)
5. Perhaps, if possible at this stage how if if $a_1, a_2, ..., a_n$ are ideals and $\cap_i a_i$ is contained in the prime ideal p, then p contains one of the a_i, and if $p = \cap_i a_i$, then $p = a_i$ for on of them (shows how prime genelize numbers, so to speak).
6. Notice that we have defined what I^k is. Let's say we started with an ideal J; is there a rasonable way of defining $\sqrt{J}$? Define $\sqrt{J} = \{a \in R \mid \exists n \in \mathbb{N}_{>0},\ a^n \in J\}$. Show that $\sqrt{J}$ is an ideal.

9.5 Ring of Fractions: Localization

In this section, we'll take a moment to discuss how we can take any ring with no zero divisors, and extend the ring to one in which every element has a multiplicative inverse! More generally, given any non zero-divisor element $d \in R$ (where R will have to be commutative), we can construct a ring

R_d which will where $d^{-1} \in R_d$. This concept was applied in the field of *algebraic geometry* for the purposes of looking at a ring "locally". What this means and how to think about this consruction "geometrically" will be discussed much later in part V in section 23.4. In this section, the full generality of this construction is not particularly useful. What's more interesting at the moment is that this gives us a way to add units to our ring!

We'll start with a practical example by using $\mathbb{Z}$ to construct $\mathbb{Q}$! Taking $\mathbb{Z}$, we can make an equivalence class

$$(a, b) \sim (c, d) \Leftrightarrow ad = bc$$

where $b, c \neq 0$. Then $[(a, b)]$ represents a fraction $\frac{a}{b}$, with the point of the equivalence relation is to make all fractions with similar divisors be the same:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \cdots$$

I'll stick with fraction notation for these equivalence relationships. We can see that we can define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \qquad \text{and} \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

It should be checked that these operations are independent of representative. With this construction, we can identity $\mathbb{Z}$ with $\left\{\frac{a}{1} | a \in \mathbb{Z}\right\}$ [24], showing how $\mathbb{Z}$ is a sub-ring of $\mathbb{Q}$.

With that intuition, we can ask about general commutative rings R. The problem comes when we allow $d = 0$ or be a zero divisor. Say $d \neq 0$ $bd = 0$. Then:

$$d = \frac{d}{1} = \frac{bd}{b} = \frac{0}{b} = 0$$

We'll avoid this by taking a set $R^\times$ which represents all the units of R

Theorem 9.5.1: Ring of Fractions

Let R be a commutative ring R (not necessarily with identity). Let D be any nonempty subset of R that does not contain 0, does not contain any zero-divisors and is closed under multiplication (i.e. $ab \in D$, for every $a, b \in D$)[a]. Then there is a commutative ring Q with 1 such that Q contains R as as a subring and every element of D is a unit in Q. The ring Q has the following additional properties

1. Every element of Q is of the form rd^{-1} for some $r \in R$ nd $d \in D$. In particualr if $D = R - \{0\}$ then Q is a field
2. (uniqueness of Q) The ring Q is the "smallest" ring containing R in which al elements of D become units. That is, if S is any commutative ring with identity and let $\varphi : R \to S$ be any injective ring homomorphism such that $\varphi(d)$ is a unit in S for every $d \in D$. Then there is an injective homomorphism $\Phi : Q \to S$ such that $\Phi|_R = \varphi$. In other words, any ring containing an isomorphic copy of R in which which all the elements of D become units must also contain an isomrophic copy of Q.

[a]such a set is called a *semi-group*

[24]meaning construct a isomorphism

***Proof*:**
We'll start like we started with $\mathbb{Z}$. Let $F = \{(r,d) | r \in R, d \in D\}$ (d in the 2nd component because we don't want zero-divisors in the bottom, and we need the denominator to be closed to define multiplication). Then, we define the equivalence relation

$$(r,d) \sim (s,e) \text{ if and only if } re = sd$$

It is easy to show that $\sim$ is an equivalence relation [a]

We commonly denote (r,d) by $\frac{r}{d}$. Now, let

$$Q = \left\{[(a,b)] \middle| a \in R, d \in D\right\}$$

be the set of equivalence relations. Note that $\frac{r}{d} = \frac{re}{de}$ for any $e \in D$, as we would expect.

We can define addition and multiplication as

$$\begin{aligned}
[(a,b)] + [(c,d)] &= [(ad+bc, bd)] \qquad &\Rightarrow \qquad \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} \\
[(a,b)] \cdot [(c,d)] &= [(ac, bd)] \qquad &\Rightarrow \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}
\end{aligned}$$

Then the following properties need to be checked. I'll show that addition is well-defined, and the rest is not hard to show that it follows:

1. The operations are well-defined (not dependent on representatives)

   Proof. We'll show addition is well-defined. Let

   $$\frac{a}{b} = \frac{a'}{b'} \ (ab' = a'd) \qquad \frac{c}{d} = \frac{c'}{d'} \ (c'd = cd')$$

   then

   $$\frac{ad+bc}{bd} = \frac{a'd'+b'c'}{b'd'}$$

   which equivalently means:

   $$\begin{aligned}
   (ad+bc)(b'd') &= adb'd' + bcb'd' && \text{distributivity in } R \\
   &= ab'dd' + cd'bb' && \text{commutativity in } R \\
   &= ab'dd' + cd'bb' && ab' = a'b, \ cd' = c'd \\
   &= (a'd' + b'c')(bd)
   \end{aligned}$$

   thus, the two are equivalent. A similar procedure works for multiplication ☐

2. Q is an abliean group under addition, where the aditive identity is $\frac{0}{d}$ for any $d \in D$ and the additive inverse of $\frac{a}{d}$ is $\frac{-a}{d}$

3. Multiplication is associative, distributive, commutative,

4. Q has an identity $\frac{d}{d}$ for any $d \in D$

Next, we show that R can indeed be embedded into Q. Let

$$i : R \to Q \qquad r \mapsto \frac{rd}{d}$$

Since $\frac{rd}{d} = \frac{re}{e}$, this function is independent of choice of d. Since d is closed under multiplication:

$$i(rs) = \frac{rsd}{d} = \frac{rs}{1} = \frac{r}{1}\frac{s}{1} = \frac{rd}{d}\frac{sd}{d} = i(r)i(s)$$

finally, for injectivity of the inclusion, notice that

$$i(r) = 0 \Leftrightarrow \frac{rd}{d} = \frac{0}{d} \Leftrightarrow rd^2 = 0 \Leftrightarrow r = 0$$

meaning the kernel is trivial, and i is injective.

Next, we show that every element $q \in Q$ can be written as rd^{-1}. Since $d = \frac{d}{1}$, then $d^{-1} = \frac{1}{d}$ since $\frac{d}{1}\frac{1}{d} = \frac{1}{1}$. Then by definition of an element in Q, we define an elemnent as $\frac{r}{d} \in Q$, and so we have that $q = rd^{-1}$.

Note that if $D = R - \{0\}$, then every element has a multiplicative inverse, and so that makes Q into a field.

Finally, we must show this extension is unique. To do this, let's say $\varphi : R \to S$ is an injective ring-homomorphism such that $\varphi(d)$ is a unit in S. In this way, S acts as "another" extension of R (since we shows that R can be imbedded into Q). Since $q = rd^{-1}$ for every $q \in Q$, extend φ to Φ by definining

$$\Phi(rd^{-1}) = \varphi(s)\varphi(d)^{-1}$$

for all $r, d \in R$. Note that $\varphi(d)^{-1}$ is defined since $\varphi(d)$ is a unit in S. This map is well-defined. If we have $rd^{-1} = re^{-1}$, then multiplying around we get $re = rs$, so by the property of ring homomrphisms $\varphi(r)\varphi(e) = \varphi(s)\varphi(d)$ and so

$$\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1} = \Phi(se^{-1})$$

showing it is indeed well-defined. Checking Φ is a ring homomorphism is strait forward using commutativity of Q.

Last step is to show Φ is injective. There is a sneaky trick you can do to pull this off. Let's say $rd^{-1} \in \ker(\Phi)$. Then that means that $r \in \ker(\varphi)$ since we'd need $\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = 0$ and $\varphi(d)^{-1}$ cannot be zero since d is a unit. However, $\ker(\varphi)$ is trivial, so $r = 0$, but then $rd^{-1} = 0d^{-1} = 0$, meaning Φ is also injective! Thus, we forced the S to have Q embedded inside of it, meaning *any* extension of R to S would inevitably have Q. This makes Q the smallest extension of R and the unique extension of R.

[a] Reflexivity and symmetry come immediately. For transitivity, combining the two equations appropriately and realise you can factor by $d \in D$, which is not zero or a zero-divisor, which will force the equation you want to be zero, forcing the equivalence for transitivity

We'll give Q a name:

Definition 9.5.2: Ring Of Fraction

Let R be a commutative ring, D be the appropriate subset and Q be the extension as in the above theorem. Then

1. The ring Q is called the *ring of fractions* of D with respect to R and is dneoted $D^{-1}R$
2. If R is an integral domain an $D = R - \{0\}$, Q is called the *field of fractions* or *quotient field* of R.

we often want all units of a ring R. We denote all units as $\mathbb{R}^{\times}$.

Here are some examples of ring of fractions

Example 9.5.3

1. the trivial example would a a field $\mathbb{F}$, which would be it's own field of fractions.
2. the construction we've done at the beginning is a special case of our construction. So, if we take $R = \mathbb{Z}$, and take $D = \mathbb{Z} - \{0\}$, then $D^{-1}\mathbb{Z} = \mathbb{Q}$
3. Let $R = \mathbb{Z}$ and take $D = \{1\}$. Then the quotient ring would be $\mathbb{Z}$ again!
4. $R = \mathbb{Z}$, and take $D = \{19, 19^2, ...\}$. This is a valid D, since it doesn't contain 0, doesn't contain zero divisors, and is closed under multiplicationso what is $D^{-1}\mathbb{Z}$?
5. The ring $2\mathbb{Z}$ is alsmot an integral domain; it doens't have an identity. However, $(R^{\times})^{-1}(2\mathbb{Z}) = \mathbb{Q}$ as well! If you think about it, being able to divide will give us the identity, as well as $\frac{1}{2}$, which will produce the rest
6. $Z[p]$ should be elaborated on
7. $\mathbb{F}[x]$ becomes $\mathbb{F}(x)$, that is, elements of the form $\frac{p(x)}{q(x)}$ where $q(x)$ is not the zero-polynomial (Since $\mathbb{F}[x]$ is an integral domain, it's the only value we would need to worry about). Notice the square-brakets became parenthesis. In this way, we might see the motivation to write $\mathbb{Q}(\sqrt{D})$, since we are allowed to divide those values as well! The difference here being that $x, x^2, x^3, ..., x^n, ...$ are all distinct elements, non of them being an element of their original field. However, this same chain doens't appear for $Q(\sqrt{D})$: $\sqrt{D}, \sqrt{D}^2 = D$. This is because $\sqrt{D}$ is not an arbitrary indeterminate value, giving us this relation, similar to what we saw with group-rings.
8. Take $\mathbb{Z}[i]$ and $D = R^{\times} = R - \{0\}$. Then the ring of fractions is $\mathbb{Q}[i]$! Notice it's not something like $\mathbb{Z}(i)$. This comes down to how multiplication is defined in this ring, which is much more akin to multiplying complex numbers.
9. what is the field of fractions of $\mathbb{Q}[\sqrt{2}]$?
10. What if we have $R = \mathbb{Z}/4\mathbb{Z}$ and $D = \{1, 3\}$?

Theorem 9.5.4: Universal Property Of Field Of Fractions

If $h : R \to F$ is an injective homomorphism from R into a field F, then there exists a unique ring homomorphism $g : Q \to F$ which extends h

***Proof*:**
Use the uniqueness of the extension to show that $\Phi(Q)$ will be in any sub-field.

The universal property on $D^{-1}R$ will in fact still work if we generalize our construction to letting D have zero-divisors. The motivation for this generalization is something that (ref:HERE TBD). This topic will be explored in Part V, section 23.4.

(Another perspective would be that we are giving a solution to all equations of the form $bx - a = 0$, for $a, b \in R$. Thus, for $R = \mathbb{Z}$, we are doing:

$$\frac{\mathbb{Z}[x_{1/2}, x_{1/3}, ..., x_{1/p}, ...]}{(2x_{1/2} - 1, 3x_{1/3} - 1, ...,)} =: \mathbb{Q}$$

Though this way of looking at it seems more complicated, it fits better with the idea that we are adding relations to $\mathbb{Z}$. In particular, it shows that we are adding all *linear* relations (i.e. solutions) to $\mathbb{Z}$. We will in later section's show how we add non-linear solution.

Note that we construct $\mathbb{Z}$ from $\mathbb{N}$ by adding all monomial-linear polynomials. We then construct $\mathbb{Q}$ by taking all linear equations. We shall further explore in *Commutative Algebra* [8] how to think of non-monic polynomials with non-unit leading coefficients.)

9.5.5 Foreshadowing By taking the "infinite" version of polynomials, we managed to make all the non-zero-divisors become units. In this section, another way of producing using by formally inverting them. These two processes shall seem very different, however they are related, and the connection will be made in section 25.

9.5.1 Constructing $\mathbb{R}$?

In ref:HERE, we mentioned how we can think of $\mathbb{N}$ being a free-monoid on 1 element (say x), so in a sense, it was constructed from 1 element and the rules of a monoid. We then showed in ref:HERE how we can construct $\mathbb{Z}$ from $\mathbb{N}$ by adding the condition that $x^2 = x$ (i.e. $1 \cdot 1 = 1$) so that x is the multiplicative identity. We just showed how we can construct $\mathbb{Q}$ from $\mathbb{Z}$. We have also give an exercise that constructs $\mathbb{C}$ from $\mathbb{R}$. The reader might feel like the natural next step up would be to construct $\mathbb{R}$ from $\mathbb{Q}$. There is in fact a few constructions of the real numbers (two popular ones are the *Dedekind cut's construction* and the *Cauchy-limit construction*). However, it turns out that there is no "algebraic" construction of the real numbers! That is, we can't start off with some set (say $\mathbb{Q}$) and extended using binary (more generally n-ary) operations with certain rules on creating new elements from old ones (like we did for $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$)! The closest we can get is the *real closure* of $\mathbb{Q}$, denoted $\overline{\mathbb{Q}}^{\mathbb{R}}$ that is covered in section 21.1.

The fact that there isn't an algebraic construction of the reals is a fascinating result, and it will have to wait till we cover ________to fully appreciate. (ref:HERE).

Found this really interesting link!

https://math.stackexchange.com/questions/3761716/is-there-an-algebraic-way-to-construct-the-reals

It seems to go down fundamentally that an algebraic construction tries to use the elements from current object to construct the bigger object *by means of finite-tuples* (because of the way we think of functions!!), so if we were to try to do $\mathbb{Q} \rightsquigarrow \mathbb{R}$, then it would be representing elements from $\mathbb{R}$ with finite tuples from $\mathbb{Q}$. But $\mathbb{Q}$ is countable, and the number of finite tuples from $\mathbb{Q}$ is countable, but $\mathbb{R}$ has uncountably many elements. Not sure about the subtlies yet, Löwenheim-Skolem's Theorem is beyond my comprehension right now, but I think this idea is good enough for now to appreciate!

Exercise 9.5.1

1. Let F be a field of characteristic 0. Show that there exists a ring R such that $\text{Frac}(R) = F$. (If F is not characteristic 0, then there are a couple of complications. The equivalent exercise is in ref:HERE (under valuations in comm alg, since it requires valuations))

9.6 Chinese Remainder Theorem

Let's say we have a ring R and a collection of ideals $I_1, I_2, ..., I_k$. Then given

$$a_1 \bmod I_1, \cdots, a_k \bmod I_k$$

does there exist a $a \in R$ such that

$$a \equiv a_1 \bmod I_1, \cdots, a \equiv a_k \bmod I_k$$

Think about it for a moment before looking at the answer

9.6.1 Taking a moment aking a moment to think about the answer.

The answer depends on the choice of I_i. The reader should have tried this with $R = \mathbb{Z}$ and noticed it is not always possible, for example if $R = \mathbb{Z}$, $a \equiv 1 \bmod 2$, $a \equiv 2 \bmod 4$ does not have a solution.

The Chinese Remainder Theorem (CRT) is a way of finding an to this question. It's name sake most likely comes from a book *Sun Tzu Suan Ching*. It is said that it was originally developed either for accounting purposes or for dividing troops.

Before we get into the actual theorem, we'll first explore a direct application of Chinese Remainder Theorem:

Example 9.6.2: Concrete Use

simple form

$$\text{if } gcd(m, n) = 1 \text{ and } 0 \leq a < m\,, 0 \leq b < n$$

then

$$\exists!\ k,\ 0 \leq k < mn \text{ such that } k \equiv a \bmod m\,, k \equiv b \bmod n$$

Let's say: $m = 6$, $n = 5$, $a = 4$, $b = 2$. Then $k = 22$.

Proof. We have

$$k \equiv 4 \bmod 6,\ k \equiv 2 \bmod 5$$

so using the classical method, we first find

$$6 \cdot 5 = 30$$

from this, we find $30/6 = 5$ and $30/5 = 6$ to be the values such that and kno that we get the solution by solving the equivalent problem of:

$$k \equiv 5 \cdot 1 \bmod 30,\ k \equiv 6 \cdot 1 \bmod 30$$

meaning the solution is $(4 \cdot 5 \cdot 5) + (2 \cdot 6 \cdot 6) =$ □

When generalizing this result, we can no longer ask for numbers to be co-prime. The equivalent concept in rings are co-maximal ideals [25]

Definition 9.6.3: Comaximal Ideals

The ideals A and B of the ring R are said to be *comaximal* if $A + B = (1)$

The intuition on when to spot something is co-maximal is analogous to two numbers being prime. If $a, b \in \mathbb{Z}$ are prime, then by Bézout's identity, there exists numbers x, y such that $xa + yb = 1$, meaning $1 \in (a) + (b)$, meaning $(a) + (b) = \mathbb{Z}$ by proposition 9.4.10. In the next chapter, we'll generalize the concept of gcd for more general rings, meaning where the gcd can be defined, this theorem has a useful application [26]

Lemma 9.6.4: Characterizing Products of Coprime Ideals

Let $I_1, I_2, ..., I_n$ be coprime ideals. Then:

$$\bigcap_{i=1}^{n} I_i = \prod_{i=1}^{n} I_i$$

***Proof*:**

We shall do induction so consdier first when $n = 2$ so that $I, J \subseteq R$ such that $I + J = (1)$. It is always the case that $IJ \subseteq I \cap J$ so we have to show that $I \cap J \subseteq IJ$ when $I + J = (1)$. Given $I + J = (1)$, there exists $a \in I$, $b \in J$, such that $a + b = 1$. Now, if $r \in I \cap J$, then

$$r = r \cdot 1 = r(a + b) = ra + ra \in IJ$$

Since $r \in J$ and $a \in I$, $ra \in IJ$, an similalry $rb \in IJ$ since $r \in I$ and $b \in J$.

Then the inductive step follows immediately.

[25]why not co-prime ideals? We'll show how prime and maximal ideals will match in special rings in chapter 50, section 50.2, here we would usually apply such a result. However, we stay general since we can prove it in more general contexts

[26]For example, we'll use this theorem when looking at polynomial rings over fields

Lemma 9.6.5: Recursive Comaximal Of Products

Let $I_1, I_2, ..., I_n \subseteq R$ be ideals such that $I_i + I_n = (1)$ for all $i \in \{1, ..., n-1\}$. Then $(I \cdots I_{n-1}) + I_n = (1)$

***Proof*:**
By assumption, there exists $a_i \in I_k$ and $1 - a_k \in I_i$ such that

$$(1 - a_1) \cdots (1 - a_{n-1}) \in I_1 \cdots I_{n-1}$$

And

$$1 - (1 - a_1) \cdots (1 - a_{n-1}) \in I_n$$

since it is a comination of the elements $a_1, a_2, ..., a_{n-1} \in I_n$

Theorem 9.6.6: Chinese Remainder Theorem

Let $A_1, A_2, ..., A_k$ be ideals in R. The map

$$R \to R/A_1 \times \cdots \times R/A_k \qquad \text{defined by} \qquad r \mapsto (r + A_1, \cdots, r + A_k)$$

is a ring homomorphism with kernel $A_1 \cap A_2 \cap \cdots \cap A_k$. If for each $i, j \in \{1, 2, ..., k\}$ with $i \neq j$ the ideals A_i and A_j are comaximal, then this map is surjective and $A_1 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$, so

$$R/(A_1 A_2 \cdots A_k) = R/(A_1 \cap A_2 \cap \cdots \cap A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k$$

***Proof*:**
Take

$$R \to \bigoplus_{i=1}^{n} R/A_i \qquad a \mapsto (a \bmod A_1, ..., a \bmod A_n)$$

Then this is easily checked to be a ring homomorphism with kernel $A = \bigcap_i A_i$. Assuming now the ideals are comaximal, we shall show surjectivity. We shall do a proof by induction. If $n = 1$ where done, so let for let's say

$$R/(A_1 \cdots A_{n-1}) \cong R/A_1 \times \cdots \times R/A_{n-1}$$

Then by lemma 9.6.5, $(A_1 \cdots A_{n-1}) + A_n = (1)$, and hence this is reduced to two ideals. By comaximality there exists $a_1 \in A_1$, $a_2 \in A_2 \cdots A_{n-1}$ such that $1 = a_1 + a_2$. Then writting

$$x = x_2 a_1 + x_1 a_2$$

we get that $x \equiv x_1 \bmod A_1$ and $x \equiv x_2 \bmod A_2 \cdots A_n$. But then $x \mapsto (x_1 \bmod A_1, x_2 \bmod A_2 A_3 \cdots A_{n-1})$, completing the proof.

A useful consequence to remember is that if there are elements $r_i \in R$ where $r_i \bmod A_i$, then there exists an element a

$$a \equiv r_i \bmod A_i$$

The ability to solve such a system mod A_i is sometimes called the *Weak Approximation Theorem*.

Corollary 9.6.7: Chinese Remainder Theorem For Integers

Let $R = \mathbb{Z}$ $a_1, a_2, ..., a_n \in \mathbb{Z}$ be elements such that $\gcd(a_i, a_j) = 1$ for all $i \neq j$. Then if $a = a_1 \cdots a_n$, The function

$$\varphi : \mathbb{Z}/a\mathbb{Z} \to \mathbb{Z}/a_1\mathbb{Z} \times \cdots \times \mathbb{Z}/a_k\mathbb{Z}$$

given in the natural way is an isomorphism

***Proof*:**
This is an immediate consequene of the chinese remainder theorem since $\gcd(a, b) = 1$ if and only if $(a, b) = (a) + (b) = 1$

Exercise 9.6.1

1. Show that if $P + Q = R$, then for any $n, m > 0$, $P^n + Q^m = R$. Conclude the CRT can be generalized to ideals of the form $A_1^{n_1}, ..., A_k^{n_k}$.

9.7⋆ Monoidal Objects

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Maybe have this here to be able to introduce R-algebras as the monoidal objects in the R-**Mod** Category?

10

Exploring Integral Domains: Factorization and Valuation

This chapter will assume basic number theory knowledge from the preliminary chapter (section 0.2)

Integral domains are a huge overarching category of rings, being in a meaningful way the generalization of integers. As they do not having zero-divisors makes it possible to talk about partially inverting elements (by proposition 9.1.24). There are many different consequences of this result especially in the right specializations, for example divisibility, size, and even the euclidean algorithm shall all be definable on the appropriate integral domains[1]. These topics all interact greatly with each other and there is no linear way of addressing these ideas. However, books are inherently written with a total ordering and we must learn content in some order, and hence this chapter has been written with the idea of exploration the notion of divisibility (which the reader can think of the best that can be done to check invertibility if an element is not a unit). As this exploration naturally progresses, all other pertinent notions about integral domains shall become natural questions and will be answered.

The following diagram sums up the results of this chapter:

[1]There is also a more geometric interpretation, which is that integral domains will correspond to "irreducible" geometric objects (think of how $R \times S$ is no longer integral. This idea is explored in [13, Irreducibility and the Decomposition Theorem]

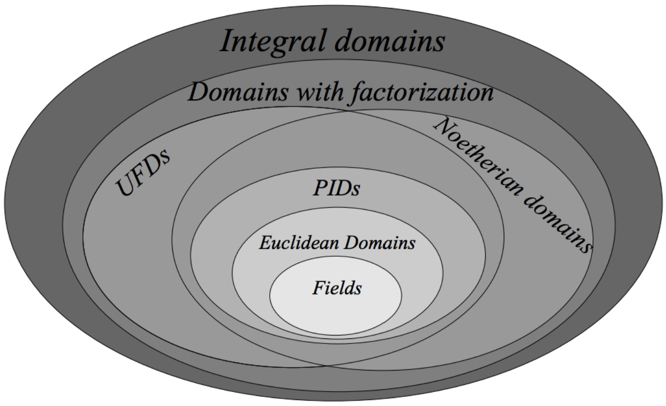


Figure 10.1: Image form Alufi's textbook

At the end of this chapter, we shall explore how these abstract notions were in fact developed to solve number-theoretic problems. We shall see our first example of finding the solution to a Diophantine equation, namely

$$x^2 + y^2 = p$$

In other words, which primes can be written as sums of two squares. As equations are the fundamental way in which we determine the properties of objects, this will be the first glimpse into the properties of prime numbers. This exploration is greatly generalized in chatper 50, section 50.4.6.

10.0.1 Note Some results in this chapter apply beyond integral domains. It will be made extra-clear when that is.

Goal

In fields, every element has an inverse. This is not so for general rings, as we have seen when studying units and zero-divisors. If not every element is a unit, there is a new idea that is introduced when learning some basic number theory: *factorization*. In this way, we can "decompose" an element into some basic building blocks. This idea of factorizing is one the reader is already familiar with: any number $x \in \mathbb{Z}$ can be factored, even uniquely, into prime numbers:

$$x = up_1^{k_1} p_2^{k_2} \cdots p_n^{k_n}$$

where $u \in \{1, -1\}$ and each p_i is a prime. Is there a generelizable notion of being factorizable in more general rings? Is this idea related to prime ideals? Some types of rings do admit an idea of factorization, or even some stronger conditions (like the euclidean algorithm being well-defined, or the factorization being unique), while other rings have a weaker condition (like the factorization not being unique or even existing). The first major focus of this chapter are these ideas.

The overarching idea in factorization is centered around the idea of divisibility like we have seen in $\mathbb{Z}$. For example:

$$2|8 \qquad 32|17728 \qquad \text{but} \qquad 3 \not| 8 \qquad 5 \not| 7$$

When an element is divisible, let's say $2\big|8$, then there exists some $x \in \mathbb{Z}$ such that $2x = 8$, that is, 8 got subdivided into other elements; in this way, factorization can be thought of as a "partial division" or "partial inverse" condition. This idea will be at the heart of our following exploration.

The result will give 3 new classes of rings and develop on 2 previously mentioned classes or rings which are very often studied.

1. *Atomic Domains* (or Domains with Factorization) are will be rings in which element can be factored into elements that can no longer be factored (they will be called irreducible elements), though not necessarily uniquely

2. *Noetherian domains* are similar to Unique Factorization domain as in every element can be factorzied, but it need not be unique. The reader may recall these as rings in which every ideal may be finitely generated.

3. *Unique Factorization Domains* (UFDs) will allow for every element to have a unique factorization, and let's us define the notion of there being a greatest common divisor between elements.

4. *Principal Ideal Domains* (PIDs) will be stronger than PIDs in that they allow for unique factorization and will have a more rigid structure. The reader may recall these as rings in which every ideal may be generated by a single element.

5. *Euclidean Domains* will have the property of the euclidean division algorithm applying to them to quickly find the greatest common divisor. These will allow for algorithmic decomposition of elements into prime elements.

Since factorization is heavily inspired by the properties of $\mathbb{Z}$, all rings in this chapter will be *integral domains* unless specified otherwise.

10.1 Factorization

We will first define more rigorously the notion of elements being factorizable

Definition 10.1.1: Divisibility

Let R be a commutative ring and $a, b \in R$ be elements of R. Then we say that a *divides* b (or b *is divisible by* a.) if there exists an element c such that

$$ca = b$$

Equivalently, $ca = b$ if and only if $b \in (a)$. Notationly, we write $a\big|b$ to mean "a divides b".

If R was not commutative, we can define left and right divisibility, though we do not cover the non-commutative case (see exercise ref:HERE for some exploration of the concept). The opposite of a being divisible by b is b being a *multiple* of a – this terminology is used much less often.

The equivalent condition with ideals are very insightful way of understanding divisibility. Here are the 4 equivlent condition's of divisibility for ease of reference:

1. $b|a$ if and only if

2. $a = bb'$ if and only if
3. $a \in (b)$ if and only if
4. $(a) \subseteq (b)$

It is unfortunate that the notation $b|a$ stuck, since you will have to swap at some point the order of the a and the b given that it's read "b divides a" and not "a is divided by b". It's a matter of getting used to.

The notation for divisibility using ideals is particularly insightful here; notice that if $u \in R$ is a unit, then for any element $r \in R$ $(r) = (ur)$. Both inclusion are easy and should be proven before continuing to read. Because of this property, it is immediate that $a\big|b$ if and only if $a\big|ub$, that is, if an element is divided (or not divided) by another, then multiplying it by a unit let's it remain divisible (or not divisible) by the other element. This is in fact something the reader is already is familiar with; notice that $6 = -1 \cdot -6$. Then $2\big|6$, and $2\big| -6$, but also $5 \nmid 6$ and $5 \nmid -6$. Thus, when we think of divisibility, it is important to think of a number being divisible "up to a unit":

Definition 10.1.2: Associative

Let R be a commutative ring and let $a, b \in R$ be two elements. Then we say that a and b are *associates* if the ideals generated by a and b are equal, that is:

$$(a) = (b)$$

It is important to check that this is equivalent to saying that a and b are associates if $a\big|b$ and $b\big|a$, showing how this links to divisibility. Notice that in the previous definition, R was not an integral domain, but just a commutative ring, which allows zero-divisors. This in fact breaks the intuition of associativity we just showed:

Example 10.1.3: associativity in non-integral domains

Let $R = \mathbb{Z}/6\mathbb{Z}$. Then $(\overline{2}) = (\overline{4})$, since

$$(\overline{2}) = \{\overline{2}, \overline{4}, \overline{0}\} = \{\overline{4}, \overline{2}, \overline{0}\} = (4)$$

Furthermore, $\overline{4} = \overline{2} \cdot \overline{2}$ but $\overline{2}$ is *not* a unit.

For this reason, restricting to integral domains let's us keep this "up to a unit" intuition for for divisibility using ideals:

Proposition 10.1.4: Associativity in Integral Domains

Let R be an integral domain. Then a and b are associates if and only if $a = ub$ for all units $u \in R$

***Proof*:**

Let $(a) = (b)$, so that for some $r, s \in R$, $ra = b$ and $a = sb$. Then

$$sra = sb = a$$

and since the cancellation property holds for integral domains, we have $sr = 1$, showing that r

and s are units.

Conversely, if $a = ub$ for all units $u \in R$, then it's clear that $(a) \subseteq (b)$ since $u \cdot b = a$ and $(a) \subseteq (b)$ since $u^{-1} \cdot a = b$.

Incidentally, notice that in example 10.1.3, the units of $\mathbb{Z}/6\mathbb{Z}$ is only $\overline{5}$ and that $\overline{2} \cdot \overline{5} = \overline{4}$, meaning the previous proposition actually holds for $\mathbb{Z}/6\mathbb{Z}$. This is true for a special subset of commutative rings usually called *rings with harmless zero-divisors* (harmless since the intuition of divisibility is preserved). These are not covered in this book, but can be explored in (reference something: here). Since the intuition for associativity holds true for all integral domains, and most work being done which includes divisibility in rings focuses on integral domains (due to the cancellation property), we shall do the same restriction here and work over integral domains rather than generalize the following proofs into much more (arguably needlessly) complex arguments to account for the case of rings with harmless zero-divisors.

With a notion of divisibility up to unit established, we can start thinking of what are the building blocks, the "fundamental elements", into which we will factorize an element:

Definition 10.1.5: Irreducible and Prime Elements

Let R be an intergral domain

1. Let $r \in R$ be a nonzero element that is not a unit. Then r is called *irreducible* in R if whenever $r = ab$, at least one of a or b must be a unit in R. Otherwise, r is said to be *reducible*.

2. The nonzero element $p \in R$ is called *prime* in R if the ideal (p) generated by p is a prime ideal. In other words, a nonzero element p is a prime if it is not a unit and whenver $p|ab$ for any $a, b \in R$ then either $p|a$ or $p|b$.

Example 10.1.6: Irreducible and Prime elements

Let $R = \mathbb{Z}$, and take $6 \in \mathbb{Z}$. Then we can see that $2 \cdot 3 = 6$. Since 2 and 3 are both not units, that means that 6 is a reducible element in $\mathbb{Z}$. On the otherhand, 2 and 3 are not reducible elements. If there existed a a, b such that $2 = ab$, then by our knowledge that 2 is a prime number (or by some elementary number theory argument), the only number that divide it are 1 and 2, so it must be that either

$$1 \cdot 2 = 2 \quad \text{or} \quad -1 \cdot -2 = 2$$

In both cases, a is a unit, and so 2 is irreducible (similarly for 3). Furthermore, 2 and -2 are associates, since $2 = -1 \cdot -2$, i.e., the differ by a unit. in $\mathbb{Z}$, every n are associates to it's negative $-n$.

As pointed out in example 9.4.23, 2 is also a prime element since (2) is a prime ideal. In fact, in the previous exercise, we used the fact that 2 is a prime integer (which corresponds to the prime ideals of $\mathbb{Z}$) in order to show it's irreducible. Though 2 is both an irreducible and a prime element in $\mathbb{Z}$, it need not be the case in general integral domains that an element is both!

As an example, take $\mathbb{F}[x]$ to be the polynomials over a field, and take $F_{0x} \subseteq \mathbb{F}[x]$ to be all the polynomials such that the coefficient of x is zero. As an example of elements in F_{0x}:

$$ax^3 + b \in F_{0x} \qquad cx^3 + dx \notin F_{0x} \qquad \forall a, b, c, d \in \mathbb{F}$$

It is easy to check that F_{0x} is an integral domain (as you should check), and so we can try to find irreducible and prime elements. Take x^6. It has two factorizations: $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$. The elements x^2 and x^3 are irreducible, but not prime.

To show x^2 and x^3 are irreducible, let's say

$$x^2 = p(x)q(x)$$

then since we're working with polynomials over an integral domain, by proposition 9.2.3:

$$\deg(x^2) = \deg(p(x)) + \deg(q(x))$$

meaning $2 = \deg(p(x)) + \deg(q(x))$. Since both $p(x)$ and $q(x)$ cannot be one (since $ax \notin F_{0x}$, for all $a \in \mathbb{F}$) then one of them has to be of degree zero, but since x^2 is monic this implies it must be 1, meaning it's a unit, meaning x^2 is irreducible. Similarly for x^3. Notice that this means that x^6 had two decompositions into irreducible factors, meaning the decomposition was not unique.

Now, take (x^2), i.e. the ideal generated by x^2. Notice that $(x^3)(x^3) \in (x^2)$ because $x^4 \cdot x^2 \in (x^2)$. However, $x^3 \notin (x^2)$. If it were, then there would exist a polynomal $p(x)$ such that $p(x)x^2 = x^3$. By the degree formula, $p(x)$ would have to be of degree 1. However, no degree 1 polynomial exists in F_{0x}. Therefore, (x^2) is not prime! This harkens back to when we talked about prime-ideals, where being prime in some way means we can work backwards. This "working backwards" let's us guarantee we decompose our number into unique elements, as we'll soon see.

Though not all irreducible elements are prime, the converse is true, that is, being "globally divisible"' is a strong enough condition that it guarantee's irreducibility:

Proposition 10.1.7: Prime Then Irreducible In Integral Domain

Let R be an integral domain. Then prime elements in R are always irreducible

***Proof*:**
Supose $(p) \subseteq R$ is a nonzero prime ideal, and that $p = ab$. Since (p) is a prime ideal, then either $a \in (p)$ or $b \in (p)$. If $b \in (p)$. Then $b = rp$ for some $r \in R$, and if $a \in (p)$ then $a = pr$ for some $r \in R$. Without loss of generality, let's say $a \in (p)$. Then

$$p = ab = prb$$

and since R is an integral domain, the cancellation property holds (proposition 9.1.24), and we get

$$1 = rb$$

implying r, b are units, and so b is a unit. The same argument applies if we chose b. Thus, p is irreducible.

Thus, being prime is stronger than being an irreducible. The natural question to ask now is whether all elements in an integral domain can be decomposed into irreducible or prime factors (if it is factored into prime factors, then automatically it is factored into irreducible factors). It might already be clear that some Integral domains we have explored have this property (ex. $\mathbb{Z}$, and if you're familiar with some manipulation of polynomials, then $\mathbb{F}[x]$). This is however not the case for all integral domains:

Example 10.1.8: Domain with no Factorization

Let $R = \overline{\mathbb{Z}} \subseteq \mathbb{C}$, where $\overline{\mathbb{Z}}$ consists of all elements in $r \in \mathbb{C}$ such there exists a monic polynomial with integer coefficients

$$a_0 + a_1 x + a_2 x^2 + \cdots + x^n \qquad a_i \in \mathbb{Z}$$

such that

$$a_0 + a_1 r + a_2 r^2 + \cdots + r^n = 0$$

In other words, $\overline{\mathbb{Z}}$ is the set of elements which are the root of a monic polynomial with integer coefficients (i.e. $\mathbb{Z}[x]$). The ring $\overline{\mathbb{Z}}$ is called the *integral closure* of $\mathbb{Z}$ over $\mathbb{C}$ and will be extensively studied in chapter 50 (algebraic number theory). For now, we just need to show that it is indeed an integral domain, and there exists elements that do not factor.

Since $\overline{\mathbb{Z}}$ is a subring of $\mathbb{C}$, we can use the subring test, that is, see if $\overline{\mathbb{Z}}$ is closed under multiplication and passes the subgroup test. Is passes the subgroup test, since for any $a, b \in \overline{\mathbb{Z}}$, there exists polynomials

$$a_0 + a_1 x + \cdots + x^n \qquad b_0 + b_1 x + \cdots + x^n$$

(where some coefficients might be zero to match the length) such that

$$a_0 + a_1 \cdot a + \cdots + a^n = 0 \qquad b_0 + b_1 \cdot b + \cdots + b^n = 0$$

then

$$(a_0 - b_0) + (a_1 - b_1)x + \cdots + x^n$$

is a polynomial that has root $a - b$. By a similar argument, $\overline{\mathbb{Z}}$ using the binomial theorem is closed under multiplication.

Note that $\overline{\mathbb{Z}}$ is an integral domain since $\mathbb{C}$ has no zero divisors (hence $\overline{\mathbb{Z}}$ can't have any either), however it is not a field. Notice that $2 \in \overline{\mathbb{Z}}$ (2 is the root of $x - 2$). However, $1/2 \notin \overline{\mathbb{Z}}$. If it were, then there would be a monic polynomial with root $1/2$. However, any monic polynomial would be of the form

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

if we plug in $1/2$, we get

$$\left(\frac{1}{2}\right)^n + a_{n-1}\left(\frac{1}{2}\right)^{n-1} + \cdots + a_1\left(\frac{1}{2}\right) + a_0$$

with a little more work (which I suggest the reader try out), this can be shown to never has a root (start with $x + b$ to see why it doesn't have a solution, and try to generalize from there).

However, $\overline{\mathbb{Z}}$ has elements (in fact all non unit elements) that do not factor! Take $r \in \overline{\mathbb{Z}}$. Then there exists integers $a_0, a_1, ..., a_{n-1} \in \mathbb{Z}$ such that

$$a_0 + a_1 r + a_2 r^2 + \cdots + a_n r^n = 0$$

Since $\overline{\mathbb{Z}} \subseteq \mathbb{C}$, the square root of any number is well-defined (see exercise ref:HERE (in Introduction to Rings)). Thus $\sqrt{r}$ is well-defined. Further, $\sqrt{r} \in \overline{\mathbb{Z}}$ since

$$a_0 + a_1\sqrt{r}^2 + a_2\sqrt{r}^4 + \cdots + a_n\sqrt{r}^{2n} = 0$$

Since $1/\sqrt{r} \notin \overline{\mathbb{Z}}$ (as similar argument to why $1/2$ is not in $\overline{\mathbb{Z}}$) it is not a unit, and hence $r = \sqrt{r}\sqrt{r}$ factors into two non-units. Now here is the key: we can repeat this process for $\sqrt{r}$, showing that

$\sqrt{r} = \sqrt{\sqrt{r}} \cdot \sqrt{\sqrt{r}}$, and so on and so forth. This process never terminates, showing that $\overline{\mathbb{Z}}$ has elements that do not factor!

For more examples, see:

https://math.stackexchange.com/questions/96790/integral-domain-that-is-not-a-factorization-domain

For this reason, we must define a special subclass of integral domains which do admit a factorization into irreducibles for every element:

Definition 10.1.9: Atomic Domains

Let R be an integral domain. Then R is an *atomic domain* (or *domain with factorization*) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ such that

$$r = q_1 q_2 \cdots q_n$$

that is, every element has a *factorization* (or *decomposition*) into irreducible

The name atomic is due to every element being decomposable into element that can no longer be subdivided, similar to atoms, where the word atom comes from the Greek *atomos* meaning “not cuttable”. Note that the q_i's need not be distinct (for example, we will show that $\mathbb{Z}$ is an atomic domain, and that $4 = 2 \cdot 2$). Usually, we are more interested in domains where the factorization into irreducible elements is unique:

Definition 10.1.10: Unique Factorization Domain (UFD)

Let R be an integral domain. Then R is a *Unique Factorization Domain* (commonly abbreviated to UFD) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ and unit u_r such that

$$r = u_q q_1 q_2 \cdots q_n$$

and if there is another set of irreducible element $q'_1, q'_2, ..., q'_m$ such that $r = u_{q'} q'_1 q'_2 \cdots q'_m$, then $n = m$, and the two products differ by a unit after some reshuffling (i.e. they are associates), that is, for appropriate $u \in R$

$$q_1 q_2 \cdots q_n = u q'_{\sigma(1)} q'_{\sigma(2)} \cdots q'_{\sigma(n)}$$

where $\sigma(i)$ is the appropriate permutation

The following corollary is stated for the sake of reference:

Corollary 10.1.11: Unique Factorization Domain $\Rightarrow$ Atomic Domain

Let R be a Unique Factorization Domain. Then R is an Atomic domain

***Proof*:**
By definition, every element in a Unique Factorization Domain has a factorization into irreducibles, and so is an Atomic Domain

Note that "unique" means unique up to unit, as we pointed out earlier. The factorization in both definitions could have had the unit dropped and just have been written $r = q_1, q_2, ..., q_n$ since it is a the result is an associate to $u^{-1}r$, and so it is harmless to drop the unit. However, I chose to keep the unit in the definition since it may lead to certain unfortunate mistakes when factoring concrete elements.

Example 10.1.12: Unique Factorization Domains

1. Every field is trivially a Unique Factorization Domain, since every nonzero element is a unit, so there are no elements for which the two properties must be verified. In some textbooks, fields are purposefully omitted as Unique Factorization Domains to omit mentioning them as exceptions in proofs. They are still considered Unique Factorization Domains in this book since they fit well in the diagram presented at the beginning of this chapter. However, in a sense, this means that in a field there is no "unique factorization", or at least it's vacuously true that all elements can be uniquely factorized since no nonzero elements are non-units.
2. As we saw in example 11.2.5, $F_{0x} \subseteq \mathbb{F}[x]$ which has x coefficient of 0 is not a Unique Factorization Domain. We saw that x^2 and x^3 are irreducible. If we take x^6, then has two factorizations: $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$, and hence the factorization is not unique.
3. The ring $\mathbb{Z}[2i]$ is *not* a Unique Factorization Domain. $4 = 2 \cdot 2 = ((-2i) \cdot 2i)$. The elements 2 and $2i$ are irreducible, but not associates, since $2 = 2 \cdot i$, but $i \notin \mathbb{Z}[2i]$. They are furthermore not prime, since $\mathbb{Z}[2i]/(2i) \cong \mathbb{Z}/4\mathbb{Z}$ which is not an integral domain (see proposition 9.4.25)
4. In the ring $\mathbb{Z}[i]$, 2 and $2i$ *are* associates. We have $i \in \mathbb{Z}[i]$, and $i \cdot i^3 = 1$ showing that i is a unit, and hence 2 and $2i$ are indeed associated. We'll soon show that $\mathbb{Z}[i]$ is also a Unique Factorization Domain (see section 10.3)
5. We shall prove in chapter 11 that if R is a UFD, then $R[x]$ is a UFD. Note how this contrasts with PID's ($\mathbb{Z}$ is a PID but $\mathbb{Z}[x]$ is not, see example 9.4.6[3]).

All of these are examples of Unique Factorization Domains. Atomic Domains which are not Unique Factorization Domains are rather difficult to find. Fortunately, there is an easy criterion that allows us to find (a subset of) common atomic domains

10.1.1 Example of Atomic Domains: Noetherian Domains

The goal of this section is to show that Noetherian domains are atomic domains, and that there exists noetherian domains which are not Unique Factorization Domain's. Let's first remember the definition of a Noetherian Domain, or more generally a Noetherian ring:

Definition 10.1.13: Noetherian Rings

Let R be ring. Then if every ideal $I \subseteq R$ is finitely generated, then R is called a *Noetherian Ring*. If R is an integral domain, we call R a *Noetherian Domain*

Most results that will be presented apply to Noetherian rings in general, and so we will work with Noetherian rings. The first thing to show is how to relate Noetherian rings to the idea of factorization. Emmy Noether discoverd in HERE a condition which let's us characterize this property[2]:

Proposition 10.1.14: Ascending Chain Condition

The following are equivalent:

1. R is a Noetherian ring
2. for every chain of ascending left-ideals of R
$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$
there exists an number m such that for all $n > m$, $I_n = I_{n+1} = \cdots$
3. Every nonempty family of left-ideals of R partially ordered by inclusion has a maximal left-ideal

Note that replacing left by right or two-sided ideals will not break the theorem. If R is an integral domain, then there is no distinction between these concepts

***Proof*:**

(1) $\Rightarrow$ (2) Let R be a Noetherian ring. If $I_1 = R$ we're done, so assume I_1 is a proper ideal of R, and consider any ascending chain of left ideals
$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \subseteq I_n \subseteq \cdots$$
Take
$$I = \bigcup_{i=1}^{\infty} I_i$$
which is an ideal for a similar reasoning as presented in the proof that every proper ideal is contained in a maximal ideal (see proposition 9.4.19). Since R is a Noetherian domain and $I \subseteq R$, then I is finitely generated. Let's say $I = (a_1, a_2, ..., a_n)$. Then these generators must each be in some ideal I_m, meaning that for all $n > m$, $I_{n+1} = I_{n+2} = \cdots$, showing that R respects the ascending chain condition

(2) $\Rightarrow$ (3) Let R respect the ascending chain condition on it's left ideals, and let $\mathcal{F}$ be a family of left-ideals of R. Pick some random $I_1 \in \mathcal{F}$. If I_1 is maximal in $\mathcal{F}$, then we'd be done, so let's say I_1 is not maximal in $\mathcal{F}$. Then there exists an I_2 such that $I_1 \subset I_2$. If I_2 is maximal in $\mathcal{F}$, then we're done. If not, then we can

[2] more on the generalization of it??

continue this process. If every element we pick is not maximal, we can create an infinitely long chain[a]

$$I_1 \subsetneq I_2 \subsetneq I_3 \cdots \subsetneq I_n \subsetneq \cdots$$

However, that contradicts the ascending chain condition, meaning for some n, the chain must stabilize. Thus, $\mathcal{F}$ must have a maximal element

(3) $\Rightarrow$ (1) Let $I \subseteq R$ be a left-ideal. Take $\mathcal{I}$ to be the set of all finitely-generated left-ideals contained in I. The set $\mathcal{I}$ is nonempty since $\{0\} \in \mathcal{I}$. It is thus a family of left-ideals of R and so by the 3rd condition, $\mathcal{I}$ admits a maximal,/Then if say I' [b]. If $I \neq I'$, then there exists an element $x \in I$ such that $x \notin I'$. But then $(I' \cup x)$ is a finitely generated left-ideal larger than I', contradicting the maximality of I'. Hence, $I = I'$, showing that I is finitely generated.

[a]technically, to create this infinitely long chain, we used the axiom of choice

[b]Note that Zorn's lemma can't be used here since it is not known whether every chain will have an upper bound in this set, their union potentially no longer being finitely generated

10.1.15 Remark This proof will become a special case of the equivalent proof for modules in section 12.3.4.

For our purposes, this let's us think of Noetherian domains in terms of the ascending chain condition. You probably already see how this might start related to Factorization domains: If every element in the ring is finitely generated, it must be decomposable into a finite number of factors. This is essentially true:

Proposition 10.1.16: A.C.C. and Factorization

Let R be an integral domain, and let r be a nonzero nonunit element of R. Then if every ascending chain of *principal ideals*

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \cdots \subseteq (r_n) \subseteq$$

stabilizes (there exists an m such that for all $l > m$, $(r_l) = (r_{l+1}) = \cdots$), then r can be factored into irreducible elements. In particular, if R respects the A.C.C. on principal ideals, then R is an atomic domain.

Re-writing these ideal using the divisibility notation, it may be more evident: if $r_1 \big| r$, $r_2 \big| r_1$, and so on eventually stabalizes, then r can be factored into irreducible elements.

***Proof*:**
Let's prove this using the contrapositive ($\neg B \Rightarrow \neg A$), so that we'll show if r does not factor into irreducible then there exists an ascending chain of principal ideals that does not stabalize.

Since r is not irreducible, then there exists two nonunit non zero element r_1, s_2 suc that $r = r_1 s_2$. Since r does not have a factorization, either r_1 or s_1 are reducible. Without loss of generality, let's say r_1 is reducible. Then $(r) \subseteq (r_1)$ since $s_2 \cdot r_1 = r$, and $r_1 = r_2 s_2$. The same being applied to r_1, we get $(r) \subseteq (r_1) \subseteq (r_2)$. Continuing on indefinitely, we get

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \cdots \subseteq (r_n) \subseteq \cdots$$

which never stabilizes since r does not factor, as we sought to show.

Notice how similar this it to Noetherian rings: In a Noetherian ring, the ascending chain condition works for *all* left/right/two-sided ideals. Therefore, it certainly works when restricted to only it's *principal ideals*. We therefore get this immediate corollary

Corollary 10.1.17: Noetherian Domain $\Rightarrow$ Atomic Domain

Let R be a Noetherian Domain. Then it is an Atomic Domain

***Proof*:**
By proposition 10.1.14, R respects the ascending chain condition on any left ideals. Since any element $r \in R$ forms a principal ideal, any ascending chain of principal ideals must stabalize. Thus, by proposition 10.1.16 r must have a factorization. Since r was arbitrary, every element has a factorization, and so R is an Atomic Domain as we sought to show

It might seem like the converse is clearly true: If R is an atomic domain, then R respects the ascending chain condition on principal ideals. P.M.Cohn made this same assertion in 1968 during his research on rings[3], however it was ultimately proven to not be the case: there are element for which there are infinitely many factorizations. Though each factorization is finite, there are factorizations for every natural number (see exercise ref:HERE, maybe?). This theorem is still useful to our endeavour, since it let's us link a finite generation property (A.C.C. forces all principal ideals to be contained in at most a finitely generated ideal[4]) to a factorization condition (a condition for how to decompose elements into simpler elements).

Unfortunately, there isn't a simple counter-example I know of until we prove the famous Hilbert's Basis Theorem in chapter 23. For the reader who wants to know the counter-example, they can either skip to Hilbert's basis theorem in *Commutative Algebra* [8, Hilbert's Basis Theorem (HBT)] (you have everything you need to know to understand it now), or take on faith that "if R is a Noetherian Domain then $R[x_1, x_2, ..., x_n]$ is a Noetherian Domain". The following is a counter-example:

Example 10.1.18: Atomic Domain $\not\Leftarrow$ Noetherian Domain
Let $R = \mathbb{Z}[x_1, x_2, ..., x_n, ...]$ be a polynomial ring over an infinite number of variables (see section 9.2). Notice that $\mathbb{Z}$ is a Noetherian Domain: As we have shown, $\mathbb{Z}$ a Principal Ideal Domain (example 9.4.6, and so any ideal of $\mathbb{Z}$ is generated by one element, and so it is Noetherian. By Hilbert's Basis Theorem, $\mathbb{Z}[x_1, x_2, ..., x_n]$ is a Noetherian Domain as well.

Now, any polynomial $f \in R$ has finitely many terms by definition, and so is also an element of the subring $\mathbb{Z}[x_{\sigma(1)}, x_{\sigma(2)}, ..., x_{\sigma(n)}]$ where $\sigma(i)$ is some appropriate permutation function so that the terms of f are in the aforementioned subring. But then it is an element of a Noetherian Domain, which means it could be factored by proposition 10.1.16. Since $\mathbb{Z}[x_{\sigma(1)}, x_{\sigma(2)}, ..., x_{\sigma(n)}] \hookrightarrow R$, the same factorization happens in R, and since f was arbitrary, R is an Atomic Domain

However, R is not a Noetherian Domain. The ideal $(x_1, x_2, ..., x_n, ...)$ of all of the terms of R is not finitely generated (it does not respect the ascending chain condition).

This means that in the image presented in the beginning of the chapter, the strict inclusion of Noetherian Domains in Atomic Domains is justified. Similarly, Being an Atomic Domain does not

[3] P.M. Cohn, Bézout rings and their subring; Proc. Camb. Phil.Soc. 64 (1968) 251–264
[4] assuming the Axiom of Choice

mean the factorization is unique

Example 10.1.19: Atomic Domain $\not\Leftarrow$ Unique Factorization Domain
Take the ring F_{0x} from example 10.1.6. Take any element $f \in F_{0x}$, and form a ascending chain of principal ideals. For the sake of contradiction, let's say the chain never stabilizes: $(f) \subsetneq (f_1) \subseteq (f_2) \subsetneq \cdots \subseteq (f_n) \subsetneq \cdots$. Then since $(f_i) \subsetneq (f_j)$, and every degree zero element is a unit (recall that $F_{0x} \subseteq \mathbb{F}[x]$ for a field $\mathbb{F}$) then the degree of f_j must be *less* than that of f_i. This means we have a decreasing chain:

$$\deg(f) > \deg(f_1) > \deg(f_2) > \cdots > \deg(f_n) > \cdots$$

Notice importantly that this chain *must* terminate since there are no polynomials of degree 0. Since f was arbitrary, every element has a factorization and so it is an Atomic Domain.

However, as we have shown in example 10.1.6, $x^6 = x^2x^2x^2 = x^3x^3$ are two factorizations of x^6 where x^2 and x^3 are not associates, showing it is not a Unique Factorization Domain.

Notice the ring F_{0x} is not a Noetherian Domain in the previous example. Using Hilbert's Basis Theorem, we can also find a Noetherian Domain which is not a UFD:

Example 10.1.20: Noetherian Domain $\not\Rightarrow$ Unique Factorization Domain
Let

$$R = \frac{\mathbb{C}[x,y,z,w]}{(xw - yz)}$$

be the polynomial ring over 4 variables quotiented by the ideal $(xw - yz)$ [a]. By Hilbert's Basis Theorem, $\mathbb{C}[x,y,z,w]$ is a Noetherian ring. By proposition 9.4.8, $\frac{\mathbb{C}[x,y,z,w]}{(xw-yz)} = R$ is a noetherian ring. It is also an integral domain, since $(xw - yz)$ is a prime ideal (see exercise ref:HERE. It is therefore a Noetherian Domain (and hence an Atomic Domain).

However, it is not a Unique Factorization Domain. The element x, y, z, w are all irreducible, and not associates. Since $xw - yz = 0$, we have $r = xw = yz$ has two distinct factorization, showing that R is not a Unique Factorization Domain.

[a] In [13, Algebraic Sets and the Zariski Topology], we will see this ring be called the *quadratic cone in* $\mathbb{A}^4$

The converse is not true either: being a Unique Factorization Domain does not imply it is a Noetherian Domain. In fact, the ring $\mathbb{Z}[x_1, x_2, ..., x_n, ...]$ from example 10.1.18 is also a Unique Factorization Domain that is not a Noetherian Domain. This proof relies a theorem similar to Hilbert's Basis Theorem (in particular a generalization of theorem 11.2.3 on page 363). However, this theorem requires much longer build-up, and so for now the reader can take it on faith that such a ring is a counter-example (see exerise ref:HERE for a way to prove this using gcd domains).

Thus, we have shown (or claimed) that

$$\begin{array}{rclrcl}
\text{Noetherian Domain} & \Rightarrow & \text{Atomic Domain} & \text{Atomic Domain} & \not\Rightarrow & \text{Noetherian Domain} \\
\text{UFD} & \Rightarrow & \text{Atomic Domain} & \text{Atomic Domain} & \not\Rightarrow & \text{UFD} \\
\text{UFD} & \not\Rightarrow & \text{Noetherian Domain} & \text{Noetherian Domain} & \not\Rightarrow & \text{UFD}
\end{array}$$

This completes some of the layers in the diagram presented at the beginning of this chapter.

:exercise

Exercise 10.1.1

1. Let R be a finitely generated ring. Does that imply R is Noetheria
2. Show that $\mathbb{C}[x, y, z, w]/(xw - yz)$ is an integral domain.
3. Show that if $M \subseteq R$ is a maximal ideal then M is Noetherian. In fact, if the prime ideals are finitely generated, then R is Noetherian.

10.1.2 Unique Factorization Domains

We know continue exploring Atomic Domains in which the factorization of each element is unique (up to unit). As a reminder, here is the definition of a UFD:

Definition 10.1.21: Unique Factorization Domain (UFD)

Let R be an integral domain. Then R is a *Unique Factorization Domain* (commonly abbreviated to UFD) if for every non-unit nonzero element $r \in R$, there exists irreducible elements $q_1, q_2, ..., q_n$ such that

$$r = q_1 q_2 \cdots q_n$$

and if there is another set of irreducible element $q'_1, q'_2, ..., q'_m$ such that $r = q'_1 q'_2 \cdots q'_m$, then $n = m$, and the two products differ by a unit after some reshuffling (i.e. they are associates), that is, for appropriate $u \in R$

$$q_1 q_2 \cdots q_n = u q'_{\sigma(1)} q'_{\sigma(2)} \cdots q'_{\sigma(n)}$$

where $\sigma(i)$ is the appropriate permutation

Perhaps the most intuitive way of thinking about elements in Unique Factorization Domains is by linking to every element of the UFD a multiset (a set that allow's multiplicity) of irreducible factors. Units and 0 are both associated to empty multisets. The multiplication structure can also be thought of as forming a free abelian monoid structure with all prime elements as generators.

Lemma 10.1.22: Irreducible and Multisets

Let R be a Unique Factorization Domain and let $a, b, c \in R$ be nonzero elements of R. Then

1. $(a) \subseteq (b)$ if and only if the multiset of irreducible elements of a is contained in the multiset of irreducible elements of b
2. a and b are associates if and only if their corresponding multisets are equal
3. if $a = bc$, then the corresponding multiset of a is the union of the multisets of b and c

This lemma essentially shows that working with elements of a Unique Factorization Domain comes down to working with their corresponding multiset, making the analysis of their structure more often than not being more set theoretical or number theoretical. As an example of this, Unique Factorization Domains make it very easy to define the concept of *greatest common divisor*.

Greatest Common Divisor

The set of sets of multi-sets thus has a natural lattice paralleled on rings given by the divisibility. In fact, meet and joins are well-defined[5]:

Definition 10.1.23: Greatest Common Divisors In Commutative Rings

Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. Then the *greatest common divisor* of a and b is a nonzero element d such that

1. $d|a$ and $d|b$,
2. if $d'|a$ and $d'|b$ then $d'|d$

A greatest common divisor of a and b will be denoted $gcd(a, b)$ or (a, b)

Notice that in the definition, R was a commutative ring, since divisibility was defined for commutative rings in general. It should be noted that the gcd of two elements is unique *up to associativity.* it is helpful to put the definition of gCD in ideal-interpretation of divisibility:

Definition 10.1.24: GCD Using Principal Ideals

Let R be a commutative ring and $(a), (b) \subseteq R$. Then $\gcd(a, b)$ is an ideal $I \subseteq R$ such that $(a, b) \subseteq I$ and I is the smallest ieal that contains (a, b). In other words

$$\gcd(a, b) = (a) + (b)$$

As an exercise, see that this does indeed match the previous definition of gcd, and test it in $\mathbb{Z}$. There do exist rings for which the gcd between two elements do not exist (in example 10.1.6, take the ring R_{0x} from the second part of the example. Prove that the $\gcd(x^5, x^6)$ does not exist, see exercise ref:HERE). For Unique Factorization Domains, every element has a gcd:

Proposition 10.1.25: GCD in Unique Factorization Domain

let R be a Unique Factorization Domain. Then for any $a, b \in R$, $\gcd(a, b)$ exists. Furthermore, given

$$a = up_1^{n_1} p_2^{n_2} \cdots p_k^{n_k} \qquad b = vp_1^{m_1} p_2^{m_2} \cdots p_k^{m_k}$$

is the factorization of a and b, that is, every p_i is irreducible, u and v are associates, every p_i is not associates to p_j, and some n_i's or m_i's can be zero. Then the gcd (up to a unit) is:

$$\gcd(a, b) = p_1^{\min(n_1, m_1)} p_2^{\min(n_2, m_2)} \cdots p_k^{\min(n_k, m_k)}$$

***Proof*:**
This is an almost immediate consequence of using the factorization of a and b. Let d equal the above equation. Then clearly $d \big| a$ an $d \big| b$. If there is some c that also divides a and b ($c \big| a$ and $c \big| b$), then $a = ca'$ and $b = cb'$ for appropriate a' and b'. By lemma 10.1.22, c must be contained

[5] see section 0.3.1

in both the multisets of a and b, and therefore

$$c = wp_1^{\ell_1} p_2^{\ell_2} \cdots p_k^{\ell_k}$$

where w is a unit and $\ell_i \leq n_i$ and $\ell_i \leq m_i$. Then $\ell_i \leq \min(n_i, m_i)$ for all i, and so $(c) \subseteq (d)$, and by lemma 10.1.22 $c \big| d$, as we sought to show

10.1.26 Relation Between Lattice's and UFDs It should be noted that there are integral domains which are not Unique Factorization Domain's but do have the concept of gcd defined for every element. The class of all integral domains which have the concept of gcd defined for every element is broader than Unique Factorization Domain's and are not always Noetherian, and so such domains need their own name and are called *gcd domains*. In a sense, gcd domains are a generalization of Unique Factorization Domains: A gcd domain is Unique Factorization Domain if and only if the gcd domain satisfies the ascending chain condition on principal ideals (similarly to how Atomic domains do). However, a gcd domain isn't always an Atomic domain, and hence fits a bit awkwardly in our current hierarchy. In essence, this is because gcd domains hold information on information between elements, while atomic domains hold element-wise information.

For future purposes, gcd domains do fit into an important hierarchy of domains: *normal domains* shall be introduced in *Commutative Algebra* [8, Integral Elements], and all gcd domains are normal domains:

$$\text{normal domains} \supseteq \text{gcd domains} \supseteq \text{UFD's}$$

The definition of GCD's using ideals suggests that the GCD, when it exists, ought to be unique in general, and indeed it is:

Proposition 10.1.27: Uniqueness Of Greatest Common Divisor (In Integral Domain)

Let R be an integral domain. If (d) and (d') are both greatest common divisors of then $(d) = (d')$.

***Proof*:**
If d and d' are greatest common divisors, then they must each divide each other (condition 2 of definition 10.1.23), and so we'll get $(d') \subseteq (d)$ and $(d') \supseteq (d)$, meaning $(d') = (d)$,

Just like for the integers, we may also define the dual of the GCD, the least common multiple. In the following, the LCM will be defined in terms of principle ideals, the similar definition should be translated into elements:

Definition 10.1.28: Least Common Multiple (lCM)

Let R be a commutative ring and $(a), (b) \subseteq R$. Then the *least common muliple* is the ideal I such that $I \subseteq (a)$ and $I \subseteq (b)$ and is the largest such ideal. Namely

$$\operatorname{lcm}((a), (b)) = (a) \cap (b)$$

Properties of UFD's

As we have shown in example 10.1.6 and 10.1.20 being an irreducible element in an integral domain does not imply that the element is a prime element. This was because prime elements are "stronger" than irreducible in that they not only have to be able to be divisible, but they have to be "globally divisible" for any factorization. However, since factorization is unique in a Unique Factorization Domain, it should come as no surpise that there is no distinction between prime elements and irreducible elements:

Proposition 10.1.29: Prime If And Only If Irreducible In UFD

Let R be a Unique Factorization Domain. Then a nonzero element is a prime if and only if it is irreducible

***Proof*:**
The fact that prime elements are irreducible was proven in proposition 10.1.7, so we'll show the converse. Let a be an irreducible element. Then a is not a unit, and if $a = bc$ then either b or c is a unit. Now, assume $bc \in (a)$ so that $d \cdot a = bc$. Then $(bc) \subseteq (a)$, and by lemma 10.1.22, the set of the irreducible factors of a (i.e. a itself) is contained in the set of irreducible factors of b and/or c. Therefore, it must be that a is a factor of b, or c, or both. If it's factors are within the factors of b, then $b \in (a)$, and if it's factors are within the factors of c, then $c \in (a)$. If in both, then $b, c \in (a)$. In any case, at least one of b or c are in (a), showing that (a) is prime.

Thus, if we see every irreducible element being prime as representing "global" irreducibility (or divisibility), then if we further assume the A.C.C. on principle ideals, we see that this ought to fully characterize UFDs. The following theorem demonstrate exactly this:

Theorem 10.1.30: Unique Factorization Domain Equivalence

Let R be an integral domain. Then R is a Unique Factorization Domain if and only if

1. The ascending chain condition for principal ideal holds in R
2. Every irreducible element is a prime element

***Proof*:**

($\Rightarrow$) Let R be a Unique Factorization Domain. Then by proposition 10.1.29 every irreducible element is a prime element, and by corrollary 10.1.11 R is an atomic domain, and so by proposition 10.1.16 R respects the ascending chain condition (this can also be proven directly using the corresponding multisets to elements)

($\Leftarrow$) Let R satisfy the A.C.C. for principal ideals and let every irreducible be prime. By proposition 10.1.16, R is an Atomic domain. It remains to show that R factors uniquely. To show this, let n be a nonunit and

$$n = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$$

be two factorizations of n into irreducibles. Then since $q_2 q_3 \cdots q_n \cdot q_1 = p_1 p_2 \cdots p_n$, we have $p_1 p_2 \cdots p_n \in (q_1)$. Since q_1 is irreducible, by the 2nd condition it is a prime, and so for some i, $p_i \in (q_1)$. Without loss of generality, we can say $p_1 \in (q_1)$ by just rearranging and relabelling the factors. Since $p_1 \in (q_1)$, then $u_1 \cdot q_1 = p_1$. Since p_1 is irreducible and q_1 is not a unit, necessarily u_1 is a unit. Then, we can re-write the equation as

$$u_1 q_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$$

Cancelling out q_1 from each side and replacing p_2 by $u_1 p_2$ produces us

$$p_2 p_3 \cdots p_n = q_2 q_3 \cdots q_m$$

from here, we can continue this process for the rest of the p_i's, where we will end up with

$$u_1 u_2 \cdots u_n q_1 q_2 \cdots q_m = q_1 q_2 \cdots q_m$$

leting $u = u_1 u_2 \cdots u_n$, we get $uq_1 q_2 \cdot q_n = q_1 q_2 \cdots q_n$. Note that $n = m$ since if not then at some point the remaininng part of the left hand side will be equal to 1, a contradiction to n not being a unit. Thus, the two factorization are equal up to associates, as we sought to show.

Notice how the uniqueness of the factorization allowed us to make the A.C.C. for principal ideals a defining feature of Unique Factorization Domains, in contrast to ho the A.C.C. for principal ideals was insufficient to define Atomic Domains.

This theorem is also very useful in easily remembering how Unique Factorization Domain's and Noetherian Domains are different: If we start with a domain with the A.C.C. for principal ideals (i.e. the types of Atomic domains we ill work with) strengthen the A.C.C. condition to work on all ideals, we get a Noetherian domain. Conversely, if we are working with a Noetherian domain and we loosen the A.C.C. condition to principal ideals but add that all irreducible are primes, then we get a Unique Factorization Domain.

10.1.3 Important UFD: Principle Ideal Domains

Principal Ideal Domains are stronger than Noetherian Domains in the sense that every ideal is generated by a single element (rather than finitely many element for Noetherian Domains). They are therefore immediately Atomic Domains by proposition 10.1.16. By theorem 10.1.30, if a Principal Ideal Domain also has that every irreducible element is prime, then it is automatically a Unique Factorization Domain. Are irreducible elements prime in a Principal Ideal Domain? The answer is yes; hence PIDs form a subclass of UFDs. Take a moment to think about why irreducible elements are prime in PIDs before proceeding to the next proposition (hint: can you prove prime ideals *must* be maximal ideals in a PID?). From this proof, there will now be a new set of examples for Unique Factorization Domains similarly to how all Noetherian domains are an easy set of examples for Atomic domains

10.1.31 Taking a moment

As the hint suggests, we first prove the following important intermediary result:

Proposition 10.1.32: In PID, Prime Ideal iff Maximal Ideal

Let R be a Principal Ideal Domain and take a nonzero ideal $(r) \subseteq R$. Then (r) is a prime ideal if and only if it is a maximal ideal.

Intuitively, since

***Proof*:**
Let R be a Principal Ideal Domain. If an ideal of R is maximal, by corollary 9.4.26 that ideal is prime. Thus we'll prove the converse.

Let $(p) \subseteq R$ be a prime ideal, and assume that $I \supseteq (p)$ is an ideal that contains (p). We must show that $I = (p)$ or $I = R$. Since R is a Principal Ideal Domain, there exists an $m \in R$ such that $(m) = I$. Since $(m) \supseteq (p)$, $p \in (m)$, and so $p = rm$. Since p is a prime element, then either $r \in (p)$ or $m \in (p)$. If $m \in (p)$, then $(m) = (p)$, fulfilling the $I = (p)$ case. If $r \in (p)$ then $r = sp = ps$ and so

$$p = rm = psm$$

And by the cancellation property of integral domains, $1 = sm$, and so $1 \in (m)$ implying that $(m) = R$, fulfilling the $I = R$ condition. That means that (p) fulfills the conditions of being a maximal ideal, as we sought to show.

The main idea is that any ideal requries only 1 element to repersent it, which as it will turn out will be key to why elements will factor uniquely.

Proposition 10.1.33: Prime If And Only If Irreducible In PID

Let R be a Principal Ideal Domain. Then a nonzero element is a prime if and only if it is irreducible. Along with the fact that Principal Ideal Domain are Noetherian Domains (and hence respecting the A.C.C. on principal ideals) we have

$$\text{PID} \Rightarrow \text{UFD}$$

***Proof*:**
Since a Principal Ideal Domain is an integral domain, we've just shown in proposition 10.1.7 that prime elements are irreducible, so we'll show the converse.

Let's say p is an irreducible element of R. To show it's prime, we must how (p) is a prime ideal. Since we're working in a Principal Ideal Domain, by 10.1.32, it's equivalent to show that (p) is a maximal ideal. So, for the sake of contradiction, let's assume (p) is not maximal, and so there exists a strictly larger ideal that is the maximal ideal containing (p) $(p) \subsetneq (m) \neq R$ for appropriate $m \in R$. Then

$$p = am$$

for some $a \in R$. Since p is irreducible, then either a or m is a unit in R. If M is a unit, then $m^{-1}m = 1 \in (m)$ implying $(m) = (1) = R$, but then it's not a maximal ideal, since maximal ideals are proper ideals. Conversely, if a is a unit, then by proposition p and m are associates, meaning $(p) = (m)$. But since (m) was maximal, that makes (p) maximal. Thus, (p), by proposition 10.1.32, (p) is a prime ideal.

Therefore, since all Principal Ideal Domain are Noetherian Domains and Noetherian respect the A.C.C. on principal ideals, by thereom 10.1.30, all Principal Ideal Domain are Unique Factorization Domains, as we sought to show.

There is also a very simple proof I like I found in Eisenbud: (Eisenbud) We know that if R is a Unique Factorization Domain, then it respects the A.C.C. on principal ideals. Use that to find a maximal ideal, which must be generated by one element since it's a Principal Ideal Domain, but then it must be that for one of the elements in the chain, it plateau's, and so it fulfills the ascending chain condition, and so it is a Unique Factorization Domain.

Corollary 10.1.34: $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)

The integers ($\mathbb{Z}$) as a ring are a Unique Factorization Domain.

***Proof*:**
As shown in example ref:HERE (under defn of pid), $\mathbb{Z}$ is a Principal Ideal Domain, and so by proposition 10.1.32 is a Unique Factorization Domain.

The converse of this theorem is not true, that is if an integral domain is a Unique Factorization Domain, this does not imply that it is a PID, justifying the strict inclusion of the image at the very beginning. Recall from example 9.4.6 that the ideal $(2, x)$ from the ring $\mathbb{Z}[x]$ is not a principal ideal. However, the ring $\mathbb{Z}[x]$ is a Unique Factorization Domain. This is rather difficult to show directly: if we have any any polynomial $p(x) \in \mathbb{Z}[x]$, how can we guarantee it factors and that it's unique up to units, that is that $\mathbb{Z}[x]$ respects the ascending chian conditions on principal ideals and that all irreducibles are primes? On page 363, we shall show this result.

(Also something about begging the question now on finding qualities that allow for PIDs? For example, in $\mathbb{Z}$, we figured out it's a pid because for every number, we can make it a linear combination of other numbers. Is that so for every pid?)

([perhaps make this an exercise?] As a final comparison between UFDs and PIDs, the Chinese Remainder Theorem (CRT). Recall that in a pid, $\gcd(a, b) = 1$ if and only if $(a, b) = (1)$. This is *not* the case for UFD's, take for example the natural map

$$\mathbb{Z}[x] \to \mathbb{Z}[x]/(2) \times \mathbb{Z}[x](x)$$

Then this map is *not* surjective, though $\gcd(2, x) = 1$, even when the kernel, $(2x)$, is still the product of the two ideals.)

Proposition 10.1.35: Sufficient Condition for Greatest Common Divisor

If a and b are nonzero elements in the commutative ring R such that the ideal gnerated by a and b is a principal ideal (d), then d is a greatest common divisor of a and b

***Proof*:**
p.274

This proposition explains the (a, b) notation for the gcd of a and b.

Note that an integral domain in which every ieal generated by 2 elements (a, b) is a principal ideal is called a *Bézout Domain* (more generally, if every *finitely generated* ideal is principal, then we have a

Bézout Domain. For Principal Ideal Domains, every ideal is principal). Note that it's not a *necessary* condition.

For future reference, we also prove the following[6]:

Proposition 10.1.36: UFD And Maximal, Then PID

Let R be a Unique Factorization Domain where every prime ideal is maximal. Then R is a Principal Ideal Domain

***Proof*:**
Let R be a Unique Factorization Domain where every prime ideal is maximal. First, note that if $I \subseteq R$ is prime, then I is principle. Indeed, for some $r \in I$, then (up to unit) we have $r = p_1, p_2, ..., p_n$. By primality, we have that $p_i \in I$. Since p_i is prime, by assumption it is maximal, and furthermore $(p_i) \subseteq I$. But then $(p_i) = I$.

Now, let's N be the set of all non-principal ideals. For the sake of contradiction let's say it's non-empty. Then N is partially ordered by inclusion and every chain has an upper bound: if $J = \bigcup_i J_i$, then if $J = (x)$, then $x \in J_i$ for some i and $(x) = J$; contradicting non-principality.

Hence, by Zorn's lemma there exists a maximal element $\mathfrak{n} \in N$. Then by what we pointed out earlier $\mathfrak{n}$ cannot be prime, or else it would be principal. Therefore there exists $a, b \in R$, $a, b \notin \mathfrak{n}$ but $ab \in \mathfrak{n}$. Consdier $\mathfrak{n} + (a)$ and $\mathfrak{n} + (b)$. By maximality of $\mathfrak{n}$ hese must be principal: $\mathfrak{n} + (a) = (u)$ and $\mathfrak{n} + (b) = (v)$. But then:

$$\mathfrak{n} = \mathfrak{n} + (ab) = (\mathfrak{n} + (a))(\mathfrak{n} + (b)) = (u)(v) = (uv)$$

but that's a contradiction to $\mathfrak{n} \in N$, showing that it must have been that $N = \emptyset$, completing the proof.

(also thought this was cool here)

Exercise 10.1.2

1. Let $R[x]$ be a Principal Ideal Domain. Then R is a UFD. Conclude that if R is not a field, then $R[x]$ is not a Principal Ideal Domain!

   Since $\mathbb{Z}$ is not a field, then $\mathbb{Z}[x]$ is not a Principal Ideal Domain! In example 9.4.6[3] it was manually shown that $(2, x)$ is not a principal ideal in $\mathbb{Z}[x]$; this theorem makes it easier to prove.

   The converse is true to: If F is a field, $F[x]$ is a Principal Ideal Domain. In fact, it's even a Euclidean Domain. We'll cover this result when talking about polynomials over fields in section 11.1.

2. (Challenging) Let R be an integral domain. Then if every prime ideal is principle, R is a PID (In this way, UFDs are almost PIDs, however not all ideals are principle)

3. (If you know Complex Analysis) Show that the ring of entire functions is a gcd domain. Show that the ring of real analytic functions is a Bézout domain.

[6]This will be useful in number theory

10.2 Euclidean Domains

So far, we have been exploring divisibility in terms of being able to directly factor elements. However, it is in general difficult to establish when an element is divisible by another (try it out yourself with relatively large elements of some of the more complicated UFDs that were presented). If an element is not divisible by another element is there some way to measure to what "degree" an element is not divisible by another? Can we quantify the lack of divisibility given by the gcd.

In the integers, there is the *euclidean algorithm* that gives the answer. The existence of an algorithm to find the GCD cannot be under-estimated: If given two integers with over a thousand digits, it will be impossible to find the greatest common divisor in a reasonable amount of time, that is, before the heat death of the universe (until a faster way of factorizing is found). The problem is about our initial choice for divisors. If I say R is a Unique Factorization Domain and $x \in R$, then how would you start finding it's prime factorization? If you have some a and b such that $ab = x$, How do you start decomposing them? Even for the integers, it is in general very difficult to find prime factors of an element x[8].

For the integers, Euclid in approximately the year 300BC discovered that there is a simple algorithm for finding the GCD that quantifies exactly how divisible two numbers are. The key observation is that if $a, b \in \mathbb{Z}$, where $a < b$ (notice the user of the total order, $<$, of $\mathbb{Z}$) then we may write

$$b = ka + r$$

where $r < a$. If $a|b$, then $r = 0$. If $a \nmid b$, then r is the "torsion" or "remainder" that is left-over. In terms of the GCD, if $a|b$, then $\gcd(a, b) = a$, and if $a \nmid b$, then $\gcd(a, b) < a$ and in the "worst case scenario" $\gcd(a, b) = 1$. The problem when generalizing this to general rings is that we require a notion of "size", namely it was crucial to use the fact that $a < b$ and it is imperative that $r < a$ to have a reasonable notion of such a factorization. In this section, we shall endavour to find a function, say $v : R \to \mathbb{N}_{\geq 0}$ that will allow us to assign a natural number to each ring element so that we may recapture this notion from the integers.

Definition 10.2.1: [Ring] Valuation

Any function $v : R \to \mathbb{Z}$ with $v(0) = 0$ is called a *valuation* on the ring R. If $v(a) > 0$ for $a \neq 0$, then v is said to be a *positive valuation*.

10.2.2 Note The notion of valuation shall be defined later in these notes on fields as *real [Archimedean] valuation* or *absolute value* (definition 21.2.1) and generalized to *G-valuation* (definition 27.1.1)[a]. The notion of a G-valuation is a generalization of the absolute value, however it is often presented with the axiom that $v(0) = \infty$. Valuations can just as well be defined with $v(0) = 0$, and hence the use of the term "valuation" here is justified.

[a]More specifically non-Archimedean valuations shall be generalized to G-valuations

[8]RSA encryption is based off of the fact that prime factorization is a every difficult task if you don't know a priori what primes you are working with

10.2.3 Note There are at least two common uses of the word "valuation" in contemporary abstract algebra: an *absolute-valuation* or more commonly an *absolute value* and a G-*valuation* (definition 27.1.1). These two concepts are non-compatible[a], notably G-valuations have axiom dictating that $v(0) = \infty$ while absolute values has an axiom dictating that $v(0) = 0$. The above use of the term valuation is closer to an *absolute value*.

[a]Though an absolute value may induce a G-valuation, this is shown when the concepts are introduced

If you know some analysis, you may wonder if the valuation is related to the one introduced on $\mathbb{R}$ or $\mathbb{C}$ (or normed vector spaces more generally). These concepts are indeed related, and we shall even see more explicit connection in chapter 21$\star$.

Returning to the issue at hand, we shall require a valuation N to be defined on an integral domain R that satisfies the following property:

Definition 10.2.4: Euclidean Valuation and Euclidean Domain

Let R be an integral domain and $N : R \to \mathbb{N}_{\geq 0}$ a valuation. Then v is a *euclidean valuation* on R if for any two elements a, b of R with $b \neq 0$ there exists elements q, r in R such that

$$a = qb + r \qquad \text{with } r = 0 \text{ or } N(r) < N(b)$$

The element q is called the *quotient* and the element r the *remainder* of the division if a euclidean valuation can be defined on a ring, then the ring is said to be a *Euclidean Domain* (or posses a *division algorithm*)

The main thing we can do in euclidean domains is run the division algorithm to keep "dividing" our numbers

$$\begin{aligned} a &= q_0 b + r_0 \\ b &= q_1 r_0 + r_1 \\ r_0 &= 2_0 r_1 + r_2 \\ &\vdots \\ r_{n-1} &= q_{n+1} r_n \end{aligned}$$

Such a sequence will terminate since $N(b) > N(r_0) > \cdots > N(r_n)$, and it will eventually hit 0.

Example 10.2.5: Euclidean Domains

1. Every field with the euclidean valuation is a euclidean domain! In fact, any valuation will work!

   Proof. Let $x \in \mathbb{F}$ be an element of a field. Then since every element in a field has a unit, then
   $$x = xb + 0$$
   where $b = x^{-1}$.

   $\square$

Thus, we see that if a division rings was also commutative, then it'd clearly be a Euclidean domain.

2. In general, every unit will trivially satisfy this condition, since if $u \in R$ is a unit, then

$$u = uu^{-1} + 0$$

and so terminates instantly. So in a sense, Euclidean domains say that if we have elements that are *not* units, how many properties of divisible things can we put on them?

Note importantly that euclidean domains must be domains (meaning no zero-divisors)! So oddities with multiplying not increasing the "size" of elements will not happen.

3. Ring of fractions in general are not euclidean domains, since they might have zero-divisors. So even though rings of fractions can be thought of as a way of enlarging a ring to make it's elements more divisible, it cannot do so if it has zero-divisors.

4. The integers $\mathbb{Z}$ are a euclidean domain with $N(a) = |a|$. The division algorithm is your usual euclidean algorithm. If you take $a, b \in \mathbb{Z}$, and $a < b$ then $b = aq + r$. We see that $N(r) = |r| < |b| = N(b)$. This will always be the case with $b \neq 0$! Thus, it is a Euclidean Domain.

5. If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is also a Euclidean domain with $N(p(x)) = deg(p(x))$. The division algorithm is your usual polynomial division. The reason is simple for valuation: polynomial division *must* produce a remainder r with degree *less* than b, by definition! This makes $\mathbb{F}[x]$ into a Euclidean Domain. We'll prove this more rigorously in section 11.1.

6. Let

$$\mathbb{Z}[i] = \{a + bi | a, b \in \mathbb{Z}\}$$

And let

$$N(a + bi) = (a + bi)(a - bi) = a^2 + b^2$$

which is the usual valuation on $\mathbb{C}$. Intuitively, this is because we have subrings $\mathbb{Z}[i] \subseteq \mathbb{C}$. These are called the *Gaussian Integers*. We'll show that Gaussian integers ($\mathbb{Z}[i]$) is a Euclidean Domain.

Consider $a, b \in \mathbb{Z}[i]$. We need to show that

$$a = kb + r \qquad N(r) < N(b)$$

for appropriate k, r. Since $\mathbb{Z}[i] \subseteq \mathbb{Q}(i)$, take

$$\frac{a}{b} = \frac{ac + bd}{c^2 + d^2} + \left(\frac{bc - ad}{c^2 + d^2}\right) = r + si$$

this gives us an idea of how close a is to b by having by looking at their fraction. Let p be the closest (or one of the 2 closest) integers to r and q be the closest (or one of the 2 closest) integers to s so that $|r - p|, |s - q| \leq \frac{1}{2}$. Then we'll show that

$$a = (p + qi)b + r$$

Whatis r? Notice that $t = (r-p) + (s-q)i$, so that $r = tb$, giving us $r = a - (p+qi)b$, showing that $r \in \mathbb{Z}[i]$. With this, we can calcualte the valuations quite easily! Notice that $N(t) = (r-p)^2 + (s-q)^2$ would at most

$$N(t) \le \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

and so, by the multiplicity of this valuation (TBD move that part earlier so that I can use it)

$$N(tb) = N(t)N(b) \le \frac{1}{2}N(b) \Rightarrow N(t) < N(b)$$

showing that $\mathbb{Z}[i]$ is indeed a Euclidean Domain.

7. Another class of Euclidean domains come from polynomial rings over fields. The proof is a little more elaborate and so we shall leave it till page 358.

Notice how central defining a valuation is to showing a ring is a euclidean domain. This algorithm *almost* produces a divisibility condition, making Euclidean domains “close” to fields. However, they are not fields, since $\mathbb{Z}$ is a euclidean domain. So instead of every element having an inverse, for every pair of elements (a, b), we can find a number(s) c such that we have a notion of $c|a$ and $c|b$. If every pair of numbers has this condition, we have a Euclidean domain!

(Add here a proof of how the gcd works for Euclidean Domains)

We have not seen many examples of Euclidean domains. The following is a class of rings that shall provide a testing ground for finding more Euclidean domains and will come up a lot in chapter 50:

Example 10.2.6: Rings of Integers of Quadratic Extenions

Let D be a squarefree integer (it's prime-factorization has exactly one factor for each prime that appears in it, so $10 = 2 \cdot 5$ is square free, but $18 = 2^2 \cdot 3$ is not). Then we can define $\mathbb{Z}[\sqrt{D}] = \left\{a + b\sqrt{D} | a, b \in \mathbb{Z}\right\}$, which is a ring, even a subring, of the quadratic field $\mathbb{Q}(\sqrt{D})$ (that we defined in the first examples). If we take $D \equiv 1 \bmod 4$, then consider the slighty larger subset

$$\mathbb{Z}\left[\frac{1+\sqrt{D}}{2}\right] = \left\{a + b\frac{1+\sqrt{D}}{2} | a, b \in \mathbb{Z}\right\}$$

is also a subring. With this, define

$$\mathcal{O} = \mathcal{O}_{\mathbb{Q}(\sqrt{(D)})} = \mathbb{Z}[\omega] = \{a + b\omega | a, b \in \mathbb{Z}\}$$

where

$$\omega = \begin{cases} \sqrt{D} & \text{If } D \equiv 2, 3 \bmod 4 \\ \frac{1+\sqrt{D}}{2} & \text{If } D \equiv 1 \bmod 4 \end{cases}$$

called the *ring of integers* in the quadratic field $\mathbb{Q}(\sqrt{D})$. The idea being that the elements of $\mathcal{O}$ of the field $\mathbb{Q}(\sqrt{D})$ have many properties analogous to those of the subring of integers $\mathbb{Z}$ in the field of rational numbers. Or, if we have a “field extension”, which in a diagram we can think of as:

$$\begin{array}{ccc} \mathbb{F} & \supseteq & O_{\mathbb{F}} \\ \uparrow & & \uparrow \\ \mathbb{Q} & \supseteq & \mathbb{Z} \end{array}$$

where the arrows represent inclusions, then we want $\mathbb{F}$ to share properties that $\mathbb{Z}$ has Given an extension field $\mathbb{F}$ for $\mathbb{Q}$, we identity rings of integers $O_{\mathbb{F}}$ a subrging of F analogous to $\mathbb{Z} \subset \mathbb{Q}$

Note also that $\mathbb{Z}[\omega] = \mathcal{O}_{\sqrt{(D)}}$ is an integral domain, and that it's field of fraction is $\mathbb{Q}(\sqrt{D})$, giving us another way of thinking of ho $\mathbb{Z}[\omega]$ is sort of the "natural" integral domain to choose.

In the special case where $= -1$, we get $\mathbb{Z}[i]$, called the *Gausian integers*, used to study powers of numbers in the complex plane.

The reason for defining this is because we can define the concept of a *field norm*. Let $N : \mathbb{Q}(\sqrt{D}) \to Q$ by

$$N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \in \mathbb{Q}$$

which is non-zero if $a + b\sqrt{D} \neq 0$. This gives an idea of "size" in the field $\mathbb{Q}(\sqrt{D})$. For instance, when $D = -1$, then the valuation of $a + bi$ is $a^2 + b^2$ (as we know it from Complex Analysis!), which is the square of the length of this complex number considered as a vector in the compelx plane.

We can see that N is multiplicative, that is, $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{Q}(\sqrt{D})$ given by

$$N(a + b\omega) = (a + b\omega)(a - b\overline{\omega}) = \begin{cases} a^2 - b^2 D & \text{if } D \cong 2, 3 \bmod 4 \\ a^2 + ab + \frac{1-\sqrt{D}}{2} & \text{if } D \cong 1 \bmod 4 \end{cases}$$

where

$$\overline{\omega} = \begin{cases} -\sqrt{D} & \text{if } D \equiv 2, 3 \bmod 4 \\ \frac{1-\sqrt{D}}{2} & \text{if } D \equiv 1 \bmod 4 \end{cases}$$

It follows that $N(\alpha) \in \mathbb{Z}$, meaning we can relate properties of $\mathbb{Z}$ to the field norm!

For example, we can use this norm to characetrize the units of $\mathbb{Z}[\omega]$. If $N(\alpha) = \pm 1$, then $(a + b\omega)^{-1} = (a + b\overline{\omega})$, which is an element of $\mathbb{Z}[\omega]$, and so α is a unit. Suppose, conversely, that α is a unit in $\mathbb{Z}[\omega]$, so $\exists \beta$ such that $\alpha\beta = 1$. Then since N is multiplicative,

$$N(\alpha)N(\beta)N(\alpha\beta) = N(1) = \pm 1$$

and since $N(\alpha)$, $N(\beta)$ must be integers, they must be ± 1, and so

$$\text{the element } \alpha \text{ is a unit in } \mathcal{O} \text{ if and only if } N(\alpha) = \pm 1$$

Thus, the field norm also gives us an easy way to find the units of a given ring of integers.

We'll be using the field norm to how for certain values of D, $\mathbb{Z}[\sqrt{D}]$ will be a Euclidean Domain, and not for others (ex. we've already shown that $\mathbb{Z}[\sqrt{-1}] = \mathbb{Z}[i]$ is a Euclidean Domain).

Euclidean domain $\Rightarrow$ PID

As mentioned earlier, Euclidean Domains will be useful to factorizing elements in UFDs in which they are defined. Not all UFDs may have a valuation satisfying the euclidean algorithm, namely since being able to define such a va implies a good deal of structure on the ring. One important consequence of the existence of such a norm is that every Eucildean domain is a PID

Proposition 10.2.7: Euclidean Domain $\Rightarrow$ Pid

Given a Euclidean domain R, then every ideal of R is a principal ideal, meaning R is also a PID. In particular, if I is an ideal, then $I = (d)$, where d is any nonzero element of I of minimum norm

***Proof*:**
If I is the zero ideal, then we're done, so let $I \neq (0)$. This means that we can choose an element $d \in I$ such that $N(d)$ is a minimum element. This exists since $\{N(a)|a \in I\}$ has a minimum element by the well-ordering principle of $\mathbb{Z}$. We know show that $(d) = I$

($\subseteq$) it's clear that $(d) \subseteq I$ by the definition of ideals

($\supset$) Let $a \in I$ be any element of our ideal. Then since R is a euclidean domain, we can write a as :

$$a = qd + r$$

where $r = 0$ or $N(r) < N(d)$. Since $r = a - qd$, and $a, qd \in I$, then $r \in I$. But then, it's impossible that $N(r) < N(d)$ since $N(d)$ is already the minimal norm!! Thus $r = 0$, implying $a = qd$. But then $a = qd \in (d)$, as we sought to show

As usual, the contra-positive should be kept in mind: R is *not* an Euclidean Domain if it is *not* a PID.

Example 10.2.8: Implications

1. Since $\mathbb{Z}$ is a Euclidean domain, it is A Principal Ideal Domain. Note that we didn't show that PID $\Rightarrow$ Euclidean domain, we'll present a counter-example soon, so this was only a hint in the right direction
2. $\mathbb{Z}[x]$ is *not* a PID since $(2, x)$ is a non-principal ideal, and so not a euclidean domain.
3. $\mathbb{Q}[x]$ is a PID, and can be shown to be the euclidean domain with $N(p(x)) = \deg(p(x))$
4. $\mathbb{Z}[\sqrt{5}]$ is not a PID, and so not a Euclidean Domain

You might wonder if *all* PID's are euclidean domains. That need not be the case. To prove this, we'll first prove a another way of finding Euclidean domains:

Definition 10.2.9: Side Divisor

Let $\overline{R} = R^{\times} \cup \{0\}$ be the collection of units of R together with 0. An element $u \in R - \overline{R}$ is called a *universal side divisor* if for evry $x \in R$ there is some $z \in \overline{R}$ such that u divides $x - z$ in R, i.e., there is a type of "division algorithm" for u: every x may be written as $x = qu + z$ where z is either zero or a unit.

The existence of universal side divisors is weakening of the Euclidean condition, since we're saying that for any element $x \in R$, we can "divide" it by u and have a remainder of a unit z. These conditions are not as general as the conditions for the euclidean algorihtm (where z doesn't need to be constrained to be $N(z) = 1$). With this, we prove the following equivalent condition:

Proposition 10.2.10: Euclidean Domain Implies Universal Side Divisors

Let R be an integral domain that is not a field. If R is a euclidean domain, then there are universal side divisors in R

***Proof*:**
Suppose R is Euclidean with respect to some norm N and let u be an element $R - \overline{R}$ (which is non-empty since F is not a field), of minimal norm. For any $x \in R$ write $x = qu + r$ where r is etiher 0 or $N(r) < N(u)$. In either case the minimality of u implies $r \in \overline{R}$. Hence u is a univerasl side divisor in R

With this, we can find

Example 10.2.11: : PID but not euclidean domain
We'll take the following ring of integers.

$$\mathcal{O}_{Q(\sqrt{-19}} = \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$$

with the field norm:

$$N(a + b\sqrt{-19}) = (a + b\frac{1+\sqrt{-19}}{2})(a - b\frac{1+\sqrt{-19}}{2}) = a^2 + ab + 5b^2$$

This ring is a PID; a fact we'll show in the next chapter. However, it is not a Euclidean domain, for there does not exist a universal side divisor.

We can find that $\overline{R} = \{0, \pm 1\}$ are the unit of R union 0. Now suppose $u \in R$ is a universal side divisor. Note that if $b \neq 0$, then

$$a^2 + ab + 5b^2 = (a + \frac{b}{2})^2 + \frac{19}{4b^2} \geq 5$$

and so the smallest values of N on R are 1 (for units ± 1) and 4 (for $a = \pm 2$). Let's say $x = 2$. Then, we have

$$2 = qu + z \Rightarrow 2 = 2u + 0 \qquad \text{or} \qquad 2 = u \pm 1$$

or, written in division form, we see that $u|2 - 0$ or $u|2 \pm 1$.

Let's say $z = 0$, so $2 = qu$. Then $N(2) = 4 = N(q)N(u)$. TBD (it's a tedious example)

It is tempting to think that PIDs are close to being Euclidean domains as they share a notion of gcd and lcm, unique factorization. The following measure the degree of seperation of a Principal Ideal Domain being a Euclidean Domain:

Definition 10.2.12: Dedikend-Hasse Valuation

Define N to be the *Dedekind-Hasse valuation* if N is a positive norm and for every nonzero $a, b \in R$ either a is an element of the ideal (b) or there is a nonzero elemtn of the ideal (a, b) of norm strictly smaller than the norm of b (i.e., either b divides a in R or their exists $s, t \in R$ such that $0 < N(sa - tb) < N(b)$)

Note that R is Euclidean with respect to a positive norm N if it is always possible to satisfy the Dedekind-Hasse norm with $s = 1$, so this is indeed a weakening of our condition for Euclidean Domains.

Proposition 10.2.13: Principal Ideal Domain If And Only If Dedekind-Hasse Norm

The integral domain R is a PId if and only if R has a Dedekind-Hasse nrom

Proof:
p.281

Corollary 10.2.14: Defning Dedekind-Hasse Norm

let R be a Principal Ideal Domain. Then there exists a multiplicative Dedekin-Hasse norm.

Proof:
p.289

Example 10.2.15
Showing that $\mathbb{Z}[\sqrt{-19}]$ is a Principal Ideal Domain but not a Euclidean Domain.

10.3 Application: Sums of Primes and Gaussian Integers

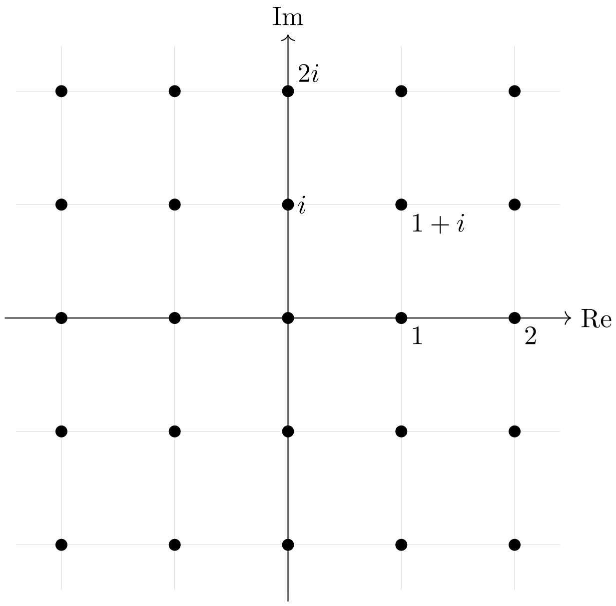
Let us use the theory we've developed to get our first result about prime numbers, namely let us classify all prime numbers that can be written as the sum of two squares $p = x^2 + y^2$. [9]

Recall that the Gaussian integers are of the form:

$$\mathbb{Z}[i] \subseteq \mathbb{C}$$

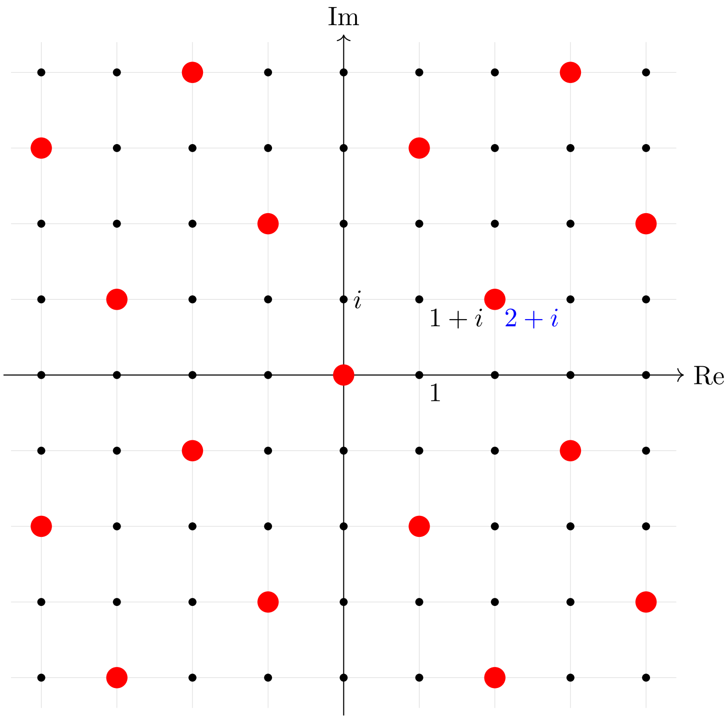
which can be thought of as the smallest ring containing $\mathbb{Z}$ and i. The Gaussian integers have a pretty geometric picture as a lattice:

[9] Naturally, a prime by definition cannot be the square of a natural number, however it can be mod q under the right condition; this is called *quadratic reciprocity* and is covered in section 50.4.6



The Gaussian Integers $\mathbb{Z}[i]$

This gives us a geometric picture of this ring! For example, we may visualize principal ideals in this grid, for example the ideal $(2+i)$ is the point:



The Gaussian Integers $\mathbb{Z}[i]$ with $(2+i)$ highlighted

Recall how we introduced the field-norm on on quadratic integers, where $\mathbb{Z}[i] = \mathbb{Z}[\sqrt{-1}]$ is an example of quadratic integers. With the above visuals, it is more clear to see that the field norm has similar geometric information to the norm presented in real or complex analysis, namely:

$$N(a+bi) = (a+bi)(a-bi) = a^2 + b^2 = \|a+bi\|^2$$

Using the fact that the codomain of the field norm is $\mathbb{N}_{\geq 0}$, we can see that the norm is in fact a euclidean valuation

Lemma 10.3.1: Gaussian Integers Are Euclidean Domain

Let $R = \mathbb{Z}[i]$ and N be the field norm. Then N is a Euclidean valuation. Furthermore, N is multiplicative so that for every $z, w \in \mathbb{Z}[i]$,

$$N(zw) = N(z)N(w)$$

***Proof*:**
Multiplicity is immediate as:

$$N(zw) = (zw)(\overline{zw}) = (z\overline{z})(w\overline{w}) = N(z)N(w)$$

The proof that it is a euclidean valuation is straight-forward if not a little verbose, and so is left as an exercise.

What could be more insightful is to understand using the lattice picture the significant of the field norm on $\mathbb{Z}[i]$ (this was demonstrated in Aluffi's textbook see [1]). Take $z, w \in \mathbb{Z}[i]$ with $w \neq 0$ nd consider the lattice given by (w). Then either z lies on a point in this lattice, in which case $z = qw$ for appropriate q, or z is inside a "box" of the lattice. Pick some points qw that is a vertex of the box containing z, and set $r = z - qw$ so that $z = qw + r$. By visual inspection, we see that the length of r is ells than that of r, namely:

$$N(r) < N(w)$$

This trick shall not work for all quadratic fields, as we shall demonstrates after finishing elaborating on Gaussian integers.

It is a good exercise to show that if u is a unit then $N(u) = \pm 1$, from which we can conclude that the only possible units are $1, -1, i, -i$. We can use the norm to also detect irreducible elements. To see this, let' say $p \in \mathbb{Z}$ is a prime. Then since $\mathbb{Z}$ is an integral domain, p is irreducible, and so $p = ab$ if and only if a or b is a unit. Let's now take $\pi \in \mathcal{O}$ such that $p = N(\pi)$. If $\pi = \alpha \cdot \beta$, then:

$$p = N(\pi) = N(\alpha)N(\beta)$$

meaning either $N(\alpha)$ or $N(\beta)$ is a unit, meaning it's value is ± 1. Thus, we have that

$$\text{If } N(\alpha) \text{ is } \pm p \text{ for a prime } p \in \mathbb{Z}, \text{ then } \alpha \text{ is irreducible (even prime) in } \mathbb{Z}[i]$$

The converse is not necessarily true: we can have a prime in $\mathbb{Z}[i]$ but it's norm is not p. Fortunately, this is a feature of primes in $\mathbb{Z}[i]$ and demonstrates a deeper insight on the structure of primes in $\mathbb{Z}[i]$:

Lemma 10.3.2: Gaussian Integers Norm of Primes

Let $\pi \subseteq \mathbb{Z}[i]$ be prime where. Then

$$N(\pi) \in \{p, p^2\}$$

for appropriate prime $p \in \mathbb{Z}$.

Fermat's theorem of sum of square (theorem 10.3.3) shall show that both p and p^2 show up as norms of primes.

***Proof*:**
Let $\pi \in \mathbb{Z}[i]$ be a prime element, that is, (π) is a prime ideal in $\mathbb{Z}[i]$. As π is not a unit, $N(\pi) \neq 1$. Then $(\pi) \cap \mathbb{Z}$ is also a prime ideal: if for $a, b \in \mathbb{Z}$, we have $ab \in (\pi)$ and $ab \in \mathbb{Z}$[a] then either a or b is an element of (π), so a or b is in $(\pi) \cap \mathbb{Z}$. Thus, there exists a prime p such that $(p) = (\pi) \cap \mathbb{Z}$.

Now, take $(p) \subseteq \mathbb{Z}[i]$. Then $(p) \subseteq (\pi)$ so that $xp = \pi$ for some $x \in \mathbb{Z}[i]$ (which may be a unit, as p may be an associate of π). Then $N(p) = p^2$, and since the norm is multiplicative

$$N(\pi) \big| N(p) = p\overline{p} = pp = p^2$$

from which it can be immediately deduced that $N(\pi) \in \{p, p^2\}$, completing the proof.

[a] if $a, b \in \mathbb{Z}$, automatically $ab \in \mathbb{Z}$

Note that irreducible does not imply prime, as it was shown with $\mathbb{Z}[\sqrt{-5}]$, however if the same trick is done as in the above proof on quadratic rings more generally it suggests that irreducible elements are associated to prime elements in $\mathbb{Z}$ via the norm map (see exercise ref:HERE). Under the right framework, this connection can be made more precise, which will be part [o]f the aim of chapter 50.

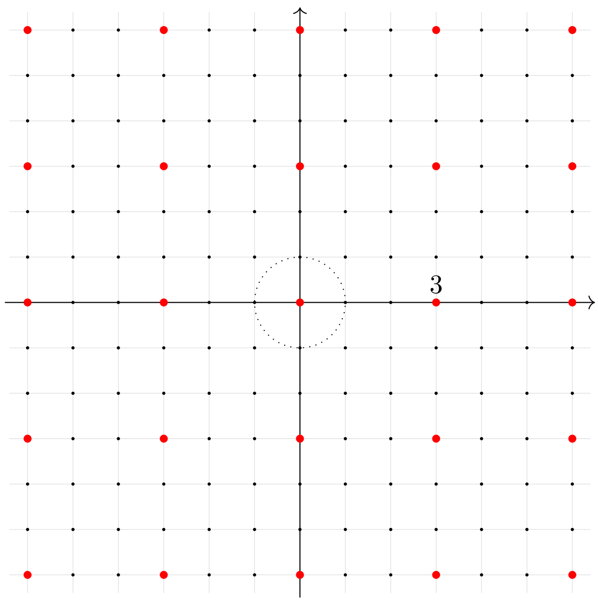
Let us find some examples of primes with norms p and p^2. Let's take $\alpha = a + bi \in \mathbb{Z}[i]$ to be prime. Then $N(\alpha) = \alpha\overline{\alpha} = a^2 + b^2 \in \{p, p^2\}$. If p is reducible in $\mathbb{Z}[i]$,then

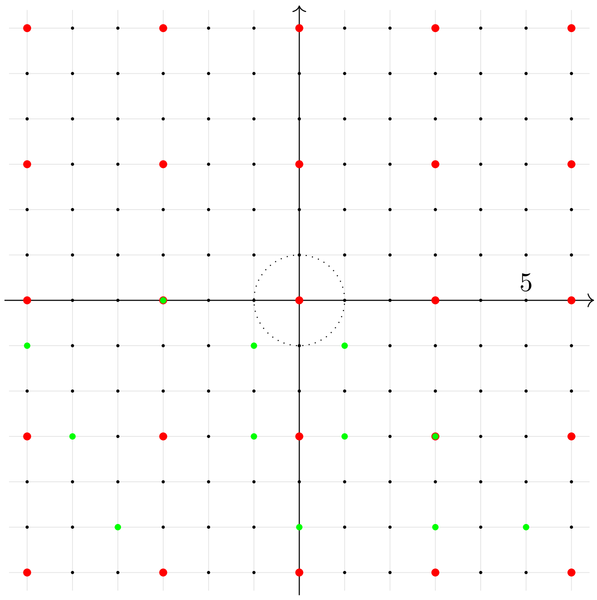
$$p = \pi\pi' \qquad \Rightarrow \qquad N(p) = N(\pi)N(\pi') = p^2$$

Thus, p will factor in $\mathbb{Z}[i]$ into two irreducibles if and only if $p = a^2 + b^2$, that is, p is the sum of two integers squares. If p is not the sum of two integer squares, it will remain irreducible. Take or example $3 \in \mathbb{Z}[i]$. Then $N(3) = 9$, and if 3 factored we would need an element $a + bi$ such that $N(a+bi) = a^2+b^2 = 3$ for natural numbers $a, b \in \mathbb{N}$. However, it can be quickly verified that no such numbers exist, and so 3 is prime in $\mathbb{Z}[i]$. For another example, take 2. Then as $2 = 1^2 + 1^2$, 2 factors in $\mathbb{Z}[i]$! In particular, $2 = (1 + i)(1 - i) = -i(1 + i)^2$ (note that $N(-i) = 1$ and so these two are associates). This will be the only case were conjugate irreducibles $a + bi$ and $a - bi$ can be associate. Let us next consider 5, so we need to see if there is an element $a + bi$ such that $N(a + bi) = 5$, and indeed:

$$N(2 + 1 \cdot i) = 2^2 + (1)^2 = 5$$

from which we find $5 = (2 + i)(2 - i)$. This factorization picture can be seen more clearly when viewing this elements via their ideals in the lattice. In the case of 3, there is no possible further refinement that can be done to the lattice, while for 5 we may further refine the lattice (without it being the entire lattice):





Let us now find a condition that shall let us immediately tell if a prime can splits in $\mathbb{Z}[i]$. First, any number in $\mathbb{Z}$ is congruent to $0, 1, 2, 3 \bmod (4)$. Therefore, the square of any number is congruent to either 0 or 1. The possible sums of square are thus $0, 1, 2 \bmod 4$. If the sum is $0, 2 \bmod 4$, the reader should check that if $p = a^2 + b^2$ then p would not be prime. Thus an odd prime (so $p > 2$) in $\mathbb{Z}$ that is the sum of two squares must be congruent to 1 mod 4 (ex. 5). Therefore, if $p \equiv 3 \bmod 4$ in $\mathbb{Z}$, then p is *not* the sum of squares, and remains irreducible (and thus prime) in $\mathbb{Z}[i]$.

Theorem 10.3.3: Fermat's Theorem on Sums Of Squares

The prime p is the sum of two integer squares, $p = a^2 + b^2$, $a, b \in \mathbb{Z}$ if and only if $p = 2$ or $p \equiv 1 \bmod 4$. Except for interchanging a and b or changing the sign of a and b, the representation of p as a sum of two squares is unique.

Furthermore, the prime elements in the Gaussian integers $\mathbb{Z}[i]$ are as follows:

1. $1 + i$ (which has norm 2)
2. the primes $p \in \mathbb{Z}$ with $p \equiv 3 \bmod 4$ (which have norm p^2)
3. $(a + bi)$, $(a - bi)$, the distinct irreducible factors of $p = a^2 + b^2 = (a + bi)(a - bi)$ for the primes $p \in \mathbb{Z}$ with $p \equiv 1 \bmod 4$ (both of which have norm p).

In other words, each prime $p \equiv 1 \bmod 4$ can be written as the sum of two squares:

$$p = a^2 + b^2$$

***Proof*:**
The case for $1+i$ has been done, so consider an odd prime $p \in \mathbb{Z}$. Then verify the following chain of implications:

$$\begin{aligned} p \text{ splits in } \mathbb{Z}[i] &\Leftrightarrow \frac{\mathbb{Z}[i]}{p\mathbb{Z}[i]} \text{ is not an integral domain} \\ &\Leftrightarrow \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2+1)} \text{ is not an integral domain} \\ &\Leftrightarrow x^2+1 \text{ is not irreducible in } \mathbb{Z}/p\mathbb{Z}[x] \\ &\Leftrightarrow x^2+1 \text{ has a root in } \mathbb{Z}/p\mathbb{Z}[x] \\ &\Leftrightarrow \exists\, n \in \mathbb{Z} \text{ such that } n^2 \equiv -1 \bmod p \end{aligned}$$

Thus, the splitting of p has been reduced to solving the congruence equation $n^2 \equiv -1 \bmod p$. To verify this condition, recall that $(\mathbb{Z}/p\mathbb{Z})^*$ is as p is prime, and it has order $p-1$. As p is odd, $p-1$ is even, so there is a generator $g \in (\mathbb{Z}/p\mathbb{Z})^*$ such that $|g| = 2\ell$ for some appropriate natural number ℓ. Then write $[n] = g^m$ for appropriate m. Now, certainly $[-1] \in (\mathbb{Z}/p\mathbb{Z})^*$ creates an (unique) order 2 subgroup, and g^ℓ also creates an order 2 subgroup, and so:

$$g^\ell = [-1]$$

Then $n^2 \equiv -1 \bmod p$ if and only if $g^{2m} = g^\ell$, that is if and only if

$$2m \equiv \ell \bmod 2\ell$$

Now this linear congruence relation is readily solvable: it is true if and only if ℓ is even, and so when $(p-1) = 2\ell$ is a multiple of 4, or shuffled around:

$$p \equiv 1 \bmod 4$$

giving us an exact condition for when p can be factored, equivalently when p can be written as the sum of two squares, completing the proof.

Lemma 10.3.4: Writing p as n^2+1

The prime number $p \in \mathbb{Z}$ divides an integer of the form n^2+1 if and only if p is either 2 or is an odd prime congruent to 1 mod 4.

***Proof*:**
By Fermat sum of square's theorem, we know that there exists an n such that $n^2+1 \equiv 0 \bmod p$, which is equivalent to saying p divides n^2+1.

Though this condition is equivalent to Fermat' sum of Square's Theorem, it is not as informative as $5 \big| 3^1 + 1 = 10$, however $3+i$ and $3-1i$ are *not* prime (for example, $3+i = (1+i)(2-i)$). There is however something very interesting that was demonstrated in this entire section: we have answered a purely number theoretic question about which primes can be expressible as sums of squares (a fact that should at first not be so obvious) by taking advantage of a lot of simple theoretical properties built-up about factorization and of cyclic groups!

10.3.5 Foreshadowing Fermat's Theorem of sums of two squares will be generalized to the factorization of any polynomial over any finite extension of a ring (theorem 50.4.7), though the strategy of writing a prime p as some polynomial expression is in general more difficult.

Exercise 10.3.1

1. I think I can do an exercise finding solutions to $y^2 = y^3 - 1$ eh? I must've made a typo here, I'll revisit

11

Exploring Polynomial Rings

When first studying algebra, we often looked to find solutions to linear and quadratic equations:

$$ax + b = 0 \qquad ax^2 + bx + c = 0$$

This naturally generalizes to polynomial of higher degree. In pre-abstract algebra, solutions to such equations where usually limited to $\mathbb{R}$, probably $\mathbb{C}$. We have now developed the abstraction to consider the value (or root) of such a polynomial for an arbitrary ring R, that is for an arbitrary polynomial in $R[x]$, does there exist an element $r \in R$ such that

$$p(r) = 0$$

As is the usual strategy in algebra, we generalize a problem into an algebraic object, in this case the polynomial ring $R[x]$, and look for interesting properties of the objects and tools we can use to manipulate the object. Such polynomials rings are paramount to study. To give an idea of their importance:

1. they arbitrarily approximate continuous functions on compact neighborhoods
2. all complex differentiable function can be locally be represented as infinite polynomials (i.e. they are analytic)
3. All relations in rings are represented by polynomial (similarly to how all relations in a group is represented by a monomial)

The last point shows that polynomial rings are used to "locally" represent rings, namely through the universal property of [commutative] universal rings (see theorem 9.2.4). It may thus seem natural to consider studying polynomials rings themselves to see what properties we may deduce from them.

One necessary limitation of the exploration of polynomial rings in chapter is that we shall not look at finding solutions to polynomials, we shall rather look at the structure of polynomial rings. The reason for this is that it is in fact a much more difficult task to find solutions requiring more build-up that starts venturing into the domain of algebraic geometry. The beginnings of this effort shall be presented in [13, Algebraic Sets and the Zariski Topology].

It must be noted that we shall further develop on polynomial rings in *Commutative Algebra* [8]. This exploration is left to that book since it is not necessary to cover some results until much later.

Note that this chapter shall use some exercises from exercise 9.3.1, most importantly:

Proposition 11.0.1: Ideals of Polynomial Ring wrt. R

Let $I \subseteq R$ be an ideal. Then:

$$(R[x])/(IR[x]) \cong (R/I)[x]$$

***Proof*:**
exercise 9.3.1[3]

11.1 Polynomial Rings over Fields

We'll now explore polynomial rings with coefficients from a field $\mathbb{F}$. The first question we may ask if what type of ring is $\mathbb{F}[x]$? It's not a field, since we already saw that x is not invertible (meaning it's also not a division ring). The next strongest ring we know is Euclidean domains. If $\mathbb{F}[x]$ is a euclidean domain, there must be a norm we define, which we'll use $N(p(x)) = \deg(p(x))$ (the usual polynomial norm). Then $\mathbb{F}[x]$ is indeed a Euclidean Domain:

Theorem 11.1.1: Field Then $\mathbb{F}[x]$ Is Euclidean Domain

Let $\mathbb{F}$ be a field. The $\mathbb{F}[x]$ is a Euclidean Domain. In particular, if $a(x)$ and $b(x)$ are two polynomials in $\mathbb{F}[x]$ with $b(x)$ nonzero, then there are *unique* $q(x)$ and $r(x)$ such that

$$a(x) = q(x)b(x) + r(x_0) \qquad \text{with} \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(b(x))$$

***Proof*:**
First, the edge cases:

If $a(x)$ is the zero polynomial, then let $q(x) = r(x) = 0$. So let $a(x)$ be a nonzero polynomial. Let $b(x)$ be an arbitrary other polynomial. If $\deg(a(x)) < \deg(b(x))$, then let $q(x) = 0$ and $r(x) = a(x)$.

Now, take the nonzero polynomial $a(x)$ such that $\deg(a(x)) > \deg(b(x))$. We'll do a proof by induction on the degree of $a(x)$ ($n = \deg(a(x))$).

For visualization, we can denote $a(x)$ and $b(x)$ as:

$$a(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \qquad b(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$$

We want to apply the induction hypothesis. Let $a'(x) = a(x) - \frac{a_n}{b_m}x^{n-m}b(x)$, which purposely defined to eliminate the monomial of degree n from $a(x)$. Then by the induciton hypothesis:

$$a'(x) = q'(x)b(x) + r(x) \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(p(x))$$

Then, we "reverse" the step of taking $a'(x)$ by doing $q(x) = q'(x) + \frac{a_n}{b_n}x^{n-m}$, which will give us

$$a(x) = q(x)b(x) + r(x) \qquad r(x) = 0 \text{ or } \deg(r(x)) < \deg(p(x))$$

showing we can factor two polynomials.

It remains to show that this factorization is unique. Suppose $q(x), r(x)$, and $q_0(x)$, $r_0(x)$ are both polynomials that satisfy the euclidean condition. Then

$$a(x) - q(x)b(x) \qquad a(x) - q_0(x)b(x)$$

are both of degree less than m. The difference between these two polynomials then:

$$b(x)(q(x) - q_0(x))$$

is also less than m. But the degree of product of polynomials over an integral domain is the sum of their degree ($m > \deg(b(x)(q(x) - q_0(x))) = \deg(b(x) + \deg(q(x) - q_0(x))$. So, $q(x) - q_0(x) = 0$, implying $q(x) = q_0(x)$. This then implies $r(x) = r_0(x)$, completing the proof of uniqueness

Corollary 11.1.2: Field Then $\mathbb{F}[x]$ PID And UFD

Let $\mathbb{F}$ be a field. The $\mathbb{F}[x]$ is a Principal Ideal domain, and a Euclidean Domain

***Proof*:**
By the previous theorem (theorem 11.1.1), we know $\mathbb{F}[x]$ is a euclidean domain. By 10.2.7, it is a Principal Ideal Domain, and since PID's are UFD's, it is also a Unique Factorization Domain

Example 11.1.3

1. $\mathbb{Q}[x]$ is a Principal Ideal Domain since $\mathbb{Q}$ is a field.
2. $\mathbb{Q}[x, y]$ is not a Principal Ideal Domain, since $\mathbb{Q}[x]$ is not a field (using the contra-positive).

One final fact we'll introduce as a result of what we talked about:

Corollary 11.1.4: Kernel Of Evaluation Map

Let $\mathbb{F}$ be a field and let α be an element of $\mathbb{F}$. Then the kernel of ev_α is the ideal $(x - \alpha)$

***Proof*:**
Let $I = \ker(\mathrm{ev}_\alpha)$. By definition of ev_α, $x - \alpha \in I$. Now, since $\mathbb{F}$ is a field, $\mathbb{F}[x]$ is a Euclidean Domain, and thus a Principal Ideal Domain, so I is a principal ideal. Let's say $I = (p)$ for some $p \in \mathbb{F}[x]$. Then by definition, $x - \alpha = up$, meaning $p | x - \alpha$. Note that p cannot have degree

higher than 2, since then $x - \alpha \notin I$. So, if p has degree one, then the two elements are associates, meaning $I = (p) = (up) = (x - \alpha)$. If p has degree 0, then it must be a constant. However, p has a root at α, so if p were a constant, it would have to be 0. But then $x - \alpha \notin (0)$ – a contradiction. Therefore $(x - \alpha) = \ker(\text{ev}_\alpha)$.

Here is also a fun result I want to put down:

Example 11.1.5: generalization of Famous infinite primes proof
Let F be a finite field. Then there are infinitely many prime elements in $F[x]$

Proof. Let's assume there is a finite amount. Then $p_1, ..., p_n$ is our list of irreducible polynomials. Note that $\deg(p_i) \geq 1$ for each polynomial. Then take $P = p_1 p_1 \cdots p_n + 1$. Then $\deg(P) \geq 1$, and so P is not a unit or 0. Therefore, P must be divisible by some p_i, meaning $p_i | P$, which implies $qp_i = P$, or equivalently $P \in (p_i)$. Next, by closure of ideals, $-p_1 p_2 \cdots p_n \in (p_i)$. Then, by closure under addition:

$$P - p_1 p_2 \cdots p_n = p_1 p_2 \cdots p_n + 1 - p_1 p_2 \cdots p_n = 1 \in (p_i)$$

meaning $up_i = 1$ for appropriate $u \in F[x]$, implying p_i is a unit! However, prime elements are *not* units. Since p_i was arbitrary, *no* p_i can divide P. But then, P will have non of the factor's in it's factorization, a contradiction to deg > 0 and the p_i's being the only irreducibles! Therefore, P must be a new prime number!

Therefore, there is an infinite number of prime numbers. □

We finish off with an application of the Chinese remainder theorem with the new fact that $k[x]$ is a Euclidean domain, and hence a UFD:

Proposition 11.1.6: Chinese Remainder On Polynomials

Let $g(x)$ be a nonconstant monic element of $k[x]$ and let

$$g(x) = f_1(x)^{n_1} f_2(x)^{n_2} \cdots f_k(x)^{n_k}$$

be its factorization into irreducibles, where the $f_i(x)$ are distinct. Then we have the following isomorphism of rings:

$$k[x]/(g(x)) \cong k[x]/(f_1(x)^{n_1}) \times k[x]/(f_2(x)^{n_2}) \times \cdots \times k[x]/(f_k(x)^{n_k})$$

***Proof*:**
This follows from the Chinese Remainder Theorem (theorem 9.6.6), since $(f_i(x)^{n_i})$ and $(f_i(x)^{n_i})$ are comaximal if $f_i(x)$ and $f_j(x)$ are distinct (they are relatively prime in the Euclidean Domain $k[x]$, hence $gcd(f_i^{n_i}, f_j^{n_j}) = 1$, and so $p(x)f_i(x)^{n_i} + q(x)f_j(x)^{n_j} = 1$, so $1 \in (f_i^{n_i}) + (f_j^{n_j})$ implying $(f_i^{n_i}) + (f_j^{n_j}) = k[x]$ by propositin 9.4.10).

Exercise 11.1.1

1. Let $R[x]$ be a Principal Ideal Domain. Show that $R[x]$ is a Euclidean Domain.

11.2 Polynomial Rings over UFD's

We've seen R is an integral domain then $R[x]$ is an integral domain. We've also seen that if R is an integral domain, than we can embed R into a field of fraction F (see section 9.5). This means that $R[x] \subseteq F[x]$. Furthermore, since F is a field, then $F[x]$ is a euclidean domain. With this, we can do lots of our computation in $F[x]$ instead of $R[x]$, at the expense of allowing fractional coefficients. The question then arises if we can use information from $F[x]$ to gain information about $R[x]$!

For example, take a polynomial $p(x) \in R[x]$. Since $p(x) \in k[x]$, and $k[x]$ is a Unique Factorization Domain, then $p(x)$ factors in $k[x]$ into irreducibles. We can then ask if $p(x)$ will factor in $R[x]$. The answer is no in general, since $R[x]$ is not always a Unique Factorization Domain, since that would imply R is a Unique Factorization Domain since the coefficients of $R[x]$ must also factor uniquely. However, if R were a unique factorization domain, then this will actually imply that $R[x]$ is Unique Factorization Domain! Hence, factoring in $k[x]$ gives information on how we factor in $R[x]$. The way we'll prove this is first by uniquely factoring in $k[x]$, then "clear denominators" to obtain unique factorization in $R[x]$!

The first thing to do is make precise the notion of comparing factorization of a polynomial in $k[x]$ and a factorization in $R[x]$

Proposition 11.2.1: Gauss's Lemma

Let R be a Unique Factorization Domain with field of fraction K and let $p(x) \in R[x]$. If $p(x)$ is reducible in $K[x]$ then $p(x)$ is reducible in $R[x]$. More precisely, if $p(x) = A(x)B(x)$ for some non constant polynomials $A(x), B(x) \in K[x]$, then there are nonzero elements $r, s \in F$ such that $rA(x) = a(x)$ and $sB(x) = b(x)$, such that $a(x), b(x) \in R[x]$ and $p(x) = a(x)b(x)$ is a factorization in $R[x]$

A concrete case that can be kept in mind is with the polynomials $p(x), q(x) \in \mathbb{Q}[x]$, then $rp(x), sq(x) \in \mathbb{Z}[x]$ for appropriates $r, s \in \mathbb{Q}$. Note that it is possible that $q(x) \in \mathbb{Q}$, i.e. $q(x)$ is a constant polynomial[1].

***Proof*:**
Let's say $p(x) \in k[x]$ and is reducible $p(x) = A(x)B(x)$. Then the coefficients of $p(x)$ are quotients from R. Since R is a Unique Factorization Domain, let's take d a number common betewen all denominators for all the coefficients of $A(x)B(x)$, so that we can obtain

$$dp(x) = a'(x)b'(x)$$

where $a'(x), b'(x)$ are now in $R[x]$ since we multiplied away the denominators with d! If d is a unit, then

$$a(x) = d^{-1}a'(x) \qquad b(x) = b'(x)$$

which completes the proof, so let's assume d is not a unit. Since $d \in R$, and R is a Unique Factorization Domain, then there is a unique factorization into irreducibles:

$$d = p_1 \cdots p_n$$

[1] In textbooks where the contra-positive of this statement is used, it is often specified that $p(x)$ is *primitive*, meaning for all principal prime $\mathfrak{p} \subseteq R$, $p(x) \notin \mathfrak{p}R[x]$; in the UFD case, this amounts to the gcd of the coefficients being 1

Since p_1 is an irreducible element, and R is a Unique Factorization Domain, then $(p_1) = p_1R[x]$ (since $R[x]$ is commutative $(a) = aR$) is prime (exercise), and so $R[x]/p_1R[x]$ is an integral domain (proposition 9.4.25) and $R[x]/p_1R[x] \cong (R/p_1R)[x]$ (exercise).

Now, reducing $dp(x) = a'(x)b'(x)$ modulo p_1, we get

$$\overline{0} = \overline{a'(x)} \cdot \overline{b'(x)}$$

Since $(R/p_1R)[x]$ is an integral domain, then either $\overline{a'(x)}$ or $\overline{b'(x)}$ is zero, say $\overline{a'(x)} = 0$. This would imply $p_1|a'(x)$, and so every coefficient of $a'(x)$ is divisible by p_1, meaning $\frac{1}{p_1}a'(x)$ also has coefficients from R! So, we can divide both sides by p_1 (that is d by p_1 and either $a'(x)$ or $b'(x)$ by p_1). But then, the factor d on the left hand side has one fewer irreducible element! We can continue this procedure till we cancel out all of the factors, leaving us with $p(x) = a(x)b(x)$, with $a(x), b(x) \in R[x]$, $a(x)$ and $b(x)$ being F-mutiples of $A(x)$ and $B(x)$!

Note that $A(x)$ and $B(x)$ aren't necessarily R-multiples, nor are they necessarily unique. For example, $x^2 \in Q[x]$ is reducible, and so reducible in $\mathbb{Z}[x]$. One factorization would be $A(x) = 2x$ and $B(x) = \frac{1}{2}x$. However, $\frac{1}{2} \notin \mathbb{Z}$: no integer multiple give's us a factorization of $A(x)$ and $B(x)$. What the theroem will tell us is that $\frac{1}{2}A(x) \in \mathbb{Z}[x]$, and $2B(x) \in \mathbb{Z}[x]$.

Since k is a field, the elements of R becomes units in $k[x]$ (the units in $k[x]$ being the non-zero constants). For example, $7x$ factors in $\mathbb{Z}[x]$ and produces two irreducible 7 and x, so $7x$ is *not* irreducible in $\mathbb{Z}[x]$. However, in $\mathbb{Q}[x]$, $7x$ is a unit 7 times an irreducible, so $7x$ *is* irreducible in $\mathbb{Q}[x]$. However, this is the *only* difference between the irreducible elements in $R[x]$ and $k[x]$

Corollary 11.2.2: Irreducibility In $R[x]$ And $k[x]$

Let R be a UFD, let $k = K(R)$ be its field of fractions, and let $p(x) \in R[x]$. Suppose the greatest common divisor of the coefficients of $p(x)$ is 1. Then $p(x)$ is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$. In particular, if $p(x)$ is a monic polynomial that is irreducible in $R[x]$, then $p(x)$ is irreducible in $K[x]$.

***Proof*:**
We'll proceed by the contrapositive: $p(x)$ is reducible in $R[x]$ if and only if $p(x)$ is reducible $K[x]$. The $(\Leftarrow)$ direction is Gauss's Lemma, so let's go $(\Rightarrow)$. Let's say $p(x)$ is reducible in $R[x]$ and the greatest comon divisor of the coefficients is 1. That means that when we reduce $p(x) = a(x)b(x)$, neither $a(x)$ or $b(x)$ can be constants! Note that polynomials which aren't constants in $K[x]$ are not units, meaning $p(x) = a(x)b(x)$ is also a factorization in $K[x]$, completing the proof!

This theorem is already incredibly powerful, it means that the task of finding the irreducible elements of $R[x]$ reduces to finding the irreducible elements of $k[x]$ (which we know is a euclidean domain) up to a multiple of R[2]. We shall later find more techniques for finding solutions to polynomials over fields, which by our above results immediately give us a way to translate these results for integral domains whose field of fraction is the given polynomial ring over a field.

Yet another result is the following which shows that extending a ring to form a polynomial ring preserves UFD!

[2]For those keen on doing algebraic geometry, this process of reducing the problem of factoring in $R[x]$ to factoring in $k[x]$ can be thought of as a form of using the stalk at 0 to find information about the elements of $R[x]$, that is it is taking advantage of local information for global results TBD

Theorem 11.2.3: R UFD If And Only If $R[x]$ UFD

R is a Unique Factorization Domain if and only if $R[x]$ is a Unique Factorization Domain

***Proof*:**
If $R[x]$ is a Unique Factorization Domain, then the constants of $R[x]$ must uniquely factor, meaning R is a Unique Factorization Domain.

So let's suppose R is a Unique Factorization Domain, and take $p(x) \in R[x]$. If $p(x)$ is a constant, then it's uniquely factorizable, so let's say $\deg(p(x)) > 0$. Let d be the g.c.d. of the coefficients of $p(x)$, so that $p(x) = dp'(x)$, meaning the g.c.d. of the coefficients of $p'(x)$ is 1. Note that this factorization is unique up to a unit multiple of d (i.e. up to a unit in R by proposition 9.2.3), and since d can be factored uniquely in R (and these are also irreducibles in $R[x]$ too), it suffices now to prove that $p'(x)$ can be uniquely factored into irreducibles in $R[x]$.

Let k be the field of fractions of R. Then since $k[x]$ is a Unique Factorization Domain (proposition 11.1.1), $p(x)$ can be factored uniquely into irreducibles in $k[x]$. By Gauss's Lemma, such a factorization implies a factorization in $R[x]$ whose factors are F-multiples of the factors in $k[x]$. Since the g.c.d. of $p'(x)$ is 1, the g.c.d. of the coefficients of each of these factors in $R[x]$ must be 1. By corollary 11.2.2, each of these factors are irreducible in $R[x]$! Hence $p(x)$ can be written as the product of irreducibles in $R[x]$.

The uniqueness of the factorization of $p(x)$ follows from the uniqueness in $k[x]$. Suppose

$$p'(x) = q_1(x)q_2(x)\cdots q_n(x) = q_1'(x)q_2'(x)\cdots q_m'(x)$$

Since the g.c.d. of the coefficients of $p'(x)$ is 1, so to it is for both factorizations. By corollary 11.2.2, each $q_i(x)$ and $q_j'(x)$ is an irreducible in $k[x]$. By uniqueness of factorization on $k[x]$, $n = m$, and the up to rearrangement, the two factorizations are the same up to associates, that is $q_i(x) = uq_i'(x)$ for a unit $u \in k[x]$. It remains to show they are associates in $R[x]$. By proposition 9.2.3, the untis of $k[x]$ are the units of F, so

$$q_i(x) = \frac{a}{b}q_i'(x)$$

for $a, b \in R$. Thus, $bq_i(x) = aq_i'(x)$. Thinking about it another way, the g.c.d. of the coefficients of the polynomial on the left hand side is b, and a on the right hand side respectively. Since in a Unique Factorization Domain, the coefficients of a nonzero polynomial is unique up to units ($a = ub$ for a unit in R), then $q_i(x) = uq_i'(x)$, and so $q_i(x)$ and $q_i'(x)$ are associates in R as well, completing the proof!

Corollary 11.2.4: Polynomial Rings in Several Variables over a UFD are UFDs

If R is a Unique Factorization Domain, then a polynomial ring in an arbitrary number of variables with coefficients in R is also a UFD

***Proof*:**
For the finitely many case, apply induction on theorem 11.2.3, since $R[x,y] = R[x][y]$, and $R[x]$ is a Unique Factorization Domain. The general case follwos fromthe definition of a polynomial ring in an arbitrar ynumber of variables as the union of polynomial rings in finitely many variables.

Example 11.2.5

1. $\mathbb{Z}[x], \mathbb{Z}[x,y]$, and so on are Unique Factorization Domains. The ring $\mathbb{Z}[x]$ now gives a good example of a Unique Factorization Domain that is *not* a Principal Ideal Domain!
2. Similarly, $\mathbb{Q}[x]$, $\mathbb{Q}[x,y]$, and so forth are Unique Factorization Domains.

Finally, we'll ask if we can weaken the condition's and still get the same result. What if R was just an integral domain? Can we follow a similar construction? Generalizing in a different direction, if R is a Unique Factorization Domain, does that imply that $R[[x]]$ is a Unique Factorization Domain? The answer to both is in general no

Example 11.2.6: Counter-Example

1. Let $R = \mathbb{Z}[2i]$, and let $p(x) = x^2 + 1$. Then the field of fractions of R is $F = \mathbb{Q}[2i]$. The polynomial $p(x)$ factors uniquely into a product of two linear facrors in $k[x] : x^2 = (x-i)(x+1)$. so in particular $p(x)$ *is* reducible in $\mathbb{Q}[2i]$. However, Neither of these factors lies in $R[x]$ (because $i \notin R$) so $p(x)$ is *irreducible* in $R[x]$. In particualr, by corollary 11.2.2, $\mathbb{Z}[2i]$ is *not* a Unique Factorization Domain! If it were, then it should've been reducible!
2. It is a rather technical project to show that if R is a UFD, then $R[[x]]$ is a UFD. The somewhat canonical counter-example is

$$R = \frac{k[x,y,z]}{(x^2 + y^3 + z^7)}$$

See (citation commented) for an in-depth explanation[a].

[a] this paper does require some algebraic knowledge introduced in part V

Exercise 11.2.1

1. Let $x^5 - x + 1 \in \mathbb{Q}[x]$. Is this polynomial irreducible? (Hint: use Gauss's Lemma)
2. Recall that if R Is a UFD, it is not necessarily the case that $R[[x]$. Show that $\mathbb{R}[[x]]$ is a UFD

11.3 Irreducibility of Polynomials

Now that we've shown R, $R[x]$, $R[x,y]$ and so forth are Unique Factorization Domains if and only if R is a Unique Factorization Domain and that irreducibility over $R[x]$ is symmetrically reflected to irreducibility over $k[x]$, we now raise the question about finding irreducible elements in such a ring (i.e. the prime elements – the "building blocks"). A polynomial $p(x)$ is reducible if we can find two polynomials of smaller degree such that $p(x) = a(x)b(x)$. A polynomial is irreducible if no such polynomials exist. In general this is difficult, which is why we look for nicer irreducibility criteria.

Throughout these proofs, we shall be taking advantage of the fact that the field of fractions is a UFD, hence the polynomial ring over the field is a UFD. We shall not even require the ring that we are using to construct the field of fraction to be a UFD, simply an integral domain. To demonstrate, consider the following key fact:

Proposition 11.3.1: Finding Linear Factors

Let R be an integral domain and let $p(x) \in R[x]$. Then $p(x)$ has a factor of degree one if and only if $p(x)$ has a root in F, i.e., there is an $r \in R$ such that $p(\alpha) = 0$

***Proof*:**
Let $p(x) = (x - r)q(x)$ for some $r \in R$, $q(x) \in R[x]$. Then certainly

$$p(r) = (r - r)q(r) = 0q(r) = 0$$

Hence, r is a root. Conversely, if $p(r) = 0$ for some $r \in R$, take the field of fractions and consider $k[x]$. Then by the division algorithm in $k[x]$ we may write

$$p(x) = q(x)(x - a) + r$$

Since $p(r) = 0$, then r must be 0. Finally, multiply $q(x)$ by an appropriate α to clear the denominators so that $\alpha q(x) = Q(x) \in R[x]$, and hence we get

$$p(x) = Q(x)(x - r)$$

as we sought to show.

Corollary 11.3.2: Maximum Number Of Roots

[Maximum Number Of Roots] Let R be an integral domain and $p(x) \in R[x]$ of degree n. Then $p(x)$ has *at most* n roots.

***Proof*:**
Let $k = K(R)$ be the field of fraction of R. Then the number of roots of $p(x) \in [R]x$ is certainly less than or equal to the number of roots in $p(x) \in k[x]$. Now, *a fortiori* k is a UFD, hence by theorem 11.2.3 $k[x]$ is a UFD, and so each root corresponds to an (irreducible) factor of degree 1. But then, f can have at most n factors of degree 1 by unique factorization, completing the proof.

Note that this statement can fail if R is not an integral domain!

Example 11.3.3: Polynomial With More Roots And Its Degree
[Polynomial With More Roots And Its Degree] Let $x^2 + x \in (\mathbb{Z}/6\mathbb{Z})[x]$. Show that this polynomial has 6 roots!

However, using proposition 11.3.1, we may show that polynomials over infinite integral domains are determined by their roots!

Corollary 11.3.4: Polynomials Detremiend By Evaluation

[Polynomials Detremiend By Evaluation] Let R be an infinite integral domain and $f, g \in R[x]$. Then $f = g$ if and only if

$$r \mapsto f(r) \qquad \text{and} \qquad r \mapsto g(r)$$

agree for all $r \in R$

***Proof*:**
Two functions agree if every $r \in R$ is a root of $f - g<$ but a nonzero polynomial over R cannot have infinitely many roots by corollary 11.3.2, hence they must be equal as we sought to show.

Continuing from this, we may ask about the irreducibility of degree 2 and 3 polynomials, which is an immediate consequence of the above results:

Proposition 11.3.5: Degree 2 Or 3 Polynomials

A polynomial of degree two or three over a field F is reducible if and only if it has a root in F.

***Proof*:**
This follow's immediately from the previous proposition, see exercise ref:HERE

The reason for this is that if we have a degree higher than 4, then we can reduce to two irreducible polynomials of degree 2, like $(x^2 + 1)(x^2 + 1)$ in $\mathbb{Q}[x]$. For higher degree polynomials, we may have more solutions depending on what type of field we are working over. For example, we shall introduce in chapter 19, section 19.5.1, the existence of field for which only linear polynomials are irreducible ($\mathbb{C}$ being a canonical example of such a field). Yet another example is that any polynomial of degree ≥ 3 over $\mathbb{R}$ is reducible[3]. For more general rings, it is in general much more difficult to determine if a polynomial is reducible. There are however some tricks that help in arbitrary situations, especially if R is a UFD (for then we may use the gcd)

This next proposition let's us limit the number of "rational" solutions of a polynomial over a UFD.

Proposition 11.3.6: Rational Roots Test

Let R be a UFD and $k = K(R)$. Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdot + a_1 x + a_0$ be a polynomial of degree n with coefficients in the UFD R. If $r/s \in k$ with $gcd(r, s) = 1$, and r/s is a root of $p(x)$ $(p(r/s) = 0)$, then r divides the constant term and s divides the leading coefficient of $p(x)$: $r \big| a_0$ and $s \big| a_n$. In particular, if $p(x)$ is a monic polynomial with integer coefficients and $p(d) \neq 0$ for all integers d dividing the constant term of $p(x)$, then $p(x)$ has no roots in $k[x]$.

[3] This is exercise ref:HeRE (closure of fields, 3rd exercise). An astute student would be able to work out this exercise

***Proof*:**
Let's say $p(r/s) = 0$. Then

$$p(r/s) = 0 = a_n(r/s)^n + a_{n-1}(r/s)^{n-1} + \cdots + a_1(r/s) + a_0$$

multiplying all by s^n, we get:

$$0 = a_n r^n + a_{n-1}r^{n-1}s + \cdots + a_1 r s^{n-1} + a_0 s^n$$

so

$$a_n r^n = s(-a_{n-1}r^{n-1} - \cdots - a_1 r s^{n-1} - a_0 s^{n-1})$$

thus, $s \big| a_n r^n$. $gcd(r, s) = 1$, then it must be that $s|a_n$. Doing the same trick for a_0:

$$a_0 s^n = r(-a_n r^{n-1} - a_{n-1}r^{n-2} - \cdots - a_1 s^{n-1})$$

then $r|a_0$.

If $a_n = 1$, then we'd have $s|1$, so $r/s = r/1 = r$. This means we only need to check the values r which divide a_0. If every value that dividse a_0 is not a root of $p(x)$, then $p(x)$ has no roots.

This gives us a really easy way of testing irreducibility of low-degree polynomials:

Example 11.3.7: Irreducibility Of Low-Degree Polynomials

1. The polynomial $p(x) = x^3 - 3x - 1$ is irreducible in $\mathbb{Z}[x]$. Since the polynomial is monic, by corollary 11.2.2, it must be reducible in $\mathbb{Q}[x]$. By the rational number test, we are given the candidates ± 1. Plugging both of them in, we get no roots of $\mathbb{Q}[x]$, and so by 11.3.5, this polynomial is irreducible, and so is irreducible in $\mathbb{Z}[x]$.

2. For any prime p, $x^2 - p$ and $x^3 - p$ are irreducible in $\mathbb{Q}[x]$. We have ± 1 and $\pm p$ as candidates. Both of them are not roots, and therefore, these polynomials are not reducible!

3. The polynomial $x^2 + 1$ is *reducible* in $(\mathbb{Z}/2\mathbb{Z})[x]$, namely $(1)^2 + 1 = 1 + 1 = 0$. It's factors are $(x + 1)(x + 1) = x^2 + 1$. The rational roots test applies since TBD.

4. The polynomial $x^2 + x + 1$ is *irreducible* in $(\mathbb{Z}/2\mathbb{Z})[x]$, since there are no roots.

5. Similarly for $x^3 + x + 1$ in $(\mathbb{Z}/2\mathbb{Z})[x]$

All the polynomials in the aforementioned examples are all of degree less than or equal to 3. This is because when factoring polynomials of degree greater than or equal to 4, it's possible that we get 2nd degree polynomials as irreducible factors (like $(x^2 + 1)(x^2 + 1)$ in $\mathbb{Q}[x]$, which has no factors).

We use another trick for these type of polynomials: We can use 11.0.1 ($R[x]/I[x] \cong (R/I)[x]$) to generalize the rational root test:

Proposition 11.3.8: Reduction Test

Let I be a proper ideal in the integral domain R and let $p(x)$ be a non-constant *monic* polynomial in $R[x]$. If the image of $p(x)$ in $(R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of smaller degree, then $p(x)$ is irreducible in $R[x]$.

***Proof*:**
For the sake of contradiction, let's supose $p(x)$ cannot be factored in $(R/I)[x]$, but $p(x)$ is reducible in $R[x]$. Then there must be two polynomials such that $p(x) = a(x)b(x)$. Note that both $a(x)$ and $b(x)$ must be monic.

Reducing the coefficients modulo I gives a factorization in $R[x]/I[x]$, but by 11.0.1, this is equivalent to $(R/I)[x]$ – contradiction our assumpiton.

The power here comes from the fact that R/I might have much simpler structure for factoring (ex. $\mathbb{Z}$ vs $\mathbb{Z}/n\mathbb{Z}$!) For example, in exercises 11.2.1, we may prove that $x^5 - x + 1$ using this trick rather quickly! So, If we reduce a polynomial into $(R/I)[x]$, and show it cannot be factored, then it is irreducible in $R[x]$ as well! However, the converse (if $p(x)$ is irreducible in $R[x]$, then $p(x)$ is irreducible in every $(R/I)[x]$) is not true.

Example 11.3.9: Limitation of reduction method

Take $x^4 + 1 \in \mathbb{Z}[x]$. This polynomial is irreducible in $\mathbb{Z}[x]$, (hint: recall that $\sqrt{2}$ is irrational), but is reducible for every prime. Verify this by considering cases of

$$p \equiv 1, 3, 5, 7 \mod 8$$

since the case of $p = 2$ can quickly be manually verified.

So, if you get that a polynomial is reducible in $(R/I)[x]$, that unfortunately does not tell you it's reducible in $R[x]$.

A special case of proposition 11.3.8 that is quite useful is called Eisenstein's Criterion (or Eisenstein-Schönenmann Criterion, since Schönenmann discovered it first)

Proposition 11.3.10: Eisenstein's Criterion

Let P be a prime ideal of the commutative ring R and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be a polynomial in $R[x]$ (where $n \geq 1$). Suppose $a_{n-1}, ..., a_0 \in P$, but $a_0 \notin P^2$. Then $p(x)$ is irreducible.

***Proof*:**
Let's take $p(x)$ like in the proposition. For the sake of contradiction, let's say $p(x)$ is reducible: $p(x) = a(x)b(x)$, where $a(x)$ and $b(x)$ are non-constant.Then by proposition 11.3.8 (the contrapositive of it), it is reducible in $(R/P)[x]$. Since every non-leading coefficient is in P:

$$\overline{p(x)} = \overline{x^n} = \overline{a(x)} \cdot \overline{b(x)}$$

Since P is prime, R/I is an integral domain, and so the constant terms of $a(x)$ and $b(x)$ must be in P, or else we'll get $\overline{x} \cdot \overline{y} = \overline{0}$. But then, multiplying those terms means $a_0 \in P^2$, a contradiction!

For the purposes of this book, the following is the most immediate use of Eisenstein's Criterion:

Corollary 11.3.11: Eisenstein's Criterion For $\mathbb{Z}[x]$

Let p be a prime in $\mathbb{Z}$ and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0$. If every a_i is divisible by p, but a_0 is not divisible by p^2, then $f(x)$ is irreducible in both $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

***Proof*:**
This is proposition 11.3.10 and using corollary 11.2.2.

Here are some examples using Eisenstein's criterion, and some more tricks when finding irreducible polynomials:

Example 11.3.12: Irreducibility Tricks

1. When solving polynomials of degree 3 or less, it'd suggest to always start with checking roots: it's the simpliest criterion.

2. The polymoial $x^4 + 10x + 5$ in $\mathbb{Z}[x]$ is irreducible, if we take $p = 5$. Since $5|10$, $5|5$, but $5^2 \nmid 5$, then Eisntein's criterion holds.

3. For any integer a such that $p\big|a$ but $p^2 \nmid a$, then $x^n - a$ is irreducible in $\mathbb{Z}[x]$. If $a = p$ a prime, then $x^n - p$ is irreducible for all positive integers n, so for $n \geq 2$, the n^{th} root of p are not rational numbers (i.e. has no roots in $\mathbb{Q}[x]$)

4. We cannot apply the same tricks we did before to the polynomial $x^4 + 1$ in $\mathbb{Z}[x]$, since nothing divides prime, and any proper ideal will not contain 1. Instead, we'll use a trick called *substitution.*

   Consider the polynomial $f(x + 1) = (x + 1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$. On this polynimal, Eisntein's criterion works, since all non-leading coefficients are divisible by 2, and $2^2 \nmid a_0$. From this, it follows that $f(x)$ is irreducible, since any factorization of $f(x)$ will be a factorization for $f(x + 1)$ (replace x by $x + 1$ in the factorization!) Remember this trick for polynomials which seem at first glance to not get an application of Eisntein's criterion.

5. (cyclotomic polynomials) Perhaps the most famous application of Eisenstein's criterion is on polynomials of the following form:

$$1 + x + x^2 + \cdots = x^{p-1} \in \mathbb{Z}[x]$$

   Eisenstein's criterion cannot be directly applied, however as before we can do the $x \mapsto x + 1$ trick to get

$$p(x+1) = 1 + (x+1) + \cdots + (x+1)^{p_1} = x^{p-1}\binom{p}{p-1}x^{p-1} + \cdots \binom{p}{2}x + \binom{p}{1}$$

   where $\binom{p}{1} = p$.It suffices to show that $p\big|\binom{p}{k}$ for $1 \leq k \leq p - 1$ to conclude the result.

6. On $\mathbb{Q}[x][X]$?

7. Notice that in the proof, we reduced to $(R/P)[x]$. If $R = \mathbb{F}$ was a field, then the only ideals are 0 and $\mathbb{F}$, meaning $(\mathbb{F}/P)[x] \cong 0[x] \cong 0$. Therefore, Einsteins criterion doesn't work will with such polynomials.

   For example, is $x^4 + 1$ reducible in $\mathbb{F}_5[x]$ ($\mathbb{F}_5$ is the field with 5 elements, and is isomorphic ot $\mathbb{Z}/p\mathbb{Z}$)? If if were over $\mathbb{Z}[x]$, we already mentioned how it is irreducible (using a tirck we which will be explained in chapter ref:HERE). It also has no roots, as can be verified by plugging in $0, ..., 4$.

   We can't use einstein's criteron, but since the degree is 4, we know if a factorization did exist, they would have to be of degree 2, that is

$$x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a + c)x^4 + (b + ac + d)x^2 + (ab + bc)x + bd$$

   note that the coefficients of x^2 are one since the coeficient of x^4 is one. We can then deduce that

$$a + c = 0, b + ac + d = 0, ad + bc = 0, bd = 1$$

   Manipulating a bit, we can get the following information:

$$\begin{aligned} c &= -a \\ b + a(-a) + d &= 0 \\ ad + b(-a) &= 0 \\ a^2 &= b + d \\ ad &= ab \end{aligned}$$

   Let's take cases: let's say $a \neq 0$. Then $b = d$, $a^2 = 2b$, $b^2 = 1$. (TBD). So, we get a contradiction.

   Let's check $a = 0$. Then $062 = 0 = b + d$, implying $b = -d$, and $b^2 = -bd = -1 = 4$, so $b = 2$. implying $d = 2$ and $a = 0$, $c = 0$. No contradiction was found! Thus, we can plug in, and factor to get

$$p(x) = x^4 + 1 = (x^2 - 3)(x^2 - 2) = (x^2 + 2)(x^2 + 3)$$

   Thus, $p(x)$ is reducible in $\mathbb{F}_5[x]$.

There are further irreducibiltiy algorithms that were programmed. One algorithm is Berlekamp Algorithm. there are others TBD.

Finally, we give a few important results that shall be very handy tools. The first tells us that factorization of a polynomial is invariant to field extensions, namely if $F \hookrightarrow E$ is an injective map, then the factor of $p(x)$ in E are the same factors (if not further reduced) in $E(x)$:

Proposition 11.3.13: Invariant Under Field Extensions

If you extend from a field F to E, then by the uniqueness of the euclidean algorithm, everything in $E[x]$ will still factor the same way as before (in $k[x]$): it cannot inroduce new solutions.

***Proof*:**
exercise

This next one is of particular interest for the structure of the multiplicative group of a field!

Theorem 11.3.14: Subgroup of Multiplicative Group is Cyclic

A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

Another way of saying this is that $F^\times$ is locally cyclic (being "locally P" means that every finite subgroup has the property P)

***Proof*:**
We will use the fundamental theorem for finitely generated abliean groups!

Let $A \subseteq F^\times$ be a finitely generated abelian group. Then:

$$A \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_k \big| n_{k-1}, n_{k-1} \big| n_{k-2}, ..., n_2 \big| n_1$. Since n_k divides the order of each of the cyclic groups in the direct product, it follows that each direct factor contains n_k elements of order dividing n_k (by prop ref:HERE). If k were greater than 1, there would therefore be a total of more than n_k such elements. But then there would be more than n_k roots of the polynomial $x^{n_k} - 1$ in the field F, contradicting proposition 11.3.1!! Hence, $k = 1$, and the group is cyclic!

Corollary 11.3.15: $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic

Let p be a prime. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of nonzero residue classes mod p is cyclic

***Proof*:**
Since $\mathbb{Z}/p\mathbb{Z}$ is a field, apply proposition 11.3.14

These corollaries will be useful when proving Kronnecker-Weber's Theorem (theorem 50.5.4):

Corollary 11.3.16: Isomorphisms Of Multiplicative Groups

Let $n \geq 2$ be an integer with factorization $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$ in $\mathbb{Z}$, where $p_1, ..., p_n$ are distinct primes. Then we have the following isomorphsism of (multiplicative) groups:

1. $(\mathbb{Z}/n\mathbb{Z})^\times = (\mathbb{Z}/p_1^{\alpha_1})^\times \times (\mathbb{Z}/p_2^{\alpha_2})^\times \times \cdots \times (\mathbb{Z}/p_n^{\alpha_n})^\times$
2. $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is the direct product of a cyclic group of order 2 and a cyclic group of order $2^{\alpha-2}$, for all $\alpha \geq 2$
3. $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is a cyclic group of order $p^{\alpha-1}(p-1)$ for all odd primes p

***Proof*:**
exercise (comment)

Exercise 11.3.1

1. Show that there are irreducible polynomial in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ for every degree.
2. Let $f(x)$ be an irreducible monic polynomial of $\mathbb{Z}[x]$. Show that $f(x)$ mod p has a root for infinitely many primes. The opposite is true too, that there are infinitely many primes for which $f(x)$ mod p does *not* have a root, but as far as I am aware that requirse some analytic number theory[4]
3. Show that $t^2 + t + 1 \in \mathbb{F}_2[t]$ is irreducible. Use this to show that $\mathbb{F}_2[t]/(t^2 + t + 1)$ is a field with 4 elements.

[4] see [17]

Part III

Modules

When we explored groups, we saw that the kernel of groups were groups (in particular a special subset of a group called the normal subgroup), and if we have an abelian group, every abelian group is the kernel of some group.

The story for rings is more complicated. As we mentioned at the beginning of the part on rings, some authors include an identity in the definition, others do not. If we include an identity, then the kernel of ring homomorphisms are not rings. Even if we take exclude identities as part of the definition, notice that practically non of the ideals we talked about were previously mentioned rings, essentially because almost all rings we talk about have an identity, but if an ideal has an identity, it is the entire ring. The cokernels is also not as well-behaved (recall that $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism). In this way, the kernel of rings and rings are very different objects, and so a lot goes into studying ideals as though they are a seperate object in ring theory. In particular, we do not focus a lot of attention of constructing rings based on other rings (similar to how we constructed polyabelian or poly-cyclic groups). Overall, the category of **Ring** and **Rng** are not the best behaved categories.

The concept of Modules can be thought of as a fix to this. Modules are more similar to abliean groups than they are to rings, for example if you have any sub-module, then the quotient will always be well-defined. Furthermore, all Rings can be thought as a module, making the study of module subsumes the study of rings. As we shall see,

We shall see that an R-module M will be an abelian group M with an ring action by R. The reason M is an abelian group and not just a set like in group actions is because it is a generalization of vector-spaces, not an addition to group actions. What I mean by this is that you have to think of the reasoning from the point of view that vector-spaces came first, and then there were many instances where the field acting on the group was too restrictive, which lead to module.

Over the 19th and 20th century, many mathematical objects were studied that all turn out to be modules. To those who want the punchline, I'll point out a few right now:

1. All abelian groups are $\mathbb{Z}$-modules
2. All linear transformations are $F[x]$-modules
3. All rings are modules (If R is a ring, then it is an R-module over itself)
4. All ideals are modules (if $I \subseteq R$ is an ideal, it is an R-module)
5. All vector-spaces over k are k-modules
6. All vector-bundles are projective modules

12

Introduction to Modules

A lot of the algebraic structures we've covered so far share a similar story: we ask how maps between them can be studied (leading us to explore sub-objects and quotient-objects), and we ask how to generate them. Modules will have the same story at the beginning. [Question: Why didn't group actions have the same story? no sub/quotient actions]. When we'll be building up modules, they'll more closely resemble groups and rings. In groups, simple groups and free groups played a bigger role than they did in ring theory. Same in Module theory (TBD).

12.1 Basic Definitions and Examples

Definition 12.1.1: [left]-Module

Let R be a ring (more generally a rng). A *left R-module* or a *left module over R* is a set M together with

1. a binary operation $+$ on M under which M is an **abelian group**, and
2. an action of R on M (that is, a map $R \times M \to M$) denoted by rm, for all $r \in R$ and for all $m \in M$ which satisfies
   (a) $(r+s)m = rm + sm$, for all $r, s \in R$, $m \in M$
   (b) $(rs)m = r(sm)$, for all $r, s \in R$, $m \in M$
   (c) $r(m+n) = rm + rn$ for all $r \in R$, $m, n \in M$

   If the rng R is a ring (i.e. has a 1), we impose the additional axiom

   (d) $1m = m$, for all $m \in M$

   And we sometimes call such a module a *unital* R-module.

Right modules are defined similarly. If the R ring is commutative, and M is an left $R-module$, we can turn it into a right $R - module$ by defining $rm = mr$. To distinguish when different rings may be acting on an abelian group M, the symbol $\curvearrowright$ (read *acts on*) may be used to show that a ring R is acting on M: $R \curvearrowright M$

Unless stated otherwise, the term "module" will reference "left-module" Note that for a module M, we usually denote what ring it is over (or what ring acts upon it) by saying it is an *[ring symbol here]-module* (ex. R-module, $F[x]$-module, $[M_{m\times n}(\mathbb{F})]$-module). We might also say R acts on M where R is the ring and M is the abelian group. We will always be working *rings* actions, however as noted in the definition, we can also work with *rng* actions. Note too that we will often say that "R acts on the module M" to mean that we define a ring-action of R on the abelian group M.

When working over rings, the distributivity condition forces M to be abelian: let $r = s = 1$:

$$(1+1)\cdot(m+n) = (1+1)\cdot m + (1+1)\cdot n = 1\cdot m + 1\cdot m + 1\cdot n + 1\cdot n = m + m + n + n$$

$$(1+1)\cdot(m+n) = 1\cdot(m+n) + 1\cdot(m+n) = 1\cdot m + 1\cdot n + 1\cdot m + 1\cdot n = m + n + m + n$$

which implies $m + n = n + m$ for all $m, n \in M$. Notice how this when working with group actions, the group acting on the set did not effect any strucutre of set, namely since the set has no additional structure. Be mindeful that when talking about a module, we will usually call M the module. For example we might say "Let $\mathbb{Z}$ act on $\mathbb{Q}$ by $r \cdot a = ra$; then $\mathbb{Q}$ has the following properties". When talking about $\mathbb{Q}$ here, we're talking about $\mathbb{Q}$ as a module, namely a $\mathbb{Z}$-module, not just as abelian group or a ring.

If R is commutative then we defined a two-sided action $r \cdot m = m \cdot r$. If R is not commutative, then defining this property will have problems with $2(b)$: Let R be non-commutative, let $r \cdot m = m \cdot r$, and take $r, s \in R$ such that $rs \neq sr$ and $rs \cdot m \neq sr \cdot m$ (an example of such a module will be presented in the examples 12.1.7). Then:

$$(rs)m = m(rs) = (mr)s = (rm)s = sr(m) = (sr)m$$

however, $sr \cdot m \neq rs \cdot m$ so this statement is not necessarily always true. Note how it may be possible that even though $rs \neq sr$ that $rs \cdot m = sr \cdot m$, for example, the trivial module structure $r \cdot m = m$ for any $r \in R$ will produce this result, however as we'll soon see in example 12.1.7, every module will always induce a *faithful* module (the word faithful being defined equivalently to the group-action case[1]), and so making a distinction between left and right modules is very important over non-commutative rings.

We now have many operators to keep track of: the group operation of M, the two rings operations of R, and the ring action of R on M. When working with modules, we often focus on the ring-action, and use group and rings structures that were previously covered.

One might ask themselves if modules and group-rings might be similar concept, boths somehow being a ring "acting" on a group. If this suspicions crossed your mind, it is good idea to check how these two structure's differ (for example, if $|G| > 1$, recall a group-ring always has zero-divisors. Find an example in the list of examples bellow that does not have this property).

Since a module is an action on an abelian group, it would make sense if there is an equivalent to a permutation representation for module's as there is for group-actions. Recall that any group action can be seen as a group-homomorphism:

$$\rho : G \to \mathrm{Sym}(S)$$

The exact same idea works for defining an R-module as an action. In order to do so, we need an a notion of an R-module homomorphism that preserves some of the structure. Before looking at the definition bellow, see if you can come up with the definition of an R-module homomorphism.

Definition 12.1.2: R-module homomorphism

Let R be a ring and let M and N be R-module A map $\varphi : M \to N$ is an *R-module homomorphism* if it respects the R-module structure of M and N, i.e.

1. $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$
2. $\varphi(r \cdot x) = r\varphi(x)$, for all $r \in R$, $m \in M$

Since M and N are both abelian groups, R-module homomorphisms are sometimes called *R-linear group homomorphisms*, where the map is R-linear since the ring actions passes through the homomorphism. Notice that both M and N both have the same ring acing on them: it is imperative that an R-module homomorphism be betewen R-modules.

Returning to the discussion on treating R-modules as an action, it is a good exercise to check ones understanding of the axioms of modules that the "axiomatic representation" of an R-module is equivalent to defining an R-module as a ring-homomorphism

$$\rho : R \to \mathrm{End}_{\mathbf{Ab}}(M)$$

where $\mathrm{End}_{\mathrm{Ab}}(M)$ is the collection of all group-homomorphisms from M to itself. Notice that $\mathrm{End}(M)$ is an [abelian] group because M is an abelian group, while the ring operation of composition, $\circ$,

[1]That is, there must exist an $m \in M$ st $r \cdot m = s \cdot m$ implies $r = s$

automatically satisfies the condition to make $\text{End}_{\mathbf{Ab}}(M)$ a ring, which should be checked (for example, $\text{id} : M \to M$ is an element of $\text{End}_{\mathbf{Ab}}(M)$, hence it has a multiplicative identity). It will sometimes useful to define an R-module as a ring homomorphsim $R \to \text{End}_{\mathbf{Ab}}(M)$, if we want to manipulate with the ring action on it (see exercise ref:HERE (how $R \times S$ can act on an R-module, easy to see with this definition))

As with rings and and with groups, we can define a category: given a ring R, the collection of R-modules and the collection of R-module homomorphisms forms a category:

Definition 12.1.3: R-Mod

The collection of all R-modules and R-module homomorphisms forms a category, which we'll denote R-**Mod** or $\mathbf{Mod}_R$.

This category has some of the nice properties that the other categories that were extensively worked in so far have:

Definition 12.1.4: Classical definition of Submodule

Let R be a ring and let M be an R-module. An R-submodule of M is a subgroup N of M which is closed under the action of ring elements, i.e., $rn \in N$, for $r \in R$, $n \in N$

Definition 12.1.5: Categorical Definition of Submodule

Let $N \subseteq M$. Then if there exists a R-module inclusion map $\varphi : N \to M$ that is also an R-module homomorphism, then N is a submodule of M

Proposition 12.1.6: The sub-module Criterion

Let R be a ring and let M be a R-module. A subset of N of M is a submodule of M if and only if

1. $N \neq \emptyset$, and
2. $x + ry \in N$ for all $r \in R$, and for all $x, y \in N$

***Proof*:**
Easy to prove both ways (recall that M is an abelian group, and for the other way, simply set the values of r and x to some appropriate values)

Example 12.1.7: Modules

1. Let R be any ring. Then $M = R$ is a left R-module with the ring multiplication as the ring action: $r \cdot s = rs$. Take a moment to think about what are the submodules of R? The answer will be revieled in the next sentence. the sub-modules of M are the left-ideals of R! This is because ideals of R are closed under multiplication of R, and the axioms of modules require closure under "scalar multiplication" by the ring elements. Is R/I for any (two-sided) ideal of R also an R-module? Take a moment to think about it. The answer is that R/I

is indeed an R-module! We will soon show that the kernel of module-homomorphisms will be submodules, showing that the axioms of a module "fix" the category of **Ring** by being closed under kernels!

From now on, if we say that R is a [left/right] R-module over itself, it will be taken that the R-action on R is [left/right] multiplication by R, with a bias towards *left* R-modules if R is non-commutative and which side of the action was not specified. We will also sometimes say "let R have the natural R-module structure" to mean the one induced by the ring operation of R.

2. Let R be a ring with 1 and let $n \in \mathbb{Z}^+$. Then since R is a group, we can define the direct product:
$$R^n = \{(a_1, ..., a_n) | a_i \in R, \text{ for all } i\}$$
We can turn R^n to an R-module by defining addition in R^n component wise, and the action being componentwise:
$$r(a_1, ..., a_n) = (ra_1, ..., ra_n)$$
Such a module R^n is called a *free module* [a]. More on these in section 12.3.2. If $R = \mathbb{R}$, then we have the usual algebraic structure of $\mathbb{R}^n$ common in analysis and linear algebra.

3. If M is an R-module, and S is a sub-ring of R, then M is automatically an S-module (i.e. the set M is also a module under the ring action with S). Since if the module axioms hold for R, i.e. the ring being closed under the ring action and the rules holding, then it will still hold for a sub-ring. In this way $\mathbb{R}$ is an $\mathbb{R}$-module, a $\mathbb{Q}$-module, and a $\mathbb{Z}$-module with the ring action being defined by regular multiplication. Notice though that the $\mathbb{R}$-submodules, $\mathbb{Q}$-submodules, and $\mathbb{Z}$-submodules differ, showing that having a different ring act on $\mathbb{R}$ greatly changes its module structure.

4. Let M be an R-module. If $I \subseteq R$ is a two-sided ideal, is it always true that M is an R/I-module? Under what condition's is it an R/I-module? The following is an exploration of when this is the case: Let I be a 2-sided ideal of R such that $am = 0$, for all $a, \in I, m \in M$. It should be checked that I is indeed an ideal (it is usually denoted $\text{Ann}_R(M)$, and called the *annihilator* of M [over R]). Then we say that M is *annihilated* by I. In this situation, we can make M into an (R/I)-module by defining an action of the quotient ring R/I on M as follows: for each $m \in M$, and $(r + I) \in R/I$, let
$$(r + I)m = rm$$
This is well defined since for any $a \in I$, $(r + a)m = rm + am = rm$. Check that the axioms of a module still hold. Furthermore, The R/I action is a *faithful* action, since if $rx = sx$, then $r = s$. To see this, consider any $x \in M$. Then since M and R are groups, define the group-homomorphism:
$$\varphi : R/I \to M \qquad \varphi(\overline{r}) = \overline{r} \cdot x$$
The function φ is indeed a group-homomorphism, since $\varphi(\overline{r} + \overline{s}) = (\overline{r} + \overline{s}) \cdot x = \overline{r} \cdot x + \overline{s} \cdot x = \varphi(\overline{r}) + \varphi(\overline{s})$. The kernel of this action is all elements such that $\varphi(\overline{r}) = \overline{r} \cdot x = (r + I) \cdot x = 0$. By definition of I,
$$I \cdot x = \{i \cdot x \mid i \in I\} = \{0\}$$
so it follows that if $\overline{r} \in \ker(\varphi)$, $\overline{r} = \overline{0}$, showing that $\ker(\varphi) = \overline{0}$ (Notice the parallell with the Orbit-Stabalizer, theorem 4.3.1)

If, conversely, if $I = \mathrm{Ann}_R(M)$, and we have defined an R/I-action on a module M, can we define an R-action on M? More generally, given two rings R and S acting on M, when can we say that the module structure they induce on M is "comparable"' in some way? One way two types of modules can be equivalent is if they are something called "Morita equivalence", which is settled in chapter 18$\star$[b])

5. What is and isn't a well defined R-module isn't always intuitive. Take for example the group ring $R = \mathbb{Z}[G]$ where $G = \{1, \sigma\}$ is a group of order 2. For the abelian group, take the module $M = \mathbb{Z}e_1 + \mathbb{Z}e_2$, where e_1, e_2 are generators for $\mathbb{R}^2$ (for example, $(1,0)$ and $(0,1)$ generate $\mathbb{R}^2$ as an abelian group), and the product $\mathbb{Z}e_i = \{ze_i | z \in \mathbb{Z}\}$.

   Define the action as:
$$\sigma(e_1) = e_1 + 2e_2 \qquad \sigma(e_2) = -e_2$$
   Then M is a $\mathbb{Z}[G]$-module! The big one to check is $r_1(r_2 m) = (r_1 r_2)m$ – the other conditions follow more easily. This is an example of how the limit on what can be defined as a module isn't easy, since an example like this one exists, checking when a module is defined can be quite useful.

6. Let R be any ring and let $F(A, R)$ be the set of all [set]-functions from A to R:
$$f \in F(A, R) \Rightarrow f : A \to R$$
   Then define the action by R to be
$$r \cdot f = rf \qquad r \cdot f(a) := rf(a) \qquad \forall a \in A$$
   since $f(a) \in R$ Band $r \in R$. Define the group operation to be the new function $f + g$ where
$$(f + g)(a) := f(a) + g(a) \qquad \forall a \in A$$
   Note that these functions are *not* R-module homomorphisms, since the set A has no structure! This R-module is is often dentoed as R^A (to symbolise all functions from A to R)[c].

   Very often in Analysis and Algebraic Geometry, the set A will be all continuous functions on a topological space X, denoted $C(X)$, and the ring R would be the real or complex numbers $\mathbb{R}$ or $\mathbb{C}$. The study of these spaces is done commonly in functional analysis and classical algebraic geometry, usually requiring some analysis knowledge. In this textbook, these types of R-module will come back when discussing free-modules in theorem 12.3.11.

7. Let X be any set and $P(X)$ the powerset.

8. (An example with a ring acting on a module in two different ways, non trivially)

[a]not dissimilar to free-groups we studied in 2.3.3

[b]For the categorically inclined, R-mod and S-mod are Morita equivalent if they are equivalent as categories. Are R-mod and R/I-mod equivalent?

[c]The reason for this notation in fact very interesting and is covered in ref:HERE (in the category theory chapter, categorification of natural numbers, see Emily Reil Category Theory in Context)

When we worked with group-actions, we can take any group G and make it act on any set S. This is not the case with modules (i.e. given some fixed ring R and abelian group M, there does not always exist a ring-homomorphism $\varphi : R \to \mathrm{End}_{\mathbf{Ab}}(M)$):

Example 12.1.8: Impossible Module

Consider $M = \mathbb{Z}$ and $R = \mathbb{Q}$. Is there always a $\mathbb{Q}$ action over $\mathbb{Z}$? If it was, then:

$$\frac{1}{2} \cdot 1 = z$$

must be an element of $\mathbb{Z}$ ($z \in \mathbb{Z}$). Then:

$$z + z = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = (\frac{1}{2} + \frac{1}{2}) \cdot 1 = 1 \cdot 1 = 1$$

as a direct consequence of the module axioms. However, this implies

$$z + z = 1$$

which is not possible for any $z \in \mathbb{Z}$ – and thus a contradiction! This is also an example where you can have a $\mathbb{Z} \subseteq \mathbb{Q}$ and a defined $\mathbb{Z}$-module, but there is no extension of M to a $\mathbb{Q}$-module for which the previous module structure is embedded within it (opposite to how if we had a $\mathbb{Q}$-module, it would automatically be a $\mathbb{Z}$-module)[a]

[a]There is in fact still a notion of extending a module along a ring map, however the original module need not embed in the result; see section 15.3, and corollary 15.3.8 for exactly when it does

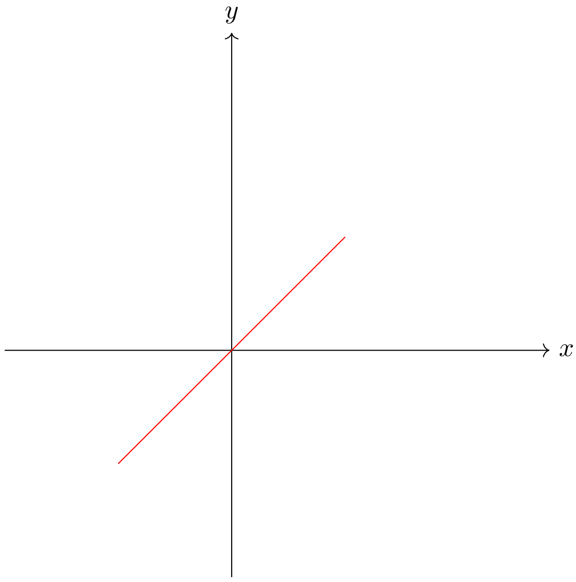
12.1.1 Vector-spaces

If the ring is a field, we give a special name to such a module:

Definition 12.1.9: Vector Space

Let M be an R-module. If $R = k$ is a field, then we call M a *vector-space* over k. An arbitrary field is often denoted by k or $\mathbb{F}$.

Due to historical reasons $\mathbb{F}$-modules homomorphisms are called *linear transformations*. The motivation for this will be more clearly seen in chapter 13, but the gist is that when $\mathbb{F} = \mathbb{R}$, the geometric intuition of the images and kernels of linear transformations will all be hyper-planes in the domain and codomain. A simple example would be $f : \mathbb{R} \to \mathbb{R} \times \mathbb{R}$ with $f(x) = (x, x)$. The resulting hyper-plane (simply called a line since it's 1 dimensional), looks like so:



If you are comfortable with linear algebra, then you are probably familiar of how simple the structure of vector-spaces are (if not, you may skip this remark: all that will be discussed here is covered in chapter 13). Essentially, a vector space V over k can be completely determined by it's dimension. For modules over a general ring R, this is sometimes far from the case: sometimes R will not even embed itself in V! However, there are ways in which we can systematically loosen the conditions on the ring k acting on V and get similar, but weaker, results. Thus, if it is intuitive to think of modules as generlizations of vector-spaces, which historically is how many mathematicians thought of modules, then these chapters on module theory can be thought of as what happens if the action on a vector-space no longer forces to give a basis.

Example 12.1.10: Vector Spaces

1. The most common example is when $R = \mathbb{R}$ and $M = R^n = \mathbb{R}^n$. This vector-space is commonly reffered to as *euclidean space* due to the classical geometrical properties that can be defined on it(ex. a notion of distance, length, angle, translation and rotation)

2. If I is a maximal ideal in a commutative ring, and $IM = 0$, then M is a vector space over R/I (since R/I is a field)

3. Take $\mathbb{F}[x]$ to be a polynomial ring over a field $\mathbb{F}$, an let $f(x)$ be any polynomial of degree $n \geq 1$. Then $V = \mathbb{F}[x]/(f(x))$ is another polynomial ring. Let $\mathbb{F}$ act on V by $a \cdot \overline{f} = \overline{af}$. This will be a vector-space. If $\mathbb{F}$ is a finite field of order p, then we can show that V has p^n.

4. (If you know some calculus). Consider $\mathbb{R}[x]$. Then it is a vector-space over $\mathbb{R}$ ($\mathbb{R}[x]$ is an abelian group, and the action is simply the multiplication in $\mathbb{R}[x]$, since $\mathbb{R} \subseteq \mathbb{R}[x]$). Recall that the derivative has the following two properties:

$$\frac{d}{dx}(p(x) + q(x)) = \frac{d}{dx}p(x) + \frac{d}{dx}q(x)$$
$$\frac{d}{dx}(cf(x)) = c\left(\frac{d}{dx}f(x)\right)$$

This shows that the derivative is in fact a linear transformation (i.e. $\mathbb{R}$-module homomorphism) from $\mathbb{R}[x]$ to itself![a]

5. If R is a ring where the only two-sided ieals are (0) and R, then R is called a *simple ring* (similarly to how simple groups are groups whose only normal subgroup is $\{0\}$ and the entire group). Then $Z(R)$ (the center of the ring) *has* to be a field (can you show this?). If $k = Z(R)$, then any simple ring is a vector-space over its center.

[a] For those who know some Linear algebra and Fourier Analysis, the "diagonalization" of the derivative is the Fourier transform of a function! For details, see the Gelfand Transformation of unbounded operators

If $R = \mathbb{F}$ is a field, we give a special name to sub-modules:

Definition 12.1.11: Subspace

If $W \subseteq V$ is a submodule of a vector space, then we call it a *subspace*

12.1.12 Remark (If you know linear algebra) When determining where a linear transformation maps, it is sufficient to know where the basis maps to. General R-modules need not have a basis well-defined on them. However, *cyclic* modules will always have a basis, since $M = R \cdot x$ for some $x \in M$. Thus, R is always a cyclic R-module over itself since $R \cdot 1 = R$. General R-modules need not have a basis, and so R as an R-module a particularly nice module.
As a consequence of this, for any R-module homomorphism: $\varphi : R \to N$, it suffices to know where $\varphi(1)$ maps to. In general, it is much harder to determine a map $\varphi : M \to N$ given where only certain elements are mapped to.

Exercise 12.1.1

1. Let $\langle m \rangle = M$ be an R-module generated by a single element (i.e. if $x \in M$, $x = a_i m$ for some $a_i \in R$)[2]. Show that $\mathrm{Ann}(M) := \{m \in M \mid \exists r \in R \text{ such that } rm = 0\}$ forms an ideal of R.
2. Let M be as in the previous question. Show that $M/\mathrm{Ann}(M) \cong HERE$ (I.e. trying to show how we can study ideals in the context of modules, and how ideals are studied as R/I R-modules
3. (practice with categories). Let A be a set and $F(A)$ be the group freely generated by A (definition 2.3.29).
    1. Prove that the collection of all vector-spaces along with linear transformations form a category. Label it k-**Vect**.
    2. Show that $F(A)$ can have a vector-space structure over k put on it such that $F : \mathbf{Set} \to k\text{-}\mathbf{Vect}$ is a functor. (Hint: See the functor defined in section 2.3.5)
4. I feel like I should add this somehow link to Ab-enriched categories
5. What is the initial object of Mod_R (if any)?

[2] We'll explore module generation in section 12.3

12.1.2 Algebras

Recall that if S is a ring, then it is always an abelian group, and there is always a unique homomorphism $\mathbb{Z} \to S$. Using this homomorphism, we can define a $\mathbb{Z}$-action given by $z \cdot s = \varphi(z)s$, and so every ring is automatically a $\mathbb{Z}$-module. More generally, any ring map $R \to S$ where R is commutative produces an R-module structure on the ring S that is compatible with the multiplication of S. To encapsulate this extra structure, we shall define a new term known as an *R-algebra* that represents a ring that also has a ring action on it.

The definition is simple if constrained to commutative rings, but there are many important cases where we must work with a non-commutative codomain (particularly in Part X), and so the above intuition must be made more precise. Given any ring homomorphism $\varphi : R \to S$, we can always define an R-module structure on S by defining

$$r \cdot s = \varphi(r)s$$

that is, the action is multiplication in S. The axioms of an R-module is immediate given the axioms of a ring and that φ is a homomorphism. Since we can multiply, we run into situations like the following: If $s_1 = r_1 \cdot x$ and $s_2 = r_2 \cdot y$, then:

$$s_1 s_2 = (r_1 \cdot x)(r_2 \cdot y)$$

Such an element is not necessarily equal to $(r_1 r_2) \cdot xy$ (similarly to how with rings, if the underlying set is not abelian, distributivity would not hold). This would be desirable, as in we would want $\varphi : R \to S$ to be a ring homomorphism that was also compatible with the R-module structure in the following way:

$$(r_1 \cdot x)(r_2 \cdot y) = (r_1 r_2) \cdot xy$$

As a commutative diagram:

$$\begin{tikzcd}
(R \times S) \times (R \times S) \arrow[r, "a \times a"] \arrow[d, "\ell"'] & S \times S \arrow[d, "m"] \\
R \times S \arrow[r, "a"'] & S
\end{tikzcd}$$

where the left map is

$$\ell\big((r_1, x), (r_2, y)\big) = (r_1 r_2,\ xy), \qquad \ell = (\mu_R \times m) \circ \sigma,$$

with σ the shuffle $\big((r_1, x), (r_2, y)\big) \mapsto \big((r_1, r_2), (x, y)\big)$.

Getting back to the homomorphism $\varphi : R \to S$ and the R-module structure on S, if we further require that R be *commutative* and that $\varphi(R) \subseteq Z(S)$ (the multiplicative center), then the left and right R-module structure will coincide (i.e. left and right multiplication produce the same module structure), and the ring operation of S will be compatible with that of R, that is

$$(r_1 r_2) \cdot (s_1 s_2) = \varphi(r_1 r_2)(s_1 s_2) = \varphi(r_1)\varphi(r_2)(s_1 s_2) = \varphi(r_1) s_1 \varphi(r_2) s_2 = (r_1 \cdot s_1)(r_2 \cdot s_2)$$

Boxing these properties we define an algebra:

Definition 12.1.13: R-algebra

Let S be a ring, R a commutative ring. Then S is called an (associative) R-algebra if S is also an R-module satisfying:

$$r \cdot (xy) = (r \cdot x)y = x(r \cdot y)$$

Equivalently, S is an R-algebra if there exists ring homomorphism $\varphi : R \to S$ where $\varphi(R) \subseteq Z(S)$. Furthermore:

1. If S is a division ring, then the R-algebra S is called a *division algebra*
2. If S is commutative, then the R-algebras S is called a *commutative algebra*

Note that S has has a natural R-module structure by forgetting the multiplication so that the action

$$r \cdot a[= a \cdot r] = \varphi(r)a$$

is the R-module structure. Unless otherwise stated, this will be the assumed R-module structure of an R-algebra S.

Algebra homomorphisms are defined similarly to how they've been defined before (can you figure it out without looking at the following definition?)

Definition 12.1.14: R-Algebra Homomorphism and Isomorphism

If A and B are R-algebra, then $\varphi : R \to S$ is an R-algebra homomorphism if φ is a ring-homomorphism and

$$r \cdot \varphi(x) = \varphi(r \cdot x)$$

If there exists an inverse function φ that is also an R-algebra homomorphism, then A and B are *algebra isomorphic*, or just *isomorphic* to each other, and we write $A \cong B$ if the context permits.

With this definition, the collection of R-algebra's along with R-algebra homomorphism forms the category R-**Alg**.

12.1.15 For the Categorically Inclined: Monoidal Object in R-module
Another way of saying that an R-algebra adds the notion of multiplication to an R-module in a "consistent" way is by noticing that R-algebra are the monoidal objects in the category of R-mod. Since all rings are $\mathbb{Z}$-modules, the discussion in section 9.7⋆ shows that rings are the monoidal objects in $\mathbb{Z}$-Mod (Hence, the category $\mathbb{Z}$-Alg is the category **Ring**).

$$\begin{tikzcd}
(R\otimes S)\otimes(R\otimes S) \arrow[r, "a\otimes a"] \arrow[d, "\cong"', "1_R\otimes\tau_{S,R}\otimes 1_S"] & S\otimes S \arrow[dd, "m"] \\
(R\otimes R)\otimes(S\otimes S) \arrow[d, "\mu_R\otimes m"'] & \\
R\otimes S \arrow[r, "a"'] & S
\end{tikzcd}$$

The word "associative" was put in parenthesis in the definition of an algebra since there is a notion of an algebra that is not associative: some author's say the object in definition 12.1.13 is an *associative algebra*, and define algebras more generally as follows: Take an R-module M, and equip it with a binary operation such that the binary operation is left and right distributive (for all $a, b, c \in M$, $a \cdot (b + c) = a \cdot b + a \cdot c$, an $(b + c) \cdot a = b \cdot a + c \cdot a$), and that it is compatible with scalars (for all $a, b \in R$ and $x, y \in M$, $(ab) \cdot (xy) = (a \cdot x)(b \cdot y)$), then M is called an R-algebra. If the binary operation is associative and an identity exists, then it would be a ring operation it will be called an associative algebra.

There is a good reason adding the prefix "associative" in the greater context of mathematics: a large domain of study in mathematics is the study of *lie groups* and *lie algebras*, which focus on understanding the structure of smooth manifolds and representations of algebraic groups. Notably, Lie algebras are *not associative*, satisfying instead a relation called the *Jacobi identity*. However, since they are non-associative, they are not *rings*. They are thus quite different objects from R-algebras as we've defined them, and so don't encompass our current discussion. Their study take up many books (references to some are HERE) which the reader with a good background in linear algebra has the algebraic background to understand[3]. For the curious, we will briefly discuss (representations of) Lie Algebra in chapter 54.

From now on, when we say algebra or R-algebra, we will mean an associative algebra as defined in definition 12.1.13.

If $R = k$ is a field, then using the terminology of vector-spaces, a k-algebra would be defining a way of multiplying vectors. It is non-trivial to always define a multiplication structure on an R-module (I suggest you try it on $\mathbb{R}^n$!). Even harder is to ask whether there is a "natural" way of creating a multiplication structure, or more broadly to make an R-module M big enough so that we can define an R-algebra in a "natural" way[4]. We will in fact define such a notion in section 16.3.

[3] Though a good basis in differential geometry would go a long way for intuition
[4] natural meaning it satisfies the appropriate universal property

Example 12.1.16: R-Algebras

1. If R has positive characteristic p ($\#(\text{im}\mathbb{Z}) = p < \infty$), then it is more natural to define a $\mathbb{Z}/p\mathbb{Z}$-algebra, namely since as a $\mathbb{Z}$-algebra the kernel of the map $\mathbb{Z} \to R$ would be $p\mathbb{Z}$, and hence by the 1st isomorphism theorem it homomorphism commutes with $\mathbb{Z}/p\mathbb{Z} \to R$.

2. If S is commutative, then any subring of $R \subseteq S$ will define R-algebra on S (take $\iota : R \hookrightarrow S$). If S is not commutative, any subring of the center will define an R-algebra.

3. Let $R[x_1, x_2, ..., x_n]$ be the polynomial ring over several variables. Then it is an R-algebra through $r \cdot p(x) = rp(x)$.

4. Let $k[G]$ be a group ring over the field k. Then $k[G]$ is naturally a k-algebra.

5. If D is a divisor ring, then $Z(D)$ is always a *field*. If $k = Z(D)$, then any division ring D has a k-algebra structure on it, which also implies it is a vector-space over k.

6. Let $R = k^n$ for some field k. Then R is a k-algebra, with multiplication done point-wise.

7. Let $R = M_n(k)$ over a field k. Then R is a k-algebra. More generally, If A is a k-algebra, then $M_n(A)$ is a k-algebra.

   Compare this example to the previous one, which we'll label $S = k^{n^2}$. As vector-spaces over k, $R \cong_k S$. However, as k-algebras, $R \ncong_{k\textbf{-Alg}} S$, since the multiplication structure is different between the two (one has nilpotent elements, the other does not).

8. If A, B are k-algebra, so is $A \times B$. By the previous part, that implies

   $$M_n(A) \times M_n(B)$$

   is also a k-algebra.

9. At the heart of real analysis is the $\mathbb{R}$-algebra $\mathbb{R}$, and at the heart of complex analysis is the $\mathbb{C}$-algebra $\mathbb{C}$. Note that $\mathbb{C}$ is also a $\mathbb{R}$-algebra.

10. Let $R = \mathbb{R}^2$. Then we know that $\mathbb{R}^2$ has point-wise multiplication, but we can also define the following:

    define complex number multiplication

    then this also forms an $\mathbb{R}$-algebra. In fact, this is $\mathbb{R}$-algebra isomorphic to $\mathbb{C}$ (over $\mathbb{R}$)!

11. (If you know linear algebra) (example with defining multiplication in vector-spaces, Also thought the "Structure Coefficients" section here was really cool! https://en.wikipedia.org/wiki/Algebra_over_a_field)

12. One might ask themselves how many $\mathbb{R}$-algebra's exist (i.e. classify all $\mathbb{R}$-algebras). A little more generally, one might ask how many finite-dimensional associatvive divison algebras there are, that is, algebra on which you can has division and asociativity, and the notion of finite-dimension is well-defined. Only $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ exist, with dimension 1, 2, and 4! This is called *Frobenius' Theorem*. In this way, the real, compelx, and quaternions are kind of special. [a]

There are a large number of areas that algebras appear in, to name a notable few:

1. (is the cross product an R-algebra) Let define a $\mathbb{R}^3$-algebra with $(a, b, c) \cdot (x, y, z) =$
2. commutative algebras in algebraic geometry
3. Banach-Algebra's in real analysis
4. C^*-algebra or $*$-algebras more generally in functional analysis
5. Though Lie algebra's are not [associative] algebras, there are some interesting intersections between the two TBD

[a] https://math.stackexchange.com/questions/368456/frobenius-from-hurwitzs-theorem

Theorem 12.1.17: Universal Property of Free R-algebras

$R[A]$ is a free commutative R-algebra over a [finite set] A

Notice how this is similar to the universal property of commutative polynomial rings. In fact, that is a special case of this theorem by taking $R = \mathbb{Z}$

***Proof*:**
Let S be any commutative R-algebra and $\alpha : A \to S$ be any set-function. Since S is an commutative R-algebra, it is accompanied by a function $f : R \to S$ defining its multiplication.

Since $A = \{a_1, a_2, ..., a_n\}$ is finite, $R[A] = R[x_1, x_2, ..., x_n]$. Similarly to how the universal property of polynomial rings was proven (theorem 9.2.4), we will extend $f : R \to S$ to $f_1 : R[x_1] \to S$ by mapping $f_1(x_1) = \alpha(a_1)$ and extending linearly. f_1 is still an R-algebra homomorphism since $f_1(n_0 + n_1x + n_2x^2 + \cdots + n_kx^k) = f(n_0) + f(n_1)\alpha(a_1) + f(n_2)\alpha(a_1)^2 + \cdots + f(n_k)\alpha(a_1)^k$, and Similarly for multiplying two polynomials (using the binomial theorem and both the additive and multiplicative linearity of f_1). We can again extend f_1 to f_2 by defining $f_2(a_2) = \alpha(a_2)$ so that $f_2 : (R[x_1])[x_2] \to S$ (usually written as $f_2 : R[x_1, x_2] \to S$) is an R-algebra homomorphism. Continuing inductively will get us an R-algebra homomorphism $F : R[x_1, x_2, ..., x_n] \to S$. Thus, picking $\iota : A \to R[x_1, x_2, ..., x_n]$ will show that $\alpha : A \to S$ clearly lifts uniquely to $F : R[A] \to S$, as we sought to show.

As a consequence of this, all R-algebras are quotient of polynomial rings!! That is, if S is a finitely generearted S algebra

$$R[x_1, x_2, ..., x_n] \twoheadrightarrow S$$

12.1.3 Abelian Groups as Modules

Some entire domains of study are done on particular modules; we have actually already studied a particular collection modules extensively! The most immediate example is that all rings are R-modules over themselves with the action defined by:

$$r \cdot x = rx$$

Then submodules are precisely the ideals of R, and so the study of all R-modules includes the study of the ring R.

In this section, we show another important classes of R-modules: let $R = \mathbb{Z}$, and let $M = A$ be *any* abelian group (finite or infinite), and write the operation of A as $+$. Make A into a $\mathbb{Z}$-module by defining the ring action as :

$$na = \begin{cases} a + a + \cdots + a \ (n \text{ times}) & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ -a - a - \cdots - a \ (n \text{ times}) & \text{if } n < 0 \end{cases}$$

Then M is a $\mathbb{Z}$-module. In fact, this ring action is the only possible ring action on M (i.e. abelian groups):

Proposition 12.1.18: Abelian Group iff $\mathbb{Z}$-module

Every abelian group is a $\mathbb{Z}$-module, and every $\mathbb{Z}$-module is an abelian group

***Proof*:**
for $\Rightarrow$, we can always define the previous ring-action and so it's always a $\mathbb{Z}$-module

For $\Leftarrow$, it is simply a consequence of the existence of 1 and the distributivity property, i.e., the ring action $\mathbb{Z}$ performs on abelian groups is "stuck" since we must play out the consequence of the axioms.

By definition

$$n \cdot 1 = n$$

and therefore

$$n \cdot 2 = n \cdot (1 + 1) = n \cdot 1 + n \cdot 1 = n + n$$

and continuity inductively

$$n \cdot k = n \cdot (1 + \cdots + 1) = n + \cdots + n$$

k times. This result is exactly the ring action we defined earlier. And so, forced by the axioms of groups and modules, it must be that every $\mathbb{Z}$ action on an abelian foms this particular $\mathbb{Z}$-module.

This proposition has a very intersting consequence: for every R-module, $R \neq \mathbb{Z}$, then it always has at least one other module structure, namely the natural $\mathbb{Z}$-module structure! An intuitive way to remember this fact is that numbers as we have learnt them earlier in our life usually have a $\mathbb{Z}$-module structure ($\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$). The natrual numbers form an abelian monoid, so this intuition doesn't directly apply to $\mathbb{N}$, but they indirectly apply since the same proof works for ring-actions on monoids (TBD?).

This example can also make us think of the class of all abelian groups as a subclass of modules. Abelian groups are also a subclass of groups. However, not every group is a module (and vice versa). The following diagram I'm about to draw shouldn't be taken as a commutative diagram, but as an "analogy":

$$\begin{tikzcd} \mathbf{Grp} & & R\text{-}\mathbf{Mod} \\ & \mathbf{Ab} \arrow[ul] \arrow[ur] & \end{tikzcd}$$

So in a sense, we are "switching gears" when studying modules. We are taking a different approach to the study of algebraic structures as compared to studying groups.

Exercise 12.1.2

1. Recall that for any $\alpha \in R$ for a ring R, $f_\alpha : R[x] \to R$ mapping $p(x) \mapsto p(a)$ is a ring-homomorphism. Is it an R-module homomorphism?

12.2 Homomorphism and Quotients of Modules

In this section we explore R-module homomorphisms and how the image of R-module homomorphisms can be represented by a particular quotient of the domain (as we've seen with groups and rings). As R-module homomorphsism are explored, they should be constrasted to rings in that they will have many nicer properties. They are in fact comparable to abelian groups; every subgroup of an abelian group is normal, and so too every submodule of a R-module will be "normal".

R-Module Homomorphisms

Definition 12.2.1: R-module homomorphism Terminology

Let R be a ring, M and N be R-module, and $\varphi : M \to N$ be a function. Then:

1. φ is called an *R-module homomorphism* if:

   (a) $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$

   (b) $\varphi(r \cdot x) = r\varphi(x)$, for all $r \in R$, $m \in M$

   If furthermore there exists a φ^{-1} such that $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \text{id}$ and φ^{-1} is an R-module homomorphism, then φ is called an *R-module isomorphism* and we write $M \cong_{\mathbf{R\text{-}Mod}} N$, or $M \cong N$ if it is unambiguous

   If $R = k$ is a field, then k-module homomorphisms are called *linear maps*, *linear functions*, or *linear transfomration*, and k-module isomorphisms are called *linear isomorphisms* (due to historical precedence)

2. Denote

   (a) $\ker(\varphi) = \{m \in M \mid \varphi(m) = 0\}$

   (b) $\text{im}(\varphi) = \varphi(M) = \{n \in N | \varphi(m) = n\}$

3. Define $\text{Hom}_R(M, N)$ be the set of all R-module homomorphisms from M into N, $\text{End}_R(M)$ the set of all R-module homomorphism from M to itself, and $\text{Aut}_R(M)$ the set of all all R-module isomorphisms from M to itself. A function $\varphi \in \text{End}_R(M)$ is called an *R-module enormophism*, and a function $\varphi \in \text{Aut}_R(M)$ is called an *R-module automorphism*.

Notice how it's key that the two structures be R-modules; If M is an R-module and N is an S-module, then $\varphi : M \to N$ is not an R-module or S-module homomrophsim unless either R has a compatible S-module structure of N has a compatible R-module structure that is preserved under φ (this is similar to how G-equivariant functions are betwen two G-sets)

Many of the properties demonstrated of group and ring homomorphisms apply for R-modules

Lemma 12.2.2: R-Module Homomorphism Equivalence

Let $\varphi : M \to N$ be a function. Then φ is an R-module homomorphism if and only if for every $r \in R$ and $x, y \in M$,

$$\varphi(x + ry) = \varphi(x) + r\varphi(y)$$

***Proof*:**
exercise

Lemma 12.2.3: R-Module Isomorphism Equivalence

Let $\varphi : M \to N$ be an R-module homomorphism. Then φ is an R-module isomorphism if and only if φ is an R-module homomorphism and bijective

***Proof*:**
Imitate the proof in proposition ref:HERE

Lemma 12.2.4: Composition Of R-Module Homomorphism/Isomorphism

Let M, N, L be R-modules. If $\varphi \in \text{Hom}_R(L, M)$ and $\psi \in \text{Hom}_R(M, N)$ then $\psi \circ \varphi \in \text{Hom}_R(L, N)$

***Proof*:**
exercise

Lemma 12.2.5: R-module Automorphism Group

Let $\text{Aut}_R(M)$ be the such set of R-module automorphism on M. Then $(\text{Aut}_R(M), \circ)$ forms a group under composition.

***Proof*:**
exercise

Example 12.2.6: Module Homomorphisms

1. if a ring is an R-module over itself with the natural ring action, then R-module homomorphisms need not be ring homomorphisms (even from R to itself), and ring homomorphisms need not be R-module homomorphisms. For example, $\mathbb{Z}$ act on itself in the natural way,

and let $\varphi(x) = 2x$. φ is not a ring homomorphism [a], but φ is an R-module homomorphism. For the example going the other way, $F[x]$ act on itself in the natural way, and let $\varphi : f(x) \mapsto f(x^2)$. Then φ is not an $F[x]$-module homomorphism [b], but it is a ring homomorphism.

2. Let R be a ring and $M = R^n$ be an R-module with the natural ring action. Then the projective map is naturally an R-module homomorphism.

3. If we take $\mathbb{R}$ to be an $\mathbb{R}$ module over $\mathbb{R}$, then $\mathbb{R}$ has no non-trivial submodules (since $\mathbb{R}$ is a field, it's only ideals are 1 and itself). We call such a module a simple or an irreducible module. Thus, the quotient structure of $\mathbb{R}$ as an $\mathbb{R}$-module is trivial. However, if we consider $\mathbb{R}$ as a $\mathbb{Q}$-module (which we can). Then $\mathbb{Q}$ is a submodule of $\mathbb{R}$ (since $\mathbb{Q}$ is clearly closed under the ring action of $\mathbb{Q}$)! $\mathbb{R}$ as a $\mathbb{Q}$-module will be an infinite-dimensional vector-space [c]. Therefore, it becomes meaningful in this situation to look at $\mathbb{R}/\mathbb{Q}$ under the $\mathbb{Q}$ action!!

4. It should be done as an exercise that if $\varphi : M \to N$ is an R-module homomorphism, then the image and kernel are both R-modules.

5. (p.346)

[a] $\varphi(xy) = 2xy \neq 2x2y = \varphi(x)\varphi(y)$
[b] $x^2 = \varphi(x) = \varphi(x \cdot 1) = x\varphi(1) = x$, a contradiction
[c] In fact, it will even be *uncountably* infinite-dimensional

Recall in remark HomGrpIfAbRmk that $(\mathrm{Hom}_{\mathbf{Set}}(A, G), +)$ (where $+$ is defined pointwise: $f + g(x) = f(x) + g(x)$) can be given a group structure if G is abelian, and is not a group if G was not abelian (i.e. it not important what the domain was for the structure of the hom-set). Since every R-module is always over an abelian group, $\mathrm{Hom}_R(M, N)$ is *always* an abelian group. In the following proposition we further the structure of $\mathrm{Hom}_R(M, N)$

Proposition 12.2.7: Properties of $\mathrm{Hom}_R(M, N)$

Let M, N and L be R-module. Then

1. $(\mathrm{Hom}_R(M, N), +)$ is an abelian group with $+$ being defined pointwise:

$$(\varphi + \psi)(m) = \varphi(m) + \psi(m) \qquad \forall \varphi, \psi \in \mathrm{Hom}_R(M, N),\ \forall m \in M$$

In particular $\varphi + \psi \in \mathrm{Hom}_R(M, N)$

2. If R is a commutative ring, then for $r \in R$ define $r\varphi$ by

$$(r\varphi)(m) = r(\varphi(m)) \qquad \text{for all } m \in M$$

Then $r\varphi \in \mathrm{Hom}_R(M, N)$, and with this action $\mathrm{Hom}_R(M, N)$ is an R-module.

3. With addition defined as above, $(\mathrm{End}_R(M), +, \circ)$ is a ring. When R is commutative, $\mathrm{End}_R(M)$ is an R-algebra.

***Proof*:**
The first statement was proven above remark HomGrpIfAbRmk.

For an R-action to be defined on $\mathrm{Hom}_R(M, N)$ we would need the following to be equal:

$$(rs\varphi)(m) = rs \cdot \varphi(m)$$
$$(r(s\varphi))(m) = r \cdot (s\varphi)(m) = sr \cdot \varphi(m)$$

showing R must be commutative. The other axioms follow quickly

Finally, $(\mathrm{End}_R(M), +, \circ)$ follows the ring axioms. If R is commutative then for ring homomorphism $f : R \to \mathrm{End}_R(M)$, since f is commutative, $f(R)$ is always abelian, and hence $f(R) \subseteq Z(\mathrm{End}_R(M))$, and so $\mathrm{End}_R(M)$ is always an R-algebra, completing the proof.

Note that $(\mathrm{Aut}_R(M), +)$ is *not* a group structure, since the identity *does not exist* (the zero homomorphism is not an automorphism)![5]

Like with groups and rings, we want to know what conditions are needed on a sub-module so that:

$$(x + N) + (y + N) = (x + y) + N \qquad r(x + N) = (rx) + N$$

are both well defined. Since Modules are already abelian groups, than M/N is already well-defined as an abelian group! What's left is to see what condition's are needed so that a submodule N can be used to define an R-module structure on the abelian group M/N. Like abelian groups, every sub-module is "normal"!

Proposition 12.2.8: sub-modules are "normal" sub-modules

Let R be a ring and let M be an R-module and let N be a submodule of M. The (additive, abeliean) quotient group M/N can be made into an R-module by definin an action of elements of R by

$$r(x + N) = (rx) + N, \qquad \text{for all } r \in R,\ x + N \in M/N$$

The natural projection $\pi : M \to M/N$ defined by $\pi(x) = x+N$ is an R-module homomorphism with kernel N.

***Proof*:**
First, M/N is already an abelian quotient group. We want the ring action to be well defined, that is

$$x + N = y + N$$

which equivalently means that $x - y \in N$. Since N is a submodule, $r(x - y) \in N$, thus

$$rx + N = ry + N$$

therefore, the R-action on M naturally induces an action on M/N. The rest of the module axioms can now e easily verified.

Next, we must show the implication goes the other way, that is, if the elements of the quotient groups are formed by the cosets $x + N$, then N is the kernel of some R-module homomorphis (i.e. the natural projection). The natural projection map π described above is already the natural

[5] semi-groups are groups without an identity. Thus, $(\mathrm{Aut}_R(M)+)$ is a natural example of a semi-group(!), with a compatible group structure with $\circ$

projection of the abelian group M into the abelian group M/N. It only remains to show that it is a R-module homomorphism:

$$\begin{aligned}\pi(rm) &= rm + N \\ &= r(m+N) && \text{(by definition of the action of } R \text{ on } M/N\text{)} \\ &= r\pi(m)\end{aligned}$$

Thus, the natural projection an R-module homomorphism, completing the proof.

As with groups and rings, R-modules also have the isomorphism theorems. Since R-modules are already abelian groups, half the proof is already done. What is left to check would be if the action still preserved under the isomorphisms. To define the 2nd isomorphism theorem, we define what it means to take sum of two modules:

Definition 12.2.9: Sum Of Modules

Let A, B be submodules of an R-module M. Then the *sum* of A and B is the set

$$A + B = \{a + b | a \in A, b \in B\}$$

As before, $A + B$ is the smallest module containing A and B (as you sould check)

Theorem 12.2.10: Isomorphism Theorems For Modules

1. Let M, N be R-modules and let $\varphi : M \to N$ be an R-module homomorphism. Then $\ker(\varphi)$ is a submodule of M and

$$M/\ker(\varphi) \cong \varphi(M)$$

2. Let A, B be submodules of the R-module M. Then

$$\frac{A+B}{B} \cong \frac{A}{A \cap B}$$

3. Let M be an R-module, and let A and B be submodules of M, with $A \subseteq B$. Then

$$(M/A)/(B/A) \cong M/B$$

4. Let N be a submodule of the R-module M. There is a bijection beetween the submodules of M which contain N and the submodules of M/N. The correspondence is give nby $A \leftrightarrow A/n$, an for all $A \supseteq N$. The correspondence commutes with the processes of taking sums and intersection (i.e., is a lattice isomorphism between the lattice of submodules of M/N an dthe lattice of submodules of M which contain N

***Proof*:**
Very similar to previous isomorphism proofs.

12.2.1 Extensions and Exactness

We conclude this section with a quick word on extensions and where they will show up. Similarly to groups (and rings), we may wonder under what conditions is a module M an extension of two modules A and B, that is $M \cong B/A$, or in terms of a short exact sequence:

$$0 \to A \to M \to B \to 0$$

Section 3.7 showed that mathematicians managed to find the building blocks of finite group extensions, known as simple groups, and classify all finite simple groups, and section 5.4 showed that semi-direct products can be represented by split exact sequences. The classification of simple modules depends on the ring that's acting, R, and is much simpler since the underlying group is abelian. Similar to groups, there exist classification project for R-modules for some particular rings R, at least four of which will be covered in this book[6].

We also endeavour to study R-modules in which the extension property works nicely. This is further explored in Part X, chapter 51. Since all modules are abelian groups, the semi-direct product is simply the direct product. However, split exact sequences are still an important concept since it gives us a way of recognizing direct products using homomorphisms. A particularly interesting collection of modules are those for which every short exact sequence is a split exact sequence, or phrased differently that every short exact sequence splits. This endeavour is taken up in chapter 17⋆, and explored in full length in Part XI.

The vocabulary for a sequence of R-module homomorphisms that represent extensions is quite important, and so we define the term here for future use:

Definition 12.2.11: Exact Chain Complex

Let $M_0, M_1, M_2, ..., M_n, ...$ is a collection of R-modules $d_i : M_i \to M_{i-1}$ be a collection of R-module homomorphisms where

$$\cdots \xleftarrow{d_0} M_0 \xleftarrow{d_1} M_1 \xleftarrow{d_2} \cdots \xleftarrow{d_n} M_n \xleftarrow{d_{n+1}} \cdots$$

and $\ker(d_i) = \operatorname{im}(d_{i-1})$. Then the tuple $(M_i, d_i)_{i=-\infty}^{\infty}$ is called an *exact chain complex*, *exact chain*, or *exact sequence*. Often, we abbreviate and write $(M_\bullet, d_\bullet)$. If the arrows are pointing the opposite direction and $\ker(d_i) = \operatorname{im}(d_{i+1})$ then the resulting tuple $(M_i, d_i)_{i=-\infty}^{\infty}$ is called an *exact cochain*. If the chain has only finite length, (i.e. only finitely many of the modules are non-zero), then the resulting exact (co)chain is called a *finite exact (co)chain*.

A single homomorphism already sits inside such a chain, once one more module is named.

Definition 12.2.12: Cokernel

Let $\varphi\colon M \to N$ be a homomorphism of R-modules. The *cokernel* of φ is the quotient module

$$\operatorname{coker}(\varphi) := N/\operatorname{im}(\varphi)$$

The kernel measures how far φ is from being injective and the cokernel how far it is from being surjective: φ is injective exactly when $\ker(\varphi) = 0$, and surjective exactly when $\operatorname{coker}(\varphi) = 0$. Both

[6] When R is a Field, a Principal Ideal Domain, a Dedekind Domain, and Semi-Simple

statements are packaged in the one exact sequence

$$0 \to \ker(\varphi) \xhookrightarrow{\iota} M \xrightarrow{\varphi} N \xrightarrow{\pi} \operatorname{coker}(\varphi) \to 0$$

which is exact at $\ker(\varphi)$ because ι is injective, at M because $\ker(\varphi)$ is by definition $\operatorname{im}(\iota)$ and is what φ kills, at N because $\ker(\pi) = \operatorname{im}(\varphi)$ is exactly what the quotient divides out, and at $\operatorname{coker}(\varphi)$ because π is surjective.

(examples and some explanation here, important because it comes back in at least the next two chaptesr)

Exercise 12.2.1

1. Let M be an R-module. Prove that $\operatorname{Hom}_R(R, M) \cong M$
2. Let R^{op} be the opposite ring of R (see exercise ref:HERE), and let $\operatorname{End}_R(R)$ be all R-module homomorphisms from R to itself. Considering $(\operatorname{End}_R(R), +, \circ)$ as a ring show that $R^{\text{op}} \cong_{\mathbf{Ring}} \operatorname{End}_R(R)$ [Hint ; think of Yoneda's lemma]

12.3 Generation of Modules

We now extend the notion of generating algebraic structures to subsets of a module. We will start by generalizing the results of section 2.3.2: we take a subset of an R-module, and permit all possible combinations and R-actions of those elements to form an R-module, or equivaelntly we take the intersection of all R-submodules containing the elements of the set.

Then, as with groups and rings, the easiest way of creating a larger module from smaller ones is to take the direct sum or direct product. Due to their simplicity, it is worth taking the time to find out under what condition's can a module be decomposed into direct products or sums (just like for groups in section 5.2). We then take the approaching of trying to define the "most free" module we can put on an arbitrary set, similar to what we've done in section 2.3.4, that is we are looking to find the universal property of free-modules. As compared to groups, we will spend more time looking at minimal generating sets. Recall that for groups, it is possible that minimal generating sets can differ in cardinatliy. With modules, we will ask if there are modules where there exists a subset $A \subseteq M$ such that A generates M and if any other subset B generated A, then $|A| \leq |B|$. It will turn out that some modules have this property, some will be close, and other will not have this property at all.

After having tackled the question of generating modules "freely" and how to recognize free modules, we will then turn our attention to finitely generated modules, and modules where all submodules are finitely generated. Similalrly to how we can classify all finitely generated abelian groups, finitely generated and Noetherian modules have more "regularly", or equivalently have less "pathological" properties.

A new approach to studying structure's of objects we will take is considering the collection of homomorphisms on modules. As we saw in proposition 12.2.7, $\operatorname{Hom}_R(M, N)$ is always an abelian group, and if R is commutative it is always R-module. Thus, the "symmetries" of an R-module are themselves and R-module, meaning we would like to explore what properties M or N can have given the properties of $\operatorname{Hom}_R(M, N)$ have as an R-module, or even better that $\operatorname{End}_R(M)$ has as an R-algebra. This exploration is very fruitful, and will be of key importance in the theory of representation (of finite groups, lie algebras, etc.). In fact, if this intrigues the reader, after reading chapter 13 and chapter 14, the reader can skip to part X and read chapter 51 (only requiring knowledge from section 19.5 for the last 2 propositions of that chapter). A particular example of the above is when $N = R$ so that we consider $\mathrm{Hom}_R(M, R)$. This module is called the *dual module* of M, and will be very important in the study of M, even outside the field of algebra[7].

How does the generation of modules compare to the generation of groups and rings? With general groups, we saw that it was very difficult to show what the elements in an arbitrarily generated group look like (it is considered one of the hardest problems in mathematics, known as the *word problem for groups*). However, if G is abelian and finitely generated, then every element can be written in a very simple form due to the fundamental theorem of finitely generated abelian groups (theorem 5.1.8) With rings, we had that the objects that we quotient by (ideals) were no longer rings, and so took the time to study generation of ideals, but we have not studied any structure theorems on rings.

With modules, the story of the exploration of module structures is much closer to that of abelian groups, namely because every module *is* an abelian group. One major advantage of studying module is that many of the previous structure's we've studied can be represented as modules: since all abelian groups are $\mathbb{Z}$-modules, the study of the structure of abelian group is subsumed in the study of the structure of modules (we prove the Fundamental Theorem of Finitely Generated Abelian Groups in greater generality in chapter 14), and since all rings are module over themselves, the study of the structure of modules is also the study of the structure of rings! Due to the fact that all module are abelian groups, we can always collect "like terms":

$$r_1a_1 + r_2a_2 + s_1a_1 = (r_1 + s_1)a_1 + a_2$$

due to this, the order of the generating elements does not affect their sum! It becomes meaningful to ask if the ring acting on a module somehow determines some (or even all!) of the structure of the module. This question is a very meaningful question, and the answer will vary depending on what type of ring is acting on the module: chapters 13, 14, and 53 will each focus modules over particular types of rings and show how the answer to this question changes depending on what type of ring is acting on the module.

12.3.1 Direct Sum of Modules

We first update the terminology from the generation of groups and rings for the generation of modules:

[7] dual modules are central to Functional analysis and Harmonic Analysis

Definition 12.3.1: Generating Modules

Let M be an left R-module and let $(N_i)_{i\in I}$ be the submodules of M where I is an indexing set.

1. The *sum* of $(N_i)_{i\in I}$ is the smallest submodule of M containing each N_i, and is the set of all finite sums of elements from N_i:

$$\sum_i N_i := \{a_1 + \cdots + a_k | a_i \in N_i \text{ for all } i\}$$

If I is finite with $|I| = k$, then it is usually written as

$$N_1 + \cdots + N_k$$

2. For any subset $A \subseteq M$, then RA is the set:

$$RA = \left\{r_1a_1 + \cdots + r_ma_m | r_1, ..., r_m \in R, a_1, ..., a_m \in A, m \in \mathbb{Z}^+\right\}$$

(where by convention $RA = \{0\}$ if $A = \emptyset$) is the smallest [left] submodule of M generated by the elements of A, an is called the *[left] submodule generated by* A. If A is finite the set $\{a_1, a_2, ..., a_n\}$ we shall write

$$Ra_1 + \cdots Ra_n$$

If N is a submodule of M (possibly $N = M$) and $N = RA$, for some subset, then A is called the *set of generators* or *generating set for* N, and we say N is generated by A and is sometimes written as $\langle A\rangle = N$.

3. A submodule N of M (possibly $N = M$) is *finitely generated* if there is some finite subset A of M such that $N = RA$.

4. A submodule N of M (possibly $N = M$) is *cyclic* if there exists an element $a \in M$ such that $N = Ra$, that is, if N is generated by one element

$$N = Ra = \{ra|\ r \in R\}$$

and is sometimes written as $\langle a\rangle = N$.

It is an easy starter exercise to show that the structures in definition 12.3.1 are indeed modules. If R is a *rng*, then it is not necessarily the case that $A \subseteq RA$.

Example 12.3.2: Example

1. Let $R = \mathbb{Z}$ and let M be any $\mathbb{Z}$-module. Take any $a \in M$. Then $\mathbb{Z}a$ is a cyclic sub-module. It is in fact a cyclic subgroup! More generally, A generates M as a $\mathbb{Z}$-module if and only if A generates M as an [abelian] group. That is, generating M through A as a group (using $+$ and $-$ in groups) will produce as many elements as generating M through A as a module (which also allows $\mathbb{Z}$-actions). Thus, the definition of a finitely generated $\mathbb{Z}$-module is identical to the definition of a finitely generated group (definition 5.1.6).

2. Let R be a ring and let $M = R$ be the [left] R-module of R acting on itself. Then R is finitely generated as a module, even if R is infinitely generated as a ring! Even better: it is cyclic, since $R \cdot 1 = R$. Thus, in terms of generation, R is the "simplest" R-module.

3. If R is an R-module over itslef, then $I \subseteq R$ is a finitely generated R-module if and only if it is a finitely generated *left-ideal*. If R was a right-module, then I would be a finitely generated *right-ideal*, and if R was commutative, I would be a finitely generated *two-sided ideal*.

4. Let M be a finitely generated R-module. Then it is not the case that every submodule of M is finitely generated! Take $P[x_1, x_2, ...] = P[X]$ (i.e. the polynomial ring over countably many coefficients) which is a $P[X]$-module over itself. Then $P[X]$ is cyclic as a $P[X]$-module, and hence finitely generaed, however $(x_2, x_4, x_6, ...) \subseteq P[X]$ is a $P[X]$-submodule of $P[X]$ (being a left-ideal), however, it is an *infinitely generated* left-ideal. Finitely generated modules in which all submodules are finitely generated will be called *Noetherian*.

5. Let R^k be an R-module with the action being pointwise multiplication. Then R^k is finitely generaed by
$$e_i = (\underbrace{0, ..., 0}_{i-1 \text{ times}}, 1, \underbrace{0, ..., 0}_{k-i \text{ times}})$$
hence, for any $(r_1, r_2, ..., r_k) \in R^k$,
$$(r_1, ..., r_k) = \sum_i r_i e_i$$

6. Here is a neat way of writing down M if $A = M/B$. Then $M \cong A' + B$, since any element in M is some element of A (namely A' is a set of coset representative when taking $b = 0$), and an element of B:
$$\frac{M}{B} = \frac{A' + B}{B} \cong A$$
Note that is in general smmimply a *subset* of M, we can define the operation on A' by taking:
$$a_1 + a_2 = a_3 \qquad a_3 \text{ is the appropriate representative from } [a_1 + a_2] \text{ that belongs in } A'$$
In this way, $A' \not\subseteq M$ for as an R-module since the binary operation is different. Note too that $M = A' + B$ does not imply that $M \cong A \oplus B$. For example, consider $A = \mathbb{Z}/6\mathbb{Z}$ with $M = \mathbb{Z}$, and $B = 6\mathbb{Z}$. Letting $A' = \{0, 1, 2, 3, 4, 5\}$, we get that as sets:
$$M = A' + B = \{0, 1, 2, 3, 4, 5\} + 6\mathbb{Z} = \mathbb{Z}$$
for example, $10 = 4 + 6$, and $101 = 5 + 96$. However, giving A' the group structure of $\mathbb{Z}/6\mathbb{Z}$:
$$\mathbb{Z} \not\cong_R (\mathbb{Z}/6\mathbb{Z}) + 6\mathbb{Z}$$
For the right hand side has torsion elements, while the left does not. For the $+$ to be upgraded to a $\oplus$, the operation of the quotient must in some natural way be compatible with the operation in M, that is A must in not only be the quotient M/B, but must also be considered as $A \subseteq M$. This is explore in chapter 17$\star$.

Naturally, we would want to find when direct sums of modules behave like direct products, or more specifically external direct sums

Definition 12.3.3: Direct Product and Sum of Modules

Let $(M_i)_{i\in I}$ be a collection of R-modules where I is an indexing set. Then the collection of tuples $(m_i)_{i\in I}$ where $m_i \in M_i$ with addition and action of R defined componentwise is called the *direct product* of $(M_i)_{i\in I}$, denoted

$$\prod_{i\in I} M_i$$

If I is finite where $|I| = n$, then it is usually denoted:

$$M_1 \times \cdots \times M_k$$

If we take the subset of $\prod_{i\in I} M_i$ where only finitely many of the terms are nonzero, then this this set is called the *external direct sum*, and is written as:

$$\bigoplus_{i\in I} M_i$$

If I is finite where $|I| = n$, then it is usually denoted:

$$M_1 \oplus M_2 \oplus \cdots \oplus M_n$$

If I is infinite and $R \neq 0$, then $\bigoplus_{i\in I} M_i \subsetneq \prod_{i\in I} M_i$.

12.3.4 Remark It might be tempting to say that:

$$M \cong N_1 \oplus N_2 \quad \Leftrightarrow \quad M = N_1 + N_2$$

however that is not the case: Take $\mathbb{Z}$ as a $\mathbb{Z}$-module over itself, $N_1 = 2\mathbb{Z}$ an $N_2 = (0)$. Then $2\mathbb{Z} + 0 = 2\mathbb{Z}$, and:

$$\mathbb{Z} \cong 2\mathbb{Z} \qquad \text{but} \qquad \mathbb{Z} \neq 2\mathbb{Z}$$

which we can see since, $1 \notin 2\mathbb{Z}$

If $M \cong A \times B$ for some modules M, A, and B, then for any $m \in M$, there exist unique pair of elements $a \in A, b \in B$ such that $m \mapsto (a, 0) + (0, b)$. Thus, any generating set (or more specifically minimal generating set) of M breaks down to a question of a generating set (or minimal generating gset) of A and B. The key here is that any element $m \in M$ can be uniquely represented by elements a and b (namely since these two modules are isomorphic).

Let's say conversely that we want to find submodules $A, B \subseteq M$ for which $M = A + B$. If a module M is finitely generated, by Zorn's lemma there must exist a minimal nonnegative integer d such that M is generated by d elements (and no less)[8]. If M was not finitely generated, this need not be the case. If M can be generated by d elements, it need not be the case that there is a unique set of generating elements, nor the case that the R-linear combination of these elements be unique:

[8]Note that minimal doesn't mean smallest! There can be multiple different $d \in \mathbb{N}$ that satisfy this property, see the discussion in section 2.3

Example 12.3.5: Non-Unique Sets And R-Linear Combinations
here

It will turn out that we almost never get a unique minimal generating set [a]. Take for example R over itself. Let $u \in R$ be a unit of R. Since u is a unit, let $v \in R$ such that $uv = 1$. Then $R \cdot 1 = R \cdot (vu) = (Rv)u = R \cdot u$ (since $u \in R$, $uv = 1 \in Rv$, and from that we can show that $(Rv)u = Ru$).

[a] If you have already taken a course in linear algebra, this is equivalent to saying that there is no unique basis

Though unique minimal generated sets are rare to come accorss, uniqe R-linear combinations are much more common. For example, taking R over itself so that $\{1\}$ generated R as an R-module. Then every R-linear combination is unique, that is

$$r \cdot 1 = s \cdot 1 \Leftrightarrow r = s$$

We can see this by taking the homomorphism $\varphi : R \to M$, $r \mapsto r \cdot 1$ and look at the kernel. This property is very desirable, since it will allow us to show that M can split into a direct sum of submodules. If the R-linear combinations are unique, then we need only concern ourselves with any minimal generating set when considering the structure of such a module!

For the rest of the chapter, we identify equivalent properties to saying R-linear combinations are unique, and identify modules which have this property.

Proposition 12.3.6: Equivalent Conditions for Linear Combinations

Let $N_1, ..., N_k$ be (not necessarily distinct) submodules of R-module M. Then the following are equivelent

1. the map $\pi : \bigoplus_{i \in I} N_i \to \sum_i N_i$ defined by

$$(n_i)_{i \in I} \mapsto \sum_i n_i$$

   is an R-module isomorphim. If $|I| < k$ is finite, we write:

$$N_1 + N_2 + \cdots + N_k \cong N_1 \times N_2 \times \cdots \times N_k$$

2. $N_j \cap (N_1 + N_2 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k) = 0$ for all $j \in \{1, 2, ..., k\}$
3. Every $x \in N_1 + N_2 + \cdots + N_k$ can be written *uniquely* in the form $a_1 + a_2 + \cdots + a_k$ with $a_i \in N_i$

***Proof*:**
For (1) implies (2). let's assume that the intersection is not empty. Then there exists some a_j such that

$$a_j = a_1 + a_2 + \cdots + a_{j-1} + a_{j+1} + \cdots + a_k$$

However, this implies that

$$\pi(0, ..., a_j, ..., 0) = \pi(a_1, a_2, ..., a_{j-1}, a_{j+1}, ..., a_k)$$

but

$$(0, ..., a_j, ..., 0) \neq (a_1, a_2, ..., a_{j-1}, a_{j+1}, ..., a_k)$$

contradiction injectivy, since π is an isomorphism

For (2) implies (3). Let's say there are two linear combinations

$$a_1 + a_2 + \cdots + a_k = b_1 + b_2 + \cdots + b_k$$

then we can isolate any jth components:

$$b_j - a_j = (b_1 - a_1) + \cdots + (b_{j-1} - a_{j-1}) + (b_{j+1} - a_{j+1}) + \cdots + (b_k - a_k)$$

which implies by our previous proposition

$$b_j - a_j \in N_j \cap (N_1 + N_2 + \cdots + N_{j-1} + N_{j+1} + \cdots + N_k) = 0$$

thus, $b_j - a_j = 0$ and so $b_j = a_j$ for every j, showing that the linear combination is unique!

For (3) implies (1), Notice that the function is surjective construction: for any element $a_1 + a_2 + \cdots + a_k$, the element $(a_1, a_2, ..., a_k)$ would map to it. What (2) does is show injectivity, since if two sums are the same, then the direct product of elements must also be the same!. This map is also easily checked to be an R-module homomorphism. For example

$$\begin{aligned}
\pi(r \cdot (a_1, a_2, ..., a_k) &= \pi(ra_1, ra_2, ..., ra_k) && (12.1)\\
&= ra_1 + ra_2 + \cdots + ra_k && (12.2)\\
&= r \cdot (a_1 + a_2 + \cdots + a_k) && (12.3)\\
&= r\pi(a_1, a_2, ..., a_k) && (12.4)
\end{aligned}$$

where line 3 comes from induction on $r \cdot (m + n) = r \cdot m + r \cdot n$, and linearity comes easily too.

This show's us how we may split up a module into "direct products" which satisfy the universal property of direct products. We now introduce a notion that is slightly stronger than the external direct product and of sums of modules:

Definition 12.3.7: Internal Direct Sum

Let $(N_i)_{i\in I}$ be a collection of submodules of M modules respecting one of the 3 condition's in proposition 12.3.6. If:

$$\sum_i N_i = M$$

then we will replace the summation symbol $\sum_i$ with $\bigoplus_i$. If the indexing set is finite, then:

$$N_1 + \cdots + N_k = N_1 \oplus \cdots \oplus N_k$$

Thus, we are *overloading* the notation of $\oplus$ to represent the fact that the sum $\sum_i N_i$ is isomorphic to $\oplus_i N_i$ *and* that $\sum_i N_i = M$

We have already shown in remark where M is an R-module that is isomorphic to a direct sum of submodules $(N_i)_{i\in I}$ but $\sum_i N_i \neq M$, namely by taking $\mathbb{Z}$ as a $\mathbb{Z}$-module and $N_1 = 2\mathbb{Z}$, $N_2 = (0)$.

Then:

$$\mathbb{Z} \cong 2\mathbb{Z} \oplus (0) \qquad \mathbb{Z} \neq 2\mathbb{Z} \oplus (0)$$

In chapter 17⋆ and 51, we will further explore how to find when a module M can be decomposed into direct products.

One case is settled immediately, and is used often enough to record. When the *ring* is a product, every module over it splits, and the splitting is forced by the two idempotents $(1,0)$ and $(0,1)$. The upshot is that module theory over $R \times S$ is nothing more than module theory over R and over S carried out side by side.

Proposition 12.3.8: Modules over a Product Ring

Let R and S be rings and put $e = (1,0)$ and $f = (0,1)$ in $R \times S$, so that e and f are central idempotents with $e + f = 1$ and $ef = 0$. Then every $(R \times S)$-module M decomposes as an internal direct sum

$$M = eM \oplus fM$$

where eM is an R-module and fM is an S-module. The assignments $M \mapsto (eM, fM)$ and $(M_1, M_2) \mapsto M_1 \oplus M_2$ are mutually inverse, and give an equivalence of categories

$$(R \times S)\textbf{-Mod} \;\simeq\; R\textbf{-Mod} \times S\textbf{-Mod}$$

Under it, a module is finitely generated, or projective, exactly when both of its components are. Freeness is *not* componentwise: $M_1 \oplus M_2$ is free over $R \times S$ if and only if M_1 and M_2 are free of the *same* rank, since a free $(R \times S)$-module is $(R \times S)^n \cong R^n \oplus S^n$.

***Proof*:**
Step 1: the decomposition. For $m \in M$ we have $m = 1 \cdot m = (e + f)m = em + fm$, so $M = eM + fM$. If $x \in eM \cap fM$, write $x = em' = fm''$; then $x = ex = e(fm'') = (ef)m'' = 0$, using that e is idempotent and $ef = 0$. So the sum is direct.

Step 2: the components are modules over the factors. The subgroup eM is stable under $R \times S$, and f acts on it as $fe = 0$, so the action factors through $(R \times S)/(0 \times S) \cong R$, making eM an R-module. Symmetrically fM is an S-module.

Step 3: the two constructions are inverse. Given R-module M_1 and S-module M_2, let $R \times S$ act on $M_1 \oplus M_2$ componentwise. Then $e(M_1 \oplus M_2) = M_1$ and $f(M_1 \oplus M_2) = M_2$, recovering the pair. Conversely Step 1 rebuilds M from (eM, fM). Both assignments send a homomorphism to its restrictions, and a map commutes with the action of e and f exactly when it preserves the two summands, so the correspondence is functorial in both directions and is an equivalence.

Step 4: the listed properties. Finite generation and projectivity are checked componentwise because the equivalence carries $R \times S$ to the pair (R, S) and direct sums to direct sums. A generating set for M projects to generating sets of eM and fM and conversely, giving finite generation; and M is a summand of a free $(R \times S)$-module exactly when each component is a summand of a free module over its factor, giving projectivity.

Freeness is the exception, and the equivalence is what shows why. A free $(R \times S)$-module of rank n corresponds to the pair (R^n, S^n), so M is free exactly when its two components are free *of a common rank.* For $R = S = k$ a field, $M = k \oplus k^2$ has both components free, of ranks 1 and 2, and is projective; but it is not free, since that would force $1 = 2$. This completes the proof.

To conclude, we re-phrase the above results in the language of universal properties. The properties of $M \oplus N$ can be described in terms of maps $\varphi : L \to M \oplus N$ where this map is uniquely determined by the maps $\varphi : L \to M$ and $\varphi : L \to N$, that is:

Proposition 12.3.9: Universal Property of Products, Modules

Let M, N be two R-modules. Then $M \oplus N$ as constructed above satisfies the universal property products: for any R-module L, if $\varphi :_M L \to M$ and $\varphi :_N L \to N$ are two R-module homomorphisms, then there exists a unique R-module homomorphism $\varphi : L \to M \oplus N$ such that the following diagram commutes:

$$\begin{tikzcd}
 & L \arrow[d, dashed, "\varphi"] \arrow[ddl, bend right, "\varphi_M"'] \arrow[ddr, bend left, "\varphi_N"] & \\
 & M \oplus N \arrow[dl, "\pi_M"'] \arrow[dr, "\pi_N"] & \\
M & & N
\end{tikzcd}$$

More generally, if M_i is a collection of R-modules for an indexing set I, for any R-module L and collection of R-module homomorphisms $\varphi_i : L \to M_i$, there exists a unique R-module homomorphism $\varphi : L \to \oplus_{i \in I} M_i$ such that all of the following diagrams commutes:

$$\begin{tikzcd}
L \arrow[d, "\varphi"'] \arrow[dr, "\varphi_i"] & \\
\oplus_{i \in I} M_i \arrow[r, "\pi_{M_i}"] & M_i
\end{tikzcd}$$

***Proof*:**
The reader should do this exercise as a good work-through of how to show a universal property, and to be comfortable with commutative diagrams for proofs in later chapters.

12.3.2 Free Modules

Given a ring R and a an arbitrary abelian group A, we can always define the trivial module via the action $r \cdot a = 0$: let's label this module $T(A)$ and call it the trivial R-module. Just like trivial groups, trivial modules offer little in structure[9]. It would be desirable if we can find a better struture on A, in particular if A represents the generating set of some R-module M, then we would like to define a "free R-module" $F(A)$ (or more particularly $F^R(A)$ to remember the ring R) which represents an R-module with no relationship, and then find a R-submodule $N \subseteq F(A)$ such that $F(A)/N \cong M$ (i.e. the submodule N woule represent adding relationships to $F(A)$). This mirrors the development of the free group from section 2.3.3 and 3.2.

Let's say we were to define a R-module $F^R(A)$ on the generating set A of M with only the bare minimum relations to make it into an R-module ($1 \cdot m = m$, $(rs) \cdot m = r \cdot (s \cdot m)$, and so forth).

[9]When dealing with groups, we also defined $\mathrm{Sym}(A)$. As a thought exercise, can you see why we cannot parallel the same idea with module? Hint: Think of $\mathbb{Z} \hookrightarrow \mathbb{Q}$

Then similarly to free groups, we would want that for any element of $x \in F^R(A)$, which is some linear combination:

$$x = \sum_{a \in B \subseteq A} r_a \cdot a \quad \text{for some subset } B \subseteq A$$

to be equal to 0 if and only if $r_a = 0$ for all a. This is similar to free groups, where any relation between generating elements can be move to the left hand side and the right hand side equals to the identity element. By proposition 12.3.6, this means that $F^R(A) = R^{|A|}$ is our desired free R-module. To see this more clearly, let's say that some $r_a \neq 0$. Then that means we have added some relation into $F^R(A)$, that is the function $\varphi : R^{|A|} \to F^R(A)$ would have a non-trivial kernel, going against the idea that $F^R(A)$ is a free-module.

If $F^R(A)$ is a free module on A, just like for free groups, any set-function from A to any R-module N, $f : A \to N$, should lift to an R-module homomorphism $F : F^R(A) \to N$, i.e. there is a map $\iota : A \hookrightarrow F^R(A)$ that injects A into $F^R(A)$ and $f = F \circ \iota$. In particular, given any arbitrary map $f : A \to N$, no matter where f maps A, whether f is injective or not, we can find a map F such that $F|_A = f$, so F must map any element of the set A exactly as how f would map it, hence F is an extension of f that has the restriction of being an R-module homomorphism. Furthermore, N is any R-module, meaning any $f : A \to N$ can be lifted to $F : F^R(A) \to N$ (see how this is not the case for an arbitrary domain M, or check out section 2.3.3 for the parallel problem in group theory). A consequence of this result is that we can determine how $F^R(A)$ maps to an R-module N simply by determining where A maps to, that is we have the universal property of free modules.

12.3.10 On Recovering A We would have to already know of the set A in order to do the above strategy. The notation $F^R(A)$ makes it seem obvious that we know about the set A, namely since we will soon construct $F^R(A)$ with respect to A, but it is not so obvious that if we are given a module M that we say is a free R-module, can we find a subset A such that $F^R(A) \cong_R M$. In fact, depending on the ring R acting on M, it will sometimes be possible to "recover" A, while other times it will be impossible[a]. We will explore this in chapter 13

[a] It will be possible to say there exists such a set using the axiom of choice, but it will be impossible to give a description of the elements

We now prove that $F^R(A) = R^{|A|}$ does indeed respect the universal property of free R-modules.

Theorem 12.3.11: Universal Property of Free Modules

For any set A, there exists a free R-module $F^R(A)$ on the set A where $F^R(A)$ satisfies the following *universal property*: If M is any R-module and $\varphi : A \to M$ is any map of sets, then there is a unique R-module homomorphism $\psi : F(A) \to M$ such that $\psi|_A = \varphi$, that is, the following diagram commutes

$$\begin{tikzcd} S \arrow[r, hook, "\text{incl.}"] \arrow[dr, "\varphi"] & F(S) \arrow[d, "\psi"] \\ & M \end{tikzcd}$$

When A is a finite set $\{a_1, a_2, ..., a_n\}$, $F^R(A) = Ra_1 \oplus \cdots + \oplus RA_n \cong R^n$, and if A has any cardinality, then:

$$F^R(A) = \bigoplus_{a \in A} R_a$$

where A is treated as an indexing set.

***Proof*:**
if $A = \emptyset$, then let $F(A) = \{0\}$. Then this trivially satisfies:

$$\begin{tikzcd} \emptyset \arrow[r, hook, "\text{incl.}"] \arrow[dr, "\varphi"] & \{0\} \arrow[d, "\Phi"] \\ & M \end{tikzcd}$$

since $\emptyset$ is included in any set, φ is the vacuous map, and $\Phi(0) = 0$ makes Φ into an R-module homomorphism which restircted to $\emptyset$ is itself again.

So let $A \neq \emptyset$. What we'll do is construct funtions with the domain as a set and the co-domain as the ring R. These functions will be the mechanism by which set A is given an R-module structure! Let $F^R(A)$ be the collection of all set functions $f : A \to R$ such that $f(a) = 0$ for all but finitely many $a \in A$:

$$F^R(A) = \{f \in \text{Hom}_{\textbf{Set}}(A, R) \mid f(a) \neq 0 \text{ for only finitely many } a \in A\}$$

Make $F^R(A)$ into an R-module by defining pointwise addition and pointwise multiplication for the action:

$$(f + a)(a) = f(a) + g(a)$$
$$(rf)(a) = r(f(a))$$

for all $a \in A$, $r \in R$, $f, g \in F^R(A)$. Notice how $F^R(A)$ is isomorphic to $R^|A|$, meaning this construction matches definition 12.3.12.

Let $\iota : A \to F^R(A)$ vis $a \mapsto f_a$, where f_a is the function which is 1 at a and zero everywhere else. We can thus identify a set $A \leftrightarrow A' \subseteq F^R(A)$ as the subset containing all of f_a. Notice that since any f has only finitely many non-zero elements, we can write any $f \in F^R(A)$ as

$$f = r_1 f_{a_1} + r_2 f_{a_2} + \cdots + r_n f_{a_n}$$

which by our identification of A' with A, represents:

$$r_1a_1 + r_2a_2 + \cdots + r_na_n$$

where f takes on the value of r_i at a_i and is zero at all other elements of A. Even better, by construction, every element of $F^R(A)$ has a *unique* expression as such a formal sum, since by construction, non of the a_i have any relation to each other, there can't be a way of cancelling them out!

To establish the universal property, of $F^R(A)$, Let's say we have a function $\varphi : A \to M$, where M is an R-module. We need to show this function raises to an R-module homomorphism $\Phi : F^R(A) \to M$. Let Φ be defined by

$$\sum_{i=1}^{n} r_1a_i \mapsto \sum_{i=1}^{n} r_i\varphi(a)$$

by the uniqueness fo the expression of the elements of $F^R(A)$ as linear combinations of a_i, we see that Φ is indeed a well dfiend R-module homomrophsim!!

check here

By definition, the restriction of $\Phi|_A$ is φ. For uniqueness of $F^R(A)$, since $F^R(A)$ is generated by A, once we know the values of an R-module homomorphism on A, tis value on every element of $F^R(A)$ are uniquely determined (by force of construction), so Φ is the unique extension of Φ to all of $F^R(A)$.

If A is finite ($\{a_1, a_2..., a_n\}$), proposition 12.3.6

$$F^R(A) = Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_n$$

Since $R \cong Ra_i$ (under $r \mapsto ra_i$) by proposition 12.3.6 again, the direct sum is isomorphic to R^n.

Definition 12.3.12: Free Module

An R-module F is said to be *free* on the subset A of F if for every nonzero element x of F, there exists a unique nonzero elements $r_1, ..., r_n$ of R and unique $a_1, ..., a_n$ in A such that $x = r_1a_1 + \cdots r_na_n$, for some $n \in \mathbb{Z}^+$, which by proposition 12.3.6 implies that:

$$F = \bigoplus_{i \in I} R$$

If R is a commutative ring, the cardinality of A is called the *free rank* of F, or just *rank* if unambiguous.

Note that A need not be finite, but the linear combinations of elements of A need to be finite (notice that we say $a_1, ..., a_n \in A$, and not $a_1, ..., a_n, ... \in A$). From now on, we will write R^n to mean the free R-module $R \oplus R \oplus \cdots \oplus R$. So $(\mathbb{Z}/2\mathbb{Z})^3$ is a free $(\mathbb{Z}/2\mathbb{Z})$-module. R^1 is a free R-module of rank 1 (to be distinguished from R which we will treat as a ring). If the superscript is infinite, say $N = |\mathbb{N}|$ or $N = |\mathbb{R}|$, then R^N is considered an infinite direct sum, not a direct product. In fact, non-trivial infinite direct product of modules (modules of the form R^N for an N of infinite cardinality) are far from being free (see exercise ref:HERE[10]). Furthermore, the notion of rank is not necessarily well-defined if R is not commutative. We shall explore more of this in chapter 13.

Example 12.3.13: Free Modules

1. Let R be any ring with identity. Then R over itself is a free-module! The element 1 will always generated all of R since 1 is a unit (and so $ab \neq 0$ for any $b \neq 0$) and $r \cdot 1 = r$, and so it's a linearly independent generating set!

2. The "freeness" is dependent on the ring you choose, as well as the module (i.e., the abelian group). So $\mathbb{Z}/n\mathbb{Z}$ over itself is free (by the logic provided above), but not over $\mathbb{Z}$ (which we already know is a well-defined module), since

$$n \cdot 3 = 0$$

   In general, one criterion for freeness is that $r \cdot a \neq 0$ for any r. This criterion is called being *torsion-free*, and if $r \cdot a = 0$, then r is said to be a *torsion element* of the R-module M. However, a module being torsion free does not imply the module is free.

3. Since $\mathbb{Z}/n\mathbb{Z}$ overt $\mathbb{Z}$ is not a free-module, it does not have a basis. If we try to choose $A \subseteq \mathbb{Z}/n\mathbb{Z}$ to be a basis, it will always have torsion elements. $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$ is also an example of a module *without* any linearly independent sets!

4. Let $\mathbb{C}[x, y]$ act on itself. Then $\mathbb{C}[x, y]$ is a free module. As stated in example 12.1.7, ideals of modules over themselves are sub-modules. Take $(x, y) \subseteq \mathbb{C}[x, y]$. Then

$$y \cdot x - x \cdot y = 0$$

   showing that (x, y) is *not* free! However, (x, y) is certainly torsion-free, since $\mathbb{C}[x, y]$ is an integral domain.

   Thus, freeness is not preserved by taking sub-modules

5. Let $\mathbb{Z}$ act on itself. Then $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$-module as we've seen in section 12.2. However, as we've seen in the 2nd example, $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$ is not free.

   Thus, freeness is not preserved when taking quotient modules.

6. Let's say $S = \{1, 2, 3\}$, and $M = \mathbb{R}^3$ (over $\mathbb{R}$). Take any $f : S \to M$. For our concrete example, let's say:

$$f(1) = (\varphi, \sqrt{2}, 4) \qquad f(2) = (0, 0, 0) \qquad f(3) = (0, 5753, 9)$$

   I chose particularly random values to show that f is indeed any set function map. Now if you consider $F^R(S)$ (over $\mathbb{R}$), we'd have some uncountable set of elements, which most importantly are all distinct! It's that last part that let's us say that

$$\psi : F^R(S) \to \mathbb{R}^3$$

   will be the unique homomorphism where $\psi|_S = f$. Consider

[10] Inspired by the link here

$$\psi(x) = \psi(\sum_{i=1}^{3} r_i s_i) = \sum_{i=1}^{3} r_i \varphi(s_i)$$

This function is indeed well defined since every x is *unique* in it's represetnation from S, since if it wasn't, and we consider

$$x = y(\sum_{i=1}^{3} r_i s_i = \sum_{i=1}^{3} r_i' s_i)$$

then it is not true in general that

$$\psi(x) = \psi(y)$$

what let's use insure that this will always be true is precicely the linear independence, or free property, of the module!

Finally, this is indeed a [unique] homomorphism. it is easy to check that

$$\psi(x + ry) = \psi(x) + r\psi(y)$$

and the uniqueness will come from free modules elements having no relation to each other (if they did, you might be able to have a second homomrophism).

7. You can have a module with linearly independent sets, but that always fail to generate the module, meaning it is not a free module. For example, $\mathbb{Z}^{\mathbb{N}}$ has a great multitude of linearly independent sets (even infinite ones), however it is not a free-module. This is a rather complicated proof, and is relegated to exercise ref:HERE in chapter ref:HERE.

8. Take $\mathbb{C} \times \mathbb{C}$ over itself. This is free module (its basis is $(1, 1)$). Then $\mathbb{C} \times \{0\}$ is a submodule, but not a free module: Any basis of $\mathbb{C} \times \{0\}$ would consist of elements of the form $(0, a)$. If A is the colleciton of all elements in the 2nd component of a basis, then:

$$\sum_{a \in A} (a, 0) \cdot (0, a) = \sum_{a \in A} (0, 0) = (0, 0)$$

meaning a non-trivial linear combination summed to zero.

9. Let R be a Principal Ideal Domain acting over itself. Then any submodule N of R is an ideal, and since it's principal $N = (a)$ for some $a \in R$. Since $(a) = Ra$, it follows that N is torsion-free (see ref:HERE). Thus, any submodule of R is torsion-free. (TBD, not sure, trying to get to if R is not principal, then (x, y) is never torsion free. Maybe move this to Mod over PID sec? exercice 6 in section 12.1 Dummit)

10. In chapter 13, we'll show that $\mathbb{R}$ is a $\mathbb{Q}$-vector space with uncountable basis! Note that even though the basis is uncountable, any element $x \in \mathbb{R}$ will still be a $\mathbb{Q}$-linear combination of *finitely many* basis elements.

11. On the other hand $\mathbb{Q}$ over $\mathbb{Z}$ is *not* a free module (i.e. $\mathbb{Q}$ is not a free abelian group). To see this, notice that for any $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$, there exists $m, n \in \mathbb{Z}$ (namely $m = bc,\ n = -da$) such that $(bc)\frac{a}{b} + (-da)\frac{c}{d} = 0$, and hence any two elements are linearly independent, meaning if $\mathbb{Q}$ was a free $\mathbb{Z}$-module, it would have to be cyclic. As we've seen in section 2.3.4, the only free cyclic $\mathbb{Z}$-module is $\mathbb{Z}$. Can you see why $\mathbb{Z} \ncong_{\mathbb{Z}} \mathbb{Q}$?

Take note of the fact that whether a module is free or not is highly dependent on what ring is acting on it. In $\mathbb{Z}$-**Mod**, $\mathbb{Z}/n\mathbb{Z}$ is certainly not a free (it is not a free group). However, we've just seen that $\mathbb{Z}/n\mathbb{Z}$ is a free-module over itself! The intuition here behind being "free" comes down to the ring-action being "free".

Corollary 12.3.14: Consequences Of Universal Property

1. If F_1 and F_2 are free modules on the same set A, there is a unique isomorphism between F_1 and F_2 which is the identity map on A
2. if F is any free R-module a free-generating set A, then $F \cong F^R(A)$. In particular, F enjoys the same universal property with respect to A as $F^R(A)$ does in theorem 12.3.11.

***Proof*:**
exercise

12.3.15 Foreshadowing Though a free R-module has no relations, it is not the case that any submodule has no relation, that is the submodule of a free module *need not be free*! In other words, though it may at first seem counter-intuitive, the process of taking a submodule can add relations (see exercsie ref:HERE)! This contrasts to how the subgroup of a free group is always free (see appendix B). There are certain rings for which the R-submodule are always free, while there are other rings where submodules are far from being free (ex. some submodules will be torsion modules, which will be far away from being free). We will explore these concepts further in chapter 13 and 17$\star$.

Measuring degree of Freeness

As we've seen, not all modules are free modules. However, we may still like to find collection of elements that are linearly independent (i.e. that are "free" of relations). We explore these notions here.

Definition 12.3.16: [Free] Rank

For any ring R, the *rank* of an R-module M is the maximum number of R-linearly independent elements of M, that is if M has rank n, there exists elements $y_1, y_2, ..., y_n$ such that

$$r_1y_1 + r_1y_2 + \cdots + r_ny_n = 0 \quad \Leftrightarrow \quad r_1 = r_2 = \cdots = r_n = 0$$

If for all $n \in \mathbb{N}$, there exists a R-linearly independent set, then M is said to be of infinite rank.

Due to the importance of the notion of rank for modules in geometry, the rank is also sometimes called the *Betti Number*[11]. The rank of a module is similar to the size of the generating set of a module, and it will in fact match exactly the idea of rank for modules over fields, but this is not necessarily the case for modules over weaker rings:

[11]The kth Betti number loosely refers to the number of k-dimensional holes ona a topological surface

Example 12.3.17: Different Rank and Size of Generators

Let $R = \mathbb{Z}[x]$ and let $M = (2, x)$ be an R-module. If we take $2 \in M$, then since R is an R an integral domain and M is an ideal of R, there does not exist a r such that

$$ra = 0$$

thus, $\{2\}$ is a linearly independent set from M, meaning it's at least rank 1. What if we two elements? Let's take $a, b \in M$. Since $a, b \in R$, in particular $-a \in R$

$$ba - ab = 0$$

meaning 2 elements will never be linearly independent. This will naturally generalize to n elements, so M is of rank 1.

However, M is generated by 2 elements, showing that the rank and the number of generators does not match.

Sometimes, authors define the minimal set of generators to be the rank of the matrix, and rank as was defined above would be called the *free rank*. Since the size of the generating sets rarely comes up outside the context of free generating sets, this book decided to stick with "rank" to mean the maximal number of linearly independent elements.

Not that it is yet to be established if there is a unique maximal rank (just like in a tree, there may be many maximal heights). For modules over integral domains, the maximal height will alwaays be unique, and under appropriate finiteness conditions it shall also be unique, however there are counter-examples over more general rings; this shall be further explore in chapter 13.

Using the notion of rank, we can see how a module might fail to be free. In the "worst case scenario", we can single out elements fail to be linearly independent:

Definition 12.3.18: Torsion Element

Let M be an R-module and $m \in M$. Then m is called a *torsion element* if m is linearly *de*pendent, in particular there exists an $r \in R$ such that

$$r \cdot m = 0$$

The collection of all torsion elements of M is denoted $\text{Tor}_R(M)$ or $\text{Tor}(M)$ is the context is clear. An element that is not torsion is called *torsion-free*.

Elements that are torsion-free can be thought of as being "locally" free. Being is a property of modules, while being torsion-free is a property of elements. If every element has this property, then we give a name to such a module:

Definition 12.3.19: Torsion-Free Modules

Let M be an R-module. Then if $\text{Tor}_R(M) = \{0\}$, then M is said to be *torsion-free* or to be a *torsion-free module.*

Being torsion free is an interesting example of the fact that it is not always enough that every element have a property for the global module to have the property. In this case, even if every element itself is free, this does not imply the module itself is free

Example 12.3.20: Torsion-free, But Not Free Module

Take $\mathbb{C}[x, y]$ over itself. Then since it's an integral domain, it is torsion free and each submodule is torsion free. However, (x, y) is not free.

Looking instead at the ring action instead of the module element, we can also find all elements of the ring that fail to be linearly dependent with *all* module elements:

Definition 12.3.21: Annihilator

Let M be an R-module. Then the set

$$\mathrm{Ann}_R(M) = \{r \in R \mid r \cdot m = 0\ \forall m \in M\}$$

is called the *annihilator* of M.

Recall that we have talked about this set in example 12.1.7(4) when discussing modules where the annihilator is trivial; such a module is given a name:

Definition 12.3.22: Faithful Module

Let M be an R-module. Then if $\mathrm{Ann}_R(M) = \{0\}$, M is said to be a *faithful R-module*, or that the ring R is acting faithfully on M

If R does not act faithfuully on M, when we may always induce a faithful action along the lines of example 12.1.7(4). The same cannot be said about non-torsion modules, instead we may ask if a module M can be split into a torsion-free part and a torsion part. Being trosion free is much more flexible than being free:

Proposition 12.3.23: Extensions And Submodules Of Torsion-Free Modules

Let M be an R-module. Then:

1. If M is torison-free, then any submodule is torsion free
2. the direct sum of torsion free modules is torsion-tree
3. if M is an extension of torsion-free modules, that is $0 \to A \to M \to B \to 0$ is a short exact sequence with A and B being torsion-free, then M is torsion-free.
4. If R is an integral domain M is free, then M is torsion-free (i.e. free modules over integral domains are torsion-free)

***Proof*:**

1. This is immediate after realising $\mathrm{Tor}_R(N) \subseteq \mathrm{Tor}_R(M)$
2. this is also almost immediate givn a similar type of reasoning as in part (1)
3. If $m \in M$, where $r \cdot m = 0$, then $r \cdot \overline{m} = \overline{0}$, but since $r \in\in \mathrm{Tor}_R(B) = \{0\}$ $r = 0$, and so $\mathrm{Tor}_R(M) = 0$.

4. Since the direct of free modules is free, and the integral domain R as a module over itself is torsion free, M is torsion-free.

In chapter 14, we shall further explore the decomposition of the torsion and torsion-free part of a module over nicer enough rings so that the decomposition is possible.

If N is not a torsion module, then $\text{Ann}(N) = 0$. It is possible that N is a torsion module, but $\text{Ann}(N) = 0$:

Example 12.3.24: Torsion Module With No Annihilators

Take

$$Z = \bigoplus_k \mathbb{Z}/2^k\mathbb{Z}$$

then Z is a $\mathbb{Z}$-torsion module, since for every $z \in \mathbb{Z}$, we can take 2^k, where we pick k relative to the largest nonzero component. However,

$$\text{Ann}(Z) = 0$$

Since if $r \in \mathbb{Z}$ was an annihilator, then $r \cdot z = 0$ for every z, but z can take any value of $\mathbb{N}$, leading us to a contradiction

If the module is finitely generated the annihilator will always be non-empty, since if we annihilate the generators, we annihilate the entire module.

Exercise 12.3.1

1. Let M be an R-module, and define $\text{Ann}_R(M) = \{r \in R \mid r \cdot m = 0, \ \forall m \in M\}$.
   1. Show that $\text{Ann}_R(M)$ is a two-sided ideal
   2. If $\text{Ann}_R(M) = 0$, then M is said to be a faithful module. Show that all free modules are faithful, but there are faithful non-free modules
2. Let R be a commutative ring and an R-module over itself. Then show that an ideal $I \subseteq R$ is a free R-module if and only if it is principal and generated by a nonzero divisor.
3. Show that $\mathbb{Z}^{\mathbb{N}}$ (a direct product of countably many $\mathbb{Z}$ is *not* a free abelian group

12.3.3 Presentations and Resolutions

We claimed that all R-modules are a quotient of a free R-module, just how all groups are quotient of free groups. We will demonstrate this here, and then explore a bit furter how we can think of relations and presentations of modules:

Proposition 12.3.25: Constructing Modules using Free Modules

Any R-module M is the quotient of a free module $F^R(A)$ for appropriate set A:

$$F^R(A)/K \cong_R M$$

***Proof*:**
Let M be an R-module and $\mathcal{F} = F^R(M)$. Define the map:

$$\varphi : \mathcal{F} \to M \qquad (r_i m_i)_{i \in M} \mapsto \sum_{i \in M} r_i m_i$$

where any element on the left hand side is a tulpe indexed by M where all but finitely many of the terms are nonzero, and hence the right hand side is well-defined. This map is evidentely surjective (since the tulpe of m in the first element maps to m), and so if K is the kernel of this map, by the first isomrophism theorem:

$$F^R(M)/K \cong_R M$$

as we sought to show.

Hence, the kernel of a free module plays the role of codifying all the relations in a module. When working with groups, we encoded these relations writting a group as either, for example:

$$\langle x, y \ : \ xy - yx = 0 \rangle \qquad \text{or} \qquad \frac{F(x,y)}{xy - yx}$$

The former notaiton being a convienient notation that can be easier to read before the introduciton of quotienting. In the next definition, we will define the presentaiton of a module. Going off of the fact that the kernel is the central object defining the relation, we incorporate this idea into the definition:

Definition 12.3.26: Presentation Of Module

Let M be an R-module. Then a *presentation* of M is a an exact sequence:

$$R^n \to R^m \to M \to 0$$

If n is finite (i.e. if $n \in \mathbb{N}$), then M is said to be *finitely presented*, or a *finitely presented module.*

Note that m and n need not be natural numbers, and that the above is *not* equivalent to

$$0 \to R^n \to R^m \to M \to 0 \tag{12.5}$$

i.e. R^n need not map injectively into R^m. The easiest way to show this is if we classify all modules for which equation (12.5) holds, which turn out to be all modules over PID's, and hence modules that are *not* over PID's (for example, $\mathbb{Z}[x]$ over itself) cannot be written down in the form presented in equation (12.5). In chapter 14, we shall show that a module over a PID shall imply the existence of a chain as given in equation (eq:modOverPidAsPresentaiton), that is it is a sufficeint condition that R is a PID. It is easier to show that it is a necessary condition; we shall show this at the end of this seciton, for now we will assume the complete result to point out some interesting observations.

It is informative to the reader to see how presentations of modules compares to presentations of groups. Note that $\mathbb{Z}$ is a Principal Ideal Domain, and so for any [finitely generated?] abelian group (i.e. any $\mathbb{Z}$-module), we have the exact sequence:

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^m \to G \to 0$$

Furthermore, any subgroup of a free group is free (see appendix B), and so more generally for any group G and appropriate sets A and B:

$$0 \to F(A) \to F(B) \xrightarrow{\varphi} G \to 0$$

where $F(A) = \ker(\varphi)$. Hence, the fact that there are modules that do not admit a chain like in equation (12.5) is a property that distinguishes groups from modules. In fact, there are even modules for which there is no finite length chain of free modules. If we interpret equation (12.5) as saying that the relations on a module is free, then if we extend the length by 1:

$$0 \to R^n \to R^m \to R^p \to M \to 0$$

can interpret the extra homomorphism as the relations of the relations, where the "relations of the relations" are free. Some modules do not have a finite length chain of this sort!! We give a name to such a chain

Definition 12.3.27: Free Resolution

Let R be a ring and M an R-module. A *free resolution* of M is an *exact* chain complex (definition 12.2.11)

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0$$

in which every F_i is a free R-module. The resolution has *finite length* k if $F_i = 0$ for every $i > k$, so that it reads

$$0 \to F_k \to F_{k-1} \to \cdots \to F_1 \to F_0 \to M \to 0$$

and M is then said to have a *finite length free resolution* of length k.

Exactness is what makes the resolution say something about M: at F_0 it says that the relations among the chosen generators are exactly the image of F_1, at F_1 that the relations among *those* relations are the image of F_2, and so on.

By definition, all free modules have a finite length free resolution of length zero since

$$0 \to R^{n_0} \to M \to 0$$

i.e. $R^{n_0} \cong M$. By the universal property of free modules, any R-module M has a free resolution (can you see why). The resolution is called a *free* resolution since we are using free-modules. If we use a different type of module, we will get a different type of resolution[12].

Example 12.3.28: Free Resolution

Let G be a finitely generated abelian group. Then by the fundamental theorem for such groups (theorem 5.1.8),

$$G \cong \mathbb{Z}^n \oplus \mathbb{Z}/n_1 Z \oplus \mathbb{Z}/n_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

As pointed out, all free-modules have a resolution of length 1, namely

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^n \to 0$$

[12] In chapter 17⋆, we will introduce projective, injective, and flat modules, and then in part XI, we will introduce projective, injective, and flat resolutions

On the other hand, any cyclic torsion group has a resolution of length 2:

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$$

These results can be combined to show that G always has a resolution of length 2. Free resolution that must be at least of length 3 will be given in part XI.

Finally, as promised earlier, we shall show that it is a necesary conditions that R-modules admitting resolutions of length two must be over PID's:

Proposition 12.3.29: Free Resolution Of Length 2, Then Pid

Let R be an integral domain such that every R-module M admits a resolution of length 2. Then R is a PID. In particular, for any resolution:

$$R^n \to M$$

There exists an R^m such that

$$0 \to R^m \to R^n \to M$$

***Proof*:**
Choose any $I \subseteq R$. Then R/I is an R-module and

$$R \to R/I$$

is an R-module homomorphism. By the property of R, the above homomorphism can be extended to a free-resolution of length 2. Since the kernel of the above homomorphism is I, we get:

$$0 \to I \to R \to R/I$$

where we must assume I is a free R-module. Since R is of rank 1, I cannot be greater than rank 1 (see exercise ref:HERE (above direct sums and homomorphisms)). But then, since I is free, I must be generated by this single element, making it free. Since this is true for an arbitrary ideal $I \subseteq R$, this implies that every ideal of R is principal making R a PID, as we sought to show.

12.3.4 Finitely Generated Modules and Noetherian Modules

As with many structures in mathematics, the "infinite" case (in our case infinitely generated objects) have different behavior than the finite case and require some different tools to analyze (possibly some analysis to use the notion of limits). We thus study the case where modules are finitely generated or all submodules are finitely generated.

Let R be a ring and let M be a left R-module. We first define what types of modules are *finitely generated*, and some consequences of being finitely generated,

Definition 12.3.30: Finitely Generated R-Module

Let M be an R-module. Then M is said to be finitely generated if there exists an $n \in \mathbb{N}_{>0}$ such that $M \cong_R R^n/K$ for some R-module K, that is there exists a map φ where

$$R^n \xrightarrow{\varphi} M \to 0$$

is exact so that $\ker(\varphi) = K$ and $R^n/K \cong_R M$.

Though we have defined finitely generated modules, we usually require the stronger condition that all submodules of a finitely generated R-module to be finitely generated (which is not true for every finitely generated modules)

Example 12.3.31: Finitely Generated But A Non-finitely Generated Submodule

Take $R = k[x_1, x_2, ..., x_n, ...]$ be the ring with infinitely many indeterminates over a field. Then R is cyclic over itself, generated by 1, but the R-submodule, i.e. ideal, $(x_1, x_2, ..., x_n, ...) \subseteq R$ is not finitely generated as an R-module.

The main inhibitor to this is We will phrase this condition a little differently and show it's equivalent to the previous statement:

Definition 12.3.32: Noetherian R-Module and Ring

Let M be a left R-module Then:

1. M is said to be *Noetherian R-module* or to satisfy the *ascending chain condition on submodules* (or A.C.C. On submodules), that is there are no infinite increasing chains of submodules, i.e., whenever
$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$
is an increasing chain of submodules of M, then there is a positive integer m such that for all $k \geq m$, $M_k = M_m$ (so the chain becomes stationary at stage m: $M_m = M_{m+1} = M_{m+2} = \cdots$)

2. The ring R is said to be Neotherian if it is Neotherian as a left-module over itself, i.e., if there are no infinite increasing chains of left-ideals in R.

One can have an analogous definition of A.C.C. for right and two-sided ideals in (possibly noncommutative) ring R. For noncommutative rings, left and right A.C.C need not relate (you may be left A.C.C but not right, and vice-versa, see exercise ref:HERE)

Theorem 12.3.33: Equivalent Conditions To Noetherian

Let R be a ring and let M be a left R-module. Then the following are equivalent

1. M is a Noetherian R-module

2. Every nonempty set of submodules of M contains a maximal element under inclusion

3. Every submodule of M (including M) is finitely generated

***Proof*:**
(1) ⇒ (2). Let S be any collection of submodules. Let's pick some sub-module $M_1 \in S$. If S is maximal, then we're done. If not, then M_1 is not maximal, meaning there exists an M_2 such that $M_1 \subseteq M_2$. If M_2 is maximal, we're done. If not, pick M_3 in a similar fashion. Keep picking sub-modules M_i. If none of them are maximal, then by the axiom of choice, we can pick an infinitely long chain of submodules

$$M_1 \subseteq M_2 \subseteq \cdots \subseteq M_k \subseteq \cdots$$

which is a contradiction to the statement that M is Noetherian.

For (2) ⇒ (3), Let M be a module. Pick $x \in M$. If $\langle x_1 \rangle = M$, then M is finitely generated. If $\langle x_1 \rangle = N_1 \subsetneq M$, then pick another element $x_2 \in M - N_1$. If $\langle x_1, x_2 \rangle = M$, we're done. If $\langle x_1, x_2 \rangle = N_2 \subsetneq M$, then pick another element $x_3 \in M - N_2$. Either this stabalizes, or we can craete an infinite chain:

$$N_1 \subseteq N_2 \subseteq \cdots \subseteq N_k \cdots$$

This is a collection of submomdules of M, and so by the 2nd condition there must exist a maximal module, say N. Since every module in this chain is finitely generated, N mut be finitely generated, and since it's maximal adding one more element to N, $N + x$ and taking it's closure, $\overline{N + x} = N'$ must make the new larger module N' equal to M. Hence, M is finitely genearted.

For (3) ⇒ (1), Take any ascending chain of modules

$$N_1 \subsetneq N_2 \subsetneq \cdots \subsetneq N_k \subsetneq \cdots$$

then $N = \bigcup_{i=1}^{\infty} N_i$ is a module, and by assumption it is finitely generated, say by $\{x_1, x_2, ..., x_n\}$. By definition of N, for each x_i, there exists a j such that $x_i \in N_j$. Since N is the union of the modules N_j, it must be that at some finite distance into the chain, a module contains all the generators. But that means the chain stabalizes.

Note that if M is a Noetherian R-module, then R need not be Noetherian. If R is noetherian, then M is a Noetherian R-module? (Use the fact that the direct sum and quotient of a noetherian ring is noetherian, an the uni prop of free modules).[13]. In proposition 9.4.18, it was shown that rings where $1 \neq 0$ have maximal ideals. Since all rings are modules over themsleves, it might seem that all rings are Noetherian modules due to the second conditions. However, this is certainly not the case, consider $R[x_1, x_2, ..., x_n, ...]$ over itself and the chain $(x_1) \subsetneq (x_1, x_2,) \subsetneq \cdots$. The subtly lies in the fact that the maximal element must lie in the collection of submodules, while the maximal element in the previous example is the union of all the submodules.

Many textbooks introduce Noetherian modules as modules where each submodule is finitely generated. The reason the A.C.C. was chosen for this textbook was for it's greater applicability in many other fields. Notions of being "Noetherian" are pervasive in mathematics, but not all structure are "Noetherian" by having every substructure be finitely generated (sometimes, the notion of substructure is not even well-defined)[14]. The A.C.C. allows us to still consider those structures.

Since Principal Ideal Domain are *a fortiori* Noetherian rings, finitely generated modules over PID's are are also Noetherian, and all the results we discussed earlier apply to such modules (including PID's over themsleves).

[13]This mirrors a more general theme in which a property of the underlying ring "becomes" the property of all modules over that ring. Can you see how we can (in a sense) call a field a "free ring"?

[14]See Noetherian topologies for an interesting example

Finite Generation vs. Finite Type

Being finitely generated as an R-module and finitely generated as an R-algebra are very differnet:

Example 12.3.34: Finite Generation: Module vs. Algebra

Take $R = k[x]$ as an k-algebra and consider the natural k-module structure. Then it is an *infinitely generated* k-module generated (for example by $\{1, x, x^2, x^3, ...\}$ however it is a *finitely generated* k-algebra generated by $\{1, x\}$.

Mo generally, contrast an finitely generated R-module M to a finitely generated R-algebra M:

$$R^{\oplus A} \twoheadrightarrow M \quad R[A] \twoheadrightarrow M$$

Definition 12.3.35: Finite Type

Let A be an R-algebra. Then A is of *finite type* (over R) if A is finitely generated as an R-algebra.

Often, instead of finite type the terms "finitely generated as an R-algebra" or even "finitely generated R'algebra" are often used interchangeably.

Exercise 12.3.2

1. Is a subring of a Noetherian ring Noetherian?
2. (If you know some complex analysis) In this exercise, we try to show how the Noetherian property is not stable by taking subrings and show demonstrate a correspondence between being Noetherian and how the zero of functions are in some way "well-behaved". Consider the following chain of subrings:

$$\begin{gathered}\mathbb{R}[x] \\ \subseteq \\ \text{Analytic functions on } \mathbb{R} \\ \subseteq \\ \text{Analytic functions on } [0,1] \\ \subseteq \\ \text{Analytic functions on } (0,1) \\ \subseteq \\ \text{Analytic functions at } 0 \\ \subseteq \\ \text{Smooth functions near } 0 \\ \subseteq \\ \text{Former Power series}\end{gathered}$$

which of these rings are Noetherian, and which are not?

3. Show that a finitely generated R-module over a Noetherian ring R is itself Noetherian
4. Let M be an R-module over an integral domain R. Then if M is of rank n, any submodule $N \subseteq M$ can be of rank at most $n \leq m$ (hint: take the ring of fractions).

12.3.5 Direct Sums and Homomorphisms

For the first time in this book, we will pay closer attention to hom-sets. The structure of hom-sets of modules will allo us to determine many important properties of modules.

Proposition 12.3.36: Direct Sums And Homomorphisms

Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be a collection of modules with I and J indexing sets. Then:

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right) = \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$$

and

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) = \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$$

***Proof*:**
let $\varphi : \oplus_{i\in I} M_i \to N$. Then for any $(\alpha_i)_{i\in I} \in \oplus_{i\in I} M_i$, there are only finitely many non-zero entries. For ease of visualization, lkets represent the element $(\alpha_i)_{i\in I}$ as

$$(\alpha_1, \alpha_2, ..., \alpha_n, 0, ...)$$

This represetnation can be confusing on two terms: we have implicitly chosen an “order” of the elements and re-labelled the indices of the finite eleents accordingly, and it makes the set I look like it is *countable*, which is not necessarily the case. These two remarks should be kept in mind while working with this represetnation of the element.

Since there are n elements, write each of them as:

$$\alpha_i = \underbrace{0 + \cdots + 0}_{i-1 \text{ times}} + \alpha_i + \underbrace{0 + \cdots + 0}_{n-i \text{ times}}$$

Now, the finiteness condition was key here so that we may use linearity inductively, so that for any $\varphi \in \mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i, N\right)$

$$\varphi((\alpha_i)_{i\in I}) = \varphi(\alpha_1) + \cdots + \varphi(\alpha_n)$$

Now, it is immediate that each function on the right hand side is naturally isomorphic to a function $\varphi_i \mathrm{Hom}_R(M_i, N)$. Labelling eahc of thes functions a φ_i, we get:

$$\varphi \to (\varphi_i)_{i\in I}$$

which is clearly an isomorphism.

The direct product case is done almost identically, except we don't need to worry about finite additivity, since for any

$$\varphi(\alpha) = (\beta_i)_{i\in I}$$

we can define:

$$\varphi_i(\alpha) = \beta_i$$

which will give us the desired isomorphism, completing the proof.

As a consequence of this, we can simplify:

$$\mathrm{Hom}_R\left(\bigoplus_i M_i, \bigoplus_j N_j\right) \cong \prod_{i,j} \mathrm{Hom}_R(M_i, N_j)$$

If you are comfortable with linear algebra, then in the case where both direct sums have only finitely many components (also known as *summands*), then if φ_{ji} represent's the component function taking in the ith component in the domain and is in the jth component of the domain, then: the there is an isomrophism mapping $\varphi \mapsto (\varphi_{ij})$ where the maps is defined as:

$$\varphi\begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \cdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix}\begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

Showing that there is a natural relation between $\mathrm{Hom}_R(\oplus_i^m M_i, \oplus_j^n N_j)$ and matrix rings, especially if $M_i = N_j$. More on that in chapter 53, section 51.

Notice how important linearity is to split up the homomorphsim in direct products in the *domain*. To make sure that no misconceptions are being formed, here is a counter-example when a function is not linear:

Example 12.3.37: Hom's And Products Not Commuting in Domain

Let $f(x, y) = xy$. Then $f(x, 0) = 0$ and $f(0, y) = 0$, showing this breakdown does not happen. This was a particualr example, not a proof of the non-existence of the isomorphism. However, we see with f that the output of f must always depend on both x and y, so any two functions f_1 and f_2 solely dependning on the x or y variable cannot be used to reconstruct f using a direct product TBD.

On the other hand, if $f : M \to A \oplus B$ was some function, then by the universal property of the product we have that $f_a : M \to A$ and $f_b : M \to B$ exists. (Think more on why this exists but the other doesn't so nicely.. think about what it means for the uni prop of the co-product and how it relates to hom-sets.. TBD)

Exercise 12.3.3

1. Show that if M is an R-module, then $\mathrm{Hom}_R(R, M) \cong M$
2. Show that $\mathrm{Hom}_R(M \oplus M, R) \cong$(want to show the case where direct product becomes matrices! i.e. bringing out the direct product makes a $M_2(R)$ matrix, whatever should be there instead of R)

3. Let $\varphi : A \to B$ be an R-module isomorphism. Show that $\Phi : \text{End}_R(A) \to \text{End}_R(B)$ where $\Phi(f) = \varphi \circ f \circ \varphi^{-1}$ is a ring isomorphism, and if R is commutative an R-algebra isomrophism.

4. show that if $I \subsetneq R$ is a proper ideal, $I \cong S$ as an R-module isomorphism $\varphi(e) = 1$ a,d $e^2 = e$, then φ is in fact a ring-isomrophism, so I has the ring structure of S (though it cannot have the ring structure of R, since it cannot contain a unit).

13

Modules over Fields – Vector Spaces and Linear Algebra I

This chapter focuses on the study of modules where the ring are fields (i.e. [unital] commutative division ring). We'll use the letter k (from German Körperor) or the symbol $\mathbb{F}$ to represent an arbitrary field. As we've mentioned in definition 12.1.9, a module over a field is called a vector-space due to the historical precedence of the terminology which is now widely accepted in and beyond mathematics (for example, the term vector-space is used in physics, biology, and chemistry). This precedence of modules over fields comes from the fact that historically, fields were studied before general rings. Many problems in physics in the 19th century turned out to requires vector-spaces (ex. solutions to ODEs formed vector spaces, many important spaces of functions formed vector-spaces, the Cartesian plane formed a vector space), leading to precedence in nomenclature for modules over fields. Mathematically, giving another name to modules over fields is not unreasonable: as we'll see, having a field act on a module is so restrictive (or powerful) that we will only need to keep track of two pieces of information to fully understand the structure and the classification of vector-spaces: the underlying field k, and a new number called the *dimension* of the vector-space[1]. No algebraic structure we've encountered requires so little data to be able to classify them (for example, the classification of finite groups required tens of thousands of paper's over almost a century, while the classification of vector-space is regularly taught in a 1st or 2nd year undergraduate class).

Due to the simplicity of vector-space structures, vector space homomorphisms (k-module homomorphisms) will be completely classified. (A further word on this before talking a bit about dual theory and geometry)

As a consequence of the historical precedence of vector-spaces, a separate vernacular was developed

[1] This is particularly true for finite dimensional vector-spaces. In the infinite dimensional case, there is usually more information present than in the finite dimensional case, and so we require more information to sufficiently classify infinite vector-spaces, see ref:HERE

Terminology over any Ring	**Terminology over Fields**
M is an R-module	M is a vector space over R
m is an element of M	m is a vector in M
N is a submodule of M	N is a subspace of M
M/N is a quotient module	M/N is a quotient space
M is a free module of rank n	M is a vector space of dimension n
M is a nonzero cyclic module	M is a 1-dimensional vector space
$\varphi : M \to N$ is an R-module homomorphism	$\varphi : M \to N$ is a (R) linear transformation
M and N are isomorphic as R-modules	M and N are isomorphic vector spaces
the subset A of M generates M	The subset A of M spans M
$M = RA$	each element of M is a linear combination of elements of A, i.e., $M = \text{Span}(A)$

for modules over fields that is now considered standard:

13.1 Vector Spaces and their Properties

Vector spaces are famous for having a very simple structure: Given a vectorspace V over k, then V is isomorphic to k^N for appropriate N. In a linear algebra class, we say that every vector space has a basis. This N will be called the *dimension* of the vector-space. In many linear algebra classes, the attention is limited to $N \in \mathbb{N}$. However, this section will allow N to be of any cardinality (ex. $|\mathbb{N}|$ or $|\mathbb{R}|$). This is to show some of the power of vector-spaces: the (algebraic) structure doesn't get more complicated when dealing with infinite dimensions[2]

Before these results are shown, for those who gain an intuition via examples, here are a couple more example's of vector-spaces

Example 13.1.1: Vector Spaces
We already gave 4 examples in example 12.1.10. We give a few more here

1. We can generlize k^n as follows: notice that we can define k^n as

$$k^n = \{f \mid f : \{1, ..., n\} \to \mathbb{R}\}$$

where addition is defined pointwise, and since k is commutative we can define a natural k-action on k^n (can you see it?). Any function $f \in k^n$ will tell us what value we should have in the component i. For example, if $f(1) = \pi$, $f(2) = 3$, an $f(3) = \sqrt{10}$, then this function is equivalent to:

$$(\pi, 3, \sqrt{10})$$

Thus, We can consider:

$$k^{\mathbb{N}} = \{f \mid f : \mathbb{N} \to \mathbb{R}\}$$

To be the $\mathbb{N}$-dimensional vector-space, or $k^{\mathbb{R}} = \{\mathbb{R} \mid f : \mathbb{R} \to \mathbb{R}\}$ to be the $\mathbb{R}$-dimensional vector space

2. Generalizing the above example, let k be a field, X a set, and $V = \text{Hom}_{\mathbf{Set}}(X, k)$ (another common notation is k^X). Then we can make V into a vector-space via point-wise operation in the codomain: for any $f, g \in V$, $c \in k$, $f + g$ is defined to be:

$$(f + g)(x) = f(x) + g(x) \qquad \forall x \in X$$

while fg and cf is defined as:

$$(fg)(x) = f(x)g(x) \qquad (cf)(x) = cf(x) \qquad \forall x \in X$$

These definitions are possible since the image of any function in V lands in a field, where all the above operations are well-defined. If we take the collection of lienar transformation from V to k (where V is a vector space), then we often denote that as V^*. Note that since k is also a ring, we can also define $(fg)(x) = f(x)g(x)$, making $\text{Hom}_{\mathbf{Set}}(X, k)$ into a k-algebra.

Often, the set X is given its own properties (a popular example are X is a locally compact Hausdorff space, or X is a smooth manifold). Then we often try to deduce more information about X given the structure of hom-set $\text{Hom}_{\mathbf{Set}}(X, k)$.

[2]On the other hand, the set-theoretical and analytic structure (if there is a topology present) gets more complicated, and arguably non-trivial, in infinite dimensions. For example, in infinite dimensions, surjective does not imply injective (similar to the proof for infinite sets), and there will exist linear transformations that are not continuous (for the appropriate topology).

3. Let $R = k[x]$ for some field x. Then $k[x]$ is a k vector-space with action $c \cdot p(x) = cp(x)$. More generally, if $R = k[x_1, x_2, ..., x_n, ...]$ is some polynomial ring over several variables, then R is a k vector-space with $c \cdot p(x_1, x_2, ..., x_n, ...) = cp(x_1, x_2, ..., x_n, ...)$.

4. Let $k = \mathbb{C}$ an $V = \mathbb{C}^n$. Then $\mathbb{C}^n$ is a complex vector space with the action:

$$z \cdot (z_1, ..., z_n) = (zz_1, ..., zz_n)$$

There is a second natural action, known as the *conjugate action*:

$$z \cdot (z_1, ..., z_n) = (\overline{z}z_1, ..., \overline{z}z_n)$$

More generally, if V is a vector-space over $\mathbb{C}$ (also known as a complex vector space), then $\overline{V}$ denote the same space with the element being conjugated before the action is applied.

5. (If you know some ODE's) Given a linear ordinary differential equation, the set of all solutions form's a vector-space. Take for example $\varphi : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$ where:

$$f \mapsto f'' + f$$

Consider the subspace V where

$$V := \{f \in C^\infty(\mathbb{R}) \mid f'' + f \equiv 0\}$$

Then using elementary ODE theory, we find that:

$$V = \operatorname{span}_{\mathbb{R}}(\{e^t, e^{-t}\})$$

The fact that all solutions to the ODE $f'' + f = 0$ is a linear combination of these two solutions is a remarkable property of *linear* ODE's. This simple property of linearity can be taken very far: in the world of PDE's (Parital Differential Equations), if you have something called a *linear PDE*, which means that if you have two solutions, then add them and/or scale the terms, the result is another solution. The power we get from this is that we can then see if this is generalizable to taking *limits* of some very simple solutions to the PDE. As a common example, let's say the solution is of the form $\cos(\alpha x)$ or $\sin(\beta x)$ for any α, β. Then under the right conditions[a], the collection of these (after being closed under addition) is *dense* in $L^2(S^1)$, meaning we can use limits to construct *any* $L^2(S^1)$ function and find another solution to our PDE. A common place this particular example is used is *Fourier Analysis*[b]

6. (If you know Differential Geometry) Let M be a smooth n-manifold and consider the collection of all $\varphi : C^\infty(M) \to \mathbb{R}$. Then the collection of all *derivations at p*[c] for some point $p \in M$, denoted $\operatorname{der}_p(M) \subseteq \operatorname{Hom}_{\mathbf{Smooth}}(C^\infty(M), \mathbb{R})$ forms a vector-space of dimension n, and is a common definition of the *tangent space* of the manifold M at the point p[d]

[a] See a textbook covering Fourier Analysis, ex. Pugh or Folland
[b] See [21]
[c] linear function where $\varphi(fg) = \varphi(f)g + f\varphi(g)$
[d] See [18, Tangent Vector as Derivation, Dimension of the Tangent Space]

Now, we shall develop a more sophisitaced notion of generating set:

Definition 13.1.2: Linear Independence And Basis

Let M be an R-module.

1. A subset $S \subseteq M$ is said to be *linearly indepenent*, or a set of *linearly independent elements*, if for any equation
$$a_1m_1 + a_2m_2 + ... + a_nm_n = 0$$
with $a_1, a_2, ..., a_n \in R$ and $m_1, m_2, ..., m_n \in S$, then $a_1 = a_2 = \cdots = a_n = 0$. If this is not the case, the elements are called *linearly dependent*. In other words, if $I = \{m_1, m_2, ..., m_n\} \subseteq M$, so that $\iota : I \to M$ is injective, then:
$$\begin{tikzcd} F^R(I) \arrow[r, "\varphi"] & M \\ I \arrow[u, "j"] \arrow[ur, "\iota"'] & \end{tikzcd} \tag{13.1}$$
commutes[a] and φ is injective. Note that I can be an arbitrary indexing set, that is for every $i \in I$, $m_i \in M$ corresponds to it, and then we would have an injection $\iota : I \to M$ which lifts to $\varphi : F^R(I) \to M$. If $R = k$, then the elements are called *linearly independent vectors*.
2. A *basis* for M is an *set* $\{a_1, a_2, ..., a_n\}$ of elements that is linearly independent and which generate sM (i.e. span M).
3. An *ordered basis* for M is an *ordered set* (a tuple) $(a_a, a_2, ..., a_n)$ of linearly independent elements which generates M (i.e. span M).

[a] The diagram commuting is the universal property of free modules

The point of an *ordered basis* is so that $A \subseteq M$ gives a *unique* isomorphism:
$$\varphi : F^R(A) \cong M$$
i.e. The φ in equation (13.1) is both injective and surjective. In particular, two ordered bases will be considered different even if one is a rearrangement of the other (hence why it's ordered).

13.1.3 Convention In this section, all bases shall implicitly be ordered basis. The convention shall be dropped when working with tensor products of vector spaces.

Note that the φ from the commutative diagram in equation (13.1) has as domain all *finite* combinations of elements of $\{m_i\}_{i \in I}$, so that when writing:
$$\sum_{i \in I} r_i m_i$$
only finitely many r_i are nonzero. If φ is injective, then $\sum_{i \in I} r_i m_i = 0$ if and only if $r_i = 0$ for all $i \in I$. Sometimes to emphasize the fact that it's ordered, a basis is called an *ordered basis*. In terms of indexing sets, the ordered elements are generalized to elements having different indexes. Notice how being linearly independent is different from proposition 12.3.6(3), where if $x \in \oplus_{i \in I} N_i$ implies

there exists unique elements $v_i \in N_i$ such that $x = \sum_i v_i$. We are adding another requirement on $\bigoplus_i v_i$, namely that the function φ in the above definition is injective.

A module M having a basis is in fact a very restricting condition:

Lemma 13.1.4: Basis iff Free

Let M be an R-module. Then M is free if and only if it admits a basis, in paritcular $B \subseteq M$ is a basis if and only if the natural homomorphism $R^{\oplus B} \to M$ is an isomorphism

***Proof*:**
This is immedaite from the definition: if $B \subseteq M$ is linearly independent and generates M, then $\varphi : R^{\oplus B} \to M$ is injective and surjective, and if $\varphi : R^{\oplus B} \to M$ is an isomorphism, then $\varphi(B)$ is a subset of M which generates it and is linearly independent.

Example 13.1.5: Linearly Dependent and Independent Sets

1. Let $M = \mathbb{Z}/n\mathbb{Z}$ be a $\mathbb{Z}$-module. Then:

$$n \cdot [1]_n = 0$$

   but naturally $n \neq 0$. More generally, if R is a ring with a zero divisor, and $(z) \subseteq R$ is the sub-module generated by the zero-divor (z), then if $zz' = 0$, $z' \cdot z = 0$,showing that the singleton z is linearly dependent.

2. Let $R = \mathbb{C}[x, y]$ be a module over itself. This ring has no zero divisors, hence the above does not apply. However, that there are still linearly dependent elements, for example:

$$y \cdot x + (-x) \cdot y = xy - xy = 0$$

   showing that x, y are *not* linearly independnet over $\mathbb{C}[x, y]$. However, if we consider $\mathbb{C}[x, y]$ over $\mathbb{C}$, then $\{x, y\}$ *are* linearly independent, since $\mathbb{C}[x, y]$ is the free $\mathbb{C}$-algebra. Hence:

$$F^{\mathbb{C}}(\{x, y\}) = \mathbb{C}^2 \to \mathbb{C}[x, y]$$

   is an injective function. However, $\{x, y\}$ is not a basis since the above homomorphsim is not surjective, since nothing maps to xy. Is there any finite that will create a surjective map?

3. If R is any ring (which includes rings with torsion elements), then $(1) = R$, and $r \cdot 1 = r1$ if and only if $r = 0$. Hence,

$$R \mapsto R$$

   is an injective homomorphism (even surjective, making it a basis). This shows that the choice of subset of the module effects whether the set if linearly independent or a basis.

   Though R has a basis when over itself, show that $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Z}$ does *not* have a basis, or even a linearly independent set.

As we saw in the above example, some modules have linearly independent sets, some don't, and others have a basis. By a simple applications of Zorn's lemma and some manipulation of definition, we can show that every module has a maximally linearly independent set:

Lemma 13.1.6: Maximally Linearly Independent Set

Let M be an R-module and let $S \subseteq M$ be a lienarly independnet set. Then there exsits a maximal linearly independent subset of M containing S.

***Proof*:**
This is an application of Zorn's lemma: Let $\mathcal{L}$ be the collection of linearly independent subset of M containing S ordered by inclusion. Since $S \in \mathcal{L}$, $\mathcal{L} \neq \emptyset$. If we take any chain $\mathcal{C} \subseteq \mathcal{L}$, and take $C = \cup\mathcal{C}$, then C is linearly independent, since it it were linearly dependent, then only finitely many linear element would be used in the equation, but then they all belong to some $C_i \in \mathcal{C}$, which by assumption is linearly independnet. Hence, by Zorn's lemma $\mathcal{L}$ contains a maximally linearly independent set containing S.

13.1.7 Trivia The Axiom of Choice and the above lemma are in fact equivalent, hence it is impossible to prove the above statement without using Zorn's lemma.

If S is a maximally linearly independent set containing $\emptyset$, we shall simply say that S is maximally linearly independent. Note that being maximally linearly independent does not guarantee a unique size or that it generates the entire module, for exmaple $\{2\} \subseteq \mathbb{Z}$ is maximally linearly independent but does not generate $\mathbb{Z}$. In the special case where $R = k$ is a field so that we have a vector-space V, then a maximally linearly independent set is indeed a basis!

Theorem 13.1.8: Vector Space has Basis

Let V be a vector space over k and let B be a maixmally lienarly independnet set. Then B is a basis of V, hence V is a free k-module. Furthermore, if S is a lieraly independent set, then S is contained in a basis B.

***Proof*:**
We need to show that B generates V. Let $v \in V$, $v \notin B$. Then by maximality of B, $B \cup \{v\}$ is linearly dependent, hence there exists $v_1, v_2, ..., v_n$ and $c_0, c_1, ..., c_n$ such that

$$c_0 v + c_1 v_1 + \cdots + c_n v_n = 0$$

with not all $c_i = 0$. Note that $c_0 \neq 0$ or else we would get $c_1 v_1 + \cdots + c_n v_n = 0$ with not all $c_i \neq 0$, contradicting linearly independence of elements in B. Since k is a field, c_0 is a unit, hence:

$$v = (-c_0^{-1} c_1) v_1 + \cdots + (-c_0^{-1} c_n) v_n$$

But then, $v \in \text{span}(B)$, showing that B generated V. By lemma 13.1.4, V is a free k-module.

Furthermore, if S is lienarly independent, then S is contained in a maximally linearly independent set, say B. But by our above reasoning B is a basis, hence S is contained in a basis, as we sought to show.

Hence, every vector-space is free. Before continuing our exploration of bases, a quick comment must be made: we can define the dual notion of *minimally spanning set* and have proved the same statment in the reverse direction namely:

Proposition 13.1.9: Vector space, Minimally Spaning then Basis

Let V be a k-vector space and let B be a minimally spanning set. Then B is a basis for V, making V a free k-module. Furthermore, if S is any spanning set,then S containins a minimal spanning set B.

***Proof*:**
exercise

Corollary 13.1.10: Subspaces Of Vector-Spaces Are Free

Let V be a vector-space and $W \subseteq V$ be a subspace. Then W is free

***Proof*:**
Let $W \subseteq V$ be a subspace. Since W is a k-module, by theorem 13.1.8, it is free.

Note that this is not true for arbitrary free-modules: there exists submodule of free modules which are *not* free[3]. In fact, since any ring R is a module over itself, then it is easy to find counter-examples; any polynomial rings over several variables will have non-free submodules, see exercise ref:HERE. Another simpler example is as follows

Example 13.1.11: Non Free Submodule

Take $\mathbb{Z}_6$ over itself, and consider the ideal (2). Show that it is not free[a].

[a]Hint: it's torsion

In later chapters, we shall see that submodules of free modules over slightly more general rings, namely PIDs, will still be free[4], however this generlization is limited in scope and does not further generlize over more general rings, as we shall discuss later.

The key in theorem 13.1.8 is that every element in a field is a unit; commutativity was not important. Thus, this same proof can work for division rings more generally[5]. Though this set is the maximal linearly independent set, since the partial order on $P(A)$ forms a lattice, there may be many different cardinalities of sets which form maximal generating sets (see exercise ref:HERE for an example where this happens for some rings and groups). We thus ask the question of when do we get bases with different cardinalites, or if there are conditions for which different basis always have the same cardinality. For vector-spaces, every basis *always* has the same cardinality! The following theorem is the key result for the proof:

[3]Though it is true for free groups, see appendix B
[4]see corollary 14.1.7
[5]And will be used in chapter 53, section 51

Theorem 13.1.12: Replacement Theorem

Assume $A = \{a_1, a_2, \cdots, a_n\}$ is a basis for V containing n elements and $B = \{b_1, b_2, ..., b_n\}$ is a set of linearly independent vectors in V. Then there is an order $a_1, a_2, ..., a_n$ st for each $k \in \{1, 2, ..., m\}$, the set $\{b_1, b_2, ..., b_k, a_{k+1}, a_{k+2}, ..., a_n\}$ is a basis of V. In other words, the elements $b_1, b_2, ..., b_m$ can be used to successively replace the elements of the basis A, and A will still be a basis.

***Proof*:**
Let the set A and B be as inthe statement of the theorem. We proceed with induction on k. If $k = 0$, we're done, so let's say $\mathcal{B}$ is a basis for V and

$$(b_1, b_2..., b_k, a_{k+1}, ..., a_n)$$

is a basis for V. Since it's a basis, it's a spanning set, and so for b_{k+1}:

$$b_{k+1} = \beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+1} a_{k+1} + \cdots + \alpha_n a_n \tag{13.2}$$

Notice not all the α_i can be zero, since then b_{k+1} is a linear combination of elements in B, but B is linearly independent. Without loss of generailty, we can re-arrange the above equation so that $\alpha_{k+1} \neq 0$. Then re-arranging, we get:

$$a_{k+1} = \frac{1}{\alpha_{k+1}}(\beta_1 b_1 + \cdots + b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n)$$

Thus, we get that:

$$\text{span}(\{b_1, ..., b_k, a_{k+1}, ..., a_n\}) = \text{span}(\{b_1, ..., b_{k+1}, a_{k+2}, ..., a_n)$$

it not suffice to show that that $\{b_1, ..., b_k, a_{k+1}, ..., a_n\}$ is linearly independent. Let's say:

$$\beta_1 b_1 + \cdots + \beta_{k+1} b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n = 0 \tag{13.3}$$

By equation (13.2), we can make equation (13.3) a linear combination of elements of $(b_1, b_2..., b_k, a_{k+1}, ..., a_n)$:

$$(\beta_1 + \beta_{k+1}\beta_1')b_1 + \cdots + (\beta_k + \beta_{k+1}\beta_k')b_k + (\beta_{k+1}\alpha_{k+1}')a_{k+1} + \cdots + (\alpha_n + \beta_{k+1}\alpha_n')a_n = 0$$

In particular, we can multily this entire equation by $\beta_{k+1}(\alpha_{k+1}')^{-1}\beta_{k+1}^{-1}$ [a] so that the coefficient of α_{k+1} is β_{k+1}. Since this is a basis by the induction hypothesis, all the coefficients must be zero, including β_{k+1}. Thus, we get

$$\begin{aligned}
0 &= \beta_1 b_1 + \cdots + 0 b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n \\
&= \beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n \\
&= \beta_1 b_1 + \cdots \beta_k b_k + 0 a_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n
\end{aligned}$$

where the last line is a basis by the induction hypothesis, and so all other coefficients must be zero, and so the set must be linearly independent, as we sought to show.

[a] The reason for writing the scalar like this is so show that the proof works even if we limit ourselves to division rings

Since the cardinality of the basis is independent of the choice of elements that make the basis, we can define it as an invariant on vector-spaces:

Definition 13.1.13: Dimension

Let V be a vector space over F. Then the cardinality of the basis is called the *dimension* of V, and is denoted $\dim_F(V)$ or $\dim(V)$ if context permits. If V is not finitely generated, then we say that V is infinite dimensional, and we write $\dim V = \infty$.

The key take-away of the Replacement Theorem is that there is a direct relation in cardinality between sets that are linearly independent and sets that are spanning:

Corollary 13.1.14: Steinitz exchange lemma

1. Suppose V has a finite basis with n elements. Any set of linearly independent vector has less than or equal to n elements. Any spanning set has greater than or equal to n elements
2. If V has some finite basis then any two bases of V have the same cardinality

***Proof*:**

1. A be the basis. If a set is linearly independent, then we can use the replacement theorem to systematically replace the basis with the elements of the linearly indepennet set. Thus, it cannot have more than n elements, since a basis has maximum n elements that are linearly independent. If a set is a spanning set, then by corollary 13.1.14, the set contains a basis, say B. Thus, $|B| \leq |A|$.

   Since B is a linearly independent, we can use the replacement theorem to replace the elements of A with B. If $|B| < |A|$, that would contradict the minimality of the set A since A is a basis, and hence a minimal spanning set. Thus, it must be that $|A| \leq |B|$.
2. This is an immediate consequence of the first part, since both both conditions have to match.

As we saw in example 13.1.5, a singleton set with a single element which has torsion is always linearly dependent. In an integral domain, we have some more power:

Corollary 13.1.15: Module Linearity

Let R be an integral domain and let M be a free R-module of rank $n < \infty$. Then any $n+1$ elements of M are R-linearly dependent, i.e., for any $y_1, y_2, ..., y_{n+1} \in M$, there are elements $r_1, r_2..., r_{n+1} \in R$ not all nonzero such that

$$r_1y_1 + r_2y_2 + \cdots + r_{n+1}y_{n+1} = 0$$

***Proof*:**
There is a trick similar to Gauss's lemma. We can embed R in its quotient field (since R is an integral domain, this is indeed an embedding), say k is its quotient field. Since M is a free module,

$$M \cong \underbrace{R \oplus R \oplus \cdots R \oplus R}_{n \text{ times}}$$

and so

$$M \subseteq \underbrace{k \oplus k \oplus \cdots \oplus k}_{n \text{ times}}$$

Since the latter is an n dimensional vector space over F, so any $n+1$ dimensional elements of M are F-linearly dependent. By clearing the denominators of the scalars (ex. multiplying through by the product of all the denominaotrs), we get an R-linear dependence realtion among $n+1$ elements of M.

An important consequence of the existence of a basis is that linear transformation $f : V \to W$ can be defined by first determining where a basis of V maps to, and then extending linearly (that is, if $v = \sum_{\beta_i \in \mathcal{B}} b_i \beta_{\in} W$, then $f(v) = f\left(\sum_{\beta_i \in \mathcal{B}} b_i \beta_i\right) = \sum_{\beta_i \in \mathcal{B}} b_i f(\beta_i)$). Note however that the choice basis is important, we shall get back to this in section 13.2, and that vector spaces may have many different bases (of the same cardinality by the replacement theorem):

Example 13.1.16: Different Basis
As we've defined in example 12.3.2the set

$$e_i = (\underbrace{0, ..., 0}_{i-1 \text{ times}}, 1, \underbrace{0, ..., 0}_{k-i \text{ times}}) \qquad 1 \leq i \leq n$$

is a basis for $\mathbb{Q}^n$. There are countably many different basis for $\mathbb{Q}^n$; can you think of a few? If you know about inner product spaces, can you think of an orthonormal [hamel][a] basis?

[a]The term orthonormal basis is usually reserved for *countable* linear combination for *Hilbert space*. Since the linear combination is not finite, it requires some analysis to define; this would be covered in a text on *Functional Analysis*. If you know about Hilbert spaces, then taking $\mathbb{C}[x]$ as the closure of $\mathbb{C}[x]$, is there an orthonormal basis for it?

13.1.1 Quotient Spaces and Subspaces

Now that we've established the struture of a vector-space, we can some questions about subspaces the structure of quotient spaces, with particular interest in finite vector-spaces to be able to take advantage of counting arguments. We already saw that subspace of vector spaces are free k-module.

Proposition 13.1.17: Dimension Of Quotient Space

Let V be a vector space over k and let W be a vector space. Then V/W is a vector space with $\dim V = \dim W + \dim V/W$. If one term is finite, all terms are finite

***Proof*:**
Certainly V/W is a vector space being a quotient k-module. How would you prove $\dim V = \dim W + \dim V/W$? Why should this be intuitive?

Naturally, since we are working with quotients, we have an associated linear transformation The following is the "homomorphism-equivalent" of the previous proposition:

Corollary 13.1.18: Rank-Nullity Theorem

Let $\varphi : V \to W$ be a linear transfomration of vector spacees over k and $\dim(V) < \infty$. Then $\ker \varphi$ is a subspace of V, $\varphi(V)$ is a subpace of W, and

$$\dim_k(V) = \dim_k(\ker(\varphi)) + \dim_k(\varphi(V))$$

***Proof*:**
follows from previous theorem

Since the dimensions are of the subspaces are invariant to basis, we can give a name to the dimension of the image and null space of a linear transformation

Definition 13.1.19: Rank And Nullity

If $\varphi : V \to W$ is a linear transformation of vector spaces over k, $\ker(\varphi)$ is called the *null space* of φ ,and the dimension of $\ker(\varphi)$ is called the *nullity* of φ. The dimension of $\varphi(V)$ is called the *rank* of φ. If $\ker(\varphi) = 0$, the transformatoin is said to be *nonsingular*.

The reasoning behind the name *nonsingular* is due to the fact that very often in (differentil) geometry, we associated to shapes a vector-space at each point (such an object is called a *vector bundle*) and if the curve "drops rank" this would correspond to a particular function having $\dim_k(\ker(\varphi)) > 0$[6]. For example, the curve $x^3 = y^2$ "drops rank" at $(0, 0)$.

Corollary 13.1.20: Properties of [finite dim.] Linear Transformations

Let $\varphi : V \to W$ be linear transfomration of vector spaces of the same finite dimenison. Then the following are equivalent

1. φ is an isomorphism
2. φ is injective, i.e., $\ker(\varphi) = 0$
3. φ is surjective, i.e. $\varphi(V) = W$
4. φ sends a basis of V to a basis in W

Notice that this proof is saying that if $f : V \to W$ is a k-homomorphism where $\dim_k(V) = \dim_k(W) = n$, then given f is injective, it is surejctive. This fails to be the case even for maps between free moduels over PIDs, for exmaple $f : \mathbb{Z} \to \mathbb{Z}$ mapping $z \mapsto 2z$ is an injective $\mathbb{Z}$-module homomorphism between free $\mathbb{Z}$-modules, but is certainly not surjective! This also fails for infinite dimensional vectors

[6] In particular, if $T_p f$ is the tangent map at p, then $T_p f$ would drop rank

spaces as will be demonsarted by a counter-example after the proof. Hence, the fact that it's sufficient to check injectivity shows how linear maps between finite vector-spaces are akin to maps between finite sets[7].

> ***Proof*:**
> Counting dimension arguments

Note that finiteness of dimension is important in the previous corrollaries: these need not hold in infinite dimension

Example 13.1.21: Infinite Dimensinoal Linear Transformations

Take $V = \ell^2$, the set of all infinite sequences of real numbers satisfying the property:

$$\left(\sum_{i \in \mathbb{N}}^{\infty} |x_i|^2\right)^{1/2} < \infty$$

Then define $f : V \to V$ via:

$$(x_1, x_2, ..., x_n, ...) \mapsto (x_1, \frac{x_2}{2}, ..., \frac{x_n}{n})$$

Then this function is injective, but not surjective. Even more is true; V and f belong to some of the nicest infinite dimensional spaces and functions (V is a Hilbert space and f is a bounded self-adjoint linear operator), and hence this property is not easily rectified.

We now take a look at subspaces. We have already established that all subspaces of a vector-spaces are free. Another importnant property is that every subspace always has a *compliment*, that is if $W \subseteq V$, then there exists a $U \subseteq V$ such that $U \cap W = \{0\}$ and $W + U = V$. This is certainly *not* true for general R-modules, simply let $M = \mathbb{Z}/4\mathbb{Z}$ be a $\mathbb{Z}$-module and take $\{0, 2\} \subseteq \mathbb{Z}/4\mathbb{Z}$. Then we would require $U = \{0, 1, 3, 4\}$, which is not a subgroup. In fact, *no* finite abelian subgroup has this property[8]. However, we may always find such a compliment for subspaces of vector-spaces, given the space is finite dimensional:

Proposition 13.1.22: Existence Of Compliment

Let V be a finite dimensional vector space and $W \subseteq V$ be a subspace. Then there exists a subspace $U \subseteq V$ such that

1. $W \cap U = \{0\}$
2. $W + U = V$

If these two conditions are satisfied, we will write $W \oplus V$ and call it the direct sums (just like in definition 12.3.3).

[7]In fact, this parallels is runs more deeply than it may at first seem, see a text on K-theory for some interesting elaboration on this notion

[8]More particularly, no torsion $\mathbb{Z}$-module has this property. In the Category of **Grp**, there are finite groups that "spilt". See section 17.2 for a generalized notion of splitting and section 5.4.2 for identifying groups that split

***Proof*:**
Let $U \subseteq V$ be a subspace. Choose basis $\{e_1, ..., e_k\}$. Then this basis can be extended to a basis $\{e_1, ..., e_k, ..., e_n\}$ for V. Define the map $\pi_U : V \to V$ given by:

$$\sum_i^n a_i e_i \mapsto \sum_i^k a_i e_i$$

This map is certainly a linear transformation, and $\text{im}(\pi) = U$. The kernel of this map is:

$$\ker(\pi_U) = \langle \sum_{i=k+1}^n a_i e_i | a_i \in k \rangle$$

Notice that $v \in \ker(\pi_U)$ if and only if $v \in \text{im}(I - \pi_U)$, namely:

$$v \in \ker(\pi_U) \Leftrightarrow \pi_U(v) = 0 \Leftrightarrow v - \pi_U(v) = v \Leftrightarrow v \in \text{im}(I - \pi_U)$$

I claim that $v = \pi_U(v) + (I - \pi_U)(v)$. Indeed if $v = \sum_i a_i e_i$:

$$\sum_i^n a_i e_i = \sum_i^k a_i e_i + \sum_{i=k+1}^n a_i e_i = \pi_U(v) + (I - \pi_U)(v)$$

and furthermore $\ker(\pi_U) \cap \text{im}(\pi_U) = 0$, hence setting $W = \ker(\pi_U)$ and $V = \text{im}(\pi_U)$ we get:

$$V = \ker(\pi_U) \oplus \text{im}(\pi_U) = U \oplus W$$

as we sought to show.

The vector space being finite dimensional is key to the proof, as this does *not* generalize to general infinite dimensional vector-spaces (it can be thought of as characterizing *Hilbert spaces*).

Corollary 13.1.23: Finite Generation of Subspace

Let $W \subseteq V$ be a subspace of a finite dimensional vector space. Then W is finitely generated

***Proof*:**
use proposition 13.1.22.

The map used in the above proof is important enough to be given a name:

Definition 13.1.24: Projection Map

Let $\pi : V \to V$ be a linear transformation $U \subseteq V$. Given $V = U \oplus W$, then the *projection map* (with respect to U) is the map $\pi_U : V \to V$ such that $\pi(v) = \pi(u + w) = u$.

Notice $\pi \circ \pi = \pi$, that is, applying the map twice gives back the same result: $\pi(\pi(v)) = \pi(v)$. This in fact characterizes such a map:

Proposition 13.1.25: Characterizing Projection Maps

Let π be linear operator over a [not necessarily finite dimensional] vector space such that $\pi \circ \pi = \pi$. Then π is a projection map.

***Proof*:**
Let $U = \text{im}(\pi)$ and $W = \text{im}(I - \pi)$. Then for any $v \in V$:

$$\pi(v) + v - \pi(v) = v$$

hence $U + W = V$. If $x \in U \cap W$. Since $x \in \text{im}(\pi)$, $x = \pi(y)$ for some $y \in V$, and so:

$$\pi(x) = \pi(\pi(y)) = \pi(y) = x$$

hence, $\pi(x) = x$. Then:

$$\begin{aligned}
x &= \pi(x) && x \in U \\
&= (I - \pi)(x) && x \in W \\
&= \pi((I - \pi)(x)) && (I - \pi)(x) = x \in U \\
&= \pi(x - \pi(x)) \\
&= \pi(x - x) && \pi(x) = x \\
&= \pi(0) \\
&= 0
\end{aligned}$$

Hence $x = 0$, and so $U \cap W = \{0\}$. With the above, we get that

$$V = U \oplus W = \ker(\pi) \oplus \text{im}(\pi)$$

as we sought to show.

Notice how the above proof did not rely on any structure of V (notably, it wasn't important that V had a basis). Thus, if $\varphi : M \to M$ is any R-module homomorphism such that $\varphi \circ \varphi = \varphi$, then

$$M = \ker(\varphi) \oplus \text{im}(\varphi)$$

For this reason, it is much more common to define projection maps as maps that satisfy the property of $\varphi^2 = \varphi$ (i.e., they are *idempotent*).

Exercise 13.1.1

1. Consider $\mathbb{R}[x]$ as an $\mathbb{R}$ module. Take $\mathbb{R}_n[x]$ to be all polynomial where $\deg(p(x)) \leq n$. Show that (evalutaion polynomials has $\left\{(x-a), (x-a)^2, ..., (x-a)^n\right\}$ as a basis). (This might at seem counter-intuitive at fisrt after doing some rings, but it's really cool!)
2. Let R be a commutative ring that is *not* a field. Show that not all R-modules are free, i.e. find a counter-example [Hint: Look back at the examples given in section 12.1]
3. (More challenging) Let R be any ring (not necessarily commutative). Show that if any left (or right) R-module is free, then R is a divison ring.

4. Show that $(x, y) \subseteq \mathbb{C}[x, y]$ is not a free $\mathbb{C}[x, y]$-submodule.

5. Show that any uncountable set of $\mathbb{C}[x]$ is linearly dependent.

6. (Right shift operator)

7. Let $R = k[x_1, ..., x_n]$ and $I = (x_1, ..., x_n)$. Show that I^m is a k-vector space with basis $\left\{x_{i_1}^{r_1} x_{i_2}^{r_2} \cdots x_{i_s}^{r_s} \mid \sum r_s = m, 1 \leq i_j \leq n\right\}$ (that is, all forms of degree m).

A lot of exercices in linear algebra are about determining if a set is linearly independent, spanning, (more?). These exercices are to practice these skills:

13.2 Linear Transformations and Matrix Representation

As mentioned at the beginning of the chapter, k-module homomorphisms are called linear transformation in the theory of vector-spaces. In this section, we will show how every linear transformation between finite dimensional vector-spaces can be represented as a matrix. Since all vector-spaces are free with an invariant basis number, when defining a linear transformation we only need to be concerned with where the basis get's mapped to. If $\varphi : V \to W$, where V and W are finite dimensional vector-spaces over k, and $\varphi \in \mathrm{Hom}_k(V, W)$. Let $\beta_V = \{v_1, v_2, ..., v_n\}$ and $\beta_W = \{w_1, w_2, ..., w_m\}$ be (ordered) bases for V and W respectively. Then we are only concerned where the v_i's map to, that is, what $\varphi(v_j) = w_j$ is. Since they map into the vector-space W, $w_j = \sum_{i=1}^{m} b_{ij} w_i$, $b_{ij} \in k$,

$$\varphi(v_j) = \sum_{i=1}^{m} b_{ij} w_i$$

Let $M_{\beta_V}^{\beta_W}(\varphi)$ be the $m \times n$ matrix whose ij entry is b_{ij} (i.e. the coefficients of w_i given v_j). This matrix is given a name:

Definition 13.2.1: Matrix Representation

Let $\varphi : V \to W$ be a linear transformation between two finite dimensional vector spaces with dimension n and M and basis β_V and β_W respecitvely. Then the *matrix representation* $M_{\beta_V}^{\beta_W}$ of φ (also written as $M_{\beta_V}^{\beta_W}(\varphi)$) is

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where each entry ij is the coefficient of the ith basis element of β_W in the linear combination of where the element v_j maps.

If a matrix A is a matrix representation of a linear transformation φ, we say that A *represents* φ *with respect to the basis* β_V, β_W. Similarly, φ would be the linear transformation represented by A with respect to the bases β_V, β_W.

With the matrix representation of φ, we can re-write $\varphi(v) = w$ as

$$\begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \tag{13.4}$$

where $v = \sum_{i=1}^{n} a_i v_i$ and $\varphi(v) = \sum_{i=1}^{m} b_i w_i$. Sometimes, the following notation is used as a shorthand for (13.4): $[\varphi(v)]_{\beta_W} = M_{\beta_V}^{\beta_W}[v]_{\beta_V}$.

Example 13.2.2: Matrix Representation

1. Let $V = \mathbb{R}^3$, $W = \mathbb{R}^2$, and take $\beta_V = \{(1,0,0),(0,1,0),(0,0,1)\}$ and $\beta_W = \{(1,0),(0,1)\}$ (i.e., the standard basis for $\mathbb{R}^3$ and $\mathbb{R}^2$). Let

$$\varphi(x,y,z) = (4x + 2z, x - y + 5z)$$

Then $\varphi(1,0,0) = (4,1)$, $\varphi(0,1,0) = (0,-1)$, and $\varphi(0,0,1) = (2,5)$. Thus, the matrix representation of φ with respect to β_V and β_W is

$$\begin{pmatrix} 4 & 0 & 2 \\ 1 & -1 & 5 \end{pmatrix}$$

Theorem 13.2.3: Matrix Representation Correspondence

Let V and W be a finite dimensional vector space over F with dimension n and m, as well as basis β_V and β_W respectively. Then the map $\Psi : \mathrm{Hom}_F(V,W) \to M_{mn}(F)$, where $\varphi \mapsto M_{\beta_V}^{\beta_W}(\varphi)$ is a [vector-space] isomorphism.

Note that the operation on $\mathrm{Hom}_F(V,W)$ if point-wise function addition and $M_{mn}(F)$ is matrix addition.

***Proof*:**
To show Ψ is F-linear, it comes down to the fact that every $\varphi \in \mathrm{Hom}_F(V,W)$ is F-linear. Since every column of $M_{\beta_V}^{\beta_W}$ represents $\varphi(v_i)$, and $(\varphi+\psi)(v) = \varphi(v) + \psi(v)$, then we get that the map Ψ is also F-linear.

The map Ψ is surjective, since given a matrix $A \in M_{mn}(F)$, the map φ that maps to it can be defined by reversing the process that was done in equation (13.4). This map is injective since if $A = B$, then $A = M_{\beta_V}^{\beta_W}(\varphi) = M_{\beta_V}^{\beta_W}(\psi) = B$. Since a basis uniquely defines a linear transformation and both representations are with respect to the same bases, it must be that $\varphi = \psi$, proving injectivity.

Thus, Ψ is an isomorphism, as we sought to show.

13.2.4 Note Different choices of bases produces different isomorphisms. There is no "natural choice" in the same way that there is no natural choice for a basis for a vector-space

Corollary 13.2.5: Dimension Of $\mathrm{Hom}_F(V, W)$

The dimension of $\mathrm{Hom}_F(V, W)$ is $\dim(V) \cdot \dim(W)$

***Proof*:**
Since $\mathrm{Hom}_F(V, W)$ is isomorphic to $M_{mn}(F)$, and $\dim(M_{mn}(F)) = mn$, then $\dim(\mathrm{Hom}_F(V, W)) = mn = \dim(V) \cdot \dim(W)$

We've shown that function addition is compatible with matrix addition. When it comes to function composition, the story becomes a little trickier, since we need the co-domain of one function to line-up with the domain of the next. Thus, to that end, let U, V, W be finite dimensional vector-spaces with dimension k, n, m and bases $\beta_U = \{u_1, u_2, ..., u_k\}$, $\beta_V = \{v_1, v_2, ..., v_n\}$, and $\beta_W = \{w_1, w_2, ..., w_m\}$ respectively. Let $\varphi : U \to V$ and $\psi : V \to W$ be two linear transformations. Their composition $\psi \circ \varphi$ is a linear fransformation from U to W, and so there exists a representation $M_{\beta_U}^{\beta_W}(\psi \circ \varphi)$ given by

$$(\psi \circ \varphi)(u_j) = \sum_{i=1}^{m} c_{ij} w_i \qquad c_{ij} \in F$$

Since φ and ψ are linear transfomation, they too have a matrix representation:

$$\varphi(u_j) = \sum_{p=1}^{n} a_{pj} v_p \qquad \psi(v_p) = \sum_{i=1}^{m} b_{ip} w_i$$

from this, we can find c_{ij} in terms of the coefficients a_{pj} and b_{ip}:

$$\begin{aligned}(\psi \circ \varphi)(u_j) &= \psi\left(\sum_{p=1}^{n} a_{pj} v_p\right) \\ &= \sum_{p=1}^{n} a_{pj} \psi(v_p) \\ &= \sum_{p=1}^{n} a_{pj} \sum_{i=1}^{m} b_{ip} w_i \\ &= \sum_{p=1}^{n} \sum_{i=1}^{m} a_{pj} b_{ip} w_i \\ &= \sum_{i=1}^{m} \sum_{p=1}^{n} a_{pj} b_{ip} w_i\end{aligned}$$

Thus, if we let $c_{ij} = \sum_{p=1}^{n} a_{pj} b_{ip}$, we get

$$= \sum_{i=1}^{m} c_{ij} w_i$$

Take a moment to look over this and think about how this would be represented as multiplying the matrix $M_{\beta_U}^{\beta} V(\varphi)$ and $M_{\beta_V}^{\beta_W}(\psi)$. In fact, the coefficents of $M_{\beta_U}^{\beta_W}(\psi \circ \varphi)$ are exactly what you get by multiplying the other two matrices! This is summed up in the following theorem:

Theorem 13.2.6: Matrix Representation And Composition

Let U, V, W be finite dimensional vector-spaces with bases β_U, β_V, β_W, of size k, m, n respectively. Let $\varphi : U \to V$ and $\psi : V \to W$ be two linear transformations. Then

$$M_{\beta_U}^{\beta_W}(\psi \circ \varphi) = M_{\beta_V}^{\beta_W}(\psi) M_{\beta_U}^{\beta_V}(\varphi)$$

Another way of seeing this theorem is by saying the following diagram commutes:

$$\begin{tikzcd}
M_{mp}(F) \times M_{mp}(F) \arrow[r] \arrow[d] & M_{mn}(F) \arrow[d] \\
\mathrm{Hom}_F(F^p, F^m) \times \mathrm{Hom}_F(R^p, R^m) \arrow[r] & \mathrm{Hom}_F(F^n, F^m)
\end{tikzcd}$$

Corollary 13.2.7: Matrix Multiplication Properties

Let $M_{mn}(F)$ be the set of $m \times n$ matrices over F. Then given appropriate matrices, matrix multiplication is associative and distributive

Note how this is different from the associativity and disctributivity of $GL_n(F)$. Namely, all the matrices of $GL_n(F)$ are square matrices. We have not yet proven that non-square matrices are associative and distributive, particularly because they do not always line up properly, and their result can pass from one set of matrices to another (ex. $A \in M_{ab}(F)$, $B \in M_{bc}(F)$, $AB = M_{ac}(F)$)

***Proof*:**
We'll show the proof for associativity – the proof for distribuity uses the same technic. Let A, B, C be matrices such that $(AB)C$ and $A(BC)$ are defined. By theorem 13.2.6, (AB) represents $S \circ T$ for appropriate linear transformation S, T. Multiplying on the right side by C, $(AB)C$, the resulting matrix represents $(S \circ T) \circ R$ for appropriate R. Similalry, $A(BC)$ is represented by $S \circ (T \circ R)$. Since the composition of functions is assosiative, Then

$$\Psi((AB)C) = (S \circ T) \circ R = S \circ (T \circ R) = \Psi(A(BC))$$

and since Ψ is injective, $(AB)C = A(BC)$. A similar proof follows for distributivity.

Corollary 13.2.8: Matrix Representation Isomorphisms

1. If β is a basis of the n-dimensional vector space V, the map $\varphi \mapsto M_\beta^\beta(\varphi)$ is a ring isomorphism and a vector-space isomorphism of $\mathrm{Hom}_F(V, V)$ (i.e. $\mathrm{End}(V)$) onto $M_n(F)$:

$$\mathrm{End}(V) \cong_{\mathbf{Ring}} M_n(F) \qquad \mathrm{End}(V) \cong_{F\text{-}\mathbf{Vect}} M_n(F)$$

2. If we restrict to automorphisms, then $\mathrm{Aut}(V) \cong GL_n(F)$ as vector-spaces[a]. In particular

$$\mathrm{Aut}_{\mathbf{Vect}}(V) \cong_{F\text{-}\mathbf{Vect}} GL_n(F)$$

[a] $\mathrm{Aut}(V)$ is not a ring

***Proof*:**
In both cases, we've already shown that it is isomorphic as F-vector-spaces, so we'll show they're isomorphic as rings

(1) Using corollary 13.2.7, we see that $M_n(F)$ is a ring (since multiplying two $n \times n$ matrices get's us an $n \times n$ matrix). We've shown this map is bijective in theorem 13.2.3. theorem 13.2.6 shows that this map also preserves the ring structure of $\text{End}(V)$, showing that the map is a ring-isomorphism.

(2) This is the same proof as (1) (Dummit addded: since it sends units to units)

Finally, we convert the powerful result of corollary 13.1.20 into for matrices, in particular we showed that if f is injective between two finite dimensional vector space of the same dimension, then it is surjective. This result is important enough that the matrix representation of such an f should be given a name

Definition 13.2.9: Nonsingular Matrix

An $m \times n$ matrix A is called *nonsingular* if $Ax = 0$ with $x \in F^n$ implies $x = 0$

For some intuition band the name, see remark ref:HERE (text under definition 13.1.19)

Proposition 13.2.10: Nonsingular Square Matrix

Let A be an $n \times n$ matrix. Then A is invertible if and only if A is nonsingular

***Proof*:**
If $\dim_k(V) = \dim_k(W)$, then $V \cong_k W$, so without loss of generailty we may say $f : V \to V$ is a linear map from V to itself. Let's say A is invertible and $Ax = 0$. then

$$0 = Ax = A^{-1}Ax = x \quad \Rightarrow \quad x = 0$$

giving us that A is nonsingular.

Conversely, let A be nonsingular. Then A represents some linear transformation $\varphi : V \to V$ for some finite dimensional vector-space V with respect to basis β and γ for the domain and codomain. Since $\varphi(x) = 0$ if and only if $x = 0$, then φ is injective, and so by corollary 13.1.20, φ is an isomorphism. Thus, there exists a linear transfomration φ^{-1} such that $\varphi^{-1} \circ \varphi = \text{id}$. By using theorem 13.2.3, let B be the representation of φ^{-1} with respect to γ and β. Using theorem 13.2.6 we have

$$AB = M_\beta^\gamma(\varphi)M_\gamma^\beta(\varphi^{-1}) = M_\gamma^\gamma(\varphi \circ \varphi^{-1}) = M_\gamma^\gamma(I) = I$$

thus $AB = I$. similarly $BA = I$. Thus, $B = A^{-1}$, as we sought to show.

(dummit here put's a throw-away definition and comment about the row rank and column rank, and that the rank of φ is the row rank of it's representation, and that the row and column rank match. Once I have a more solid grasp on whats going on here, I'll come back to it; for now my intuition relates this to kernel an co-kernel, or more generally injectivity/surjectivity test)

Exercise 13.2.1

1. We know that in general, if R is a ring and $x \in R$ is not a unit, that doesn't imply that x is a zero-divisor (ex. take the integers). Show that for the matrix ring $M_n(k)$, if $A \in M_n(k)$ is not a unit, then A is a zero-divisor.

2. Show that any polynomial (over a subfield of $\mathbb{C}$) of degree n is uniquely determined by $n+1$ points (You may freely use the fact that a polynomial of degree n can be broken down into a product of n linear terms).

3. (If you know some complex arithmetics) Let $f : \mathbb{C} \to \mathbb{C}$ be a $\mathbb{C}$-linear function. Recall that $\mathbb{C}$ is a 2 dimensional $\mathbb{R}$-vector space, and so f may be represented as a real matrix. Show that if $f(z) = \alpha z = (a+bi)z$, then the matrix representation of f in the standard basis will be of the form:
$$[f] = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

13.2.1 Change of Basis

While representing vectors in a vector space in terms of coordinates simplifies their representation, and linear transformations and their compositions can be seen as matrices and matrix multiplication, we are *always* dependent on a choice of basis (i.e. a choice of isomorphism $\varphi : k^n \to V$). As pointed out earlier, there is no "canonical" choice for a basis – there are many choices of bases which will produce isomorphic vector spaces[9]. Thus, in this section we will explore how to change from one basis to another to be able to compare representations. We will be working with vector-spaces, but we will only be using properties of free-modules meaning the theory is true for free-modules in general.

Let V be an F-vector space of dimension n. Let β_A and β_B be two basis for V, meaning we have the two isomorphissm
$$F^{\oplus \beta_A} \xrightarrow{\varphi} V \qquad F^{\oplus \beta_B} \xrightarrow{\psi} V$$

Then we will get the following isomorphism:
$$F^{\oplus \beta_A} \xrightarrow{\psi^{-1} \circ \varphi} F^{\oplus \beta_B}$$

Then by theorem 13.2.3, there is a matrix $N_{\beta_A}^{\beta_B}$ that corresponds to $(\psi^{-1} \circ \varphi)$. In particular, the jth column of $N_{\beta_A}^{\beta_B}$ is the image of the jth element of A viewed as a column vector in $f^{\oplus \beta_B}$.

If $\beta_A = (a_1, a_2, ..., a_n)$ and $\beta_B = (b_1, b_2, ..., b_n)$ then
$$(\psi^{-1} \circ \varphi)(a_j) = \sum_{i=1}^{n} n_{ij} b_i$$

We give this matrix a name:

[9] There are ways to make it "canonical" by augmenting the vector-space to have further structure which may naturally induce a basis (for example, an inner product). For such examples, see *Linear Algebra* [12].

Definition 13.2.11: Change Of Basis

Let V be a F-vector space of dimension n, and let β_A and β_B be two bases corresponding to the isomorphisms φ and ψ. Then the *change of basis matrix* is the matrix representation of $(\psi^{-1} \circ \varphi)$.
We can represent this matrix as $N_{\beta_A}^{\beta_B}$ or $N_{\beta_A}^{\beta_B}(V)$

Corollary 13.2.12: Properties Of Change Of Basis

Let $N_{\beta_A}^{\beta_B}$ and $N_{\beta_B}^{\beta_C}$ be two change of basis for a vector-space V. Then

1. $(N_{\beta_A}^{\beta_B})^{-1} = N_{\beta_B}^{\beta_A}$
2. $N_{\beta_A}^{\beta_B} N_{\beta_B}^{\beta_C} = N_{\beta_A}^{\beta_C}$

***Proof*:**
Writing out the isomorphism explicitely will immediately give the result, and so it is left as an exercise

With the change of basis for a vector space taken care of, we know check how to perform a change of basis for a matrix representation of a linear transformation. The idea behind this comes down to the following visual:

$$\begin{tikzcd}
F^{\oplus \beta_A} \arrow[dr, "\varphi_1"] \arrow[dd, "\nu_{\beta_A}^{\beta_B} = \varphi_2^{-1} \circ \varphi_1"'] & & & F^{\oplus \beta_C} \arrow[dl, "\varphi_2"'] \arrow[dd, "\mu_{\beta_C}^{\beta_D} = \psi_2^{-1} \circ \psi_1"] \\
& V \arrow[r, "\alpha"] & W & \\
F^{\oplus \beta_B} \arrow[ur, "\psi_1"'] & & & F^{\oplus \beta_D} \arrow[ul, "\psi_2"]
\end{tikzcd}$$

Let N_A^B be the matrix representation of $\varphi_2^{-1} \circ \varphi_1$ and M_C^D be the matrix representation of $\psi_2^{-1} \circ \psi_1$. If we choose the basis β_A for V and β_C for W, then we have:

$$\varphi_2^{-1} \circ \alpha \circ \varphi_1 : F^{\oplus \beta_A} \to F^{\oplus \beta_C} \tag{13.5}$$

Let P be the matrix representation of $\varphi_2^{-1} \circ \alpha \circ \varphi_1$. If we choose the basis β_B for V and β_D for W, then we have:

$$\psi_2^{-1} \circ \alpha \circ \psi_1 : F^{\oplus \beta_B} \to F^{\oplus \beta_D} \tag{13.6}$$

now with some manipulation, we can transform equation 13.6 into equation 13.5 like so

$$\psi_2^{-1} \circ \alpha \circ \psi_1 = (\mu_{\beta_C}^{\beta_D} \circ \varphi_2^{-1}) \circ \alpha \circ (\varphi_1 \circ (\nu_{\beta_A}^{\beta_B})^{-1}) = \mu_{\beta_C}^{\beta_D} \circ (\varphi_2^{-1} \circ \alpha \circ \varphi_1) \circ (\nu_{\beta_A}^{\beta_B})^{-1}$$

And using the matrix correspondence, this is equivalent to

$$M_{\beta_C}^{\beta_D} P (N_{\beta_A}^{\beta_B})^{-1} = M_{\beta_C}^{\beta_D} P N_{\beta_B}^{\beta_A} = Q$$

This should be able to be read quite intuitively: $N_{\beta_B}^{\beta_A}$ converts our vector represented by a linear combination of vectors in β_B into linear combination of vectors in β_A, then P transforms the result and outputs it as a linear combination of vectors in β_C, and finally $M_{\beta_C}^{\beta_D}$ converst the result into a linear representation of vectors in β_D. W

We sum up these results into a proposition:

Proposition 13.2.13: Change Of Basis For Linear Transformation

Let $\varphi : V \to W$ be a linear transformation from two finite dimensional vector spaces, and let P be the matrix representation of φ with respect to bassi β_A and β_C for V and W respectively. Then the matrix representing linear transformation φ with respect to basis β_B and β_D is

$$M_{\beta_C}^{\beta_D} P N_{\beta_B}^{\beta_A}$$

with the notation given above.

More generally, if M and N are any invertible matrix, they represent a change of basis, so MPN would be a change of basis (given the appropriate dimensions).

Definition 13.2.14: Equivalence

Let $P, Q \in M_{mn}(F)$. Then we say P and Q are *equivalent* if they represent the same linear transformation $\varphi : V \to W$ up to a chocie of basis. If that's the case, there exists matrices M and N such that

$$Q = MPN$$

It should be shown that equivalence is an equivalence class, and hence there is an entire equivalence class of matrices that can represent a linear transformation, where any representative of the equivalence class equates to a choice of basis for the domain and codomain. This brings up an interesting question: is there a particular representative (i.e. choices of bases for P, Q) that will make it "easier" to see what the linear transformation is doing? This is a very interesting question, and in the next section shall be explored to great success, in particular, all linear transformation can be represented in "reduced form" and which can be algorithmically found.

13.2.2 Similarity

Before proceeding, there is one more important concept that must be covered, namely the special case where $V = W$ is the same vector-space:

$$\mathrm{Hom}_R(V, V) = \mathrm{End}_R(V)$$

that is, we shall look at linear transformations from V to itself. Such endomorphisms are in fact extremely common; they can be thought of as representing the "change in state" of a system. As the reader may surmise, this means they have applications all other the physical science, however they shall amply show up in pure mathematics. In fact, chapter 14 is essentially the theoretical build-up for studying such functions.

For this discussion, we shall tacitly assume $R = k$ is a field, however for most of the discussion having R simply be an integral domain shall work. The reader should keep in mind though that many results work in broader generality want to make this more the responsibility of the author rather than the reader

We shall simply start by asserting that the vector-space V in $\mathrm{Hom}_R(V,V)$ shall be considered to be the exact same space, not two isomorphic copies of these spaces as generally is considered. In this situation, an element $\alpha \in \mathrm{Hom}_R(V,V) = \mathrm{End}_R(V)$ is usually called a *linear operator* to emphasize the fact that the domain and codomain are "identical". With this in mind, there are new questions that could be asked, such as if there exists $v \in V$ such that $\alpha(v) = \lambda v$ for some $\lambda \in R$, or if a subspace $W \subseteq V$ maps into itself.

Another important consequence of identifying the domain and codomain is that when choosing a basis, the *same* basis must be chosen for both of them (as α is considered to be mapping V to itself). This means that there is a more specific notion of equivalence as only one matrix is chosen:

Definition 13.2.15: Similarity

Let $A, B \in M_n(R)$. Then these two matrices are *similar* if they represent the same linear operator $\alpha : V \to V$ up to choice of basis for V. Equivalently, two endomorphism α, β are similar if there exists an automorphism φ such that $\beta = \varphi \circ \alpha \circ \varphi^{-1}$
Similarity is an equivalence relation[a], and so we say $A \sim B$ if and only if A and B are similar.

[a]check if you are not convinced

An immediate consequence of staring at this diagram:

$$\begin{tikzcd}
R^n \arrow[dd, "\nu^{\beta_B}_{\beta_A} = \varphi^{-1}\circ\varphi"'] \arrow[dr, "\varphi"] & & & R^n \arrow[dl, "\varphi"'] \arrow[dd, "\mu^{\beta_D}_{\beta_C} = \psi^{-1}\circ\psi"] \\
& V \arrow[r, "\alpha"] & V & \\
R^n \arrow[ur, "\psi"'] & & & R^n \arrow[ul, "\psi"]
\end{tikzcd}$$

we see that $A \sim B$ if and only if there exists an invertible $P \in \mathrm{GL}_n(R)$ such that

$$B = PAP^{-1}$$

which explains the definition for similarity of endomorphisms, though the definition for endomorphisms generalizes to non-finite cases. When working over a finite-dimensional vector spaces, then each endomorphisms has a matrix representation.

The reader may have noticed that $A \sim B$ if B is the conjugate of A via an invertible matrix. In the language of group theory, this shows that $\mathrm{GL}(V)$ naturally acts on $\mathrm{End}_R(V)$ via conjugation[10], hence $A \sim B$ iff A, B are in the same orbit in this action.

Just like for equivalence, we may ask for when two matrices are similar, or if there is a nice representation in the orbit. This question is a bit trickier because the domain and codomain have are the same, and so there is less flexibility in choosing representatives. In chapter 14, we shall show how to find a nice basis to find a nice representation of α.

13.2.3 Gaussian Elimination: Finding a Nice Basis

Let $\varphi : V \to W$ be a linear transformation, and P be some matrix representation given by choosing a basis for V and W. Then P belongs to some equivalence class representing all possible representatives

[10]Recall that conjugation is *the* natural group-action

of φ. If we multiply P on the left or right by an invertible matrix, then by by proposition 13.2.13, P remains in the same equivalence class. It would thus be nice if given P, we may simplify P as much as possible to get an "nice" looking represetnative for φ. To do this, we will find nice sets of invertible matrices that do "elementary" operations to a marix which will allow the matrix P to be put in as simple a form as possible. In particular, we will find invertible matrice that will allow us to perfom the following "elementary" operations:

1. switching two rows (resp. columns) or P. For example, let's say we want to switch the 1st and 2nd row of a matrix. This can be done by multiplying on the right (resp. left) by:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

   and the identity on the left (resp. right). In particular, we swapped the rows of the above matrix.

2. multiply all elements in a row (resp. column) of P by a unit of R by multiplying on the left (resp. right) by a matrix of the form:

$$\begin{pmatrix} u & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

   This particular matrix will scale the first row (resp. column) by a factor of u when multiplying on the left (resp. right).

3. add a multiple of one row (resp. column) to another row (resp. column) of P. For example, to add a multiple of the first row to the second,we would multiply on the right (resp. left) by:

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

   This particular matrix will add 3 copies of the 2nd row (resp. column) to the 1st row (resp. column) when multiplying on the left (resp. right).

It should be checked that these matrices are indeed invertible matrices (can you find their inverse?), and that the side on which the matrix is multiplie by effects either columns or rows. The following proposition summarizes the above discussion, generalizing it to general commutative rings R:

Proposition 13.2.16: Equivalence An Elementary Matrices

Let $P, Q \in M_{m,n}(R)$ where R is a commutative ring. Then if Q may be obtained from a sequence of elementary operations, then P an Q are equivalent.

***Proof*:**
Verify this proposition.

Due to these matrices being the transition matrices between two equivalent matrices, we give matrices that perfom elementary operations a name:

Definition 13.2.17: Elementary Matrices

Elementary matrices are matrices such that when mutliplying a matrix from the left or right side,the reuslting matrix has had an elementary operation performed on it.

Notice that proposition 13.2.16 is not an "if and only if", simple an "if". This is because it is possible that P and Q are equivalent, but the invertible matrices that "connects" the two are not the product of elementary matrices. If R has certain extra properties, then it will be the case that all invertible matrices can be written as a product of elementary matrices, as we'll explore now. To make this easier to exlpore, let's give a name to all invertible linear matrices

Definition 13.2.18: General Linear Matrices

Let R be a ring and $M_n(R)$ the set of all matrices of of dimension n over R. Then:

$$\mathrm{GL}_n(R) := (M_n(R))^\times$$

that is, $\mathrm{GL}_n(R)$ is the set of all units of $M_n(R)$, i.e. the set of all invertible matrices of dimension n over R.

The word "dimension" is used to indicate the number of rows and columns (which are both equal in this case, and hence why a single number is named). The term "general" is used since there other set of invertible matrices commonly studied, usually given a different prefix ("special", "orthogonal", "orthonormal", "unitary", and so forth). This set of invertible matrices is called "general" since it ill encompass *all* of these other sets of invertible matrices. For who may have come across these terms before, notice that this set was already mentioned in definition 1.2.30. One thing that as difficult to prove directly with matrices but easy to prove now is that $(\mathrm{GL}_n(R), \cdot)$ is associative: every invertible matrix represents a function, and function composition is associative, and hence $\cdot$ is an associative operation.

As mentioned, there can be matrices in $\mathrm{GL}_n(R)$ which are not a product of elementary matrices, that is, $\mathrm{GL}_n(R)$ need not be generated by elementary matrices. However, if $R = k$ is a field, then this is in fact the caes, and the "if" in proposition 13.2.16 becomes and "iff"

Proposition 13.2.19: General Linear Group Of Field

Let $R = k$ and $\mathrm{GL}_n(k)$ be the general linear group over a field. Then $\mathrm{GL}_n(k)$ is generated by elementary matrices.

***Proof*:**

The idea is as follows: take advantage that all entries are units. Make a_{11} 1 and then eliminate all a_{i1} and a_{1j} (i.e. make them zero). Then you have made a $(n-1)\times(n-1)$ matrix ith a 1 in the top-left. Repeat this process until you get the identity:

$$I_n = M \cdot P \cdot N$$

Since M and N are invertible, we wget:

$$(NM)^{-1} = M^{-1}I_nN^{-1} = P$$

registering all the operations will produce you product of elementary matrices, an (NM) is still the prouct of invertible matrices, and so $(NM)^{-1} = P$ is the product of invertible matrices, completing the proof.

The above method can be used more generally to find the inverse of a matrix, and we only need to multiply *on the left hand side* or the *right hand side*. The method is common enough to be given a name: *Gaussian Elimination.* [11]. Since Gaussian elimination usually reffers to operations on the left, we may think of this as a form of "row"-equivalence. However, as we saw we may as easily multiply the elementary matrices only on the right and get "column"-equivalence. This shows us that the two forms of equivalence are the same. Some authors define the notion of *row rank* and *column rank* for matrices which correspond to the dimension of a vector-space spanned by the rows and column respectfully, and proceed to show that the row and column rank are the same number. For reference, we put the followign definition in a box:

Definition 13.2.20: [matrix] Rank

Let A be a matrix. Then the *row rank* is the dimension spanned by its rows, the *column rank* is the dimension spanned by its colun. By the above remark, the row-rank and the column-rank are equal, and so the single numer is called the *rank* of the matrix. If the rank is maximal, the matrix is said to be of *full rank*.

If a matrix is full rank, then the linear map it represents is surjective.

Corollary 13.2.21: Equivalence Of Matrices

Let $R = K$ be a field and $P \in M_{m,n}(R)$ be any m trix. Then P is equivalent to:

$$\left(\begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array}\right)$$

for appropriate k, $1 \leq k \leq \min(m,n)$, here 0 is the represents that all of the entries are 0 (for the appropriate size).

***Proof*:**
This is just combining the previous results.

Thus, two matrices that have a different value of r in the above matrix will *not* be equivalent. This also shows that there are only *finitely many* equivalence classes matrix representations of linear transformation's between two finite dimensional vector-spaces, in particular there is one for each dimension$1 \leq k \leq n$. This also show's that if we look at a linear transformation in the right way (i.e. pick the right basis), a linear transformation *always* looks like it some projection eliminating certain

[11] Though Gauss has most likely invented the above method when working with matrices, it has been known under slightly different symbols and theory at least as far back as Han China

dimensions while embedding the rest of the space into the codomain! See proposition 13.1.22 to see how this intuition and the existence of compliments of subspaces work together.

To finish off this section, we can ask if the same logic can be done for the notion of similarity. Unfortunately, similarity is more complicated: since only one matrix is being picked (i.e. only one basis is being picked) the rows and column operations must be done "simultaneously". Not all hope is lost: using properties of PID's will show us how to find a good matrix representation for similar matrices in the next chapter.

Gaussian Elimination over Euclidean Domain

From the rings we explored earlier, the euclidean domain is a good start, and indeed with some cleverness a matrix over a euclidean domain can be reduced, though not intirely to an identity matrix in a sub-matrix of the original.

To demonstrate the idea, let R be a euclidean domain with valuation ν, and take any 2×2 matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

then use elementary matrices to put the element with the lowest valuation at position $(1, 1)$. Then use the euclidean algorithm to find:

$$b = aq + r$$

where either $r = 0$ or $\nu(r) < \nu(a)$. Then do the appropriate elementary operation to get:

$$\begin{pmatrix} a & r \\ c & d - aq \end{pmatrix}$$

if $r = 0$, then we have eliminated one of the components. If $r \neq 0$, then repeat the same process, moving the term with the lowest valuation to the $(1, 1)$ position using elementary matrices. Since each term has a finite valuation, it must be that eventually $r = 0$. With minor modifications, the same thing can be done in the $(2, 1)$ position, getting us:

$$\begin{pmatrix} a' & 0 \\ 0 & d' \end{pmatrix}$$

Next, notice that we may find an a', d' such that $a' \big| d'$. For if not, we may consider:

$$\begin{pmatrix} a' & d' \\ 0 & d' \end{pmatrix}$$

and repeat the process until we get to the same state, and keep repeating until either $d' = 0$ or $a' \big| d'$. Thus, we produce a 2×2 matrix where the diagonal divides each other.

From this, it is not too hard to see that this is easily generelizable to an $n \times n$ matrix:

Theorem 13.2.22: Smith Normal Form

Let R be a euclidean domain and $P \in M_{m,n}(R)$. Then P is equivalent to a matrix of the form

$$\left(\begin{array}{ccc|c} [ccc|c]d_1 & \cdots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & \cdots & d_k & 0 \\ \hline 0 & \cdots & 0 & 0 \end{array}\right)$$

where

$$d_1 \big| \cdots \big| d_k$$

The resulting matrix is called the *smith normal form*.

***Proof*:**
exercise

Corollary 13.2.23: Finding Inverse Of Matrix

Let $A \in \mathrm{GL}_n(k)$. Then A^{-1} can be found by performing Gaussian Elimination.

***Proof*:**
This is taking the strategy done in theorem 13.2.22 and keeping track of the elementary matrices used to produce the inverse. Multiplying all of them gives the desired result.

Using smith normal forms, we may prove that $\mathrm{GL}_n(R)$ is generated by elementary matrices. Unfortunately, there does exist a counter if R is a PID; there is exists a 2×2 counter-example over $\mathbb{Q}(\sqrt{19})$ (see cite:HERE, under books/math/papers/ the GL2 paper). On the other hand, we can put any matrix in $\mathrm{GL}_n(R)$ into smith normal form using the gcd and some non-elementary matrix operations. This shall be further explored in chapter 14.

13.2.4 Geometric Result: Systems of Linear Equations

Vector spaces and linear maps have part of their origins in geometry, where the term *linear* originates. The rigorous interpretation of a concept being "geometrical" is something that would be explored in a class on [differential] geometry, and so for the uses of this book geometrical shall mean "locally or globally looks like the solutions to equations":

$$\begin{aligned} f_1(\vec{x}) &= a_1 \\ f_2(\vec{x}) &= a_2 \\ &\vdots \\ f_n(\vec{x}) &= a_n \end{aligned}$$

Often, the constant is moved to the left modifying the equations to be:

$$\begin{aligned} f_1(\vec{x}) - a_1 &= 0 \\ f_2(\vec{x}) - a_2 &= 0 \\ &\vdots \\ f_n(\vec{x}) - a_n &= 0 \end{aligned}$$

This understanding of geometrical is in fact very potent, for example the reader may have already encountered many geometrical objects that are defined in this way, for example:

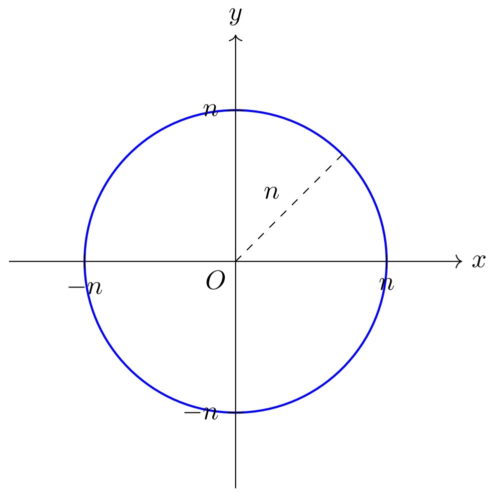


Figure 13.1: solution to $x^2 + y^2 - n^2 = 0$

For the particular explorations of this section, we shall show that vector spaces can be interpreted as the of solutions to *linear equations*:

Definition 13.2.24: Linear Equation and System of Linear Equations

Let V be a k-vector space of dimension n and $k[x_1, x_2, ..., x_n]$ be a polynomial ring over the field k with n variables. Then a *linear equation* is a polynomial, possibly over several variables, where the degree of each variable is at most 1:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

A collection of linear equations is called a *system of linear equations*.

Given a (system of) linear equation(s), we can define *linear functions*: if $\sum_i a_ix_i$ is a linear equation then we can define:

$$L_1 : k^n \to k \qquad L(x_1, x_2, ..., x_n) = a_1x_1 + a_2x_2 + \cdots a_nx_n$$

and similarly a system of linear functions is a collection of linear functions $L_1, ..., L_m$. Since these functions are linear, they can be represented by a matrix, namely given the standard basis:

$$A = [L_1] = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}$$

so that if $\vec{x}$ is an $n \times 1$ matrix with with x_i coefficients, then plugging in a value from k^n is represented as matrix multiplication by

$$A\vec{x} = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

Equivalently, any system of linear equation can be combined to form a linear equation:

$$L : k^n \to k^m \qquad L = \begin{bmatrix} L_1 \\ L_2 \\ \vdots \\ L_m \end{bmatrix} \tag{13.7}$$

Let's now say we want to find the solution to the system:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= b_n \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= b_m \end{aligned}$$

Then if $\vec{b} = (b_{k1})$, then what we are solving for is:

$$A\vec{x} = \vec{b}$$

If A is invertible and $n = m$, then the answer is simple:

$$\vec{x} = A^{-1}\vec{b}$$

where A^{-1} is found using corollary 13.2.23 (how does this relate properties of linear maps that have been explored earlier?). If A is not invertible or $n \neq m$, the next best thing that can be done is find its smith normal form giving us matrices M, N such that

$$M \cdot A \cdot N = \left(\begin{array}{cccc} [ccc|c]d_1 & \cdots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & \cdots & d_k & 0 \\ \hline 0 & \cdots & 0 & 0 \end{array}\right)$$

then if $\vec{c} = M\vec{b}$ and $y = (y_k)$ we get:

$$(M \cdot A \cdot N) \cdot \vec{y} = \vec{c}$$

and producing the system:

$$d_1 y_1 = c_1 \tag{13.8}$$
$$d_2 y_2 = c_2 \tag{13.9}$$
$$\vdots \tag{13.10}$$
$$d_r y_r = c_r \tag{13.11}$$
$$0 = c_{r+1} \tag{13.12}$$
$$\ddots \tag{13.13}$$
$$= c_m \tag{13.14}$$

where $d_i \big| c_i$. Conversely, if there is a set of linear equation of the form given in equation 13.8, then it has a solution of $d_i \big| c_i$.

Where this links to geometry is when considering the space of solution to systems of linear equations. First, notice that without loss of generality the linear equations may equal to 0, for this would simply translate the resulting set of solutions in the vector-space, hence it suffices to consider:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n &= 0 \end{aligned}$$

That is, it suffices to consider the kernel of a system of linear equations, equivalently the kernel of L as defined in equation (13.7).

Now, to endeavor exploring the shape of the kernel of linear functions, let's start with $k = \mathbb{R}$, $L : \mathbb{R}^3 \to \mathbb{R}$, and always use the standard basis. It is easy to visualize $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3$, and higher dimensional analogues can be fathomed. Naturally, if $L(x) = 0$, then $\ker = \mathbb{R}^3$. It is more interesting if L is non-trivial, which for linear functions implies L is surjective. Then by the Rank-Nullity Theorem $\dim \ker = 3 - 1 = 2$, i.e. it is a plane. A plane that is one-dimension less than the ambient space (in this case $\mathbb{R}^3$) is called a *hyper-plane*. Since it is a subspace, it must contain 0 and hence passes through the origin. Since the plane is two dimension, it is spanned by two vectors. Using Gaussian elimination, we can deduce it is spanned by two vectors:

$$\left\{ \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \right\}$$

Example 13.2.25: Finding A Hyperplane

Take $L(x, y, z) = 3x + 2y - 3z$. Find a basis for the kernel. The kernel when embedded in $\mathbb{R}^3$ looks like so:

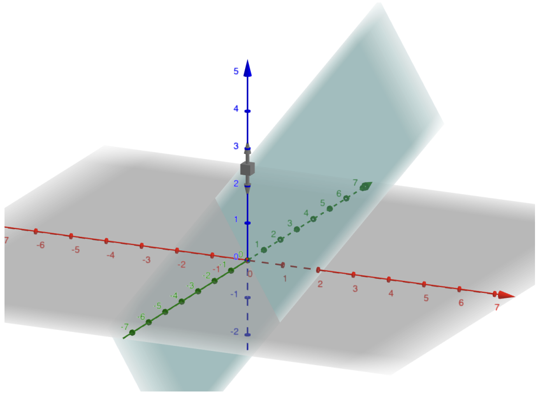


Figure 13.2: $3x + 2y - 3z = 0$

Notice that since the image is one dimensional, the hyper-plane is defined by only one equation, however it is a two dimensional object. More generally, for a surjective linear function $L : \mathbb{R}^n \to \mathbb{R}$, the hyperplane shall be $(n-1)$ dimensional and be defined by a single equation.

Next, let's consider the case where $L : \mathbb{R}^3 \to \mathbb{R}^2$, that is there are two linear equations $L_1 : \mathbb{R}^3 \to \mathbb{R}$ and $L_2 : \mathbb{R}^3 \to \mathbb{R}$. Let's again assume that L is surjective, or full-rank, for if L was one dimension less it would be equivalent to the case of $L : \mathbb{R}^3 \to \mathbb{R}$. Then by the Rank-Nullity theorem we have that $\ker L = 3 - 2 = 1$. Intuitively, this is because there are more restrictions for $L(x) = 0$, namely both L_1 and L_2 must equal 0. Geometrically, this can be thought of as the *intersection* of the kernel of L_1 and the kernel of L_2. Since L is full rank, the intersection is not parallel (the geometric word for this is that the intersection is *transversal*). More generally, $L : \mathbb{R}^n \to \mathbb{R}^2$ would have $\dim \ker L = n - 2$, and would be defined by two linear equations.

Extended the pattern further, given a full-rank linear function $L : \mathbb{R}^n \to \mathbb{R}^m$, the resulting plane given by the kernel would be $n - m$ dimensional and would represent the intersection of m hyper-plane of dimension $n - 1$, each new hyper-plane restricting the dimension of the kernel. The orientation of the kernel is completely dependent on the linear map in question. Nothing in these results required to work specifically with $k = \mathbb{R}$, it was simply convenient for visual purposes. These results can be summarized into the following:

Theorem 13.2.26: Classification Of Planes

Let V be a k-vector-space of dimension n. Then the sub-spaces of V correspond to the kernel of full-rank linear maps $L : V \to k^m$, where $1 \leq m \leq n$.

***Proof*:**
Fill in the blanks from the above discussion

Given that L is surjective, notice that as m increases in size, the dimension of the kernel decreases in size; in a sense it is the "co-dimension" of the domain. This observation is not lost on mathematicians, and the notion of codimension is quite important in geometry: the dimension of an object represents the degree of freedom an object may have within a space (ex. a hyper-plane is $n - 2$ dimensional) while the codimension represents the degree of restriction an object may have within a space (ex. a hyper-plane is 1 codimensional).

(Also, a geometric interpretation of first is iso theorem: when solving system of Lin. Eq, look for kernel. But notice the solution to $Ax + c = b$ is just an affine translation of the kernel! All elementary row operations fix the kernel and permute the cosets, when the elementary operations generate $\mathrm{GL}_n(R)$.)

Example 13.2.27: Exercices

Something with transpose. Perhaps even with conjugate? Get started with $\langle A, A*\rangle$ stuff?

Exercise 13.2.2

1. (from dummit and foote) (note that chagne of absis for a lienar transformation from V ot itself is the same as conjugation by some element of the group $GL(V)$ of nonsingualr linear transformation of V to V. In particular , the relation "similarity" is an equivalence relation whose equivalence classes are the orbits of $GL(V)$ acting by conjugation on $\mathrm{Hom}_F(V, V)$.)

13.2.5 Use Case: Reading Presentations

Recall from section 12.3.3 that a presentation of a [finitely generated] R-module can be represented as an exact sequence:

$$R^n \to R^M \to M \to 0$$

Using the above results, we can more easily read the presentation. Let $\varphi : R^n \to R^m$. Then by exactness, we have that $M \cong \mathrm{coker}(\varphi)$. Hence, if we find the image of φ, we found the relations of M, and and if R is a field of euclidean domain, this is task is simplified using Gaussian elimination:

Example 13.2.28: Finding A Presentation

Let $R = \mathbb{Z}$ and let φ be represented by the matrix:

$$\begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 5 & 9 \end{pmatrix}$$

which means that $\varphi : \mathbb{Z}^3 \to \mathbb{Z}^2$ and this matrix determines an abelian group G (since abelian groups have free resolution of length at most 2 as mentioned in section 12.3.3). To find G, let us first simplify the above matrix. The reader should check that the matrix reduces to:

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \\ 0 & 0 \end{pmatrix}$$

Hence, G is isomorphic to the cokenrel of $\varphi : \mathbb{Z}^2 \to \mathbb{Z}^3$ that maps $(1, 0)$ to $(1, 0, 0)$ and $(0, 1)$ to $(0, 3, 0)$. In particular, since the image is $\mathbb{Z} \oplus 3\mathbb{Z} \oplus 0$, we get

$$G \cong \mathrm{coker}\varphi \cong \frac{\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}}{\mathbb{Z} \oplus 3\mathbb{Z} \oplus 0} \cong (\mathbb{Z}/3\mathbb{Z}) \oplus \mathbb{Z}$$

Showing how a matrix can uniquely represent an abelian group! In the next chapter, we shall show how a matrix can more generally represent a finitely generated PID.

13.2.29 Remark Notice that we may eliminate first row and column if the leading term is 1 (or more generally a unit) since in the quotient the summand shall be completely quotiented.

13.3 Dual Spaces and Forms

Let's say you are studying some mathematical object, say the set of linear transformations. We may want to find numerical properties. For example of linear transformations, we can quantify the size of a matrix, or give some numerical value to similarity, or the dimension of the image. We can define a function from $L(V)$ to a field, say $\mathbb{R}$, that shall give us this quantification. The collection of all these functions are called the *functionals* on $L(V)$. Depending on what type of property we want want the invariances that are being sought after, we will have different types of functionals. For example, when working in analysis, we shall often want our functional to be continuous or perhaps bounded in some way [12]. In Algebra, we shall want our functionals to be homomorphisms, and depending on what other property we want we shall add more properties to our functionals (ex. it shall be important in [13, Affine Space and Regular Functions] that the functional preserves solutions to a special set of polynomial equations).

The set of functionals on a vector space V can be thought of as containing all possible quantifiable "information" about a space (with the limits of the information provided by the properties of the functionals). Most of the time, the space of functionals is larger than the original space[13]. In some special cases, the spaces of all functionals is isomorphic to the space over which we are taking the functionals (as we shall see, all linear functionals on a finite dimensional vector space V is isomorphic to V). In this special case, we can think of our space being characterized by the property the functionals represent! This realization was the origin of *dual theory*, which we shall start exploring in this section. Dual theory is normally explored in earnest in Real and Functional analysis, however it shall show up in some important ways in [13, Coordinate Rings], and certain key functionals shall give a lot of information on linear transformations, namely the trace and the determinant.

13.3.1 Dual Spaces

In mathematics, there is frequently two "opposing" or "dual" ways of describing objects: Let's say X is our object of study

1. internally: Using properties internal to your object to describe that object. Often, maps *to* the object will give us internal information ($f : A \to X$ for some appropriate object A). This can may also be thought of as local information

2. externally: Using some exterior property (ex. be it by putting it an ambient space, or a sapce in which the object X "resonates" or "interacts" with[14]). Ofen, maps *from* the object will give us external information ($f : X \to A$ for osme appropriate object A). This can also often be thought as global information

[12] See Lebesgue Spaces in [19] for more details
[13] This is a guiding principle behind many generalizations in Mathematics, Distribution Theory being the prototypical example
[14] If you know Fourier Analysis, think of phase space

As an example of this, when we were working with group theory, we found two way of looking at all possible groups:

1. Using quotients of free groups, we managed to make normal subgroups of free groups represent relations between elements. In the process of doing this, we defined a surjective homomorphism:
$$\varphi : F(A) \to G$$
2. Using Cayley's theorem, we manage to embed any group into a subgroup of S_A. In the process of doing this, we defined an invective map
$$\psi : G \to S_A$$

Notice how the first one defined us what's happening between elements in a sort of "local" way, while in the 2nd example we defined it by using properties of the larger group S_A and how elements of G embed into S_A. Another example of this principle can be seen with the universal property of products and coproducts of groups. In a sense, the universal property of products only requires the product group to have groups A and B "locally" in the most efficient way, leading to the direct product of A and B, while the corpoduct group require it to have A and B "globally" in the most efficient way, leading to the free product of A and B.

Often when studying an object X, we will study if from these two perspectives: from maps going *into* X and maps going *out of* X. One remarkable phenomena that occurs often in mathematics is that there is often a [contravariant] *dual object* in which the maps *into* and *out of* X are naturally associated to maps *out of* and *into* a dual object X^*. Note the reversal in the directions of the maps! It is often very fruitful to look at objects from the perspective of both X and X^* to gain more information.

To demonstrate a simple but powerful example of "dualness" in mathematics (or the pervasiveness of dualness), we require knowledge of fields outisde of algebra. Usually, I (the author of this book) would decide to leave out any such example to insure no prerequisites are required to learn the material. However, I have decided to break this rule here for those who know a bit of topology since it is an example so powerful that it changed my view of duality, and requires elementary topology knowledge. The following example is *not* necessary for understanding any material of this book and can be considered motivation for dual spaces and dual theory in general for those who know some topology (if you need a refresher on topology, see appendix C). In this example, we will take a moment and venture outisde of algebra and into the realm of analysis and and geometry to give an interesting example.We will give a common phenomena in which "algebraic" object have "geometric" objects as their dual. In particular, unlike the duality of a group in its represetnation in terms of the quotient of a free group or an embedding into S_G, both of which are algebraic ways of looking at groups, we will show that dual objects are sometimes found in other fields of mathematics, allowing us to connect these two fields of mathematics proof-strategies and techniques!

First, define a *Boolean Algebra* to be a $\mathbb{F}_2$-algebra (where $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$). Since $\mathbb{F}_2$ is a field, a Boolean Algebra B is module-isomorphic to $\mathbb{F}_2^n$ for some cardinality n. For $0, 1$ in a Boolean algebra B:

$$0 + 0 = 0 \quad 0 + 1 = 1 + 0 = 0 \quad 1 + 1 = 0 \qquad 0 \cdot 0 = 0 \cdot 1 = 1 \cdot 0 = 0 \quad 1 \cdot 1 = 0$$

We usually also define the additional operation of $\neg$:

$$\neg(x) = 1 - x$$

If you have done some computer science or logic, then you might notice that 1 and 0 can represent the values of *true* and *false* respectively, while $+$ and $\cdot$ represent XOR and AND respectively (hence the name Boolean to this algebra). This algebra underlines most of the logic behind computer science, making us want to find properties of boolean algebras. One way we may study these properties is by looking for maps going *into* a boolean algebra B. Dually, by a result known as *Stone Duality*, we can instead study all the maps *out of* a totally disconnected[15] Hausdorff spaces B^*! Thus, the "dual" of Boolean Algebra is all continuous maps from B^* to $\mathbb{R}$! More generally, locally compact[16] topological spaces (which we can consider as a geometric object) are the dual of a type of algebra known as a C^*-algebras[17] (which we consider as algebraic objects): this is known as *Gelfand Duality*[18]. The study of these objects that are related to each other by "duality" is called *Dual Theory*.

Coming back to algebra and the study of vector-spaces, we will see the very beginnings of Dual Theory by studying the simplest case of dual objects, *dual vector-spaces*. The way we will define the dual of a vector-space will be through "external" means. When we defined vector spaces using a basis, that is a "local" property. In a sense, we showed that the map $f : F(A) \to V$ can define a vector space where A is a basis. What we'll now do is the *dual* equivalent: we'll define a vector space by mapping V into a different space.

The question arrises as to what space might we might we map to? When working with groups, it was not immediately obvious that S_A would be the right group to map. Fortunately for us, there is an "obvious" space to map to: for any vector-sapce over k, we are always given two algebraic objects, namely the abelian group V and the field k. Since k is always a vector-sapce over itself,we can consider linear maps $f : V \to k$! The collection of all such maps is called the *dual space* of V.

Definition 13.3.1: Dual Space

Let V be a vector space over k. Then $V^* = \mathrm{Hom}_k(V, k)$ is the space of all linear transformations from V to f, and is called the *dual space* of V.

By proposition 12.2.7, $\mathrm{Hom}_k(V, k)$ is a vector-space over k (if you're not convinced, it is recommended to reprove this before continuing). The notation $V^\vee$ is also often used to define the dual space and will be used interchangibly. Elements inside the dual space V^* are sometimes called *covectors*, paralleling to how elements inside a vector space V is called *vectors*. In physics and analysis, there is seomtimes a different convention for how to write the evalutation of if $f \in V^*$ at $x \in V$: instead of writing $f(x)$, to mean "evaluate f at the point x", it is sometimes written $\langle f, x\rangle$.

Elements of V^* are given a name:

[15] A topological space in which the only connected sets are singletons

[16] a topological space in which for any point x, there exists a compact neighbourhood

[17] A Banach Algebra over $\mathbb{C}$ with a map known as an involution satisfying the properties of an adjoins

[18] This duality requires a good deal more topology and analysis than is part of the scope of this book. However, in [13, The Zariski Topology], we will see the Zariski Duality which studies a geometric-algebraic duality without the need for much analysis or topology

Definition 13.3.2: Linear Functionals

Let V^* be the dual space of V. Then $f \in V^*$ is called a *linear functional* or *linear form*.

Example 13.3.3: Linear Functional and Dual Spaces

1. (If you know some calculus) Perhaps the most famous linear functional is $\int_a^b \_\_\_ dx : L^1(\mathbb{R}) \to \mathbb{R}$, where $L^1(\mathbb{R})$ will denote the set of all (either Riemann or Lebesgue)[a] integrable functions $f : \mathbb{R} \to \mathbb{R}$ where $\int_{\mathbb{R}} |f| dx < \infty$ (so that the integral is well-defined). Note that $L^1(\mathbb{R})$ is a vector-space with addition and scalar multiplication defined pointwise since the codomain is $\mathbb{R}$:

$$f + g \Leftrightarrow (f+g)(a) = f(a) + (g)$$
$$c \cdot f \Leftrightarrow (c \cdot f)(a) = cf$$

Now, $\int_a^b \_\_\_ dx$ takes in a function $f : \mathbb{R} \to \mathbb{R}$, an gives th number $\int_a^b f dx \in \mathbb{R}$. By basic properties of integarls, we know that:

$$\int_a^b (f+g)dx = \int_a^b f dx + \int_a^b g dx$$
$$\int_a^b cf dx = c \int_a^b f dx \qquad \forall a \in \mathbb{R}$$

and hence $\int_a^b \_\_\_ dx$ is indeed a linear functional! Hence $\int_a^b \_\_\_ dx \in L^1(\mathbb{R})^*$, for any $a, b \in \mathbb{R}$[b]
The study of Analysis focuses a lot on the study of the properties of these functionals, and would be covered in many Analysis courses. For textbooks, see ref:HERE.

2. Let $C([a,b])$ be the collection of all continuous maps from $[a,b]$ to $\mathbb{R}$, $f : [a,b] \to \mathbb{R}$, with addition and scalar multiplication defined pointwise. Then define: $\mathrm{ev}_x : C([a,b]) \to \mathbb{R}$ to be

$$f \mapsto f(x)$$

Then it can be shown that ev_x is a linear transformation, called the *evaluation map*, and is thus a linear functional. More generally, as long as the codomain is a field $\mathbb{R}$, then the evaluation map is a linear functional.

3. A more advanced case of the above example occurs if we consider the set $C^1(\mathbb{R})$ of all differentiable functions[c] forms a vector-space with point-wise addition and scalar multiplicaiton. The operator $D_a : C^1(\mathbb{R}) \to \mathbb{R}$ takes in a differentiable function f and gives back $Df(a)$ (which in first year calculus is more commonly denoted $f'(a)$).

4. (If you know geometry or analysis) The following examples are not examples of dual spaces as we've defined them, but the generlization of them. In this generlization, the dual space we've defined would be called a *linear* dual space, or the dual space of in the category of $\mathbf{Vect}_k$. More generally, if X is an object in a category, then given a *dualizing* object D (often it is $\mathbb{R}$, $\mathbb{C}$, $\mathbb{C}^\times$, $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{R}/\mathbb{Z}$ depending on the context), then $\mathrm{Hom}_C(X, D)$ is the dual space. In the following example, let D be either $\mathbb{R}$ or $\mathbb{C}$. Let's say X is either a vector-space

over $\mathbb{R}$ or $\mathbb{C}$ (you should have function spaces in mind) or a manifold over $\mathbb{R}$ or $\mathbb{C}$. Then $C_0(X)$, $C^k(X)$, $L^p(X)$, an so forth are all subspace of the dual space X^* in the appropriate category.

Another place in which [ring] funcitonals shall become of paramount importance is in [13, Affine Space and Regular Functions] and [13, The Dictionary], where a duality is set up between algebraic and geometric objects.

[a]For the sake of this example, it doesn't matter, though L^1 usually denote *Lebesuge* measurable functions where $\int |f| < \infty$

[b](if you know some analysis) one classical result known as Riesz Representation Theorem (for measures) is that all functionals $f \in L^1(\mathbb{R})^*$ will be of the form $f \mapsto \int fg$ where $g \in L^\infty(\mathbb{R})$ and the essential support of g is bounded.

[c]in particular, continuously differentiable functions. This distinction is of no importance for this example

In the above examples, all the vector-spaces are infinite dimensional. The vector-space being infinite dimensional will in fact greatly complicate the structure of dual spaces[19]. In general, more structure is added to a vector-space (ex. a norm) to have better "control" of infinite dimensional vector-spaces. Such discussion is left to a course on functional analysis[20]. We will mainly focus on the theory of finite dimensional vector spaces, since the results of this theory is in it of itself quite fruitful and provides a good basis for its study of dual spaces more generally.

Given a basis for V, it straight-forward to find a basis for $V^\vee$:

Definition 13.3.4: Dual Basis

Let $\beta = \{v_1, v_2, ..., v_n\}$ be a basis for the finite dimensional vector-space V. Then define $v_i^* \in V^*$ to be

$$v_i^*(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

for $1 \leq i, j \leq n$. Let β^* be the set of dual-basis elements

Since β^* has the word basis in it's name, you might guess that it is ineed a basis for the dual space:

Proposition 13.3.5: Dual Basis

Let $\beta*$ be defined as in definition 13.3.4. Then β^* is a basis for V^*

***Proof*:**
exercise

As a consequence, V^* has the same dimension as V

Proposition 13.3.6: Reflectivity Of Finite Dimensional Vector Spaces

Let V be a finite dimensional vector space. Then $V \cong V^*$.

[19]This is one of the analytic properties that was discussed in the footnote in page 427

[20]Give references to books HERE, ex. Folland

***Proof*:**
By proposition 13.3.5, since V and V^* are of the same dimension, $V \cong V^*$.

An intuitive way of viewing the reflectively product to reference the intuition presented in the introduction of this section mentioning how functional represent *information*. Each linear function provides some information about a vector-space (prototypical examples would be the projection functions $\pi_{\beta,i} : V \to k$ picking a component of a vector given a choice of basis β). The type of information provided is restricted by the properties of the functions on V, in this case the co-domain is always k, and the functional is always linear. In finite dimension, these functions can be completely characterized by elements in V by the existence of the above isomorphism. The choice of the isomorphism is still an important factor to consider, since there is not natural basis or extra information that allows for favoring one isomorphism over another.

Since the structure of V isomorphic to V^*, it would be interesting if each map $f : V \to W$ can be associated to an appropriate "dual" function f^* where composition is respected, hopefully giving the existence of the functor $(-)^*$ from Vect_k to itself. To start, there is a notion of a dual function:

Definition 13.3.7: Dual map (Transpose)

Let $f : V \to W$ be a linear transformation. Then the *transpose* of f, denoted f^*, is defined as:

$$f^* : W^* \to V^* \qquad f^*(\varphi) = \varphi \circ f$$

Unfortunately, $(-)^*$ can't so easily be defined as each choice of isomorphism $V \cong V^*$ for each vector-space $V \in \mathbf{Vect}_k$ may not be compatible, see exercise ref:HERE. Note that the transpose of a function insures that maps that used to "out of" the space V will now "go into" the space V^* (and similarly for maps that go into V). Thus, this definition matches with our goal for V^* to be the dual of V, and furthermore is independent of any choice of basis. The name transpose comes from the matrix representation of f^*

Proposition 13.3.8: Matrix Representation Of Dual Map

Let $f : V \to W$ be a linear transformation and $A = (a_{ij})$ be the matrix representation of f given some basis β and γ for V and W respectively. Then if $f^* : W^* \to V^*$ is the dual map of f, where W^* and V^* have as basis β^* and γ^*, then the change of basis matrix A^* is:

$$A^* = [f]_\beta^\gamma = [f^*]_{\gamma^*}^{\beta^*} = (a_{ji}) = A^t$$

Intuitively, in the matrix representation, each entry a_{ij} transposed with entry a_{ji}. For this reason, the map f^* is sometimes called the *transpose* of f.

***Proof*:**
exercise

As a corollary, if A is a nonsingular map $f : V \to W$, then the matrix representation for the map $(f^{-1})^* : V^* \to W^*$ is $(A^{-1})^t$.

Similarly to how the kernel of a linear transformation f provides the needed information to deduce the structure of the image of f, we get the dual equivalent for f^*, given an appropriate "dual" for the kernel:

Definition 13.3.9: Annihilator Of Dual Map

$U \subseteq V$ be a subspace of V. Then the *annihilator* of U is:

$$U^o := \{f \in V^* \mid f(U) = 0\}$$

Note that $V^0 = \{0\}$ and $\{0\}^0 = V$. Another common notation is $U^\perp$, which is pronounced "U perp", which shows how U^0 is related to the orthogonal complement of U. That relation needs an inner product, and is carried out in *Linear Algebra* [12].

Lemma 13.3.10: Properties Of Annihilator

Let $U \subseteq V$ be a subspace. Then:

1. $U^0 \subseteq V^*$ is a subspace
2. if $\dim(V) < \infty$, then $\dim(V) = \dim(U) + \dim(U^0)$

***Proof*:**
exercise

Proposition 13.3.11: Properties Of Dual Maps

Let V and W be (not necessarily finite dimensional) vector space over k, $f \in \mathrm{Hom}_k(V, W)$ so that $f^* \in \mathrm{Hom}_k(W^*, V^*)$ is its dual map. Then:

1. $\ker(f^*) = (\mathrm{im}(f))^0$ (implying f^* injective if and only if f surjective)
2. $\mathrm{im}(f^*) \subseteq (\ker(f))^0$ is a subspace
3. If $\dim(V), \dim(W) < \infty$, then $\mathrm{im}(f^*) = (\ker(f))^0$ (implying f^* surjective if and only if f injective)

***Proof*:**
(prove these instead of exercise? The adjoint generalization is in *Linear Algebra* [12].)

The first point in proposition 13.3.11 is particularly useful in that in infinite dimensions, f being injective *does not* imply f is surjective. In the right situation, checking the injectivity of a linear transformation is *much* easier than checking surjectivity, and so checking the injectivity of f^* is often advantageous.

The next topic we will cover might at first seem abstract, however it is an important demonstration of the concept of *naturality*. Since we have taken $\mathrm{Hom}_k(V, k)$ and have shown it is a vector-space, we can take the dual of it again:

Definition 13.3.12: Double Dual

Let V be a vector-space and V^* it's dual. Then the *double-dual* of V is the dual of V^*, denoted V^{**}

The double dual has the property of being *naturally isomorphic* to V. To explain the idea of a natural isomorphism, we will show how V is *not* naturally isomorphic to V^*:

Example 13.3.13: Double Dual Motivation

Let V be a finite-dimensional vector space over k and $V^* = \text{Hom}(V, \mathbb{F})$. If $\beta = \{e_1, ..., e_n\}$ is a basis for V, and $\beta^* = \{e_1^*, ..., e_n^*\}$ is the dual basis for V^* where:

$$e_j^*(e_i) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Then the maps $e_i \mapsto e_i^*$ are bijective and linear, meaning V and V^* are isomorphic ($V \cong V^*$).

We can further consider $V^{**} = \text{Hom}(\text{Hom}(V, \mathbb{F}), \mathbb{F})$ being the set of all linear maps mapping all elements form $\text{Hom}(V, \mathbb{F})$ to $\mathbb{F}$. We can in similar fashion make an isomorphism $V^* \cong V^{**}$ letting us conclude that $V \cong V^{**}$.

This whole construction started with a choice of basis form V, which need not be unique, and so it's not a "property" of V. However, we can still define an isomorphism between V and V^{**} *without* any reference to a basis. For any $v \in V$, we can define a map

$$(f \mapsto f(v)) : V^* \to \mathbb{F} \qquad (f \mapsto f(v))(g) = g(v)$$

that is, every functional $g \in V^*$ evaluates the value and gets us $g(v)$. This map is sometimes called the evaluation function. Let's switch up the notation for simplicity:

$$ev_v : V^* \to \mathbb{F} \qquad ev_v(g) = g(v)$$

where ev_v stands for evaluate at v. Now, defining map $V \to \text{Hom}(V^*, \mathbb{F})$, $v \mapsto ev_v$ is an *isomorphism*. Bijectivity is left to check, for linearity, let φ be the function in question, so that for $k \in \mathbb{F}$ and $v, w \in V$:

$$ev(k \cdot v) = ev(kv) = ev_{kv} = k \cdot ev_v = kev(v)$$

$$ev(v + w) = ev_{v+w} = ev_v + ev_w = ev(v) + ev(w)$$

where kv represents the vector we get when applying the ring action on v. Note that we essentially used the lineary of the object we are mapping to in order to prove linearity.

Thus, we defined this isomorphism with *no* reference to a basis on V. We can't do the same trick for $V \to V^*$, since if we try to do $v \mapsto f(v)$, it is ambiguous which f we are talking about.

The intuition behind the double-dual having a natural isomorphism is that there is more information to work with; the reader should ponder on what extra information has been provided.

13.3.2 Prelude to Adjoints of Vector Space

must also foreshadow Yoneda's lemma here and also foreshadow, if possible (potentially in a titledbox) Z-valued Points.

Now that we have the basic structure of (finite dimensional) dual spaces, we may return to the discussion of defining V internally or externally. Let's say that you have been given $\text{Hom}_k(V, k)$, but not the structure of V. Is the information in $\text{Hom}_k(V, k)$ sufficient to recover the structure of V?

The answer is almost: the following proposition shows how much information we can recover:

Proposition 13.3.14: Recovering V From it's Dual

Let V be some vector-space and V^* it's dual. Then if $f(v) = f(w)$ for all $f \in V^*$, then

$$v = w$$

***Proof*:**
Exercise. This can be proven with and without the use of a basis for V.

13.3.3 Functor of Points for Vector Spaces

Another way to bring in the intuition from the introduction is to take the dual of a functional $V \to k$, namely to think of elements of V as functions $k \to V$. The reader should check that all linear transformations $k \to V$ are of the form:

$$t \to tx$$

for some $x \in V$. Represent these transformations as $g_x : k \to V$. Taking the dual of g_x we get:

$$\begin{aligned} g_x^*(\varphi) &= \varphi \circ g_x \\ &\equiv \varphi(g_x(t)) && \forall t \in k \\ &= \varphi(tx) && \forall t \in k \\ &= t\varphi(x) && \forall t \in k \\ &\equiv g_{\varphi(x)} \end{aligned}$$

Hence g_x^* maps φ to $g_{\varphi(x)} : k \to k$ where $t \mapsto t\varphi(x)$. With the identification between elements of V and the linear transformations $k \to V$, we may interpret this as a functional $V^* \to k$ where:

$$\varphi \mapsto \varphi(x)$$

Given $V \cong V^*$, then if β is the basis defining the isomorphism, each $\varphi \in V^*$ is associated to a vector $v \in V$, namely the vector for which:

$$\begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \end{pmatrix} \mapsto \sum_i v_i \beta_i$$

where $\sum_i$ is a finite sum. Then the linear functional $V \to k$ is defined by:

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \end{pmatrix} \mapsto \sum_i v_i x_i$$

where again $\sum_i$ is a finite sum since only finitely many v_i (and x_i) are nonzero Let's label this functional f_x. The element $x \in V$ may be interpreted, functionally, as $t \to tx$ or $v \to \sum_i v_i x_i$. Geometrically, the first function can loosely be thought of as defining a magnitude and direction. This is *not* a correct interpretation in infinite dimensions in general, since a magnitude requires a norm and a bare vector space carries none, however it is valid in finite dimension[21]. The second function can is the "dual" notion, where instead of focusing on x, the focus is on how x manipulates every other element. The reader may want to ponder how this relates to the "internal" and "external" way of describing object mentioned at the beginning of section 13.3. This also shows that V always *injects* into V^*.

As mentioned, it is not always the case that $V \cong V^*$. This immediately falls apart if V is infinite dimensional. To demonstrate this, let us consider the duality presented above where each $x \in V$ is associated with the functional f_x. The key is that given any basis β for V, $[x]_\beta$ has only finitely many components. Hence, the following valid function $F \in V^*$ is not associated to any f_x:

$$[v]_\beta \to \sum_i v_i$$

where the sum is finite since v when represented as an infinite tuple must only have finitely many nonzero components. The functional F is associated to the sequence $(1, 1, ..., 1, ...)$, namely the sequence where 1 is in each position. No f_x can be associated to F since by construction f_x, when represented as an infinite tuple, must have all but finitely position zero. This immediately shows that V^* is much larger, and hence contains much more information than the elements of V can represent. In fact, V^* contains even more interesting functionals. Consider:

$$(v_1, v_2, ..., v_n, ...) \mapsto (0, v_1, ..., v_{n-1}, ...)$$

Notationally, this may be represented as $f(\beta_i) = \beta_{i+1}$ given a basis β for V. This is called he *right shift operator*. This linear transformation can only be defined since there are infinite dimensions, hence each component may shift over by 1.

There are in fact so many linear functionals on infinite dimensional vector spaces that the study of such spaces is almost always accompanied with additional structure to bring some more regularity to the elements. The core problem stems from the extra flexibility working with infinitely many dimensions entails. Hence, some form of "finiteness" is usually imposed via the introduction of a *norm*, or a particular powerful norm called an *inner product*. In the case where there is an inner product on a vector space V, the isomorphism $V \cong V^*$ is recovered where the isomorphism is an inner-product space isomorphism. The inner product gives even more information. As shown in exercise ref:HERE, $(-)^*$ is not a functor since there is no choice of basis that shall naturally allow for composition of linear transformations. In the case of there being an inner product, this issue will be resolved, namely for each matrix representation $[L^*] = [L]^t$, there shall exist a linear transformation L^t for which $L^t : W \to V$ is a linear transformation that shall make $(-)^*$ (resp. $(-)^t$) a functor! This new linear transformation L^t shall be called the *adjoint* of L. The exploration of these notions is very fruitful even in finite dimensions, and it is carried out in *Linear Algebra* [12].

Exercise 13.3.1

1. Some things with infinite dimensions

2. Let $U \subseteq V$ be a subspace. Show that $U^{00} \cong U$. If $\widehat{(\_\_\_)}$ is the natural embedding of V into V^{**}, then $\hat{U} = U^{00}$

3. Let U_1, U_2, V be finitely dimensional vector spaces. Show that:

    1. $(U_1 + U_2)^0 = U_1^0 \cap U_2^0$

[21]This interpretation holds true in Hilbert Spaces

2. $(U_1 \cap U_2)^0 = U_1^0 + U_2^0$
3. $U^{00} = U$
4. If $W \subseteq V$, then $(V/W)^* = W^*$

4. Consider the category whose only object is a vector-space V and morphisms are all linear transformations from V to itself. Choosing a basis for V, show that $(-)^*$ is a contravariant-functor.

13.3.4 Multilinear Forms

Studying linear functionals allow us to give some scalar value to vectors. However, this is sometimes not enough: it is very common in mathematics that one of the two scenarios occur (we will see examples of these in this and coming section[22])

1. We want to study many functional on spaces simultaneously (ex. both V and V^*)
2. We require many copies of the same space to have enough "data" to define the desired functional

In this section, we elaborate on the theory allowing us to do just that, give some examples elaborating the above intuitions, and in the coming section we will see how these are of great use in defining some invariants for linear transformations and bringing in geometry into vector-spaces:

Definition 13.3.15: Bilinear and Multilinear Maps

Let V and W be vector spaces over k. Then a map $\varphi : V \times W \to k$ is called *bilinear* if

1. for any $v \in V$, the map $\varphi(v, \cdot) : W \to k$ mapping $w \mapsto \varphi(v, w)$ is linear
2. for any $w \in W$, the map $\varphi(\cdot, w) : V \to k$ mapping $v \mapsto \varphi(v, w)$ is linear.

More generally, if $V_1, V_2, ..., V_n$ are vector-space sover k, then $\varphi : V_1 \times V_2 \times \cdots \times V_n \to k$ is *multilinear* (or n-linear if the number of variables is known) if for all i, $1 \leq i \leq n$, given fixed $v_1, v_2, ..., v_{i-1}, v_{i+1}, ..., v_n$, the map

$$\varphi(v_1, v_2, ..., v_{i-1}, \cdot, v_{i+1} ..., v_n)$$

is linear. Two important types of multilinear maps are:

- (Symmetric) for any $v_i, v_j \in V$:

$$\varphi(v_1, ..., v_i, ..., v_j, ..., v_n) = \varphi(v_1, ..., v_j, ..., v_i, ..., v_n)$$

- (Alternating) for any $v_i, v_j \in V$:

$$\varphi(v_1, ..., v_i, ..., v_j, ..., v_n) = -\varphi(v_1, ..., v_j, ..., v_i, ..., v_n)$$

[22] There are also many examples in differential geometry, such as covariant/contravariant/mixed tensor fields on a manifold

13.3.16 WARNING Since we are now dealing with multilinear maps, these are *no longer morphisms in the category of* Vect_k! Hence, multilinear maps are a different object of study. In chapter 15, we will show how we can study multilinear maps via linear algebra.

Example 13.3.17: Bilinear and Multilinear Maps

1. Let $\varphi : V \times V^* \to k$ be defined by $(v, f) \mapsto f(v)$. Then φ is bilinear
2. Let $V = \mathbb{R}^n$ be a vector-space over $\mathbb{R}$ and take $\varphi : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ to be defined as:
$$\varphi\left(\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}\right) = \sum_{i=1}^{n} x_i y_i$$
Then φ is bilinear, and is called the *dot product* or *scalar product* of two vectors $x, y \in \mathbb{R}^n$
3. Let $\mathbb{R}$ be a vector-space over itself, and take $V : \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} \to \mathbb{R}$. Then the map that multiplies all the elements:
$$V(x_1, x_2, ..., x_n) = x_1 x_2 \cdots x_n$$
is a n-linear. As a paticualr example, consider $\varphi : \mathbb{R} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and imagine any 3-tulpe (x_1, x_2, x_3) represents the length, width, and height of a shape. Then $V(x_1, x_2, x_3) = x_1 x_2 x_3$ will give the volume of a rectangular prism (or a cube if $x_1 = x_2 = x_3$). The fact that multiplication produces a natural multi-linear map means that multilienar maps show up a lot in geometry. it is for this reason that multilinear maps are sometimes called *volume forms*. We will see more of these in section 16.3.
4. (If you know some calculus) Let $V = C([0, 1], \mathbb{R})$ be the set of all continuous functions from $[0, 1]$ to $\mathbb{R}$. Then define $\varphi : V \times V \to \mathbb{R}$ to be:
$$\varphi(f, g) = \int_0^1 f(t)g(t)\, dt$$
Then φ is bilinear. This is the infinite dimensional equivalent of the dot product.

In the rest of this section, we will focus on Bilinear forms: the results can easily be generlized using induction to Mutilinear forms. The first thing worth trying to find is wether there is still a reasonable notion of matrix representaiton of bilinear forms, since they are no longer linear maps. The following definition and lemma give us jut that:

Definition 13.3.18: Bilinear Form Matrix Representation

Let V and W be finite dimensional vector spaces over k of dimension n and m and with basis β, γ respectively. Let $\varphi : V \times W \to k$ a bilinear form. Then define the *matrix representation* of φ to be:

$$[\varphi]_{\beta,\gamma} := (\varphi(\beta_i, \gamma_j))_{\substack{1\leq i\leq n \\ 1\leq j\leq m}}$$

Lemma 13.3.19: Bilinear Form Matrix Representation

Let V and W be finite dimensional vector spaces over k of dimension n and m and with basis β, γ respectively. Let $\varphi : V \times W \to k$ a bilinear form. Then

$$\varphi(v, w) = [v]_\beta^t [\varphi]_{\beta,\gamma} [w]_\gamma$$

***Proof*:**
Write $v = \sum_{i=1}^n a_i\beta_i$ and $w = \sum_{j=1}^m b_j\gamma_j$, so that $[v]_\beta$ has entries a_i and $[w]_\gamma$ has entries b_j. Bilinearity of φ expands each argument separately:

$$\varphi(v, w) = \varphi\Big(\sum_i a_i\beta_i,\ \sum_j b_j\gamma_j\Big) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j\, \varphi(\beta_i, \gamma_j)$$

On the other side, the (i, j) entry of $[\varphi]_{\beta,\gamma}$ is $\varphi(\beta_i, \gamma_j)$, so the matrix product is $[v]_\beta^t [\varphi]_{\beta,\gamma} [w]_\gamma = \sum_i \sum_j a_i\, \varphi(\beta_i, \gamma_j)\, b_j$. The two sums agree term by term.

Definition 13.3.20: Orthogonal

Let $\varphi : V \times V \to k$ be a bilinear form and $S \subseteq V$ a *subset*. Then the set:

$$S^\perp := \{y \in V \mid \varphi(x, y) = 0 \text{ for every } x \in S\}$$

is called the *orthogonal complement* of S. If $\varphi(x, y) = 0$, then x is *perpendicular* to y, written $x \perp y$. If $x \perp y$ for all $y \in S$, then $x \perp S$.

The orthogonal complement is always a subspace, whatever the subset S was.

Lemma 13.3.21: The Orthogonal Complement Is a Subspace

Let $\varphi\colon V \times V \to k$ be a bilinear form and $S \subseteq V$ any subset. Then $S^\perp$ is a subspace of V, and $S^\perp = (\operatorname{span} S)^\perp$.

***Proof*:**
Linearity of φ in its second argument gives $\varphi(x, 0) = 0$ for every x, so $0 \in S^\perp$. If $y, y' \in S^\perp$ and $a \in k$, then for every $x \in S$

$$\varphi(x, y + ay') = \varphi(x, y) + a\,\varphi(x, y') = 0 + a \cdot 0 = 0$$

so $y + ay' \in S^\perp$, and $S^\perp$ is a subspace.

For the second claim, $S \subseteq \operatorname{span} S$ forces $(\operatorname{span} S)^\perp \subseteq S^\perp$. Conversely let $y \in S^\perp$ and let $x = \sum_{i=1}^m a_i x_i$ with $x_i \in S$; linearity in the first argument gives $\varphi(x, y) = \sum_i a_i \varphi(x_i, y) = 0$, so $y \in (\operatorname{span} S)^\perp$, as we sought to show.

Taking S to be all of V produces the subspace on which the form carries no information at all, and the form deserves a name according to whether that subspace is trivial.

Definition 13.3.22: The Radical and Nondegeneracy

Let $\varphi\colon V \times V \to k$ be a bilinear form. The *radical* of φ is the subspace

$$\operatorname{rad}(\varphi) := V^\perp = \{y \in V \mid \varphi(x, y) = 0 \text{ for every } x \in V\}$$

The form φ is *nondegenerate* if $\operatorname{rad}(\varphi) = 0$, and *degenerate* otherwise.

For a symmetric form the radical is the same computed on either side, so no ambiguity arises from having used the second argument. The radical is detected by the matrix of the form in any single basis, which is the form in which nondegeneracy is checked in practice.

Lemma 13.3.23: Nondegeneracy and the Matrix of the Form

Let V be finite dimensional with basis β, let $\varphi\colon V \times V \to k$ be a bilinear form, and let $A = [\varphi]_{\beta,\beta}$ be its matrix (definition 13.3.18). Then

$$\operatorname{rad}(\varphi) = \left\{y \in V \;\middle|\; A^t [y]_\beta = 0\right\}$$

and consequently φ is nondegenerate if and only if A is invertible. The criterion does not depend on which basis β was chosen, since the left-hand side does not.

***Proof*:**
By lemma 13.3.19, $\varphi(x, y) = [x]_\beta^t A [y]_\beta$ for all $x, y \in V$. Fix y. Then $y \in \operatorname{rad}(\varphi)$ means $[x]_\beta^t A [y]_\beta = 0$ for every $x \in V$, and as x ranges over V the coordinate vector $[x]_\beta$ ranges over all of k^n. A vector $u \in k^n$ satisfies $w^t u = 0$ for every $w \in k^n$ if and only if $u = 0$, so the condition is $A[y]_\beta = 0$; transposing the defining relation gives the displayed form. Hence $\operatorname{rad}(\varphi) = 0$ if and only if A has trivial kernel, which for a square matrix is invertibility.

The final sentence needs no comparison of bases: $\operatorname{rad}(\varphi)$ was defined without reference to any basis, so if A is invertible for one choice of β the radical is zero, and then the same equivalence run backwards makes the matrix invertible for every other choice, completing the proof.

Exercise 13.3.2

1. Let $f\colon V \times W \to k$ be a bilinear form on a pair of spaces, and write $W^\perp := \{v \in V \mid f(v, w) = 0 \text{ for all } w \in W\}$ and $V^\perp := \{w \in W \mid f(v, w) = 0 \text{ for all } v \in V\}$ for its two kernels. Show that f descends to a well-defined bilinear form $\bar{f}\colon V/W^\perp \times W/V^\perp \to k$, and that $\bar{f}$ is non-degenerate in each variable.

13.4 Linear Operators Invariants: Trace and Determinant

In this section, we will go over to common functionals on $M_n(k)$ (equivalently, $\text{End}_k(V)$ for an n-dimensional vector-space) that give more insight on linear transformations. A good chunk of the following theory works well over more general rings, and so we will sometimes work over a general commutative ring R.

The first functional on $M_n(k)$ we will define is called the *trace* and will allow us to detect matrices are similar to each other (i.e. the two are related by a change of basis):

Definition 13.4.1: Trace

Let $M_n(R)$ be the collection of all $n \times n$ matrices over k. Then $\text{tr} : M_n(R) \to R$ is the map:

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii}$$

The trace is studied for the following very useful properties:

Proposition 13.4.2: Properties of Trace

Let $A, B \in M_n(R)$. Then:

1. $\text{tr}(A) = \text{tr}(A^t)$
2. $\text{tr}(AB) = \text{tr}(BA)$

***Proof*:**
computational exercise

As a consequence, if A and B are equivalent so that there exists a P such that $A = PBP^{-1}$, then:

$$\text{tr}(A) = \text{tr}(PBP^{-1}) = \text{tr}(P(BP^{-1})) = \text{tr}((BP^{-1})P) = \text{tr}(BPP^{-1}) = \text{tr}(B)$$

Hence the trace is invariant under change of basis, making it a good tool to study linear maps without needing to pick a basis

Definition 13.4.3: Trace Of Linear Map

Let $f : V \to W$ be a linear map and let β, γ be basis for V and W respectively. Then define $\text{tr}(f)$ to be

$$\text{tr}(f) := \text{tr}([f]_\beta^\gamma)$$

Corollary 13.4.4: Trace Of Dual

Let $f : V \to W$ and $f^* : W^* \to V^*$ be the dual map of f. Then:

$$\text{tr}(f) = \text{tr}(f^*)$$

***Proof*:**
exercise

(check this out to: geometric interpretation of trace)

The next map is a very common multilinear form that will be used extensively: this form will also be independnet of basis, but with the extra information giving a very clear geometric picture of how a linear transformation effects a 1-cube in $\mathbb{R}^n$:

Definition 13.4.5: Determinant

Let R be a commutative ring and $A = (a_{ij}) \in M_n(R)$ a square matrix. then the *determinant* of A is a map $\det : M_n(R) \to R$:

$$\det(A) := \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} a_{i\sigma(i)} \in R$$

Proposition 13.4.6: Properties Of The Determinant

Let R be a commutative ring and $A \in M_n(R)$. Then:

1. $\det(I_n) = 1$.
2. $\det(A) = \det(A^t)$, here $(a_{ij})^t = (a_{ji})$.
3. (multilinearity) det is R-linear in each column separately: if the j-th column of A is $cu + c'u'$ for $c, c' \in R$ and column vectors u, u', and if A_u, $A_{u'}$ denote the matrices agreeing with A outside column j and having u, respectively u', in column j, then

$$\det(A) = c\det(A_u) + c'\det(A_{u'})$$

   By (2) the same holds for rows.
4. (alternating) If two columns of A are equal then $\det(A) = 0$; the same holds for rows. Combined with (3), if one column is a scalar multiple of another then $\det(A) = 0$.
5. (permuting rows) If A_τ denotes the matrix whose i-th row is the $\tau(i)$-th row of A, for $\tau \in S_n$, then $\det(A_\tau) = (-1)^\tau \det(A)$.

***Proof*:**
All five are read off the defining sum.

1. For $A = I_n$ the product $\prod_i \delta_{i\sigma(i)}$ vanishes unless $\sigma(i) = i$ for every i, so only $\sigma = \mathrm{id}$ contributes, and it contributes 1.
2. By definition $\det(A^t) = \sum_\sigma (-1)^\sigma \prod_i a_{\sigma(i)i}$. Re-indexing the product by $j = \sigma(i)$ gives $\prod_i a_{\sigma(i)i} = \prod_j a_{j\sigma^{-1}(j)}$, and $(-1)^\sigma = (-1)^{\sigma^{-1}}$ since a permutation and its inverse have the same parity. As σ runs over S_n so does σ^{-1}, so the two sums agree term for term.
3. In the term indexed by σ, exactly one factor is drawn from column j, namely $a_{\sigma^{-1}(j)\,j}$.

Each term is therefore R-linear in column j with the other columns held fixed, and a sum of such terms is too.

4. Suppose columns $j \neq k$ of A are equal, and let $\tau = (j\ k)$. Since $a_{ij} = a_{ik}$ for all i, one has $a_{i\,\tau(m)} = a_{im}$ for every i and m, and hence
$$\prod_i a_{i\,\tau\sigma(i)} = \prod_i a_{i\sigma(i)}$$
while $(-1)^{\tau\sigma} = -(-1)^{\sigma}$. The assignment $\sigma \mapsto \tau\sigma$ is an involution of S_n without fixed points, so it partitions S_n into pairs whose two terms are negatives of one another. The sum is therefore 0. Note that this argument cancels terms exactly and so does not require 2 to be invertible in R. The statement for rows follows by (2), and the statement for proportional columns by pulling the scalar out with (3).

5. By definition $\det(A_\tau) = \sum_\sigma (-1)^\sigma \prod_i a_{\tau(i)\sigma(i)}$. Substituting $j = \tau(i)$ turns the product into $\prod_j a_{j\,\sigma\tau^{-1}(j)}$. Writing $\rho = \sigma\tau^{-1}$, so that $(-1)^\sigma = (-1)^\rho(-1)^\tau$, and letting ρ run over S_n as σ does,
$$\det(A_\tau) = (-1)^\tau \sum_{\rho \in S_n} (-1)^\rho \prod_j a_{j\rho(j)} = (-1)^\tau \det(A)$$
completing the proof.

The determinant is multiplicative, and this holds for all square matrices over a commutative ring, not only invertible ones. The proof is a direct expansion of the defining sum, with proposition 13.4.6(4) killing every term that is not indexed by a permutation.

Proposition 13.4.7: Multiplicativity of the Determinant

Let R be a commutative ring and let $A, B \in M_n(R)$. Then
$$\det(AB) = \det(A)\det(B)$$

***Proof*:**
Write $A = (a_{ij})$ and $B = (b_{ij})$, so that $(AB)_{ij} = \sum_k a_{ik}b_{kj}$. Expanding the defining sum,
$$\det(AB) = \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik} b_{k\sigma(i)} \right)$$
Expanding the product over i amounts to choosing an index k for each i, that is, a function $f\colon \{1, \dots, n\} \to \{1, \dots, n\}$, so
$$\det(AB) = \sum_f \left(\prod_{i=1}^{n} a_{if(i)} \right) \left(\sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} b_{f(i)\sigma(i)} \right)$$
the sum being over all n^n such functions. For fixed f, let B_f be the matrix whose i-th row is the $f(i)$-th row of B, so that $(B_f)_{ij} = b_{f(i)j}$. The inner sum is exactly $\det(B_f)$.

If f is not injective, say $f(i) = f(i')$ with $i \neq i'$, then B_f has two equal rows and $\det(B_f) = 0$ by proposition 13.4.6(4). Only the injective f survive, and an injective self-map of a finite set is a permutation, so the sum runs over $f = \tau \in S_n$. For such an f, B_τ is B with its rows permuted by τ, so $\det(B_\tau) = (-1)^\tau \det(B)$ by proposition 13.4.6(5). Therefore

$$\det(AB) = \sum_{\tau \in S_n} \left(\prod_{i=1}^{n} a_{i\tau(i)} \right) (-1)^\tau \det(B) = \left(\sum_{\tau \in S_n} (-1)^\tau \prod_{i=1}^{n} a_{i\tau(i)} \right) \det(B) = \det(A)\det(B)$$

as we sought to show.

A block diagonal matrix has the determinant one would hope for. This is the computation that makes determinants of matrices assembled out of smaller ones tractable.

Proposition 13.4.8: Determinant of a Block Diagonal Matrix

Let R be a commutative ring and let $M \in M_n(R)$ be block diagonal with square diagonal blocks $B_1, \ldots, B_m$, so that $M_{uv} = 0$ whenever u and v lie in different blocks. Then

$$\det(M) = \prod_{t=1}^{m} \det(B_t)$$

***Proof*:**
Index the rows and columns of M by pairs (t, u), where t names the block and u the position inside it, so that $M_{(t,u),(t',u')}$ is $(B_t)_{uu'}$ when $t = t'$ and 0 otherwise. A permutation σ of the index set contributes a nonzero term to the defining sum only if σ maps each block into itself, since otherwise some factor $M_{(t,u),\sigma(t,u)}$ has $t' \neq t$ and vanishes. Such a σ is precisely a tuple $(\sigma_1, \ldots, \sigma_m)$ of permutations of the individual blocks, acting as $\sigma(t, u) = (t, \sigma_t(u))$.

These block permutations move disjoint sets of indices, so σ is the product of the commuting permutations they determine and $(-1)^\sigma = \prod_t (-1)^{\sigma_t}$, the sign being a homomorphism. Hence

$$\det(M) = \sum_{(\sigma_1,\ldots,\sigma_m)} \prod_{t=1}^{m} (-1)^{\sigma_t} \prod_u (B_t)_{u\sigma_t(u)} = \prod_{t=1}^{m} \left(\sum_{\sigma_t} (-1)^{\sigma_t} \prod_u (B_t)_{u\sigma_t(u)} \right) = \prod_{t=1}^{m} \det(B_t)$$

the middle step being the distributive law, since the tuples range independently over the factors, completing the proof.

Determinents will have the incredible property of being able to detect nonsingualr matrice. To see this, let's take a moment study the determinent of serval maps of the form $f : \mathbb{R}^2 \to \mathbb{R}^2$ (HERE, show the geometric intuition)

With this geometric intuition, we will now proceed with how to capture it algebraically:

Proposition 13.4.9: Elementary Matrices And Determinant

Let A be a matrix then

1. (sitching rows) $\det(A') = -\det(A)$
2. (adding copies of one ro to another) $\det(A') = \det(A)$
3. (multiplying one ro by a unit c) $\det(A') = c\det(A)$

***Proof*:**
here

Proposition 13.4.10: Invertability And Determinants

Let R be a commutative ring. Then:

1. $A \in M_n(R)$ is invertible if and only if $\det(A)$ is a unit in R
2. the determinante is a homomorphism $\det : \mathrm{GL}_n(R) \to R^\times$, that is, for any $A, B \in \mathrm{GL}_n(R)$:
$$\det(A \cdot B) = \det(A) \cdot \det(B)$$

***Proof*:**
(if $R = k$ is a field)

(words here)

***Proof*:**
(if R is a general commutative ring)

We will finish off this section an application of determinents to solve systems of linear equations. This will become particularly useful for a few proofs in the book[23].

Proposition 13.4.11: Cramer's Rule

Let R be a commutative ring, $A \in M_n(R)$, and $\det(R)$ be a unit of R. Let $A^{(j)}$ be the matrix obtined by replacing the jth column of A by the column vector b. Then:

$$x_j = \det(A)^{-1}\det(A^{(j)})$$

***Proof*:**

(Also really want to put down the "matrix invertible if determant maps to units" result)

The final topic covered in this section will be that of defining the *volume*:

[23] In particular, proposition 26.1.4

Definition 13.4.12: Volume

Let V be an inner product space and $e = (e_1, e_2, ..., e_n)$ an orthonormal basis for V. Then if $v = (v_1, v_2, ..., v_n)$ is a set of vectors given by a change of basis matrix $v = Ae$, *volume* of v is

$$\mathrm{vol}(v) = |\det(A)|$$

The absolute value around $\det(A)$ is to eliminate orientation information. Geometrically, this can be seen as the volume of a parallellapided when $v_1, v_2, ..., v_n$ are linearly independent, namely if

$$P = \left\{ \sum a_i v_1 \;\middle|\; 0 \leq x_i < 1 \right\}$$

Then the intuitive notion of volume of P agrees with our definition. The case where $v = (e_1, e_2, ..., e_n)$ shows that the volume of a vector of unit lengths is always 1.

Exercise 13.4.1

1. Let $A \in M_n(k)$, $k \subseteq \mathbb{C}$. Show that $\det(\overline{A}) = \overline{\det(A)}$ where we define $\overline{A}$ component-wise to be $(\overline{A})_{ij} = \overline{a_{ij}}$ with $\overline{a_{ij}}$ being the complex conjugate $(\overline{x + iy} = x - iy)$
2. Let $A \in M_n(k)$. Define the *adjucate* of A to be $\mathrm{Ajd}(A) = C^T$ where C is the cofactor expansion matrix of A. show that if $\det(A) \neq 0$, then $A^{-1} = \det(A)^{-1}\,\mathrm{Adj}(A)$.
3. (If you konw multi-variable calculus) Treating $M_n(\mathbb{R})$ as $\mathbb{R}^{n^2}$, show that the dervative of the determinant map $\det : M_n(\mathbb{R}) \to \mathbb{R}$ at the identity I is the trace tr: $(\det)'(I) = \mathrm{tr}$. This gives a surprisingly interesting link between the two concepts, and is important in the beginning of the study of *Lie Algebras* of many matrix groups such as $\mathrm{GL}_n(k)$, $\mathrm{SL}_n(k)$, $O_n(k)$, and so forth, where $k \in \{\mathbb{R}, \mathbb{C}\}$.
4. More generally, show that $\det : M_n(\mathbb{R}) \to \mathbb{R}$ has a derivative at any point A, namely: $(D_A \det)(X) = \mathrm{tr}(\mathrm{Adj}(A^t)X)$.
5. Let R be an integral domain and $f : M \to N$ a R-module homomorphism between free R-modules of rank n, m respectively (note M, N satisfy the IBN property). Choosing a basis for M, N and a matrix A representing f, show that if $\det(A) = 0$ Then A is not invertible. Show furthermore A is not linearly dependent.
6. Take the same setup as in question (13.4.1 (5)). Show that the converse is not true: $\det(A) \neq 0$ does *not* imply A is invertible (hint: consider $R = \mathbb{Z}$).

13.5 Eigenvectors and Characteristic Polynomial

After having found the trace and determinant, the reader may wonder if there are more invariant functionals on $\mathrm{End}_k(V)$ for an n-dimensional vector space. As it turns out, these are both special cases of a collection of n invariant given by the coefficients of the following polynomial:

Definition 13.5.1: Characteristic Polynomial

Let M be a free R-module and let $\alpha \in \mathrm{End}_R(M)$. Let $I \in \mathrm{End}_R(M)$ denote the identity operator. Then the characteristic polynomial is defined to be:

$$P_\alpha(t) := \det(tI - \alpha) \in \mathrm{End}_R(M)$$

which when α has a basis representation A, then $P_\alpha(t)$ is an element of $R[t]$, R-polynomials with indeterminate t.

The origins of this seemingly complicated looking polynomial comes from its link to eigenvectors and eigenvalues (namely scalars and vector of a linear operator that satisfy $\alpha(v) = \lambda v$ for appropriate $\lambda \in R$ and $v \neq 0$) which shall be explored soon. Some important properties are established first:

Proposition 13.5.2: Properties Of The Characteristic Polynomial

Let $P_\alpha(t)$ be the characteristic polynomial of the linear operator $\alpha : M \to M$ for a free R-module of rank n. Then:

1. $P_\alpha(t)$ is monic of degree n
2. The coefficient of t^{n-1} term is $-\mathrm{tr}(\alpha)$
3. The constant term is $(-1)^n \det(\alpha)$
4. If α and β are similar, then $P_\alpha = P_\beta$

***Proof*:**

1. Expanding the definition immediately reveals the result. Let $A = (a_{ij})$ be a representative of α given some choice of basis. Then:

$$P_\alpha(t) = \det \begin{pmatrix} t - a_{11} & -a_{12} & \cdots & -a_{1n} \\ t - a_{21} & t - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & t - a_{n2} & \cdots & t - a_{2n} \end{pmatrix}$$

2. Choose A as above. Then notice that when expanding $P_\alpha(t)$, the values of the t^{n-1} term only come from terms on the diagonal, namely we get:

$$\sum_i^n t \cdot t \cdot \cdots \cdot t \cdot a_{ii} \cdot t \cdot \cdots \cdot t \cdot t$$

giving us the desired result

3. Setting $t = 0$ and using multi-linearity gives the desired result
4. Assuming $\alpha \sim \beta$, there exists a ρ such that

$$\alpha = \rho \circ \beta \circ \rho^{-1}$$

Then by using properties of functions, we see that:

$$tI - \alpha = \rho \circ (tI - \beta) \circ \rho^{-1}$$

But then, the determinant of both sides are the same. Choosing a matrix representation P for ρ and B for $tI - \beta$ we see that by proposition 13.4.10

$$\det(PBP^{-1}) = \det(P)\det(B)\det(P)^{-1} = \det(B)\det(P)\det(P)^{-1} = \det(B)$$

hence

$$P_\alpha(t) = \det(tI - \alpha) = \det(\rho \circ (tI - \beta) \circ \rho^{-1}) = \det(tI - \beta) = P_\beta(t)$$

completing the proof.

Hence, the characteristic polynomial is of the form:

$$t^n - (\mathrm{tr}\alpha)t^{n-1} + \cdots + (-1)^n(\det \alpha)$$

As far as I am aware, the other coefficients of this polynomial don't have a special name like the trace and determinant, however on the other hand these coefficients are special since they always appear in any dimension (they are equal when $n = 1$ for trivial reasons, the determinant is zero if α is not surjective, and the trace is zero if α is a commutator, see exercise ref:HERE not done yet see this). Furthermore, note that all coefficients are invariant under similarity, meaning that if the trace or determinant between two linear operators differ, they are *not* similar! Unfortunately, the converse does not hold:

Example 13.5.3: Similar Matrices

Let:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

These each have a characteristic polynomial of $t^2 - 4$. However, they are not similar: Notice that A is the identity, and conjugating A by any invertible matrix P would give the identity, and hence never B.

As mentioned earlier, the characteristic polynomial naturally arises when studying eigenvector and values. Let us start by formally defining these terms:

Definition 13.5.4: Eigenvalue and Spectrum

Let M be a free R-module over a commutative ring R and let $\alpha \in \mathrm{End}_R(M)$ be a linear operator on M. Then a scalar $\lambda \in R$ is called an *eigenvalue* of α if there exists a nonzero $v \in M$ such that

$$\alpha(v) = \lambda v$$

The collection of all eigenvalues is called the *spectrum* of α.

The existence of eigenvalues of an operator α means that in some subspace $N \subseteq M$ the transformation $\alpha|_N$ is simply scaling. Over $\mathbb{C}$[24], we shall see that under a suitable choice of basis that scaling represents one of the two possible behaviors of a linear operator, in particular a linear operator can

[24]so that there is no "algebraic obstruction", an intuition that shall be covered in great detail in chapter 19

only either scale or *shear*. In example 13.5.3, matrix B scaled the first basis vector and sheared the second basis vector. Scaling is the easier of the two behaviours as the algebraic invariant for detecting scaling is the characteristic polynomial, while detecting and characterizing shearing requires more complicated invariants: shearing shall be explored in chapter 14, section 14.3.

Eigenvalues are extremely important, having a wide range of applications in quantum mechanics (the term *spectrum of an atom* comes from finding the spectrum of an endomorphism) functional analysis (K-theory has its origin in the spectrum of infinite linear operators), differential equations (in finding a basis to the solutions of an ODE and in defining the Fourier Transform), to number theory and Field theory (as we shall see later in the book), and so much more! Due to the prominence of eigenvalues, some may claim that the study of endomorphisms of vector-spaces is one of the fundamental concepts of mathematics or the sciences. In *Algebraic Geometry* [3] it is shown that any commutative ring has a spectrum associated to (though by the definition of a spectrum of a ring, the connection shall not be obvious[25]).

The eigenvalue is invariant under similarity, which the reader should attempt to prove before looking at the following: Letting α, β be similar using ρ ($\alpha = \rho \circ \beta \circ \rho^{-1}$) and
$beta(v) = \lambda v$, then $\alpha \circ \rho = \rho \circ \beta$ and

$$\alpha(\rho(v)) = (\rho \circ \beta \circ \rho^{-1})(\rho(v)) = \rho \circ \beta(v) = \rho(\lambda v) = \lambda \rho(v)$$

since $\rho(v) \neq 0$ if $\lambda \neq 0$ (namely, it is invertible), we see that λ is an eigenvalue of α if and only if λ is an eigenvalue of β.

Now, as promised, the following shows why the characteristic polynomial is a natural concept based on its invariance in similarity:

Proposition 13.5.5: Eigenvalues And Characteristic Polynomial

Let $P_\alpha(t)$ be the characteristic polynomial of the linear operator α. Then λ is an eigenvalue if and only if λ is a root of $P_\alpha(t)$. In other words, the set of roots of $P_\alpha(t)$ is exactly the spectrum of α.

***Proof*:**
Using the properties of determinants:

$$\begin{aligned} \lambda \text{ is an eigenvalue} &\Leftrightarrow \exists v \neq 0 \text{ s.t. } \alpha(v) = \lambda v \\ &\Leftrightarrow \exists v \neq 0 \text{ s.t. } (\lambda I - \alpha)(v) = 0 \\ &\Leftrightarrow (\lambda I - \alpha) \text{ is not injective} \\ &\Leftrightarrow \det(\lambda I - \alpha) = 0 \\ &\Leftrightarrow P_\alpha(\lambda) = 0 \end{aligned}$$

as we sought to show.

[25] The curious reader may checkout EYNKTA functional analysis where this connection is fully fleshed out, as the connection is fundamentally a functional analysis result and would be too far out of the scope of an algebra textbook to be covered

Example 13.5.6: Eigenvalues And Characteristic Polynomials

1. If α is not injective, then 0 is an eigenvalue, otherwise 0 is not an eigenvalue.
2. Take
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
This matrix has a characteristic polynomial of $t^2 + 1$. If $R = \mathbb{R}$, then this characteristic polynomial has *no* roots. Similarly, the linear operator this matrix represents has no eigenvalues. Geometrically, this matrix can be seen as rotating $\mathbb{R}^2$ by $\pi/2$ degrees, which is why it has no eigenvalues over $\mathbb{R}$. If $R = \mathbb{C}$, then the polynomial has two roots, namely $i, -i$. The reader should verify that these are eigenvalues of a linear operator represented by this matrix. Intuitively, multiplying by i can be thought of as representing rotation by $\pi/2$ degree's which is why it is an eigenvalue.
3. Take
$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$
This matrix has a characteristic polynomial of $t^2 - 2$. If $R = \mathbb{R}$, then this polynomial has roots $\pm\sqrt{2}$, while if $R = \mathbb{Q}$, this polynomial has no roots, which equivalently implies that there is no eigenvalues when $R = \mathbb{Q}$. Geometrically, this transformation can be seen as reflecting the plane across a line passing through the origin with an irrational slope. Since the slope is irrational, there is no nonzero irrational scalar to a vector, and hence cannot be a rational eigenvalue.

Geometrically, eigenvectors correspond to the linear operator scaling in the appropriate direction. In the simplest case, an n-dimensional linear operator α can be represented in a basis for which $[\alpha]$ just scales appropriately in each direction. We take a moment to explore these type of matrices.

Since roots have the notion of multiplicity (the number of roots sharing the same value), it is natural to contemplate the multiplicity of eigenvalues and see what information this gives us:

Definition 13.5.7: Algebraic Multiplicity

Let λ be an eigenvalue of a linear operator α over a finitely generated free module. Then the *algebraic multiplicity* is its multiplicity as a root of $P_\alpha(t)$.

Example 13.5.8: Algebraic Multiplicity

Let M be a finitely generated free module over an integral domain R.

1. the identity operator I on M has a single eigenvalue, namely 1, which has algebraic multiplicity equal to the rank of M.
2. (give a non-trivial example)

Notice that the sum of the algebraic multiplicities is bounded by the rank of the free module M (resp. the dimension of the vector-space V). If R is *algebraically closed* (meaning every polynomial of degree n over R has exactly n roots[26]), then the sum of the algebraic multiplicities of the eigenvalues always equals the rank of M. The prototypical algebraically closed field is $\mathbb{C}$, the complex numbers.

The definition has been careful to call the multiplicity "algebraic". There is another notion of multiplicity which corresponds to the geometric behavior of multiplicity, namely there is a special subspace that is linked to each eigenvalue:

Definition 13.5.9: Eigenspace And Eigenvector

Let λ be an eigenvalue associated to the linear operator α over a finitely generated free R-module M. Then the sub-module

$$E_\lambda = \ker(\lambda I - \alpha)$$

is called the *eigenspace* associated to λ, and the elements of E_λ are called *eigenvectors* and represent the elements of M such that

$$\alpha(v) = \lambda v$$

Technically, it is more correct to call these spaces "eigen-modules" rather than eigenspace, however the study of eigen-modules over vector-spaces greatly precedes the generalization to eigen-modules over R-modules, giving us our current vocabulary.

Definition 13.5.10: Geometric Multiplicity

Let E_λ be an eigenspace corresponding to an eigenvalue λ. Then the rank of E_λ is the *geometric multiplicity* of λ.

It should be shown that the eigenspace is invariant under similarity, meaning it a special subspace that is invariant under α (we shall later call spaces that are invariant under an operator α-stable, where α is the operator in question).

Example 13.5.11: Geometric And Algebraic Multiplicity

Take

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Then the characteristic polynomial is $(t-1)^2$, and hence 1 is an eigenvalue with algebraic multiplicity 2. To find it's geometric multiplicity, by considering

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 + v_2 \\ v_2 \end{pmatrix}$$

we see that the right hand side is equal to $1v$ if and only if $v_2 = 0$, hence the eigenspace is

$$\operatorname{span} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

which is 1 dimensional, hence the geometric multiplicity is 1.

[26]See definition 19.5.1 and the lemma under it for more details

A result that should stand out to the reader given the above example is the following

Proposition 13.5.12: Algebraic And Geometric Multiplicity

Let λ be an eigenvalue associated to the linear operator α over a finitely generated free R-module M. Then the geometric multiplicity is always less than or equal to the algebraic multiplicity

***Proof*:**
exercise

One of the main results comes in the algebraic multiplicities add up to the rank of the free module and when the algebraic multiplicity matches the geometric multiplicity. Then there exists a basis P making α a bunch of scalars, namely:

Theorem 13.5.13: Diagonalizable

Let α be a linear operator over a finitely generated free R-module M. Then if the algebraic multiplicities of the eigenvalues sum to n and the algebraic multiplicities of each eigenvalue equal to their respective geometric multiplicities, there exists an invertible matrix $P \in \mathrm{GL}_n(R)$ such that for any representation A of α:

$$A = P \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} P^{-1}$$

where $\lambda_1, \lambda_2, ..., \lambda_n$ are the eigenvalues of α, and α (resp. A) is said to be *diagonalizable*. If $m = V$ is a vector-space, α is diagonalizable if and only if there exists a basis for V consisting of eigenvectors.

***Proof*:**
exercise.

Definition 13.5.14: Diagonalizable

Let α be a linear operator over a finitely generated free R-module M. Then if α admits a representation whose only nonzero entries lie on the diagonal, α is said to be *diagonalizable*. Similarly, a matrix that is similar to a diagonal matrix is said to be *diagonalizable*.

If α is diagonalizable, then M can be expressed in the following useful decomposition:

Theorem 13.5.15: Spectral Decomposition

Let $\alpha : M \to M$ be diagonalizable. Then M admits the following decomposition into direct sums

$$M \cong \bigoplus_{\text{eigenvalue } \lambda} E_\lambda$$

***Proof*:**
exercise

For future reference, let's give a name to the above decomposition:

Definition 13.5.16: Spectral Decomposition

Let $\alpha : M \to M$ be diagonalizable. Then the decomposition in the above theorem is called the *spectral decomposition* of M.

Given the spectral decomposition of M, α can simply be thought of as scaling each eigenspace E_λ by the eigenvalue, making the "action" of α on the space M completely transparent!

If the algebraic multiplicity does not sum to the rank of M or the geometric multiplicity does not match the algebraic multiplicity, then the matrix is *not* diagonalizable. If the algebraic multiplicity does not sum to the rank of M, then if we are working over field[27] we can take something called the *algebraic closure* of the field, which as the name implies means it is *algebraically closed*. To rigorously go over the algebraic closure of a field is too much work at this stage, and so will just be taken for granted when it is brought up, the curious reader may checkout part IV chapter 19, section 19.5.1. With the reader's current knowledge, most of the section can be parsed.

If the geometric multiplicity does not match the algebraic multiplicity, then it is a harder to find a form. However, in the next chapter, we shall see that there exists a representation that is *almost* diagonalizable: it shall be diagonalizable with possibly a 1 in the super-diagonal (the position right above the diagonal). In fact, *all* square matrices will be able to have this almost diagonal form! This shall be the aforementioned *shearing* behaviour that is the only other behaviour besides scaling that a matrix shall represent when choosing an appropriate basis.

13.6 Geometry and Inner Product

A vector space over an arbitrary field carries no notion of length and no notion of angle, and none can be manufactured from the field axioms alone. Producing one takes two things a general field does not have: an ordering, so that a quantity can be called nonnegative, and a conjugation, so that $\langle v, v\rangle$ can be forced to be real. The real and complex numbers give both, and over them the whole geometric apparatus follows, namely inner products and the Cauchy-Schwarz inequality, orthonormal bases and the Gram-Schmidt process, orthogonal complements and projections, the Riesz representation of a linear functional, adjoints, and the spectral theorem for normal and self-adjoint operators.

That development belongs to a book about $\mathbb{R}$ and $\mathbb{C}$ rather than to one about modules over a general ring, and it is carried out in *Linear Algebra* [12]. Nothing later in this book depends on it.

[27] more generally, over integral domains, since the ring of fractions of an integral domains is a field

14

Modules Over PIDs: Linear Algebra II

Modules over fields have some of the simplest possible structures. Because they are always free, they admit a trivial short exact sequence:

$$0 \to R^n \to M \to 0$$

In the language of free resolutions (definition 12.3.27), every module over a field admits a free resolution of length 0 (recall indexing starts at 0). It is natural to want to study modules that admit a free resolution of length 1. Specifically, if R is a ring and M is a finitely generated module with a surjection

$$R^n \to M \to 0$$

we ask when the kernel is also free, meaning there exists an R^m such that

$$0 \to R^m \to R^n \to M \to 0$$

is exact. As was foreshadowed in section 12.3.3, all finitely generated modules over PIDs admit free resolutions of length 1, and hence they shall be the focus of this chapter. Having a free resolution of length 0 (i.e., being free) implied the existence of a basis, and even over PIDs, matrix representations of linear transformations admitted Smith Normal Forms. A free resolution of length 1 offers slightly less rigid structural results; however, it remains incredibly powerful, introducing us to *Jordan Canonical Forms* and *Rational Canonical Forms* and expands the number of modules for which we have a structure theorem. These forms will provide a crucial new way of representing linear transformations, giving us deep invariants and information (namely ref:HERE).

The reader may naturally ask if it is interesting to further push the length of the free resolution and explore modules that admit a free resolution of length 2, 3, or n. While modules over polynomial rings always admit a finite length resolution (Hilbert's Syzygy Theorem; theorem 57.2.5), most interesting examples of such modules become a good deal more complex and require a build-up in algebraic geometry and homological algebra. For the interested reader, these lengths are intimately linked to the *Krull dimension* of a commutative ring[1]. The essential concepts for understanding this link will be covered later in this book (specifically, [13, Dimension] and chapter 57 section 57.2), and further reading can be done in [CITATION].

Goal

The goal of this section is to show finitely generated modules over PID's have the following structure:

$$M \cong R^k \oplus R/(a_1) \oplus \cdots \oplus R/(a_m)$$

We'll define the terms a_m and how to find k soon. Essentially, M shall be split into a part which is "as linearly dependent as possible", and the rest of M will be the elements which cannot be linearly independent (over R). These "rest of elements" of M can be identified with the simplest quotient rings of R, in particular a cyclic module.

Since PID's are UFD's, we will be able to find a 2nd representation given appropriate prime elements:

$$M \cong R^k \oplus R/(p_1) \oplus \cdots \oplus R/(p_n)$$

where p_i will turn out to represent prime elements of R (and the k will be the same).

A consequence of this would be the proof of the Fundamental Theorem of finitely Generated abelian groups (theorem 5.1.8). Since $\mathbb{Z}$ is a finitely generated Principal Ideal Domain, if M is a $\mathbb{Z}$-module (and so an abelian group), it will be a consequence of the following theoretical build-up that

$$M \cong \mathbb{Z}^k \oplus \mathbb{Z}/n_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

Another consequence will be the ability to classify all linear maps from a vector-space to itself (up to similarity). This means that the structure of $M_n(k)$ and $\mathrm{GL}_n(k)$ will be eminently understandable, having huge consequence in representation theory (part X) and many areas beyond algebra (Differential Geometry, K-theory, Optimization, ODEs, and so forth). In particular, if V is a vector space over $\mathbb{F}$ and T is a linear transformation, then like in example ref:HERE, we can form an $F[x]$-module over V where representing T with the action:

$$p(x) \cdot v = (a_n x^n + a_{n-1}x^{n-1} + \cdots a_1 x + a_0) \cdot v = a_n T(v)^n + \cdots + a_1 T(v) = a_0$$

then we can use the fact that $F[x]$ is a Principal Ideal Domain to decompose T into a nicer representation. Since every linear transformation can be represented as a matrix, then we can also decompose matrices in this fashion (which will be called Rational Canonical form and Jordan Canonical Form).

Note that finiteness is an important condition: we shall go over an example where if you have an infinitely generated Principal Ideal Domain over a module, then the decomposition is not unique, among other issues [2]

[1]The link is not perfect; often some additional conditions must be placed on the ring, such as R needing to be a regular local ring or Cohen–Macaulay

[2]wikipedia: Structure theorem for finitely generated modules over a principal ideal domain, Non-finitely generated modules

14.1 Over PID, Submodule of Free Module is Free

Theorem 13.1.12 used the fact that a field k was composed of only units (and 0) to show dimension of a vector-space was well-defined. Not all free R-modules have this property. Fortunately, this will not be an issue for PID's as the following much more general result shows:

Lemma 14.1.1: Invariant Basis Well-defined Over Commutative Ring

Let R be a nonzero commutative ring and let $n, m \in \mathbb{N}$. If $R^n \cong R^m$ as R-modules, then $n = m$

This result can be proven for arbitrary cardinalities, which is left as an exercise. In particular a vector space over a field has a well-defined dimension, finite or not.

***Proof*:**
As R is commutative, there exists a maximal ideal, say $\mathfrak{m}$. Then it is easy to verify that if $R^n \cong R^m$ then:

$$(R/\mathfrak{m})^n \cong (R/\mathfrak{m})^m$$

this is then a vector-space which by theorem 13.1.12 implies $n = m$, completing the proof.

Definition 14.1.2: Invariant Basis Number

Let R be a ring. Then R has the *invariant basis number property* (IBN property) if $R^n \cong R^m$ if and only if $n = m$.

A free R-module where R has the IBN property has the notion of *rank* well-defined.

If R is not commutative, this need not be the case:

Example 14.1.3: Ring Without IBN Property
Let $V = k^{\oplus \mathbb{N}}$ and $R = \mathrm{End}_k(V)$. Prove that:

$$R \cong R^4$$

Thus, there is no ambiguity for the notion of rank when working with free-modules over commutative rings.

Recall that a sub-module of a free module need not be free. For example, if we take $\mathbb{C}[x, y]$ over itself, then it is free with rank 1, but the sub-module (ideal) (x, y) is not free. If it were of rank 1[3], then (x, y) would be generated by a single element, but it has been shown earlier this is not possible. However, if R is a PID, then just like for modules over fields, submodules of free modules over a PID are free. For free modules, this was because every module over a field is free, however there do exist modules over PID's that are not free (consider any torsion abelian group). To prove this result, we shall show that if F is a free module over a PID and $M \subseteq F$, we can iteratively split off a cyclic direct summand $\langle y \rangle$ from $M \subseteq F$, which shall result in separating the free part from the torsion part. Furthermore, y can be chosen in such a way that it is a scalar multiple of a basis element of F: $y = ax$ for appropriate $a \in R$; think of submodules of $\mathbb{Z}^n$ as a canonical example of this behavior:

[3]Why can't the rank be ≥ 1? hint: use field of fractions

Lemma 14.1.4: Splitting Modules Over PID

Let R be a PID, F a finitely generated free module over R, and $M \subseteq F$ be a nonzero submodule. Then there exists elements $a \in R$, $x \in F$, $y \in M$, and submodules $F' \subseteq F$ and $M' \subseteq M$ such that

1. $y = ax$
2. $M' = F' \cap M$
3. $F = \langle x \rangle \oplus F'$, $M = \langle y \rangle \oplus M'$

***Proof*:**
Since $F \cong R^n$, it is perhaps tempting to think that we can our factor may be one of the components of R^n, however this implicitly chooses a particular representation (i.e. a basis for the isomorphism $\cong$). Since we don't know that M is free yet, we cannot know how to pick a basis representation of F that matches well with a basis representation of M, we cannot pick a basis β for F in such a way that the free submodule M is a projection. In that case, the splitting in the lemma would be obtained by choosing a component, or in terms of functions using the projection map:

$$\pi(r_1, r_2, ..., r_n) = r_1$$

Instead, we shall employ the following trick that has been used in theorem ref:HERE to make it choice-independent: we shall take the see of all possible homomorphism $\varphi : F \to R$ and use Zorn's lemma to choose a maximal element.

Thus, define the following collection of ideals

$$\mathscr{F} = \{\varphi(M) \mid \varphi \in \mathrm{Hom}_R(F, R)\} = \{\varphi(M) \mid \varphi \in F^*\}$$

This set contain non-trivial functions since the projections $\pi_k \in \mathscr{F}$ and M, F are nonzero. Thus since R is a PID, namely it is Noetherian, by theorem 12.3.33 there exists a $\alpha : F \to R$ such that $\alpha(M)$ is maximal. Since $\mathscr{F}$ is nontrivial, it must be that $\alpha(M) \neq 0$. Since R is a PID, there exists an $a \in R$ such that $\alpha(M) = a$. Thus, there exists a $y \in M$ such that $\alpha(y) = a$.

We shall see that this y and a are the desired values. To find the desired x, we shall show that a divides $\varphi(y)$ for all $\varphi \in F^*$. To see this, let b_φ be the element such that

$$(b_\varphi) = (a, \varphi(y))$$

Then there exists $r, s \in R$ such that

$$b = ra + s\varphi(y)$$

Using this, define the homomorphism $\psi := r\alpha + s\varphi$. Since $a \in (b)$, $\alpha(M) \subseteq (b)$, and by maximality $\psi(M) \subseteq \alpha(M)$, hence $\alpha(M) = \psi(M)$. But then

$$(a) = (b) = (a, \varphi(y))$$

namely, $\varphi(y) \in (a)$, hence a divides $\varphi(y)$. Now, since F is a finitely generated free module, $F \cong R^n$. Given this representation of F, we may write:

$$y = (s_1, s_2, ..., s_n)$$

and define the projection $\pi_i : F \to R$, $\pi_i(y) = s_i$. Since $\pi_i \in \mathscr{F}$, a divides s_i for each s_i, namely there exists $r_1, ..., r_n$ such that $s_i = ar_i$. Letting x be the element associated to the element

$$(r_1, ..., r_n)$$

in the isomorphism, we get that

$$y = ax$$

Given us our desired x.

Let's now show that x, y have the desired properties. First, notice that

$$a = \alpha(y) = \alpha(ax) = a\alpha(x)$$

Since R is an integral domain, $\alpha(x) = 1$. Next, let $F' = \kappa(\alpha)$ and $M' = F' \cap M$. Notice that for every $z \in f$,

$$z = \alpha(z)x + (z - \alpha(z)x)$$

which by linearity:

$$\alpha(z - \alpha(z)x) = \alpha(z) - \alpha(z)\alpha(x) = \alpha(z) - \alpha(z) = 0$$

Thus,

$$z - \alpha(z)x \in \ker \alpha$$

implying that $F = \langle x \rangle + F'$. Next, take $rx \in \langle x \rangle \cap F'$. Then $\alpha(rx) = 0$, and by linearity $r\alpha(x) = 0$, but $\alpha(x) = 1$ so $r = 0$, and so $\langle x \rangle \cap F' = 0$, which by definition implies:

$$F = \langle x \rangle \oplus F'$$

Giving part of the 3rd claim. Finally, if $z \in M$, then $a \big| \alpha(z)$, and indeed

$$\alpha(z) \in \alpha(M) = (a)$$

Writing $\alpha(z) = ca$, we get $\alpha(z)x = cax = cy$. Splitting z as we've done before, notice that:

$$z - \alpha(z)x = z - cy \in M \cap F' = M'$$

and so, repeating the same logic as before, we get:

$$M = \langle y \rangle \oplus M'$$

completing the last part of the proof.

With this, we may prove one of the main results:

Theorem 14.1.5: Submodule Of Free Module Over PID

Let R be a PID, let F be a free R-module of finite rank n, and let $M \subseteq F$. Then M is free, of rank at most n

***Proof*:**
If $M = 0$, we are done, so let $M \neq 0$. by lemma 14.1.4 there exists a $y_1 \in M$ and a $M^1 \subseteq M$ such that

$$M = \langle y_1 \rangle \oplus M^1$$

If $M^1 = 0$, we are done. Otherwise, since $M^1 \subseteq F$, we may keep applying lemma 14.1.4 producing a y_2 and M^2. Since the set $\{y_1, y_2, ..., y_m\}$ is linearly independent and F has finite rank, this process must terminate, leaving us with a $M^m = 0$ for some $m \leq n$, at which point:

$$M = \langle y_1 \rangle \oplus \langle y_2 \rangle \oplus \cdots \oplus \langle y_m \rangle$$

as we sought to show.

The finite rank hypothesis is not needed for the conclusion, only for this proof: what makes the argument stop is that a linearly independent subset of F cannot have more than n elements. Removing the hypothesis costs exactly one thing, namely a well-ordering in place of the counter m, and the following proof is the same argument with that substitution made. It is worth reading the finite case first, because the well-ordering conceals how simple the mechanism is: peel off one basis vector, split what the submodule contributes in that coordinate, and recurse.

Theorem 14.1.6: Submodule Of Free Module Over PID, General Case

Let R be a PID, let F be a free R-module of any rank, and let $M \subseteq F$ be a submodule. Then M is free

***Proof*:**
Let $(e_\lambda)_{\lambda \in \Lambda}$ be a basis of F and well-order Λ, which is possible by theorem A.2.4. For each λ write

$$F_{<\lambda} = \bigoplus_{\mu < \lambda} Re_\mu, \qquad F_{\leq\lambda} = \bigoplus_{\mu \leq \lambda} Re_\mu,$$

and set $M_{<\lambda} = M \cap F_{<\lambda}$ and $M_{\leq\lambda} = M \cap F_{\leq\lambda}$.

Step 1: one coordinate at a time. Every $x \in F_{\leq\lambda}$ has a unique expression $x = y + re_\lambda$ with $y \in F_{<\lambda}$ and $r \in R$, so

$$\pi_\lambda \colon F_{\leq\lambda} \to R, \qquad \pi_\lambda(y + re_\lambda) = r$$

is a well defined R-module homomorphism with $\ker(\pi_\lambda) = F_{<\lambda}$. Its image on $M_{\leq\lambda}$ is a submodule of R, that is an ideal, and R is a principal ideal domain, so

$$\pi_\lambda(M_{\leq\lambda}) = (a_\lambda) \qquad \text{for some } a_\lambda \in R$$

Let $\Lambda' = \{\lambda \in \Lambda \mid a_\lambda \neq 0\}$, and for $\lambda \in \Lambda'$ choose $x_\lambda \in M_{\leq\lambda}$ with $\pi_\lambda(x_\lambda) = a_\lambda$.

Step 2: the splitting at λ. If $a_\lambda = 0$ then $M_{\leq\lambda} \subseteq \ker(\pi_\lambda)$, so $M_{\leq\lambda} = M_{<\lambda}$. If $a_\lambda \neq 0$ then

$$M_{\leq\lambda} = M_{<\lambda} \oplus Rx_\lambda$$

For the sum: given $m \in M_{\leq\lambda}$, write $\pi_\lambda(m) = ra_\lambda$; then $\pi_\lambda(m - rx_\lambda) = 0$, so $m - rx_\lambda \in M \cap F_{<\lambda} = M_{<\lambda}$. For directness: if $rx_\lambda \in M_{<\lambda}$ then $0 = \pi_\lambda(rx_\lambda) = ra_\lambda$, and R is a domain with $a_\lambda \neq 0$, so $r = 0$. The same computation shows $Rx_\lambda \cong R$ is free of rank one.

Step 3: the x_λ span M. Put $N = \sum_{\lambda \in \Lambda'} R x_\lambda \subseteq M$ and suppose $N \neq M$. Every element of F has finite support, so every $m \in M$ lies in $M_{\leq \lambda}$ for λ the largest index in its support; hence the set

$$S = \{\lambda \in \Lambda \mid M_{\leq \lambda} \not\subseteq N\}$$

is nonempty, and being a subset of a well-ordered set it has a least element λ_0. Choose $m \in M_{\leq \lambda_0}$ with $m \notin N$. By Step 2, $m = m' + r x_{\lambda_0}$ with $m' \in M_{<\lambda_0}$, where $r = 0$ in the case $a_{\lambda_0} = 0$. Now m' has finite support contained in $\{\mu \mid \mu < \lambda_0\}$, so $m' \in M_{\leq \mu}$ for some $\mu < \lambda_0$, and $M_{\leq \mu} \subseteq N$ by minimality of λ_0. Then $m = m' + r x_{\lambda_0} \in N$, a contradiction. So $N = M$.

Step 4: the x_λ are independent. Suppose $r_1 x_{\lambda_1} + \cdots + r_n x_{\lambda_n} = 0$ with $\lambda_1 < \cdots < \lambda_n$ in Λ'. Each x_{λ_i} with $i < n$ lies in $F_{\leq \lambda_i} \subseteq F_{<\lambda_n} = \ker(\pi_{\lambda_n})$, so applying π_{λ_n} leaves $0 = \pi_{\lambda_n}(r_n x_{\lambda_n}) = r_n a_{\lambda_n}$, whence $r_n = 0$. Repeating with λ_{n-1}, and so on down, gives $r_i = 0$ for every i.

So $(x_\lambda)_{\lambda \in \Lambda'}$ is a basis of M, and M is free.

The finite case does not require the divisibility result proved above. Using it, we can get a more precise result giving us something similar to a smith normal form for PIDs:

Corollary 14.1.7: Basis For Submodules Of Free Modules

Let R be a PID, F a finitely-generated free R-module, and $M \subseteq F$. Then there exists a basis $(x_1, x_2, ..., x_n)$ for F and nonzero elements $a_1, a_2, ..., a_m \in R$ with $m \leq n$ such that

$$(a_1 x_1, ..., a_m x_m)$$

is a basis for M. Furthermore, $a_1 | a_2 | \cdots | a_m$

***Proof*:**
By theorem 14.1.5 M is free and by inductively applying lemma 14.1.4 the basis for M can be chosen so that it is of the form in the lemma.

To show the coefficients divide each other, the end of the above proof shows that $a_1 \big| a_2$ since then the argument can be repeated inductively.

Corollary 14.1.8: If PID, Then Free Resolution Of Length 2

Let R be a PID. Then every finitely generated R-module M and every surjective homomorphism:

$$R^{m_0} \xrightarrow{\pi_0} M \to 0$$

there exists a free R-module R^{m_1} and a homomorphism $\pi_1 : R^{m_1} \to R^{m_0}$ such that the sequence

$$0 \to R^{m_1} \xrightarrow{\pi_1} R^{m_0} \xrightarrow{\pi_0} M \to 0$$

is exact. Along with proposition 12.3.29, we get that R is a PID if and only if every R-module admits a free resolution of length 2.

***Proof*:**
Since R^{m_0} is an R-module over a PID, $\ker(\pi_0)$ is is free by theorem 14.1.5, and since R^{m_0} is finitely generated, $\ker(\pi_0) \cong R^{m_1}$ for some $m_1 \in \mathbb{N}$. Hence choose the natural embedding $\pi_1 : \ker(\pi_0) \hookrightarrow R^{m_0}$ giving the desired exact sequence, completing the proof.

Since all modules over PID's admit a free resolution of length 2, they are characterized by a matrix (up to equivalence), namely if

$$0 \to R^m \xrightarrow{\varphi} R^n \xrightarrow{\pi_0} M \to 0$$

Then $M = \mathrm{coker}(\varphi)$, and by exactness we know that φ injects into R^n (or, if the reader rather not use exactness, φ injects since $R^m = \ker(\pi_0)$). Then by corollary 14.1.7, given a basis $(x_1, x_2, ..., x_n)$ of R^n, there exists nonzero elements $a_1, ..., a_m \in R$ where $a_1 \big| a_2 \big| \cdots \big| a_m$ and $(a_1x_1, a_2x_2, ..., a_mx_m)$ forms a basis for $R^m \cong \ker(\pi_0)$. Hence, M is presented by the matrix:

$$\begin{pmatrix} a_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_m \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

By remark (readingPresentationsAndUnits),any row/column which contains an a_i that is a unit may be removed. Without loss of generality, let's say that $a_1, a_2, ..., a_m$ are non-units. Then:

$$M \cong (R/a_1) \oplus \cdots (R/a_m) \oplus R^{(n-m)}$$

Notice what just happened: a finitely generated module over M is isomorphic to a rather simple structure, that is such modules lend themselves to a classification theorem! The following theorem summarizes this result and proves a couple more important facts:

Theorem 14.1.9: Invariant Factor Form of Modules over PID's

Let R be a Principal Ideal Domain and let M be a finitely generated R-module. Then:

1. M is isomorphic to a direct sum of finitely many cyclic modules. More precisely,

$$M \cong_R R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

   for some integer $r \geq 0$ and nonzero element $a_1, a_2, ..., a_m$ of R which are not units in R and which satisfy the divisibility relations

$$a_1 \mid a_2 \mid \cdots \mid a_m$$

2. M is torsion-free if and only if M is free

3. In the decomposition of (1)

$$\text{Tor}(M) \cong_R R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m)$$

   in particular M is a torsion module if and only if $r = 0$ and in this case the annihilator of M is the ideal (a_m).

In ref:HERE, we shall prove the uniqueness of this decomposition up to isomorphism, allowing us to say it is *the* direct sum of finitely many cyclic modules.

***Proof*:**
Let's first state more succinctly the above analysis to prove the existence of part (1). Let M be a finitely generated module over a PID R, say $x_1, x_2, ..., x_n$ is a generating set of minimal cardinality. Then given R^n with basis $b_1, b_2, ..., b_n$:

$$\pi : R^n \to M \qquad \pi(b_i) = x_i$$

is a surjective R-module homomorphism with kernel $\ker(\pi)$ which is free by theorem 14.1.5. Then by corollary 14.1.7 choose a basis $y_1, y_2, ..., y_n$ for R^n such that $a_1y_1, a_2y_2, ..., a_my_m$ is a basis for $\ker(\pi)$ with $a_1 \big| a_2 \big| \cdots \big| a_m$. Then:

$$M \cong R^n / \ker(\pi) = (y_1R \oplus y_2R \oplus \cdots \oplus y_nR)/(a_1y_1R \oplus a_2y_R2 \oplus \cdots \oplus a_my_mR))$$

which the reader may verify (exercise ref:HERE) that is isomorphic to:

$$R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}$$

This immediately gives us the other two conclusions:

(2) Since $R/(a_i)$ is torsion and the direct product of torsion modules is torsion, then M is torsion free if and only if $M \cong R^n$, which implies it's free (hence being "element-wise" free implies being "globally" free)

(3) This follows from the fact that the direct product of torsion modules is torsion, hence

$$\text{Tor}_R(M) \cong_R R/(a_1) \oplus \cdots \oplus R/(a_m)$$

In order to prove the uniqueness of this decomposition, we shall show how we can systematically "embed" parts of M into a vector-space and take advantage of the invariant basis number (IBN) of finitely dimensional isomorphic vector spaces, that is if two vector-spaces are isomorphic they are the same dimension.

First, since R is a PID the annihilator of M is a principal ideal. We start by finding the annihilator of M when it is a torsion module:

Lemma 14.1.10: Annihilator of Module Over PID

Let M be a finitely generated torsion module over a PID R with decomposition:

$$R/(a_1) \oplus \cdots R/(a_m)$$

with $a_1 \big| \cdots \big| a_m$. Then $\text{Ann}_R(M) = (a_m)$.

***Proof*:**
$0 \neq r \in \text{Ann}_R(M)$ so that

$$0 = r \cdot (1, ..., 1) = (r, ..., r)$$

which by the given decomposition implies:

$$r \equiv 0 \text{ mod}\,(a_m) \qquad \Leftrightarrow \qquad r \in (a_m)$$

Hence, $\text{Ann}_R(M) \subseteq (a_m)$. Conversely, if $r \in (a_m)$ and $y \in M$, decomposing y so that

$$y = (y_1, ..., y_m) \qquad y_i \in R/(a_i)$$

since $r \in (a_m) \subseteq (a_i)$, we see that for each a_i:

$$ry_i = 0$$

hence $ry = 0$, and so $r \in \text{Ann}_R(M)$, as we sought to show.

Next, since R is a UFD, we may find a decomposition, we have $a_m = up_1^{k_1}p_2^{k_2}\cdots p_r^{k_r}$. Since $\gcd(p_i^{k_i}, p_j^{k_j}) = 1$, by exercise ref:HERE (rings, under CRT)

$$(p_i^{k_i}) + (p_j^{k_j}) = (1)$$

Hence, by the Chinese Remainder Theorem: $R/(a_m) \cong R/(p_1^{k_1}) \oplus \cdots \oplus R/(p_r^{k_r})$ Doing this to every a_i gives the following decomposition:

Definition 14.1.11: Elementary Divisor Form Of Modules Over PID's

Let M be a finitely generated torsion module over a PID R. Then given the above decomposition, the isomorphism:

$$M \cong R^r \oplus R/(p_1^{k_1}) \oplus \cdots \oplus R/(P_s^{k_s})$$

is called a[a] elementary divisor form of M with each (not necessarily unique) p_i being primes are R and are called the *elementary divisors.*

[a]we can't yet say "the"

Given M is torsion, since M can be decomposed into modules of the form $R/(p_k^{k_i})$, we may group together these modules to form a decomposition:

$$M \cong N_1 \oplus \cdots N_s$$

where the elements of N_i are annihilated by a power of a distinct prime p_i.

Theorem 14.1.12: Primary Decomposition Theorem

Let R be a PID and let M be a nonzero torsion R-module (not necessarily finitely generated) with nonzero annihilator a. Suppose the factorization of a into distinct primer powers in R is

$$a = up_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$$

and let $N_i = \{x \in M | p_i^{\alpha_i} x = 0\}$, $1 \leq i \leq n$. Then N_i is a submodule of M with annihilator $p_i^{\alpha_i}$ and is the submodule of M of all elements annihilated by some power of p_i. Thus, we get

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_n$$

If M is finitely generated, then each N_i is the direct sum of finitely many cyclic modules whose annihilators are divisors of $p_i^{\alpha_i}$

***Proof*:**
The finite case has already been proved. For the general case, modify the CRT to modules and take advantage of the fact that ideals of the form $(p_i^{k_i})$ are coprime (since R is a PID) to get the desired result

We give a name to the N_i modules

Definition 14.1.13: p_i-Primary Components

The submodules N_i in the previous theorem are called *p_i-primary components* of M.

With this, we come to the key result:

Lemma 14.1.14: Embedding Module over PID into Vector Space

Let R be a PID and let p be a prime in R. Let F denote the field $R/(p)$. Then:

1. if $M = R^t$, then $M/pM \cong F^t$
2. if $M = R/(a)$ where a is a nonzero element of R, then

$$M \cong \begin{cases} F & \text{if } p \text{ divides } a \text{ in } R \\ 0 & \text{if } p \text{ does not divides } a \text{ in } R \end{cases}$$

3. if $M = R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_k)$ where each a_i is divisible by p, then $M/pM \cong F^k$

***Proof*:**

1. Let $M = R^t$. Define the map $\varphi : R^t \to (R/(p))^t$ by:

$$(r_1, ..., r_t) \mapsto (r_1 \bmod (p), ..., r_t \bmod (p))$$

Then certainly $\ker(\varphi) = pR^t = pM$, φ is surjective, and thus $M/pM \cong F^t$

2. Let $M = R/(a)$. Let $\pi : R \to R/(a)$ be the quotient map. Then

$$\pi\left((p)\right) = pM = p(R/(a)) = \left((p) + (a)\right)/(a)$$

Since R is a PID,

$$(p) + (a) = \gcd(a, p)$$

If p divides a, then $\gcd(a, p) = p$, and is 1 otherwise. Applying the appropriate isomorphism theorems gives the desired result.

3. Let $M = R/(a_1) \oplus \cdots \oplus R/(a_k)$. This follows the same procedure as theorem 14.1.9 with the modification that each a_i is divisible by p, giving us $M/pM \cong F^k$

Theorem 14.1.15: Fundamental Theorem of Uniqueness of Factors

Let R be a Principal Ideal Domain. Then two finitely generated R-modules M_1 and M_2 are isomorphic if and only if they have the same rank and the same list of invariant factors (resp. list of elementary divisors).

***Proof*:**
If M_1 and M_2 have the same rank and list of invariant factors (resp. Elementary divisors), then they are clearly isomorphic, hence consider the converse: assume $M_1 \cong_R M_2$. Then any isomorphism must map torsion to torsion, and hence $\mathrm{Tor}_R(M_1) \cong_R \mathrm{Tor}_R(M_2)$. Then:

$$R^{t_1} \cong_R M_1/\mathrm{Tor}_R(M_1) \cong_R M_2/\mathrm{Tor}_R(M_2) \cong_R R^{t_2}$$

that is $R^{t_1} \cong_R R^{t_2}$. Choosing any prime $p \in R$, we may apply lemma 14.1.14 $F^{t_1} \cong_F F^{t_2}$ (can you see why this is an F-module isomorphism where $F = R/(p)$?). Then as we saw before,

$t_1 = t_2$, giving the invariant rank.

Hence, we may quotient by the free-part of each module and show that $\mathrm{Tor}_R(M_1) \cong_R \mathrm{Tor}_R(M_2)$ implies the invariant factors (resp. elementary divisors) are the same. Starting by showing the elementary divisors are the same, it suffices to show that for any fixed prime p, the elementary divisors which are a power of p are the same for M_1 and M_2. Hence, we may reduce to the case where $M_1 \cong_R M_2$ have annihilators which are a power of p.

We do this by induction on the power of p for the annihilator of $M - 1$. If the power is 0, then $M_1 = M_2 = 0$ and hence they are clearly isomorphic. Thus ,assume M_1 (thus M_2) have nontrivial elementary divisors. Say that the elementary divisors of M_1 and M_2 are:

$$M_1 : \underbrace{p, p, ..., p}_{a \text{ times}}, p^{k_1}, p^{k_2}, ..., p^{k_s}$$
$$M_2 : \underbrace{p, p, ..., p}_{b \text{ times}}, p^{r_1}, p^{r_2}, ..., p^{r_t}$$

Then the submodules pM_1, pM_2 would eliminate the p elementary divisors and reduce each k_i, r_i by 1 (i.e. they would have power $k_i - 1$ and $r_i - 1$). Since $M_1 \cong_R M_2$, $pM_1 \cong_R pM_2$. By the induction hypothesis, the elementary divisors of pM_1 are the same as those of pM_2, that is $s = t$ and $k_i - 1 = r_i - 1$, and hence $k_i = r_i$. What's left is to show that $a = b$. By what we've just shown, we have the following isomorphism

$$M_1/pM_1 \cong_R M_2/pM_2$$

which without the fact that the elementary divisors of pM_1 and pM_2 are the same would not necessarily be true (recall that $\mathbb{Z} \cong_{\mathbb{Z}} 2\mathbb{Z}$ but $\mathbb{Z}/2\mathbb{Z} \ncong_{\mathbb{Z}} 2\mathbb{Z}/2\mathbb{Z}$). Then by lemma 14.1.14 we have $F^{a+s} \cong_F F^{b+t}$ implying $a + s = b + t$ which since $s = t$ we get $a = b$, giving us that the elementary divisors match!

Finally, for the invariant factors, invariant factors $a_1 \big| \cdots \big| a_m$ and $b_1 \big| \cdots \big| b_n$ for M_1 and M_2 respectively, take the set of elementary factors for M_1 and notice that a_m is the product of the largest of the prime powers among these elementary divisors, a_{m_1} is the product of the largest prime powers once the factors of a_m have been removed, and so forth. Doing the same procedure to the set of b's, we see that these must produce the same elements, and hence the invariant factors must match, completing the proof.

Finally, we can now talk about *the* invariant factor form and *the* elementary divisor form of a finitely generated R-module M for R a PID.

Corollary 14.1.16: Computation Shortcut For Invariant Factors

Let R be a Principal Ideal Domain and let M be a finitely generated R-module. Then:

1. The elementary divisors of M are the prime power factors of the invariant factors of M.
2. The largest invariant factor of M is the product of the largest of the distinct prime powers among the elementary divisors of M, the next largest invariant factor is the product of the largest of the distinct prime powers among the remaining elementary divisors of M, and so on

***Proof*:**
Fallows by the uniqueness given from 14.1.15

Corollary 14.1.17: The Fundamental Theorem of f.g. Abelian Groups

Let G be a finitely generated group G. Then:

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \mathbb{Z}_{n_s}$$

for some integers $r, n_1, ..., n_s$ with the following conditions:

1. $r \geq 0$ and $n_j \geq 2$ for all j, and
2. $n_{i+1} | n_i$ for $1 \leq i \leq s-1$

***Proof*:**
Take $R = \mathbb{Z}$ in theorem 14.1.9. Note that the invariant factors are listed in reverse order in Chapter 5 for ease of computation.

Summary

Any finitely generated module over a PID is fully characterised by a number $n \in \mathbb{N}$ and a choice of non-units ideals $(a_1), ..., (a_n)$ such that

$$(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_n)$$

If M is torsion, (a_n) is the annihilator of M. The elements $a_1, a_2, ..., a_n$ may be combined to a single ideal that holds all the information about he module, the ideal has a name:

Definition 14.1.18: Characteristic Ideal

Let M be a finitely generated module over a PID. Then the collection of invariant factors forms an ideal:

$$(a_1, ..., a_m)$$

known as the *characteristic ideal* of M.

The name "characteristic ideal" is given due to its link to the characteristic polynomial explored in section 13.5.

14.2 Application: Linear Algebra revisited

With the classification established, it can be put to good use in better understanding linear operators between vector spaces. Recall in section 13.2.2 that it was mentioned that chapter 14 is essentially the theoretical build-up for studying linear operators. We now make good on this promise. In this section, we define a $k[x]$-module structure on vector-spaces given by a linear operator α, and we shall define a new invariant similar to the characteristic polynomial known as the *minimal polynomial.*

To define this new module, we'll have to juggle lot of components. Let k be a field, let x be an indeterminate and let R be the polynomial ring $k[x]$. Let V be an k-module (i.e. a vector space over k). Notice that we're using the same field for V and for $k[x]$. Finally, let α be a linear operator on V, that is a linear operator from V to V where we consider the domain and codomain to be the same object. We shall extend the k-module V (i.e. a vector-space) into an $k[x]$-module using α. To accomplish this, we'll take advantage of nice properties of linear operators. First, define

$$\alpha^0 = I \qquad\qquad \alpha^n = \alpha \circ \alpha \circ \cdots \circ \alpha \qquad (n \text{ times})$$
$$a_n\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 \in \mathrm{End}_k(V)$$

where I is the identity map from V to V and $\circ$ is function composition (which is well-defined since α maps to V). We'll now define the ring action of $k[x]$; let $p(x) \in k[x]$ be a polynomial. Then define an action of the ring element $p(x)$ on the module element b by:

$$\begin{aligned} p(x) \cdot v &= (a_n\alpha^n + \cdots + a_1\alpha + a_0)(v) \\ &= a_n\alpha^n(v) + \cdots + a_1\alpha(v) + a_0 v \end{aligned}$$

That is, $p(x)$ acts by substituting the linear operator α for x in $p(x)$ and applying the resulting linear operator to v. One should check that these rules satisfy the module axioms that make V into an $k[x]$-module. Note that k is a subring of $k[x]$, and this new module with constant polynomials acting on it is simply the structure of the vector space. Thus the definition of $k[x]$-module *extends* the action of k to an action of the larger ring $k[x]$.

The key is that $k[x]$ is a PID when k is a field (recall theorem 11.1.1 and its corollaries) and so we have:

$$V \cong_{k[x]} (k[x])^n \oplus (k[x]/(f_1)) \oplus \cdots \oplus (k[x]/(f_m))$$

Since V is finite k-dimensional, the action $k[x] \curvearrowright V$ must have rank 0, since if it had rank $n \geq 1$, then the invariant factor $(k[x])^n$ would be nonzero, and it is k-isomorphic to:

$$k[x] \cong k \oplus k \oplus \cdots \oplus k \oplus \cdots$$

which would make V infinite k-dimensional. Thus,

$$V \cong_{k[x]} (k[x]/(f_1)) \oplus \cdots \oplus (k[x]/(f_m))$$

Since the action of $k[x]$ is completely dependent on α, the above proposition may be interpreted as there being a special basis for V that will make the action of α on V rather transparent!

Example 14.2.1: Actions of this Module

1. If $\alpha = 0$, that is, the function maps every element to the identity in V (recall that V is an [abelian] group). Then

   $$p(x)v = a_n \cdot 0 + \cdots + a_1 \cdot 0 + a_0 v = a_0 v$$

   Thus, with such a linear operator, the $k[x]$-module structure is just like the k-module structure.

2. Another simple example is $\alpha = I$, that is, the identity. Thus

$$p(x)v = a_n v + \cdots + a_0 v = (a_n + \cdots + a_0)v$$

   Or simply the sum of coefficient of $p(x)$ act on v

3. Here's a more specific example. Take V to be the affine n-space k^n and let α be the "shift operator"

$$\alpha(x_1, ..., x_n) = (x_2, x_3, ..., x_n, 0)$$

   Since V is a vector-space, let e_i be a choice of basis vectors. Then applying this linear operator to only the basis, we get:

$$\alpha^k(e_i) = \begin{cases} e_{i-k} & \text{if } i > k \\ 0 & \text{if } i \le k \end{cases}$$

   So that if $m < n$

$$(a_m x^m + a_{m-1}x^{m-1} + \cdots + a_0)e_n = (0, 0, ..., 0, a_m, a_{m-1}, ..., a_0)$$

4. under what condition is $W \subseteq V$ a $k[x]$-submodule of V with the $k[x]$-action? Since $k[x]$ extends the actions of k, it is necessary that W be a subspace. By the subspace criterion we have that it is necessary that $\alpha(W) \subseteq W$. Not all subspaces satisfy this conditions, if they do they are called α-stable subspaces.

Now, notice that we may take the annihilator of the $k[x]$-module V, that is all elements $p(x) \in k[x]$ such that $p(\alpha) = 0$ for all $\alpha \in V$. Since $k[x]$ is a PID, the annihilator is a principal ideal. This ideal shall be important in contexts where R need not be a field, and hence we define it more generally:

Definition 14.2.2: Annihilator Ideal and Minimal Polynomial

Let M be a finitely generated free R-module where R is an integral domain, and $\alpha \in \mathrm{End}_R(M)$. Then the ideal:

$$\mathfrak{M}_\alpha = \mathrm{Ann}_R(M)$$

is called the *annihilator ideal* of α. If K is the field of fraction of R, then α can be viewed as an element of $\mathrm{End}_K(K^n)$ and $\mathfrak{M}_\alpha^{(K)} \subseteq k[x]$ is the annihilator of α over K, where since $k[x]$ is a PID we get:

$$\mathfrak{M}_\alpha^{(K)} = (m_\alpha) \qquad m_\alpha \in k[x]$$

the resulting polynomial is called the *minimal polynomial* of α

Recall that $k[x]$ is a PID if and only if k is a field, hence it is natural to work in the setting of $R = k$ is a field to use the above module structure and the classification theorem. The term "minimal polynomial" is because it is the smallest polynomial (with respect to divisibility) that solves the equation $f(\alpha) = 0$. The reader may ask if the characteristic polynomial shares the same property, that is if

$$P_\alpha(\alpha) = 0$$

This shall indeed be the case and shall be called the *Cayley-Hamilton Theorem* and will be proven in due time[4] (the reader may try to use definition 14.1.18 to find a connection between these two polynomials and hence a proof of the above equation). The annihilator ideal shares the same nice invariance properties as the characteristic polynomial:

Lemma 14.2.3: Annihilator Ideal And Similarity

Let α be similar to β. Then $\mathfrak{M}_\alpha = \mathfrak{M}_\beta$

***Proof*:**
Let $\rho \in \mathrm{GL}_R(M)$ such that $\beta = \rho \circ \alpha \circ \rho^{-1}$. Then since when composing conjugates:

$$\beta^k = \rho \circ \alpha^k \circ \rho^{-1}$$

we get that for any $f(x) \in R[x]$,

$$f(\beta) = \rho \circ f(\alpha) \circ \rho^{-1}$$

Hence, $f(\alpha) = 0$ if and only if $f(\beta) = 0$, as we sought to show.

The converse is not necessarily, the case: if $\mathfrak{M}_\alpha = \mathfrak{M}_\beta$, it does not imply that $\alpha \sim \beta$:

Example 14.2.4: Same Annihilators, Not Similar
(put a matrix in JcF that makes this evident. Also ask if then we can condlue that the char poly and min poly is enough to deduce similarity for dimension $n \leq 3$)

On the other hand, going back to example 13.5.3, we see that $x - 1$ is the annihilator of the identity matrix, while it is not the annihilator of:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Coming back to the $k[x]$-module on V, we may use it to finally have a converse to establishing two linear operators are similar:

Lemma 14.2.5: Similarity And New Module Structure

Let α, β be linear opereators on a free R-module M. Then the corresponding $R[x]$-module structures on M are isomorphic if and only if $\alpha \sim \beta$

***Proof*:**
Let M_α, M_β be the corresponding $R[x]$-modules given by α, β respectively. Assume $\alpha \sim \beta$ so that there exists a $\rho \in \mathrm{GL}(M)$ such that

$$\beta = \rho \circ \alpha \circ \rho^{-1} \Rightarrow \beta \circ \rho = \rho \circ \alpha$$

I claim that $\rho : M_\alpha \to M_\beta$ is an $R[x]$-linear map. It is certainly a R-linear map, hence to see if

[4]the curious reader may see exercise ref:HERE to prove this result with judicious application of Cramer's rule

it $R[x]$ linear consider the action of x in M_α and M_β. Then:

$$\rho(x \cdot v) = \rho \circ \alpha(v) = \beta \circ \rho(v) = x \cdot \rho(v)$$

which then may be generalized to show that it is indeed $R[x]$-linear. Furthermore, ρ is certanly an invertible $R[x]$-linear map (being bijective and an $R[x]$-module homomorphism) and hence $M_\alpha \cong_{R[x]} M_\beta$, as we sought to show.

Corollary 14.2.6: Similarity And Isomorphism Classes

Let α, β be two linear operators on a free R-module M. Then there is a one-to-one correspondence between the similarity classes of R-linear operators of a M and the isomorphism classes of of $R[x]$-module structures on M.

***Proof*:**
an immediate consequence of the above. If M has finite rank, then it is a bijection.

Exercise 14.2.1

1. Show that the $k[x]$-module on V can be constructed using the universal property of polynomials

14.3 Rational and Jordan Canonical Form

From now on, $R = F$ is a field and V is an n dimensional vector-space over F. We shall use the above results to find anice representation of a lienar operator α, namely we shall find a matrix representation for the action of x $R[x] \curvearrowright V$.

Take the following decomposition of V:

$$V \cong k[x]/(a_1(x)) \oplus k[x]/(a_2(x)) \oplus \cdots \oplus k[x]/(a_m(x))$$

where $a_1(x), a_2(x), ..., a_m(x)$ are polynomial in $F[x]$ of degree at least 1 with the divisibility condition

$$a_1(x)|a_2(x)|\cdots|a_m(x)$$

These invariant factors are determined up to units, which in $F[x]$ is F, meaning we can safely say they are all monic. Since the annihilator of V is the ideal $(a_m(x))$ (lemma 14.1.10), we get:

$$a_1(x)|a_2(x)|\cdots|m_A(x)$$

we will make a proposition of this for easy reference:

Proposition 14.3.1: Finding Minimal Polynomial

The minimal polynomial of $m_T(x)$ is the largest invariant factor of V. All the invariant vactors of V divide $m_T(x)$

***Proof*:**
direct application of (theroem 14.1.9)

Be mindful that since everything divides $m_A(t)$, $a_i(x)|m_A(x)$, then $(m_A(x)) \subseteq (a_i(x))$, so it's the *smallest* ideal of all the invariant factors. This should make sense, as since everything divides it, all of the other ideals should be bigger. Be careful then when thinking of the word "minimal", for it's actually the *biggest* polynomial relative to all the invariant factors. It is smallest when it comes to being the *smallest* polynomial which can annihilate V over $F[x]$ (i.e $m_T(T) \equiv 0$) All the other invariant factors will not manage to do this.

To calculate V with respect to $F[x]/(a_i)$ we will do the following:

For every $F[x]/(a_i)$, we can treat it as a vector-space over $F[x]$, and get a basis

$$1, \overline{x}, \overline{x}^2, ..., \overline{x}^{k-1}$$

where $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 x + b_0$ is any monic polynomial in $F[x]$ and $\overline{x} = x \bmod (a(x))$.

Next recall that:

$$V \cong F[x]/(a_1(x)) \oplus F[x]/(a_2(x)) \oplus \cdots \oplus F[x]/(a_m(x))$$

Applying T on the left-side get's transalte to multiplying by x on the right-side.

$$T(V) \cong x \cdot F[x]/(a_1(x)) \oplus x \cdot F[x]/(a_2(x)) \oplus \cdots \oplus x \cdot F[x]/(a_m(x))$$

we we concentrate on on $F[x]/(a_i(x))$ and it's basis we get

$$1 \mapsto \overline{x},\ \overline{x} \mapsto \overline{x}^2,\ \overline{x}^2 \mapsto \overline{x}^2,\ \cdots \overline{x}^{k-1} \mapsto \overline{x}^k = -b_0 - b_1\overline{x} - \cdots - b_{k-1}\overline{x}^{k-1}$$

where we got the last value since

$$a(\overline{x}) = 0 \Rightarrow \overline{x}^k + b_{k-1}\overline{x}^{k-1} + \cdots + b_1\overline{x} + b_0 = 0$$

With respect to the aformentioned basis, the linear transformation T (i.e. the matrix multiplication by x) is therefore

$$\begin{pmatrix} 0 & 0 & \cdots & \cdots & \cdots & -b_0 \\ 1 & 0 & \cdots & \cdots & \cdots & -b_1 \\ 0 & 1 & \cdots & \cdots & \cdots & -b_2 \\ 0 & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & \cdots & 1 & -b_{k-1} \end{pmatrix}$$

If we we do this for $F[x]/(m_A(x))$, then we can already see that if we do $m_A(V)$, we will annihilate V

Example 14.3.2: Sanity Check
Let's say $m_T(x) = x^5 + x^4 + 5x^3 - 2x^2 + 3$. Then, we we just consider $F[x]/(m_T(x))$, and think of $T(V)$ as multiplying the basis of $F[x](m_T(x))$ by x, we get the matrix:

$$\begin{pmatrix} 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

If we do $m_T(V)$ we get (test)

$$\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^5+\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^4+5\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^3-2\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}^2$$
$$+3\begin{pmatrix}0&0&0&-3\\1&0&0&2\\0&1&0&-5\\0&0&1&-1\end{pmatrix}$$
$$=\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

that is, m_T indeed annihilated V, as we sought to show. This doesn't tell us yet what the other invariant factors should be, or even how to find the minimal polynomial, but it does confirm that matrices of this type do what we would hope they do.

we give this matrix a name:

Definition 14.3.3: Companion Matrix

Let $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 + b_0$ be any monic polynomial in $F[x]$. The *companion matrix* of $a(x)$ is the $k \times k$ matrix whith 1's down the first subdiagonal, $-b_0, -b_1, .., -b_{k-1}$ down the alst column and zeros elsewhere. The companion matrix of $a(x)$ will be denoted $C_{a(x)}$

We can do this process to each cyclic module $F[x]/(a_i(x)$. Let β_i be the elements of V that map to the basis of $F[x]/(a_i(x))$. Then by definitin, the linear transformation T acts on β_i by the companion matrix for $a_i(x)$, as we have seen by how the right hand side is multiplies by x. If we take the union of every β_i,$\beta = \cup_i \beta_i$, then this forms a basis for V (with over $F[x]$) since the sum on the RHs is a direct sum, and with respect to this basis the linear transformation T has as matrix the *direct sum* of the companion matrices, which by ref:HERE [5] we saw can be represented by:

$$\begin{pmatrix}C_{a_1(x)} & & & \\ & C_{a_2(x)} & & \\ & & \ddots & \\ & & & C_{a_m(x)}\end{pmatrix}$$

As we can see, this matrix is determined by the invariant factors of $F[x]$-module V, in fact is uniquely determined by theorem 14.1.15. Thus, we can give this matrix a name:

[5] showing this eventually using the universal property?

Definition 14.3.4: Rational Cannonical Form

1. A matrix is said to be in *rational canonical form* if it is the direct sum of companion matrices for monic polynomials $a_1(x), a_2(x) ..., a_m(x)$ of degree at least one with $a_1(x)|a_2(x)| \cdots |a_m(x)$. The polynomial $a_i(x)$ are called the *invariatn factors* of the martrix. Such a matrix is also said to be a *block diagonal* matrix with blocks the companion matrice for $a_i(x)$
2. A *rational canonical form* for a linear transformation T is a matrix represetning T which is in rational canonical form

Note that we proved that the RCF is unique, meaning if

1. we are given monic polynomial $a_1(x), a_2(x), ..., a_m(x)$ of degree at least 1 such that $a_i(x)|a_{i+1}(x)$ for all i
2. We have a linear transformation $T : V \to V$, with a matrix representation A
3. we have a basis β of V such that $[A]_\beta$ is the direct sum of companion matrices
4. Then V is the direct sum of T-stable subspaces D_i, one for each $b_i(x)$, such that the amtrix of T on each D_i is the compantion matrix of $b_i(x)$. Some more words

which prove the following result:

Theorem 14.3.5: Rational Canonical Form

Let V be a finite dimensional vector sapce over the field F and let T be a linear transformation of V

1. There is a basis for V with respect to which the matrix T is in rational canonical form, i.e., is a block diagonal matrix whose diagonal blocks are the companion matrices for monic polynomials $_1(x), a_2(x), ..., a_m(x)$ of degree at least one with $a_1(x)|a_2(x)| \cdots |a_m(x)$
2. The factors $a_1, ..., a_m$ are called the *invariant factors* of α.

Corollary 14.3.6: RCF And Similarity

Let $\alpha, \beta : V \to V$ be two linear on a finitely dimensional vector space V. Then $\alpha \sim \beta$ if they have the same RCF.

***Proof*:**
combine everything we did earlier.

why it's called rational: if we go in a bigger field, the values don't change. In this way it's rational Here is a theorem linking to that

Corollary 14.3.7: RCF Over Extended Fields

Let A and B be two $n \times n$ matrices over a field F and suppose F is a subfield of the field K ($F \subseteq K$, or sometimes written K/F) Then

1. The RCF of A is the same whether it is computed over K or over F. The minimal and characteristic polynomials and the invariatn factors of A are the same whether A is considered as a matrix over F or as a matrix over K.

2. The matrices A and B are similar over K if and only if they are similar over F, i.e., there exists an invertible $n \times n$ matrix p with entries from K such that $B = P^{-1}AP$ if and only if there exists an (in general different) invertible $n \times n$ matrix Q with entries from F such that $B = Q^{-1}AQ$

***Proof*:**
Let M be the rational canonical form of a matrix A over the field F ($\subseteq K$). Since M is in rational canonical form, by 14.3.5, it is unique. Thus, if we were to extend the field to K, then we will get the same matrix. Thus, the invariant factors are the same as well. In particular, since the minimal polynomial is the largest invariant factor of A, it also does not depend on the field over which A is viewed.

Lemma 14.3.8: Characteristic Polynomial For Block Matrix

Let $a(x) \in F[x]$ be any monic polynomial

1. The characteristic polynomial is the product of the invariant factors:

$$P_\alpha(x) = a_1(x)a_2(x)\cdots a_m(x)$$

2. If M is the block diagonal matrix

$$M = \begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_k \end{pmatrix}$$

given by the direct sum of matrices $A_1, A_2, ..., A_k$, then the characteristic polynomial of M is the product of the characteristic polynomial of $A_1, A_2, ..., A_k$

***Proof*:**
A direct application of the determinant formula.

This gives the promised Cayley-Hamilton Theorem:

Proposition 14.3.9: Cayley-Hamilton Theorem

Let A be an $n \times n$ matrices over the field F. Then the minimal polynomial of A divides the characteristic polynomial of A.

***Proof*:**
This is immediate by the above lemma.

This will come back in Field theory where finding a polynomial which is used to represent an extension will exactly be the polynomial found using Cayley-Hamilton (build up, and eventually ref:HERE into field theory)

14.3.1 Converting a square matrix to Rational Canonical Form

Let A be an $n \times n$ matix with entries from a field F

1. First, diagonalize the matrix $xI - A$ over $F[x]$ (same process as described when finding Eigenspaces in ref:HERE), while keeping track of the process so that you can apply them to step (2) (you can do them on the same time). The matrix should reach the Smith Normal from form theorem ref:HERE

   Let $d_1, d_2, ..., d_m$ be the degrees ofthe monic nonconstant polynomials $a_1(x), ..., a_m(x)$ that are in the respective diagonals

2. Let P' be an $n \times n$ identity matrix. apply the operations from (1) to P'.

3. Once you have diagonalize $xI - A$ to the Smith Normal Form, the first $n - m$ columns of the matrix P' should be 0, and the remaining m columns of P' are nonzero. For each $i = 1, 2, ...m$, multiply the i^{th} nonzero column of P' successively by $A^0 = I, A^1, A^2, ..., A^{d_i - 1}$, where d_i is the integer in (1) aboce and use the resulting column vectors (in this order) as the next d_i column of an $n \times n$ matrix $P>$ Then $P^{-1}AP$ is in rational canonical form (whose diagonal blocks are the companion matrices for the polynomials $a_1(x), ..., a_m(x)$ in (1))

Example 14.3.10: Finding Rational Canonical Form

1. Let the following 3×3 matrices be over $\mathbb{Q}$:

$$A = \begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 0 & -4 & 85 \\ 1 & 4 & -30 \\ 0 & 0 & 3 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}$$

   We can see that each matrix has the same characteirstic polynomial

$$c_A(x) = c_B(x) = c_C(x) = (x-2)^2(x-3)$$

   Since the minimal and characteristic polynomial have the same roots, the only possibilities for the minimal polynomials are $(x-2)(x-3)$ or $(x-2)^2(x-3)$. Since there are only two options, we it will either be one minimal polynomial or the other. Thus, with some computation, we see that $(A - 2I)(A - 3I) = 0$, $(B - 2I)(B - 3I) \neq 0$ (the 1,1-entry is nonzero) and $(C-2I)(C-3I) \neq 0$ (the 1, 2-entry is nonzero). Thus, we get

$$m_A(x) = (x-2)(x-3), \qquad m_B(x) = m_C(x) = (x-2)^2(x-3)$$

since other invariant factors would need to divide $m_B(c)/m_C(x)$, there must be no other invariatn facors for B and C. Since the invariant factors for A divide the minimal polynomial and have as a product the characteristic polynomial, we see that A has for invariant factors the polynomials, $x-2, (x-2)(x-3) = x^2-5x+6$. (For 2×2 and 3×3 matrices the determination of the characteristic and minimal polynomials determine all the invariant factors, due to ________).

Thus, we can conclude that B and C are similar, and both are not simialr to A. The rational canonical forms are(note $(x-2)^2(x-3) = x^3-7x^2+16x-12$)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 12 \\ 10 & 0 & -16 \\ 0 & 1 & 7 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 12 \\ 1 & 0 & -16 \\ 0 & 1 & 7 \end{pmatrix}$$

2. In example (1), we managed to conclude the rational canonical form from the minimal polynomials, which we can always do for 3×3 matries and lower. However, that is not the case for matries of higher degree. Before doing such an example, let's practice by going the long way on matrix A from example (1).

   (a) We first start by reducint $xI - A$ in $\mathbb{Q}[x]$

   $$xI - A = \begin{pmatrix} x-2 & 2 & -14 \\ 0 & x-3 & 7 \\ 0 & 0 & x-2 \end{pmatrix}$$

   which we can do by

   here

   so, our final result is

   $$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-2 & 0 \\ 0 & 0 & x^2-5x+6 \end{pmatrix}$$

   This gives us the invariant factors of $x-2$ and $x^2-5x+6 = (x-2)(x-3)$ for this matrix. Now, let V be a 3 dimensional vector space over $\mathbb{Q}$ with standard basis e_1, e_2, e_3 and let T be the corresponding linear operator (which defines the action of x on V):

   $$\begin{aligned} xe_1 &= T(e_1) = 2e_1 \\ xe_2 &= T(e_2) = -2e_1 + 3e_2 \\ xe_3 &= T(e_3) = 14e_1 - 7e_2 + 2e_3 \end{aligned}$$

   taking the row operations from earlier

   here

   and applying them to this basis we'll get

   here

with a final result of

$$[-e_1 - (x-3)(e_2 e_1), e_3 + 7(e_2 - e_1), e_2 - e_1]$$

and pplying the result of T from eraliaer, we'll get

$$[0, -7e_1 + 7e_2 + e_3, -e_1 + e_2]$$

which correspond to the elements 1, $x-2$, and $x^2 - 5x + 6$ we found earlier. Since the first invariant factor has degree 0, we can ignore it. So, we have $f_1 = -7e_1 + 7e_2 + e_3$, and $f_2 = -e_1 + e_2$. These are the $\mathbb{Q}[x]$-module generators for the two cyclic factors of V in its invariant factor decomposition as $\mathbb{Q}[x]$-modules. For the corresponding $\mathbb{Q}$-vector space bases for theset two factros, since $x-2$ has degree 1, and $x^2 - 5x + 6$ has degree 2, we get f_1, f_2, xf_2 as a basis, where $xf_2 = T(f_2) = -4e_1 + 3e_2$, giving us

$$P = \begin{pmatrix} -7 & -1 & -4 \\ 7 & 1 & 3 \\ 1 & 0 & 0 \end{pmatrix}$$

which we can use to find P^{-1} and get

$$P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix}$$

which is the same result we had in (1).

(b) We can also convert strait into Rational cannonical form. Given the operation for the diagonalization of A, we use them on the identity matrix:

$$here$$

Since we have $d_1 = 1$ and $d_2 = 2$, we can take the 2nd row the following for P:

$$\begin{pmatrix} -7 \\ 7 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} A \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix}$$

which gives the same P as before.

Exercise 14.3.1

1. Let$A \in M_n(k)$ be a square matrix. Then A is similar to its trnaspose (hint: is transpose similarity invariant?)

14.4 Jordan Canonical Form

The RCF was found by taking using the invariant factors. Let us now find a form based on the elementary divisors when the characteristic polynomial factors completely. If the characteristic polynomial does not factor completely, then the field can be enlarged so that every polynomial factors completely (as mentioned in section 13.5 the avid reader may checkout section 19.5.1 for more details on algebraic closure). As we've outlined earlier, the similarity of two linear transformations is independent of the base field, and hence this pursuit does lead to a useful matrix form for deducing similarity.

Given a $\alpha \in \mathrm{End}_k(V)<$ we may obtain the elementary divisor decomposition of the correspondign $k[x]$-module to be:

$$V = \bigoplus_{i,j} \frac{k[x]}{p_i(x)^{r_{ij}}}$$

Then it is an immediate consequence that the characteristic polynomial $P_\alpha(x)$ is equal to the product:

$$\prod_{i,j} p_i(x)^{r_{ij}}$$

This gives us the following results:

Lemma 14.4.1: Elementary Divisors And Characteristic Polynomial

Let α be a linear operators on a finitely dimensional vector space V, and assume its characteristic polynomial $P_\alpha(x)$ factor completely so aht

$$P_\alpha(x) = \prod_i^s (x - \lambda_i)^{m_i}$$

where λ_i are the distinct eigenvalues and m_i are teh algebraic multiplicities. Then:

$$p_i(x) = (t - \lambda_i) \qquad m_i = \sum_j r_{ij}$$

And the minimal polynomial is:

$$m_\alpha(x) = \prod_{i=1}^s (x - \lambda_i)^{\max_j \{r_{ij}\}}$$

***Proof*:**
This is an exercise in putting together what was covered so far.

With this, let's start finding the matrix representation of α. Just like before, let's tart with the cyclic factor, $F[x]/((x-\lambda)^k)$. We'll purposefully be choosing our basis so that our representation is particularly simple:

$$(\overline{x} - \lambda)^{k-1}, (\overline{x} - \lambda)^{k-2}, ..., (\overline{x} - \lambda), (\overline{x} - \lambda^)0 = 1$$

this is indeed a basis, since expanding out, simplifying, we get these elements to be a linear combination of $\overline{x}^{k-1}, \overline{x}^{k-2}, ..., \overline{x}, 1$, meaning they do form a basis. Then, we can associate the linear operator T to

x, and get

$$\begin{cases} (x-\lambda)^{r-1} \mapsto x(x-\lambda)^{r-1} = \lambda(x-\lambda)^{r-1} + (x-\lambda)^r = \lambda(x-\lambda)^{r-1} \\ (x-\lambda)^{r-2} \mapsto x(x-\lambda)^{r-2} = (x-\lambda)^{r-1} + \lambda(x-\lambda)^{r-1} \\ \vdots \\ 1 \mapsto x = (x-\lambda) + \lambda \end{cases}$$

which in matrix form is be represented as:

$$\begin{pmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

where the blank spots are all zero. We give a name to this matrix

Definition 14.4.2: Jordan Block

The $k \times k$ matrix with λ along the main diagonal and 1 along the first superdiagonal depicted avoce is called the *$k \times k$ elementary Jordan matrix with eigenvalue* λ, or the *Jordan block of size k with eigenvalue* λ

As before, we may combine all of these to get a full representation of α:

Definition 14.4.3: Jordan canonical Form (JCF)

Let α be a linear operator on a finitely dimensional vector space V. Then it's *Jordan Canonical Form* (JCF) is the combination of all the Jordan blocks:

$$\left(\begin{array}{ccc} [c|c|c] J_{\lambda_1,(r_{1,j})} & & \\ \hline & \ddots & \\ \hline & & J_{\lambda_s,(r_{s,j})} \end{array}\right)$$

where $(x-\lambda_i)^{r_{ij}}$ are the elementary divisors of α.

14.4.4 An ambiguity in Uniqueness While the RCF is is completely unique up to isomorphism, since there is no unique way of arranging the eigenvalues, there is no unique way of arranging the Jordan blocks (unlike the invariant blocks which are arranged by the divisibility of the invariant factors)

So far the Jordan canonical form has only been described. What has not yet been said is when an operator *has* one: unlike the rational canonical form, which exists over any field, the Jordan form requires the relevant polynomials to factor into linear pieces. The elementary divisor decomposition of chapter 14 settles the question, and the proof also explains where the superdiagonal 1s come from.

Theorem 14.4.5: Existence of the Jordan Canonical Form

Let k be a field, let V be a finite dimensional k-vector space and let α be a linear operator on V whose minimal polynomial factors into linear factors over k (which holds for every α when k is algebraically closed). Then there is a basis of V in which the matrix of α is in Jordan canonical form, and the multiset of Jordan blocks appearing is uniquely determined by α. Equivalently, a matrix $A \in M_n(k)$ whose minimal polynomial splits over k is similar to a matrix in Jordan canonical form, unique up to the order of its blocks.

***Proof*:**
Regard V as a $k[x]$-module with x acting as α. It is finitely generated, being finite dimensional over k, and it is torsion, since the minimal polynomial $m(x)$ annihilates it. As $k[x]$ is a principal ideal domain, definition 14.1.11 gives a decomposition

$$V \cong \bigoplus_{i=1}^{s} k[x]/\big(p_i(x)^{r_i}\big)$$

with each p_i irreducible, and the multiset of the $p_i^{r_i}$ is uniquely determined by V; there is no free part because V is torsion.

Step 1: the p_i are linear. The polynomial $m(x)$ annihilates V, hence annihilates each summand $k[x]/(p_i^{r_i})$, and therefore $p_i^{r_i}$ divides $m(x)$. By hypothesis $m(x)$ is a product of linear factors, and $k[x]$ is a unique factorization domain, so every irreducible factor of m is linear. Hence each p_i is linear, say $p_i = x - \lambda_i$.

Step 2: each summand is a Jordan block. Fix a summand $W = k[x]/\big((x-\lambda)^r\big)$ and, for $1 \le j \le r$, let $e_j \in W$ be the class of $(x-\lambda)^{r-j}$. These r elements form a k-basis of W, since $1, (x-\lambda), \dots, (x-\lambda)^{r-1}$ is a k-basis of $k[x]/\big((x-\lambda)^r\big)$: the polynomials $(x-\lambda)^d$ for $0 \le d < r$ have distinct degrees $0, 1, \dots, r-1$, so they are a basis of the space of polynomials of degree less than r, which maps isomorphically onto the quotient. Writing $x = \lambda + (x-\lambda)$ and setting $e_0 := 0$, which is consistent because $(x-\lambda)^r = 0$ in W,

$$\alpha(e_j) = x \cdot (x-\lambda)^{r-j} = \lambda(x-\lambda)^{r-j} + (x-\lambda)^{r-j+1} = \lambda e_j + e_{j-1}$$

So the matrix of $\alpha|_W$ in the ordered basis $(e_1, \dots, e_r)$ has λ in every diagonal entry and 1 in every entry of the first superdiagonal, which is the Jordan block $J_{\lambda,r}$ of definition 14.4.2.

Step 3: assembly. Concatenating the bases of the summands gives a basis of V in which the matrix of α is block diagonal with the blocks of Step 2, which is the Jordan canonical form of definition 14.4.3. Uniqueness of the multiset of blocks is the uniqueness of the elementary divisors, since the block $J_{\lambda,r}$ corresponds to the elementary divisor $(x-\lambda)^r$, as we sought to show.

Only the order of the blocks is undetermined, which is the ambiguity recorded in box 14.4.4.

An interesting proeprty the JCF gives us is that it distinguishes algebraic and geometric multiplicity of eigenvalues:

Proposition 14.4.6: Geometric An Algebraic Multiplicity Of Eigenvalues Using JCF

Let α be a linear operator on a finitely dimensional vector space V. Then the geometric multiplicity of an eigenvalue λ of α equals the number of Jordan blocks corresponding to λ in the Jordan canonical form of α, and the algebraic multiplicity equals the number of times the eigenvalue shows up in the Jordan canonical form.

***Proof*:**
The fact that the algebraic multiplicity of the eigenvalue corresponds to the number or times it shows up in the JCF is immediate, hence let's show the number of Jordan blocks equasl the geometric multiplicity. Let:

$$J =: \begin{pmatrix} \lambda & 1 & \cdots & 0 & 0 \\ 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

be a Jordan block and $v = \begin{pmatrix} v_1 \\ \vdots \\ v_r \end{pmatrix}$ be an eigenvector corresponding to λ. Then:

$$\lambda \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{pmatrix} = J \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{pmatrix} = \begin{pmatrix} \lambda v_1 + v_2 \\ \lambda v_2 + v_3 \\ \vdots \\ \lambda v_r \end{pmatrix}$$

which solving gives us:

$$v_2 = \cdots = v_r = 0$$

Hence, the eigenspace of λ is generated by:

$$v = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

completing the proof.

Corollary 14.4.7: Diagonalizability And Minimal Polynomial

Assume the characteristic polynomial of $\alpha \in \mathrm{End}_k(V)$ factors completely over k. Then α is diagonalizable iff the minimal polynomial of α has no multiple roots

***Proof*:**
exercise

Naturally, the diagonalizability of a matrix also depends on the base field, for example the rotation by $\pi/2$ degrees for $\mathbb{R}^2$ is not diagonalizable, while it is diagonalizable over $\mathbb{C}$

14.4.1 Changing From one Canonical Form to Another

see D& F

14.4.2 Computation: Convert to JCF

HERE

14.5 The Jordan-Chevalley Decomposition

A matrix in Jordan canonical form is the sum of two pieces of very different character: its diagonal, which scales, and its superdiagonal of 1s, which shears. These are the two behaviours promised in chapter 13, and the two pieces even commute, since within each Jordan block the diagonal piece is the scalar matrix λI. What theorem 14.4.5 does not tell us is whether this splitting belongs to the operator or to the basis: perform the same surgery on the matrix of α in some other basis, and the two pieces need not commute and need not agree with the pieces the Jordan basis produces. The goal of this section is to show that the splitting is intrinsic. Every operator whose characteristic polynomial factors completely is the sum of a diagonalizable operator and a nilpotent operator that commute, the pair is unique, and both are polynomial expressions in the operator itself with zero constant term. The polynomial clause is the strongest part of the statement: anything that commutes with α commutes with both pieces, and any subspace that α preserves both pieces preserve. First, names for the two behaviours.

Definition 14.5.1: Semisimple and Nilpotent Operators

Let V be a finite dimensional vector space over a field k and let $\alpha \in \mathrm{End}_k(V)$. Then α is said to be *semisimple* if it is diagonalizable in the sense of definition 13.5.14[a], and α is said to be *nilpotent* if $\alpha^m = 0$ for some $m \geq 1$. The smallest such m is called the *index* of the nilpotent operator α.

[a] over a field where polynomials need not factor completely, the right definition is instead that the minimal polynomial of α is a product of distinct irreducible factors, and corollary 14.4.7 shows the two definitions agree whenever the characteristic polynomial factors completely. Everything in this section happens in that setting, so no distinction arises.

Two lemmas about diagonalizable operators are needed, and both are useful well beyond this section. The first says that diagonalizability survives restriction.

Lemma 14.5.2: Restriction of a Diagonalizable Operator

Let $\alpha \in \mathrm{End}_k(V)$ be diagonalizable with distinct eigenvalues $\mu_1, ..., \mu_r$, and let $W \subseteq V$ be an α-stable subspace (example 14.2.1). Then

$$W = (W \cap E_{\mu_1}) \oplus \cdots \oplus (W \cap E_{\mu_r})$$

and the restriction $\alpha|_W$ is diagonalizable.

***Proof*:**
By theorem 13.5.15, $V = E_{\mu_1} \oplus \cdots \oplus E_{\mu_r}$, hence any $w \in W$ decomposes as $w = w_1 + \cdots + w_r$ with $w_j \in E_{\mu_j}$, and the content of the lemma is that each w_j lies in W. Fix j and consider the polynomial

$$\pi_j(t) = \prod_{l \neq j} \frac{t - \mu_l}{\mu_j - \mu_l} \in k[t]$$

whose denominators are nonzero because the μ_l are distinct. For $v \in E_{\mu_l}$ we have $\alpha v = \mu_l v$, so every polynomial in α acts on E_{μ_l} as its value at μ_l, and π_j takes the value 0 at each μ_l with $l \neq j$ and the value 1 at μ_j. Therefore

$$\pi_j(\alpha)w = \sum_{l=1}^{r} \pi_j(\mu_l) w_l = w_j$$

Since W is α-stable, it is stable under every power of α and under linear combinations of those powers, so $\pi_j(\alpha)W \subseteq W$ and $w_j \in W$. This proves that W is the sum of the subspaces $W \cap E_{\mu_j}$, and the sum is direct because the E_{μ_j} are independent. Concatenating bases of the $W \cap E_{\mu_j}$ now gives a basis of W consisting of eigenvectors of $\alpha|_W$, which is therefore diagonalizable by theorem 13.5.13, as we sought to show.

The second lemma removes the one ambiguity a diagonalizable operator carries, namely *which* basis diagonalizes it: commuting operators can be made to share one.

Lemma 14.5.3: Simultaneous Diagonalization

Let V be a finite dimensional vector space over a field k and let $\mathcal{F} \subseteq \mathrm{End}_k(V)$ be any family of diagonalizable operators that commute pairwise. Then there is a single basis of V consisting of common eigenvectors of every member of $\mathcal{F}$, so that every member of $\mathcal{F}$ is diagonal in one shared basis. Such a family is called *simultaneously diagonalizable.*

***Proof*:**
We induct on $\dim V$. If every member of $\mathcal{F}$ is a scalar multiple of the identity, then any basis of V diagonalizes all of $\mathcal{F}$ at once, and this in particular settles $\dim V \leq 1$. Otherwise, fix some $\alpha \in \mathcal{F}$ that is not a scalar. By theorem 13.5.15,

$$V = E_{\mu_1} \oplus \cdots \oplus E_{\mu_r}$$

where $\mu_1, ..., \mu_r$ are the distinct eigenvalues of α. Here $r \geq 2$: if α had a single eigenvalue μ, a diagonalizing basis would represent α by the matrix μI, making $\alpha = \mu\mathrm{id}$ a scalar. Consequently each eigenspace satisfies $0 < \dim E_{\mu_j} < \dim V$.

Every $\beta \in \mathcal{F}$ preserves every E_{μ_j}: for $v \in E_{\mu_j}$,

$$\alpha(\beta v) = \beta(\alpha v) = \mu_j(\beta v)$$

so $\beta v \in E_{\mu_j}$. By lemma 14.5.2 each restriction $\beta|_{E_{\mu_j}}$ is diagonalizable, and for fixed j the restrictions commute pairwise because the members of $\mathcal{F}$ do. The inductive hypothesis applied to E_{μ_j} produces a basis of E_{μ_j} of common eigenvectors of all the restrictions. Concatenating these bases over $j = 1, ..., r$ gives a basis of V, and each of its vectors is a common eigenvector of every member of $\mathcal{F}$, completing the proof.

Lemma 14.5.4: Sums and Products of Commuting Semisimple or Nilpotent Operators

Let V be a finite dimensional vector space over a field k and let $a, b \in \mathrm{End}_k(V)$ commute.

1. If a and b are semisimple, then $a + b$, $a - b$ and ab are semisimple
2. If a and b are nilpotent, then $a + b$, $a - b$ and ab are nilpotent

***Proof*:**
(1). By lemma 14.5.3 applied to the family $\{a, b\}$, there is a basis in which both a and b are diagonal, and sums, differences and products of diagonal matrices are diagonal.

(2). Say $a^u = 0$ and $b^v = 0$. Since $ab = ba$, the binomial theorem applies and

$$(a \pm b)^{u+v-1} = \sum_{i=0}^{u+v-1} \binom{u+v-1}{i} (\pm 1)^{u+v-1-i}\, a^i b^{u+v-1-i}$$

In each term either $i \geq u$ or $u + v - 1 - i \geq v$, so every term vanishes and $a \pm b$ is nilpotent. Finally, $(ab)^u = a^u b^u = 0$, using commutativity again, completing the proof.

The remaining input is a decomposition of V into pieces on which α is a scalar plus a nilpotent, and it has already been proven in this chapter under another name. When α is not diagonalizable, the eigenspaces of theorem 13.5.15 are too small to fill out V, and the correct enlargement is to relax "killed by $\alpha - \lambda\mathrm{id}$" to "killed by a power of $\alpha - \lambda\mathrm{id}$".

Definition 14.5.5: Generalized Eigenspace

Let V be a finite dimensional vector space over a field k, let $\alpha \in \mathrm{End}_k(V)$ and let $\lambda \in k$. Then the subspace

$$V_\lambda = \left\{ v \in V \middle| (\alpha - \lambda\mathrm{id})^m v = 0 \text{ for some } m \geq 1 \right\}$$

is called the *generalized eigenspace* of α at λ, and its nonzero elements are called *generalized eigenvectors*.

The eigenspace of definition 13.5.9 is the $m = 1$ part of V_λ, and $V_\lambda \neq 0$ exactly when λ is an eigenvalue: if $(\alpha - \lambda\mathrm{id})^m v = 0$ with $v \neq 0$ and m minimal, then $(\alpha - \lambda\mathrm{id})^{m-1} v$ is an eigenvector for λ. The generalized eigenspaces are exactly the primary components of theorem 14.1.12 for the $k[x]$-module structure that α places on V, so the machinery of this chapter already decomposes V into them.

Corollary 14.5.6: Generalized Eigenspace Decomposition

Let V be a nonzero finite dimensional vector space over a field k and let $\alpha \in \mathrm{End}_k(V)$ have characteristic polynomial

$$P_\alpha(x) = \prod_{i=1}^{r} (x - \lambda_i)^{m_i}$$

factoring completely over k, with $\lambda_1, ..., \lambda_r$ distinct. Then

$$V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r} \qquad \text{where} \qquad V_{\lambda_i} = \ker(\alpha - \lambda_i \mathrm{id})^{m_i}$$

and the restriction of $\alpha - \lambda_i\mathrm{id}$ to V_{λ_i} is nilpotent.

***Proof*:**
Regard V as a $k[x]$-module with x acting as α, as in the proof of theorem 14.4.5. It is a nonzero torsion module, and its annihilator is generated by the minimal polynomial $m_\alpha(x)$ (definition 14.2.2), which by lemma 14.4.1 factors as

$$m_\alpha(x) = \prod_{i=1}^{r} (x - \lambda_i)^{\nu_i}$$

where $\nu_i = \max_j r_{ij}$ and $m_i = \sum_j r_{ij}$ in terms of the elementary divisor exponents, so that $1 \le \nu_i \le m_i$ for every i. Theorem 14.1.12 applied to this factorization yields

$$V = N_1 \oplus \cdots \oplus N_r$$

where N_i is the kernel of $(\alpha - \lambda_i\mathrm{id})^{\nu_i}$ and equally the set of all elements killed by *some* power of $\alpha - \lambda_i\mathrm{id}$, which is the definition of V_{λ_i}. The sandwich

$$\ker(\alpha - \lambda_i\mathrm{id})^{\nu_i} \subseteq \ker(\alpha - \lambda_i\mathrm{id})^{m_i} \subseteq V_{\lambda_i} = \ker(\alpha - \lambda_i\mathrm{id})^{\nu_i}$$

obtained from $\nu_i \le m_i$ forces all three subspaces to coincide, and the middle description is the one in the statement. Finally, $(\alpha - \lambda_i\mathrm{id})^{\nu_i}$ vanishes on V_{λ_i} by construction, so the restriction of $\alpha - \lambda_i\mathrm{id}$ to V_{λ_i} is nilpotent, as we sought to show.

Everything is now assembled.

Theorem 14.5.7: Jordan-Chevalley Decomposition

Let V be a nonzero finite dimensional vector space over a field k and let $\alpha \in \mathrm{End}_k(V)$ be an operator whose characteristic polynomial factors into linear factors over k, which holds for every α when k is algebraically closed, with no restriction on the characteristic of k. Then:

1. there exist $s, n \in \mathrm{End}_k(V)$ with
$$\alpha = s + n \qquad sn = ns$$
where s is semisimple and n is nilpotent

2. there are polynomials $p(t), q(t) \in k[t]$ with zero constant term such that $s = p(\alpha)$ and $n = q(\alpha)$. In particular, s and n commute with every operator that commutes with α, and every α-stable subspace of V is s-stable and n-stable

3. the pair (s, n) is unique: if $\alpha = s' + n'$ with s' semisimple, n' nilpotent and $s'n' = n's'$, then $s' = s$ and $n' = n$

The expression $\alpha = s + n$ is called the *Jordan-Chevalley decomposition* (or *additive Jordan decomposition*) of α, and s and n are called the *semisimple part* and the *nilpotent part* of α.

***Proof*:**
We first build the polynomial p, then verify that $s = p(\alpha)$ and $n = \alpha - s$ have every property claimed, and prove uniqueness last. Throughout, write
$$P_\alpha(x) = \prod_{i=1}^{r} (x - \lambda_i)^{m_i}$$
with $\lambda_1, ..., \lambda_r$ the distinct eigenvalues of α.

Step 1: constructing p by interpolation. We seek a polynomial $p(t) \in k[t]$ satisfying the congruences
$$p(t) \equiv \lambda_i \bmod (t - \lambda_i)^{m_i} \qquad 1 \le i \le r$$
together with $p(0) = 0$, which is the congruence $p(t) \equiv 0 \bmod (t)$. For $i \neq j$ the ideals $\big((t-\lambda_i)^{m_i}\big)$ and $\big((t - \lambda_j)^{m_j}\big)$ of $k[t]$ are comaximal: $t - \lambda_i$ and $t - \lambda_j$ are non-associate primes of $k[t]$, so $\gcd\big((t - \lambda_i)^{m_i}, (t - \lambda_j)^{m_j}\big) = 1$, and a Bézout identity in the Euclidean domain $k[t]$ places 1 in the sum of the two ideals, exactly as in the proof of proposition 11.1.6. The congruence at 0 needs care, and there are two cases according to whether 0 is an eigenvalue.

If no λ_i is 0, then $t = t - 0$ is a monic linear polynomial distinct from every $t - \lambda_i$, so by the same argument the ideal (t) is comaximal with each $\big((t - \lambda_i)^{m_i}\big)$. The map of the Chinese Remainder Theorem (theorem 9.6.6) is then surjective onto the product of the $r + 1$ quotients, and choosing the preimage of the tuple $(\lambda_1, ..., \lambda_r, 0)$ produces a $p(t)$ satisfying all $r + 1$ congruences at once. The last congruence says that t divides $p(t)$, which is exactly the vanishing of the constant term.

If instead $\lambda_{i_0} = 0$ for some necessarily unique i_0, the congruence at 0 cannot be imposed as a separate condition, and does not need to be: the ideals (t) and $\big(t^{m_{i_0}}\big)$ are not comaximal, so theorem 9.6.6 does not apply to the enlarged list of $r + 1$ ideals, but the i_0-th congruence already reads
$$p(t) \equiv \lambda_{i_0} = 0 \bmod \big(t^{m_{i_0}}\big)$$

Hence we impose only the original r congruences, whose moduli are pairwise comaximal as shown above, and theorem 9.6.6 again provides a solution $p(t)$. Since $m_{i_0} \geq 1$, divisibility by $t^{m_{i_0}}$ implies divisibility by t, and the constant term of p vanishes in this case too, this time as a consequence of the eigenvalue congruence at $\lambda_{i_0} = 0$ rather than as an extra condition.

In both cases the result is a polynomial $p(t)$ with $p(0) = 0$ such that $(t - \lambda_i)^{m_i}$ divides $p(t) - \lambda_i$ for every i. Set

$$s = p(\alpha) \qquad q(t) = t - p(t) \qquad n = q(\alpha) = \alpha - s$$

and note that $q(0) = 0 - p(0) = 0$ as well.

Step 2: s is semisimple and n is nilpotent. By corollary 14.5.6, $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_r}$ with $V_{\lambda_i} = \ker(\alpha - \lambda_i \mathrm{id})^{m_i}$. Fix i and write $p(t) - \lambda_i = h_i(t)\,(t - \lambda_i)^{m_i}$. Evaluating at α and applying to $v \in V_{\lambda_i}$,

$$(s - \lambda_i \mathrm{id})v = h_i(\alpha)(\alpha - \lambda_i \mathrm{id})^{m_i} v = 0$$

so s acts on V_{λ_i} as the scalar λ_i. A basis of V adapted to the direct sum decomposition therefore consists of eigenvectors of s, and s is semisimple. On V_{λ_i} the operator $n = \alpha - s$ acts as $\alpha - \lambda_i \mathrm{id}$, whose m_i-th power kills V_{λ_i}. Setting $m = \max_i m_i$, the power n^m kills every summand and hence all of V, so n is nilpotent.

This proves clauses (1) and (2): s and n are polynomials in α with zero constant term by Step 1, so each is a linear combination of *positive* powers of α. Such operators commute with α, with each other, and with every operator commuting with α, and they carry any α-stable subspace into itself.

Step 3: uniqueness. Let $\alpha = s' + n'$ with s' semisimple, n' nilpotent and $s'n' = n's'$. Then s' commutes with $s' + n' = \alpha$, and so does n'. An operator commuting with α commutes with every power of α and hence with every polynomial in α, so s' and n' commute with s and with n. Rearranging $s + n = s' + n'$ gives

$$s - s' = n' - n$$

and all four operators commute pairwise. By lemma 14.5.4, the left side is semisimple and the right side is nilpotent, so the operator $\delta = s - s' = n' - n$ is both. Being nilpotent, δ has no eigenvalue besides 0: if $\delta v = \mu v$ with $v \neq 0$ and $\delta^m = 0$, then $0 = \delta^m v = \mu^m v$ forces $\mu = 0$. Being semisimple, δ is represented by a diagonal matrix in some basis, and the diagonal entries of a diagonal matrix are eigenvalues, hence all 0. So $\delta = 0$, that is $s' = s$ and $n' = n$, as we sought to show.

In a Jordan basis the decomposition becomes visible: Step 2 makes s act as λ_i on each generalized eigenspace, which is precisely the diagonal of the Jordan canonical form of theorem 14.4.5, and $n = \alpha - s$ is what remains, the superdiagonal of 1s. Uniqueness then says that the recipe "keep the diagonal of the JCF" computes the semisimple part correctly, while the first example below shows that the same recipe applied in any other basis computes nothing of the sort.

Example 14.5.8: Computing a Jordan-Chevalley Decomposition

1. Take $k = \mathbb{Q}$ and

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \qquad \text{so that} \qquad P_A(t) = (t-1)(t-2)$$

Splitting off the diagonal gives a diagonal piece D and a nilpotent piece N with

$$DN = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = ND$$

so $A = D + N$ is *not* the Jordan-Chevalley decomposition of A. Indeed, the minimal polynomial of A has no multiple roots, so A is already diagonalizable by corollary 14.4.7 and its decomposition is $s = A$ and $n = 0$. The recipe of the proof agrees: 0 is not an eigenvalue, and the congruences

$$p \equiv 1 \bmod (t-1) \qquad p \equiv 2 \bmod (t-2) \qquad p \equiv 0 \bmod (t)$$

are solved by $p(t) = t$, so $s = p(A) = A$.

2. Now take

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \qquad \text{so that} \qquad P_B(t) = (t-1)^2(t-2)$$

Again 0 is not an eigenvalue, so the proof imposes the congruences

$$p \equiv 1 \bmod (t-1)^2 \qquad p \equiv 2 \bmod (t-2) \qquad p \equiv 0 \bmod (t)$$

The moduli multiply to the degree 4 polynomial $t(t-1)^2(t-2)$, so there is a unique solution of degree at most 3, namely

$$p(t) = t^3 - 3t^2 + 3t$$

All three congruences can be checked directly: $p(t)$ has no constant term, $p(t) - 1 = (t-1)^3$, and $p(2) = 8 - 12 + 6 = 2$. To compute $s = p(B)$, note that on the upper 2×2 Jordan block J the identity $p(t) = (t-1)^3 + 1$ gives $p(J) = (J-I)^3 + I = I$, since $(J-I)^2 = 0$, while on the lower block $p(2) = 2$. Hence

$$s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \qquad n = B - s = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and a direct multiplication confirms $sn = ns$, both products being equal to n.

One clause of the theorem deserves a closing word. Interpolation without the congruence at 0 would already produce commuting semisimple and nilpotent parts, so the zero constant term looks like a free refinement, and Step 1 of the proof shows that it costs nothing. It is nevertheless the clause that transfers the strongest properties of α to s and n. A polynomial in α with constant term $c \neq 0$ acts as $c\mathrm{id}$ plus a combination of positive powers of α, and the summand $c\mathrm{id}$ carries any nonzero subspace onto itself, never into a smaller one. With the constant term gone that obstruction disappears: whenever $W' \subseteq W$ are subspaces with $\alpha(W) \subseteq W'$, every positive power of α also carries W into W', hence so do s and n. The semisimple and nilpotent parts therefore satisfy every containment of the form $\alpha(W) \subseteq W'$ that α satisfies, of which stability is the special case $W' = W$. This stronger inheritance is invisible at the level of a single matrix, and it becomes essential the moment operators are constrained to lie in a subspace of $\mathrm{End}_k(V)$ that does not contain the identity.

15

Tensor Products Of Modules: Linear Algebra III

The results of chapter 13 and chapter 14 describe modules and the linear maps between them, and several of the objects those chapters produced are not linear maps. The determinant of section 13.4 takes n column vectors and is linear in each one separately, not in all of them together. An inner product (section 13.6) takes two vectors and is linear in each. The consequence is that such maps are not morphisms in Vect_k, so no theorem about morphisms applies to them. Multiplication in a ring is the same phenomenon at its most basic. The map $R \times R \to R$ sending (a, b) to ab is additive in each variable when the other is held fixed, and it is not additive in the pair: the product $(a + a')(b + b')$ is $ab + ab' + a'b + a'b'$, not $ab + a'b'$.

This chapter removes the obstruction by changing the domain rather than the map in a natural way. From two modules M and N over a ring R it constructs a single module $M \otimes_R N$, together with a bilinear map $M \times N \to M \otimes_R N$ through which every bilinear map out of $M \times N$ factors, uniquely, as a homomorphism out of $M \otimes_R N$. A question about bilinear maps on $M \times N$ then becomes a question about homomorphisms on $M \otimes_R N$, where the machinery of the previous two chapters applies. Iterating the construction on a single module produces the tensor algebra $\mathcal{T}(M)$ and its two quotients, the symmetric algebra $\mathcal{S}(M)$ and the exterior algebra $\bigwedge(M)$. The determinant reappears in section 16.4 as the map an endomorphism induces on the top exterior power.

15.1 The Tensor Product of Modules

Suppose for a moment that the module $M \otimes_R N$ has been built, along with a map $\iota \colon M \times N \to M \otimes_R N$ whose value at (m, n) is written $m \otimes n$. Asking that ι be additive in each variable forces

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n, \qquad m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$$

giving relations any construction must impose. The condition governing scalars needs more care. Take R commutative first, with M, N and the target L all R-modules. A map $\varphi\colon M \times N \to L$ that is R-linear in each variable satisfies

$$\varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn)$$

so a scalar may be moved from one argument to the other. Over a non-commutative ring that same conditin is nearly vacuous. Suppose M, N and L are left R-modules and φ is R-linear in each variable. Evaluating $\varphi(rm, sn)$ two ways, once by extracting s from the second argument and once by extracting r from the first, gives

$$(sr)\varphi(m, n) = s\varphi(rm, n) = \varphi(rm, sn) = r\varphi(m, sn) = (rs)\varphi(m, n)$$

so that $(rs - sr)\varphi(m, n) = 0$ for every $m \in M$ and $n \in N$. Any such φ is annihilated by every commutator in R, and over a ring with few central elements this leaves almost nothing.

The repair is to weaken the condition in two places. The target drops from an R-module to an abelian group, so that "pulling out a scalar" is no longer available and cannot be contradicted. And the two modules are taken on opposite sides, M a right R-module and N a left one, so that the surviving requirement $\varphi(mr, n) = \varphi(m, rn)$ moves a scalar across the comma without ever moving it outside. This is the condition the construction can enforce.

Definition 15.1.1: R-Balanced Map

Let R be a ring, let M be a right R-module, let N be a left R-module, and let L be an abelian group written additively. A map $\varphi\colon M \times N \to L$ is *R-balanced*, or *middle linear over R*, if for all $m, m_1, m_2 \in M$, all $n, n_1, n_2 \in N$ and all $r \in R$,

$$\begin{aligned}
\varphi(m_1 + m_2, n) &= \varphi(m_1, n) + \varphi(m_2, n) \\
\varphi(m, n_1 + n_2) &= \varphi(m, n_1) + \varphi(m, n_2) \\
\varphi(mr, n) &= \varphi(m, rn)
\end{aligned}$$

The third axiom is the only one that mentions R at all, and it is weaker than linearity in either variable, since it never asserts what happens to a scalar that is pulled out of the map, namely as L has no module structure.

Example 15.1.2: R-Balanced Maps

1. Let R be commutative and let M, N, L be R-modules, with $mr := rm$ making M a right R-module. Every map that is R-linear in each variable is R-balanced: $\varphi(mr, n) = \varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn)$. So over a commutative ring the balanced condition is implied by, and strictly weaker than, bilinearity.

2. For any ring R and any right R-module M, the action map $M \times R \to M$ sending (m, r) to mr is R-balanced, where R is viewed as a left module over itself. Additivity in each variable is a module axiom, and the third axiom is associativity of the action: $(ms)r = m(sr)$.

3. Take $R = \mathbb{Z}$. Every $\mathbb{Z}$-module is an abelian group and every abelian group is a $\mathbb{Z}$-module (section 12.1), so a $\mathbb{Z}$-balanced map is exactly a map additive in each variable: the third axiom, $\varphi(mk, n) = \varphi(m, kn)$ for $k \in \mathbb{Z}$, already follows from the first two. Balanced maps over $\mathbb{Z}$ therefore impose no condition beyond biadditivity.

4. The determinant of a 2×2 matrix, read as a map $F^2 \times F^2 \to F$ sending the pair of columns $\big((a,b),(c,d)\big)$ to $ad - bc$, is F-bilinear (section 13.4) and hence F-balanced. It is one of the maps this chapter exists to bring inside linear algebra.

The construction now writes down the freest object carrying such a map. Start with a group in which the pairs (m,n) satisfy nothing whatsoever, then impose the three axioms of definition 15.1.1 by hand. Write $F^{\mathbb{Z}}(A)$ for the free $\mathbb{Z}$-module on a set A (theorem 12.3.11), that is, the free abelian group with basis A.

Applied to the underlying *set* of $M \times N$, this group is enormous. If $R = \mathbb{R}$, $M = \mathbb{R}^2$ and $N = \mathbb{R}^3$, the basis of $F^{\mathbb{Z}}(M \times N)$ is the whole of $\mathbb{R}^2 \times \mathbb{R}^3$, so distinct pairs (m,n) and (m',n') are independent generators even when m and m' differ by a scalar. Nothing that holds in M or in N survives into $F^{\mathbb{Z}}(M \times N)$; every relation the tensor product satisfies has to be put in by quotienting.

Definition 15.1.3: Tensor Product

Let R be a ring, let M be a right R-module and let N be a left R-module. Let $K \subseteq F^{\mathbb{Z}}(M \times N)$ be the subgroup generated by all elements of the three forms

$$\begin{aligned}
&(m_1 + m_2,\, n) - (m_1, n) - (m_2, n)\\
&(m,\, n_1 + n_2) - (m, n_1) - (m, n_2)\\
&(mr,\, n) - (m, rn)
\end{aligned}$$

taken over all $m, m_1, m_2 \in M$, all $n, n_1, n_2 \in N$ and all $r \in R$. The *tensor product of M and N over R* is the quotient group

$$M \otimes_R N := F^{\mathbb{Z}}(M \times N)/K$$

The coset of the basis element (m,n) is written $m \otimes n$ and is called a *simple tensor*; the subscript R is dropped when the ring is clear from context.

Killing K is exactly what makes the three defining identities hold in the quotient, so that for all $m, m_i \in M$, $n, n_i \in N$ and $r \in R$,

$$\begin{gathered}
(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n \qquad m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2\\
mr \otimes n = m \otimes rn
\end{gathered} \tag{15.1}$$

These three are the only relations available. Anything else true in $M \otimes_R N$ has to be derived from them, and the next lemma is the derivation used most often.

Lemma 15.1.4: Simple Tensors Generate

Let M be a right R-module and N a left R-module. Then

1. $0 \otimes n = 0$ and $m \otimes 0 = 0$ for all $m \in M$, $n \in N$;
2. every element of $M \otimes_R N$ is a finite sum $\sum_{i=1}^{k} m_i \otimes n_i$ of simple tensors;
3. two group homomorphisms $M \otimes_R N \to L$ that agree on every simple tensor are equal.

***Proof*:**

1. By the first identity in (15.1) applied to $m_1 = m_2 = 0$, one has $0 \otimes n = (0+0) \otimes n = 0 \otimes n + 0 \otimes n$, and subtracting $0 \otimes n$ from both sides in the group $M \otimes_R N$ gives $0 \otimes n = 0$. The argument for $m \otimes 0$ is the same using the second identity.

2. The group $F^{\mathbb{Z}}(M \times N)$ is generated by its basis, so every element of it is a finite sum $\sum_i k_i(m_i, n_i)$ with $k_i \in \mathbb{Z}$, and the quotient map is surjective, so every element of $M \otimes_R N$ is a finite sum $\sum_i k_i(m_i \otimes n_i)$. It therefore suffices to see that an integer multiple of a simple tensor is again a simple tensor. For $k \geq 0$ induction on k using the first identity in (15.1) gives $k(m \otimes n) = (km) \otimes n$, the base case $k = 0$ being part (1). For $k < 0$ it suffices to treat $k = -1$, and
$$(-m) \otimes n + m \otimes n = (-m + m) \otimes n = 0 \otimes n = 0$$
gives $(-m) \otimes n = -(m \otimes n)$, so $k(m \otimes n) = (km) \otimes n$ for every $k \in \mathbb{Z}$.

3. Let ψ, ψ' be two such homomorphisms. The set on which two group homomorphisms agree is a subgroup of the domain, and by (2) the simple tensors generate $M \otimes_R N$, so that subgroup is everything, completing the proof.

Part (2) says that every element of the tensor product is a finite sum of simple tensors; it does not say that every element *is* a simple tensor, and that distinction is the most common source of error in computations with tensor products; box 15.1.6 exhibits an element that is not simple.

The definition of $M \otimes_R N$ was engineered to make ι balanced and to impose nothing further. The following theorem is the statement that it succeeded, and it is the only property of the tensor product used from here on: every later result in this chapter is proved from it rather than from the quotient construction.

Theorem 15.1.5: Universal Property of the Tensor Product

Let R be a ring, let M be a right R-module and let N be a left R-module. The map
$$\iota\colon M \times N \to M \otimes_R N, \qquad \iota(m,n) = m \otimes n$$
is R-balanced, and it is universal among such maps: for every abelian group L and every R-balanced map $\varphi\colon M \times N \to L$ there is a unique group homomorphism $\psi\colon M \otimes_R N \to L$ with $\varphi = \psi \circ \iota$, that is, making
$$\begin{tikzcd} M \otimes_R N \arrow[r, "\psi"] & L \\ M \times N \arrow[u, "\iota"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$
commute.

***Proof*:**
That ι is R-balanced is precisely the three identities (15.1), which hold because the corresponding elements of $F^{\mathbb{Z}}(M \times N)$ were placed in K.

Existence. Regard $M \times N$ as a set and φ as a map of sets into the abelian group L, which is a $\mathbb{Z}$-module. By the universal property of free modules (theorem 12.3.11, applied with $R = \mathbb{Z}$) there is a group homomorphism $\tilde{\varphi}\colon F^{\mathbb{Z}}(M \times N) \to L$ with $\tilde{\varphi}\big((m,n)\big) = \varphi(m,n)$ on each basis element. Evaluating $\tilde{\varphi}$ on the three families of generators of K gives

$$\begin{aligned}
\tilde{\varphi}\big((m_1+m_2,n) - (m_1,n) - (m_2,n)\big) &= \varphi(m_1+m_2,n) - \varphi(m_1,n) - \varphi(m_2,n) = 0\\
\tilde{\varphi}\big((m,n_1+n_2) - (m,n_1) - (m,n_2)\big) &= \varphi(m,n_1+n_2) - \varphi(m,n_1) - \varphi(m,n_2) = 0\\
\tilde{\varphi}\big((mr,n) - (m,rn)\big) &= \varphi(mr,n) - \varphi(m,rn) = 0
\end{aligned}$$

each vanishing by the corresponding axiom of definition 15.1.1. A subgroup is generated by any set whose elements it contains together with their sums and inverses, and $\ker\tilde{\varphi}$ is a subgroup containing all three families, so $K \subseteq \ker\tilde{\varphi}$. Since $F^{\mathbb{Z}}(M \times N)$ is abelian, K is normal in it, so theorem 3.1.14 applies and yields a unique map $\psi\colon F^{\mathbb{Z}}(M \times N)/K \to L$ with $\tilde{\varphi} = \psi \circ \pi$, where π is the quotient map. It is a group homomorphism, since for $x, y \in F^{\mathbb{Z}}(M \times N)$,

$$\psi\big(\pi(x)+\pi(y)\big) = \psi\big(\pi(x+y)\big) = \tilde{\varphi}(x+y) = \tilde{\varphi}(x) + \tilde{\varphi}(y) = \psi(\pi(x)) + \psi(\pi(y))$$

and π is surjective. Reading the identity $\tilde{\varphi} = \psi \circ \pi$ on the basis element (m,n) gives $\psi(m \otimes n) = \varphi(m,n)$, which is $\varphi = \psi \circ \iota$.

Uniqueness. A homomorphism ψ' with $\varphi = \psi' \circ \iota$ satisfies $\psi'(m \otimes n) = \varphi(m,n) = \psi(m \otimes n)$ for every simple tensor, so $\psi' = \psi$ by lemma 15.1.4, completing the proof.

The theorem is what converts the tensor product from a construction into a tool. To produce a homomorphism out of $M \otimes_R N$ one never returns to cosets of a free group: one writes down a balanced map on $M \times N$, checks three identities, and reads off the homomorphism. To prove two homomorphisms out of $M \otimes_R N$ equal, one checks them on simple tensors. Both moves are used repeatedly below, starting immediately.

15.1.6 Warning: Not Every Tensor Is Simple Let F be a field, let $V = F^2$ with standard basis e_1, e_2, and consider

$$t = e_1 \otimes e_1 + e_2 \otimes e_2 \in V \otimes_F V$$

Then t is not a simple tensor. Suppose $t = v \otimes w$. For linear functionals $f, g \in V^*$ (section 13.3) the map $V \times V \to F$ sending (x,y) to $f(x)g(y)$ is F-bilinear, hence F-balanced, so by theorem 15.1.5 below it induces a homomorphism $\theta_{f,g}\colon V \otimes_F V \to F$ with $\theta_{f,g}(x \otimes y) = f(x)g(y)$. Evaluating on both expressions for t,

$$f(v)g(w) = \theta_{f,g}(t) = f(e_1)g(e_1) + f(e_2)g(e_2)$$

Taking (f,g) to be (e_1^*, e_1^*), then (e_2^*, e_2^*), then (e_1^*, e_2^*) gives in turn $v_1w_1 = 1$, $v_2w_2 = 1$ and $v_1w_2 = 0$, where $v = v_1e_1 + v_2e_2$ and $w = w_1e_1 + w_2e_2$. The first two force v_1 and w_2 to be nonzero, contradicting the third.

A consequence worth recording is that the pair $(M \otimes_R N, \iota)$ is pinned down by the theorem, so the arbitrary choices in definition 15.1.3 leave no trace.

Corollary 15.1.7: Uniqueness of the Tensor Product

Let T be an abelian group and $j: M \times N \to T$ an R-balanced map with the property that for every abelian group L and every R-balanced $\varphi: M \times N \to L$ there is a unique homomorphism $\psi: T \to L$ with $\varphi = \psi \circ j$. Then there is a unique isomorphism $\theta: M \otimes_R N \to T$ with $\theta \circ \iota = j$.

***Proof*:**
Applying theorem 15.1.5 to the balanced map j gives a homomorphism $\theta: M \otimes_R N \to T$ with $\theta \circ \iota = j$, and applying the hypothesis on T to the balanced map ι gives $\eta: T \to M \otimes_R N$ with $\eta \circ j = \iota$. Then $\eta \circ \theta$ is an endomorphism of $M \otimes_R N$ satisfying $(\eta \circ \theta) \circ \iota = \iota$, and so is the identity; by the uniqueness clause of theorem 15.1.5 applied to $\varphi = \iota$, the two agree. Symmetrically $\theta \circ \eta = \mathrm{id}_T$, using the uniqueness clause assumed for T. Hence θ is an isomorphism, and it is the unique one satisfying $\theta \circ \iota = j$ by that same uniqueness clause, as we sought to show.

So far $M \otimes_R N$ is only an abelian group, and definition 15.1.1 was designed to make that unavoidable: nothing in the data of a right module M and a left module N produces an R-action on the tensor product. A module structure has to come from somewhere else, and the smallest extra hypothesis that gives one is a second, compatible action on one of the factors.

Definition 15.1.8: (S, R)-Bimodule

Let R and S be rings. An abelian group M is an (S, R)-*bimodule* if it is a left S-module, a right R-module, and the two actions commute:

$$s(mr) = (sm)r \qquad \text{for all } s \in S,\ r \in R,\ m \in M$$

Example 15.1.9: Bimodules

1. Any ring R is an (R, R)-bimodule over itself, with both actions given by ring multiplication; the compatibility axiom is associativity, $s(mr) = (sm)r$.
2. Let $f: R \to S$ be a ring homomorphism. Then S is an (S, R)-bimodule: the left S-action is multiplication in S, and the right R-action is $s \cdot r := sf(r)$. Compatibility reads $s'(sf(r)) = (s's)f(r)$, which is associativity in S. This is the structure that makes extension of scalars along f possible, and it is taken up in section 15.3.
3. As a special case of (2), let $I \subseteq R$ be a two-sided ideal and $f: R \to R/I$ the quotient map. Then R/I is an $(R/I, R)$-bimodule.
4. Let R be commutative and M an R-module. Setting $mr := rm$ makes M an (R, R)-bimodule, since the compatibility axiom $s(mr) = (sm)r$ reads $(sr)m = (rs)m$. This is called the *standard* bimodule structure on M, and it is the structure meant whenever a module over a commutative ring is treated as a bimodule without further comment.

Proposition 15.1.10: Module Structure on the Tensor Product

Let M be an (S,R)-bimodule and let N be a left R-module. Then $M \otimes_R N$ is a left S-module under the action determined by

$$s \cdot (m \otimes n) = (sm) \otimes n$$

on simple tensors.

***Proof*:**
Fix $s \in S$ and consider $\lambda_s \colon M \times N \to M \otimes_R N$ defined by $\lambda_s(m,n) = (sm) \otimes n$. It is R-balanced. Additivity in the first variable holds because $s(m_1 + m_2) = sm_1 + sm_2$ and $\otimes$ is additive in its first argument; additivity in the second is immediate. For the third axiom, the bimodule condition gives $s(mr) = (sm)r$, so

$$\lambda_s(mr, n) = \big(s(mr)\big) \otimes n = \big((sm)r\big) \otimes n = (sm) \otimes rn = \lambda_s(m, rn)$$

the middle equality being the third relation of (15.1). By theorem 15.1.5 there is therefore a unique group endomorphism Λ_s of $M \otimes_R N$ with $\Lambda_s(m \otimes n) = (sm) \otimes n$. Define $s \cdot x := \Lambda_s(x)$ for $x \in M \otimes_R N$; this is additive in x because Λ_s is a homomorphism.

The remaining module axioms are equalities between group homomorphisms $M \otimes_R N \to M \otimes_R N$, so by lemma 15.1.4 each may be checked on simple tensors alone. On $m \otimes n$,

$$\begin{aligned}
\Lambda_{s+s'}(m \otimes n) &= \big((s+s')m\big) \otimes n = (sm + s'm) \otimes n \\
&= (sm) \otimes n + (s'm) \otimes n = (\Lambda_s + \Lambda_{s'})(m \otimes n) \\
\Lambda_{ss'}(m \otimes n) &= \big((ss')m\big) \otimes n = \big(s(s'm)\big) \otimes n \\
&= \Lambda_s\big((s'm) \otimes n\big) = (\Lambda_s \circ \Lambda_{s'})(m \otimes n) \\
\Lambda_1(m \otimes n) &= (1m) \otimes n = m \otimes n
\end{aligned}$$

using in each case only a left S-module axiom for M. Hence $\Lambda_{s+s'} = \Lambda_s + \Lambda_{s'}$, $\Lambda_{ss'} = \Lambda_s \circ \Lambda_{s'}$ and $\Lambda_1 = \mathrm{id}$, which are exactly the axioms making $M \otimes_R N$ a left S-module, completing the proof.

The proof is a template for everything that follows: a map is defined on simple tensors, shown balanced, promoted to a homomorphism by theorem 15.1.5, and its properties are then checked on simple tensors by lemma 15.1.4. No later argument in this chapter returns to the free group.

Specializing to a commutative ring recovers the situation the chapter opened with, in which both factors and the target are modules over the same ring and the universal property is stated for bilinear maps. That case needs the module-theoretic form of bilinearity, which definition 13.3.15 gave only for vector spaces with target the base field.

Definition 15.1.11: R-Bilinear Map

Let R be a commutative ring and let M, N and L be R-modules. A map $\varphi \colon M \times N \to L$ is *R-bilinear* if for each fixed $n \in N$ the map $\varphi(-,n) \colon M \to L$ is an R-module homomorphism, and for each fixed $m \in M$ the map $\varphi(m,-) \colon N \to L$ is an R-module homomorphism.

Theorem 15.1.12: Universal Property Of Tensor Product (Over Commutative Rings)

Let R be a commutative ring and let M, N be R-modules, each carrying its standard bimodule structure. Then $M \otimes_R N$ is an R-module, the map $\iota\colon M \times N \to M \otimes_R N$ is R-bilinear, and for every R-module L and every R-bilinear map $\varphi\colon M \times N \to L$ there is a unique R-module homomorphism $\psi\colon M \otimes_R N \to L$ with $\varphi = \psi \circ \iota$.

***Proof*:**
The R-module structure is proposition 15.1.10 applied with $S = R$, using the standard bimodule structure on M from example 15.1.9(4); explicitly, $r(m \otimes n) = (rm) \otimes n$. That ι is R-bilinear now follows from (15.1) together with this action: it is additive in each variable, $r(m \otimes n) = (rm) \otimes n$ by definition of the action, and $r(m \otimes n) = (mr) \otimes n = m \otimes rn$ by the third relation.

Let φ be R-bilinear. By example 15.1.2(1) it is R-balanced, so theorem 15.1.5 gives a unique group homomorphism ψ with $\varphi = \psi \circ \iota$. It remains to see that ψ is R-linear. Fix $r \in R$. The two maps $x \mapsto \psi(rx)$ and $x \mapsto r\psi(x)$ are group homomorphisms $M \otimes_R N \to L$, and on a simple tensor

$$\psi\big(r(m \otimes n)\big) = \psi\big((rm) \otimes n\big) = \varphi(rm, n) = r\varphi(m, n) = r\psi(m \otimes n)$$

so they agree on all simple tensors and hence are equal by lemma 15.1.4. Thus ψ is R-linear. Uniqueness is inherited: an R-module homomorphism satisfying $\varphi = \psi \circ \iota$ is in particular a group homomorphism satisfying it, and there is only one of those, as we sought to show.

Two computations show what the construction does to familiar modules, and they point in opposite directions. The first says that tensoring against the ring itself changes nothing; the second says that tensoring two nonzero modules can annihilate both.

Example 15.1.13: Tensoring Against the Ring
Let R be a ring and N a left R-module. Then $R \otimes_R N \cong N$ as left R-modules, where R carries its (R, R)-bimodule structure from example 15.1.9(1).

The action map $\mu\colon R \times N \to N$ with $\mu(r, n) = rn$ is R-balanced: additivity in each variable is a module axiom, and $\mu(rs, n) = (rs)n = r(sn) = \mu(r, sn)$. By theorem 15.1.5 it induces a group homomorphism $\bar{\mu}\colon R \otimes_R N \to N$ with $\bar{\mu}(r \otimes n) = rn$. In the other direction let $\nu\colon N \to R \otimes_R N$ be $\nu(n) = 1 \otimes n$, which is additive. Then $\bar{\mu}(\nu(n)) = \bar{\mu}(1 \otimes n) = n$, so $\bar{\mu} \circ \nu = \mathrm{id}_N$. For the other composite, on a simple tensor

$$\nu\big(\bar{\mu}(r \otimes n)\big) = \nu(rn) = 1 \otimes rn = 1 \cdot r \otimes n = r \otimes n$$

the third equality being the third relation of (15.1) read backwards. So $\nu \circ \bar{\mu}$ agrees with the identity on simple tensors and is the identity by lemma 15.1.4. Finally $\bar{\mu}$ is R-linear, since $\bar{\mu}\big(s(r \otimes n)\big) = \bar{\mu}\big((sr) \otimes n\big) = (sr)n = s(rn) = s\,\bar{\mu}(r \otimes n)$ on simple tensors.

Example 15.1.14: Tensor Products of Cyclic Groups

1. $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$. Since $3 \equiv 1 \pmod 2$, one has $\bar{a} = 3\bar{a}$ in $\mathbb{Z}/2\mathbb{Z}$ for every a, and therefore

$$\bar{a} \otimes \bar{b} = (3\bar{a}) \otimes \bar{b} = \bar{a} \otimes (3\bar{b}) = 0$$

using $3\bar{b} = \bar{0}$ in $\mathbb{Z}/3\mathbb{Z}$. So every simple tensor vanishes, and by lemma 15.1.4(2) the whole group is zero.

2. $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$. The same manipulation is unavailable here, and in fact $\bar{1} \otimes \bar{1} \neq 0$. To see it, the multiplication map $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ sending $(\bar{a}, \bar{b})$ to $\overline{ab}$ is $\mathbb{Z}$-balanced, so it induces Φ with $\Phi(\bar{1} \otimes \bar{1}) = \bar{1} \neq 0$; a homomorphism sends 0 to 0, so $\bar{1} \otimes \bar{1} \neq 0$. Conversely every simple tensor is $\bar{a} \otimes \bar{b}$ with $a, b \in \{0, 1\}$, hence is 0 or $\bar{1} \otimes \bar{1}$, so by lemma 15.1.4(2) the group $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is cyclic, generated by $t = \bar{1} \otimes \bar{1}$. That generator satisfies $2t = (2\bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0$, so it has order dividing 2, and it is nonzero, so its order is exactly 2. Hence $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, with Φ the isomorphism.

The contrast between the two computations is the first real content of the construction. Both $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are nonzero, and their tensor product is not; so $\otimes_R$ is not a construction that remembers its inputs. The mechanism is visible in the computation: the relation $mr \otimes n = m \otimes rn$ lets a scalar be moved from a factor where it acts invertibly to one where it acts as zero. section 15.3 turns this from an obstruction into a tool, and the systematic account of which sequences of modules survive tensoring is the subject of section 17.5.

Example 15.1.15: Tensors as Multilinear Maps

Outside algebra, and particularly in physics and in analysis, the word "tensor" usually names a multilinear map rather than an element of a tensor product. In that convention a k-tensor on a vector space V over F *is* a multilinear map $T\colon V^k \to F$, and the tensor product of a k-tensor T and an ℓ-tensor S is defined directly by the formula

$$(T \otimes S)(x_1, \dots, x_k, y_1, \dots, y_\ell) = T(x_1, \dots, x_k) \cdot S(y_1, \dots, y_\ell)$$

which is again multilinear, now in $k + \ell$ arguments.

The two usages agree when V is finite dimensional. A multilinear map $V^k \to F$ is the same thing as a linear functional on $V^{\otimes k}$, which is the universal property of theorem 15.2.11 below, and for finite-dimensional V the space of such functionals is identified with $(V^*)^{\otimes k}$ by the basis computation of corollary 15.2.6; under that identification the displayed formula is the image of $T \otimes S$ in the sense of definition 15.1.3. The reason for the analyst's convention is that a multilinear map has a domain and a target, so conditions like continuity, differentiability, or being C^r can be imposed on it; an element of an abstract quotient group cannot carry such conditions until it is realized as a function. This is the form in which tensors appear in *Calculus on Manifolds* [6].

Exercise 15.1.1

1. Let R be a ring and M a right R-module, with R carrying its (R, R)-bimodule structure. Show that $M \otimes_R R \cong M$, and identify the isomorphism on simple tensors.

2. Let $m, n \geq 1$ and let $d = \gcd(m, n)$. Show that

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$$

(Suggestion: since $\mathbb{Z}$ is a principal ideal domain, $(d) = (m) + (n)$ by definition 10.1.24, so $d = um + vn$ for some $u, v \in \mathbb{Z}$; use this to show first that d annihilates the left-hand side.)

3. Let R be a ring, M a right R-module and N a left R-module, and suppose there is an $r \in R$

with $Mr = 0$ and $rN = N$. Show that $M \otimes_R N = 0$. Deduce that $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$ for every finite abelian group A, and that $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$.

4. Let R be commutative, let M be generated as an R-module by a set X and let N be generated by a set Y. Show that $M \otimes_R N$ is generated as an R-module by the simple tensors $x \otimes y$ with $x \in X$ and $y \in Y$. Give an example showing that these need not be independent.

5. Let M be a right R-module, N a left R-module and L an abelian group. Show that $\varphi \mapsto \psi$ is a bijection from the set of R-balanced maps $M \times N \to L$ onto $\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, L)$, and that it commutes with composition by a homomorphism $L \to L'$.

6. Let R be commutative, let A, B, M be R-modules and let $f \colon A \to M$ and $g \colon B \to M$ be R-linear. Show that $(a, b) \mapsto f(a) + g(b)$ is an R-module homomorphism $A \oplus B \to M$, and exhibit f and g for which it is not R-bilinear. Conclude that $A \oplus B$ and $A \otimes_R B$ solve different universal problems.

15.2 Properties of the Tensor Product

With one exception, named where it occurs, no proof below returns to the free abelian group of section 15.1. Everything is derived from theorem 15.1.5 and lemma 15.1.4, by the two moves those results give: to produce a map out of a tensor product, write down a balanced map and invoke the universal property; to prove two maps out of a tensor product equal, check them on simple tensors. What follows records how $\otimes_R$ interacts with each construction the previous chapters gave, in the order in which the proofs depend on one another: homomorphisms, direct sums, free modules, and then the two structural isomorphisms, commutativity and associativity, that make an unbracketed $M_1 \otimes \cdots \otimes M_n$ meaningful. The section closes with the ring structure carried by a tensor product of algebras.

proposition 15.1.10 produced a left module structure from a second action on the left-hand factor. A second action on the right-hand factor does the mirror-image job, and the two are independent, so both can be imposed at once. The right action is what theorem 15.2.8 needs, since there the middle factor of a triple product is acted on from both sides.

Proposition 15.2.1: Two-Sided Module Structure on a Tensor Product

Let R, S, T be rings, let M be an (S, R)-bimodule and let N be an (R, T)-bimodule. Then $M \otimes_R N$ is an (S, T)-bimodule, with actions determined on simple tensors by

$$s \cdot (m \otimes n) = (sm) \otimes n \qquad \text{and} \qquad (m \otimes n) \cdot t = m \otimes (nt)$$

***Proof*:**
The left S-module structure is proposition 15.1.10, which uses only that M is an (S, R)-bimodule and N a left R-module.

For the right T-action, fix $t \in T$ and consider $\rho_t \colon M \times N \to M \otimes_R N$ given by $\rho_t(m, n) = m \otimes (nt)$. It is R-balanced: additivity in m is additivity of $\otimes$ in its first argument; additivity in n follows because $n \mapsto nt$ is additive and $\otimes$ is additive in its second argument; and for the third axiom the (R,T)-bimodule condition on N gives $r(nt) = (rn)t$, so

$$\rho_t(mr, n) = (mr) \otimes (nt) = m \otimes r(nt) = m \otimes (rn)t = \rho_t(m, rn)$$

the middle equality being the third relation of (15.1). By theorem 15.1.5 there is a unique group endomorphism P_t of $M \otimes_R N$ with $P_t(m \otimes n) = m \otimes nt$; set $x \cdot t := P_t(x)$, which is additive in x because P_t is a homomorphism.

The remaining right-module axioms are equalities of group homomorphisms, so by lemma 15.1.4(3) each may be checked on simple tensors:

$$\begin{aligned}
P_{t+t'}(m \otimes n) &= m \otimes n(t+t') = m \otimes (nt + nt') = m \otimes nt + m \otimes nt' = (P_t + P_{t'})(m \otimes n)\\
P_{tt'}(m \otimes n) &= m \otimes n(tt') = m \otimes (nt)t' = P_{t'}(m \otimes nt) = (P_{t'} \circ P_t)(m \otimes n)\\
P_1(m \otimes n) &= m \otimes n1 = m \otimes n
\end{aligned}$$

The second line gives $x(tt') = (xt)t'$, which is the associativity axiom for a *right* action, the order of composition being reversed exactly as it should be. Finally the two actions commute: $x \mapsto s(xt)$ and $x \mapsto (sx)t$ are group homomorphisms, and on a simple tensor both give $(sm) \otimes (nt)$, so they agree by lemma 15.1.4(3), completing the proof.

Homomorphisms of the two factors induce a homomorphism of the tensor product, and the assignment respects composition. This is what makes $\otimes_R$ usable inside diagrams rather than only on individual modules.

Theorem 15.2.2: The Tensor Product of Two Homomorphisms

Let $\varphi\colon M \to M'$ be a homomorphism of right R-modules and let $\psi\colon N \to N'$ be a homomorphism of left R-modules.

1. There is a unique group homomorphism $\varphi \otimes \psi\colon M \otimes_R N \to M' \otimes_R N'$ with

   $$(\varphi \otimes \psi)(m \otimes n) = \varphi(m) \otimes \psi(n)$$

   for all $m \in M$ and $n \in N$.

2. If M and M' are (S,R)-bimodules and φ is also a homomorphism of left S-modules, then $\varphi \otimes \psi$ is a homomorphism of left S-modules. In particular, if R is commutative and all four modules carry their standard structures, then $\varphi \otimes \psi$ is R-linear.

3. $\mathrm{id}_M \otimes \mathrm{id}_N = \mathrm{id}_{M \otimes_R N}$, and if $\lambda\colon M' \to M''$ and $\mu\colon N' \to N''$ are further homomorphisms of right and left R-modules respectively, then

   $$(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$$

***Proof*:**

1. The map $M \times N \to M' \otimes_R N'$ sending (m,n) to $\varphi(m) \otimes \psi(n)$ is R-balanced. Additivity in each variable follows from additivity of φ and ψ together with additivity of $\otimes$ in each argument. For the third axiom, φ is a homomorphism of right R-modules and ψ of left

R-modules, so

$$\varphi(mr) \otimes \psi(n) = \big(\varphi(m)r\big) \otimes \psi(n) = \varphi(m) \otimes r\psi(n) = \varphi(m) \otimes \psi(rn)$$

the middle equality being the third relation of (15.1) in $M' \otimes_R N'$. theorem 15.1.5 now gives the homomorphism, and its uniqueness.

2. Both $x \mapsto (\varphi\otimes\psi)(sx)$ and $x \mapsto s(\varphi\otimes\psi)(x)$ are group homomorphisms $M\otimes_R N \to M'\otimes_R N'$, so by lemma 15.1.4(3) it suffices to compare them on a simple tensor, where

$$(\varphi\otimes\psi)\big(s(m\otimes n)\big) = (\varphi\otimes\psi)\big((sm)\otimes n\big) = \varphi(sm)\otimes\psi(n) = \big(s\varphi(m)\big)\otimes\psi(n) = s\big(\varphi(m)\otimes\psi(n)\big)$$

using S-linearity of φ in the third step and proposition 15.1.10 in the fourth. When R is commutative and the structures are standard, an R-module homomorphism is S-linear for $S = R$, so the statement applies with $S = R$.

3. All three claims are equalities of group homomorphisms out of $M \otimes_R N$, so lemma 15.1.4(3) reduces each to a check on simple tensors. There $(\mathrm{id} \otimes \mathrm{id})(m \otimes n) = m \otimes n$, and

$$(\lambda \otimes \mu)\big((\varphi \otimes \psi)(m \otimes n)\big) = (\lambda \otimes \mu)\big(\varphi(m) \otimes \psi(n)\big) = \lambda(\varphi(m)) \otimes \mu(\psi(n))$$

which is the value of $(\lambda \circ \varphi) \otimes (\mu \circ \psi)$ on $m \otimes n$, completing the proof.

Parts (1) and (3) say precisely that fixing a left R-module N makes $- \otimes_R N$ a functor from right R-modules to abelian groups, and symmetrically for $M \otimes_R -$. The question of how this functor treats exact sequences is left open here and settled in section 17.5; the computation $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$ of example 15.1.14 is already a warning that it does not treat them well.

The next two results compute a tensor product one summand at a time.

Theorem 15.2.3: Tensor Products of Direct Sums

Let M, M' be right R-modules and let N, N' be left R-modules. There are unique group isomorphisms

$$(M \oplus M') \otimes_R N \cong (M \otimes_R N) \oplus (M' \otimes_R N)$$
$$M \otimes_R (N \oplus N') \cong (M \otimes_R N) \oplus (M \otimes_R N')$$

determined on simple tensors by $(m, m')\otimes n \mapsto (m\otimes n, m'\otimes n)$ and $m\otimes(n, n') \mapsto (m\otimes n, m\otimes n')$ respectively. If M and M' are (S, R)-bimodules, the first is an isomorphism of left S-modules; if R is commutative and all modules carry their standard structures, both are isomorphisms of R-modules.

***Proof*:**
Consider $\alpha\colon (M \oplus M') \times N \to (M \otimes_R N) \oplus (M' \otimes_R N)$ given by $\alpha\big((m, m'), n\big) = (m \otimes n, m' \otimes n)$. It is R-balanced, each axiom holding in both coordinates at once; for the third,

$$\alpha\big((m, m')r, n\big) = \big(mr \otimes n,\ m'r \otimes n\big) = \big(m \otimes rn,\ m' \otimes rn\big) = \alpha\big((m, m'), rn\big)$$

since the action on $M \oplus M'$ is componentwise. By theorem 15.1.5 there is a group homomorphism $\bar{\alpha}$ on $(M \oplus M') \otimes_R N$ with the stated values.

For the inverse, the maps $\beta_1\colon M \times N \to (M \oplus M') \otimes_R N$ with $\beta_1(m,n) = (m,0) \otimes n$ and $\beta_2\colon M' \times N \to (M \oplus M') \otimes_R N$ with $\beta_2(m',n) = (0,m') \otimes n$ are R-balanced, because $m \mapsto (m,0)$ and $m' \mapsto (0,m')$ are homomorphisms of right R-modules; so they induce group homomorphisms $\bar{\beta}_1$ and $\bar{\beta}_2$. Define

$$\beta\colon (M \otimes_R N) \oplus (M' \otimes_R N) \to (M \oplus M') \otimes_R N, \qquad \beta(x,y) = \bar{\beta}_1(x) + \bar{\beta}_2(y)$$

which is a group homomorphism.

On a simple tensor,

$$\beta\big(\bar{\alpha}((m,m') \otimes n)\big) = (m,0) \otimes n + (0,m') \otimes n = \big((m,0) + (0,m')\big) \otimes n = (m,m') \otimes n$$

so $\beta \circ \bar{\alpha} = \mathrm{id}$ by lemma 15.1.4(3). In the other direction, lemma 15.1.4(2) applied in each summand shows that $(M \otimes_R N) \oplus (M' \otimes_R N)$ is generated as a group by the elements $(x,0)$ and $(0,y)$ with x and y simple tensors, so it is enough to evaluate $\bar{\alpha} \circ \beta$ on those. On $(m \otimes n, 0)$,

$$\bar{\alpha}\big(\bar{\beta}_1(m \otimes n)\big) = \bar{\alpha}\big((m,0) \otimes n\big) = (m \otimes n,\ 0 \otimes n) = (m \otimes n,\ 0)$$

by lemma 15.1.4(1), and the computation on $(0, m' \otimes n)$ is the same. Hence $\bar{\alpha} \circ \beta = \mathrm{id}$ and $\bar{\alpha}$ is an isomorphism.

When M, M' are (S,R)-bimodules, so is $M \oplus M'$, and S-linearity of $\bar{\alpha}$ is again an equality of group homomorphisms checked on simple tensors, where both sides give $\big((sm) \otimes n, (sm') \otimes n\big)$. Uniqueness in all cases is lemma 15.1.4(3).

The second isomorphism is the mirror image of the first: interchange the two factors throughout, replacing each appeal to additivity in the first argument of $\otimes$ by additivity in the second, and proposition 15.1.10 by proposition 15.2.1. No new step enters, completing the proof.

Theorem 15.2.4: Arbitrary Direct Sums Commute with Tensor Products

Let $\{M_i\}_{i \in I}$ be a family of right R-modules and let N be a left R-module. Then there is a unique group isomorphism

$$\left(\bigoplus_{i \in I} M_i\right) \otimes_R N \cong \bigoplus_{i \in I} (M_i \otimes_R N)$$

sending $(m_i)_i \otimes n$ to $(m_i \otimes n)_i$. If each M_i is an (S,R)-bimodule it is an isomorphism of left S-modules, and if R is commutative it is an isomorphism of R-modules.

***Proof*:**
Write $M = \bigoplus_{i \in I} M_i$, whose elements are the families $(m_i)_i$ with $m_i = 0$ for all but finitely many i. The assignment $\alpha\big((m_i)_i, n\big) = (m_i \otimes n)_i$ lands in the direct sum rather than the product, since $m_i = 0$ forces $m_i \otimes n = 0$ by lemma 15.1.4(1), so all but finitely many entries vanish. It is R-balanced for the same componentwise reason as in theorem 15.2.3, and so induces $\bar{\alpha}$ on $M \otimes_R N$.

Let $\kappa_j\colon M_j \to M$ be the inclusion of the j-th summand, a homomorphism of right R-modules. Each $\beta_j\colon M_j \times N \to M \otimes_R N$ given by $\beta_j(m,n) = \kappa_j(m) \otimes n$ is therefore R-balanced, inducing

$\bar{\beta}_j$. Define

$$\beta\colon \bigoplus_{i\in I}(M_i \otimes_R N) \to M \otimes_R N, \qquad \beta\big((x_i)_i\big) = \sum_{i\in I} \bar{\beta}_i(x_i)$$

a finite sum because $(x_i)_i$ has finite support, and a group homomorphism because each $\bar{\beta}_i$ is. On a simple tensor, writing $(m_i)_i = \sum_i \kappa_i(m_i)$ as the finite sum it is,

$$\beta\big(\bar{\alpha}((m_i)_i \otimes n)\big) = \sum_i \kappa_i(m_i) \otimes n = \left(\sum_i \kappa_i(m_i)\right) \otimes n = (m_i)_i \otimes n$$

using additivity of $\otimes$ in its first argument, so $\beta \circ \bar{\alpha} = \mathrm{id}$. Conversely $\bigoplus_i (M_i \otimes_R N)$ is generated as a group by the elements $\kappa'_j(x)$ with x a simple tensor of $M_j \otimes_R N$, where κ'_j is the inclusion of the j-th summand, and

$$\bar{\alpha}\big(\beta(\kappa'_j(m \otimes n))\big) = \bar{\alpha}\big(\kappa_j(m) \otimes n\big) = \kappa'_j(m \otimes n)$$

since every entry of $\kappa_j(m)$ other than the j-th is zero and therefore contributes $0 \otimes n = 0$. Hence $\bar{\alpha} \circ \beta = \mathrm{id}$. The bimodule claims and uniqueness are as in theorem 15.2.3, completing the proof.

Theorem 15.2.4 decomposes a direct sum in the *left* argument, and theorem 15.2.3 handles two summands on either side. The remaining case, an arbitrary direct sum in the right argument, is what a free presentation of a module needs, so it is recorded separately.

Corollary 15.2.5: Arbitrary Direct Sums in the Right Argument

Let M be a right R-module and $\{N_i\}_{i\in I}$ a family of left R-modules. Then there is a unique group isomorphism

$$M \otimes_R \left(\bigoplus_{i\in I} N_i\right) \cong \bigoplus_{i\in I} (M \otimes_R N_i)$$

sending $m \otimes (n_i)_i$ to $(m \otimes n_i)_i$. If M is an (S,R)-bimodule it is an isomorphism of left S-modules.

***Proof*:**
Write $N = \bigoplus_{i\in I} N_i$. The assignment $\alpha\big(m, (n_i)_i\big) = (m \otimes n_i)_i$ lands in the direct sum rather than the product, since $n_i = 0$ for all but finitely many i and $m \otimes 0 = 0$ by lemma 15.1.4(1). It is additive in each variable componentwise, and R-balanced because

$$\alpha\big(mr, (n_i)_i\big) = (mr \otimes n_i)_i = (m \otimes rn_i)_i = \alpha\big(m, r \cdot (n_i)_i\big)$$

so it induces $\bar{\alpha}\colon M \otimes_R N \to \bigoplus_i (M \otimes_R N_i)$.

For the inverse, let $\kappa_j\colon N_j \to N$ be the inclusion of the j-th summand, a map of left R-modules. Each $\beta_j\colon M \times N_j \to M \otimes_R N$ given by $\beta_j(m,n) = m \otimes \kappa_j(n)$ is R-balanced, hence induces $\bar{\beta}_j$ on $M \otimes_R N_j$, and these assemble into a map β on the direct sum.

The composites are the identity on simple tensors, which suffices by lemma 15.1.4. Writing $(n_i)_i = \sum_{j\in F} \kappa_j(n_j)$ over the finite set F of nonzero entries,

$$\beta\big(\bar{\alpha}(m \otimes (n_i)_i)\big) = \sum_{j\in F} m \otimes \kappa_j(n_j) = m \otimes \sum_{j\in F} \kappa_j(n_j) = m \otimes (n_i)_i$$

and conversely $\bar{\alpha}\big(\beta(\kappa'_j(m \otimes n))\big) = \bar{\alpha}\big(m \otimes \kappa_j(n)\big) = \kappa'_j(m \otimes n)$, every entry of $\kappa_j(n)$ other than the j-th being zero and so contributing $m \otimes 0 = 0$. The S-linearity claim holds because the left S-action on both sides is $s \cdot (m \otimes n) = (sm) \otimes n$ by proposition 15.2.1, which $\bar{\alpha}$ respects, completing the proof.

Applying theorem 15.2.4 to a free module, where every summand is a copy of the ring, computes the tensor product of two free modules completely. This is the result that makes tensor products calculable in linear algebra: it says that a basis of M and a basis of N determine a basis of $M \otimes_R N$, indexed by pairs.

Corollary 15.2.6: Tensor Products of Free Modules

Let R be a commutative ring, let M be a free R-module with basis $\{m_i\}_{i \in I}$ and let N be a free R-module with basis $\{n_j\}_{j \in J}$. Then $M \otimes_R N$ is a free R-module with basis

$$\{m_i \otimes n_j\}_{(i,j) \in I \times J}$$

In particular $R^s \otimes_R R^t \cong R^{st}$, and if V, W are finite-dimensional vector spaces over a field F then

$$\dim_F(V \otimes_F W) = \dim_F(V) \cdot \dim_F(W)$$

***Proof*:**
Since $\{m_i\}$ is a basis, $M = \bigoplus_{i \in I} Rm_i$ with each $Rm_i \cong R$ as an R-module, the isomorphism carrying m_i to 1 (theorem 12.3.11). Applying theorem 15.2.4, and then example 15.1.13 to each summand after transporting it along $Rm_i \cong R$ by theorem 15.2.2, gives R-module isomorphisms

$$M \otimes_R N \cong \bigoplus_{i \in I} \big(Rm_i \otimes_R N\big) \cong \bigoplus_{i \in I} N$$

under which $m_i \otimes n$ corresponds to the family whose i-th entry is n and whose other entries are 0. Writing $N = \bigoplus_{j \in J} Rn_j \cong \bigoplus_{j \in J} R$ in the same way and substituting,

$$M \otimes_R N \cong \bigoplus_{i \in I} \bigoplus_{j \in J} R \cong \bigoplus_{(i,j) \in I \times J} R$$

and following $m_i \otimes n_j$ through the three isomorphisms shows that it corresponds to the element of $\bigoplus_{I \times J} R$ with 1 in position (i, j) and 0 elsewhere. Those elements form the standard basis of $\bigoplus_{I \times J} R$, and an isomorphism of R-modules carries a basis to a basis, so $\{m_i \otimes n_j\}$ is a basis of $M \otimes_R N$.

The two special cases follow by counting: $I \times J$ has st elements when $|I| = s$ and $|J| = t$, and the dimension of a vector space is the cardinality of any basis, as we sought to show.

This closes the identification promised in example 15.1.15: for finite-dimensional V, a basis of V^* and one of W^* produce a basis of $V^* \otimes_F W^*$, matching the basis of the space of bilinear forms on $V \times W$ under the correspondence of theorem 15.1.12.

Two structural isomorphisms remain. Both are needed before an expression such as $M_1 \otimes M_2 \otimes M_3$ can be written without brackets.

Proposition 15.2.7: Commutativity of the Tensor Product

Let R be a commutative ring and let M, N be R-modules with their standard structures. Then there is a unique R-module isomorphism

$$\tau\colon M \otimes_R N \to N \otimes_R M, \qquad \tau(m \otimes n) = n \otimes m$$

***Proof*:**
The map $M \times N \to N \otimes_R M$ sending (m, n) to $n \otimes m$ is R-bilinear: it is additive in each variable, and for $r \in R$,

$$n \otimes (rm) = (nr) \otimes m = r(n \otimes m)$$

using the third relation of (15.1) and the description of the action in theorem 15.1.12, with the same computation in the other variable. So theorem 15.1.12 gives a unique R-module homomorphism τ with $\tau(m \otimes n) = n \otimes m$. The same construction with the roles of M and N exchanged gives $\tau'\colon N \otimes_R M \to M \otimes_R N$ with $\tau'(n \otimes m) = m \otimes n$, and both composites fix every simple tensor, so both are the identity by lemma 15.1.4(3), as we sought to show.

Theorem 15.2.8: Associativity of the Tensor Product

Let M be a right R-module, let N be an (R, T)-bimodule and let L be a left T-module. Then there is a unique isomorphism of abelian groups

$$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$$

carrying $(m \otimes n) \otimes l$ to $m \otimes (n \otimes l)$. If M is an (S, R)-bimodule it is an isomorphism of left S-modules; if $R = T$ is commutative and all three modules carry their standard structures, it is an isomorphism of R-modules.

***Proof*:**
By proposition 15.2.1, $M \otimes_R N$ is a right T-module with $(m \otimes n)t = m \otimes (nt)$, and $N \otimes_T L$ is a left R-module with $r(n \otimes l) = (rn) \otimes l$. Both tensor products in the statement are therefore defined.

Step 1: a homomorphism for each l. Fix $l \in L$. The map $M \times N \to M \otimes_R (N \otimes_T L)$ sending (m, n) to $m \otimes (n \otimes l)$ is R-balanced. It is additive in m; it is additive in n because $n \mapsto n \otimes l$ is additive and $\otimes$ is additive in its second argument; and

$$(mr) \otimes (n \otimes l) = m \otimes r(n \otimes l) = m \otimes \big((rn) \otimes l\big)$$

which is the third axiom. By theorem 15.1.5 there is a unique group homomorphism $\theta_l\colon M \otimes_R N \to M \otimes_R (N \otimes_T L)$ with $\theta_l(m \otimes n) = m \otimes (n \otimes l)$.

Step 2: assembling over l. The map $(M \otimes_R N) \times L \to M \otimes_R (N \otimes_T L)$ sending (x, l) to $\theta_l(x)$ is T-balanced. It is additive in x because θ_l is a homomorphism. For additivity in l, both $\theta_{l+l'}$ and $\theta_l + \theta_{l'}$ are group homomorphisms out of $M \otimes_R N$, and on a simple tensor both give $m \otimes (n \otimes l) + m \otimes (n \otimes l')$, so they are equal by lemma 15.1.4(3). For the third axiom, $x \mapsto \theta_l(xt)$ and $x \mapsto \theta_{tl}(x)$ are group homomorphisms, and on a simple tensor

$$\theta_l\big((m \otimes n)t\big) = \theta_l(m \otimes nt) = m \otimes \big((nt) \otimes l\big) = m \otimes \big(n \otimes (tl)\big) = \theta_{tl}(m \otimes n)$$

the third equality being the third relation of (15.1) inside $N \otimes_T L$. So theorem 15.1.5 gives a group homomorphism Θ on $(M \otimes_R N) \otimes_T L$ with $\Theta\big((m \otimes n) \otimes l\big) = m \otimes (n \otimes l)$.

Step 3: the inverse. The same two steps run with the outer factors exchanged. Fixing $m \in M$, the map $N \times L \to (M \otimes_R N) \otimes_T L$ sending (n, l) to $(m \otimes n) \otimes l$ is T-balanced, giving $\eta_m \colon N \otimes_T L \to (M \otimes_R N) \otimes_T L$ with $\eta_m(n \otimes l) = (m \otimes n) \otimes l$; and $(m, y) \mapsto \eta_m(y)$ is then R-balanced by the same three checks. This yields a group homomorphism Ξ on $M \otimes_R (N \otimes_T L)$ with $\Xi\big(m \otimes (n \otimes l)\big) = (m \otimes n) \otimes l$.

Step 4: the composites. By lemma 15.1.4(2) every element of $(M \otimes_R N) \otimes_T L$ is a finite sum of terms $x \otimes l$ with $x \in M \otimes_R N$, and each such x is itself a finite sum of simple tensors, so by additivity of $\otimes$ in its first argument every element is a finite sum of terms $(m \otimes n) \otimes l$. Since $\Xi \circ \Theta$ is a group homomorphism fixing each such term, it is the identity; the same argument on the other side gives $\Theta \circ \Xi = \mathrm{id}$.

The S-linearity of Θ, when M is an (S, R)-bimodule, is once more an equality of group homomorphisms verified on the generating terms $(m \otimes n) \otimes l$, where both sides give $(sm) \otimes (n \otimes l)$; and uniqueness holds because a homomorphism is determined by its values on those terms, completing the proof.

15.2.9 Foreshadowing: Monoidal Structure on R-Mod For a commutative ring R, the three isomorphisms

$$(M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L), \qquad R \otimes_R M \cong M, \qquad M \otimes_R N \cong N \otimes_R M$$

of theorem 15.2.8, example 15.1.13 and proposition 15.2.7 are not three unrelated facts. They are the associator, the unit isomorphism and the braiding of a structure called a *symmetric monoidal category*, carried by R-Mod with $\otimes_R$ as its product and R as its unit. In that language the R-algebras of definition 12.1.13 are exactly the monoid objects of R-Mod, and the compatibility conditions the three isomorphisms satisfy with one another are the subject of a coherence theorem. The structure is developed in *Category Theory* [7]; nothing in this chapter depends on it.

Associativity licenses the notation $M_1 \otimes_R \cdots \otimes_R M_n$: define it to mean the left-bracketed product $(\cdots(M_1 \otimes_R M_2) \otimes_R \cdots) \otimes_R M_n$, and write $m_1 \otimes \cdots \otimes m_n$ for the corresponding element. Repeated application of theorem 15.2.8 produces an isomorphism from any other bracketing onto this one matching the corresponding simple tensors, which is what makes the bracket-free notation harmless. With that in place the universal property extends from two factors to n.

Definition 15.2.10: Multilinear Maps

Let R be a commutative ring and let $M_1, \ldots, M_n$ and L be R-modules with their standard structures. A map $\varphi \colon M_1 \times \cdots \times M_n \to L$ is *R-multilinear*, or *n-linear*, if for each index i and each choice of fixed elements $m_k \in M_k$ for $k \neq i$, the map

$$M_i \to L, \qquad m \mapsto \varphi(m_1, \ldots, m_{i-1}, m, m_{i+1}, \ldots, m_n)$$

is an R-module homomorphism. For $n = 2$ and $n = 3$ one says *bilinear* and *trilinear* rather than 2-linear and 3-linear.

This is definition 13.3.15 with the target allowed to be any R-module rather than the base field, and with any number of arguments; for $n = 2$ it is definition 15.1.11.

Theorem 15.2.11: Universal Property for Multilinear Maps

Let R be a commutative ring and let $M_1, \ldots, M_n$ and L be R-modules. For every R-multilinear map $\varphi\colon M_1 \times \cdots \times M_n \to L$ there is a unique R-module homomorphism

$$\psi\colon M_1 \otimes_R \cdots \otimes_R M_n \to L, \qquad \psi(m_1 \otimes \cdots \otimes m_n) = \varphi(m_1, \ldots, m_n)$$

***Proof*:**
Induction on n. The case $n = 1$ is the identity and $n = 2$ is theorem 15.1.12, so suppose $n \geq 3$ and the statement holds for $n - 1$. Write $P = M_1 \otimes_R \cdots \otimes_R M_{n-1}$, so that by definition of the left-bracketed product the n-fold tensor product is $P \otimes_R M_n$.

First note that P is generated as an R-module by the simple tensors $m_1 \otimes \cdots \otimes m_{n-1}$: this holds for $n - 1 = 1$ trivially, and the inductive step is lemma 15.1.4(2) applied to $P' \otimes_R M_{n-1}$ together with additivity of $\otimes$ in its first argument.

Fix $m_n \in M_n$. The map $(m_1, \ldots, m_{n-1}) \mapsto \varphi(m_1, \ldots, m_n)$ is $(n-1)$-linear, since fixing all but one of its arguments fixes all but one argument of φ. By the inductive hypothesis there is a unique R-module homomorphism $\psi_{m_n}\colon P \to L$ with $\psi_{m_n}(m_1 \otimes \cdots \otimes m_{n-1}) = \varphi(m_1, \ldots, m_n)$.

The map $P \times M_n \to L$ sending (x, m_n) to $\psi_{m_n}(x)$ is R-bilinear. Linearity in x holds because ψ_{m_n} is an R-module homomorphism. For linearity in m_n, the maps $\psi_{m_n + m_n'}$ and $\psi_{m_n} + \psi_{m_n'}$ are R-module homomorphisms out of P agreeing on every $m_1 \otimes \cdots \otimes m_{n-1}$, by linearity of φ in its last argument, and those generate P; so the two are equal. The same argument gives $\psi_{rm_n} = r\psi_{m_n}$.

theorem 15.1.12 therefore gives a unique R-module homomorphism ψ on $P \otimes_R M_n$ with $\psi(x \otimes m_n) = \psi_{m_n}(x)$, and evaluating at $x = m_1 \otimes \cdots \otimes m_{n-1}$ gives the stated formula. Uniqueness follows because the simple tensors $m_1 \otimes \cdots \otimes m_n$ generate $M_1 \otimes_R \cdots \otimes_R M_n$, by the generation statement above and lemma 15.1.4(2), completing the proof.

One fact about $\otimes_R$ cannot be obtained from the universal property, because it is a statement about how an equation in $M \otimes_R N$ arises rather than about maps out of it. Lemma 15.1.4(2) says every element is a finite sum of simple tensors; the lemma below says that when such a sum vanishes, the vanishing already occurs inside a finitely generated piece of M. Its proof is the one place in this section that returns to the construction in definition 15.1.3.

Lemma 15.2.12: Vanishing Is Witnessed Finitely

Let R be a ring, M a right R-module and N a left R-module, and suppose

$$\sum_{i=1}^{k} m_i \otimes n_i = 0 \qquad \text{in } M \otimes_R N$$

Then there is a *finitely generated* submodule $M_0 \subseteq M$ containing $m_1, \ldots, m_k$ such that already

$$\sum_{i=1}^{k} m_i \otimes n_i = 0 \qquad \text{in } M_0 \otimes_R N$$

The same holds with the roles of M and N exchanged.

***Proof*:**
By definition 15.1.3 the group $M \otimes_R N$ is $F^{\mathbb{Z}}(M \times N)/K$, so the hypothesis says that the element $\sum_i (m_i, n_i)$ of $F^{\mathbb{Z}}(M \times N)$ lies in K. Since K is generated by the three displayed forms, there are finitely many generators $g_1, \ldots, g_t$ of K and integers $c_1, \ldots, c_t$ with

$$\sum_{i=1}^{k} (m_i, n_i) = \sum_{j=1}^{t} c_j \, g_j$$

Each g_j mentions only finitely many elements of M: a generator of the first form mentions m, m' and $m + m'$, one of the second mentions a single m, and one of the third mentions mr and m. Let $S \subseteq M$ be the finite set consisting of $m_1, \ldots, m_k$ together with every element of M mentioned by some g_j, and let M_0 be the submodule of M generated by S. It is finitely generated and contains each m_i.

Now form $F^{\mathbb{Z}}(M_0 \times N)$ and the subgroup K_0 generated by the three forms of definition 15.1.3 taken over M_0, N and R, so that $M_0 \otimes_R N = F^{\mathbb{Z}}(M_0 \times N)/K_0$. Every g_j lies in $F^{\mathbb{Z}}(M_0 \times N)$, since all of its M-coordinates belong to $S \subseteq M_0$; and g_j is a generator of K_0, because M_0 is a submodule and so is closed under the addition and the R-action that the three forms use. Hence the displayed identity holds in $F^{\mathbb{Z}}(M_0 \times N)$ and exhibits $\sum_i (m_i, n_i)$ as an element of K_0, which is to say $\sum_i m_i \otimes n_i = 0$ in $M_0 \otimes_R N$.

The statement with M and N exchanged is the same argument applied to the second coordinate.

The lemma does not say that $M_0 \otimes_R N \to M \otimes_R N$ is injective, and in general it is not. What it says is that a *relation* that holds downstairs already holds upstairs, which is what makes it possible to reduce a question about an arbitrary module to the finitely generated case. Theorem 17.5.7 is the use it was recorded for.

A tensor product of quotients of a commutative ring is again a quotient, and the ideal one quotients by is the sum. This is the computation behind most small examples.

Proposition 15.2.13: Tensor Product of Quotient Rings

Let R be a commutative ring and let I, J be ideals of R. Then there is an R-module isomorphism
$$R/I \otimes_R R/J \cong R/(I+J), \qquad \overline{a} \otimes \overline{b} \mapsto \overline{ab}$$

***Proof*:**
The map $R/I \times R/J \to R/(I+J)$ sending $(\overline{a}, \overline{b})$ to $\overline{ab}$ is well defined: if $a - a' \in I$ and $b - b' \in J$ then
$$ab - a'b' = (a-a')b + a'(b-b') \in I + J$$
It is R-bilinear, so theorem 15.1.12 induces an R-module homomorphism Φ with $\Phi(\overline{a} \otimes \overline{b}) = \overline{ab}$.

In the other direction set $\Psi(\overline{r}) = \overline{r} \otimes \overline{1}$. Throughout, a bar in the left-hand factor denotes a class in R/I and a bar in the right-hand factor a class in R/J. To see that Ψ is well defined, let $r - r' \in I + J$ and write $r - r' = a + b$ with $a \in I$ and $b \in J$, so that
$$\overline{r - r'} \otimes \overline{1} = \overline{a} \otimes \overline{1} + \overline{b} \otimes \overline{1}$$
The first term vanishes because $\overline{a} = \overline{0}$ already in R/I. For the second, moving the scalar b across by the third relation of (15.1),
$$\overline{b} \otimes \overline{1} = (b \cdot \overline{1}) \otimes \overline{1} = \overline{1} \otimes (b \cdot \overline{1}) = \overline{1} \otimes \overline{b} = 0$$
since $\overline{b} = \overline{0}$ in R/J. So Ψ is well defined, and it is R-linear.

The composite $\Phi \circ \Psi$ sends $\overline{r}$ to $\overline{r}$. For the other composite,
$$\Psi\big(\Phi(\overline{a} \otimes \overline{b})\big) = \overline{ab} \otimes \overline{1} = (\overline{a} \cdot b) \otimes \overline{1} = \overline{a} \otimes b\overline{1} = \overline{a} \otimes \overline{b}$$
so $\Psi \circ \Phi$ fixes every simple tensor and is the identity by lemma 15.1.4(3), as we sought to show.

Taking $R = \mathbb{Z}$, $I = (m)$ and $J = (n)$ recovers $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$, since $(m) + (n) = (\gcd(m,n))$ by definition 10.1.24; that is the second exercise of section 15.1, and both computations of example 15.1.14 are the cases $(m,n) = (2,3)$ and $(m,n) = (2,2)$.

When the two factors are R-algebras rather than bare modules, their multiplications combine into one on the tensor product. This is the construction that will produce the tensor algebra in section 16.3, and it is the source of the coproduct description that closes this section.

Theorem 15.2.14: Tensor Product of R-Algebras

Let R be a commutative ring and let A and B be R-algebras (definition 12.1.13). Then there is a unique R-bilinear multiplication on $A \otimes_R B$ satisfying
$$(a \otimes b)(a' \otimes b') = (aa') \otimes (bb')$$
and it makes $A \otimes_R B$ into an R-algebra with identity $1_A \otimes 1_B$. If A and B are both commutative, so is $A \otimes_R B$.

***Proof*:**
Step 1: the multiplication exists. The map $A \times B \times A \times B \to A \otimes_R B$ sending (a, b, a', b') to $(aa') \otimes (bb')$ is R-multilinear: in each of its four arguments the other three are fixed, and the resulting map is a composite of multiplication in A or in B, which is R-linear in each variable by definition 12.1.13, with $\otimes$, which is R-linear in each argument by theorem 15.1.12. By theorem 15.2.11 it induces an R-module homomorphism

$$\mu'\colon A \otimes_R B \otimes_R A \otimes_R B \to A \otimes_R B, \qquad \mu'(a \otimes b \otimes a' \otimes b') = (aa') \otimes (bb')$$

By theorem 15.2.8 there is an isomorphism $(A \otimes_R B) \otimes_R (A \otimes_R B) \cong A \otimes_R B \otimes_R A \otimes_R B$ matching $(a \otimes b) \otimes (a' \otimes b')$ with $a \otimes b \otimes a' \otimes b'$; composing it with μ' gives an R-module homomorphism μ on $(A \otimes_R B) \otimes_R (A \otimes_R B)$ with $\mu\big((a \otimes b) \otimes (a' \otimes b')\big) = (aa') \otimes (bb')$. Setting $xy := \mu(x \otimes y)$ defines a map $(A \otimes_R B) \times (A \otimes_R B) \to A \otimes_R B$ which is R-bilinear, being the composite of the R-bilinear ι with the R-linear μ, and which satisfies the displayed formula on simple tensors. It is the unique such: two R-bilinear maps agreeing on pairs of simple tensors induce, by theorem 15.1.12, two homomorphisms on $(A \otimes_R B) \otimes_R (A \otimes_R B)$ agreeing on the simple tensors $(a \otimes b) \otimes (a' \otimes b')$, which generate it, so they coincide.

Step 2: the ring axioms. Distributivity is additivity of the multiplication in each variable, which is part of bilinearity. For associativity, both $(x, y, z) \mapsto (xy)z$ and $(x, y, z) \mapsto x(yz)$ are R-trilinear, being built from bilinear maps, so by theorem 15.2.11 each corresponds to a homomorphism on the triple tensor product, and the two agree if and only if they agree on triples of simple tensors. There

$$\big((a\otimes b)(a'\otimes b')\big)(a''\otimes b'') = \big((aa')a''\big)\otimes\big((bb')b''\big) = \big(a(a'a'')\big)\otimes\big(b(b'b'')\big) = (a\otimes b)\big((a'\otimes b')(a''\otimes b'')\big)$$

by associativity in A and in B. For the identity, $x \mapsto (1_A \otimes 1_B)x$ and the identity map are both additive and agree on simple tensors, so they are equal by lemma 15.1.4(3), and likewise on the other side.

Step 3: the R-algebra structure. The R-module structure is the one $A \otimes_R B$ already carries, and the compatibility $r(xy) = (rx)y = x(ry)$ required by definition 12.1.13 is R-bilinearity of the multiplication. Finally, if A and B are commutative then $(x, y) \mapsto xy$ and $(x, y) \mapsto yx$ are R-bilinear and agree on pairs of simple tensors, since $(aa') \otimes (bb') = (a'a) \otimes (b'b)$, so they are equal by the uniqueness argument of Step 1, completing the proof.

Example 15.2.15: Tensoring Two Copies of $\mathbb{C}$ over $\mathbb{R}$
The base ring matters, and comparing $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ with $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ shows how much. The first is $\mathbb{C}$ itself by example 15.1.13. The second has $\mathbb{R}$-dimension 4 by corollary 15.2.6, since $\mathbb{C}$ is free of rank 2 over $\mathbb{R}$, and it is not even a domain:

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$$

as $\mathbb{R}$-algebras. To see this, define $\Phi\colon \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C} \times \mathbb{C}$ on simple tensors by

$$\Phi(z \otimes w) = (zw,\ z\overline{w})$$

This is well defined because $(z, w) \mapsto (zw, z\overline{w})$ is $\mathbb{R}$-bilinear, complex conjugation being $\mathbb{R}$-linear, so theorem 15.1.12 applies. It is a ring homomorphism: on simple tensors

$$\Phi\big((z\otimes w)(z'\otimes w')\big) = \Phi(zz'\otimes ww') = \big(zz'ww',\ zz'\overline{ww'}\big) = \big(zw\cdot z'w',\ z\overline{w}\cdot z'\overline{w'}\big) = \Phi(z\otimes w)\Phi(z'\otimes w')$$

and both $(x, y) \mapsto \Phi(xy)$ and $(x, y) \mapsto \Phi(x)\Phi(y)$ are $\mathbb{R}$-bilinear, so agreeing on simple tensors makes them equal. Also $\Phi(1 \otimes 1) = (1, 1)$.

For surjectivity, compute Φ on three elements:

$$\Phi(1 \otimes 1) = (1, 1), \qquad \Phi(i \otimes 1) = (i, i), \qquad \Phi(i \otimes i) = (i \cdot i,\ i\bar{i}) = (-1, 1)$$

The image of Φ is both an $\mathbb{R}$-subspace and a subring. It contains $(1, 1) - (-1, 1) = (2, 0)$ and $(1, 1) + (-1, 1) = (0, 2)$, hence $(1, 0)$ and $(0, 1)$; and being closed under multiplication it then contains $(1, 0)(i, i) = (i, 0)$ and $(0, 1)(i, i) = (0, i)$. Those four elements form an $\mathbb{R}$-basis of $\mathbb{C} \times \mathbb{C}$, so Φ is surjective. Both sides have $\mathbb{R}$-dimension 4, and a surjective linear map between vector spaces of equal finite dimension is an isomorphism.

So a tensor product of two fields over a common subfield need not be a field, and need not even be an integral domain: $\mathbb{C} \times \mathbb{C}$ has the zero divisors $(1, 0)$ and $(0, 1)$.

The last result of this section reads the tensor product of algebras backwards, as the answer to a universal problem rather than as a construction. Recall that in R-Mod the direct sum $M \oplus N$ is simultaneously the product and the coproduct of M and N; this is a feature of *finite* families only, since for an infinite index set the product $\prod_i M_i$ and the coproduct $\bigoplus_i M_i$ differ, the latter consisting of the finitely supported families. In commutative R-algebras the two come apart even for two factors. The product is the direct product $A \times B$ with componentwise operations, and the coproduct is the tensor product.

Lemma 15.2.16: Generation of $A \otimes_R B$ as an Algebra

Let R be a commutative ring and A, B be R-algebras. Then $A \otimes_R B$ is generated as an R-algebra by the set $\{a \otimes 1 | a \in A\} \cup \{1 \otimes b | b \in B\}$.

***Proof*:**
By theorem 15.2.14, $(a \otimes 1)(1 \otimes b) = (a \cdot 1) \otimes (1 \cdot b) = a \otimes b$, so every simple tensor is a product of two elements of the stated set. By lemma 15.1.4(2) every element of $A \otimes_R B$ is a finite sum of simple tensors, completing the proof.

Theorem 15.2.17: Universal Property of the Coproduct in R-Alg

Let R be a commutative ring and let A and B be commutative R-algebras, with structure maps $i_A\colon R \to A$ and $i_B\colon R \to B$. Then the maps

$$\iota_A\colon A \to A \otimes_R B, \quad a \mapsto a \otimes 1 \qquad \text{and} \qquad \iota_B\colon B \to A \otimes_R B, \quad b \mapsto 1 \otimes b$$

are R-algebra homomorphisms, and $(A \otimes_R B, \iota_A, \iota_B)$ is the coproduct of A and B in the category of commutative R-algebras: for every commutative R-algebra Z and every pair of R-algebra homomorphisms $f\colon A \to Z$ and $g\colon B \to Z$ there is a unique R-algebra homomorphism $h\colon A \otimes_R B \to Z$ with $h \circ \iota_A = f$ and $h \circ \iota_B = g$:

$$\begin{tikzcd}
R \arrow[r, "i_B"] \arrow[d, "i_A"'] & B \arrow[d, "\iota_B"] \arrow[ddr, bend left, "g"] & \\
A \arrow[r, "\iota_A"] \arrow[drr, bend right, "f"'] & A \otimes_R B \arrow[dr, dashed, "h"] & \\
 & & Z
\end{tikzcd}$$

***Proof*:**
That ι_A is an R-algebra homomorphism is immediate from theorem 15.2.14: it is additive and R-linear because $\otimes$ is R-linear in its first argument, multiplicative because $(a \otimes 1)(a' \otimes 1) = (aa') \otimes 1$, and unital because $\iota_A(1_A) = 1_A \otimes 1_B$. The same holds for ι_B.

Existence. Define $\Phi\colon A \times B \to Z$ by $\Phi(a, b) = f(a)g(b)$. It is R-bilinear: it is additive in each variable because f and g are additive and multiplication in Z distributes, and for $r \in R$,

$$\Phi(ra, b) = f(ra)g(b) = \big(rf(a)\big)g(b) = r\big(f(a)g(b)\big) \qquad \Phi(a, rb) = f(a)\big(rg(b)\big) = r\big(f(a)g(b)\big)$$

using R-linearity of f and of g and the compatibility $r(xy) = (rx)y = x(ry)$ of definition 12.1.13 in Z. By theorem 15.1.12 there is a unique R-module homomorphism $h\colon A \otimes_R B \to Z$ with $h(a \otimes b) = f(a)g(b)$.

The map h is multiplicative. Both $(x, y) \mapsto h(xy)$ and $(x, y) \mapsto h(x)h(y)$ are R-bilinear, and on a pair of simple tensors

$$\begin{aligned} h\big((a \otimes b)(a' \otimes b')\big) &= h(aa' \otimes bb') = f(aa')g(bb') \\ &= f(a)f(a')g(b)g(b') = f(a)g(b)f(a')g(b') \\ &= h(a \otimes b)\, h(a' \otimes b') \end{aligned}$$

the fourth equality using that Z is commutative. Two R-bilinear maps agreeing on pairs of simple tensors are equal, by the argument in Step 1 of theorem 15.2.14, so h is multiplicative. It is unital, since $h(1_A \otimes 1_B) = f(1)g(1) = 1$, and therefore an R-algebra homomorphism. Finally $h(\iota_A(a)) = h(a \otimes 1) = f(a)g(1) = f(a)$ and likewise $h \circ \iota_B = g$, so the diagram commutes.

Uniqueness. Suppose h' is an R-algebra homomorphism with $h' \circ \iota_A = f$ and $h' \circ \iota_B = g$. Then h and h' agree on every $a \otimes 1$ and every $1 \otimes b$. By lemma 15.2.16 those elements generate $A \otimes_R B$ as an R-algebra, and two R-algebra homomorphisms agreeing on a generating set are equal, since the set on which they agree is a subalgebra. Hence $h' = h$, completing the proof.

Every commutative ring is an algebra over $\mathbb{Z}$ in exactly one way, which turns the theorem into a statement about rings with no base ring chosen in advance.

Corollary 15.2.18: Universal Property of the Coproduct in CRing

Let A and B be commutative rings. Then $A \otimes_{\mathbb{Z}} B$, with $\iota_A(a) = a \otimes 1$ and $\iota_B(b) = 1 \otimes b$, is the coproduct of A and B in the category **CRing** of commutative rings.

***Proof*:**
By proposition 9.1.29 there is exactly one ring homomorphism $\mathbb{Z} \to A$ for every ring A, so every commutative ring carries exactly one $\mathbb{Z}$-algebra structure, and its image lies in the centre because A is commutative. Moreover every ring homomorphism $\varphi\colon A \to B$ of commutative rings is automatically a $\mathbb{Z}$-algebra homomorphism: the two ring homomorphisms $\varphi \circ i_A$ and i_B from $\mathbb{Z}$ to B coincide by that same uniqueness. Hence the category of commutative $\mathbb{Z}$-algebras and **CRing** have the same objects and the same morphisms, and theorem 15.2.17 applied with $R = \mathbb{Z}$ gives the claim, completing the proof.

The asymmetry between the two constructions is worth recording. The product $A \times B$ is formed with no reference to a base ring, while the coproduct is formed over one, and in **CRing** the base is forced to be the initial object $\mathbb{Z}$. Over a general base R, the object $A \otimes_R B$ solves the coproduct problem for R-algebras, which is the same as a pushout of $A \leftarrow R \rightarrow B$ in commutative rings. Reversing every arrow turns a pushout of rings into a pullback of spaces, and this is the computation that makes $\otimes_R$ the algebraic form of the fibre product of affine schemes in *EYNTKA Algebraic Geometry* [3].

Exercise 15.2.1

1. Show that $\varphi \otimes \mathrm{id}$ need not be injective when φ is. (Take $\varphi\colon \mathbb{Z} \to \mathbb{Z}$ to be multiplication by 2 and tensor with $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}$.) Conclude that the functor of theorem 15.2.2 does not preserve injections.

2. Let R be commutative, let M be an R-module and let I be an ideal of R. Show that $M \otimes_R R/I \cong M/IM$, and recover proposition 15.2.13 as the case $M = R/J$.

3. Show that every element of $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ is a simple tensor, and deduce that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}$. (Note that example 15.1.13 does not apply: the left factor is $\mathbb{Q}$, not the base ring $\mathbb{Z}$.)

4. Let R be commutative and A an R-algebra. Show that the isomorphism $A \otimes_R R \cong A$ of example 15.1.13 is an isomorphism of R-algebras.

5. Using the isomorphism of example 15.2.15, find the two nontrivial idempotents of $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ explicitly as combinations of simple tensors, and verify directly that each squares to itself.

6. Let R be commutative and let M, N be finitely generated R-modules, generated by s and t elements respectively. Show that $M \otimes_R N$ is generated by st elements. Show also that the bound is attained but not always sharp, by computing a case where $M \otimes_R N$ needs fewer.

7. Let F be a field and let V, W be finite-dimensional F-vector spaces. Show that
$$V^* \otimes_F W \to \mathrm{Hom}_F(V, W), \qquad f \otimes w \mapsto \big(v \mapsto f(v)w\big)$$
is an isomorphism of F-vector spaces, and identify where finite-dimensionality is used.

8. Combine theorem 15.2.8 and proposition 15.2.7 into an isomorphism $(A \otimes_R B) \otimes_R (A \otimes_R B) \cong (A \otimes_R A) \otimes_R (B \otimes_R B)$ carrying $(a \otimes b) \otimes (a' \otimes b')$ to $(a \otimes a') \otimes (b \otimes b')$. Use it, together with the maps $A \otimes_R A \to A$ and $B \otimes_R B \to B$ induced by multiplication, to give a second construction of the multiplication of theorem 15.2.14.

15.3 Extension of Scalars

The previous two sections held the ring fixed and varied the modules. This one does the opposite. Given a ring homomorphism $f : R \to S$, there are two ways to move a module between the two rings, and they are not symmetric: passing from S to R needs nothing, while passing from R to S requires a construction. That construction is the tensor product, applied to the bimodule of example 15.1.9(2), and everything in this section is a consequence of theorem 15.1.5 rather than a new construction.

The cheap direction first.

Definition 15.3.1: Restriction of Scalars

Let $f : R \to S$ be a ring homomorphism and let L be a left S-module. The *restriction of scalars* of L along f is the left R-module L_R with the same underlying abelian group and action

$$r \cdot l := f(r)\, l$$

An S-module homomorphism is in particular an R-module homomorphism $L_R \to L'_R$, so restriction of scalars sends S-modules to R-modules and S-linear maps to R-linear maps.

The module axioms hold because f is a ring homomorphism, exactly as in the construction of an R-algebra from a ring map (section 12.1.2); nothing is created and nothing is destroyed, only forgotten. The other direction cannot work this way. An R-module N carries no S-action to forget, and there is in general no way to define one on the same underlying set: if $R = \mathbb{Z}$, $S = \mathbb{Q}$ and $N = \mathbb{Z}/2\mathbb{Z}$, then any $\mathbb{Q}$-action on N would make multiplication by 2 invertible on a group killed by 2. A new module has to be built, and the tensor product builds it.

Definition 15.3.2: Extension of Scalars

Let $f : R \to S$ be a ring homomorphism and let N be a left R-module. Regard S as an (S, R)-bimodule as in example 15.1.9(2), with left action given by multiplication in S and right action $s \cdot r := sf(r)$. The *extension of scalars* of N along f is the left S-module

$$S \otimes_R N$$

whose S-action, given by proposition 15.1.10, is determined on simple tensors by $s' \cdot (s \otimes n) = (s's) \otimes n$.

No verification is owed here: the module structure was proved once in proposition 15.1.10 and the only input is the bimodule structure on S, which is example 15.1.9(2). The relation that does the work is the third one in (15.1), which now reads

$$sf(r) \otimes n = s \otimes rn$$

so a scalar of R may be moved out of the module and into S, where it becomes invertible whenever $f(r)$ is.

Example 15.3.3: Extension of Scalars of a Free Module
Let $f: R \to S$ be a ring homomorphism. Then

$$S \otimes_R R^n \cong S^n$$

as left S-modules, and more generally $S \otimes_R R^{(I)} \cong S^{(I)}$ for a free module on any index set I. Indeed $R^{(I)} = \bigoplus_{i \in I} R$, so theorem 15.2.4 applied on the right-hand factor gives

$$S \otimes_R \bigoplus_{i \in I} R \cong \bigoplus_{i \in I} \big(S \otimes_R R\big) \cong \bigoplus_{i \in I} S$$

the second isomorphism being example 15.1.13 in each summand. Following a basis element e_i through, $1 \otimes e_i$ corresponds to the i-th standard generator of $S^{(I)}$. So extension of scalars carries a free R-module of rank n to a free S-module of the same rank: a basis survives, and only the ring it is a basis over changes.

Free modules therefore lose nothing. In general something is lost, and the map that measures the loss is the following.

The assignment $\iota: N \to S \otimes_R N$ with $\iota(n) = 1 \otimes n$ is additive, and it is R-linear when $S \otimes_R N$ is regarded as an R-module by restriction:

$$\iota(rn) = 1 \otimes rn = 1 \cdot f(r) \otimes n = f(r)\big(1 \otimes n\big) = r \cdot \iota(n)$$

using the third relation and then the definition of the S-action. It need not be injective. For $R = \mathbb{Z}$, $S = \mathbb{Q}$ and N any finite abelian group, $\mathbb{Q} \otimes_{\mathbb{Z}} N = 0$ by the third exercise of section 15.1, so ι is the zero map on a nonzero module.

The next theorem says that although N may not sit inside $S \otimes_R N$, every R-linear map from N into an S-module factors through it, uniquely and S-linearly. It is the defining property of extension of scalars and the reason the construction is the right one.

Theorem 15.3.4: Universal Property of Extension of Scalars

Let $f: R \to S$ be a ring homomorphism, let N be a left R-module and let $\iota: N \to S \otimes_R N$ be the R-module homomorphism $\iota(n) = 1 \otimes n$. Then for every left S-module L and every R-module homomorphism $\varphi: N \to L_R$ there is a unique S-module homomorphism $\psi: S \otimes_R N \to L$ with $\varphi = \psi \circ \iota$, that is, making

$$\begin{tikzcd} S \otimes_R N \arrow[r, "\psi"] & L \\ N \arrow[u, "\iota"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

commute.

***Proof*:**
Existence. The map $S \times N \to L$ sending (s, n) to $s\varphi(n)$ is R-balanced. Additivity in s holds because the S-action on L distributes, and additivity in n because φ is additive. For the third axiom, the right R-action on S is $s \cdot r = sf(r)$ and the R-action on L_R is $r \cdot l = f(r)l$, so

$$(s \cdot r)\varphi(n) = \big(sf(r)\big)\varphi(n) = s\big(f(r)\varphi(n)\big) = s\big(r \cdot \varphi(n)\big) = s\,\varphi(rn)$$

the last equality because φ is R-linear. By theorem 15.1.5 there is a unique group homomorphism $\psi\colon S \otimes_R N \to L$ with $\psi(s \otimes n) = s\varphi(n)$.

It is S-linear. Fix $s' \in S$; the maps $x \mapsto \psi(s'x)$ and $x \mapsto s'\psi(x)$ are group homomorphisms, and on a simple tensor

$$\psi\big(s'(s \otimes n)\big) = \psi\big((s's) \otimes n\big) = (s's)\varphi(n) = s'\big(s\varphi(n)\big) = s'\psi(s \otimes n)$$

so they agree by lemma 15.1.4(3). Finally $\psi(\iota(n)) = \psi(1 \otimes n) = 1 \cdot \varphi(n) = \varphi(n)$, so $\varphi = \psi \circ \iota$.

Uniqueness. As an S-module, $S \otimes_R N$ is generated by the image of ι, since $s \otimes n = s(1 \otimes n) = s\,\iota(n)$ and every element is a finite sum of simple tensors by lemma 15.1.4(2). An S-module homomorphism is determined by its values on a generating set, and any ψ' with $\varphi = \psi' \circ \iota$ agrees with ψ on $\iota(N)$, so $\psi' = \psi$, completing the proof.

Restated without diagrams, the theorem says that R-linear maps out of N into an S-module are the same thing as S-linear maps out of $S \otimes_R N$. That is an adjunction, and it is worth naming as one, because it is the form in which the construction is used downstream.

Corollary 15.3.5: Extension of Scalars Is Left Adjoint to Restriction

Let $f\colon R \to S$ be a ring homomorphism. For every left R-module N and every left S-module L the map

$$\mathrm{Hom}_S\big(S \otimes_R N,\, L\big) \to \mathrm{Hom}_R\big(N,\, L_R\big), \qquad \psi \mapsto \psi \circ \iota$$

is an isomorphism of abelian groups. A pair of functors related this way is called an *adjunction*, and this one is expressed by saying that $S \otimes_R -$, from left R-modules to left S-modules, is *left adjoint* to restriction of scalars. The general theory is [7, Adjoint Functors]; only the displayed bijection is used below.

***Proof*:**
The map is additive, since $(\psi + \psi') \circ \iota = \psi \circ \iota + \psi' \circ \iota$, and it lands in $\mathrm{Hom}_R(N, L_R)$ because ι is R-linear and ψ is S-linear, hence R-linear after restriction. It is surjective by the existence half of theorem 15.3.4 and injective by the uniqueness half: if $\psi \circ \iota = \psi' \circ \iota$ then both are maps induced by the same φ, so they coincide, as we sought to show.

Two computations show what the construction does. The first is the standard way a real vector space is turned into a complex one.

Example 15.3.6: Complexification
Take $R = \mathbb{R}$, $S = \mathbb{C}$ and f the inclusion. For a real vector space V the *complexification* is the complex vector space

$$V_{\mathbb{C}} := \mathbb{C} \otimes_{\mathbb{R}} V$$

Every element of $V_{\mathbb{C}}$ has the form $1 \otimes v + i \otimes w$ for unique $v, w \in V$. A simple tensor already does, since for $z = x+iy$ with $x, y \in \mathbb{R}$ the third relation of (15.1) gives $z \otimes v = x \otimes v + iy \otimes v = 1 \otimes xv + i \otimes yv$; and a sum of two such collapses to one by additivity of $\otimes$ in its second argument. If V has $\mathbb{R}$-basis $e_1, \dots, e_n$ then $V_{\mathbb{C}}$ has $\mathbb{C}$-basis $1 \otimes e_1, \dots, 1 \otimes e_n$ by example 15.3.3, so

$$\dim_{\mathbb{C}}(V_{\mathbb{C}}) = \dim_{\mathbb{R}}(V)$$

A linear map $\varphi\colon V \to W$ extends to $\mathrm{id}_{\mathbb{C}} \otimes \varphi\colon V_{\mathbb{C}} \to W_{\mathbb{C}}$ by theorem 15.2.2, and on the bases above it has the same matrix as φ; the entries are simply now read as complex numbers. This is what licenses the standard move of computing eigenvalues of a real matrix over $\mathbb{C}$.

One warning. Complexifying a space that is already complex does not return it. Reading $\mathbb{C}$ itself as a real vector space and complexifying gives $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ by example 15.2.15, which has $\mathbb{C}$-dimension 2, not 1. The construction depends on the ring the tensor product is taken over, not only on the two spaces.

Example 15.3.7: Induced Representations

Let K be a field, let G be a finite group and let $H \leq G$ be a subgroup. The group rings $K[H]$ and $K[G]$ of definition 9.2.19 are related by the inclusion $K[H] \hookrightarrow K[G]$, so extension of scalars along it carries $K[H]$-modules to $K[G]$-modules. Since a $K[G]$-module is the same data as a representation of G over K, this manufactures a representation of G from one of H.

The dimension is forced. Choosing coset representatives $g_1, \dots, g_k$ for G/H, where $k = [G : H]$, gives $K[G] = \bigoplus_{i=1}^{k} g_i K[H]$ as a right $K[H]$-module, so $K[G]$ is free of rank k over $K[H]$. If V is a representation of H of dimension d, then theorem 15.2.4 gives

$$K[G] \otimes_{K[H]} V \cong \bigoplus_{i=1}^{k} \big(g_i K[H] \otimes_{K[H]} V\big) \cong V^{\oplus k}$$

as K-vector spaces, so the induced representation has dimension kd. Concretely, $\mathbb{Z}/3\mathbb{Z}$ sits inside S_3 with index 2, so a 10-dimensional representation of $\mathbb{Z}/3\mathbb{Z}$ induces a 20-dimensional representation of S_3.

The construction is the *induced representation*, and its theory, including Frobenius reciprocity, which is corollary 15.3.5 read in this setting, is developed in *Core Representation Theory* [9].

Returning to the map ι, the universal property settles exactly when it is injective, and the answer identifies what part of N can survive into any S-module at all.

Corollary 15.3.8: When the Unit Is Injective

Let $f\colon R \to S$ be a ring homomorphism, let N be a left R-module and let $\iota\colon N \to S \otimes_R N$ be as above. Then every R-module homomorphism from N into (the restriction of) a left S-module vanishes on $\ker \iota$. Consequently N embeds as an R-submodule of some left S-module if and only if ι is injective, in which case $N \cong \iota(N) \subseteq S \otimes_R N$.

***Proof*:**
Let L be a left S-module and $\theta\colon N \to L_R$ an R-module homomorphism. By theorem 15.3.4 there is an S-module homomorphism ψ with $\theta = \psi \circ \iota$, so θ kills $\ker \iota$.

If ι is injective it is itself such an embedding, with image $\iota(N)$. If conversely some $\theta\colon N \to L_R$ is injective, then $\ker \iota \subseteq \ker \theta = 0$ by the previous paragraph, so ι is injective, completing the proof.

For a finite abelian group N and $\mathbb{Z} \hookrightarrow \mathbb{Q}$ this recovers a fact already known: ι is the zero map, so no nonzero element of N survives into any $\mathbb{Q}$-vector space, which is another way of saying that a torsion element cannot live in a space where every nonzero integer acts invertibly.

The adjunction of corollary 15.3.5 is one instance of a general relationship between $\otimes$ and Hom. Stating it here rather than alongside its applications keeps every property of the tensor product in one place; it is used in section 17.5 to compare tensoring with the Hom functors, and in *Homological Algebra* [10] to produce derived functors.

Theorem 15.3.9: Hom-Tensor Adjunction

Let R and S be rings, let A be a right R-module, let B be an (R,S)-bimodule and let C be a right S-module. Give $\mathrm{Hom}_S(B,C)$ the right R-action $(g \cdot r)(b) := g(rb)$. Then there is an isomorphism of abelian groups

$$\mathrm{Hom}_S\big(A \otimes_R B,\, C\big) \cong \mathrm{Hom}_R\big(A,\, \mathrm{Hom}_S(B,C)\big)$$

sending φ to the map $a \mapsto \big(b \mapsto \varphi(a \otimes b)\big)$.

***Proof*:**
First, the objects are what the statement claims. Every right R-module is a $(\mathbb{Z}, R)$-bimodule, so proposition 15.2.1, applied with $\mathbb{Z}$ in the role of the left-hand ring and S in the role of the right-hand one, makes $A \otimes_R B$ a right S-module with $(a \otimes b)s = a \otimes (bs)$. The stated action on $\mathrm{Hom}_S(B,C)$ is a right action, since $\big((g \cdot r) \cdot r'\big)(b) = (g \cdot r)(r'b) = g(rr'b) = \big(g \cdot (rr')\big)(b)$.

The forward map. Given $\varphi \in \mathrm{Hom}_S(A \otimes_R B, C)$, define $\Lambda(\varphi)(a)(b) = \varphi(a \otimes b)$. For fixed a the map $\Lambda(\varphi)(a)$ is S-linear: it is additive because $\otimes$ is additive in its second argument, and

$$\Lambda(\varphi)(a)(bs) = \varphi\big(a \otimes bs\big) = \varphi\big((a \otimes b)s\big) = \varphi(a \otimes b)\, s = \Lambda(\varphi)(a)(b)\, s$$

using S-linearity of φ. The map $\Lambda(\varphi)$ is R-linear: it is additive because $\otimes$ is additive in its first argument, and

$$\Lambda(\varphi)(ar)(b) = \varphi(ar \otimes b) = \varphi(a \otimes rb) = \Lambda(\varphi)(a)(rb) = \big(\Lambda(\varphi)(a) \cdot r\big)(b)$$

the second equality being the third relation of (15.1). Finally Λ is additive in φ, directly from the definition.

The backward map. Given $\Psi \in \mathrm{Hom}_R(A, \mathrm{Hom}_S(B,C))$, the map $A \times B \to C$ sending (a,b) to $\Psi(a)(b)$ is R-balanced: it is additive in a because Ψ is additive, additive in b because each $\Psi(a)$ is, and

$$\Psi(ar)(b) = \big(\Psi(a) \cdot r\big)(b) = \Psi(a)(rb)$$

which is the third axiom. By theorem 15.1.5 there is a unique group homomorphism $\Gamma(\Psi)\colon A \otimes_R B \to C$ with $\Gamma(\Psi)(a \otimes b) = \Psi(a)(b)$, and it is S-linear because $x \mapsto \Gamma(\Psi)(xs)$ and $x \mapsto \Gamma(\Psi)(x)s$

are group homomorphisms agreeing on simple tensors, where both give $\Psi(a)(bs)$. Again Γ is additive.

The two are mutually inverse. For $\Psi \in \mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C))$ and all $a \in A$, $b \in B$,

$$\Lambda\big(\Gamma(\Psi)\big)(a)(b) = \Gamma(\Psi)(a \otimes b) = \Psi(a)(b)$$

so $\Lambda(\Gamma(\Psi))(a) = \Psi(a)$ for every a, giving $\Lambda \circ \Gamma = \mathrm{id}$. In the other direction, $\Gamma(\Lambda(\varphi))$ and φ are group homomorphisms out of $A \otimes_R B$ agreeing on every simple tensor, since $\Gamma(\Lambda(\varphi))(a \otimes b) = \Lambda(\varphi)(a)(b) = \varphi(a \otimes b)$, so they are equal by lemma 15.1.4(3). Hence Λ is an isomorphism of abelian groups, completing the proof.

Taking $B = S$, viewed as an (R, S)-bimodule along $f \colon R \to S$, and using $\mathrm{Hom}_S(S, C) \cong C_R$ recovers corollary 15.3.5 in its right-handed form: extension of scalars is the case of the Hom-tensor adjunction in which the bimodule is the ring itself.

Exercise 15.3.1

1. Let $f \colon R \to S$ be a ring homomorphism and let $I \subseteq R$ be an ideal. Show that $S \otimes_R R/I \cong S/f(I)S$ as S-modules, where $f(I)S$ is the ideal of S generated by $f(I)$.
2. Let A be a finitely generated abelian group, so that $A \cong \mathbb{Z}^r \oplus T$ with T finite by chapter 14. Show that $\mathbb{Q} \otimes_{\mathbb{Z}} A \cong \mathbb{Q}^r$ as $\mathbb{Q}$-vector spaces, so that extension of scalars along $\mathbb{Z} \hookrightarrow \mathbb{Q}$ destroys exactly the torsion and $r = \dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} A)$.
3. Let $R \subseteq S$ be an inclusion of rings and suppose S is free as a right R-module with a basis containing 1. Show that $\iota \colon N \to S \otimes_R N$ is injective for every left R-module N. Where does the argument fail for $\mathbb{Z} \subseteq \mathbb{Q}$?
4. Viewing S as an (R, S)-bimodule along $f \colon R \to S$, show that $\mathrm{Hom}_R(S, L)$ is a left S-module for every left R-module L, and that
$$\mathrm{Hom}_R\big(M_R,\, L\big) \cong \mathrm{Hom}_S\big(M,\, \mathrm{Hom}_R(S, L)\big)$$
for every left S-module M. (So restriction of scalars has a right adjoint as well as a left one; this is *coextension of scalars*.)
5. Let V be a real vector space. Show that $V_{\mathbb{C}} \cong V \oplus V$ as real vector spaces, and describe the complex structure that this decomposition carries in terms of the two summands.
6. Let $f \colon R \to S$ and $g \colon S \to T$ be ring homomorphisms. Show that extension of scalars along $g \circ f$ agrees with extension along f followed by extension along g, that is $T \otimes_S (S \otimes_R N) \cong T \otimes_R N$ naturally in N.
7. Show that theorem 15.3.9 is natural in C: for an S-module homomorphism $h \colon C \to C'$, the two ways of passing from $\mathrm{Hom}_S(A \otimes_R B, C)$ to $\mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C'))$ agree.

15.4 Application: The Kronecker Product

chapter 13 and chapter 14 computed with matrices: a linear map became an array once bases were chosen, and questions about the map became questions about the array. Nothing in this chapter has yet been written down that way. The apparatus for doing so is already in place: corollary 15.2.6 says that bases of V and W determine a basis of $V \otimes_R W$, and theorem 15.2.2 says that maps of the factors determine a map of the tensor product. Putting the two together produces a matrix, and it has a name.

Definition 15.4.1: Kronecker Product

Let R be a commutative ring, let $A = (a_{ij}) \in M_{m\times n}(R)$ and let $B \in M_{p\times q}(R)$. The *Kronecker product* $A \otimes B \in M_{mp\times nq}(R)$ is the block matrix

$$A \otimes B := \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}$$

Equivalently, indexing rows by pairs (i, k) with $1 \le i \le m$, $1 \le k \le p$ and columns by pairs (j, l) with $1 \le j \le n$, $1 \le l \le q$, both ordered lexicographically,

$$(A \otimes B)_{(i,k),(j,l)} = a_{ij}b_{kl}$$

The lexicographic indexing is part of the definition rather than a convenience: a different ordering of the pairs produces a matrix conjugate to this one by a permutation, which is the content of proposition 15.4.3(4) below. Every statement in this section is made with respect to that ordering.

Proposition 15.4.2: The Kronecker Product Represents the Tensor Product of Maps

Let R be a commutative ring and let V, V', W, W' be free R-modules with ordered bases $(e_1, \ldots, e_n)$, $(e'_1, \ldots, e'_m)$, $(f_1, \ldots, f_q)$ and $(f'_1, \ldots, f'_p)$ respectively. Let $\varphi\colon V \to V'$ be represented by A and $\psi\colon W \to W'$ by B. Give $V \otimes_R W$ the basis $(e_j \otimes f_l)$ and $V' \otimes_R W'$ the basis $(e'_i \otimes f'_k)$ of corollary 15.2.6, each ordered lexicographically. Then $\varphi \otimes \psi$ is represented by $A \otimes B$.

***Proof*:**
By theorem 15.2.2 and the bilinearity of $\otimes$,

$$(\varphi \otimes \psi)(e_j \otimes f_l) = \varphi(e_j) \otimes \psi(f_l) = \left(\sum_{i=1}^{m} a_{ij}e'_i\right) \otimes \left(\sum_{k=1}^{p} b_{kl}f'_k\right) = \sum_{i=1}^{m}\sum_{k=1}^{p} a_{ij}b_{kl}\,(e'_i \otimes f'_k)$$

By definition 13.2.1 the entry of the representing matrix in row (i, k) and column (j, l) is the coefficient of $e'_i \otimes f'_k$ in the image of $e_j \otimes f_l$, which the display gives as $a_{ij}b_{kl}$. That is the entry prescribed by definition 15.4.1, completing the proof.

Every property of the Kronecker product below is therefore a property of $\otimes$ read in coordinates, and the two viewpoints are used interchangeably from here on.

Proposition 15.4.3: Arithmetic of the Kronecker Product

Let R be a commutative ring.

1. (mixed product rule) $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ whenever AC and BD are defined.
2. $I_m \otimes I_p = I_{mp}$.
3. If $A \in M_m(R)$ and $B \in M_p(R)$ are invertible then so is $A \otimes B$, with $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.
4. There is a permutation matrix $P \in M_{mp}(R)$, depending only on m and p, with
$$P(A \otimes B)P^{-1} = B \otimes A$$
for all $A \in M_m(R)$ and $B \in M_p(R)$.

***Proof*:**

1. Both sides are indexed by pairs, and summing over the intermediate index pair (r, s),
$$\begin{aligned}\big((A \otimes B)(C \otimes D)\big)_{(i,k),(j,l)} &= \sum_r \sum_s a_{ir} b_{ks}\, c_{rj} d_{sl} \\ &= \left(\sum_r a_{ir} c_{rj}\right)\left(\sum_s b_{ks} d_{sl}\right) = (AC)_{ij}(BD)_{kl}\end{aligned}$$
the middle step using commutativity of R to separate the two sums. The right-hand side is the $(i, k), (j, l)$ entry of $(AC) \otimes (BD)$.
2. $(I_m \otimes I_p)_{(i,k),(j,l)} = \delta_{ij}\delta_{kl}$, which is 1 if $(i, k) = (j, l)$ and 0 otherwise.
3. By (1) and (2), $(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I_m \otimes I_p = I_{mp}$, and the same on the other side.
4. Both $\{(i, k)\}$ and $\{(k, i)\}$ index sets of size mp, so the slot-exchanging bijection $\pi(i, k) = (k, i)$ between them has a permutation matrix P, determined by $Pe_{(i,k)} = e_{(k,i)}$ on standard basis vectors. For any M this gives $\big(PMP^{-1}\big)e_{\pi(y)} = PMe_y = \sum_x M_{xy} e_{\pi(x)}$, that is $\big(PMP^{-1}\big)_{\pi(x),\pi(y)} = M_{x,y}$. Applying this with $M = A \otimes B$,
$$\big(P(A \otimes B)P^{-1}\big)_{(k,i),(l,j)} = (A \otimes B)_{(i,k),(j,l)} = a_{ij} b_{kl} = b_{kl} a_{ij} = (B \otimes A)_{(k,i),(l,j)}$$
again using commutativity, completing the proof.

Part (4) is proposition 15.2.7 in coordinates: the isomorphism $V \otimes W \cong W \otimes V$ becomes, once bases are fixed on both sides, conjugation by the permutation that exchanges the two index slots. The matrix P is the reason the ordering convention in definition 15.4.1 is harmless.

The trace and the determinant of a Kronecker product are determined by those of the factors. Both formulas hold over an arbitrary commutative ring.

Theorem 15.4.4: Trace and Determinant of a Kronecker Product

Let R be a commutative ring, $A \in M_m(R)$ and $B \in M_p(R)$. Then

$$\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B) \qquad \text{and} \qquad \det(A \otimes B) = \big(\det A\big)^p \big(\det B\big)^m$$

***Proof*:**
Trace. The diagonal entries of $A \otimes B$ are those indexed by $(i,k) = (j,l)$, that is by $i = j$ and $k = l$, and there the entry is $a_{ii}b_{kk}$. Hence

$$\operatorname{tr}(A \otimes B) = \sum_{i=1}^{m}\sum_{k=1}^{p} a_{ii}b_{kk} = \left(\sum_{i=1}^{m} a_{ii}\right)\left(\sum_{k=1}^{p} b_{kk}\right) = \operatorname{tr}(A)\operatorname{tr}(B)$$

Determinant. By the mixed product rule and proposition 15.4.3(2),

$$A \otimes B = (AI_m) \otimes (I_pB) = (A \otimes I_p)(I_m \otimes B)$$

so $\det(A \otimes B) = \det(A \otimes I_p)\det(I_m \otimes B)$ by proposition 13.4.7. The second factor is immediate: $I_m \otimes B$ is block diagonal with m diagonal blocks, each equal to B, so $\det(I_m \otimes B) = (\det B)^m$ by proposition 13.4.8.

For the first factor, proposition 15.4.3(4) gives a permutation matrix P with $P(A \otimes I_p)P^{-1} = I_p \otimes A$. Determinants are unchanged by conjugation, since proposition 13.4.7 gives $\det(P)\det(P^{-1}) = \det(PP^{-1}) = \det(I) = 1$ and hence $\det(PMP^{-1}) = \det(M)$. Now $I_p \otimes A$ is block diagonal with p diagonal blocks each equal to A, so

$$\det(A \otimes I_p) = \det(I_p \otimes A) = (\det A)^p$$

by proposition 13.4.8 again. Multiplying the two factors gives the claim, as we sought to show.

Note the asymmetry of the exponents: each determinant is raised to the size of the *other* factor. A quick check is $A = B = I$, where both sides are 1, and a sharper one is $m = p = 2$ with $\det A = \det B = 2$, giving $\det(A \otimes B) = 16$ rather than 4.

Rank behaves as simply, though here a field is needed: the argument passes through a complement, and over a general commutative ring a submodule need not be a direct summand.

Proposition 15.4.5: Rank of a Kronecker Product

Let F be a field, $A \in M_{m\times n}(F)$ and $B \in M_{p\times q}(F)$. Then

$$\operatorname{rank}(A \otimes B) = \operatorname{rank}(A) \cdot \operatorname{rank}(B)$$

***Proof*:**
Let $\varphi\colon V \to V'$ and $\psi\colon W \to W'$ be the linear maps represented by A and B, so that $A \otimes B$ represents $\varphi\otimes\psi$ by proposition 15.4.2, and rank is the dimension of the image (definition 13.1.19).

Choose complements $V' = \operatorname{im}\varphi \oplus U$ and $W' = \operatorname{im}\psi \oplus U'$, which exist because F is a field.

Applying theorem 15.2.3 in each variable,

$$V' \otimes_F W' \cong \left(\mathrm{im}\varphi \otimes \mathrm{im}\psi\right) \oplus \left(\mathrm{im}\varphi \otimes U'\right) \oplus \left(U \otimes \mathrm{im}\psi\right) \oplus \left(U \otimes U'\right)$$

and under this decomposition the natural map $\mathrm{im}\varphi \otimes_F \mathrm{im}\psi \to V' \otimes_F W'$ is the inclusion of the first summand, hence injective.

The image of $\varphi \otimes \psi$ is exactly that summand. It is contained in it: every element of $V \otimes_F W$ is a finite sum of simple tensors by lemma 15.1.4(2), and $(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w)$ lies in $\mathrm{im}\varphi \otimes \mathrm{im}\psi$. It contains it, because those same elements $\varphi(v) \otimes \psi(w)$ span $\mathrm{im}\varphi \otimes \mathrm{im}\psi$, again by lemma 15.1.4(2) applied there. Therefore

$$\mathrm{rank}(\varphi \otimes \psi) = \dim_F \left(\mathrm{im}\varphi \otimes_F \mathrm{im}\psi\right) = \dim_F(\mathrm{im}\varphi) \cdot \dim_F(\mathrm{im}\psi) = \mathrm{rank}(\varphi)\mathrm{rank}(\psi)$$

the middle equality being corollary 15.2.6, as we sought to show.

Eigenvalues multiply as well. One half of the statement needs nothing at all.

Proposition 15.4.6: Eigenvectors of a Kronecker Product

Let F be a field, $A \in M_m(F)$, $B \in M_p(F)$, and suppose $Av = \lambda v$ and $Bw = \mu w$ with $v \neq 0$ and $w \neq 0$. Then $v \otimes w \neq 0$ and

$$(A \otimes B)(v \otimes w) = \lambda\mu\,(v \otimes w)$$

so $\lambda\mu$ is an eigenvalue of $A \otimes B$.

***Proof*:**
Reading $A \otimes B$ as $\varphi \otimes \psi$ by proposition 15.4.2 and applying theorem 15.2.2,

$$(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w) = (\lambda v) \otimes (\mu w) = \lambda\mu\,(v \otimes w)$$

using bilinearity to extract both scalars. For $v \otimes w \neq 0$, extend v to a basis of F^m and w to a basis of F^p; then $v \otimes w$ is one of the basis elements of $F^m \otimes_F F^p$ furnished by corollary 15.2.6, and a basis element is nonzero, completing the proof.

That accounts for the products of eigenvalues that come from actual eigenvectors. The full spectrum, counted with multiplicity, needs a triangular form, and theorem 14.4.5 gives one exactly when the minimal polynomials split.

Theorem 15.4.7: Spectrum of a Kronecker Product

Let F be a field and let $A \in M_m(F)$ and $B \in M_p(F)$ have minimal polynomials that split into linear factors over F, which holds for all A and B when F is algebraically closed. List the eigenvalues of A as $\lambda_1, \ldots, \lambda_m$ and those of B as $\mu_1, \ldots, \mu_p$, each repeated according to its algebraic multiplicity. Then the characteristic polynomial of $A \otimes B$ is

$$\prod_{i=1}^{m} \prod_{k=1}^{p} (x - \lambda_i \mu_k)$$

so the eigenvalues of $A \otimes B$, with algebraic multiplicity, are the mp products $\lambda_i \mu_k$.

***Proof*:**
Two general facts are used repeatedly, so both are recorded first.

The determinant of an upper triangular matrix is the product of its diagonal entries. In the defining sum a term $\prod_i t_{i\sigma(i)}$ is nonzero only if $i \leq \sigma(i)$ for every i, and since $\sum_i \big(\sigma(i) - i\big) = 0$ this forces $\sigma(i) = i$ for all i; only the identity term survives. Consequently an upper triangular T has characteristic polynomial $\prod_i (x - t_{ii})$, since $xI - T$ is again upper triangular.

Similar matrices have the same characteristic polynomial, because $\det\big(xI - PMP^{-1}\big) = \det\big(P(xI - M)P^{-1}\big) = \det(xI - M)$ by proposition 13.4.7, applied over the commutative ring $F[x]$.

Now by theorem 14.4.5 there are invertible P and Q with $A = PJ_AP^{-1}$ and $B = QJ_BQ^{-1}$, where J_A and J_B are in Jordan canonical form and hence upper triangular. Combining the two facts, the characteristic polynomial of A is $\prod_i (x - (J_A)_{ii})$, so the diagonal entries of J_A are exactly the eigenvalues $\lambda_1, \ldots, \lambda_m$ of A listed with algebraic multiplicity, and likewise for J_B and the μ_k.

By the mixed product rule and proposition 15.4.3(3),

$$A \otimes B = \big(PJ_AP^{-1}\big) \otimes \big(QJ_BQ^{-1}\big) = (P \otimes Q)\,(J_A \otimes J_B)\,(P \otimes Q)^{-1}$$

so $A \otimes B$ is similar to $J_A \otimes J_B$ and the two have the same characteristic polynomial.

The matrix $J_A \otimes J_B$ is upper triangular for the lexicographic ordering. Its $(i,k),(j,l)$ entry is $(J_A)_{ij}(J_B)_{kl}$, which vanishes unless $i \leq j$ and $k \leq l$; and if (i,k) comes strictly after (j,l) lexicographically then either $i > j$, or $i = j$ and $k > l$, and in both cases one of the two factors is 0. Its diagonal entries are $(J_A)_{ii}(J_B)_{kk} = \lambda_i \mu_k$.

So $J_A \otimes J_B$ is upper triangular with diagonal entries $\lambda_i \mu_k$, and by the first recorded fact its characteristic polynomial is $\prod_{i,k}(x - \lambda_i \mu_k)$, as we sought to show.

Example 15.4.8: Computing with the Kronecker Product
Take $R = \mathbb{Q}$ and

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

so that $\det A = -2$, $\mathrm{tr} A = 5$, $\det B = 1$ and $\mathrm{tr} B = 0$. Assembling the blocks $a_{ij}B$,

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & 2 \\ -1 & 0 & -2 & 0 \\ 0 & 3 & 0 & 4 \\ -3 & 0 & -4 & 0 \end{pmatrix}$$

Its diagonal entries are all 0, so its trace is $0 = 5 \cdot 0 = \mathrm{tr}(A)\mathrm{tr}(B)$, as theorem 15.4.4 requires. That theorem also predicts

$$\det(A \otimes B) = (\det A)^2 (\det B)^2 = (-2)^2 \cdot 1^2 = 4$$

Both A and B are invertible, so both have rank 2, and proposition 15.4.5 predicts $\mathrm{rank}(A \otimes B) = 4$, consistent with the nonzero determinant.

Replacing A by the rank one matrix

$$A' = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

gives $\mathrm{rank}(A' \otimes B) = 1 \cdot 2 = 2$, so $A' \otimes B$ is a 4×4 matrix of rank 2; correspondingly $\det(A' \otimes B) = (\det A')^2 (\det B)^2 = 0$. Note that the rank drops to 2 rather than to 3: a Kronecker product cannot have rank 3 at all, since by proposition 15.4.5 its rank is a product of two integers each at most 2.

One identification remains, and it explains a phenomenon recorded without explanation in section 15.1. For finite-dimensional spaces the tensor product of a dual space with another space is the space of linear maps between them.

Theorem 15.4.9: $V^* \otimes W$ and Linear Maps

Let F be a field and let V, W be finite-dimensional F-vector spaces. Then

$$\Theta \colon V^* \otimes_F W \to \mathrm{Hom}_F(V, W), \qquad \Theta(f \otimes w)(v) = f(v)\, w$$

is an isomorphism of F-vector spaces. Under Θ:

1. the simple tensors correspond exactly to the homomorphisms of rank at most one;
2. the least number of simple tensors whose sum is a given $t \in V^* \otimes_F W$ equals $\mathrm{rank}\Theta(t)$.

***Proof*:**

The map $(f, w) \mapsto \big(v \mapsto f(v)w\big)$ is F-bilinear, so theorem 15.1.12 gives the F-linear map Θ.

Let $(e_1, \ldots, e_n)$ be a basis of V with dual basis $(e_1^*, \ldots, e_n^*)$ (section 13.3) and $(w_1, \ldots, w_p)$ a basis of W. By corollary 15.2.6 the np elements $e_i^* \otimes w_k$ form a basis of $V^* \otimes_F W$. Their images are the maps $e_j \mapsto \delta_{ij} w_k$, whose matrices with respect to the two bases are the np elementary matrices having a single 1 in position (k, i); those form a basis of $\mathrm{Hom}_F(V, W)$. An F-linear map carrying a basis onto a basis is an isomorphism, so Θ is one.

(1). The image of $\Theta(f \otimes w)$ is contained in Fw, so its rank is at most one. Conversely, let θ have rank at most one. If $\theta = 0$ it is $\Theta(0 \otimes 0)$. Otherwise $\mathrm{im}\theta = Fw$ for some $w \neq 0$, and writing $\theta(v) = f(v)w$ defines a map $f \colon V \to F$ which is linear because θ is and w is a basis of its image;

then $\theta = \Theta(f \otimes w)$.

(2). If $t = \sum_{j=1}^{r} f_j \otimes w_j$ then $\Theta(t) = \sum_j \Theta(f_j \otimes w_j)$ is a sum of r maps of rank at most one. The image of a sum is contained in the sum of the images, and $\dim(X+Y) \le \dim X + \dim Y$, so $\operatorname{rank}\Theta(t) \le r$. Conversely let $\operatorname{rank}\Theta(t) = r$ and choose a basis $u_1, \ldots, u_r$ of $\operatorname{im}\Theta(t)$. Every v has $\Theta(t)(v) = \sum_{j=1}^{r} g_j(v) u_j$ for unique scalars $g_j(v)$, and each $g_j \colon V \to F$ so defined is linear because $\Theta(t)$ is and the u_j are independent. Then $\Theta\big(\sum_j g_j \otimes u_j\big) = \Theta(t)$, and Θ is injective, so $t = \sum_{j=1}^{r} g_j \otimes u_j$ is a sum of r simple tensors, completing the proof.

Part (2) settles the question left open in box 15.1.6. Taking $W = V$ of dimension n, the element $\sum_{i=1}^{n} e_i^* \otimes e_i$ corresponds under Θ to the identity map, which has rank n; so it needs exactly n simple tensors and is not itself simple once $n \ge 2$. The witness exhibited there, $e_1 \otimes e_1 + e_2 \otimes e_2$, is this element for $n = 2$ under the identification of V with V^* by the dual basis, and the ad hoc argument given there is replaced by a count of rank.

Exercise 15.4.1

1. Show that $A \otimes B$ and $B \otimes A$ are always similar, and give explicit 2×2 matrices A and B for which they are not equal.

2. Show that $(A \otimes B)^k = A^k \otimes B^k$ for every $k \ge 1$, and deduce that if $A^r = 0$ then $(A \otimes B)^r = 0$ for every B. Conclude that a Kronecker product with a nilpotent factor is nilpotent, with index at most that of the factor.

3. Deduce from the mixed product rule that $A \mapsto A \otimes I_p$ is an injective ring homomorphism $M_m(R) \to M_{mp}(R)$, and that its image commutes elementwise with the image of $B \mapsto I_m \otimes B$.

4. Identify the step in the proof of proposition 15.4.5 that requires a field. Then show it genuinely fails over $\mathbb{Z}$: with $\varphi \colon \mathbb{Z} \to \mathbb{Z}$ multiplication by 2 and ψ the identity on $\mathbb{Z}/2\mathbb{Z}$, show that the natural map $\operatorname{im}\varphi \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is not injective.

5. Let V and W be arbitrary vector spaces over F, not assumed finite dimensional. Show that Θ of theorem 15.4.9 is still injective and that its image is exactly the set of homomorphisms of finite rank. Deduce that Θ is an isomorphism if and only if V or W is finite dimensional.

6. Show that under the isomorphism $V^* \otimes_F V \cong \operatorname{End}_F(V)$ of theorem 15.4.9, the trace of definition 13.4.3 corresponds to the evaluation map determined by $f \otimes v \mapsto f(v)$. (This is the basis-free description of the trace.)

15.5 Summary

Everything in this chapter rests on one universal property: bilinear maps out of $M \times N$ are homomorphisms out of a single module $M \otimes_R N$. The tables below state what follows from it, each entry pointing at the result that establishes it.

Representing objects and adjunctions

Each class of map has a representing object, and in every case it is a tensor product. The representing object is determined up to unique isomorphism by its property (corollary 15.1.7), so the particular construction used in definition 15.1.3 is not part of the data.

Maps	On	Represented by	Reference
R-balanced, into an abelian group	$M \times N$	$M \otimes_R N$	theorem 15.1.5
R-bilinear, into an R-module	$M \times N$	$M \otimes_R N$	theorem 15.1.12
R-multilinear	$M_1 \times \cdots \times M_n$	$M_1 \otimes_R \cdots \otimes_R M_n$	theorem 15.2.11
R-linear, into an S-module	N	$S \otimes_R N$	theorem 15.3.4
Pairs of R-algebra maps	A, B	$A \otimes_R B$	theorem 15.2.17

The second and third rows require R commutative and the standard structures of example 15.1.9(4); the fifth requires A and B commutative. Two facts are used constantly with these: simple tensors generate, so a homomorphism out of a tensor product is determined by its values on them (lemma 15.1.4), and not every element is itself simple (box 15.1.6).

Two of these are adjunctions, which is the form in which downstream books invoke them.

Left adjoint	Right adjoint	Reference
$S \otimes_R -$	restriction of scalars along $R \to S$	corollary 15.3.5
$- \otimes_R B$	$\mathrm{Hom}_S(B, -)$	theorem 15.3.9

The first is the case $B = S$ of the second.

Module structure

The tensor product of a right module and a left module is only an abelian group. Any further structure comes from a second action on one of the factors.

Hypothesis	Structure on $M \otimes_R N$	Reference
M right R-, N left R-module	abelian group only	definition 15.1.3
M an (S, R)-bimodule	left S-module	proposition 15.1.10
M an (S, R)-, N an (R, T)-bimodule	(S, T)-bimodule	proposition 15.2.1
R commutative, standard structures	R-module	theorem 15.1.12
A, B R-algebras	R-algebra	theorem 15.2.14

What is preserved, and what is not

Statement	Reference
$R \otimes_R N \cong N$	example 15.1.13
$(\bigoplus_i M_i) \otimes_R N \cong \bigoplus_i (M_i \otimes_R N)$	theorem 15.2.4
Free $\otimes$ free is free, on the basis $\{m_i \otimes n_j\}$	corollary 15.2.6
$M \otimes_R N \cong N \otimes_R M$	proposition 15.2.7
$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$	theorem 15.2.8
$(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$	theorem 15.2.2
$\mathrm{rank}(A \otimes B) = \mathrm{rank}(A)\mathrm{rank}(B)$ over a field	proposition 15.4.5

The first, fourth and fifth rows are the unit, commutativity and associativity constraints of a symmetric monoidal structure on R-Mod (box 15.2.9).

Fails	**Witness**	**Reference**
Being nonzero	$\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$	example 15.1.14
Injectivity of a map	multiplication by 2 against $\mathbb{Z}/2\mathbb{Z}$	section 17.5
Containing the original module	ι has a kernel	corollary 15.3.8
Being an integral domain	$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$	example 15.2.15

The second row is the one with consequences beyond this chapter: which modules D make $D \otimes_R -$ preserve injections is the definition of flatness, taken up in section 17.5.

In coordinates

Once bases are fixed, corollary 15.2.6 turns each of the constructions above into a matrix identity.

Object	**Coordinate form**	**Reference**
$\varphi \otimes \psi$	the Kronecker product $(a_{ij}B)$	proposition 15.4.2
$\mathrm{tr}(A \otimes B)$	$\mathrm{tr}(A)\mathrm{tr}(B)$	theorem 15.4.4
$\det(A \otimes B)$	$(\det A)^p(\det B)^m$	theorem 15.4.4
Eigenvalues of $A \otimes B$	the products $\lambda_i\mu_k$	theorem 15.4.7
$\mathrm{Hom}_F(V, W)$	$V^* \otimes_F W$	theorem 15.4.9

Computations worth reusing

Identity	**Reference**
$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$	proposition 15.2.13
$R/I \otimes_R R/J \cong R/(I + J)$	proposition 15.2.13
$S \otimes_R R^n \cong S^n$	example 15.3.3
$V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$, with $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$	example 15.3.6
$A \otimes_{\mathbb{Z}} B$ is the coproduct in **CRing**	corollary 15.2.18

16

Multilinear Algebra

chapter 15 represented the bilinear maps on a product of two modules. Fixing a single module M and allowing more arguments produces further classes of map on M^k: the k-multilinear maps, those left unchanged by permuting the arguments, and those that vanish whenever two arguments coincide. Each class has its own representing object, and the three objects are quotients of one another.

Summing over k turns each family of representing objects into a ring. The tensor algebra $\mathcal{T}(M)$, the symmetric algebra $\mathcal{S}(M)$ and the exterior algebra $\bigwedge(M)$ are graded by the number of arguments, and each carries a universal property of its own, now for homomorphisms into an algebra rather than into a module. The determinant returns at the end as the map an endomorphism induces on the top exterior power, recovering section 13.4 from a universal property rather than from the defining sum.

16.1 Multilinear, Symmetric and Alternating Maps

The k-multilinear maps on M already have a representing object. definition 15.2.10 defined the class and theorem 15.2.11 produced the module through which every such map factors: an R-linear map out of the k-fold tensor product of M with itself is the same thing as a k-multilinear map on M^k.

Two subclasses of these maps occur constantly, and both are cut out by requiring the map to be insensitive to something. A symmetric map does not see the order of its arguments; an alternating map vanishes on a repetition among them. Each subclass is again represented by a module, and in each case that module is a quotient of the k-fold tensor product by exactly the elements the subclass is required to kill. That is the pattern of this section, applied three times.

Definition 16.1.1: Tensor Powers

Let R be a commutative ring and M an R-module. For $k \geq 1$ the *k-th tensor power* of M is

$$\mathcal{T}^k(M) := \underbrace{M \otimes_R M \otimes_R \cdots \otimes_R M}_{k \text{ factors}}$$

bracketed to the left as in section 15.2, with $\mathcal{T}^0(M) := R$ and $\mathcal{T}^1(M) = M$. Its elements are called *k-tensors*.

In this notation theorem 15.2.11 reads: the map $M^k \to \mathcal{T}^k(M)$ sending $(m_1, \ldots, m_k)$ to $m_1 \otimes \cdots \otimes m_k$ is k-multilinear, and every k-multilinear map on M^k factors through it uniquely by an R-module homomorphism. Nothing further is needed about $\mathcal{T}^k(M)$ here; the two quotients are what this section constructs.

Definition 16.1.2: Symmetric Multilinear Maps

Let R be a commutative ring and let M, N be R-modules. A k-multilinear map $\varphi\colon M^k \to N$ is *symmetric* if

$$\varphi\big(m_{\sigma(1)}, \ldots, m_{\sigma(k)}\big) = \varphi(m_1, \ldots, m_k)$$

for every $\sigma \in S_k$ and all $m_1, \ldots, m_k \in M$.

Definition 16.1.3: Alternating Multilinear Maps

With R, M, N as above, a k-multilinear map $\varphi\colon M^k \to N$ is *alternating* if

$$\varphi(m_1, \ldots, m_k) = 0 \qquad \text{whenever } m_i = m_j \text{ for some } i \neq j$$

The determinant of definition 13.4.5, read as a function of the n columns of a matrix, is the standard example of an alternating n-multilinear map $(R^n)^n \to R$: it is multilinear in the columns and vanishes when two of them agree, which is proposition 13.4.6(3) and (4). A symmetric bilinear form, such as an inner product (section 13.6), is the standard example of a symmetric map.

Alternating maps are often introduced by a sign rule rather than by the vanishing condition. The two agree except in characteristic 2, and the direction that always holds is worth isolating, since it is what makes the exterior power computable.

Lemma 16.1.4: Equivalence Of Alternating Map

Let R be a commutative ring and let $\varphi\colon M^k \to N$ be k-multilinear.

1. If φ is alternating then for every $\sigma \in S_k$,

$$\varphi\big(m_{\sigma(1)}, \ldots, m_{\sigma(k)}\big) = (-1)^\sigma\, \varphi(m_1, \ldots, m_k)$$

2. If 2 is a unit in R, the converse holds: a k-multilinear map satisfying the displayed sign rule is alternating.

***Proof*:**
(1). For $\sigma \in S_k$ write $\varphi^\sigma(m_1, \dots, m_k) := \varphi(m_{\sigma(1)}, \dots, m_{\sigma(k)})$. Substituting shows $(\varphi^\sigma)^\tau = \varphi^{\tau\sigma}$, so if the identity $\varphi^\sigma = (-1)^\sigma \varphi$ holds for two permutations it holds for their product:

$$\varphi^{\tau\sigma} = (\varphi^\sigma)^\tau = \big((-1)^\sigma\varphi\big)^\tau = (-1)^\sigma\varphi^\tau = (-1)^\sigma(-1)^\tau\varphi = (-1)^{\tau\sigma}\varphi$$

Since the transpositions generate S_k, it therefore suffices to treat a transposition $\tau = (a\ b)$ with $a < b$.

Fix all arguments except those in positions a and b, and let $f(x, y)$ denote the value of φ with x in position a and y in position b. Then f is bilinear, and $f(x, x) = 0$ for every x because φ is alternating. Expanding,

$$0 = f(x + y,\, x + y) = f(x, x) + f(x, y) + f(y, x) + f(y, y) = f(x, y) + f(y, x)$$

so $f(y, x) = -f(x, y)$, which is exactly the claim for τ.

(2). Suppose the sign rule holds and $m_i = m_j$ with $i \neq j$. Applying the rule to the transposition $\tau = (i\ j)$, the left-hand side is φ evaluated at the same arguments, since interchanging two equal entries changes nothing, while the right-hand side is $-\varphi(m_1, \dots, m_k)$. Hence $2\,\varphi(m_1, \dots, m_k) = 0$, and multiplying by the inverse of 2 gives $\varphi(m_1, \dots, m_k) = 0$, completing the proof.

The hypothesis in (2) cannot be dropped. Over $R = \mathbb{Z}/2\mathbb{Z}$ every sign $(-1)^\sigma$ equals 1, so the sign rule says only that the map is symmetric; and the multiplication map $\varphi\colon \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ with $\varphi(x, y) = xy$ is symmetric and bilinear but has $\varphi(\overline{1}, \overline{1}) = \overline{1} \neq \overline{0}$, so it is not alternating. In characteristic 2 the vanishing condition of definition 16.1.3 is strictly stronger, and it is the one the exterior power is built from.

Each subclass is now represented, by quotienting $\mathcal{T}^k(M)$ by the elements a map in the subclass is obliged to kill.

Definition 16.1.5: Symmetric Powers

Let $C^k(M) \subseteq \mathcal{T}^k(M)$ be the submodule generated by all elements

$$m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)} \qquad \sigma \in S_k,\ m_i \in M$$

The *k-th symmetric power* of M is $\mathcal{S}^k(M) := \mathcal{T}^k(M)/C^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 m_2 \cdots m_k$. Since $C^0(M) = C^1(M) = 0$, one has $\mathcal{S}^0(M) = R$ and $\mathcal{S}^1(M) = M$.

Theorem 16.1.6: Universal Property of the Symmetric Power

Let R be a commutative ring and M an R-module. The map

$$\varsigma\colon M^k \to \mathcal{S}^k(M), \qquad \varsigma(m_1, \dots, m_k) = m_1 m_2 \cdots m_k$$

is symmetric and k-multilinear, and it is universal among such maps: for every R-module N and every symmetric k-multilinear $\varphi\colon M^k \to N$ there is a unique R-module homomorphism $\overline{\varphi}\colon \mathcal{S}^k(M) \to N$ with $\varphi = \overline{\varphi} \circ \varsigma$.

***Proof*:**
The map ς is the composite of the universal k-multilinear map of theorem 15.2.11 with the quotient map $\pi\colon \mathcal{T}^k(M) \to \mathcal{S}^k(M)$, hence k-multilinear. It is symmetric because the difference $m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}$ lies in $C^k(M)$ by construction and so maps to 0.

Let φ be symmetric and k-multilinear. By theorem 15.2.11 there is a unique R-module homomorphism $\psi\colon \mathcal{T}^k(M) \to N$ with $\psi(m_1 \otimes \cdots \otimes m_k) = \varphi(m_1, \dots, m_k)$. On a generator of $C^k(M)$,

$$\psi\big(m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}\big) = \varphi(m_1, \dots, m_k) - \varphi(m_{\sigma(1)}, \dots, m_{\sigma(k)}) = 0$$

precisely because φ is symmetric. A homomorphism vanishing on a generating set vanishes on the submodule it generates, so $C^k(M) \subseteq \ker\psi$ and ψ factors through the quotient, giving $\overline{\varphi}$ with $\overline{\varphi} \circ \varsigma = \varphi$.

For uniqueness, the products $m_1 \cdots m_k$ generate $\mathcal{S}^k(M)$, being the images of the simple tensors, which generate $\mathcal{T}^k(M)$; two homomorphisms agreeing on a generating set are equal, as we sought to show.

Definition 16.1.7: Exterior Powers

Let $A^k(M) \subseteq \mathcal{T}^k(M)$ be the submodule generated by all simple tensors

$$m_1 \otimes \cdots \otimes m_k \qquad \text{with } m_i = m_j \text{ for some } i \neq j$$

The *k-th exterior power* of M is $\bigwedge^k(M) := \mathcal{T}^k(M)/A^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 \wedge \cdots \wedge m_k$, the operation being called the *wedge* or *exterior product*. Since $A^0(M) = A^1(M) = 0$, one has $\bigwedge^0(M) = R$ and $\bigwedge^1(M) = M$.

Theorem 16.1.8: Universal Property of the Exterior Power

Let R be a commutative ring and M an R-module. The map

$$\alpha\colon M^k \to \bigwedge^k(M), \qquad \alpha(m_1, \dots, m_k) = m_1 \wedge \cdots \wedge m_k$$

is alternating and k-multilinear, and it is universal among such maps: for every R-module N and every alternating k-multilinear $\varphi\colon M^k \to N$ there is a unique R-module homomorphism $\overline{\varphi}\colon \bigwedge^k(M) \to N$ with $\varphi = \overline{\varphi} \circ \alpha$.

***Proof*:**
As before α is k-multilinear, being the composite of the universal k-multilinear map with the quotient map, and it is alternating because a simple tensor with a repeated entry lies in $A^k(M)$ by construction.

Given an alternating k-multilinear φ, let $\psi\colon \mathcal{T}^k(M) \to N$ be the homomorphism given by theorem 15.2.11. Each generator of $A^k(M)$ is a simple tensor $m_1 \otimes \cdots \otimes m_k$ with $m_i = m_j$ for some $i \neq j$, and ψ sends it to $\varphi(m_1, \dots, m_k)$, which is 0 because φ is alternating. So $A^k(M) \subseteq \ker\psi$ and ψ factors through $\bigwedge^k(M)$. Uniqueness holds because the elements $m_1 \wedge \cdots \wedge m_k$ generate

$\bigwedge^k(M)$, completing the proof.

The sign rule now transfers from maps to elements.

Proposition 16.1.9: Anticommutativity of the Wedge Product

Let M be an R-module. In $\bigwedge^k(M)$,

$$m_1 \wedge \cdots \wedge m_k = 0 \quad \text{whenever } m_i = m_j \text{ for some } i \neq j$$
$$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = (-1)^\sigma \, m_1 \wedge \cdots \wedge m_k$$

for every $\sigma \in S_k$. In particular $m \wedge m' = -\, m' \wedge m$ for all $m, m' \in M$.

***Proof*:**
The first identity is the statement that the generators of $A^k(M)$ die in the quotient. The second is lemma 16.1.4(1) applied to the alternating map α of theorem 16.1.8, whose values are precisely the elements $m_1 \wedge \cdots \wedge m_k$. Taking $k = 2$ and σ the transposition gives the last claim, completing the proof.

Both $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ have been built as quotients of $\mathcal{T}^k(M)$. They can also be found inside it, as submodules, provided $k!$ is invertible. The comparison rests on an action of the symmetric group.

Proposition 16.1.10: The Symmetric Group Acts on Tensor Powers

Let M be an R-module and $k \geq 1$. For each $\sigma \in S_k$ there is a unique R-module homomorphism $\sigma \cdot (-)$ of $\mathcal{T}^k(M)$ with

$$\sigma \cdot (m_1 \otimes \cdots \otimes m_k) = m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(k)}$$

and these assemble into a left action of S_k on $\mathcal{T}^k(M)$ by R-module automorphisms.

***Proof*:**
For fixed σ the assignment $(m_1, \ldots, m_k) \mapsto m_{\sigma^{-1}(1)} \otimes \cdots \otimes m_{\sigma^{-1}(k)}$ is k-multilinear, since each argument appears in exactly one slot of the output, so theorem 15.2.11 gives the homomorphism and its uniqueness.

For the action law, evaluate on a simple tensor. The i-th slot of $\tau \cdot (m_1 \otimes \cdots \otimes m_k)$ holds $m_{\tau^{-1}(i)}$, so the i-th slot of $\sigma \cdot \big(\tau \cdot (m_1 \otimes \cdots \otimes m_k)\big)$ holds $m_{\tau^{-1}(\sigma^{-1}(i))} = m_{(\sigma\tau)^{-1}(i)}$, which is the i-th slot of $(\sigma\tau) \cdot (m_1 \otimes \cdots \otimes m_k)$. Two homomorphisms agreeing on the simple tensors are equal, so $\sigma \cdot (\tau \cdot z) = (\sigma\tau) \cdot z$ for all z. The identity permutation acts as the identity, so each σ acts invertibly, completing the proof.

The inverse in the formula is what makes the assignment a left action rather than a right one; without it the composition law would come out reversed.

Definition 16.1.11: Symmetric and Alternating Tensors

Let M be an R-module and $k \geq 1$. A tensor $z \in \mathcal{T}^k(M)$ is

1. a *symmetric k-tensor* if $\sigma \cdot z = z$ for every $\sigma \in S_k$;
2. an *alternating k-tensor* if $\sigma \cdot z = (-1)^\sigma z$ for every $\sigma \in S_k$.

These form submodules of $\mathcal{T}^k(M)$, written $\mathcal{T}^k_{\text{sym}}(M)$ and $\mathcal{T}^k_{\text{alt}}(M)$, each being an intersection of kernels of R-module homomorphisms.
The *symmetrization* and *antisymmetrization* operators on $\mathcal{T}^k(M)$ are

$$\operatorname{Sym}(z) := \sum_{\sigma \in S_k} \sigma \cdot z \qquad \text{and} \qquad \operatorname{Alt}(z) := \sum_{\sigma \in S_k} (-1)^\sigma \, \sigma \cdot z$$

Proposition 16.1.12: Symmetric and Alternating Tensors When $k!$ Is Invertible

Let M be an R-module, $k \geq 1$, and suppose $k!$ is a unit in R. Then $\frac{1}{k!}\operatorname{Sym}$ and $\frac{1}{k!}\operatorname{Alt}$ are idempotent endomorphisms of $\mathcal{T}^k(M)$ with images $\mathcal{T}^k_{\text{sym}}(M)$ and $\mathcal{T}^k_{\text{alt}}(M)$, and they induce isomorphisms of R-modules

$$\mathcal{S}^k(M) \xrightarrow{\sim} \mathcal{T}^k_{\text{sym}}(M) \qquad \text{and} \qquad \bigwedge^k(M) \xrightarrow{\sim} \mathcal{T}^k_{\text{alt}}(M)$$

In particular $\mathcal{T}^k(M) = \mathcal{T}^k_{\text{sym}}(M) \oplus \ker \operatorname{Sym}$, and likewise for the alternating part.

***Proof*:**
The symmetric case. Reindexing the sum shows $\tau \cdot \operatorname{Sym}(z) = \operatorname{Sym}(z)$ for every τ, since $\sigma \mapsto \tau\sigma$ permutes S_k; so Sym lands in $\mathcal{T}^k_{\text{sym}}(M)$. If z is already symmetric then every term of the sum equals z, so $\operatorname{Sym}(z) = k!\,z$ and $\frac{1}{k!}\operatorname{Sym}$ restricts to the identity there. Hence $\frac{1}{k!}\operatorname{Sym}$ is idempotent with image exactly $\mathcal{T}^k_{\text{sym}}(M)$, and an idempotent endomorphism splits its domain as image plus kernel.

Writing $\rho = \sigma^{-1}$, the generator $m_1 \otimes \cdots \otimes m_k - m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}$ of $C^k(M)$ is $z - \rho \cdot z$ for the simple tensor z, and $\operatorname{Sym}(z - \rho \cdot z) = \operatorname{Sym}(z) - \operatorname{Sym}(z) = 0$ by the reindexing above. So $\frac{1}{k!}\operatorname{Sym}$ kills $C^k(M)$ and induces $\Theta \colon \mathcal{S}^k(M) \to \mathcal{T}^k_{\text{sym}}(M)$.

Its inverse is the composite $\mathcal{T}^k_{\text{sym}}(M) \hookrightarrow \mathcal{T}^k(M) \xrightarrow{\pi} \mathcal{S}^k(M)$. Indeed, for a simple tensor z one has $\pi(\sigma \cdot z) = \pi(z)$, since $z - \sigma \cdot z \in C^k(M)$, so

$$\pi\left(\tfrac{1}{k!}\operatorname{Sym}(z)\right) = \tfrac{1}{k!} \sum_{\sigma \in S_k} \pi(\sigma \cdot z) = \pi(z)$$

and both sides are homomorphisms, so this holds for all z; that is one composite. In the other order, $\frac{1}{k!}\operatorname{Sym}$ is the identity on $\mathcal{T}^k_{\text{sym}}(M)$, which is the second. So Θ is an isomorphism.

The alternating case. The same reindexing gives $\tau \cdot \operatorname{Alt}(z) = (-1)^\tau \operatorname{Alt}(z)$, so Alt lands in $\mathcal{T}^k_{\text{alt}}(M)$; and if z is alternating then $(-1)^\sigma \sigma \cdot z = (-1)^\sigma (-1)^\sigma z = z$ for every σ, so $\operatorname{Alt}(z) = k!\,z$. Idempotence and the splitting follow as before.

That Alt kills $A^k(M)$ needs the pairing argument. Let $z = m_1 \otimes \cdots \otimes m_k$ have $m_a = m_b$ with

$a \neq b$ and set $\tau = (a\ b)$. Then $\tau \cdot z = z$, because the two interchanged entries are equal. Pairing each σ with $\sigma\tau$ gives $(\sigma\tau) \cdot z = \sigma \cdot (\tau \cdot z) = \sigma \cdot z$ while $(-1)^{\sigma\tau} = -(-1)^{\sigma}$, so the two terms cancel and $\operatorname{Alt}(z) = 0$. Hence $\frac{1}{k!} \operatorname{Alt}$ induces a map on $\bigwedge^k(M)$.

For the inverse, let $\pi_A : \mathcal{T}^k(M) \to \bigwedge^k(M)$ be the quotient map. By proposition 16.1.9, $\pi_A(\sigma \cdot z) = (-1)^{\sigma}\pi_A(z)$ on simple tensors, so

$$\pi_A\left(\tfrac{1}{k!}\operatorname{Alt}(z)\right) = \tfrac{1}{k!}\sum_{\sigma \in S_k}(-1)^{\sigma}(-1)^{\sigma}\pi_A(z) = \pi_A(z)$$

using $(-1)^{\sigma}(-1)^{\sigma} = 1$; and $\frac{1}{k!}\operatorname{Alt}$ is the identity on $\mathcal{T}^k_{\mathrm{alt}}(M)$. So the two maps are mutually inverse, as we sought to show.

Defining an invariant element by averaging over a finite group, which is what $\frac{1}{k!}\operatorname{Sym}$ does, is a technique that recurs whenever the order of the group is invertible in the ring; it is the mechanism behind Maschke's theorem ([9, Maschke's Theorem]), where averaging over a finite group produces an invariant complement to a subrepresentation. When $k!$ is not invertible the comparison genuinely fails, and the exercises give a witness.

Exercise 16.1.1

1. Show that $C^k(M)$ is already generated by the elements $z - \tau \cdot z$ with z a simple tensor and τ a transposition, so that symmetry of a multilinear map need only be checked on transpositions.
2. Let R have characteristic 2. Show that a k-multilinear map satisfies the sign rule of lemma 16.1.4 if and only if it is symmetric, and exhibit a symmetric trilinear map over $\mathbb{Z}/2\mathbb{Z}$ that is not alternating.
3. Show that if M is generated by fewer than k elements then $\bigwedge^k(M) = 0$.
4. Let $\varphi\colon M \to N$ be a surjection of R-modules. Show that the induced maps $\mathcal{S}^k(M) \to \mathcal{S}^k(N)$ and $\bigwedge^k(M) \to \bigwedge^k(N)$ are surjective. (Use theorem 15.2.2 on $\mathcal{T}^k$ first, then check that generators map to generators.)
5. Take $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z}$. Compute $\mathcal{T}^2(M)$, $\mathcal{S}^2(M)$ and $\bigwedge^2(M)$.
6. Show that $\operatorname{Alt} \circ \operatorname{Sym} = 0$ on $\mathcal{T}^k(M)$ for every $k \geq 2$, over any commutative ring.
7. Continuing the previous exercise with $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z}$, where $2!$ is not a unit, compute $\mathcal{T}^2_{\mathrm{alt}}(M)$ and compare it with $\bigwedge^2(M)$. Conclude that the invertibility hypothesis in proposition 16.1.12 cannot be dropped: the two modules are not merely joined by a different isomorphism, they are not isomorphic at all.

16.2 Graded Rings and Graded Modules

The tensor powers of definition 16.1.1 come with a bookkeeping device that has so far gone unused. Associativity (theorem 15.2.8) identifies $\mathcal{T}^k(M) \otimes_R \mathcal{T}^l(M)$ with $\mathcal{T}^{k+l}(M)$, so a k-tensor times an l-tensor is a $(k+l)$-tensor: the index adds. A ring carrying an index of this kind is called graded, and the three algebras of section 16.3 are all built by taking a graded ring apart and putting it back together modulo relations that respect the index. Grading is therefore developed first, in its own right.

Definition 16.2.1: Graded Ring

A ring S is a *graded ring* if it comes with a decomposition

$$S = \bigoplus_{k \geq 0} S_k$$

of its underlying additive group into subgroups such that

$$S_k S_l \subseteq S_{k+l} \qquad \text{for all } k, l \geq 0$$

An element of S_k is *homogeneous of degree* k, and S_k is the *homogeneous component of* S *of degree* k. Every $s \in S$ has a unique expression $s = \sum_k s_k$ with $s_k \in S_k$ and all but finitely many s_k zero; the s_k are the *homogeneous components* of s.

The base ring plays no role in the definition, and the letter S is used for the graded ring throughout to keep it distinct from a ring R that may be acting. Note also that S_k is only required to be an additive subgroup: the multiplicative structure is carried entirely by the containments $S_k S_l \subseteq S_{k+l}$. What those containments force is the following.

Proposition 16.2.2: The Degree-Zero Part

Let $S = \bigoplus_{k \geq 0} S_k$ be a graded ring. Then

1. $1 \in S_0$, and S_0 is a subring of S;
2. each S_k is an S_0-module, and S is an S_0-algebra when S_0 lies in the centre of S;
3. $S_+ := \bigoplus_{k>0} S_k$ is a two-sided ideal of S, and $S/S_+ \cong S_0$ as rings.

***Proof*:**

1. Write $1 = \sum_k e_k$ with $e_k \in S_k$. For $x \in S_l$ the product $x \cdot 1 = \sum_k x e_k$ has $x e_k \in S_{l+k}$, so comparing homogeneous components of degree l on both sides of $x = x \cdot 1$ gives $x = x e_0$. Additivity extends this from homogeneous x to all of S, and the same argument on the other side gives $e_0 x = x$. So e_0 is a two-sided identity, and an identity is unique, so $1 = e_0 \in S_0$. Now $S_0 S_0 \subseteq S_0$ makes S_0 closed under multiplication, it is an additive subgroup by construction, and it contains 1; so it is a subring.

2. $S_0 S_k \subseteq S_k$ and $S_k S_0 \subseteq S_k$, so each S_k is a module over the subring S_0 on either side, and S is their direct sum. The algebra statement is definition 12.1.13, whose hypothesis is exactly that the image of S_0 is central.

3. S_+ is an additive subgroup, and for $x \in S_l$ and $y \in S_k$ with $k > 0$ one has $xy \in S_{k+l}$ with $k + l > 0$; additivity extends this to all of S, on both sides, so S_+ is a two-sided ideal. The projection $S \to S_0$ sending s to its degree-zero component is additive, sends 1 to 1 by (1), and is multiplicative because the degree-zero component of a product st is $s_0 t_0$, every

other contribution $s_k t_l$ with $k + l = 0$ and $k, l \geq 0$ being absent. Its kernel is exactly S_+, completing the proof.

Note that $S_i \subseteq S_j$ is not to be expected for $i < j$: the components are summands, not a filtration. In $R[x]$ graded by degree, $S_1 = Rx$ and $S_2 = Rx^2$ are disjoint apart from 0.

Definition 16.2.3: Graded Modules, Ideals and Homomorphisms

Let $S = \bigoplus_{k \geq 0} S_k$ and $T = \bigoplus_{k \geq 0} T_k$ be graded rings.

1. A left S-module M is *graded* if it comes with a decomposition $M = \bigoplus_{k \geq 0} M_k$ into additive subgroups with $S_k M_l \subseteq M_{k+l}$ for all k, l. Elements of M_k are *homogeneous of degree k*.

2. An ideal $I \subseteq S$ is a *graded ideal*, or a *homogeneous ideal*, if

$$I = \bigoplus_{k \geq 0} (I \cap S_k)$$

   Writing $I_k := I \cap S_k$, a graded ideal is in particular a graded S-module.

3. A ring homomorphism $\varphi \colon S \to T$ is *graded* if $\varphi(S_k) \subseteq T_k$ for every k.

Example 16.2.4: Graded Rings

1. Any ring S is graded with $S_0 = S$ and $S_k = 0$ for $k > 0$. This is the *trivial* grading, and it shows the definition imposes nothing by itself.

2. For any ring R, the polynomial ring $R[x]$ is graded by degree:

$$R[x] = \bigoplus_{k \geq 0} Rx^k$$

   the degree-k component being the free R-module on the single monomial x^k, and $Rx^k \cdot Rx^l \subseteq Rx^{k+l}$. It is worth being precise about what the components are, since the powers $(x)^k$ of the ideal (x) are *not* a grading: they are nested, $(x) \supseteq (x)^2 \supseteq \cdots$, so their sum is not direct.

3. More generally $R[x_1, \ldots, x_n]$ is graded with k-th component the free R-module on the monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n = k$; these are the *homogeneous polynomials* of degree k. Here $S_0 = R$, and S_+ is the ideal $(x_1, \ldots, x_n)$ of polynomials with zero constant term.

4. Let R be commutative and $I \subseteq R$ an ideal. The *Rees algebra* of R with respect to I is

$$\mathrm{Rees}_R(I) := \bigoplus_{k \geq 0} I^k = R \oplus I \oplus I^2 \oplus \cdots$$

   with multiplication induced by that of R. It is graded because $I^k I^l = I^{k+l}$, so in particular $I^k I^l \subseteq I^{k+l}$. The summands are indexed copies of the ideals I^k, kept formally separate; the sum inside R would not be direct, for the same reason as in (2).[a]

[a]Rees algebras are the algebraic form of a blow-up in algebraic geometry, and are taken up alongside the Artin-Rees lemma in *Commutative Algebra* [8, Artin-Rees Lemma].

Quotienting by a graded ideal is exactly the operation that keeps the grading.

Proposition 16.2.5: Quotient of a Graded Ring by a Graded Ideal

Let $S = \bigoplus_k S_k$ be a graded ring and $I \subseteq S$ a graded ideal, and write $I_k = I \cap S_k$. Then S/I is a graded ring with

$$(S/I)_k \cong S_k/I_k$$

and the quotient map $S \to S/I$ is a graded ring homomorphism.

***Proof*:**
Let $\pi\colon S \to S/I$ be the quotient map and let $(S/I)_k := \pi(S_k)$, an additive subgroup. These generate S/I additively, since π is surjective and the S_k generate S. Their sum is direct: if $\sum_k \pi(s_k) = 0$ with $s_k \in S_k$ and almost all zero, then $\sum_k s_k \in I$, so every $s_k \in I$ by definition 16.2.3(2), whence $\pi(s_k) = 0$ for each k. The containment $(S/I)_k(S/I)_l \subseteq (S/I)_{k+l}$ follows by applying π to $S_kS_l \subseteq S_{k+l}$. So $S/I = \bigoplus_k (S/I)_k$ is a graded ring, and $\pi(S_k) = (S/I)_k$ says exactly that π is graded.

For the displayed isomorphism, the restriction $\pi|_{S_k}\colon S_k \to (S/I)_k$ is a surjective homomorphism of additive groups with kernel $S_k \cap I = I_k$, so $S_k/I_k \cong (S/I)_k$ by theorem 3.1.14, completing the proof.

The same bookkeeping applies to a graded homomorphism, whose kernel and image decompose degree by degree.

Proposition 16.2.6: Kernel and Image of a Graded Homomorphism

Let $\varphi\colon S \to T$ be a graded ring homomorphism and write $\varphi_k := \varphi|_{S_k}\colon S_k \to T_k$. Then

$$\ker\varphi = \bigoplus_{k\geq 0} \ker\varphi_k \qquad \text{and} \qquad \varphi(S) = \bigoplus_{k\geq 0} \varphi(S_k)$$

so $\ker\varphi$ is a graded ideal of S and $\varphi(S)$ is a graded subring of T.

***Proof*:**
If $s \in \ker\varphi$ has homogeneous components s_k, then $\sum_k \varphi(s_k) = 0$ with $\varphi(s_k) \in T_k$, so each $\varphi(s_k) = 0$ by uniqueness of the decomposition in T; hence $s \in \bigoplus_k \ker\varphi_k$. The reverse inclusion is clear, and the displayed equality shows $\ker\varphi$ is graded.

For the image, $\varphi(S) = \sum_k \varphi(S_k)$ because φ is additive and the S_k span S, and the sum is direct because $\varphi(S_k) \subseteq T_k$ and the T_k are independent. Closure under multiplication is $\varphi(S_k)\varphi(S_l) \subseteq \varphi(S_{k+l})$, which holds because φ is multiplicative, completing the proof.

Those two results give the last of several equivalent ways to recognize a graded ideal.

Proposition 16.2.7: Characterizing Graded Ideals

Let $S = \bigoplus_k S_k$ be a graded ring and $I \subseteq S$ an ideal. The following are equivalent.

1. I is a graded ideal.
2. For every $s \in I$, each homogeneous component of s lies in I.
3. I is generated by homogeneous elements.
4. I is the kernel of a graded ring homomorphism out of S.

***Proof*:**
(1) ⇒ (2). If $I = \bigoplus_k (I \cap S_k)$ then every $s \in I$ is a finite sum $\sum_k t_k$ with $t_k \in I \cap S_k$. That expression is also the decomposition of s into homogeneous components, which is unique, so each component $s_k = t_k$ lies in I.

(2) ⇒ (3). By (2) the set H of homogeneous elements of I is contained in I, and every $s \in I$ is the sum of its homogeneous components, all of which lie in H. So H generates I.

(3) ⇒ (1). Let I be generated by homogeneous elements h_α, say $h_\alpha \in S_{d_\alpha}$. An element $s \in I$ is a finite sum $s = \sum_\alpha a_\alpha h_\alpha$ with $a_\alpha \in S$. Decomposing each $a_\alpha = \sum_j a_{\alpha,j}$ into homogeneous components,

$$s = \sum_\alpha \sum_j a_{\alpha,j} h_\alpha$$

and each $a_{\alpha,j} h_\alpha$ is homogeneous of degree $j + d_\alpha$ and lies in I. Collecting the terms of each fixed degree therefore exhibits the homogeneous components of s as elements of I, so $s \in \bigoplus_k (I \cap S_k)$. The reverse inclusion is automatic, giving (1).

(1) ⇒ (4). By proposition 16.2.5 below, S/I is a graded ring and the quotient map is a graded homomorphism; its kernel is I.

(4) ⇒ (2). Let $\varphi\colon S \to T$ be graded with kernel I, and let $s \in I$ have homogeneous components s_k. Then $0 = \varphi(s) = \sum_k \varphi(s_k)$ with $\varphi(s_k) \in T_k$, and the decomposition of 0 into homogeneous components of T is unique, so every $\varphi(s_k) = 0$ and every $s_k \in I$, completing the proof.

For prime ideals the grading cuts the work down: only homogeneous elements need to be tested.

Corollary 16.2.8: Prime Graded Ideals

Let S be a graded commutative ring and $I \subseteq S$ a graded ideal with $I \neq S$. Then I is prime if and only if for all *homogeneous* $f, g \in S$ with $fg \in I$, either $f \in I$ or $g \in I$.

***Proof*:**
If I is prime the condition holds, since it is a special case of the definition.

Conversely, suppose the condition holds and let $f, g \in S$ with $fg \in I$ but $f \notin I$ and $g \notin I$. By proposition 16.2.7(2) some homogeneous component of f lies outside I, and only finitely many components are nonzero, so there is a largest index a with $f_a \notin I$; choose b likewise for g.

Consider the homogeneous component of fg in degree $a + b$, namely $\sum_{i+j=a+b} f_i g_j$. In any term with $i > a$ one has $f_i \in I$ by maximality of a, so $f_i g_j \in I$; and in any term with $i < a$ one has $j = a + b - i > b$, so $g_j \in I$ and again $f_i g_j \in I$. Every term except $f_a g_b$ therefore lies in I. Since $fg \in I$ and I is graded, the whole degree-$(a + b)$ component lies in I by proposition 16.2.7(2), and subtracting the other terms gives $f_a g_b \in I$. As f_a and g_b are homogeneous, the hypothesis forces $f_a \in I$ or $g_b \in I$, contradicting the choice of a and b. So I is prime, completing the proof.

Finally, the operations on ideals of definition 9.4.1 preserve gradedness.

Proposition 16.2.9: Arithmetic of Homogeneous Ideals

Let S be a graded commutative ring and let $I, J \subseteq S$ be graded ideals. Then each of

$$I + J, \qquad IJ, \qquad I \cap J, \qquad \sqrt{I}$$

is a graded ideal.

***Proof*:**
Throughout, proposition 16.2.7 is used to switch between the definition and the two convenient criteria.

$I + J$ and IJ. By criterion (3) each of I and J has a generating set of homogeneous elements, say H_I and H_J. Then $H_I \cup H_J$ generates $I + J$ and consists of homogeneous elements, and the products $\left\{hh' \middle| h \in H_I,\ h' \in H_J\right\}$ generate IJ and are homogeneous, since $S_k S_l \subseteq S_{k+l}$. Criterion (3) applies to both.

$I \cap J$. Let $s \in I \cap J$. Each homogeneous component of s lies in I by criterion (2) applied to I, and in J by the same criterion applied to J, hence in $I \cap J$. Criterion (2) gives the claim.

$\sqrt{I}$. That $\sqrt{I}$ is an ideal is proposition 9.4.3. For criterion (2), let $s \in \sqrt{I}$ and argue by induction on the number of nonzero homogeneous components of s. If that number is 0 there is nothing to prove. Otherwise let d be the largest degree with $s_d \neq 0$, and choose n with $s^n \in I$. The homogeneous component of s^n in the top degree nd is s_d^n, because every other product of n components has total degree strictly less than nd. Since $s^n \in I$ and I is graded, criterion (2) gives $s_d^n \in I$, so $s_d \in \sqrt{I}$. Then $s - s_d \in \sqrt{I}$, as $\sqrt{I}$ is an ideal, and it has one fewer nonzero homogeneous component, so the inductive hypothesis puts all of its components in $\sqrt{I}$. Those together with s_d are the components of s, completing the proof.

Exercise 16.2.1

1. Show that a graded ring S has $S_0 \neq 0$ unless $S = 0$, and give an example of a graded ring in which $S_1 = 0$ but $S_2 \neq 0$.
2. Show that an ideal of $R[x_1, \ldots, x_n]$ is graded if and only if it is generated by homogeneous polynomials, and use this to show that $(1 + x) \subseteq \mathbb{Z}[x]$ is not graded.
3. Show that $(y - x^2) \subseteq k[x, y]$ is not a graded ideal for the standard grading, while $(y,\, y - x^2)$ is. (For the second, identify the ideal.)
4. Let S be a graded ring. Show that S_+ is the unique largest graded ideal I with $I \cap S_0 = 0$, and that S is generated as an S_0-algebra by any set of homogeneous elements whose images generate S_+/S_+^2 as an S_0-module.

5. Show that a graded ideal I is radical if and only if $s_d^n \in I$ implies $s_d \in I$ for every homogeneous s_d and every $n \geq 1$.
6. Show that $I = (x^2 - y^2) \subseteq k[x, y]$ is a graded ideal that is not prime, and exhibit homogeneous $f, g \notin I$ with $fg \in I$. (By corollary 16.2.8 such a homogeneous witness must exist whenever a graded ideal fails to be prime.)

16.3 The Tensor, Symmetric and Exterior Algebras

section 16.1 produced one module for each degree k. Adding them up produces a ring. The multiplication is concatenation of tensors, which raises the degree by exactly the amount it should, so the result is a graded ring in the sense of definition 16.2.1; and the sum acquires a universal property that the individual degrees do not have, one about homomorphisms into an algebra rather than into a module. The symmetric and exterior algebras are then quotients of it, and the last thing to check is that quotienting the whole and quotienting degree by degree agree.

Definition 16.3.1: Tensor Algebra

Let R be a commutative ring and M an R-module. The *tensor algebra* of M is the R-module

$$\mathcal{T}(M) := \bigoplus_{k \geq 0} \mathcal{T}^k(M) = R \oplus M \oplus \mathcal{T}^2(M) \oplus \cdots$$

equipped with the multiplication determined on homogeneous elements by the composite

$$\mathcal{T}^k(M) \times \mathcal{T}^l(M) \xrightarrow{\iota} \mathcal{T}^k(M) \otimes_R \mathcal{T}^l(M) \xrightarrow{\sim} \mathcal{T}^{k+l}(M)$$

where the second map is the associativity isomorphism of theorem 15.2.8 when $k, l \geq 1$, and is example 15.1.13 when $k = 0$ or $l = 0$, and extended to all of $\mathcal{T}(M)$ by additivity in each variable.

On simple tensors the multiplication is concatenation,

$$(m_1 \otimes \cdots \otimes m_i)(m_1' \otimes \cdots \otimes m_j') = m_1 \otimes \cdots \otimes m_i \otimes m_1' \otimes \cdots \otimes m_j'$$

which is what makes it computable. Note that this is not the multiplication of theorem 15.2.14, which was $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$ and required both factors to be algebras already; here M is only a module, and the product of two elements of M lands not in M but one degree higher.

Theorem 16.3.2: $\mathcal{T}(M)$ is an R-algebra

Let R be a commutative ring and M an R-module.

1. $\mathcal{T}(M)$ is an associative R-algebra with identity $1 \in R = \mathcal{T}^0(M)$, and it is graded, with $\mathcal{T}^k(M)\mathcal{T}^l(M) \subseteq \mathcal{T}^{k+l}(M)$. It contains M as the homogeneous component of degree 1.
2. (universal property) For every R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ there is a unique R-algebra homomorphism $\Phi\colon \mathcal{T}(M) \to A$ with $\Phi|_M = \varphi$:

$$\begin{tikzcd} \mathcal{T}(M) \arrow[r, "\Phi"] & A \\ M \arrow[u, "\iota"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

***Proof*:**

1. The multiplication is R-bilinear on each pair of homogeneous components, being a composite of the R-bilinear ι with an R-module isomorphism, and it is extended biadditively, so it distributes over addition and commutes with the R-action; that is the compatibility definition 12.1.13 asks for. The containment $\mathcal{T}^k(M)\mathcal{T}^l(M) \subseteq \mathcal{T}^{k+l}(M)$ holds by construction, which is the grading.

   For associativity, both $(x, y, z) \mapsto (xy)z$ and $(x, y, z) \mapsto x(yz)$ are R-trilinear, so by theorem 15.2.11 each is determined by its values on triples of simple tensors, where both are the concatenation

   $$m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j \otimes m''_1 \otimes \cdots \otimes m''_h$$

   since concatenation of finite sequences is associative. For the identity, $1 \in R = \mathcal{T}^0(M)$ acts on $\mathcal{T}^k(M)$ through the isomorphism $R \otimes_R \mathcal{T}^k(M) \cong \mathcal{T}^k(M)$ of example 15.1.13, which is the identity map. Finally $\mathcal{T}^1(M) = M$ by definition 16.1.1.

2. Fix A and φ. For each $k \geq 1$ the map

   $$M^k \to A, \qquad (m_1, \ldots, m_k) \mapsto \varphi(m_1)\varphi(m_2)\cdots\varphi(m_k)$$

   is k-multilinear: φ is R-linear and multiplication in A is R-linear in each variable by definition 12.1.13. So theorem 15.2.11 gives a unique R-module homomorphism $\Phi_k\colon \mathcal{T}^k(M) \to A$ with $\Phi_k(m_1 \otimes \cdots \otimes m_k) = \varphi(m_1)\cdots\varphi(m_k)$. Let $\Phi_0\colon R \to A$ be the structure map of the R-algebra A, and let $\Phi := \sum_k \Phi_k$ on the direct sum, which is a well-defined R-module homomorphism because every element has only finitely many nonzero components.

   Φ is multiplicative: both $(x, y) \mapsto \Phi(xy)$ and $(x, y) \mapsto \Phi(x)\Phi(y)$ are R-bilinear, and on a pair of simple tensors both give $\varphi(m_1)\cdots\varphi(m_i)\varphi(m'_1)\cdots\varphi(m'_j)$, so they agree by theorem 15.1.12. It sends 1 to 1 because Φ_0 does. Restricted to $\mathcal{T}^1(M) = M$ it is φ.

   For uniqueness, $\mathcal{T}(M)$ is generated as an R-algebra by M: the simple tensors $m_1 \otimes \cdots \otimes m_k$ are products of elements of M, and they generate $\mathcal{T}^k(M)$ as an R-module by lemma 15.1.4. An R-algebra homomorphism is determined by its values on an algebra generating set, so any Φ' agreeing with φ on M equals Φ, completing the proof.

Part (2) says that $\mathcal{T}(-)$ converts modules into algebras in the freest possible way: prescribing an algebra homomorphism out of $\mathcal{T}(M)$ needs no more than prescribing a module homomorphism out of M. In categorical language $\mathcal{T}(-)$ is left adjoint to the functor forgetting the multiplication of an R-algebra, which is the same shape of statement as corollary 15.3.5; the general theory is [7, Adjoint Functors].

The two quotients of interest are cut out by degree-two relations. Before taking them, one compatibility has to be settled, since section 16.1 defined $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ by quotienting $\mathcal{T}^k(M)$ by a *submodule*, while the algebras are obtained by quotienting $\mathcal{T}(M)$ by an *ideal*. The two operations agree, but that is a statement to prove rather than to observe.

Lemma 16.3.3: The Defining Ideals Are Graded, Degree by Degree

Let R be a commutative ring and M an R-module. Let $\mathcal{C}(M) \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by all $m \otimes m' - m' \otimes m$ with $m, m' \in M$, and let $\mathcal{A}(M) \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by all $m \otimes m$ with $m \in M$. Then both are graded ideals, and for every $k \geq 0$

$$\mathcal{C}(M) \cap \mathcal{T}^k(M) = C^k(M), \qquad \mathcal{A}(M) \cap \mathcal{T}^k(M) = A^k(M)$$

where $C^k(M)$ and $A^k(M)$ are the submodules of definition 16.1.5 and definition 16.1.7.

***Proof*:**
Both ideals are generated by homogeneous elements of degree 2, so both are graded by proposition 16.2.7(3), and it remains to identify the degree-k parts. Write $\mathcal{C}(M)_k$ for $\mathcal{C}(M) \cap \mathcal{T}^k(M)$.

The symmetric case, $C^k(M) \subseteq \mathcal{C}(M)_k$. By definition $C^k(M)$ is generated by the differences $z - \sigma \cdot z$ for simple tensors z and $\sigma \in S_k$. Since adjacent transpositions generate S_k, and

$$z - (\tau_1 \cdots \tau_r) \cdot z = \sum_{j=1}^{r} \Big((\tau_{j+1} \cdots \tau_r) \cdot z - (\tau_j \tau_{j+1} \cdots \tau_r) \cdot z \Big)$$

telescopes with each summand of the form $w - \tau \cdot w$ for a simple tensor w and an adjacent transposition τ, it suffices to treat $\tau = (i\ i+1)$. There

$$z - \tau \cdot z = (m_1 \otimes \cdots \otimes m_{i-1})\big(m_i \otimes m_{i+1} - m_{i+1} \otimes m_i\big)(m_{i+2} \otimes \cdots \otimes m_k)$$

a product in $\mathcal{T}(M)$ of an element of $\mathcal{T}^{i-1}(M)$, a generator of $\mathcal{C}(M)$, and an element of $\mathcal{T}^{k-i-1}(M)$, hence an element of $\mathcal{C}(M)$; and it is homogeneous of degree k.

The symmetric case, $\mathcal{C}(M)_k \subseteq C^k(M)$. An element of $\mathcal{C}(M)$ is a finite sum of products $x\,(m \otimes m' - m' \otimes m)\,y$ with $x, y \in \mathcal{T}(M)$. Expanding x and y into homogeneous components and then into simple tensors, and keeping only the terms of total degree k, every such term is of the form $w - \tau \cdot w$ for a simple tensor $w \in \mathcal{T}^k(M)$ and an adjacent transposition τ, by the display above read backwards. Each lies in $C^k(M)$, and $C^k(M)$ is a submodule, so the sum does too.

The alternating case. The inclusion $\mathcal{A}(M)_k \supseteq A^k(M)$ needs slightly more, since a generator of $A^k(M)$ is a simple tensor whose repeated entries need not be adjacent. Note first that for $u, v \in M$,

$$u \otimes v + v \otimes u = (u+v) \otimes (u+v) - u \otimes u - v \otimes v \in \mathcal{A}(M)$$

so interchanging two adjacent factors of a simple tensor changes it by an element of $\mathcal{A}(M)$, up to sign. Given $z = m_1 \otimes \cdots \otimes m_k$ with $m_a = m_b$ and $a < b$, repeatedly interchanging adjacent factors moves m_b into position $a+1$, so that $z \equiv \pm z'$ modulo $\mathcal{A}(M)$, where z' is a simple tensor with two equal adjacent entries. Then z' factors as a product of an element of $\mathcal{T}^{a-1}(M)$, the generator $m_a \otimes m_a$, and an element of $\mathcal{T}^{k-a-1}(M)$, so $z' \in \mathcal{A}(M)$ and hence $z \in \mathcal{A}(M)$. Being homogeneous of degree k, it lies in $\mathcal{A}(M)_k$.

The reverse inclusion $\mathcal{A}(M)_k \subseteq A^k(M)$ follows as in the symmetric case: the degree-k part of a product $x\,(m \otimes m)\,y$ expands into simple tensors each having two equal adjacent entries, and those are generators of $A^k(M)$, completing the proof.

Corollary 16.3.4: The Quotients Are Graded

With the notation above, $\mathcal{T}(M)/\mathcal{C}(M)$ and $\mathcal{T}(M)/\mathcal{A}(M)$ are graded rings, and their homogeneous components of degree k are $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ respectively.

***Proof*:**
By lemma 16.3.3 both ideals are graded, so proposition 16.2.5 applies and gives graded quotients with degree-k component $\mathcal{T}^k(M)/(\mathcal{C}(M) \cap \mathcal{T}^k(M))$, which is $\mathcal{T}^k(M)/C^k(M) = \mathcal{S}^k(M)$ by the same lemma and definition 16.1.5; the alternating case is identical, completing the proof.

Definition 16.3.5: Symmetric Algebra

The *symmetric algebra* of an R-module M is the quotient R-algebra

$$\mathcal{S}(M) := \mathcal{T}(M)/\mathcal{C}(M)$$

where $\mathcal{C}(M)$ is the two-sided ideal generated by the elements $m \otimes m' - m' \otimes m$ for $m, m' \in M$. By corollary 16.3.4 it is graded with $\mathcal{S}(M)_k = \mathcal{S}^k(M)$, and the product of the images of $m_1, \ldots, m_k$ is written $m_1 m_2 \cdots m_k$.

Theorem 16.3.6: The Symmetric Algebra Is the Free Commutative Algebra

Let R be a commutative ring and M an R-module. Then $\mathcal{S}(M)$ is a commutative R-algebra, and for every *commutative* R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ there is a unique R-algebra homomorphism $\Phi\colon \mathcal{S}(M) \to A$ restricting to φ on $M = \mathcal{S}^1(M)$.

***Proof*:**
The algebra $\mathcal{S}(M)$ is generated by the image of M, since $\mathcal{T}(M)$ is generated by M and the quotient map is surjective. The images of any two elements of M commute, because $m \otimes m' - m' \otimes m \in \mathcal{C}(M)$. An associative algebra generated by a set of pairwise commuting elements is commutative: the set of elements commuting with everything in the generating set is a subalgebra, so it is the whole algebra, and then the same argument applied a second time gives commutativity outright. So $\mathcal{S}(M)$ is commutative.

Let A be commutative and $\varphi\colon M \to A$ an R-module homomorphism. By theorem 16.3.2 there is a unique R-algebra homomorphism $\Psi\colon \mathcal{T}(M) \to A$ with $\Psi|_M = \varphi$. On a generator of $\mathcal{C}(M)$,

$$\Psi(m \otimes m' - m' \otimes m) = \varphi(m)\varphi(m') - \varphi(m')\varphi(m) = 0$$

because A is commutative. An algebra homomorphism vanishing on a set of generators of a two-sided ideal vanishes on the ideal, so $\mathcal{C}(M) \subseteq \ker \Psi$ and Ψ factors through $\mathcal{S}(M)$, giving Φ. Uniqueness holds because $\mathcal{S}(M)$ is generated as an R-algebra by the image of M, as we sought to show.

Definition 16.3.7: Exterior Algebra

The *exterior algebra* of an R-module M is the quotient R-algebra

$$\bigwedge(M) := \mathcal{T}(M)/\mathcal{A}(M)$$

where $\mathcal{A}(M)$ is the two-sided ideal generated by the elements $m \otimes m$ for $m \in M$. By corollary 16.3.4 it is graded with $\bigwedge(M)_k = \bigwedge^k(M)$, and the image of $m_1 \otimes \cdots \otimes m_k$ is written $m_1 \wedge \cdots \wedge m_k$.

Proposition 16.3.8: The Exterior Algebra Is the Free Algebra on Squares Zero

Let R be a commutative ring and M an R-module. For every R-algebra A and every R-module homomorphism $\varphi\colon M \to A$ satisfying $\varphi(m)^2 = 0$ for all $m \in M$, there is a unique R-algebra homomorphism $\Phi\colon \bigwedge(M) \to A$ restricting to φ on $M = \bigwedge^1(M)$.

***Proof*:**
As in theorem 16.3.6, theorem 16.3.2 gives $\Psi\colon \mathcal{T}(M) \to A$ with $\Psi|_M = \varphi$, and $\Psi(m \otimes m) = \varphi(m)^2 = 0$ kills every generator of $\mathcal{A}(M)$, hence the ideal. The induced Φ is unique because $\bigwedge(M)$ is generated as an R-algebra by the image of M, completing the proof.

The wedge product is not commutative, and it is not anticommutative either; the sign depends on the degrees. proposition 16.1.9 gave the rule for elements of degree 1, and the general rule follows from it.

Proposition 16.3.9: Graded Commutativity of the Exterior Algebra

Let $a \in \bigwedge^p(M)$ and $b \in \bigwedge^q(M)$. Then

$$a \wedge b = (-1)^{pq}\, b \wedge a$$

In particular elements of odd degree anticommute and elements of even degree are central among homogeneous elements; the identity $ab = -ba$ fails as soon as one of the two degrees is even.

***Proof*:**
Both $(a,b) \mapsto a \wedge b$ and $(a,b) \mapsto (-1)^{pq} b \wedge a$ are R-bilinear on $\bigwedge^p(M) \times \bigwedge^q(M)$, so by theorem 15.1.12 it suffices to compare them on pairs of products of degree-one elements, which generate each factor. Take $a = m_1 \wedge \cdots \wedge m_p$ and $b = m'_1 \wedge \cdots \wedge m'_q$. Moving m'_1 leftwards past each of $m_1, \ldots, m_p$ in turn costs a factor of -1 each time by proposition 16.1.9, so p sign changes; doing this for each of the q elements of b in turn gives $(-1)^{pq}$ in total and rearranges $a \wedge b$ into $b \wedge a$. Hence $a \wedge b = (-1)^{pq} b \wedge a$.

For the last sentence, take $p = q = 1$ to get $m \wedge m' = -m' \wedge m$, and $p = 2$, $q = 1$ to get $a \wedge m = m \wedge a$, which is not $-m \wedge a$ unless $2a \wedge m = 0$, completing the proof.

Example 16.3.10: Tensor Algebras

1. Let $R = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$. Then $(\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Q}/\mathbb{Z}) = 0$ by the third exercise of section 15.1, so $\mathcal{T}^k(M) = 0$ for every $k \geq 2$ and
$$\mathcal{T}(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Q}/\mathbb{Z}$$
Addition is componentwise, and the multiplication is
$$(r, \overline{p})(s, \overline{q}) = (rs,\ r\overline{q} + s\overline{p})$$
since the product of two degree-one elements lands in $\mathcal{T}^2 = 0$. This is the *square-zero extension* of $\mathbb{Z}$ by $\mathbb{Q}/\mathbb{Z}$: the degree-one part is an ideal whose square vanishes.

2. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$. Then $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ by proposition 15.2.13, and inductively $\mathcal{T}^k(M) \cong \mathbb{Z}/n\mathbb{Z}$ for every $k \geq 1$, so
$$\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus \cdots \cong \mathbb{Z}[x]/(nx)$$
the isomorphism sending the degree-k generator to x^k; the relation $nx = 0$ records that the degree-one part is killed by n, and every higher degree inherits it.

3. Take $M = \mathbb{Q}$ as a module over itself. Then $\mathcal{T}^k(\mathbb{Q}) \cong \mathbb{Q}$ for every k by example 15.1.13, so $\mathcal{T}(\mathbb{Q}) \cong \mathbb{Q}[x]$ as graded $\mathbb{Q}$-algebras. Here the tensor algebra happens to be commutative, because M has rank one; for a module of rank at least two it never is, as section 16.4 will show.

Exercise 16.3.1

1. Show that $\mathcal{T}(M)$ is commutative if and only if $\mathcal{S}(M) = \mathcal{T}(M)$, and that this happens if and only if $m \otimes m' = m' \otimes m$ for all $m, m' \in M$.

2. Show that $\mathcal{S}(M)$ satisfies the universal property of theorem 16.3.6 with respect to R-algebra homomorphisms into non-commutative algebras as well, provided the image of M is required to consist of pairwise commuting elements.

3. Let $I \subseteq \mathcal{T}(M)$ be the two-sided ideal generated by $\mathcal{T}^{\geq d}(M) := \bigoplus_{k \geq d} \mathcal{T}^k(M)$ for some $d \geq 1$. Show that $I = \mathcal{T}^{\geq d}(M)$, that it is graded, and identify the quotient $\mathcal{T}(M)/I$ as an R-module.

4. Show that $\bigwedge(M)$ is commutative when $2 = 0$ in R, and give an example over $\mathbb{Z}$ where it is not. (For the example, use the determinant to produce an alternating bilinear map on $\mathbb{Z}^2$ and deduce that $e_1 \wedge e_2$ has infinite order.)
5. Let $\varphi\colon M \to N$ be an R-module homomorphism. Show that it induces graded R-algebra homomorphisms $\mathcal{T}(\varphi)$, $\mathcal{S}(\varphi)$ and $\bigwedge(\varphi)$ between the corresponding algebras, and that each is surjective when φ is.
6. Show that $\mathcal{S}(M \oplus N) \cong \mathcal{S}(M) \otimes_R \mathcal{S}(N)$ as R-algebras, using theorem 16.3.6 together with theorem 15.2.17, which identifies the right-hand side as the coproduct of commutative R-algebras.

16.4 Bases, Ranks and the Determinant

Nothing so far has required M to be free. When it is, all three graded algebras become computable: a basis of M determines a basis of each homogeneous component, and the ranks are the three counts one would guess from thinking of the components as words, unordered words, and sets. The exterior case then delivers the determinant, which reappears not as the defining sum of definition 13.4.5 but as the action of an endomorphism on a rank-one module.

Theorem 16.4.1: Bases of the Tensor Powers

Let R be a commutative ring and let M be a free R-module with basis $\{e_i\}_{i\in I}$. Then $\mathcal{T}^k(M)$ is free with basis
$$\left\{\, e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_k} \;\middle|\; (i_1, \ldots, i_k) \in I^k \,\right\}$$
indexed by all k-tuples. If I is finite of size n, then $\mathcal{T}^k(M)$ is free of rank n^k.

***Proof*:**
Induction on k. For $k = 0$ and $k = 1$ the claim is $\mathcal{T}^0(M) = R$ and $\mathcal{T}^1(M) = M$. For the inductive step, $\mathcal{T}^k(M) = \mathcal{T}^{k-1}(M) \otimes_R M$ by definition 16.1.1, and $\mathcal{T}^{k-1}(M)$ is free on the $(k-1)$-tuples by hypothesis, so corollary 15.2.6 makes $\mathcal{T}^k(M)$ free on the products of a basis element of each factor, which are exactly the k-tuples. The rank count is $|I^k| = n^k$, completing the proof.

In particular $\mathcal{T}(M)$ is the free associative R-algebra on the set $\{e_i\}$: its elements are the R-linear combinations of words in the e_i, multiplied by concatenation. For $|I| \geq 2$ it is not commutative, since $e_1 \otimes e_2$ and $e_2 \otimes e_1$ are distinct basis elements, which settles the question raised in example 16.3.10(3).

The symmetric algebra is the polynomial ring, and it is quickest to identify the whole algebra at once and read off the components afterwards.

Theorem 16.4.2: The Symmetric Algebra of a Free Module

Let R be a commutative ring and M free with basis $e_1, \ldots, e_n$. Then there is an isomorphism of graded R-algebras

$$\mathcal{S}(M) \;\cong\; R[x_1, \ldots, x_n]$$

carrying e_i to x_i. Consequently $\mathcal{S}^k(M)$ is free with basis the monomials

$$e_{i_1} e_{i_2} \cdots e_{i_k}, \qquad 1 \le i_1 \le i_2 \le \cdots \le i_k \le n$$

and has rank $\binom{n+k-1}{k}$.

***Proof*:**
The polynomial ring is a commutative R-algebra, so the R-module homomorphism $M \to R[x_1, \ldots, x_n]$ sending e_i to x_i induces, by theorem 16.3.6, an R-algebra homomorphism

$$\Phi \colon \mathcal{S}(M) \longrightarrow R[x_1, \ldots, x_n], \qquad \Phi(e_i) = x_i$$

Conversely the universal property of polynomial rings (theorem 9.2.4) gives an R-algebra homomorphism $\Psi \colon R[x_1, \ldots, x_n] \to \mathcal{S}(M)$ with $\Psi(x_i) = e_i$, since $\mathcal{S}(M)$ is commutative by theorem 16.3.6.

Each composite is an R-algebra endomorphism fixing a set of algebra generators, hence the identity: $\mathcal{S}(M)$ is generated by the image of M, and $R[x_1, \ldots, x_n]$ by the x_i. So Φ is an isomorphism. It is graded because it sends the degree-one generators e_i to the degree-one generators x_i and both algebras are generated in degree one, so it carries a product of k generators to a product of k generators.

Restricting to degree k, $\mathcal{S}^k(M) = \mathcal{S}(M)_k$ by corollary 16.3.4, and the degree-k component of $R[x_1, \ldots, x_n]$ is free on the monomials of total degree k (example 16.2.4(3)). Their preimages are the products $e_{i_1} \cdots e_{i_k}$ with indices weakly increasing, one for each monomial. Counting monomials of degree k in n variables is the standard problem of distributing k identical items into n labelled boxes, whose answer is $\binom{n+k-1}{k}$, completing the proof.

The exterior powers need a separate argument, because there is no ambient ring to compare against. Spanning is immediate; independence is produced by exhibiting, for each proposed basis element, an alternating map that detects it. Those maps are minors, and the fact that a determinant is alternating is proposition 13.4.6. There is no circularity: det is the sum of definition 13.4.5, defined without any reference to exterior algebra, and it is only in theorem 16.4.7 below that the two are identified.

Theorem 16.4.3: Bases of the Exterior Powers

Let R be a commutative ring and M free with basis $e_1, \ldots, e_n$. Then $\bigwedge^k(M)$ is free with basis

$$\{\, e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k} \mid 1 \le i_1 < i_2 < \cdots < i_k \le n \,\}$$

so it has rank $\binom{n}{k}$; in particular $\bigwedge^k(M) = 0$ for $k > n$, and $\bigwedge^n(M)$ is free of rank one on $e_1 \wedge \cdots \wedge e_n$. The whole exterior algebra $\bigwedge(M)$ is then free of rank 2^n.

***Proof*:**

Spanning. By theorem 16.4.1 the tensors $e_{i_1} \otimes \cdots \otimes e_{i_k}$ span $\mathcal{T}^k(M)$, so their images span $\bigwedge^k(M)$. If two indices coincide the image is 0 by proposition 16.1.9; otherwise the same proposition rearranges the factors into strictly increasing order up to a sign. So the displayed set spans.

Independence. Fix a strictly increasing $I = (i_1 < \cdots < i_k)$ and define

$$\varphi_I\colon M^k \to R, \qquad \varphi_I(v_1, \ldots, v_k) = \det\Big((v_j)_{i_a} \Big)_{1 \le a, j \le k}$$

the determinant of the $k \times k$ matrix whose (a, j) entry is the i_a-th coordinate of v_j in the basis $e_1, \ldots, e_n$. Each coordinate is R-linear in v_j, and the determinant is R-linear in each column by proposition 13.4.6(3), so φ_I is k-multilinear; and if $v_j = v_{j'}$ for $j \neq j'$ then the matrix has two equal columns, so φ_I vanishes by proposition 13.4.6(4). Thus φ_I is alternating, and theorem 16.1.8 induces $\overline{\varphi_I}\colon \bigwedge^k(M) \to R$.

Evaluate $\overline{\varphi_I}$ on a proposed basis element $e_J = e_{j_1} \wedge \cdots \wedge e_{j_k}$ with J strictly increasing. The matrix in question has (a, b) entry δ_{i_a, j_b}. If $I = J$ it is the identity matrix and the determinant is 1 by proposition 13.4.6(1). If $I \neq J$ then, both being strictly increasing k-tuples, some i_a lies outside J, and then row a is zero, so the determinant is 0. Hence $\overline{\varphi_I}(e_J) = \delta_{IJ}$.

Now suppose $\sum_J c_J e_J = 0$ is a relation among the proposed basis elements, with $c_J \in R$. Applying $\overline{\varphi_I}$ gives $c_I = 0$ for every I. So the set is independent, and with spanning it is a basis.

Counts. The strictly increasing k-tuples from $\{1, \ldots, n\}$ are the k-element subsets, of which there are $\binom{n}{k}$; for $k > n$ there are none, so $\bigwedge^k(M) = 0$, and for $k = n$ there is exactly one. Summing over k gives $\sum_{k=0}^n \binom{n}{k} = 2^n$, completing the proof.

Example 16.4.4: Exterior Algebras of Small Vector Spaces

Let F be a field and V an F-vector space.

1. If $\dim V = 1$ with basis v, then theorem 16.4.3 gives $\bigwedge^0(V) = F$, $\bigwedge^1(V) = V$, and $\bigwedge^i(V) = 0$ for every $i \ge 2$, since there are no strictly increasing i-tuples from a one-element set. So

$$\bigwedge(V) = F \oplus V$$

with every product of two elements of V equal to zero.

2. Now let $\dim V = 2$ with basis v, v'. The only strictly increasing pairs give $\bigwedge^2(V) = F\,(v \wedge v')$, and $\bigwedge^i(V) = 0$ for $i \ge 3$. Computing a general product of two vectors,

$$\begin{aligned}(av + bv') \wedge (cv + dv') &= ac\,(v \wedge v) + ad\,(v \wedge v') + bc\,(v' \wedge v) + bd\,(v' \wedge v') \\ &= (ad - bc)\, v \wedge v'\end{aligned}$$

using $v \wedge v = v' \wedge v' = 0$ and $v' \wedge v = -v \wedge v'$. The coefficient is the determinant of the matrix with those columns, which is theorem 16.4.7 in the case $n = 2$.

3. If $\dim V = 3$ with basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, then $\bigwedge^2(V)$ has rank $\binom{3}{2} = 3$, the same as V itself, and

$$v \wedge w = (v_1 w_2 - v_2 w_1)(\mathbf{e}_1 \wedge \mathbf{e}_2) + (v_2 w_3 - v_3 w_2)(\mathbf{e}_2 \wedge \mathbf{e}_3) + (v_1 w_3 - v_3 w_1)(\mathbf{e}_1 \wedge \mathbf{e}_3)$$

Identifying the basis of $\bigwedge^2(V)$ with that of V turns $\wedge$ into the cross product of vector calculus. The coincidence is exactly $\binom{3}{2} = 3$ and happens in no other dimension, which is why the cross product is a three-dimensional phenomenon.

4. Let $R = \mathbb{Z}[x, y]$ and $I = (x, y)$, both regarded as R-modules. The inclusion $\varphi\colon I \hookrightarrow R$ induces $\bigwedge^2(\varphi)\colon \bigwedge^2(I) \to \bigwedge^2(R)$. Now R is free of rank one, so $\bigwedge^2(R) = 0$ by theorem 16.4.3, while $x \wedge y \neq 0$ in $\bigwedge^2(I)$. For the latter, every $f \in I$ has zero constant term, so its coefficients f_x and f_y of x and y are well defined; the map $(f, g) \mapsto f_x g_y - f_y g_x$ into $R/I = \mathbb{Z}$ is R-bilinear, because multiplying f by $h \in R$ scales both coefficients by $h(0, 0)$, and it is alternating. So theorem 16.1.8 induces $\bigwedge^2(I) \to \mathbb{Z}$ sending $x \wedge y$ to 1. Hence $\bigwedge^2(\varphi)$ is not injective even though φ is: the exterior power of an injection need not be an injection, which is the same failure of exactness recorded for $\otimes$ in section 15.5.

Exterior powers turn direct sums into a graded tensor product, in the same way symmetric powers turn them into an ordinary one. The formula below is what makes the exterior algebra computable on a module built from simpler pieces, and it is the tool behind the fact that a stably free module of rank one is free.

Theorem 16.4.5: Exterior Powers of a Direct Sum

Let R be a commutative ring and M, N two R-modules. Then for every $k \geq 0$ there is an isomorphism of R-modules

$$\bigwedge^k(M \oplus N) \;\cong\; \bigoplus_{i+j=k} \bigwedge^i(M) \otimes_R \bigwedge^j(N)$$

natural in M and N, carrying $\alpha \otimes \beta$ to $\alpha \wedge \beta$ when $\bigwedge M$ and $\bigwedge N$ are read inside $\bigwedge(M \oplus N)$.

***Proof*:**
Step 1: a twisted product on the tensor product. Put $T = \bigwedge(M) \otimes_R \bigwedge(N)$, graded by total degree, so that T_k is the right-hand side. Define a multiplication on homogeneous elements by

$$(\alpha \otimes \beta) \cdot (\alpha' \otimes \beta') = (-1)^{\deg \beta \, \deg \alpha'} (\alpha \wedge \alpha') \otimes (\beta \wedge \beta')$$

and extend bilinearly. This is well defined, both wedge products being bilinear, and it is associative: comparing the two bracketings of a triple product, each produces $(\alpha \wedge \alpha' \wedge \alpha'') \otimes (\beta \wedge \beta' \wedge \beta'')$ with sign exponent $\deg \beta(\deg \alpha' + \deg \alpha'') + \deg \beta' \deg \alpha''$. The element $1 \otimes 1$ is a unit, so T is an R-algebra.

Step 2: a map out of $\bigwedge(M \oplus N)$. Consider the R-module map $\varphi\colon M \oplus N \to T$ sending (m, n) to $m \otimes 1 + 1 \otimes n$, landing in degree one. Then

$$\varphi(m, n)^2 = (m \wedge m) \otimes 1 + m \otimes n + (-1)^{1 \cdot 1} m \otimes n + 1 \otimes (n \wedge n) = 0$$

the two middle terms coming from $(m \otimes 1)(1 \otimes n)$ and $(1 \otimes n)(m \otimes 1)$ with signs $(-1)^0$ and $(-1)^1$, and the outer terms vanishing since $v \wedge v = 0$. So proposition 16.3.8 gives a unique R-algebra map $\Phi\colon \bigwedge(M \oplus N) \to T$ restricting to φ.

Step 3: a map back. The inclusions $M \hookrightarrow M \oplus N$ and $N \hookrightarrow M \oplus N$ induce algebra maps $\iota_M\colon \bigwedge M \to \bigwedge(M \oplus N)$ and ι_N, again by proposition 16.3.8, since the composites $M \to \bigwedge(M \oplus N)$

and $N \to \bigwedge(M \oplus N)$ square to zero. The assignment $(\alpha, \beta) \mapsto \iota_M(\alpha) \wedge \iota_N(\beta)$ is R-bilinear, hence induces $\Psi\colon T \to \bigwedge(M \oplus N)$ on the tensor product, and Ψ respects the twisted multiplication: moving $\iota_N(\beta)$ past $\iota_M(\alpha')$ costs exactly $(-1)^{\deg \beta \deg \alpha'}$ by proposition 16.3.9.

Step 4: the two composites are the identity. Both $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are algebra maps, and on degree-one elements they are the identity: $\Psi(\Phi(m, n)) = \Psi(m \otimes 1) + \Psi(1 \otimes n) = m + n = (m, n)$, and similarly in the other direction. Since $\bigwedge(M \oplus N)$ is generated as an algebra by its degree-one part, the uniqueness clause of proposition 16.3.8 makes $\Psi \circ \Phi$ the identity; and T is generated by the elements $m \otimes 1$ and $1 \otimes n$, which gives the other. Both maps preserve total degree, so restricting to degree k gives the displayed isomorphism, as we sought to show.

Corollary 16.4.6: A Stably Free Module of Rank One Is Free

Let R be a commutative ring and P a finitely generated projective R-module with $P_\mathfrak{p} \cong R_\mathfrak{p}$ for every prime $\mathfrak{p}$. If $P \oplus R^n \cong R^{n+1}$ for some $n \geq 0$, then $P \cong R$.

***Proof*:**

First, $\bigwedge^i(P) = 0$ for $i \geq 2$. Exterior powers are quotients of tensor powers, so they commute with localization; hence $\left(\bigwedge^i P\right)_\mathfrak{p} \cong \bigwedge^i(P_\mathfrak{p}) \cong \bigwedge^i(R_\mathfrak{p})$, which vanishes for $i \geq 2$ by theorem 16.4.3. A module vanishing at every prime is zero.

Now apply theorem 16.4.5 with $M = P$ and $N = R^n$ in degree $n+1$. The summand indexed by (i, j) vanishes unless $i \leq 1$ and $j \leq n$, the latter by theorem 16.4.3; since $i + j = n + 1$, the only survivor is $(i, j) = (1, n)$. Hence

$$R \cong \bigwedge^{n+1}(R^{n+1}) \cong \bigwedge^{n+1}(P \oplus R^n) \cong \bigwedge^{1}(P) \otimes_R \bigwedge^{n}(R^n) \cong P$$

using theorem 16.4.3 for the two free top powers, as we sought to show.

The top exterior power of a free module of rank n is free of rank one, so an endomorphism of M acts on it by multiplication by a scalar. That scalar is the determinant, and this is the basis-free description of it.

Theorem 16.4.7: The Determinant as the Top Exterior Power

Let R be a commutative ring, let M be free of rank n, and let $\varphi\colon M \to M$ be an R-module homomorphism with matrix A with respect to some basis. Then

$$\bigwedge^{n}(\varphi)\colon \bigwedge^{n}(M) \to \bigwedge^{n}(M)$$

is multiplication by $\det(A)$. Consequently $\det(A)$ does not depend on the chosen basis, and for endomorphisms φ, ψ of M,

$$\det(\varphi \circ \psi) = \det(\varphi)\det(\psi)$$

***Proof*:**
Let $e_1, \dots, e_n$ be the basis and $A = (a_{ij})$, so $\varphi(e_j) = \sum_i a_{ij} e_i$. By theorem 16.4.3 the module $\bigwedge^n(M)$ is free of rank one on $\omega := e_1 \wedge \dots \wedge e_n$, so it suffices to compute $\bigwedge^n(\varphi)(\omega) = \varphi(e_1) \wedge \dots \wedge \varphi(e_n)$.

Expanding each factor and using multilinearity of the wedge,

$$\varphi(e_1) \wedge \dots \wedge \varphi(e_n) = \sum_f \left(\prod_{j=1}^n a_{f(j)\,j} \right) e_{f(1)} \wedge \dots \wedge e_{f(n)}$$

the sum running over all functions $f\colon \{1, \dots, n\} \to \{1, \dots, n\}$. If f is not injective then two factors coincide and the wedge vanishes by proposition 16.1.9; an injective self-map of a finite set is a permutation, so only $f = \sigma \in S_n$ contributes, and there $e_{\sigma(1)} \wedge \dots \wedge e_{\sigma(n)} = (-1)^\sigma \omega$ by the same proposition. Therefore

$$\bigwedge^n(\varphi)(\omega) = \left(\sum_{\sigma \in S_n} (-1)^\sigma \prod_{j=1}^n a_{\sigma(j)\,j} \right) \omega = \det(A^t)\, \omega = \det(A)\, \omega$$

the last step by proposition 13.4.6(2). Since $\bigwedge^n(\varphi)$ is defined from φ alone, with no basis involved, the scalar it multiplies by is an invariant of φ; so any two bases give matrices with the same determinant.

Finally $\bigwedge^n$ is functorial by theorem 15.2.2, so $\bigwedge^n(\varphi \circ \psi) = \bigwedge^n(\varphi) \circ \bigwedge^n(\psi)$, and composing two multiplications by scalars multiplies the scalars, completing the proof.

The multiplicativity just obtained is proposition 13.4.7 again, now with a reason attached rather than a computation: determinants multiply because $\bigwedge^n$ is a functor into rank-one modules, and endomorphisms of a rank-one free module compose by multiplying their scalars. The Leibniz sum of definition 13.4.5 is what the scalar looks like once a basis is fixed.

Corollary 16.4.8: Invertibility and the Top Exterior Power

Let M be free of rank n over a commutative ring R and φ an endomorphism of M. Then φ is an isomorphism if and only if $\det(\varphi)$ is a unit of R.

***Proof*:**
If φ is an isomorphism with inverse ψ, then $\det(\varphi)\det(\psi) = \det(\mathrm{id}_M) = 1$ by theorem 16.4.7 and proposition 13.4.6(1), so $\det(\varphi)$ is a unit. The converse is proposition 13.4.10(1) applied to the matrix of φ, completing the proof.

Exercise 16.4.1

1. Let M be free of rank n and let $v_1, \dots, v_k \in M$ be linearly dependent. Show that $v_1 \wedge \dots \wedge v_k = 0$ in $\bigwedge^k(M)$.
2. Let F be a field with $-1 \neq 1$ and V a finite-dimensional F-vector space. Show that $V \otimes_F V \cong \mathcal{S}^2(V) \oplus \bigwedge^2(V)$, and check the dimensions against theorem 16.4.2 and theorem 16.4.3.

3. Verify that $\mathcal{T}^k$, $\mathcal{S}^k$ and $\bigwedge^k$ of a free module of rank n have ranks n^k, $\binom{n+k-1}{k}$ and $\binom{n}{k}$, and show that for $k = 2$ the last two add up to the first. Then show the analogous identity fails for $k = 3$ by computing all three ranks at $n = 2$.
4. Let M be free of rank n and φ an endomorphism. Show that $\bigwedge^k(\varphi)$ acts on the basis of theorem 16.4.3 by the $k \times k$ minors of the matrix of φ, and deduce that φ has rank less than k if and only if $\bigwedge^k(\varphi) = 0$ when R is a field.
5. Let M be free of finite rank n. Show that $\bigwedge(M)$ is finitely generated as an R-module, of rank 2^n, while $\mathcal{S}(M)$ is not finitely generated as an R-module unless $M = 0$.
6. Show that theorem 16.4.3 needs freeness. Take $R = \mathbb{Z}$ and $M = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, compute $\mathcal{T}^2(M)$ using theorem 15.2.3, identify the submodule $A^2(M)$, and conclude that $\bigwedge^2(M) \cong \mathbb{Z}/2\mathbb{Z}$. Compare with what $\binom{2}{2} = 1$ would predict for a free module of rank 2.

16.5⋆ The Koszul Complex

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

The exterior powers of a free module assemble into a complex, and when the ring elements defining it are well enough behaved that complex is exact. The result is the standard source of free resolutions in commutative algebra, and it is the first place where the exterior algebra does work that has nothing to do with volume or orientation.

Nothing later in this book uses it. It is here because the books that do use it are downstream: *Commutative Algebra* [8] and *Homological Algebra* [10] both build on the complex constructed below, and this is the last point in the prerequisite chain at which the exterior powers are available and the statement can be made.

Two definitions are needed, and neither has appeared so far.

Definition 16.5.1: Regular Sequence

Let R be a commutative ring. A sequence $a_1, \ldots, a_n \in R$ is a *regular sequence* if

1. a_1 is not a zero divisor in R, and for each $i \geq 2$ the image of a_i is not a zero divisor in $R/(a_1, \ldots, a_{i-1})$;
2. $(a_1, \ldots, a_n) \neq R$.

An ideal generated by a regular sequence is called a *regular ideal*.

Condition (1) is the one that does the work, and it says the a_i impose genuinely new conditions one after another: each is a non-zero-divisor on the quotient by the previous ones. In $R = k[x_1, \ldots, x_n]$ the variables themselves form a regular sequence, since $k[x_1, \ldots, x_n]/(x_1, \ldots, x_{i-1}) \cong k[x_i, \ldots, x_n]$ is a domain in which $x_i \neq 0$. Note that an initial segment $a_1, \ldots, a_m$ of a regular sequence is again one, directly from the definition; this is what makes induction on n available below.

The object the complex will produce is a free resolution in the sense of definition 12.3.27: an exact chain complex $\cdots \to F_1 \to F_0 \to M \to 0$ with every F_i free. Every module admits one, since a module is a quotient of a free module and the kernel is again a module; the content of a particular resolution is that it is short, explicit, or both. Over a principal ideal domain chapter 14 produced very short ones, because a submodule of a free module over a PID is free, so the process stops after one step. Over a general ring the process need not stop, and the Koszul complex is an explicit finite resolution available whenever the defining elements form a regular sequence.

The differential is contraction against a linear functional, and it exists for the same reason everything in this chapter exists: it is an alternating map, so theorem 16.1.8 builds it.

Proposition 16.5.2: Contraction Against a Functional

Let R be a commutative ring, F an R-module and $\lambda \in \operatorname{Hom}_R(F, R)$. For each $r \geq 1$ there is a unique R-module homomorphism

$$d_r \colon \bigwedge^{r}(F) \longrightarrow \bigwedge^{r-1}(F)$$

with

$$d_r(v_1 \wedge \cdots \wedge v_r) = \sum_{j=1}^{r} (-1)^{j-1} \lambda(v_j)\, v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_r$$

where $\widehat{v_j}$ means that the factor is omitted. Setting $d_0 = 0$, these satisfy the graded Leibniz rule

$$d(u \wedge w) = d(u) \wedge w + (-1)^p\, u \wedge d(w) \qquad \text{for } u \in \bigwedge^{p}(F)$$

***Proof*:**
Existence. The right-hand side, read as a function of $(v_1, \dots, v_r)$, is r-multilinear: each v_j appears once in each term, linearly, since λ is linear and $\wedge$ is multilinear.

It is alternating. Suppose $v_a = v_b =: v$ with $a < b$. Any term with $j \notin \{a, b\}$ still contains both v_a and v_b among its factors, so it vanishes by proposition 16.1.9. The two remaining terms are

$$(-1)^{a-1}\lambda(v)\, W_a \qquad \text{and} \qquad (-1)^{b-1}\lambda(v)\, W_b$$

where W_a omits the factor in position a and W_b the one in position b. In W_a the surviving copy of v sits in position $b-1$, and in W_b it sits in position a; moving it from one to the other is $b-1-a$ interchanges of adjacent factors, so $W_b = (-1)^{b-1-a} W_a$ by proposition 16.1.9. The sum of the two terms is therefore

$$\big((-1)^{a-1} + (-1)^{b-1}(-1)^{b-1-a}\big)\lambda(v) W_a = \big((-1)^{a-1} + (-1)^{a}\big)\lambda(v) W_a = 0$$

So the map is alternating, and theorem 16.1.8 gives d_r and its uniqueness.

Leibniz rule. Both sides are R-bilinear in (u, w), so it suffices to check them on products of degree-one elements, which generate each factor. For $u = v_1 \wedge \cdots \wedge v_p$ and $w = v_{p+1} \wedge \cdots \wedge v_{p+q}$, the defining formula for $d(u \wedge w)$ has $p+q$ terms. The first p of them are exactly the terms of $d(u) \wedge w$; the last q are the terms of $u \wedge d(w)$, except that in $d(u \wedge w)$ the term omitting v_{p+j} carries the sign $(-1)^{p+j-1}$ while in $u \wedge d(w)$ it carries $(-1)^{j-1}$. The two differ by $(-1)^p$, which is the stated factor, completing the proof.

Definition 16.5.3: Koszul Complex

Let R be a commutative ring and $a_1, \ldots, a_n \in R$. Let $F = R^n$ with basis $e_1, \ldots, e_n$ and let $\lambda\colon F \to R$ be the R-module homomorphism with $\lambda(e_i) = a_i$. The *Koszul complex* $K_\bullet(a_1, \ldots, a_n)$ is

$$0 \longrightarrow \bigwedge^n(F) \xrightarrow{d_n} \bigwedge^{n-1}(F) \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_2} \bigwedge^1(F) \xrightarrow{d_1} \bigwedge^0(F) = R$$

with d_r the contraction of proposition 16.5.2. On basis elements,

$$d_r\big(e_{i_1} \wedge \cdots \wedge e_{i_r}\big) = \sum_{j=1}^{r} (-1)^{j-1} a_{i_j}\, e_{i_1} \wedge \cdots \wedge \widehat{e_{i_j}} \wedge \cdots \wedge e_{i_r}$$

Proposition 16.5.4: The Koszul Complex Is a Complex

With the notation above, $d_{r-1} \circ d_r = 0$ for every $r \geq 1$, so $\operatorname{im}(d_r) \subseteq \ker(d_{r-1})$.

***Proof*:**
Both composites are R-linear, so it suffices to evaluate on a basis element $e_I = e_{i_1} \wedge \cdots \wedge e_{i_r}$ with $i_1 < \cdots < i_r$. Applying d_r and then d_{r-1} produces a sum of terms indexed by ordered pairs of distinct positions (j, k), the term being $\pm\, a_{i_j} a_{i_k}$ times the wedge with both factors omitted. Fix $j < k$ and collect the two terms that omit those two positions.

Removing position j first contributes the sign $(-1)^{j-1}$; in the shortened wedge the factor originally at position k now sits at position $k-1$, so removing it contributes $(-1)^{k-2}$, for a total of $(-1)^{j+k-3} = (-1)^{j+k-1}$. Removing position k first contributes $(-1)^{k-1}$, and the factor at position j is unmoved, so removing it contributes $(-1)^{j-1}$, for a total of $(-1)^{j+k-2} = (-1)^{j+k}$. The two signs are opposite and the coefficients $a_{i_j} a_{i_k}$ agree because R is commutative, so the two terms cancel. Every term cancels in such a pair, completing the proof.

Exactness is where the hypothesis on the a_i enters, and it is the reason regular sequences were defined.

Theorem 16.5.5: Exactness of the Koszul Complex

Let R be a commutative ring and let $a_1, \ldots, a_n$ be a regular sequence, with $I = (a_1, \ldots, a_n)$. Then $\ker(d_r) = \operatorname{im}(d_{r+1})$ for every $r \geq 1$, and $R/\operatorname{im}(d_1) = R/I$. Consequently

$$0 \longrightarrow \bigwedge^n(F) \longrightarrow \cdots \longrightarrow \bigwedge^1(F) \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

is a finite free resolution of R/I, of length n.

***Proof*:**
The statement about $R/\mathrm{im}(d_1)$ holds for any a_i: $d_1 = \lambda$ has image $\lambda(F) = (a_1, \dots, a_n) = I$. Freeness of each term is theorem 16.4.3. So only exactness at $r \geq 1$ is at issue, and it is proved by induction on n.

Base case $n = 1$. The complex is $0 \to R \xrightarrow{a_1} R$, and exactness at degree 1 says that multiplication by a_1 is injective, which is the statement that a_1 is not a zero divisor.

Inductive step. Let $n \geq 2$, let $F' \subseteq F$ be free on $e_1, \dots, e_{n-1}$ and let $K'_\bullet$ be the Koszul complex of $a_1, \dots, a_{n-1}$ on F', with differentials d'. An initial segment of a regular sequence is regular, so the inductive hypothesis applies to K'.

By theorem 16.4.3 every basis element of $\bigwedge^r(F)$ either involves e_n or does not, which gives

$$\bigwedge^r(F) = \bigwedge^r(F') \;\oplus\; \Big(\bigwedge^{r-1}(F') \wedge e_n\Big)$$

so every element is uniquely $x + y \wedge e_n$ with $x \in \bigwedge^r(F')$ and $y \in \bigwedge^{r-1}(F')$. The Leibniz rule of proposition 16.5.2, together with $d(e_n) = a_n$, computes the differential in these coordinates:

$$d_r\big(x + y \wedge e_n\big) = \Big(d'_r(x) + (-1)^{r-1} a_n\, y\Big) + \Big(d'_{r-1}(y)\Big) \wedge e_n \tag{16.1}$$

Suppose now $r \geq 1$ and $d_r(x + y \wedge e_n) = 0$. Comparing the two summands,

$$d'_{r-1}(y) = 0 \qquad \text{and} \qquad d'_r(x) + (-1)^{r-1} a_n y = 0$$

Producing v with $d'_r(v) = y$. If $r \geq 2$ then $r - 1 \geq 1$, so the inductive hypothesis gives exactness of K' at degree $r-1$ and $y = d'_r(v)$ for some $v \in \bigwedge^r(F')$. If $r = 1$ then $y \in \bigwedge^0(F') = R$ and the first equation is vacuous, so a different argument is needed. There the second equation reads $\lambda'(x) + a_n y = 0$ with $\lambda' = d'_1$, so $a_n y \in (a_1, \dots, a_{n-1})$. Because $a_1, \dots, a_n$ is a regular sequence, a_n is not a zero divisor on $R/(a_1, \dots, a_{n-1})$, so $y \in (a_1, \dots, a_{n-1})$; writing $y = \sum_{i<n} c_i a_i$ and setting $v = \sum_{i<n} c_i e_i \in \bigwedge^1(F')$ gives $d'_1(v) = y$.

Producing u. From $d'_r(x) = -(-1)^{r-1} a_n y = (-1)^r a_n\, d'_r(v)$ one gets

$$d'_r\big(x - (-1)^r a_n v\big) = 0$$

and $r \geq 1$, so the inductive hypothesis gives $x - (-1)^r a_n v = d'_{r+1}(u)$ for some $u \in \bigwedge^{r+1}(F')$.

Conclusion. Apply (16.1) in degree $r + 1$ to $u + v \wedge e_n$:

$$d_{r+1}\big(u + v \wedge e_n\big) = \Big(d'_{r+1}(u) + (-1)^r a_n v\Big) + d'_r(v) \wedge e_n = x + y \wedge e_n$$

So every cycle in degree r is a boundary, which is exactness, completing the proof.

Example 16.5.6: The Koszul Complex of the Variables
Take $R = k[x_1, \dots, x_n]$ over a field k and the regular sequence $x_1, \dots, x_n$, so that $I = (x_1, \dots, x_n)$ and $R/I \cong k$. The theorem produces a free resolution of k by free R-modules of ranks $\binom{n}{k}$, of total length n:

$$0 \to R \to R^n \to R^{\binom{n}{2}} \to \cdots \to R^n \to R \to k \to 0$$

For $n = 1$ this is $0 \to k[x] \xrightarrow{x} k[x] \to k \to 0$, which is the observation that x is not a zero divisor. For $n = 2$ it is

$$0 \to R \xrightarrow{\binom{-x_2}{x_1}} R^2 \xrightarrow{(x_1 \quad x_2)} R \to k \to 0$$

and exactness in the middle says precisely that a pair (f, g) with $fx_1 + gx_2 = 0$ is a multiple of $(-x_2, x_1)$: the only relations among x_1 and x_2 are the obvious ones. Regularity is exactly what makes that true, and it fails for a non-regular sequence, as the exercises show.

The construction is the algebraic origin of several later ones. In *Homological Algebra* [10] the Koszul complex resolves the residue field in the proof of the Hilbert syzygy theorem, in *Commutative Algebra* [8] its homology measures how far a sequence is from being regular, and the same complex reappears in intersection theory, where the exactness above expresses that n hypersurfaces meeting in the expected dimension impose independent conditions.

Exercise 16.5.1

1. Show that x, y is a regular sequence in $k[x, y]$, and that x, x is not. Write out the Koszul complex for x, x and exhibit an element of $\ker(d_1)$ that is not in $\operatorname{im}(d_2)$, so that theorem 16.5.5 genuinely needs its hypothesis.
2. Show that a permutation of a regular sequence need not be regular, using $a_1 = x$, $a_2 = y(1-x)$, $a_3 = z(1 - x)$ in $k[x, y, z]$. (Check that a_1, a_2, a_3 is regular but a_2, a_3, a_1 is not.)
3. Show that $d_n \colon \bigwedge^n(F) \to \bigwedge^{n-1}(F)$ is injective for a regular sequence, and identify $\bigwedge^n(F)$ and the image of d_n explicitly when $n = 2$.
4. Show that the Koszul complex of $a_1, \ldots, a_n$ is, up to isomorphism, the tensor product over R of the n two-term complexes $0 \to R \xrightarrow{a_i} R \to 0$, in the sense that its degree-r term is the direct sum of the products of r of the left-hand copies with $n-r$ of the right-hand ones. (Only the module structure is asked for, not the differential.)
5. Write out the Koszul complex of x, y, z in $k[x, y, z]$ explicitly, giving the matrices of d_3, d_2 and d_1 with respect to the bases of theorem 16.4.3, and verify $d_2 d_3 = 0$ and $d_1 d_2 = 0$ by direct computation.

16.6 Summary

One module M, four classes of map on its powers, and four objects representing them. The tables below state what was established, each entry pointing at the result that establishes it.

The representing objects

Each class of multilinear map on M^k is represented by a module, and each representing object is a quotient of the tensor power by exactly what the class is required to kill.

Maps on M^k	**Represented by**	**Universal property**	**Rank**
k-multilinear	$\mathcal{T}^k(M)$	theorem 15.2.11	n^k
symmetric k-multilinear	$\mathcal{S}^k(M)$	theorem 16.1.6	$\binom{n+k-1}{k}$
alternating k-multilinear	$\bigwedge^k(M)$	theorem 16.1.8	$\binom{n}{k}$

The ranks are for M free of rank n, and are theorem 16.4.1, theorem 16.4.2 and theorem 16.4.3 respectively. In particular $\bigwedge^k(M) = 0$ once $k > n$, and $\bigwedge^n(M)$ is free of rank one, which is what makes the determinant a scalar.

The graded algebras

Summing the powers over k turns each family into a ring, graded in the sense of definition 16.2.1, and each acquires a universal property about homomorphisms into an algebra.

Algebra	**Construction**	**Universal for maps into**	**Reference**
$\mathcal{T}(M)$	$\bigoplus_k \mathcal{T}^k(M)$	any R-algebra	theorem 16.3.2
$\mathcal{S}(M)$	$\mathcal{T}(M)/\mathcal{C}(M)$	commutative R-algebras	theorem 16.3.6
$\bigwedge(M)$	$\mathcal{T}(M)/\mathcal{A}(M)$	algebras where $\varphi(m)^2 = 0$	proposition 16.3.8

That the two routes to $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ agree, one quotienting $\mathcal{T}^k(M)$ by a submodule and the other quotienting $\mathcal{T}(M)$ by an ideal, is lemma 16.3.3 and corollary 16.3.4. For M free of rank n the three algebras are the free associative algebra on n generators, the polynomial ring $R[x_1, \ldots, x_n]$ (theorem 16.4.2), and a free module of rank 2^n (theorem 16.4.3).

Graded rings

Statement	**Reference**
$1 \in S_0$; S_0 is a subring; S_+ is an ideal with $S/S_+ \cong S_0$	proposition 16.2.2
Graded $\Leftrightarrow$ components in I $\Leftrightarrow$ homogeneous generators $\Leftrightarrow$ a graded kernel	proposition 16.2.7
S/I is graded, with $(S/I)_k \cong S_k/I_k$	proposition 16.2.5
$\ker\varphi$ and $\operatorname{im}\varphi$ decompose degree by degree	proposition 16.2.6
Primality is testable on homogeneous elements	corollary 16.2.8
$I + J$, IJ, $I \cap J$ and $\sqrt{I}$ are graded	proposition 16.2.9

Signs

The exterior algebra is governed by signs, and three separate statements are involved, which are easy to conflate.

Statement	**Reference**
Alternating $\Rightarrow$ the sign rule; the converse needs 2 invertible	lemma 16.1.4
$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = (-1)^\sigma m_1 \wedge \cdots \wedge m_k$	proposition 16.1.9
$a \wedge b = (-1)^{pq} b \wedge a$ for $a \in \bigwedge^p$, $b \in \bigwedge^q$	proposition 16.3.9
$\mathcal{S}^k(M)$ and $\bigwedge^k(M)$ sit inside $\mathcal{T}^k(M)$ when $k!$ is invertible	proposition 16.1.12

In characteristic 2 the first row is strict: a symmetric bilinear form satisfies the sign rule vacuously without being alternating, so the vanishing condition of definition 16.1.3 is the one the exterior power is built from.

The determinant, and what fails

Statement	**Reference**
$\bigwedge^n(\varphi)$ is multiplication by $\det(\varphi)$	theorem 16.4.7
det is independent of the basis, and multiplicative	same
φ is an isomorphism $\Leftrightarrow$ $\det(\varphi)$ is a unit	corollary 16.4.8
$\bigwedge^k$ does not preserve injections	example 16.4.4(4)
$\bigwedge^k(M)$ need not be free when M is not	theorem 16.4.3

The second row is the conceptual reading of multiplicativity: determinants multiply because $\bigwedge^n$ is a functor landing in rank-one modules, where composition is multiplication of scalars. The elementary proof from the defining sum is proposition 13.4.7.

The exterior powers of a free module also assemble into a complex, which resolves $R/(a_1, \ldots, a_n)$ whenever the a_i are regular. That construction is section 16.5$\star$, marked as skippable for a reader of this book, and is the form in which *Commutative Algebra* [8] and *Homological Algebra* [10] take up the exterior algebra.

17⋆

Projective, Injective and Flat Modules

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

Given two R-modules A and C, which modules M fit into a short exact sequence

$$0 \to A \to M \to C \to 0\,?$$

This is the *extension problem.* Section 3.6 asked it for groups and left it open, and theorem 3.7.9 is exactly the reason it matters: the Jordan-Hölder theorem pins down the composition factors of a finite group without saying how they are assembled. For modules there is always at least one answer, $M = A \oplus C$, and for $A = \mathbb{Z}$ and $C = \mathbb{Z}/n\mathbb{Z}$ there are exactly two.

Exactness itself is the tool. A short exact sequence records how M is assembled from A and C, and the three constructions already available ($\mathrm{Hom}_R(D, -)$, $\mathrm{Hom}_R(-, D)$ and $- \otimes_R D$) each turn such a sequence into one that is exact everywhere except at a single spot. Whether the failure at that spot vanishes depends on the module D, and the modules for which it does are the projective, injective and flat modules of the title. What measures the failure when it does not vanish is a module $\mathrm{Ext}^1_R(C, A)$, whose elements turn out to be in bijection with the answers to the question above.

17.1 Exact Sequences, Splitting, and the Extension Problem

Definition 12.2.11 already said what it means for a chain of R-module homomorphisms to be exact: the image of each map is the kernel of the next. Two degenerate cases fix how short an exact sequence can be before it stops carrying information.

Proposition 17.1.1: Injective and Surjective Maps in Exact Sequences

Let A, B and C be R-modules.

1. The sequence $0 \to A \xrightarrow{\psi} B$ is exact at A if and only if ψ is injective.
2. The sequence $B \xrightarrow{\varphi} C \to 0$ is exact at C if and only if φ is surjective.

***Proof*:**

1. The only homomorphism $\alpha\colon 0 \to A$ sends $0 \mapsto 0$, so $\mathrm{im}(\alpha) = 0$. Exactness at A says $\mathrm{im}(\alpha) = \ker(\psi)$, which is therefore the assertion $\ker(\psi) = 0$, and a homomorphism is injective exactly when its kernel is trivial.
2. The only homomorphism $\beta\colon C \to 0$ sends every c to 0, so $\ker(\beta) = C$. Exactness at C says $\mathrm{im}(\varphi) = \ker(\beta) = C$, which is the assertion that φ is surjective.

Consequently $0 \to A \xrightarrow{f} B \to 0$ is exact precisely when f is both injective and surjective, so A and B are isomorphic and nothing has been learned. The first length that carries information is one longer.

Definition 17.1.2: Short Exact Sequence

A *short exact sequence* of R-modules is an exact sequence

$$0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$$

In this situation B is called an *extension of C by A*.

Unwinding proposition 17.1.1, the displayed sequence is short exact if and only if ψ is injective, φ is surjective and $\mathrm{im}(\psi) = \ker(\varphi)$. The first isomorphism theorem (theorem 12.2.10) then gives $C \cong B/\psi(A)$, while ψ identifies A with the submodule $\psi(A)$. An extension of C by A is therefore a module B containing a copy of A whose quotient by that copy is C, packaged in a form that names the two maps along with the three modules.

17.1.3 The Two Extension Conventions Definition 3.6.3 called G an extension of A by B when $G/A \cong B$, which is the order used in classical group theory and is the reverse of the order fixed above. The convention attached to short exact sequences is the one used for the rest of this chapter: the module being extended is the quotient C, and the module extending it is the submodule A. Both orders are current in the literature, and the safe reading is always the sequence itself.

Example 17.1.4: Short Exact Sequences

1. For any A and C the sequence
$$0 \to A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \to 0, \qquad \iota(a) = (a, 0), \quad \pi(a, c) = c$$
is short exact, since ι is injective, π is surjective, and $\mathrm{im}(\iota) = A \oplus 0 = \ker(\pi)$. Every pair of

modules therefore admits at least one extension.

2. Any homomorphism $\varphi\colon B \to C$ yields

$$0 \to \ker(\varphi) \hookrightarrow B \xrightarrow{\varphi} \mathrm{im}(\varphi) \to 0$$

and when φ is surjective the right-hand term is C itself.

3. Let M be an R-module and S a generating set for M. The homomorphism $\varphi\colon F(S) \to M$ extending the inclusion of S (theorem 12.3.11) is surjective, so with $K = \ker(\varphi)$,

$$0 \to K \hookrightarrow F(S) \xrightarrow{\varphi} M \to 0$$

is short exact. Every module is thus a quotient of a free module, and $F(S)$ is an extension of M by the module K of relations holding among the chosen generators.

Exactness is not confined to modules, nor to length three.

Example 17.1.5: The Centre and the Outer Automorphisms
Section 4.7.2 described the centre $Z(G)$ of a group and its outer automorphism group $\mathrm{Out}(G)$ as opposite ends of one construction, and exactness states this precisely. Let $\sigma\colon G \to \mathrm{Aut}(G)$ send g to conjugation by g, so that $\mathrm{im}(\sigma) = \mathrm{Inn}(G)$ (definition 4.7.3) and $\ker(\sigma) = Z(G)$, and let ρ be the quotient map by $\mathrm{Inn}(G)$, which is normal in $\mathrm{Aut}(G)$ by proposition 4.7.8. Then

$$1 \longrightarrow Z(G) \hookrightarrow G \xrightarrow{\sigma} \mathrm{Aut}(G) \xrightarrow{\rho} \mathrm{Out}(G) \longrightarrow 1$$

is exact: at $Z(G)$ because the inclusion is injective, at G because $\ker(\sigma) = Z(G)$, at $\mathrm{Aut}(G)$ because $\ker(\rho) = \mathrm{Inn}(G) = \mathrm{im}(\sigma)$, and at $\mathrm{Out}(G)$ because ρ is surjective. Cutting the sequence in the middle recovers theorem 4.7.4, the isomorphism $G/Z(G) \cong \mathrm{Inn}(G)$.

Example 17.1.4(1) makes $A \oplus C$ available for every pair. It is not always the only answer.

Example 17.1.6: Two Extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$
Fix $n \geq 2$. Two short exact sequences of $\mathbb{Z}$-modules with the same outer terms are

$$0 \to \mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0 \qquad \text{and} \qquad 0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{p} \mathbb{Z}/n\mathbb{Z} \to 0$$

where $\iota(z) = (z, \overline{0})$ and $\pi(z, \overline{a}) = \overline{a}$, while $\cdot n$ is multiplication by n and p is reduction modulo n. The first is short exact by example 17.1.4(1). The second is short exact because multiplication by n is injective on $\mathbb{Z}$, reduction is surjective, and $\mathrm{im}(\cdot n) = n\mathbb{Z} = \ker(p)$.

The two middle terms are not isomorphic, since $\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ has an element of order n and $\mathbb{Z}$ has no element of finite order other than 0. So $\mathbb{Z}/n\mathbb{Z}$ has at least two genuinely different extensions by $\mathbb{Z}$.

The same phenomenon occurs among finite modules. Both

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\overline{a} \mapsto \overline{2a}} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}/2\mathbb{Z} \to 0$$

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\iota} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

are short exact (for the first, the injection has image $\{\overline{0}, \overline{2}\}$, which is exactly the kernel of reduction modulo 2), and $\mathbb{Z}/4\mathbb{Z} \not\cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

17.1.1 Maps Between Short Exact Sequences

Two extensions of the same pair can differ, and there is more than one reasonable sense in which two of them are the same.

Definition 17.1.7: Homomorphism and Equivalence of Short Exact Sequences

Let

$$0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0 \qquad \text{and} \qquad 0 \to A' \xrightarrow{\psi'} B' \xrightarrow{\varphi'} C' \to 0$$

be short exact sequences of R-modules.

1. A *homomorphism of short exact sequences* from the first to the second is a triple (α, β, γ) of R-module homomorphisms making

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "\psi"] \arrow[d, "\alpha"'] & B \arrow[r, "\varphi"] \arrow[d, "\beta"'] & C \arrow[r] \arrow[d, "\gamma"'] & 0 \\ 0 \arrow[r] & A' \arrow[r, "\psi'"'] & B' \arrow[r, "\varphi'"'] & C' \arrow[r] & 0 \end{tikzcd}$$

   commute, that is $\beta \circ \psi = \psi' \circ \alpha$ and $\gamma \circ \varphi = \varphi' \circ \beta$.

2. The homomorphism is an *isomorphism of short exact sequences* if α, β and γ are all isomorphisms, and the extensions B and B' are then called *isomorphic*.

3. The two sequences are *equivalent* if $A = A'$ and $C = C'$ and there is an isomorphism as in (2) with $\alpha = \mathrm{id}_A$ and $\gamma = \mathrm{id}_C$. The extensions B and B' are then called *equivalent* extensions.

Equivalence is strictly stronger than isomorphism, and the gap between them is what the classification in section 17.6 counts. An isomorphism is free to rebuild the outer terms by any automorphisms it likes; an equivalence holds the outer terms fixed and asks only that the middles be matched compatibly. Two extensions of one fixed pair can accordingly be isomorphic as diagrams while remaining inequivalent as extensions.

Example 17.1.8: Homomorphisms, Isomorphisms and Equivalences

1. Let $n|m$ with $n, m > 1$ and put $k = m/n$. Take the second sequence of example 17.1.6 on top and, beneath it, the sequence built from $\ell(\overline{a}) = \overline{na}$ and reduction modulo n:

$$\begin{tikzcd} 0 \arrow[r] & \mathbb{Z} \arrow[r, "\cdot n"] \arrow[d, "\alpha"'] & \mathbb{Z} \arrow[r, "p"] \arrow[d, "\beta"'] & \mathbb{Z}/n\mathbb{Z} \arrow[r] \arrow[d, "\mathrm{id}"'] & 0 \\ 0 \arrow[r] & \mathbb{Z}/k\mathbb{Z} \arrow[r, "\ell"'] & \mathbb{Z}/m\mathbb{Z} \arrow[r, "p'"'] & \mathbb{Z}/n\mathbb{Z} \arrow[r] & 0 \end{tikzcd}$$

   with α and β the reductions modulo k and modulo m. The bottom row is short exact: ℓ is well defined because $a \equiv a' \pmod{k}$ gives $na \equiv na' \pmod{m}$, injective because $na \equiv 0$

(mod $m = nk$) gives $a \equiv 0 \pmod{k}$, and $\mathrm{im}(\ell) = n\mathbb{Z}/m\mathbb{Z} = \ker(p')$. Both squares commute, since $\ell(\alpha(a)) = \overline{na} = \beta(na)$ and $p'(\beta(z)) = \overline{z} = \mathrm{id}(p(z))$.

2. Let p be a prime and, for each $c \in \{1, \dots, p-1\}$, let

$$\mathcal{E}_c\colon \qquad 0 \to \mathbb{Z}/p\mathbb{Z} \xrightarrow{\psi} \mathbb{Z}/p^2\mathbb{Z} \xrightarrow{\varphi_c} \mathbb{Z}/p\mathbb{Z} \to 0, \qquad \psi(\overline{x}) = \overline{px}, \quad \varphi_c(\overline{y}) = \overline{cy}$$

   Each is short exact: ψ is well defined and injective because $px \equiv 0 \pmod{p^2}$ forces $x \equiv 0 \pmod{p}$; φ_c is surjective because c is invertible modulo p; and $\mathrm{im}(\psi) = p\mathbb{Z}/p^2\mathbb{Z} = \ker(\varphi_c)$.

   Any two of them are isomorphic. Taking $\alpha = \mathrm{id}$, $\beta = \mathrm{id}$ and γ multiplication by $c'c^{-1}$ makes both squares of definition 17.1.7 commute, and γ is an automorphism of $\mathbb{Z}/p\mathbb{Z}$ because $c'c^{-1}$ is a unit modulo p.

   No two of them are equivalent. Every endomorphism β of $\mathbb{Z}/p^2\mathbb{Z}$ is multiplication by some t, so $\beta \circ \psi = \psi$ forces $tp \equiv p \pmod{p^2}$, that is $t \equiv 1 \pmod{p}$. Then $\varphi_{c'} \circ \beta$ is multiplication by $c't \equiv c' \pmod{p}$, and requiring it to equal φ_c forces $c' \equiv c \pmod{p}$.

   So $\mathcal{E}_1, \dots, \mathcal{E}_{p-1}$ are $p-1$ pairwise inequivalent extensions of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$, all isomorphic to one another, and none of them is $\mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$.

3. The two extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ in example 17.1.6 are not even isomorphic, since an isomorphism of short exact sequences carries B to B' isomorphically and the two middle terms are not isomorphic.

A homomorphism of short exact sequences constrains its own three components: the two outer maps already determine a great deal about the middle one.

Proposition 17.1.9: Short Five Lemma

Let (α, β, γ) be a homomorphism of short exact sequences, in the notation of definition 17.1.7. Then

1. if α and γ are injective, so is β;
2. if α and γ are surjective, so is β;
3. if α and γ are isomorphisms, so is β, and the two sequences are isomorphic.

***Proof*:**
This is the first diagram chase in the book and is written out element by element. A chase is an algorithm with three moves: push an element along an available arrow, use commutativity of a square to identify the two routes around it, and use exactness of a row to move *backwards* along an arrow once the element is known to lie in that arrow's image. The bookkeeping is done by redrawing the relevant square with elements in place of modules, and $\mapsto$ in place of $\to$.

Step 1: β is injective. Let $b \in B$ with $\beta(b) = 0$; the goal is $b = 0$. Chase b around the right-hand square:

$$\begin{tikzcd} b \arrow[r, mapsto, "\varphi"] \arrow[d, mapsto, "\beta"'] & \varphi(b) \arrow[d, mapsto, "\gamma"] \\ 0 \arrow[r, mapsto, "\varphi'"'] & 0 \end{tikzcd}$$

Down then right gives $\varphi'(\beta(b)) = \varphi'(0) = 0$, and right then down gives $\gamma(\varphi(b))$. The square commutes, so $\gamma(\varphi(b)) = 0$; and γ is injective, so $\varphi(b) = 0$, that is $b \in \ker(\varphi)$.

Exactness is what makes the next move available. The top row is exact at B, so $\ker(\varphi) = \operatorname{im}(\psi)$, and there is an $a \in A$ with $\psi(a) = b$. Knowing only that φ kills b, the chase has produced a preimage in A, which is where the hypothesis on α can be used. Now chase a around the left-hand square:

$$\begin{tikzcd} a \arrow[r, mapsto, "\psi"] \arrow[d, mapsto, "\alpha"'] & b \arrow[d, mapsto, "\beta"] \\ \alpha(a) \arrow[r, mapsto, "\psi'"'] & 0 \end{tikzcd}$$

Commutativity gives $\psi'(\alpha(a)) = \beta(\psi(a)) = \beta(b) = 0$. The map ψ' is injective by proposition 17.1.1 applied to the bottom row, so $\alpha(a) = 0$, and α is injective by hypothesis, so $a = 0$. Therefore $b = \psi(a) = \psi(0) = 0$.

Step 2: β is surjective. Let $b' \in B'$; the goal is to produce a preimage under β. Both surjectivity hypotheses give a pullback. Since γ is surjective there is a $c \in C$ with $\gamma(c) = \varphi'(b')$, and since φ is surjective by proposition 17.1.1 there is a $b \in B$ with $\varphi(b) = c$. Chasing b gives

$$\begin{tikzcd} b \arrow[r, mapsto, "\varphi"] \arrow[d, mapsto, "\beta"'] & c \arrow[d, mapsto, "\gamma"] \\ \beta(b) \arrow[r, mapsto, "\varphi'"'] & \varphi'(b') \end{tikzcd}$$

where the bottom right entry is computed by commutativity:

$$\varphi'(\beta(b)) = \gamma(\varphi(b)) = \gamma(c) = \varphi'(b')$$

Hence $\varphi'(b' - \beta(b)) = 0$, so $b' - \beta(b) \in \ker(\varphi') = \operatorname{im}(\psi')$ by exactness of the bottom row, and $b' - \beta(b) = \psi'(a')$ for some $a' \in A'$. Since α is surjective, $a' = \alpha(a)$ for some $a \in A$, and the left square gives

$$b' - \beta(b) = \psi'(\alpha(a)) = \beta(\psi(a))$$

Therefore $b' = \beta(b) + \beta(\psi(a)) = \beta\big(b + \psi(a)\big)$, so b' lies in the image of β.

Step 3. If α and γ are isomorphisms then Steps 1 and 2 both apply, so β is injective and surjective, hence an isomorphism, and (α, β, γ) is an isomorphism of short exact sequences by definition 17.1.7(2).

Neither chase used the R-action, only the underlying additive groups, and the single subtraction in Step 2 can be written multiplicatively as $b'\beta(b)^{-1}$. The proposition and its proof therefore hold verbatim for short exact sequences of groups, abelian or not.

The next result relates two short exact sequences by splicing their kernels and cokernels into one long exact sequence. Its proof runs the same three moves, and from here on a chase states which move it is making rather than tracking every element.

Proposition 17.1.10: Mini Snake Lemma

Let (α, β, γ) be a homomorphism of short exact sequences, in the notation of definition 17.1.7. Then there is an exact sequence of R-modules

$$0 \to \ker(\alpha) \xrightarrow{\overline{\psi}} \ker(\beta) \xrightarrow{\overline{\varphi}} \ker(\gamma) \xrightarrow{d} \operatorname{coker}(\alpha) \xrightarrow{\overline{\psi}'} \operatorname{coker}(\beta) \xrightarrow{\overline{\varphi}'} \operatorname{coker}(\gamma) \to 0$$

in which $\overline{\psi}$ and $\overline{\varphi}$ are the restrictions of ψ and φ, the maps $\overline{\psi}'$ and $\overline{\varphi}'$ are induced by ψ' and φ' on cokernels, and d is the *connecting homomorphism* constructed in the proof.

***Proof*:**
The whole statement is read off the following diagram, in which the two middle rows are the given ones and the outer rows are the sequence being constructed. The connecting homomorphism d is the one arrow not drawn: it runs from the top right corner $\ker(\gamma)$ to the bottom left corner $\operatorname{coker}(\alpha)$, and the route taken to build it (down the right column, left along the bottom row, then up the left column) is the snake the lemma is named for.

$$\begin{tikzcd}
 & \ker(\alpha) \arrow[r, "\overline{\psi}"] \arrow[d, hook] & \ker(\beta) \arrow[r, "\overline{\varphi}"] \arrow[d, hook] & \ker(\gamma) \arrow[d, hook] & \\
0 \arrow[r] & A \arrow[r, "\psi"] \arrow[d, "\alpha"] & B \arrow[r, "\varphi"] \arrow[d, "\beta"] & C \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & A' \arrow[r, "\psi'"] \arrow[d, two heads] & B' \arrow[r, "\varphi'"] \arrow[d, two heads] & C' \arrow[r] \arrow[d, two heads] & 0 \\
 & \operatorname{coker}(\alpha) \arrow[r, "\overline{\psi}'"'] & \operatorname{coker}(\beta) \arrow[r, "\overline{\varphi}'"'] & \operatorname{coker}(\gamma) &
\end{tikzcd}$$

Step 1: the four induced maps exist. If $a \in \ker(\alpha)$ then $\beta(\psi(a)) = \psi'(\alpha(a)) = 0$, so ψ carries $\ker(\alpha)$ into $\ker(\beta)$; the same computation with the right square carries $\ker(\beta)$ into $\ker(\gamma)$. Dually $\psi'(\operatorname{im}(\alpha)) \subseteq \operatorname{im}(\beta)$, because $\psi'(\alpha(a)) = \beta(\psi(a))$, so ψ' descends to $\overline{\psi}' : \operatorname{coker}(\alpha) \to \operatorname{coker}(\beta)$; the same computation gives $\overline{\varphi}'$.

Step 2: construction of d. Let $c \in \ker(\gamma)$. Pull back along φ, which is surjective, to get $b \in B$ with $\varphi(b) = c$. Commutativity of the right square gives $\varphi'(\beta(b)) = \gamma(\varphi(b)) = \gamma(c) = 0$, so $\beta(b) \in \ker(\varphi') = \operatorname{im}(\psi')$ by exactness of the bottom row. Since ψ' is injective there is a *unique* $a' \in A'$ with $\psi'(a') = \beta(b)$. Define

$$d(c) := a' + \operatorname{im}(\alpha) \in \operatorname{coker}(\alpha)$$

The three steps just taken, namely pulling back along φ, crossing by β and pulling back along ψ', are the three segments of the route described above.

Step 3: d is a well defined homomorphism. The only choice made was the lift b. If $\varphi(b_1) = c$ as well then $b_1 - b \in \ker(\varphi) = \operatorname{im}(\psi)$, say $b_1 - b = \psi(a)$, so

$$\psi'(a_1') - \psi'(a') = \beta(b_1) - \beta(b) = \beta(\psi(a)) = \psi'(\alpha(a))$$

and injectivity of ψ' gives $a_1' - a' = \alpha(a) \in \operatorname{im}(\alpha)$. The two answers agree in $\operatorname{coker}(\alpha)$. Additivity and R-linearity follow because a lift of $c_1 + rc_2$ may be taken to be $b_1 + rb_2$, and ψ' and β are homomorphisms.

Step 4: exactness, at each of the six interior spots.

At $\ker(\alpha)$*.* The map $\overline{\psi}$ is a restriction of the injective ψ, so it is injective.

At $\ker(\beta)$*.* Since $\varphi \circ \psi = 0$ the composite $\overline{\varphi} \circ \overline{\psi}$ is zero, giving $\operatorname{im}(\overline{\psi}) \subseteq \ker(\overline{\varphi})$. Conversely let $b \in \ker(\beta)$ with $\varphi(b) = 0$. Exactness of the top row gives $b = \psi(a)$, and $\psi'(\alpha(a)) = \beta(\psi(a)) = \beta(b) = 0$, so $\alpha(a) = 0$ by injectivity of ψ'. Thus $a \in \ker(\alpha)$ and $b = \overline{\psi}(a)$.

At $\ker(\gamma)$*.* If $c = \overline{\varphi}(b)$ with $b \in \ker(\beta)$, then b is an admissible lift of c in Step 2 and $\beta(b) = 0 = \psi'(0)$, so $d(c) = 0 + \operatorname{im}(\alpha) = 0$. Conversely suppose $d(c) = 0$, so the element a' of Step 2 is $\alpha(a)$ for some $a \in A$. Then $\beta(b) = \psi'(\alpha(a)) = \beta(\psi(a))$, so $b - \psi(a) \in \ker(\beta)$, while $\varphi(b - \psi(a)) = \varphi(b) = c$. Hence $c = \overline{\varphi}(b - \psi(a))$.

At $\operatorname{coker}(\alpha)$*.* With c, b and a' as in Step 2,

$$\overline{\psi}'(d(c)) = \psi'(a') + \operatorname{im}(\beta) = \beta(b) + \operatorname{im}(\beta) = 0$$

so $\operatorname{im}(d) \subseteq \ker(\overline{\psi}')$. Conversely let $a' + \operatorname{im}(\alpha)$ lie in $\ker(\overline{\psi}')$, meaning $\psi'(a') = \beta(b)$ for some $b \in B$. Then $\gamma(\varphi(b)) = \varphi'(\beta(b)) = \varphi'(\psi'(a')) = 0$, so $c := \varphi(b)$ lies in $\ker(\gamma)$, and b is a lift of c exhibiting $d(c) = a' + \operatorname{im}(\alpha)$.

At $\operatorname{coker}(\beta)$*.* Since $\varphi' \circ \psi' = 0$ the composite $\overline{\varphi}' \circ \overline{\psi}'$ is zero. Conversely let $b' + \operatorname{im}(\beta)$ lie in $\ker(\overline{\varphi}')$, so $\varphi'(b') = \gamma(c)$ for some $c \in C$, and write $c = \varphi(b)$ using surjectivity of φ. Then $\varphi'(b' - \beta(b)) = \gamma(c) - \gamma(\varphi(b)) = 0$, so $b' - \beta(b) = \psi'(a')$ for some a', and $b' + \operatorname{im}(\beta) = \overline{\psi}'(a' + \operatorname{im}(\alpha))$.

At $\operatorname{coker}(\gamma)$*.* The map $\overline{\varphi}'$ is induced by the surjective φ', so it is surjective.

The name records that this is the three-module case. The general snake lemma replaces each of the three modules in a row by a whole chain complex and each kernel and cokernel by homology, and the six-term sequence above becomes a long exact sequence continuing in both directions. That version is [10, Snake Lemma]. The connecting homomorphism is what turns a short exact sequence into a longer one, and section 17.6 uses it to build the long exact sequences of Ext.

17.1.2 Split Sequences

Among all extensions of C by A, the direct sum $A \oplus C$ carries no information beyond its two ends. Example 17.1.6 showed that other extensions exist, so the ones that reduce to the direct sum deserve a name.

Definition 17.1.11: Split Short Exact Sequence

A short exact sequence $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ of R-modules *splits* if $\psi(A)$ has a complement in B, that is if there is a submodule $C' \subseteq B$ with

$$B = \psi(A) \oplus C'$$

The extension B is then called a *split extension* of C by A.

The complement is automatically a copy of C, so the definition need not demand it. If $B = \psi(A) \oplus C'$, then φ restricted to C' is injective because $C' \cap \ker(\varphi) = C' \cap \psi(A) = 0$, and surjective because $\varphi(B) = \varphi(\psi(A)) + \varphi(C') = \varphi(C')$. Hence $B \cong \psi(A) \oplus C \cong A \oplus C$.

Definition 17.1.12: Section and Retraction

Let $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ be a short exact sequence of R-modules.

1. A *section* of φ is an R-module homomorphism $\mu \colon C \to B$ with $\varphi \circ \mu = \mathrm{id}_C$. It is also called a *splitting homomorphism* for the sequence.
2. A *retraction* of ψ is an R-module homomorphism $\lambda \colon B \to A$ with $\lambda \circ \psi = \mathrm{id}_A$.

Proposition 17.1.13: Splitting Criteria

For a short exact sequence $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ of R-modules the following are equivalent.

1. The sequence splits.
2. φ has a section $\mu \colon C \to B$.
3. ψ has a retraction $\lambda \colon B \to A$.

***Proof*:**
(1) $\Rightarrow$ (2). Let $B = \psi(A) \oplus C'$. As noted above φ restricts to an isomorphism $\varphi|_{C'} : C' \to C$, so its inverse followed by the inclusion $C' \subseteq B$ is a homomorphism $\mu \colon C \to B$, and $\varphi(\mu(c)) = c$ for every $c \in C$.

(2) $\Rightarrow$ (1). Let μ be a section and put $C' := \mu(C)$, a submodule of B. For any $b \in B$,

$$b = \big(b - \mu(\varphi(b))\big) + \mu(\varphi(b))$$

and $\varphi(b - \mu(\varphi(b))) = \varphi(b) - \varphi(b) = 0$, so the first summand lies in $\ker(\varphi) = \psi(A)$ while the second lies in C'. The sum is direct: if $\mu(c) \in \psi(A) = \ker(\varphi)$ then $0 = \varphi(\mu(c)) = c$, so $\mu(c) = 0$. Hence $B = \psi(A) \oplus C'$.

(1) $\Rightarrow$ (3). Let $B = \psi(A) \oplus C'$ and let $\pi_{\psi(A)}$ be the projection onto the first summand. The map ψ is an isomorphism onto $\psi(A)$ by proposition 17.1.1, so $\lambda := \psi^{-1} \circ \pi_{\psi(A)}$ is defined, and $\lambda(\psi(a)) = \psi^{-1}(\psi(a)) = a$.

(3) $\Rightarrow$ (1). Let λ be a retraction and put $C' := \ker(\lambda)$. For any $b \in B$,

$$b = \psi(\lambda(b)) + \big(b - \psi(\lambda(b))\big)$$

where the first summand lies in $\psi(A)$ and the second lies in C', since $\lambda(b-\psi(\lambda(b))) = \lambda(b)-\lambda(b) = 0$. The sum is direct: if $b = \psi(a)$ and $\lambda(b) = 0$ then $a = \lambda(\psi(a)) = \lambda(b) = 0$, so $b = 0$. Hence $B = \psi(A) \oplus C'$.

Example 17.1.14: Split and Non-split Sequences

1. The sequence $0 \to A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \to 0$ splits, with section $\mu(c) = (0, c)$.
2. Of the two sequences in example 17.1.6, the first splits and the second does not. A section of the second would be a nonzero homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}$, and the image z of $\overline{1}$ would satisfy $nz = 0$, forcing $z = 0$ and hence the whole map to be zero.
3. The sequence $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ does not split, since a splitting would give $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, and only the left-hand side has an element of order 4.
4. Over $R = k[x]$ the sequence

$$0 \to (x) \hookrightarrow k[x] \to k[x]/(x) \to 0$$

does not split. A section would embed $k[x]/(x)$ as a nonzero submodule of $k[x]$ annihilated by x, and $k[x]$ is an integral domain, so no nonzero element is annihilated by x.

Splitting is a property of the sequence alone, and it is carried along by any isomorphism of sequences; equivalence is not needed.

Proposition 17.1.15: Splitting Is an Isomorphism Invariant

Let (α, β, γ) be an isomorphism of short exact sequences, in the notation of definition 17.1.7. Then the top sequence splits if and only if the bottom one does.

***Proof*:**
Suppose the top sequence splits and let $\mu\colon C \to B$ be a section of φ, which exists by proposition 17.1.13. Set

$$\mu' := \beta \circ \mu \circ \gamma^{-1} \colon C' \to B'$$

a homomorphism because γ^{-1} is one. Commutativity of the right square reads $\varphi' \circ \beta = \gamma \circ \varphi$, so

$$\varphi' \circ \mu' = \varphi' \circ \beta \circ \mu \circ \gamma^{-1} = \gamma \circ \varphi \circ \mu \circ \gamma^{-1} = \gamma \circ \gamma^{-1} = \mathrm{id}_{C'}$$

Hence μ' is a section of φ' and the bottom sequence splits, again by proposition 17.1.13.

For the converse, note that $(\alpha^{-1}, \beta^{-1}, \gamma^{-1})$ is an isomorphism of short exact sequences from the bottom one to the top one: composing $\beta \circ \psi = \psi' \circ \alpha$ with β^{-1} on the left and α^{-1} on the right gives $\psi \circ \alpha^{-1} = \beta^{-1} \circ \psi'$, and the right square is handled the same way. Applying the first paragraph to it gives the converse.

The vocabulary is now sufficient to state the question this chapter opened with. Given R-modules A and C, write $\mathcal{E}(C, A)$ for the set of equivalence classes of extensions of C by A, in the sense of definition 17.1.7(3). This set is never empty, and all split extensions lie in a single class: if μ is a section of φ, then $B = \psi(A) \oplus \mu(C)$ by the proof of proposition 17.1.13, so $\beta(a, c) := \psi(a) + \mu(c)$ is an isomorphism $A \oplus C \to B$, both ψ and μ being injective. It satisfies $\beta(\iota(a)) = \psi(a)$ and $\varphi(\beta(a, c)) = c = \pi(a, c)$, so $(\mathrm{id}_A, \beta, \mathrm{id}_C)$ is an equivalence with the direct-sum sequence. By example 17.1.8(2) the set can have more than one element.

The *extension problem* for modules asks for $\mathcal{E}(C, A)$. Section 17.6 answers it by constructing a module $\mathrm{Ext}^1_R(C, A)$ together with a bijection $\mathrm{Ext}^1_R(C, A) \leftrightarrow \mathcal{E}(C, A)$ carrying 0 to the class of the split extension.

Exercise 17.1.1

1. For each of the following short exact sequences of $\mathbb{Z}$-modules, decide whether it splits. Exhibit a section when one exists, and prove that none exists otherwise.

    1. $0 \to \mathbb{Z} \xrightarrow{\cdot 3} \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$
    2. $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with the evident injection and projection
    3. $0 \to \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with $\overline{a} \mapsto \overline{3a}$ and reduction modulo 3
    4. $0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$, with $\overline{a} \mapsto \overline{3a}$ and reduction modulo 3

2. Let p be a prime. Show that every extension of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$ has a middle term of order p^2, hence isomorphic to $\mathbb{Z}/p^2\mathbb{Z}$ or to $\mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$ by theorem 5.1.8, and that in the second case the sequence splits. In the first case, show the sequence is equivalent to one of the $\mathcal{E}_c$ of example 17.1.8(2); you may use that $\mathbb{Z}/p^2\mathbb{Z}$ has a unique subgroup of order p. Conclude that $\mathcal{E}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$ has exactly p elements.

3. Let $m, n \geq 1$. Take both rows of definition 17.1.7 to be $0 \to \mathbb{Z} \xrightarrow{\cdot m} \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \to 0$ and take α, β and γ all to be multiplication by n. Verify that this is a homomorphism of short exact sequences, identify all six terms of the sequence of proposition 17.1.10 in terms of $g = \gcd(m, n)$, and compute the connecting homomorphism d explicitly. Then specialise to $m = 4$ and $n = 6$.

4. Let (α, β, γ) be a homomorphism of short exact sequences.

    1. Show that if β is injective then so is α.
    2. Show that if β is surjective then so is γ.
    3. Give an example in which β is an isomorphism while α is not surjective and γ is not injective. *Hint*: one of the two rows may end in the zero module.

5. Show that a short exact sequence $0 \to A \to B \xrightarrow{\varphi} C \to 0$ of R-modules splits whenever C is a free module. *Hint*: choose a basis of C, lift each basis element along φ, and use theorem 12.3.11.

6. Show that if $0 \to A \xrightarrow{\psi} B \xrightarrow{\varphi} C \to 0$ splits then for every R-module D there are isomorphisms of abelian groups
$$\mathrm{Hom}_R(D, B) \cong \mathrm{Hom}_R(D, A) \oplus \mathrm{Hom}_R(D, C)$$
$$\mathrm{Hom}_R(B, D) \cong \mathrm{Hom}_R(A, D) \oplus \mathrm{Hom}_R(C, D)$$

Hint: use the section and the retraction given by proposition 17.1.13.

7. *(The five lemma.)* Consider a commutative diagram of R-modules

$$\begin{tikzcd}
A_1 \arrow[r] \arrow[d, "f_1"'] & A_2 \arrow[r] \arrow[d, "f_2"'] & A_3 \arrow[r] \arrow[d, "f_3"'] & A_4 \arrow[r] \arrow[d, "f_4"'] & A_5 \arrow[d, "f_5"'] \\
B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5
\end{tikzcd}$$

with exact rows. Prove that if f_2 and f_4 are isomorphisms, f_1 is surjective and f_5 is injective, then f_3 is an isomorphism. *Hint*: run the two chases of proposition 17.1.9 one column further in each direction.

17.2 Projective Modules

Example 17.1.14 produced short exact sequences that do not split, so the extension problem has content. The instrument for measuring the failure is already available: fix an R-module D, apply $\mathrm{Hom}_R(D, -)$ to a short exact sequence, and see how much of the sequence survives.

Throughout, ψ_* denotes the map induced on $\mathrm{Hom}_R(D, -)$ by a homomorphism ψ, reserving the primes of definition 17.1.7 for the second row of a diagram.

Proposition 17.2.1: Left Exactness of $\mathrm{Hom}_R(D, -)$

Let D be an R-module.

1. For every R-module homomorphism $\psi\colon L \to M$, the assignment

$$\psi_*\colon \mathrm{Hom}_R(D, L) \to \mathrm{Hom}_R(D, M), \qquad f \mapsto \psi \circ f$$

is a homomorphism of abelian groups.

2. If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then

$$0 \to \mathrm{Hom}_R(D, L) \xrightarrow{\psi_*} \mathrm{Hom}_R(D, M) \xrightarrow{\varphi_*} \mathrm{Hom}_R(D, N)$$

is exact, for every D. In this situation one says that $\mathrm{Hom}_R(D, -)$ is *left exact*.

Note what part (2) does *not* claim: that φ_* is surjective, so that the sequence may be closed off with a $\to 0$ on the right. That claim is false in general, and the whole section grows out of the failure.

***Proof*:**
(1) For $f, g \in \mathrm{Hom}_R(D, L)$ and $d \in D$,

$$\big(\psi \circ (f + g)\big)(d) = \psi\big(f(d) + g(d)\big) = \psi(f(d)) + \psi(g(d)) = \big(\psi \circ f + \psi \circ g\big)(d)$$

using that ψ is a homomorphism, so $\psi_*(f + g) = \psi_*(f) + \psi_*(g)$.

(2) *Exactness at* $\mathrm{Hom}_R(D, L)$, which by proposition 17.1.1 is the injectivity of ψ_*. Suppose $\psi \circ f = 0$. Then $\psi(f(d)) = 0$ for every $d \in D$, so $f(d) \in \ker(\psi) = 0$ because ψ is injective, and $f = 0$.

Exactness at $\operatorname{Hom}_R(D, M)$, that is $\operatorname{im}(\psi_*) = \ker(\varphi_*)$.

($\subseteq$) For $f \in \operatorname{Hom}_R(D, L)$ we have $\varphi_*(\psi_*(f)) = \varphi \circ \psi \circ f$, and $\varphi \circ \psi = 0$ because $\operatorname{im}(\psi) = \ker(\varphi)$. So $\varphi_*(\psi_*(f)) = 0$.

($\supseteq$) Let $F \in \ker(\varphi_*)$, so $\varphi \circ F = 0$. For each $d \in D$ we have $\varphi(F(d)) = 0$, hence $F(d) \in \ker(\varphi) = \operatorname{im}(\psi)$, so there is an $l \in L$ with $\psi(l) = F(d)$; and l is unique because ψ is injective. Define $F' \colon D \to L$ by letting $F'(d)$ be that unique l, so that $\psi \circ F' = F$ by construction.

It remains to check that F' is a homomorphism, which is the only step here with content. For $d, e \in D$ and $r \in R$,

$$\psi\big(F'(d) + rF'(e)\big) = \psi(F'(d)) + r\psi(F'(e)) = F(d) + rF(e) = F(d + re) = \psi\big(F'(d + re)\big)$$

and ψ is injective, so $F'(d + re) = F'(d) + rF'(e)$. Hence $F' \in \operatorname{Hom}_R(D, L)$ and $F = \psi_*(F') \in \operatorname{im}(\psi_*)$.

Now the failure, computed before anything is named after it. The sequence at hand is the non-split extension of example 17.1.6 with $n = 2$.

Example 17.2.2: $\operatorname{Hom}_R(D, -)$ Need Not Be Right Exact

Take $R = \mathbb{Z}$, the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

and $D = \mathbb{Z}/2\mathbb{Z}$. Then $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0$: a homomorphism h sends $\overline{1}$ to some $z \in \mathbb{Z}$ with $2z = h(\overline{2}) = h(\overline{0}) = 0$, so $z = 0$. On the other hand $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, since such a homomorphism is determined by the image of $\overline{1}$ and both choices are legitimate. So applying $\operatorname{Hom}_{\mathbb{Z}}(D, -)$ gives

$$0 \to 0 \xrightarrow{(\cdot 2)_*} 0 \xrightarrow{\pi_*} \mathbb{Z}/2\mathbb{Z}$$

which is exact, as proposition 17.2.1 promised, and which cannot be extended by $\to 0$ on the right, since π_* is the zero map into a nonzero group.

The obstruction has a name in the language of diagrams. The element of $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ that is not hit is $\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$, and saying it is not hit is saying that the dashed arrow below does not exist:

$$\begin{tikzcd}
 & \mathbb{Z}/2\mathbb{Z} \arrow[dl, dashed, "g\,?"'] \arrow[d, "\mathrm{id}"] & \\
\mathbb{Z} \arrow[r, "\pi"'] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Indeed the only homomorphism $g \colon \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ is the zero map, and $\pi \circ 0 = 0 \neq \mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$.

A homomorphism out of D that fails to lift is exactly what stops $\operatorname{Hom}_R(D, -)$ from being exact. The modules for which this never happens are the subject of this section.

Definition 17.2.3: Projective Module

An R-module P is *projective* if the functor $\mathrm{Hom}_R(P, -)$ is exact, that is if for every short exact sequence $0 \to L \to M \to N \to 0$ of R-modules the sequence

$$0 \to \mathrm{Hom}_R(P, L) \to \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N) \to 0$$

is again short exact.

By proposition 17.2.1 everything in that sequence is automatic except surjectivity on the right, so the definition asks for one thing only: that every homomorphism out of P lift along every surjection. The theorem below makes that precise and adds two further descriptions, one about splitting and one about free modules.

Theorem 17.2.4: Equivalencies for Projective Modules

Let P be an R-module. The following are equivalent.

1. P is projective.
2. *(Lifting.)* For all R-modules M and N, if $\varphi\colon M \to N$ is surjective, then every $f \in \mathrm{Hom}_R(P, N)$ has a lift $g \in \mathrm{Hom}_R(P, M)$ with $\varphi \circ g = f$:

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "f"] & \\ M \arrow[r, "\varphi"'] & N \arrow[r] & 0 \end{tikzcd}$$

3. *(Splitting.)* Every short exact sequence $0 \to L \to M \to P \to 0$ of R-modules splits.
4. *(Summand.)* P is isomorphic to a direct summand of a free R-module.

***Proof*:**

(1) $\Rightarrow$ (2). Let $\varphi\colon M \to N$ be surjective and put $L = \ker(\varphi)$, so that $0 \to L \to M \xrightarrow{\varphi} N \to 0$ is short exact by example 17.1.4(2). Condition (1) makes $\varphi_*\colon \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$ surjective, so a given f equals $\varphi_*(g) = \varphi \circ g$ for some g.

(2) $\Rightarrow$ (1). Proposition 17.2.1 gives exactness at the first two spots of the sequence in definition 17.2.3, and (2) applied to the surjection φ says precisely that φ_* is surjective.

(2) $\Rightarrow$ (3). Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} P \to 0$ be short exact. Then φ is surjective, so applying (2) to $f = \mathrm{id}_P$ produces $\mu \in \mathrm{Hom}_R(P, M)$ with $\varphi \circ \mu = \mathrm{id}_P$. That is a section of φ, so the sequence splits by proposition 17.1.13.

(3) $\Rightarrow$ (4). By example 17.1.4(3), applied with $S = P$ as a generating set, there is a short exact sequence

$$0 \to K \hookrightarrow F \xrightarrow{\pi} P \to 0$$

with F free. By (3) it splits, so $F = K \oplus C'$ for some submodule C', and $\pi|_{C'} : C' \to P$ is an isomorphism by the remark following definition 17.1.11. Hence P is isomorphic to the direct summand C' of F.

(4) $\Rightarrow$ (2). Suppose $F = P' \oplus K$ with F free and $P' \cong P$; replacing P by P', we may assume P itself is a summand, and write $\iota\colon P \to F$ for the inclusion and $\rho\colon F \to P$ for the projection, so $\rho \circ \iota = \mathrm{id}_P$.

Let $\varphi\colon M \to N$ be surjective and $f \in \mathrm{Hom}_R(P, N)$. Let $(e_s)_{s\in S}$ be a basis of F. For each s the element $f(\rho(e_s))$ lies in N, so by surjectivity of φ we may choose $m_s \in M$ with $\varphi(m_s) = f(\rho(e_s))$. By theorem 12.3.11 there is a unique homomorphism $G\colon F \to M$ with $G(e_s) = m_s$ for every s, and then $\varphi \circ G$ and $f \circ \rho$ are homomorphisms $F \to N$ agreeing on a basis, hence equal.

Now set $g := G \circ \iota\colon P \to M$. It is a homomorphism, and

$$\varphi \circ g = \varphi \circ G \circ \iota = f \circ \rho \circ \iota = f \circ \mathrm{id}_P = f$$

which is the required lift.

17.2.5 Summand, Not Sum Condition (4) says direct **summand** of a free module, which is weaker than being a free module. A direct summand of M is any submodule admitting a complement, so if $M = A \oplus B \oplus C$ then

$$A,\ B,\ C,\ A \oplus B,\ A \oplus C,\ B \oplus C,\ A \oplus B \oplus C$$

are all direct summands of M, and only the last is M itself. Reading (4) as "direct sum" would collapse the notion of projective onto the notion of free, and the two are genuinely different, as the two examples below show.

Corollary 17.2.6: Free Modules Are Projective

Every free R-module is projective.

***Proof*:**
A free module F is a direct summand of itself, with complement 0, so theorem 17.2.4(4) applies.

The converse fails, and it fails for a reason worth seeing twice. Both witnesses come from splitting a ring into a direct sum of ideals; the summands are then projective by construction and too small to be free.

Example 17.2.7: Projective But Not Free

1. Let $R = \mathbb{Z}/6\mathbb{Z}$. The Chinese remainder theorem gives a ring isomorphism $R \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, which is in particular an isomorphism of R-modules, so the ideal $P = \overline{3}R = \{\overline{0}, \overline{3}\}$ is a direct summand of R, with complement $\overline{4}R = \{\overline{0}, \overline{4}, \overline{2}\}$. Since R is free of rank one over itself, P is projective by theorem 17.2.4(4).

   It is not free. A free R-module of finite rank k has 6^k elements and P has 2, while a free module of infinite rank is infinite; and $2 \neq 6^k$ for every $k \geq 0$.

2. Let F be a field, $n \geq 2$, and $R = M_n(F)$ the ring of $n \times n$ matrices. Let $C = F^n$ be the module of column vectors, a left R-module by matrix multiplication. Writing E_{jj} for the

matrix with a 1 in position (j,j) and zeros elsewhere,

$$R = \bigoplus_{j=1}^{n} RE_{jj}$$

since RE_{jj} consists of the matrices supported in column j and every matrix is the sum of its columns. Each RE_{jj} is isomorphic to C as a left R-module, by reading off column j. So C is a direct summand of the free module R and is projective.

It is not free. Counting dimensions over F, a free R-module of finite rank k has F-dimension kn^2, and one of infinite rank is infinite dimensional; meanwhile $\dim_F C = n$, which is finite, and $n = kn^2$ has no solution $k \in \mathbb{Z}_{\geq 0}$ once $n \geq 2$.

Condition (4) is structural but awkward to verify, since it requires producing a free module and a complement. The following reformulation is the working criterion, because it asks only for elements of A and homomorphisms out of A, both of which can be exhibited.

Proposition 17.2.8: Dual Basis Lemma

An R-module A is projective if and only if there exist a family $(a_i)_{i\in I}$ of elements of A and a family $(f_i)_{i\in I}$ of homomorphisms $f_i \in \operatorname{Hom}_R(A, R)$ such that, for every $a \in A$,

1. $f_i(a) = 0$ for all but finitely many $i \in I$, and
2. $a = \sum_{i\in I} f_i(a)\, a_i$.

Such a pair of families is called a *dual basis* for A. It may be taken finite exactly when A is finitely generated.

***Proof*:**
($\Rightarrow$) Let A be projective. By example 17.1.4(3) there is a free module F with basis $(e_i)_{i\in I}$ and a surjection $\pi\colon F \to A$, and by theorem 17.2.4(3) the sequence $0 \to \ker(\pi) \to F \xrightarrow{\pi} A \to 0$ splits, so proposition 17.1.13 gives $\sigma\colon A \to F$ with $\pi \circ \sigma = \mathrm{id}_A$.

Put $a_i := \pi(e_i)$, and let $f_i(a)$ be the i-th coordinate of $\sigma(a)$ in the basis (e_i), so that $\sigma(a) = \sum_i f_i(a)e_i$. Each f_i is a homomorphism $A \to R$, being σ followed by a coordinate function, and (1) holds because an element of F has only finitely many nonzero coordinates. Applying π,

$$a = \pi(\sigma(a)) = \pi\Big(\sum_i f_i(a)e_i\Big) = \sum_i f_i(a)\pi(e_i) = \sum_i f_i(a)a_i$$

which is (2).

($\Leftarrow$) Given the two families, let F be free on a basis $(e_i)_{i\in I}$ and define $\pi\colon F \to A$ by $\pi(e_i) = a_i$, using theorem 12.3.11. Define $\sigma\colon A \to F$ by $\sigma(a) = \sum_i f_i(a)e_i$, which is a well defined element of F by (1) and is a homomorphism because each f_i is. Then (2) says exactly that $\pi(\sigma(a)) = a$, so $\pi \circ \sigma = \mathrm{id}_A$. In particular π is surjective, and σ is a section of π, so $0 \to \ker(\pi) \to F \xrightarrow{\pi} A \to 0$ splits by proposition 17.1.13 and A is a direct summand of F by definition 17.1.11. Theorem 17.2.4(4) makes A projective.

For the last sentence: if the families are finite then (2) exhibits $a_1, \ldots, a_n$ as generators of A. Conversely if A is generated by n elements, take $F = R^n$ in the forward direction, so that I is finite.

Chapter 15 produced an isomorphism $V^* \otimes_F W \cong \mathrm{Hom}_F(V, W)$ for finite dimensional vector spaces (theorem 15.4.9). Over a general commutative ring the same map exists, and the dual basis says exactly when it is an isomorphism.

Corollary 17.2.9: Evaluation and Finitely Generated Projectives

Let R be a commutative ring, A an R-module, and $A^* := \mathrm{Hom}_R(A, R)$. Let

$$\theta \colon A^* \otimes_R A \to \mathrm{End}_R(A), \qquad \theta(f \otimes b)(a) = f(a)b$$

Then A is finitely generated and projective if and only if θ is an isomorphism, and in fact if and only if θ is surjective.

***Proof*:**
The map θ exists because $(f, b) \mapsto \big(a \mapsto f(a)b\big)$ is R-bilinear, so theorem 15.1.12 produces it.

Suppose first that θ is surjective. Then $\mathrm{id}_A = \theta(t)$ for some $t \in A^* \otimes_R A$, and every element of a tensor product is a finite sum of simple tensors, say $t = \sum_{i=1}^n f_i \otimes a_i$. Unwinding, $a = \mathrm{id}_A(a) = \sum_{i=1}^n f_i(a)a_i$ for every $a \in A$. That is a finite dual basis, so A is finitely generated and projective by proposition 17.2.8.

Conversely let A be finitely generated and projective, with a finite dual basis $(a_i, f_i)_{i=1}^n$ from proposition 17.2.8. Two observations do the work. The first is that

$$g = \sum_{i=1}^n g(a_i)\, f_i \qquad \text{for every } g \in A^*$$

since evaluating the right-hand side at a gives $\sum_i g(a_i)f_i(a) = g\big(\sum_i f_i(a)a_i\big) = g(a)$, using R-linearity of g.

Surjectivity. Given $\alpha \in \mathrm{End}_R(A)$, put $t = \sum_i f_i \otimes \alpha(a_i)$. Then for every a,

$$\theta(t)(a) = \sum_i f_i(a)\, \alpha(a_i) = \alpha\Big(\sum_i f_i(a)a_i\Big) = \alpha(a)$$

so $\theta(t) = \alpha$.

Injectivity. Let $t = \sum_{j=1}^m g_j \otimes b_j$ satisfy $\theta(t) = 0$. Expanding each g_j by the displayed identity and collecting,

$$t = \sum_j \Big(\sum_i g_j(a_i)f_i\Big) \otimes b_j = \sum_i f_i \otimes \Big(\sum_j g_j(a_i)b_j\Big) = \sum_i f_i \otimes \theta(t)(a_i) = 0$$

where the middle step moves the scalars $g_j(a_i) \in R$ across the tensor sign, which is legitimate because the tensor product is over R.

Example 17.2.10: Projective Modules

Throughout, proposition 12.3.36 is what lets Hom_R be computed one summand at a time.

1. If F is a field then every F-module is free (theorem 13.1.8), so every vector space is projective.

2. Any ring R is free of rank one over itself and hence projective. Directly: $\mathrm{Hom}_R(R, M) \cong M$ by $h \mapsto h(1)$, so applying $\mathrm{Hom}_R(R, -)$ to a short exact sequence returns that sequence, which is exact.

3. No nonzero finite abelian group is a projective $\mathbb{Z}$-module. A projective is a direct summand of a free module by theorem 17.2.4(4), hence a submodule of one; a free $\mathbb{Z}$-module has no nonzero element of finite order; and every nonzero element of a nonzero finite group has finite order.

   For $\mathbb{Z}/n\mathbb{Z}$ with $n \geq 2$ the same conclusion is visible in the functor. Applying $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -)$ to $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$ gives

$$0 \to 0 \to 0 \to \mathbb{Z}/n\mathbb{Z}$$

   since $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ by the argument of example 17.2.2 and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$. The last map is not surjective.

4. $\mathbb{Q}/\mathbb{Z}$ is not a projective $\mathbb{Z}$-module, being a nonzero torsion module and so not a submodule of a free one. Correspondingly

$$0 \to \mathbb{Z} \hookrightarrow \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$$

   does not split: a section would exhibit $\mathbb{Q}/\mathbb{Z}$ as a submodule of $\mathbb{Q}$, and $\mathbb{Q}$ has no nonzero element of finite order while $\mathbb{Q}/\mathbb{Z}$ is nonzero and torsion.

5. $\mathbb{Q}$ is not a projective $\mathbb{Z}$-module either, and the reason is different: it is torsion-free, so the argument above does not apply. Instead, *every* homomorphism $h\colon \mathbb{Q} \to \mathbb{Z}$ is zero. For if $h(x) \neq 0$ for some x, then for every $n \geq 1$

$$h(x) = h\Big(n \cdot \frac{x}{n}\Big) = n\, h\Big(\frac{x}{n}\Big)$$

   so n divides $h(x)$ for every $n \geq 1$, forcing $h(x) = 0$.

   Consequently every homomorphism from $\mathbb{Q}$ into a free $\mathbb{Z}$-module $F = \bigoplus_{i \in I} \mathbb{Z}$ is zero, since composing with each coordinate projection gives a homomorphism $\mathbb{Q} \to \mathbb{Z}$. So $\mathbb{Q}$ is not isomorphic to a submodule of a free $\mathbb{Z}$-module, in particular not to a direct summand of one, and theorem 17.2.4(4) fails.

6. Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ be short exact with N projective, and write $X^* = \mathrm{Hom}_R(X, R)$. Then

$$0 \to N^* \xrightarrow{\varphi^*} M^* \xrightarrow{\psi^*} L^* \to 0$$

   is short exact, where $\varphi^*(h) = h \circ \varphi$ and $\psi^*(g) = g \circ \psi$.

   What makes this work is *not* that the modules are projective as arguments of $\mathrm{Hom}_R(-, R)$; projectivity of L, M and N says nothing directly about a contravariant Hom. It is that N projective forces the sequence to split (theorem 17.2.4(3)), and a split sequence is preserved by any additive functor. Explicitly, let μ be a section of φ and λ a retraction of ψ (proposition 17.1.13). Then λ^* is a section of ψ^*, since $\psi^*(\lambda^*(g)) = g \circ \lambda \circ \psi = g$, so ψ^* is surjective; and μ^* is a retraction of φ^*, since $\mu^*(\varphi^*(h)) = h \circ \varphi \circ \mu = h$, so φ^* is injective. Exactness in the middle holds because $\psi^*\varphi^*(h) = h \circ \varphi \circ \psi = 0$, and conversely a $g \in M^*$ with $g \circ \psi = 0$ kills $\psi(L) = \ker(\varphi)$ and so factors through $M/\ker(\varphi) \cong N$ by theorem 12.2.10.

7. If P_1 and P_2 are projective then so is $P_1 \oplus P_2$: writing $F_1 = P_1 \oplus K_1$ and $F_2 = P_2 \oplus K_2$ with F_i free, the module $F_1 \oplus F_2$ is free and contains $P_1 \oplus P_2$ as a direct summand with complement $K_1 \oplus K_2$.

Over one class of rings the gap between projective and free closes entirely.

Proposition 17.2.11: Projective Modules Over a PID

Let R be a principal ideal domain. Then every projective R-module is free. Together with corollary 17.2.6, projective and free coincide over a PID.

***Proof*:**
Let P be projective. By theorem 17.2.4(4) there is a free R-module F and a submodule $P' \subseteq F$ with $P \cong P'$, namely a direct summand. By theorem 14.1.6 the submodule P' is free, and hence so is P.

No finiteness hypothesis is needed, which is exactly what theorem 14.1.6 was proved for; the finite rank case (theorem 14.1.5) would give the statement only for finitely generated P. Example 17.2.7 shows that some hypothesis on R is unavoidable, since $\mathbb{Z}/6\mathbb{Z}$ and $M_n(F)$ are not domains. A second class where the two notions agree is the local rings, where every finitely generated projective module is free; that result belongs with the theory of local rings and is proved in *Commutative Algebra* [8, Projective Module over Local Ring].

Two different presentations of the same module by projectives differ by no more than a free summand. This is the result that makes an invariant computed from a projective resolution independent of the resolution chosen, and section 17.6 depends on it.

Proposition 17.2.12: Schanuel's Lemma

Let M be an R-module and let

$$0 \to K_1 \hookrightarrow P_1 \xrightarrow{\pi_1} M \to 0 \qquad \text{and} \qquad 0 \to K_2 \hookrightarrow P_2 \xrightarrow{\pi_2} M \to 0$$

be short exact sequences with P_1 and P_2 projective. Then

$$K_1 \oplus P_2 \cong K_2 \oplus P_1$$

***Proof*:**
Let
$$X = \{(p_1, p_2) \in P_1 \oplus P_2 \mid \pi_1(p_1) = \pi_2(p_2)\}$$
which is a submodule of $P_1 \oplus P_2$, being the kernel of the homomorphism $(p_1, p_2) \mapsto \pi_1(p_1) - \pi_2(p_2)$. Let q_1 and q_2 be the two coordinate projections restricted to X.

q_2 is surjective with kernel K_1. Given $p_2 \in P_2$, surjectivity of π_1 produces $p_1 \in P_1$ with $\pi_1(p_1) = \pi_2(p_2)$, and then $(p_1, p_2) \in X$ maps to p_2. Its kernel consists of the pairs of the form $(p_1, 0)$ that lie in X, and $(p_1, 0) \in X$ says $\pi_1(p_1) = \pi_2(0) = 0$, so the kernel is $K_1 \oplus 0$, which is isomorphic to K_1.

So $0 \to K_1 \to X \xrightarrow{q_2} P_2 \to 0$ is short exact, and P_2 is projective, so it splits by theorem 17.2.4(3), giving $X \cong K_1 \oplus P_2$ by the remark following definition 17.1.11. Exchanging the roles of the two sequences, the same argument applied to q_1 gives $X \cong K_2 \oplus P_1$.

Exercise 17.2.1

1. For each pair, decide whether the module is projective over the indicated ring, and justify the answer.
    1. $\mathbb{Z}/3\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    2. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/4\mathbb{Z}$
    3. $2\mathbb{Z}$ over $\mathbb{Z}$
    4. $\mathbb{Z}[x]/(x)$ over $\mathbb{Z}[x]$
    5. $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
2. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $P = \overline{3}R$, the projective module of example 17.2.7(1). Exhibit a dual basis for P consisting of a single pair (a_1, f_1), and verify both conditions of proposition 17.2.8 directly. Then do the same for the complement $\overline{4}R$.
3. Show that the evaluation map θ of corollary 17.2.9 fails to be surjective in each of the following cases, and identify which hypothesis fails.
    1. $R = \mathbb{Z}$ and $A = \mathbb{Q}$
    2. $R = \mathbb{Z}$ and $A = \bigoplus_{k=1}^{\infty} \mathbb{Z}$
4. Take $R = \mathbb{Z}$, $P = \mathbb{Z}$, $M = \mathbb{Z}$, $N = \mathbb{Z}/6\mathbb{Z}$ with φ the reduction map, and $f\colon \mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ the homomorphism with $f(1) = \overline{5}$. Determine every lift g of f along φ, and show that the set of lifts is a coset of $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, 6\mathbb{Z})$ inside $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z})$.
5. Let $(P_i)_{i \in I}$ be a family of R-modules. Show that $\bigoplus_{i \in I} P_i$ is projective if and only if every P_i is projective. *Hint*: for one direction, a direct summand of a direct summand is a direct summand.
6. Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules.
    1. Show that if L and N are projective then so is M.
    2. Give an example with M projective and L not projective. *Hint*: a ring of the form $\mathbb{Z}/p^2\mathbb{Z}$ has a non-projective ideal.
7. Call an R-module M *finitely presented* if there is an exact sequence $R^s \xrightarrow{\alpha} R^t \to M \to 0$ for some $s, t \in \mathbb{N}$. Use proposition 17.2.12 to show that if M is finitely presented then *every* surjection $\varphi\colon R^r \to M$ from a finitely generated free module has finitely generated kernel. Deduce that for a finitely generated M the converse holds too, so that finite presentation may be tested on any one such surjection. *Hint*: the kernel of $R^t \to M$ is $\mathrm{im}(\alpha)$, and a direct summand of a finitely generated module is finitely generated.

17.3 Injective Modules

Section 17.2 fixed a module D and asked what $\operatorname{Hom}_R(D,-)$ does to a short exact sequence. The other slot of Hom_R asks the same question with the arrows reversed: fix D and apply $\operatorname{Hom}_R(-,D)$, which turns a homomorphism $M \to N$ into a homomorphism $\operatorname{Hom}_R(N,D) \to \operatorname{Hom}_R(M,D)$ by composition. A short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ therefore produces

$$0 \longrightarrow \operatorname{Hom}_R(N,D) \xrightarrow{\varphi^*} \operatorname{Hom}_R(M,D) \xrightarrow{\psi^*} \operatorname{Hom}_R(L,D) \longrightarrow 0$$

with the outer terms of the original sequence exchanged. The superscript stars record that the assignment reverses arrows, in contrast to the lower stars of section 17.2.

Proposition 17.3.1: Left Exactness of $\operatorname{Hom}_R(-,D)$

Let D be an R-module.

1. For every R-module homomorphism $\varphi\colon M \to N$, the assignment
$$\varphi^*\colon \operatorname{Hom}_R(N,D) \to \operatorname{Hom}_R(M,D), \qquad f \mapsto f \circ \varphi$$
is a homomorphism of abelian groups.

2. If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then
$$0 \to \operatorname{Hom}_R(N,D) \xrightarrow{\varphi^*} \operatorname{Hom}_R(M,D) \xrightarrow{\psi^*} \operatorname{Hom}_R(L,D)$$
is exact, for every D. In this situation one says that $\operatorname{Hom}_R(-,D)$ is *left exact*, the word being applied to the sequence as displayed, after the reversal.

As with proposition 17.2.1, part (2) does not claim that the last map is surjective. The two propositions fail at opposite ends of the original sequence: $\operatorname{Hom}_R(D,-)$ can miss homomorphisms into the *right* hand term N, and $\operatorname{Hom}_R(-,D)$ can miss homomorphisms out of the *left* hand term L.

***Proof*:**
(1) For $f, g \in \operatorname{Hom}_R(N,D)$ and $m \in M$,
$$\big((f+g)\circ\varphi\big)(m) = f(\varphi(m)) + g(\varphi(m)) = \big(f\circ\varphi + g\circ\varphi\big)(m)$$
so $\varphi^*(f+g) = \varphi^*(f) + \varphi^*(g)$.

(2) *Exactness at* $\operatorname{Hom}_R(N,D)$, which is the injectivity of φ^*. Suppose $f \circ \varphi = 0$. Every $n \in N$ is $\varphi(m)$ for some m, since φ is surjective, so $f(n) = f(\varphi(m)) = 0$ and $f = 0$.

Exactness at $\operatorname{Hom}_R(M,D)$, that is $\operatorname{im}(\varphi^*) = \ker(\psi^*)$.

($\subseteq$) For $f \in \operatorname{Hom}_R(N,D)$, $\psi^*(\varphi^*(f)) = f\circ\varphi\circ\psi = 0$, since $\varphi\circ\psi = 0$ by $\operatorname{im}(\psi) = \ker(\varphi)$.

($\supseteq$) Let $g \in \operatorname{Hom}_R(M,D)$ with $g\circ\psi = 0$, so g vanishes on $\operatorname{im}(\psi) = \ker(\varphi)$. Let $\pi\colon M \to M/\ker(\varphi)$ be the quotient map. By theorem 12.2.10 the map g factors as $g = \overline{g}\circ\pi$ for a unique $\overline{g}\colon M/\ker(\varphi) \to D$, and φ induces an isomorphism $\overline{\varphi}\colon M/\ker(\varphi) \to N$ with $\overline{\varphi}\circ\pi = \varphi$. Put $f := \overline{g}\circ\overline{\varphi}^{-1} \in \operatorname{Hom}_R(N,D)$. Then
$$\varphi^*(f) = f\circ\varphi = \overline{g}\circ\overline{\varphi}^{-1}\circ\overline{\varphi}\circ\pi = \overline{g}\circ\pi = g$$

so $g \in \operatorname{im}(\varphi^*)$.

The failure at the other end is visible in the same sequence that example 17.2.2 used, with the roles of the two slots exchanged.

Example 17.3.2: $\operatorname{Hom}_R(-, D)$ Need Not Be Right Exact

Take $R = \mathbb{Z}$, the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

and $D = \mathbb{Z}/2\mathbb{Z}$. A homomorphism out of $\mathbb{Z}$ is determined by the image of 1, so $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, and likewise $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. Applying $\operatorname{Hom}_{\mathbb{Z}}(-, D)$ gives

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi^*} \mathbb{Z}/2\mathbb{Z} \xrightarrow{(\cdot 2)^*} \mathbb{Z}/2\mathbb{Z}$$

Here $\pi^*(\mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}) = \pi \neq 0$, so π^* is injective and therefore, both groups having two elements, bijective. And $(\cdot 2)^*$ is the zero map: for $g \in \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ and $x \in \mathbb{Z}$,

$$\big(g \circ (\cdot 2)\big)(x) = g(2x) = 2\,g(x) = 0$$

because every element of $\mathbb{Z}/2\mathbb{Z}$ is killed by 2. The displayed sequence is thus exact, as proposition 17.3.1 promised, and it cannot be closed off with a $\to 0$ on the right, since $(\cdot 2)^*$ has image 0 inside a group of order two.

In the language of diagrams, the homomorphism not in the image of $(\cdot 2)^*$ is π itself, and saying it is not in the image says that π does not *extend* along the injection $\cdot 2$:

$$\begin{tikzcd} 0 \arrow[r] & \mathbb{Z} \arrow[r, hook, "\cdot 2"] \arrow[d, "\pi"'] & \mathbb{Z} \arrow[dl, dashed, "F\,?"] \\ & \mathbb{Z}/2\mathbb{Z} & \end{tikzcd}$$

Any such F would satisfy $F(2x) = 2F(x) = 0$ while $\pi(x) \neq 0$ for odd x.

The contrast with section 17.2 is worth naming precisely, because the two words are easy to swap. A projective module is one out of which every homomorphism *lifts* along a surjection. An injective module is one into which every homomorphism *extends* along an injection.

Definition 17.3.3: Injective Module

An R-module Q is *injective* if the functor $\operatorname{Hom}_R(-, Q)$ is exact, that is if for every short exact sequence $0 \to L \to M \to N \to 0$ of R-modules the sequence

$$0 \to \operatorname{Hom}_R(N, Q) \to \operatorname{Hom}_R(M, Q) \to \operatorname{Hom}_R(L, Q) \to 0$$

is again short exact.

Theorem 17.3.4: Injective Modules Equivalences

Let Q be an R-module. The following are equivalent.

1. Q is injective.
2. *(Extension.)* For all R-modules L and M, if $\psi\colon L \to M$ is injective, then every $f \in \operatorname{Hom}_R(L, Q)$ has an extension $F \in \operatorname{Hom}_R(M, Q)$ with $F \circ \psi = f$:

$$\begin{tikzcd} 0 \arrow[r] & L \arrow[r, hook, "\psi"] \arrow[d, "f"'] & M \arrow[dl, dashed, "F"] \\ & Q & \end{tikzcd}$$

3. *(Splitting.)* Every short exact sequence $0 \to Q \to M \to N \to 0$ of R-modules splits. Equivalently, whenever Q is a submodule of an R-module M, it is a direct summand of M.

***Proof*:**

(1) $\Rightarrow$ (2). Let $\psi\colon L \to M$ be injective and put $N = M/\psi(L)$, so that $0 \to L \xrightarrow{\psi} M \to N \to 0$ is short exact by proposition 17.1.1 and theorem 12.2.10. Condition (1) makes $\psi^*\colon \operatorname{Hom}_R(M, Q) \to \operatorname{Hom}_R(L, Q)$ surjective, so a given f is $\psi^*(F) = F \circ \psi$ for some F.

(2) $\Rightarrow$ (1). Proposition 17.3.1 gives exactness at the first two spots of the sequence in definition 17.3.3, and (2) applied to the injection ψ says precisely that ψ^* is surjective.

(2) $\Rightarrow$ (3). Let $0 \to Q \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ be short exact, so ψ is injective. Applying (2) to $f = \mathrm{id}_Q$ gives $\lambda \in \operatorname{Hom}_R(M, Q)$ with $\lambda \circ \psi = \mathrm{id}_Q$, a retraction of ψ, so the sequence splits by proposition 17.1.13.

(3) $\Rightarrow$ (2). Let $\psi\colon L \to M$ be injective and $f \in \operatorname{Hom}_R(L, Q)$. The construction that follows glues Q and M along L. Let

$$W = \big\{ \big(f(l), -\psi(l)\big) \;\big|\; l \in L \big\} \subseteq Q \oplus M$$

which is a submodule, being the image of the homomorphism $l \mapsto (f(l), -\psi(l))$, and set $X = (Q \oplus M)/W$. Write $\iota\colon Q \to X$ for $q \mapsto \overline{(q, 0)}$ and $j\colon M \to X$ for $m \mapsto \overline{(0, m)}$, both homomorphisms.

The map ι is injective: if $(q, 0) \in W$ then $q = f(l)$ and $\psi(l) = 0$ for some l, and ψ is injective, so $l = 0$ and $q = 0$. Thus $\iota(Q)$ is a submodule of X isomorphic to Q, and by (3) it is a direct summand, say $X = \iota(Q) \oplus C$. Let $\rho\colon X \to Q$ be the projection onto the first summand followed by ι^{-1}, so that $\rho \circ \iota = \mathrm{id}_Q$.

Put $F := \rho \circ j\colon M \to Q$. For $l \in L$ the element $(f(l), -\psi(l))$ lies in W, so $\overline{(0, \psi(l))} = \overline{(f(l), 0)}$ in X, that is $j(\psi(l)) = \iota(f(l))$. Hence

$$F(\psi(l)) = \rho\big(j(\psi(l))\big) = \rho\big(\iota(f(l))\big) = f(l)$$

so $F \circ \psi = f$.

The name records condition (2): the property concerns extending along injections, exactly as *projective* records lifting along surjections. Condition (3) is the concrete reading: an injective module cannot be enlarged without splitting off, so it is as far from the inside of a bigger module as a submodule can be.

For projective modules theorem 17.2.4 offered a fourth description, as a direct summand of a free module, which reduces every question to free modules. Nothing that simple is available here, and the classification of injective modules over a general ring is genuinely harder. What replaces it is a test that only has to be run on the ideals of R.

Theorem 17.3.5: Baer's Criterion

Let R be a ring and Q an R-module.

1. Q is injective if and only if, for every left ideal I of R, every R-module homomorphism $g\colon I \to Q$ extends to an R-module homomorphism $G\colon R \to Q$.
2. Suppose R is a principal ideal domain. Then Q is injective if and only if $rQ = Q$ for every *nonzero* $r \in R$.

The hypothesis $r \neq 0$ in (2) is not cosmetic: $0 \cdot Q = 0$, so without it the condition would fail for every nonzero module.

***Proof*:**
(1) ($\Rightarrow$) The inclusion $I \hookrightarrow R$ is injective, so theorem 17.3.4(2) extends any $g\colon I \to Q$ to $G\colon R \to Q$.

($\Leftarrow$) Assume the extension property for ideals. It suffices to verify theorem 17.3.4(2), so let $\psi\colon L \to M$ be injective and $f \in \mathrm{Hom}_R(L, Q)$. Replacing L by $\psi(L)$ and f by $f \circ (\psi|^{\psi(L)})^{-1}$, we may assume L is a submodule of M and ψ the inclusion.

Step 1: a poset of partial extensions. Let

$$\mathcal{L} = \{(L', f') \mid L \subseteq L' \subseteq M,\ f' \in \mathrm{Hom}_R(L', Q),\ f'|_L = f\}$$

ordered by $(L', f') \leq (L'', f'')$ when $L' \subseteq L''$ and $f''|_{L'} = f'$. This is a partial order, and $\mathcal{L}$ is nonempty because $(L, f) \in \mathcal{L}$.

Step 2: chains have upper bounds. Let $\{(L_i, f_i)\}_{i \in \mathcal{I}}$ be a chain. The union $\mathfrak{L} = \bigcup_i L_i$ is a submodule of M, since any two of its elements lie in a common L_i by totality of the chain. Define $\mathfrak{f}\colon \mathfrak{L} \to Q$ by $\mathfrak{f}(x) = f_i(x)$ for any i with $x \in L_i$; this is unambiguous because the chain is compatible, and it is a homomorphism because any two elements lie in a common L_i. Then $(\mathfrak{L}, \mathfrak{f}) \in \mathcal{L}$ bounds the chain.

Step 3: a maximal element. By Zorn's lemma (theorem A.2.3) there is a maximal $(M', F) \in \mathcal{L}$.

Step 4: $M' = M$. Suppose not, and choose $m \in M$ with $m \notin M'$. Set

$$I = \{r \in R \mid rm \in M'\}$$

which is a left ideal of R: it contains 0; it is closed under addition because M' is; and if $r \in I$ and $s \in R$ then $(sr)m = s(rm) \in M'$. Define $g\colon I \to Q$ by $g(r) = F(rm)$, which is a homomorphism since F is and since $r \mapsto rm$ is. By hypothesis g extends to $G\colon R \to Q$.

Now let $M'' = M' + Rm$, a submodule of M strictly containing M', and define

$$F' \colon M'' \to Q, \qquad F'(m' + rm) = F(m') + G(r)$$

This is well defined. If $m_1' + r_1 m = m_2' + r_2 m$ then $(r_1 - r_2)m = m_2' - m_1' \in M'$, so $r_1 - r_2 \in I$ and therefore

$$G(r_1) - G(r_2) = G(r_1 - r_2) = g(r_1 - r_2) = F\big((r_1 - r_2)m\big) = F(m_2') - F(m_1')$$

which rearranges to $F(m_1') + G(r_1) = F(m_2') + G(r_2)$. That F' is a homomorphism is immediate from the formula, and taking $r = 0$ gives $F'|_{M'} = F$, so in particular $F'|_L = f$ and $(M'', F') \in \mathcal{L}$.

Since $m \in M''$ and $m \notin M'$, the pair (M'', F') is strictly larger than (M', F), contradicting maximality. Hence $M' = M$ and F is the required extension.

(2) ($\Rightarrow$) Let $x \in Q$ and $r \in R$ with $r \neq 0$. Define $g \colon (r) \to Q$ by $g(sr) = sx$. This is well defined because R is an integral domain: if $sr = s'r$ with $r \neq 0$ then $s = s'$. It is a homomorphism, and by part (1) it extends to $G \colon R \to Q$. Then

$$x = g(r) = G(r) = r\,G(1)$$

so $x \in rQ$. As x was arbitrary, $rQ = Q$. Only the domain hypothesis was used here, not that ideals are principal, so this direction holds over any integral domain.

($\Leftarrow$) Assume $rQ = Q$ for every nonzero r, and verify the criterion of part (1). Let I be an ideal of R and $g \colon I \to Q$ a homomorphism. Since R is a principal ideal domain, $I = (r)$ for some r. If $r = 0$ then $I = 0$ and $G = 0$ extends g. If $r \neq 0$, then $g(r) \in Q = rQ$, so there is a $y \in Q$ with $ry = g(r)$. Define $G \colon R \to Q$ by $G(s) = sy$, a homomorphism. For any element sr of I,

$$G(sr) = sry = s\,g(r) = g(sr)$$

so $G|_I = g$. By part (1), Q is injective.

Corollary 17.3.6: Quotients of Injectives Over a PID

Let R be a principal ideal domain, Q an injective R-module and $N \subseteq Q$ a submodule. Then Q/N is injective.

***Proof*:**
For $r \neq 0$ we have $rQ = Q$ by theorem 17.3.5(2), hence

$$r(Q/N) = (rQ + N)/N = (Q + N)/N = Q/N$$

and theorem 17.3.5(2) applies again in the other direction.

The condition appearing in theorem 17.3.5(2) is worth a name of its own, both because it is the only practical test for injectivity over $\mathbb{Z}$ and because it is the source of every concrete injective module in this chapter.

Definition 17.3.7: Divisible Group

Let R be an integral domain. An R-module M is *divisible* if $rM = M$ for every nonzero $r \in R$; equivalently, if for every $x \in M$ and every nonzero $r \in R$ there is a $y \in M$ with $ry = x$.
For $R = \mathbb{Z}$ this says that an abelian group G is divisible when every $x \in G$ is n times some element of G, for every $n \geq 1$. Such a G is called a *divisible group*.

In this language, theorem 17.3.5(2) reads: over a principal ideal domain, injective and divisible are the same condition. Over an integral domain that is not principal only half survives, namely that injective modules are divisible; the reverse implication used principality when it wrote $I = (r)$, and it genuinely fails in general.

Example 17.3.8: Divisible Groups

1. $\mathbb{Q}$ is divisible: given $\frac{a}{b} \in \mathbb{Q}$ and $n \geq 1$, the rational $\frac{a}{nb}$ satisfies $n \cdot \frac{a}{nb} = \frac{a}{b}$. So $\mathbb{Q}$ is an injective $\mathbb{Z}$-module. More generally the field of fractions of any integral domain R is a divisible R-module.

2. Any quotient of a divisible module is divisible, since $r(M/N) = (rM + N)/N = M/N$. In particular $\mathbb{Q}/\mathbb{Z}$ is divisible, hence an injective $\mathbb{Z}$-module. Note that $\mathbb{Q}/\mathbb{Z}$ is a torsion group, so divisibility does not force torsion-freeness; and in a torsion group the element y with $ry = x$ is not unique.

3. Fix a prime p and let
$$\mathbb{Z}_{p^\infty} = \{[x] \in \mathbb{Q}/\mathbb{Z} \mid p^n[x] = 0 \text{ for some } n \geq 0\} = \left\{ \left[\tfrac{a}{p^n}\right] \in \mathbb{Q}/\mathbb{Z} \;\middle|\; a \in \mathbb{Z},\ n \geq 0 \right\}$$
the p-primary part of $\mathbb{Q}/\mathbb{Z}$, called the *Prüfer p-group*[a]. The two descriptions agree: $p^n \left[\frac{a}{p^n}\right] = [a] = 0$ gives one inclusion, and if $p^n[x] = 0$ with $[x] = \left[\frac{c}{d}\right]$ in lowest terms then $d \big| p^n$, so $[x]$ has the displayed form. It is a subgroup, being the set of elements killed by some power of p.
It is divisible. Let $[x] = \left[\frac{a}{p^n}\right]$ and let $m \geq 1$, and write $m = p^k u$ with u coprime to p. Since u is invertible modulo p^{n+k}, choose v with $uv \equiv 1 \pmod{p^{n+k}}$, say $uv = 1 + p^{n+k}w$, and set $y = \left[\frac{av}{p^{n+k}}\right] \in \mathbb{Z}_{p^\infty}$. Then
$$my = \left[\frac{p^k uav}{p^{n+k}}\right] = \left[\frac{a(1+p^{n+k}w)}{p^n}\right] = \left[\frac{a}{p^n} + ap^k w\right] = [x]$$
since $ap^k w \in \mathbb{Z}$. So $\mathbb{Z}_{p^\infty}$ is an injective $\mathbb{Z}$-module.

4. A direct product $\prod_i M_i$ of divisible modules is divisible, since the required y may be chosen coordinatewise, and so is a direct sum. Conversely if $M \oplus N$ is divisible then so are M and N: given $x \in M$ and $r \neq 0$, some (y, z) has $r(y, z) = (x, 0)$, and then $ry = x$.

5. No nonzero finite abelian group is divisible: if $|G| = m$ then $mg = 0$ for every g, so $mG = 0 \neq G$. Combined with theorem 5.1.8, no nonzero finitely generated abelian group is divisible, since a free summand $\mathbb{Z}$ is not divisible either.

[a] It is also $\varinjlim \mathbb{Z}/p^k\mathbb{Z}$ along the inclusions.

Before collecting examples of injective modules, one closure property is needed, and it is the one that fails to be symmetric with the projective case.

Proposition 17.3.9: Summands and Products of Injectives

Let R be a ring.

1. A direct summand of an injective R-module is injective.
2. A direct product $\prod_{i\in I} Q_i$ of R-modules is injective if and only if every Q_i is injective. In particular a finite direct sum of injective modules is injective.

***Proof*:**
(1) Let $Q = E \oplus C$ with Q injective, and let $\iota\colon E \to Q$ and $\rho\colon Q \to E$ be the inclusion and projection, so $\rho \circ \iota = \mathrm{id}_E$. Given an injection $\psi\colon L \to M$ and $f \in \mathrm{Hom}_R(L, E)$, the map $\iota \circ f$ extends to some $H \in \mathrm{Hom}_R(M, Q)$ by theorem 17.3.4(2). Then $F := \rho \circ H$ satisfies $F \circ \psi = \rho \circ \iota \circ f = f$, so theorem 17.3.4(2) holds for E.

(2) ($\Leftarrow$) Let every Q_i be injective, let $\psi\colon L \to M$ be injective and let $f \in \mathrm{Hom}_R(L, \prod_i Q_i)$. Each component $\pi_i \circ f$ extends to $F_i \in \mathrm{Hom}_R(M, Q_i)$, and $F := (F_i)_i$ satisfies $\pi_i \circ F \circ \psi = \pi_i \circ f$ for every i, hence $F \circ \psi = f$.

($\Rightarrow$) Each Q_j is a direct summand of $\prod_i Q_i$, with complement $\prod_{i\neq j} Q_i$, so part (1) applies.

Example 17.3.10: Injective Modules

1. $\mathbb{Z}$ is not an injective $\mathbb{Z}$-module: it is not divisible, since $2y = 1$ has no solution in $\mathbb{Z}$. Equivalently, the sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ does not split (example 17.1.14(2)), so theorem 17.3.4(3) fails. Since $\mathbb{Z}$ is free, free does not imply injective, in contrast with corollary 17.2.6.

2. $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and $\mathbb{Z}_{p^\infty}$ are injective $\mathbb{Z}$-modules, by example 17.3.8 and theorem 17.3.5(2). So is $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$, by proposition 17.3.9(2).

3. No nonzero finitely generated $\mathbb{Z}$-module is injective, by example 17.3.8(5).

4. If $R = F$ is a field then every F-module is injective. Indeed every F-module is free (theorem 13.1.8), and any short exact sequence of F-modules $0 \to Q \to M \to N \to 0$ has N free, hence projective, hence the sequence splits by theorem 17.2.4(3); theorem 17.3.4(3) follows. Over a field, projective, free and injective all coincide.

5. $\mathbb{Z}/2\mathbb{Z}$ is not an injective $\mathbb{Z}/4\mathbb{Z}$-module. It occurs as the ideal $(\overline{2}) \subseteq \mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/4\mathbb{Z}$ is indecomposable as a module over itself, since its only proper nonzero submodule is $(\overline{2})$ and a complement would have to be a second one. So $(\overline{2})$ is not a direct summand and theorem 17.3.4(3) fails. Compare exercise set 17.2.1, question 1: the same module is not projective either.

17.3.1 Embedding a Module in an Injective

Example 17.1.4(3) says every module is a quotient of a free, hence of a projective, module. The dual statement is that every module is a submodule of an injective one. It is true, but it is not formal: there is no dual of a free module to hand, and the proof goes through $\mathbb{Z}$ and back.

Lemma 17.3.11: $\mathbb{Z}$-module Submodules of Injective Modules

Every $\mathbb{Z}$-module is a submodule of an injective $\mathbb{Z}$-module.

***Proof*:**
Let M be a $\mathbb{Z}$-module and A a set of generators, so that theorem 12.3.11 gives a surjection $\pi\colon \bigoplus_{a\in A}\mathbb{Z} \to M$ from the free $\mathbb{Z}$-module on A. Writing $K = \ker(\pi)$, theorem 12.2.10 gives $M \cong \bigoplus_{a\in A}\mathbb{Z}/K$.

Now $\bigoplus_{a\in A}\mathbb{Z}$ sits inside $V := \bigoplus_{a\in A}\mathbb{Q}$, the direct sum of copies of $\mathbb{Q}$ indexed by A, which is divisible by example 17.3.8(1) and (4), hence injective. Since $K \subseteq \bigoplus_{a\in A}\mathbb{Z} \subseteq V$, the quotient V/K is defined, and it is divisible by example 17.3.8(2), hence injective. Finally

$$M \cong \bigoplus_{a\in A}\mathbb{Z}/K \subseteq V/K$$

exhibits M as a submodule of an injective $\mathbb{Z}$-module.

Passing from $\mathbb{Z}$ to an arbitrary ring is done by a change of rings, and the mechanism is the adjunction of theorem 15.3.9. The proof below writes the correspondence out explicitly rather than invoking naturality.

Lemma 17.3.12: Inducing Injective R-modules

Let $f\colon S \to R$ be a homomorphism of commutative rings and let Q be an injective S-module. Regard R as an S-module through f, and give

$$\mathrm{Hom}_S(R, Q)$$

the R-module structure $(r \cdot h)(x) := h(xr)$. Then $\mathrm{Hom}_S(R, Q)$ is an injective R-module.

***Proof*:**
Write $f_*(N)$ for an R-module N regarded as an S-module through f, so that $s \cdot n = f(s)n$.

Step 1: the correspondence. For an R-module N define

$$\alpha \mapsto \widetilde{\alpha}, \quad \widetilde{\alpha}(n) = \alpha(n)(1) \qquad \text{and} \qquad \beta \mapsto \widehat{\beta}, \quad \widehat{\beta}(n)(r) = \beta(rn)$$

between $\mathrm{Hom}_R\big(N, \mathrm{Hom}_S(R, Q)\big)$ and $\mathrm{Hom}_S\big(f_*(N), Q\big)$. These are the two directions of theorem 15.3.9 applied with N in the role of A, the ring R in the role of the bimodule B and Q in the role of C, using $N \otimes_R R \cong N$; the verifications are short enough to give directly.

That $\widetilde{\alpha}$ is S-linear: $\alpha(n) \in \mathrm{Hom}_S(R, Q)$ is S-linear on R, whose S-action is $s \cdot x = f(s)x$, so

$\alpha(n)(f(s)) = s\,\alpha(n)(1)$; combining with R-linearity of α,

$$\widetilde{\alpha}(s \cdot n) = \alpha(f(s)n)(1) = \big(f(s) \cdot \alpha(n)\big)(1) = \alpha(n)(f(s)) = s\,\widetilde{\alpha}(n)$$

That $\widehat{\beta}(n)$ is S-linear: $\widehat{\beta}(n)(s \cdot r) = \beta(f(s)rn) = \beta(s \cdot (rn)) = s\,\widehat{\beta}(n)(r)$. That $\widehat{\beta}$ is R-linear: $\widehat{\beta}(r'n)(r) = \beta(rr'n) = \widehat{\beta}(n)(rr') = \big(r' \cdot \widehat{\beta}(n)\big)(r)$.

The two assignments are mutually inverse:

$$\widehat{\widetilde{\alpha}}(n)(r) = \widetilde{\alpha}(rn) = \alpha(rn)(1) = \big(r \cdot \alpha(n)\big)(1) = \alpha(n)(r), \qquad \widetilde{\widehat{\beta}}(n) = \widehat{\beta}(n)(1) = \beta(n)$$

and both are visibly compatible with restriction along a submodule inclusion $L \subseteq N$, in the sense that $\widetilde{\alpha|_L} = \widetilde{\alpha}|_L$.

Step 2: the extension property. Verify theorem 17.3.4(2) for $\mathrm{Hom}_S(R, Q)$. Let $L \subseteq M$ be R-modules and $\alpha \in \mathrm{Hom}_R(L, \mathrm{Hom}_S(R, Q))$. Then $\widetilde{\alpha} \in \mathrm{Hom}_S(f_*(L), Q)$, and $f_*(L) \subseteq f_*(M)$ is still an inclusion of S-modules, since restriction of scalars changes only the action. As Q is an injective S-module, theorem 17.3.4(2) gives $\beta \in \mathrm{Hom}_S(f_*(M), Q)$ with $\beta|_L = \widetilde{\alpha}$.

Put $A := \widehat{\beta} \in \mathrm{Hom}_R(M, \mathrm{Hom}_S(R, Q))$. Then

$$A|_L = \widehat{\beta|_L} = \widehat{\widetilde{\alpha}} = \alpha$$

so A extends α, and $\mathrm{Hom}_S(R, Q)$ is injective.

Corollary 17.3.13: Inducing Injectives Using Divisible Groups

Let R be a commutative ring and D a divisible abelian group. Then $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

***Proof*:**
The group D is an injective $\mathbb{Z}$-module by theorem 17.3.5(2), and there is a unique ring homomorphism $\mathbb{Z} \to R$, so lemma 17.3.12 applies with $S = \mathbb{Z}$.

Lemma 17.3.14: R-module Submodules of Injective Modules

Let R be a commutative ring and M an R-module. Then M is a submodule of an injective R-module.

***Proof*:**
By lemma 17.3.11 there is an injective $\mathbb{Z}$-module D with $M \subseteq D$ as $\mathbb{Z}$-modules; by theorem 17.3.5(2) the group D is divisible. Now compose three injections of R-modules:

$$M \xrightarrow{\sim} \mathrm{Hom}_R(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, D)$$

The first sends m to the map $x \mapsto xm$, and is an isomorphism of R-modules for the action $(r \cdot h)(x) = h(xr)$, its inverse being $h \mapsto h(1)$. The second is the inclusion of the R-linear maps among the additive ones, which respects that action. The third is composition with $M \subseteq D$, injective because a map into M is zero exactly when it is zero into D. By corollary 17.3.13 the target $\mathrm{Hom}_{\mathbb{Z}}(R, D)$ is an injective R-module.

Definition 17.3.15: Injective Resolution

Let R be a ring and M an R-module. An *injective resolution* of M is an exact chain complex (definition 12.2.11)

$$0 \longrightarrow M \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

in which every Q^i is injective. It has *finite length* k if $Q^i = 0$ for every $i > k$.

Corollary 17.3.16: Existence of Injective Resolutions

Every module over a commutative ring admits an injective resolution.

***Proof*:**
Embed M in an injective Q^0 by lemma 17.3.14, embed the quotient Q^0/M in an injective Q^1, embed $Q^1/(Q^0/M)$ in an injective Q^2, and continue. The composites $Q^i \to Q^i/(\text{previous image}) \hookrightarrow Q^{i+1}$ have image the kernel of the next map by construction, which is exactness.

17.3.2 Injective Hulls

Every module embeds in an injective one, but lemma 17.3.14 produces a wasteful embedding: nothing stops Q from being far larger than M. Asking for the smallest such Q turns out to have a clean answer, and the condition that pins it down is that M should meet every nonzero piece of Q.

Definition 17.3.17: Essential Extension

Let $M \subseteq E$ be R-modules. Then M is *essential* in E, and E is an *essential extension* of M, if

$$N \cap M \neq 0 \qquad \text{for every nonzero submodule } N \subseteq E$$

Equivalently, for every $x \in E$ with $x \neq 0$ there is an $r \in R$ with $rx \in M$ and $rx \neq 0$.

The two formulations agree because Rx is a nonzero submodule whenever $x \neq 0$, and any nonzero N contains such an Rx.

Lemma 17.3.18: Essential Extensions Compose

Let $M \subseteq E \subseteq F$ be R-modules with M essential in E and E essential in F. Then M is essential in F.

***Proof*:**
Let $N \subseteq F$ be a nonzero submodule. Then $N \cap E \neq 0$ because E is essential in F, and $N \cap E$ is a nonzero submodule of E, so $(N \cap E) \cap M \neq 0$ because M is essential in E. That intersection is contained in $N \cap M$.

Theorem 17.3.19: Universal Property of Injective Hulls

Let R be a commutative ring and M an R-module.

1. There is an injective R-module E containing M as an essential submodule. Such an E is called an *injective hull*, or *injective envelope*, of M, and is written $E(M)$.
2. If E and E' are both injective hulls of M, there is an isomorphism $E \to E'$ restricting to the identity on M.
3. If $M \subseteq Q$ with Q injective, there is an injective homomorphism $E(M) \to Q$ restricting to the identity on M, and its image is a direct summand of Q. So $E(M)$ is the smallest injective module containing M.

***Proof*:**
(1) By lemma 17.3.14 choose an injective Q with $M \subseteq Q$.

Step 1: a maximal essential extension inside Q. Let

$$\Sigma = \{E \mid M \subseteq E \subseteq Q \text{ and } M \text{ is essential in } E\}$$

partially ordered by inclusion. It contains M, so it is nonempty. A chain $\{E_i\}$ in Σ has upper bound $\bigcup_i E_i$, which is a submodule of Q containing M, and M is essential in it: a nonzero x in the union lies in some E_i, and $Rx \cap M \neq 0$ there. By Zorn's lemma (theorem A.2.3) there is a maximal $E \in \Sigma$.

Step 2: a maximal complement. Let $\mathcal{C}$ be the set of submodules $C \subseteq Q$ with $C \cap E = 0$, ordered by inclusion. It contains 0, and a chain's union again meets E trivially, so Zorn's lemma gives a maximal $C \in \mathcal{C}$.

Step 3: E is essential in Q/C. First, $E \cap C = 0$, so E maps injectively into Q/C. Let X/C be a nonzero submodule of Q/C, so $C \subsetneq X \subseteq Q$. By maximality of C we have $X \cap E \neq 0$; choose $0 \neq e \in X \cap E$. Then $e \notin C$, since $E \cap C = 0$, so the class $e + C$ is a nonzero element of X/C lying in the image of E.

Step 4: E is a direct summand of Q. The sum $E + C$ is direct, so there is a projection $p\colon E \oplus C \to E \subseteq Q$. Since $E \oplus C$ is a submodule of Q and Q is injective, theorem 17.3.4(2) extends p to a homomorphism $g\colon Q \to Q$ with $g|_E = \mathrm{id}_E$ and $g(C) = 0$.

Then $\ker(g) \cap E = 0$, because g is the identity on E, and $C \subseteq \ker(g)$; so maximality of C forces $C = \ker(g)$. Hence g induces an isomorphism $Q/C \to \mathrm{im}(g)$, carrying the image of E onto E itself. By Step 3, E is essential in $\mathrm{im}(g)$, and M is essential in E, so M is essential in $\mathrm{im}(g)$ by lemma 17.3.18. Thus $\mathrm{im}(g) \in \Sigma$ and $E \subseteq \mathrm{im}(g)$, and maximality of E gives $E = \mathrm{im}(g)$.

Now $g\colon Q \to Q$ has image E and restricts to the identity on E, so $g \circ g = g$. An idempotent endomorphism splits its domain: every q is $g(q) + (q - g(q))$ with the first term in $\mathrm{im}(g)$ and the second in $\ker(g)$, and an element of $\mathrm{im}(g) \cap \ker(g)$ is $g(y)$ with $0 = g(g(y)) = g(y)$. Therefore $Q = \mathrm{im}(g) \oplus \ker(g) = E \oplus C$.

Step 5. By proposition 17.3.9(1) the summand E of the injective module Q is injective, and M is essential in E by construction.

(2) Let E and E' be injective hulls of M. Since E' is injective and $M \subseteq E$, theorem 17.3.4(2) extends the inclusion $M \hookrightarrow E'$ to a homomorphism $h\colon E \to E'$. Then $\ker(h) \cap M = 0$, because

h is injective on M, and M is essential in E, so $\ker(h) = 0$ and h is injective.

The image $h(E) \cong E$ is injective, so $h(E)$ is a direct summand of E' by theorem 17.3.4(3), say $E' = h(E) \oplus X$. Since $M = h(M) \subseteq h(E)$, the submodule X meets M trivially, and M is essential in E', so $X = 0$. Hence h is an isomorphism, and it is the identity on M by construction.

(3) Let $M \subseteq Q$ with Q injective. Extending the inclusion $M \hookrightarrow Q$ along $M \subseteq E(M)$ gives $h\colon E(M) \to Q$ with $h|_M$ the inclusion. As in (2), $\ker(h) \cap M = 0$ and M essential in $E(M)$ force h to be injective, and $h(E(M)) \cong E(M)$ is injective, hence a direct summand of Q by theorem 17.3.4(3).

Combining (1) and (3): every injective module containing M contains a copy of $E(M)$ over M, and $E(M)$ is injective, so $E(M)$ is a smallest one. Exercise set 17.3.1 below computes $E(\mathbb{Z}) = \mathbb{Q}$ and $E(\mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}_{p^\infty}$.

One consequence of injectivity is worth recording before leaving the section, because it is what makes duality work for abelian groups at all. Section 13.3 defined the dual of a vector space as $V^* = \operatorname{Hom}_k(V, k)$ and found $V \cong V^{**}$ naturally in finite dimensions. Over $\mathbb{Z}$ the same recipe collapses: $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ for every $n \geq 2$, by the argument of example 17.2.2, so dualizing against $\mathbb{Z}$ destroys every torsion group. Replacing $\mathbb{Z}$ by the injective module $\mathbb{Q}/\mathbb{Z}$ repairs it, since a homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ sends $\overline{1}$ to an element killed by n, and those form the cyclic group $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, so

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$$

The module $\mathbb{Q}/\mathbb{Z}$ is the *dualizing module* for abelian groups. Injectivity is exactly what makes this work: it is what keeps $\operatorname{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ exact, so that the dual of a submodule is a quotient of the dual. The isomorphism displayed above is not natural, there being no canonical generator of a cyclic group, but the double dual is; the systematic development, in which these homomorphisms become characters, is [9, Character Theory].

Exercise 17.3.1

1. For each pair, decide whether the module is injective over the indicated ring, and justify the answer.

    1. $\mathbb{Q}$ over $\mathbb{Z}$
    2. $\mathbb{Z}/6\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    3. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/4\mathbb{Z}$
    4. $\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$
    5. $F[x]$ over $F[x]$, for a field F

2. Let $g\colon 6\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ be the homomorphism of $\mathbb{Z}$-modules determined by $g(6) = \left[\frac{1}{4}\right]$. Find every extension $G\colon \mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ of g, and say how many there are. Then explain why theorem 17.3.5 guarantees at least one without any computation.

3. Essential extensions.

    1. Show that $\mathbb{Z}$ is essential in $\mathbb{Q}$ as a $\mathbb{Z}$-module.
    2. Show that $\mathbb{Z}/p\mathbb{Z}$ is essential in $\mathbb{Z}/p^2\mathbb{Z}$.
    3. Show that $\mathbb{Z} \oplus 0$ is *not* essential in $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

4. Deduce from (a) that $E(\mathbb{Z}) \cong \mathbb{Q}$.

4. Show that every nonzero subgroup of $\mathbb{Z}_{p^\infty}$ contains an element of order p, and deduce that $\mathbb{Z}/p\mathbb{Z}$ is essential in $\mathbb{Z}_{p^\infty}$. Conclude that $E(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}_{p^\infty}$ as $\mathbb{Z}$-modules. *Hint*: the subgroups of $\mathbb{Z}_{p^\infty}$ are totally ordered by inclusion.

5. Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules.

   1. Show that if L is injective the sequence splits, so $M \cong L \oplus N$.
   2. Show that if L and N are injective then so is M.
   3. Give an example with M injective and N not injective. *Hint*: take $R = \mathbb{Z}/4\mathbb{Z}$, and use example 17.3.10(5). Why does no such example exist over $\mathbb{Z}$?

6. *(The dual of Schanuel's lemma.)* Let

$$0 \to M \to Q_1 \to C_1 \to 0 \qquad \text{and} \qquad 0 \to M \to Q_2 \to C_2 \to 0$$

   be short exact sequences of R-modules with Q_1 and Q_2 injective. Prove that $C_1 \oplus Q_2 \cong C_2 \oplus Q_1$. *Hint*: run the proof of proposition 17.2.12 backwards, replacing the submodule of $P_1 \oplus P_2$ by the quotient of $Q_1 \oplus Q_2$ by the copy of M embedded as $m \mapsto (\iota_1(m), -\iota_2(m))$.

7. Compute injective resolutions.

   1. Show that $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is an injective resolution of $\mathbb{Z}$ as a $\mathbb{Z}$-module, of length 1.
   2. Show that multiplication by p on $\mathbb{Z}_{p^\infty}$ is surjective with kernel the subgroup of order p, and deduce that $0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}_{p^\infty} \xrightarrow{\cdot p} \mathbb{Z}_{p^\infty} \to 0$ is an injective resolution of $\mathbb{Z}/p\mathbb{Z}$, again of length 1.
   3. Conclude that every $\mathbb{Z}$-module has an injective resolution of length at most 1. *Hint*: corollary 17.3.6.

17.4⋆ Divisible Groups

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Example 17.3.8 exhibited $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and the Prüfer groups $\mathbb{Z}_{p^\infty}$ as divisible, and theorem 17.3.5(2) made each of them an injective $\mathbb{Z}$-module. This section proves that the list is complete: every divisible group is a direct sum of copies of those, in essentially one way. Since divisible and injective coincide over $\mathbb{Z}$, that is a full classification of the injective $\mathbb{Z}$-modules.

Everything here is about abelian groups, so *group* means *abelian group* throughout, written additively, and $\mathrm{Tor}(G)$ denotes the torsion subgroup of definition 12.3.18.

A subgroup of a divisible group need not be divisible: $\mathbb{Z} \subseteq \mathbb{Q}$ is the standard example. It is therefore not obvious that a group has a largest divisible subgroup, and the first thing to establish is that it does.

Definition 17.4.1: Divisible Subgroup

Let G be an abelian group. Write dG for the subgroup of G generated by all the divisible subgroups of G, that is, the set of finite sums $x_1 + \cdots + x_k$ with each x_i lying in some divisible subgroup. If $dG = 0$, the group G is called *reduced*.

Lemma 17.4.2: dG Is the Largest Divisible Subgroup

Let G be an abelian group. Then dG is divisible, and every divisible subgroup of G is contained in dG.

***Proof*:**
Let $x \in dG$ and $n \geq 1$. Write $x = x_1 + \cdots + x_k$ with $x_i \in D_i$ for divisible subgroups $D_i \subseteq G$. Each D_i is divisible, so there is a $y_i \in D_i$ with $ny_i = x_i$. Then $y := y_1 + \cdots + y_k$ lies in dG and

$$ny = ny_1 + \cdots + ny_k = x_1 + \cdots + x_k = x$$

so $n\,dG = dG$. The second claim holds by construction, since every divisible subgroup is one of the D_i available in the generating set.

Lemma 17.4.3: Splitting Off the Divisible Part

Let G be an abelian group. Then there is a subgroup $H \subseteq G$ with

$$G = dG \oplus H$$

and every such H is reduced, and isomorphic to G/dG.

***Proof*:**
By lemma 17.4.2 the subgroup dG is divisible, hence an injective $\mathbb{Z}$-module by theorem 17.3.5(2), so the short exact sequence

$$0 \to dG \hookrightarrow G \to G/dG \to 0$$

splits by theorem 17.3.4(3). This gives $G = dG \oplus H$ for some subgroup H, with $H \cong G/dG$ by the remark following definition 17.1.11.

Such an H is reduced. The subgroup dH is a divisible subgroup of H, hence a divisible subgroup of G, so $dH \subseteq dG$ by lemma 17.4.2. Also $dH \subseteq H$. Therefore $dH \subseteq dG \cap H = 0$.

So every abelian group is a divisible group plus a reduced one, and the two halves do not interact. Classifying the divisible groups is thus a genuine piece of the classification of abelian groups, and it is the piece that has a clean answer. The route splits G once more, into its torsion part and the rest.

Lemma 17.4.4: The Torsion Subgroup of a Divisible Group

Let G be a divisible abelian group. Then $\mathrm{Tor}(G)$ is divisible, and

$$G \cong \mathrm{Tor}(G) \oplus G/\mathrm{Tor}(G)$$

***Proof*:**
Let $x \in \mathrm{Tor}(G)$ and $n \geq 1$. Since G is divisible there is a $y \in G$ with $ny = x$. Since x is a torsion element there is an $m \geq 1$ with $mx = 0$, and then

$$(mn)y = m(ny) = mx = 0$$

so y is a torsion element and $y \in \mathrm{Tor}(G)$. Hence $n\mathrm{Tor}(G) = \mathrm{Tor}(G)$ for every $n \geq 1$, and $\mathrm{Tor}(G)$ is divisible.

Being divisible, $\mathrm{Tor}(G)$ is an injective $\mathbb{Z}$-module by theorem 17.3.5(2), so the short exact sequence $0 \to \mathrm{Tor}(G) \to G \to G/\mathrm{Tor}(G) \to 0$ splits by theorem 17.3.4(3), and the complement is isomorphic to $G/\mathrm{Tor}(G)$.

The two factors are now treated separately. The quotient turns out to carry a structure it had no right to expect.

Lemma 17.4.5: Structure of $G/\mathrm{Tor}(G)$

Let G be a divisible abelian group. Then $G/\mathrm{Tor}(G)$ carries a unique structure of $\mathbb{Q}$-vector space compatible with its addition, and consequently

$$G/\mathrm{Tor}(G) \cong \bigoplus_{i \in I} \mathbb{Q}$$

for some index set I.

***Proof*:**
Write $A = G/\mathrm{Tor}(G)$.

Step 1: A is torsion-free. This holds for every abelian group G. If $n\overline{x} = \overline{0}$ with $n \geq 1$, then $nx \in \mathrm{Tor}(G)$, so $m(nx) = 0$ for some $m \geq 1$; then $(mn)x = 0$ with $mn \geq 1$, so $x \in \mathrm{Tor}(G)$ and $\overline{x} = \overline{0}$.

Step 2: A is divisible, being a quotient of the divisible group G (example 17.3.8(2)).

Step 3: division is unambiguous. If $ny = ny'$ with $n \geq 1$ then $n(y - y') = 0$, and A is torsion-free by Step 1, so $y = y'$. Combined with Step 2: for every $x \in A$ and $n \geq 1$ there is exactly one $y \in A$ with $ny = x$.

Step 4: the action. For $\frac{a}{b} \in \mathbb{Q}$ written with $b \geq 1$, and $x \in A$, define $\frac{a}{b} \cdot x$ to be the unique $y \in A$ with $by = ax$, which exists and is unique by Step 3.

This does not depend on the representative. If $\frac{a}{b} = \frac{a'}{b'}$ with $b, b' \geq 1$, so $ab' = a'b$, let y and y' be the elements produced by the two fractions. Then

$$bb'y = b'(by) = b'ax = a'bx = b(b'y') = bb'y'$$

and $bb' \geq 1$, so $y = y'$ by Step 3.

Step 5: the axioms. Write $y = \frac{a}{b}x$, $u = \frac{c}{d}x$ and $y_i = \frac{a}{b}x_i$, so $by = ax$, $du = cx$ and $by_i = ax_i$.

- Additivity in x: $b(y_1 + y_2) = ax_1 + ax_2 = a(x_1 + x_2)$, so $\frac{a}{b}(x_1 + x_2) = y_1 + y_2$.
- Additivity in the scalar: $bd(y + u) = d(by) + b(du) = dax + bcx = (ad + bc)x$, so $\frac{a}{b}x + \frac{c}{d}x = \frac{ad+bc}{bd}x$.
- Associativity: let $v = \frac{a}{b}u$, so $bv = au$. Then $bd\,v = d(bv) = dau = a(du) = acx$, so $\frac{a}{b}\left(\frac{c}{d}x\right) = \frac{ac}{bd}x$.
- Unit: $1 \cdot x$ is the unique y with $y = x$.

So A is a $\mathbb{Q}$-vector space, and the structure is unique because Step 3 leaves no choice about what $\frac{a}{b} \cdot x$ can be.

Step 6. By theorem 13.1.8 the vector space A has a basis, indexed by some set I, so $A \cong \bigoplus_{i \in I} \mathbb{Q}$.

The torsion factor is where the Prüfer groups appear, and the argument has three stages: break the torsion into primary pieces, check each piece is still divisible, and identify a divisible p-group.

Lemma 17.4.6: Structure of $\mathrm{Tor}(G)$

Let G be a divisible abelian group. Then

$$\mathrm{Tor}(G) \cong \bigoplus_{p \text{ prime}} \bigoplus_{i \in I_p} \mathbb{Z}_{p^\infty}$$

for index sets I_p.

***Proof*:**

Write $T = \mathrm{Tor}(G)$, which is divisible by lemma 17.4.4. For a prime p let

$$T_p = \left\{ x \in T \;\middle|\; p^k x = 0 \text{ for some } k \geq 0 \right\}$$

the *p-primary component*, a subgroup of T since $p^k x = p^l y = 0$ gives $p^{k+l}(x - y) = 0$.

Step 1: $T = \bigoplus_p T_p$. This holds for any torsion group. For the sum, let $x \in T$ have order $n = p_1^{n_1} \cdots p_k^{n_k}$ with the p_i distinct primes and $n_i \geq 1$, and set $q_i = n/p_i^{n_i}$. No prime divides every q_i, since p_j does not divide q_j, so $\gcd(q_1, \ldots, q_k) = 1$ and there are integers h_i with

$$h_1 q_1 + h_2 q_2 + \cdots + h_k q_k = 1 \tag{17.1}$$

Put $x_i = h_i q_i x$. Then $p_i^{n_i} x_i = h_i n x = 0$, so $x_i \in T_{p_i}$, and multiplying (17.1) by x gives $x = x_1 + \cdots + x_k$.

For directness, suppose $x \in T_p \cap \sum_{q \neq p} T_q$. Then $p^k x = 0$ for some k, so the order of x is a power of p. On the other hand $x = y_1 + \cdots + y_r$ with $y_j \in T_{q_j}$ for primes $q_j \neq p$, and each y_j has order a power of q_j, so the order of x divides a product of powers of primes different from p. A number that is both a power of p and coprime to p is 1, so $x = 0$.

Step 2: each T_p is divisible. Let $x \in T_p$ and $n \geq 1$. Since T is divisible there is a $y \in T$ with $ny = x$. Decompose $y = \sum_q y_q$ by Step 1. Then $x = ny = \sum_q n y_q$ with $n y_q \in T_q$, and x lies in

T_p, so uniqueness of the decomposition forces $ny_q = 0$ for $q \neq p$ and $ny_p = x$. As $y_p \in T_p$, this shows $nT_p = T_p$.

Step 3: a nonzero divisible p-group contains a copy of $\mathbb{Z}_{p^\infty}$. Let D be a nonzero divisible group in which every element has order a power of p, and pick $g \neq 0$, of order p^k with $k \geq 1$. Then $x_1 := p^{k-1}g$ has order exactly p. Using divisibility choose x_2 with $px_2 = x_1$, then x_3 with $px_3 = x_2$, and so on.

Each x_r has order exactly p^r: repeatedly applying $px_{i+1} = x_i$ gives $p^{r-1}x_r = x_1 \neq 0$ and $p^r x_r = px_1 = 0$. Also $x_r = px_{r+1} \in \langle x_{r+1}\rangle$, so the cyclic groups $\langle x_1\rangle \subseteq \langle x_2\rangle \subseteq \cdots$ form a chain, and $Z := \bigcup_r \langle x_r\rangle$ is a subgroup of D.

Define $\theta\colon Z \to \mathbb{Z}_{p^\infty}$ by $\theta(ax_r) = \left[\frac{a}{p^r}\right]$. It is well defined: if $ax_r = bx_s$ with $r \leq s$, then $x_r = p^{s-r}x_s$, so $(ap^{s-r} - b)x_s = 0$, so p^s divides $ap^{s-r} - b$, and therefore $\left[\frac{a}{p^r}\right] = \left[\frac{ap^{s-r}}{p^s}\right] = \left[\frac{b}{p^s}\right]$. It is a homomorphism by the same computation, surjective because every element of $\mathbb{Z}_{p^\infty}$ has the form $\left[\frac{a}{p^r}\right]$ (example 17.3.8(3)), and injective because $\theta(ax_r) = 0$ says $p^r \big| a$, which says $ax_r = 0$. So $Z \cong \mathbb{Z}_{p^\infty}$.

Step 4: a divisible p-group is a direct sum of copies of $\mathbb{Z}_{p^\infty}$. Let D be such a group and let $\mathcal{T}$ be the set of families of subgroups of D, each member of each family isomorphic to $\mathbb{Z}_{p^\infty}$, whose sum inside D is direct. Order $\mathcal{T}$ by inclusion of families. It contains the empty family. A chain has the union of its families as an upper bound, which is again in $\mathcal{T}$ because directness is a condition on finitely many members at a time, and any finitely many members of the union already lie in one family of the chain. By Zorn's lemma (theorem A.2.3) there is a maximal family $\{Z_j\}_{j\in J}$; put $H = \bigoplus_{j\in J} Z_j \subseteq D$.

The group H is divisible, being a direct sum of divisible groups (example 17.3.8(4)), hence injective, so $D = H \oplus K$ for some K by theorem 17.3.4(3). The complement K is divisible: given $x \in K$ and $n \geq 1$, divisibility of D gives $d = h + k$ with $nd = x$, and comparing components in $H \oplus K$ gives $nh = 0$ and $nk = x$. Every element of K has order a power of p, so if $K \neq 0$ then Step 3 gives a subgroup $Z \subseteq K$ isomorphic to $\mathbb{Z}_{p^\infty}$; since $H \cap K = 0$, the family $\{Z_j\}_{j\in J} \cup \{Z\}$ is again in $\mathcal{T}$ and strictly larger, contradicting maximality. So $K = 0$ and $D = H$.

Applying Steps 2 and 4 to each T_p and assembling with Step 1 gives the statement.

Theorem 17.4.7: Classification of Divisible Groups

Let G be a divisible abelian group. Then there are index sets I and I_p, one for each prime p, with

$$G \cong \left(\bigoplus_{p \text{ prime}} \bigoplus_{i\in I_p} \mathbb{Z}_{p^\infty}\right) \oplus \bigoplus_{i\in I} \mathbb{Q}$$

The cardinalities of these index sets are determined by G, namely

$$|I| = \dim_{\mathbb{Q}}\big(G/\mathrm{Tor}(G)\big) \qquad \text{and} \qquad |I_p| = \dim_{\mathbb{F}_p} G[p], \quad \text{where } G[p] = \{g \in G \mid pg = 0\}$$

Since divisible and injective agree over $\mathbb{Z}$ (theorem 17.3.5(2)), this classifies the injective $\mathbb{Z}$-modules.

***Proof*:**
Existence. By lemma 17.4.4, $G \cong \mathrm{Tor}(G) \oplus G/\mathrm{Tor}(G)$. Lemma 17.4.6 identifies the first summand and lemma 17.4.5 the second.

The invariants. Suppose G is written as displayed. Each $\mathbb{Z}_{p^\infty}$ is a torsion group and each copy of $\mathbb{Q}$ is torsion-free, so the torsion subgroup of the right-hand side is exactly the first bracket, and

$$G/\mathrm{Tor}(G) \cong \bigoplus_{i \in I} \mathbb{Q}$$

which is a $\mathbb{Q}$-vector space with a basis indexed by I. Since G determines $\mathrm{Tor}(G)$ and hence $G/\mathrm{Tor}(G)$ up to isomorphism, and the cardinality of a basis is an invariant of a vector space (lemma 14.1.1 and the remark following it), $|I|$ is determined by G.

For the primary invariants, note that $G[p]$ is killed by p and so is a vector space over $\mathbb{F}_p$, and that forming $G[p]$ commutes with direct sums, an element of a direct sum being killed by p exactly when each of its components is. Now

$$\mathbb{Q}[p] = 0, \qquad \mathbb{Z}_{q^\infty}[p] = 0 \text{ for } q \neq p, \qquad \mathbb{Z}_{p^\infty}[p] = \left\{ \left[\tfrac{a}{p}\right] \;\middle|\; a \in \mathbb{Z} \right\} \cong \mathbb{Z}/p\mathbb{Z}$$

The first holds because $\mathbb{Q}$ is torsion-free; the second because every element of $\mathbb{Z}_{q^\infty}$ has order a power of q, and no nonzero such order is divisible by p; the third because $p\left[\frac{a}{p^n}\right] = 0$ forces $p^{n-1} \big| a$, so the element is $\left[\frac{b}{p}\right]$ for some b. Therefore

$$G[p] \cong \bigoplus_{i \in I_p} \mathbb{Z}/p\mathbb{Z}$$

an $\mathbb{F}_p$-vector space with a basis indexed by I_p. As G determines $G[p]$, the cardinality $|I_p|$ is determined by G.

Exercise 17.4.1

1. Compute dG, the largest divisible subgroup, for each of the following.
   1. $G = \mathbb{Z}$
   2. $G = \mathbb{Q}$
   3. $G = \mathbb{Q} \oplus \mathbb{Z}$
   4. $G = \mathbb{Z}/6\mathbb{Z}$
   5. $G = \mathbb{Q}/\mathbb{Z}$
2. Decide which of the following are divisible, and justify each answer.
   1. $\mathbb{R}$ under addition
   2. $\mathbb{R}/\mathbb{Z}$ under addition
   3. $\mathbb{C}^\times$ under multiplication
   4. $\mathbb{Z}[1/p]$ under addition, for a prime p

5. $\mathbb{Q}^\times$ under multiplication

3. For each of the following divisible groups, compute the invariants $|I|$ and $|I_p|$ of theorem 17.4.7.
    1. $\mathbb{Q}/\mathbb{Z}$
    2. $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$
    3. $\mathbb{Z}_{p^\infty} \oplus \mathbb{Z}_{p^\infty}$
    4. $\mathbb{R}$ under addition. *Hint*: compare cardinalities.

4. Show that $\mathbb{Z}_{p^\infty} \cong \mathbb{Z}[1/p]/\mathbb{Z}$, and that the proper subgroups of $\mathbb{Z}_{p^\infty}$ are exactly the finite cyclic groups $\langle \left[\frac{1}{p^n}\right] \rangle \cong \mathbb{Z}/p^n\mathbb{Z}$ for $n \geq 0$, totally ordered by inclusion. Deduce that $\mathbb{Z}_{p^\infty}$ has no maximal subgroup.

5. Show that every homomorphism $f \colon G \to H$ of abelian groups satisfies $f(dG) \subseteq dH$, and that $G \mapsto dG$ is therefore a functor $\mathbf{Ab} \to \mathbf{Ab}$. *Hint*: the image of a divisible group is divisible.

6. Show that a nonzero divisible group has no maximal subgroup. *Hint*: if M were maximal, G/M would be a simple abelian group, hence $\mathbb{Z}/p\mathbb{Z}$ for some prime p; but G/M is divisible. Deduce that no nonzero divisible group is finitely generated, recovering example 17.3.8(5).

17.5 Flat Modules

The third functor available on a short exact sequence is the tensor product. Fix a right R-module D and apply $D \otimes_R -$, which theorem 15.2.2 turns into a covariant functor from left R-modules to abelian groups, landing in left S-modules when D is an (S, R)-bimodule. Everything below is stated for $D \otimes_R -$; the functor $- \otimes_R D$ on right modules behaves identically, with left and right exchanged throughout.

This functor fails at the opposite end from the two in section 17.2 and section 17.3. Both Hom functors are left exact and can lose surjectivity; tensoring is *right* exact and can lose injectivity. The reason is the Hom-tensor adjunction of theorem 15.3.9: $D \otimes_R -$ is a left adjoint, and left adjoints preserve the constructions that sit at the right-hand end of a sequence.

Theorem 17.5.1: $D \otimes_R -$ Is Right Exact

Let D be a right R-module and let

$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$

be a short exact sequence of left R-modules. Then

$$D \otimes_R L \xrightarrow{1\otimes\psi} D \otimes_R M \xrightarrow{1\otimes\varphi} D \otimes_R N \to 0$$

is an exact sequence of abelian groups, and of left S-modules when D is an (S, R)-bimodule. The map $1 \otimes \psi$ need not be injective.

***Proof*:**
Surjectivity of $1 \otimes \varphi$. By lemma 15.1.4(2) every element of $D \otimes_R N$ is a finite sum $\sum_i d_i \otimes n_i$. Each n_i is $\varphi(m_i)$ for some m_i, since φ is surjective, so the element is the image of $\sum_i d_i \otimes m_i$.

$\operatorname{im}(1 \otimes \psi) \subseteq \ker(1 \otimes \varphi)$. By theorem 15.2.2 the composite $(1 \otimes \varphi) \circ (1 \otimes \psi)$ is $1 \otimes (\varphi \circ \psi)$, and $\varphi \circ \psi = 0$ because $\operatorname{im}(\psi) = \ker(\varphi)$.

$\ker(1 \otimes \varphi) \subseteq \operatorname{im}(1 \otimes \psi)$. Write $I = \operatorname{im}(1 \otimes \psi)$. By the previous paragraph $1 \otimes \varphi$ kills I, so it factors as

$$\overline{\varphi}\colon (D \otimes_R M)/I \to D \otimes_R N, \qquad \overline{\varphi}\big(\overline{d \otimes m}\big) = d \otimes \varphi(m)$$

by theorem 12.2.10. It suffices to produce an inverse, for then $\overline{\varphi}$ is injective, which says exactly that $\ker(1 \otimes \varphi) = I$.

Define $\beta\colon D \times N \to (D \otimes_R M)/I$ by $\beta(d, n) = \overline{d \otimes m}$ for any m with $\varphi(m) = n$. This does not depend on the choice: if $\varphi(m) = \varphi(m')$ then $m - m' \in \ker(\varphi) = \operatorname{im}(\psi)$, say $m - m' = \psi(l)$, and then

$$d \otimes m - d \otimes m' = d \otimes \psi(l) = (1 \otimes \psi)(d \otimes l) \in I$$

The map β is R-balanced in the sense of definition 15.1.1: it is additive in d because $\otimes$ is; it is additive in n because a lift of $n + n'$ may be taken to be $m + m'$; and $\beta(dr, n) = \overline{dr \otimes m} = \overline{d \otimes rm} = \beta(d, rn)$, since rm is a lift of rn and by the third relation of (15.1).

By theorem 15.1.5 there is a homomorphism $\overline{\beta}\colon D \otimes_R N \to (D \otimes_R M)/I$ with $\overline{\beta}(d \otimes n) = \overline{d \otimes m}$. Both composites fix every simple tensor, hence are the identity by lemma 15.1.4(3), so $\overline{\varphi}$ is an isomorphism.

Now the failure at the left, computed before anything is named after it, on the same sequence that example 17.2.2 and example 17.3.2 used.

Example 17.5.2: $D \otimes_R -$ Need Not Be Left Exact

1. Take $R = \mathbb{Z}$, the short exact sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$, and $D = \mathbb{Z}/2\mathbb{Z}$. Since $A \otimes_{\mathbb{Z}} \mathbb{Z} \cong A$ for every A, and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ by proposition 15.2.13, applying $D \otimes_{\mathbb{Z}} -$ gives

   $$\mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \otimes (\cdot 2)} \mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \otimes \pi} \mathbb{Z}/2\mathbb{Z} \to 0$$

   The first map is zero: for $\overline{a} \in \mathbb{Z}/2\mathbb{Z}$ and $x \in \mathbb{Z}$,

   $$\overline{a} \otimes 2x = \overline{a} \cdot 2 \otimes x = \overline{0} \otimes x = 0$$

   So the sequence is exact where theorem 17.5.1 promises and fails to be exact at the left, the first map being zero on a group of order two.

2. The sharpest version replaces the middle term. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is injective, but $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$, since

   $$\overline{a} \otimes q = \overline{a} \otimes 2 \cdot \tfrac{q}{2} = \overline{a} \cdot 2 \otimes \tfrac{q}{2} = 0$$

   for every q. So $1 \otimes \iota$ maps $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ to the zero group, and injectivity fails as badly as it can.

Proposition 17.5.3: Conditions for $A \otimes_R -$ to Be Exact

Let A be a right R-module. The following are equivalent.

1. For every short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ of left R-modules, the sequence
$$0 \to A \otimes_R L \xrightarrow{1\otimes\psi} A \otimes_R M \xrightarrow{1\otimes\varphi} A \otimes_R N \to 0$$
is short exact.
2. For every injective homomorphism $\psi\colon L \to M$ of left R-modules, the map $1 \otimes \psi\colon A \otimes_R L \to A \otimes_R M$ is injective.

***Proof*:**

(1) $\Rightarrow$ (2). Given ψ injective, put $N = M/\psi(L)$, so that $0 \to L \xrightarrow{\psi} M \to N \to 0$ is short exact by proposition 17.1.1 and theorem 12.2.10. Condition (1) makes $1 \otimes \psi$ injective.

(2) $\Rightarrow$ (1). Theorem 17.5.1 gives exactness at $A \otimes_R M$ and at $A \otimes_R N$, and (2) gives injectivity of $1 \otimes \psi$, which is exactness at $A \otimes_R L$ by proposition 17.1.1.

Definition 17.5.4: Flat Module

A right R-module A is *flat* if it satisfies the equivalent conditions of proposition 17.5.3, that is if the functor $A \otimes_R -$ is exact. A left R-module is *flat* when the corresponding statement holds for $- \otimes_R A$.

The side matters only because $\otimes_R$ does. Over a commutative ring the two notions coincide, by proposition 15.2.7, and the qualification can be dropped.

Corollary 17.5.5: Projective Modules Are Flat

Every free R-module is flat, and more generally every projective R-module is flat.

***Proof*:**
First, R itself is flat: the map $R \otimes_R L \to L$ sending $r \otimes l$ to rl is an isomorphism with inverse $l \mapsto 1 \otimes l$, and under it $1 \otimes \psi$ becomes ψ, so $1 \otimes \psi$ is injective whenever ψ is.

Next let $F = \bigoplus_{i\in I} R$ be free. By theorem 15.2.4 there are isomorphisms $F \otimes_R L \cong \bigoplus_{i\in I}(R \otimes_R L) \cong \bigoplus_{i\in I} L$, and similarly for M, carrying $1 \otimes \psi$ to the map $\bigoplus_i \psi$ acting coordinatewise. A direct sum of injective maps is injective, so F is flat.

Finally let P be projective. By theorem 17.2.4(4) there is a free module F and a module K with $F \cong P \oplus K$. By theorem 15.2.3,
$$F \otimes_R L \cong (P \otimes_R L) \oplus (K \otimes_R L)$$
compatibly with M in place of L, and under these identifications $1_F \otimes \psi$ is $(1_P \otimes \psi) \oplus (1_K \otimes \psi)$. A direct sum of maps is injective only if each summand is, so injectivity of $1_F \otimes \psi$ forces injectivity of $1_P \otimes \psi$.

So free implies projective implies flat, and the second implication is strict in a way the first is not obviously: example 17.5.8 exhibits a flat $\mathbb{Z}$-module that is not projective. Over a principal ideal domain there is a criterion for flatness as concrete as Baer's is for injectivity.

Proposition 17.5.6: Flat Implies Torsion-Free

Let R be an integral domain and A a flat R-module. Then A is torsion-free.

***Proof*:**
Let $r \in R$ be nonzero. Multiplication by r is an injective R-module homomorphism $R \to R$, since R is a domain, so flatness makes $1 \otimes (\cdot r)\colon A \otimes_R R \to A \otimes_R R$ injective. The map $A \otimes_R R \to A$ sending $a \otimes r$ to ar is an isomorphism, with inverse $a \mapsto a \otimes 1$, and it carries $1 \otimes (\cdot r)$ to multiplication by r on A. Hence $ra = 0$ forces $a = 0$, and A has no nonzero torsion element.

Theorem 17.5.7: Flat Equals Torsion-Free Over a PID

Let R be a principal ideal domain and A an R-module. Then A is flat if and only if A is torsion-free.

***Proof*:**
One direction is proposition 17.5.6, which needs only that R is a domain.

For the converse let A be torsion-free, let $\psi\colon L \to M$ be an injective homomorphism, and let $\xi \in A \otimes_R L$ satisfy $(1 \otimes \psi)(\xi) = 0$. Write $\xi = \sum_{i=1}^k a_i \otimes l_i$ using lemma 15.1.4(2), so that

$$\sum_{i=1}^k a_i \otimes \psi(l_i) = 0 \qquad \text{in } A \otimes_R M$$

Step 1: pass to a finitely generated piece. By lemma 15.2.12 there is a finitely generated submodule $A_0 \subseteq A$ containing $a_1, \dots, a_k$ with

$$\sum_{i=1}^k a_i \otimes \psi(l_i) = 0 \qquad \text{in } A_0 \otimes_R M$$

Step 2: A_0 is flat. A submodule of a torsion-free module is torsion-free, so A_0 is a finitely generated torsion-free module over a principal ideal domain, hence free by theorem 14.1.9(2), hence flat by corollary 17.5.5.

Step 3: conclude. The map $1_{A_0} \otimes \psi\colon A_0 \otimes_R L \to A_0 \otimes_R M$ is therefore injective, and Step 1 says it kills the element $\sum_i a_i \otimes l_i$ of $A_0 \otimes_R L$. So that element is zero in $A_0 \otimes_R L$, and its image under the map $A_0 \otimes_R L \to A \otimes_R L$ induced by the inclusion is ξ. Hence $\xi = 0$, and $1 \otimes \psi$ is injective.

Example 17.5.8: Flat Modules

Throughout, $R = \mathbb{Z}$ unless stated otherwise, so theorem 17.5.7 applies and flat means torsion-free.

1. $\mathbb{Z}$ is flat, being free. For $n \geq 2$ the module $\mathbb{Z}/n\mathbb{Z}$ is not flat, having $\overline{1}$ as a torsion element; example 17.5.2(1) is that failure for $n = 2$, seen directly.

2. $\mathbb{Q}$ is a flat $\mathbb{Z}$-module, being torsion-free. It is *not* projective, by example 17.2.10(5). So the implication projective $\Rightarrow$ flat is strict, and flatness is genuinely weaker than the other two conditions of this chapter.

3. $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module (example 17.3.10(2)) but is not flat, being a nonzero torsion group. So injective does not imply flat.

4. $\mathbb{Q} \oplus \mathbb{Z}$ is torsion-free, hence flat. It is not projective: a direct summand of a projective module is projective, being a summand of a summand of a free module, and $\mathbb{Q}$ is not projective. It is not injective either, since a direct summand of an injective module is injective by proposition 17.3.9(1) and $\mathbb{Z}$ is not injective (example 17.3.10(1)).

5. A direct sum $\bigoplus_i A_i$ is flat if and only if every A_i is, by the computation in the proof of corollary 17.5.5 applied to theorem 15.2.4.

6. Over a field every module is free (theorem 13.1.8), hence flat. Over a field, free, projective, injective and flat all coincide.

7. Let R be a principal ideal domain and $D \subseteq R \setminus \{0\}$ a multiplicatively closed set containing 1. The ring of fractions $D^{-1}R$ of section 9.5 sits inside the field of fractions of R, which is torsion-free as an R-module, so $D^{-1}R$ is a torsion-free and therefore flat R-module. That localization is flat holds over any commutative ring, and is taken up in *Commutative Algebra* [8, Localization Is Exact, Characterization of Localization].

Corollary 17.5.9: Tensor Products of Projective Modules

Let R be a commutative ring and let P and Q be projective R-modules. Then $P \otimes_R Q$ is projective.

***Proof*:**

Choose P' and Q' with $P \oplus P' = F$ and $Q \oplus Q' = G$ free, using theorem 17.2.4(4). Then $F \otimes_R G$ is free by corollary 15.2.6, and theorem 15.2.3 expands it as

$$F \otimes_R G \cong (P \otimes_R Q) \oplus (P \otimes_R Q') \oplus (P' \otimes_R Q) \oplus (P' \otimes_R Q')$$

So $P \otimes_R Q$ is a direct summand of a free module and theorem 17.2.4(4) applies.

A ring homomorphism $R \to S$ making S flat as an R-module is called a *flat extension*, and the condition is what allows a construction performed over R to be transported to S without losing exactness. Example 17.5.8(7) is the first example. Flat extensions, the local criterion that detects flatness one prime at a time, and the geometric reading of flatness as a family of fibres varying continuously all belong with commutative algebra and are developed in *Commutative Algebra* [8].

Exercise 17.5.1

1. Decide whether each module is flat over the indicated ring, and justify.
    1. $\mathbb{Z}/6\mathbb{Z}$ over $\mathbb{Z}$
    2. $\mathbb{Q}$ over $\mathbb{Z}$
    3. $\mathbb{Z}[1/2]$ over $\mathbb{Z}$
    4. $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$
    5. $k[x]/(x)$ over $k[x]$, for a field k
2. Let $m, n \geq 1$ and $d = \gcd(m, n)$. Using $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$ (proposition 15.2.13), show directly that $\mathbb{Z}/2\mathbb{Z}$ is not a flat $\mathbb{Z}$-module, by tensoring the injection $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, $\overline{a} \mapsto \overline{2a}$, with $\mathbb{Z}/2\mathbb{Z}$ and computing both sides.
3. Show that a direct sum $\bigoplus_{i \in I} A_i$ of right R-modules is flat if and only if every A_i is flat.
4. Let R be a principal ideal domain.
    1. Show that every submodule of a flat R-module is flat.
    2. Show that a quotient of a flat R-module need not be flat, by exhibiting a surjection from a flat $\mathbb{Z}$-module onto a non-flat one.
5. Let R be a commutative ring and let A and B be flat R-modules. Show that $A \otimes_R B$ is flat. *Hint*: theorem 15.2.8 identifies $(A \otimes_R B) \otimes_R L$ with $A \otimes_R (B \otimes_R L)$.
6. Show that every element of $M \otimes_R N$ lies in the image of $M_0 \otimes_R N_0 \to M \otimes_R N$ for some finitely generated submodules $M_0 \subseteq M$ and $N_0 \subseteq N$. *Hint*: lemma 15.1.4(2). Explain why this is weaker than lemma 15.2.12, and why the weaker statement would not suffice in the proof of theorem 17.5.7.
7. Assemble the three strictness witnesses of this chapter into one picture. Exhibit $\mathbb{Z}$-modules, or modules over a ring of your choice, showing that each of the implications

$$\text{free} \Rightarrow \text{projective} \Rightarrow \text{flat}$$

is strict, and that neither implication holds between injective and flat in either direction.

17.6 Ext and Tor

Section 17.1 asked for $\mathcal{E}(C, A)$, the set of equivalence classes of extensions of C by A, and everything since has been about the three functors that a short exact sequence can be pushed through. Each is exact at one end and can fail at the other. Writing $0 \to A \to B \to C \to 0$ for a short exact sequence and D for a fixed module, proposition 17.2.1, proposition 17.3.1 and theorem 17.5.1 say

$$\begin{gathered} 0 \to \operatorname{Hom}_R(D, A) \to \operatorname{Hom}_R(D, B) \to \operatorname{Hom}_R(D, C) \\ 0 \to \operatorname{Hom}_R(C, D) \to \operatorname{Hom}_R(B, D) \to \operatorname{Hom}_R(A, D) \\ A \otimes_R D \to B \otimes_R D \to C \otimes_R D \to 0 \end{gathered}$$

and in each case the missing end is missing for a reason: something is there, and it is not zero. This section names it. The construction is given in full and the computations below are unconditional, but three of the theorems are stated without proof, because their proofs need machinery that this book does not build. Where that happens it is said explicitly, and the reason is given.

17.6.1 Complexes and Their Homology

Definition 12.2.11 defined an *exact* chain complex. What is needed here is the same object without the exactness, together with a measure of how far from exact it is.

Definition 17.6.1: Chain Complex and Homology

A *chain complex* of R-modules is a family $(M_i)_{i\in\mathbb{Z}}$ together with homomorphisms $d_i\colon M_i \to M_{i-1}$ satisfying $d_i \circ d_{i+1} = 0$ for every i, equivalently $\mathrm{im}(d_{i+1}) \subseteq \ker(d_i)$. Its *$i$-th homology* is the quotient module

$$H_i(M_\bullet) := \ker(d_i)/\mathrm{im}(d_{i+1})$$

A *cochain complex* is the same data with the arrows reversed, $d^i\colon M^i \to M^{i+1}$ and $d^{i+1}\circ d^i = 0$, and its *i-th cohomology* is $H^i(M^\bullet) := \ker(d^i)/\mathrm{im}(d^{i-1})$.

So a complex is exact at M_i exactly when $H_i(M_\bullet) = 0$, and the homology records the failure one degree at a time. This is the whole of the apparatus needed below; the theory of complexes proper is *Homological Algebra* [10].

17.6.2 The Construction

Fix an R-module M and choose a projective resolution of it, which exists by example 17.1.4(3) and corollary 17.2.6: an exact complex

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$$

with every P_i projective. Deleting M leaves the complex $P_\bullet$, which is no longer exact at P_0; its homology there is M itself. Now apply one of the three functors to $P_\bullet$ and take homology.

Definition 17.6.2: Tor

Let R be a ring, M a right R-module with a chosen projective resolution $P_\bullet$, and N a left R-module. The complex $P_\bullet \otimes_R N$ has differentials $d_i \otimes 1$ and satisfies $(d_i \otimes 1)(d_{i+1} \otimes 1) = (d_i d_{i+1}) \otimes 1 = 0$ by theorem 15.2.2, so it is a chain complex. Define

$$\mathrm{Tor}_i^R(M,N) := H_i\big(P_\bullet \otimes_R N\big) = \ker(d_i \otimes 1)/\mathrm{im}(d_{i+1} \otimes 1)$$

Definition 17.6.3: Ext

With M, $P_\bullet$ as above and N any R-module, the family $\mathrm{Hom}_R(P_i, N)$ with the maps d_{i+1}^* is a cochain complex, since $d_{i+1}^* d_i^* = (d_i d_{i+1})^* = 0$. Define

$$\mathrm{Ext}_R^i(M,N) := H^i\big(\mathrm{Hom}_R(P_\bullet, N)\big) = \ker(d_{i+1}^*)/\mathrm{im}(d_i^*)$$

17.6.4 Two Meanings of Tor The symbol Tor now carries two jobs. Definition 12.3.18 wrote $\mathrm{Tor}_R(M)$, or $\mathrm{Tor}(M)$, for the torsion submodule of M, and definition 17.6.2 writes $\mathrm{Tor}_i^R(M, N)$ for the module just constructed. The subscript decides: an *integer* index means the functor of this section, a *ring* means the torsion submodule, and the functor always takes two arguments. The collision is standard in the literature and is not an accident of notation: proposition 17.6.7 shows that $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ is the subgroup of A killed by n, so the derived functor really is measuring torsion.

The definitions depend on a choice, and nothing so far says the answer does not. It does not, and that is the first of the three deferred theorems.

Theorem 17.6.5: The Derived Functors Are Well Defined

Let R be a ring and M, N be R-modules.

1. $\mathrm{Tor}_i^R(M, N)$ and $\mathrm{Ext}_R^i(M, N)$ do not depend on the choice of projective resolution of M, up to isomorphism.
2. For Tor, a resolution by flat modules (definition 17.5.4) may be used in place of a projective one, with the same result.
3. $\mathrm{Tor}_i^R(M, N)$ may equally be computed from a projective resolution of the second argument N, and $\mathrm{Ext}_R^i(M, N)$ from an *injective* resolution (definition 17.3.15) of N, in both cases giving the same module.

The proofs are omitted here, and the reason is a genuine gap rather than an economy: all three rest on the comparison theorem, which lifts a map of modules to a map of resolutions uniquely up to chain homotopy, and on the fact that chain homotopic maps induce the same map on homology. Neither statement is available in this book, both being about complexes rather than modules, and they are proved in *Homological Algebra* [10, Unique Projection Resolution, Chain-Homotopy and Cohomology]. Proposition 17.2.12 is the degree-one instance of the comparison theorem: two projective presentations of the same module differ by a free summand, which is why the degree-one answer cannot depend on the presentation.

Every computation below fixes one explicit resolution, so each is a complete calculation on its own terms; theorem 17.6.5 is what upgrades it from a statement about that resolution to a statement about M.

Proposition 17.6.6: Degree Zero

Let M and N be R-modules. Then

$$\mathrm{Tor}_0^R(M, N) \cong M \otimes_R N \qquad \text{and} \qquad \mathrm{Ext}_R^0(M, N) \cong \mathrm{Hom}_R(M, N)$$

***Proof*:**

Let $P_\bullet \to M$ be a projective resolution. Exactness at P_0 says $P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$ is exact.

For Tor, apply $- \otimes_R N$. By theorem 17.5.1 the sequence $P_1 \otimes N \to P_0 \otimes N \to M \otimes N \to 0$ is exact, so $\varepsilon \otimes 1$ identifies $(P_0 \otimes N)/\mathrm{im}(d_1 \otimes 1)$ with $M \otimes_R N$. Since $d_0 = 0$ in the deleted complex, $\ker(d_0 \otimes 1) = P_0 \otimes N$ and the left-hand side is exactly $\mathrm{Tor}_0^R(M, N)$.

For Ext, apply $\mathrm{Hom}_R(-, N)$. By proposition 17.3.1 the sequence $0 \to \mathrm{Hom}_R(M, N) \xrightarrow{\varepsilon^*} \mathrm{Hom}_R(P_0, N) \xrightarrow{d_1^*} \mathrm{Hom}_R(P_1, N)$ is exact, so ε^* is an isomorphism onto $\ker(d_1^*)$. Since $d_0^* = 0$, that kernel is exactly $\mathrm{Ext}_R^0(M, N)$.

Proposition 17.6.7: Ext and Tor of a Cyclic Group

Let $n \geq 1$ and let A be an abelian group. Write $A[n] = \{a \in A \mid na = 0\}$. Then, as abelian groups,

$$\mathrm{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong \begin{cases} A[n] & i = 0 \\ A/nA & i = 1 \\ 0 & i \geq 2 \end{cases} \qquad \mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong \begin{cases} A/nA & i = 0 \\ A[n] & i = 1 \\ 0 & i \geq 2 \end{cases}$$

***Proof*:**

The sequence $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \to 0$ of example 17.1.6 is a projective resolution of $\mathbb{Z}/n\mathbb{Z}$, since $\mathbb{Z}$ is free: take $P_0 = P_1 = \mathbb{Z}$, $d_1 = \cdot n$, and $P_i = 0$ for $i \geq 2$. The deleted complex is

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to 0$$

concentrated in degrees 1 and 0.

Every homomorphism $\mathbb{Z} \to A$ is determined by the image of 1, so $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, A) \cong A$ by $h \mapsto h(1)$, and under this identification $d_1^* = (\cdot n)^*$ is multiplication by n on A, since $(h \circ (\cdot n))(1) = h(n) = n\,h(1)$. Likewise $\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A$ by $x \otimes a \mapsto xa$, and $d_1 \otimes 1$ becomes multiplication by n. Both functors therefore turn the deleted complex into

$$0 \to A \xrightarrow{\cdot n} A \to 0$$

with A in degrees 1 and 0 for Tor, and in degrees 0 and 1 for Ext, the arrows of a cochain complex running the other way.

Reading off homology: the kernel of multiplication by n is $A[n]$ and its cokernel is A/nA, and every other module in the complex is zero. For Tor this gives $\mathrm{Tor}_1 = \ker(\cdot n) = A[n]$ and $\mathrm{Tor}_0 = A/nA$; for Ext it gives $\mathrm{Ext}^0 = \ker(\cdot n) = A[n]$ and $\mathrm{Ext}^1 = A/nA$. All higher terms vanish because $P_i = 0$ for $i \geq 2$.

The two degree-zero answers agree with proposition 17.6.6: $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ is the set of elements of A killed by n, and $\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A/nA$ by proposition 15.2.13. Specialising the degree-one answers,

$$\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z},\, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}, \qquad \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}, \qquad \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z},\, \mathbb{Q}) = 0$$

the last because $\mathbb{Q} = n\mathbb{Q}$.

Corollary 17.6.8: Vanishing

Let $i \geq 1$.

1. If P is projective then $\operatorname{Ext}^i_R(P, N) = 0$ and $\operatorname{Tor}^R_i(P, N) = 0$ for every N.
2. If F is flat then $\operatorname{Tor}^R_i(M, F) = 0$ for every M.

***Proof*:**
(1) A projective module is its own projective resolution: take $P_0 = P$, $\varepsilon = \mathrm{id}_P$ and $P_i = 0$ for $i \geq 1$, which is exact. The deleted complex has a single nonzero term in degree 0, so every functor applied to it has vanishing homology in degrees $i \geq 1$.

(2) Let $P_\bullet \to M$ be a projective resolution, so the complex $\cdots \to P_1 \to P_0 \to M \to 0$ is exact. Since F is flat, applying $- \otimes_R F$ preserves that exactness, by proposition 17.5.3 applied at each spot. So $P_\bullet \otimes_R F$ is exact except in degree 0, which is to say $\operatorname{Tor}^R_i(M, F) = 0$ for $i \geq 1$.

Part (2) is the sense in which flatness is the condition that makes Tor disappear, just as projectivity kills Ext in the first argument. The dual statement, that $\operatorname{Ext}^i_R(M, Q) = 0$ for Q injective, needs theorem 17.6.5(3): compute Ext from an injective resolution of Q, which may be taken to be Q itself.

17.6.3 What They Repair

Theorem 17.6.9: Long Exact Sequences

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of R-modules and let D be an R-module. Then there are homomorphisms δ_i, δ'_i and ∂_i, called *connecting homomorphisms*, making the following sequences exact.

$$0 \to \operatorname{Hom}_R(D, A) \to \operatorname{Hom}_R(D, B) \to \operatorname{Hom}_R(D, C) \xrightarrow{\delta_0} \operatorname{Ext}^1_R(D, A) \to \cdots$$
$$\cdots \to \operatorname{Ext}^i_R(D, A) \to \operatorname{Ext}^i_R(D, B) \to \operatorname{Ext}^i_R(D, C) \xrightarrow{\delta_i} \operatorname{Ext}^{i+1}_R(D, A) \to \cdots$$

$$0 \to \operatorname{Hom}_R(C, D) \to \operatorname{Hom}_R(B, D) \to \operatorname{Hom}_R(A, D) \xrightarrow{\delta'_0} \operatorname{Ext}^1_R(C, D) \to \cdots$$
$$\cdots \to \operatorname{Ext}^i_R(C, D) \to \operatorname{Ext}^i_R(B, D) \to \operatorname{Ext}^i_R(A, D) \xrightarrow{\delta'_i} \operatorname{Ext}^{i+1}_R(C, D) \to \cdots$$

$$\cdots \to \operatorname{Tor}^R_i(C, D) \xrightarrow{\partial_i} \operatorname{Tor}^R_{i-1}(A, D) \to \operatorname{Tor}^R_{i-1}(B, D) \to \operatorname{Tor}^R_{i-1}(C, D) \to \cdots$$
$$\cdots \to A \otimes_R D \to B \otimes_R D \to C \otimes_R D \to 0$$

These are the sequences the section exists to produce: each short exact sequence, which the three functors truncate, is restored to an exact sequence by the derived functors, running off to infinity in the direction the functor was failing. The proofs are again omitted, and again for a reason that can be named: the connecting homomorphisms are built by the snake lemma applied to a short exact sequence of *complexes* rather than of modules, which is the general form of proposition 17.1.10, and that is developed in *Homological Algebra* [10, Snake Lemma].

Notice how the first sequence recovers a fact already proved. If D is projective then $\operatorname{Ext}^1_R(D, A) = 0$ by corollary 17.6.8, so the first sequence reads $0 \to \mathrm{Hom}_R(D, A) \to \mathrm{Hom}_R(D, B) \to \mathrm{Hom}_R(D, C) \to 0$, which is definition 17.2.3. In the same way $\mathrm{Tor}_1^R(C, D) = 0$ for D flat turns the third into the short exact sequence of proposition 17.5.3(1). The derived functors do not replace the three module classes; they say what happens when a module is not in them.

17.6.4 The Extension Problem, Answered

Section 17.1 defined $\mathcal{E}(C, A)$ and showed it has at least one element, the class of the split extension, and sometimes more.

Theorem 17.6.10: Ext^1 Classifies Extensions

Let R be a ring and let A and C be R-modules. Then there is a bijection

$$\mathrm{Ext}^1_R(C, A) \longleftrightarrow \mathcal{E}(C, A)$$

between the underlying set of $\mathrm{Ext}^1_R(C, A)$ and the set of equivalence classes of extensions of C by A, under which 0 corresponds to the class of the split extension $A \oplus C$. The addition of $\mathrm{Ext}^1_R(C, A)$ transports to an operation on $\mathcal{E}(C, A)$, the *Baer sum*, making it an abelian group.

This is the theorem the chapter opened with, and it is the last of the three whose proof is omitted. Half of the construction is short, and it uses the following description of Ext^1, which also isolates why the theorem is plausible.

Proposition 17.6.11: Ext^1 as a Cokernel

Let $0 \to K \hookrightarrow P_0 \xrightarrow{\varepsilon} M \to 0$ be exact with P_0 projective, and let N be any R-module. Then

$$\mathrm{Ext}^1_R(M, N) \cong \mathrm{coker}\Big(\mathrm{Hom}_R(P_0, N) \xrightarrow{\text{restrict}} \mathrm{Hom}_R(K, N)\Big)$$

***Proof*:**
Extend the presentation to a projective resolution by choosing a projective P_1 with a surjection $d\colon P_1 \twoheadrightarrow K$ and setting $d_1 = \iota \circ d$, where $\iota\colon K \to P_0$ is the inclusion, and continuing with a resolution of $\ker(d)$.

Since d is surjective, $d^*\colon \mathrm{Hom}_R(K, N) \to \mathrm{Hom}_R(P_1, N)$ is injective by proposition 17.3.1, and its image is $\ker(d_2^*)$: the sequence $P_2 \to P_1 \xrightarrow{d} K \to 0$ is exact, so applying the same proposition identifies $\mathrm{Hom}_R(K, N)$ with the kernel of $\mathrm{Hom}_R(P_1, N) \to \mathrm{Hom}_R(P_2, N)$. Under this identification $\mathrm{im}(d_1^*)$ becomes the image of the restriction map $\mathrm{Hom}_R(P_0, N) \to \mathrm{Hom}_R(K, N)$, because $d_1 = \iota \circ d$ gives $d_1^* = d^* \circ \iota^*$. Hence

$$\mathrm{Ext}^1_R(M, N) = \ker(d_2^*)/\mathrm{im}(d_1^*) \cong \mathrm{Hom}_R(K, N)/\mathrm{im}(\iota^*)$$

which is the stated cokernel.

Now the construction. An extension $0 \to A \to B \xrightarrow{\beta} C \to 0$ together with a projective presentation $0 \to K \to P \xrightarrow{\varepsilon} C \to 0$ produce, by projectivity of P applied to the surjection β, a map $P \to B$ over

C; it carries K into A, and the class of the resulting $K \to A$ in the cokernel of proposition 17.6.11 is the invariant attached to the extension. What is genuinely work is that the assignment is well defined, bijective and additive, and that is deferred to *Homological Algebra* [10], which owes the full correspondence.

The theorem closes the loop with example 17.1.8(2), where the extensions of $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{Z}/p\mathbb{Z}$ were counted by hand. Proposition 17.6.7 gives

$$\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z},\, \mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})\big/p(\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$$

which has p elements, matching the $p-1$ pairwise inequivalent nonsplit extensions found there together with the split one. Likewise $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ says there are exactly n equivalence classes of extensions of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$, while example 17.1.6 exhibited only two non-isomorphic middle terms. There is no conflict: equivalence is finer than isomorphism, exactly as definition 17.1.7 warned, and Ext^1 counts the finer relation.

Finally, $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}) = 0$ says every extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Q}$ splits, which is theorem 17.3.4(3) for the injective module $\mathbb{Q}$; and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z}, A) = A[n]$ says the failure of $-\otimes \mathbb{Z}/n\mathbb{Z}$ to preserve an injection is measured by the n-torsion, which is theorem 17.5.7 read one module at a time.

Exercise 17.6.1

1. Using proposition 17.6.7, compute each of the following.
   1. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$
   2. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$
   3. $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}_{p^\infty})$, for a prime p

2. Use theorem 17.6.10 and proposition 17.6.7 to say how many equivalence classes of extensions of $\mathbb{Z}/4\mathbb{Z}$ by $\mathbb{Z}$ there are. Then show that the possible middle terms are $\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}$ and $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, so that strictly fewer isomorphism types occur than there are classes. *Hint*: the extension attached to k has middle term $(\mathbb{Z} \oplus \mathbb{Z})/\langle (4, -k) \rangle$.

3. Show directly from definition 17.6.3 that $\operatorname{Ext}^i_R(M, N) = 0$ for all $i \geq 1$ whenever M admits a projective resolution of length 0, and that $\operatorname{Ext}^i_R(M, N) = 0$ for all $i \geq 2$ whenever M admits one of length 1. Deduce that $\operatorname{Ext}^i_{\mathbb{Z}}(M, N) = 0$ for every $i \geq 2$ and every pair of abelian groups M, N. *Hint*: theorem 14.1.6.

4. Let R be a principal ideal domain. Show that $\operatorname{Tor}^R_1(M, N) = 0$ whenever either argument is torsion-free, and give an example over $R = \mathbb{Z}$ where $\operatorname{Tor}^{\mathbb{Z}}_1(M, N) \neq 0$.

5. Verify the first long exact sequence of theorem 17.6.9 by hand in one case: take $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ and $D = \mathbb{Z}/2\mathbb{Z}$, compute every term using proposition 17.6.7, and check that the sequence you obtain is exact. Identify the connecting map δ_0.

6. Show that $\operatorname{Ext}^1_R(M, N) = 0$ for every N if and only if M is projective. *Hint*: one direction is corollary 17.6.8; for the other, take a projective presentation $0 \to K \to P \to M \to 0$ and use the description of Ext^1 as $\operatorname{coker}\big(\operatorname{Hom}_R(P, N) \to \operatorname{Hom}_R(K, N)\big)$ with $N = K$.

7. Explain why $\operatorname{Tor}_R(M)$ and $\operatorname{Tor}^R_1(M, N)$ are not the same kind of object, and compute both for $M = \mathbb{Z}/6\mathbb{Z}$ over $R = \mathbb{Z}$, taking $N = \mathbb{Z}/4\mathbb{Z}$ in the second.

17.7 Summary

One short exact sequence, three functors that truncate it, three classes of module that stop the truncation, and two families of modules that measure it when it happens. The tables below state what was established, each entry pointing at the result that establishes it.

Short exact sequences

Statement	Reference
$0 \to A \to B$ exact $\Leftrightarrow$ injective; $B \to C \to 0$ exact $\Leftrightarrow$ surjective	proposition 17.1.1
Short five lemma: α, γ injective (surjective) $\Rightarrow \beta$ is	proposition 17.1.9
Mini snake lemma: kernels and cokernels splice into one exact sequence	proposition 17.1.10
Splits $\Leftrightarrow$ a section exists $\Leftrightarrow$ a retraction exists	proposition 17.1.13
Splitting is carried along by any isomorphism of sequences	proposition 17.1.15
Equivalence of extensions is strictly finer than isomorphism	example 17.1.8(2)

The last row is what makes the extension problem a counting problem rather than a classification of middle terms, and it is what theorem 17.6.10 counts.

The three functors against exactness

Each functor below is exact at one end of a short exact sequence and can fail at the other. Naming the modules for which it does not fail is what the chapter is for. The first and third are covariant and the second contravariant.

Functor	Always	Exact when	Reference
$\mathrm{Hom}_R(D, -)$	left exact	D projective	proposition 17.2.1, definition 17.2.3
$\mathrm{Hom}_R(-, D)$	left exact	D injective	proposition 17.3.1, definition 17.3.3
$D \otimes_R -$	right exact	D flat	theorem 17.5.1, definition 17.5.4

For the contravariant row, *left exact* refers to the sequence after the outer terms have been exchanged, which is how proposition 17.3.1 states it. The two Hom functors can lose surjectivity; tensoring can lose injectivity. The failures are at opposite ends because $D \otimes_R -$ is a left adjoint and $\mathrm{Hom}_R(D, -)$ its right adjoint (theorem 15.3.9).

The three classes

Class	Equivalent descriptions	Reference
Projective P	maps out of P lift along surjections; every $0 \to L \to M \to P \to 0$ splits	theorem 17.2.4
	P is a direct summand of a free module; P has a dual basis	proposition 17.2.8
Injective Q	maps into Q extend along injections; every $0 \to Q \to M \to N \to 0$ splits	theorem 17.3.4
	every $g\colon I \to Q$ on a left ideal extends to R	theorem 17.3.5
Flat A	$1 \otimes \psi$ is injective for every injective ψ	proposition 17.5.3

Projective modules have a structural description and injective modules do not, which is why Baer's criterion exists: it replaces a quantifier over all modules by one over the ideals of R.

How the classes relate, and where the implications are strict

Statement	Witness	Reference
free $\Rightarrow$ projective $\Rightarrow$ flat	(always)	corollary 17.2.6, corollary 17.5.5
projective $\not\Rightarrow$ free	$\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}/6\mathbb{Z}$; F^n over $M_n(F)$	example 17.2.7
flat $\not\Rightarrow$ projective	$\mathbb{Q}$ over $\mathbb{Z}$	example 17.5.8(2)
injective $\not\Rightarrow$ flat	$\mathbb{Q}/\mathbb{Z}$ over $\mathbb{Z}$	example 17.5.8(3)
flat $\not\Rightarrow$ injective	$\mathbb{Z}$ over $\mathbb{Z}$	example 17.3.10(1)
free $\not\Rightarrow$ injective	$\mathbb{Z}$ over $\mathbb{Z}$	example 17.3.10(1)
Projective, injective and flat pass to direct summands	(always)	proposition 17.3.9

Over particular rings the classes collapse.

Ring	Collapse	Reference
A field	free = projective = injective = flat	example 17.3.10(4)
A PID	projective = free	proposition 17.2.11
A PID	injective = divisible	theorem 17.3.5(2)
A PID	flat = torsion-free	theorem 17.5.7
An integral domain	injective $\Rightarrow$ divisible, and not conversely	theorem 17.3.5(2)

Resolutions and hulls

Every module is a quotient of a projective one and a submodule of an injective one, so both kinds of resolution exist. The injective side needed real work, having no dual of a free module to hand.

Statement	**Reference**
Every module is a quotient of a free module	example 17.1.4(3)
Every module is a submodule of an injective module	lemma 17.3.14
Injective resolutions exist	corollary 17.3.16
Every module has an injective hull, unique over the module	theorem 17.3.19
Two projective presentations differ by a free summand	proposition 17.2.12
Divisible groups: $\bigoplus_p \bigoplus_{I_p} \mathbb{Z}_{p^\infty} \oplus \bigoplus_I \mathbb{Q}$, invariants determined	theorem 17.4.7

The last row is the classification of injective $\mathbb{Z}$-modules, since divisible and injective agree over $\mathbb{Z}$. It is the one starred division of the chapter, and nothing outside it depends on it.

Ext, Tor, and the answer

Statement	**Reference**
$\mathrm{Tor}_i^R(M,N) = H_i(P_\bullet \otimes_R N)$ and	definition 17.6.2,
$\mathrm{Ext}_R^i(M,N) = H^i(\mathrm{Hom}_R(P_\bullet, N))$	definition 17.6.3
$\mathrm{Tor}_0 = M \otimes_R N$ and $\mathrm{Ext}^0 = \mathrm{Hom}_R(M,N)$	proposition 17.6.6
$\mathrm{Ext}_{\mathbb{Z}}^i(\mathbb{Z}/n\mathbb{Z}, A) = A[n],\ A/nA,\ 0$ and	proposition 17.6.7
$\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = A/nA,\ A[n],\ 0$	
$\mathrm{Ext}^i(P,N) = 0$ for P projective; $\mathrm{Tor}_i(M,F) = 0$ for F flat	corollary 17.6.8
Independent of the resolution; flat resolutions suffice for Tor	theorem 17.6.5
Long exact sequences in each variable	theorem 17.6.9
$\mathrm{Ext}_R^1(C,A) \leftrightarrow \mathcal{E}(C,A)$, with 0 the split class	theorem 17.6.10

The last row answers the question the chapter opened with, and the row above it is what turns the answer into a computation: $\mathrm{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ has p elements, which is the count example 17.1.8(2) arrived at by hand.

Three of those statements are proved elsewhere. Theorem 17.6.5 and theorem 17.6.9 need the comparison theorem and chain homotopies, and theorem 17.6.10 needs the full construction of the correspondence. The first two are [10, Unique Projection Resolution, Chain-Homotopy and Cohomology], and the third is deferred there with the correspondence. That book takes up the three module classes of this chapter in the setting of an abelian category and takes the derived functors as its subject.

18⋆

Morita Theory

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The rings R and $M_n(R)$ agree in almost nothing. One may be commutative; the other is not, once $n \geq 2$. Their unit groups are unrelated, and so are their underlying additive groups. Yet every statement about left R-modules translates into a statement about left $M_n(R)$-modules and back again, because the two module categories are equivalent. Chapter 12 met this phenomenon, named it *Morita equivalence*, and deferred it. This chapter settles it.

The result is Morita's theorem: every equivalence R-**Mod** $\to S$-**Mod** is given by tensoring with a bimodule, and the bimodules that occur are exactly those that are finitely generated, projective, and large enough to reach every module. Most of what the proof needs is already available. Chapter 15 constructed the tensor product of bimodules and proved the Hom-tensor adjunction (theorem 15.3.9); chapter 17⋆ gave projective modules and the dual basis lemma (proposition 17.2.8). One notion is new, and section 18.1 isolates it before section 18.2 proves the theorem and section 18.3 works out what an equivalence preserves and what it destroys.

Nothing in Part IV uses this chapter, which is why it is marked optional. Its payoff is further out: Morita equivalence is the reduction that makes the representation theory of finite-dimensional algebras tractable, in *Core Representation Theory* [9], and the invariance that makes K_0 computable, in *Algebraic K-Theory* [4].

18.1 Generators and Progenerators

An equivalence of categories carries an object to one with the same categorical description, so the first question is which description singles out R among left R-modules. Three properties do, and each is phrased purely in terms of maps: R is projective (definition 17.2.3), it is finitely generated, and every left R-module is a quotient of a direct sum of copies of R. The first two have names already.

The third does not.

Definition 18.1.1: Generator

Let R be a ring. A left R-module P is a *generator* of R-**Mod** if for every left R-module M there is an index set I and a surjection

$$P^{(I)} := \bigoplus_{i\in I} P \twoheadrightarrow M$$

The module R is a generator: any M is the image of the map $R^{(M)} \to M$ sending the basis element indexed by m to m. So definition 18.1.1 asks of P exactly what makes that argument work for R.

Checking the definition directly means quantifying over every module and every index set, which is unworkable. The next definition gathers the images of all maps $P \to R$ into a single subset of R, and theorem 18.1.4 shows that this one subset already decides the question.

Definition 18.1.2: Trace Ideal

Let P be a left R-module. The *trace ideal* of P is

$$\mathrm{Tr}(P) := \sum_{f\in \mathrm{Hom}_R(P,R)} f(P) \subseteq R$$

the additive subgroup of R generated by the elements $f(p)$ for $f \in \mathrm{Hom}_R(P,R)$ and $p \in P$.

Lemma 18.1.3: The Trace Ideal Is Two-Sided

For every left R-module P, the set $\mathrm{Tr}(P)$ is a two-sided ideal of R.

***Proof*:**
It is an additive subgroup by construction, so only the two multiplications need checking, and since $\mathrm{Tr}(P)$ is generated additively by the elements $f(p)$ it is enough to check them on such an element.

For left multiplication, let $r \in R$. Then $r\,f(p) = f(rp)$ because f is R-linear, and $rp \in P$, so $r\,f(p) \in f(P) \subseteq \mathrm{Tr}(P)$.

For right multiplication, fix $a \in R$ and define $g\colon P \to R$ by $g(x) = f(x)a$. This g is additive, and for $r \in R$ and $x \in P$,

$$g(rx) = f(rx)a = \big(r f(x)\big)a = r\big(f(x)a\big) = r\,g(x)$$

so $g \in \mathrm{Hom}_R(P,R)$. Hence $f(p)a = g(p) \in \mathrm{Tr}(P)$, completing the proof.

Theorem 18.1.4: Characterizations of a Generator

Let P be a left R-module. The following are equivalent.

1. P is a generator of R-**Mod**.
2. $\operatorname{Tr}(P) = R$.
3. R is isomorphic to a direct summand of P^n for some finite $n \geq 1$.
4. The functor $\operatorname{Hom}_R(P, -)\colon R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ is faithful.

***Proof*:**
(1) $\Rightarrow$ (2). Apply definition 18.1.1 to $M = R$, giving a surjection $\pi\colon P^{(I)} \to R$. Write $\iota_i\colon P \to P^{(I)}$ for the inclusion of the i-th summand and set $f_i := \pi \circ \iota_i \in \operatorname{Hom}_R(P, R)$. Since π is surjective there is $x \in P^{(I)}$ with $\pi(x) = 1$, and an element of a direct sum has only finitely many nonzero coordinates, so $x = \sum_{i \in F} \iota_i(p_i)$ for a finite $F \subseteq I$ and elements $p_i \in P$. Then

$$1 = \pi(x) = \sum_{i \in F} f_i(p_i) \in \operatorname{Tr}(P)$$

By lemma 18.1.3 the set $\operatorname{Tr}(P)$ is an ideal, and an ideal containing 1 is the whole ring, so $\operatorname{Tr}(P) = R$.

(2) $\Rightarrow$ (3). By (2) there are $f_1, \ldots, f_n \in \operatorname{Hom}_R(P, R)$ and $p_1, \ldots, p_n \in P$ with $\sum_{i=1}^n f_i(p_i) = 1$. Define

$$\varphi\colon P^n \to R, \quad \varphi(x_1, \ldots, x_n) = \sum_{i=1}^n f_i(x_i), \qquad \sigma\colon R \to P^n, \quad \sigma(r) = (rp_1, \ldots, rp_n)$$

Both are R-linear: φ because each f_i is, and σ because the action on P^n is coordinatewise. For every $r \in R$,

$$\varphi(\sigma(r)) = \sum_{i=1}^n f_i(rp_i) = r \sum_{i=1}^n f_i(p_i) = r$$

so σ is a section of φ. In particular φ is surjective, and proposition 17.1.13 applied to $0 \to \ker(\varphi) \to P^n \xrightarrow{\varphi} R \to 0$ shows that this sequence splits, so $P^n \cong \sigma(R) \oplus \ker(\varphi)$ with $\sigma(R) \cong R$.

(3) $\Rightarrow$ (4). Let $f\colon M \to N$ be a nonzero map of left R-modules; the claim is that $f_* = \operatorname{Hom}_R(P, f)$ is nonzero. Choose $m \in M$ with $f(m) \neq 0$ and let $g\colon R \to M$ be the R-linear map $g(r) = rm$, so $f(g(1)) = f(m) \neq 0$ and hence $f \circ g \neq 0$. By (3) there are R-linear maps $\iota\colon R \to P^n$ and $\rho\colon P^n \to R$ with $\rho \circ \iota = \mathrm{id}_R$. Then

$$(f \circ g \circ \rho) \circ \iota = f \circ g \circ (\rho \circ \iota) = f \circ g \neq 0$$

so $f \circ g \circ \rho\colon P^n \to N$ is nonzero. A map out of a finite direct sum vanishes if and only if its restriction to every summand vanishes, so there is a coordinate inclusion $\kappa_j\colon P \to P^n$ with $f \circ (g \circ \rho \circ \kappa_j) \neq 0$. Setting $h := g \circ \rho \circ \kappa_j \in \operatorname{Hom}_R(P, M)$ gives $f_*(h) \neq 0$.

(4) $\Rightarrow$ (1). Let M be a left R-module and put

$$T := \sum_{h \in \operatorname{Hom}_R(P, M)} h(P) \subseteq M$$

a submodule of M, being a sum of the submodules $h(P)$. Let $q\colon M \to M/T$ be the quotient map. Every $h \in \mathrm{Hom}_R(P, M)$ has $h(P) \subseteq T$, so $q \circ h = 0$; that is, q_* is the zero map on $\mathrm{Hom}_R(P, M)$. Since $\mathrm{Hom}_R(P, -)$ is faithful and q_* agrees with the map induced by the zero homomorphism, it follows that $q = 0$, hence $M/T = 0$ and $T = M$. The map

$$P^{(\mathrm{Hom}_R(P,M))} \to M$$

whose restriction to the summand indexed by h is h therefore has image $T = M$, so it is surjective. As M was arbitrary, P is a generator, completing the proof.

Condition (2) is the one to check in practice, since it trades a quantifier over all modules for a computation inside R. Condition (3) is the form section 18.2 uses: it presents R as a retract of a finite power of P, which is what allows an argument about R to be transported to P. Condition (4) is the categorical reading, that P is a generator exactly when no nonzero map between modules is invisible to P.

Definition 18.1.5: Progenerator

A left R-module P is a *progenerator* if it is finitely generated, projective, and a generator of R-**Mod**.

These are the three properties of R listed at the start of the section, so R is a progenerator. Theorem 18.2.8 will show that the progenerators are precisely the modules that can take its place.

Example 18.1.6: Progenerators and Two Near Misses

1. For any ring R and any $n \geq 1$, the module R^n is a progenerator. It is free, hence projective by corollary 17.2.6, and finitely generated; the coordinate projection $R^n \to R$ is surjective, so $\mathrm{Tr}(R^n) = R$.

2. *Finitely generated and projective, but not a generator.* Let $R = \mathbb{Z}/6\mathbb{Z}$ and $e = \overline{3}$, which satisfies $e^2 = \overline{9} = \overline{3} = e$. Then $R = Re \oplus R(1-e)$, so $P := Re = \{\overline{0}, \overline{3}\}$ is a direct summand of R, hence projective by theorem 17.2.4(4), and it is generated by one element. Every $f \in \mathrm{Hom}_R(P, R)$ satisfies
$$f(e) = f(e \cdot e) = e\, f(e) \in eR$$
and R is commutative here, so $f(P) = f(Re) = Rf(e) \subseteq Re$, giving $\mathrm{Tr}(P) = Re \neq R$. By theorem 18.1.4 the module P is not a generator. Concretely, every element of a direct sum of copies of P is annihilated by $\overline{2}$, while $\overline{1} \in R$ is not, so no such sum surjects onto R.

3. *A generator that is not projective.* Let $R = \mathbb{Z}$ and $P = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. The projection onto the first coordinate lies in $\mathrm{Hom}_{\mathbb{Z}}(P, \mathbb{Z})$ and has image $\mathbb{Z}$, so $\mathrm{Tr}(P) = \mathbb{Z}$ and P is a generator by theorem 18.1.4. It is not projective: over the principal ideal domain $\mathbb{Z}$ projective and free coincide (proposition 17.2.11), and P has the nonzero torsion element $(0, \overline{1})$, so it is not free.

The two near misses show that neither condition in definition 18.1.5 implies the other, so both are genuine requirements on the bimodule that Morita's theorem produces.

Exercise 18.1.1

1. Let P be a left R-module and $n \geq 1$. Show that $\mathrm{Tr}(P^n) = \mathrm{Tr}(P)$, and deduce that P is a generator if and only if P^n is.
2. Show that every nonzero free left R-module is a generator of R-**Mod**, and that any module with a nonzero free direct summand is a generator.
3. Let $R = \mathbb{Z}/6\mathbb{Z}$. Decide, for each of the R-modules $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, whether it is a progenerator, justifying each answer using theorem 18.1.4.
4. Let P be a projective left R-module. Using the dual basis lemma (proposition 17.2.8), show that $\mathrm{Tr}(P)$ is idempotent, that is $\mathrm{Tr}(P)\,\mathrm{Tr}(P) = \mathrm{Tr}(P)$.
5. Let P be a generator of R-**Mod**. Show that $\mathrm{Hom}_R(P, M) = 0$ implies $M = 0$. Then show that this consequence does not characterize generators, by exhibiting a ring R and a non-generator P for which it still holds.

18.2 Morita's Theorem

The statement to be proved compares two module categories, so it needs the vocabulary for comparing categories. Functors are available (definition 8.0.4); the maps between functors are not, and the notion of two categories being "the same" rests on them.

Definition 18.2.1: Natural Transformation and Natural Isomorphism

Let $F, G\colon \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta\colon F \Rightarrow G$ assigns to each object C of $\mathcal{C}$ a morphism $\eta_C\colon F(C) \to G(C)$ of $\mathcal{D}$ such that for every morphism $f\colon C \to C'$ of $\mathcal{C}$ the square

$$\begin{tikzcd} F(C) \arrow[r, "\eta_C"] \arrow[d, "F(f)"'] & G(C) \arrow[d, "G(f)"] \\ F(C') \arrow[r, "\eta_{C'}"'] & G(C') \end{tikzcd}$$

commutes. It is a *natural isomorphism*, written $F \cong G$, if every η_C is an isomorphism.

Definition 18.2.2: Equivalence of Categories

An *equivalence* between categories $\mathcal{C}$ and $\mathcal{D}$ is a pair of functors $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. The functor G is a *quasi-inverse* of F, and $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*.

An equivalence is bijective on morphism sets: F induces a bijection $\mathrm{Hom}_{\mathcal{C}}(C, C') \to \mathrm{Hom}_{\mathcal{D}}(FC, FC')$, with inverse obtained by applying G and conjugating by the natural isomorphism $GF \cong \mathrm{id}$. Every property of an object stated purely in terms of morphisms therefore transfers across an equivalence, which is what makes theorem 18.1.4(4) the useful characterization of a generator.

Definition 18.2.3: Morita Equivalence

Rings R and S are *Morita equivalent* if R-**Mod** and S-**Mod** are equivalent categories.

Now fix a left R-module P. Two rings act on it: R on the left, and its own endomorphism ring. Writing endomorphisms on the right makes the second action a right action, and that is the convention the whole section uses.

Proposition 18.2.4: The Bimodules Attached to a Module

Let P be a left R-module and set $S := \mathrm{End}_R(P)^{\mathrm{op}}$ and $Q := \mathrm{Hom}_R(P, R)$.

1. P is an (R, S)-bimodule under $p \cdot s := s(p)$.
2. Q is an (S, R)-bimodule under $(s \cdot q)(p) := q(s(p))$ and $(q \cdot r)(p) := q(p)r$.

***Proof*:**

1. Multiplication in S is $s_1 s_2 = s_2 \circ s_1$, so $p \cdot (s_1 s_2) = (s_2 \circ s_1)(p) = s_2(s_1(p)) = (p \cdot s_1) \cdot s_2$, and $p \cdot \mathrm{id}_P = p$; the action is additive in each variable because endomorphisms are additive. Compatibility with the left action is R-linearity of s:

$$(rp) \cdot s = s(rp) = r\, s(p) = r(p \cdot s)$$

2. First, $s \cdot q$ and $q \cdot r$ lie in Q: the map $p \mapsto q(s(p))$ is a composite of R-linear maps, and $(q \cdot r)(ap) = q(ap)r = a\, q(p)\, r = a\, (q \cdot r)(p)$. The left action is an action because

$$\big((s_1 s_2) \cdot q\big)(p) = q\big((s_1 s_2)(p)\big) = q\big(s_2(s_1(p))\big) = (s_2 \cdot q)(s_1(p)) = \big(s_1 \cdot (s_2 \cdot q)\big)(p)$$

using $(s_1 s_2)(p) = s_2(s_1(p))$ from the opposite multiplication. The right action is one because $(q \cdot (r_1 r_2))(p) = q(p) r_1 r_2 = ((q \cdot r_1) \cdot r_2)(p)$. They commute:

$$\big((s \cdot q) \cdot r\big)(p) = (s \cdot q)(p)\, r = q(s(p))\, r = (q \cdot r)(s(p)) = \big(s \cdot (q \cdot r)\big)(p)$$

completing the proof.

With P an (R, S)-bimodule and Q an (S, R)-bimodule, the two tensor products $P \otimes_S Q$ and $Q \otimes_R P$ are an (R, R)-bimodule and an (S, S)-bimodule respectively (proposition 15.2.1). Each carries an evaluation map to the ring on the corresponding side, and the entire theorem is the assertion that both are isomorphisms.

Proposition 18.2.5: The Morita Maps

With P, S and Q as in proposition 18.2.4, there are well-defined maps

$$\mu\colon P\otimes_S Q\to R,\quad \mu(p\otimes q)=q(p),\qquad \tau\colon Q\otimes_R P\to S,\quad \tau(q\otimes p)=\bigl[x\mapsto q(x)p\bigr]$$

the first a homomorphism of (R,R)-bimodules and the second of (S,S)-bimodules, and they satisfy the *associativity identities*

$$\mu(p\otimes q)\,p'=p\cdot\tau(q\otimes p'),\qquad q\cdot\mu(p\otimes q')=\tau(q\otimes p)\cdot q'$$

for all $p,p'\in P$ and $q,q'\in Q$.

***Proof*:**
Well-definedness. The assignment $(p,q)\mapsto q(p)$ is additive in each variable and satisfies $\mu(p{\cdot}s,q)=q(s(p))=(s\cdot q)(p)=\mu(p,s\cdot q)$, so it is S-balanced and induces μ on $P\otimes_S Q$. For τ, the map $x\mapsto q(x)p$ is R-linear since $q(ax)p=a\,q(x)p$, so it is an element of S; the assignment is additive in each variable and

$$\tau(q\cdot r,p)(x)=(q\cdot r)(x)p=q(x)rp=\tau(q,rp)(x)$$

so it is R-balanced and induces τ.

Bimodule linearity. For μ: $\mu(rp\otimes q)=q(rp)=r\,q(p)$ and $\mu(p\otimes q\cdot r)=(q\cdot r)(p)=q(p)r$. For τ, recall $s_1s_2=s_2\circ s_1$ in S. On the left,

$$\tau(s\cdot q\otimes p)(x)=(s\cdot q)(x)\,p=q(s(x))\,p=\bigl(\tau(q\otimes p)\circ s\bigr)(x)=\bigl(s\,\tau(q\otimes p)\bigr)(x)$$

and on the right, using R-linearity of s,

$$\tau(q\otimes p\cdot s)(x)=q(x)\,s(p)=s\bigl(q(x)p\bigr)=\bigl(s\circ\tau(q\otimes p)\bigr)(x)=\bigl(\tau(q\otimes p)\,s\bigr)(x)$$

The identities. For the first, $\mu(p\otimes q)p'=q(p)p'$, while $\tau(q\otimes p')$ is the endomorphism $x\mapsto q(x)p'$, so $p\cdot\tau(q\otimes p')=\tau(q\otimes p')(p)=q(p)p'$. For the second, evaluate both sides at $x\in P$:

$$\bigl(q\cdot\mu(p\otimes q')\bigr)(x)=q(x)\,q'(p),\qquad \bigl(\tau(q\otimes p)\cdot q'\bigr)(x)=q'\bigl(\tau(q\otimes p)(x)\bigr)=q'\bigl(q(x)p\bigr)=q(x)\,q'(p)$$

as we sought to show.

The two maps measure exactly the two conditions of definition 18.1.5, one each.

Lemma 18.2.6: What the Morita Maps Detect

With notation as above:

1. $\mathrm{im}\mu=\mathrm{Tr}(P)$, so μ is surjective if and only if P is a generator.
2. τ is surjective if and only if P is finitely generated and projective.

***Proof*:**

1. The image of μ is the additive subgroup generated by the elements $q(p)$ for $q \in \operatorname{Hom}_R(P, R)$ and $p \in P$, which is definition 18.1.2 verbatim. The equivalence with being a generator is theorem 18.1.4, (1) $\Leftrightarrow$ (2), since a subgroup of R equals R exactly when the map onto it is surjective.

2. ($\Rightarrow$) If τ is surjective, then $\mathrm{id}_P = \tau(t)$ for some $t \in Q \otimes_R P$, and every element of a tensor product is a finite sum of simple tensors, say $t = \sum_{i=1}^n q_i \otimes p_i$. Evaluating at $x \in P$,

$$x = \mathrm{id}_P(x) = \sum_{i=1}^n q_i(x)\, p_i$$

which is a finite dual basis, so P is finitely generated and projective by proposition 17.2.8.

($\Leftarrow$) If P is finitely generated and projective, proposition 17.2.8 gives a finite dual basis $(p_i, q_i)_{i=1}^n$, and the displayed identity says $\tau\big(\sum_i q_i \otimes p_i\big) = \mathrm{id}_P$. The image of τ is a two-sided ideal of S, being the image of an (S, S)-bimodule map, and an ideal containing the identity is the whole ring, completing the proof.

Surjectivity of the two maps is therefore exactly the progenerator hypothesis. What is not yet clear is injectivity, and the next lemma is the reason the theorem needs no separate argument for it: each map is forced to be injective by its own surjectivity, through the associativity identities that tie it to the other.

Lemma 18.2.7: Surjective Forces Bijective

If μ is surjective then μ is injective. If τ is surjective then τ is injective.

***Proof*:**
Suppose μ is surjective and let $t = \sum_{i=1}^m p_i \otimes q_i$ lie in $\ker \mu$, so that $\sum_i q_i(p_i) = 0$. Surjectivity gives elements $a_j \in P$ and $b_j \in Q$ with $1 = \sum_{j=1}^k \mu(a_j \otimes b_j)$. Then

$$t = \sum_i \Big(\sum_j \mu(a_j \otimes b_j)\Big) p_i \otimes q_i = \sum_{i,j} \big(\mu(a_j \otimes b_j)\, p_i\big) \otimes q_i = \sum_{i,j} \big(a_j \cdot \tau(b_j \otimes p_i)\big) \otimes q_i$$

using the first associativity identity of proposition 18.2.5. Moving the element of S across the tensor and applying the second identity,

$$t = \sum_{i,j} a_j \otimes \big(\tau(b_j \otimes p_i) \cdot q_i\big) = \sum_{i,j} a_j \otimes \big(b_j \cdot \mu(p_i \otimes q_i)\big) = \sum_j a_j \otimes \Big(b_j \cdot \sum_i \mu(p_i \otimes q_i)\Big)$$

and the inner sum is $\mu(t) = 0$, so $t = 0$. The argument for τ is the same with the roles of the two identities and of (P, R) and (Q, S) interchanged, completing the proof.

Theorem 18.2.8: Morita's Theorem

Let P be a progenerator in R-**Mod**, and set $S = \mathrm{End}_R(P)^{\mathrm{op}}$ and $Q = \mathrm{Hom}_R(P, R)$. Then

$$\mu\colon P \otimes_S Q \xrightarrow{\sim} R, \qquad \tau\colon Q \otimes_R P \xrightarrow{\sim} S$$

are isomorphisms of bimodules, and the functors

$$Q \otimes_R -\colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}, \qquad P \otimes_S -\colon S\text{-}\mathbf{Mod} \to R\text{-}\mathbf{Mod}$$

are mutually quasi-inverse equivalences. In particular R and S are Morita equivalent.

***Proof*:**
Since P is a generator, μ is surjective by lemma 18.2.6(1); since P is finitely generated and projective, τ is surjective by lemma 18.2.6(2). Both are then injective by lemma 18.2.7, hence bijective, and they are bimodule maps by proposition 18.2.5.

For the functors, let N be a left S-module. Associativity of the tensor product (theorem 15.2.8) and the isomorphism just established give

$$Q \otimes_R \big(P \otimes_S N\big) \cong \big(Q \otimes_R P\big) \otimes_S N \cong S \otimes_S N \cong N$$

and symmetrically, for a left R-module M,

$$P \otimes_S \big(Q \otimes_R M\big) \cong \big(P \otimes_S Q\big) \otimes_R M \cong R \otimes_R M \cong M$$

Each isomorphism in these chains is natural in the module argument, since theorem 15.2.8 is natural and the remaining two steps are the unit isomorphisms of the tensor product. The two chains are therefore natural isomorphisms $(Q\otimes_R -)\circ(P\otimes_S -) \cong \mathrm{id}$ and $(P\otimes_S -)\circ(Q\otimes_R -) \cong \mathrm{id}$, which is definition 59.5.5, completing the proof.

The theorem manufactures an equivalence from every progenerator. The converse says that this is the only source: an equivalence cannot arise any other way, and the progenerator producing it is recovered as the image of the ring.

Theorem 18.2.9: Every Equivalence Is Tensoring with a Progenerator

Let $F\colon R$-**Mod** $\to S$-**Mod** be an equivalence with quasi-inverse G. Put $Q := F(R)$ and $P := G(S)$. Then Q carries a right R-action making it an (S, R)-bimodule, P carries a right S-action making it an (R, S)-bimodule,

$$P \otimes_S Q \cong R \ \text{ as } (R, R)\text{-bimodules}, \qquad Q \otimes_R P \cong S \ \text{ as } (S, S)\text{-bimodules}$$

and $F \cong Q \otimes_R -$. Moreover P is a progenerator in R-**Mod**.

***Proof*:**
Step 1: the right action on Q. For $r \in R$ let $\rho_r\colon R \to R$ be $x \mapsto xr$, which is left R-linear. Since $\rho_{r_1 r_2} = \rho_{r_2} \circ \rho_{r_1}$ and F is a functor, $F(\rho_{r_1 r_2}) = F(\rho_{r_2}) \circ F(\rho_{r_1})$. Setting $q \cdot r := F(\rho_r)(q)$ therefore defines a right R-action on Q, and it commutes with the left S-action because each

$F(\rho_r)$ is a map of S-modules. The same construction gives the right S-action on P.

Step 2: F preserves direct sums and cokernels. An equivalence is bijective on morphism sets, as recorded after definition 59.5.5, and it is essentially surjective, every object of S-**Mod** being isomorphic to $F(G(N))$. A cocone is a direct sum, or a cokernel, exactly when a certain map of morphism sets is bijective for every test object; bijectivity on morphism sets transports the condition forward and essential surjectivity gives every test object on the far side. So F carries direct sums to direct sums and cokernels to cokernels.

Step 3: $F \cong Q \otimes_R -$. For a left R-module M and $m \in M$ let $\lambda_m : R \to M$ be $r \mapsto rm$. The assignment $(q, m) \mapsto F(\lambda_m)(q)$ is additive in each variable and R-balanced, since $\lambda_{rm} = \lambda_m \circ \rho_r$ gives $F(\lambda_{rm})(q) = F(\lambda_m)(q \cdot r)$. It therefore induces $\theta_M : Q \otimes_R M \to F(M)$, and θ is natural in M because $f \circ \lambda_m = \lambda_{f(m)}$ for any $f : M \to M'$.

At $M = R$ the map θ_R is the isomorphism $Q \otimes_R R \cong Q = F(R)$, since $\lambda_r = \rho_r$. Both $Q \otimes_R -$ and F preserve direct sums, the first by corollary 15.2.5 and the second by Step 2, so $\theta_{R^{(I)}}$ is an isomorphism for every index set I. Now take any M and a free presentation $R^{(J)} \to R^{(I)} \to M \to 0$, which exists because every module is a quotient of a free one and the kernel of that quotient is again such a quotient. Both functors are right exact, again by theorem 17.5.1 and Step 2, so the rows of

$$\begin{tikzcd}
Q \otimes_R R^{(J)} \arrow[r] \arrow[d, "\theta_{R^{(J)}}"'] & Q \otimes_R R^{(I)} \arrow[r] \arrow[d, "\theta_{R^{(I)}}"'] & Q \otimes_R M \arrow[r] \arrow[d, "\theta_M"] & 0 \\
F(R^{(J)}) \arrow[r] & F(R^{(I)}) \arrow[r] & F(M) \arrow[r] & 0
\end{tikzcd}$$

are exact and the square commutes by naturality. The two left vertical maps are isomorphisms, and a cokernel is determined up to unique isomorphism by the map it is taken of, so θ_M is an isomorphism.

Step 4: the bimodule isomorphisms. By Step 3 applied to F and to G, the composite $GF \cong P \otimes_S (Q \otimes_R -) \cong (P \otimes_S Q) \otimes_R -$ using theorem 15.2.8. Since $GF \cong \mathrm{id}$, evaluating at R gives $(P \otimes_S Q) \otimes_R R \cong R$, that is $P \otimes_S Q \cong R$; the isomorphism is one of (R, R)-bimodules because the natural isomorphism $GF \cong \mathrm{id}$ commutes with the maps ρ_r, whose images generate the right action. Symmetrically $Q \otimes_R P \cong S$.

Step 5: P is a progenerator. From $P \otimes_S Q \cong R$: the tensor product is generated as an abelian group by simple tensors, so the map $\bigoplus_{q \in Q} P \to P \otimes_S Q$ sending the summand indexed by q to $p \mapsto p \otimes q$ is surjective, exhibiting R as a quotient of a direct sum of copies of P. Hence P is a generator. From $Q \otimes_R P \cong S$: the element of $Q \otimes_R P$ corresponding to 1_S is a *finite* sum $\sum_{i=1}^n q_i \otimes p_i$, and unwinding the isomorphism as in lemma 18.2.6(2) yields $x = \sum_i q_i(x) p_i$ for all $x \in P$, a finite dual basis. So P is finitely generated and projective by proposition 17.2.8, completing the proof.

Taken together the two theorems say that Morita equivalence is not an exotic relation to be checked category by category: R and S are Morita equivalent precisely when $S \cong \mathrm{End}_R(P)^{\mathrm{op}}$ for some progenerator P over R. The next example is the case that motivated the whole discussion.

Example 18.2.10: A Ring and Its Matrix Rings

Let R be any ring and $n \geq 1$, and take $P = R^n$, the module of columns. By example 18.1.6(1) it is a progenerator. Its endomorphism ring is computed one summand at a time: an R-linear map $R \to R$ is right multiplication by its value at 1, so $\mathrm{End}_R(R) \cong R^{\mathrm{op}}$, and therefore

$$\mathrm{End}_R(R^n) \cong M_n\big(\mathrm{End}_R(R)\big) \cong M_n(R^{\mathrm{op}}) \cong M_n(R)^{\mathrm{op}}$$

the last isomorphism being transposition, which reverses the order of multiplication. Hence

$$S = \mathrm{End}_R(R^n)^{\mathrm{op}} \cong M_n(R)$$

and theorem 18.2.8 gives an equivalence R-**Mod** $\simeq M_n(R)$-**Mod**. Unwinding it, $Q = \mathrm{Hom}_R(R^n, R) \cong R^n$ as rows, the equivalence R-**Mod** $\to M_n(R)$-**Mod** sends M to the column space M^n with $M_n(R)$ acting by matrix multiplication, and its quasi-inverse sends N to eN where $e = E_{11}$ is the idempotent matrix with a single 1 in the top-left entry.

Nothing about R is recovered by $M_n(R)$ beyond its module category: for $n \geq 2$ the ring $M_n(R)$ is never commutative, so a commutative R is Morita equivalent to noncommutative rings. Section 18.3 makes precise which properties survive.

Exercise 18.2.1

1. Let P be a progenerator over R and $S = \mathrm{End}_R(P)^{\mathrm{op}}$. Show that $\mathrm{Hom}_R(P, -)$ and $Q \otimes_R -$ are naturally isomorphic as functors R-**Mod** $\to S$-**Mod**, where $Q = \mathrm{Hom}_R(P, R)$.
2. Show that Morita equivalence is an equivalence relation on rings.
3. Show that if R and S are Morita equivalent, then so are $M_n(R)$ and $M_n(S)$ for every $n \geq 1$.
4. Show that a commutative ring R and the matrix ring $M_n(R)$ have isomorphic centres for every $n \geq 1$, and explain why example 18.2.10 does not contradict this.
5. Let D be a division ring. Show that every progenerator over D is isomorphic to D^n for some $n \geq 1$, and deduce that the rings Morita equivalent to D are exactly the matrix rings $M_n(D)$.
6. Give an example of two rings that are Morita equivalent but not isomorphic, and an example of two rings that are not Morita equivalent even though both are commutative and have the same cardinality.
7. Let P be a progenerator over R. Show that P is also a progenerator when regarded as a right S-module, where $S = \mathrm{End}_R(P)^{\mathrm{op}}$.

18.3 Morita Invariants and the Basic Algebra

Example 18.2.10 showed that R and $M_n(R)$ have the same module category while differing in nearly every element-level respect. A property of a ring that transfers across every Morita equivalence is called a *Morita invariant*, and the question this section answers is which properties those are. The test is uniform: a property is a Morita invariant when it can be stated in terms of the module category alone, and it fails to be one as soon as a single equivalence moves it.

Commutativity is the basic failure. For $n \geq 2$ the ring $M_n(R)$ is noncommutative whatever R is, so no property implying commutativity survives. What does survive is the part of the multiplication that is visible categorically, and theorem 18.3.2 identifies it exactly: the center.

Proposition 18.3.1: The Center as Endomorphisms of the Identity

Let R be a ring. Sending $c \in Z(R)$ to the family of maps $\eta^c_M \colon M \to M$, $\eta^c_M(m) = cm$, is a ring isomorphism from $Z(R)$ (definition 9.1.8) to the ring of natural transformations $\mathrm{id}_{R\textbf{-Mod}} \Rightarrow \mathrm{id}_{R\textbf{-Mod}}$, under pointwise addition and composition.

***Proof*:**
The map is well defined. For $c \in Z(R)$ each η^c_M is R-linear, since $\eta^c_M(rm) = c(rm) = (cr)m = (rc)m = r(cm) = r\,\eta^c_M(m)$, which is exactly where centrality is used. It is natural: for $f \colon M \to N$ and $m \in M$,

$$f\big(\eta^c_M(m)\big) = f(cm) = c\,f(m) = \eta^c_N\big(f(m)\big)$$

Injectivity. If $\eta^c = \eta^{c'}$ then in particular $c = \eta^c_R(1) = \eta^{c'}_R(1) = c'$.

Surjectivity. Let η be a natural transformation of the identity and put $c := \eta_R(1)$. For $r \in R$, R-linearity of η_R gives $\eta_R(r) = \eta_R(r \cdot 1) = r\,\eta_R(1) = rc$. Naturality applied to the right multiplication $\rho_r \colon R \to R$, $x \mapsto xr$, which is a map of left R-modules, gives $\eta_R(xr) = \eta_R(x)r$; at $x = 1$ this reads $\eta_R(r) = cr$. Comparing the two expressions, $rc = cr$ for every $r \in R$, so $c \in Z(R)$.

It remains to see that $\eta = \eta^c$ on every module, not only on R. For $m \in M$ let $\lambda_m \colon R \to M$ be $r \mapsto rm$. Naturality gives

$$\eta_M(m) = \eta_M\big(\lambda_m(1)\big) = \lambda_m\big(\eta_R(1)\big) = \lambda_m(c) = cm = \eta^c_M(m)$$

Ring structure. Addition is pointwise on both sides. For multiplication, $\eta^c_M\big(\eta^{c'}_M(m)\big) = c(c'm) = (cc')m = \eta^{cc'}_M(m)$, so composition corresponds to multiplication in $Z(R)$, and $\eta^1 = \mathrm{id}$, completing the proof.

The right-hand side of proposition 18.3.1 mentions only the category R-**Mod**, never the ring R. That is the whole content of the next theorem.

Theorem 18.3.2: The Center Is a Morita Invariant

If R and S are Morita equivalent, then $Z(R) \cong Z(S)$ as rings.

***Proof*:**
Let $F \colon R\textbf{-Mod} \to S\textbf{-Mod}$ be an equivalence with quasi-inverse G, and let $\alpha \colon GF \cong \mathrm{id}$ and $\beta \colon FG \cong \mathrm{id}$ be the natural isomorphisms of definition 59.5.5. Given a natural transformation η of $\mathrm{id}_{R\textbf{-Mod}}$, define

$$\Phi(\eta)_N := \beta_N \circ F\big(\eta_{G(N)}\big) \circ \beta_N^{-1} \colon N \to N \qquad (N \in S\textbf{-Mod})$$

Each component is a composite of S-linear maps. It is natural in N: for $g \colon N \to N'$, naturality of β gives $g \circ \beta_N = \beta_{N'} \circ FG(g)$, and naturality of η at $G(g)$ gives $\eta_{G(N')} \circ G(g) = G(g) \circ \eta_{G(N)}$, so applying F and conjugating,

$$g \circ \Phi(\eta)_N = \beta_{N'} \circ F\big(G(g) \circ \eta_{G(N)}\big) \circ \beta_N^{-1} = \beta_{N'} \circ F\big(\eta_{G(N')} \circ G(g)\big) \circ \beta_N^{-1} = \Phi(\eta)_{N'} \circ g$$

So Φ carries natural transformations of $\mathrm{id}_{R\textbf{-Mod}}$ to natural transformations of $\mathrm{id}_{S\textbf{-Mod}}$. It preserves addition because F is additive on morphism sets and conjugation is additive, and it preserves composition because the conjugating isomorphisms cancel in the middle:

$$\Phi(\eta)_N \circ \Phi(\eta')_N = \beta_N \circ F\big(\eta_{G(N)}\big) \circ F\big(\eta'_{G(N)}\big) \circ \beta_N^{-1} = \beta_N \circ F\big(\eta_{G(N)} \circ \eta'_{G(N)}\big) \circ \beta_N^{-1}$$

The same construction applied to G gives Ψ in the opposite direction, and $\Psi\Phi = \mathrm{id}$ follows from α being a natural isomorphism by the same cancellation. Hence Φ is a ring isomorphism, and proposition 18.3.1 on each side identifies its source and target with $Z(R)$ and $Z(S)$, completing the proof.

Two consequences follow at once, and together they say that Morita equivalence is a coarser relation than isomorphism on noncommutative rings while collapsing to isomorphism on commutative ones.

Corollary 18.3.3: Commutative Rings Are Morita Rigid

Let R and S be commutative rings. Then R and S are Morita equivalent if and only if $R \cong S$.

***Proof*:**
If $R \cong S$ then the isomorphism induces an equivalence of module categories by restriction of scalars along it. Conversely, if R and S are Morita equivalent then $Z(R) \cong Z(S)$ by theorem 18.3.2, and commutativity gives $Z(R) = R$ and $Z(S) = S$ by definition 9.1.8, so $R \cong S$, as we sought to show.

Example 18.3.4: Morita Equivalence Is Coarser than Isomorphism
Take R any ring and $n \geq 2$. Then R and $M_n(R)$ are Morita equivalent by example 18.2.10 but not isomorphic whenever R is commutative, since $M_n(R)$ is not commutative: the matrix units E_{11} and E_{12} satisfy $E_{11}E_{12} = E_{12}$ while $E_{12}E_{11} = 0$. Consistently with theorem 18.3.2, the center is preserved: $Z(M_n(R))$ consists of the scalar matrices cI with $c \in Z(R)$, so $Z(M_n(R)) \cong Z(R)$.

By corollary 18.3.3 no two non-isomorphic commutative rings are Morita equivalent, so every instance of the phenomenon involves a noncommutative ring.

The remaining business of the section is the construction that produces those equivalences in practice. Example 18.1.6(2) showed that Re is projective and finitely generated for an idempotent e, and computed its trace ideal in a special case; the general computation is what decides when Re is a progenerator.

Definition 18.3.5: Full Idempotent

An idempotent $e \in R$, so $e^2 = e$, is *full* if $ReR = R$, that is if the two-sided ideal generated by e is the whole ring.

Theorem 18.3.6: The Corner Ring of a Full Idempotent

Let $e \in R$ be an idempotent. Then

1. Re is a finitely generated projective left R-module with $\mathrm{Tr}(Re) = ReR$, so Re is a progenerator if and only if e is full;
2. $\mathrm{End}_R(Re)^{\mathrm{op}} \cong eRe$, the *corner ring* of e.

Consequently, if e is full then R and eRe are Morita equivalent.

***Proof*:**

1. Since $e^2 = e$, the element $1 - e$ is also idempotent and $R = Re \oplus R(1-e)$: every r decomposes as $r = re + r(1-e)$, and if x lies in the intersection, written both as $x = re$ and as $x = r'(1-e)$, then multiplying on the right by e gives $x = re \cdot e = xe = r'(1-e)e = 0$, the last step because $(1-e)e = e - e^2 = 0$. So Re is a direct summand of R, hence projective by theorem 17.2.4(4), and it is generated by the single element e.

   For the trace ideal, take $f \in \mathrm{Hom}_R(Re, R)$. Then $f(e) = f(e \cdot e) = e\,f(e)$, so $f(e) \in eR$, and since every element of Re has the form re,
   $$f(Re) = f(Re \cdot e) = R\,f(e) \subseteq ReR$$
   Conversely, for each $a \in R$ the map $f_a\colon Re \to R$ given by $f_a(x) = xa$ is left R-linear with image Rea. Summing over $a \in R$ gives $\mathrm{Tr}(Re) \supseteq \sum_a Rea = ReR$. Hence $\mathrm{Tr}(Re) = ReR$, and theorem 18.1.4 makes Re a generator exactly when $ReR = R$. Combined with the previous paragraph, Re is a progenerator exactly when e is full.

2. A map $\varphi \in \mathrm{End}_R(Re)$ is determined by $\varphi(e)$, since $\varphi(re) = r\varphi(e)$, and $\varphi(e) = \varphi(e \cdot e) = e\,\varphi(e)$ shows $\varphi(e) \in eRe$ (it already lies in Re). Conversely each $x \in eRe$ gives $\varphi_x(re) = rex = rx$, which is left R-linear and lands in Re because $x = ex'e \in Re$. The two assignments are mutually inverse and additive.

   For the multiplication, let $\varphi(e) = ebe$ and $\psi(e) = eae$. Then
   $$(\varphi \circ \psi)(e) = \varphi(eae) = \varphi\big((ea) \cdot e\big) = (ea)\,\varphi(e) = ea\,ebe = (eae)(ebe) = \psi(e)\,\varphi(e)$$
   using $e^2 = e$ in the last-but-one step. So $\varphi \mapsto \varphi(e)$ reverses multiplication, giving $\mathrm{End}_R(Re) \cong (eRe)^{\mathrm{op}}$ and therefore $\mathrm{End}_R(Re)^{\mathrm{op}} \cong eRe$.

For the final claim, if e is full then Re is a progenerator by (1), so theorem 18.2.8 applied to $P = Re$ gives an equivalence between R-**Mod** and $\mathrm{End}_R(Re)^{\mathrm{op}}$-**Mod** $= eRe$-**Mod** by (2), completing the proof.

Theorem 18.3.6 is the form in which Morita theory is used downstream. Passing from R to eRe for a full idempotent shrinks the ring while keeping the module category, and choosing e well makes the smaller ring markedly more tractable. The matrix case is the choice $e = E_{11}$ in $R = M_n(A)$, where $eRe \cong A$ recovers example 18.2.10. In the representation theory of finite-dimensional algebras the same construction selects one idempotent from each isomorphism class of indecomposable projectives and produces the *basic algebra*, over which every algebra's representation theory can be read; that development, which needs the semisimple structure theory this book does not carry, is in *Core Representation Theory* [9].

Example 18.3.7: The Corner Ring of a Non-Full Idempotent

Fullness is not automatic. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $e = \overline{3}$, as in example 18.1.6(2). Then $ReR = Re = \{\overline{0}, \overline{3}\}$, which is not R, so e is not full. The corner ring is $eRe = \{\overline{0}, \overline{3}\} \cong \mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/6\mathbb{Z}$ is not Morita equivalent to $\mathbb{Z}/2\mathbb{Z}$: both are commutative and not isomorphic, so corollary 18.3.3 rules it out. The construction still produces a ring, but without fullness it produces the wrong one.

One invariant is worth recording separately, because it is the reason *Algebraic K-Theory* [4] cares about this section at all.

Corollary 18.3.8: Finitely Generated Projectives Correspond

Let $F\colon R\textbf{-Mod} \to S\textbf{-Mod}$ be an equivalence. Then F restricts to a bijection between the isomorphism classes of finitely generated projective R-modules and those of finitely generated projective S-modules, and this bijection preserves direct sums.

***Proof*:**

Projectivity is preserved: M is projective exactly when $\mathrm{Hom}_R(M, -)$ carries surjections to surjections (definition 17.2.3), F is bijective on morphism sets, and F carries surjections to surjections because a map is surjective exactly when its cokernel vanishes and F preserves cokernels by theorem 18.2.9, Step 2.

Being finitely generated is preserved because $F \cong Q \otimes_R -$ with $Q \otimes_R R \cong Q$ (theorem 18.2.9), so F carries R^n to Q^n and a summand of R^n to a summand of Q^n. That Q^n is itself finitely generated over S is theorem 18.2.9 applied with the roles of F and G exchanged, which makes $Q = F(R)$ a progenerator in $S\textbf{-Mod}$; the same argument then runs backwards through G. Direct sums are preserved by Step 2 of the same proof, and the quasi-inverse G gives the inverse bijection on isomorphism classes, completing the proof.

Since K_0 of a ring is built from the isomorphism classes of finitely generated projectives under direct sum, corollary 18.3.8 says that K_0 cannot distinguish Morita equivalent rings. That statement, and the group it is about, belong to *Algebraic K-Theory* [4].

Exercise 18.3.1

1. Show that the center of $M_n(R)$ consists of the scalar matrices rI with $r \in Z(R)$, and deduce $Z(M_n(R)) \cong Z(R)$ directly, without using theorem 18.3.2.
2. Decide which of the following are Morita invariants, with proof or counterexample: being a division ring; being commutative; being an integral domain (definition 9.1.21); having exactly two two-sided ideals.
3. Show that if R is commutative then an idempotent $e \in R$ is full if and only if $e = 1$, and give a noncommutative ring with a full idempotent $e \neq 1$.
4. Let $R = M_2(\mathbb{Z})$ and $e = E_{11}$. Verify directly that e is full, identify Re and eRe explicitly, and describe the equivalence $R\textbf{-Mod} \simeq \mathbb{Z}\textbf{-Mod}$ it produces.

5. Show that if e and f are idempotents of R with $Re \cong Rf$ as R-modules, then $eRe \cong fRf$ as rings.

6. Let R be a ring in which $1 = e_1 + \cdots + e_n$ for orthogonal idempotents e_i, meaning $e_i e_j = 0$ for $i \neq j$. Show that $R \cong \bigoplus_i Re_i$ as left R-modules, and that e_1 is full whenever $Re_1 \cong Re_i$ for every i.

Part IV

Fields

When studying classical algebra, a lot of time and energy is dedicated to solving equations. Famously, for any equation of the form $ax^2 + bx + c = 0$ has solution

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

which was known since at least the time of ancient Egypt (though not in the current algebraic form[1])After a long gap, Niccolò Tartaglia discovers the general equation for the cubic equation in terms by using an "imaginary" number i as a trick to solve the problem of $\sqrt{-n}$ coming up (it will take some more time before i is considered a number and not simply a tool to solve equations). Using this equation, a solution for quartic equation with respect to x was also found. The mission was on in finding a solution to the quintic equation. However, over 200 years, no such equation was found. Abel and Galois have in fact proved that only using $+, -, \times, \div$ and $\sqrt{\_}$, it is *impossible* to isolate x!

This is the origin of the study of field theory and Galois theory, and will be the driving motivation in this part of the book, however the study of fields have in modern times moves past this and are now used as a primary tool in the study of algebraic number theory. In chapter 50, we will use the language of field theory as the language of algebraic number theory.

[1]It was a geometric proof involving fitting squares of different sizes together

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Field Theory

A reminder that a field F is a unital commutative division ring (definition 9.1.11). Another way of thinking of a field is that it's a set with two binary operations both acting like abelian groups. Yet another way of seeing them is that $F^\times = F - \{0\}$ is an abelian group. In this chapter, we'll be mostly thinking about fields as unital commutative divsion rings, in particular, we're going to develop how the trivial ideal structure of fields (proposition 9.4.10) lets us uniquely define some extensions of a field. We will denote $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ and usually use the letters k and K to refer to fields.

Furthermore, the complex numbers $\mathbb{C}$ will be referenced for it's nice algebraic properties (ex. $i^2 = 1$ or $z = a + bi$ for any $z \in \mathbb{C}$). If the reader is unfamiliar with these notion, there are many resources online explaining these concepts[1].

19.1 Properties of Fields

First, all fields are rings, and so we can form the category **Fld** with fields as objects and ring homomorphisms as morphisms. We will use the term field homomorphsisms sometimes since the objects of study are fields with the knowledge that they are just ring homomorphisms. In general, the term ring homomorphism will be reserved for when rings will be brought into the discussion as will sometimes happen. Since the objects are fields, it is important to remember that all homomorphisms must be *injective* (see corollary 9.4.13)

[1] A famous book explaining these notions is *Althor's Complex Analysis*, in which the first chapter explores complex numbers

Proposition 19.1.1: Field Homomorphisms

Let $\varphi : k \to K$ be a ring homomorphism of fields. Then φ is either identically 0 or is injective, so that the image of φ is either 0 or isomorphic to k. Since the set 0 is not a field (by definition, $0 \neq 1$), φ in **Fld** can only be injective.

***Proof*:**
This is proposition 9.4.10 (2) updated so that we only care about fields.

Thus, If a homomorphism between fields $\varphi : k \to K$ exists, $\varphi(k) \cong k$. In this way, K can be thought of as extending k:

Definition 19.1.2: Field Extension

Let k and K be fields. If there exists a homomorphism $\varphi : k \to K$ (which as we've shown must be injective), then we say that K *extends* k, or K is a *field extension* of k. We will denote this by $k \subseteq K$, K/k, or

$$\begin{array}{c} F \\ | \\ K \end{array}$$

the field from which we are extending is sometimes called the *base field* or *ground field*.

Note that since fielsd are commutative, field extension $\varphi : k \to F$ makes F a k-algebra.

Example 19.1.3: Field Extensions

1. For any field F, F/F is trivially a field extension, since the identity map is a field homomorphism.

2. Perhaps the most famous field extension is that of $\mathbb{C}/\mathbb{R}$, since $\mathbb{R}$ is directly embedded into $\mathbb{C}$. A perhaps easy mistake to make when first learning about field extensions is that $\mathbb{R}^2/\mathbb{R}$ is not a field extension since $\mathbb{R}^2$ is simply not a field. On the other hand, $\mathbb{C}/\mathbb{Q}$ is also a field extension, since the natural inclusion map $\iota : \mathbb{Q} \to \mathbb{C}$ is a field homomorphism. Similarly, $\mathbb{R}/\mathbb{Q}$ is a field extension.

3. In example 9.1.19, we saw that $\mathbb{Q}(\sqrt{D})$ is a field where D is not a perfect square (i.e. there does not an element $d \in \mathbb{Q}$ s.t. $d^2 \in \mathbb{Q}$). For a concrete examples, let's say $D = 2$. Then $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is a field extension. If $D = -1$, then we usually write $\sqrt{-1}$ as i, and we get that $\mathbb{Q}(i)/\mathbb{Q}$ is a field extension. We also have that $\mathbb{C}/\mathbb{Q}(i)$ is a field extension, with the map $\iota : \mathbb{Q}(i) \to \mathbb{C}$ given by mapping $a + bi \mapsto a + bi$. Notice that the map $a + bi \mapsto a - bi$ is also a valid field homomorphism, showing that there need not be a unique map.

   We will soon find a way to define $\mathbb{Q}(\sqrt{D})$ without needing to reference set-builder notation.

4. Let k be a field and $k(x) = \text{Frac}(k[x])$ be the field of fractions of $k[x]$. Then $k(x)/k$ is a field extension.

The notation K/k may be read as "K over k". It does *not* represent the quotient K by k. This can be consider an overload of notation, but since the kernel of a ring homomorphism between fields must *always* be trivial, it is safe to see that K/k would not represent a quotient of the field K by k.

Another thing we can observe is that every field homomorphism $\varphi : F \to K$ can be extended into a ring homomorphism $f^{\varphi} : F[x] \to K[x]$ through

$$\sum_i a_i x^i \mapsto \sum_i \varphi(a_i) x^i$$

the fact that this is a ring homomorphisms comes straight from the properties of φ. Thus, we can consider this process to be a functor $f^{—} : \mathbf{Fld} \to \mathbf{Ring}$.

If F/k is any field extension, then we can naturally make F into a *vector-space over* k by making k act on F by multiplication (since $k \subseteq F$, this action is well-defined). Since every field has a prime subfield, every field is at least a vector-space over it's prime-subfield. The dimension of the vector-space is given a name:

Definition 19.1.4: Degree

The *degree* (or *relative degree* or *index*) of a field of extension F/k, denoted $[F : k]$, is the dimension of F as a vector-space over k (i.e. $[F : k] = \dim_k(F)$). The extension is said to be *finite* if $[F : k]$ is finite and is said to be *infinite* otherwise.

Definition 19.1.5: Finite Extension

Let F/k be a field extension. Then F/k is called a *finite extension* if $[F : k]$ is finite, and an *infinite extension* otherwise.

In example 19.1.3, $\mathbb{C}/\mathbb{R}$ is an extension of degree 2, the basis of $\mathbb{C}$ over $\mathbb{R}$ consiss of $\{1, i\}$, and hence a finite extension. The extension $k(x)/k$ is an extension if infinite degree: If it were finite, then any rational function can be written as a sum of a finite number of polynomials up to a constant $c \in k$, but then this collection must have a highest degree term, say n. Then any polynomial of degree $n + 1$ cannot be written as a linear combination of the basis – a conradiction. Hence, it is an infinite extension. Note too that as was demonstrated in theorem 13.1.8, every vector-space has a basis, and so any element $x \in F$ can be represented as a finite linear combination:

$$a_1\beta_1 + a_2\beta_2 + \cdots + a_n\beta_n \qquad a_i \in k,\ \beta_i \in \mathcal{B}$$

We will soon show how to find such a basis.

Since F/k can also be represented as a vector-space over k, we might ask if any field homomorphism/isomorphism can is also a linear transformation. The answer is yes, if we are a bit careful. Let's say that F/k and L/k are two fields extending k, and $\varphi : F \to L$ is a field homomorphism/isomorphism. We will look for a condition under which φ is also a k-vector space homomorphism. Linearly comes for free. For homogeneity of the scalar, we see that for any $r \in k$ and $m \in F$:

$$\varphi(rm) = \varphi(r)\varphi(m) \overset{!}{=} r\varphi(m)$$

where $\overset{!}{=}$ is the condition we need so that φ is a k-vector space homomorphism. In particular, this conditions say's that for any element $r \in k$, $\varphi(r) = r$. This is stronger than saying φ

maps k to itself (i.e. $\varphi(k) = k$), rather it is saying that φ is the *identity* on k. Since such field homomorphisms/isomorphsisms are also vector-space homomorphisms, they are given a name:

Definition 19.1.6: Fixed Homomorphisms

Let F/k and L/k be two field extensions of k and $\varphi : F \to L$ be a field homomorphism (resp. isomorphism). Then if φ is the identity on k, φ is said to be a *k-homomorphsm* (resp. *k-isomorphism*). If there exists a k-isomorphisms $A \to B$, then we'll say that A is k-isomorphic to B and write $A \cong_k B$.

Note that a fixed homomorhism is in fact a k-algebra homomorphism: if L/k, then there is a natural k-module structure on L given by multiplication.

As was shown in section 13.2, a linear transformation is determined by where it maps it's basis to. Hence, if we have a k-homomorphism, then can deduce where the element map (or resp. construct a k-homomrophism) by finding where the basis maps to (resp. by only choosing where the basis maps to). Even better, this linear transformation respects the multiplicative structure of F (that is $\varphi(ab) = \varphi(a)\varphi(b)$), that if a basis β can be generated multiplicatively and additively from a set S, $\langle S\rangle = \beta$, then a k-homomorphism is only determined by where it sends S. Even better, this makes any k-homomorphism a k-algebra homomrophism, and a field F that extends k (F/k) is a k-algebra.

One more word of warning is in order: Let's say we find two field extensions A/k, B/k such that $A \cong_k B$. Then this implies that $\dim_k A = \dim_k B$, however it does not imply $A = B$. This might seem obvious, however it is an mistake that is easy to make after working with field extensions for awhile (this nuance will be particularly important in theorem 19.5.13).

In most of this and the next chapter, we will focus on finite extensions. Some results for extensions of infinite degree will arise as a natural generalization of the results of finite extensions, however finiteness is used in some key proofs to construct bijections (most importantly, in proposition 20.1.12, TBD). We will touch on infinite extensions in TBD (don't yet know when they will become central, i.e. when transcendental degrees will come in play).

Characteristic

The following is is perhaps the simplest invariant of a field k (that is, if $\varphi : k \to K$ is a homomorphism, the property we're about to discuss is preserved). Every field has a multiplicative identity $1 \in k$. Since k is a field, it is also a ring, and hence there exists a unique ring homomorphism from $\mathbb{Z}$ to k mapping 1 to 1 (see proposition 9.1.29):

$$\mathbb{Z} \to k,\ n \mapsto \begin{cases} \underbrace{1 + 1 \cdots + 1}_{n \text{ times}} & n > 0 \\ -(\underbrace{1 + 1 \cdots + 1}_{n \text{ times}}) & n < 0 \\ 0 & n = 0 \end{cases}$$

We will use multiplication notation $n \cdot 1$ as a shorthand [2]. Recall $\operatorname{im}(\mathbb{Z})$ of the natural homomorphism is always the smallest subring of k (exercise ref:HERE (ex:ringCatExercise)). Recall further from exercise ref:HERE (ibid) the following definition:

[2]Note that we can also have defined it as a $\mathbb{Z}$-module on k

Definition 19.1.7: Characteristic

The *characteristic* of a field F, dentoed $ch(F)$, is the order of $\mathrm{im}(\mathbb{Z})$. If $|\mathrm{im}(\mathbb{Z})| < \infty$, we say that F has a *finite characteristic*, and characteristic 0 otherwise. In other words, it is defined as the smallest positive integer n such that $n \cdot 1 = 0$ if such a n exists, and is defined to be 0 otherwise

If n exists, then n needs to be prime. If it weren't, then

$$n = n \cdot 1 = (a \cdot 1)(b \cdot 1) = 0$$

meaning either $a \cdot 1$ or $b \cdot 1$ must be zero-divisors – a contradiction since a field has no zero-divisors! Notice that we only need to have an integral domain for the previous proposition, and that the characteristic of the integral on an integral domain will be the same as it's field of fractions. Since $\mathbb{Z}/p\mathbb{Z}$ is a field (since $\mathbb{Z}/p\mathbb{Z}^\times = \mathbb{Z}/p\mathbb{Z} - \{0\}$, see exercise ref:HERE (ex:introToRings)). However, $\mathbb{Z}$ is not a field. Using theorem 9.5.1, we know the smallest field containing $\mathbb{Z}$ would be $\mathbb{Q}$! In both cases, the field this map injects to must be the *smallest* field that contains 1 (which can also be thought of as the field *generated* by 1 in k)

Definition 19.1.8: Prime Subfield

The *prime subfield* of a field k is the subfield k generated by the multiplicative identity 1_F of k. It is (isomorphic to) either $\mathbb{Q}$ (if $\mathrm{ch}(k) = 0$) or k_p (if $\mathrm{ch}(k) = p$).

19.1.9 Note A shorthand of writing the characteristic of a field is the following: if $p \cdot 1 = 0$, then it is sometimes written $p = 0$ to mean a field with characteristic p.

Notice that for a field k of characteristic n ($n \geq 0$), $k/\mathbb{F}_p$ (resp. $k/\mathbb{Q}$) is always a field extension, since we can always map $\mathbb{F}_p$ or $\mathbb{Q}$ to k through the identity map.

Note that for fields of characteristic 0, field homomorphism are automatically a vector-space homomorphism over $\mathbb{Q}$: linearity holds automatically, and for any scalar $c \in \mathbb{Q}$:

$$\varphi(cx) = \varphi(c)\varphi(x) \stackrel{!}{=} c\varphi(x)$$

where $\stackrel{!}{=}$ comes from how a field homomorphism must map $\mathbb{Q}$ to itself.

The claim at the beginning of this section should now be checked, namely that if $\mathrm{ch}(k) = n$, $n \geq 0$, and $\varphi : k \to K$ is a homomorphism, then $\mathrm{ch}(K) = n$ (see exercise ref:HERE). Since charactersitic is preserved, we can take the subcategory of $\mathbf{Fld}_0$ or $\mathbf{Fld}_p$ where we restric to fields of characteristic p or 0. We won't really do this, however it is inisghtful that doing so will now make these categories have initial objects, namely the respective prime subfield (can you prove this? Can you see why **Fld** has no initial objects?). More generally, since homomorphisms can only be injective, we can take $\mathbf{Fld}_k$ to be the category ring homomorphisms as morphisms and all fields that extend k as objects! In most of the following studies, we will fix a field k and study extensions from it, and so field theory can be thought of as studying particular $\mathbf{Fld}_k$ categories

Torsion in Fields

As we have shown in theorem 11.3.14, $F^\times$ is locally cyclic. We recall the theorem here for convinience:

Theorem 19.1.10: $F^\times$ is Locally Cyclic

A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

Note that this does not mean that for every element $a \in F$, there exists a power n such that $a^n = 1$. For exmaple, in $\mathbb{R}$, no power of 2 will result in $2^n = 1$. If this is the case for an element, we will give it a name:

Definition 19.1.11: Root of Unity

Let F be a field and $r \in F$. If there exists an $n \in \mathbb{N}$ such that $r^n = 1$, then we call r an *nth root of unity*.

Example 19.1.12

In $\mathbb{C}$, all elements of the form $e^{\frac{2\pi ik}{n}}$ for $0 \leq k \leq n-1$ are nth roots of unity. In fact, they are all the roots of unity of $\mathbb{C}$. The real numbers $\mathbb{R}$ have only 2 roots of unity of degree 1 and 2 respectively: ± 1.

Notice that not all $e^{\frac{2\pi ik}{n}}$ are generators for the cyclic subgroup they belong to. The generators of these cyclic subgroups are given a name:

Definition 19.1.13: Primitive Root Of Unity

Let F be a field. Then a *primitive nth root of unity* is a generator of the cyclic multiplicative subgroup of all nth roots of unity.

We will return to the study of the roots of unity later; they will be the "building blocks" for [finitely generated] abelian groups in field theory (and Representation theory)[3]

Exercise 19.1.1

1. (Do some exercises showing relations beween roots of unity and other identites? Ex. I think $\zeta_8 + \zeta_8^7 = \sqrt{2}$. Important later. Also elaborate on $\mathbb{C}$ for all future purposes, since it comes up a lot)
2. (Analysis Problem) Show that the roots of unity are dense in $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$.

Generating Fields

Let's say $k \subseteq F$ is a subfield of F, and take some subset $A \subseteq F$. We can construct a field using k and A in the same way as we have done all the way back in section 2.3.2: Take all subfield of F containing k and A, and intersect them. Similarly to section 2.3.2, this is equivalent to "closing" the field k with A by taking all possible combinations. There is always at least exists the field K with

[3] Remember by the fundamental theory of finitely generated abelian groups that any finitely generated abelian group is the direct product of a free part $\mathbb{Z}^r$ and a torsion part? Notice how the roots of unity can represent any cyclic subgroup?

this which contains this property. Since the intersection of fields is a field (this should be done as an exercise), then we can intersect all fields with this property to get the smallest such subfield.

Definition 19.1.14: Field Generated By Elements Over F

Let F/k be a field extension and S be a set of elements of F. Then the smallest subfield of F containing both k and the elements of S is denoted $k(S)$ and is called *the field generated by S over k*. If S is finite so that $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$, we usually denote $k(S)$ by $k(\alpha_1, \alpha_2, ..., \alpha_n)$ and call it the field *generated by $\alpha, \alpha_2, ..., \alpha_n$ over k*.

If we have a field extension F/k, and F can be generated by a single element, then we call that a simple extension:

Definition 19.1.15: Simple Extension

Let F/k be a field extension. Then if there exists an element $\alpha \in F$ such that $F = k(\alpha)$, we say that F is a *simple extension*, and call α the *primitive element* of the extension. The field $k(\alpha)$ is sometimes called "k adjoin α".

Be mindful to not confuse the terms "finite generation over k", "finite type k-algebra" [4], or "finite degree over k" (for example, $k(t)/k$ is a simple extension, but certainly not finite as was discussed under the definition of degree).

A similar definition works if we broaden to rings: instead of taking the intersection of all fields containing α and F, we take all rings containing α and F. It is easy to generlize the proof that the intersection of fields is a field to rings. We will usually denote this smallest ring $F[S]$, and if $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is finite, then $F[S] = F[\alpha_1, \alpha_2, ..., \alpha_n]$.

Proposition 19.1.16: Characterizing $F(S)$ and $F[S]$

Let K/F be a field extensions, $S \subseteq K$, and consider both $F(S)$ and $F[S]$. Then:

1. $F[S]$ can be characterized as the set of all finite linear combinations with coefficients in K of finite products of powers of elements of S, that is, polynomials with variables taken from S
2. $F(S)$ can be characterized as the set of all $ab^{-1} \in K$ with $a, b \in F[S]$ with $b \neq 0$. Furthermore, $F(S) \cong \text{Frac}(F[S])$

Note that in the "polynomials" mentioned above, the varaibles S are *not* indeterminates, since they might have relations in K (for example, it might be that for some $s \in S$, $s^2 + 1 = 0$)

***Proof*:**
These proofs are similar to those of section 2.3.2 with appropriate modifications for the added operation of multiplication and inversing.

[4] It is important to point out that there is actually a deep connection between those two concepts which we'll explore in part V. As foreshadowing, try seeing if you can think of a finitely generated field extension F/k that is not a finite-type k-algebra (or vice-versa). The answer to this question is given by Zariski's Lemma

To close off this section, we introduce a generalization to notion of the smallest field K/k generated by some elements to the smallest field generated with all the elements of another field:

Definition 19.1.17: Composite Field

Let K_1 and K_2 be two subfield of K. Then the *composite field* of K_1 and K_2, denoted K_1K_2, is the smallest subfield of K containing both K_1 and K_2. Similarly, the composite of any collection of subfields of K is the smallest subfield containign all the subfields.

Note that like in section 2.3.2, we can describe K_1K_2 as the intersection of all subfields contianing K_1 and K_2 (or any number of subfields).

Example 19.1.18: Composite Fields

The composite field $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt[3]{2})$ is $\mathbb{Q}(\sqrt[6]{2})$, since thie field contains both these fields, and conversely any field containing both $\sqrt{2}$ and $\sqrt[3]{2}$ contains their quotient, namely $\sqrt[6]{2}$

Exercise 19.1.2

1. Sometimes, we want to talk about field extensions without an underlying field. Let E, F be two fields of the same characteristic, say p. Show that $E \otimes_{\mathbb{F}_p} F$ (resp. $E \otimes_{\mathbb{Q}} F$) as a $\mathbb{F}_p$ (resp. $\mathbb{Q}$)-algebra is a field if and only if $[EF : \mathbb{F}_p] = [E : \mathbb{F}_p][F : \mathbb{F}_p]$ (resp. $[EF : \mathbb{Q}] = [E : \mathbb{Q}][F : \mathbb{Q}]$)

19.2 Field Extensions

We now continue our exploration of field extensions. As we saw in proposition 19.1.16, any field K/F such that $K = F(S)$ is the field of fractions of all "polynomials" with variables S and coefficients F. In particular, these polynomials have "solutions" or "roots" characterizing their relations. Similarly to how generators in a group where characterized by a monomial, (for example, D_4 is characterized with the generators a, b satisfying the monomails $ab^{-1}ab = 1$ and $a^2 = b^4 = 1$), generators in fields are characterized by polynomial relationships (for example, $\mathbb{C}/\mathbb{R}$ is characterized by being generated by $\mathbb{R}$ and i where $i^2 = -1$). Just like how any relation of elements within a group can be shifted so all the elements are on the left hand side (ex. writing $a^{-1}bab = 1$ instead of $ab = b^{-1}a$), we can do the same with polynomial relationships (ex. $i^2 + 1 = 0$). In proposition 19.2.24, we will show why it only matters to characterize polynomials, and not rational polynomials (i.e. elements of $p(S)/q(S)$)

As we've seen in chapter 11, every polynomial $p(x) \in F[x]$ can be broken down into irreducible polynomials. We will start finding an easy way of extending any field k which will give us a large class of field and field extensions to work with using irreducible polynomials (recall that x^2+1 has no solutions in $\mathbb{R}$, but does in $\mathbb{C}$, so is there an extension from $\mathbb{R}$ to $\mathbb{C}$?). We will find that the method used to construct fields will almost create "universal fields" (the same way free groups are sort of the universal groups since any group is the quotient of a free group). This endevaour to create a universal property will be very close in obtaining a universal property for fields, however it will fail in one important way. Due to that failure, it will be important carry around some more information to take into account the lake of universality. In fact, the lack of universal property may be seen less as a failure and more as a peek into the further structure that underly field extensions. Figuring all of this out is the goal of this section

Field Extensions using Polynomial

If we take our field to be $\mathbb{R}$, then $x^2 + 1 = 0$ does not have a solution in $\mathbb{R}$. The question then is whether there is a larger field K for which $x^2 + 1 = 0$ *has* a solution. More generally, given any irreducible polynomial $p(x) \in k[x]$, does there exists a field K such that K contains a root of $p(x)$[5]?. The answer is yes! We actually have all the tools already to answer this question!! I suggest to take a moment and try to find the larger field K which satisfies this!!

19.2.1 Taking a moment

Theorem 19.2.2: Kronecker's Theorem

Let k be a field and let $p(x) \in k[x]$ be an irreducible polynomial. Then there exists a field K containing an isomorphic copy of k in which $p(x)$ has a root. Identifying k with this isomorphic copy through i shows that there exists an extension of k in which $p(x)$ has a root. Furthermore, if L/k is a field extension in which $p(x)$ has a root, then there exists a k-homomorphism $j : K \to L$ such that

$$\begin{tikzcd} K \arrow[r, hook, "j"] & L \\ k \arrow[u, hook, "\iota"] \arrow[ur, hook, "i"'] & \end{tikzcd}$$

commutes. Due to this, the field K produced in the following construction is sometimes called the *Kronecker field for* $p(x)$ *over* k.

If in the Kronecker construction we use a *reducible* polynomials, then the resulting object is no longer a field (it contains zero-divisors), but remains to be a k-algebra.

***Proof*:**
Consider

$$K = k[x]/(p(x))$$

Since $k[x]$ is a Principal Ideal Domain (corrollary 11.1.2) and since $p(x)$ is irreducible (so prime by proposition 10.1.33), then $p(x)$ is maximal (proposition 10.1.32). Then $k[x]/(p(x))$ is a field by proposition 9.4.19. If we take the cannonical projection $\pi : k[x] \to K(= k[x]/(p(x)))$, then we can restrict it to F $(\pi|_F : F \to K)$ to get a homomorphism from F to K. Since this cannot be identically 0 or else K would be the zero field, so by proposition 19.1.1, F maps injectively into K, so $\varphi(F)(\cong F)$ is an isomorphic copy of F contained in K. We identify F with its isomorphic image in K and view K as a *subfield* of K. If $\overline{x} = \pi(x)$ denotes the image of x in the quotient K, then

$$\begin{aligned} p(\overline{x}) &= \overline{p(x)} && \text{since } \pi \text{ is a homomorphism} \\ &= p(x) \bmod (p(x)) && \text{in } k[x]/(p(x)) \\ &= 0 \end{aligned}$$

so that K does indeed contain a root of the polynomial $p(x)$! Thus, K is an extension of F in

[5]We can assume that $p(x)$ is irreducible in $F[x]$, since if any factors of $p(x)$ has a root, then so does $p(x)$

which the polynomial $p(x)$ has a root

For the second assertion, let's say L/k has a root to $p(x)$, say $p(\alpha) = 0$. Then the evaluation map $\mathrm{ev}_\alpha : k[x] \to L$ (which we have proved is a ring-homomorphism in theorem 9.2.4), $g(x) \mapsto g(\alpha)$ must map $p(x)$ to 0. Hence, $p(x) \in \ker(\mathrm{ev}_\alpha)$ meaning $(p(x)) \subseteq \ker(\mathrm{ev}_\alpha)$, and by maximality of $(p(x))$ $(p(x)) = \ker(\mathrm{ev}_\alpha)$. Since ideals correspond one-to-one with homomorphisms, we have a homomorphism:

$$j : \frac{k[x]}{(p(x))} \to L$$

and since $\frac{k[x]}{(p(x)} \cong K$, we have:

$$j : K \to L$$

Furthermore, for any $r \in k$, $j(r) = r$. Thus j clearly commutes in the aformentioned diagram, completing the proof.

From now on, when we talk about irreducible polynomial, we will specifically mean *non-linear irreducible polynomials*, since linear irreducible polynomials produce trivial field extensions (i.e. k being extended to sovle for $x - \alpha \in k[x]$ is just k) unless stated otherwise.

The field extension to the Kronecker field K/k is *almost* universal with respect to finding roots to the polynomial $p(x)$. If it were universal, then if there was another field extension F/k such that F has a root of the irreducible polynomial $p(x) \in k[x]$, there would exist a *unique* homomorphism $j : F \to K$ such that the diagarm commutes. However, the homomorphism j need not be unique.

Example 19.2.3: Lack Of Uniqueness

Take $K = \mathbb{Q}[x]/(x^2 - 2)$ and $\mathbb{R}$. Then the polynomial $x^2 - 2$ has a root in $\mathbb{R}$, hence we know there exists a homomorphism:

$$K = \mathbb{Q}[x]/(x^2 - 2) \hookrightarrow \mathbb{R}$$

However, there exists two homomorphisms mapping K to $\mathbb{R}$. Since K extends $\mathbb{Q}$, any homomorphism can be represented by where the roots map: we can map $1 \mapsto 1$ and $\overline{x} \mapsto \sqrt{2}$, or map $1 \mapsto 1$ and $\overline{x} \mapsto -\sqrt{2}$. Both of these are valid homomorphisms, and so adjoints are not universal with respect to roots.

The universality can be recovered if we keep track of which root we want to map to (ex. $\sqrt{2}$), however we would have to make sure that there is only 1 homomorphism given some root (how do we know in more general cases that there can't be 2 homomorphisms mapping $\overline{x}$ to the same root?). This concern is solved soon in theorem 19.2.27.

Since the aforementioned diagram commutes, notice that for any $r \in k$:

$$r \overset{!}{=} i(r) = j(\iota(r)) = j(r)$$

where ι is the natural inclusion into K, and the image of i is identified with the domain, justifying the $\overset{!}{=}$. Thus, j must *always* fix the base field, i.e. j is a k-homomorphism/isomorphism.

You might be concerned about *which* root of an irreducible polynomial does $\overline{x}$ represent in the previous theorem. For example, $x^2 - 2$ is irreducible over $\mathbb{Q}$, with $\pm\sqrt{2}$. So which root should we pick to represent $\overline{x}$? In fact, we can pick $\overline{x}$ to be represented *any* root of this polynomial: the field extending to contain $\sqrt{2}$ or $-\sqrt{2}$ will be isomorphic. More generally, if $p(x)$ is any irreducible polynomial, then $k[x]/(p(x))$ can be representing *any* choice of the possible roots of $p(x)$[6] and extending k to contain that root. To make the choice concrete, we can choose a particular root of $p(x)$, say α (which we define by saying α satisfies some polynomial equation), and isomorphically identify $K \xrightarrow{\sim} k(\alpha)$ (which we will show in more detail in theroem 19.2.8).

Note that to find the root of a polynomial, we required that polynomial to be irreducible. This technique will not obtain us a root of $(x^2-2)(x^3-2) = x^6 - 5x^2 + 6$, namely becaue $(x^6 - 5x^2 + 6)$ is not even a prime ideal (the resulting quotient contains zero-divisors). In section 19.5.2, we will explore how to find extensions for which *any* polynomial has all of its roots, essentially by finding the roots of every irreducible polynomial and progressively extending our field by adjoning roots.

Note that we haven't shown a way to "find" the root. For example, if we were looking at irreducible polynomial over $\mathbb{R}$ and we pick a root $\alpha \in \mathbb{C}$ that satisfies the polynomial equation, we don't have a computational way of finding it:

Example 19.2.4: Existence, but not construction

Let's say say F is a field extension $F/\mathbb{Q}$. Consider the irreducible polynomial $x^3 - 3x - 1$. over $\mathbb{Q}$ (it has no rationals roots by prosposition 11.3.5). Thus, $F = \mathbb{Q}[x]/(x^3 - 3x - 1)$ is a field which has a root! The fact that all the roots of $x^3 - 3x - 1$ are real (and hence being able to identify F with a subfield of $\mathbb{R}$) can be considerd as taken for granted for now; we will have the tools to show that by the end of section 20.1.1. Visually, if we plot $x^3 - 3x - 1$, notice that all the roots are real, and hence $\mathbb{Q}$ can only by roots of $x^3 - 3x - 1$ which are real.

Let's say α is the root, so $\alpha \in \mathbb{R}$. We might hope that we can find which α in $\mathbb{R}$ satisfies the polynomial given that we know α is a root, however there is no *algebraic* way of finding this root! Using Newton-Raphsons approximation method, we can find that it is a real number $x \approx 1.97938...$. Finding these roots is a sub-branch of numerical analysis known as *root-finding algorithms*

For most common roots, we have already developped a symbol to represent them, namely $\sqrt[n]{m}$ is the root of $x^n - m$. More generally, if we extend $\mathbb{Q}[x]/(x^n - m)$ to solve another polynomial of a similar form, say $x^k - l$:

$$\frac{(\mathbb{Q}[x]/(x^n - m))[y]}{(y^k - l)}$$

we can "nest" as many square roots as we want. Since adding and multiplying is closed in a field, any extension of such polynomials produce elements of the form:

$$\sqrt[3]{\frac{3+\sqrt[9]{40}}{\sqrt[8]{\sqrt{2}\sqrt{10}}}}$$

However, most more complicated polynomials are not given a special symbol, a notable exception being the golden ratio φ which is the root of $x^2 - x - 1$[7]. In section 20.2.4, we will further explore extensions which only add radical elements.

With the extension of a field through an irreducible polynomial constructed, it would be advantageous to be able to easily work with it's elements. Since K is a subfield of k, we can describe K as a k-vector-space!

[6] What it means to "choose" a root here is a bit ambiguous, since we have yet to say there exists a field that makes $p(x)$ factor into linear terms. Such a field is constructed in section 19.5.1

[7] Though, this is not much of an exception since $\varphi = \frac{1+\sqrt{5}}{2}$

Theorem 19.2.5: Basis For Kronecker Field

Let $p(x) \in F[x]$ be an irreducible polynomial of degree n over the field F and let K be the field $F[x]/(p(x))$. Let $\theta = x \bmod (p(x)) \in K$. Then the elements

$$1, \theta, \theta^2, ..., \theta^{n-1}$$

are a basis for K as a vector space over F, so the degree of the extension is n, i.e., $[K : F] = n$. Thus

$$K = \left\{a_0 + a_1\theta + a_2\theta^2 + \cdots + a_{n-1}\theta^{n-1} | a_0, a_1, ..., a_{n-1} \in F\right\}$$

consists of all polynomials of degree $< n$ in θ. In particular, all field extensions of this form have finite degree.

***Proof*:**
Take $K = F[x]/(p(x))$, and consider $a(x) \in F[x]$ to be any polynomial. Then, by the Euclidean algorithm, we get that

$$a(x) = q(x)p(x) + r(x) \qquad q(x), r(x) \in F[x] \text{ with } \deg(r(x)) < n$$

then in K, we get that $[a(x)] = [r(x) + q(x)p(x)] = [r(x)]$, so any polynomial can be represented by it's residue polynomial, which will always be of degree $< n$. Thus, $1, \theta, \theta^2, ..., \theta^{n-1}$ (the image of $1, x, x^2, ..., x^{n-1}$) span the quotient. It remains to show they are linearly independent to conclude they are a basis.

If they weren't, then for some $b_i \in F$ (not all 0)

$$b_0 + b_1\theta + \cdots + b_{n-1}\theta^{n-1} = 0$$

that is

$$b_0 + b_1 x + \cdots + b_{n-1}x^{n-1} \equiv 0 \bmod (p(x))$$

meaning $p(x) \big| b_0 + b_1 x + \cdots + b_{n-1}x^{n-1}$. However, that's a contradiction, since the degree of $p(x)$ is higher than the polynomial on the right hand side (which is at most $n-1$)! Thus, these must be linearly independent. Since they also span, they fulfill the condition of a basis.

Since a basis is generated as a k-algebra by θ, any k-homomorphism $\varphi : k(\theta) \to F$ is determined by where it maps θ, making the map particularly easy to determine if we determine where θ maps to (the criterion for which we do in corollary 19.2.29).

As a reminder of arithmetics in $F[x]/(p(x))$:

Corollary 19.2.6: Shortcut On Arithmetics In K

Let K be as in theorem 19.2.2, and let $a(\theta), b(\theta) \in K$ be two polynomials of degree $< N$ in θ. Then addition in K is defined simply bu usual polynomial addition and multiplicatin in K is defined by

$$a(\theta)b(\theta) = r(\theta)$$

where $f(x)$ is the remainder (of degree $< n$) obtaiend after dividing the polynomial $a(x)b(x)$ by $p(x)$ in $F[x]$

***Proof*:**
An exercice in quotient ring of polynomials. Addition multiplication is done as in any quotient ring:

1. addition is usual addition of polynomials, since it doesn't change the degree
2. For multiplication, one can multiply the terms, then find a representative in the coset with degree $< n$

Note that what is described in corrollary 19.2.6 works in any quotient field, including over polynomials which are *not* irreducible. However, the result is not a field – not even an integral domain [8].

Example 19.2.7: Field Extension Through Irreducible Polynomials

1. Let $F = \mathbb{R}$ and $p(x) = x^2 + 1$. Then we know that $x^2 + 1$ does not have a root in $\mathbb{R}$. Thus, consider:
$$K = \mathbb{R}[x]/(x^2 + 1)$$
is an extension field (of degree 2) that contains a root to this polynomial. Let $\theta = \overline{x}$. By theorem 19.2.5, $K/\mathbb{R}$ has degree 2 and any element $t \in K$ can be written as:
$$t = a + b\theta \qquad a, b \in \mathbb{R}$$
If you are comfortable with the arithmetics of $\mathbb{C}$, show that:
$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$$
2. Let $F = \mathbb{Q}$ and $p(x) = x^2 + 1$. Then by substitution with Eisntein's Criteron, $x^2 + 1$ is irreducible, hence $K = \mathbb{Q}[x]/(x^2 + 1)$ is a field extension of $\mathbb{Q}$. Since the base field is now $\mathbb{Q}$, any element $t \in K$ is of the form:
$$t = a + b\theta \qquad a, b \in \mathbb{Q}$$
3. Let $F = \mathbb{Q}$ and $p(x) = x^2 - 2$. Then $x^2 - 2$ is irreducible by Eisenstein's, so $F = \mathbb{Q}[x]/(x^2 - 2)$ is a field extension of $\mathbb{Q}$. Letting $K = \mathbb{Q}[x](x^2 + 1)$, show that:
$$F \ncong K$$
that is, extending by different roots of different irreducible polynomial produces distinct fields.

[8] Recall that if $p(x) = a(x)b(x)$, $a(x)$, $b(x)$ not being units, then in $F[x]/(p(x))$, $\overline{a(x)}, \overline{b(x)}$ are both nonzero, but $\overline{a(x)} \cdot \overline{b(x)} = \overline{0}$, meaning $F[x]/(p(x))$ has zero-divisors

4. Let $F = \mathbb{F}_2$ (the finite field with 2 elements) and $p(x) = x^2 + x + 1$. Then $p(x)$ is irreducible since $0^2 + 0 + 1 = 1^1 + 1 + 1 = 1$. Hence, $K = \mathbb{F}_2[x]/(x^2 + x + 1)$ is irreducible of degree 2. Can you make the multiplication table for this field?

Classifying Simple and Transcendental Extensions

Let's now turn our attention to classifying field extensions, given $k \subseteq K$ where we consider k the base field. If $A \subseteq K$, then we already have a notion of field extensions: $k \subseteq k(A)$. We will focus on the special cae where $A = \{\alpha\}$, that is, simple extension with $\alpha \in K$. As an example of this, imagine taking $\mathbb{R} \subseteq \mathbb{C}$, $\alpha = i$, and considering $\mathbb{R}(i)$. Then it is a known fact that $i^2 + 1 = 0$. We might ask if in general, given some $\alpha \in K$, can we find a polynomial that sovles it? In this way, we can characterize the field $k(\alpha)$ without needing to resort to taking the intersection of all subfields of K containing F and α. The answer is in fact, half the time. All simple extensions can be characterized by either being isomorphic to the rational functions over the base field or in terms of a quotient of a polynomial ring, giving us an easy classification of simple extensions:

Theorem 19.2.8: Classifying Simple Extensions

Let K/k be a field, $\alpha \in K$, and $k(\alpha)$. Consider $\mathrm{ev}_\alpha : k[x] \to k(\alpha)$ mapping $f(x) \mapsto f(\alpha)$. Then if ev_α is injective, $k(\alpha)$ is an infinite extension and $k(\alpha) \cong k(x)$, and if ev_α is not injective, then there exists a polynomial $p(x)$ such that $k[x]/(p(x)) \cong k(\alpha)$. Furthermore, $p(x)$ is the polynomial of *minimal degree* that has α as a root.

Note that in this theorem, we're saying that *any* field over K extending k in which $p(x)$ contains a root in K and is irreducible in k contains a subfield isomorphic to the extension of k as we constructed in theorem 19.2.2 and that this field is (up to isomorphism) the smallest extension of k containing such a root.

***Proof*:**
Take the natural homomorphism

$$\mathrm{ev}_\alpha : F[x] \to F(\alpha)(\subseteq K) \qquad a(x) \mapsto a(\alpha)$$

which maps the polynomial $a(x)$ to the polynomial evaluated at α. Note that $\mathrm{ev}_\alpha(F[x])$ is an integral domain, since $F(\alpha)$ is integral domain.

If ev_α is injective, then $F[x]$ can be identified with an subring of $F(\alpha)$ that is an integral domain. By the universal property of field of fraction (theorem 9.5.4), ev_α uniquely extends to a homomorphism

$$\varphi : F(x) \to F(\alpha)$$

Since $\varphi(F(x))$ is a field containing F and α, $\varphi(F(x)) \supseteq F(\alpha)$, and since φ is a function $\varphi(F(x)) \subseteq F(\alpha)$, hence $\varphi(F(x)) = F(\alpha)$. To find the degree $[F(\alpha) : F]$, notice that $1[= \alpha^0], \alpha, \alpha^2, \ldots$ are all linearly independent over F in $F(\alpha)$ (since $1[x^0], x, x^2, \ldots$ are all linearly independent all linearly indepndent over F in $F(x)$), hence any basis must at least contain countably many linearly independent elements, and hence the degree must be infinite.

On the otherhand, if ev_α is not injective, then $\ker(\mathrm{ev}_\alpha) \neq 0$ and ev_α is clearly surjective. Since ev_α is surjective and $F(\alpha)$ is a field, $\ker(\mathrm{ev}_\alpha)$ must be a maximal ideal, and since $F[x]$ is a euclidean domain (and hence a PID), any polynomial of minimal degree in $\ker(\mathrm{ev}_\alpha)$ can represent the ideal, that is $(m_{\alpha,k}(x)) = \ker(\text{ev}_\alpha)$ for some minimal degree polynomial $m_{\alpha,k}(x) \in \ker(\text{ev}_\alpha)$. In fact, this minimal polynomial must be unique up to unit: if there where two minimal polynomials where $(m_{\alpha,k}) = (m'_{\alpha,k})$, then they must be equal up to unit, i.e. up to a multiple of k. Hence, we have that

$$\frac{k[x]}{(m_{\alpha,k}(x))} \cong F(\alpha)$$

By theorem 19.2.5, $[F(\alpha) : F] = \deg m_{\alpha,k}(x)$, and so the extension is finite, completing the proof.

This theorem motivates the following vocabulary:

Definition 19.2.9: Algebraic and Transcendental Elements

Let F be a field and let K be an extension of F. Then:

1. The element $\alpha \in K$ is said to be *algebraic* over F if α is a root of some nonzero polynomial $f(x) \in F[x]$.

2. If α is not algebraic over F (i.e., is not the root of any nonzero polynomial with coefficeints in F) then α is said to be *transcendental over F*.

Dealing with more general transcendental extensions is a straight-forward generlization of the previous thoerem; it naturally generlizes for $k(\alpha_1, ..., \alpha_n) \subseteq K$ where each α_i is transcendental. We first define the following term for the corollary:

(are all monomorphisms injective? I replaced the word monomorphism with injective in my notes for definition 19.2.10)

Definition 19.2.10: Algebraically Independent

Let R be a k-algebra. Then $\alpha_1, ..., \alpha_n \in R$ are *algebraically independent* over k if the k-algebra homomorphism ev_k (i.e., the evaluation map) which fixes k:

$$\text{ev}_k : k[x_1, ..., x_n] \to R \qquad \text{ev}_k(x_i) = \alpha_i$$

is injective. The set is called *algebraically dependent* otherwise.

The definition is stated for a finite list, and that is the only case a polynomial relation can see: a relation involves finitely many indeterminates however large the set. So the notion extends to arbitrary subsets without any new content.

Definition 19.2.11: Algebraically Independent Set

Let K/k be a field extension. A subset $S \subseteq K$ is *algebraically independent* over k if every finite list of distinct elements of S is algebraically independent over k in the sense of definition 19.2.10. The empty set is algebraically independent.

The analogy with linear algebra now runs further than it first appears. An algebraically independent set behaves like a linearly independent one, a set over which K is algebraic behaves like a spanning one, and the two notions meet in an object that deserves the name basis.

Definition 19.2.12: Transcendence Basis

Let K/k be a field extension. A subset $B \subseteq K$ is a *transcendence basis* of K over k if B is algebraically independent over k and K is an algebraic extension of $k(B)$.

Existence follows the same route as for vector-space bases: take a maximal independent set. Maximality is exactly what forces the algebraicity.

Theorem 19.2.13: Existence of a Transcendence Basis

Let K/k be a field extension and let $S \subseteq K$ be a subset with K algebraic over $k(S)$. Let $A \subseteq S$ be algebraically independent over k. Then there is a transcendence basis B of K/k with $A \subseteq B \subseteq S$. In particular, taking $S = K$ and $A = \emptyset$, every field extension has a transcendence basis.

***Proof*:**
Order by inclusion the collection of algebraically independent subsets of S containing A; it is nonempty, containing A. A chain in this collection has its union as an upper bound, and that union is algebraically independent because a dependence relation involves finitely many elements and would therefore already lie in some member of the chain. By Zorn's lemma there is a maximal such subset B.

It remains to see that K is algebraic over $k(B)$. Let $s \in S$. If $s \in B$ this is clear, and otherwise $B \cup \{s\}$ is a subset of S containing A that strictly contains B, so by maximality it is algebraically dependent: some finite list of distinct elements of $B \cup \{s\}$ satisfies a nontrivial polynomial relation over k. That list must involve s, since B is independent, so collecting the relation as a polynomial in s with coefficients in $k[B]$ gives a nonzero polynomial over $k(B)$ killing s; hence s is algebraic over $k(B)$.

So every element of S is algebraic over $k(B)$, whence $k(S)$ is algebraic over $k(B)$, and K is algebraic over $k(S)$ by hypothesis. Algebraicity is transitive in towers, so K is algebraic over $k(B)$ and B is a transcendence basis, as we sought to show.

The substance is that the size of a transcendence basis does not depend on the basis chosen. As in linear algebra, this rests on an exchange argument comparing an independent set against a set over which everything is algebraic.

Lemma 19.2.14: Independent Sets Are No Larger Than Algebraic Generating Sets

Let K/k be a field extension, let $S = \{s_1, \dots, s_n\} \subseteq K$ be finite with K algebraic over $k(S)$, and let $A \subseteq K$ be algebraically independent over k. Then A is finite with $|A| \leq n$.

***Proof*:**
It suffices to show that no $n+1$ distinct elements of A can be algebraically independent. Suppose $a_1, \dots, a_{n+1} \in A$ are distinct. The argument replaces the s_j by the a_i one at a time, and the

claim to be maintained after r steps is that, after renumbering the s_j, the field K is algebraic over $k(a_1, \ldots, a_r, s_{r+1}, \ldots, s_n)$.

Step 1: the base case and the replacement. For $r = 0$ the claim is the hypothesis. Suppose it holds for some $r < n + 1$, so K is algebraic over $F_r = k(a_1, \ldots, a_r, s_{r+1}, \ldots, s_n)$. Then a_{r+1} is algebraic over F_r, so there is a nonzero polynomial relation

$$f\left(a_{r+1};\, a_1, \ldots, a_r,\, s_{r+1}, \ldots, s_n\right) = 0$$

with f a nonzero polynomial over k in which the first slot occurs. The relation must involve at least one s_j with $j > r$: otherwise it exhibits a nontrivial algebraic relation among $a_1, \ldots, a_{r+1}$ over k, contradicting the independence of A. Renumber so that s_{r+1} occurs. Reading the relation as a nonzero polynomial in s_{r+1} over $k(a_1, \ldots, a_{r+1}, s_{r+2}, \ldots, s_n)$ shows s_{r+1} is algebraic over that field, so F_r is algebraic over $F_{r+1} = k(a_1, \ldots, a_{r+1}, s_{r+2}, \ldots, s_n)$, and by transitivity K is too. This advances the claim to $r + 1$.

Step 2: the contradiction. Running the replacement n times gives that K is algebraic over $k(a_1, \ldots, a_n)$. Then a_{n+1} is algebraic over $k(a_1, \ldots, a_n)$, which is a nontrivial algebraic relation among distinct elements of A, contradicting independence. Hence A has at most n elements, completing the proof.

Theorem 19.2.15: Transcendence Degree Is Well-Defined

Let K/k be a field extension admitting a finite transcendence basis. Then every transcendence basis of K/k is finite, and any two have the same number of elements.

***Proof*:**
Let B be a finite transcendence basis, of size n, and let B' be any transcendence basis. Since K is algebraic over $k(B)$ and B' is algebraically independent, lemma 19.2.14 gives that B' is finite with $|B'| \leq n$. Exchanging the roles of B and B', which is legitimate because B' is now known to be finite and K is algebraic over $k(B')$, gives $n \leq |B'|$. Hence $|B'| = n$, as we sought to show.

Definition 19.2.16: Transcendence Degree

Let K/k be a field extension admitting a finite transcendence basis. The *transcendence degree* $\operatorname{trdeg}_k K$ is the common number of elements of its transcendence bases. An extension is algebraic exactly when $\operatorname{trdeg}_k K = 0$, and $\operatorname{trdeg}_k k(x_1, \ldots, x_n) = n$ for indeterminates x_i.

Transcendence degree adds along towers, which is what makes it usable as a notion of dimension. The proof is the statement that a transcendence basis of the top over the middle, together with one of the middle over the bottom, is a transcendence basis of the top over the bottom.

Proposition 19.2.17: Transcendence Degree Is Additive in Towers

Let $k \subseteq K \subseteq L$ be fields with L/k admitting a finite transcendence basis. Then

$$\operatorname{trdeg}_k L = \operatorname{trdeg}_k K + \operatorname{trdeg}_K L$$

***Proof*:**
Let B be a transcendence basis of K/k and C a transcendence basis of L/K; both are finite by lemma 19.2.14, since L is algebraic over $k(B \cup C)$ once the two are combined. The sets B and C are disjoint, an element of C being transcendental over $K \supseteq k(B)$.

Algebraicity. L is algebraic over $K(C)$, and K is algebraic over $k(B)$, so $K(C)$ is algebraic over $k(B)(C) = k(B \cup C)$; by transitivity L is algebraic over $k(B \cup C)$.

Independence. Suppose a nontrivial polynomial relation over k holds among distinct elements of $B \cup C$. Collect it as a polynomial in the elements of C with coefficients in $k[B] \subseteq K$. Those coefficients are not all zero, since the relation is nontrivial and B is algebraically independent over k; so the relation is a nontrivial algebraic relation among elements of C over K, contradicting the independence of C over K.

So $B \cup C$ is a transcendence basis of L/k, and counting gives $\operatorname{trdeg}_k L = |B| + |C| = \operatorname{trdeg}_k K + \operatorname{trdeg}_K L$, completing the proof.

Corollary 19.2.18: Classifying Finitely Generated Transcendental Extensions

Let K/k be a field extensions such that $K = k(\alpha_1, ..., \alpha_n)$ where each element is trancsendental. Then if $k(x_1, ..., x_n)$ is the field of rational functions over the indeterminates x_i[a], then, the set $\alpha_1, ..., \alpha_n$ is algebraically independent, and:

$$k(x_1, ..., x_n) \cong_k k(\alpha_1, ..., \alpha_n)$$

[a] sometimes called *formal variables*

The case for arbitrarily many transcendental elements is the same proof with more book-keeping on notation.

***Proof*:**
This is an immediate generalization of theorem 19.2.8

This shows that in terms of fields, algebraically independent sets don't offer much to our analysis of classification; they act more like a basis with essentially no relation, a resemblance made precise in definition 19.2.12 below. This is not to say that they don't offer any use in Galois theory, where we will correspond group for transcendental extensions, but in terms of field theory their structure is very simple. Thus, we will take more time stuyding algebraically dependent sets, especially since we can use theorem 19.2.2 to study study irreducible polynomials that aquire a root.

Going back to the minimal polynomial that we found in the proof,

Definition 19.2.19: Minimal Polynomial

Let $k(\alpha)$ be a finite simple extension as in theorem 19.2.8, and $m_{\alpha,k}(x)$ be the polynomial given in the proof. Then $m_{\alpha,k}(x)$ is called the *minimal polynomial* of α over k .

it can be shown that $m(x)$ is unique *up to base field* by taking advantage of the euclidean algorithm for $F[x]$ (see exercise ref:HERE), and we have an easy way of linking minimal polynomial between fields:

Proposition 19.2.20

If L/F is an extension of fields and α is algebraic over both F and L, then $m_{\alpha,L}(x)$ divides $m_{\alpha,F}(x)$ in $L[x]$.

***Proof*:**
Since both $m_{\alpha,L}(x)$ and $m_{\alpha,F}(x)$ are polynomials in $L[x]$, we can divide $m_{\alpha,F}(x)$ by $m_{\alpha,L}(x)$ by each other. If we consider:

$$m_{\alpha,L}(x) = m_{\alpha,F}(x)g(x) + r(x)$$

then since α is a root:

$$0 = m_{\alpha,L}(\alpha) = m_{\alpha,F}(\alpha)g(\alpha) + r(\alpha) = 0g(x) + r(\alpha)$$

which implies $r(\alpha) = 0$, and has degree smaller than $m_{\alpha,F}(x)$ – a contradiction to the minimality of the degree of $m_{\alpha,F}(x)$ unless $r(x) = 0$. Thus:

$$m_{\alpha,L}(x) = m_{\alpha,F}(x)g(x) \qquad g(x) \in L[x]$$

However, since $m_{\alpha,L}(x)$ is the minimal polynomial over $L[x]$ that has a root α, it must be that $g(x) = l \in L$. However, if we proceed with the same proof dividing $m_{\alpha,F}(x)$ by $m_{\alpha,L}(x)$:

$$m_{\alpha,F}(x) = m_{\alpha,L}(x)g(x) \qquad g(x) \in L[x]$$

then this does not contradict minimality, showing that $m_{\alpha,L}$ divides $m_{\alpha,F}$ in $L[x]$.

to obtain (using the euclidean algorithm):

$$m_{\alpha,F}(x) = m_{\alpha,L}(x)g(x) + r(x)$$

where $\deg r(x) < \deg m_{\alpha,F}(x)$. Then:

$$0 = m_{\alpha,L}(\alpha) = m_{\alpha,F}(\alpha)g(\alpha) + r(\alpha) = 0g(x) + r(\alpha)$$

which implies $r(\alpha) = 0$, and has degree smalelr than $m_{\alpha,F}(x)$ – a contradiction to the minimality of the degree of $m_{\alpha,F}(x)$ unless $r(x) = 0$. Thus:

$$m_{\alpha,L}(x) = m_{\alpha,F}(x)g(x) \qquad g(x) \in L[x]$$

completing the proof.

The miniaml polynomial will always be irreducible: If it was reducible, then $m_{\alpha,k}(x) = g(x)h(x)$ would produce a polynomial where either $g(\alpha) = 0$ or $h(\alpha) = 0$, contradicting the minimality of $m_{\alpha,k}(x)$. To find the minimal polynomial we can do (ex. 20 in the book), it is surprisngly linked to finding a particular characteritic polynomial!!

Proposition 19.2.21: Finding Minimal Polynomial

Let F/k be a finite field extension. Then there exists a minimal polynomial $m_{\alpha,k}$ such that $m_{\alpha,k}(\alpha) = 0$.

***Proof*:**
Let F/k be a finite field extension.

1. For any $\alpha \in F$, prove that α acting by left multiplication on F is an k-linear transfomration of F, label it L_α.
2. Show α is a root of the characteristic polynomial for L_α. This gives an effective procedure for determining an equation of degree n satisfied by an element α in the extension of k of degree n.

Alternitavely, we can use what find a Gröbner bases(see section 23.3)

Example 19.2.22: Minimal Polynomials

1. The minimal polynomial for $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$ and $\sqrt{2}$ is of degree 2 over $\mathbb{Q}$: $[\mathbb{Q}(\sqrt{2}), \mathbb{Q}] = 2$
2. The minimal polynomial for $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2$ and $\sqrt[3]{2}$ is of degree 3 over $\mathbb{Q}$: $[\mathbb{Q}(\sqrt[3]{2}), \mathbb{Q}] = 3$
3. similarly, for any $n > 1$, the polynomial $x^n - 2$ is irreducible (by Eisenstein's Criterion).
4. The minimal polynomial and the degree of an element α are dependent on the base field! For example, if $F = \mathbb{R}$, then the element $\sqrt[n]{2}$ has degree 1, with minimal polynomial
$$m_{\sqrt[n]{2},\mathbb{R}}(x) = x - \sqrt[n]{2}$$
5. the minimal polynomial for the golden ratio φ is $x^2 - x - 1$.
6. In general, most algebraic elements do not have a special symbol representing them, and so we would have to say that some $\alpha \in K$ has minimal polynomial (say) $x^5 - x - 1$

As a result of theorem 19.2.8, it also makes sense to ask the context of the existence of an irreducible polynomial what is the degree of an α over F: given $p(x)$ is the polynomial for which α is a root, then it's the degree of $\dim_F F[x]/(p(x)) = \dim_F F(\alpha)$. We may write $\deg \alpha$ (or $\deg_F \alpha$) for the degree of α (over F). If $\deg \alpha = 1$, then $\alpha \in F$.

Combining the classification of finite simple extensions with corrollary 19.2.6, we get to easily represent $F(\alpha)$

Corollary 19.2.23: Representing $F(\alpha)$

Suppose in theorem 19.2.8 that $p(x)$ is of degree n. Then

$$F(\alpha) = \left\{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} | a_0, a_1, ..., a_{n-1} \in F\right\} \subseteq K$$

Proposition 19.2.24: Finite Simple Extensions Elements

Let $k(\alpha)/k$ be a finite simple extension. Then:

$$k[\alpha] = k(\alpha)$$

***Proof*:**
exercise

This justifies why we only need to care about polynomials, and not rational polynomials.

Example 19.2.25: Simple Extension

1. In $\mathbb{Q}(\sqrt{2})$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$, and we can write $\mathbb{Q}(\sqrt{2}) = \left\{a + b\sqrt{2} \;\middle|\; a, b \in \mathbb{Q}\right\}$. We could also have chosen the other root of $x^2 - 2$, namely $-\sqrt{2}$. However, these two fields are isomrophic: $\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}(-\sqrt{2})$, showing they are algebraically indistinguishable.
2. In $\mathbb{Q}(\sqrt[3]{2})$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt[3]{2}) = 3$, and we can write $\mathbb{Q}(\sqrt[3]{2}) = \left\{a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2 \;\middle|\; a, b, c \in \mathbb{Q}\right\}$.
3. Over $\mathbb{R}$, $x^3 - 2$ has no more solutions than the last example. However, over $\mathbb{C}$, there are two more solutions:
$$\zeta_2 = \sqrt[3]{2}\left(\frac{-1 + i\sqrt{3}}{2}\right) \qquad \zeta_3 = \sqrt[3]{2}\left(\frac{-1 - i\sqrt{3}}{2}\right)$$
Thus, $\mathbb{Q}(\zeta_2)$ and $\mathbb{Q}(\zeta_3)$ are field extensions.

Thus, we have fully classified all simple extensions of a field and dealt with transcendental extensions. In following sections, we will show how to classify fields generated by finitely many elements (ex. $k(\alpha, \beta)$) and infinitely many elements. Before continuing into that endeavour, isomorphisms between fields should be given some more attention to be more cear on how to classify fields.

The fields in example 19.2.25 need not be isomorphic (as fields) if the roots solve different polynomials, meaning different irreducible polynomials will produce different extensions:

Example 19.2.26: $\mathbb{Q}(\sqrt{2}) \ncong \mathbb{Q}(\sqrt{5})$

Take $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$. Let's say φ is an isomorphism. Then we know that

$$\varphi(1) = 1$$

since 1 is the only element of order 1 in both fields. Then:

$$\varphi(\sqrt{2}) = a + b\sqrt{5}$$

for some non-zero value in the codomain. Then

$$\varphi(2) = a^2 + 2ab\sqrt{5} + 5b^2$$

but we also know that

$$\varphi(1) = 1 \qquad \varphi(2) = \varphi(1+1) = \varphi(1) + \varphi(1) = 1 + 1 = 2$$

so

$$2 = a^2 + 2ab\sqrt{5} + 5b^2 \qquad a, b \in \mathbb{Q}$$

If $a = b = 0$, $2 = 0$, a contradiction. If $a, b \neq 0$, then:

$$\frac{2 - a^2 - 5b^2}{2ab} = \sqrt{5}$$

implying that $\sqrt{5}$ is a rational number, a contradiction. If $a = 0, b \neq 0$, then:

$$\frac{\sqrt{2}}{\sqrt{5}} = b$$

but b is a rational number, a contraidction. If $a \neq 0, b = 0$, then:

$$\sqrt{2} = a$$

but a is a rational number, a contradiction. Thus, these two fields cannot be isomorphic.

However, notice that the extension of to $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(-\sqrt{2})$ were isomorphic, and so was all the extensions of $\mathbb{Q}$ by roots of $x^3 - 2$:

$$\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}\left(\sqrt[3]{2}\left(\frac{-1 + i\sqrt{3}}{2}\right)\right) \cong \mathbb{Q}\left(\sqrt[3]{2}\left(\frac{-1 - i\sqrt{3}}{2}\right)\right)$$

even though those 3 fields are not equal. In particular, that means that non of these fields contain *all* of the roots of $x^3 - 2$. One of the large motifs in the following study of extensions is to find extensions in which for any irreducible (or more generally any polynomial) will contain *all* of the roots.

19.2.1 Isomorphism of Fields Extensions

As we have seen in theorem 19.2.2, If we know that an irreducible polynomial $p(x) \in k[x]$ has a root in a field F. Then there exists a [not necessarily unique] isomorphism φ such that the following diagram commutes:

$$\begin{tikzcd} k[x]/(p(x)) \arrow[r, "\varphi"] & F \\ k \arrow[u, hook, "\iota"] \arrow[ur, hook, "\iota"'] & \end{tikzcd}$$

The fact that φ is not unique gives raise to several interesting results. As we have shown in example 19.2.3, $\mathbb{Q}(\sqrt{2})$ has two homomorphisms into $\mathbb{R}$. In fact, since $-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, we have that:

$$\begin{tikzcd} \mathbb{Q}(\sqrt{2}) \arrow[r, "\varphi"] & \mathbb{Q}(-\sqrt{2}) \\ \mathbb{Q} \arrow[u, hook, "\iota"] \arrow[ur, hook, "\iota"'] & \end{tikzcd}$$

If we extend the bottom to be the identity map form k to k, then we get a square:

$$\begin{tikzcd} \mathbb{Q}(\sqrt{2}) \arrow[r, "\sim"] & \mathbb{Q}(-\sqrt{2}) \\ \mathbb{Q} \arrow[u] \arrow[r, "\mathrm{id}"] & \mathbb{Q} \arrow[u] \end{tikzcd}$$

Showing the fields with the two roots of $x^2 - 2$ are isomorphic. In fact,$-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$, and so this is in fact a field automorphism (though in general it doesn't have to be, like $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt[3]{2}\zeta_3)$). We have claimed earlier than two fields that have two different roots of the same irreducible polynomial adjoined to them are algebraically indistinguishable. This shows it in the case for $x^2 - 2$ over $\mathbb{Q}$, and the following theorem shows the generlizatoin of this result. Note how by keeping track of the root we have in fact recovered a sense of uniqueness:

Theorem 19.2.27: Extending Isomorphisms to Adjoint Fields

Let $k_1 \subseteq F_1 = k_1(\alpha_1)$, $k_2 \subseteq F_2 = k_2(\alpha_2)$ be simple extensions and let $\varphi : k_1 \to k_2$ be an isomorphism of fields. Let $p(x) \in k_1[x]$ be an irreducible polynomial and let $p'(x) \in k_2[x]$ be the irredcibble polynomial obtained by applying the map φ to the coefficients of $p(x)$. Let α_1 be a root of $p(x)$ (in some extension of k_1) and let α_2 be a root of $p'(x)$ (in some extension k_2). Then there is an isomorphism:

$$\sigma : k_1(\alpha_1) \to k_2(\alpha_2) \qquad \alpha_1 \mapsto \alpha_2$$

mapping α_1 to α_2 and extending φ, .i.e., such that σ restricted to F is the isomorphism φ.

In other words, if we have

$$\begin{tikzcd} k_1(\alpha) & k_2(\beta) \\ k_1 \arrow[u, hook] \arrow[r, "\sim"] & k_2 \arrow[u, hook] \end{tikzcd}$$

and $\iota(m_{\alpha,k_1})(x)) = m_{\beta,k_2}(x)$ (i.e. it maps the coefficients of the first minimal polynomial to the second one), then we can *uniquely* extend ι to an isomorphism j between the two adjoint fields:

$$\begin{tikzcd} k_1(\alpha) \arrow[r, dashed, "\sim"] & k_2(\beta) \\ k_1 \arrow[u, hook] \arrow[r, "\sim"] & k_2 \arrow[u, hook] \end{tikzcd}$$

such that $j(\alpha) = \beta$. Note that j being an extension of ι means the elements of k_1 are fixed in j.

***Proof*:**
Let $\varphi : k_1 \to k_2$ be an isomorphism of fields. The map φ induces a ring isomorphism:

$$f^{\varphi} : k_1[x] \to k_2[x]$$

defined by applying φ to the coefficeints of a polynomial in $k_1[x]$. Let $p(x) \in k_1[x]$ be an irreducible polynomial, and let $p'(x) \in k_2[x]$ be the polynomial obtained by applying the map φ to the coefficients of $p(x)$, i.e., the image of $p(x)$ under f^{φ}. The isomorphism f^{φ} maps the maximal ideal $(p(x))$ to the ideal $(p'(x))$, so the ideal is also maximal, which shows that $p'(x)$ is also irreducible in $k_2[x]$.

Taking the quotinet by these ideals of the map, we get the following isomorphism of fields

$$k_1[x]/(p(x)) \to k_2[x]/(p'(x))$$

By theorem 19.2.8, the left side is isomorphic to $k_1(\alpha_1)$, and the right side to $k_2(\alpha_2)$, so composing these maps will produce a map for σ! It should also be clear now that the restriction of this isomorphism to k_1 is φ.

This isomorphism will come back big time when we address galois theory. It is sometimes represented in as in the following diagram:

$$\begin{array}{lccc} \sigma : & k_1(\alpha_1) & \xrightarrow{\cong} & k_2(\alpha_2) \\ & | & & | \\ \varphi : & k_1 & \xrightarrow{\cong} & k_2 \end{array}$$

If we let $\mathrm{id} : k \to k$ be our k-isomorphism, and $m_{\alpha,k}(x)$ be our minimal polynomial with roots α, β, then $\varphi : k(\alpha) \xrightarrow{\sim} k(\beta)$ is also an isomorphism, showing that the roots of a minimal polynomials are algebraically indistinguishable. Furthermore, we can upgrade Kronecker's Theorem so that it gives us a universaly property: instead of L being a field that has a root of $p(x)$, upgrade that to $\alpha \in L$ being a root of $p(x)$. Then there exists a *unique* homomorphism mapping $\overline{x}$ to α (with uniqueness established by the above theorem).

Through this theorem, we get another important insight. If we have a non-trivial finite field extension $k(\alpha)/k$ which has roots of a minimal polynomial $m_{k,\alpha}(x)$, then if any other roots of $m_{k,\alpha}(x)$ are in $k(\alpha)$, there is a k-automorphisms of $k(\alpha)$ (can you see why a k-automorphism is determined by where it sends the roots?). This means we can find elements of the group $\mathrm{Aut}_k(k(\alpha))$ (where the subscript means we fix k), and hence the structure of $k(\alpha)$ [9]. More conretely, $k(\alpha)/k$ represents adding a new element α to the field k which satisfies an algebraic relation $m_{\alpha,k}(\alpha)$ (a polynomial). Finding information about the new field $k(\alpha)$ by looking at the k-automorphisms is essentially studying the properties of the polynomial $m_{\alpha,k}(x)$.

Even better, we will show that this characterizes *all* k-automorphisms, giving us a complete description of these symmetries. These symmetries are actually at the heart of studying fields in galois theory. When Galois originally formulated his theory, he looked different ways to permute the roots and noticed that there are often limitation to the permutations of the roots since the roots may be related

[9] Recall that we already used the automorphism group to deduce information about groups, for example $\mathrm{Out}(G) \cong_{\mathbf{Grp}} \mathrm{Aut}(G)$ if and only if G is abelian

to each other[10]. For example if α is a root of $x^3 - x - 1$, then the other two roots are:

$$\frac{1}{1-\alpha} \qquad 1 + \frac{1}{\alpha}$$

Similarly, if ζ is a root of $x^n - 1$, then the other roots of $x^n - 1$ are $\zeta^2, \zeta^3, ..., \zeta^{n-1}, 1$. Thus, we take some time to define our terms more precisely, and elaborate further what the automorphisms look like:

Definition 19.2.28: Automorphism and fixed Element

Let K be a field and $\text{Aut}(K)$ be the collection of automorphisms (isomorphisms from k to itself). Then:

1. An auotmorphism $\sigma \in \text{Aut}(K)$ is said to *fix* an element $\alpha \in K$ if $\sigma\alpha = \alpha$. If F is a subset of K, then an automorphism σ is said to *fix* F if it fixes all the elements of F

$$\sigma a = a \qquad \forall a \in F$$

2. If F/k be an extension of fields, then $\text{Aut}(F/k)$ or $\text{Aut}_k(F)$ is the collection of all automorphsism of F which fix k.

the previous theorem gives us the following link between $\text{Aut}_k(F)$ and F given that F is finite and simple:

Corollary 19.2.29: Order of Fixed Automorphism Group

Let F/k be a simple finite extension (so $F = k(\alpha)$ for some $\alpha \in F$), and let $m_{k,\alpha}(x) \in k[x]$ be the minimal polynomial of α. Then $|\text{Aut}_k(F)|$ is equal to the number of *distinct* roots of $m_{k,\alpha}(x)$ in F. In particular,

$$|\text{Aut}_k(F)| \leq [k(\alpha) : k] \qquad |\text{Aut}_k(F)| = [k(\alpha) : k]_d$$

where $[k(\alpha) : k]_d$ represents the number of *distinct* roots $k(\alpha)$ contains of the minimal polynomial $m_{k,\alpha})(x)$. Equality holds if and only if $m_{k,\alpha}(x)$ factors over F as a product of *distinct linear* polynomials

This result will be called the *fixed root criterion* when we cover Galois theory.

***Proof*:**
Let $\sigma \in \text{Aut}_k(F)$ be an automorphism fixing k (so for any $t \in k$, $\sigma(t) = t$). Then since every element $t \in F = k(\alpha)$ can be written as a polynomial $t = f(\alpha) \in k[\alpha]$ (recall that $k(\alpha)$ is finite, and so has a basis of the form $\alpha, \alpha^2, ..., \alpha^n$ for some appropriat n), and σ fixes k, all the elements are determined by where α maps to, that is $\sigma(\alpha)$. Then since σ is an automorphism (hence ring homomorphism), given

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

[10]Later we shall see that the roots are almost never related to each other, that is most Galois groups are S_n, but it is a good insight to notice that the roots of small polynomials are often related

is the minimal polynomial with coefficients from k so that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then, applying any automorphism σ to this polynomial will get us

$$\begin{aligned} \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) &= \sigma(0) \\ \sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + \sigma(a_0) &= 0 && \sigma \text{ fixes } a_i \text{ for each } i \end{aligned}$$

since $a_i \in k$ for each i, showing that $\sigma(\alpha)$ is also a root of the minimal polynomial, i.e. σ must map α to some other root of $m_{k,\alpha}$.

As α is the generator, all roots are determined by where ' maps to, hence there can be at most $\deg(m_{k,\alpha}(x))$ automorphisms.

Conversely, for any root β of $p(x)$ in $k(\alpha)$, $\beta \in k(\alpha)$, we can take the isomorphism $\mathrm{id} : k \to k$, which by theorem 19.2.27, there exists a unique lift to a root $k(\alpha) \to k(\beta)$ that fixes k. Since $k(\alpha) \cong k(\beta)$, then we can use this isomorphism to define an automorphism from $k(\alpha)$ to $k(\alpha)$, showing us that $\mathrm{Aut}_k(F)$ must have at least as many elements as there are roots of the minimal polynomial in F. If F contains all the roots of $p(x)$ and each root of $p(x)$ is distinct, then $|\mathrm{Aut}_k(F)| = [F : k]$, completing the proof.

Interpreting the above example as a group actions on the set of roots R_F of $m_{\alpha,k}(x)$ in F, $\mathrm{Aut}_k(F) \curvearrowright R_F$, we have that $\mathrm{Aut}_k(F)$ acts *faithfully and transitively* on the set of roots of $p(x)$ in F (the group $\mathrm{Aut}_k(F)$ can be identified with some transitive subgroup of the symmetric group acting on the roots of $p(x)$ in F). This correlation hints at the connection beween fields and groups at the heart of the galois theory, since any collection of automorphisms will always form a group[11]. The main impediment to directly translating this result of more general [finite] field extensions[12] is that the minimal polynomial might not factor into *linear* and *distinct* terms over the simple extension $k(\alpha)/k$[13]. For example, in $\mathbb{Q}(\sqrt[3]{2})$, the polynomial $x^3 - 1$ doesn't fully factor. The next two sections will encompass finding field extensions in which these conditions are met. In particular, we shall take some time to find fields in which:

1. All polynomials factor (splitting fields)
2. If $f(x) \in k[x]$ is irreducible, then in F/k, either $f(x)$ factors completely or $f(x)$ remains irreducible (normal extensions)
3. Given a polynomial $f(x)$, F/k factors completely (Splitting field)
4. If $f(x)$ is irreducible in k, then in F/k, $f(x)$ factors into linear terms of degree 1 (Separable extensions)

As a final word on relating isomorphisms and extensions, note that if $\dim_k A = \dim_k B$, this does not imply $[B : A] = 1$ or that $A \cong_k B$:

[11] If you are not convinced, now is good time to check this
[12] infinite extension are more finicky to deal with, and are relegated to chapter ref:HERE
[13] It is possible that a irreducible polynomial factors linearly, but the factors are not distinct, see section 19.6

Example 19.2.30: Nuance On Dimension

Let $A = \mathbb{Q}(\sqrt{3})$ and $B = \mathbb{Q}(\zeta_3\sqrt{3})$. Then the two are of the same dimension, both extend $\mathbb{Q}$ and $A \cong_k B$, but $[B : A]$ is not even well defined. To visualize:

$$\begin{tikzcd}
\mathbb{Q}(\sqrt{3}) \arrow[rr, phantom, "\cong_k"] & & \mathbb{Q}(\zeta_3\sqrt{3}) \\
& \mathbb{Q} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

Exercise 19.2.1

1. Let $\text{ch}(k) = n$, $n \geq 0$, and $\varphi : k \to K$ be a homomorphism. Then $\text{ch}(K) = n$.
2. Show that the intersection of any collection of subfields of a field is a field. Is the same true for the union? What if the fields where given a partial order through inclusion?
3. (personal idea, not sure if valid yet) Want to give another intuition using the fact that $R[x]$ is a free monoid, so that in a sense, we can get a "presenation" when we do $R[x]/p(x)$. If $p(x)$ is also irreducible then we get a field as a result.
4. Let $N \subseteq M_{2\times 2}(\mathbb{R})$ be a subset where
$$N = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \;\middle|\; a, b \in \mathbb{R} \right\}$$
Then $N \cong \mathbb{C}$.
5. Show that If Let E/k and F/k be field extensions. Show that if E/k is finite, then so is EF/F. Show that if E/k and F/k are finite, so is EF/F.
6. Suppose that $f(x) \in k[x]$ where $f(x)$ factors into $f(x) = p_1(x)p_2(x)\cdots p_n(x)$ where $p_i(x) \neq p_j(x)$. Show that $k[x]/(f(x))$ is a product of fields (Hint: Chinese Remainder Theroem). If one of the terms had multiplicity greater than 1, would the result still be a product of fields?
7. (If you know Calculus) The $\mathbb{R}$-algebra $\mathbb{R}[x]/(x^2)$ is sometimes the *dual numbers*. Let any element of this degree 2 extension of $\mathbb{R}$ be written in the form $a + b\epsilon$. If $p'(x)$ is the derivtive of $p(x)$, show that $p(a + b\epsilon) = P(a) + bP'(a)\epsilon$, that is, the derivitve "automatically" shows up.
8. Let m, n be square free and $n \neq m$. Show that $\mathbb{Q}(\sqrt{n}) \not\cong_{\mathbb{Q}} (\sqrt{m})$.
9. Let ζ_n be a root of unity where n is odd. Show that $\mathbb{Q}(\zeta_n) \cong_{\mathbb{Q}} \mathbb{Q}(\zeta_{2n})$

19.3 Algebraic Extensions

We'll now focus some attention on the algebraic elements of an extension. We first give name to elements for which all elements are algebraic or non-algebraic:

Definition 19.3.1: Algebraic and Transcendental Extensions

Let F be a field and let K be an extension of F. Then:

1. The element $\alpha \in K$ is said to be *algebraic* over F if α is a root of some nonzero polynomial $f(x) \in F[x]$.
2. If α is not algebraic over F (i.e., is not the root of any nonzero polynomial with coefficeints in F) then α is said to be *transcendental over* F.
3. The extension K/F is said to be *algebraic* or an *algebraic extension* if every element of K is algebraic over F.
4. The extension K/F is said to be *transcendental* or a *transcendental extension* if every element of K not in F is transcendental over F.

Note that if an element α is algebraic over a field F, then it is algebraic over *any* extension field L of F (if $f(x)$ having α as a root has coefficients in F, then it also has coefficients in L). Note too that the polynomial for which α must be a root need not be irreducible, however it is an immediate consequence that if α is the root of a polynomial $p(x) = (a_1(x))^{n_1} \cdots (a_k(x))^{n_k}$, it is a root for some irreducible polynomial $a_i(x)$.

Example 19.3.2: Algebraic extensions

1. Take $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$. Then every element in $\mathbb{Q}(\sqrt{2})$ is a solution to a polynomial over $\mathbb{Q}$, thus $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is algebraic: any element $a + b\sqrt{2}$ is the root of $\left(\frac{x-a}{b}\right)^2 - 2$. If you take $\mathbb{Q}(\sqrt{2})$ as a vector-space over $\mathbb{Q}$, then we see that 1 and $\sqrt{2}$ generate $\mathbb{Q}(\sqrt{2})$. Note that $\sqrt{2}$ is not in $\mathbb{Q}$, but but $\sqrt{2}$ is algebraic over $\mathbb{Q}$).

   More generally, is it true that any finite simple extension $F(\alpha)$ of F is an algebraic extension? Is every element of $F(\alpha)$ the root of a polynomial with coefficients in F?

2. $\mathbb{C}$ as an $\mathbb{R}$-vector space is generated by 1 and i. Both of these are root to polynomials over $\mathbb{R}$, in particular $x - 1$ and $x^2 + 1$. It is a famous result that any $c \in \mathbb{C}$ is a solution of a polynomial over $\mathbb{R}$ (i.e., with coefficients from $\mathbb{R}$). Thus, $\mathbb{C}/\mathbb{R}$ is algebraic.

   The degree of $[\mathbb{C} : \mathbb{R}] = 2$

3. Consider $\mathbb{C}$ again. For any $\alpha \in \mathbb{C}$, is there a polynomial $p(x) \in \mathbb{C}[x]$ for which $p(\alpha) = 0$? In a sense, we're asking if $\mathbb{C}$ is algebraic over itself. [a]

4. For a moment, take for granted that there is no polynomial in $\mathbb{Q}$ such that π is a root (TBD? appendix?). Then π is *transcendental* over $\mathbb{Q}$, and so $\mathbb{R}/\mathbb{Q}$ is *not* an algebraic extension.

   Here is an idea to ponder: What is $[\mathbb{R} : \mathbb{Q}]$? The answer will be given in example 19.3.15

[a] The answer is yes! We call a field such that every element of it has a root over itself *algebraically closed*

In these examples, the algebraic extensions all had a finite degree. Is it possible to have an algebraic extension with infinite degree? If K is algebaric over F, then every element of K is a roof a polynomial over F? What about the converse: If K/F is finite, is K algebraic over F? (take a moment to ponder here)

Proposition 19.3.3: degree of Algebraic Extension

The element α is algebraic over F if and only if the simple extension $F(\alpha)/F$ is finite. More precisely, if α is an element of an extension of degree n over F, then α satisfies a polynomial of degree at most n over F and if α satisfies a polynomial of degree n over F, then the degree of $F(\alpha)$ over F is at most n.

***Proof*:**
If α is algebraic over F (so for some $p(x) \in F[x]$, $p(\alpha) = 0$), then the degree of the extension $F(\alpha)/F$ is the degree of the minimal polynomial, making it an extension of finite degree at most n.

Conversly, let α be an element of degee n over F (ex. $[F(\alpha) : F] = n$). Then by 19.2.5, we know that $n+1$ elements will be linearly dependent:

$$c_0 + c_1\alpha + c_2\alpha^2 + \cdots + c_n\alpha^n = 0$$

with $c_0, c_2, ..., c_n \in F$ which are not all 0. Thus, α must be the root of some non-zero polynomial with coefficients from F (of degree $\leq n$), showing α is algebraic over F.

Corollary 19.3.4: Finite Extension Then Algebraic Extension

If the extension K/F is finite, then it is algebaric

***Proof*:**
let K/F be finite and take any $\alpha \in K$. Consider the simple extension $F(\alpha)$. Consider K as an F-vector space, then since the extension is finite, $\dim_F K = n$. Now, $F(\alpha) \subseteq K$ would be a subspace of K ($F(\alpha)$ also being an F-vector sapce), so

$$\dim_F F(\alpha) \leq \dim_F K \qquad \text{or} \qquad [F(\alpha) : F] \leq [K : F]$$

showing the extension is finite, and so by proposition 19.3.3, α is algebraic over F. since α was an artibrary element of K, this means that K is algebaic over F, as we sought to show.

Though not important to us in this chapter, we also have the following characterisation of an algegraic extension more useful in algebraic number theory:

Corollary 19.3.5: Algebraic Extension and Algebras

K/F is algebraic if and only if every sub F-algebra is a field

***Proof*:**
exercise

Note that the converse of corrollary 19.3.4 is not true: If K is algebraic over F, this does not mean that K/F has finite degree

Example 19.3.6: Algebraic and infinite degree

We require corollary 19.3.14 which is introduced in a couple of pages, but we can still state the case with the added assumption here to get the idea:

Take $\mathbb{C}/\mathbb{Q}$, and take the set $\overline{\mathbb{Q}}$ to be the set of all algebraic elements over $\mathbb{Q}$. Then by the aforementioned corollary, this is a subfield of $\mathbb{C}$. Since $\sqrt[n]{2}$ is algebraic over $\mathbb{Q}$ (root of the polynomial $x^n - 2$), we have

$$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = n$$

for every n. Since $\mathbb{Q}(\sqrt[n]{2}) \subseteq \overline{\mathbb{Q}}$, it follows that $\overline{\mathbb{Q}}$ is an algebraic extension of infinite degree.

However, a weaker version of this direction does have a converse, and we will build up to it now. Let's first show that the finite extension of a finite extension remains a finite extension (if you like, the composition of finite extensions is a finite extension)

Theorem 19.3.7: Multiplicity of Finite Extensions (Tower Law)

Let $k \subseteq E \subseteq F$ be fields. Then

$$[F : k] = [F : E][E : k]$$

that is, extension degree multiply each other, where if one side of the equation is infinite, so is the other.

***Proof*:**

The infinite case is clear, so we will deal with the finite case. First, if F is finite dimensional over k, then the subspace E is also finite dimensional. Furthermore, any linear combination of elements of F over the field k naturally is a linear combination over the larger field E. Thus, if $[F : k]$ is finite, so is $[F : E]$ and $[E : k]$.

Conversely, suppose $[F : E]$ and $[E : k]$ are both finite. Let $(e_1, e_2, ..., e_m)$ be a basis for E over k and $(f_1, f_2, ..., f_n)$ be a basis for F over E. I claim that

$$(e_1f_1, e_1f_2, ..., e_1f_n, e_2f_1, ..., e_mf_n)$$

is a basis for F over k.

Let $g \in F$. Then for appropriate $d_i \in E$,

$$g = \sum_j d_j f_j$$

Next, for each d_j, for appropriate $c_{ij} \in k$, we have

$$d_j = \sum_i c_{ij} e_i$$

Thus, we have:

$$g = \sum_i \sum_j c_{ij} e_i f_j$$

Therefore, the products $e_i f_j$ span F, showing that F must at most be finite dimensional.

To show they form a basis, assume

$$\sum_{i,j} \lambda_{ij} e_i f_j = 0$$

Then:

$$\sum_j \left(\sum_i \lambda_{ij} e_i \right) f_j = 0$$

which, since the the elemnts f_j are linearly independent over E, we have that $\sum_i \lambda_{ij} e_i = 0$. But then, the elements e_i are linearly indepenent over e_i,and so each $\lambda_{ij} = 0$, showing that the original linear combination is also linearly independent, as we sought to show.

Notice how this corollary mirrors Lagranges theorem with respect to the order of groups. This parralel is actually quite a deep one which will capture our attention in the next chapter. The following corollary deepends this connection:

Corollary 19.3.8: Degree Extension Divisibility

Suppose L/F is a finite extension and let K be any subfield of L contianing F, $F \subseteq K \subseteq L$. Then $[K : F]$ divides $[L : F]$

***Proof*:**
Since

$$[L : F] = [L : K][K : F]$$

this result holds.

This result gives us a similar power that Lagrange's theorem gives us

Example 19.3.9: Use Of Tower Law

1. We can quickly rule out that $\sqrt{2} \notin \mathbb{Q}(\sqrt[n]{2})$ if n is odd. To see this, if $\sqrt{2} \in \mathbb{Q}(\sqrt[n]{2})$, then $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[n]{2})$. Then by what we've just shown:

$$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$$

   However, that would imply that an even number divides an odd numbers – a contradiction.

2. More generally, we can conclude the following: If K/k is an extension of odd degree and $\alpha \in K$, then α can be written as a polynomial in α^2, that is $\alpha = p(\alpha^2) = \sum_i k_i(\alpha^2)^i$ for $k_i \in k$.

   To see this, we first notice that through simple set theory we must have

$$k \subseteq k(\alpha^2) \subseteq k(\alpha)$$

   and that $k(\alpha)/k(\alpha^2)$. To find the degree d of this extension, notice that over $k(\alpha^2)$, we get that $x^2 - \alpha^2 \in k(\alpha^2)[x]$, so $d \leq 2$. On the other hand, d must divide $[k(\alpha) : k]$ which is odd, forcing us to have $d = 1$. Thus $k(\alpha^2) = k(\alpha)$, and so $\alpha \in k(\alpha^2)$.

3. Let's say K/k and $[K : k]$ is prime. Then K contains *no* fields that contain k. This is similar to how all finite groups of prime order are simple.

4. Take $\mathbb{Q} \subseteq \mathbb{R}$ and consider $\mathbb{Q}/(x^3 - 3x - 1) \cong \mathbb{Q}(\alpha)$ as in example 19.2.4. Since $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, and $2 \not| 3$, then $\mathbb{Q}(\sqrt{2}) \not\subseteq \mathbb{Q}(\alpha)$, so $\sqrt{2} \notin \mathbb{Q}(\alpha)$! It is much more difficult to directly show $\sqrt{2}$ cannot be written as a $\mathbb{Q}$-linear combination of $1, \alpha, \alpha^2$.

5. More in Dummit and Foote?

Continuing to strive towards a partial converse, we define the following generlization of a simple extension. The key feature about finitely generated fields is that they can be decomposed into simple extensions:

Lemma 19.3.10: Recursive Definition Of Extension Field

$F(\alpha, \beta) = (F(\alpha))(\beta)$, i.e., the field generated over F by α and β is the field generated by β over the field $F(\alpha)$ generated by α.

***Proof*:**
This proof takes complete advantage that by definition, field extensions are the *smallest* field containing F and α (or α and β)

The field $F(\alpha, \beta)$ is the smallest field to contain F, α, β, so $F(\alpha, \beta) \subseteq (F(\alpha))(\beta)$. Conversely, $F((\alpha))(\beta)$ is the smallest field to contain $F(\alpha)$ and β. Since $F(\alpha) \subseteq F(\alpha, \beta)$ and $\beta \in F(\alpha, \beta)$, $(F(\alpha))(\beta) \subseteq F(\alpha, \beta)$, implying

$$F(\alpha, \beta) = (F(\alpha))(\beta)$$

With this, we can break down any finitely generated:

$$K = F(\alpha_1, \alpha_2, ..., \alpha_k) = (F(\alpha_1, \alpha_2, ..., \alpha_{k-1}))(\alpha_k) = ((\cdots F(\alpha_1))(\alpha_2)) \cdots (\alpha_n)$$

Thus, we get the following chain of extensions:

$$F = F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq F_k = K$$

where

$$F_{i+1} = F_i(\alpha_{i+1}) \qquad i = 0, 1, ..., k-1$$

Importantly, this reduces everything down to lots of simple extensions! Now, suppose that the elements $\alpha_1, \alpha_2, ..., \alpha_k$ are algebraic over F of degrees $n_1, n_2, ..., n_k$

$$[F(\alpha_1) : F] = n_1, \; [F(\alpha_2) : F] = n_2 \cdots [F(\alpha_k) : F] = n_k$$

By proposition 19.2.20, we can replace F with F_i. Then these can all be characterized in terms of simple extensions, meaning we can use proposition 19.3.3. The relative extension degree $[F_{i+1} : F_i]$ is equal to the degree of the minimal polynomoial of α_{i+1} over F_i, which is at most n_{i+1} (and is equal to n_{i+1} if and only if the minimal polynomial of α_{i+1} over F remains irreducible over F_i). So, by the Tower Law:

$$[K : F] = [F_k : F_{k-1}][F_{k-1} : F_{k-2}] \cdots [F_1 : F_0] \leq [F(\alpha_1) : F][F(\alpha_2) : F] \cdots [F(\alpha_k) : F] \tag{19.1}$$

And so $[K : F]$ must be a *finite* extension of degree less than or equal to $n_1 n_2 \cdots n_k$! We sum up this entire discussion of finite degrees in the following theorem:

Theorem 19.3.11: Degree of Algebraic Extension

The extension K/F is finite if and only if K is generaed by a finite number of algebraic elements over F. More precisely, a field generated over F by a finite number of algebraic elements of degrees $n_1, n_2, ..., n_k$ is algegraic of degree $\leq n_1, n_2, ..., n_k$

Before, we hade

$$\alpha \text{ algebraic over } F \Leftrightarrow F(\alpha)/F \text{ finite}$$

and when we try to generlize, we loose one side of the implication

$$K \text{ algebraic over } F \Leftarrow K/F \text{ finite}$$

with the counter example 19.3.6 for $\Rightarrow$.

We now restricted the hypothesis to give us the if and only if again:

$$K \text{ generated by finite number of algebraic elements over } F \Leftrightarrow K/F \text{ finite}$$

***Proof*:**
Let the extension K/F be finite, so $a_1, a_2, ..., a_n$ forms a basis for K over F. Then since $F(a_i) \subseteq K$ as a F-vector-space, $[F(a_i) : F] \leq [K : F]$, meaning it has finite degree, and so by 19.3.3, a_i is algebraic. Since a_i was arbitrary, this means that every a_i is algebraic. Thus K is generated by a finite number of algebraic elements over F.

For the converse, let's say K is generated by a finite number of algebraic elements over F. Let

$$\alpha_1, \alpha_2, ..., \alpha_n$$

all be algebraic over F and generate K. Since they generated K, $K = F(\alpha_1, \alpha_2, ..., \alpha_n)$. Then by what we've seen in equation (19.1), we have that K/F is finite.

Using what we've found, it is easier to compute the degree of non-simple extensions:

Example 19.3.12: Computing Degree of non-simple Extension

1. The extension $\mathbb{Q}(\sqrt[6]{2}, \sqrt{2})$ is $\mathbb{Q}(\sqrt[6]{2})$ since $\sqrt{2} \in \mathbb{Q}(\sqrt[6]{2})$. Another way of showing this is that the degree d of $\sqrt{2}$ over $\mathbb{Q}(\sqrt[6]{2})$ is 1. Thus, even though the degree of $\sqrt{2}$ is 2 and the degree of $\sqrt[6]{2}$ is 6, the degre of $\mathbb{Q}(\sqrt[6]{2}, \sqrt{2})$ is 6, not 12.

2. Consider $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. We'll proceed to find the degree of this extensions over $\mathbb{Q}$. Since $\sqrt{3}$ is of degree 2 over $\mathbb{Q}$, then the degree of the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}(\sqrt{2})$ is at most 2 ($\sqrt{3}$ can at most add another dimension), and *is* 2 if and only if $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$. To see if it is, let's suppose it is reducible. Since $x^2 - 3$ is of degree 2, by proposition 11.3.5, it is irreducible if and only if it has a root, that is, if $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$.

   If $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$, $\sqrt{3} = a + b\sqrt{2}$. From here, it's a lot of manipulation (squaring both sides and checking cases). It will turn out that every possibility wil make either $\sqrt{2}$, $\sqrt{3}$ or $\sqrt{6}$ rational – a contradiction.

Thus, $\deg_{\mathbb{Q}(\sqrt{2})} \mathbb{Q}(\sqrt{2}, \sqrt{3}) = 2$, so

$$\deg_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}, \sqrt{3}) = 4$$

and we can find the basis for it by "closing" the set $1, \sqrt{2}, \sqrt{3}$ to produce $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$, giving us

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \left\{a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} | a, b, c, d \in \mathbb{Q}\right\}$$

Corollary 19.3.13: Algebraic Elements are Closed

Suppose α and β are algebraic over F. Then $\alpha \pm \beta$, $\alpha\beta$, α/β (for $\beta \neq 0$, are all algebraic (since α/β is algebraic $1/\alpha$ is also algebraic)

***Proof*:**
All these elements are part of $F(\alpha, \beta)$, which is finite over F by theorem 19.3.11, and so are algebraic by corollary 19.3.4 (in particular, $F(\alpha + \beta)$ is a simple extension, and since $\alpha + \beta \in F(\alpha, \beta)$(and $F \subseteq F(\alpha, \beta)$), we have that $F(\alpha + \beta) \subseteq F(\alpha, \beta)$ by definition, so it is a finite extension, so by corollary 19.3.4 every element will be a root of a polynomial over F).

Corollary 19.3.14: Algebraic Elements Form a Field

Let L/k be an arbitrary extension. Then the collection of elements of k that are algebraic over F form a subfield K of F.

***Proof*:**
By the previous corollary (19.3.13), we get that the collection is closed under $+$ and $\times$. From here, it is easy to verify any extra axiom we want.

Example 19.3.15: Algebraic Closure

1. Consider the extension $\mathbb{C}/\mathbb{Q}$, and let $\overline{\mathbb{Q}}$ represent the subfield of $\mathbb{C}$ containing all elements that are algebraic over $\mathbb{Q}$. Then $\sqrt[n]{2} \in \overline{\mathbb{Q}}$ for all $n \in \mathbb{N}$ (taking the positive roots), so $[\overline{\mathbb{Q}} : \mathbb{Q}] \geq n$ for every $n > 1$. Thus, $\overline{\mathbb{Q}}$ is an *infinite* algebraic extension of $\mathbb{Q}$, called the field of *algebraic numbers*. We have now proved what we have assumed in example 19.3.6.

2. Take $\mathbb{R}$ and the subfield $\overline{\mathbb{Q}}$ consisting of all algebraic elements over $\mathbb{Q}$, and consider $\mathbb{R} \cap \overline{\mathbb{Q}}$. This set is countable: $\mathbb{Q}$ is countable, and $\mathbb{Q}[x]$ consists of the countable union of polynomials of degree n ($S_0 \cup S_1 \cup \cdots$ where S_n contains all polynomials of degree n), so $\mathbb{Q}[x]$ is countable. $\overline{\mathbb{Q}}$ must have a cardinality at least less than $\mathbb{Q}[x]$, and so is also countable.

   Since $\mathbb{R}$ is uncountable, we see that there must be elements in $\mathbb{R}$ which are *not* algebraic (i.e. transcendental) over $\mathbb{Q}$. In fact, the vast majority of elements are transcendental. The subfield $\overline{\mathbb{Q}} \cap \mathbb{R}$ is a *proper* subfield of $\mathbb{R}$.

   Though the majority of elements[a] are transcendental, it is extremely difficult to show that an element *is* transcendental. For example, it was proved that $\pi = 3.1415926...$ was transcendental in 1880, and $e = 2.171828...$ is transcendental in 1870. Proving these are transcendental is very involved. Even combining these elements is tricky to prove: e^π was proven to be transcendental in 1934, however as of writing this book, it is unknown whether π^e is transcendental. In fact, it is also not known whether $\pi + e$ or πe are transcendental.

We actually have a small list of numbers which we *know* to be transcendental, even though the vast majority of them are transcendental [b]. As of writting this, there are approximately 24 classes of known transcendental numbers, including numbers like e^a for an algebraic number a, a^b where a is albebraic $\neq 0, 1$ and b is irrational (ex. $2^{\sqrt{2}}$), $\sin(a)$ for an algebraic number a, and so on (commented here link to wikipedia, since it was not liking some characters in it)

[a] In terms of probability, 100% of the elements

[b] If you've taken some statistics/measure theory, then if $\mathbb{A}$ are the algebraic numbers over $\mathbb{Q}$, $\int \chi_{\mathbb{A}} = 0$, i.e., the chances of picking an algebraic number from the set of real number is 0!

We can extend the results of theorem 19.3.11 a bit further to algebraic extensions (finite or infinite)

Theorem 19.3.16: Composing Algebraic extensions

Let $k \subseteq E \subseteq F$ be field extensions. Then F/k is algebraic if and only if F/E and E/k are algebraic.

***Proof*:**
If F/k is algebraic, then *a fortiori* F/E is algebraic (If $\alpha \in F$ are all roots of some polynomial $k[x]$ and $k \subseteq E$, then they are certainly roots in $E[x]$). Since $E \subseteq F$, *a priori* E/k.

For the other way, assume E/k and F/E are algebraic, and consider $\alpha \in F$. Since F/E is algebraic, there exists a $p(x) \in E[x]$ such that

$$p(\alpha) = \alpha^n + e_{n-1}\alpha^{n-1} + \cdots + e_1\alpha + e_0 = 0$$

We thus have that α is algebraic over the subfield $k(e_0, ..., e_{n-1}) \subseteq E$. Hence, by theorem 19.3.11

$$k(e_1, ..., e_{n-1}) \subseteq k(e_1, ..., e_{n-1}, \alpha)$$

is a finite extension. Furthermore, by theorem 19.3.11 again, we have that

$$k \subseteq k(e_1, ..., e_{n-1})$$

is also a finite extension, since each e_i is in E, so is algebraic over k by assumption that E/k is algebaric. Hence, by the Tower Law,

$$k \subseteq k(e_1, ..., e_n, \alpha)$$

is a finite extension as well, and hence α is algebraic over k. Since α was arbitrary, F is algebraic over k, as we sought to show.

To close off this section, we show how our previous discussion applies to composite fields:

Proposition 19.3.17: Degree Of Composite Field

Let K_1 and K_2 be two finite extensions of a field F contained in K. Then

$$[K_1K_2 : F] \leq [K_1 : F][K_2 : F]$$

with equality if and only if an F-basis for one of the fields remains linearly independend over the over field. If $\alpha_1, \alpha_2, ..., \alpha_n$ and $\beta_1, \beta_2, ..., \beta_n$ are bases for K_1 and K_2 over F respectively, then the elements $\alpha_i\beta_j$ for $i = 1, 2, ..., n$ and $j = 1, 2, ..., m$ span K_1K_2 over F

***Proof*:**
exercise

summing up the result in a diagram:

$$\begin{tikzcd} & K_1K_2 & \\ K_1 \arrow[ur, dash, "\leq n"] & & K_2 \arrow[ul, dash, "\leq m"'] \\ & F \arrow[ul, dash, "n"] \arrow[ur, dash, "m"'] & \end{tikzcd}$$

Corollary 19.3.18: Degree of Composite Field and Equality

Suppose that $[K_1 : F] = n$, $[K_2 : F] = m$ in proposition 19.3.17, where n and m are relatively prime $(n, m) = 1$. Then $[K_1K_2 : F] = [K_1 : F][K_2 : F] = nm$

***Proof*:**
In general, the extension is divisible by n and m since K_1 and K_2 are subfields of K_1K_2 (so divisible by its least common multiple). Since $(m, n) = 1$, then $[K_1K_2 : F]$ is divisible by nm. Since by proposition 19.3.17, $[K_1K_2 : F] \leq nm$, we get $[K_1K_2 : F] = nm$

Exercise 19.3.1

1. (practice quadratic extensions here, see Dummit and Foote example)
2. Show that If Let E/k and F/k be field extensions. Show that if E/k is algebraic, then so is EF/F. Show that if E/k and F/k are algebraic, so is EF/F.

19.4 Historical Detour: Straightedge and Compass Construction

For millennia in mathematical history, mathematicians as far back as ancient greeks tried to construct shapes or compute numbers through geometric means by using a straight edge and compass. For example, using a straight edge and compass, there is a way to compute sums, products, quotient (which would have been known as ratios to the ancient greeks), and square-roots. Geometry, mathematicians have figured out how to double the area of the square, take exactly half the angle, and navigate using the stars [14]. These all conformed with the rules of Euclid's geometry, and so straight-edge and compass constructions seemed like a powerful algorithm for geoemtry and arithmetics. However, a few problems were open since at least since the times of the ancient greeks around 400 BC, namely:

1. Given a square, using a straight-edge and compass, can we find a circle with the same area
2. Given a cube, using a straight-edge and compass, can we find another cube that's double it's volume
3. Given an angle, using a straight-edge and compass, can we triscect the angle (i.e. split the angle into 3 equa angles)

These problems don't have much use practically. However the simpliciy in their statement, the ubiquity of using rulers and compasses in many fields of engeneering, navigation, and so on, as well as the belief that the foundation of the world's geometry is inherently euclidean (in particular, that geometric reality followed euclidean rules and were provable using euclid's simple axioms), made these problems the central attention of a huge plethora of mathematicians for over 2 millennia.

With the new tools of field theory, we can characterize the types of numbers a straight-edge and compass can give us:

1. using a straight-dge construction, the 4 basic operations $(+, -, \times, \text{div})$ can be derivd. Thus, it will form a field.
2. the compass constructions allows the performance of square-roots. Translating this into the cartesian plane, it will be the intersction of a straight-line and a circle, which will sometimes gives polynomials which will have solutions *not* inside the base field, and so we would use an algebraic extension. Such polynomials will always be of degree 2. Hence, if we keep trying to extend these fields, we will always get multiples of 2 as the degree of the extension

From this comes the theorem that feels so mind-blowing that I can state after literal millenia of people trying to solve this problem:

Theorem 19.4.1: Straightedge And Compass Imposibilities

The 3 classical greek straight-edge and compass problems:

1. **Doubling the square** Using only straight edge and compasss, construct a cube with precisely twice hte volume of a given cube
2. **Triscecting an Angle** Given a straightedge and compass, triseced any given angle θ
3. **Squaring the circle** Given a straightedge and compass, construct a square whose area is precisely the area of a given circle.

are all answered in the negative

[14] and an additional instrument to measure the angle between the person and the horizon, and the north star

***Proof*:**
Comes down to the aforementioned intuition and a bit more: Dummit and Foote p. 553

19.5 Closures of Fields

Through theorem 19.2.2, we saw that if we have a irreducible polynomial $f(x) \in k[x]$, we can find a larget field which gives $f(x)$ a root: $f(x) = f'(x)(x - \alpha)$. However, we have only established result for 1 polynomial, and for only 1 root. It would be nice if the set of all roots of every polynomials formed a field, and even better if that field existed for any field k and was unique (up to isomorphism). In such a field, every polynomial would *fully factor* or *split* into linear terms. Through corollary 19.3.14, we found that given K/k the set of algebraic elements $\alpha \in K$ over k forms a field, namely the set of elements from the larger field K which are algebraic over k. In example (19.3.15) (1), we where implicitely using the fact that every polynomial over $\mathbb{C}$ reduces to linear terms, allowing us to say that $\overline{\mathbb{Q}}$ to be algebraically closed. It would be much better if the definition would be independent of any larger field, that is, find a field $\overline{k}$ such that the only irreducible polynomials in $\overline{k}[x]$ are of degree 1. Furthermore, if $\overline{k}$ existed for every field and was unique, we can call it *the* algebraic closure of k. This is the first thing proven in this section. Then, we take a closer look at fields for which only some subset $S \subseteq k[x]$ have polynomials that split. These fields will be at the heart of Galois theory, for as we saw in corollary 19.2.29, the group of k-automorphism $\mathrm{Aut}_k(F)$ depends on the number of roots of a polynomial in F. Thus, we take some time constructing such a field.

19.5.1 Algebraic Closure

We start off this section by reminding the reader what an algebraically closed field is:

Definition 19.5.1: Algebraically Closed

Let K be a field. Then K is called algebraically closed if every polynomial in $K[x]$ factors into a product of linear terms.

There are many equivalent ways of defining algebraic closure:

Lemma 19.5.2: Algebraic Closure Equivalences

Let K be a field. then the following are equivalent:

1. K ia algebraically closed
2. every maximal ideal in $K[x]$ is of the form $(x - c)$ for all $c \in K$
3. There does not exist nontrivial algebraic extensions of K
4. If L/K is some field extension, and $\alpha \in L$ is algebraic over K, then $\alpha \in K$

***Proof*:**
exercise

These show that being an algebraically closed field K means that there are no more algebraic extensions L/K, besides trivial ones. Given some field k, there can be many field extensions that are algebraically closed. For example, $\overline{\mathbb{Q}}/\mathbb{Q}$, $\mathbb{C}/\mathbb{Q}$, and $\mathbb{C}(t)/\mathbb{Q}$ are all field extensions of $\mathbb{Q}$ to algebraically closed fields. However, the first extensions seems in some way more simple or "minimal" than the other in that it is defined through only algebraic elements (recall form example (19.3.15)(1) that all the elements of $\overline{\mathbb{Q}}$ are algebraic over $\mathbb{Q}$), while $\mathbb{C}$ contains transcendental elements (we've mentioend in example (19.3.15)(2) how π and e are not algebraic).

Definition 19.5.3: Algebraic Closure

Let k be any field. Then an *algebraic closure* of k is a field $\overline{k}$ where $\overline{k}/k$ is an *algebaric extension* and $\overline{k}$ is algebraically closed.

Soon, the statement can be re-written with "the algebraic closure" and "the field" instead of "an algebraic closure" and "a field". The field $\overline{k}$ will be the smallest field in which every nonconstant polynomial $p(x) \in k[x]$ will split into linear factors, and since $\overline{k}$ is algebraically closed, the subfield of $\overline{k}$ that contains all the roots of all non-constant polynomials in $k[x]$ is $\overline{k}$ itself. To show the existence of such a field, we first prove that there exists a field extension where every polynomial in $k[x]$ has *at least* one root, and then recursively apply the theorem to get a long chain of field extensions, each containing at least one more root to every polynomial.

Lemma 19.5.4: Adding One Root

Let k be a field. Then there exists a field K such that every non-constant polynomial of $k[x]$ admits at least 1 root.

The following proof is credited to Emil Artin

***Proof*:**

Let $\mathcal{F}$ be a set that is in bijection with all non-constant monic polynomials $F \subseteq k[x]$: $\mathcal{F} = \{t_f\}_{f \in F}$, and let $k[\mathcal{F}]$ be a polynomial ring with indeterminates $t_f \in \mathcal{F}$. Let $I \subseteq k[\mathcal{F}]$ be the set of all polynomials generated by all $f(t_f)$ (that is, the polynomial f that takes in the indeterminate t_f that it is in bijective correspondence).

I claim that I is a proper ideal of $k[\mathcal{F}]$. Indeed, if it weren't, then we would have:

$$1 = \sum_i^n a_i f(t_{f_i})$$

with $a_i \in k[\mathcal{F}]$. The identity cannot be possible, for that would lead to imply $1 = 0$. Let's take $k \subseteq F$ where each polynomial $f_1(x), ..., f_n(x)$ has a root $\alpha_1, ..., \alpha_n$ (apply theorem 19.2.2 n times). Then viewing the previous identity to be in $F[\mathcal{F}]$ we get:

$$1 = \sum_i^n a_i f(t_{f_i}) = \sum_i^n a_i f(\alpha_i) = \sum_i^n a_i \cdot 0 = 0$$

a contradiction.

Now, since I is proper, it is contained in a maximal ideal m (proposition 9.4.18). Thus, we have the field extension:

$$k \subseteq K := \frac{k[\mathcal{F}]}{m}$$

By construction every non-constant monic (and so every nonconstant) polynomial $f(x) \in k[x]$ must have a root in K, namely $\overline{t_f}$

Note that we had to use Zorn's lemma to find the maximal ideal m. Very commonly, a statement using Zorn's lemma is equivalent to it (ex. instead of using Zorn's lemma to prove the existence of a maximal ideal, we can use the existence of a maximal ideal to prove Zorn's lemma). We might ask if the existence if the previous result is also equivalent to Zorn's lemma, however it turns out that this is not the case (it is apparently due to the *compactness theorem of first order logic*, which is known to be weaker than the axiom of choice)

We can now find a field K_1 in which every polynomial in $k[x]$ has at least 1 root. We can continue this process by applying the theroem on K_1, and defining the field K_2 for which every polynomial of K_1 has at least 1 root in $K_1[x]$ (and every polynomial of degree 2 in $k[x]$ must not split into linear factors). Continuing this process, we get a chain:

$$k \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

Now, take $L = \bigcup_i K_i$[15]. L can easily be turned into a field through the operation of $a + b$ and $a \cdot b$ where we choose K_i in which $a, b \in K_i$.

Lemma 19.5.5

Let L be defined as above. Then L is algebraically closed

***Proof*:**
Let $f(x) \in L[x]$ be a nonconstant polynomial. Then since $f(x)$ has finitely many coefficients, there exists some field K_i for some i which has all the coefficients of $f(x)$. Thus, $f(x)$ has a root in $K_{i+1} \subseteq L$: $f(x) = f'(x)(x - \alpha)$. Since $f'(x) \in L[x]$ (the coefficients of $f'(x)$ must be in some K_j, $j \geq i$), this process repeats until f is completely split into linear terms. Since this is true for arbitrary $f(x) \in L[x]$, L is algebraically closed, completing the proof.

Thus, for any field k, there exists an algebreaically closed field L such that L/k. Note that L need not be algebraic closure, since L/k need not be an algebraic extension. However, with an algebraically closed field, we can very simply construct an algebraic closure of k:

Theorem 19.5.6: Existence Of Algebraic Closure

Let k be a field and L/k be a field extension where L is algebraically closed. Then

$$\overline{k} := \{\alpha \in L \mid \alpha \text{ is algebraic over } k\}$$

is an algebraic closure of k

Note how this mimicked $\overline{\mathbb{Q}}$ in example (19.3.15)(2) assuming $\mathbb{C}$ is algebraically closed.

[15]This is an example of a *direct limit*

***Proof*:**
First, by corrollary 19.3.14, $\overline{k}$ is a field, and *a prior* $\overline{k}$ is an algebraic extension of k. To show that k is algeraically closed, let α be algebraic over $\overline{k}$. Then

$$k \subseteq \overline{k} \subseteq \overline{k}(a)$$

is a composition of albegraic extensions, hence $\overline{k}(a)$ is algebraic (theorem 19.3.16), thus α is algebraic over k. But then by definition of $\overline{k}$, $\alpha \in \overline{k}$. Thus, by lemma 19.5.2$\overline{k}$ is algebraically closed, completing the proof

Next, we strive for uniqueness. As we establisehd in theorem 19.2.2, being a field extension is not a universal property, and so the best we an strive for is the existence of a homomorphism, though not necesarrily a unique one. We first establish a lemma:

Lemma 19.5.7: Embedding Into Algebraic Extension

Let L/k be any field extension where L is algebraically closed over k. Let F/k be an algebraic extensions. Then there exists a [not necessarily unique] homomorphism $i : F \to L$ fixing k.

Note that *a-priori*, L need not have any relation to F (ex. we don't know if F is a subset of L, they might only intersect on k).

***Proof*:**
This is a Zorn's lemma argument similar to Baer's Criterion (theorem 17.3.5). Let S be the set of homomorphisms of the form

$$i_K : K \to L$$

The set S is nonempty since $i_k : k \to L \in S$. Define a patial orer on S $i_K \preceq i_{K'}$ if $K \subseteq K' \subseteq L$ and $i_{K'}|_K = i_K$. To apply Zorn's lemma, we need to show that every chain has an upper bound. Let $C \subseteq S$ be a chain, and let K_C be the union of all the domains of elements $i_K \in C$. Then K_C is clearly a field (lemma ref:HERE), and for every $\alpha \in K_C$ define $i_{K_C}(\alpha)$ to be $i_K(\alpha)$ for appropriate K where $\alpha \in K$. Then i_{K_C} is clearly well-defined, not being dependent on the choice of K, and so we have a homomorphism $i_{K_C} : K_C \to L$ where $i_K \preceq i_{K_C}$ for all $i_K \in C$, showing that i_{K_C} is indeed an upper bound for C. Thus, by Zorn's lemma, Z has an upper bound $i_G : G \to L$ where $k \subseteq G \subseteq L$.

I claim that $G = F$, which will prove the statement. For the sake of contradiction, let's assume this is not the case, so that there exists an element $\alpha \in F \setminus G$. Consider $G(\alpha)/G$. Since $\alpha \in F$ is algebraic over k, α is algebraic over G, and since $\alpha \notin G$, it is the root of an irreducible polynomial $g(x) \in G[x]$. Let $H = i_G(G)$. Then as discussed in the first pages of section 19.2 i_G induces a ring homomorphism

$$f^{i_G} : G[x] \to H[x]$$

Let $h(x) = f^{i_G}(g(x))$. Then $h(x)$ is irreducible over H, and since L is algebraically closed, it must have a root $\beta \in L$.

We are now setup in a situation to apply theorem 19.2.27: we have an isomorphism $i_G : G \to H$ and two simples extensions $G(\alpha), H(\beta)$ with α and β being roots of $g(x)$ and $h(x)$ respectively (with $h(x) = i_G(g(x))$). Thus, i_G lifts to an isomorphism:

$$i_{G(\alpha)} G(\alpha) \to H(\beta) \subseteq L$$

which sends α to β. Then $i_G \preceq i_{G(\alpha)}$, contradicting the maximality of i_G.

Thus, it must be that $G = F$, and so $i_F : F \to L$ is a field homomorphism, as we sought to show.

The fact that the isomorphism is not unique will become important in representing algebraic closures. We first finish off by showing that the algebraic closure is indeed unique up to isomorphism (given a fixed field k)

Theorem 19.5.8: Uniqueness Of Algebraic Closure

Let $k \subseteq \overline{k_1}$ and $k \subseteq \overline{k_2}$ be two algebraic closures of k. Then $\overline{k_1} \cong \overline{k_2}$ where the isomorphism extends the identity on k, or equivalently k_1 is isomorphic to k_2 as extensions. In terms of a commutative diagram, in the following

$$\begin{tikzcd} k_1 \arrow[rr, "\sim"] & & k_2 \\ & k \arrow[ul, hook] \arrow[ur, hook] & \end{tikzcd}$$

there always exists an isomorphism from k_1 to k_2

the second condition for the isomorphism is so that the two fields represent the algebraic closure over the common field k. Sometimes, such a isomorphism (resp. homomorphism) is called a k-isomorphism (resp. k-homomorphism)

***Proof*:**

Since $\overline{k_1}/k$ is algebraic and $\overline{k_2}$ is algebraically closed over k, by lemma 19.5.7, there exists a homomorphism $i : \overline{k_1} \to \overline{k_2}$. This homomorphism is injective, being a field homomorphism, and surjective since otherwise, $\overline{k_1} \subseteq \overline{k_2}$ would be a non-trivial algebraic extension, contradicting lemma 19.5.2 since $\overline{k_1}$ is algebraically closed. Thus, i must be an isomorphism as we sought to show.

The fact that the isomorphism is not necessarily unique has some important consequences: There is no "cannonical" way of representing the algebraic closure. At first, it might seem harmless that the isomorphism is not unique, as long as the two field are isomorphic. However, this means there is no real cannonical "choice" when it comes to representing algebraic closures.

Take for example $\overline{\mathbb{R}}$, which we know to be $\mathbb{C}$. We can write $\overline{\mathbb{R}}$ as $\mathbb{R}[i]$, which for many mathematicians is "the" algebraic closure of $\mathbb{R}$. However, if we go over to an electrical engineer, we will see that they have also constructed an algebraic closure for $\mathbb{R}$ to represent electrical charge and circuits where they use the letter j (since i is already used for something else), and write it as $\mathbb{R}[j]$. We might think it to be harmless to make the isomorphism $1 \mapsto 1$ and $i \mapsto j$, however the electrical engineers have decided that electrical charge should be represented by a -1. Does this means we should instead have

$$1 \mapsto 1 \qquad i \mapsto -j?$$

Mathematically, there is no good reason to pick one representation over another. In other words, if one person constructs a splitting field L for k and another person constructs a algebraic closure L'

for k, then we know that we get:

$$\begin{tikzcd} k_1 \arrow[rr, "\sim"] & & k_2 \\ & k \arrow[ul, hook] \arrow[ur, hook] & \end{tikzcd}$$

however, there is no "natural" way to associate an element of L to an element of L'. This often means that we should keep track of the construction of the splitting field we have done, since if we don't we might not be able to communicate to other mathematicians who have done a different construction (how can you compare your work if you can't correspond the results you have come up with with the other mathematicians).

The idea there there is no natural choice of isomorphism can be made more rigorous if we ask it in the language of category theory. If we map every field to it's algebraic closure $k \to \overline{k}$, does this define a functor? In order for it to define a functor, we would need that every map $k \to l$ of field defines a map of algebraic closures

$$\begin{tikzcd} k \arrow[r] \arrow[d] & \overline{k} \arrow[d, dashed] \\ l \arrow[r] & \overline{l} \end{tikzcd}$$

However, it is not clear what map should be chosoen here[16]. There is in fact a way to reduce this ambiguity, which I don't fully undertand yet intuity, but it was mentioned in Richard Borcherds video: here.

19.5.2 Splitting Field and Normal Extensions

Having established that every field has an algebraic closure, meaning we can unambigurously talk about roots of polynomials, we now return to analyzing algebraic extensions. We start off by zoom in on fields that only split some subset of polynomials $S \subseteq k[x]$. In the following sections, we will be working over the case where S is finite [17]. Since S is finite, say $S = \{f_1(x), ..., f_n(x)\}$, then having each $f_i(x)$ split is equivalent to having $f_1(x)f_2(x) \cdots f_n(x)$, thus, we usually have $S = \{f(x)\}$ be one polynomial

Definition 19.5.9: Splitting Field

Let k be a field and $f(x) \in k[x]$ be a polynomial where $\deg(f(x)) = d$. Then a *splitting field* for $f(x)$ over k is a minimal extension F/k such that

$$f(x) = c\prod_i^d (x - \alpha_i)$$

i.e. $f(x)$ splits completely in $F[x]$, and $f(x)$ does not factor in any proper subfield of F.

Furthermore, by minimality, $F = k(\alpha_1, ..., \alpha_n)$ (i.e. F is generated by the roots of $f(x)$ in F)

[16] perhaps you can force through a construction using the axiom of choice, but it is certainly not a natural choice
[17] The infinite case of this I'm not yet sure is not studied TBD

The fact that a splitting field exists is immediate from the fact that $\overline{k}$ exists, namely if $f(x)$ is a polynomial, then $F \subseteq \overline{k}$ generated by the roots of $f(x)$ in $\overline{k}$. Furthermore, the splitting field is in fact unique (up to isomorphism), justifying replacing any "a splitting field" in the definition to "the splitting field"

Theorem 19.5.10: Uniqueness Of Splitting Field

Let k be a field and let $f(x) \in k[x]$. Then the splitting field F for $f(x)$ over k is unique up to isomorphism, and $[F : k] \leq (\deg(f))!$ (i.e. the factorial of the degree of f). Furthermore, if $i : k \to k'$ is an isomorphism, $f(x) \in k[x]$ and $g(x) = i(f(x))$, then i extends to an isomorphism from the splitting field of $g(x)$ over k' to the splitting field of $f(x)$ over k:

$$\begin{tikzcd} F \arrow[r, dashed, "\sim"] & F' \\ k \arrow[u, hook] \arrow[r, "\sim"] & k' \arrow[u, hook] \end{tikzcd}$$

***Proof*:**
We will first construct a splitting field explicitely with $[F : k] \leq \deg(f)$, and then use that field to show uniqueness. For existence, this is an application of induction on the degree of the polynomial. Degree 1 is trivial, covering the base-case, so let's say the splitting field was constructed for all fields and all polynomials of degree $\deg(f) - 1$. Let $f(x)$ be some polynomial that is not yet fully factored into linear terms; let $q(x)$ be some irreducible non-linear factor over k. Then

$$k \subseteq F' := \frac{k[x]}{(q(x))}$$

is an extension of degree $\deg q \leq \deg f$ in which $q(x)$ has a root (and so $f(x)$ has a root), say α, and so $q(x) = q'(x)(x - \alpha)$. Then we can define the polynomial $h(x) = f(x)/(x - \alpha) \in F'[x]$ of degree $\deg(f) - 1$. Thus, by the induction hypothesis, there exists a splitting field F for $h(x)$ over F' and

$$[F : F'] \leq (\deg(f) - 1)!$$

Since the factors of $f(x)$ over F are $(x - \alpha)$ and the factors of $h(x)$, it follows that F is a splitting field for $f(x)$, and

$$[F : k] = [F : F'][F' : k] \leq (\deg(f) - 1)!(\deg(f)) = (\deg(f))!$$

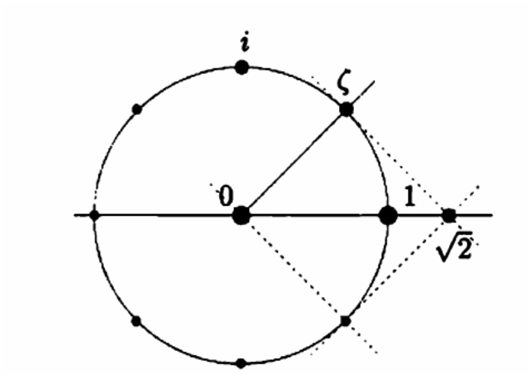
To prove that an isomorphism $i : k \to k'$ extends to isomorphisms of splitting fields as stated, let F be the splitting field for $f(x)$, and consider $k \to k' \subseteq \overline{k'}$. Since the extension $k \subseteq F$ is algebraic, by lemma 19.5.7 there exists a k-homomorphism $i : F \to \overline{k'}$. Since F is generated over k by the roots of $f(x)$ in F, we have

$$i(F) = k'(\alpha_1, ..., \alpha_d) \subseteq \overline{k'}$$

where the α_i's are the roots of $g(x) = i(f(x))$ in $\overline{k'}$. Therefore, $L := i(F)$ is independent of the given isomorphism i and the splitting field F. Applying this to i and to the identity $k' \to k'$ completes the proof (since both F and G are isomorphic to L)

Example 19.5.11: Splitting Fields

1. The splitting field of $x^2 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$, since the two roots are $\pm\sqrt{2}$ and $-\sqrt{2} \in \mathbb{Q}(\sqrt{2})$. The splitting field of $(x-2)^n$ is also $\mathbb{Q}(\sqrt{2})$ for all $n \in \mathbb{N}_{>0}$, since we only need to add the root $\sqrt{2}$ (resp. $-\sqrt{2}$) for this polynomial to split.
2. Splitting field for $(x^2-2)(x^2-3)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Note that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is equal to the composite field $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3})$, i.e. $\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.
3. More generally, If $f(x)$ is a polynomial over k such that
$$f(x) = \prod g_i(x)$$
where each $g_i(x)$ is irreduible over k, then the splitting field of $f(x)$ is the composite field of the splitting fields for each $g_i(x)$ since each splitting field is the smallest field containing all the roots of g_i, and the composite field is the smallest field containing all the splitting fields.
4. splitting field of $x^3 - 2$ over $\mathbb{Q}$ is not $\mathbb{Q}(\sqrt[3]{2})$, since we are missing two roots, namely:
$$\sqrt[3]{2}\left(\frac{-1+i\sqrt{3}}{2}\right) \qquad \sqrt[3]{2}\left(\frac{-1-i\sqrt{3}}{2}\right)$$
Thus, the splitting field is $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$.
5. The splitting field of x^2+1 over $\mathbb{Q}$ is $\mathbb{Q}(i)$, and the splitting field of the same polynomial x^2+1 over $\mathbb{R}$ is $\mathbb{C}$, showing that the splitting field is dependent on the base field.
6. Let F be the splitting field of $x^8 - 1$ over $\mathbb{Q}$. With some knowledge of complex analysis, we see that all the roots are 8th root of unit, that is,
$$\zeta = e^{\frac{2\pi i}{8}}$$
is a generator for the roots (i.e. the primite elements), and so $F = \mathbb{Q}(\zeta)$. Visually, these roots look like so:



In fact, ζ is also the root of x^4+1 (since $\zeta^4 = -1$), and so is the root of the smaller irreducible polynomial x^4+1 over $\mathbb{Q}$. There is no smaller minimal polynomial for which ζ is a root (since smaller powers of ζ is not in $\mathbb{R}$) and so:
$$[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$$

showing that the degree of the splitting field is much less than $8! = 40'320$. To better undersatnd this field, notice that $\zeta^2 = i$, so $\sqrt{2} = \zeta + \zeta^7$. Thus:

$$\mathbb{Q}(i, \sqrt{2}) \subseteq \mathbb{Q}(\zeta)$$

Conversely, $\zeta = \frac{\sqrt{2}}{2}(1+i) \in \mathbb{Q}(i, \sqrt{2})$, showing that:

$$\mathbb{Q}(i, \sqrt{2}) \supseteq \mathbb{Q}(\zeta)$$

showing that they are in fact equal.

As an exercise, show that $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(i, \sqrt{2})) = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$

7. Let $p(x) = x^4 - 1$. this polynomial in fact factors over $\mathbb{Q}$:

$$x^4 - 1 = (x-1)(x+1)(x^2+1)$$

and so the splitting field of $x^4 - 1$ is the splitting field of $x^2 + 1$, which is $\mathbb{Q}(i)$

8. Let $p(x) = x^4 + 2$. This polynomial is irreducible by Eisenstein. Given $\zeta = e^{\frac{2\pi i}{8}}$, then the roots of $p(x)$ are:

$$\sqrt[4]{2}\zeta,\ \sqrt[4]{2}\zeta^3,\ \sqrt[4]{2}\zeta^5,\ \sqrt[4]{2}\zeta^7$$

and so the splitting field is $K = \mathbb{Q}(\sqrt[4]{2}\zeta,\ \sqrt[4]{2}\zeta^3,\ \sqrt[4]{2}\zeta^5,\ \sqrt[4]{2}\zeta^7)$. We can simply the adjoint elements with the following observations:

$$K \subseteq \mathbb{Q}(\zeta, \sqrt[4]{2}) \subseteq \mathbb{Q}(i, \sqrt{2}, \sqrt[4]{2}) = \mathbb{Q}(i, \sqrt[4]{2})$$

so $K \subseteq \mathbb{Q}(i, \sqrt[4]{2})$. For the $\supseteq$ direaction, notice that:

$$\sqrt{2} = \frac{(\sqrt[4]{2}\zeta)^3}{\sqrt[4]{2}\zeta^3} \in K \qquad i = \frac{(\sqrt[4]{2})^2}{\sqrt{2}} \in K$$

So $\zeta = \frac{\sqrt{2}}{2}(1+i) \in K$ and $\sqrt[4]{2} = \frac{\sqrt[4]{2}\zeta}{\zeta} \in K$, showing that $K \supseteq \mathbb{Q}(i, \sqrt[4]{2})$. With a little bit of computation, we see that:

$$[\mathbb{Q}(i, \sqrt[4]{2})] = 8$$

From Dummite and Foote. There are in fact a ton of good example that I can put here, check them out! Overall, small variation in polynomials can create quite large differences in splitting fields (as we've seen with x^4+2, x^4+1, x^4-1). This shows how precise the tools of field theory are in the study of polynomials.

Given a polynomial $f(x)$ over k, we can find a field F in which $f(x)$ factor completely over F. In fact, we have also gained a slightly stronger result for free: if $g(x)$ is irreducible in k has a root in F, then it in fact *also* splits. The idea is that the automorphism group acts transitively on all the roots of an irreducible polynomials (recall the discussion under corollary 19.2.29). If all the roots are inside the field, then the automorphism group will shuffle around all the roots within the field. Hence, if we are given that a single root is inside a field, and that the image of each automorphism in the automorphism group is inside the field, we would have that all the roots are within the field. As the action of the automorphism group $\mathrm{Aut}_k(F) \curvearrowright R_F$ is a evaluation action $\sigma \cdot \alpha = \sigma(\alpha)$ (often called the *galois conjugate* or just *conjugate)*, another way of saying it is that the field is closed under conjugation of the automorphism group. To understand this phenomenon better, we give a name to fields with this property:

Definition 19.5.12: Normal Extension

let F/k be a field extension. Then F/k is called a *normal extension* if for every polynomial $f(x) \in k[x]$, $f(x)$ has a root if and only if $f(x)$ splits as a product of linear factors over F

Some authors like to say that F/k needs to also be *algebraic*. This so that we need not worry about any transcendental elements in the following proofs. It is no harm on making F/k possibly not algebraic, since we work with finite extensions in the majority of cases

A normal extension is *not necessarily* the algebraic closure since not necessarily all irreducible polynomials have a root, however the irreducible polynomials over the base field become the polynomials that split completely in the normal extension, constant polynomials, and irreducible polynomials that have no roots in the normal extension (or, they can be thought of as polynomials that have yet to have a root added to the extension).

The name normal is given to this type of extension is due to the fact that most of the nice properties of automorphism groups we have discussed in corollary 19.2.29 apply to normal extensions[18]

Theorem 19.5.13: Finite Normal Extension Equivalences

Let F/k be a field extension. Then the following are equivalent:

1. F is finite and normal
2. F is the splitting field of some polynomial $f(x) \in k[x]$
3. For all k-homomorphisms, $F \xrightarrow{\varphi} \overline{F}$ where $\varphi(k) = k$, $\varphi(F) = F$

***Proof*:**
We'll first show $2 \Leftrightarrow 3$ since it's simple. For $2 \Rightarrow 3$, let F be the splitting fifeld for some polynomial $f(x) \in k[x]$, then if $F \xrightarrow{\varphi} \overline{F}$ is a k-homomorpism, $F \xrightarrow{\varphi} \varphi(F)$ is a k-isomorphism. Hence, φ must map roots of $f(x)$ to roots of $f(x)$: If α is a root of $f(x)$, then we know that $m_{\alpha,k}(x)\big|f(x)$, and by corollary 19.2.29, $\varphi(\alpha)$ is another root of $m_{\alpha,k}$, and hence $f(x)$. Since F is a splitting field, it contains all the roots of $f(x)$, and so $\varphi(\alpha) \in F$. Since the k-homomorphism is determined by where it maps the roots of the generators of F (namely the roots of $f(x)$), $\varphi(F) \subseteq F$. Since φ must be injective, $F \subseteq \varphi(F)$, so $F = \varphi(F)$.

For $3 \Rightarrow 2$, since all k-homomorphisms $F \xrightarrow{\varphi} \overline{F}$ map $\varphi(F) = F$. If F/k is purely transcendental (i.e., no elements of $\alpha \in F - k$ is the the root of a polynomial), Then then clearly $\varphi(F) = F$ or else φ would not be injective. Thus, if F/k is not purely transcendental, choose an algebraic element $\alpha \in F$ over k, so $m_{\alpha,k}(x)$ is it's minimal polynomial. Since φ is defined by where it maps roots of minimal polynomials over $k[x]$, $\varphi(\alpha)$ is another root of $m_{\alpha,k}(x)$. Since $\varphi(F) = F$, $\varphi(\alpha) \in F$. Using corollary 19.2.29, we can see that there are as many automorphisms as distinct roots of $m_{\alpha,k}(x)$, and so every root of $m_{\alpha,k}(x)$ is in F. Since this is true for every algebraic element, we see that $\varphi(F) = F$.

[18] In fact, normal subgroups got their name from normal extensions! The connection will be made in the chapter 20 when discussing Galois Theory in theorem 20.1.23

Next, we show $1 \Rightarrow 2$. Assume F is finite and normal. Since it's finite, $F = k(\alpha_1, \alpha_2, ..., \alpha_n)$, where each α_i is algebraic over k. Next, let $m_i(x)$ be the minimal polynomial of α_i. Since F is normal, and each m_i has at least one root (namely α_i), $m_i(\alpha)$ splits. Then $f(x) = m_1(x) \cdots m_n(x)$ is a polynomial that splits over F. But then, F is the smallest field in which $f(x)$ splits (since $F = k(\alpha_1, ..., \alpha_n)$), and so F is the splitting field for $f(x)$.

Finally, we show $2 \Rightarrow 1$: assume F is the splitting field for some polynomial $f(x) \in k[x]$, and assume that $p(x) \in k[x]$ is an irreducible polynomial over k such that F contains a root α for $p(x)$. If $p(x)$ has degree less than or equal to 2, we're done, so let's say $\deg p(x) \geq 3$. Since $F \subseteq \overline{k}$, let $\beta \in \overline{k}$ be a different root of $p(x)$. We want to show that $\beta \in F$. This will show that F is normal, and by theorem 19.5.10, F must be finite.

Since $\mathrm{id} : k \to k$ is a field homomorphism, by theorem 19.2.27, there is an isomorphism $i : k(\alpha) \to k(\beta)$ sending $\alpha \mapsto \beta$. Note that $F(\beta) \subseteq \overline{k}$. Visually, we can imagine all of this in the commutative diagram:

$$\begin{tikzcd}
& k(\alpha) \arrow[r, hook] \arrow[dd, hook, "\iota"] & F \arrow[r, hook] & \overline{k} \\
k \arrow[ur, hook] \arrow[dr, hook] & & & \\
& k(\beta) \arrow[r, hook] & F(\beta) \arrow[r, hook] & \overline{k}
\end{tikzcd}$$

Note that we cannot draw the identity map from $\overline{k}$ to $\overline{k}$, since ι maps $\alpha \mapsto \beta$. Now, notice that F is the *splitting field* $f(x)$ over $k(\alpha)$. Indeed, F contains all the roots of $p(x)$ and is generated over k (and hence $k(\alpha)$) by the roots of $p(x)$. Similarly, $F(\beta)$ is the splitting field for $f(x)$ over $k(\beta)$. Then by the uniqueness of the splitting field (theorem 19.5.10), ι extends to an isomorphism $\iota' : F \to F(\beta)$. Now, since ι restricts to the identity on k, ι' is an isomorphism of k-vector spaces, in particular $\dim_k F = \dim_k F(\beta)$ (since splitting fields are always finite extensions). In particular, we get to take advantage of facts about liner algebra, in particular that if the dimensions of two vector-spaces are the same, then any injective linear transformation is in fact an isomrophism.

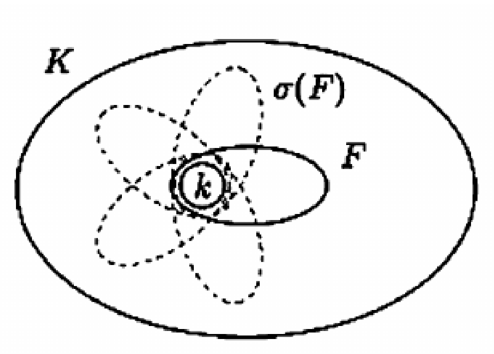
Now, we also have the natural inclusion map $\iota : F \to F(\beta)$ which is both a k-homomorphism and a k-vector spaces. Since F and $F(\beta)$ are k-vector spaces of the same [finite!] dimension, i must *also* be an isomorphism, i.e.

$$[F(\beta) : F] = 1$$

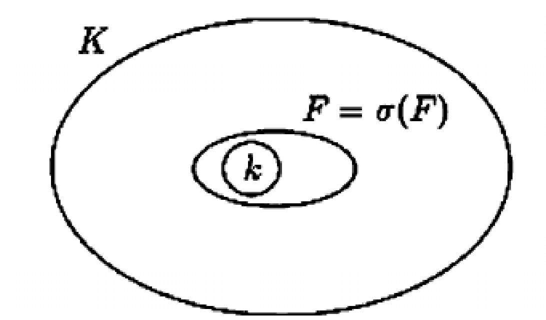
Thus, $\beta \in F$, completing the proof.

Note that it can be proven that F is normal (not necessarily finite) if and only if for any k-homomorphism $\varphi : F \to \overline{F}$, $\varphi(F) = F$. The proof naturally generlizes from the proof of $2 \Leftrightarrow 3$ since we can apply the results to each irreducible polynomial and their roots.

19.5.14 Remark Notice how condition (3) of theorem 19.5.13 allows us to talk about "the" embedding of a Normal extension F/k into $\overline{k}$, since $\varphi(F) = F$ for any F. Recall that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal, and that we have shown that $\mathbb{Q}(\sqrt[3]{2})$ is isomorphic to 3 subfields of $\overline{\mathbb{Q}}$ which do not all overlap:



However, if the extensoin is normal, then there is a notion of "the" embedding since they all overlap:



In this way, having a [algebraic][a] normal extension F/k can be thought of as being more "precise" than a non-normal extension, since any other isomorphic copy of F is just an automorphism of F.

[a] This is to ensure that $F \subseteq \overline{k}$. The algebraic elements of F is also what we care about when we look at k-homomorphisms $F \to \overline{k}$

(there is also a cool proof the professor gave in class, the Jan 20th lecture)

Example 19.5.15: Consequences

Let's say $p(x)$ is an irreducible polynomial over $\mathbb{Q}$ with a complex root that may be expressed as a polynomial in i and $\sqrt[4]{2}$ over $\mathbb{Q}$. Then *all* roots of $p(x)$ may be expressed as a polynomial in i and $\sqrt[4]{2}$. As we have shown in example 19.5.11, $\mathbb{Q}(\sqrt[4]{2}, i)$ is a splitting field over $\mathbb{Q}$, and hence a normal extension.

Unlike algebraic and finite extensions, normal extensions are more finicky when it comes to composing normal extensions:

Proposition 19.5.16: Tower Law On Normal Extensions

If $L/K/k$ and L/k is normal, then L/K is normal

***Proof*:**
Let $f(x) \in K[x]$ be irreducible with a root $\alpha \in L$. Then $f(x) = m_{\alpha,K}(x)$ is the minimal polynomial over K. Note that

$$m_{\alpha,K}(x)|m_{\alpha,k}(x)$$

And since $m_{\alpha,k}(x)$ splits over L (since L/k is normal), $m_{\alpha,K}$ splits over L, as we sought to show.

Note that this result has little wiggle room:

Example 19.5.17: Low Tolerance for Normal Extension

Consider $L/K/k$ where:

$$\begin{array}{c} \mathbb{Q}(\sqrt[3]{2}, \zeta_3) \\ | \\ \mathbb{Q}(\sqrt[3]{2}) \\ | \\ \mathbb{Q} \end{array}$$

where σ_3 is one of the two other roots of $x^3 - 2$. Then while L/k is normal, K/k is *not* normal here (we have yet to add enough roots). Furthermore, if L/K and K/k is normal, this does not imply that L/k is normal. For example:

$$\begin{array}{c} \mathbb{Q}(\sqrt[4]{2}) \\ | \\ \mathbb{Q}(\sqrt{2}) \\ | \\ \mathbb{Q} \end{array}$$

To see this, notice that each extension is of degree 2, and hence is normal (exercise). However, $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not normal, since 2 of the roots of $x^4 - 2$ are complex, while $\mathbb{Q}(\sqrt[4]{2})$ is contained in the reals, and hence cannot have all the roots.

Notice how this behavior is mimicked by normal subgroups.

Definition 19.5.18: Normal Closure

Let K/k be an algebraic extenion. Then the *normal closure* of K given k is an algebraic extension N/K such that

1. N/k is normal
2. If $N \supseteq N' \supseteq k$, and N'/k is normal, then $N = N'$

Proposition 19.5.19: Existence Of Normal Closure

Let K/k be a finite extension. Then there exists a unique normal closure N up to isomorphism fixing k.

Note the finiteness condition given. It is true for algebraic ones too, but we did not yet prove that

***Proof*:**
Since K is finite, let $K = k(\alpha_1, \alpha_2, ..., \alpha_n)$ where each $\alpha_i \in K$ is algebraic over k. Let

$$f(x) = \prod m_{\alpha_i,k}(x) \in k[x]$$

and let N be the splitting field of $f(x)$ over K. Then $[N : K] < \infty$. But then, N is the splitting field of $f(x)$ over k as $K = k(\alpha_1, \alpha_2, ..., \alpha_n)$. But this implies N/k is normal.

Finally, let's say $k \subseteq N' \subseteq N$ and N'/k is normal. Then by normality, N' has to split each $m_{\alpha_i,k}(x)$, and so splits $f(x)$, and so N/k *is* a normal closure.

Exercise 19.5.1

1. (Splitting Field proof using induction)
2. Show that If Let E/k and F/k be field extensions. Show that if E/k is normal, then so is EF/F. Show that if E/k and F/k are normal, so is EF/F.
3. Show that every polynomial of $\mathbb{R}[x]$ of degree ≥ 3 is reducible. Show that every irreducible polynomial is linear or is of degree two with negative discriminant, that is if $ax^2 + bx + c = 0$, then $b^2 - 4ac = 0$.
4. Let L/K be an algebraic field extension, and $\varphi_K : K \to \overline{F}$ be a homomorphism to an algebraically closed field. Then φ_K extends to a homomorphism $\varphi_L : L \to \overline{F}$ (use Zorn's lemma and a proof similar to lemma 19.5.7 with Zorn being applied to (E, φ_E) where E/K is a subextension of L/K and $\varphi_E : E \to \overline{F}$ extends φ_K)

19.6 Separable Extensions

We've now shown that there exists a field K over F such that a polynomial $f(x)$ splits (or a field $\overline{F}$ such that all polynomials in $F[x]$ split). A pattern has been occuring which you might have not noticed: every time we split an irreducible polynomial completely into linear factors, each factor was *unique*, that is, we had that the multiplicity of every linear term was 1. Does that need to be the case for any polynomial over any field? The answer is no:

Example 19.6.1: Higher Multiplicity

Let p be a prime number, and take the field $\mathbb{F}_p(t)$. Consider the polynomial:

$$p(x) = x^p - t \in \mathbb{F}_p(t)[x]$$

Then $p(x)$ is irreducible: by Eisenstein's criterion, it is irreducible in $\mathbb{F}_p[t][x]$ (since (t) is prime in $\mathbb{F}_p[t]$), and hence it is prime in $\mathbb{F}_p(t)[x]$ by corollary 11.2.2. Let u be a root of $p(x)$ in the extension $L/\mathbb{F}_p(t)$ (L can be the algebraic closure, or the splitting field). Then notice that:

$$x^p - t = (x-u)^p$$

Simply expand the right hand side of the equation. Thus, u has multiplicity p, i.e., the minimal polynomial over $\mathbb{F}_p(t)$ for $u \in L$ vanishes p times at u[a].

[a] $p(x)$ must be the irreducible polynomial, since if the degree was smaller then $p(x)$ would be reducible

Thus, in general, a polynomial $f(x) \in k[x]$ can factorize over a splitting field as:

$$f(x) = (x-\alpha_1)^{n_1}(x-\alpha_2)^{n_2}\cdots(x-\alpha_k)^{n_k}$$

where $\alpha_1, \alpha_2, ..., \alpha_k$ are distinct elements in the splitting fields, and $n_1, n_2, ..., n_k$ is the multiplicity of every root. However, this seems to be a rare occurence given the fields we have worked with. In fact, most field we will work with do not have this property. It will also become important in Galois theory that a polynomial would have to to linear polynomials with multiplicity 1[19]; for those who are curious, see what hapens to the automorphism group $\text{Aut}_k(F)$ of the splitting field F if $(x-\alpha)^n$ is the linear factorization of $f(x)$ over k. Hence, we give a name to polynomials that factor with multiplicity 1:

Definition 19.6.2: Separable Polynomial

A polynomial over k is called *separable* if it factors into linear terms with multiplicity 1 in it's splitting field. Equivalently, it has no multiple roots (i.e., all its' roots are distinct). A polynomial which is not separable is called *inseparable*.

My intuition behind the name is that the roots cannot be completely "separated" from each other if the polynomial is *inseparable*. In Algebraic Geometry, polynomials that have roots with multiplicity higher than 1 (say multiplicity m) shall correspond to a collection of points that shall all be arbitrarily close to each other while not being equal, ostensibly being "inseparable"[20]. Naturally, the choice of splitting field does not affect whether a polynomial is separable since all splitting fields for a polynomial are isomorphic, and isomorphisms preserves roots (theorem 19.2.27). Even better, we can use *any* field in which the chosen polynomial $f(x)$ splits into linear factors (since there will be a homomorphism from the splitting field F to the field K in which $f(x)$ splits). For our purposes, that means that it is just as good to work over the algebraic closure $\overline{k}$.

[19](the correspondence Galois group is trivial otherwise)
[20]See the plane with two origins for an example of such a geometric object

Definition 19.6.3: Separable Extension

Let F/k be a field extension and $\alpha \in F$. Then:

1. The element α is called *seprable* if α is algebraic and the minimal polynomial corresponding to α is seperable. Otherwise, the element α is called *inseparable*
2. F/K is called a *separable extension* if every element $\alpha \in F$ is separable, and an *inseparable extension* otherwise.

19.6.4 Remark Note that it is not immediately obvious that if α is separable over k that $k(\alpha)/k$ is a separable extension (what if $1+\alpha$ is inseparable?). We will answer this question after we develop more tools to work with separable elements.

Using the fact that separability is independent of the field in which it splits, we can use a clever trick in finding whether a polynomial is separable using a tool usually defined in calculus:

Definition 19.6.5: Derivative

Given a polynomial $f(x) \in k[x]$ where

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

then its *derivative* or *formal derivative* is the image of the function $D_x : k[x] \to k[x]$ where:

$$D_x f(x) = n a_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + 2a_2 x + a_1$$

or more succinctly

$$D_x\left(\sum_i^n a_i x^i\right) = \sum_i^n i a_i x^{i-1} \tag{19.2}$$

Notice that this derivative acts exactly like the derivative seen in a regular calculus course [21]. In a calculus course, we derive this formula using a limit construction. However, if we just think about the derivative (restricted to polynomials) as a function, then we get a purely algebraic definition of the derivative. In fact, the derivative (and even its generalization with he external operator on differential forms) can be studied purely algebraically; see [3] a deeper exploration of these ideas[22].

Note that the i in equation (19.2) is well-defined since we always have the unique homomorphism $\iota : \mathbb{Z} \to k[x]$ since $\mathbb{Z}$ is initial in **Ring**, and so i is the element associated with $\iota(i)$. The *linearity* and *Leibniz property* of the derivative can both be proven with just this definition:

$$D_x(f(x) + g(x)) = D_x f(x) = D_x g(x)$$
$$D_x(f(x)g(x)) = (D_x f(x))g(x) + f(x)(D_x g(x))$$

We will often abbreviate notation and write $f'(x) := D_x f(x)$ when it is not ambiguous.

[21] A good resources to learn this would be Spivak's Analysis

[22] For example, algebraic differential forms shall be useful to determine the separability of algebraic maps, which is the generalization of the idea that an extension is separable if each element has a nonzero derivative

Proposition 19.6.6: Separable Polynomial Test

A polynomial $f(x)$ has multiple roots α if and only if α is also a root of $D_x f(x)$, i.e., $f(x)$ and $D_x f(x)$ are both divisible by the minimal polynomial for α. In particular, $f(x)$ is separable if and only if it is relatively prime to its derivative

$$(f(x), f'(x)) = 1$$

Noet how $f(x)$ is *not necessarily irreducible*. This means that the derivative test is incredibly powerful, telling us that if $(f(x), f'(x)) = 1$, then any irreducible factor of $f(x)$ but all be distinct and have distinct roots.

***Proof*:**
First, let's say $f(x)$ is inseparable so that it has multiple roots in a splitting field F:

$$f(x) = (x - \alpha)^n g(x)$$

where $n \geq 2$ and $g(x) \in F[x]$. Then:

$$f'(x) = n(x - \alpha)^{n-1} g(x) + (x - \alpha)^n g'(x)$$

showing that $\gcd(f(x), f'(x)) \neq 1$ in $F[x]$. By corrollary 11.2.2 to Gauss's Lemma, $\gcd(f(x), f'(x)) \neq 1$ in $k[x]$
Conversely, Assume that $f(x)$ and $f'(x)$ share a root (say $\alpha \in \overline{k}$), so that $\gcd(f(x), f'(x)) \neq 1$ in $k[x]$. Since $f(x)$ has a root, we can write it as:

$$f(x) = (x - \alpha) h(x)$$

for $h(x) \in k[x]$ (or $\overline{k}[x]$?). Then the derivative is:

$$f'(x) = h(x) + (x - \alpha) h'(x)$$

Since $f(x)$ and $f'(x)$ share a root, it must be that $(x - \alpha) \big| h(x)$. But then:

$$(x - \alpha)^2 \big| f(x)$$

showing that $f(x)$ has multiple roots, hence inseparable, as we sought to show.

Using this theorem, we can see that $p(x) = x^p - t$ is inseparable since $p'(x) = px^{p-1} = 0$ (since the characteristic of $\mathbb{F}_p(t)$ is p). Thus, $\gcd(x^p - t, 0) = x^p - t \neq 1$. In fact, this captures a bigger result:

Proposition 19.6.7: Separability and irreducible polynomial

Let k be a field and $f(x) \in k[x]$ be an *inseparable and irreducible* polynomial. Then

$$f'(x) = 0$$

***Proof*:**
Since $f(x)$ is inseparable, by proposition 19.6.6, $f(x)$ and $f'(x)$ share a common irreducible factor $q(x)$, that is $f(x) = q(x)g(x)$ for non-constant $g(x) \in k[x]$. However, f is irreducible, and so $q(x)$ must be an associate of $f(x)$ (and hence be of equal degree). However, that implies that it has degree higher than $f'(x)$ if $f'(x) \neq 0$. Since $q(x)\big|f'(x)$, the only possibility left is that $f'(x) = 0$, as we sought to show.

Corollary 19.6.8: Separability Over Characteristic 0

let k be a field of characteristic 0 and $f(x) \in k[x]$ an irreducible polynomial. Then $f(x)$ is seperable.

***Proof*:**
Since f is irreducible, it must have degree at least 2, and so $f'(x) \neq 0$, showing it is separable by proposition 19.6.7

Thus, every algebraic extension of a characteristic 0 field is a separable extension, and fields in which an irreducible polynomial is inseperable must have characteristic p. Even better, given an inseperable polynomial:

$$f(x) = \sum_{i}^{n} a_i x^i$$

since $f'(x) = 0$:

$$f'(x) = \sum_{i}^{n} i a_i x^{i-1} = 0$$

thus, $ia_i = 0$ for all i, so ia_i is a multiple of p. If i itself is a multiple of p, then certainly $ia_i = 0$, and so it must be that $a_i = 0$ for all indicies that are *not* multiples of p. Therefore, we have reduced inseparable polynomials to polynomials of the form:

$$f(x) = a_0 + a_p x^p + a_{2p} + x^{2p} + \cdots + a_{np} x^{np} = \sum_{k=1}^{n} a_k (x^p)^k$$

Thus, an inseparable polynomial $f(x)$ can be thought of as a polynomial in x^p, $f(x) = g(x^p)$. Using this, we can define the following homomorphism which will be useful to characterize fields which have inseparable polynomials:

Definition 19.6.9: Frobenius Homomorphism

Let k be a field such that $p = \mathrm{ch}(k) > 0$. Then the *Frobenius homomorphism* is a map $k \to k$ defined by

$$x \mapsto x^p$$

It should be verified that this map is indeed a field homomorphism. the Frobenius must be injective, being a homomorphism between fields, however it need not be surjective. Using this homomorphism, we will find all fields in which being separable is identical to being algebraic:

Definition 19.6.10: Perfect Field

Let k be a field. Then k is called a *perfect field* if all irreducible polynomials in $k[x]$ are seperable, that is, every algebraic extension is a separable extension.

Proposition 19.6.11: Frobenius Homomorphism And Separability

Let k be a field. Then k is perfect if and only if either:

1. if $\mathrm{ch}(k) = 0$
2. if $\mathrm{ch}(k) > 0$ and the Frobenius homomorphism is surjective.

***Proof*:**
If $\mathrm{ch}(k) = 0$, then by corollary 19.6.8, any irreducible polynomial is already seperable, so let's say $\mathrm{ch}(k) = p$ for some prime p.

($\Leftarrow$) By our discussion from earlier, we know an inseparable irreducible polynomial must be of the form:

$$f(x) = \sum_{i=1}^{n} a_i (x^p)^i$$

Now, since the Frobenius homomorphism is surjective, for everry a_i, there exists a b_i such that $b_i \mapsto a_i$, that is $(b_i)^p = a_i$. Thus, using the fact that this map is indeed a ring homomorphism, we get:

$$f(x) = \sum_{i=1}^{n} b_i^p (x^p)^i = \sum_{i=1}^{n} (b_i x^i)^p = \left(\sum_{i=1}^{n} b_i x^i \right)^p = g(x)^p$$

where $g(x) = \sum_{i=1}^{n} b_i x^i$. But then $f(x)$ has become reducible! This contradicts our assumption of the irreducibility of $f(x)$, showing us that no such polynomial can exist.

($\Rightarrow$) Take the contra-positive, that $\mathrm{ch}(k) \neq 0$ (say $\mathrm{ch}(k) = p$) and that the Frobenius homomorphism is not surjective. Let's say that φ is the Frobenius homomorphism and that $t \in k \setminus \varphi(k)$. I claim that $x^p - t$ does not split into linear factors. Indeed, emulating what we have done in example 19.6.1, we get $x^p - t$ is irreducible, and that $(x-u)^p = x^p - t$

Here is another one: Suppose that T has characteristic $p > 0$, and that $x \mapsto x^p$ is not an automorphism. Since a ring homomorphism from a field into itself is injective or zero, then $x \mapsto x^p$ is not surjective.

Let $a \in T$ be such that $a \neq b^p$ for every $b \in T$. Let $f(x) = x^p - a$. We prove that f is irreducible and inseparable. If α is a root of $x^p - a$, then $x^p - a = (x-a)^p$, so α is a multiple root of f (with multiplicity p), hence f is inseparable. Now, let $g(x)$ be an irreducible factor of $f(x)$. Note that $\alpha \notin T$, otherwise $a = \alpha^p$, contrary to the assumption. Then $g(x) = (x-\alpha)^k$ for some $1 < k \leq p$. Using the binomial theorem, we have

$$g(x) = (x-\alpha)^k = x^k - k\alpha x^{-1} + \cdots + (-\alpha)^k$$

Therefore, $k\alpha \in T$. Since $\alpha \notin T$, then $k = p$, so $g = f$. Hence, f is irreducible.

Corollary 19.6.12: Finite Fields Are Perfect

let k be a finite field. Then k is perfect

***Proof*:**
Since a finite field must have a $\mathrm{ch}(k) = p > 0$, the Frobenius homomorphism is well-defined. Furthermore, Since k is a field, the homomorphism is injective. But an injective map between two finite sets is surjective, hence the Frobenius homomorphism is surjective, and so by proposition 19.6.11 k is perfect.

Thus, we've shown that over most fields we have worked with so far, they are separable. Inseparable extensions do not behave so well (as we'll see, their galois group is [almost] trivial, and hence we get little information about them through galois theory), and so more focus is given in understanding the more common subclass of seperable extensions of a field.

19.6.1 Separability and Embeddings

It should be noted that there *are* separable extensions even if a field is not perfect. For example $x^6 - tx^3 + t \in \mathbb{F}_3(t)[x]$ is irreducible over $\mathbb{F}_3(t)$ and factors into unique linear terms in the splitting field, however the field is not perfect since $x^6 - x^3 + 1$ is irreducible and inseparable. Therefore, we need a better way of finding separable polynomial without relying on nice properties of the ground field. This will be part of the goal of this section.

To characterize separable extensions without relying on k being perfect, we will take advantage that a k-homomorphism is determined by where it maps the roots of minimal polynomials of elements of F. In this way, we get an idea of how many distinct roots a minmial polynomial contains. Recall that we have been able to characterize normal fields through k-homomorphisms: if $\sigma : F \to \overline{k}$ is a k-homomorphism and F/k is normal, then $\sigma(F) = F$, which as we talked about in remark 19.5.14 shows that normal extensions shows that we can characterize all k-homomorphisms as in fact being k-automorphisms (in the sense that all isomorphic copies of F are just symmetries of F). In general, separable extensions need not be normal, and hence they might have multiple embeddings into $\overline{k}$. However, understanding all possible k-homomorphisms still gives us a lot of information about our field as was just pointed out. Studying these embeddings will give us some more information about separable extensions, and will allow us to answer the question of whether an extension $k(\alpha)/k$ is separable if α is separable over k, and to see that any algebraic extension can further be broken down into a separable extension followed by a "purely inseparable" extension: F_{sep}/k and F/F_{sep}.

Definition 19.6.13: Separable Degree

Let F/k be a field extension. Then the number of k-homomorphism $F \to \overline{k}$ called the *separable degree* and is denoted

$$[F : k]_s = \mathrm{Hom}_k(F, \overline{k}) = \mathrm{Emb}_k(F)$$

where $\mathrm{Emb}_k(F)$ is shortand for all possible k-embeddings of F into the algebraic closure of F [a]

[a] Note that for any two algebraic closures $\overline{k}_1, \overline{k}_2$, $\mathrm{Hom}(F, \overline{k}_1) \cong \mathrm{Hom}(F, \overline{k}_2)$, making the definition independent of particular algebraic closure, and hence the notation $\mathrm{Emb}_k(F)$ unambiguous

For the curious reader who would want to see the connection between separability and the separable degree and wants a hint before looking at the solution, they may want to think about what happens to the root, since the following theorem is a generalization of theorem 19.2.27.

19.6.14 moment to ponder

Proposition 19.6.15: Separable Degree of Simple Extensions

Let $k \subseteq k(\alpha)$ be a simple algebraic extension. Then $[k(\alpha) : k]_s$ is equal to the number of distincts roots in $\overline{k}$ of the minimal polynomial of α. In particular,

$$[k(\alpha) : k]_s \leq [k(\alpha) : k]$$

with equality if and only if α is separable over k

Notice how this proposition gives us a way to check the separability of the element α without any appeal to k (i.e., without knowing whether k is perfect or not). In fact more is true: $[k(\alpha) : k]_s \mid [k(\alpha) : k]$. The proof of this fact is a bit more laborious than just showing the inequality since it requires the notion of *separable closure*, which is simply the intersection of all fields that are separable, and so we will leave it as an exercise (I think I'll actually incorporate this material).

***Proof*:**
This proof will be very similar to that of corrollary 19.2.29. Consider the collection of k-homomorpihism $\{\iota_r : k(\alpha) \to \overline{k}\}$. By corrollary 19.2.29, we can associate each of these homomorphisms with where a root r of the minimal polynomial $m_{\alpha,k}(x)$ maps to, i.e. $\iota_r(\alpha) = r$. We will start by showing that this correspondence is bijective:

$$\{\iota_r : k(\alpha) \to \overline{k}\} \leftrightarrow \{r\}$$

Injectivity comes easily since ι_r fixes k, and so it is completely determined by where α maps to, which necessarily is a root of the minimal polynomial for α. To show surjectivity, consider $\beta \in \overline{k}$ which is some root of the minimal polynomial $m_{\alpha,k}(x)$. Then by lemma 19.5.7, $k(\beta) \subseteq \overline{k}$ exists, and by theorem 19.2.27, there exists a unique isomorphism $k(\alpha) \xrightarrow{\sim} k(\beta)$. Hence, combining with $k(\beta) \subseteq \overline{k}$, we get an extension $k(\alpha) \subseteq \overline{k}$ that maps α to the root β, showing surjectivity.

Thus, we have establisehd a bijection between the roots of the minimal polynomial and the number of embeddings of a simple algebraic extension. Since the degree of α is the degree of the minimal polynomial, which is less than or equal to the number of roots the minimal polynomial contains, we have that:

$$[k(\alpha) : k]_s \leq [k(\alpha) : k]$$

If furthermore $k(\alpha)$ is separable, then the number of roots is the degree of the polynomial, and hence:

$$[k(\alpha) : k]_s = [k(\alpha) : k]$$

completing the proof.

Corollary 19.6.16: Tower Law For Finite Separable Extensions

Furthermore, if $F/E/k$ is a sequence of algebraic extensions, $[F:k]_s$ is finite if and only if $[F:E]_s$ and $[E:k]_s$ is finite, and furthermore:

$$[F:k]_s = [F:E]_s[E:k]_s$$

***Proof*:**
We'll show that if $[E:k]_s$ or $[F:E]_s$ are infinite, so must $[F:k]_s$, and then show that if they are both finite, so must $[F:k]_s$ be. Let $\varphi : E \to \overline{k}$ be a k-homomorphism. Then since F/E exists and is algebraic over E (and hence k), by lemma 19.5.7, there exists an exension of φ to $\Phi : F \to \overline{k}$ fixing E (i.e. $\Phi|_E = \varphi$). Furthermore, an extension F into $\overline{E} = \overline{k}$ fixing E extends *a fortiori* the identity on k. Thus, if either $[E:k]_s$ or $[F:E]_s$ is infinite, so must $[F:k]_s$ be.

Conversely if $[F:E]_s$ and $[E:k]_s$ is finite, we'll show $[F:k]_s$ is finite and the multiplicity formula holds. This is almost immediate: for any embedding $F \to \overline{k}$ that extends k, we can can first find all extensions from k to E fixing k, of which there are $[E:k]_s$ possible ways of doing so, and then and then extend any chosen embedding from F to $\overline{E} = \overline{k}$ ($F \to \overline{E}$) fixing E in $[F:E]_s$ different ways, giving us a total of $[F:E]_s[E:k]_s$ total possible extensions, completing the proof.

Proposition 19.6.17: Finite Separable Extension Equivalence

Let F/k be a finite extension. Then $[F:k]_s \leq [F:k]$, and the following are equivalent:

1. $F = k(\alpha_1, ..., \alpha_n)$ where each α_i, is separable over k
2. F/k is a separable extension
3. $[F:k]_s = [F:k]$

***Proof*:**
We'll first generlize proposition 19.6.15 for finite extensions to show that $[F:k]_s \leq [F:k]$. Since F/k is finite, F is finitely generated over k, so let $F = k(\alpha_1, ..., \alpha_n)$. Using proposition 19.6.15, corollary 19.6.16, and theorem 19.3.7:

$$\begin{aligned}
[F:k]_s &= [k(\alpha_1, ..., \alpha_{n-1})(\alpha_n) : k(\alpha_1, ..., \alpha_{n-1})]_s \cdots [k(\alpha_1) : k]_s \\
&\leq [k(\alpha_1, ..., \alpha_{n-1})(\alpha_n) : k(\alpha_1, ..., \alpha_{n-1})] \cdots [k(\alpha_1) : k] \\
&= [F:k]
\end{aligned}$$

showing the inequality

(1 $\Rightarrow$ 3) Since each α_i is separable over k, it is separable over $k(\alpha_1, ..., \alpha_{i-1})$ (the minimal polynomial of α_i over the intermediate field divides the minimal polynomial over the ground field), and so the inequalites in the previous argument are in fact all equalities by corollary 19.6.16

(3 ⇒ 2) Let $[F:k]_s = [F:k]$, take $\alpha \in F$, and consider $k \subseteq k(\alpha) \subseteq F$. Since F is a finite extension, by corollary 19.6.16,

$$[F:k(\alpha)]_s[k(\alpha):k]_s = [F:k]_s = [F:k] = [F:k(\alpha)][k(\alpha):k]$$

Thus, we have two numbers, say a, b such that $a \leq A$, $b \leq B$ and $ab = AB$. This forces $a = A$ and $b = B$, and so we can cancel $[F:k(\alpha)]_s$ and $[F:k(\alpha)]$ on both sides to get:

$$[k(\alpha):k]_s = [k(\alpha):k]$$

which, by proposition 19.6.15 shows that α is separable. Since $\alpha \in F$ was arbitrary, this shows that all elements of F are separable, and hence F/k is a separable extension.

(2 ⇒ 1) Since F/k is separable and F is finite, $F = k(\alpha_1, ..., \alpha_n)$, and each α_i must be separable by definition.

We can finally answer the question posed in remark ref:RemarkSeparableEx: if α is separable over k, then so is $1+\alpha$, α^2, and any other combination of α, since $k(\alpha)/k$ is separable.

Proposition 19.6.18: Tower Law For Separable Extensions

Let $F/E/k$ be a sequence of algebraic extensions. Then F/k is separable if and only if F/E and E/k is separable.

***Proof*:**
Suppose first that F/k is separable. Then E/k is separable, since every element of E lies in F. And for $\alpha \in F$ the minimal polynomial $m_{\alpha,E}$ divides $m_{\alpha,k}$ in $E[x]$, because $m_{\alpha,k}$ lies in $E[x]$ and has α as a root; a divisor of a polynomial with no repeated roots has none either, so $m_{\alpha,E}$ is separable and F/E is a separable extension.

Conversely, let F/E and E/k both be separable and take $\alpha \in F$. The point is to replace E, which may be enormous, by the finite piece of it that the minimal polynomial of α sees. Write

$$m_{\alpha,E}(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0 \qquad c_j \in E$$

and put $K = k(c_0, ..., c_{n-1})$. Each c_j lies in E and so is separable over k, whence K/k is a finite separable extension by proposition 19.6.17.

Now $m_{\alpha,E}$ has all its coefficients in K, so α is algebraic over K and $m_{\alpha,K}$ divides $m_{\alpha,E}$ in $K[x]$. The latter is separable because F/E is, so $m_{\alpha,K}$ is separable and $K(\alpha)/K$ is a finite separable extension. Multiplying separable degrees along $K(\alpha)/K/k$ with corollary 19.6.16, and ordinary degrees with theorem 19.3.7,

$$[K(\alpha):k]_s = [K(\alpha):K]_s\,[K:k]_s = [K(\alpha):K]\,[K:k] = [K(\alpha):k]$$

using proposition 19.6.17 on each of the two factors. That same proposition, read backwards, makes $K(\alpha)/k$ separable, so α is separable over k. As $\alpha \in F$ was arbitrary, F/k is separable, as we sought to show.

Note that along with the fact that an extension is normal (implying the minimal polynomial splits), we get a bijection between the roots of the minimal polynomial and the automorphisms fixing the base field. For this reason, this will be one of the definitions (out of many equivalent definitions) of a galois extension.

Exercise 19.6.1

1. Let $K/\mathbb{Q}$ be a finite extension. Show that there exists an injective ring homomorphism $K \hookrightarrow \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ for appropriate r_1, r_2. This is called the *Archimedean embedding* and the codomain is called the *Minkowski space*; both are taken up in [5, the Archimedean embedding].

19.6.2 Separability and Simple Extensions

Finally, we show that all finite-degree separable fields (of infinite cardinality) are in fact simple. This means that when we were working with the finite extension F/k, we can always find an α where $F = k(\alpha)$. We will show the case for fields of finite cardinality in the next section.

Proposition 19.6.19: Algebraic And Simple Extensions

Let F/k be an algebraic extension. Then F is simple if and only if the number of distinct intermediate fields $k \subseteq E \subseteq F$ is finite

***Proof*:**
Let F/k be simple, so that $F = k(\alpha)$. Since F is algebraic, let $m_{\alpha,k}(x)$ be the minimal polynomial over k. Embed $k(\alpha)$ into an algebraic closure $\overline{k}$. If E is an intermediary field $k \subseteq E \subseteq F$, then $k(\alpha) = F = E(\alpha)$ is also a simple algebraic extension. Denote $m_{E,\alpha}(x)$ to be the minimal polynomial for $E(\alpha)$ over E. Since $m_{k,\alpha}(x) \in E[x]$ for all E and $m_{k,\alpha}(\alpha) = 0$, we know that each $m_{E,\alpha}(x)$ is a factor of $m_{k,\alpha}(x)$.

I claim that E is determined by $m_{E,\alpha}(x)$. Since $m_{k,\alpha}(x)$ has finitely many factors in $\overline{k}$, this will prove there are only finitely many intermediate fields, completing one direction of the proof.

Proceeding with proving the claim, I will show that E is generated (over k) by the coefficients of $m_{E,\alpha}(x)$. To see this, let E' be the set generated by k and the coefficients of $m_{E,\alpha}(x)$. Then $m_{E,\alpha}(x) \in E'[x]$, and since $m_{E,\alpha}(x)$ is irreducible in E, it is irreducible in E', and in fact minimal. Note too that $E'(\alpha) = k(\alpha) = E(\alpha)$, so $[k(\alpha) : E'] = \deg(m_{E,\alpha}(x)) = [k(\alpha) : E]$

Now, we have $E' \subseteq E \subseteq k(\alpha)$, therefore by the tower law:

$$\deg(m_{E,\alpha}(x)) = [k(\alpha) : E'] = [k(\alpha) : E][E : E'] = \deg(m_{E,\alpha}(x))[E : E']$$

Thus, it must be that $[E : E'] = 1$, hence $E = E'$, showing that E is indeed completely determined by $m_{E,\alpha}(x)$, hence there can only be finiely many intermediate fields.

Conversely, let's assume there are finitely many intermediate field $k \subseteq E \subseteq F$. Then it must be that the extension F/k is finite, or else we could construct an infinite sequene of sub-extensions $k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2) \subseteq \cdots F$. Thus, F/k is algebraic. We will deal with the case of F being a finite field in corrollary 19.7.15. Thus, for now, assume k is a field of infinite cardinality. We have to show that every $F = k(\alpha_1, ..., \alpha_n)$ is in fact simple.

Arguing inductively, it suffices to show that $k(\alpha, \beta)$ is simple. For every $c \in k$, we have:

$$k \subseteq k(c\alpha + \beta) \subseteq k(\alpha, \beta)$$

Since there are only finitely many intermediate fields, it must be that for some two $c, c' \in k$, $c \neq c$:

$$k(c\alpha + \beta) = k(c'\alpha + \beta)$$

But then:

$$\alpha = \frac{(c'\alpha + \beta) - (c\alpha + \beta)}{c' - c} \in k(c\alpha + \beta)$$
$$\beta = (c\alpha + \beta) - c\alpha \in k(c\alpha + \beta)$$

Thus, $k(\alpha, \beta) \subseteq k(c\alpha + \beta)$. Since $k(c\alpha + \beta) \subseteq k(\alpha, \beta)$, we have an equality, showing that it is indeed simple, completing the proof.

Theorem 19.6.20: Primitive Element Theorem

Let F/k be a finite separable extension. Then F is also a *simple* extension.

***Proof*:**
Arguing inductively, we may take $F = k(\alpha, \beta)$ with α and β separable over k (and hence algebraic). As before, the finite case will be dealt with in corrollary 19.7.15.

Let I be the number of k-homomorphisms $\iota : F \to \overline{k}$ (so $|I| = [F : k]_s$). Choose two $\iota, \iota' \in I$ where $\iota \neq \iota$. If x is an indeterminate, then the polynomials

$$\iota(\alpha)x + \iota(\beta) \quad \iota'(\alpha)x + \iota'(\beta)$$

are different (or else $\iota'(\alpha) = \iota(\alpha)$ and $\iota'(\beta) = \iota(\beta)$ so ι, ι' would act in the same way on the whole of $k(\alpha, \beta)$ forcing $\iota = \iota'$)

Therefore, define:

$$f(x) = \prod_{\iota \neq \iota'} ((\iota(\alpha)x + \iota(\beta)) - (\iota'(\alpha)x + \iota'(\beta))) \in \overline{k}[x]$$

The polynomial $f(x)$ cannot be identically 0 since k is infinite. Thus, take $c \in k$ such that $f(c) \neq 0$. So, a distinct map $\iota \in I$ will map $\gamma = c\alpha + \beta$ to distinct elements $\iota(\alpha)c + \iota(\beta)$ (recall $\iota(c) = c$). Since $|I| = [F : k]_s$, and each $\iota(\gamma)$ is a root of the minimal polynomial of γ over k, we have:

$$[F : k]_s \leq [k(\gamma) : k] \leq [F : k]$$

By assumption, the extension is separable, so we have equalities above:

$$[F : k]_s = [k(\gamma) : k] = [F : k]$$

Thus:

$$F = k(\gamma)$$

as we sought to show.

Using this, we can extend corollary 19.2.29 from simple to finite extensions:

Corollary 19.6.21: Order of $\mathrm{Aut}_k(F)$ Of Finite Extension

Let F/k be a finite seprable extension. Then:

$$|\mathrm{Aut}_k(F)| \leq [F : k]$$

with equlaity if and only if F/k is a finite separable normal extension.

***Proof*:**
By theorem 19.6.20, if K is finite and separable, it is simple, and so the inequality immediately holds by corollary 19.2.29. Equality holds if the minimal polynomial splits, which it does over it's splitting field, which is a normal extension by theorem 19.5.13. Conversely, if K/F is normal, then $f(x)$ splits completly in F and has distinct roots since α is separable over k, establishing equalitly once again, completing the proof.

Note that the separability condition is necessary:

Example 19.6.22: Nonseparable Nonsimple Extension
Let $F = \mathbb{F}_p(u, v)$ be the field with two indeterminate variables over $\mathbb{F}_p$, and subfield $k = \mathbb{F}_p(s, t)$ such that $s = u^p$ and $t = v^p$. Then F/k is an algebraic extension; the minimal polynomial of u over k is $x^p - s$, and the minimal polynomial of v over $k(u)$ is $y^p - t$. Thus:

$$[F : k] = p^2$$

Arguing like before, As c ranges in k, we get the intermediate fields:

$$k \subseteq k(cu + v) \subseteq F$$

If any two chocies c, c' get give the same intermediate field, then we can deduce $k(u, v) = k(cu + v)$ as we've argued earlier. This would then give us:

$$[k(cu + v) : k] = p^2$$

However, the minimal polynomial of $cu + v$ has degree p:

$$(cu + v)^p = c^p + st \in k(s, t) = k$$

Thus, since $k = \mathbb{F}_p(s, t)$ is infinite, there are infinitely many intermediate fields, and it follows that F is *not* a simple extension of k by proposition 19.6.19

As a final result, we state the following that shall be handy for [13, Rational Maps and Birational Classification]:

Corollary 19.6.23: Lüroh's Theorem

Let $k(t)/k$ be a simple extension and $E \subseteq k(t)$ an intermediary field containing k. Then there exists a polynomial g such that
$$E = k(g(t))$$

***Proof*:**
Exercise. Sketch of proof: use the primitive element theorem, use Gauss's lemma, and find the minimal polynomial of the primitive element.

Exercise 19.6.2

1. Show that If E/k and F/k be field extensions, E/k is separable, then so is EF/F. Show that if E/k and F/k are separable, so is EF/F.
2. (exercise on automatic differentiation? $k[x, \epsilon]/(\epsilon^2)$)
3. Let F/k is simple so that $F = k(\alpha)$. then if L is the splitting field for a [not necessarily separable] minimal polynomial $m_{\alpha,k}(x)$, $L = \prod_{\sigma \in \mathrm{Emb}_k(F)} \sigma(F)$. If F is normal, then $L = F$, and if F separable, we have an idea of the number of σ exists, namely the degree of $k(\alpha)$ over k)

19.6.3 Important Functional: Resultant and Discriminant

One very useful functional on $R[x]$ that allows us to detect when a polynomial has a root even R need not contain the roots is the following:

Theorem 19.6.24: Resultant

Let R be an integral domain and $f(x), g(x) \in R[x]$ be two polynomials:

$$\begin{aligned} f(x) &= a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \\ g(x) &= b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0 \end{aligned}$$

where $a_n, b_m \neq 0$ so that their degree is the highest order term. Then f and g have a common root if and only if:

$$\operatorname{Res}(f,g) := \det \begin{pmatrix} a_n & 0 & \cdots & 0 & b_m & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{m-1} & b_m & \cdots & 0 \\ a_{n-2} & a_{n-1} & \cdots & 0 & b_{m-2} & b_{m-1} & \ddots & 0 \\ \vdots & \vdots & \ddots & a_n & \vdots & \vdots & \ddots & b_m \\ a_0 & a_1 & \cdots & \vdots & b_0 & b_1 & \cdots & \vdots \\ 0 & a_0 & \ddots & \vdots & 0 & b_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1} & \vdots & \vdots & \ddots & b_1 \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{pmatrix} = 0$$

is zero.

***Proof*:**
Let $f, g \in R[x]$ be (not necessarily monic) polynomials with degree n and m respectively. If either $f = 0$ or $g = 0$, there is nothing to prove, so assume both are nonzero. Let $\mathcal{P}_n$, $\mathcal{P}_m$ be the free R-modules of polynomial up to degree strictly less than n and strictly less than m respectively; then $\mathcal{P}_n, \mathcal{P}_m$ have dimension (rank)[a] n, m, so $\mathcal{P}_n \times \mathcal{P}_m$ has dimension $n + m$. Define:

$$\varphi : \mathcal{P}_m \times \mathcal{P}_n \to \mathcal{P}_{n+m} \qquad \varphi(p, q) = fp + gq$$

As R is an integral domain, $\deg(fp + gq) = n + m$, otherwise it may be of a smaller degree and we may get unwanted zero-polynomials. The function φ is linear as:

$$\begin{aligned} \varphi((p_1, q_1) + (p_2, q_2)) &= \varphi(p_1 + q_2, q_1 + q_2) \\ &= f(p_1 + p_2) + g(q_1 + q_2) \\ &= fp_1 + fp_2 + gq_1 + gq_2 \\ &= (fp_1 + gq_1) + (fp_2 + gq_2) \\ &= \varphi(p_1, q_1) + \varphi(p_2, q_2) \end{aligned}$$

Hence, this transformation has a matrix representation. Choosing the (ordered) basis $(x^{n-1}, x^{n-2}, ..., x, 1)$ for $\mathcal{P}_n$ and the (ordered) basis $(y^{m-1}, y^{m-2}, ..., y, 1)$ for $\mathcal{P}_m$, we get the matrix representation:

$$\begin{pmatrix} a_n & 0 & \cdots & 0 & b_m & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{m-1} & b_m & \cdots & 0 \\ a_{n-2} & a_{n-1} & \cdots & 0 & b_{m-2} & b_{m-1} & \ddots & 0 \\ \vdots & \vdots & \ddots & a_n & \vdots & \vdots & \ddots & b_m \\ a_0 & a_1 & \cdots & \vdots & b_0 & b_1 & \cdots & \vdots \\ 0 & a_0 & \ddots & \vdots & 0 & b_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1} & \vdots & \vdots & \ddots & b_1 \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{pmatrix}$$

Note that $f \notin \mathcal{P}_n$ and $g \notin \mathcal{P}_m$ or else there would always exist $\varphi(-g, f) = -fg + fg = 0$, which would ruin the following argument using the determinant. Let M represent this matrix.

If f, g share a common $\alpha \in \overline{K}$ where $K = \text{Frac}(R)$, then there exists an $h \in \overline{K}[x]$ such that $h(x)f_0(x) = f(x)$ and $h(x)g_0(x) = g(x)$ for appropriate factors $f_0, g_0 \in \overline{K}[x]$. Then simply take $\varphi(-g_0, f_0) = 0$ so $\det(M) = 0$ (see exercise 13.4.1(13.4.1 (5)))

Conversely, suppose $\det(M) = 0$. Then there exists p_0, q_0 such that

$$fp_0 + gq_0 = 0 \qquad \Rightarrow \qquad fp_0 = -gq_0 \tag{19.3}$$

We'll use this to show that f, g are not relatively prime. Suppose $(f, g) = 1$ so that there exists s, t such that $ft + sg = 1$. Then $ftp_0 + sgp_0 = p_0$. Substituting $fp_0 = -gq_0$ from equation (19.3) we get:

$$gq_0p_0 + sgp_0 = g(q_0p_0 + sp_0) = p_0$$

As R is an integral domain, $g \big| p_0$ so $p_0 = gh$. Similarly, $q_0 = fr$.

Both of these cause a contradiction. For example, considering $p_0 = gh$, then:

$$m > \deg(p_0) = \deg(gh) = \deg(g) + \deg(h) = m + \epsilon$$

where $\epsilon \in \mathbb{N}$. Thus (f, g) is *not* relatively prime. Then if $K = \text{Frac}(R)$ we have that $\overline{K}[x]$ is a PID and it must be that there exists a unit $h \in \overline{K}[x]$

$$(h) = (f, g)$$

showing f, g have a common root.

[a]Note that as this is over finite dimensions and over commutative rings, the rank is invariant and hence there is no issue when we use the specific rank later in the proof

Definition 19.6.25: Resultant

The determinant in theorem 19.6.24 $\text{Res}(f, g)$ is called the *resultant*. If the base-ring is not clear, we often denote it as $\text{Res}_R(f, g)$ for the desired ring R, and if the indeterminate element is not clear we may denote it as $\text{Res}_x(f, g)$.

The form above is useful as it can be defined over an integral domain R and is better suited for computational purposes as no field extensions are required. The following is often useful when strictly working over algebraically closed fields:

Proposition 19.6.26: Characterizing Resultant using Algebraically Closed Fields

Let $f, g \in R[x]$ be (not necessarily monic) polynomial over an integral domain R, and let $\overline{K}$ be the algebraic closure of $K = \text{Frac}(R)$. Then if $\alpha_1, ..., \alpha_n$ are the roots of f and $\beta_1, ..., \beta_m$ the roots of g respectively:

$$\text{Res}(f, g) = a_n^m b_n^m \prod_{i.j} (\alpha_i - \beta_j)$$

***Proof*:**
exercise.

One of the more important application is to find whether a polynomial has a multiple root:

Definition 19.6.27: Discriminant

Let $f \in R[x]$ be a polynomial over an integral domain. Then:

$$\Delta(f) = \text{disc}(f) = \frac{(-1)^{n(n-2)/2}}{a_n} \text{Res}(f, f')$$

If f is monic,

$$\Delta(f) = \text{disc}(f) = (-1)^{n(n-2)/2} \text{Res}(f, f')$$

Proposition 19.6.28: Discriminant As A Product

Let $f \in R[x]$ be polynomial over an integral domain, $K = \text{Frac}(R)$, and let L/K be the splitting field of f. Then if $\alpha_1, ..., \alpha_n$ are the roots of f,

$$\Delta(f) = \prod_{1 \leq i \leq j \leq n} (\alpha_i - \alpha_j)^2$$

***Proof*:**
exercise

As is immediate after proposition 19.6.6:

Proposition 19.6.29: Environment Title

Let $f \in \mathbb{R}[x]$ be a polynomial. Then if $\Delta(f) = 0$, f has a multiple root in the splitting field.

***Proof*:**
immediate.

here)

Separability has a numerical measure that is often more useful than the definition. A finite extension F/k makes F a finite-dimensional k-vector space, so every element of F acts on it by multiplication and has a trace in the sense of definition 13.4.3. That assignment is a k-linear form on F, and the symmetric bilinear form it induces detects separability exactly.

Definition 19.6.30: Trace of a Finite Field Extension

Let F/k be a finite field extension and let $x \in F$. Multiplication by x is a k-linear endomorphism $m_x \colon F \to F$, and the *trace* of x is

$$\operatorname{tr}_{F/k}(x) \;:=\; \operatorname{tr}(m_x) \in k$$

the trace of that endomorphism in the sense of definition 13.4.3. The *trace form* of F/k is the symmetric k-bilinear form

$$\langle x, y\rangle \;:=\; \operatorname{tr}_{F/k}(xy)$$

Proposition 19.6.31: The Trace Form of a Separable Extension Is Nondegenerate

Let F/k be a finite separable field extension. Then the trace form of F/k is nondegenerate: if $x \in F$ satisfies $\operatorname{tr}_{F/k}(xy) = 0$ for all $y \in F$, then $x = 0$. Equivalently, for any k-basis $e_1, \ldots, e_n$ of F there is a dual basis $e_1^*, \ldots, e_n^*$ with $\operatorname{tr}_{F/k}(e_i e_j^*) = \delta_{ij}$.

***Proof*:**
Step 1: reduce to a computation in one element. By theorem 19.6.20 the extension is simple, say $F = k(\alpha)$ with α of degree n over k and minimal polynomial m. Then $1, \alpha, \ldots, \alpha^{n-1}$ is a k-basis of F, and it suffices to show that the Gram matrix $M = \big(\operatorname{tr}_{F/k}(\alpha^{i+j})\big)_{0\le i,j\le n-1}$ of the trace form in that basis is invertible: by lemma 13.3.23 a bilinear form on a finite-dimensional space is nondegenerate (definition 13.3.22) exactly when its matrix in a basis is invertible, and the criterion is independent of which basis is used.

Step 2: traces are power sums of the roots. In the basis $1, \alpha, \ldots, \alpha^{n-1}$ the matrix of m_α is the companion matrix of m, so the characteristic polynomial of m_α is m itself. Let $\alpha_1, \ldots, \alpha_n$ be the roots of m in a splitting field; they are distinct because F/k is separable, so m is a separable polynomial (definition 19.6.3, proposition 19.6.7). The eigenvalues of the companion matrix are exactly the α_r, so those of m_α^j are the α_r^j, and therefore

$$\operatorname{tr}_{F/k}(\alpha^j) \;=\; \operatorname{tr}\big(m_\alpha^j\big) \;=\; \sum_{r=1}^{n} \alpha_r^j$$

for every $j \ge 0$.

Step 3: a Vandermonde determinant. Let $V = (\alpha_r^i)_{0\le i\le n-1,\,1\le r\le n}$ be the Vandermonde matrix of the roots. By Step 2,

$$M_{ij} \;=\; \sum_{r=1}^{n} \alpha_r^i \alpha_r^j \;=\; \big(VV^{\mathsf T}\big)_{ij}$$

so $M = VV^{\mathsf T}$ and $\det M = (\det V)^2 = \prod_{r<s}(\alpha_r - \alpha_s)^2$. The roots are distinct, so this is nonzero

and M is invertible. The dual basis is then obtained by inverting M, completing the proof.

The dual basis is what makes the trace form a finiteness tool: an element whose products with a fixed basis all have trace in a subring is pinned inside the free module spanned by the dual basis. That is the mechanism behind the finiteness of integral closure in *Commutative Algebra* [8, Finiteness of Integral Closure], and behind the finiteness of the ring of integers of a number field in *Algebraic Number Theory* [5].

Both of those applications need one more thing before the trace form can be brought to bear: the extension in question has to be separable over the subfield chosen as a base. A transcendence basis makes K algebraic over $k(B)$, but says nothing about separability, and in characteristic p the two can come apart. Over a perfect field they can always be reconciled.

Definition 19.6.32: Separating Transcendence Basis

Let K/k be a field extension. A transcendence basis B of K/k is *separating* if K is a separable algebraic extension of $k(B)$. The extension K/k is *separably generated* if it admits one.

Theorem 19.6.33: Perfect Base Fields Give Separating Transcendence Bases

Let k be a perfect field and let $K = k(a_1, \dots, a_n)$ be a finitely generated field extension. Then some subset of $\{a_1, \dots, a_n\}$ is a separating transcendence basis of K over k. In particular every finitely generated extension of a perfect field is separably generated.

***Proof*:**
If $\operatorname{char} k = 0$ every algebraic extension is separable (corollary 19.6.8), so any transcendence basis extracted from the generators by theorem 19.2.13 is separating. Assume $\operatorname{char} k = p > 0$ and induct on n, the case $n = 0$ being trivial.

Step 1: dispose of the independent case. If $a_1, \dots, a_n$ are algebraically independent over k they form a transcendence basis and $K = k(a_1, \dots, a_n)$ is separable over itself, so the basis is separating. Otherwise there is a nonzero $f \in k[X_1, \dots, X_n]$ with $f(a_1, \dots, a_n) = 0$, and we may take f irreducible, replacing it by an irreducible factor that still vanishes.

Step 2: perfectness forces a nonzero partial derivative. Suppose every X_i occurred in f only through p-th powers, so that $f = \sum_\alpha c_\alpha X^{p\alpha}$. Since k is perfect, every c_α is a p-th power, say $c_\alpha = d_\alpha^p$, and then

$$f = \sum_\alpha d_\alpha^p X^{p\alpha} = \Big(\sum_\alpha d_\alpha X^\alpha\Big)^p$$

using that raising to the p-th power is a ring homomorphism in characteristic p (proposition 19.6.11). That contradicts irreducibility of f. So some variable occurs in f outside a p-th power, which is to say $\partial f/\partial X_i \neq 0$ for some i; fix such an i.

Step 3: the i-th generator is separable over the others. Put $F = k(a_1, \dots, \widehat{a_i}, \dots, a_n)$, the subfield generated by the remaining generators. Viewing f as a polynomial in X_i with coefficients in $k[X_1, \dots, \widehat{X_i}, \dots, X_n]$, it is nonconstant in X_i (otherwise $\partial f/\partial X_i = 0$) and irreducible, so up to a unit it is the minimal polynomial of a_i over F. Its derivative in X_i is nonzero and of smaller degree, hence not divisible by the minimal polynomial, so the minimal polynomial has no repeated root by proposition 19.6.6. Thus a_i is separable algebraic over F, and $K = F(a_i)$ is a finite separable extension of F.

Step 4: induct. The field F is generated over k by $n-1$ elements, so by induction some subset B of $\{a_1, \dots, \widehat{a_i}, \dots, a_n\}$ is a separating transcendence basis of F/k. Then K is algebraic over F, which is algebraic over $k(B)$, so K is algebraic over $k(B)$ and B is a transcendence basis of K/k. Separability passes up the tower $k(B) \subseteq F \subseteq K$ by proposition 19.6.18, both steps being separable, so $K/k(B)$ is separable and B is separating, completing the proof.

The hypothesis is not decorative. Over an imperfect field the theorem fails: if $t \in k$ is not a p-th power, the extension $k(u)$ with $u^p = t$ adjoined to a rational function field produces a finitely generated extension with no separating transcendence basis, and the trace form degenerates accordingly.

Exercise 19.6.3

1. Show that if f, g are polynomials of degree n, m respectively then
$$\begin{aligned}(x^{m-1}f(x)) \wedge (x^{m-2}f(x)) \wedge \cdots \wedge f(x) \wedge (x^{n-1}g(x) \wedge \cdots \wedge g(x) \\ = \operatorname{Res}(f,g) x^{n+m-1} \wedge x^{n+m-2} \wedge \cdots \wedge 1\end{aligned}$$
2. Let $f, g \in R[x]$. Show that $\operatorname{Res}(f,g) = 0$ iff f, g have a common divisor in $\operatorname{Frac}(R)$.
3. Show that $\operatorname{Res}(f(x+a), g(x+b)) = \operatorname{Res}(f(x), g(x))$ for $a, b \in R$.
4. Show that $\Delta(fg) = \Delta(f)\Delta(g)\operatorname{Res}(f,g)^2$.
5. Let $f(x) = ax^2 + bx + c$. Show that $\Delta(f) = b^2 - 4ac$.
6. (a bunch about the resultant under ring homomorphism!)
7. (There are a ton of cool exercises that can be put here! Importantly, how the resultant changes when the polynomials change)
8. (on homogeneous polynomials; and homogeneity of Res in general)
9. (A note on how it does *not* generalize nicely to multiple polynomials.

19.7 Application: Cyclotomic Extensions and Finite Fields

In this section, we use the tools given to us to gain some important examples of field extensions and analyse the corresponding $\operatorname{Aut}_k(F)$, and generalize theorem 19.6.20 to show finite separable extensions of finite fields are in fact simple.

We start by using the tools that were developed to study the important class of polynomials of the form $x^n - 1$; such extensions will be called *Cyclotomic extensions*. These polynomials represent the "torsion units" and shall show up everywhere when considering solutions to equations as they often add the necessary "symmetry" to find all the solutions to an algebraic equation. They are also important for the continuation of Field Theory to Kummer Theory, which is the study of the case where $\mathrm{Aut}_k(F)$ is an abelian group, and they form an important first step towards Fermat's Last theorem through their application to solutions of Elliptic curves[23].

After having classified Cyclotomic extensions, we shall finish the section with the study of extensions of finite fields. As it turns out, all finite extensions will be splitting fields of polynomials of the form $x(x^{p^d}-1)$, meaning they will have a very simple structure and automorphism group.

19.7.1 Cyclotomic Polynomials and Fields

We start off with some reminders:

Definition 19.7.1: Roots Of Unity Reminder

We will denote $\zeta_n = e^{\frac{2\pi i}{n}}$, and the cyclic subgroup they generated to be μ_n. Note that:

$$\zeta_n^n = e^{\frac{2\pi i k}{k}} = e^0 = 1$$

If $\langle \zeta_n^k \rangle = \mu_n$, then this element is called the *primitve root of unity.*

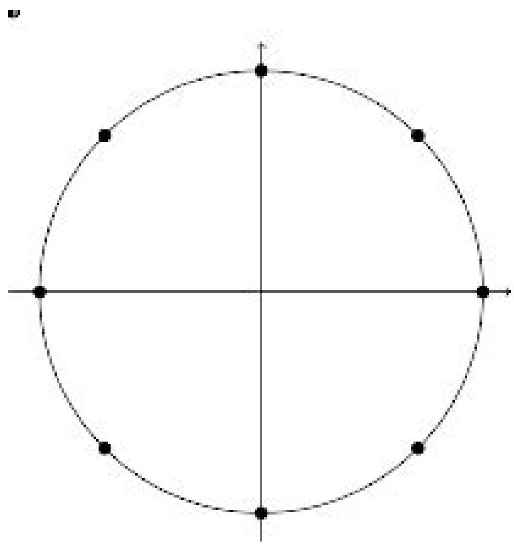
The element ζ_n is always a root of $x^n - 1$, and each ζ_n^k, $1 \leq k \leq n$ where $\gcd(n,k) = 1$ consist of the rest of the roots of $x^n - 1$, for

$$(\zeta_n^k)^n = 1$$

and if $\gcd(n,k) = \ell$, then:

$$(\zeta_n^k)^\ell = 1$$

It is an easy mistake to think that $\zeta_2 = i$, however $\zeta_2 = -1$ and $\zeta_4 = i$. These elements always form a group under multiplication, which we labeled μ_n. This group always lies on the unit circle, for example:



[23]if $\mathrm{ch}(F) \notin 2,3$, then an elliptic curve is a polynomial of the form $y^2 = x^3 + ax + b$, $a, b \in F$. A more technical definition is given when $\mathrm{ch}(F) \in \{2,3\}$ that would be presented in a course on elliptic curves

If you know about the euler representation of complex numbers:

$$re^{i\theta} = r(\cos(\theta) + i\sin(\theta))$$

Then $e^{\frac{2\pi i}{n}} = \zeta_n = \cos(2\pi/n) + i\sin(2\pi/n)$. Other identities we get are:

1. $\zeta_3 = \frac{-1+i\sqrt{3}}{2}$
2. $\zeta_5 = \frac{\sqrt{5}-1}{4} + \frac{\sqrt{10+2\sqrt{5}}}{4}i$
3. $\zeta_6 = \frac{1+i\sqrt{3}}{2}$ (to not be confused with ζ_3)
4. $\zeta_8 = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

These can be deduced using the results from section 19.4. In paticular Gauss proved that primitive nth roots of unity ζ_n can be constructed using the square-roots, additions, subtraction, multiplication, and division of rational numbers, which we saw implies that n is a power of two, or in our case we would require that n is the product of two discint *fermat primes*.

Every root of unity of a root of $x^n - 1$ for some n, however this polynomial is not irreducible, namely $x^n - 1 = (x-1)p(x)$. The polynomial $p(x)$ will turn out to be irreducible, and hence is given a name:

Definition 19.7.2: Cyclotomic Polynomials

Let:

$$\Phi_n(x) = \prod_{\zeta \text{ primitive } n\text{th root of } 1} (x - \zeta) = \prod_{\substack{1 \le m \le n \\ \gcd(m,n)=1}} (x - \zeta_n^m)$$

Then the resulting polynomial is called the *nth cyclotomic polynomial*

As we have seen in the section on number theory (section 0.2), there are $\varphi(n)$ primitive roots of 1, where φ is Euler's totient function. The term cyclotomic comes from the greek *cyclo* (circle) and *tomos* (divide), owing to how the roots of unity divide the circle into regular sections. Though not obvious, this polynomial has rational coefficients and is irreducible over $\mathbb{Q}$ (which we'll show).

Example 19.7.3: Cyclotomic Polynomials

In the case of $n = p$ is a prime, it is simple to see it's irreducible. Every non-identity element of μ_p is a generator, i.e. every pth root of 1 is primitive except 1. In fact, 1 is a root of $x^p - 1$ ($1^p - 1 = 0$), and so we get:

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$$

which is irreducible over $\mathbb{Q}$ using Eisenstein's trick along with shifting (noting that p divides $\binom{n}{k}$). Furthermore, $\mu_p = C_p = \mathbb{Z}/p\mathbb{Z}$.

Aside: for those who know Taylor series and the exponential function, notice how close $\Phi_p(x)$ resembles the pariatal sum for the taylor expansion of e^x:

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^p}{p!}$$

the main difference being the coefficient of the partial taylor expansion of e^x being $1/n!$. This reflects on the "rotation" nature of e^x, as well as it's link to roots of unity. However, note that e^x has no roots (even when expended to $\mathbb{C}$, since $e^z \cdot e^{-z} = 1$). Hence, though the partial expansion may have root, even roots that resemble the roots of unity, taylor expansion, the "infinite polynomial representation" need not have roots!

For ease of reference, we record what happens when n is not prime:

Lemma 19.7.4: Cyclotomic Polynomial divisor Representation

For all $n > 0$,

$$x^n - 1 = \prod_{1 \leq d \mid n} \Phi_d(x)$$

***Proof*:**
For all ζ_n^k, $0 \leq k \leq n-1$, $(\zeta_n^k) - 1 = 0$. Hence, $x^n - 1$ splits into products

$$x^n - 1 = \prod_{k}^{n} (x - \zeta_k)$$

If $n = dk$ then $1 = \zeta_n^n = (\zeta_n^d)^k$, hence every dth root of unity is an nth root of unity. In particular, every primitive dth root of unity is an nth root of unity. Conversely, if $\zeta_n^d \in \mu_n$, then ζ_n^d forms cyclic subgroup, say μ_d where d is a divisor of n. But then ζ_n^d is a primitive dth root of unity for some divisor d. This establishes:

$$x^n - 1 = \prod_{1 \leq d \mid n} \left(\prod_{\substack{\zeta \text{ primitive} \\ d\text{th root of } 1}} (x - \zeta) \right) = \prod_{1 \leq d \mid n} \Phi_d(x)$$

as we sought to show.

Corollary 19.7.5: Integer Coefficients

The cyclotomic polynomials $\Phi_n(x)$ have integer coefficients

***Proof*:**
This is a proof by [strong] induction. The case of $\Phi_1(x) = x - 1$ is immediate, so assume that for $m < n$, Φ_m is a monic polynomial with integer coefficients. Then:

$$f(x) = \prod_{\substack{1 \leq d \mid n \\ d < n}} \Phi_d(x)$$

is a monic polynomial with degree $< n$. Since $f(x), x^n - 1 \in \mathbb{Z}[x]$ are two monic polynomials with integer ceofficients, we may use the euclidean algorithm to find $q(x), r(x) \in \mathbb{Z}[x]$, $\deg(r(x)) < \deg(f(x))$ or $r(x) = 0$ where

$$x^n - 1 = f(x)q(x) + r(x)$$

Then by lemma 19.7.4 we have

$$x^n - 1 = f(x)\Phi_n(x)$$

Hence, we get

$$f(x)(q(x) - \Phi_n(x)) = r(x)$$

But that implies $\deg(r(x)) > \deg(f(x))$, a contradiction. Thus $r(x) = 0$, giving us $f(x)q(x) = f(x)\Phi_n(x)$, which implies $\Phi_n(x) = q(x) \in \mathbb{Z}[x]$, as we sought to show.

Example 19.7.6: Cyclotomic Polynomial

1. Let $p(x) = x^4 - 1 = \Phi_1(x)\Phi_2(x)\Phi_4(x)$. Then:

$$\Phi_4(x) = \frac{x^4 - 1}{x^2 - 1} = x^2 + 1$$

2. Let $p(x) = x^6 - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_6(x)$. Then:

$$\Phi_6(x) = \frac{x^6 - 1}{(x^2 - 1)(x^2 + x + 1)} = x^2 - x + 1$$

3. Let $p(x) = x^{12} - 1 = \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x)$. Then:

$$\Phi_{12}(x) = \frac{x^{12} - 1}{(x^6 - 1)(x^2 + 1)} = x^4 - x^2 + 1$$

It might seem that all the coefficients are 0 or ± 1. However, the first counter-example to this problem is when $n = 105$, and in general the coefficients can be as large as we might want

Proposition 19.7.7: Irreducibility Of Cyclotomic Polynomials

For all $n \in \mathbb{N}_{>0}$, $\Phi_n(x) \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$

Note that since $\Phi_n(x)$ is monic, by Gauss's lemma, irreducibility in $\mathbb{Q}[x]$ is equivalen to ireducibility in $\mathbb{Z}[x]$.

***Proof*:**
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Definition 19.7.8: Cyclotomic Field

The splitting field $\mathbb{Q}(\zeta_n)$ for the polynomial $x^n - 1$ over $\mathbb{Q}$ is the nth *cyclotomic field*

Proposition 19.7.9: Cyclotomic Field Automorphism

$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

***Proof*:**
First, the cardinality of both are the same since cyclotomic fields are a splitting field with degree $\varphi(n)$ (Euler's Totient) while $(\mathbb{Z}/n\mathbb{Z})^\times$ has order $\varphi(n)$ as well. Define

$$j : (\mathbb{Z}/n\mathbb{Z})^\times \to \mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \qquad [m]_n \mapsto (\zeta_n \mapsto \zeta_n^m)$$

This map is certainly independent of representative and sends roots of polynomials of $m_{\zeta_n,\mathbb{Q}}(x)$ to themselves making it an automorphism.

19.7.2 Finite Fields

Let F be a finite field of characteristic p. Then as we remarked under definition 19.1.8,

$$F/\mathbb{F}_p$$

where $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Since F is finite, let $d = [F : \mathbb{F}_p]$. Since F is a d-dimensional vector-space over $\mathbb{F}_p$, $F \cong_{\mathbf{Vect}} \mathbb{F}_p^d$, hence the order of $|F|$ is a power of p: $|F| = p^d$.

With everything we've learned, we can now classify all finite fields. Interestingly, we get essentially the best possible result we might ask for when classifying objects: for every order p^d, we get a unique field up to isomorphism:

Theorem 19.7.10: Classifying Finite Fields

Let $q = p^d$ be a power of the prime number p. Then the polynomial $x^q - x$ is separable over $\mathbb{F}_p$, and the splitting field of the polynomial $x^q - x$ over $\mathbb{F}_p$ is a field with exactly q elements. Conversely, let F be a field with exactly q elements. Then F is a splitting field for $x^q - x$ over $\mathbb{F}_p$

Hence, all finite extension of finite fields is abelian, and even better cyclic.

***Proof*:**
Let F be the splitting field of $x^q - x$ over $\mathbb{F}_p$. Then necessarily F is finite, and hence it's perfect by corollary 19.6.12, and hence it's a separable extension, meaning F contains q distinct roots of $x^q - x$. We need to show that every element of F is a root (i.e. $|F| = q$). Let R be the set of roots of $x^q - x$ from F. Then I claim that R is a field. Since F is the smallest field containing all the roots, it follows that $R = F$.

We simply need to show it's a subring, and since all nonzero elements are units, it is a field. For any $a, b \in R$, since $a^q - a = 0$, $b^q - b = 0$, then $a^q = a$ and $b^q = b$. Then using the binomial theorem and the characteristic of F, we get:

$$(a - b)^q = a^q + (-1)^q b^q = a - b$$

where $(-1)^q = -1$ if p is odd, and $(-1)^q = 1 = -1$ if $p = 2$. Furthermore, if $b \neq 0$, then:

$$(ab^{-1})^q = a^q (b^q)^{-1} = ab^{-1}$$

showing that R is closed under both operations, hence it is a subfield, hence $R = F$

On the other hand, let F be a field with q elements. Then $|F^\times| = q - 1$, and so by Lagrange

Theorem, for any $0 \neq a \in F^\times$, $|a|\big|q-1$ Thus:

$$a \neq 0 \Rightarrow a^{q-1} = 1 \Rightarrow a^q - a = 0$$

showing that a *must* be a root of $x^q - x$. Furthermore, $0^q - 0 = 0$, showing that $x^q - x$ has q roots in F (i.e. all elements of F). But then, F is the splitting field of $x^q - x$ by the previous logic, as we sought to show.

Corollary 19.7.11: Unique Finite Field For Every Order

For every $q = p^d$, p a prime and $d \in \mathbb{N}_{>0}$, there exists a unique field of order q

***Proof*:**
this follows from theorem 19.7.10 and the uniqueness of the splitting field (theorem 19.5.10)

Definition 19.7.12: Galois Field

Let F be a field of order $q = p^d$. Then F is called the *Galois Field* of order q, and we usually denote it as $\mathbb{F}_q$ or $\mathbb{F}_{p^d}$.

Example 19.7.13: Galois Fields

let us show hat x^4+1 is reducible over *any* field $\mathbb{F}_p$, and hence over any finite field $F/\mathbb{F}_p$. First, when $p = 2$, $x^4 + 1 = (x+1)^4$, hence this is certainly the case when $p = 2$. If p is odd, I claim $x^4+1\big|x^{p^2} - x$. Indeed, $n^2 \equiv 1 \mod 8$ for any odd number, thus $8\big|(p^2-1)$, hence $x^8 - 1\big|x^{p^2-1} - x$ (exercise). Thus:

$$(x^4+1)\big|(x^8-1)\big|\big|(x^{p^2-1}-x)\big|(x^{p^2}-x)$$

Thus, x^4+1 factors completely in the splitting field of $x^{p^2} - x$, that is in $\mathbb{F}_{p^2}$. Thus, if α is a root of x^4+1, :

$$\mathbb{F}_p \subseteq \mathbb{F}_p(\alpha) \subseteq \mathbb{F}_{p^2}$$

Thus $[\mathbb{F}_p(\alpha) : \mathbb{F}_p]\big|[\mathbb{F}_{p^2} : \mathbb{F}_p] = 2$, thus α has either degree 1 or 2 over $\mathbb{F}_p$. But then, the minimal polynomial must be a factor of x^4+1 of degree 1 or 2, thus x^4+1 must be reducible.

Corollary 19.7.14: Finite Field Extensions

Let p be a prime and let $d \leq e$ be positive integers. Then there is an extension $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$ if and only if $d\big|e$. If $d\big|e$, then there is exactly one such extension, in the sense that $\mathbb{F}_{p^e}$ conains a unique copy of $\mathbb{F}_{p^d}$.

Furthermore, all extensions $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$ are simple

***Proof*:**
First, let's show that extensions are only dependent on the divisibility of the order. Let's say there is an extension $\mathbb{F}_P \subseteq \mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e}$. Then $[F_{p^d} : \mathbb{F}]$ divides $[\mathbb{F}_{p^e} : \mathbb{F}]$, meaning $d\big|e$.

Conversely, let's say $d \leq e$ such that $d\big|e$. Since:

$$p^e - 1 = (p^d - 1)((p^d)^{\frac{e}{d}-1} + (p^d)^{\frac{e}{d}-\frac{e}{d}})$$

we see that $p^d - 1$ divides $p^e - 1$, and so $x^{p^d-1} - 1$ divides $x^{p^e-1} - 1$ (Aluffi gave this as exercsie V.2.13). Therefore:

$$(x^{p^d} - x)\big|(x^{p^e} - x)$$

Since $\mathbb{F}_{p^e}$ is a splitting field for the second polynomial by theorem 19.7.10, it must contain a unique copy of the splitting field for the first plynomial (exercsie 4.5 in Aluffi, p. 442).

Finally, to show the extension is simple, recall that any finite field is necessarily cyclic, as we recalled in theorem 19.1.10 ($F^\times$ is locally cyclic, and since F is finite, $F^\times$ is cyclic). Let's say $\alpha \in F_{p^e}$ generated the multiplicative group of $\mathbb{F}_{p^e}$. Then clearly, α will generated $\mathbb{F}_{p^e}$ over any subfield, so if $d\big|e$, $\mathbb{F}_{p^e} = \mathbb{F}_{p^d}(\alpha)$, showing that $\mathbb{F}_{p^e}/\mathbb{F}_{p^e}$ is simple, completing the proof.

Corollary 19.7.15: Separable Finite Extensions Of Finite Fields

Let k be a finite field and F/k a finite separable extension. Then F is a simple extension.

***Proof*:**
Most of the proof of this was already done by proposition 19.6.19. If k is a finite field, then so is F, and the extensions are automatically simple by corollary 19.7.14.

Using these results, we can be more gain a lot of information about polynomials over finite fields:

Corollary 19.7.16: Finite Fields And Irreducible Polynomials

Let F be a finite field. Then for all integers $n \geq 1$, there exists irreducible polynomials of degree n in $F[x]$

***Proof*:**
Let $n \geq 1$ and $F = \mathbb{F}_{p^d}$. Then

$$\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^{dn}}$$

is a finite simple extension, say $\mathbb{F}_{p^{dn}} = \mathbb{F}_{p^d}(\alpha)$. Then sine

$$[\mathbb{F}_{p^{dn}} : \mathbb{F}_{p^d}] = n$$

and α has a minimal polynomial over $\mathbb{F}_{p^d} = F$, we see that we can find an irreducible polynomial of degree n in $F[x]$.

Note how this proof doesn't necessarily generalize for infinite fields, since we don't have so much control over the degree of the extension (worth pondering over).

Corollary 19.7.17: Factorizing $x^{q^n} - x$

here p. 443 aluffi

***Proof*:**

Example 19.7.18: Factorizing Polynomials over Finite Fields
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Another result that finiteness brings is that the structure of $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$ is very simple. We already know that any finite field is perfect, hence all algebraic extensions are separable, and so

$$|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| = d$$

However, we can say even more about the structure of the group

Proposition 19.7.19: Structure of Aut Of Finite Fields

Let $\mathbb{F}_{p^d}/\mathbb{F}_p$ be a field extension of degree d. Then $\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$ is cyclic, and generated by the Frobenius isomorphism.

***Proof*:**
Let $\varphi : \mathbb{F}_{p^d} \to \mathbb{F}_{p^d}$ be the map such that $x \mapsto x^p$. Since finite fields are perfect, φ is surjective, and hence an isomorphism. For any $x \in \mathbb{F}_p$, by Fermat's theorem (theorem 10.3.3)

$$x^p \bmod p$$

and so φ is in fact a $\mathbb{F}_p$-isomorphism, meaning $\varphi \in \text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})$. Since $|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| = d$, we need to show that $|\varphi| = d$.

Since φ is in a finite group, there exists an e such that $\varphi^e = \text{id}$, so $x^{p^e} = x$ for all $x \in \mathbb{F}_{p^d}$. Then $x^{p^e} - x$ has p^d roots in $\mathbb{F}_{p^d}$ (since φ makes all elements of $\mathbb{F}_{p^d}$ satisfy this relation). Then since the number of roots of a polynomial cannot exeed the highes degree, we have that $p^d \leq p^e$, which implies $d \leq e$. Thus, in terms of order of groups, we get that:

$$|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})| \leq |\varphi|$$

but by Lagrange's theorem, $|\varphi|\,\big|\,|\text{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d})|$,

This result is rather interesting: it means that given $(\mathbb{Z}/m\mathbb{Z})^*$ we have a natural relation $p \bmod m$ associated to the Frobenius morphism $x \mapsto x^p$. This relation to prime being modded and the Frobenius homomorphism shall go far in algebraic number theory (chapter 50) and will fully be realized in class field theory (see [17]).

(A word on a way to construct an algebraic closure of finite fields, and on how the category of finite fields seems to be close to another known category)

To close off this section, we saw that all finite extensions of finite fields is a cyclic extension. This has not been shown for finite extension for $\mathbb{Q}$. However, it turns out that this is *almost* the case, namely all finite extensions of $\mathbb{Q}$ embed into a cyclic extensions. This requires a lot more build-up, and is easier to prove in a course on Class Field Theory[24]. These notes shall indeed prove this fact not using class field theory as the “reward” for the material covered in chapter 50 (see theorem 50.5.4).

[24] See CITE:HERE for my book on it

Exercise 19.7.1

1. Let $f(x)$ be an irreducible polynomial of degree n over $\mathbb{F}_p$. Show that any field extension of $f(x)$ splits and is isomorphic to $x^{p^n} - x$
2. Let $k = \mathbb{F}_p$. Show that $f(x^p) = f(x)^p$.
3. Let L be a finite division algebra (or "noncommutative field") with center $\mathbb{F}_p$. Show that L is a field (hint: use class equation, theorem 4.6.3)

20

Galois Theory

For eons, mathematicians looked for relations between coefficients and roots of a polynomial. The answer is “trivial” for linear polynomials, given:

$$ax + b = 0 \qquad \text{with solution } \alpha \text{, then} \quad \alpha = -ba^{-1}$$

For a quadratic $x^2 + bx + c$ with roots α and β, then $\alpha + \beta = -b$ and $\alpha\beta = c$[1], and each individual root is given by the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

More generally, a lot of focus was put into finding the roots of polynomials given their coefficients[2], eventually finding the cubic and quartic equation relating the coefficients to roots of cubic and quartic polynomials respectively. However, the effort to generalize to higher degree polynomials brought limited results, in particular even for the next degree (the quintic) it was unclear how its roots can written in terms of it’s coefficients and the arithmetic operations of $+, -, \times, \div$ and $\sqrt{-}$. It wasn’t until the turn of the 18th century when serious progress has been made on this front. The key insight was to treat the coefficients of a polynomials as determining a dynamical system (that is, the dynamic of how the roots of a polynomial change as we move around the coefficients). In chapter 19, the right context to consider is the algebraic closure of a field; for pedagogical purposes we stick to $\overline{\mathbb{Q}}$ or $\mathbb{C}$ namely since this is the theories starting point and generalizations to $\mathbb{F}_{q^n}$ and other fields/rings is the natural progression.

To demonstrate these dynamics, consider $x^2 + x - 1 = 0$. This equation (by using the quadratic

[1] See section 20.2.3 for much more detail about these relations

[2] This is a little anachronistic, for when the quadratic formula was first discovered in Babylonia and more succinctly by Brahmagupta, the “modern” notion of polynomial and variables where not used, but rather it was shown by relating areas of circles and rectangles

equations) has solutions:

$$x_1 = \frac{-1+\sqrt{5}}{2} x_2 = \frac{-1-\sqrt{5}}{2}$$

If the b and c coefficients are swapped to get $x^2 - x + 1 = 0$, then the solutions are:

$$x_1 = \frac{1+i\sqrt{3}}{2} x_2 = \frac{1-i\sqrt{3}}{2}$$

In other words if (a, b, c) represents the coefficients, then the permutation action $(2\ 3) \in S_3$ does not leave the roots invariant, that is this is *not* invariant to coefficient permutation. Note these dynamics are "discrete", but they are the shadow of a continuous one. Let $\mathcal{P}_n$ denote the space of monic degree-n complex polynomials with distinct roots, that is, $\mathbb{C}^n$ with the discriminant locus removed. Sending a polynomial to its unordered set of roots exhibits the space of *ordered* root tuples as an n-sheeted covering of $\mathcal{P}_n$, so a *loop* of coefficients returns the root set to itself while permuting the individual roots. This is a homomorphism

$$\pi_1(\mathcal{P}_n) \longrightarrow S_n$$

and the permutation above is one of its values: the discrete action is what a continuous loop leaves behind. The group $\pi_1(\mathcal{P}_n)$ is the braid group B_n, and for a generic polynomial the image of this map is all of S_n, the same S_n that will shortly appear as a Galois group. That a fundamental group turns up here is not an accident; see box 20.0.1.

Now, given a general $ax^2 + bx + c = 0$, factoring as $(x - \alpha)(x - \beta)$, then by commutativity:

$$(x - \alpha)(x - \beta) = (x - \beta)(x - \alpha)$$

In other words, applying an S_2 action on the roots leaves the coefficients invariant[3 4].

The first immediate introduction of dynamics into polynomials is by applying group-actions to the roots. The realization that links roots to coefficients is that, given we can completely split a polynomial:

$$x^n + \cdots + \alpha_1 x + \alpha_0 = (x - a_1)(x - a_2)\cdots(x - a_n)$$

each coefficient of the original polynomial can be represented as a *symmetric function* (see section 20.2.3).

$$\begin{aligned}
\alpha_{n-1} &= -(a_1 + a_2 + \cdots + a_n) \\
\alpha_{n-2} &= a_1 a_2 + a_1 a_3 + \cdots + a_1 a_n + a_2 a_3 + \cdots a_{n-1} a_n \\
&\cdots \\
\alpha_0 &= (-1)^n a_1 a_2 \cdots a_n
\end{aligned}$$

the signs alternating, exactly as in the quadratic case above where $\alpha + \beta = -b$ and $\alpha\beta = c$. These form a natural relationship between the roots and the coefficients. It was noticed that these equations are invariant under permutation action of S_n; if $A = \{A_0, ..., A_{n-1}\}$ represent the above n equations and the action is given by $(r\ s) \cdot a_r = a_s$, then $\sigma \cdot A_i = A_i$ for each A_i and σ, hence the action $S_n \curvearrowright A$ is the trivial action. Thus, there is always a natural group-action.

[3] Concretely, take $x^2 - c$, whose roots are $\pm\sqrt{c}$. As c travels once around the circle $c = re^{i\theta}$, with θ running from 0 to 2π, the roots $\pm\sqrt{r}e^{i\theta/2}$ trade places: a single loop of the coefficient around the discriminant realises the transposition in S_2

[4] As a consequence, there is no continuous function that can take in coefficients and return roots

As any roots of polynomial induce such equations and induce this trivial action, these don't give enough information about polynomials. It shall be shown that roots of a generic polynomial satisfy no further equation, and in these cases there is nothing more that can be done to understand a polynomial via a group-action on its roots. However, there are polynomials whose roots shall have *more* relations the roots satisfy, limiting what group can act on them (i.e. a subgroup of S_n will act trivially, but S_n will not). This chapter shall make rigorous this realization and translate it into the modern language of field theory.

This realization almost managed to show the general quintic is unsolvable:

> "The quintic was almost proven to have no general solutions by radicals by Paolo Ruffini in 1799, whose key insight was to use permutation groups, not just a single permutation. His solution contained a gap, which Cauchy considered minor, though this was not patched until the work of the Norwegian mathematician Niels Henrik Abel, who published a proof in 1824, thus establishing the Abel-Ruffini theorem."[5]

This theory was established with the aim to show there is an unsolvable quintic, but not how to find whether a given quintic is solvable. This is where the work of Évariste Galois comes in. He noticed that the key insight was to work with all possible relationships of the roots of the polynomial, assuming all the roots exist.

Modern uses of Galois Theory

Galois groups are central in both algebraic geometry and number theory, where they serve different purposes in each domain. The galois group of a polynomial:

1. establishes the algebraic relations between the roots and creates a "covering space" that has some similar properties to the covering spaces seen in algebraic topology (see remark 20.0.1),
2. establishes the invariance necessary to find a solution over a lower field, in particular if L/K is a field extension and a root of an $L[x]$ polynomial has a root in K, then the root is invariant under the $\mathrm{Gal}_K(L)$-action.

From action on roots to field theory

Take the polynomial:

$$x^4 - 2 = 0$$

which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion (proposition 11.3.10). Adjoining $\sqrt[4]{2}$, $\mathbb{Q}(\sqrt[4]{2})$, the polynomial splits to:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x^2 + \sqrt{2})$$

Adjoining $\sqrt[4]{2}i$, the polynomial splits completely:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - \sqrt[4]{2}i)(x + \sqrt[4]{2}i) \tag{20.1}$$

Already, there is some interesting phenomena at play, for example, another way of getting this splitting is by adjoining $\sqrt[4]{2}$ and then adjoining i (or adjoining i then $\sqrt[4]{2}$). In fact, by just adjoining

[5]taken from Wikipedia for Galois Theory

$\sqrt[4]{2}+i$, we immediately get this splitting. Indeed, certainly $\mathbb{Q}(\sqrt[4]{2}+i) \subseteq \mathbb{Q}(\sqrt[4]{2}, i)$. Conversely, to show $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, letting $a = \sqrt[4]{2}+i$, then:

$$\begin{aligned} a - i &= \sqrt[4]{2} \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ a^4 - 4a^3 i - 6a^2 + 4ai + 1 &= 2 \\ \Leftrightarrow\ i(4a - 4a^3) &= 1 - a^4 + 6a^2 \\ \Leftrightarrow\ i &= \frac{1 - a^4 + 6a^2}{4a - 4a^3} \end{aligned}$$

showing that $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, and so $a - i = \sqrt[4]{2} \in \mathbb{Q}(\sqrt[4]{2}+i)$. More generally for any finite separable extension $K/\mathbb{Q}$, such an element always exists and is constructible by the primitive element theorem (theorem 19.6.20).

Let us understand the equations the roots form given by symmetric functions. As $x^4 - 2 = x^4 + 0x^3 + 0x^2 + 0x - 2$, the only meaningful relation obtained by multiplying out the roots in (20.1) is:

$$(\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) = -2 \qquad \text{equiv.} \qquad (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) + 2 = 0$$

This equation is invariant under the action of S_4, namely if

$$(a_1, a_2, a_3, a_4) = (\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}, -\sqrt[4]{2}i)$$

represent the 4 roots, and $A_0 = (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) + 2 = 0$, then:

$$\sigma \cdot A_0 = A_0 \qquad \forall \sigma \in S_4$$

However, there are more equations that can be formed with these roots (using the operations of a field). With the labelling above, the first and third roots are negatives of one another, as are the second and fourth:

$$\begin{aligned} A_1:\ a_1 + a_3 &= \sqrt[4]{2} + (-\sqrt[4]{2}) = 0 \\ A_2:\ a_2 + a_4 &= \sqrt[4]{2}i + (-\sqrt[4]{2}i) = 0 \end{aligned}$$

These equations are *not* invariant under the permutation action of S_4:

$$(1\ 2) \cdot A_1 \neq A_1 \qquad (\text{concretely: } (1\ 2) \cdot (a_1 + a_3) = a_2 + a_3 = \sqrt[4]{2}i - \sqrt[4]{2} \neq 0)$$

A subgroup of S_4 must be chosen, namely notice that $D_4 \cong \langle (1\ 2\ 3\ 4), (1\ 3) \rangle \subseteq S_4$[6] keeps them invariant. It does not fix each one separately: the four-cycle carries A_1 to A_2 and back. What it preserves is the *pairing* $\{\{1,3\},\{2,4\}\}$ of the roots into negative pairs, and so the pair of equations as a set:

$$D_4 \cdot A_0 = A_0 \qquad D_4 \cdot \{A_1, A_2\} = \{A_1, A_2\}$$

Indeed D_4 is precisely the stabiliser of that pairing inside S_4, and it has order $8 = [\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}]$, the first hint that this is the right subgroup. Are there any other equations that we can be considered

[6] The reader may recall exercise set 1.3.1 asks to show S_n is generated by a two cycle and an n-cycle; it was important the 2-cycle was an *adjacent* cycle

which would further restrict the choice of invariant permutation action? To answer this, the above is re-framed in the language of field extensions.

Let us say f is a polynomial over $\mathbb{Q}$ and that α is a root of f. Then the algebraic extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains *all* possible polynomial equations that can be generated with elements from $\mathbb{Q}$ and α (recall $\mathbb{Q}(\alpha) = \mathbb{Q}[\alpha]$, see proposition 19.2.24). If any of these polynomials happen to be another root of f, that is $p(\alpha) = \beta$ where $f(\beta) = 0$ and $p(x) \in \mathbb{Q}[x]$, then $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains this relationship between the roots. Let's say one of these relationships is represented as :

$$P(x) = p(x) - \beta \text{ so that } P(\alpha) = p(\alpha) - \beta = 0$$

Then a $\mathbb{Q}$-automorphism (in this case nothing more than a field automorphism) would have to preserve this relationship (it must map roots to roots). Hence, the set of $\mathbb{Q}$-automorphism captures the relationships between α and all possible roots that are $\mathbb{Q}$-polynomials of α. If $K/\mathbb{Q}$ is the splitting field of f, then the $\mathbb{Q}$-automorphisms of K would be restricted to automorphism that must satisfy all possible relationships between roots. Given $K/\mathbb{Q}$ is a splitting field for f, as any $\mathbb{Q}$-automorphisms σ is determined by where the roots map to, it is associated to a permutation in S_n. If there were no relationships (you need to add n roots to split f), then the automorphism group would be isomorphic to S_n. If there is any other type of group, that means there are some relationships between the roots (just like the example of $\mathbb{Q}(\sqrt[4]{2} + i)/\mathbb{Q}$).

The automorphism group of the splitting field above is called the *galois group* and it is the aim of this chapter define the galois group for any field extension L/K, to be able to find some galois groups, to find how frequent certain galois groups are, to show the correspondence between field extension and galois groups, and answer some long-pending mathematical questions such as the solvability of [general] quintic using the newly developed theory.

Another way of thinking about galois group is they are a rough classification tool for field extensions: if two field extensions have different Galois groups they cannot be the same extensions. It is similar in roughness to the fundamental group $\pi_1(X)$ as is often encountered in algebraic topology. To see this roughness in classification, as the splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$ of $x^4 - 2$ is equal to $\mathbb{Q}(\sqrt[4]{2} + i)$, and the minimal polynomial of $\sqrt[4]{2} + i$ is:

$$x^8 + 4x^6 + 2x^4 + 28x^2 + 1$$

then they share the same restrictions on their roots as they have the same galois group, but they certainly also have many differences (for example, these two polynomials have different discriminants, see definition 19.6.27).

20.0.1 Galois Group as Fundamental Group

(Assuming strong familiarity with algebraic geometry and algebraic topology) Under the right framework in algebraic geometry, galois groups can be interpreted as *étale fundamental group* for $\operatorname{Spec} K$,

$$\pi_1^{\text{ét}}(\operatorname{Spec}(K)) \cong \operatorname{Gal}_K(\overline{K})$$

The name *fundamental group* is well-deserved for if X is a connected $\mathbb{C}$-scheme of finite type (i.e. “essentially” a connected variety) and $X(\mathbb{C})$ is its analytification, then:

$$\pi_1^{\text{ét}}(X) \cong \pi_1^{\text{top}}\widehat{(X(\mathbb{C})})$$

where $\pi_1^{\text{top}}(-)$ is the usual fundamental group and $\widehat{(-)}$ is the profinite completion with respect to its finite-index normal subgroups. See [3] for the details.

20.1 Galois Extension and Galois Group

First, let's the recall the definition of an automorphism and define a new term:

Definition 20.1.1: Automorphism and fixed Element

1. An isomorphism σ of F with itself is called an *automorphism* of F. The collection of automorphisms of F is denoted $\text{Aut}(F)$. If $\alpha \in F$, we write $\sigma\alpha$ for $\sigma(\alpha)$
2. An auotmorphism $\sigma \in \text{Aut}(F)$ is said to *fix* an element $\alpha \in F$ if $\sigma\alpha = \alpha$. If k is a subset of F, then an automorphism σ is said to *fix* F if it fixes all the elements of F
$$\sigma a = a \qquad \forall a \in k$$
3. Let F/k be an extension of fields. Then $\text{Aut}(F)$ is the collection of automorphissm of F, and $\text{Aut}_k(F)$ is collection of automorphsism of F which fix F.

Example 20.1.2: Automorphism and Fixed F

The identity map id is always an automorphism.

Furthermore, the prime field of any field K is always fixed. If P is the prime-field of F, then for any $\sigma \in \text{Aut}(K)$, $\sigma(1) = 1$, since $\sigma(0) = 0$. Thus, $\sigma(a) = a$ for every $a \in P$. This means that $\mathbb{Q}$ and $\mathbb{F}_p$ have only trivial automorphisms

$$\text{Aut}(\mathbb{Q}) = \{\text{id}\}\,, \text{Aut}(\mathbb{F}_p) = \{\text{id}\}$$

and if P is a prime field of F, then

$$\text{Aut}_P(F) = \text{Aut}(F)$$

Next, we recall a very useful criterion for finding automorphisms of algebraic extensions K/F

Proposition 20.1.3: Fixed Roots Criterion

Let F/k be a field extension and let $\alpha \in F$ be algebraic over k. Then for any $\sigma \in \text{Aut}(F/k)$, $\sigma\alpha$ is a root of the minimal polynomial for α over k, i.e., $\text{Aut}(F/k)$ permutes the roots of irreducible polynomials. Equivalently, any polynomial whith coefficeints in k having α as a root also has $\sigma\alpha$ as a root.
Furthermore, $\text{Aut}_k(k(\alpha))$ acts *transitively* on the set of roots of $m_{k,\alpha}(x)$ (i.e. if β_1, β_2 are roots of $m_{k,\alpha}(x)$, then there exists a $\sigma \in \text{Aut}_k(k(\alpha))$ such that $\sigma\beta_1 = \beta_2$)

***Proof*:**
Let
$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$
be the minimal polynomial with coefficients from F so that
$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

then, applying any automorphism σ to this polynomial will get us

$$\sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) = \sigma(0)$$
$$\sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + \sigma(a_0) = 0 \qquad \sigma \text{ fixes } a_i \text{ for each } i$$

since $a_i \in F$ for each i, showing that $\sigma(\alpha)$ is also a root of the minimal polynomial

Transitivty is left as an exercise (recall the appropriate theorem)

Example 20.1.4: Automorphism Groups

1. Take $K = \mathbb{Q}(\sqrt{2})$. Then $\text{Aut}(\mathbb{Q}(\sqrt{2})) = \text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$. If $\sigma \in \text{Aut}(\mathbb{Q}(\sqrt{2}))$, then by the fixed root criterion, it must be that

$$\sigma(\sqrt{2}) = \pm\sqrt{2}$$

and since σ must fix everything else in place, this means that we get

$$\sigma(a + b\sqrt{2}) = a \pm b\sqrt{2}$$

in other words, we've determined were every other element must map to! Thus, we have the maps

$$\sqrt{2} \mapsto \sqrt{2} \qquad \sqrt{2} \mapsto \sqrt{-2}$$

If we let the first map be represetned by 1 and the second by σ, then we get

$$\text{Aut}(\mathbb{Q}(\sqrt{2})) = \{1, \sigma\} = C_2$$

that is, $\text{Aut}(\mathbb{Q}(\sqrt{2}))$ is a cyclic group of order 2.

2. Let $K = \mathbb{Q}(\sqrt[3]{2})$. By theorem 19.2.5, every element of $\mathbb{Q}$ can be written as

$$a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2$$

If we consider any $\sigma \in \text{Aut}(\mathbb{Q}(\sqrt[3]{2}))$, then we get

$$a + b\sigma(\sqrt[3]{2}) + c\sigma(\sqrt[3]{2}) \qquad a, b, c \in \mathbb{Q}$$

and since $\mathbb{Q}(\sqrt[3]{2})$ has non of it's other roots (recall example 19.3.12). Then that means that the only automorphism is the identity, meaning

$$\text{Aut}(\mathbb{Q}(\sqrt[3]{2})) = \{1\}$$

3. Now let $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$, be the splitting field of $x^3 - 2$ in example (19.5.11), which unlike the previous example will insure we have interesting automorphisms since K contains all the roots of $x^3 - 2$. Let's label these roots $\sqrt[3]{2}, \omega, \omega^2$. Then:

$$[\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) : \mathbb{Q}(\sqrt[3]{2})] : [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6$$

We know that $\mathbb{Q}(\sqrt[3]{2})$ contains one root of $x^3 - 2$ over $\mathbb{Q}$ ($\sqrt[3]{2} \in \mathbb{Q}(\sqrt[3]{2})$), and that $\mathbb{Q}(\sqrt[3]{2}, \omega)$ also contains the other two roots over $\mathbb{Q}$, and in particular over $\mathbb{Q}(\sqrt[3]{2})$ ($\omega, \omega^2 \in \mathbb{Q}(\sqrt[3]{2}, \omega)$).

Next, Since the degree of K is 6, we know it has a basis consisting of 6 elements, namely:

$$1,\ \sqrt[3]{2},\ \sqrt[3]{4},\ \omega,\ \omega\sqrt[3]{2},\ \omega\sqrt[3]{4}$$

Since any $\mathbb{Q}$-automorphism of K is a $\mathbb{Q}$-linear automorphism on K, it suffices to characterise these automorphisms by where by map these basis elements. By the fixed roots criterion, a $\mathbb{Q}$-automorphism σ on K can map $\sqrt[3]{2}$ to:

$$\sqrt[3]{2},\ \omega\sqrt[3]{2},\ \omega^2\sqrt[3]{2}$$

and map ω to:

$$\omega,\ \omega^2$$

Using this information, we can carry through computation's to see that $\mathrm{Aut}_{\mathbb{Q}}(K)$ is S_3. (I think we will encounter theorems that will make this easier later)

4. More on computing automorphisms groups here: here

Since any field homomorphism must be injective, $\mathrm{Aut}_k(-)$ can in fact respect composition, making it a [contravariant] functor[7]

Lemma 20.1.5: Field Automorphism Is Functorial

let $\mathbf{Field}_k$ be all fields such that any $F \in \mathbf{Field}_k$ has k as a subfield: $k \subseteq F$. Then $\mathrm{Aut}_k(-) : \mathbf{Field}_k \to \mathbf{Grp}$ sending F to $\mathrm{Aut}_k(F)$ and $F \to L$ to $\mathrm{Aut}_k(L) \to \mathrm{Aut}_k(F)$ via $\varphi \mapsto \varphi|_F$ is a contraviant functor

***Proof*:**
simple definition pushing.

Though $\mathrm{Aut}_k(-)$ being functorial is quite powerful and a fact that shall be often used, it actually perserves even more structure, namely the latice of groups and interemetiate fields is [contravariantly] preserved. The following propositions builds up to this result. Since $\mathrm{Aut}_k(F)$ has a relatively easy to find structure, any connections between the structure of $\mathrm{Aut}_k(F)$ (which is a finite group if $[F : k] < \infty$) and F/k would allow us to use the full power of group theory (and better still finite group theory) to analise field extensions. We endevaour to deepen this connection in this chapter. One of the most important connection is that intermediate fields $k \subseteq E \subseteq F$ correspond to subgroups of $\mathrm{Aut}(F/k)$! This is called the *Galois correspondence*, and is what we will be building up for the rest of this section. We start by showing how subgroups of $\mathrm{Aut}_k(F)$ correspond to an intermediate field $k \subseteq E \subseteq F$, and conversely intermediate extension $k \subseteq E \subseteq F$ correspond a subgroup fo $\mathrm{Aut}_k(F)$

Proposition 20.1.6: Fixing F with automorphisms

Let $H \leqslant \mathrm{Aut}_k(F)$ be a subgroup of automorphisms of F (or $H \subseteq \mathrm{Aut}_k(F)$ more generally). Then the collection F^H of elements of F fixed by all the elements of H is a subfield of F extending k, $k \subseteq F^H \subseteq F$. Conversely, Let E be an intermediate field $k \subseteq E \subseteq F$. Then $\mathrm{Aut}_E(F) \leqslant \mathrm{Aut}_k(F)$.

[7] Note that $\mathrm{Aut}_{\mathbf{Grp}}(-)$ is *not* a functor; there are counter-examples, see exercise ref:HERE

***Proof*:**
Let F^H be the collection of fixed elements by element $h \in H$. Then by definition, every element of k is fixed by H, so F^H is nonempty and $k \subseteq F^H$. Next, for any $a, b \in F^H$

$$h(a \pm b) = h(a) \pm h(b) = a \pm b \qquad h(ab) = h(a)h(b) = ab \qquad h(a^{-1}) = h(a)^{-1} = a^{-1}$$

that is, F^H is closed under the operations of F, and so $F^H \subseteq F$, showing that $k \subseteq F^H \subseteq F$.

For the other direction, let $\mathrm{Aut}_E(F)$ be the set of automorphisms that fix E. This subset is nonempty since $\mathrm{id}_F \in \mathrm{Aut}_E(F)$. If $\sigma \in \mathrm{Aut}_E(F)$, clearly $\sigma^{-1} \in \mathrm{Aut}_E(F)$. Finally, for any $\sigma, \tau \in \mathrm{Aut}_E(F)$, given $e \in E$,

$$\sigma \circ \tau^{-1}(e_1) = \sigma(\tau^{-1}(e)) = \sigma(e) = e$$

showing that $\sigma \circ \tau^{-1} \in \mathrm{Aut}_E(F)$, an so $\mathrm{Aut}_E(F) \leqslant \mathrm{Aut}_k(F)$, completing the proof.

Definition 20.1.7: Fixed Field

Let $H \subseteq \mathrm{Aut}_k(F)$. Then the subfield of F fixed by all the elements of H is called the *fixed field* of H, and is denoted F^H.

Example 20.1.8: Fixed Field

1. The fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}))$ is $\mathbb{Q}$. As we have seen in example 20.1.4.
2. The fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ is $\mathbb{Q}(\sqrt[3]{2})$, since $\mathrm{Aut}_{\mathbb{Q}}(\sqrt[3]{2}) = \{\mathrm{id}\}$. If we add the other roots of $x^3 - 2$, then the fixed field of $\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}, \omega))$ is $\mathbb{Q}$.

Using this vocabulary, we can make the following correspondence betweeen F/k and its intermediate fields with $\mathrm{Aut}_k(F)$ and it's subgroups:

Proposition 20.1.9: Galois Correspondence

Let $k \subseteq F$ be a fields extension. Then there exists a correspondence:

$$\left\{\begin{matrix}\text{Intermediate fields } E \\ k \subseteq E \subseteq F\end{matrix}\right\} \underset{F^H}{\overset{\mathrm{Aut}_E(F)}{\rightleftarrows}} \left\{\begin{matrix}\text{subgroups } H \\ H \subseteq \mathrm{Aut}_k(F)\end{matrix}\right\}$$

sending an intermediate field E to the subgroup $\mathrm{Aut}_E(F)$, and a subgroup H of $\mathrm{Aut}_k(F)$ to the fixed field F^H. Furthermore, we have the following properies:

1. The correspondence is inclusion reversing, that is:
   - if $k \subseteq E_1 \subseteq E_2 \subseteq F$ are two subfield of F extending k , then $\mathrm{Aut}_{E_2}(F) \leqslant \mathrm{Aut}_{E_1}(F)$
   - if $H_1 \leqslant H_2 \leqslant \mathrm{Aut}_k(F)$ are two subgroups of automorphsism with associated fixed fields F^{H_1} and F^{H_2} respectively, then $F^{H_2} \subseteq F^{H_1}$.
2. For all subgroups $H \leqslant \mathrm{Aut}_k(F)$ and intermediate fields $k \subseteq E \subseteq F$
   - $E \subseteq F^{\mathrm{Aut}_E(F)}$
   - $H \subseteq \mathrm{Aut}_{F^H}(F)$
3. If E_1E_2 is the smallest subfield of F containing the two intermediate the E_1 and E_2, and $\langle G_1, G_2\rangle$ denotes the smallest subgroup of $\mathrm{Aut}_k(F)$ containing G_1 and G_2, then:

$$\mathrm{Aut}_{E_1E_2}(F) = \mathrm{Aut}_{E_1}(F) \cap \mathrm{Aut}_{E_2}(F)$$
$$F^{\langle G_1, G_2\rangle} = F^{G_1} \cap F^{G_2}$$

***Proof*:**
The Galois correspondence is an immediate consequence of proposition 20.1.6.

For the inclusion-reversal, consider the extension, $k \subseteq E_1 \subseteq E_2 \subseteq F$. Then $\mathrm{Aut}_{E_2}(F)$ *a fortiori* fixes all the elements of E_1, since $E_1 \subseteq E_2$, and so can only be *smaller* as the number of elements in Aut_{E_2} depend on the number of roots that can be premuted (as shown in the root permutation Criterion). A similar argument can be made for $H_1 \leqslant H_2 \leqslant \mathrm{Aut}_k(F)$.

For the subsets, Take the intermediate field E, $k \subseteq E \subseteq F$. Then $F^{\mathrm{Aut}_E(F)}$ is the set of elements That are fixed by the automorphisms that fix E. Then it is immediate that $E \subseteq F^{\mathrm{Aut}_E(F)}$. Similarly for $H \subseteq \mathrm{Aut}_{F^H}(F)$.

For the minimality statements, If E_1E_2 is the smallest of F containing both E_1 and E_2, then the $\subseteq$ inclusion is immediate (since $\mathrm{Aut}_{E_1}(F) \cap \mathrm{Aut}_{E_2}(F)$ contains only elements that fix E_1 and E_2), while the $\supseteq$ comes from the fact that $\mathrm{Aut}_{E_1}(F)$ and $\mathrm{Aut}_{E_2}(F)$ are in themselves the minimal subgroups of $\mathrm{Aut}_k(F)$ that fix all elements of E_1 and E_2. A similar argument can be made for the second assertion.

It is natural to ask whether the galois correspondence is bijective. Before showing when it is, we will show that it is possible that it is not:

Example 20.1.10: non-bijective Galois Correspondence

Consider $Q(\sqrt[3]{2})/\mathbb{Q}$. Then we know that $[\mathbb{Q}(\sqrt{3})/\mathbb{Q}] = 3$, which is prime, and so the only intermediate fields are $\mathbb{Q}$ and $\mathbb{Q}(\sqrt[3]{2})$. For $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$, recall that $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, and that $\sqrt[3]{2}$ is the only root of the minimal polynomial within $\mathbb{R}$. Thus, since automorphisms of fields permute roots, $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))$ can only have one element:

$$\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2})) = \{e\}$$

Hence, we have:

$$\{\mathbb{Q}, \mathbb{Q}(\sqrt[3]{2})\} \rightleftarrows \{\,\{e\}\,\} = \{\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))\}$$

showing that the function from intermediate fields to subgroups of the automorphism group is not injective with respect to intermediate fields in general[a] (for every intermediate field, we may map it to the same subgroup of $\text{Aut}_k(F)$). In this case, $\mathbb{Q} \subseteq F^{\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))}$ and $\mathbb{Q}(\sqrt[3]{2}) \subseteq F^{\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}))}$

[a] Or similarly, the correspondence is not surjective with respect to subgroups of $\text{Aut}_k(F)$

This examples shows that it possible that $E \subsetneq F^{\text{Aut}_E(F)}$. Fortunately, in the finite case, we always have that for any $H \leqslant \text{Aut}_k(F)$, there exists an intermediate field (namely F^H) such that $H = \text{Aut}_{F^H}(F)$ (and hence, the correspondence is at least always surjective with respect to intermediate fields). To prove this result we need the following lemma which shows that the field F^H is essentially as nice as a field as we want. The high-level idea is that automorphisms in $H = \text{Aut}_E(F)$ fix E, and F^H is *defined* to be the field that is fixed by H:

Lemma 20.1.11: Properties Of Fixed Field

Let F/k be a finite extension, and let $H \leqslant \text{Aut}_k(F)$. Then F/F^H is normal and separable. It is furthermore finite and simple.

Note that we are not given that F^H or k is a perfect field, and so separability is a powerful result. Similarly, we do not know beforehand that F is a normal extension of k, yet F is a normal extension of F/F^H. Note too that we can take $H = \{e\}$ so that $F/F^{\{e\}}$ has all the aforementioned properies. As we've seen in the previous example, it is not necessarily the case that $F^{\{e\}} = k$. (in the previous example, we had $F^{\{e\}} = \mathbb{Q}(\sqrt[3]{2})$)

***Proof*:**

First, F/F^H is finite since F/k is finite, so by the tower law:

$$[F : k] = [F : F^H][F^H : k]$$

Next, to show F/F^H is a separable extension, we'll show any $\alpha \in F$ is separable over F^H. To do so, we use the following clever trick: Let $\alpha \in F$ be any element of F (possibly an element of F^H). We'll construct the minimal polynomial of α, and show that it is separable. Take $\sigma \in H$. Since F is finite over k, α is the root of some minimal polynomial, $m_{\alpha,k}(x)$. By the fixed roots criterion, $h\alpha$ must be a root of the minimal polynomial $m_{\alpha,k}(x)$ of α over k (which might also imply $h\alpha = \alpha$ if no other root are in F). Since $m_{\alpha,k}(x)$ has only finitely many roots, the H-orbit of α (i.e. all possibles distinct images of α under all possible $\sigma \in H$) consists of finitely many elements $\alpha_1, ..., \alpha_n$. Now, the group H acts on the H-orbit by premuting its elements. Therefore, every element of H keeps the polynomial:

$$q_\alpha(x) = (x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n)$$

fixed (where $\alpha_1 = \alpha$). In other words, the coefficients of $q_\alpha(x)$ *must* be in the fixed field F^H: $q_\alpha(x) \in F^H[x]$! Next, we show that $q_\alpha(x) = m_{\alpha,F^H}(x)$, which will show that the minimal polynomial corresponding to α is separable, showing α is separable. To see this, let $g(x)$ be any other polynomial having α as a root. Then:

$$g(\alpha) = b_m\alpha^m + \cdots + b_1\alpha + b_0 = 0$$

and by definition, any $\sigma \in H$ will map α to another root α_i, so:

$$g(\alpha_i) = b_m\alpha_i^m + \cdots + b_1\alpha_i + b_0 = 0$$

meaning $g(x)$ is divisible by $x - \alpha_i$ for every i, and hence $q_\alpha(x)\big|g(x)$. Since the minimal polynomial of α over F^H also has α as a root, we have that $q_\alpha(x)\big|m_{\alpha,f^H}(x)$, and as we've shown before $m_{\alpha,F^H}(x)\big|q_\alpha(x)$. Thus, $q_\alpha(x) = m_{\alpha,F^H}(x)$ up to a constant. Thus, α is separable and since α was arbitrary, F/F^H is a separable extension (Note that it's possible that $\alpha \in F^H$, in which case $q_\alpha(x) = x - \alpha$, which is certainly the minimal polynomial of α over F^H and is separable).

Next, since F/F^H is finite and separable, then by theorem 19.6.20, it simple. Let's say $F = F^H(\beta)$, so $F^H(\beta)/F^H$ is simple. Let $q_\beta(x)$ be define as we've done above (note that $q_\beta(x) = m_{\beta,F^H}(x)$ and that it is not possible that $q_\beta(x) = x - \beta$ unless $F = F^H$)

Finally, since $q_\beta(x) \in F^H[x]$ splits in F, and F is generated over F^H (namely, F is generated by F and β), F is the splitting field of $q_\alpha(x)$ over F^H, and so $F^H \subseteq F$ is a normal extension by theorem 19.5.13, completing the proof (notice again how finiteness was important for this part of the proof)

(checkout other proofs of this that do not require using the fact it's a simple extension, since the primitive roots theorem is sometimes proven using Galois Theory)

This proposition provides us with a couple more results: The polynomial $q_\alpha(x)$ (say of degree n) can be expanded to:

$$q_\alpha(x) = x^n - s_1x^{n-1} + s_2x^{n-1} - \cdots + (-1)^{n-1}s_{n-1}x + (-1)^n s_n$$

where each s_i are functions $s_i(\alpha_1, ..., \alpha_n)$ and represent:

$$s_1(\alpha_1, ..., \alpha_n) = \alpha_1 + \alpha_2 + \cdots + \alpha_n = \sum_i \alpha_i$$

$$s_2(\alpha_1, ..., \alpha_n) = \sum_{i<j} \alpha_i\alpha_j$$

$$s_3(\alpha_1, ..., \alpha_n) = \sum_{i<j<k} \alpha_i\alpha_j\alpha_k$$

$$\vdots$$

$$s_n(\alpha_1, ..., \alpha_n) = \alpha_1\alpha_2\cdots\alpha_n$$

Since $q_\alpha(x) \in F^H$, then each of these coefficients are in F^H. Furthermore, $\deg(q_\alpha(x)) \leq |H|$

Proposition 20.1.12: $H = \mathrm{Aut}_{F^H}(F)$

Let F/k be a *finite* extension, and let H be a subgroup of $\mathrm{Aut}_k(F)$. Then $|H| = [F : F^H]$ and

$$H = \mathrm{Aut}_{F^H}(F)$$

In particular, the Galois correspondence is surejctive for a finite field (for any subgroup $H \leqslant \mathrm{Aut}_k(F)$, there exists an intermediate field (namely F^H) such that $H = \mathrm{Aut}_{F^H}(F)$)

Intuitively, we want to show that $\mathrm{Aut}_{F^H}(F)$ *only* fixes F^H, and so $H = \mathrm{Aut}_{F^H}(F)$.

***Proof*:**
Let $H \leqslant \mathrm{Aut}_k(F)$. By proposition 20.1.9, we have that $H \leqslant \mathrm{Aut}_{F^H}(F)$, so $|H| \leq |\mathrm{Aut}_{F^H}(F)|$. Since $\mathrm{Aut}_{F^H}(F)$ is finite, verifying that $\geq$ holds will complete the proof.

To see this, recall that we've shown in lemma 20.1.11 that $F = F^H(\alpha)$ for some $\alpha \in F$. Thus, by corollary 19.2.29, $|\mathrm{Aut}_{F^H}(F)|$ equals the number of distinct roots of $m_{\alpha,F^H}(\alpha)$. Though we do not know the minimal polynomial $m_{\alpha,F^H}(x)$, we do know that α is a root of the polynomial $q_\alpha(x)$ defined in lemma 20.1.11. By construction, the number of roots of $q_\alpha(x)$ must be less than or equal $|H|$, giving us:

$$|\mathrm{Aut}_{F^H}(F)| \leq |H|$$

And since $|\mathrm{Aut}_{F^H}(F)| \geq |H|$, we get $|\mathrm{Aut}_{F^H}(F)| = |H|$ and since $H \subseteq \mathrm{Aut}_{F^H}(F)$, then $H = \mathrm{Aut}_{F^H}(F)$, giving the desired equality.

Finally, to show that $[F : F^H] = H$, since F/F^H is finite, separable, and normal, by corollary 19.6.21, $[F : F^H] = |\mathrm{Aut}_{F^H}(F)|$, completing the proof.

A particular consequence of this is that $\mathrm{Aut}_{F^{\mathrm{Aut}_E(F)}}(F) = \mathrm{Aut}_E(F)$, since the intermediate field that maps to $\mathrm{Aut}_{F^{\mathrm{Aut}_E(F)}}(F)$ is $F^{\mathrm{Aut}_E(F)}$, and so the group that it maps to is $\mathrm{Aut}_E(F)$.

Note that F/k being a *finite* extension was necesary for the proof. The Galois correspondence is in fact not necessarily surjective in the infinite case. This correspondence can be recovered given a suitable topology known as the *Krull topology* on $\mathrm{Aut}_k(F)$, which is used to limit the galois correspondence to *closed* subgroups of the automorphism group, which will regain surjectivity. More on this in appendix HERE (or another chapter?)

20.1.1 Galois Correspondence

We finally have reached a point where we can define a Galois extension. the following theorem is sometime also called the *Galois Extension Equivalences*, but it will be called by it's more famous name:

Theorem 20.1.13: Fundamental Theorem of Galois Theory I

Let F/k be a finite field extension. Then the following are equivalent:

1. F is the splitting field of a separable polynomial $f(x) \in k[x]$ over k
2. F/k is separable and normal
3. $|\text{Aut}_k(F)| = [F : k]$
4. $k = F^{\text{Aut}_k(F)}$ is the fixed field of $\text{Aut}_k(F)$
5. F/k is separable, and if K/F is an algebraic extension and $\sigma \in \text{Aut}_k(K)$, then $\sigma(F) = F$
6. The Galois correspondence for F/k is a bijection

***Proof*:**
Many of these have already been shown to be equivalent. We have shown:

- $1 \Leftrightarrow 2$ in theorem 19.5.13
- $2 \Leftrightarrow 3$ in corollary 19.6.21
- $3 \Leftrightarrow 4$ follows from proposition 20.1.12, since $[F : F^{\text{Aut}_k(F)}] = |\text{Aut}_k(F)|$, and since $k \subseteq F^{\text{Aut}_k(F)}$, we have that

$$k = F^{\text{Aut}_k(F)} \text{ if and only if } |\text{Aut}_k(F)| = [F : k]$$

- $2 \Leftrightarrow 5$ follows from the 3rd equivalence of theorem 19.5.13 along with the fact that finite separable extensions are simple (theorem 19.6.20)

Thus, we'll show $6 \Rightarrow 4$ and $1 \Rightarrow 6$, which will complete the chain of implications.

Starting with $6 \Rightarrow 4$, assume the galois correspondence is bijective, and consider $E = F^{\text{Aut}_k(F)}$. By proposition 20.1.12, $\text{Aut}_{F^{\text{Aut}_k(F)}}(F) = \text{Aut}_k(F)$, and so E and $F^{\text{Aut}_k(F)}$ map to the same subgroup: $\text{Aut}_k(F) = \text{Aut}_E(F)$. Since the correlation is bijective, it must be that $k = E$, implying 4.

Next, we'll show $1 \Rightarrow 6$. Since F/k is a finite extension, by proposition 20.1.12, we already know the Galois correspondence is surjective (i.e., has a right-inverse), so it suffice to show it's injective, or equivalently that it has a left-inverse. To that end, we need to show that every intermediate field E equals the fixed field of the corresponding subgroup $\text{Aut}_E(F)$, that is $E = F^{\text{Aut}_E(F)}$. So, pick some E such that $k \subseteq E \subseteq F$. Since F is the splitting field for some separable polynomial $f(x) \in k[x] \subseteq E[x]$, we see that F is the splitting field of the separable polynomial $f(x)$ over E (i.e., condition (1) holds for F/E too). Since $1 \Leftrightarrow 4$, then $E = F^{\text{Aut}_E}(F)$, thus $1 \Rightarrow 6$ completing the proof.

It can be helpful to have the following visual to keep track of where intermediate fields are with respect to their galois group:

$$\begin{aligned} \mathrm{Aut}_E(F) &= G \\ E &= F^G \end{aligned}$$

Definition 20.1.14: Galois Extension

Let F/k be a finite extension. Then F is called *Galois* or F/k is called a *Galois extension* if it satisfies any of the conditions in theorem 20.1.13.

Note that we have limited the definition to *finite galois extensions*, due to the requirement to bring in more tools to study the infinite equivalence. Nothing is lost in doing so, since infinite galois extensions will naturally generalize and improve upon the results we'll present. From now on, we will take galois extension to mean finite galois extension.

Corollary 20.1.15: Galois Extension Of Char 0

Let F/k be an extension of characteristic 0. Then F is galois over k if and only if F is finite and normal over k

Note that the finite condition was only added since we are focusing on finite Galois extensions.

***Proof*:**
If F is galois then it is *a priori* normal. Conversely, if F is finite and normal, then by corollary 19.6.8, any algebraic extension of F is separable, and hence by theorem 20.1.13, it is Galois.

Definition 20.1.16: Galois Group

Let F/k be a Galois extension. Then the corresponding automrophsim group $\mathrm{Aut}_k(F)$ is called the *galois group* of the extension. We will denote it by $\mathrm{Gal}_k(F)$ or $\mathrm{Gal}(F/k)$

The symbol $\mathrm{Gal}_k(F)$ is simply to emphasize we are working over a Galois extension, and doesn't reflect any additional structrue or data given to $\mathrm{Aut}_k(F)$. Furthermore, (word on $\mathrm{Gal}_k(F)$ acting on $k[x]$ or $F[x]$?)

Definition 20.1.17: Galois Group Of A Polynomial

Let $f(x)$ be a separable polynomial over $k[x]$. Then the *galois group* $\mathrm{Gal}_k(f(x))$ is the galois group of the splitting field of $f(x)$.

Definition 20.1.18: Abelian and Cyclic Extensions

Let F/k be a galois extension. Then F is called a *cyclic* (resp. *abelian*) extension if it's corresponding Galois group is cyclic (resp. abelian)
More generally, if the Galois group G has some property (ex. nilpotent, solvable, free), then the respetive extension is said to have that same property (ex. nilpotent extension, solvable extension, free extension)

Note that any vocabulary already used (like simple extension) will be prioritized over the group theory vocabulary (a simple extension is not related to simple group)

The equivalent conditions for Galois extensions gives us many details on results we have encountered earlier:

20.1.19 Remark

1. For any $H \leqslant \mathrm{Aut}_k(F)$, F/F^H is a Galois extension, and every intermediate field $k \subseteq E \subseteq F$ appears as a fixed field for a unique $H \leqslant \mathrm{Aut}_k(F)$ (If F/k is not Galois, this is not the case, like for $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$). This also means that by the 4th condition of Galois extension equivlences:
$$F^H = F^{\mathrm{Aut}_{F^H}(F)}$$
and since every intermediate field appears as a fixed field, we get that:
$$E = F^{\mathrm{Aut}_E(F)}$$

2. Let F/k be any finite extension. Then $|\mathrm{Aut}_k(F)|$ divides $[F : k]$. Taking any $G = \mathrm{Aut}_k(F)$, then
$$[F : k] = [F : F^G][F^G : k] \overset{!}{=} |\mathrm{Aut}_{F^{\mathrm{Aut}_k(F)}}(F)|[F^G : k] = |\mathrm{Aut}_k(F)|[F^G : k]$$
where $\overset{!}{=}$ comes from theorem 20.1.13(3). If F/k is Galois, then $F^G = k$, meaninig $|\mathrm{Aut}_k(F)|[F^G : k] = |\mathrm{Aut}_k(F)|[k : k] = |\mathrm{Aut}_k(F)|$.

3. Let F/k be a Galois extension. Then $|\mathrm{Aut}_k(F)|$ divides $|\mathrm{Aut}_{F^H}(F)|$. Taking any $H \leqslant \mathrm{Aut}_k(F)$ and $G = \mathrm{Aut}_k(F)$, then
$$|\mathrm{Aut}_k(F)| = [F : k] = [F : F^H][F^H : k] = |\mathrm{Aut}_{F^H}(F)|[F^H : k]$$
Switching up the notation, this means that $|G| = |H|[F^H : k]$, meaning $[F^H : k]$ is the index of H in G: $[F^H : k] = [G : H]$. The fact that it equals to the index means that it corresponds bijectively with the cosets $g\mathrm{Aut}_k(F^H)$ where $g \in \mathrm{Aut}_k(F)$, however it is yet to be shown how these two sets can correspond, only that they are the same size. In theroem 20.1.23, we'll see when this set of cosets forms a group.

4. It may be really difficult to figure out if we have found all intermediate fields of an infinite in general, namely since in general, there are infinitely many possible choices. However, it is relatively easy to find the subgroups of a finite group! Hence, using finite group theory allows us to bring in the powerful tools having that finiteness brings (including Sylow's theorems).

5. Note how the 5th condition is almost the condition for an extension being normal, where just also assumed that the extension is separable. Note that the 5th condition still works if the extension is not separable. This can motivate the idea that we can somhow make separability a superfluous condition given some appropriate generalization, and indeed we can. If we replace Galois Group with an appropriate *algebraic group*, then we can recover the bijective correspondence[a]

6. Given the base field is perfect, for any extension F/k, we can always extend it to its normal closure $N/F/k$, which is normal over k, and so is Galois. Hence, many field theory problems over $\mathbb{Q}$ can be translated "wlog" to Galois theory

[a] algebraic groups are schemes (with with a group structure), for which one of the motivations for their invention was to distinguish roots higher multiplicities at a point

If $E = F^H$, the last observation can be seen visually as:

$$\begin{array}{ccc} F & & \\ \Big| & \} & |H|=[F:E] \\ E & & \\ \Big| & \} & |G:H|=[E:k] \\ k & & \end{array}$$

Example 20.1.20: Galois Groups

1. Let $K = \mathbb{Q}(\sqrt{2})/\mathbb{Q}$. Then since $[K : \mathbb{Q}] = 2$. and $|\mathrm{Aut}(K/\mathbb{Q})| = 2$ (as we've seen in example 20.1.4), then K is galois with galois group

$$\mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{1, \sigma\} \cong \mathbb{Z}/2\mathbb{Z}$$

With the identity automorphism and

$$a + b\sqrt{2} \mapsto a - b\sqrt{2}$$

2. More generally, let K be any quadratic extension of F ($K = F(\sqrt{D})$). If $\mathrm{ch}(K) \neq 2$, then K is Galois.

3. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* Galois, since $|\mathrm{Aut}(\sqrt[3]{2})| = 1 < 3 = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}]$. Notice that $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2})$ is not a fixed field, since the only fixed field is $\mathbb{Q}(\sqrt[3]{2})$.

4. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ be an extension of $\mathbb{Q}$. Then since K is the splitting field of $(x^2-2)(x^2-3)$, it Galois. To determine its Galois group, we know from example (19.3.12) that $[K : \mathbb{Q}] = 4$, giving us the amount of automorphisms to look for. Since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, by the fixed roots criterion, we see that any automorphism $\sigma \in \mathrm{Aut}_{\mathbb{Q}}(K)$ must map $\sqrt{2}$ to $\pm\sqrt{2}$, and since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, we see that any automorphism $\tau \in \mathrm{Aut}_{\mathbb{Q}}(K)$ must map $\sqrt{3}$ map $\pm\sqrt{3}$.
Going through defining a σ and a τ by permuting $\sqrt{2}$ and $\sqrt{3}$ we see that:

$$\mathrm{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

Next, we can use proposition 20.1.9 to find the associated fixed field given the subgroups of $\text{Gal}(K/\mathbb{Q})$. Using some arithmetic, we see that:

$$\begin{array}{cc} \{1\} & \mathbb{Q}(\sqrt{2},\sqrt{3}) \\ \\ \{1,\sigma\} & \mathbb{Q}(\sqrt{3}) \\ \\ \{1,\sigma\tau\} & \mathbb{Q}(\sqrt{6}) \\ \\ \{1,\tau\} & \mathbb{Q}(\sqrt{2}) \\ \\ \{1,\sigma,\tau,\sigma\tau\} & \mathbb{Q} \end{array}$$

which exhausts all possible intermediate fields of $\mathbb{Q}(\sqrt{2},\sqrt{3})/\mathbb{Q}$

5. Take the splitting field of x^4-2 (say we choose adjoints $\mathbb{Q}(i,\sqrt[4]{2})$). Since $\text{ch}(\mathbb{Q})=0$, the polynomial must be separable, and hence $\mathbb{Q}(i,\sqrt[4]{2})$ must be Galois. Then we know that:

$$[\mathbb{Q}(i,\sqrt[4]{2}):\mathbb{Q}] = [\mathbb{Q}(i,\sqrt[4]{2}):\mathbb{Q}(\sqrt[4]{2})][\mathbb{Q}(\sqrt[4]{2}):\mathbb{Q}] = 4\cdot 2 = 8$$

Telling us the Galois group has 8 elements. Since $i^n\sqrt[4]{2}$ are the 4 roots of x^4-2, we see that

$$\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2})) \leqslant S_4$$

Since $|\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))| = 2\cdot 4$, $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2}))$ has a Sylow 2-subgroup of S_4. By Sylow's 2nd theorem (theorem 4.8.5) this group then *must* be D_8.

This makes finding the automorphisms quite easily. Notice that the automorphism σ:

$$\begin{cases} \sqrt[4]{2} \longrightarrow i\sqrt[4]{2} \\ \\ i \longrightarrow i \end{cases}$$

is cyclic of order 4, and the automorphism τ:

$$\begin{cases} \sqrt[4]{2} \longrightarrow \sqrt[4]{2} \\ \\ i \longrightarrow -i \end{cases}$$

is cyclic of order 2, and that:

$$\sigma\tau = \tau^{-1}\sigma$$

Which gives us that $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(i,\sqrt[4]{2})) \cong D_8$! We correspond the subgroups with the intermediate fields in example 20.1.22.

6. As we've seen with cyclotomic fields, by proposition 19.7.9,

$$\mathrm{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^{\times}$$

giving us a Galois group for every $n \in \mathbb{N}_{>0}$.

7. By theorem 19.7.10, all finite field are Galois, since they are the splitting field over the separable polynomial $x^{p^d} - x$. As we've seen in proposition 19.7.19, the galois group is cyclic of order d:

$$\mathrm{Aut}_{\mathbb{F}_p}(\mathbb{F}_{p^d}) \cong \mathbb{Z}/d\mathbb{Z}$$

and is generated by the frobenius k-automorphism $x \mapsto x^p$

8. (An example of a normal, but not separable extension, is $F_2(x^{1/2})/F_2(x)$ (since $x^2 - t$ is inseparable over $\mathbb{F}_2(x)$), and so it's not Galois)

The correspondence by theorem 20.1.13 is in fact even stronger: we can correspond the intermediate lattice to the subgroup lattice, and we can (Something about how the III gal fund thm feels like it's defininga group-action). The first extension of the theorem is to give the correspondence between the lattices of subgroups of the galois group and of the field extensions. To show two lattices corresond, we need to show that if A_1 and A_2 are two objects, then A_1A_2 (i.e. the smallest object containing both A_1 and A_2) corresponding to B_1B_2, or if the relation is contravariant, A_1A_2 corresponds to $B_1 \cap B_1$. This is what we show with subgroups and intermediate fields:

Theorem 20.1.21: Fundamental Theorem Of Galois Theory II

Let F/k be a Galois extension. Then the Galois correspondence is an inclusion-reversing isomorphism between the lattice of intermediate subfield of F/k with the lattice of subgroups of $\mathrm{Aut}_k(F)$.

In particular, If E_1, E_2 are intermediate fields and G_1, G_2 are the correspondong subgroups of $\mathrm{Aut}_k(F)$, then $E_1 \cap E_2$ coresponds to $\langle G_1, G_2\rangle$ and E_1E_2 corresponds to $G_1 \cap G_2$

***Proof*:**
From theorem 20.1.13, we get the bijection between the intermediate fields and the subgroups of $\mathrm{Gal}_k(F)$, and from the 3rd property of the Galois Correspondence (proposition 20.1.9), we get that:

$$\mathrm{Aut}_{E_1E_2}(F) = G_1 \cap G_2 \qquad F^{\langle G_1, G_2\rangle} = E_1 \cap E_2$$

completing the proof.

We usually write the lattice of fields with k on the bottom and F on top, while the lattice of groups usualy has $\{e\}$ on top and $\mathrm{Aut}_k(F)$ on bottom to more easily associate the two lattices:

$$
\begin{tikzcd}
F \arrow[d, dash] & \{e\} \arrow[d, dash] \\
E \arrow[d, dash] & \mathrm{Aut}_E(F) \arrow[d, dash] \\
k & \mathrm{Aut}_k(F)
\end{tikzcd}
$$

Some authors mark the Galois group though their own edges:

$$
\begin{tikzcd}
F \arrow[d, dash, "\mathrm{Aut}_E(F)"'] \arrow[dd, dash, bend left, "\mathrm{Aut}_k(F)"] \\
E \arrow[d, dash] \\
k
\end{tikzcd}
$$

Example 20.1.22: Lattice Correspondence

1. As we've shown in example (20.1.20)(3), where we corresponded the subgroups of $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3}))$ with the intermediate fields $\mathbb{Q} \subseteq E \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$, we found that $\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3})) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ and intermediate fields for each subgroup. We can make this result more precise. If the subgroup lattice is:

$$
\begin{tikzcd}
& \langle (0,0) \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle (1,0) \rangle \arrow[dr, dash] & \langle (1,1) \rangle \arrow[d, dash] & \langle (0,1) \rangle \arrow[dl, dash] \\
& \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} &
\end{tikzcd}
$$

then the corresponding intermediate field lattice is:

$$
\begin{tikzcd}
& \mathbb{Q}(\sqrt{2}, \sqrt{3}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash] & \mathbb{Q}(\sqrt{6}) \arrow[d, dash] & \mathbb{Q}(\sqrt{3}) \arrow[dl, dash] \\
& \mathbb{Q} &
\end{tikzcd}
$$

We can futhermore deduce that there are *no* further intermediate extensions $k \subseteq E \subseteq F$.

2. As we've shown in example (20.1.20)(4), $x^4 - 2$ has splitting field $\mathbb{Q}(i, \sqrt[4]{2})$ with Galois group

$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(i, \sqrt[4]{2})) = D_8$. Then:

$$\begin{tikzcd}[column sep=small]
 & & & & D_8 = \langle a, b\rangle \arrow[dlll, dash] \arrow[d, dash] \arrow[drrr, dash] & & & & \\
 & \langle a^2, b\rangle \arrow[dl, dash] \arrow[d] \arrow[drrr, dash] & & & \langle a\rangle \arrow[d, dash] & & & \langle a^2, ab\rangle \arrow[dlll, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\langle b\rangle \arrow[drrrr, dash] & \langle a^2 b\rangle \arrow[drrr, dash] & & & \langle a^2\rangle \arrow[d, dash] & & & \langle ab\rangle \arrow[dlll, dash] & \langle a^3 b\rangle \arrow[dllll, dash] \\
 & & & & \{e\} & & & &
\end{tikzcd}$$

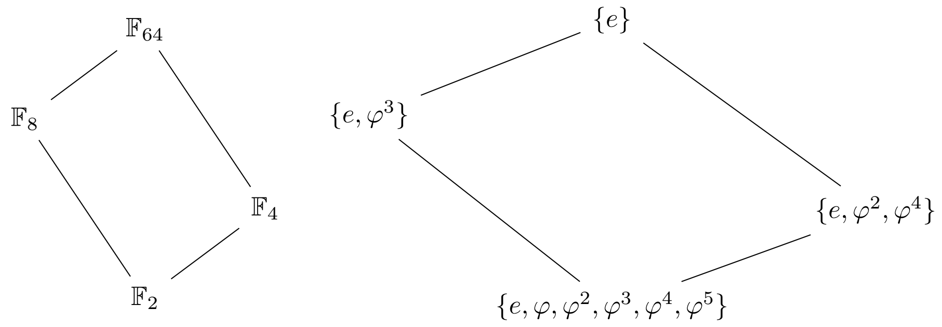
with corresponding intermediate field lattice:

$$\begin{tikzcd}[column sep=small]
 & & & & \mathbb{Q} \arrow[dlll, dash] \arrow[d, dash] \arrow[drrr, dash] & & & & \\
 & \mathbb{Q}(\sqrt[4]{4}) \arrow[dl, dash] \arrow[d] \arrow[drrr, dash] & & & \mathbb{Q}(i) \arrow[d, dash] & & & \mathbb{Q}(\sqrt[4]{4}i) \arrow[dlll, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt[4]{2}) \arrow[drrrr, dash] & \mathbb{Q}(\sqrt[4]{2}i) \arrow[drrr, dash] & & & \mathbb{Q}(i, \sqrt[4]{4}) \arrow[d, dash] & & & \mathbb{Q}(\sqrt[4]{2}(1+i)) \arrow[dlll, dash] & \mathbb{Q}(\sqrt[4]{2}(1-i)) \arrow[dllll, dash] \\
 & & & & \mathbb{Q}(i, \sqrt[4]{2}) & & & &
\end{tikzcd}$$

3. Take $\mathbb{F}_{64}/\mathbb{F}_2$, which is a degree 6 extension with Galois group $\mathrm{Aut}_{\mathbb{F}_2}(\mathbb{F}_{64}) \cong \mathbb{Z}/6\mathbb{Z}$. the subgroup lattice is:

$$\begin{tikzcd}
 & \{0\} \arrow[dl, dash] \arrow[ddr, dash] & \\
\{0, 3\} \arrow[ddr, dash] & & \\
 & & \{0, 2, 4\} \arrow[dl, dash] \\
 & \mathbb{Z}/6\mathbb{Z} &
\end{tikzcd}$$

which corresponds to:



The next result comes from recalling that if $k \subseteq E \subseteq F$ is a series of field extensions where F/k is normal, this does not imply E/k is normal (as we have discussed in proposition 19.5.16 and example 19.5.17). Intuitively, this can be thought of as E having some but not all the required roots so that F is normal (ex. $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \varphi)$). However, it is sometimes the case that the intermediate fields are normal (ex. $Q \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$). We may ask when more generally is an extension E/k also normal, or more generally Galois? The following theorem answers that question:

Theorem 20.1.23: Fundamental Theorem Of Galois Theory III

Let F/k be a galois extension, and let E be an intermediate field. Then E/k is Galois if and only if $\mathrm{Aut}_E(F)$ is *normal* in $\mathrm{Aut}_k(F)$. If this is the case, then

$$\mathrm{Aut}_k(E) = \mathrm{Aut}_k(F)/\mathrm{Aut}_E(F)$$

Hence, *normal* subgroups arise from *normal* extensions. In fact, the term "normal" originate in field theory, and the term normal subgroup was coined after it's field theory counterpart.

The proof essentially comes down to a couple of key observations about the cosets of $\mathrm{Aut}_k(F)/\mathrm{Aut}_E(F)$ and how we can surmise information about them through group actions! We have already observed in remark 20.1.19 that E/k has the same cardinality as $\{g\mathrm{Aut}_E(F) \mid g \in \mathrm{Aut}_k(F)\}$. We will make this correspondence more precise.

(We have seen in theorem 19.5.13 that E is normal if and only if $\iota(E) = E$ for all ι. We are building up the consequences of this I think)

Let F/K be a Galois extension. Identify $F \subseteq \overline{k}$ within some algebraic closure of $\overline{k}$. Let $k \subseteq E \subseteq F$ be some intermediate field, and consider:

$$I = \left\{\iota : E \to \overline{k} \;\middle|\; \iota \text{ is a } k\text{-homomorphism } \right\} = \mathrm{Hom}_k(E, \overline{k})$$

We can equivalently see E/k as an arbitrary extension of k where for some $\iota \in I$, $\iota(E) \subseteq F$.

Claim 1: for all $\iota \in I$, $\iota(E) \subseteq F$

Proof. Let ι be a $\iota \in I$ such that $\iota(E) \subseteq F$. Since E is separable (since $\iota(E) \subseteq F$ and F is separable) and finite, is is simple by theorem 19.6.20, Thus $E = k(\alpha)$ for some $\alpha \in \overline{k}$ and α is the root some minimal polynomial over k, $m_{\alpha,k}(x)$. Since $\iota(\alpha)$ determines ι, looking at where it maps is all that matters. Since $\iota(\alpha)$ is *some* root of the minimum polynomial $m_{\alpha,k}(x)$, and F is normal, F contains *all* roots of the minimal polynomial. But then, for any ι, $\iota(E) \subseteq F$, as we sought to show. □

Notice that

Claim 2: $\mathrm{Aut}_k(F)$ acts on I: for $g \in \mathrm{Aut}_k(F)$ and $\iota \in I$, $g \circ \iota \in I$

Proof. Since $\iota(E) \subseteq F$, and $g : F \to F$, we can intermret ι as $\iota : E \to F$ and compose the functions □

claim 3: $\mathrm{Aut}_k(F)$ is transitive on I (i.e. on $\mathrm{Hom}_k(E, \overline{k})$)

Proof. We need to show that for any $\iota_1, \iota_2 \in I$, $g \circ \iota_1 = \iota_2$. This follows from uniqueness of the splitting field. Since $\iota_1(E)$ and $\iota_2(E)$ each contain k and are each contained in F, and F is a splitting field over k for some polynomial, say $f(x)$ (since F is Galois, it follows from theorem 20.1.13). Hence, *a fortiori*, F must be the splitting field of $f(x)$ over $\iota_1(E)$ and over $\iota_2(E)$. Note that:

$$\iota_2 \circ \iota_1^{-1} : \iota_1(E) \to \iota_2(E)$$

is an k-isomorphism, and hence by the uniqueness of the splitting field, this isomorphism is extended to a k-isomorphism:

$$g : F \to F$$

By construction, $g|_{\iota_1(E)} = \iota_2 \circ \iota_1^{-1}$, or equivalently:

$$g \circ \iota_1 = \iota_2$$

as we sought to show. □

Thus, the galois group $\mathrm{Gal}_k(F)$ always acts transitively on $\mathrm{hom}_k(E, \overline{k})$ (do we know how it linkes corresponding embeddings more precisely? Is that the III thm?)

Now, choose an $\iota \in I$ and identify E with the intermediate field $k \subseteq \iota(E) \subseteq F$. Then $\mathrm{Aut}_E(F)[= \mathrm{Aut}_{\iota(E)}(F)]$ is a subgroup of $\mathrm{Aut}_k(F)$ of elements that restrict to the identity on $E[= \iota(E)]$. More particularly, $\mathrm{Aut}_E(F)$ consists of the *stabalizers of* I (i.e. for all $g \in \mathrm{Aut}_E(F)$, $g \circ \iota = \iota$). Thus, by the Orbit-Stabalizer Thoerem, we can deduce the following result:

Let $G = \mathrm{Aut}_k(F)$. Then consdering I as a G-set, I is isomorphic to the set of left-cosets of $\mathrm{Aut}_E(F)$ in $\mathrm{Aut}_k(F)$

20.1.24 key Isomorphism

$$\mathrm{hom}_k(E, \overline{k}) \quad \cong_G \quad \{g\mathrm{Aut}_E(F) \mid g \in \mathrm{Aut}_k(F)\}$$

(Aluffi says he views this statement as a key statement in Galois Theory, so it's worth pondering over it more. I believe it's becasue the bijection is not just a bijection, but also a G-isomorphism, and we see that through this result.)

***Proof*:**
of theorem 20.1.23

Let $G = \text{Aut}_k(F)$ and $I = \{\iota : E \to \overline{k} \mid \iota \text{ is a } k\text{-homomorphism}\}$. Then as we've just seen, I is a G-set, and it's stabalizer is $\text{Aut}_E(F)$. By proposition ref:HERE (I don't think I've proven this, it's proposition II.9.12 in Aluffi), if ι_1, ι_2 are both in the stabalizer, they are conjugates: there exists a $g \in \text{Aut}_k(F)$ such that $g \circ \iota_1 \circ g^{-1} = \iota_2$, in particular, ι and id are conjugates for any $\iota \in I$. Thus, if $\text{Aut}_E(F)$ is normal:

$$\text{Aut}_{\iota(E)}(F) = \text{Aut}_E(F)$$

which, by the bijectivity of the Galois correspondence for F/k, we have that $\iota(E) = E$. But then E/k satisfies the 5th equivalence of Galois Extensions in theorem 20.1.13, meaning E/k is Galois

Conversely, if E/k is galois, then $\iota(E) = E$ for all ι (again by the 5th equivalence of Galois extensions in theorem 20.1.13). Define the homomorphism ρ which takes $g \in \text{Aut}_k(F)$ and restricts it to $E[= \iota(E)]$:

$$\rho : \text{Aut}_k(F) \to \text{Aut}_k(E)$$

This homomorphism is surjective (since the G-action acts transitively on I), and $\ker(\rho)$ is all automorphism that restrict to the identity on E, that is, $\text{Aut}_E(F)$. But then, we know that the kernel of a group homomorphism is normal, and by the 1st isomorphism theorem for groups:

$$\text{Aut}_k(F)/\text{Aut}_E(F) \cong \text{Aut}_k(E)$$

completing the proof.

("It sends subgroup to subgroups by conjugation", i.e. if E is associatd to $H = \text{Aut}_E(F)$, then for any $\sigma \in \text{Aut}_k(F)$, $\sigma(E)$ is associated to $\text{Aut}_{\sigma(E)}(F) = \sigma H \sigma^{-1}$)

(There is another really powerful result. Let's say $F/E/k$ is a sequence of extensions, F/k and E/k both Galois (and hence F/E is Galois. Then the map

$$\text{Gal}_E(F) \times \text{Gal}_k(E) \to \text{Gal}_k(F) \qquad (\sigma, \tau) \mapsto \sigma \circ \tau$$

is is well defined and bijective (namely, since the identity must map to the identity, and since E/k is normal $\text{Gal}_k(F)$ is determined by what happens to E, and then fixing E, what happens to F. Thus, constructing Galois groups is much easier in this case!!! Note, however, that this is a homomorphism if and only if $E \cap F = k$. Once I read through Dummit's and foote on Composite fields, this will be incorporated))

Corollary 20.1.25: Isomorphism Of Categories

Let $\textbf{Fld}_k^E$ be the collection of fields F such that $k \subseteq F \subseteq E$ whose morphisms are k-homomorphisms and $\mathcal{O}_G$ where $G = \text{Aut}_k(E)$ be the category whose objects are left G-sets G/H for every $H \subseteq G$ and the morphisms are G-equivariant maps $G/H \to G/K$. Then as categories, $\mathcal{O}_G^{\text{op}} \cong \textbf{Flb}_k^E$

***Proof*:**
see p.21 of Category theroy in Context if you need some help

(more in dummit and foote on Galois Composite fields! Really cool!)

Example 20.1.26: Application To Lattices
(Draw two latties, and show the normal extensions)

20.1.27 Foreshadowing Can give the connection between Galois group/extensions to Algebraic geometry with regular covers and deck transformations

Exercise 20.1.1

1. Let $L = \mathbb{Q}(\zeta_p)$ where p is an odd prime. Show there exists a subfield $K \subseteq L$ such that $K = \mathbb{Q}(\sqrt{p'})$ where
$$p' = \begin{cases} p & p \equiv 1 \bmod 4 \\ -p & p \equiv 2, 3 \bmod 4 \end{cases}$$

2. Let $K/\mathbb{Q}$ be an odd Galois Extension, Show that L must be totally real.

20.1.2 Composite Fields

We now take what we proved earlier and do two things with it: we extend the results for composite fields and we classify all cyclic Galois Extensions (given an appropriate restriction on the base field) In the previous chapters, there has consistenly been a question working with composite fields (exercise ref:HERE), in particular, it was asked to be shown that:

> If E/k and F/k are field extensions, and E/k is (finite/algebraic/normal/separable resp.), then EF/F is (finite/algebraic/normal/separable resp). If E/k and F/k are (finite/algebraic/normal/separable resp.) then EF/k is (finite/algebraic/normal/separable resp.)

Since [finite] Galois extensions are finite, normal, and separable, we see immediately that if E/k is Galois, then EF/F is Galois, and if E/k and F/k are Galois, so is EF/k. The natural question to ask is what's the galois group of a composite field and of EF/F:

Proposition 20.1.28: Galois Composite Fields I

Suppose E/k is Galois and F/k is any field extension. Then EF/F is Galois, and $\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$. Visually, we see that:

$$\begin{tikzcd}
& KF \arrow[dl, dash] \arrow[dr, dash, "G"] & \\
F \arrow[dr, dash, "G"'] & & K \arrow[dl, dash] \\
& F \cap K \arrow[d] & \\
& k &
\end{tikzcd}$$

The reader should ponder why does F/k need to be finite for this theorem?

***Proof*:**
Using the Galois extension equivalences, we can quickly prove that EF/k is Galois, namely since E/k is Galois, it is the splitting field of a separable polynomial $p(x) \in k[x] \in F[x]$. The roots of $p(x)$ generate E over k ,and so they generate EF over F, i.e., EF is the splitting field of $p(x)$ over F, thus EF/F is a Galois Extension

To show that $\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$, we'll first show we can embedd $\mathrm{Aut}_F(EF)$ into $\mathrm{Aut}_k(E)$, and then find that the subgroup of $\mathrm{Aut}_k(E)$ associated to this embedding is $\mathrm{Aut}_{E\cap F}(E)$. For the embedding, consider first any F-automorphism $\sigma : EF \to EF$ (and hence k-automorphism). Then we can restrict σ to E, giving us $\sigma : E \to \sigma(E)$. Since E is Galois, $\sigma(E) = E$. Thus, any element of $\mathrm{Aut}_F(EF)$ get's mapped to an element of $\mathrm{Aut}_k(E)$, that is, we have a natural group homomorphism:

$$\rho : \mathrm{Aut}_F(EF) \to \mathrm{Aut}_k(E)$$

This homomorphism is injective: If the only element that maps to the identity is the identity, then we're done. Indeed, if $\sigma \in \mathrm{Aut}_k(E)$ is the identity on E, since any element $\mathrm{Aut}_F(EF)$ is the identity on F, the coresponding σ in $\mathrm{Aut}_F(EF)$ must be the identity on EF, showing injectivity.

Thus, $\mathrm{Aut}_F(EF)$ embeds into $\mathrm{Aut}_k(E)$, and so it is associated with some $H \leqslant \mathrm{Aut}_k(E)$. By proposition 20.1.12, $H = \mathrm{Aut}_{E^H}(E)$. It suffices to show that $E^H = E \cap F$. For the $\supseteq$ direction, since every $\sigma \in \mathrm{Aut}_F(EF)$ fixes F, the restriction $\rho(\sigma)$ fixes $E \cap F$, hence $E^H \supseteq E \cap F$. For the $\subseteq$ direction, take $\alpha \in E^H$. Then $\alpha \in E$. By definition, $F(\alpha)$ is in the fixed field of $\mathrm{Aut}_F(EF)$, which is F sinc F is Galois over EF/F (1st Galois Theorem (20.1.13)(4)). Thus, $\alpha \in F$, and so $E^H \subseteq E \cap F$. Therefore, $H = E \cap F$, and:

$$\mathrm{Aut}_F(EF) \cong \mathrm{Aut}_{E\cap F}(E)$$

as we sought to show

Example 20.1.29: Composite Fields Determining Abelian Extensions

1. As we have seen in section 19.7.1, if ζ_n is the nth root of unity, $\mathbb{Q}(\zeta_n)$ is the splitting field of $x^n - 1$ over $\mathbb{Q}$, and hence is Galois since $\text{ch}(\mathbb{Q}) = 0$. We have shown in proposition ref:HERE that $\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

   Now, let k be any field of characteristic 0. Then the splitting field K/k for $x^n - 1$ is the composite of k and $\mathbb{Q}(\zeta)$, namely $K = k(\zeta)$. Since $\mathbb{Q}/(\zeta)/\mathbb{Q}$ is Galois and $k/\mathbb{Q}$ is a field extension, by proposition 20.1.28, we get that:

$$\text{Aut}_k(k(\zeta)) \cong \text{Aut}_{\mathbb{Q}(\zeta)\cap k}(\mathbb{Q}(\zeta)) \leqslant \text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\zeta)) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

   showing that $\text{Aut}_k(k(\zeta))$ is some subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$. in particular, *any* such extension *must* be abelian.

   The fact that every extension of the form $k(\zeta)$ is abelian is quite powerful. In fact, they are relatively easy to classify. The study of Galois extensions that are abelian(which also must contain n distinct roots of unity for some $n \in \mathbb{N}$) is called *Kummer Theory*. We touch a bit of Kummer theory in the following, since we need it for showing no quintic is unsolvable, and will later show that all finite abelian group show up as a field extension of $\mathbb{Q}$ in section 20.4.

2. Generalizing the results from the exercises, we know that if E/k and F/k are Galois, so is EF/k. Can you find a way to link $\text{Gal}_k(EF)$ with $\text{Gal}_k(E)$ and $\text{Gal}_k(F)$? (Hint: Consider $\text{Gal}_k(EF) \to \text{Gal}_k(E) \times \text{Gal}_k(F)$, $\sigma \mapsto (\sigma|_E, \sigma|_F)$)

Corollary 20.1.30: Order of Galois Composite

Let E/k be a galois extension and F/k be any finite extension. Then:

$$[EF : k] = \frac{[E : k][F : k]}{[E \cap F : k]}$$

***Proof*:**
exercise

Note that this formula does not hold in general if niether of the extensions is Galois.

Example 20.1.31: Failure of Formula with no Galois Extension

Let $k = \mathbb{Q}$, $E = \mathbb{Q}(\sqrt[3]{2})$, and $F = \mathbb{Q}(\zeta_3\sqrt[3]{2})$), since $EF = \mathbb{Q}(\zeta_3\sqrt[3]{2}), \sqrt{3}$ (with minimal polynomial $x^3 - 2$, and hence degree 6 extension) and:

$$6 = [\mathbb{Q}(\zeta_3\sqrt[3]{2})] \neq \frac{[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}][\mathbb{Q}(\zeta_3\sqrt[3]{2}) : \mathbb{Q}]}{[\mathbb{Q} : \mathbb{Q}]} = 9$$

Using this, we may answer the arguably more intesesting questions of the composite and intersection of fields being galois over the same base field:

Proposition 20.1.32: Galois Composite Fields II

Let $E/k, F/k$ be galois extensions of F. Then:

1. $E \cap F$ is galois over k
2. EF is galois over k. It's galois group is isomorphic to:

$$G = \{(\sigma, \tau) \mid \sigma|_{E\cap F} = \tau|_{E\cap F}\} \leqslant \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$$

3. If $E \cap F = k$, then:

$$\mathrm{Gal}_k(EF) \cong \mathrm{Gal}_k(E) \times \mathrm{Gal}_k(F)$$

***Proof*:**
D&F p.593

Corollary 20.1.33: Galois Closure

Let F/k be a finite separable extension. Then there exists a minimal finite galois extension E/k such that $E/F/k$.

***Proof*:**
Simply take the composite of the splitting fields of the minimal polynomials for the basis of F over k (which is separable since F/k is separable), nad intersect all the galois extensions of k containing F.

Definition 20.1.34: Galois Closure

Let F/K be a finite separable extension. Then the smallest extension $E/F/k$ such that E/k is galois is called the *galois closure* of F over k.

Exercise 20.1.2

1. Show there exists fields E, F such that $[EF : F] \neq [F : E \cap F]$

20.1.3 A bit of Character Theory and Kummer Theory

Let F/K be a galois extension. Then each $\sigma \in \mathrm{Gal}_K(F)$ is also a K-linear isomorphism $F \to F$ of n-dimensional K-vector spaces. Thus, does σ have eigenvalues? Can it be diagonalized? How does each $\sigma \in \mathrm{Gal}_K(F)$ relate to each other? To start:

Lemma 20.1.35: Eigenvalues Are Roots of Unity

Let F/K be a galois extension and $\sigma \in \mathrm{Gal}_K(F)$. Then if σ has an eigenvalue λ, then λ is a root of unity

***Proof*:**
Let σ be as in the statement and assume $\sigma(\alpha) = \lambda\alpha$ where $\lambda \in K$. Then as $\mathrm{Gal}_K(F)$ is a finite group, the exists an n such that $\sigma^n = \mathrm{id}$. Thus:

$$\alpha = \mathrm{id}(\alpha) = \sigma^n(\alpha) = \lambda^n\alpha$$

Then $\lambda^n = 1$, hence λ is a root of unity.

Thus, σ is diagonalizable if K contains enough roots of unity; such an extension is called a *Kummer extension* (see definition 20.4.2, the definition is delayed due to the need to consider the behaviour over different characteristic which is covered in section 20.4).

Next, we would like to know how each automorphism relates to each other. We shall use a concept from Character theory to frame the discussion:

Definition 20.1.36: Group Character

Let G be a group and L a field. Then a *[linear] character* (denoted χ) is a homomorphism from G to the multiplicative group of L:

$$\chi : G \to L^\times$$

Example 20.1.37: Group Character

For our purposes, any field homomorphism $\sigma : F \to K$ induces a character $\chi_\sigma : F^\times \to K^\times$ (since σ maps the multiplicative group of F into the multiplicative group of K). Note that χ_σ captures essentially all the mapping information of σ, since χ only omits mapping 0 from F to 0 from K, but we can simply deduce that. Thus, if we are working with χ_σ, we are showing that we do not want to remember that the additive structure is preserved.

Definition 20.1.38: Character Linearly Independent

The characters $\chi_1, ..., \chi_n$ of G are *linearly independent* over L if they are linearly independent as functions on G, i.e. for any $a_i \in L$

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_n\chi_n = 0 \quad \text{if and only if} \quad a_i = 0, 1 \leq i \leq n$$

Note that the expression on the left hand side of the above equality is a large function, and hence the it means that characters are linearly independent if for all $g \in G$:

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_n\chi_n(g) = 0 \quad \text{if and only if} \quad a_i = 0, 1 \leq i \leq n$$

Theorem 20.1.39: Artin's Lemma

Let $\chi_1, ..., \chi_n$ be distinct characters of G with values in L. Then they are linearly independent over L

***Proof*:**
Suppose that the characters are linearly dependent, that is, there exists (possibly many) sets of coefficients coefficients $a_1, ..., a_n \in L$ where not all are zero and

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_n\chi_n = 0$$

To construct a minimality argument, choose a set of not-all-zero coefficients of minimal cardinally that satisfies the above equation (such coefficients exists since these character are linearly dependent). Thus, for any $g \in G$:

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_n\chi(g) = 0 \tag{20.2}$$

Let g_0 be an element of G such that $\chi_1(g_0) \neq \chi_n(g_0)$ (which exists since $\chi_1 \neq \chi_n$). Since equation (20.2) holds for every element of G, we have that:

$$a_1\chi_1(g_0g) + a_2\chi_2(g_0g) + \cdots + a_n\chi_n(g_0g) = 0 \tag{20.3}$$

i.e.

$$a_1\chi_1(g_0)\chi_1(g) + a_2\chi_2(g_0)\chi_2(g) + \cdots + a_n\chi_n(g_0)\chi_n(g) = 0 \tag{20.4}$$

Now, multiplying equation (20.2) by $\chi_n(g_0)$ and subtracting by equation (20.4), we get:

$$[\chi_n(g_0) - \chi_1(g_0)]a_1\chi_1(g) + \cdots + [\chi_n(g_0) - \chi_{n-1}(g_0)]a_{n-1}\chi_{n-1}(g) = 0$$

which holds for all $g \in G$. However, the first coefficient is nonzero, and this is a relation with fewer nonzero coefficients than the minimal choice we have made earlier – a contradiction.

Corollary 20.1.40: linear independence of Automorphisms (Dedekind's Lemma)

Let $\sigma_1, \sigma_2, ..., \sigma_n$ be distinct embeddings of K into L. Then they are linearly independent as functions on K. In particular, distinct automorphism of a field K are linearly independent as functions on K.

***Proof*:**
This is an immediate consequence of theorem 20.1.39

Independence of automorphisms determines the structure of a Galois extension as a module over its Galois group. A finite Galois extension L/k with group G is a k-vector space of dimension $|G|$ carrying a G-action, so it is a $k[G]$-module of the right size to be free of rank one, and the normal basis theorem says that it is. Equivalently, L has a k-basis permuted simply transitively by G, in contrast with the bases $1, \alpha, \ldots, \alpha^{n-1}$ coming from a primitive element, on which G acts in no controlled way.

Theorem 20.1.41: Normal Basis Theorem

Let L/k be a finite Galois extension with group G, and put $n = [L : k] = |G|$. There exists $x \in L$ whose G-orbit $\{\sigma(x) \mid \sigma \in G\}$ is a k-basis of L. Such a basis is called a *normal basis*, and its existence says exactly that

$$L \cong k[G]$$

as $k[G]$-modules, so that L is free of rank one over the group ring.

***Proof*:**
Write $G = \{\sigma_1, \ldots, \sigma_n\}$ with $\sigma_1 = 1$. The two statements are equivalent: a k-basis of the form $\{\sigma(x) \mid \sigma \in G\}$ is precisely a $k[G]$-basis of L consisting of the single element x.

The criterion used throughout is this. For $x \in L$ the orbit $\{\sigma(x) \mid \sigma \in G\}$ is a k-basis if and only if

$$\det\left(\sigma_i^{-1}\sigma_j(x)\right)_{i,j} \neq 0$$

Indeed, suppose the determinant is non-zero and $\sum_j c_j\sigma_j(x) = 0$ with $c_j \in k$. Applying σ_i^{-1} and using that the c_j lie in k, hence are fixed, gives $\sum_j c_j\sigma_i^{-1}\sigma_j(x) = 0$ for every i. So the matrix annihilates the vector (c_j) and therefore $c = 0$. As there are n elements in the orbit and $\dim_k L = n$, independence is enough. The converse also holds, though it is not needed below: writing $v_j = \sigma_j(x)$ and $\tau_i = \sigma_i^{-1}$, which runs over G, the matrix is $(\tau_i(v_j))$, and if $v = wa$ for a k-basis w and $a \in \mathrm{GL}_n(k)$ then $\det(\tau_i(v_j)) = \det(\tau_i(w_l))\det(a)$, non-zero because $\det(\tau_i(w_l)) \neq 0$ by corollary 20.1.40. It is not the matrix of a k-linear map: for $x \in k^\times$ every entry equals x and the determinant vanishes for $n \geq 2$, so independence of automorphisms is doing the work here too.

Case 1: k infinite. Fix a k-basis $w_1, \ldots, w_n$ of L and consider

$$g(X_1, \ldots, X_n) = \det\Big(\textstyle\sum_l X_l\,\sigma_i^{-1}\sigma_j(w_l)\Big)_{i,j} \in L[X_1, \ldots, X_n]$$

This polynomial is not zero. By corollary 20.1.40 the automorphisms in G are linearly independent over L. Hence the $n \times n$ matrix $\left(\sigma(w_l)\right)_{\sigma,l}$ is invertible over L: an L-linear relation among its rows gives $\sum_\sigma a_\sigma\sigma$ vanishing on the k-basis w, hence, that map being k-linear, vanishing on all of L. Choose $(\xi_l) \in L^n$ with $\sum_l \xi_l\,\sigma(w_l) = 1$ for $\sigma = 1$ and 0 for every other $\sigma \in G$. Substituting $X_l = \xi_l$ turns the (i,j) entry into 1 when $\sigma_i^{-1}\sigma_j = 1$, that is when $i = j$, and into 0 otherwise, so $g(\xi) = \det(I_n) = 1$.

Now g has coefficients in L, not in k, so the familiar statement about polynomials over the field one substitutes from does not apply directly. What is needed is the infinite-subset form: *a non-zero polynomial in $L[X_1, \ldots, X_n]$ cannot vanish on S^n for an infinite subset $S \subseteq L$.* This follows by induction on n, the case $n = 1$ being that a non-zero one-variable polynomial has finitely many roots (corollary 11.3.2): writing $g = \sum_d g_d(X_1, \ldots, X_{n-1})X_n^d$ with some $g_{d_0} \neq 0$, induction gives $s \in S^{n-1}$ with $g_{d_0}(s) \neq 0$, and then $g(s, X_n)$ is a non-zero polynomial in one variable, so it has a non-root in S. Applying this with $S = k$, which is infinite, gives $t_1, \ldots, t_n \in k$ with $g(t) \neq 0$. Put $x = \sum_l t_l w_l$. Then $\det(\sigma_i^{-1}\sigma_j(x)) = g(t) \neq 0$ and the criterion applies.

Case 2: k finite. Then L/k is an extension of finite fields, so G is cyclic, generated by the Frobenius σ. Regard L as a module over the polynomial ring $k[T]$ with T acting as σ. Because $\sigma^n = 1$, the minimal polynomial of σ as a k-linear operator on L divides $T^n - 1$, so it has degree at most n. It has degree exactly n: a relation $\sum_{i<n} c_i\sigma^i = 0$ with $c_i \in k$ not all zero would be a non-trivial L-linear dependence among the distinct automorphisms $1, \sigma, \dots, \sigma^{n-1}$, which corollary 20.1.40 forbids.

So the minimal polynomial of σ has degree $n = \dim_k L$. It divides the characteristic polynomial, which is monic of the same degree by proposition 14.3.9, so the two are equal, and σ then admits a cyclic vector. That last step is theorem 14.3.5 together with proposition 14.3.1, which identifies the minimal polynomial as the largest invariant factor, and lemma 14.3.8, which makes the characteristic polynomial the product of the invariant factors: equality of degrees forces a single invariant factor, so $L \cong k[T]/(a_1)$ is cyclic over $k[T]$. Concretely there is $x \in L$ with $L = \operatorname{span}_k\{x, \sigma x, \dots, \sigma^{n-1}x\}$. That spanning set has n elements in an n-dimensional space, so it is a basis, and it is the G-orbit of x, completing the proof.

Proposition 20.1.42: Galois And Cyclic Extensions

Let F/k be an extension of degree m. Assume that k contains a primitive mth root of 1 and $\operatorname{char}(k)$ does not divide m. Then F/k is Galois and cyclic if and only if $F = k(\delta)$ with $\delta^m \in k$.

***Proof*:**
Let $\zeta \in k$ be a primitive m-th root of 1.

($\Leftarrow$) Let's say $F = k(\delta)$ with $\delta^m = c \in k$. Then all m roots of the polynomial $x^m - c$ are in F, in particular,

$$c, \zeta\delta, \zeta^2\delta, ..., \zeta^{m-1}\delta \in F$$

Furthermore, F must be generated by them. Hence F is the splitting field of the separable polynomial $x^m - c$ in k (separable since $\operatorname{ch}(k)$ does not divide m). Thus, F is Galois

Next, to see that $\operatorname{Aut}_k(F)$ is cyclic, note that for any $\varphi \in \operatorname{Aut}_k(F)$, it is determined by $\varphi(\delta)$, which his necessarily a root of $x^m - c$, and so $\varphi(\delta) = \zeta^i\delta$ for some i determined up to a multiple of m. Then this can be used to define an isomorphism of $\operatorname{Aut}_k(F)$ with $\mathbb{Z}/m\mathbb{Z}$.

($\Rightarrow$) Conversely, assume that F/k is Galois and $\operatorname{Aut}_k(F)$ is cyclic (say, generated by φ). Since the automorphisms φ^i $(1 \le i \le m-1)$ are pairwise distinct, by corollary 20.1.40, they are linearly independent. Therefore, there exists an $\alpha \in F$ such that

$$\delta := \alpha + \zeta^{-1}\varphi(\alpha) + \cdots + \zeta^{-(m-2)}\varphi^{m-2}(\alpha) + \zeta^{-(m-1)}\varphi^{m-1}(\alpha) \neq 0$$

Then note that :

$$\varphi(\delta) := \varphi(\alpha) + \zeta^{-1}\varphi^2(\alpha) + \cdots + \zeta^{-(m-2)}\varphi^{m-1}(\alpha) + \zeta^{-(m-1)}\alpha = \zeta\delta$$

since $\zeta \in k$ so $\varphi(\zeta) = \zeta$. Then δ is *not* fixed by any nonidentity element of $\operatorname{Aut}_k(F)$, that is $\operatorname{Aut}_{k(\delta)}(F) = \{e\}$, so $k(\delta) = F$. Furthermore:

$$\varphi(\delta^m) = \varphi(\delta)^m = \zeta^m\delta^m = \delta^m$$

so, δ^m is fixed by φ, and hence by every element of $\operatorname{Aut}_k(F)$. Since F/k is Galois, it must be that $\delta^m \in k$, completing the proof.

20.1.43 Kummer Theory This proposition shall have an alternative proof (theorem 20.4.6) in section 20.4.

Proposition 20.1.44: Kummer Extension and Eigenvalue

Let $F = K(\sqrt[n]{a})$ and let K have enough roots of unity and $\text{char}(K) \nmid n$. Let σ be a generator for $\text{Gal}_K(F)$. Then $\sqrt[n]{a}$ is the eigenvector of σ.

***Proof*:**
exercise.

Exercise 20.1.3

1. Let $k(t)/k$ be an infinite extension. Determine $\text{Aut}_k(k(t))$
2. Show that for all n, there is an extension F of $\mathbb{Q}$ such that $F/\mathbb{Q}$ is not Galois. Similarly, show that for all n, there exists an extension L such that $\text{Gal}_{\mathbb{Q}}(L) \cong \mathbb{Z}/n\mathbb{Z}$ (Hint: show that for all n, there exists a p such that $p \equiv 1 \bmod n$)

20.2 Application of Galois Theory

20.2.1 Fundamental Theorem of Algebra

(Some build-up with properties of $\mathbb{C}$ that are given a exercsie 5.23)

Theorem 20.2.1: Fundamental Theorem of Algebra

$\mathbb{C}$ is Algebraically closed

***Proof*:**
Take $f(x) \in \mathbb{C}[x]$ to be a nonconstant polynomial. If $f(x)$ is irreducible, then since conjugation is a field-homomorphism, $f(x) \to \overline{f(x)}$, then $f(x)\overline{f(x)}$ is also irreducible. Since $f(x)\overline{f(x)} \in \mathbb{R}[x]$, we can assume without loss of generality that all coefficents are real. Let F be the splitting field of $f(x)$ over $\mathbb{R}$, and embed $F \subseteq \overline{\mathbb{R}}$ (we want to show that $\overline{\mathbb{R}} = \mathbb{C}$).

Now, consider the extension:

$$\mathbb{R} \subseteq F(i)$$

this extension is Galois: it is the splitting field of the square-free part of $f(x)(x^2 + 1)$ (any irreducible polynomial over $\mathbb{R}$ of degree ≥ 2 will have such a square free part, as should be verified TBD). Let $G = \text{Aut}_{\mathbb{R}}(F(i))$ By Sylow's Theorem, G has a 2-sylow subgroup H (even if $2 \nmid |G|$, since then it would be $\{e\}$). Then the index $[G : H]$ is oddand so by the Galois correspondence, it corresponds to a finite separable extension $\mathbb{R} \subseteq E$ with $[E : \mathbb{R}]$ odd. Since every polynomial of odd degree has a root by elementary calculus [a], it must be that $\mathbb{R} \subseteq E$ is *trivial* (since TBD, was exercsie VII.5.23 in Aluffi which stated that if $k[x]$ had no irreducible polynomials of degree $n > 0$, then it has no separable extensions). Thus, $[G : H] = 1$, proving that G is a 2-group. Since $\mathbb{C} = \mathbb{R}(i) \subseteq F(i)$, the Galois group ofthe Galois extension $|C \subseteq F(i)$ is a subgroup of G, and therefore it is a 2-group: $|\mathrm{Aut}_{\mathbb{C}}(F(i))| = 2^n$ for some $n \geq 0$.

To finish the proof, we must show that $n = 0$. By proposition ref:HERE, for any $1 \leq m \leq n$, a group of order p^n contains a subgroup of order p^m. For this case, we have that $\mathrm{Aut}_{\mathbb{C}}(F(i))$ contains a subgroup of order 2^{n-1}. However, by exercise 5.23 again, $\mathbb{C}$ contains *no* quadratic exensions of $\mathbb{C}$, since there are no irreducible polynomials in $\mathbb{C}[x]$.

Thus, $n = 0$, so $\mathrm{Aut}_{\mathbb{C}}(F(i))$ is trivial, and so $F(i) = \mathbb{C}$, in particular $\mathbb{C}$ contains the roots of $f(x)$.

[a]Namely, the intermediate value theorem

This proof can also be accomplished without Sylow's theorem's by HERE. However, since the tools were developped, it is good practice to use them.

Corollary 20.2.2: No Field Structure On $\mathbb{R}^n$, $n \geq 3$

There does not exist a field structure on $\mathbb{R}^n$ $(n \geq 3)$ such that $\mathbb{R}$ includes into $\mathbb{R}^n$ (i.e. $\mathbb{R}^n$ is a field extension of $\mathbb{R}$)

***Proof*:**
For the sake of contradiction, assume $\mathbb{R} \subseteq \mathbb{R}^n$ is a field extension for $n \geq 3$. Since $\mathbb{R}^n$ is an n-dimensional vector-space, it is a degree n algebraic extension. Hence, by lemma 19.5.7, there is an (injective) homomorphism $\mathbb{R}^n \to \mathbb{C}$ fixing $\mathbb{R}$. However, the dimension of the right hand side is greater than 2, and hence no such homomorphism exists, a contradiction.

Note that we had to be careful in how we state the above corollary. It is not true that there is no field structure on $\mathbb{R}^n$ $(n \geq 3)$. Since $\mathbb{R}^n$ and $\mathbb{R}$ have the same cardinality, we may (by the axiom of choice) find a bijection from $\mathbb{R}^n$ to $\mathbb{R}$. Then we may put a field structure on $\mathbb{R}$ through the bijection. However, the field structure on $\mathbb{R}^n$ would certainly not have $\mathbb{R}$ as a subfield, and it impossible to determine the product of any two elements[8].

Theorem 20.2.3: Frobenius Theorem

There are only 3 finite-dimensional associative division algebras, namely

1. $\mathbb{R}$, the real numbers
2. $\mathbb{C}$, the complex numbers
3. $\mathbb{H}$, the Quaternions

If we drop associativity, there is furthermore:

4. $\mathbb{O}$, the Octonions

[8]loosely speaking, if we would be able to explicitly find a bijection from $\mathbb{R}^n$ to $\mathbb{R}$, but that would be a "countable process", but $\mathbb{R}^n$ and $\mathbb{R}$ is uncountable

***Proof*:**
This is beyond this book; for a proof see for example [20, Bott Periodicity]. A high level summary is the existence of a division algebras on $\mathbb{R}^n$ is tied to whether or not the tangent bundle S^{n-1} is parallelizable (which as it turns out only TS^0, TS^1, TS^3, TS^7 are parallelizable). This result is dependent on the classical result in algebraic topology known as Bott Periodicity.

As an exercise, show that $\mathbb{H}$ is not a $\mathbb{C}$-algebra. The quaternions are also the Clifford algebra of the negative definite form $\langle -1, -1 \rangle$ on $\mathbb{R}^2$, which is example 22.7.6 in section 22.7.

20.2.2 Construction of Regular n-gon

20.2.3 Symmetric Functions

If $t_1, ..., t_n$ are indeterminates, then the polynomial:

$$P_n(x) := (x - t_1)(x - t_2) \cdots (x - t_n) \in \mathbb{Z}[t_1, ..., t_n][x]$$

is universal in the sense thati every polynomial of degree n with coefficeints in (say) an intergral domain may be obtained from $P_n(x)$ given a suitable choice for $t_1, ..., t_n$, choices form a possibly larger ring. Expanding this polynomial, we get that:

$$P_n(x) = x^n - s_1(t_1, ..., t_n)x^{n-1} + \cdots + (-1)^n s_n(t_1, ..., t_n)$$

where each s_i are the *elementary functions.* If $n = 3$, then:

$$\begin{aligned} s_1(t_1, t_2, t_3) &= t_1 + t_2 + t_3 \\ s_2(t_1, t_2, t_2) &= t_1t_2 + t_1t_3 + t_2t_3 \\ s_3(t_1, t_2, t_3) &= t_1t_2t_3 \end{aligned}$$

The function s_n are called symmetric since they are invariant under permutation (essentially, since $P_n(x)$ is invariant under permutation). The name "elementary" comes because of the following theorem

Theorem 20.2.4: Fundamental Theorem On Symmetric Functions

Let K be a field and let $\varphi \in K(t_1, ..., t_n)$. Then φ is invariant under permutation (i.e. symmetric) if and only if it is a rational function (with coefficient in K) of the elementary symmetric functions $s_1, ..., s_n$.

The functions s_i are viewed as elements of $K[t_1, ..., t_n]$ which we can do unambiguously since $\mathbb{Z}$ is initial in **Ring**.

***Proof*:**
Let $F = K(t_1, ..., t_n)$ and $k = K(s_1, ..., s_n)$ be the subfield generated by the elementary symmetric functions over K. Then F is a splitting field of the separable polynomial $P_n(x)$ over k, i.e. F is Galois and:

$$|\mathrm{Aut}_k(F)| = [F : k] \leq n!$$

Notice that the symmetric group S_n acts (faithfully) on F by permuting $t_1, ..., t_n$ and the action is the identity on $k = K(s_1, ..., s_n)$ since each s_i is symmetric. Thus, S_n can be identified with some subgroup of $\mathrm{Aut}_k(F)$. However,

$$n! = |S_n| \leq |\mathrm{Aut}_k(F)| \leq n!$$

showing that $\mathrm{Aut}_k(F) \cong S_n$

Now, since F/k is Galois, we get $k = F^{S_n}$. This in fact completes the rpoof: if $\varphi \in K(t_1, .., t_n)$, then φ is invariant under the S_n action on $t_1, ..., t_n$ if and only if $\varphi \in k \in K(s_1, ..., s_n)$, as we sought to show.

The theorem above is about rational functions of finitely many variables. Downstream work with formal characteristic-class calculi needs the same statement for formal power series over $\mathbb{Q}$, which does not follow from it, so it is proved here alongside.

The elementary symmetric functions are one basis for the symmetric polynomials, and for counting problems they are the wrong one: the classes that arise geometrically are indexed by partitions, and expressing them in the s_i hides that indexing behind a determinant. The Schur polynomials are the basis adapted to it, and they are recorded here because [11] builds its degeneracy-locus formulas out of them.

Definition 20.2.5: Schur Polynomial

A *partition* $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0)$ of length at most n is a weakly decreasing tuple of non-negative integers; write $|\lambda| = \sum_i \lambda_i$. Put $\delta = (n-1, n-2, \ldots, 1, 0)$. The *Schur polynomial* attached to λ in the variables $t_1, \ldots, t_n$ is the bialternant

$$s_\lambda(t_1, \ldots, t_n) \;=\; \frac{\det\left(t_i^{\lambda_j + n - j}\right)_{1\leq i,j\leq n}}{\det\left(t_i^{n-j}\right)_{1\leq i,j\leq n}}$$

the denominator being the Vandermonde determinant $\prod_{i<j}(t_i - t_j)$.

Proposition 20.2.6: Schur Polynomials Are a Basis

Each s_λ is a symmetric polynomial, homogeneous of degree $|\lambda|$, and the s_λ for λ of length at most n form a $\mathbb{Z}$-basis of the ring of symmetric polynomials in $t_1, \ldots, t_n$. For the single-row partition $\lambda = (m)$ one has $s_{(m)} = h_m$, the complete homogeneous symmetric polynomial, and for the single-column partition $\lambda = (1^m)$ one has $s_{(1^m)} = s_m$, the m-th elementary symmetric polynomial.

***Proof*:**
The numerator of definition 20.2.5 is alternating: transposing two variables transposes two rows of the matrix and changes the sign of the determinant. The Vandermonde is alternating for the same reason, so the quotient is symmetric. It is a polynomial because an alternating polynomial vanishes on every hyperplane $t_i = t_j$ and is therefore divisible by each $t_i - t_j$, hence by their product, these being pairwise coprime. Degrees subtract to $|\lambda| + \binom{n}{2} - \binom{n}{2} = |\lambda|$.

For the basis claim, order partitions lexicographically. The monomial t^λ occurs in s_λ with coefficient 1 and every other monomial of s_λ is lexicographically smaller, so the transition matrix from $\{s_\lambda\}$ to the monomial symmetric functions $\{m_\lambda\}$ is unitriangular over $\mathbb{Z}$, hence invertible; and the m_λ are a $\mathbb{Z}$-basis by inspection. The two special cases are the standard evaluations of the bialternant, the second being the observation that $\det(t_i^{\lambda_j+n-j})$ for $\lambda = (1^m)$ factors as the Vandermonde times s_m.

Theorem 20.2.7: Jacobi–Trudi Formulas

For a partition λ with conjugate partition λ',

$$s_\lambda = \det\left(h_{\lambda_i - i + j}\right)_{1\le i,j\le \ell}, \qquad s_\lambda = \det\left(s_{\lambda'_i - i + j}\right)_{1\le i,j\le \ell'}$$

where h_m is the complete homogeneous and s_m the elementary symmetric polynomial, with the conventions $h_0 = s_0 = 1$ and $h_m = s_m = 0$ for $m < 0$, and ℓ, ℓ' are the lengths of λ and λ'.

***Proof Cited*:**
Not reproduced here (QC #3(a)). The clean proof is the Lindström–Gessel–Viennot lattice-path argument: both determinants count families of non-intersecting paths, and the sign-reversing involution on intersecting families cancels every off-diagonal term. That is combinatorics rather than algebra, and belongs with the lattice-path methods of the combinatorics volume; the identity is recorded here because this is where the symmetric functions live.

What the identity is for: it converts a partition-indexed basis element into a determinant in classes one can compute with. In [11] the h_m and s_m become Chern classes of a bundle, and the determinant becomes the Giambelli and Thom–Porteous formulas.

Lemma 20.2.8: Symmetric Power Series in the Chern Roots

Let $\Phi \in \mathbb{Q}[[a_1,\dots,a_r]]$ be invariant under every permutation of $a_1,\dots,a_r$. Then there is a unique $\widetilde{\Phi} \in \mathbb{Q}[[y_1,\dots,y_r]]$ with

$$\Phi(a_1,\dots,a_r) = \widetilde{\Phi}\big(e_1(a),\dots,e_r(a)\big)$$

If Φ is homogeneous of degree d, then $\widetilde{\Phi}$ is a polynomial, isobaric of weight d when y_i is assigned weight i.

***Proof*:**
Existence. Each homogeneous component of Φ is symmetric, and an isobaric polynomial of weight d in the y_i contributes only in degree d after substituting $y_i = e_i$; so it suffices to treat a symmetric homogeneous polynomial P of degree d and assemble the results over d. Order the monomials $a^\nu = a_1^{\nu_1}\cdots a_r^{\nu_r}$ lexicographically, and let a^ν be the largest monomial occurring in a nonzero P, with coefficient λ. Then $\nu_1 \ge \nu_2 \ge \cdots \ge \nu_r$: if $\nu_j < \nu_{j+1}$ for some j, the transposed monomial also occurs in P, by symmetry, and is lexicographically larger. The product

$$e_1^{\nu_1-\nu_2} e_2^{\nu_2-\nu_3} \cdots e_{r-1}^{\nu_{r-1}-\nu_r} e_r^{\nu_r}$$

is symmetric and homogeneous of degree $\sum_j \nu_j = d$, and its lexicographic leading monomial is a^ν with coefficient 1, since the leading monomial of e_k is $a_1 a_2 \cdots a_k$ and the exponent of a_j in the displayed product is $\sum_{k \geq j}(\nu_k - \nu_{k+1}) = \nu_j$. Subtracting λ times it from P leaves a symmetric homogeneous polynomial of degree d whose leading monomial is strictly smaller. There are finitely many monomials of degree d, so the process terminates and expresses P as an isobaric polynomial of weight d in the e_i.

Uniqueness. It suffices to show that $e_1, \ldots, e_r$ are algebraically independent, since a difference of two representations of Φ is a power series R with $R(e_1, \ldots, e_r) = 0$, whose isobaric components of distinct weights land in distinct degrees in the a_j and so vanish separately. Let $Q = \sum_m \kappa_m y^m$ be a nonzero polynomial. The leading monomial of $e^m = e_1^{m_1} \cdots e_r^{m_r}$ is $a_1^{m_1+\cdots+m_r} a_2^{m_2+\cdots+m_r} \cdots a_r^{m_r}$, and the assignment $m \mapsto (m_1 + \cdots + m_r,\, m_2 + \cdots + m_r,\, \ldots,\, m_r)$ is injective, so distinct exponent vectors m produce distinct leading monomials. Among the finitely many m with $\kappa_m \neq 0$, take the one whose leading monomial is lexicographically largest. Every monomial of every other term is strictly smaller than it, so it survives in $Q(e_1, \ldots, e_r)$ with coefficient $\kappa_m \neq 0$, and $Q(e_1, \ldots, e_r) \neq 0$, as we sought to show.

One important result that was proven in the previous theorem is noted for future reference:

Lemma 20.2.9: S_n Extension

Let K be a field, $t_1, ..., t_n$ indeterminates, and let $s_1, ..., s_n$ be the elementary symmetric functions on $t_1, ..., t_n$. Then the extension:

$$K(s_1, ..., s_n) \subseteq K(t_1, ..., t_n)$$

is Galois, with Galois group S_n

From this, we can prove that every finite group is represented as a galois group:

Theorem 20.2.10: Finite Group Represented by Galois Group

Let G be a finite group. Then there exists a Galois extension F/k such that $\mathrm{Aut}_k(F) \cong G$

***Proof*:**
exercsie

Note that it is not certain if every finite group G appears as an extension of F over $\mathbb{Q}$. This is called the *inverse Galois Problem*. more here! Can also checkout Aluffi p.473

Corollary 20.2.11

Let D be the discriminant of the field extension. Then:

$$K(s_1, ..., s_n)(\sqrt{D}) \subseteq K(t_1, ..., t_n)$$

is Galois, with Galois group A_n.

20.2.4 Solvability By Radicals

We are finally in a position to prove that an arbitrary degree 5 polynomial cannot be written in terms of it's coefficients using the operations $+, -, \times, \div, \sqrt{\quad}$. Is is famously known, any quadratic polynomial over k ($\text{char}(k) \neq 2$) can be written in terms of:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

and similar equations exists for degree 3 and 4 polynomials, though they get horrendously messy.

20.2.12 Goal There exists irreducible polynomials of degree $n \geq 5$ that are unsolvable

To avoid complications, we will assume in this section that $\text{char}(k) = 0$.

Definition 20.2.13: Radical Extension

Let F/k be a field extension. Then if $F = k(\alpha_1, \alpha_2, ..., \alpha_n)$ where

$$k \subseteq k(\alpha_1) \subseteq k(\alpha_1, \alpha_2), \subseteq \cdots \subseteq k(\alpha_1, ..., \alpha_n)$$

where for each $i \in \{1, ..., n\}$, there exists an n_i such that $\alpha^{n_i} \in k(\alpha_1, ..., \alpha_{i-1})$, then F/k is called a *radical extension*

Elements of radical extensions are easy to picture: if $k = \mathbb{Q}$, then simply nest as many nth-roots as you please, for example:

$$\sqrt[7]{3 + \sqrt[2]{19}^5 \sqrt[11]{3} + 2}$$

it is also clear by definition that a radical extension is the composition of cyclic exensions provided that the appropriate roots of 1 are in k. The next result insures us that any [separable] radical extension is contained in a radical extensions that is *Galois*:

Lemma 20.2.14: Separable And Galois Radical Extensions

Let F/k be a separable radical extension. Then F is conatiend in a Galois radical extension

***Proof*:**
Let F/K be a separable radical extension. Then F is a separable finite extension of K, and so by the primitive root theorem (theorem 19.6.20), there exists an $\alpha \in F$ such that $F = k(\alpha)$. Let L be the splitting field of $k(\alpha)$. Then L is equal to the composite of all the field $\sigma(F)$ for every σ that embeds F into $\overline{k}$:

$$L = \sigma_1(F) \cdots \sigma_n(F)$$

Then the composition of radical extensions is radical, showing that L is radical. Thus, L is the splitting field of a separable polynomial, and so it is Galois, completing the proof.

Lemma 20.2.15: Solvability In Char 0

Let k be a field such that $\text{char}(k) = 0$, and let $f(x) \in k[x]$ be an irreducible polynomial. Then the roots of $f(x)$ can be written in terms of field operations and radicals if and only if the splitting field fo $f(x)$ is contined in a Galois Radical extension

***Proof*:**

($\Leftarrow$) If the splitting field $f(x)$ is contained in a radical extension (not necessarily Galois), then *a fortiori* the roots can be written using field operations and radicals.

($\Rightarrow$) Conversely, assume the roots of the irreducible polynomial $f(x)$ can be written in terms of field operations and radicals . Then any formula for a root t_1 of $f(x)$ can be turned into a formula for t_i by applying an element σ_i of $\text{Aut}_k(\overline{k})$ sending t_1 to t_i (this automorphism exists since $f(x)$ is irreducible). Thus, if the formula exists for one root, a formula exists for all roots. Thus, $k(t_1), ..., k(t_n)$ are each radical extensions and

$$L = k(t_1)k(t_2)\cdots k(t_n)$$

is a radical extension that contains all the roots of $f(x)$. Thus, by minimality of the splitting field, it is contained in a radical extension of k. By lemma 20.2.14, it must also be contained in a galois radical extension, as we sought to show.

Thus, we can characterize when a polynomial is solvable by radicals by when it is contained in a Galois radical extension. Since the extension is Galois, we have a corresponding Galois group to analyze. We will soon see that the appropriate type of group that is needed for $f(x)$ to be solvable is one we have encountered a long time ago:

Definition 20.2.16: Solvable Extension

Let F/K be a Galois extension. Then F/K is a *solvable extension* if $\text{Aut}_K(F)$ is a solvable group

Recall that if G is a solvable group and finite, then G can be decomposed into a subnormal series with cyclic factors:

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots G_n = \{1\}$$

notice how this parrallels the fact that any radical extension is a sequence of cyclic extensions.

Lemma 20.2.17: Relating Radical And Solvable Extensions

Let F/K be a galois extension with $\text{char}(k) = 0$. If F/K has enough roots of 1, then F/K is radical if and only if it is solvable

What is meant by “enough roots of 1” is that K contains primitive mth roots of 1 for a common multiple m of the order of the corresponding cyclic factors.

***Proof*:**
This is a generalization of proposition 20.1.41.

Let F/K be a radical extension, so that:

$$K \subseteq k(\delta_1) \subseteq \cdots \subseteq K(\delta_1, ..., \delta_r) = F$$

where $\delta_i^{m_i} \in K(\delta_1, ..., \delta_{i-1})$. By assumption, K contains primitive m_ith roots of 1 for each i. Next, consider the corresponding subgroups of $G = \mathrm{Aut}_k(F)$:

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_r = \{e\}$$

where $G_i = \mathrm{Aut}_{K(\delta_1,...,\delta_i)}(F)$. Next, note That F is Galois over each intermediate field (by proposition 19.5.16), and each extension:

$$K(\delta_1, ..., \delta_{i-1}) \subseteq K(\delta_1, ..., \delta_i)$$

satisfies proposition 20.1.41, and hence each G_i is cyclic. By theorem 20.1.23, each G_i is normal (since the automorphism group is abelian) in the previous G_{i-1} with a cyclic quotient, and hence G is solvable.

Conversely, the same argument works in reverse using the converse implication in proposition 20.1.41 to show that an extension with enough roots of 1 is radical.

We were dependent on each extension containing enough roots of unity. In the following, we show that we can drop them:

Proposition 20.2.18: Solvable In Char 0

Let K be a field where $\mathrm{char}(K) = 0$, and let F/K be a Galois extension. Then F/K is solvable if and only if it is containd in a Galois Radical extension

Technically, we only need that $\mathrm{char}(K)$ not divide the relevant degrees of the extensions, however for our purposes it is safe to assume $\mathrm{char}(K) = 0$.

***Proof*:**

($\Rightarrow$) Assume $\mathrm{Aut}_K(F)$ is solvable. Let M be the common multiple of the cyclic quotients and let ζ be a mth root of 1 in an algebraic closure $\overline{K}$ of K. The splitting field $K(\zeta)$ of $x^M - 1$ is Galois over K since $\mathrm{char}(K) = 0$. By proposition 20.1.28, the extension $K(\zeta) \subseteq F(\zeta)$ is also Galois, with Galois group isomorphic to $\mathrm{Aut}_{K(\zeta)\cap F}(F)$. Since $K(\zeta) \cap F$ is Galois over K (Aluffi says exercise 6.13), this subgroup is normal, and it follows (corollary IV.3.13) that $\mathrm{Aut}_{K(\zeta)}(F(\zeta))$ is solvable.

Since $K(\zeta) \subseteq F(\zeta)$ has enough roots of 1, by lemma 20.2.17, it is radical. Thus, so is $K \subseteq F(\zeta)$ since $K \subseteq K(\zeta)$ is by trivially radical. Thus, F/K is contained in a radical extension.

($\Leftarrow$) Conversely, assume that F/K is Galois and contained in a radical Galois extension L/K. Let M be a common multiple of the order of the cyclic factors in this extensions,and let ζ be a primitive mth root of 1. The composite $L(\zeta)$ is Galois and radical over $K(\zeta)$ (show that if E/K is radical and T/K is any extension, then ET/K is a radical extension), and it has enough roots of 1.

By lemma 20.2.17, $L(\zeta)/K(\zeta)$ is solvable. Then we actually have that $L(\zeta)/K$ is solvable. Indeed, this extension is Galois since it's a composition of galois (exercise 6.13). Applying the fundamental theorem to:

$$K \subseteq K(\zeta) \subseteq L(\zeta)$$

to established that $\mathrm{Aut}_K(K(\zeta))$ is isomorphic to the quotient of $\mathrm{Aut}_K(L(\zeta))$ by its normal subgroup $\mathrm{Aut}_{K(\zeta)}(L(\zeta))$. Both $\mathrm{Aut}_{K\zeta)}(L(\zeta))$ and $\mathrm{Aut}_K(K(\zeta))$ are solvable (being abelian), and it follows that $\mathrm{Aut}_K(L(\zeta))$ is solvable.

Since F is an intermediate field and Galois over T, by the fundamental theorem, $\mathrm{Aut}_K(F)$ is a homomorphic image of $\mathrm{Aut}_K(L(\zeta))$. This shows that $\mathrm{Aut}_K(F)$ is a homomorphic image of a solvable group, hence it is itself solvable, and so we're done.

Corollary 20.2.19: Galois Criterion

Let K be a field such that $\mathrm{char}(K) = 0$ and let $f(x) \in K[x]$ be an irreducible polynomial. Then f(x) is solvable by radicals if and only if its galois group is solvable

***Proof*:**
This is an immediate consequence of lemma 20.2.15 and proposition 20.2.18.

Theorem 20.2.20: Abel's Theorem

Let $f(x)$ be a generic polynomial of degree 5 or higher. Then it admits no solutions by radicals

***Proof*:**
By lemma 20.2.9, the Galois group of the generic polynomial of degree n is S_n. The group S_n is not solvable for $n \geq 5$ (see ref:HERE), and hence the statement follows from the Galois Criterion.

Furthermore, we know that S_3 and S_4 *are* solvable, with subnormal series:

$$\{e\} \trianglelefteq \mathbb{Z}/2\mathbb{Z} \trianglelefteq D_6 = S_3$$

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4$$

hence all degree less than or equal to 4 polynomial have a root using radicals.

Exercise 20.2.1

1. Let $\alpha \in F$ for an algebraic extension $F/\mathbb{Q}$. In particular, we may think of F as being embedded in $\mathbb{C}$ so that $\alpha = a + bi$. Does there always exist an N such that $\alpha^N \in \mathbb{R}$?

20.3 Galois Groups of Polynomials

In this section, we shall focus on techniques to compute the galois group of polynomial. So far, much of the focus was on showing what properties can be deduced given properties of the galois group, there have not been many technics to find galois groups (for example, can you come up with a non-solvable degree 5 polynomial?). Hence, this section is a compilation of interesting facts and techniques useful for working with galois groups of poylnomials.

Lemma 20.3.1: Transitivity Of Galois Group

Let $f(x) \in k[x]$ be a separable irreucible polynomial of degree n. Then $\mathrm{Gal}_k(f(x))$ acts transitively on the set of roots of $f(x)$ in $\overline{k}$. In particular, $\mathrm{Gal}_k(f(x))$ may be identified with a *transitive* subgroup of the symmetric group S_n.

***Proof*:**
exercise

By corrollary 19.2.29, given α a root of $m_{\alpha,k}(x)$, then the elements of $\mathrm{Gal}_k(m_{\alpha,k}(x))$ are determined by where this root maps. This motivates the following definition:

(In general)

$$\mathrm{Gal}_k(fh) \leqslant \mathrm{Gal}_k(f) \times \mathrm{Gal}_k(h) \leqslant S_{n_1} \times S_{n_2} \leqslant S_n$$

(but more relations can be added: here)

This gives us the limits of the possible subgroups of galois groups:

1. The only possible group for degree 2 irreducible polynomials is: $\mathbb{Z}/2\mathbb{Z}$
2. The only possible groups for degree 3 irreducible polynomials is:
$$\mathbb{Z}/3\mathbb{Z} \cong A_3, S_3$$
3. The only possible groups for degree 4 irreducible polynomials is:
$$S_4, A_4, D_8, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}$$

Definition 20.3.2: Galois Conjugate

Let $\alpha \in L$ be an algebraic element of L over k. Then the set of roots of the minimal polynomial $m_{\alpha,k}(x)$ is called the *galois conjugates*, *algebraic conjugates*, or *conjugate elements* of α over k.

The motivation comes from the case of $\alpha \in \mathbb{C}$ which are algebraic over $\mathbb{R}$. In this case, the minimal polynomial is either $x - \alpha$ (in which case it has one root), or it is of degree 2 with galois conjugate $\overline{\alpha}$. We thus see that the galois conjugate is the generalization of the complex conjugate.

For the next result, we recall the following result of the discriminant;

Definition 20.3.3: Discriminant Reminder

Let $f(x) \in k[x]$ be a separable polynomial with roots $\alpha_1, ..., \alpha_n$ in its splitting field. Then the *discriminant* of $f(x)$ is the element $D = \Delta^2$ where

$$\Delta = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)$$

Every permutation of the roots fixes D, so D is fixed by the whole Galois group, therefore $D \in k$. Odd permutations move Δ (if $\mathrm{ch}(k) \neq 2$) and even permutations fix it. Tus, Δ is fixed by the Galois group G (i.e. $\Delta \in k$) if and only if $G \subseteq A_n$

Lemma 20.3.4: Discriminant And Galois Group

Let k be a field where $\mathrm{ch}(k) \neq 2$ and let $f(x) \in k[x]$ be a separable polynomial with discriminant D. Then the Galois group o f $f(x)$ is contaiend in the alternating group A_n if and only if D is a quare in k

Example 20.3.5: Discriminant

1. Let $f(x) = x^3 + ax^2 + bx + c$ be a general monic cubic. By shifting by $a/3$, we get

   $$f(x - a/3) = x^3 + px + q$$

   for appropriate p, q. Then it can be computed that:

   $$D = -4p^3 - 27q^2$$

   Then by our above lemma, if D is a square then the galois group is S_3, and A_3 otherwise. For example, the galois group of $x^3 - 2$ is S_3 since $D = -108$ is not a square in $\mathbb{Q}$, while $x^3 - 3x + 1$ has galois group $A_3 \cong \mathbb{Z}/3\mathbb{Z}$ since $D = 81 = 9^2$.

2. Let $f(x) \in \mathbb{Q}[x]$ be of degree p for prime p with $p - 2$ real roots and 2 non-real complex roots. Then $\mathrm{Gal}_{\mathbb{Q}}(f(x)) = S_p$. Prove this, then show that the galois group of $x^5 - 5x - 1$ over $\mathbb{Q}$ is S_5.

Other useful discriminants are:

$$\begin{aligned}
\mathrm{D}(x^3 + ax + b) &= -4a^3 - 27b^2 \\
\mathrm{D}(x^4 + ax + b) &= -27a^4 + 256b^3 \\
\mathrm{D}(x^4 + ax^2 + b) &= 16b(a^2 - 4b)^2
\end{aligned}$$

(a word on the inverse galois problem? Sovled for every finite abelian group?)

Exercise 20.3.1

1. Let $K/\mathbb{Q}$ be Galois. Show that if $\mathrm{disc}(K)$ is square-free, then K has no intermediate fields

20.4 Kummer Theory and Artin-Schreier Extensions

As we saw an section 20.2.4, if a field extension F/K is a radical extension, then the Galois group is solvable. The converse is not necessarily true:

Example 20.4.1: Solvable, But Not Radical Extension

Consider $K = \mathbb{Q}(\sqrt{5})$ and $M = K(\zeta_3)$; recall the minimal polynomial of ζ_3 is $x^2 + x + 1$. Then $[M : K] = 2$, and it is easy to see then that if M/K were a radical extension then $M = K[\sqrt{\alpha}]$ for some $\alpha \in K$ but no such element can exist.

By doing multiple examples like these, the reader would notice that this seems to be more of a problem of there being a lack of roots of unity in K to insure there are enough solutions to $x^n - a$. In fact, there is no additional issues that arise when the characteristic of K is 0, and the issues in finite characteristic only happen for certain bad primes. We thus explore this converse: that under these mild extra conditions, if the galois group is solvable then the extension is a radical extension. As solvable groups are towers of cyclic groups, it shall suffice to show that if the galois groups in the tower of fields is cyclic, each corresponding field extension is given by a radical $\sqrt[n]{\alpha}$. An interesting consequence of this is that the galois groups of Kummer extensions shall be characterized by certain units of the base field. This shall in fact be true for all abelian extensions (not just Kummer extensions), but the proof of this result is a lot more complicated and was one of the crowning achievements 20th century number theory proved in 1930; the result is called the *Artin Reciprocity* and its proof can be found in [17].

Definition 20.4.2: Kummer Extensions and G-Extensions

Let L/K be a field extension. Then L/K is called a *Kummer Extension* if:

1. for some n, K contains μ_n (the roots of $x^n - 1$)
2. L/K is an Abelian extension with exponent n[a]

If in particular we want extensions with Galois groups G, we shall call such extensions G-extensions.

[a] The least common multiple of the orders of the elements

Example 20.4.3: Kummer Extensions

1. Any quadratic extension of $\mathbb{Q}$ is a Kummer extension of degree 2, since $\mathbb{Q}$ have $1, -1$ and has abelian extension. More generally, any multi-quadratic extension is a Kummer extension of degree 2
2. There is no Kummer extensions of degree 3 over $\mathbb{Q}$ since $\mathbb{Q}$ does not have a 3rd root of unity, since it's complex.
3. More generally if K contains μ_n (meaning the characteristic of K does not divide n), then adjoining any nth root of $a \in K$, $L = K(\sqrt[n]{a})$ is a Kummer extension (of degree m dividing n, since the minimal polynomial divides $x^n - a$)

So let's now say that L/K is a Kummer $\mathbb{Z}/n\mathbb{Z}$-extension where the characteristic of K does not divide n. In order to show $L = K(\sqrt[n]{a})$, we must find an appropriate $a \in K^*$. Let us find some necessary conditions. As $\mathrm{Gal}_K(L) = \mathbb{Z}/n\mathbb{Z}$, there is a generator σ such that $\sigma^n = 1$. Then given $x^n = a$, we have the roots:

$$\sqrt[n]{a},\ \sqrt[n]{a}\zeta_n, ..., \sqrt[n]{a}\zeta_n^{n-1}$$

note all these roots of unity are distinct and in K by the characteristic condition. Thus, by field theory we know that σ must map $\sqrt[n]{a}$ to another one of these roots, and that this conditions characterizes σ; without loss of generality we can say $\sigma(\sqrt[n]{a}) = \zeta\sqrt[n]{a}$. Importantly, interpreting L as a K-vector space, $\sqrt[n]{a}$ becomes an *eigenvector* of σ. Note that as $\sigma^n = 1$, the possible eigenvalues must be $1, \zeta_n, ..., \zeta_n^{n-1}$. Thus if σ is diagonalizable, we can find such an a such that $a^n = \alpha \in K$. Exercise 20.4.1(20.4.1 (1)) gives a very intuitive computational approach to finding this. We present the following more abstract way of finding it that works more generally so that we can use it later in number theory and cohomology theory in this book:

Theorem 20.4.4: Hilbert's 90th Theorem

Let K be a field of characteristic 0 and L/K be an abelian extension such that

$$\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$$

If σ generate $\mathrm{Gal}_K(L)$, then:

1. For $\beta \in L^\times$ such that $\mathrm{Nm}_{L/K}(\beta) = 1$, there exists an $\alpha \in L^\times$ such that
$$\beta = \frac{\sigma(\alpha)}{\alpha}$$
2. Let $\beta \in L$ such that $\mathrm{tr}(\beta) = 0$. Then there exists an $\alpha \in L$ such that
$$\beta = \alpha - \sigma(\alpha)$$

***Proof*:**
Since L/K is a finite separable extension, pick t so that $L = K(t)$. Let

$$S = \beta + \beta\sigma(\beta) + \beta\sigma(\beta)\sigma^2(\beta) + \cdots = \sum_{i=1}^{n-1}\prod_{j}^{i}\sigma^j(\beta)$$

If $S \neq 0$, then:

$$\frac{\sigma(S)}{S} = \beta^{-1} \implies \frac{\sigma(S^{-1})}{S^{-1}} = \beta$$

Thus, we would need to guarantee such a nonzero S exists.

To see this, notice that solving $\beta = \frac{\alpha}{\sigma(\alpha)}$ is equivalent to $\beta\sigma(\alpha) = \alpha$. Notice that

$$\beta\sigma(-) : L \to L$$

mapping $\alpha \mapsto \beta\sigma(\alpha)$ is a K-linear map. If we show that it has an eigenvector with eigenvalue 1, then we solved our problem, for if α is such an eigenvector then (if we represent our function as f)

$$\alpha = f(\alpha) = \beta\sigma(\alpha)$$

We shall verify this by extending scalars from K to L. Recall by corollary 14.3.7 that extending scalars does not change the eigenvectors. We may thus extend our map to:

$$\begin{aligned} \mathrm{id} \otimes_K \beta\sigma(-) : L \otimes_K L &\to L \otimes_K L \\ \ell \otimes_K \ell' &\mapsto \ell \otimes_K \beta\sigma(\ell') \end{aligned}$$

Since L/K is finite and separable, there exists a γ such that $L = K(\gamma)$. Then γ has minimal polynomial $m_{\gamma,K}(x) \in K[x]$, and we have the isomorphisms:

$$(L \otimes_K L) \cong \left(L \otimes_K \frac{K[x]}{m_{\gamma,K}(x)} \right) \cong \frac{L[x]}{m_{\gamma,K}(x)} \cong L^n$$

Since any $\ell' \in L$ can be written as $\ell' = p(\gamma)$ for some polynomial $p(x) \in K[x]$, we have:

$$(\ell \otimes_K p(\gamma)) \mapsto \ell(p(\gamma), p(\sigma(\gamma)), ..., p(\sigma^{n-1}(\gamma)))$$

Thus, the map $\beta\sigma(-)$ becomes interpreted as $L^n \to L^n$ as:

$$\ell(p(\gamma), ..., p(\sigma^{n-1}(\gamma))) \mapsto \ell\left(\beta p(\sigma^{n-1}(\gamma)), \sigma(\beta p(\gamma)), ..., \sigma^{n-1}(\beta p(\sigma^{n-2}(\gamma)))\right)$$

or perhaps a bit more clearly:

$$(\ell_1, \ell_2, ..., \ell_n) \mapsto (\beta\ell_n, \sigma(\beta\ell_1), ..., \sigma^{n-1}(\beta\ell_{n-1}))$$

Since $\mathrm{Nm}_{L/k}(\beta) = 1$,

$$\left(1, \sigma(\beta), \sigma(\beta)\sigma^2(\beta), ..., \sigma(\beta)\sigma^2(\beta)\cdots\sigma^{n-1}(\beta)\right)$$

is an eigenvector with eigenvalue 1, hence our original transformation had an eigenvector of eigenvalue 1, as we sought to show.

For the trace argument, say $\beta \in L$ such that $\mathrm{tr}(\beta) = 0$. Take

$$S = \beta 2\sigma(\beta) + 3\sigma^2(\beta) + \cdots n\sigma^{n+1}(\beta)$$

Then:

$$\sigma(S) = \sigma(\beta) + 2\sigma^2(\beta) + \cdots + n\sigma^n(\beta)$$

since $\sigma^n = \mathrm{id}$, $\sigma^n(\beta) = \beta$, so we get:

$$S - \sigma(S) = -n\beta$$

Thus, set $\alpha = -\beta/n$, completing the proof.

20.4.5 Foreshadowing In part XI, we shall reformulate this theorem in purely Homological terms, namely it says that $H^1(G, L^*) = 0$

We now look at some properties of Kummer extensions. First, by some Galois theory, we see that if $\mathbb{Q}(\zeta_n) \subseteq K$, $L = K(\alpha)$ where $\alpha^n = \beta \in K$, then L/K is an abelian Galois extension, for:

$$\varphi : \mathrm{Gal}_K(L) \xrightarrow{\sim} \mu_n \qquad \tau \mapsto \frac{\tau(\alpha)}{\alpha}$$

which is well-defined, and any nth root of β gives the same isomorphism. Note how important $\mu_n \subseteq K$ is for this isomorphism to work! This is certainly not true in general (is $\mathbb{Q}(\sqrt[n]{2})/\mathbb{Q}$ galois for $n > 2$?). Using Hilbert's 90th problem, the converse of this statement is also true:

Theorem 20.4.6: Primitive Element For Cyclic Kummer Extension

Let $\text{char}(K) = 0$, L/K be a Galois extension and $\text{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$. Assume K contains μ_n, the group of roots of unity generated by ζ_n. Then there exists an $\alpha \in L^\times$ such that

$$L = K(\alpha) \qquad \alpha^n \in K^\times$$

***Proof*:**
Let ζ_n be a primitive nth root of unity. Recall that $\text{Nm}_{L/K}(\zeta_n) = \zeta_n^n = 1$. Thus, by Hilbert's 90th, there exists $\alpha \in L$ such that $\sigma(\alpha) = \zeta_n \alpha$. Since σ is a generator, we see that α is not fixed by any σ, hence:

$$L = K(\alpha)$$

Furthermore,

$$\sigma(\alpha^n) = \sigma(\alpha)^n = \zeta_n^n \alpha^n = \alpha^n$$

Hence, each σ fixes α^n, meaning $\alpha^n = \beta \in K$, hence

$$L = k(\sqrt[n]{\beta})$$

as we sought to show.

With the existence established, let us now established to what extent it is unique. Say $K(\sqrt[n]{a}) \cong K(\sqrt[n]{b})$. Then we know $\sqrt[n]{a}$ and $\sqrt[n]{b}$ are both eigenvectors, but they may have different eigenvalues. This is not an issue as we may raise one of them to the appropriate power, say $\sqrt[n]{b}$ so that the eigenvalue is the same root of unity as $\sqrt[n]{a}$ and it is still of order n (which we can do as $\gcd(n, n-1) = 1$). We may thus simply assume they have the same eigenvalue. But then:

$$\sigma\left(\frac{\sqrt[n]{a}}{\sqrt[n]{b}}\right) = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

and hence it is fixed for all σ^i, and hence is in K. We thus get the following correspondence:

Corollary 20.4.7: Cyclic Kummer Extension Correspondence

Let $\mu_n \subseteq K$. Then we have the following bijective correspondence

$$\left\{\begin{matrix} L/K \\ \text{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z} \end{matrix}\right\} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \left\{\begin{matrix} G \subseteq K^\times/(K^\times)^n \\ G \cong \mathbb{Z}/n\mathbb{Z} \end{matrix}\right\}$$

***Proof*:**
First finding ψ, pick $\overline{\beta} \in K^\times/(K^\times)^n$. Pick $\beta \in K^\times$ which represents $\overline{\beta}$. Then define

$$\psi(\langle\overline{\beta}\rangle) = K(\sqrt[n]{\beta}) = L$$

this is well-defined for if we have any other coset representative, $\beta' = \beta\gamma^n$, $K(\sqrt[n]{\beta}) = K(\sqrt[n]{\beta'})$.

Certainly $\mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$ and $m \big| n$. We need to show that $n = m$. For the sake of contradiction, suppose not. Then if σ generates $\mathrm{Gal}_K(L)$, then if we let $\alpha = \sqrt[n]{\beta}$, we have

$$\sigma(\alpha) = \zeta_m \alpha \implies \sigma(\alpha^m) = \zeta_m^m \alpha^m = \alpha^m$$

hence $\alpha^m \in K^\times$. Then $\beta = \alpha^n = (\alpha^m)^{\frac{n}{m}} \in (K^\times)^{n/m}$, a contradiction.

Conversely, to define φ, suppose we have an L where $\langle \sigma \rangle = \mathrm{Gal}_K(L)$. By theorem 20.4.6, pick $\alpha \in L^\times$ such that $\sigma(\alpha)/\alpha = \zeta_n$. Then α is well-defined up to being an element of $K^\times$. In particular, $[\alpha^n] \in K^\times/(K^\times)^n$ is well-defined. Thus, take the map:

$$\varphi(L) = \langle [\alpha^n] \rangle \subseteq K^\times/(K^\times)^n$$

It remains to show these maps are inverses of each other (important exercise!)

Corollary 20.4.8: Pontryagin Dual For Kummer Extensions

Let $\mathbb{Q}(\zeta_n) \subseteq K$, and $K_n \subseteq \overline{K}$ the maximal finite Kummer extension, that is the maximal finite abelian extension with $\mathrm{ord}(\mathrm{Gal}_K(K_n)) \big| n$ (every element of the group has order that divides n). Then there exists a natural isomorphism:

$$\mathrm{Gal}_K(K_n) \cong \mathrm{Hom}(K^\times/(K^\times)^n, \mu_n)$$

$$\sigma \mapsto \left(\overline{\beta} \mapsto \frac{\sigma(\sqrt[n]{\beta})}{\sqrt[n]{\beta}} \right)$$

***Proof*:**
exercise

We can do all cyclic extensions at once by taking advantage of the infinite galois theory developed earlier

Theorem 20.4.9: Kummer Pairing

Let K contain all roots of unity, let M/K be the extension of K by all possible roots, and let μ_n be all nth roots of unity. Then there exists a map:

$$\varphi : \mathrm{Gal}_K(M) \times (K^*/(K^*)^n) \to \mu_n \qquad \varphi(\sigma, a) = \frac{\sigma(a)}{a}$$

***Proof*:**
exercise.

Definition 20.4.10: Kummer Pairing

Let K contain all roots of unity, let M/K be the extension of K by all possible roots, and let μ_n be all nth roots of unity. Then the map

$$\varphi : \mathrm{Gal}_K(M) \times (K^*/(K^*)^n) \to \mu_n \qquad \varphi(\sigma, a) = \frac{\sigma(a)}{a}$$

is called the *Kummer Pairing*.

This shows that the galois group and the units are almost dual to each other. A point was made to put μ_n instead of $\mathbb{Z}/n\mathbb{Z}$; these two groups are group isomorphic, however there as a different group-action on μ_n, namely the galois group acts on it non-trivially. The fact there is a non-trivial action is often called the *Tate Twist*.

Artin-Schreier Extensions

If $\mathrm{char}(K)\big|n$, then $x^n - a$ has multiple roots and hence the extension is not Galois, and furthermore if L/k is cyclic, all eigenvalues of σ are 1, loosing the method to find the element α. Note that there do exist such galois extensions, for example $\mathbb{F}_{p^p}/\mathbb{F}_p$, and hence this is a valid problem.

The key is to use generalized eigenvalues.

I'll definitely type all of this up, I'll leave it blank for now but it is pretty neat to finish off the correspondence between solvable groups and field extensions

Definition 20.4.11: Artin-Schreier Extensions

Let L/K be a cyclic extension of order n where $\mathrm{char}(K)\big|n$. Then L is an *Artin-Schreier* extension $L = K(\alpha)$ where $\sigma(\alpha) = \alpha + 1$ and $\alpha^p - \alpha \in K$ and α is the solution to $x^p - x - a$ (as compared to $x^n - a$ for Kummer extensions).

There is a similar Artin-Schreier pairing as well.

Exercise 20.4.1

1. Show that we can find α such that $L = K(\alpha)$ where L is a Kummer $\mathbb{Z}/n\mathbb{Z}$-extension showing all elements can be written as the sum of eigenvalues. (hint: `https://www.youtube.com/watch?v=UaeJNQ5x17g&list=PL8yHsr3EFj53Zxu3iRGMYL_89GDMvdkgt&index=15`)

21⋆

More Structures on Fields

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The most common fields the reader will encounter are $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$. Each of these fields are natural in some way; $\mathbb{Q}$ is initial in $\mathbf{Fld}_0$, $\mathbb{R}$ is the smallest complete ordered field[1], and $\mathbb{C}$ is the smallest complete and algebraically closed field. A key property all three of these fields share is that there is a notion of *size*. For $\mathbb{Q}$ and $\mathbb{R}$, there is even the stronger notion of *ordering* for all the elements. For $\mathbb{R}$ and $\mathbb{C}$, there is a notion of *differentiability*. In this chapter, we explore these properties

Recall that an order can be defined on a set (definition 0.1.14). In this section, we shall explore fields with orders, or fields for which these is a function that maps to an ordered field.

This chapter shall also incorporate some topology from chapter C as it is beneficial for a deeper understanding of the subjects presented in this chapter.

21.1 Ordered and Real Fields

Definition 21.1.1: Ordered Field

Let k be a field. Then k is a *totally ordered field* (or *ordered field* for short) if there exists a (total) order $\leq$ on k such that for all $x, y, z \in k$:

1. $x < y$, then $x + z < y + z$
2. If $z > 0$, then if $x < y$, $zx < zy$

If a field k admits a total ordering, then k is said to be a *formally real field.*

[1] More technically, it is the unique Dedekind-complete ordered field up to isomorphism

Famously, $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. The reader should prove that $\mathbb{C}$ is *not* an ordered field.

Proposition 21.1.2: Properties Of Ordered Fields

Let k be an ordered field and $x \in k$. Then

1. $x > 0$ if and only if $-x < 0$ (similarly, $x < 0$ if and only if $-x > 0$)
2. $x > y$ if and only if $x - y > 0$
3. $x < y$ if and only if $-x > -y$
4. $x > y > 0$, then $y^{-1} > x^{-1} > 0$
5. If $x < y$ and $z < 0$, then $xz = (-x)(-z) > (-y)(-z) = yz$
6. $x^2 > 0$ (hence $1 > 0$ always since $1 = 1^2 > 0$)

Furthermore, ordered fields have characteristic zero

***Proof*:**
All of these should be done as an exercise. For the last one, note that

$$0 < 1 < 1 + 1 < 1 + 1 + 1 < \cdots$$

Note how this proposition immediately shows that $\mathbb{C}$ is not ordered since $-1 = i^2 \not> 0$, and that it can be shown that all finite fields are not ordered. In fact, this is a characterizing property of non-ordered fields

Proposition 21.1.3: Artin-Schreier Conditions

Let k be a field. Then k can be an ordered field. Then the following are equivalent:

1. k is an ordered field
2. -1 is *not* the sum of squares
3. 0 is *not* a (non-zero) sum of square

***Proof*:**

$(1 \Rightarrow 2)$ see previous proposition.

$(2 \Leftrightarrow 3)$ If $-1 = \sum_i^n x_i^2$, then $0 = 1 + \sum_i^n x_i^2 = 1^2 + \sum_i^n x_i^2$. Conversely, if $0 = \sum_i^n x_i^2$ for nonzero x_i, then

$$-x_n^2 = \sum_i^{n-1} x_i^2 \qquad \Leftrightarrow \qquad -1 = \sum_i^{n-1} (\frac{x_i}{x_n})^2$$

Then by proposition 21.1.2, squares of elements cannot be negative, however $-1 < 0$.

(2 ⇔ 1) Assume -1 is not a sum of squares. Let S be the set of all elements that are (non-zero) sums of squares. Notably, $0, -1 \notin S$, S is closed under addition, and since k is commutative S is also closed under multiplication. Furthermore, it is a multiplicative subgroup of $k \setminus \{0\})$ for if $s \in S$ then:

$$\frac{1}{s} = \frac{s}{s^2} = \frac{\sum_i x_i^2}{s^2} = \sum_i \frac{x_i^2}{s^2} = \sum_i \left(\frac{x_i}{s}\right)^2$$

Now take the collection of sets that include S and that are closed under addition, multiplication, and don't contain -1 and 0. These sets are closed under inclusion and are certainly closed under chains, hence by Zorn's lemma there exists a maximal set $M \subseteq k$ that is closed under addition is a multiplicative subgroup of $k \setminus \{0\}$, which also cannot contain $-1, 0 \in M$. Then $M, \{0\}$, and $-M$ must be disjoint. I claim that

$$k = -M \cup \{0\} \cup M$$

Given this is proven, k then must be ordered, where $x < y$ iff $y - x \in M$. Suppose $0 \neq a \in k$ such that $-a \notin M$. Define:

$$M' = \{x + ay \mid x, y \in M \cup \{0\}, (x \neq 0) \vee (x \neq y)\}$$

Certainly $M \subseteq M'$ and M' is closed under addition, and is closed under multiplication for if $x, y, z, t \in M \cup \{0\}$, Then:

$$(x + ay)(z + at) = (xz + a^2yt) + a(yz + xt)$$

where $(xz + a^2yt), yz + xt \in M \cup \{0\}$ (note that $a^2 \in S \subseteq M'$). Furthermore, it should be clear that $1 \in M'$ $1 \in S \subseteq M \subseteq M'$) and $0 \notin M'$ (or else $a \in -M$). Finally, if $t \in M'$, then:

$$t^{-1} = \frac{t}{t^2} = \frac{x}{t^2} + a\frac{y}{t^2} \in M'$$

since $t^2 \in M$. Thus, M' is a multiplicative subgroup of $k \setminus \{0\}$, and hence by maximality $M = M'$, giving $a \in M$.

Since any ordered field is characteristic zero, it must contain a subfield

$$Q = \left\{ \frac{m1}{n1} \,\middle|\, m, n \in \mathbb{Z} \right\}$$

field-isomorphic to $\mathbb{Q}$. In fact, Q is order-field isomorphic, for $n1 > 0$ in Q when $n > 0$ in $\mathbb{Z}$, and $m1/n1 > 0$ in Q when $m/n > 0$ in $\mathbb{Q}$, and so $a1/b1 > c1 > d$ in Q iff $a/b > c/d$ in $\mathbb{Q}$. Thus, we usually identify Q with $\mathbb{Q}$ as an ordered subfield. A property that we may take for granted that an ordered field may have is the following:

Definition 21.1.4: Archimedean Field

Let k be an ordered field. Then k is an *Archimedean field* if every positive element of k is less than a positive integer

Though it may seem like all fields have this property, there do exists field without it, most notably the *hyperreals numbers* which can be thought of as the "order completion" of the reals (think of adding an element ϵ st $0 < \epsilon < x$ for all $x > 0$ and a ω st $0 < x < \omega$ for all $x > 0$). Fortunately, all the ordered fields we shall work with will be Archimedean due to the following classification result:

Theorem 21.1.5: Classification Of Archimedean Fields

Let k be an Archimedean field. Then k is isomorphic to a field:

$$\mathbb{Q} \subseteq k' \subseteq \mathbb{R}$$

***Proof*:**
Grillet p.233

Let us focus on formally real fields. If k is formally real, then so is $k(x)$. For an algebraic extension

Proposition 21.1.6: Algebraic Extension Of Formally Real Field

Let k be a formally real field. Then if $\alpha^2 \in k$ such that $\alpha^2 > 0$, $k(\alpha)$ is a formally real field

***Proof*:**
exercise

If an algebraic extension of a formally real field is formally real, then it is called a *real extension*.

Proposition 21.1.7: Odd Extension Of Formally Real Field

Let k be a formally real field. Then every finite extension of odd degree is a formally real field.

***Proof*:**
Grillet p.234

Using this, we can find the real closure of a formally real field:

Definition 21.1.8: Closed Real Field

Let k be a formally real field. Then k is said to be *closed* if there is no real algebraic extension L/k.

Theorem 21.1.9: Detecting Real Closure

Let k be a formally real field. Then k is real closed if and only if:

1. every positive element of k is a square in k
2. every polynomial of odd degree has a root in k

From these condition, we get that $\overline{k} = k(i)$ where $i^2 = -1$.

***Proof*:**
Grillet p.235

Due to this, a real closed field can be made into an ordered field in a unique way, namely $x \leq y$ if and only if $y - x = a^2$ for some $a \in k$. It should also be clear that the field of all algebraic real numbers is certainly a real closed field.

Corollary 21.1.10: Irreducible Elements In Real Closed Field

Let k be a real closed field. Then $f \in k[x]$ is irreducible if and only if f has degree 1 or f has degree 2 with no roots

***Proof*:**
exercise

Finally, we have:

Theorem 21.1.11: Real Closure

Every formally real field has a real closure

***Proof*:**
Grillet p.236

The reader should verify that any two real closures of k are order-field isomorphic.

Finally, we state without proof the following characterization of real-closed fields that show a naturalness to the choice of $\mathbb{R}$

Theorem 21.1.12: Artin-Schreier Theorem

Let $k \neq \overline{k}$ be a field. Then the following are equivalent:

1. k is real closed
2. $[\overline{k} : k]$ is finite
3. there is an upper bound for the degrees of irreducible polynomials in $k[x]$

> ***Proof*:**
> Grillet p.236-7

A variant of the theorem is worth recording, also without proof. Let G_K denote the absolute Galois group of a field K, and let $\sigma \in G_K$ be an involution, so that $\sigma \neq \mathrm{id}$ and $\sigma \circ \sigma = \mathrm{id}$. Then the fixed field $\overline{K}^{\sigma}$ is real closed. The two statements carry the same content, since the fixed field of an involution sits at degree 2 below $\overline{K}$ and theorem 21.1.12 then applies to it.

21.1.1 The Fields That Carry a Geometry

A vector space over an arbitrary field has no length and no angle, and section 13.6 recorded what producing them takes: an ordering, so that $\langle v, v\rangle$ can be called nonnegative, and a conjugation, so that $\langle v, v\rangle$ can be forced into the ordered field underneath. Both were described there as structure that $\mathbb{R}$ and $\mathbb{C}$ happen to carry. The results of this section say where that structure comes from, and how little freedom there was in it.

Whether a field can support a notion of positivity at all is decided by the field itself. By proposition 21.1.3, k admits a total order exactly when -1 is not a sum of squares, a condition stated in the ring operations alone with no order mentioned.

One step further up, the order and the conjugation both stop being choices. Let k be real closed. Every positive element of k is a square by theorem 21.1.9, and every nonzero square is positive by proposition 21.1.2(6), so the positive elements are exactly the nonzero squares. Hence $x \leq y$ holds if and only if $y - x$ is a square, and the order is recovered from the field operations rather than imposed on them. The same theorem gives $\overline{k} = k(i)$. Since -1 is not a square in a formally real field this extension has degree 2; it is the splitting field of $x^2 + 1$ over k, hence normal, and separable because an ordered field has characteristic zero (proposition 21.1.2). So $k(i)/k$ is Galois, $\mathrm{Gal}(\overline{k}/k)$ has order 2, and its one nontrivial element is $a + bi \mapsto a - bi$. A real closed field therefore comes with a unique order and a unique conjugation, both determined by k.

Corollary 21.1.10 gives the third property. Over a real closed k every irreducible polynomial has degree 1 or 2, so a characteristic polynomial factors into linear and quadratic pieces over k itself and not only over $\overline{k}$.

Theorem 21.1.12 says that no other configuration of this shape exists. If $k \neq \overline{k}$ and $[\overline{k} : k]$ is finite, then k is real closed and the degree is 2. In particular no field has an algebraic closure of degree 3: a field sits at finite distance below its algebraic closure in exactly one way, and that way is the real closed one.

What none of this pins down is $\mathbb{R}$. The field of real algebraic numbers is real closed as well and satisfies every condition above, so the separation between the two is order-completeness rather than an algebraic property. Theorem 21.1.5 gives the algebraic half of that separation by placing every Archimedean field between $\mathbb{Q}$ and $\mathbb{R}$. The geometry built on the order and the conjugation (inner products, orthogonality, adjoints, the spectral theorem) is carried out over $\mathbb{R}$ and $\mathbb{C}$ in *Linear Algebra* [12].

21.2 Absolute Values on General Fields

The reader [ought to have] showed that $\mathbb{C}$ is not an ordered field. The next best thing is some notion of "size". check this! Intuitively, size is a totally ordered concept, and hence the absolute value shall be a function that will capture the intuition of size with codomain $\mathbb{R}$:

Definition 21.2.1: Absolute Value

Let k be a field. Then an *absolute value real valuation* on k is a function $v : k \to \mathbb{R}$ whose image is denoted $|x|_v$ such that

1. (positivity) $|x|_v \geq 0$ for all $x \in k$ and $|x|_v = 0$ if and only if $x = 0$
2. (multiplicative) $|xy|_v = |x|_v|y|_v$
3. (triangle inequality) $|x + y|_v \leq |x|_v + |y|_v$

If the absolute value is unambiguous, $|x|_v$ will be denoted $|x|$. If the image of v is a finitely generated abelian group, then $| - |_v$ is called a *discrete absolute value*

21.2.2 Note The reader should be careful with the definition *real valuation* as there is a different concept called *valuation* that is not a generalization o specialization of the above (see definition 27.1.1).

If the reader is familiar with norms, then this can be thought of as the field-equivalent as norms are defined on vector-spaces. In fact, taking a field to be a vector-space over itself, the existence of a absolute value induces a norm. If the reader is familiar with some Lebesgue space theory, they may recall Hölder's inequality stating that:

$$\|fg\|_1 \leq \|f\|_p\|g\|_q \qquad \frac{1}{p} + \frac{1}{q} = 1$$

which contrasts with the strict equality result given above. Recall that equality happens if and only if $\alpha|f|^p = \beta|g|^q$ for $\alpha, \beta \neq 0$, that is they are "power multiples" of each other. This is certainly the case for fields. Furthermore, there is only one absolute value instead of the 3 norms present in Hölder's inequality. A norm that satisfies the 2nd condition (on an algebra!) is called a *multiplicative norm*, and implies certain geometric invariances in the space.

Example 21.2.3: Absolute Values

1. The usual absolute values on $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ all satisfy the above definition
2. For any field k, there always exists the trivial valuation

$$|x|_t = 1 \qquad x \neq 0$$

   This real valuation shall be an example of a *nonarchimedean* real valuation.

3. Let k be any field and take $k(x)$, the simple transcendental extension of k. Then there are

two interesting valuations that may be defined. The first is called the infinity valuation:

$$|f/g|_\infty = \begin{cases} 2^{\deg f - \deg g} & f \neq 0 \\ 0 & f = 0 \end{cases}$$

and the 0 valuation:

$$|f/g|_0 = \begin{cases} 2^{\operatorname{ord} f - \operatorname{ord} g} & f \neq 0 \\ 0 & f = 0 \end{cases}$$

where $\operatorname{ord}(f)$ for a polynomial $f(x) = a_0 + a_1 x_1 + \cdots + a_n x^n$ to be the smallest $k \geq 0$ such that $a_k \neq 0$. The choice of 2 was arbitrary in both and can be replaced with any positive constant.

From the 0-valuation, we may define:

$$\left|\frac{f(x)}{g(x)}\right|_a = \left|\frac{f(x-a)}{g(x-a)}\right|_0$$

These valuations when restricted to k are all trivial.

With positivity and the triangle inequality, the absolute value can define a metric (see chapter C) and hence a metric topology on a field:

$$d(x,y) = |x-y|_v$$

For the p-adic case, the idea behind this metric is that $|x-y|$ is small is $x \equiv y \mod p^n$ for a large n, that is a large power of p divides the difference.

Proposition 21.2.4: Equivalent Topoloies

Let v, w be two absolutes values on a field k. Then the following are equivalent:

($\Rightarrow$) v, w induce the same topology on k

($\Rightarrow$) Given $x \in k$, $|x|_v < 1$ if and only if $|x|_w < 1$

($\Rightarrow$) There exists a $n \in k$ such that $|x|_w = |x|_v^n$ for all $x \in k$

***Proof*:**
exercise

The following can be htougt of as the CRT for real valutions

Proposition 21.2.5: Approximation Theorem

Let $|-|_1, ..., |-|_n$ be pairwise inequivalent valuations for a field K, and let $a_1, ..., a_n \in K$ be some choice elements. Then for every $\epsilon > 0$,there exists a $x \in K$ such that

$$|x - a_i|_i < \epsilon \qquad \forall 1 \leq i \leq n$$

***Proof*:**
exercise, use the negation of the above proposition (or Neukrich p.118)

As mentioned earlier, the trivial valuation is non-Archimedean. The following show that non-Archimedean valuations can be thought of as absolute values induced by *ultra-metrics*

Definition 21.2.6: Archimedean Valuation

Let v be a real valuation on a field k. Then if for all $n \in \mathbb{N}$, there exists a $x \in k$ such that

$$|n|_v \leq |x|_v$$

the valuation is called an *Archimedean valuation*. On the other hand, if there exists an x such that for all $n \in \mathbb{N}$:

$$|n|_v \leq |x|_v \qquad \forall n \in \mathbb{N}$$

Then v is called a *non-Archimedean valuation*.

Proposition 21.2.7: Characterizing Non-Archimedean Valuation

Let v be a real valuation on a field k. Then the following are equivalent:

($\Rightarrow$) v is non-Archimedean

($\Rightarrow$) $|n|_v \leq 1$ for all $n \in \mathbb{N}$

($\Rightarrow$) $|n|_v \leq 1$ for all $n \in \mathbb{Z}$

($\Rightarrow$) for all $x, y \in k$,
$$|x+y|_v \leq \max(|x|_v, |y|_v)$$

The final inequality upgrading the triangle inequality is called the *strong triangle inequality* or *ultra-metric inequality*.

Note that if $|x| \neq |y|$ in an ultra-metric , then $|x+y| = \max(|x|, |y|)$!

***Proof*:**

(1) $\Rightarrow$ (2) Let v be a non-Archimedean valuation. For the sake of contradiction, suppose that for some $n \in \mathbb{N}$, $|n|_v > 1$. Then by multiplicity we may have an $|n|_v$ with an arbitrarily large valuation:

$$|n^k| = |n|_v^k \xrightarrow{k \to \infty} \infty$$

Which would imply there is no $x \in k$ such that $|n|_v \leq |x|_v$ for all $n \in \mathbb{N}$.

(2) $\Rightarrow$ (3) show that $|-1| = |1|$ and use this to conclude.

(3) ⇒ (4) Take:

$$\begin{aligned}|x+y|^n &= \left|\sum_k^n \binom{n}{k} x^k y^{n-k}\right| \\ &\leq \sum_k^n \left|\binom{n}{k}\right| |x|^k |y|^{n-k} \\ &\leq \sum_k^n |x|^k |y|^{n-k} \\ &\leq (n+1)\max(|x|^n, |y|^n)\end{aligned}$$

Hence

$$|x+y| \leq (n+1)^{1/n}\max(|x|,|y|)$$

As this is true for all n, and $(n+1)^{1/n} \xrightarrow{n\to\infty} 1$, we get the desired result

(4) ⇒ (1) by induction we get:

$$|n| = |1+1+\cdots+1| \leq |1| = 1$$

giving the desired result.

Corollary 21.2.8: Values On different characteristic

Let k be a finite with characteristic $p \neq 0$. Then there does not exist an Archimedean valuation on k.

If k is a finite field, the only valuation is the trivial valuation.

***Proof*:**
exercise.

In the case of characteristic 0, given $k = \mathbb{Q}$, all the absolute values may be classified. Define p-valuation to be:

$$v_p(m/n) = \begin{cases} p^{-k} & \frac{m}{n} = p^k \frac{m'}{n'} \neq 0,\ p \nmid m', n' \\ 0 & \frac{m}{n} = 0 \end{cases}$$

Then

Theorem 21.2.9: Classification Of Absolute Values On Rationals

Let $k = \mathbb{Q}$. Then every non-trivial valuation on k is either a p-valuation or the usual absolute value.

This result is due to Ostrowski.

***Proof*:**
Let v be a nonarchemidian valuation. Then $|n|_v \leq 1$ for all $n \in \mathbb{Z}$. If $|n|_v = 1$ for all $0 \neq n \in \mathbb{Z}$ then

$$|m/n|_v|n|_v = |m|_v = 1$$

which can be sued to show that $|x|_v = 1$ for all $x \in \mathbb{Q}$. Thus, the following set is non-empty:

$$P = \{x \in \mathbb{Z} \mid |n|_v < 1\}$$

This can be shown to be an prime ideal of $\mathbb{Z}$ (exercise), and since $\mathbb{Z}$ is a PID we have that $P = (p)$ for appropriate prime $p \in \mathbb{Z}$. Let

$$c = |p|_v < 1$$

Now, if

$$\frac{m}{n} = p^k \frac{m'}{n'} \neq 0$$

where $p \nmid m', n'$, then by definition of the prime ideal:

$$|m'|_v = |n'|_v = 1 \qquad \Rightarrow \qquad n\left|\frac{m}{n}\right| = c^k$$

This valuation is not equivalent to v_p. Note that v_p and v_q are not equivalent as that would imply $|p|_v = 1$ (exercise).

The Archimedean is left as a real-analysis exercise, or see Grillet p.242,

In *Commutative Algebra* [8] (valuation rings and DVRs), many of these concepts shall be generalized to more general valuations, namely we have been working with *real* valuations, however there are more generally G-valuations for totally ordered groups G.

In an analysis class, the absolute value is used to define the completion of a ring. If k is a field with an absolute value, then the completion $\widehat{k}$ of k is the collection of Cauchy-sequences under the appropriate quotient where k is dense in $\widehat{k}$, and the absolute value n extends naturally to $\widehat{v}$ in $\widehat{k}$.

21.2.1 Extending Valuation

Eventually here

21.2.2 Ostrowski Theorem

Show that if $(K, |-|)$ is a field with an Archimedean absolute value, then

$$\widehat{K} \underset{\sim}{\in} \{\mathbb{R}, \mathbb{C}\}$$

Neukirch p.124

21.3 Derivation on Field

Derivations are the algebraic form of differentiation, and the module of relative differential forms packages every derivation of a ring into a single universal one. The construction is pure commutative algebra and belongs here rather than in any geometry book: it is used by [13] for the canonical class of a curve, by [3] for the sheaf of differentials, and by [2] for invariant differentials on a group variety.

Definition 21.3.1: A-derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. Then an A-derivation of B into M is a map $d : B \to M$ such that

1. $d(b + b') = d(b) + d(b')$
2. $d(bb') = d(b)b' + bd(b')$
3. $d(a) = 0$ for all $a \in A$

Definition 21.3.2: Module of Relative Differential Forms

Let A be a commutative ring, B an A-algebra. Then a module of relative differential form is a B-module $\Omega_{B/A}$ with an A-derivation $d : B \to \Omega_{B/A}$ which satisfies the universal property of relative differential forms: if M is another B-module and $d' : B \to M$ an A-derivation, then there is a unique B-module homomorphism $f : \Omega_{B/A} \to M$ such that the following diagram commutes:

$$\begin{tikzcd} B \arrow[r, "d"] \arrow[dr, "d'"'] & \Omega_{B/A} \arrow[d, "f"] \\ & M \end{tikzcd}$$

Certainly $\Omega_{B/A}$ exists via the usual construction: take the free B-module F generated by the primitives $d(b)$ for $b \in B$ and quotient by the submodule generated by the relationships:

$$d(b + b') - d(b) - d(b') \quad \text{and} \quad d(bb') - d(b)b' - bd(b') \quad \text{and} \quad d(a)$$

Then the map $d : B \to \Omega_{B/A}$ is given by $b \mapsto d(b)$. Then it is an exercise in algebra to see that this satisfies the universal property and $(\Omega_{B/A}, d)$ is unique up to unique isomorphism. The following gives an alternative insightful construction:

Proposition 21.3.3: Natural Differential Structure

Let B be an A-algebra, $\delta : B \otimes_A B \to B$ the diagonal map given by $f(b \otimes b') = bb'$, and let $I = \ker \delta$. Then I/I^2 has the structure of a left B-module, and we can define a map

$$d : B \to I/I^2 \qquad d(b) = 1 \otimes b - b \otimes 1 \bmod I^2$$

In which case $(I/I^2, d)$ is a module of relative differential for B/A.

***Proof*:**
Write $\delta\colon B \otimes_A B \to B$ for the multiplication map and $I = \ker\delta$. Give I/I^2 the B-module structure coming from the left factor, $b\cdot(x\otimes y) = (bx)\otimes y$; the two possible structures agree modulo I^2, since $b\otimes 1 - 1\otimes b \in I$.

d is an A-derivation. Additivity is clear. For the Leibniz rule, compute in $B\otimes_A B$:

$$d(bb') - b\,d(b') - b'\,d(b) = \big(1\otimes bb' - bb'\otimes 1\big) - b\big(1\otimes b' - b'\otimes 1\big) - b'\big(1\otimes b - b\otimes 1\big)$$

which expands to $-(1\otimes b - b\otimes 1)(1\otimes b' - b'\otimes 1)$, a product of two elements of I and so zero in I/I^2. And $d(a) = 1\otimes a - a\otimes 1 = 0$ for $a\in A$, the tensor being over A.

Universality. Let $d'\colon B\to M$ be an A-derivation. Define $B\otimes_A B\to M$ by $x\otimes y\mapsto x\,d'(y)$, which is A-bilinear and hence well defined. It kills I^2: for $u = \sum x_i\otimes y_i$ and $v = \sum x'_j\otimes y'_j$ in I, meaning $\sum x_iy_i = 0$ and $\sum x'_jy'_j = 0$, expanding uv and applying the Leibniz rule for d' leaves $\big(\sum x_iy_i\big)\big(\sum x'_jd'(y'_j)\big) = 0$. So it descends to $f\colon I/I^2\to M$, and $f(d(b)) = f(1\otimes b - b\otimes 1) = d'(b) - b\,d'(1) = d'(b)$ since $d'(1) = 0$. Uniqueness holds because the $d(b)$ generate I/I^2: any $\sum x_i\otimes y_i\in I$ equals $\sum x_i(1\otimes y_i - y_i\otimes 1)$ modulo nothing, using $\sum x_iy_i = 0$.

So $(I/I^2, d)$ has the universal property of definition 21.3.2, completing the proof.

Example 21.3.4: Relative Differential

1. Take $A=\mathbb{R}$ and $B = \mathbb{R}[x]$. Then $I = \ker\delta$. Then as $\mathbb{R}[x]$ is an integral domain, it can be shown that any element is a linear combination of elements of the form:
$$f(x)\otimes g(x) - f(x)g(x)\otimes 1 \in I$$
Then it should be verified any element in I can be written as:
$$f(x)(1\otimes g(x) - g(x)\otimes 1)$$
The reader should then verify that I/I^2 is generated by $f'(x)(x\otimes 1 - 1\otimes x)$[a]. Let dx represent this element. Notice that this is the image of $d(x)$. Now, let's consider:
$$d(f(x) = d\left(\sum_i a_ix^i\right) = \sum_i a_i d(x^i)$$
Hence, let's compute $d(x^i)$. For $i = 2$, we get:
$$d(x^2) = d(x\cdot x) = xd(x) - xd(x) = 2xdx$$
Repeating this pattern, we can see that:
$$d(x^n) = nx^{n-1}dx$$
2. For example, taking $B = \mathbb{R}[x_1, ..., x_n]$, Then $\Omega_{B/A}$ is a free B-module generated by $dx_1, ..., dx_n$.
3. In proposition 21.3.10, we shall see that if E/F is a finitely generated separable extension (of characteristic 0), then
$$\Omega_{B/A} = \bigoplus_{b\in T} Bdb$$
where T is a transcendental A-basis for B.

4. What if we take $B = A[\epsilon]/(\epsilon^2)$?
5. Let $A = \mathbb{R}$ and $B = C^\infty(M)$ for some smooth n-manifold M. Then $C^\infty(M)$. Then $\Omega_{C^\infty(M)/\mathbb{R}}$ is a locally free $C^\infty(M)$-module (based off of the patches of M), consisting of the space of sections of the cotangent bundle T^*M.

[a]Hint: what is $(1 \otimes x^2 - x^2 \otimes 1)$. Repeat for higher powers

Proposition 21.3.5: Base Change For Differentials

Let A', B be A-algebras and $B' = B \otimes_A A'$. Then:

$$\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$$

Furthermore, if S is a multiplicative subset of of S, then:

$$\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$$

***Proof*:**
Both statements are checked against the universal property of definition 21.3.2.

For the base change, a B'-module M receives an A'-derivation from B' exactly when it receives an A-derivation from B that is A'-linear on the second factor, since $B' = B \otimes_A A'$ is generated over A' by the image of B. So $\mathrm{Der}_{A'}(B', M) \cong \mathrm{Der}_A(B, M)$ naturally, and representing both sides gives $\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$.

For the localization, an A-derivation $d\colon B \to M$ into an $S^{-1}B$-module extends uniquely to $S^{-1}B$ by the quotient rule $d(b/s) = \big(s\,d(b) - b\,d(s)\big)/s^2$, which is forced by the Leibniz rule applied to $b = (b/s) \cdot s$. So $\mathrm{Der}_A(S^{-1}B, M) \cong \mathrm{Der}_A(B, M)$ for such M, and representing gives $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$, completing the proof.

Proposition 21.3.6: Differentials and Exact Sequence I

Let $A \to B \to C$ be a (not necessarily exact nor a complex) sequence of rings. Then there is an exact sequence of C-modules:

$$\Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

***Proof*:**
The maps are the natural ones: $db \otimes c \mapsto c\,db$ and $dc \mapsto dc$, the latter making sense because an A-derivation of C is in particular one killing the image of A.

Exactness is checked by applying $\mathrm{Hom}_C(-, M)$ for an arbitrary C-module M and using definition 21.3.2, which turns the sequence into

$$0 \to \mathrm{Der}_B(C, M) \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M)$$

This is exact: a derivation of C over A kills the image of B exactly when it is a derivation over B, which is exactness at the middle term and injectivity on the left. Since $\mathrm{Hom}_C(-, M)$ is left exact and the resulting sequence is exact for every M, the original sequence is exact at $\Omega_{C/A}$

and at $\Omega_{C/B}$, completing the proof.

Proposition 21.3.7: Differentials and Exact Sequence II

Let B be an A-algebra, I be an ideal of B, and $C = B/I$. Then there is a natural exact sequence of C-modules

$$I/I^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to 0,$$

where for any $b \in I$, if $\bar{b}$ is its image in I/I^2, then $\delta\bar{b} = db \otimes 1$. In particular, I/I^2 has a natural structure of a C-module, and δ is a C-linear map, even though it is defined via the derivation d.

***Proof*:**
First, δ is well defined and C-linear. For $b, b' \in I$ the Leibniz rule gives $d(bb') = b\,db' + b'\,db$, which lies in $I\,\Omega_{B/A}$ and so vanishes in $\Omega_{B/A} \otimes_B C$; hence δ kills I^2. C-linearity follows because multiplying b by an element of I again lands in I^2.

For exactness, apply $\mathrm{Hom}_C(-, M)$ as in proposition 21.3.6. The sequence becomes

$$0 \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M) \to \mathrm{Hom}_C(I/I^2, M)$$

An A-derivation of B into M, where M is a C-module and so killed by I, descends to $C = B/I$ exactly when it vanishes on I; and its restriction to I is precisely the composite with δ. That is exactness at the middle and injectivity on the left, and as before it holds for every M, giving the stated sequence, completing the proof.

Corollary 21.3.8: Finite Generation of Differential Forms

If B is a finitely generated A-algebra or a localization of a finitely generated A-algebra, then $\Omega_{B/A}$ is a finitely generated B-module

***Proof*:**
As B is a finitely generated A-algebra, it is the quotient of a polynomial ring. Hence, it follows from our direct computations from example 21.3.4 and, proposition 21.3.7, and the localization result comes from proposition 21.3.5.

For finite extensions the module of differentials sees exactly one thing, namely separability, and it sees it sharply: a nonzero derivation exists precisely when some element of the extension fails to be separable. The mechanism is already at work on a simple extension $k(\alpha)$, where a k-derivation is pinned down by its value on α and that value is constrained only through the single scalar $f'(\alpha)$.

Lemma 21.3.9: Derivations Detect Separability

Let F/k be a finite field extension. Then the following are equivalent:

1. F/k is a separable extension
2. $\Omega_{F/k} = 0$
3. the only k-derivation of F into F is the zero map

***Proof*:**

($1 \Rightarrow 2$) Let M be any F-module and $D : F \to M$ a k-derivation. Since F/k is finite and separable, theorem 19.6.20 gives $F = k(\beta)$ for some $\beta \in F$, and β is separable over k by definition 19.6.3. Write $g(x) = m_{\beta,k}(x) = \sum_j c_j x^j$. Additivity and the Leibniz rule give $D(\beta^j) = j\beta^{j-1}D(\beta)$, and D kills each $c_j \in k$, so

$$0 = D\big(g(\beta)\big) = \sum_j c_j D(\beta^j) = g'(\beta)D(\beta)$$

Now β is a root of the separable polynomial g, hence not a multiple root, so $g'(\beta) \neq 0$ by proposition 19.6.6. As $g'(\beta)$ is a unit in the field F, this forces $D(\beta) = 0$. Every element of F is a polynomial in β with coefficients in k by proposition 19.2.24, so additivity and the Leibniz rule propagate $D(\beta) = 0$ to $D = 0$. Thus $\mathrm{Der}_k(F, M) = 0$ for every F-module M.

Apply this to $M = \Omega_{F/k}$: the universal derivation d of definition 21.3.2 is itself zero. Both the identity and the zero map $\Omega_{F/k} \to \Omega_{F/k}$ are F-module homomorphisms f with $f \circ d = d$, so the uniqueness clause of definition 21.3.2 forces them to agree, giving $\Omega_{F/k} = 0$.

($2 \Rightarrow 3$) By definition 21.3.2 the k-derivations of F into F are in bijection with the F-module homomorphisms $\Omega_{F/k} \to F$, and the zero module admits only the zero homomorphism.

($3 \Rightarrow 1$) Take the contrapositive: assume F/k is inseparable and produce a nonzero k-derivation of F into F. Since F/k is finite, write $F = k(\alpha_1, ..., \alpha_n)$ and set $F_0 = k$ and $F_i = F_{i-1}(\alpha_i)$, so that $F_n = F$.

First, some α_i is inseparable over F_{i-1}. If instead every α_i were separable over F_{i-1}, then $[F_i : F_{i-1}]_s = [F_i : F_{i-1}]$ for each i by proposition 19.6.15, and multiplying these together using corollary 19.6.16 on the left and theorem 19.3.7 on the right would give $[F : k]_s = [F : k]$, whence F/k is separable by proposition 19.6.17, contrary to assumption. So let i be the *largest* index for which α_i is inseparable over F_{i-1}, and abbreviate

$$E = F_{i-1} \qquad \alpha = \alpha_i \qquad L = F_i = E(\alpha)$$

By maximality of i, each α_j with $j > i$ is separable over F_{j-1}; running the same product over the tower $L = F_i \subseteq \cdots \subseteq F_n = F$ gives $[F : L]_s = [F : L]$, so F/L is a finite separable extension, again by proposition 19.6.17.

A nonzero E-derivation of L. Let $f(x) = m_{\alpha,E}(x)$. It is irreducible, and it is inseparable because α is inseparable over E (definition 19.6.3), so proposition 19.6.7 gives $f'(x) = 0$. Reading off the leading term of f', this already forces $\mathrm{ch}(k) = p > 0$ and $f \in E[x^p]$, exactly as in the discussion preceding definition 19.6.9; in characteristic 0 there is nothing to prove, since corollary 19.6.8 rules out inseparable irreducibles outright. Regard L as an $E[x]$-module through the evaluation map $\mathrm{ev}_\alpha : E[x] \to L$ of theorem 19.2.8, whose kernel is (f), and consider

$$\partial : E[x] \to L \qquad \partial(h) = h'(\alpha)$$

with h' the formal derivative of definition 19.6.5. It is additive, it kills E, and the Leibniz rule for the formal derivative gives

$$\begin{aligned}\partial(h_1h_2) = (h_1h_2)'(\alpha) &= h_1'(\alpha)h_2(\alpha) + h_1(\alpha)h_2'(\alpha)\\ &= \partial(h_1)h_2(\alpha) + h_1(\alpha)\partial(h_2)\end{aligned}$$

so ∂ is an E-derivation in the sense of definition 21.3.1. It kills the ideal (f): for any $h \in E[x]$,

$$\partial(fh) = (fh)'(\alpha) = f'(\alpha)h(\alpha) + f(\alpha)h'(\alpha) = 0$$

because $f' = 0$ and $f(\alpha) = 0$. Hence ∂ factors through ev_α as an additive map $D : L \to L$, and the displayed Leibniz identity descends, since both sides depend only on the values $h_1(\alpha), h_2(\alpha)$. So D is an E-derivation of L into L with

$$D(\alpha) = \partial(x) = 1$$

and in particular $D \neq 0$.

Extending D across F/L. Since F/L is finite and separable, theorem 19.6.20 gives $F = L(\beta)$ with β separable over L. Put $g(x) = m_{\beta,L}(x)$; as in $(1 \Rightarrow 2)$, $g'(\beta) \neq 0$ by proposition 19.6.6. For $h(x) = \sum_j c_jx^j \in L[x]$ write $h^D(x) = \sum_j D(c_j)x^j$; expanding a product coefficientwise and applying the Leibniz rule for D to each term gives

$$(h_1h_2)^D = h_1^Dh_2 + h_1h_2^D$$

Set

$$t = -\frac{g^D(\beta)}{g'(\beta)} \in F$$

and regard F as an $L[x]$-module through evaluation at β. Define

$$\Delta : L[x] \to F \qquad \Delta(h) = h^D(\beta) + h'(\beta)\,t$$

Then Δ is additive, and combining the two product rules just recorded,

$$\begin{aligned}\Delta(h_1h_2) &= h_1^D(\beta)h_2(\beta) + h_1(\beta)h_2^D(\beta) + \big(h_1'(\beta)h_2(\beta) + h_1(\beta)h_2'(\beta)\big)t\\ &= \Delta(h_1)h_2(\beta) + h_1(\beta)\Delta(h_2)\end{aligned}$$

On a constant $c \in L$ we get $\Delta(c) = D(c)$, so Δ vanishes on E and a fortiori on k. By the choice of t,

$$\Delta(g) = g^D(\beta) + g'(\beta)t = 0$$

and therefore $\Delta(gh) = \Delta(g)h(\beta) + g(\beta)\Delta(h) = 0$ for every $h \in L[x]$, since $g(\beta) = 0$. So Δ kills the kernel (g) of evaluation at β and descends, by theorem 19.2.8, to an additive map $\widetilde{D} : F \to F$; as before the Leibniz identity descends with it. Thus $\widetilde{D}$ is a k-derivation of F into F restricting to D on L, and

$$\widetilde{D}(\alpha) = D(\alpha) = 1 \neq 0$$

So F carries a nonzero k-derivation into itself, which is the contrapositive of $(3 \Rightarrow 1)$, completing the proof.

Note that the primitive element theorem is used only where it is legitimate, on the separable extension F/L; it says nothing about F/k itself, which by example 19.6.22 need not be simple once separability fails. The whole content of $(3 \Rightarrow 1)$ is that an inseparable step in a tower is a step at which a derivation may be defined freely, and the remaining steps are separable and so carry the derivation along uniquely.

Proposition 21.3.10: Differentials and Field Extensions

Let K/k be a finitely generated field extension. Then:

$$\dim_K \Omega_{K/k} \geq \operatorname{trdeg} K/k$$

where equality holds iff K is separably generated over k.

***Proof Key ideas*:**
This is the separability criterion for field extensions in differential form, and both implications reduce to the exact sequences above: for a separably generated extension a transcendence basis gives a basis of Ω, and conversely a nonzero differential of a purely inseparable generator is killed by the p-th-power relation. A full account is in [15, chapter II.8]; the statement is recorded here because [13] and [3] both use it to detect smoothness.

The finite algebraic case is the one this book uses most often, and it is not left to a citation: lemma 21.3.9 proves that a finite extension K/k has $\Omega_{K/k} = 0$ exactly when it is separable. Note that the transcendence-degree statement above genuinely needs the finitely generated hypothesis, whereas the finite case does not.

Proposition 21.3.11: Local Ring and Relative Differential

Let B be a local ring containing a field k isomorphic to $B/\mathfrak{m}$. Then given the map δ from proposition 21.3.7:

$$\delta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\cong} \Omega_{B/k} \otimes_B k$$

***Proof*:**
By proposition 21.3.7 $\operatorname{coker}(\delta) = 0$, so δ is surjective. To show that δ is injective, we shall show the following dual vector space map is surjective:

$$\delta' : \operatorname{Hom}_k(\Omega_{B/k} \otimes k, k) \to \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$$

The term on the left is isomorphic to $\operatorname{Hom}_B(\Omega_{B/k}, k)$, which by definition of the differentials, can be identified with the set $\operatorname{Der}_k(B, k)$ of k-derivations of B to k. If $d : B \to k$ is a derivation, then $\delta'(d)$ is obtained by restricting to $\mathfrak{m}$, and noting that $d(\mathfrak{m}^2) = 0$. Now, to show that δ' is surjective, let $h \in \operatorname{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$. For any $b \in B$, we can write $b = \lambda + c$, $\lambda \in k$, $c \in \mathfrak{m}$, in a unique way. Define $db = h(\bar{c})$, where $\bar{c} \in \mathfrak{m}/\mathfrak{m}^2$ is the image of c. Then one verifies immediately that d is a k-derivation of B to k, and that $\delta'(d) = h$. Thus δ' is surjective, as required.

We finish off this brief summary of the essential results about relative differentials by showing the relation between regular local rings and relative differentials. We require the following lemma:

Lemma 21.3.12: Characterizing Free Modules using Residue and Quotient Field

Let A be a Noetherian local integral domain, with residue field k and quotient field K. Then if M is a finitely generated A-module and $\dim_k M \otimes_A k = \dim_k M \otimes_A K = r$, then M is free of rank r.

***Proof*:**
Since $\dim_k M \otimes k = r$, by Nakayama's lemma M can be generated by r elements. Hence, there is a surjective map $\varphi : A^r \to M$. Let R be its kernel. Then we obtain an exact sequence

$$0 \to R \otimes K \to K^r \to M \otimes K \to 0,$$

As $\dim_K M \otimes K = r$, we have $R \otimes K = 0$. But R is torsion-free, so $R = 0$, and M is isomorphic to A^r, completing the proof.

22⋆

Bilinear, Quadratic and Alternating Forms over a Field

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

A bilinear form is a piece of data attached to a vector space, and section 13.3 said what that data is: a matrix, once a basis is fixed. It did not say when two such data are the same. Two forms should count as the same when some change of basis carries one to the other, and the matrix statement of that relation is $A \mapsto P^t AP$ rather than the $A \mapsto P^{-1}AP$ that governs linear operators. The two relations are genuinely different, and the invariants of the first are not the eigenvalues.

This chapter classifies bilinear forms over a field up to that relation, in the two cases that occur: the symmetric forms, where the answer is subtle and field-dependent, and the alternating ones, where it is not. The answer is controlled by a single group built from the field, the square classes $k^\times/(k^\times)^2$, and this is why the subject sits after Galois theory: for a field of characteristic other than 2 that group is exactly what Kummer theory at $n = 2$ (section 20.4) classifies. Over $\mathbb{C}$ the group is trivial and the classification collapses to the rank; over $\mathbb{R}$ it has two elements and one further invariant appears; over a general field it can be arbitrarily complicated, and the bookkeeping of what survives is the Witt group.

22.1 Quadratic Forms and Symmetric Bilinear Forms

The forms of interest are the symmetric ones, and over a field where 2 is invertible a symmetric bilinear form carries exactly the same information as the single-variable function $x \mapsto \varphi(x, x)$. That function is easier to write down and is what one meets in practice, so the correspondence between the two is recorded first. It fails in characteristic 2, and the failure is not a technicality to be waved past: it is why the whole theory below carries the hypothesis $\operatorname{char} k \neq 2$.

Definition 22.1.1: Quadratic Form over a Field

Let k be a field and V a finite-dimensional k-vector space. A *quadratic form* on V is a function $q\colon V \to k$ such that

1. $q(\lambda v) = \lambda^2 q(v)$ for all $\lambda \in k$ and $v \in V$;
2. the function $b_q\colon V \times V \to k$ defined by
$$b_q(x, y) := q(x + y) - q(x) - q(y)$$
is bilinear.

The form b_q is the *polar form* of q. It is symmetric, since the defining expression is unchanged by exchanging x and y.

Two consequences of the definition are used constantly and are worth extracting at once. Setting $y = x$ in the polar form and applying (1) gives
$$b_q(x, x) = q(2x) - 2q(x) = 4q(x) - 2q(x) = 2q(x)$$
so the polar form recovers q after division by 2, and only then. And if φ is any symmetric bilinear form, the function $q_\varphi(x) := \varphi(x, x)$ satisfies both conditions above, with $b_{q_\varphi} = 2\varphi$ by the same computation. The two constructions are therefore mutually inverse exactly when 2 is invertible.

Proposition 22.1.2: Quadratic Forms and Symmetric Bilinear Forms Correspond

Let k be a field with $\operatorname{char} k \neq 2$ and let V be a finite-dimensional k-vector space. The assignments
$$\varphi \longmapsto q_\varphi, \qquad q \longmapsto \tfrac{1}{2} b_q$$
are mutually inverse bijections between the set of symmetric bilinear forms on V and the set of quadratic forms on V.

***Proof*:**
Step 1: the assignments are well defined. Let φ be symmetric bilinear and put $q_\varphi(x) = \varphi(x, x)$. Then $q_\varphi(\lambda v) = \varphi(\lambda v, \lambda v) = \lambda^2 \varphi(v, v) = \lambda^2 q_\varphi(v)$, which is condition (1). For condition (2), expanding by bilinearity and using symmetry,
$$b_{q_\varphi}(x, y) = \varphi(x + y, x + y) - \varphi(x, x) - \varphi(y, y) = 2\varphi(x, y)$$
which is bilinear. So q_φ is a quadratic form and $b_{q_\varphi} = 2\varphi$. Conversely, if q is a quadratic form then b_q is symmetric bilinear by definition 22.1.1, hence so is $\frac{1}{2} b_q$, using $\operatorname{char} k \neq 2$ to divide.

Step 2: the composites are the identity. Starting from a symmetric φ, Step 1 gives $\frac{1}{2} b_{q_\varphi} = \frac{1}{2} \cdot 2\varphi = \varphi$. Starting from a quadratic form q and writing $\varphi = \frac{1}{2} b_q$, the computation recorded before the proposition gives $b_q(x, x) = 2q(x)$, so
$$q_\varphi(x) = \varphi(x, x) = \tfrac{1}{2} b_q(x, x) = \tfrac{1}{2} \cdot 2q(x) = q(x)$$
Both composites are the identity, so the assignments are mutually inverse bijections, as we sought to show.

Because of the proposition, the words *quadratic form* and *symmetric bilinear form* are used interchangeably below whenever $\operatorname{char} k \neq 2$, and disc, orthogonality and the matrix of a form are attached to either at will. The next example shows what is lost when the hypothesis is dropped, and it is the reason the hypothesis is carried explicitly rather than assumed silently.

Example 22.1.3: Polarization Fails in Characteristic Two

Let $k = \mathbb{F}_2$ and $V = k^2$ with coordinates (x, y).

The map $q \mapsto b_q$ is not injective. Take $q(x, y) = x^2$. Then $q(\lambda v) = \lambda^2 q(v)$ holds, and

$$b_q\big((x, y), (x', y')\big) = (x + x')^2 - x^2 - x'^2 = 2xx' = 0$$

since $2 = 0$ in k. So q and the zero form have the same polar form, yet $q(1, 0) = 1 \neq 0$. More generally $b_q(v, v) = 2q(v) = 0$ for every quadratic form in characteristic 2, so every polar form is alternating and the diagonal information in q is invisible to b_q. Throughout this example *alternating* means the vanishing condition of definition 16.1.3, that $\varphi(v, v) = 0$ for every v. In characteristic 2 that is strictly stronger than the sign rule appearing in definition 13.3.15, since there $-1 = 1$ makes the sign rule say nothing beyond symmetry.

The map $q \mapsto b_q$ is not surjective either. The form $\varphi\big((x, y), (x', y')\big) = xx'$ is symmetric bilinear, but $\varphi\big((1, 0), (1, 0)\big) = 1 \neq 0$, so φ does not vanish on the diagonal and is therefore not the polar form of anything.

Orthogonal bases can fail to exist. Let $\psi\big((x, y), (x', y')\big) = xy' + x'y$, with matrix $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ in the standard basis. It is symmetric, nondegenerate, and nonzero, since $\psi(e_1, e_2) = 1$. But $\psi(v, v) = 2xy = 0$ for every v. Were w_1, w_2 an orthogonal basis, the matrix of ψ in it would be $\operatorname{diag}\big(\psi(w_1, w_1), \psi(w_2, w_2)\big) = \operatorname{diag}(0, 0)$, forcing $\psi = 0$ and contradicting $\psi(e_1, e_2) = 1$. So ψ admits no orthogonal basis at all. Contrast $\varphi(u, v) = xx'$ above, which is already diagonal in the standard basis: failing to be a polar form and failing to be diagonalizable are different failures, and ψ exhibits the second.

All three failures trace to the division by 2 in proposition 22.1.2. In characteristic 2 the quadratic and the bilinear theories separate, and the correct objects are quadratic forms together with their alternating polar forms, and that theory is not developed here.

The classification question can now be posed precisely. Fixing a basis turns a form into a matrix (definition 13.3.18), and changing the basis changes the matrix in a way that must be pinned down before any invariant can be extracted.

Theorem 22.1.4: Change of Basis for the Matrix of a Form

Let V be a finite-dimensional k-vector space with bases $\beta = (\beta_1, \dots, \beta_n)$ and $\beta' = (\beta'_1, \dots, \beta'_n)$, and let $\varphi\colon V \times V \to k$ be a bilinear form. Let P be the $n \times n$ matrix whose j-th column is $[\beta'_j]_\beta$, the coordinate vector of β'_j in the basis β (this is the change of basis matrix of definition 13.2.11). Then P is invertible, $[v]_\beta = P[v]_{\beta'}$ for every $v \in V$, and

$$[\varphi]_{\beta'} = P^t\, [\varphi]_\beta\, P$$

***Proof*:**
The columns of P are the coordinate vectors of a basis, hence linearly independent, so P is invertible. Writing $v = \sum_j c_j \beta'_j$, that is $[v]_{\beta'} = (c_j)_j$, linearity of the coordinate map gives

$$[v]_\beta = \Big[\sum_j c_j \beta'_j\Big]_\beta = \sum_j c_j [\beta'_j]_\beta = P[v]_{\beta'}$$

which is the second claim. For the third, write $A = [\varphi]_\beta$. By lemma 13.3.19, for all $v, w \in V$

$$\varphi(v,w) = [v]_\beta^t A [w]_\beta = \big(P[v]_{\beta'}\big)^t A \big(P[w]_{\beta'}\big) = [v]_{\beta'}^t \big(P^t AP\big)[w]_{\beta'}$$

Taking $v = \beta'_i$ and $w = \beta'_j$ makes the coordinate vectors the standard basis vectors e_i and e_j, so the displayed expression is the (i,j) entry of P^tAP. On the other hand that value is $\varphi(\beta'_i, \beta'_j)$, which is the (i,j) entry of $[\varphi]_{\beta'}$ by definition 13.3.18. The two matrices agree entrywise, completing the proof.

Definition 22.1.5: Congruent Matrices

Two matrices $A, B \in M_n(k)$ are *congruent* if $B = P^tAP$ for some invertible $P \in M_n(k)$. Congruence is an equivalence relation: it is reflexive by $P = I$, symmetric because $A = (P^{-1})^t BP^{-1}$, and transitive because $(PQ)^t A(PQ) = Q^t(P^tAP)Q$.

Theorem 22.1.4 says that the matrices of one form in the various bases of V form exactly one congruence class, so a property of matrices is a property of the form precisely when it is congruence-invariant. Similarity, the relation $A \mapsto P^{-1}AP$ appropriate to linear operators, is a different relation, and the invariants of a form are not those of an operator: the determinant, for instance, is not congruence-invariant, but the next definition extracts from it something that is.

Definition 22.1.6: The Square Classes of a Field

Let k be a field. The squares $(k^\times)^2 = \{c^2 \mid c \in k^\times\}$ form a subgroup of the abelian group $k^\times$, and the quotient

$$k^\times/(k^\times)^2$$

is the group of *square classes* of k. Every element has order dividing 2, so it is a vector space over $\mathbb{F}_2$. The class of $a \in k^\times$ is written $a\,(k^\times)^2$, and two elements lie in the same class exactly when their ratio is a square.

The square classes are where the classification lives, and their size measures how far the field is from having all square roots. For $k = \mathbb{C}$, or any algebraically closed field, every element is a square and the group is trivial. For $k = \mathbb{R}$ it has order 2, the classes being the positive and the negative reals. For $k = \mathbb{Q}$ it is infinite. The relevance to forms is immediate from theorem 22.1.4.

Definition 22.1.7: Discriminant of a Nondegenerate Form

Let φ be a nondegenerate symmetric bilinear form (definition 13.3.22) on a finite-dimensional k-vector space V, and let β be any basis. The *discriminant* of φ is the square class

$$\operatorname{disc}(\varphi) := \det\big([\varphi]_\beta\big)\,(k^\times)^2 \in k^\times/(k^\times)^2$$

Proposition 22.1.8: The Discriminant Is Well Defined and Is an Invariant

The class $\operatorname{disc}(\varphi)$ of definition 22.1.7 does not depend on the basis β, and congruent forms have equal discriminants.

***Proof*:**
Nondegeneracy makes $\det([\varphi]_\beta) \neq 0$ by lemma 13.3.23, so the class is taken in $k^\times$ and the definition makes sense. If β' is a second basis then $[\varphi]_{\beta'} = P^t[\varphi]_\beta P$ by theorem 22.1.4, and the multiplicativity of the determinant together with $\det(P^t) = \det(P)$ gives

$$\det\big([\varphi]_{\beta'}\big) = (\det P)^2 \det\big([\varphi]_\beta\big)$$

The two determinants differ by the square $(\det P)^2$ of the nonzero scalar $\det P$, hence lie in the same square class. The same computation, read as a statement about congruent matrices rather than about two bases, gives the second claim, as we sought to show.

22.1.9 Convention: the discriminant is unsigned, and the Witt group will twist it
Definition 22.1.7 takes the determinant with no sign attached. That is the right convention here and the wrong one for the Witt group, and the two sections genuinely differ, so the choice is recorded rather than left implicit.
At a fixed rank n the factor $(-1)^{n(n-1)/2}$ is a constant and carries no information, while multiplicativity over orthogonal sums, $\operatorname{disc}(\varphi \perp \psi) = \operatorname{disc}(\varphi)\operatorname{disc}(\psi)$, holds exactly as stated and is what the classification over a finite field uses.
The Witt group of section 22.6 identifies φ with $\varphi \perp \langle 1, -1\rangle$, and the unsigned discriminant does not survive that identification: adding $\langle 1, -1\rangle$ multiplies the determinant by -1. The *signed* discriminant $(-1)^{n(n-1)/2}\det$ does survive, since the rank rises by two and the sign changes to compensate; the price is that it is multiplicative only up to $(-1)^{mn}$. Each section therefore takes the convention its own problem needs, and section 22.6 states the twist when it introduces it.

The discriminant is the first invariant of the subject that is invisible over $\mathbb{R}$ and $\mathbb{C}$: there the square class group has at most two elements, so the discriminant carries at most one bit, whereas over a general field it is as large as $k^\times/(k^\times)^2$. The notion of sameness it is invariant under deserves its own name.

Definition 22.1.10: Isometry and the Orthogonal Group

Let (V, φ) and (V', φ') be finite-dimensional k-vector spaces equipped with symmetric bilinear forms. An *isometry* $f\colon (V, \varphi) \to (V', \varphi')$ is a linear isomorphism $f\colon V \to V'$ with

$$\varphi'\big(f(x), f(y)\big) = \varphi(x, y) \qquad \text{for all } x, y \in V$$

The two forms are *isometric*, written $\varphi \cong \varphi'$, if such an f exists. The isometries of (V, φ) to itself form a group under composition, the *orthogonal group* $O(\varphi)$ of the form.

Choosing bases turns an isometry into a congruence: if f is an isometry and β is a basis of V, then the matrix of φ' in the basis $f(\beta)$ is $[\varphi]_\beta$, so two forms on spaces of the same dimension are isometric exactly when their matrices are congruent. The classification of forms up to isometry and the classification of symmetric matrices up to congruence are therefore one problem in two languages. The chapter's opening example is a form the reader has already met.

Example 22.1.11: The Trace Form as a Quadratic Form

Let F/k be a finite separable extension with $\operatorname{char} k \neq 2$. The trace form $\langle x, y\rangle = \operatorname{tr}_{F/k}(xy)$ of definition 19.6.30 is a symmetric k-bilinear form on F, viewed as a k-vector space, and it is nondegenerate by proposition 19.6.31. Its associated quadratic form under proposition 22.1.2 is

$$q(x) = \operatorname{tr}_{F/k}(x^2)$$

so F carries a canonical quadratic form built from nothing but its field structure, and definition 22.1.7 attaches to it a square class in $k^\times/(k^\times)^2$.

Take $k = \mathbb{Q}$ and $F = \mathbb{Q}(\sqrt{d})$ with d a squarefree integer, $d \neq 1$. In the basis $\beta = (1, \sqrt{d})$ one computes $\operatorname{tr}(1) = 2$, since multiplication by 1 is the identity of a two-dimensional space. Next $\operatorname{tr}(\sqrt{d}) = 0$, since multiplication by $\sqrt{d}$ sends $1 \mapsto \sqrt{d}$ and $\sqrt{d} \mapsto d$, so its matrix in that basis has both diagonal entries zero. Finally $\operatorname{tr}(d) = 2d$. Hence

$$[\langle\,,\rangle]_\beta = \begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix}, \qquad \det = 4d$$

and since 4 is a square,

$$\operatorname{disc} = 4d\,(\mathbb{Q}^\times)^2 = d\,(\mathbb{Q}^\times)^2$$

The discriminant of the trace form recovers the square class of d, which is precisely the datum distinguishing the fields $\mathbb{Q}(\sqrt{d})$ from one another. The invariant is doing real work already: over $\mathbb{Q}$ the square class group is infinite, so this single class separates infinitely many non-isometric forms, where over $\mathbb{R}$ or $\mathbb{C}$ it could not have separated more than two.

Exercise 22.1.1

1. Let q be a quadratic form with polar form b_q. Prove the parallelogram identity $q(x+y)+q(x-y) = 2q(x) + 2q(y)$ directly from definition 22.1.1, using no hypothesis on the characteristic. Then explain why the identity carries no information when $\operatorname{char} k = 2$, and say what that has to do with example 22.1.3.

2. Exhibit two matrices in $M_2(\mathbb{Q})$ that are congruent but not similar. Then show that when the

change of basis matrix P of theorem 22.1.4 satisfies $P^t = P^{-1}$ the two relations produce the same matrix, so congruence and similarity can differ only through changes of basis that are not of this kind.

3. Let φ and ψ be nondegenerate symmetric bilinear forms on V and W, and let $\varphi \perp \psi$ denote the form on $V \oplus W$ given by $(\varphi \perp \psi)\big((v,w),(v',w')\big) = \varphi(v,v') + \psi(w,w')$. Show that $\varphi \perp \psi$ is nondegenerate and that $\operatorname{disc}(\varphi \perp \psi) = \operatorname{disc}(\varphi)\operatorname{disc}(\psi)$.
4. Let $F = \mathbb{F}_{q^2}$ and $k = \mathbb{F}_q$ with q odd. Compute the trace form of F/k in a basis of your choosing and determine its discriminant as an element of $k^\times/(k^\times)^2$. Is it the trivial class?
5. Let φ be a symmetric bilinear form on V with radical $R = \operatorname{rad}(\varphi)$ (definition 13.3.22). Show that φ descends to a well-defined form $\bar{\varphi}$ on V/R, that $\bar{\varphi}$ is nondegenerate, and that $\operatorname{disc}(\bar{\varphi})$ is therefore defined even though $\operatorname{disc}(\varphi)$ is not.

22.2 Diagonalization and Orthogonal Bases

A symmetric matrix is congruent to a diagonal one, and the diagonal entries are visible invariants only up to squares. Both halves of that sentence are proved here, and together they reduce the classification problem to a question about the square class group of the field.

Theorem 22.2.1: Every Symmetric Form Has an Orthogonal Basis

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of finite dimension n, and let φ be a symmetric bilinear form on V. Then V has a basis $w_1, \dots, w_n$ with $\varphi(w_i, w_j) = 0$ whenever $i \neq j$. Equivalently, every symmetric matrix over k is congruent to a diagonal matrix.

***Proof*:**
Induct on n. For $n = 0$ there is nothing to prove, and for $n = 1$ any basis is orthogonal because the condition $i \neq j$ is vacuous.

Step 1: either φ vanishes identically or some vector is non-isotropic. Suppose $\varphi(v,v) = 0$ for every $v \in V$. Then for all $x, y \in V$,

$$0 = \varphi(x+y, x+y) = \varphi(x,x) + 2\varphi(x,y) + \varphi(y,y) = 2\varphi(x,y)$$

and since $\operatorname{char} k \neq 2$ this forces $\varphi(x,y) = 0$, so φ is the zero form and every basis is orthogonal. This is the only place the characteristic hypothesis is used, and it cannot be dropped: the form ψ of example 22.1.3 is a nonzero symmetric form over $\mathbb{F}_2$ with $\psi(v,v) = 0$ for every v, so it falsifies the implication just proved, and that example shows it admits no orthogonal basis at all.

Otherwise choose $w_1 \in V$ with $a_1 := \varphi(w_1, w_1) \neq 0$.

Step 2: split off w_1. Let $W = \{w_1\}^\perp$, a subspace by lemma 13.3.21. Every $x \in V$ decomposes as

$$x = \frac{\varphi(w_1, x)}{a_1} w_1 + \left(x - \frac{\varphi(w_1, x)}{a_1} w_1\right)$$

and the bracketed vector lies in W, since applying $\varphi(w_1, -)$ to it gives $\varphi(w_1, x) - \frac{\varphi(w_1,x)}{a_1} a_1 = 0$. So $V = kw_1 + W$. The sum is direct: if $\lambda w_1 \in W$ then $0 = \varphi(w_1, \lambda w_1) = \lambda a_1$, and $a_1 \neq 0$ forces $\lambda = 0$. Hence $V = kw_1 \oplus W$ and $\dim W = n - 1$.

Step 3: induct. The restriction $\varphi|_{W \times W}$ is a symmetric bilinear form on W, so by the inductive hypothesis W has an orthogonal basis $w_2, \ldots, w_n$. Then $w_1, w_2, \ldots, w_n$ is a basis of V by Step 2, and it is orthogonal: $\varphi(w_i, w_j) = 0$ for $2 \leq i \neq j \leq n$ by the choice of the w_i, and $\varphi(w_1, w_j) = 0$ for $j \geq 2$ because $w_j \in W = \{w_1\}^\perp$.

For the matrix formulation, a symmetric $A \in M_n(k)$ is the matrix of a symmetric form on k^n in the standard basis, and an orthogonal basis for that form has diagonal matrix, and by theorem 22.1.4 the two matrices are congruent, completing the proof.

In an orthogonal basis the form is determined by the n scalars $a_i = \varphi(w_i, w_i)$, and the standard notation records exactly that: write

$$\varphi \cong \langle a_1, a_2, \ldots, a_n \rangle$$

for a form admitting an orthogonal basis with those diagonal entries, so that the associated quadratic form is $q(x) = a_1 x_1^2 + \cdots + a_n x_n^2$ in the corresponding coordinates. The notation is an assertion about an isometry class, not about a chosen basis, and the next proposition says how far the entries themselves are from being invariants.

Proposition 22.2.2: Rescaling the Entries of a Diagonalization

Let $\operatorname{char} k \neq 2$. For any $c_1, \ldots, c_n \in k^\times$,

$$\langle a_1, \ldots, a_n \rangle \cong \langle a_1 c_1^2, \ldots, a_n c_n^2 \rangle$$

so each diagonal entry is well defined only up to multiplication by a nonzero square. A diagonal entry that is zero stays zero, and the number of zero entries is the dimension of the radical, hence is an invariant.

***Proof*:**
Let $w_1, \ldots, w_n$ be an orthogonal basis realizing $\langle a_1, \ldots, a_n \rangle$ and put $w_i' = c_i w_i$. Each c_i is nonzero, so $w_1', \ldots, w_n'$ is again a basis, and it is orthogonal because $\varphi(w_i', w_j') = c_i c_j \varphi(w_i, w_j) = 0$ for $i \neq j$. Its diagonal entries are $\varphi(w_i', w_i') = c_i^2 a_i$, which gives the displayed isometry. Since $c_i^2 \neq 0$, the entry $c_i^2 a_i$ vanishes exactly when a_i does.

For the last claim, in an orthogonal basis a vector $x = \sum_i x_i w_i$ lies in $\operatorname{rad}(\varphi)$ if and only if $\varphi(w_j, x) = a_j x_j = 0$ for every j, that is if and only if $x_j = 0$ for every j with $a_j \neq 0$. So the radical is spanned by those w_j with $a_j = 0$, and its dimension is the number of zero entries, which is independent of the diagonalization because the radical is, completing the proof.

Corollary 22.2.3: The Discriminant from a Diagonalization

Let $\varphi \cong \langle a_1, \ldots, a_n \rangle$ be nondegenerate, so every $a_i \neq 0$. Then

$$\operatorname{disc}(\varphi) = a_1 a_2 \cdots a_n \, (k^\times)^2$$

***Proof*:**
In an orthogonal basis the matrix of φ is $\operatorname{diag}(a_1, \dots, a_n)$, whose determinant is $a_1 \cdots a_n$. Apply definition 22.1.7, whose independence of the basis is proposition 22.1.8. Consistently with proposition 22.2.2, replacing each a_i by $a_i c_i^2$ multiplies the product by $(c_1 \cdots c_n)^2$ and so does not change the class, as we sought to show.

Two fields where the square class group is completely known illustrate how much of the classification the discriminant sees. The first is the degenerate case, in the sense that the group is trivial and nothing beyond the rank survives.

Example 22.2.4: Forms over an Algebraically Closed Field
Let k be algebraically closed with $\operatorname{char} k \neq 2$. Every $a \in k^\times$ has a square root, so $(k^\times)^2 = k^\times$ and the square class group of definition 22.1.6 is trivial. Given any symmetric form, diagonalize it by theorem 22.2.1 as $\langle a_1, \dots, a_r, 0, \dots, 0\rangle$ with the $a_i \neq 0$, and apply proposition 22.2.2 with $c_i = a_i^{-1/2}$ to obtain

$$\varphi \cong \langle \underbrace{1, \dots, 1}_{r}, \underbrace{0, \dots, 0}_{n-r} \rangle$$

So a form over an algebraically closed field is determined up to isometry by n and r, and r is the rank of any matrix of φ. The discriminant of a nondegenerate such form is the class of 1 and carries no information, as it must, the group having one element.

Example 22.2.5: Forms over a Finite Field
Let $k = \mathbb{F}_q$ with q odd. The group $\mathbb{F}_q^\times$ is cyclic by theorem 11.3.14, of even order $q - 1$, so the squaring map has kernel $\{\pm 1\}$ of order 2 and image of index 2. Hence

$$\mathbb{F}_q^\times / (\mathbb{F}_q^\times)^2 \cong \mathbb{Z}/2\mathbb{Z}$$

with the two classes the nonzero squares and the non-squares. Unlike the algebraically closed case, the discriminant now carries a genuine bit, and it obstructs isometries that the rank alone would permit.

Take $q = 5$, where the squares in $\mathbb{F}_5^\times = \{1, 2, 3, 4\}$ are $1 = 1^2$ and $4 = 2^2$, so the non-squares are 2 and 3. The forms $\langle 1, 1\rangle$ and $\langle 1, 2\rangle$ on $\mathbb{F}_5^2$ are both nondegenerate of rank 2, but by corollary 22.2.3 their discriminants are the classes of 1 and of 2 respectively, which are distinct in $\mathbb{F}_5^\times/(\mathbb{F}_5^\times)^2$. By proposition 22.1.8 an isometry would force those classes to agree, so

$$\langle 1, 1\rangle \ncong \langle 1, 2\rangle$$

Rank does not classify forms over $\mathbb{F}_5$. Whether rank together with the discriminant does classify them is a genuine question, and it is answered affirmatively in section 22.6 once enough is known about which values a form represents.

Exercise 22.2.1

1. Exhibit two different orthogonal bases for the same form on $\mathbb{Q}^2$ whose diagonal entries are not related by the rescaling of proposition 22.2.2 entry by entry, and reconcile this with that proposition.

2. Let $k = \mathbb{R}$. Determine the square class group, and show that every nondegenerate form is isometric to $\langle 1, \dots, 1, -1, \dots, -1 \rangle$. Explain why the discriminant alone does not classify these, and identify the invariant that does. (This book does not prove that the counts are well defined; that is Sylvester's law of inertia, and it needs an order on the scalars.)

3. Over $k = \mathbb{Q}$, show that $\langle 1, 1 \rangle$ and $\langle -1, -1 \rangle$ have the same rank and the same discriminant and are nevertheless not isometric, by asking which nonzero rationals each represents. Conclude that rank and discriminant together do not classify forms over $\mathbb{Q}$, in contrast with example 22.2.4. Show also that $\langle 1, 1 \rangle \cong \langle 2, 2 \rangle$ over $\mathbb{Q}$, so a difference of diagonal entries is not by itself evidence of non-isometry.

4. Let $\varphi \cong \langle a_1, \dots, a_n \rangle$ be nondegenerate over k and let $c \in k^\times$ be a value represented by φ, meaning $c = q(v)$ for some $v \in V$. Show that $\varphi \cong \langle c, b_2, \dots, b_n \rangle$ for some b_i, and deduce that the set of represented values is a union of square classes.

22.3 Alternating Forms, the Linear Darboux Theorem, and Lagrangians

Section 22.1 and section 22.2 treated the symmetric forms, where the classification runs through the square classes of k and is not settled by the rank alone. The alternating forms are the second class of bilinear forms for which a classification is available, and there the answer does not depend on the field: a single integer, the codimension of the radical, determines the form up to isometry, and neither the statement nor its proof carries a hypothesis on the characteristic.

The classification is proved in the shape of a basis in which the matrix of the form is one fixed block matrix. That basis then makes visible the subspaces on which an alternating form vanishes identically. When the form is nondegenerate such a subspace has dimension at most half that of the ambient space, and the ones attaining the bound are the Lagrangian subspaces.

Definition 22.3.1: Alternating Bilinear Form

Let k be a field, with no hypothesis on its characteristic, and let V be a finite-dimensional k-vector space. A bilinear form $\varphi\colon V \times V \to k$ (definition 13.3.15) is *alternating* if

$$\varphi(v, v) = 0 \qquad \text{for every } v \in V$$

This is the vanishing condition of definition 16.1.3 read for a 2-multilinear map, and not the sign rule listed in definition 13.3.15. The two conditions differ in characteristic 2, and throughout this section *alternating* means the displayed vanishing condition.

Two alternating forms (V, φ) and (V', φ') are *isometric* if there is a linear isomorphism $f\colon V \to V'$ with $\varphi'\big(f(x), f(y)\big) = \varphi(x, y)$ for all $x, y \in V$. This is the condition of definition 22.1.10 with the hypothesis that the two forms be symmetric dropped, and it is used in that generality below.

One expansion carries the whole section. For $x, y \in V$ the vanishing condition applied to $x + y$ gives

$$0 = \varphi(x + y, x + y) = \varphi(x, x) + \varphi(x, y) + \varphi(y, x) + \varphi(y, y) = \varphi(x, y) + \varphi(y, x)$$

so every alternating form satisfies

$$\varphi(y,x) = -\varphi(x,y) \qquad \text{for all } x, y \in V \tag{22.1}$$

which is lemma 16.1.4(1) for the transposition in S_2, its hypotheses holding here because a field is a commutative ring and both V and k are k-modules. The converse needs 2 to be invertible, and that is lemma 16.1.4(2): over a field with $\operatorname{char} k \neq 2$ a bilinear form satisfying (22.1) is alternating, since setting $y = x$ there gives $2\varphi(x,x) = 0$. Over such a field, therefore, *alternating* and *skew-symmetric* name the same forms, and a form that is both symmetric and alternating is zero: symmetry and (22.1) together give $\varphi(x,y) = -\varphi(x,y)$, hence $2\varphi(x,y) = 0$, hence $\varphi(x,y) = 0$. The classification below and the one in section 22.2 therefore overlap only in the zero form. In characteristic 2 neither statement survives, and $-1 = 1$ makes (22.1) say only that φ is symmetric.

The condition is visible in the matrix of the form. Fix a basis $\beta = (\beta_1, \ldots, \beta_n)$ of V and let $A = (a_{ij}) = [\varphi]_\beta$ be the matrix of definition 13.3.18. If φ is alternating then $a_{ii} = \varphi(\beta_i, \beta_i) = 0$ and, by (22.1), $a_{ji} = -a_{ij}$. Conversely, suppose A has zero diagonal and satisfies $A^t = -A$. For $v = \sum_i x_i \beta_i$, lemma 13.3.19 gives

$$\varphi(v,v) = \sum_{i,j} x_i x_j a_{ij} = \sum_i x_i^2 a_{ii} + \sum_{i<j} x_i x_j \big(a_{ij} + a_{ji}\big) = 0$$

so φ is alternating. Both conditions on A are needed: in characteristic 2 the matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $A^t = -A$ and has a nonzero diagonal entry.

Example 22.3.2: The Determinant Form, and the Overlap in Characteristic Two

1. Let $V = k^2$ and
$$\varphi(x,y) = x_1 y_2 - x_2 y_1$$
which is the determinant of the matrix with columns x and y. It is bilinear in each column and $\varphi(x,x) = x_1 x_2 - x_2 x_1 = 0$, so it is alternating. Its matrix in the standard basis is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, of determinant 1, so φ is nondegenerate by lemma 13.3.23.

2. Over $k = \mathbb{F}_2$ the form in (1) reads $\varphi(x,y) = x_1 y_2 + x_2 y_1$, which is exactly the form ψ of example 22.1.3. That form is symmetric, nonzero and alternating at once, which is possible only because $\operatorname{char} k = 2$; the example also shows it admits no orthogonal basis, so the conclusion of theorem 22.2.1 genuinely fails for it.

3. Still over $\mathbb{F}_2$, the form $\varphi(x,y) = x_1 y_1$ is symmetric, hence satisfies (22.1) because $-1 = 1$, but $\varphi(e_1, e_1) = 1 \neq 0$. So the sign rule does not imply the vanishing condition when 2 is not invertible, and the hypothesis in lemma 16.1.4(2) cannot be dropped.

The proof of theorem 22.2.1 split off one dimension at a time, and the entry it left behind was pinned down only up to a square (proposition 22.2.2). For an alternating form there is no such entry to leave behind, since $\varphi(v,v) = 0$ for every v, and the smallest piece that carries any information is two-dimensional. On such a piece the one scalar available can be scaled to 1 outright, which is why no square classes enter the answer. The resulting normal form deserves a name before it is proved to exist.

Definition 22.3.3: Symplectic Basis and Symplectic Space

Let k be a field, with no hypothesis on its characteristic, let V be a finite-dimensional k-vector space, and let φ be an alternating bilinear form on V (definition 22.3.1). A *symplectic basis* for φ is a basis

$$e_1, f_1, e_2, f_2, \ldots, e_m, f_m$$

of V, necessarily of even cardinality $2m$, such that $\varphi(e_i, f_i) = 1$ for every i, while

$$\varphi(e_i, f_j) = 0 \quad (i \neq j), \qquad \varphi(e_i, e_j) = 0, \qquad \varphi(f_i, f_j) = 0$$

for all i and j. A pair (V, φ) with φ alternating and nondegenerate (definition 13.3.22) is called a *symplectic space*.

Example 22.3.4: The Standard Symplectic Space k^{2m}

Let k be any field and let $e_1, f_1, \ldots, e_m, f_m$ be the standard basis of k^{2m}, listed in that order. Define

$$\varphi\Big(\sum_i (x_i e_i + y_i f_i),\ \sum_i (x_i' e_i + y_i' f_i)\Big) = \sum_{i=1}^{m} (x_i y_i' - y_i x_i')$$

Each summand is the form of example 22.3.2(1) in the coordinate pair (x_i, y_i), so φ is bilinear and vanishes when its two arguments agree, hence is alternating. Evaluating on basis vectors gives $\varphi(e_i, f_i) = 1$, while $\varphi(e_i, f_j) = 0$ for $i \neq j$ and $\varphi(e_i, e_j) = \varphi(f_i, f_j) = 0$ for all i and j, because the i-th summand involves only the i-th coordinate pair. So the standard basis is symplectic. The matrix of φ in this basis is block diagonal with m blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so its determinant is 1 and φ is nondegenerate by lemma 13.3.23.

Listing the same basis as $e_1, \ldots, e_m, f_1, \ldots, f_m$ changes the matrix to $\left(\begin{smallmatrix} 0 & I_m \\ -I_m & 0 \end{smallmatrix}\right)$. The two orderings differ by a permutation matrix P, so by theorem 22.1.4 the matrices differ by $A \mapsto P^t A P$ and the determinant is unchanged, being multiplied by $(\det P)^2 = 1$. This second ordering identifies k^{2m} with $k^m \oplus k^m$ and writes the form as

$$\varphi\big((x, y), (x', y')\big) = x^t y' - y^t x' \tag{22.2}$$

for column vectors $x, y, x', y' \in k^m$, which is the shape used for computations below.

A degenerate example is obtained by padding. On k^{2m+r} with basis $e_1, f_1, \ldots, e_m, f_m, z_1, \ldots, z_r$, extend φ by declaring $\varphi(z_j, -) = \varphi(-, z_j) = 0$ for every j. The result is alternating, and its radical is $\operatorname{span}(z_1, \ldots, z_r)$. To see this, write $v = \sum_i (x_i e_i + y_i f_i) + \sum_j c_j z_j$; then $\varphi(e_i, v) = y_i$ and $\varphi(f_i, v) = -x_i$ and $\varphi(z_j, v) = 0$, and since φ is linear in its first argument, v lies in the radical (definition 13.3.22) exactly when every x_i and every y_i vanishes.

Every alternating form is one of the forms just written down. This is the classification, and it holds over every field.

Theorem 22.3.5: Linear Darboux Theorem

Let k be a field, with no hypothesis on its characteristic, let V be a k-vector space of finite dimension n, and let φ be an alternating bilinear form on V (definition 22.3.1).

1. There are integers $m, r \geq 0$ with $n = 2m + r$ and a basis

$$e_1, f_1, \ldots, e_m, f_m, z_1, \ldots, z_r$$

   of V such that $\varphi(e_i, f_i) = 1$ for every i, while every other value of φ on an ordered pair of basis vectors is 0 except $\varphi(f_i, e_i) = -1$.

2. For any such basis, $\operatorname{rad}(\varphi) = \operatorname{span}(z_1, \ldots, z_r)$. Hence $r = \dim \operatorname{rad}(\varphi)$ and $2m = n - \dim \operatorname{rad}(\varphi)$, so both integers are determined by φ and do not depend on the basis chosen in (1).

3. φ is nondegenerate if and only if $r = 0$, and in that case $n = 2m$ is even and the basis of (1) is a symplectic basis (definition 22.3.3). In particular a symplectic space has even dimension.

4. Let φ' be an alternating form on a k-vector space V' with $\dim V' = \dim V$. Then φ and φ' are isometric if and only if $\dim \operatorname{rad}(\varphi) = \dim \operatorname{rad}(\varphi')$. In particular, for each even n there is exactly one nondegenerate alternating form on an n-dimensional space up to isometry, and none at all when n is odd.

***Proof*:**

1. *Step 1: a pair e, f with $\varphi(e, f) = 1$.* Assume $\varphi \neq 0$ and choose $u, w \in V$ with $c := \varphi(u, w) \neq 0$. Put $e := u$ and $f := c^{-1}w$, so that $\varphi(e, f) = c^{-1}\varphi(u, w) = 1$, and $\varphi(f, e) = -1$ by (22.1). The two are linearly independent: if $\alpha e + \beta f = 0$ then applying $\varphi(e, -)$ gives $\alpha\varphi(e, e) + \beta\varphi(e, f) = \beta = 0$, and applying $\varphi(f, -)$ gives $\alpha\varphi(f, e) = -\alpha = 0$. Write $P = \operatorname{span}(e, f)$, a two-dimensional subspace.

   Step 2: $V = P \oplus P^{\perp}$. The set $P^{\perp}$ is a subspace and equals $\{e, f\}^{\perp}$, both by lemma 13.3.21 applied to the subset $\{e, f\}$, whose span is P. Given $v \in V$, set

$$v' := v - \varphi(e, v)\, f + \varphi(f, v)\, e$$

   Then, using $\varphi(e, e) = \varphi(f, f) = 0$, $\varphi(e, f) = 1$ and $\varphi(f, e) = -1$,

$$\varphi(e, v') = \varphi(e, v) - \varphi(e, v)\cdot 1 + \varphi(f, v)\cdot 0 = 0 \qquad \varphi(f, v') = \varphi(f, v) - \varphi(e, v)\cdot 0 + \varphi(f, v)\cdot(-1) = 0$$

   so $v' \in P^{\perp}$, and $v = \big(\varphi(e, v)f - \varphi(f, v)e\big) + v'$ exhibits v in $P + P^{\perp}$. The sum is direct: if $\alpha e + \beta f \in P^{\perp}$ then $\varphi(e, \alpha e + \beta f) = \beta = 0$ and $\varphi(f, \alpha e + \beta f) = -\alpha = 0$. Hence $V = P \oplus P^{\perp}$ and $\dim P^{\perp} = n - 2$.

   Step 3: induction on n. For $n = 0$ take the empty basis and $m = r = 0$. Let $n \geq 1$ and assume the statement for all spaces of smaller dimension. If $\varphi = 0$, any basis of V serves with $m = 0$ and $r = n$. Otherwise Steps 1 and 2 produce $V = P \oplus P^{\perp}$ with $\dim P^{\perp} = n - 2$.

The restriction of φ to $P^\perp \times P^\perp$ is bilinear and satisfies the vanishing condition, so it is alternating, and the inductive hypothesis gives a basis $e_2, f_2, \ldots, e_m, f_m, z_1, \ldots, z_r$ of $P^\perp$ with the stated table of values. Set $e_1 := e$ and $f_1 := f$. The concatenated list is a basis of V by Step 2. Its values are as claimed: within $P^\perp$ by the inductive hypothesis, on the pair (e_1, f_1) by Step 1, and for a pair with one entry in $\{e_1, f_1\}$ and the other in $P^\perp$ because $\varphi(e_1, v) = \varphi(f_1, v) = 0$ for $v \in P^\perp$ by the definition of $P^\perp$, whence also $\varphi(v, e_1) = \varphi(v, f_1) = 0$ by (22.1). Counting the list gives $n = 2m + r$.

2. Let $v = \sum_i a_i e_i + \sum_i b_i f_i + \sum_j c_j z_j$. Since φ is linear in its first argument, $v \in \operatorname{rad}(\varphi)$ holds if and only if $\varphi(x, v) = 0$ for x running over the basis. The table of values in (1) gives

$$\varphi(e_i, v) = b_i, \qquad \varphi(f_i, v) = -a_i, \qquad \varphi(z_j, v) = 0$$

for all i and j. So $v \in \operatorname{rad}(\varphi)$ if and only if every a_i and every b_i vanishes, that is if and only if $v \in \operatorname{span}(z_1, \ldots, z_r)$. Those z_j are part of a basis, hence independent, so $\dim \operatorname{rad}(\varphi) = r$, and $2m = n - r$ follows from $n = 2m + r$. The radical was defined in definition 13.3.22 without reference to a basis, so r and m are the same for every basis as in (1).

3. By definition 13.3.22, φ is nondegenerate exactly when $\operatorname{rad}(\varphi) = 0$, which by (2) happens exactly when $r = 0$. Then $n = 2m$, and the list of (1) is a basis satisfying the conditions of definition 22.3.3.

4. Suppose first that $\dim \operatorname{rad}(\varphi) = \dim \operatorname{rad}(\varphi')$. By (2) the two forms have bases as in (1) with the same m and the same r, since $\dim V = \dim V'$. Let $f\colon V \to V'$ be the linear map carrying each basis vector of V to the correspondingly named basis vector of V'. It is an isomorphism, being a bijection on bases, and the two bilinear maps $(x, y) \mapsto \varphi'(f(x), f(y))$ and $(x, y) \mapsto \varphi(x, y)$ agree on every ordered pair of basis vectors by construction, hence agree everywhere by bilinearity. So f is an isometry in the sense of definition 22.3.1.

   Conversely let f be an isometry. Its inverse is one as well: putting $x = f^{-1}(x')$ and $y = f^{-1}(y')$ in the defining identity gives $\varphi\big(f^{-1}(x'), f^{-1}(y')\big) = \varphi'(x', y')$. Now let $v \in \operatorname{rad}(\varphi)$ and let $x' \in V'$. Since f is onto, $x' = f(x)$ for some x, and then $\varphi'(x', f(v)) = \varphi(x, v) = 0$. Hence $f\big(\operatorname{rad}(\varphi)\big) \subseteq \operatorname{rad}(\varphi')$, and the same argument for f^{-1} gives $f^{-1}\big(\operatorname{rad}(\varphi')\big) \subseteq \operatorname{rad}(\varphi)$, that is $\operatorname{rad}(\varphi') \subseteq f\big(\operatorname{rad}(\varphi)\big)$. So f restricts to an isomorphism between the two radicals, and their dimensions agree.

   For the last sentence, take φ and φ' nondegenerate of the same dimension n: both radicals are zero, so they are isometric, and the standard form of example 22.3.4 gives one such φ when n is even. When n is odd, (3) forbids a nondegenerate alternating form outright, completing the proof.

The symmetric case behaves differently over the very same field. Example 22.2.5 exhibits two nondegenerate symmetric forms of rank 2 over $\mathbb{F}_5$ that are not isometric, the discriminant separating them, while part (4) makes any two nondegenerate alternating forms on a space of the same dimension isometric, over $\mathbb{F}_5$ or over any other field. No invariant of the discriminant's kind can exist for alternating forms. Indeed, if φ is nondegenerate alternating with symplectic basis β_0, then $[\varphi]_{\beta_0}$ is block diagonal with m blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ by definition 22.3.3, so $\det[\varphi]_{\beta_0} = 1$. Any other basis β gives

$[\varphi]_\beta = P^t[\varphi]_{\beta_0}P$ by theorem 22.1.4, whence

$$\det[\varphi]_\beta = (\det P)^2$$

So the determinant of every matrix of φ is a nonzero square, and the square class that definition 22.1.7 extracts in the symmetric case is trivial here for every form and separates nothing.[1]

In odd dimension the theorem is a nonexistence statement: there is no nondegenerate alternating form at all, and every alternating form on a space of odd dimension n has a radical of odd dimension $n - 2m$. Symmetric forms are unconstrained in this respect, since $\langle 1, 1, 1\rangle$ has the identity as its matrix and is nondegenerate in dimension 3 by lemma 13.3.23.

The basis of theorem 22.3.5 exhibits two subspaces of dimension m on which φ vanishes identically, namely $\operatorname{span}(e_1, \ldots, e_m)$ and $\operatorname{span}(f_1, \ldots, f_m)$. Subspaces with that property have a name, and it is given for an arbitrary bilinear form since the symmetric case needs it too.

Definition 22.3.6: Totally Isotropic Subspace

Let k be a field, with no hypothesis on its characteristic, let V be a finite-dimensional k-vector space, and let φ be a bilinear form on V. A subspace $W \subseteq V$ is *totally isotropic* for φ if

$$\varphi(x, y) = 0 \qquad \text{for all } x, y \in W$$

equivalently if $W \subseteq W^\perp$ in the notation of definition 13.3.20. The shorter term *isotropic subspace* is used below for the same condition.

Example 22.3.7: Totally Isotropic Subspaces in Two Standard Forms

1. Let φ be alternating on V over any field. Every line $W = kv$ is totally isotropic, since $\varphi(\lambda v, \mu v) = \lambda\mu\, \varphi(v, v) = 0$. So an alternating form has isotropic subspaces in abundance, whatever the field.

2. In the standard symplectic space k^{2m} of example 22.3.4, the subspaces $E = \operatorname{span}(e_1, \ldots, e_m)$ and $F = \operatorname{span}(f_1, \ldots, f_m)$ are totally isotropic, since $\varphi(e_i, e_j) = 0$ and $\varphi(f_i, f_j) = 0$ for all i, j and the condition is bilinear in the two arguments. Each has dimension m, half of $\dim k^{2m}$.

3. Let $k = \mathbb{R}$ and let $\varphi = \langle 1, 1\rangle$ on $\mathbb{R}^2$, that is $\varphi(x, y) = x_1y_1 + x_2y_2$, with associated quadratic form $q(x) = x_1^2 + x_2^2$ under proposition 22.1.2. If W is totally isotropic and $v \in W$, then $q(v) = \varphi(v, v) = 0$, and a sum of two real squares vanishes only when both do, so $v = 0$. The only totally isotropic subspace is 0. So the abundance in (1) is a feature of the alternating case and not of bilinear forms in general.

The subspaces in item (2) have exactly half the dimension of the ambient space, and item (1) produces nothing larger there. Half is in fact the maximum whenever the form is nondegenerate, for symmetric and alternating forms alike, and the bound follows from a dimension count for orthogonal

[1]The name of the theorem is by analogy with Darboux's theorem in symplectic geometry, which states that a closed nondegenerate differential 2-form on a smooth manifold admits, around each point, coordinates in which it is the constant form displayed in example 22.3.4. That theorem is about manifolds and is not proved in this book; the statement here is its pointwise linear-algebra counterpart.

complements. Degeneracy destroys the bound at once: the zero form on any V is alternating, and V itself is then totally isotropic.

Proposition 22.3.8: The Dimension Bound for Isotropic Subspaces

Let k be a field, with no hypothesis on its characteristic, let V be a k-vector space of finite dimension n, and let φ a nondegenerate bilinear form on V (definition 13.3.22).

1. For every subspace $W \subseteq V$,

$$\dim W + \dim W^{\perp} = n$$

2. If W is totally isotropic (definition 22.3.6) then $\dim W \leq \frac{1}{2}n$, with equality if and only if $W = W^{\perp}$.

3. Assume in addition that φ is alternating, so that $n = 2m$ is even by theorem 22.3.5(3). Then every totally isotropic subspace of V is contained in a totally isotropic subspace of dimension m. Such subspaces exist, and the maximal totally isotropic subspaces of V are exactly those of dimension m.

***Proof*:**

1. Let $d = \dim W$ and define $\Psi\colon V \to W^*$ by $\Psi(y)(x) = \varphi(x, y)$ for $x \in W$. Each $\Psi(y)$ is linear on W and Ψ is linear in y, both by bilinearity of φ. By definition 13.3.20, $\Psi(y) = 0$ says exactly that $y \in W^{\perp}$, so $\ker \Psi = W^{\perp}$.

   Ψ is surjective. First, the map $\Theta\colon V \to V^*$ with $\Theta(y) = \varphi(-, y)$ has kernel $\operatorname{rad}(\varphi)$, which is 0 by definition 13.3.22 and the nondegeneracy hypothesis. The dual basis of proposition 13.3.5 is a basis of V^*, so $\dim V^* = \dim V$, while corollary 13.1.18 gives $\dim V = \dim \ker \Theta + \dim \Theta(V) = \dim \Theta(V)$. Hence the image $\Theta(V)$ is a subspace of V^* of the full dimension, so $\Theta(V) = V^*$. Second, every $g \in W^*$ extends to V: choose a basis $w_1, \ldots, w_d$ of W, extend it to a basis $w_1, \ldots, w_n$ of V, and let $\tilde{g} \in V^*$ be the linear map with $\tilde{g}(w_i) = g(w_i)$ for $i \leq d$ and $\tilde{g}(w_i) = 0$ for $i > d$; then $\tilde{g}$ and g are linear maps on W agreeing on a basis of W, so $\tilde{g}|_W = g$. Combining, given $g \in W^*$ pick y with $\Theta(y) = \tilde{g}$; then $\Psi(y) = \tilde{g}|_W = g$.

   Now corollary 13.1.18 applied to Ψ gives $n = \dim \ker \Psi + \dim \Psi(V) = \dim W^{\perp} + \dim W^*$, and $\dim W^* = \dim W$ again by proposition 13.3.5, this time for the space W.

2. If W is totally isotropic then $W \subseteq W^{\perp}$ by definition 22.3.6, so $\dim W \leq \dim W^{\perp} = n - \dim W$ by (1), which rearranges to $2 \dim W \leq n$. Equality holds exactly when $\dim W = \dim W^{\perp}$, and a subspace contained in another of the same finite dimension equals it, so equality is equivalent to $W = W^{\perp}$.

3. Let W be totally isotropic with $d = \dim W < m$. By (1), $\dim W^{\perp} = 2m - d > d = \dim W$, and $W \subseteq W^{\perp}$, so there is a vector $v \in W^{\perp}$ with $v \notin W$. Put $W' = W + kv$, of dimension $d + 1$. It is totally isotropic: for $w, w' \in W$ and $\lambda, \mu \in k$,

$$\varphi(w + \lambda v,\ w' + \mu v) = \varphi(w, w') + \mu\, \varphi(w, v) + \lambda\, \varphi(v, w') + \lambda\mu\, \varphi(v, v)$$

and the four terms vanish in turn because W is totally isotropic, because $v \in W^{\perp}$, because $\varphi(v, w') = -\varphi(w', v) = 0$ by (22.1) and $v \in W^{\perp}$, and because φ is alternating. Repeating the step $m - d$ times produces a totally isotropic subspace of dimension m containing W.

Existence follows by applying this to $W = 0$, or directly from theorem 22.3.5(3): the span of $e_1, \dots, e_m$ in a symplectic basis is totally isotropic of dimension m. Finally, a totally isotropic subspace of dimension m is maximal among totally isotropic subspaces, since any subspace properly containing it has dimension exceeding the bound in (2); and one of dimension $d < m$ is not maximal, since the construction above properly enlarges it. So maximality is equivalent to having dimension m, as we sought to show.

Part (3) fails for symmetric forms: example 22.3.7(3) gives a nondegenerate symmetric form of rank 2 over $\mathbb{R}$ whose only totally isotropic subspace is 0, so there the maximal one has dimension 0 rather than 1. The alternating case is uniform because $\varphi(v, v) = 0$ holds for every v by definition: enlarging an isotropic W needs a vector of $W^{\perp}$ outside W on which the form vanishes, and the vanishing is automatic, so the dimension count in (1) gives the vector by itself. The subspaces attaining the bound carry a name.

Definition 22.3.9: Lagrangian Subspace

Let k be a field, with no hypothesis on its characteristic, and let (V, φ) be a symplectic space of dimension $2m$, so φ is alternating and nondegenerate (definition 22.3.3). A *Lagrangian subspace* of (V, φ) is a totally isotropic subspace $L \subseteq V$ with $\dim L = m$.

By proposition 22.3.8(2) this is the same as requiring $L = L^{\perp}$, and by proposition 22.3.8(3) it is the same as requiring L to be maximal among the totally isotropic subspaces of V. That same part guarantees Lagrangian subspaces exist in every symplectic space.

Example 22.3.10: Lagrangian Subspaces of the Standard Symplectic Space

Work in $k^{2m} = k^m \oplus k^m$ with the form (22.2), $\varphi\big((x, y), (x', y')\big) = x^t y' - y^t x'$.

1. For $S \in M_m(k)$ let $L_S = \{(x, Sx) \mid x \in k^m\}$, the graph of S. It is a subspace of dimension m, the map $x \mapsto (x, Sx)$ being linear and injective. For $x, x' \in k^m$,

$$\varphi\big((x, Sx), (x', Sx')\big) = x^t S x' - (Sx)^t x' = x^t S x' - x^t S^t x' = x^t\big(S - S^t\big)x'$$

   Taking x and x' to be the i-th and j-th standard basis vectors of k^m turns the right-hand side into the (i, j) entry of $S - S^t$, so the expression vanishes for all x, x' if and only if $S = S^t$. Hence L_S is Lagrangian exactly when S is symmetric. Taking $S = 0$ recovers $E = \operatorname{span}(e_1, \dots, e_m)$.

2. Conversely, every Lagrangian L with $L \cap F = 0$, where $F = \{(0, y) \mid y \in k^m\}$, is of this form. The projection $\pi(x, y) = x$ restricts to a linear map $L \to k^m$ with kernel $L \cap F = 0$, hence injective, and $\dim L = m$ forces it to be bijective by corollary 13.1.18. Let σ be its inverse and write $\sigma(x) = (x, Sx)$, where S is linear because σ is. Then $L = L_S$, and S is symmetric by (1).

3. For $m = 1$ every line in k^2 is Lagrangian: a line is totally isotropic by example 22.3.7(1) and has dimension $1 = m$. Over $k = \mathbb{F}_q$ the lines of k^2 number $(q^2 - 1)/(q - 1) = q + 1$, so

there are $q+1$ Lagrangians. For $m \geq 2$ having dimension m is not sufficient: inside k^4 the subspace $\operatorname{span}(e_1, f_1)$ has dimension $2 = m$ and is not totally isotropic, since $\varphi(e_1, f_1) = 1$.

Items (1) and (2) together put the Lagrangians meeting F trivially in bijection with the symmetric $m \times m$ matrices, the assignment $S \mapsto L_S$ being injective because L_S determines Sx as the second component of its element with first component x. The symmetric matrices form a k-vector space of dimension $\binom{m+1}{2}$. The two counts agree where they overlap: for $m = 1$ every 1×1 matrix is symmetric, so over $\mathbb{F}_q$ that bijection accounts for q Lagrangians, and the one remaining line, F itself, brings the total to the $q+1$ of item (3).

Exercise 22.3.1

1. Over $k = \mathbb{F}_2$, show that every symmetric bilinear form on k^2 satisfies the sign rule $\varphi(x, y) = -\varphi(y, x)$. Then determine which of the eight symmetric bilinear forms on $\mathbb{F}_2^2$ are alternating in the sense of definition 22.3.1, and say how the count is consistent with lemma 16.1.4(2).

2. Let $\operatorname{char} k \neq 2$ and let φ be a nondegenerate alternating form on a k-vector space of dimension n, with matrix A in some basis. Using $\det(A^t) = \det(A)$ and $A^t = -A$, show directly that n is even. Then explain which step of your argument fails over $\mathbb{F}_2$, and why the conclusion nevertheless still holds there.

3. Let k be any field. Show that a bilinear form φ on k^n is alternating if and only if there are scalars a_{ij} for $1 \leq i < j \leq n$ with
$$\varphi(x, y) = \sum_{i<j} a_{ij}\big(x_i y_j - x_j y_i\big)$$
and that the a_{ij} are then determined by φ. Deduce that over $k = \mathbb{F}_q$ there are exactly $q^{n(n-1)/2}$ alternating forms on k^n.

4. Let $\operatorname{char} k \neq 2$. Show that every bilinear form φ on V is uniquely $\varphi = s + a$ with s symmetric and a alternating, and identify s and a in terms of φ and $(x, y) \mapsto \varphi(y, x)$. Then show the statement fails over $\mathbb{F}_2$ by exhibiting a bilinear form on $\mathbb{F}_2^2$ that is not such a sum.

5. Let (V, φ) be a symplectic space of dimension $2m$ and let L be a Lagrangian subspace with basis $e_1, \ldots, e_m$. Show that there exist $f_1, \ldots, f_m \in V$ making $e_1, f_1, \ldots, e_m, f_m$ a symplectic basis of (V, φ). (Suggestion: for each i apply proposition 22.3.8(1) to the span of the e_j with $j \neq i$ to produce u_i with $\varphi(e_j, u_i) = 1$ when $i = j$ and 0 otherwise, then replace u_i by $u_i + \sum_j c_{ij} e_j$ for a suitable choice of scalars, noting that this leaves the previous values unchanged.) Deduce that for any two Lagrangian subspaces L and L' of (V, φ) there is an isometry of (V, φ) carrying L onto L'.

6. Let φ be an alternating form on V with radical $R = \operatorname{rad}(\varphi)$. Show that $\bar{\varphi}(x+R, y+R) := \varphi(x, y)$ is a well-defined alternating form on V/R, that it is nondegenerate, and that $\dim V/R$ is even. Compare the resulting statement with theorem 22.3.5(2).

7. Let (V, φ) be a symplectic space of dimension $2m$. Show that there is a chain of subspaces $0 = W_0 \subset W_1 \subset \cdots \subset W_m$ with $\dim W_i = i$ and each W_i totally isotropic, and that W_m is then a Lagrangian subspace. Show also that no such chain can be continued to a totally isotropic W_{m+1}.

22.4 Isotropy, the Hyperbolic Plane, and Witt Decomposition

A diagonalization records the values a form takes on n chosen vectors and says nothing about the vectors the form kills. This section organizes a symmetric form around those vectors instead, the v with $\varphi(v,v)=0$. Two cases are extreme. A form may have no such vector other than 0; or it may be spanned by two of them paired against each other, and then it is a single two-dimensional form which is the same over every field. The theorem of this section splits an arbitrary symmetric form into pieces of exactly those two kinds together with its radical, and so isolates the one summand in which the arithmetic of k is still visible.

The number of two-dimensional pieces and the isometry class of the remaining piece are determined by the form, but the argument for that is Witt cancellation and is deferred to section 22.5. What is available here without it is a statement about the subspaces on which the form vanishes identically: the largest dimension such a subspace can have is an invariant by construction, and it counts the two-dimensional pieces in a decomposition built from a subspace attaining it.

Every statement below breaks V into mutually orthogonal subspaces, and the notation for that is fixed first. It has already been used in box 22.1.9.

Definition 22.4.1: Orthogonal Sum of Forms

Let k be a field.

1. Let φ and ψ be bilinear forms on finite-dimensional k-vector spaces V and W. Their *orthogonal sum* is the bilinear form $\varphi \perp \psi$ on $V \oplus W$ given by

   $$(\varphi \perp \psi)\big((v,w),(v',w')\big) := \varphi(v,v') + \psi(w,w')$$

   It is symmetric when φ and ψ are.

2. Let φ be a bilinear form on V and let $U_1, U_2 \subseteq V$ be subspaces with $V = U_1 \oplus U_2$ and $\varphi(u_1,u_2)=0$ for all $u_1 \in U_1$ and $u_2 \in U_2$. Then V is the *internal orthogonal sum* of U_1 and U_2, written $V = U_1 \perp U_2$.

Both constructions iterate to finitely many summands.

The two halves of the definition describe one situation from outside and from inside. If $V = U_1 \perp U_2$ internally, then $(u_1,u_2) \mapsto u_1+u_2$ is an isometry from $\varphi|_{U_1} \perp \varphi|_{U_2}$ to φ in the sense of definition 22.1.10, because expanding $\varphi(u_1+u_2,\, u_1'+u_2')$ by bilinearity leaves only $\varphi(u_1,u_1')$ and $\varphi(u_2,u_2')$, the two cross terms being zero by hypothesis. Concatenating a basis β of U_1 with a basis γ of U_2 gives a basis of V in which every entry $\varphi(\beta_i,\gamma_j)$ vanishes, so by definition 13.3.18 the matrix of φ is block diagonal with blocks $[\varphi|_{U_1}]_\beta$ and $[\varphi|_{U_2}]_\gamma$. The determinant of an orthogonal sum is therefore the product of the determinants of its summands, and by lemma 13.3.23 the sum is nondegenerate exactly when both summands are. The diagonal notation of section 22.2 is an orthogonal sum of lines,

$$\langle a_1,\dots,a_n\rangle = \langle a_1\rangle \perp \cdots \perp \langle a_n\rangle$$

so theorem 22.2.1 says exactly that every symmetric form is an orthogonal sum of one-dimensional forms.

Definition 22.4.2: Isotropic and Anisotropic Vectors

Let k be a field, let V be a finite-dimensional k-vector space, let φ be a symmetric bilinear form on V, and write $q(v) := \varphi(v, v)$ for the associated quadratic form. A nonzero vector $v \in V$ is *isotropic* if $q(v) = 0$, and *anisotropic* if $q(v) \neq 0$. The form φ is *isotropic* if V contains an isotropic vector, and *anisotropic* otherwise, that is if

$$q(v) = 0 \implies v = 0$$

The zero vector is excluded deliberately: $q(0) = 0$ for every form, so admitting it would make every form isotropic. An anisotropic form is automatically nondegenerate, since a nonzero $v \in \mathrm{rad}(\varphi)$ (definition 13.3.22) satisfies $\varphi(v, x) = 0$ for every x and in particular $q(v) = 0$, which is an isotropic vector. Restricting an anisotropic form to a subspace leaves it anisotropic, since the defining implication is inherited by subsets, and hence leaves it nondegenerate.

A subspace on which φ vanishes identically, $\varphi(w, w') = 0$ for all $w, w' \in W$, is *totally isotropic* (definition 22.3.6). When $\mathrm{char}\, k \neq 2$ it suffices to check the quadratic form on W: if $q(w) = 0$ for every $w \in W$, then for $w, w' \in W$,

$$0 = q(w + w') = q(w) + 2\varphi(w, w') + q(w') = 2\varphi(w, w')$$

and dividing by 2 gives $\varphi(w, w') = 0$. A totally isotropic subspace is thus one all of whose nonzero vectors are isotropic.

Example 22.4.3: Isotropy of Diagonal Forms

Whether a form is isotropic depends on the field and not only on the diagonal entries.

1. $\langle 1, 1\rangle$ over $\mathbb{R}$ is anisotropic: $x^2 + y^2 = 0$ forces $x^2 = -y^2$, and a sum of two nonnegative reals vanishes only when both do.

2. $\langle 1, 1\rangle$ over $\mathbb{F}_5$ is isotropic. The vector $(2, 1)$ is nonzero and $2^2 + 1^2 = 5 = 0$ in $\mathbb{F}_5$. Over $\mathbb{F}_3$ the squares are 0 and 1, so $x^2 + y^2$ takes only the values $0, 1, 2$ and vanishes only at $x = y = 0$: the same diagonal entries give an anisotropic form.

3. $\langle 1, -1\rangle$ is isotropic over every field, since $q(1, 1) = 1 - 1 = 0$ and $(1, 1) \neq 0$.

4. Let d be a squarefree integer with $d \neq 1$ and let $F = \mathbb{Q}(\sqrt{d})$. By example 22.1.11 the trace form of $F/\mathbb{Q}$ is $\langle 2, 2d\rangle$ in the basis $(1, \sqrt{d})$, with quadratic form $q(x, y) = 2x^2 + 2dy^2$. A zero with $y = 0$ forces $x = 0$, so an isotropic vector has $y \neq 0$, and dividing $x^2 = -dy^2$ by y^2 gives $-d = (x/y)^2$. The trace form is therefore isotropic exactly when $-d$ is a square in $\mathbb{Q}$. A positive squarefree integer is a rational square only when it equals 1, so this happens for $d = -1$ alone: of all the quadratic fields, only $\mathbb{Q}(i)$ has isotropic trace form, and $q(1, 1) = 2 - 2 = 0$ exhibits the vector.

Item (3) is the smallest isotropic form, and it is worth presenting in coordinates that exhibit its two isotropic directions at once. In place of one distinguished isotropic vector it carries two, paired against each other, and that pairing is what keeps the form nondegenerate.

Definition 22.4.4: The Hyperbolic Plane

Let k be a field and let φ be a symmetric bilinear form on a k-vector space U. A *hyperbolic pair* in U is a pair of vectors $e, f \in U$ with

$$\varphi(e,e) = \varphi(f,f) = 0, \qquad \varphi(e,f) = 1$$

The form φ is a *hyperbolic plane* if $\dim U = 2$ and U has a basis which is a hyperbolic pair, so that the matrix of φ in that basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. A hyperbolic plane is written H. A subspace of (V, φ) is *a hyperbolic plane in* V when the restriction of φ to it is one, and the orthogonal sum of m copies,

$$mH := \underbrace{H \perp \cdots \perp H}_{m}$$

is the *hyperbolic space* of rank $2m$, with $0H := 0$.

Writing H for *the* hyperbolic plane is legitimate because any two are isometric: if (e, f) and (e', f') are hyperbolic pairs spanning U and U', then the linear isomorphism carrying $e \mapsto e'$ and $f \mapsto f'$ gives the two forms the same matrix in the corresponding bases, and by lemma 13.3.19 a form is recovered from its matrix in a basis, so the isomorphism is an isometry. That matrix has determinant $-1 \neq 0$, so a hyperbolic plane is nondegenerate by lemma 13.3.23, and its discriminant (definition 22.1.7) is the square class of -1.

Example 22.4.5: The Hyperbolic Plane as a Diagonal Form

Let $\operatorname{char} k \neq 2$ and let H have hyperbolic pair (e, f). Put

$$x := e + \tfrac{1}{2} f, \qquad y := e - \tfrac{1}{2} f$$

Then $x + y = 2e$ and $x - y = f$, and since $2 \neq 0$ these recover e and f, so (x, y) is again a basis of H. Expanding by bilinearity and using $\varphi(e,e) = \varphi(f,f) = 0$ together with $\varphi(e,f) = 1$,

$$q(x) = 2 \cdot \tfrac{1}{2}\varphi(e,f) = 1, \qquad q(y) = -2 \cdot \tfrac{1}{2}\varphi(e,f) = -1$$

$$\varphi(x,y) = -\tfrac{1}{2}\varphi(e,f) + \tfrac{1}{2}\varphi(f,e) = 0$$

so (x, y) is an orthogonal basis and

$$H \cong \langle 1, -1 \rangle$$

The product of the diagonal entries is -1, which agrees through corollary 22.2.3 with the determinant computed above. Over $\mathbb{R}$ this is the indefinite plane $x^2 - y^2$. Over a field containing an i with $i^2 = -1$, proposition 22.2.2 applied with $c_1 = 1$ and $c_2 = i$ gives $\langle 1, -1 \rangle \cong \langle 1, 1 \rangle$; combined with example 22.2.4, every nondegenerate binary form over an algebraically closed field of characteristic other than 2 is a hyperbolic plane.

The hyperbolic plane was assembled from a pair of isotropic vectors. The next result runs the construction in reverse: a single isotropic vector of a nondegenerate form always has a partner, and the plane the two span splits off as an orthogonal summand. The splitting is what makes the construction iterable, and iterating it is the proof of the decomposition theorem.

Proposition 22.4.6: Every Isotropic Vector Lies in a Hyperbolic Plane

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let φ be a nondegenerate symmetric bilinear form on V. Let $v \in V$ be isotropic. Then there is an $f \in V$ such that (v, f) is a hyperbolic pair, and, writing $H_v := \operatorname{span}(v, f)$,

$$V = H_v \perp H_v^{\perp}$$

where H_v is a hyperbolic plane in V and $\varphi|_{H_v^{\perp}}$ is nondegenerate. In particular a nondegenerate symmetric form is anisotropic if and only if V contains no hyperbolic plane.

***Proof*:**
Step 1: producing the partner. An isotropic vector is nonzero, so $v \notin \operatorname{rad}(\varphi) = 0$ and there is a $w_0 \in V$ with $\varphi(v, w_0) \neq 0$. Replacing w_0 by $w := \varphi(v, w_0)^{-1} w_0$ gives $\varphi(v, w) = 1$. Set

$$f := w - \frac{q(w)}{2}\, v$$

which uses $\operatorname{char} k \neq 2$ to divide. Using $q(v) = \varphi(v, v) = 0$,

$$\varphi(v, f) = \varphi(v, w) - \frac{q(w)}{2}\varphi(v, v) = 1$$

and, expanding by bilinearity and symmetry,

$$\varphi(f, f) = \varphi(w, w) - 2 \cdot \frac{q(w)}{2}\varphi(w, v) + \frac{q(w)^2}{4}\varphi(v, v) = q(w) - q(w) = 0$$

so (v, f) is a hyperbolic pair. The two vectors are linearly independent: a relation $f = \lambda v$ would give $1 = \varphi(v, f) = \lambda\varphi(v, v) = 0$, and $v \neq 0$. Hence $\dim H_v = 2$, the matrix of $\varphi|_{H_v}$ in the basis (v, f) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and H_v is a hyperbolic plane in V, nondegenerate by lemma 13.3.23.

Step 2: the splitting. Let $x \in V$ and put

$$x' := x - \varphi(x, f)\, v - \varphi(x, v)\, f$$

Using the values of φ on the pair (v, f),

$$\varphi(x', v) = \varphi(x, v) - \varphi(x, f)\varphi(v, v) - \varphi(x, v)\varphi(f, v) = \varphi(x, v) - \varphi(x, v) = 0$$

$$\varphi(x', f) = \varphi(x, f) - \varphi(x, f)\varphi(v, f) - \varphi(x, v)\varphi(f, f) = \varphi(x, f) - \varphi(x, f) = 0$$

so x' is orthogonal to v and to f, hence to their span by lemma 13.3.21, that is $x' \in H_v^{\perp}$. Since $x - x' \in H_v$, this gives $V = H_v + H_v^{\perp}$. The sum is direct: an element $u \in H_v \cap H_v^{\perp}$ lies in H_v and is orthogonal to every element of H_v, hence lies in $\operatorname{rad}(\varphi|_{H_v})$, which is zero by Step 1. The two summands are orthogonal by the definition of $H_v^{\perp}$ (definition 13.3.20), so $V = H_v \perp H_v^{\perp}$ in the sense of definition 22.4.1.

Step 3: the complement is nondegenerate. Let $x \in \operatorname{rad}\big(\varphi|_{H_v^{\perp}}\big)$, so $\varphi(x, y) = 0$ for every $y \in H_v^{\perp}$. Since $x \in H_v^{\perp}$ and φ is symmetric, $\varphi(x, y) = 0$ for every $y \in H_v$ as well. By Step 2 each element of V is a sum of one vector of each kind, so $\varphi(x, -)$ vanishes on V and $x \in \operatorname{rad}(\varphi) = 0$.

Step 4: the equivalence. If φ is isotropic, Steps 1 to 3 produce a hyperbolic plane in V. Conversely, a hyperbolic plane in V contains a vector e with $\varphi(e,e) = 0$ and $e \neq 0$, the latter because $\varphi(e,f) = 1$ forbids $e = 0$, so φ is isotropic. Hence V contains a hyperbolic plane precisely when φ is isotropic, completing the proof.

The proposition converts one-dimensional information, the existence of a single vector with $q(v) = 0$, into a two-dimensional summand of known isometry type. What is left of φ is the restriction $\varphi|_{H_v^\perp}$, again nondegenerate and of dimension two smaller, so the exchange can be repeated until no isotropic vector remains. In dimension two there is nothing left to repeat, and the statement becomes a criterion.

Example 22.4.7: Binary Forms and Hyperbolic Planes

Let $\operatorname{char} k \neq 2$ and let $\varphi \cong \langle a, b\rangle$ be a nondegenerate form of rank 2, so that $a, b \in k^\times$ by proposition 22.2.2. If φ is isotropic then proposition 22.4.6 puts a hyperbolic plane H_v inside V, and $\dim H_v = 2 = \dim V$ forces $H_v = V$, so $\varphi \cong H$. A nondegenerate binary form is therefore either anisotropic or a hyperbolic plane.

The entries determine which of the two occurs. An isotropic vector for $q(x, y) = ax^2 + by^2$ has both coordinates nonzero, since $y = 0$ forces $ax^2 = 0$ and hence $x = 0$, and symmetrically $x = 0$ forces $y = 0$. Multiplying $ax^2 = -by^2$ by b and dividing by x^2 gives

$$-ab = \left(\frac{by}{x}\right)^2$$

Conversely, if $-ab = c^2$ then $c \neq 0$ because $ab \neq 0$, and the nonzero vector (b, c) satisfies $ab^2 + bc^2 = ab^2 - ab \cdot b = 0$. So $\langle a, b\rangle$ is a hyperbolic plane exactly when $-ab$ is a square, which by corollary 22.2.3 says exactly that $\operatorname{disc}(\varphi)$ is the class of -1. For instance $\langle 1, 1\rangle$ over $\mathbb{F}_5$ has $-ab = -1 = 4 = 2^2$, so it is the hyperbolic plane, whereas over $\mathbb{F}_3$ the element $-1 = 2$ is not a square and the form is anisotropic, matching example 22.4.3.

Iterating proposition 22.4.6 splits a hyperbolic plane off a nondegenerate form as long as one isotropic vector can be found, and the process stops only when none can. A degenerate form has to be handled first: the nonzero vectors of its radical are isotropic and orthogonal to all of V, so no partner exists for them and proposition 22.4.6 does not apply. Splitting the radical off and then iterating gives the decomposition.

Theorem 22.4.8: Witt Decomposition

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. Then there are an integer $m \geq 0$, subspaces $H_1, \ldots, H_m \subseteq V$ each of which is a hyperbolic plane in V, and a subspace $V_{\mathrm{an}} \subseteq V$ on which φ is anisotropic, such that

$$V = \operatorname{rad}(\varphi) \perp H_1 \perp \cdots \perp H_m \perp V_{\mathrm{an}}$$

Writing $r = \dim \operatorname{rad}(\varphi)$, this reads

$$\varphi \cong \langle \underbrace{0, \ldots, 0}_{r} \rangle \perp mH \perp \varphi|_{V_{\mathrm{an}}}$$

***Proof*:**
Step 1: splitting off the radical. Write $R := \operatorname{rad}(\varphi)$ and choose a basis of R. By theorem 13.1.8 it extends to a basis of V; let V' be the span of the added vectors, so that $V = R \oplus V'$. Every element of R is orthogonal to every element of V, in particular to every element of V', so $V = R \perp V'$ in the sense of definition 22.4.1, and $\varphi|_R = 0$.

The restricted form $\varphi|_{V'}$ is nondegenerate. Let $x \in V'$ satisfy $\varphi(x, y) = 0$ for all $y \in V'$. Since also $\varphi(x, y) = 0$ for all $y \in R$, and $V = R + V'$, the map $\varphi(x, -)$ vanishes on V, so $x \in R \cap V' = 0$.

Step 2: splitting hyperbolic planes off a nondegenerate form. It suffices to prove that a nondegenerate symmetric form ψ on a space U decomposes as $U = H_1 \perp \cdots \perp H_m \perp U_{\text{an}}$ with each H_i a hyperbolic plane in U and $\psi|_{U_{\text{an}}}$ anisotropic; applying this to $\varphi|_{V'}$ and appending Step 1 gives the theorem, because every summand of V' is orthogonal to R.

Induct on $\dim U$. If ψ is anisotropic, take $m = 0$ and $U_{\text{an}} = U$. This covers $\dim U = 0$, where there is nothing to decompose, and $\dim U = 1$: an isotropic vector v would span U, and then $\psi(v, \lambda v) = \lambda\psi(v, v) = 0$ for every λ would put v in $\operatorname{rad}(\psi)$, contradicting nondegeneracy. Otherwise ψ has an isotropic vector, and proposition 22.4.6 gives a hyperbolic plane $H_1 \subseteq U$ with

$$U = H_1 \perp H_1^{\perp}$$

and $\psi|_{H_1^{\perp}}$ nondegenerate. Since $\dim H_1^{\perp} = \dim U - 2$, the inductive hypothesis applies to $H_1^{\perp}$ and gives $H_1^{\perp} = H_2 \perp \cdots \perp H_m \perp U_{\text{an}}$ with each H_i a hyperbolic plane and $\psi|_{U_{\text{an}}}$ anisotropic. Every one of these subspaces lies in $H_1^{\perp}$ and is therefore orthogonal to H_1, so the displayed decomposition of U refines to $U = H_1 \perp H_2 \perp \cdots \perp H_m \perp U_{\text{an}}$.

Step 3: the diagonal reading. Take the chosen basis of R, a hyperbolic pair in each H_i, and an orthogonal basis of V_{an}, which exists by theorem 22.2.1. Their concatenation is a basis of V, and by the block-diagonal description following definition 22.4.1 the matrix of φ in it is the block sum of the zero matrix of size r, m copies of $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and a diagonal matrix. That is the asserted isometry, completing the proof.

The three summands answer three different questions about φ. The radical is where the form carries no information at all, and its dimension is already known to be an invariant by proposition 22.2.2. The hyperbolic part carries no information beyond its rank, since any two hyperbolic planes over k are isometric, whatever k is. Everything that distinguishes φ from other forms of the same rank is therefore concentrated in $\varphi|_{V_{\text{an}}}$, and how large that summand can be is a property of the field. Over an algebraically closed field it has dimension at most 1: a two-dimensional subspace of an anisotropic space is nondegenerate, hence a hyperbolic plane by example 22.4.5, hence isotropic, which contradicts anisotropy. This recovers example 22.2.4, where the classification collapsed to the rank. Over a finite field the bound is one step weaker, and a counting argument gives it.

Example 22.4.9: Isotropy over a Finite Field
Let $k = \mathbb{F}_q$ with q odd.

Every equation $ax^2 + by^2 = c$ with $a, b \in k^\times$ and $c \in k$ has a solution. As recorded in example 22.2.5, the squaring map on $\mathbb{F}_q^\times$ has kernel $\{\pm 1\}$, which has two elements because q is odd, so there are $(q-1)/2$ nonzero squares and the set $\left\{x^2 \;\middle|\; x \in k\right\}$ has $(q+1)/2$ elements. Multiplication by a is a bijection of k, so

$$S := \{ax^2 \mid x \in k\}, \qquad T := \{c - by^2 \mid y \in k\}$$

each have $(q+1)/2$ elements. Their sizes add to $q+1 > q = |k|$, so S and T are not disjoint, and a common element gives x, y with $ax^2 = c - by^2$.

Every nondegenerate form of rank at least 3 *over* $\mathbb{F}_q$ *is isotropic.* Diagonalize by theorem 22.2.1 as $\varphi \cong \langle a_1, \dots, a_n \rangle$ in an orthogonal basis $w_1, \dots, w_n$; nondegeneracy makes every a_i nonzero by the last claim of proposition 22.2.2. With $n \geq 3$, solve $a_1x^2 + a_2y^2 = -a_3$ by the previous paragraph and set $v := xw_1 + yw_2 + w_3$, which is nonzero because its third coordinate is 1. Then $q(v) = a_1x^2 + a_2y^2 + a_3 = 0$.

Consequently, in the Witt decomposition of a form over $\mathbb{F}_q$ the anisotropic summand has dimension at most 2. Both values occur: $\langle 1 \rangle$ is anisotropic of rank 1, and if $\epsilon \in \mathbb{F}_q^\times$ is a non-square then $\langle 1, -\epsilon \rangle$ has $-ab = \epsilon$ a non-square, so it is anisotropic of rank 2 by example 22.4.7.

Theorem 22.4.8 produces some number m of hyperbolic planes, and nothing in its proof says that a different sequence of choices produces the same number. A quantity that is visibly independent of all choices is available instead, and the rest of the section relates the two.

Definition 22.4.10: The Witt Index

Let k be a field, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. The *Witt index* of φ is

$$\nu(\varphi) := \max\{\dim W \mid W \subseteq V \text{ a totally isotropic subspace}\}$$

The maximum exists: the zero subspace is totally isotropic, so the set of dimensions is nonempty, and it is bounded above by $\dim V$.

The Witt index is an invariant of φ by construction, since it is defined without reference to a basis or a decomposition, and isometries carry totally isotropic subspaces to totally isotropic subspaces of the same dimension. It is zero exactly when φ is anisotropic, since a one-dimensional totally isotropic subspace is the span of an isotropic vector. The next result computes it against theorem 22.4.8.

Proposition 22.4.11: The Witt Index Counts Hyperbolic Planes

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space of dimension n and let φ be a nondegenerate symmetric bilinear form on V.

1. Let $W \subseteq V$ be totally isotropic with $\dim W = d$. Then there are hyperbolic planes $H_1, \ldots, H_d$ in V, with hyperbolic pairs (e_i, f_i) such that $e_1, \ldots, e_d$ is a basis of W, and a subspace U with $\varphi|_U$ nondegenerate, such that

$$V = H_1 \perp \cdots \perp H_d \perp U$$

   In particular $2d \leq n$, which is also proposition 22.3.8(2) applied to the totally isotropic W.

2. In the situation of (1), the form $\varphi|_U$ is anisotropic if and only if W is maximal among the totally isotropic subspaces of V.

3. $\nu(\varphi) \leq n/2$; some decomposition as in theorem 22.4.8 has exactly $\nu(\varphi)$ hyperbolic planes; and every such decomposition has at most $\nu(\varphi)$ of them.

***Proof*:**

1. Induct on d. For $d = 0$ take $U = V$, and there is nothing to prove. Let $d \geq 1$ and choose a basis $w_1, \ldots, w_d$ of W. Since W is totally isotropic and $w_1 \neq 0$, the vector w_1 is isotropic, so proposition 22.4.6 gives an f_1 making (w_1, f_1) a hyperbolic pair, with $H_1 = \operatorname{span}(w_1, f_1)$ a hyperbolic plane, $V = H_1 \perp H_1^\perp$ and $\varphi|_{H_1^\perp}$ nondegenerate.

   Consider the linear map $\lambda\colon W \to k$ defined by $\lambda(x) = \varphi(x, f_1)$. It is nonzero, since $\lambda(w_1) = \varphi(w_1, f_1) = 1$, so $\ker \lambda$ has dimension $d - 1$. Moreover $\ker \lambda = W \cap H_1^\perp$: every $x \in W$ satisfies $\varphi(x, w_1) = 0$ because W is totally isotropic and $w_1 \in W$, so x lies in $H_1^\perp$ exactly when in addition $\varphi(x, f_1) = 0$, using lemma 13.3.21 to reduce orthogonality to H_1 to orthogonality to the two spanning vectors.

   Set $W_1 := W \cap H_1^\perp$, a totally isotropic subspace of $H_1^\perp$ of dimension $d - 1$. The inductive hypothesis, applied to the nondegenerate form $\varphi|_{H_1^\perp}$, gives hyperbolic planes $H_2, \ldots, H_d$ in $H_1^\perp$ with hyperbolic pairs (e_i, f_i) whose vectors $e_2, \ldots, e_d$ form a basis of W_1, and a subspace U with $\varphi|_U$ nondegenerate and $H_1^\perp = H_2 \perp \cdots \perp H_d \perp U$. All of these lie in $H_1^\perp$ and are therefore orthogonal to H_1, so $V = H_1 \perp H_2 \perp \cdots \perp H_d \perp U$. Finally $w_1 \notin W_1$, because $\lambda(w_1) = 1$ while λ vanishes on W_1, so $w_1, e_2, \ldots, e_d$ are d linearly independent vectors of W and hence a basis of it; setting $e_1 := w_1$ gives the statement. The dimension count is $n = 2d + \dim U \geq 2d$.

2. Write $\mathcal{H} := H_1 \perp \cdots \perp H_d$, so $V = \mathcal{H} \oplus U$ and $W = \operatorname{span}(e_1, \ldots, e_d)$ by (1).

   Suppose first that $\varphi|_U$ is anisotropic, and let $W' \supseteq W$ be totally isotropic. Take $x \in W'$ and write $x = \sum_i (a_i e_i + b_i f_i) + u$ with $u \in U$. For each j the vector e_j lies in $W \subseteq W'$, so $\varphi(x, e_j) = 0$; on the other hand the only summand of x not orthogonal to e_j is $b_j f_j$, whence $\varphi(x, e_j) = b_j$. So every b_j vanishes and $x = \sum_i a_i e_i + u$. Expanding $q(x)$ by bilinearity, the terms $\varphi(e_i, e_j)$ vanish for all i, j and the terms $\varphi(e_i, u)$ vanish because $e_i \in \mathcal{H}$ and $u \in U$, leaving $q(x) = q(u)$. As $x \in W'$ is isotropic or zero, $q(u) = 0$, so $u = 0$ by anisotropy and $x \in W$. Hence $W' = W$ and W is maximal.

Suppose instead that $\varphi|_U$ is isotropic, say $u \in U$ is nonzero with $q(u) = 0$. Then $W + ku$ is totally isotropic: expanding $\varphi(w + au, w' + a'u)$ leaves $\varphi(w, w') = 0$, the cross terms $\varphi(w, u) = 0$ and $\varphi(u, w') = 0$ since $W \subseteq \mathcal{H}$ and $u \in U$, and $aa'q(u) = 0$. It strictly contains W, because $u \notin \mathcal{H} \supseteq W$, so W is not maximal. The two implications together give the stated equivalence.

3. Applying (1) to a totally isotropic subspace of dimension $\nu(\varphi)$ gives $2\nu(\varphi) \leq n$. That subspace is maximal, being of largest dimension, so (2) makes the corresponding U anisotropic and

$$V = H_1 \perp \cdots \perp H_{\nu(\varphi)} \perp U$$

is a decomposition as in theorem 22.4.8, with radical zero and exactly $\nu(\varphi)$ hyperbolic planes. Conversely, let $V = H_1' \perp \cdots \perp H_m' \perp V_{\text{an}}$ be any such decomposition, with hyperbolic pair (e_i', f_i') in H_i'. The vectors $e_1', \ldots, e_m'$ lie in distinct summands of a direct sum and are nonzero, hence are linearly independent, and their span is totally isotropic: $\varphi(e_i', e_j') = 0$ for $i \neq j$ by orthogonality of the summands and $\varphi(e_i', e_i') = 0$ by the definition of a hyperbolic pair. So $m \leq \nu(\varphi)$, completing the proof.

Part (3) does not say that every decomposition has $\nu(\varphi)$ hyperbolic planes. It produces one that does and bounds all the others by that number, and no argument given above closes the gap: comparing two decompositions of the same space requires an isometry between them, and producing one is what Witt cancellation gives. Section 22.5 proves it, and with it every decomposition of a nondegenerate φ in theorem 22.4.8 has $m = \nu(\varphi)$ and an anisotropic summand determined up to isometry. Nondegeneracy is needed there: the Witt index of a degenerate form counts its radical as well, and running the same statement on a complement of the radical gives $m = \nu(\varphi) - \dim \operatorname{rad}(\varphi)$ instead, by exercise 22.4.1 (4). What is established here is that $\nu(\varphi)$ bounds the number of hyperbolic planes in any decomposition, and that the bound is attained when φ is nondegenerate. Over $\mathbb{R}$ that number can be computed directly from a diagonalization.

Example 22.4.12: The Witt Index over the Reals

Let φ be nondegenerate on a real vector space V of dimension n. By theorem 22.2.1 and proposition 22.2.2 applied with $c_i = |a_i|^{-1/2}$, there is a basis $u_1, \ldots, u_p, w_1, \ldots, w_r$ of V, orthogonal for φ, with $q(u_i) = 1$ and $q(w_j) = -1$ and $p + r = n$.

Pairing the vectors two at a time, $\operatorname{span}(u_i, w_i)$ carries the form $\langle 1, -1 \rangle$, which is a hyperbolic plane by example 22.4.5. Doing this for $i \leq \min(p, r)$ and collecting the remaining $|p - r|$ basis vectors, which all have the same value of q, gives

$$\varphi \cong \min(p, r)\, H \perp \langle \pm 1, \ldots, \pm 1 \rangle$$

with $|p - r|$ entries in the last summand, all equal to $+1$ if $p > r$ and all to -1 if $r > p$. That summand is anisotropic, a sum of $|p - r|$ squares being zero only when each is.

The Witt index is $\min(p, r)$. It is at least that, by the span of the $u_i + w_i$ for $i \leq \min(p, r)$, which are linearly independent and satisfy $q(u_i + w_i) = 1 - 1 = 0$ and $\varphi(u_i + w_i, u_j + w_j) = 0$ for $i \neq j$. For the reverse bound let W be totally isotropic and let $P := \operatorname{span}(u_1, \ldots, u_p)$. A nonzero $x \in P$ has $q(x) = \sum_i x_i^2 > 0$, so x is not isotropic and $W \cap P = 0$, forcing $\dim W \leq n - p = r$. The same argument with $N := \operatorname{span}(w_1, \ldots, w_r)$, where $q(y) = -\sum_j y_j^2 < 0$ for $y \neq 0$, gives $\dim W \leq n - r = p$. Hence $\nu(\varphi) = \min(p, r)$, in agreement with proposition 22.4.11(3).

Since $\nu(\varphi)$ and n depend only on φ, the unordered pair $\{p, r\}$ does too. Which of the two is p is not settled by this argument, and that remaining bit is Sylvester's law of inertia, which needs the order on $\mathbb{R}$ in a way the present chapter does not use.

Exercise 22.4.1

1. Let $\operatorname{char} k \neq 2$ and let $a \in k^\times$. Show that $\langle 1, -a\rangle$ is isotropic if and only if a is a square in k, and that in that case $\langle 1, -a\rangle \cong H$. Using that $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is infinite, deduce that $\mathbb{Q}$ carries infinitely many pairwise non-isometric anisotropic binary forms.

2. Let $\operatorname{char} k \neq 2$. Show that the hyperbolic plane represents every scalar: for each $c \in k$ there is $x \in H$ with $q(x) = c$. Deduce that a nondegenerate isotropic form on any V represents every element of k.

3. Let $\operatorname{char} k \neq 2$ and let $\varphi \cong \langle a, b\rangle$ be nondegenerate of rank 2. Using example 22.4.7, decide for which pairs (a, b) over $k = \mathbb{Q}$ the form is a hyperbolic plane, and exhibit one anisotropic and one hyperbolic example.

4. Let φ be a symmetric bilinear form on V with radical $R = \operatorname{rad}(\varphi)$, and let $\bar{\varphi}$ be the induced nondegenerate form on V/R from exercise 22.1.1 (5). Show that $\nu(\varphi) = \dim R + \nu(\bar{\varphi})$.

5. Let φ be nondegenerate on V and let $W \subseteq V$ be totally isotropic. Show that W is maximal among totally isotropic subspaces if and only if every isotropic vector of $W^\perp$ already lies in W.

6. Show that $\nu(\varphi \perp \psi) \geq \nu(\varphi) + \nu(\psi)$ for symmetric bilinear forms φ and ψ, and give an example over $\mathbb{R}$ where the inequality is strict.

7. Let q be odd and let φ be a nondegenerate symmetric bilinear form of odd rank n over $\mathbb{F}_q$. Show that $\nu(\varphi) = (n-1)/2$, so the Witt index over a finite field is forced by the rank alone when the rank is odd.

8. Let $V = \mathbb{F}_7^3$ with φ the standard dot product, so $q(x, y, z) = x^2+y^2+z^2$. Verify that $v = (3, 2, 1)$ is isotropic, run the construction in the proof of proposition 22.4.6 with $w = (0, 0, 1)$ to produce a partner f, compute a spanning vector of $H_v^\perp$, and write out the resulting decomposition $V = H_v \perp \langle a\rangle$. Is a a square in $\mathbb{F}_7$?

22.5 Witt Cancellation and Witt's Extension Theorem

Theorem 22.4.8 splits a form into an anisotropic part, a sum of hyperbolic planes and the radical, and each step of that construction rests on a choice: which isotropic vector to complete to a hyperbolic plane, and which complement to continue in. Nothing proved so far forces two runs of the construction to return the same number of hyperbolic planes, or isometric anisotropic parts. This section proves that they do. The result that delivers it is a cancellation law for orthogonal sums: a nondegenerate summand common to both sides of an isometry may be deleted. The section then proves Witt's extension theorem, which says that an isometry between two subspaces of a nondegenerate space is always the restriction of an isometry of the whole space. Both rest on one construction, the reflection in an anisotropic vector, which this book has not yet defined.

Throughout, $q(x) = \varphi(x, x)$ is the quadratic form attached to a symmetric bilinear form φ by proposition 22.1.2, and $\varphi \perp \psi$ is the orthogonal sum of definition 22.4.1. One further fact about that operation is used repeatedly below: the bijections $\big(v, (w, x)\big) \mapsto \big((v, w), x\big)$ and $(v, w) \mapsto (w, v)$ leave the defining expression unchanged, so they are isometries, and an orthogonal sum of several forms may be regrouped and reordered freely without changing its isometry class.

Every theorem below produces an isometry of a space with itself, so those isometries need a name.

Definition 22.5.1: The Orthogonal Group of a Form

Let k be a field, let V be a finite-dimensional k-vector space and let φ be a symmetric bilinear form on V. The *orthogonal group* of φ is

$$O(\varphi) = \{\sigma\colon V \to V \mid \sigma \text{ is an isometry of } (V,\varphi) \text{ with itself}\}$$

where isometry is meant in the sense of definition 22.1.10, and the operation is composition of maps.

That this is a group takes nothing beyond the definition: id_V is an isometry; a composite of linear bijections preserving φ is again a linear bijection preserving φ; and if σ is an isometry then so is σ^{-1}, since $\varphi(\sigma^{-1}x, \sigma^{-1}y) = \varphi(\sigma\sigma^{-1}x, \sigma\sigma^{-1}y) = \varphi(x,y)$ for all $x, y \in V$.

The construction used throughout the section takes an anisotropic vector (definition 22.4.2) and produces the isometry that negates it and fixes everything orthogonal to it. Over $\mathbb{R}$ with the dot product this is the familiar mirror reflection in the hyperplane $\{v\}^{\perp}$, and the formula below is the one from that setting, written so that it makes sense over any field in which $q(v)$ is invertible.

Definition 22.5.2: Reflection in an Anisotropic Vector

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $v \in V$ be anisotropic, that is $q(v) = \varphi(v,v) \neq 0$. The *reflection in v* is the map $\tau_v\colon V \to V$ given by

$$\tau_v(x) = x - \frac{2\varphi(v,x)}{\varphi(v,v)}\, v$$

The denominator is nonzero precisely because v is anisotropic, which is why the construction is available for anisotropic vectors and for no others.

The properties that make τ_v useful are all direct computations with the defining formula.

Lemma 22.5.3: Reflections Are Involutive Isometries

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $v \in V$ be anisotropic. Then:

1. $V = kv \oplus \{v\}^{\perp}$;
2. τ_v is linear, $\tau_v(v) = -v$, and $\tau_v(x) = x$ for every $x \in \{v\}^{\perp}$;
3. τ_v preserves φ and satisfies $\tau_v \circ \tau_v = \mathrm{id}_V$; in particular $\tau_v \in O(\varphi)$ and $\tau_v^{-1} = \tau_v$;
4. $\tau_{\lambda v} = \tau_v$ for every $\lambda \in k^{\times}$;
5. $\det \tau_v = -1$. In particular $\tau_v \neq \mathrm{id}_V$, so τ_v has order exactly 2 in $O(\varphi)$.

***Proof*:**
Write $a = q(v) = \varphi(v, v) \neq 0$ throughout. The set $\{v\}^\perp$ is a subspace by lemma 13.3.21.

1. Every $x \in V$ decomposes as
$$x = \frac{\varphi(v,x)}{a}\, v + \left(x - \frac{\varphi(v,x)}{a}\, v\right)$$
and applying $\varphi(v, -)$ to the bracketed vector gives $\varphi(v,x) - \frac{\varphi(v,x)}{a}\, a = 0$, so that vector lies in $\{v\}^\perp$ and $V = kv + \{v\}^\perp$. The sum is direct: if $\lambda v \in \{v\}^\perp$ then $0 = \varphi(v, \lambda v) = \lambda a$, and $a \neq 0$ forces $\lambda = 0$.

2. Linearity holds because $x \mapsto \varphi(v,x)$ is linear and τ_v subtracts a scalar multiple of the fixed vector v. Substituting $x = v$ gives $\tau_v(v) = v - \frac{2a}{a} v = v - 2v = -v$, and if $\varphi(v,x) = 0$ then the subtracted term vanishes, so $\tau_v(x) = x$.

3. For $x, y \in V$, expanding by bilinearity and symmetry,
$$\varphi\big(\tau_v(x), \tau_v(y)\big) = \varphi(x,y) - \frac{2\varphi(v,y)}{a}\varphi(x,v) - \frac{2\varphi(v,x)}{a}\varphi(v,y) + \frac{4\varphi(v,x)\varphi(v,y)}{a^2}\, a$$
The last three terms are $-2c$, $-2c$ and $+4c$ with $c = \varphi(v,x)\varphi(v,y)/a$, so they cancel and $\varphi(\tau_v(x), \tau_v(y)) = \varphi(x,y)$. For the involution, first
$$\varphi\big(v, \tau_v(x)\big) = \varphi(v,x) - \frac{2\varphi(v,x)}{a}\, a = -\varphi(v,x)$$
so
$$\tau_v\big(\tau_v(x)\big) = \tau_v(x) - \frac{2\varphi(v, \tau_v(x))}{a}\, v = x - \frac{2\varphi(v,x)}{a}\, v + \frac{2\varphi(v,x)}{a}\, v = x$$
Hence τ_v is a linear bijection preserving φ, which is an isometry of (V, φ) with itself in the sense of definition 22.1.10, so $\tau_v \in O(\varphi)$ by definition 22.5.1 and $\tau_v^{-1} = \tau_v$.

4. For $\lambda \neq 0$ the vector λv is anisotropic, since $q(\lambda v) = \lambda^2 a \neq 0$, and
$$\tau_{\lambda v}(x) = x - \frac{2\varphi(\lambda v, x)}{\lambda^2 a}\, \lambda v = x - \frac{2\lambda^2\varphi(v,x)}{\lambda^2 a}\, v = \tau_v(x)$$

5. By (1) choose a basis $v, z_2, \ldots, z_n$ of V with $z_2, \ldots, z_n$ a basis of $\{v\}^\perp$. By (2) the matrix of τ_v in this basis is $\operatorname{diag}(-1, 1, \ldots, 1)$, whose determinant is -1. Since $\operatorname{char} k \neq 2$ this is not 1, so $\tau_v \neq \mathrm{id}_V$, and $\tau_v^2 = \mathrm{id}_V$ from (3) then makes its order exactly 2, completing the proof.

Example 22.5.4: Reflections of $\langle 1, 1\rangle$ over $\mathbb{Q}$

Let $k = \mathbb{Q}$ and let φ be the form on $V = \mathbb{Q}^2$ with matrix the identity in the standard basis e_1, e_2, so $q(x_1, x_2) = x_1^2 + x_2^2$ and $\varphi \cong \langle 1, 1\rangle$.

Take $v = e_1$, with $q(e_1) = 1$. Then $\tau_{e_1}(x) = x - 2\varphi(e_1, x)e_1 = x - 2x_1 e_1$, so $\tau_{e_1}(e_1) = -e_1$ and $\tau_{e_1}(e_2) = e_2$, and the matrix of τ_{e_1} is $\operatorname{diag}(-1, 1)$.

Take instead $v = (1,1)$, with $q(v) = 2 \neq 0$. Here $\tau_v(x) = x - \frac{2\varphi(v,x)}{2} v = x - \varphi(v,x)v$, so

$$\tau_v(e_1) = (1,0) - (1,1) = (0,-1), \qquad \tau_v(e_2) = (0,1) - (1,1) = (-1,0)$$

and the matrix of τ_v is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. Both matrices have determinant -1 and square to the identity, as lemma 22.5.3 requires.

Their product is

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

an isometry of determinant 1 whose square is $-\mathrm{id}$ and whose fourth power is id. So $O(\langle 1,1\rangle)$ over $\mathbb{Q}$ contains an element of order 4, which is not a reflection: reflections have order 2 and determinant -1. Products of reflections are therefore strictly more general than reflections, and corollary 22.5.10 will say how much more general.

Reflections are used here for one property: they move a vector to any other vector of the same nonzero value under q. That is how every theorem below produces the isometry it needs, and it takes at most two reflections.

Lemma 22.5.5: Two Reflections Move a Vector to Any Other of the Same Value

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space with a symmetric bilinear form φ, and let $u, w \in V$ satisfy $q(u) = q(w) \neq 0$. Then there is a $\sigma \in O(\varphi)$ with $\sigma(u) = w$, and σ can be taken to be a product of at most two reflections.

***Proof*:**
Expanding both terms by bilinearity and using $q(w) = q(u)$,

$$q(u-w) + q(u+w) = \big(q(u) - 2\varphi(u,w) + q(w)\big) + \big(q(u) + 2\varphi(u,w) + q(w)\big) = 4q(u)$$

Since $\operatorname{char} k \neq 2$, the scalar $4 = 2^2$ is nonzero, so $4q(u) \neq 0$ and $q(u-w)$ and $q(u+w)$ do not both vanish.

Case 1: $q(u-w) \neq 0$. Put $v = u - w$, an anisotropic vector. Then

$$q(v) = q(u) - 2\varphi(u,w) + q(w) = 2\big(q(u) - \varphi(u,w)\big) = 2\varphi(v,u)$$

the last equality because $\varphi(v,u) = \varphi(u,u) - \varphi(w,u) = q(u) - \varphi(u,w)$. Hence $2\varphi(v,u)/q(v) = 1$, and

$$\tau_v(u) = u - \frac{2\varphi(v,u)}{q(v)} v = u - v = w$$

so $\sigma = \tau_v$ works, and it lies in $O(\varphi)$ by lemma 22.5.3.

Case 2: $q(u-w) = 0$. Then $q(u+w) = 4q(u) \neq 0$ by the displayed identity. The vectors u and $-w$ satisfy $q(-w) = q(w) = q(u) \neq 0$ and $q\big(u - (-w)\big) = q(u+w) \neq 0$, so Case 1 applied to the pair $u, -w$ gives $\tau_{u+w}(u) = -w$. Also $q(w) \neq 0$, so τ_w is defined, and $\tau_w(-w) = -\tau_w(w) = w$ by lemma 22.5.3(2). Therefore

$$(\tau_w \circ \tau_{u+w})(u) = \tau_w(-w) = w$$

and $\sigma = \tau_w \circ \tau_{u+w}$ is a product of two reflections, lying in $O(\varphi)$ because $O(\varphi)$ is closed under composition (definition 22.5.1), as we sought to show.

Both cases occur. In example 22.5.4 the vectors $u = e_1$ and $w = e_2$ have $q(u) = q(w) = 1$ and $q(u - w) = q(1, -1) = 2 \neq 0$, so Case 1 applies and the single reflection $\tau_{(1,-1)}$ carries e_1 to e_2; indeed $\tau_{(1,-1)}(e_1) = e_1 - \frac{2\cdot 1}{2}(1, -1) = (0, 1)$. Case 2 is not an artifact of the proof: over $\mathbb{Q}$ with $\varphi \cong \langle 1, 1, -1\rangle$ the vectors $u = e_1$ and $w = (1, 1, 1)$ both have $q = 1$ while $q(u-w) = q(0, -1, -1) = 0$, and there no single reflection suffices: an equality $\tau_v(u) = w$ reads $u - w = \frac{2\varphi(v,u)}{q(v)} v$, and $w \neq u$ makes the coefficient nonzero, so $q(u - w)$ would be a nonzero square multiple of $q(v) \neq 0$ and in particular nonzero, contradicting $q(u - w) = 0$.

Lemma 22.5.5 carries no nondegeneracy hypothesis on φ, which matters immediately: the cancellation theorem is proved for arbitrary φ_1 and φ_2, whose radicals are not controlled, and only the summand being cancelled is required to be nondegenerate. The proof reduces that summand to one dimension, where the lemma does the work.

Theorem 22.5.6: Witt Cancellation

Let k be a field with $\operatorname{char} k \neq 2$. Let ψ be a *nondegenerate* symmetric bilinear form on a finite-dimensional k-vector space, and let φ_1, φ_2 be symmetric bilinear forms on finite-dimensional k-vector spaces. If

$$\psi \perp \varphi_1 \cong \psi \perp \varphi_2$$

then $\varphi_1 \cong \varphi_2$.

***Proof*:**
Step 1: reduction to a one-dimensional ψ. Let ψ live on U with $\dim U = m$. By theorem 22.2.1 the space U has an orthogonal basis $u_1, \dots, u_m$, so $\psi \cong \langle c_1, \dots, c_m\rangle$ with $c_i = \psi(u_i, u_i)$. By the last claim of proposition 22.2.2 the number of indices with $c_i = 0$ is $\dim \operatorname{rad}(\psi)$, which is 0 because ψ is nondegenerate; hence every $c_i \neq 0$. Regrouping the orthogonal sum,

$$\psi \perp \varphi_j \cong \langle c_1\rangle \perp \big(\langle c_2, \dots, c_m\rangle \perp \varphi_j\big) \qquad (j = 1, 2)$$

So if the theorem holds whenever ψ is one-dimensional and nondegenerate, then cancelling $\langle c_1\rangle$ from the hypothesis gives $\langle c_2, \dots, c_m\rangle \perp \varphi_1 \cong \langle c_2, \dots, c_m\rangle \perp \varphi_2$, and $m-1$ further cancellations, formally an induction on m with the case $m = 0$ reading $\varphi_1 \cong \varphi_2$, give the theorem. It therefore suffices to treat $\psi = \langle c\rangle$ with $c \in k^\times$.

Step 2: the one-dimensional case. Let φ_j live on W_j and realize $\langle c\rangle \perp \varphi_j$ on $V_j = ke_j \oplus W_j$, where $q(e_j) = c$, $\varphi(e_j, W_j) = 0$, and the form restricted to W_j is φ_j. Write $\varphi^{(j)}$ for the form on V_j. First,

$$W_j = \{e_j\}^\perp \text{ inside } V_j$$

since $W_j \subseteq \{e_j\}^\perp$ by construction, and conversely if $x = \lambda e_j + y$ with $y \in W_j$ satisfies $\varphi^{(j)}(e_j, x) = 0$, then that value is λc, and $c \neq 0$ forces $\lambda = 0$.

Let $f\colon V_1 \to V_2$ be an isometry. Put $u = f(e_1) \in V_2$. Then $q^{(2)}(u) = q^{(1)}(e_1) = c \neq 0$ and $q^{(2)}(e_2) = c$, so lemma 22.5.5 applied inside $(V_2, \varphi^{(2)})$ gives $\sigma \in O(\varphi^{(2)})$ with $\sigma(u) = e_2$. The composite $g = \sigma \circ f\colon V_1 \to V_2$ is again an isometry, and $g(e_1) = e_2$.

Step 3: g carries W_1 onto W_2. For $x \in V_1$,

$$\varphi^{(2)}\big(e_2, g(x)\big) = \varphi^{(2)}\big(g(e_1), g(x)\big) = \varphi^{(1)}(e_1, x)$$

so $x \in \{e_1\}^\perp$ if and only if $g(x) \in \{e_2\}^\perp$. Since g is bijective, g maps $\{e_1\}^\perp$ onto $\{e_2\}^\perp$, which by Step 2 says $g(W_1) = W_2$. The restriction $g|_{W_1} : W_1 \to W_2$ is a linear isomorphism preserving the forms, that is an isometry $\varphi_1 \cong \varphi_2$, completing the proof.

The theorem is what makes the decomposition of theorem 22.4.8 independent of the choices made in producing it.

Corollary 22.5.7: The Anisotropic Part and the Witt Index Are Invariants

Let k be a field with $\operatorname{char} k \neq 2$ and let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space. Suppose

$$\varphi \cong \varphi_1 \perp m_1 H \qquad \text{and} \qquad \varphi \cong \varphi_2 \perp m_2 H$$

with φ_1 and φ_2 anisotropic and $m_1, m_2 \geq 0$. Then $m_1 = m_2$ and $\varphi_1 \cong \varphi_2$.

***Proof*:**
By definition 22.4.4 a hyperbolic plane H has a basis e, f with $q(e) = q(f) = 0$ and $\varphi(e, f) = 1$, so its matrix in that basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, of determinant $-1 \neq 0$; by lemma 13.3.23 the form H is nondegenerate. The matrix of an orthogonal sum in the union of two bases is block diagonal with the two matrices as blocks, so its determinant is the product of theirs, and mH is nondegenerate for every $m \geq 0$ by the same lemma. Also e is a nonzero isotropic vector of H.

Assume without loss of generality that $m_1 \leq m_2$. Regrouping the orthogonal sum,

$$\varphi_1 \perp m_1 H \cong \varphi \cong \varphi_2 \perp m_2 H \cong \big(\varphi_2 \perp (m_2 - m_1)H\big) \perp m_1 H$$

Since $m_1 H$ is nondegenerate, theorem 22.5.6 cancels it from the two ends and gives

$$\varphi_1 \cong \varphi_2 \perp (m_2 - m_1)H$$

If $m_2 > m_1$, the right-hand form has the nonzero isotropic vector e of any one of its hyperbolic summands, and the preimage of e under the isometry is nonzero and isotropic for φ_1, since an isometry is injective and preserves q; that contradicts the hypothesis that φ_1 is anisotropic. Hence $m_1 = m_2$, the term $(m_2 - m_1)H$ is the form on the zero space, and $\varphi_1 \cong \varphi_2$, as we sought to show.

So the Witt index of definition 22.4.10 deserves its name. Proposition 22.4.11(3) computes $\nu(\varphi)$ as the number of hyperbolic planes in the decomposition produced by the proof of theorem 22.4.8, and corollary 22.5.7 says that number is the same in every decomposition of that shape; so $m = \nu(\varphi)$ in *any* of them, and the anisotropic part is determined up to isometry by φ alone. Two nondegenerate forms $\varphi \cong \varphi_1 \perp mH$ and $\varphi' \cong \varphi'_1 \perp m'H$, with φ_1 and φ'_1 anisotropic, are therefore isometric if and only if $m = m'$ and $\varphi_1 \cong \varphi'_1$: if $\varphi \cong \varphi'$ then φ has the two decompositions $\varphi \cong \varphi_1 \perp mH$ and $\varphi \cong \varphi'_1 \perp m'H$, so corollary 22.5.7 gives $m = m'$ and $\varphi_1 \cong \varphi'_1$, while conversely $m = m'$ and $\varphi_1 \cong \varphi'_1$ give $\varphi \cong \varphi_1 \perp mH \cong \varphi'_1 \perp m'H \cong \varphi'$. That biconditional is the reduction the classification of forms over a given field uses.

Cancellation deletes a summand from both sides of an isometry that is already given. The extension theorem is the statement that isometries are plentiful: any isometry defined on a subspace already extends to the whole space, with no compatibility condition beyond preserving the form. The subspace is allowed to be degenerate, and that is the substance of the theorem; for a subspace containing an anisotropic vector, the reduction to a smaller space is the argument already used twice above, whereas a totally isotropic subspace gives that argument nothing to work with. The following lemma is what repairs that case: it manufactures, for an isotropic vector inside a totally isotropic subspace, a partner making it half of a hyperbolic plane, chosen orthogonal to the rest of the subspace.

Lemma 22.5.8: Hyperbolic Partners for Isotropic Vectors

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space V, let $U \subseteq V$ be a subspace with $\varphi(x, y) = 0$ for all $x, y \in U$, let $u \in U$ be nonzero, and let $U_1 \subseteq U$ be a subspace with $U = ku \oplus U_1$. Then there exists $u^* \in V$ with

$$q(u^*) = 0, \qquad \varphi(u, u^*) = 1, \qquad \varphi(u^*, x) = 0 \text{ for all } x \in U_1$$

***Proof*:**
Let $u_1, \dots, u_m$ be a basis of U_1, so that $u, u_1, \dots, u_m$ is a basis of U and in particular a linearly independent family in V. Consider the linear map

$$\Phi \colon V \to k^{m+1}, \qquad \Phi(y) = \big(\varphi(u, y), \varphi(u_1, y), \dots, \varphi(u_m, y)\big)$$

Suppose Φ were not surjective. Its image is then a proper subspace of k^{m+1}, hence contained in the kernel of some nonzero linear functional (extend a basis of the image to a basis of k^{m+1} and take the member of the dual basis of proposition 13.3.5 attached to a vector outside the image): there are $\lambda_0, \dots, \lambda_m \in k$, not all zero, with $\lambda_0\varphi(u, y) + \sum_{i=1}^{m} \lambda_i\varphi(u_i, y) = 0$ for every $y \in V$. Bilinearity rewrites this as $\varphi(z, y) = 0$ for every $y \in V$, where $z = \lambda_0 u + \sum_i \lambda_i u_i$. As φ is symmetric, this says $z \in \operatorname{rad}(\varphi)$ (definition 13.3.22), which is zero by nondegeneracy; and $u, u_1, \dots, u_m$ being linearly independent then forces every $\lambda_j = 0$, contradicting the choice of the λ_j. So Φ is surjective.

Choose $y \in V$ with $\Phi(y) = (1, 0, \dots, 0)$, so $\varphi(u, y) = 1$ and $\varphi(u_i, y) = 0$ for each i; by bilinearity the latter gives $\varphi(x, y) = 0$ for every $x \in U_1$. Since $\operatorname{char} k \neq 2$ the scalar $q(y)/2$ is defined, and we may set

$$u^* := y - \frac{q(y)}{2}\, u$$

Because $u \in U$ and φ vanishes identically on U, we have $q(u) = 0$ and $\varphi(u, x) = 0$ for $x \in U_1$. Hence

$$\varphi(u, u^*) = \varphi(u, y) - \frac{q(y)}{2} q(u) = 1 - 0 = 1$$

$$q(u^*) = q(y) - 2 \cdot \frac{q(y)}{2}\varphi(y, u) + \frac{q(y)^2}{4} q(u) = q(y) - q(y) + 0 = 0$$

using $\varphi(y, u) = \varphi(u, y) = 1$ by symmetry. Finally, for $x \in U_1$ both $\varphi(y, x) = 0$ and $\varphi(u, x) = 0$, so $\varphi(u^*, x) = 0$, completing the proof.

Theorem 22.5.9: Witt's Extension Theorem

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a nondegenerate symmetric bilinear form on a finite-dimensional k-vector space V, let $U \subseteq V$ be a subspace, and let $f\colon U \to V$ be an injective linear map with

$$\varphi\big(f(x), f(y)\big) = \varphi(x, y) \qquad \text{for all } x, y \in U$$

Then there is an $F \in O(\varphi)$ with $F|_U = f$.

***Proof*:**
Induct on $\dim V$. If $\dim V = 0$ then $U = 0$ and $F = \mathrm{id}_V$ serves. So let $\dim V = n \geq 1$ and assume the theorem for every nondegenerate symmetric bilinear form on a space of dimension smaller than n. If $U = 0$, take $F = \mathrm{id}_V$; assume $U \neq 0$.

Step 1: reduction to the case where φ does not vanish identically on U. Suppose $\varphi(x, y) = 0$ for all $x, y \in U$. Choose $u \in U$ nonzero and a subspace U_1 with $U = ku \oplus U_1$, and let $u_1, \ldots, u_m$ be a basis of U_1. Lemma 22.5.8 gives $u^* \in V$ with $q(u^*) = 0$, $\varphi(u, u^*) = 1$ and $\varphi(u^*, U_1) = 0$.

Put $u' = f(u)$ and $U_1' = f(U_1)$. Since f is injective and preserves φ, the subspace $f(U)$ satisfies $\varphi(x, y) = 0$ for all $x, y \in f(U)$, the vector u' is nonzero, and $f(U) = ku' \oplus U_1'$. A second application of lemma 22.5.8, to the subspace $f(U)$ with the vector u' and the complement U_1', gives $u'^* \in V$ with $q(u'^*) = 0$, $\varphi(u', u'^*) = 1$ and $\varphi(u'^*, U_1') = 0$.

Because $\varphi(u, u^*) = 1$ while $\varphi(u, x) = 0$ for every $x \in U$, the vector u^* is not in U, so $\tilde{U} := U \oplus ku^*$ has $u, u^*, u_1, \ldots, u_m$ as a basis. Define a linear map $\tilde{f}\colon \tilde{U} \to V$ by $\tilde{f}|_U = f$ and $\tilde{f}(u^*) = u'^*$.

The map $\tilde{f}$ preserves φ. By bilinearity it is enough to compare φ on pairs of basis vectors with φ on the pairs of their images, and the six comparisons are:

$$q(u) = 0 = q(u'), \qquad q(u^*) = 0 = q(u'^*), \qquad \varphi(u, u^*) = 1 = \varphi(u', u'^*)$$

$$\varphi(u, u_i) = 0 = \varphi\big(f(u), f(u_i)\big), \qquad \varphi(u^*, u_i) = 0 = \varphi(u'^*, f(u_i)), \qquad \varphi(u_i, u_j) = \varphi\big(f(u_i), f(u_j)\big)$$

the fourth and sixth because f preserves φ, the fifth by the two orthogonality conditions on u^* and u'^*.

The map $\tilde{f}$ is injective. Let $x = \lambda u + \mu u^* + w$ with $w \in U_1$ and $\tilde{f}(x) = 0$, that is $\lambda u' + \mu u'^* + f(w) = 0$. Applying $\varphi(u', -)$ and using $\varphi(u', u') = 0$, $\varphi(u', u'^*) = 1$ and $\varphi(u', f(w)) = \varphi(f(u), f(w)) = \varphi(u, w) = 0$ gives $\mu = 0$. Then $f(\lambda u + w) = 0$, so $\lambda u + w = 0$ by injectivity of f, so $\lambda = 0$ and $w = 0$ because $U = ku \oplus U_1$.

Finally $q(u + u^*) = q(u) + 2\varphi(u, u^*) + q(u^*) = 2 \neq 0$, so φ does not vanish identically on $\tilde{U}$. Any $F \in O(\varphi)$ with $F|_{\tilde{U}} = \tilde{f}$ satisfies $F|_U = f$, so it suffices to prove the theorem for the pair $(\tilde{U}, \tilde{f})$, which is of the kind treated next.

Step 2: the case where φ does not vanish identically on U. Some $u \in U$ has $q(u) \neq 0$: otherwise q vanishes on U and, for $x, y \in U$, $0 = q(x + y) = q(x) + 2\varphi(x, y) + q(y) = 2\varphi(x, y)$, which with $\operatorname{char} k \neq 2$ makes φ vanish identically on U.

Then $q(f(u)) = q(u) \neq 0$, so lemma 22.5.5 gives $\sigma \in O(\varphi)$ with $\sigma(f(u)) = u$. The map $f' := \sigma \circ f$ is again an injective linear map $U \to V$ preserving φ, and $f'(u) = u$.

Set $V_1 = \{u\}^\perp$ and $U' = U \cap V_1$. Four facts about these:

1. $V = ku \oplus V_1$ with $\varphi(u, V_1) = 0$, by lemma 22.5.3(1) applied to (V, φ) and the anisotropic vector u; in particular $\dim V_1 = n - 1$.
2. $U = ku \oplus U'$, by the same part of the same lemma applied to $(U, \varphi|_U)$, whose orthogonal complement of u inside U is $U \cap \{u\}^\perp = U'$.
3. $\varphi|_{V_1}$ is nondegenerate: if $y \in V_1$ satisfies $\varphi(y, V_1) = 0$, then also $\varphi(y, u) = 0$, so $\varphi(y, V) = 0$ by (1) and bilinearity, whence $y \in \mathrm{rad}(\varphi) = 0$.
4. $f'(U') \subseteq V_1$: for $x \in U'$, $\varphi(u, f'(x)) = \varphi(f'(u), f'(x)) = \varphi(u, x) = 0$.

So $f'|_{U'} : U' \to V_1$ is an injective linear map from a subspace of V_1 into V_1 preserving $\varphi|_{V_1}$, and $\dim V_1 = n - 1$. The inductive hypothesis gives $G \in O(\varphi|_{V_1})$ with $G|_{U'} = f'|_{U'}$.

Step 3: assembling F. Define $F' : V \to V$ by $F'(\lambda u + y) = \lambda u + G(y)$ for $\lambda \in k$ and $y \in V_1$; this is well defined and bijective by (1) and the bijectivity of G. It preserves φ: for $\lambda, \mu \in k$ and $y, z \in V_1$, using $G(y), G(z) \in V_1$ and $\varphi(u, V_1) = 0$ twice,

$$\varphi\big(\lambda u + G(y), \mu u + G(z)\big) = \lambda\mu\, q(u) + \varphi\big(G(y), G(z)\big) = \lambda\mu\, q(u) + \varphi(y, z) = \varphi(\lambda u + y, \mu u + z)$$

So $F' \in O(\varphi)$. It restricts to f' on U: writing $x \in U$ as $x = \lambda u + x'$ with $x' \in U'$ by (2),

$$F'(x) = \lambda u + G(x') = \lambda u + f'(x') = \lambda f'(u) + f'(x') = f'(x)$$

Set $F = \sigma^{-1} \circ F'$, an element of $O(\varphi)$. Then $F|_U = \sigma^{-1} \circ f' = \sigma^{-1} \circ \sigma \circ f = f$, completing the proof.

The hypothesis that φ be nondegenerate on the ambient space is not removable, and it enters the proof twice: in Step 1 through lemma 22.5.8, and in Step 2 as fact (3), where $\mathrm{rad}(\varphi) = 0$ is what makes $\varphi|_{V_1}$ nondegenerate and so licenses the inductive hypothesis. What goes wrong without it is direct: an isometry of V carries $\mathrm{rad}(\varphi)$ onto itself, so a form-preserving injection sending a vector outside the radical into the radical extends to no isometry, and exercise 22.5.1 (6) exhibits one. Nondegeneracy of the *subspace* U, on the other hand, is nowhere assumed, and Step 1 is the work of doing without it.

Taking U to be a line already extends lemma 22.5.5 past its restriction to anisotropic vectors. Let φ be nondegenerate, let $c \in k$, and let v, v' be nonzero vectors with $q(v) = q(v') = c$. The map $f : kv \to V$ with $f(\lambda v) = \lambda v'$ is linear and injective, and $\varphi(f(\lambda v), f(\mu v)) = \lambda\mu\, q(v') = \lambda\mu\, q(v) = \varphi(\lambda v, \mu v)$, so theorem 22.5.9 extends it to an isometry of V carrying v to v'. So $O(\varphi)$ acts transitively on the nonzero vectors of each fixed value of q, including the value 0, where no reflection was available. The last consequence counts how many reflections it takes to build an arbitrary isometry.

Corollary 22.5.10: Reflections Generate the Orthogonal Group

Let k be a field with $\operatorname{char} k \neq 2$ and let φ be a nondegenerate symmetric bilinear form on a k-vector space V of finite dimension n. Then every $\sigma \in O(\varphi)$ is a product of at most $2n$ reflections τ_v in anisotropic vectors $v \in V$, where the empty product is read as id_V. In particular $O(\varphi)$ is generated by its reflections. The bound $2n$ is not the sharp one; see the discussion after the proof.

***Proof*:**
Induct on n. For $n = 0$ the group $O(\varphi)$ is trivial and id_V is the empty product. Let $n \geq 1$ and assume the statement for nondegenerate forms on spaces of dimension $n - 1$.

First, V contains an anisotropic vector: if $q(x) = 0$ for every $x \in V$, then for all $x, y \in V$

$$0 = q(x + y) = q(x) + 2\varphi(x, y) + q(y) = 2\varphi(x, y)$$

and $\operatorname{char} k \neq 2$ makes φ the zero form, so $\operatorname{rad}(\varphi) = V \neq 0$, contradicting nondegeneracy. Fix such a u, with $q(u) \neq 0$.

Let $\sigma \in O(\varphi)$. Then $q(\sigma(u)) = q(u) \neq 0$, so lemma 22.5.5, applied to the pair $\sigma(u)$ and u, gives $\pi \in O(\varphi)$ with $\pi(\sigma(u)) = u$ and π a product of at most two reflections. Put $\rho = \pi \circ \sigma$, so $\rho(u) = u$.

Set $V_1 = \{u\}^\perp$. As in the proof of theorem 22.5.9, $V = ku \oplus V_1$ with $\varphi(u, V_1) = 0$ by lemma 22.5.3(1), $\dim V_1 = n - 1$, and $\varphi|_{V_1}$ is nondegenerate because a vector of V_1 orthogonal to V_1 is also orthogonal to u, hence to V, hence zero. Moreover $\rho(V_1) = V_1$: for $y \in V_1$, $\varphi(u, \rho(y)) = \varphi(\rho(u), \rho(y)) = \varphi(u, y) = 0$, so $\rho(V_1) \subseteq V_1$, and ρ is injective, so the inclusion is an equality by dimension count.

By the inductive hypothesis $\rho|_{V_1}$ is a product $\tau_{v_1}^{V_1} \cdots \tau_{v_r}^{V_1}$ of $r \leq 2(n-1)$ reflections of $(V_1, \varphi|_{V_1})$, with each $v_i \in V_1$ anisotropic. Each v_i is anisotropic in V as well, and the reflection τ_{v_i} of V satisfies $\tau_{v_i}(u) = u$, since $\varphi(v_i, u) = 0$, and restricts on V_1 to $\tau_{v_i}^{V_1}$, the defining formula being the same. Hence $\tau_{v_1} \cdots \tau_{v_r}$ and ρ agree on u and on V_1, so they agree on $V = ku \oplus V_1$, and ρ is a product of at most $2(n - 1)$ reflections of V.

Finally $\sigma = \pi^{-1} \circ \rho$, and π^{-1} is a product of at most two reflections, since each reflection is its own inverse by lemma 22.5.3(3) and the inverse of $\tau_v \tau_w$ is $\tau_w \tau_v$. So σ is a product of at most $2 + 2(n - 1) = 2n$ reflections, as we sought to show.

The sharp form of this count is the Cartan-Dieudonné theorem, proved in Artin's *Geometric Algebra*: over a field of characteristic other than 2, every isometry of an n-dimensional nondegenerate space is a product of at most n reflections. The bound proved above loses a factor of two at Case 2 of lemma 22.5.5, where two reflections are used in place of one. Recovering the sharp bound requires a separate analysis of the isometries σ for which $q(x - \sigma(x)) = 0$ for every $x \in V$, so that no single reflection returns $\sigma(x)$ to x. Such isometries exist: on $H \perp H$, with the two hyperbolic bases e_1, f_1 and e_2, f_2, the map fixing e_1 and e_2 and sending $f_1 \mapsto f_1 + e_2$ and $f_2 \mapsto f_2 - e_1$ is one. For them the reflection count is governed by $\dim \operatorname{im}(\sigma - \mathrm{id}_V)$ rather than by n. That analysis is not carried out in this book.

Exercise 22.5.1

1. Let φ be a nondegenerate symmetric bilinear form on V with $\dim V = n \geq 1$ and $\operatorname{char} k \neq 2$, and let $\sigma \in O(\varphi)$. Show that if A is the matrix of φ and S the matrix of σ in one common basis, then $S^t AS = A$ (use lemma 13.3.19), and deduce $\det \sigma = \pm 1$ with the help of lemma 13.3.23. Using $\det \tau_v = -1$ from lemma 22.5.3, deduce that $SO(\varphi) := \{\sigma \in O(\varphi) \mid \det \sigma = 1\}$ is a subgroup of index exactly 2, and that the parity of the number of reflections in a factorization of σ given by corollary 22.5.10 does not depend on the factorization.

2. Let $\operatorname{char} k \neq 2$, let φ be a symmetric bilinear form on V, and let $v, w \in V$ be anisotropic. Show that $\tau_v = \tau_w$ if and only if $kv = kw$, so that $v \mapsto \tau_v$ is a bijection from the set of anisotropic lines of V onto a set of involutions in $O(\varphi)$. Then count the reflections in $O(\langle 1, 1\rangle)$ over $\mathbb{F}_5$, by listing the six lines of $\mathbb{F}_5^2$ and deciding which are anisotropic.

3. Show that the hypothesis that ψ be nondegenerate can be removed from theorem 22.5.6: for arbitrary symmetric bilinear forms over a field with $\operatorname{char} k \neq 2$, $\psi \perp \varphi_1 \cong \psi \perp \varphi_2$ implies $\varphi_1 \cong \varphi_2$. Reduce, using theorem 22.2.1, to cancelling $\langle 0\rangle$, and for that case use that an isometry carries the radical of one form onto the radical of the other, together with the description of the radical in an orthogonal basis given by proposition 22.2.2.

4. Over $k = \mathbb{Q}$, show that $\langle 1, 1\rangle$ and $\langle -1, -1\rangle$ are not isometric by asking which nonzero rationals each represents. Deduce from theorem 22.5.6 that $\langle 1, 1, 5\rangle \ncong \langle -1, -1, 5\rangle$, and check that these two forms have the same rank and the same discriminant, so that neither invariant of section 22.1 could have separated them.

5. Let φ be nondegenerate on V with $\operatorname{char} k \neq 2$, and recall the totally isotropic subspaces of definition 22.3.6. Call one *maximal* if no strictly larger subspace of V is totally isotropic. Using theorem 22.5.9, show that any two maximal totally isotropic subspaces of V have the same dimension. Then explain why lemma 22.5.5 gives nothing about isotropic vectors, by exhibiting two nonzero isotropic vectors u, w of $H \perp H$ with $q(u - w) = q(u + w) = 0$.

6. Let $\operatorname{char} k \neq 2$ and let $V = k^3$ carry the form whose matrix in the standard basis is $\operatorname{diag}(1, -1, 0)$. Show that $f\colon k(e_1 + e_2) \to V$ with $f(e_1 + e_2) = e_3$ is an injective linear map preserving the form, and that no isometry of V restricts to it. Conclude that the nondegeneracy hypothesis in theorem 22.5.9 cannot be dropped.

7. Let $\operatorname{char} k \neq 2$ and let $a \in k^\times$. Show that $\langle a, -a\rangle \cong H$, by exhibiting an isotropic vector and applying proposition 22.4.6. Then show that the isometry class of $\langle a, -a\rangle$ does not depend on a, even though the square class of a may vary.

8. Let $\operatorname{char} k \neq 2$ and let V be four-dimensional with basis e_1, f_1, e_2, f_2 and the symmetric bilinear form determined by $\varphi(e_1, f_1) = \varphi(e_2, f_2) = 1$, all other products of two basis vectors being zero, so that $V \cong H \perp H$. Define σ by $\sigma(e_i) = e_i$, $\sigma(f_1) = f_1 + e_2$ and $\sigma(f_2) = f_2 - e_1$. Show that $\sigma \in O(\varphi)$, that $x - \sigma(x)$ lies in the totally isotropic subspace spanned by e_1 and e_2 for every $x \in V$, and hence that $q(x - \sigma(x)) = 0$ for every x. Deduce that no reflection τ_v satisfies $\tau_v(\sigma(x)) = x$ for any x with $\sigma(x) \neq x$, so that Case 2 of lemma 22.5.5 is unavoidable for this σ.

22.6 The Witt Group

Theorem 22.4.8 writes a nondegenerate form as an anisotropic form orthogonal to a number of hyperbolic planes, and corollary 22.5.7 says that the anisotropic part is determined up to isometry. Two forms with isometric anisotropic parts therefore differ only in how many hyperbolic planes are attached to them, and this section discards that number. What remains is an abelian group under $\perp$, and a commutative ring once the tensor product of forms is available.

The construction needs no group completion. The inverse of a class is again the class of a form, because $\varphi \perp (-\varphi)$ is a sum of hyperbolic planes, and that computation is the first item below. The group is then computed for the two families of fields whose square classes are known: the algebraically closed fields, where it has two elements, and the finite fields, where it has four. The finite case answers the question left open at the end of example 22.2.5.

Throughout, k is a field with $\operatorname{char} k \neq 2$ and every form is a symmetric bilinear form on a finite-dimensional k-vector space, identified with its quadratic form through proposition 22.1.2 whenever that is convenient. The orthogonal sum $\varphi \perp \psi$ is definition 22.4.1, and everything used below about it is established there: the block-diagonal Gram matrix, the multiplicativity $\det(\varphi \perp \psi) = \det(\varphi)\det(\psi)$, that the sum is nondegenerate exactly when both summands are, and that $\perp$ descends to isometry classes and is commutative and associative up to isometry. Write $-\varphi$ for the form $(x, y) \mapsto -\varphi(x, y)$ on V. In the diagonal notation of section 22.2,

$$\langle a_1, \dots, a_m\rangle \perp \langle b_1, \dots, b_n\rangle = \langle a_1, \dots, a_m, b_1, \dots, b_n\rangle$$

and $-\langle a_1, \dots, a_n\rangle = \langle -a_1, \dots, -a_n\rangle$. Finally mH denotes the orthogonal sum of m copies of the hyperbolic plane H (definition 22.4.4), with $0H$ the zero form on the zero space.

The identification that drives everything below is that a form and its negative add up to nothing but hyperbolic planes.

Lemma 22.6.1: A Form and Its Negative Sum to Hyperbolic Planes

Let k be a field with $\operatorname{char} k \neq 2$.

1. For every $a \in k^\times$ one has $\langle a, -a\rangle \cong H$. In particular $H \cong \langle 1, -1\rangle$, so $\det H = -1$.
2. If φ is a nondegenerate form of rank n on V, then $\varphi \perp (-\varphi) \cong nH$.

***Proof*:**

1. By example 22.4.7 a nondegenerate binary form $\langle a, b\rangle$ is a hyperbolic plane exactly when $-ab$ is a square in $k^\times$. Here $b = -a$, so $-ab = a^2$ is a square and $\langle a, -a\rangle \cong H$. Taking $a = 1$ gives $H \cong \langle 1, -1\rangle$, whose matrix in an orthogonal basis is $\operatorname{diag}(1, -1)$, of determinant -1.
2. Diagonalize $\varphi \cong \langle a_1, \dots, a_n\rangle$ by theorem 22.2.1. Nondegeneracy forces every $a_i \neq 0$: by the last claim of proposition 22.2.2 the number of zero entries is the dimension of the radical, which is 0 here by definition 13.3.22. Then $-\varphi \cong \langle -a_1, \dots, -a_n\rangle$, so

$$\varphi \perp (-\varphi) \cong \langle a_1, \dots, a_n, -a_1, \dots, -a_n\rangle \cong \langle a_1, -a_1\rangle \perp \cdots \perp \langle a_n, -a_n\rangle$$

the second isometry being the permutation of an orthogonal basis that places each a_i next to its negative. Applying part (1) to each summand gives $\varphi \perp (-\varphi) \cong nH$, as we sought to show.

Every form of rank n is isometric to a form on k^n, obtained by transporting it along a choice of basis, so the isometry classes of nondegenerate forms over k constitute a set rather than a proper class. That set carries $\perp$ as a commutative associative operation with the zero form as a neutral element, and the lemma says that after hyperbolic planes are discarded every element acquires an inverse.

Definition 22.6.2: Witt Equivalence and the Witt Group

Let k be a field with $\operatorname{char} k \neq 2$. Two nondegenerate symmetric bilinear forms φ and ψ over k, possibly on spaces of different dimensions, are *Witt equivalent*, written $\varphi \sim \psi$, if there are integers $m, n \geq 0$ with

$$\varphi \perp mH \cong \psi \perp nH$$

The *Witt group* $W(k)$ is the set of equivalence classes, the class of φ being written $[\varphi]$, equipped with the operation

$$[\varphi] + [\psi] := [\varphi \perp \psi]$$

That $\sim$ is an equivalence relation and that the operation is well defined are proved in theorem 22.6.3.

Two small computations show what the relation does. Over any k one has $\langle 1 \rangle \perp \langle -1 \rangle \cong H$ by lemma 22.6.1, so $\langle 1, -1 \rangle \sim 0$: the hyperbolic plane is Witt trivial, which is the whole content of the definition. Consequently $\langle 1, 1, -1 \rangle \sim \langle 1 \rangle$, taking $m = 0$ and $n = 1$ against $\psi = \langle 1 \rangle$, since $\langle 1 \rangle \perp H \cong \langle 1, 1, -1 \rangle$. Forms of different ranks can therefore be Witt equivalent, and rank is not an invariant of a Witt class.

Theorem 22.6.3: The Witt Classes Form an Abelian Group

Let k be a field with $\operatorname{char} k \neq 2$. Witt equivalence is an equivalence relation on nondegenerate forms over k, the operation of definition 22.6.2 is well defined on classes, and $W(k)$ is an abelian group whose neutral element is the class $[0] = [H]$ of the zero form and in which $-[\varphi] = [-\varphi]$.

***Proof*:**
Step 1: $\sim$ is an equivalence relation. Reflexivity is the case $m = n = 0$, and symmetry is immediate from the symmetry of $\cong$. For transitivity, suppose $\varphi \perp mH \cong \psi \perp nH$ and $\psi \perp pH \cong \chi \perp qH$. Adding pH to the first isometry and nH to the second, which is legitimate because orthogonal sums of isometric forms are isometric, and rearranging summands by commutativity and associativity of $\perp$, gives

$$\varphi \perp (m+p)H \cong \psi \perp (n+p)H \cong \chi \perp (q+n)H$$

so $\varphi \sim \chi$.

Step 2: the operation is well defined. Let $\varphi \sim \varphi'$ and $\psi \sim \psi'$, say $\varphi \perp mH \cong \varphi' \perp m'H$ and $\psi \perp nH \cong \psi' \perp n'H$. Taking the orthogonal sum of the two isometries and rearranging summands,

$$(\varphi \perp \psi) \perp (m+n)H \cong (\varphi' \perp \psi') \perp (m'+n')H$$

so $\varphi \perp \psi \sim \varphi' \perp \psi'$. Both $\varphi \perp \psi$ and $\varphi' \perp \psi'$ are nondegenerate, since orthogonal sums of nondegenerate forms are, so the sum of two classes is again a class.

Step 3: the group axioms. Commutativity and associativity of $+$ are those of $\perp$, transported to classes by Step 2. The zero form 0 on the zero space is nondegenerate, its radical being the zero space, and $\varphi \perp 0 = \varphi$, so $[0]$ is neutral. Taking $m = 0$ and $n = 1$ in definition 22.6.2 gives $H \perp 0H \cong 0 \perp 1H$, that is $H \sim 0$, so $[H] = [0]$. Finally, if φ has rank n then $\varphi \perp (-\varphi) \cong nH$ by lemma 22.6.1, and $nH \sim 0$ directly from definition 22.6.2, taking the two integers there to be 0 and n, so

$$[\varphi] + [-\varphi] = [\varphi \perp (-\varphi)] = [nH] = [0]$$

which exhibits $[-\varphi]$ as the inverse of $[\varphi]$, completing the proof.

The group is built from forms alone: no formal differences are introduced, and no group completion of the monoid of isometry classes is taken, because the negative of a form is a form and lemma 22.6.1 makes the two cancel. What the construction has cost is the ability to see the rank, since φ and $\varphi \perp H$ have become equal. The next result says that nothing else was lost, and it is where Witt cancellation enters: cancellation plays no part in the group axioms above, and its role is to identify the classes with objects one can list.

Proposition 22.6.4: Each Witt Class Contains Exactly One Anisotropic Form

Let k be a field with $\operatorname{char} k \neq 2$ and let φ, ψ be nondegenerate forms over k, with Witt decompositions $\varphi \cong \varphi_{\mathrm{an}} \perp m_\varphi H$ and $\psi \cong \psi_{\mathrm{an}} \perp m_\psi H$ as in theorem 22.4.8. Then

$$\varphi \sim \psi \quad \text{if and only if} \quad \varphi_{\mathrm{an}} \cong \psi_{\mathrm{an}}$$

Consequently $[\varphi] \mapsto \varphi_{\mathrm{an}}$ is a bijection from $W(k)$ to the set of isometry classes of anisotropic forms over k, and $[\varphi] = 0$ if and only if $\varphi \cong m_\varphi H$.

***Proof*:**
First note that an anisotropic form is nondegenerate: a nonzero vector v of the radical satisfies $\varphi(v, v) = 0$, hence is an isotropic vector (definition 22.4.2), so the radical of an anisotropic form is zero and definition 13.3.22 applies. Both φ_{an} and ψ_{an} are therefore nondegenerate, as is H, which is what the cancellation theorem requires.

Step 1: isometric anisotropic parts give Witt equivalence. If $\varphi_{\mathrm{an}} \cong \psi_{\mathrm{an}}$ then

$$\varphi \perp m_\psi H \cong \varphi_{\mathrm{an}} \perp (m_\varphi + m_\psi)H \cong \psi_{\mathrm{an}} \perp (m_\varphi + m_\psi)H \cong \psi \perp m_\varphi H$$

so $\varphi \sim \psi$ by definition 22.6.2.

Step 2: Witt equivalence gives isometric anisotropic parts. Suppose $\varphi \perp mH \cong \psi \perp nH$. Substituting the two Witt decompositions and writing $a = m_\varphi + m$ and $b = m_\psi + n$,

$$\varphi_{\mathrm{an}} \perp aH \cong \psi_{\mathrm{an}} \perp bH$$

This is exactly the situation corollary 22.5.7 settles: two Witt decompositions of isometric forms have isometric anisotropic parts and equal hyperbolic counts. Applying it gives $a = b$ and $\varphi_{\mathrm{an}} \cong \psi_{\mathrm{an}}$.

Step 3: the two consequences. The two steps say exactly that $[\varphi] \mapsto \varphi_{\mathrm{an}}$ is well defined and injective on $W(k)$; it is surjective because an anisotropic form χ is nondegenerate with $\chi_{\mathrm{an}} = \chi$ and $m_\chi = 0$. For the last claim, $[\varphi] = [0]$ holds if and only if φ_{an} is isometric to the zero form on the zero space, that is if and only if $\varphi \cong m_\varphi H$, completing the proof.

So $W(k)$ is the set of anisotropic forms over k, given an addition that is not orthogonal sum: to add two anisotropic forms, take their orthogonal sum and discard the hyperbolic planes that appear in it. The integer m_φ discarded in the process is the Witt index of definition 22.4.10, the two agreeing by proposition 22.4.11(3), and the pair (anisotropic part, Witt index) is a complete isometry invariant of a nondegenerate form, while the class in $W(k)$ retains only the first half of that pair. The first field to compute is one over which every anisotropic form has rank at most one.

Example 22.6.5: The Witt Group of an Algebraically Closed Field

Let k be algebraically closed with $\operatorname{char} k \neq 2$; the case $k = \mathbb{C}$ is the one to keep in mind. Fix $c \in k$ with $c^2 = -1$, which exists because $x^2 + 1$ has a root.

No anisotropic form has rank 2 or more. Indeed, by example 22.2.4 a nondegenerate form of rank n is isometric to $\langle 1, \ldots, 1 \rangle$, and if $n \geq 2$ the vector $v = (c, 1, 0, \ldots, 0)$ is nonzero and satisfies

$$q(v) = c^2 + 1 = -1 + 1 = 0$$

so the form is isotropic. The anisotropic forms are therefore the zero form and the rank one form $\langle 1 \rangle$, the latter being anisotropic because $x^2 = 0$ forces $x = 0$. By proposition 22.6.4,

$$W(k) = \{\, [0],\ [\langle 1 \rangle] \,\}$$

has two elements. Its group law is forced: $2[\langle 1 \rangle] = [\langle 1, 1 \rangle]$, and $\langle 1, 1 \rangle \cong \langle 1, -1 \rangle$ by proposition 22.2.2, since $-1 \cdot c^2 = 1$ rescales the second entry by the square c^2, so $\langle 1, 1 \rangle \cong H$ by lemma 22.6.1 and $2[\langle 1 \rangle] = 0$. Hence $W(k) \cong \mathbb{Z}/2\mathbb{Z}$, the isomorphism sending $[\varphi]$ to the rank of φ modulo 2.

Addition is not the only operation available on forms. The tensor product of the underlying spaces carries a form built from the two given ones, and it is compatible with $\perp$ and with Witt equivalence, which upgrades $W(k)$ from a group to a ring. In diagonal coordinates the operation is multiplication of the diagonal entries in all pairs, which is what makes it computable.

Proposition 22.6.6: The Tensor Product of Forms and the Witt Ring

Let k be a field with $\operatorname{char} k \neq 2$, let φ be a symmetric bilinear form on V and ψ one on W.

1. There is exactly one bilinear form $\varphi \otimes \psi$ on $V \otimes_k W$ satisfying

$$(\varphi \otimes \psi)(v \otimes w,\ v' \otimes w') = \varphi(v, v')\,\psi(w, w')$$

   for all $v, v' \in V$ and $w, w' \in W$, and it is symmetric.

2. If $\varphi \cong \langle a_1, \ldots, a_m \rangle$ and $\psi \cong \langle b_1, \ldots, b_n \rangle$, then $\varphi \otimes \psi$ is isometric to the diagonal form whose mn entries are the products $a_i b_j$. In particular $\varphi \otimes \psi$ is nondegenerate when φ and ψ are, and isometric inputs produce isometric outputs.

3. The rule $[\varphi] \cdot [\psi] := [\varphi \otimes \psi]$ is well defined and makes $W(k)$ a commutative ring with identity $[\langle 1 \rangle]$.

***Proof*:**

1. Fix bases $v_1, \ldots, v_m$ of V and $w_1, \ldots, w_n$ of W. By corollary 15.2.6 the mn vectors $v_i \otimes w_j$ form a basis of $V \otimes_k W$. Let M be the $mn \times mn$ matrix indexed by pairs, with entry $\varphi(v_i, v_{i'})\psi(w_j, w_{j'})$ in position $\big((i,j),(i',j')\big)$, and define $B(x, y) := [x]^t M [y]$ using coordinates in that basis. Taking coordinates is linear and matrix multiplication is bilinear in the two coordinate vectors, so B is a bilinear form.

   To see that B has the stated values, write $v = \sum_i x_i v_i$, $v' = \sum_{i'} x'_{i'} v_{i'}$, $w = \sum_j y_j w_j$ and $w' = \sum_{j'} y'_{j'} w_{j'}$. The map $(v, w) \mapsto v \otimes w$ is bilinear by theorem 15.1.12, so $v \otimes w = \sum_{i,j} x_i y_j\,(v_i \otimes w_j)$, which is the coordinate vector of $v \otimes w$ in the chosen basis. Hence

$$\begin{aligned} B(v \otimes w, v' \otimes w') &= \sum_{i,j,i',j'} x_i y_j x'_{i'} y'_{j'}\, \varphi(v_i, v_{i'})\psi(w_j, w_{j'}) \\ &= \Big(\sum_{i,i'} x_i x'_{i'} \varphi(v_i, v_{i'})\Big)\Big(\sum_{j,j'} y_j y'_{j'} \psi(w_j, w_{j'})\Big) \\ &= \varphi(v, v')\,\psi(w, w') \end{aligned}$$

   the last line by bilinearity of φ and of ψ. For uniqueness, the simple tensors span $V \otimes_k W$ by lemma 15.1.4, and two bilinear forms agreeing on all pairs of vectors from a spanning set agree everywhere, by expanding both arguments. Symmetry follows the same way: the bilinear forms $(x, y) \mapsto B(x, y)$ and $(x, y) \mapsto B(y, x)$ agree on pairs of simple tensors because φ and ψ are symmetric, hence agree.

2. Choose the bases in (1) to be orthogonal bases realizing the two diagonalizations, which exist by theorem 22.2.1, so that $\varphi(v_i, v_{i'}) = 0$ for $i \neq i'$ with $\varphi(v_i, v_i) = a_i$, and likewise for ψ. Then for $(i, j) \neq (i', j')$,

$$(\varphi \otimes \psi)(v_i \otimes w_j,\ v_{i'} \otimes w_{j'}) = \varphi(v_i, v_{i'})\psi(w_j, w_{j'}) = 0$$

   since at least one of the two factors has distinct indices, while the value at $(i, j) = (i', j')$ is $a_i b_j$. So the basis $\{v_i \otimes w_j\}$ is orthogonal for $\varphi \otimes \psi$ with the mn diagonal entries $a_i b_j$, which is the claim. If all a_i and all b_j are nonzero then so are all a_ib_j, so the matrix of $\varphi\otimes\psi$ in this basis is diagonal with nonzero entries, hence invertible, and lemma 13.3.23 gives nondegeneracy. Finally, if $\varphi' \cong \varphi$ and $\psi' \cong \psi$, then $\varphi' \cong \langle a_1, \ldots, a_m\rangle$ and $\psi' \cong \langle b_1, \ldots, b_n\rangle$ for the very same entries, since the diagonal notation asserts an isometry class and not a chosen basis. Running the argument just given for those diagonalizations exhibits $\varphi' \otimes \psi'$ and $\varphi \otimes \psi$ as isometric to one and the same diagonal form, so isometric inputs produce isometric outputs. The multiset $\{a_ib_j\}$ is not itself an invariant of the pair of isometry classes, being pinned down only up to multiplying its entries by squares, but the isometry class it names is.

3. Write $\varphi \cong \langle a_1, \ldots, a_m\rangle$, $\psi \cong \langle b_1, \ldots, b_n\rangle$ and $\chi \cong \langle c_1, \ldots, c_p\rangle$, all nondegenerate. By (2) the products a_ib_j and b_ja_i agree, giving $\varphi\otimes\psi \cong \psi\otimes\varphi$; the products $(a_ib_j)c_l$ and $a_i(b_jc_l)$ agree, giving associativity; the entries of $\varphi\otimes(\psi \perp \chi)$ are the a_ib_j together with the a_ic_l, which are the entries of $(\varphi\otimes\psi) \perp (\varphi\otimes\chi)$, giving distributivity over $\perp$; and $\langle 1\rangle \otimes \varphi \cong \varphi$ because the products $1 \cdot a_i$ are the a_i.

   It remains to check that the rule respects Witt equivalence. First, $\langle a\rangle \otimes H \cong \langle a\rangle \otimes \langle 1, -1\rangle \cong \langle a, -a\rangle \cong H$, using lemma 22.6.1 at both ends and (2) in the middle, so by distributivity

$$\psi \otimes H \cong (\langle b_1\rangle \otimes H) \perp \cdots \perp (\langle b_n\rangle \otimes H) \cong nH$$

   Now suppose $\varphi \sim \varphi'$, say $\varphi \perp pH \cong \varphi' \perp p'H$. Tensoring with ψ and using that isometric inputs give isometric outputs, then expanding by distributivity and the previous display,

$$(\varphi\otimes\psi) \perp pnH \cong (\varphi'\otimes\psi) \perp p'nH$$

   so $\varphi\otimes\psi \sim \varphi'\otimes\psi$. Commutativity gives the same conclusion in the second variable, so the product of two classes does not depend on the representatives. The ring axioms are the isometries established in the previous paragraph, read on classes, completing the proof.

$W(k)$ is called the *Witt ring* when the multiplication is in play, and the group and the ring have the same underlying set. The multiplication restricts usefully to rank one forms: $\langle a\rangle \otimes \langle b\rangle \cong \langle ab\rangle$ by part (2), so the square classes of k act on $W(k)$ through $[\langle a\rangle]$, and each $[\langle a\rangle]$ is a unit since $\langle a\rangle \otimes \langle a\rangle \cong \langle a^2\rangle \cong \langle 1\rangle$ by proposition 22.2.2.

Two invariants from section 22.1 can now be tested against Witt equivalence. Rank fails, as the computation after definition 22.6.2 shows, but rank modulo 2 survives because a hyperbolic plane has rank 2. The discriminant also fails, and for a reason worth isolating: $\det(\varphi \perp H) = -\det(\varphi)$ by lemma 22.6.1(1), so $\operatorname{disc}(\varphi \perp H)$ and $\operatorname{disc}(\varphi)$ differ by the class of -1, which is nontrivial in most fields. A sign attached to the determinant repairs this, and box 22.1.9 announced it.

Definition 22.6.7: The Signed Discriminant

Let φ be a nondegenerate symmetric bilinear form of rank n over a field k with $\operatorname{char} k \neq 2$. The *signed discriminant* of φ is the square class

$$d_\pm(\varphi) := (-1)^{n(n-1)/2} \operatorname{disc}(\varphi) \in k^\times/(k^\times)^2$$

with disc as in definition 22.1.7. For $n = 0$ the determinant of the empty matrix is 1, so $d_\pm$ of the zero form is the trivial class.

The definition depends on nothing but the isometry class of φ, since the rank does not and disc does not by proposition 22.1.8. For a diagonal form the sign multiplies the product of the entries: $d_\pm\langle a_1, \dots, a_n\rangle = (-1)^{n(n-1)/2} a_1 \cdots a_n$ modulo squares, by corollary 22.2.3. Thus $d_\pm\langle a\rangle = a$, $d_\pm\langle a, b\rangle = -ab$ and $d_\pm\langle a, b, c\rangle = -abc$.

Proposition 22.6.8: Rank Modulo Two and the Signed Discriminant Are Witt Invariants

Let k be a field with $\operatorname{char} k \neq 2$.

1. The assignment $e\big([\varphi]\big) := \dim \varphi + 2\mathbb{Z}$ is a well defined surjective ring homomorphism $e\colon W(k) \to \mathbb{Z}/2\mathbb{Z}$.
2. For nondegenerate φ of rank m and ψ of rank n,
$$d_\pm(\varphi \perp \psi) = (-1)^{mn}\, d_\pm(\varphi)\, d_\pm(\psi)$$
3. Witt equivalent forms have equal signed discriminants, so $d_\pm$ is a well defined map $W(k) \to k^\times/(k^\times)^2$.
4. Restricted to the classes of even rank, $d_\pm$ is a group homomorphism from $\big(\ker e, +\big)$ to $k^\times/(k^\times)^2$.

***Proof*:**

1. If $\varphi \perp mH \cong \psi \perp nH$ then comparing dimensions gives $\dim \varphi + 2m = \dim \psi + 2n$, so $\dim \varphi \equiv \dim \psi \pmod 2$ and e is well defined. It is additive because $\dim(\varphi \perp \psi) = \dim \varphi + \dim \psi$, and multiplicative because $\dim(V \otimes_k W) = \dim V \cdot \dim W$ by corollary 15.2.6. It sends the identity $[\langle 1\rangle]$ to $1 + 2\mathbb{Z}$, which also gives surjectivity.
2. Both sides are computed from determinants and ranks. The rank of $\varphi \perp \psi$ is $m + n$ and its determinant is $\det(\varphi)\det(\psi)$, by the block form recorded at the start of this section. So the two sides differ only in the power of -1, and
$$\frac{(m+n)(m+n-1)}{2} - \frac{m(m-1)}{2} - \frac{n(n-1)}{2} = \frac{2mn}{2} = mn$$
after expanding the numerator as $m^2 + 2mn + n^2 - m - n - m^2 + m - n^2 + n$. Raising -1 to that difference gives the stated factor.
3. By lemma 22.6.1(1) the hyperbolic plane has rank 2 and determinant -1, so $d_\pm(H) = (-1)^{2(2-1)/2} \cdot (-1) = (-1)(-1) = 1$ in $k^\times/(k^\times)^2$. Applying (2) with $\psi = H$, where $n = 2$ makes $(-1)^{mn} = 1$,
$$d_\pm(\varphi \perp H) = d_\pm(\varphi)\, d_\pm(H) = d_\pm(\varphi)$$
Iterating, $d_\pm(\varphi \perp mH) = d_\pm(\varphi)$ for every $m \geq 0$. Now if $\varphi \perp mH \cong \psi \perp nH$, then the two sides have equal rank and equal discriminant, the latter by proposition 22.1.8, hence equal signed discriminant; combining the three equalities gives $d_\pm(\varphi) = d_\pm(\psi)$.
4. If m and n are both even then $(-1)^{mn} = 1$, so (2) reads $d_\pm(\varphi \perp \psi) = d_\pm(\varphi) d_\pm(\psi)$. Since addition in $W(k)$ is induced by $\perp$ and the target group is written multiplicatively, this is the homomorphism property on $\ker e$, as we sought to show.

The restriction in (4) is not an artifact of the proof. Over $\mathbb{Q}$ one has $d_\pm\langle 1\rangle = 1$ while $d_\pm\big(\langle 1\rangle \perp \langle 1\rangle\big) = d_\pm\langle 1,1\rangle = -1$, and -1 is not a square in $\mathbb{Q}$, so $d_\pm$ is not additive on all of $W(\mathbb{Q})$. The subgroup where it is additive has a name.

Definition 22.6.9: The Fundamental Ideal

Let k be a field with $\operatorname{char} k \neq 2$. The *fundamental ideal* of $W(k)$ is

$$I(k) := \ker\big(e\colon W(k) \to \mathbb{Z}/2\mathbb{Z}\big)$$

the set of Witt classes of even rank. It is an ideal of the Witt ring, being the kernel of the ring homomorphism of proposition 22.6.8(1), and $W(k)/I(k) \cong \mathbb{Z}/2\mathbb{Z}$.

The classes $[\langle 1,-a\rangle]$ for $a \in k^\times$ lie in $I(k)$ and have signed discriminant $-1\cdot(1\cdot(-a)) = a$, so $d_\pm$ maps $I(k)$ onto the whole square class group. Rank modulo 2 and the signed discriminant are thus two homomorphisms out of $W(k)$ and out of $I(k)$ respectively, and over a finite field they turn out to separate every class from every other.

Proposition 22.6.10: Rank and Discriminant Classify Forms over a Finite Field

Let q be an odd prime power and fix a non-square $\epsilon \in \mathbb{F}_q^\times$.

1. For all $a, b \in \mathbb{F}_q^\times$ and all $c \in \mathbb{F}_q$ there exist $x, y \in \mathbb{F}_q$ with $ax^2 + by^2 = c$.
2. Every nondegenerate form over $\mathbb{F}_q$ of rank at least 3 is isotropic, and every nondegenerate form of rank at least 2 represents every element of $\mathbb{F}_q^\times$.
3. Two nondegenerate forms over $\mathbb{F}_q$ are isometric if and only if they have the same rank and the same discriminant.
4. Up to isometry the anisotropic forms over $\mathbb{F}_q$ are exactly the zero form, $\langle 1\rangle$, $\langle\epsilon\rangle$ and $\langle 1,-\epsilon\rangle$.

***Proof*:**

1. This is proved in example 22.4.9, by the counting argument on $S = \{ax^2 \mid x \in \mathbb{F}_q\}$ and $T = \{c - by^2 \mid y \in \mathbb{F}_q\}$, each of size $(q+1)/2$ in a set of size q.
2. The first claim is example 22.4.9, which deduces it from (1). For the second, let φ be nondegenerate of rank $n \geq 2$ and diagonalize it by theorem 22.2.1 as $\varphi \cong \langle a_1,\dots,a_n\rangle$ in an orthogonal basis $w_1,\dots,w_n$, every a_i being nonzero by the last claim of proposition 22.2.2. Given $c \in \mathbb{F}_q^\times$, part (1) gives $x, y \in \mathbb{F}_q$ with $a_1x^2 + a_2y^2 = c$. Put $v := xw_1 + yw_2$; then $q(v) = a_1x^2 + a_2y^2 = c$, which is nonzero, so $v \neq 0$ and φ represents c.
3. Isometric forms have the same rank and, by proposition 22.1.8, the same discriminant. For the converse, induct on the common rank n. For $n = 0$ both forms are the zero form on the zero space. For $n = 1$, write $\varphi \cong \langle a\rangle$ and $\psi \cong \langle b\rangle$; equality of the discriminants means $b = ac^2$ for some $c \in \mathbb{F}_q^\times$ by corollary 22.2.3 and definition 22.1.6, and then $\langle a\rangle \cong \langle b\rangle$ by proposition 22.2.2.

   Let $n \geq 2$. Diagonalizing φ produces a vector v with $c := \varphi(v,v) \neq 0$, and by (2) the form

ψ also represents c, say $\psi(v', v') = c$. A vector of nonzero square splits off: for x in the space of φ,

$$x = \frac{\varphi(v,x)}{c}\, v + \left(x - \frac{\varphi(v,x)}{c}\, v\right)$$

where applying $\varphi(v, -)$ to the bracket gives $\varphi(v,x) - \frac{\varphi(v,x)}{c} c = 0$, so the bracket lies in $\{v\}^{\perp}$ (definition 13.3.20), a subspace by lemma 13.3.21; and if $\lambda v \in \{v\}^{\perp}$ then $\lambda c = 0$, so $\lambda = 0$. Hence the space is the direct sum of kv and $\{v\}^{\perp}$, these are orthogonal, and $\varphi \cong \langle c\rangle \perp \varphi'$ with φ' the restriction to $\{v\}^{\perp}$. The same argument gives $\psi \cong \langle c\rangle \perp \psi'$. Both φ' and ψ' have rank $n-1$ and are nondegenerate, since $c\det(\varphi') = \det(\varphi) \neq 0$ and likewise for ψ. Their discriminants agree: in the square class group, $\operatorname{disc}(\varphi') = c^{-1}\operatorname{disc}(\varphi) = c^{-1}\operatorname{disc}(\psi) = \operatorname{disc}(\psi')$. By the inductive hypothesis $\varphi' \cong \psi'$, and taking the orthogonal sum with the identity of $\langle c\rangle$ gives $\varphi \cong \psi$.

4. An anisotropic form has rank at most 2 by (2). The zero form is anisotropic vacuously. In rank 1, every $\langle a\rangle$ is anisotropic, and by (3) the isometry class is determined by the discriminant, which is the class of a; the two square classes of $\mathbb{F}_q$ are those of 1 and ϵ by example 22.2.5. In rank 2, the form $\langle a, b\rangle$ is isotropic exactly when $-ab$ is a square: given $ax^2 + by^2 = 0$ with $(x,y) \neq (0,0)$, the case $y = 0$ would force $ax^2 = 0$ and hence $x = 0$, so $y \neq 0$ and $-ab = a^2(x/y)^2$ is a square; conversely if $-ab = t^2$ then $(x,y) = (t,a)$ is nonzero and $at^2 + ba^2 = a(t^2 + ab) = 0$. So $\langle a,b\rangle$ is anisotropic exactly when $-ab$ is a non-square, that is when $\operatorname{disc}\langle a,b\rangle = ab$ is the class of $-\epsilon$. By (3) that single condition on the discriminant pins down one isometry class in rank 2, and $\langle 1, -\epsilon\rangle$ is in it, completing the proof.

Part (3) is the affirmative answer promised at the end of example 22.2.5: over $\mathbb{F}_q$ the rank and the discriminant are a complete system of invariants, so the two forms $\langle 1,1\rangle$ and $\langle 1,2\rangle$ exhibited there over $\mathbb{F}_5$ fail to be isometric for the only reason available. Part (4) turns proposition 22.6.4 into a count.

Example 22.6.11: The Witt Group of a Finite Field

Let q be odd and $\epsilon \in \mathbb{F}_q^{\times}$ a non-square. By proposition 22.6.4 the elements of $W(\mathbb{F}_q)$ correspond to isometry classes of anisotropic forms, and proposition 22.6.10(4) lists four of them, so

$$W(\mathbb{F}_q) = \{\, [0],\ [\langle 1\rangle],\ [\langle \epsilon\rangle],\ [\langle 1, -\epsilon\rangle]\,\}$$

has exactly four elements. Which group of order four it is depends on q modulo 4, through whether -1 is a square.

The case $q \equiv 1 \pmod 4$. Then -1 is a square, say $-1 = c^2$: the group $\mathbb{F}_q^{\times}$ is cyclic by theorem 11.3.14, of order $q-1$, which is divisible by 4 in this case, so it has an element of order 4, whose square has order 2 and hence equals -1, the unique element of order 2. For any $a \in \mathbb{F}_q^{\times}$, proposition 22.2.2 gives $\langle a\rangle \cong \langle ac^2\rangle = \langle -a\rangle$, so every form satisfies $\varphi \cong -\varphi$ after diagonalizing, and

$$2[\varphi] = [\varphi] + [-\varphi] = 0$$

by theorem 22.6.3. A group of order 4 in which every element has order dividing 2 is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

The case $q \equiv 3 \pmod 4$. Then -1 is a non-square, so $\epsilon = -1$ is an admissible choice, and $\langle 1,1\rangle$ is anisotropic by the criterion in the proof of proposition 22.6.10(4), since $-1 \cdot 1 = -1$ is a non-square. Hence $2[\langle 1\rangle] = [\langle 1,1\rangle] \neq 0$ by proposition 22.6.4. On the other hand $\langle 1,1,1,1\rangle$ and

$2H \cong \langle 1,-1,1,-1\rangle$ both have rank 4 and discriminant the class of 1, so they are isometric by proposition 22.6.10(3), giving $4[\langle 1\rangle] = [2H] = 0$. So $[\langle 1\rangle]$ has order 4 in a group of order 4, and $W(\mathbb{F}_q) \cong \mathbb{Z}/4\mathbb{Z}$.

The invariants separate the four classes. Writing each class through its anisotropic representative and applying proposition 22.6.8, the rank modulo 2 and the signed discriminant take the values

	$[0]$	$[\langle 1\rangle]$	$[\langle \epsilon\rangle]$	$[\langle 1,-\epsilon\rangle]$
e	0	1	1	0
$d_\pm$	1	1	ϵ	ϵ

the last entry because $d_\pm\langle 1,-\epsilon\rangle = -(1\cdot(-\epsilon)) = \epsilon$. The four pairs are distinct, so a Witt class over a finite field is determined by its rank modulo 2 together with its signed discriminant.

A worked instance. Take $q = 5$. The squares in $\mathbb{F}_5^\times$ are 1 and 4, so $\epsilon = 2$ is a non-square and $-1 = 4 = 2^2$ is a square, putting this in the first case: $W(\mathbb{F}_5) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, with anisotropic representatives 0, $\langle 1\rangle$, $\langle 2\rangle$ and $\langle 1,-2\rangle = \langle 1,3\rangle$. The two forms of example 22.2.5 land in different classes: $\langle 1,1\rangle \cong \langle 1,-1\rangle \cong H$ is Witt trivial, while $[\langle 1,2\rangle] = [\langle 1\rangle] + [\langle 2\rangle]$ has signed discriminant $-2 = 3$, a non-square, so it is not the class of 0.

Exercise 22.6.1

1. Take $k = \mathbb{F}_7$. List the square classes, determine the four anisotropic forms over $\mathbb{F}_7$ up to isometry, and identify $W(\mathbb{F}_7)$ as an abstract group. Which of the four classes is $3[\langle 1\rangle]$?
2. Let $\operatorname{char} k \neq 2$ and let φ be nondegenerate over k. Show that $[\varphi] = 0$ in $W(k)$ if and only if $\varphi \cong mH$ for some $m \geq 0$, and deduce that a nondegenerate form of odd rank is never Witt trivial. Conclude that $W(k)$ has at least two elements for every such k.
3. Let φ and ψ be nondegenerate forms over k with $[\varphi] = [\psi]$ and $\dim\varphi = \dim\psi$. Show that $\varphi \cong \psi$. Give an example over some field showing that the hypothesis on the dimensions cannot be dropped.
4. Suppose -1 is a square in k, where $\operatorname{char} k \neq 2$. Show that $\varphi \cong -\varphi$ for every form over k, hence that $2[\varphi] = 0$ for every $[\varphi] \in W(k)$, so that $W(k)$ is a vector space over $\mathbb{F}_2$. Check that example 22.6.5 is an instance.
5. Show that $a \mapsto [\langle a\rangle]$ induces a well defined group homomorphism from $k^\times/(k^\times)^2$ to the group of units of the Witt ring, and that it is injective for every field k with $\operatorname{char} k \neq 2$. (For injectivity, identify the anisotropic part of $\langle a\rangle$.)
6. Show that $d_\pm \colon I(k) \to k^\times/(k^\times)^2$ is surjective for every k with $\operatorname{char} k \neq 2$. Then compute $I(\mathbb{F}_q)$ for q odd and show that $d_\pm$ is an isomorphism of groups $I(\mathbb{F}_q) \xrightarrow{\sim} \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^2$.
7. Show that $I(k)$ is generated as an abelian group by the classes $[\langle 1,-a\rangle]$ with a ranging over $k^\times$. (Start by expressing $[\langle a\rangle] - [\langle 1\rangle]$ as one of these generators up to sign.)
8. Show that the form $\langle 1,1,\dots,1\rangle$ of rank n is anisotropic over $\mathbb{Q}$ for every n, and deduce that the classes $n[\langle 1\rangle]$ for $n \geq 0$ are pairwise distinct, so that $W(\mathbb{Q})$ is infinite and $[\langle 1\rangle]$ has infinite order. Contrast this with example 22.6.11.

22.7 The Clifford Algebra

Every construction so far has compared one form with another. This section attaches to a single form an associative algebra, built so that the form becomes the operation of squaring: inside it the equation $v^2 = q(v)$ holds for every $v \in V$. That equation is meaningless in V, where there is no product, so the object carrying it must be an algebra containing V, and section 16.3 is where algebras containing a given vector space are constructed.

The construction below is a quotient of $\mathcal{T}(V)$ by one family of relations, and for $q = 0$ it is the exterior algebra. What is proved here is that the quotient has dimension 2^n rather than collapsing, that maps out of it are prescribed by their restriction to V, and that although the $\mathbb{Z}$-grading of $\mathcal{T}(V)$ does not survive, its reduction modulo 2 does.

Definition 16.3.7 formed $\bigwedge(V)$ by killing the elements $v \otimes v$ of $\mathcal{T}(V)$, which forces $v^2 = 0$ in the quotient. Killing $v \otimes v - q(v)$ instead forces $v^2 = q(v)$. Assume $q \neq 0$ and choose v with $q(v) \neq 0$. The element $v \otimes v - q(v)$ is then a sum of a nonzero term of degree 2 and a nonzero term of degree 0, so it is not homogeneous, and the ideal generated by these elements is not a graded ideal in the sense of definition 16.2.3: were it graded, the two homogeneous components of $v \otimes v - q(v)$ would lie in it separately, so the invertible scalar $q(v)$, and with it 1, would lie in it and the quotient would be 0, contradicting theorem 22.7.5. For $q \neq 0$ the images of the $\mathcal{T}^d(V)$ therefore do not make the quotient a graded ring (definition 16.2.1); only in the excluded case $q = 0$, where the ideal is the one of definition 16.3.7 and is graded, does the $\mathbb{Z}$-grading survive (example 22.7.2). Proposition 22.7.7 says exactly how much of the grading is left in general.

Definition 22.7.1: Clifford Algebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let q be a quadratic form on V (definition 22.1.1). Let $I(q) \subseteq \mathcal{T}(V)$ be the two-sided ideal of the tensor algebra (definition 16.3.1) generated by the elements

$$v \otimes v - q(v) \cdot 1 \qquad (v \in V)$$

where $q(v) \cdot 1$ lies in $\mathcal{T}^0(V) = k$. The *Clifford algebra* of q is the quotient k-algebra

$$C(q) := \mathcal{T}(V)/I(q)$$

and $\iota\colon V \to C(q)$ denotes the composite of the inclusion $V = \mathcal{T}^1(V) \subseteq \mathcal{T}(V)$ with the projection $\pi\colon \mathcal{T}(V) \to C(q)$.

Write φ for the symmetric bilinear form attached to q by proposition 22.1.2, so that $\varphi(v, v) = q(v)$. Two identities in $C(q)$ are used in everything below. The first is the defining relation

$$\iota(v)^2 = q(v)$$

which holds because π is an algebra homomorphism, so that $\iota(v)^2 = \pi(v \otimes v) = \pi\big(q(v) \cdot 1\big) = q(v)$, the middle equality holding because the two arguments differ by a generator of $I(q)$. The second identity is obtained by applying the first to $v + w$. Bilinearity of φ gives $q(v + w) = \varphi(v + w, v + w) = q(v) + 2\varphi(v, w) + q(w)$, while expanding the square in $C(q)$ gives

$$\iota(v + w)^2 = \big(\iota(v) + \iota(w)\big)^2 = \iota(v)^2 + \iota(v)\iota(w) + \iota(w)\iota(v) + \iota(w)^2$$

using that ι is linear. Equating the two and cancelling $\iota(v)^2 = q(v)$ and $\iota(w)^2 = q(w)$ leaves

$$\iota(v)\iota(w) + \iota(w)\iota(v) = 2\varphi(v,w) \tag{22.3}$$

Since $\operatorname{char} k \neq 2$, the right-hand side vanishes exactly when v and w are orthogonal (definition 13.3.20), so orthogonality of vectors becomes anticommutation of their images. This is the mechanism that turns a choice of orthogonal basis, which theorem 22.2.1 always gives, into a presentation of $C(q)$ by generators and relations.

Example 22.7.2: The Clifford Algebra of the Zero Form

Let k be a field with $\operatorname{char} k \neq 2$ and take $q = 0$ on a k-vector space V of dimension n. The generators of $I(0)$ are the elements $v \otimes v - 0 = v \otimes v$ for $v \in V$, which are precisely the generators of the ideal $\mathcal{A}(V)$ of definition 16.3.7. The two ideals therefore coincide, and

$$C(0) = \mathcal{T}(V)/\mathcal{A}(V) = \bigwedge(V)$$

not merely up to isomorphism but as the same quotient. Its dimension is

$$\dim_k \bigwedge(V) = \sum_{r=0}^{n} \binom{n}{r} = 2^n$$

since $\bigwedge^r(V)$ is free of rank $\binom{n}{r}$ by theorem 16.4.3 and vanishes for $r > n$. Note that nothing in definition 22.7.1 asked q to be nondegenerate, and for $V \neq 0$ the zero form has radical all of V.

The definition determines the algebra but leaves two questions open: how large it is, and how to produce a homomorphism out of it. The second is answered first, and the answer then gives the first, because the way to bound the dimension of $C(q)$ from below is to map it onto something of known dimension.

Proposition 22.7.3: The Universal Property of the Clifford Algebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a finite-dimensional k-vector space, and let q be a quadratic form on V.

1. For every k-algebra A and every k-linear map $f\colon V \to A$ satisfying $f(v)^2 = q(v)\cdot 1_A$ for all $v \in V$, there is a unique k-algebra homomorphism $F\colon C(q) \to A$ with $F \circ \iota = f$.
2. Let q' be a quadratic form on V', with associated symmetric bilinear forms φ and φ', and let $g\colon (V,\varphi) \to (V',\varphi')$ be an isometry (definition 22.1.10). Then there is a unique k-algebra isomorphism $C(g)\colon C(q) \to C(q')$ with $C(g)\big(\iota(v)\big) = \iota'\big(g(v)\big)$ for all $v \in V$. In particular isometric forms have isomorphic Clifford algebras.

***Proof*:**

1. *Existence.* Take $R = k$ and $M = V$ in theorem 16.3.2(2): its hypotheses ask for a k-algebra and a k-module homomorphism from V into it, which are A and f. So it gives a unique k-algebra homomorphism $\Psi\colon \mathcal{T}(V) \to A$ with $\Psi|_V = f$. On a generator of $I(q)$,

$$\Psi\big(v \otimes v - q(v)\cdot 1\big) = \Psi(v)\Psi(v) - q(v)\Psi(1) = f(v)^2 - q(v)\cdot 1_A = 0$$

by the hypothesis on f. Every element of $I(q)$ is a finite sum of products $x\gamma y$ with $x, y \in \mathcal{T}(V)$ and γ one of those generators, and $\Psi(x\gamma y) = \Psi(x)\Psi(\gamma)\Psi(y) = 0$, so $I(q) \subseteq \ker\Psi$. Hence

$$F\big(x + I(q)\big) := \Psi(x)$$

is well defined: two representatives of the same class differ by an element of $I(q)$, on which Ψ vanishes. It is k-linear, multiplicative and unital because Ψ is and because $\pi\colon \mathcal{T}(V) \to C(q)$ is a surjective algebra homomorphism with $F \circ \pi = \Psi$. Finally $F \circ \iota = \Psi|_V = f$.

Uniqueness. The algebra $C(q)$ is generated as a k-algebra by $\iota(V)$: the simple tensors $v_1 \otimes \cdots \otimes v_m$ generate $\mathcal{T}^m(V)$ as a k-module by lemma 15.1.4, hence together with 1 they span $\mathcal{T}(V)$, and π is surjective and sends $v_1 \otimes \cdots \otimes v_m$ to $\iota(v_1)\cdots\iota(v_m)$. Now if F_1, F_2 are algebra homomorphisms out of $C(q)$ agreeing on $\iota(V)$, the set on which they agree is closed under sums, products and scalars and contains 1, so it is a subalgebra containing a generating set, hence all of $C(q)$. So F is unique.

2. An isometry satisfies $\varphi'(g(v), g(v)) = \varphi(v, v)$, that is $q'(g(v)) = q(v)$. So the k-linear map $f := \iota' \circ g\colon V \to C(q')$ satisfies

$$f(v)^2 = \iota'\big(g(v)\big)^2 = q'\big(g(v)\big) = q(v)$$

and part (1) gives a unique algebra homomorphism $C(g)$ with $C(g) \circ \iota = \iota' \circ g$. The inverse map g^{-1} is again an isometry, since applying $\varphi'(g(-), g(-)) = \varphi(-,-)$ to the vectors $g^{-1}(x), g^{-1}(y)$ gives $\varphi'(x, y) = \varphi(g^{-1}x, g^{-1}y)$, so the same construction produces $C(g^{-1})\colon C(q') \to C(q)$. The composite $C(g^{-1}) \circ C(g)$ is an algebra homomorphism $C(q) \to C(q)$ carrying $\iota(v)$ to $\iota(g^{-1}g(v)) = \iota(v)$, and the identity of $C(q)$ carries $\iota(v)$ to $\iota(v)$ as well. Both therefore solve the problem of part (1) for $A = C(q)$ and $f = \iota$, whose hypothesis $\iota(v)^2 = q(v)$ holds, so uniqueness forces $C(g^{-1}) \circ C(g) = \mathrm{id}_{C(q)}$. Exchanging the roles of g and g^{-1} gives $C(g) \circ C(g^{-1}) = \mathrm{id}_{C(q')}$, so $C(g)$ is an isomorphism, completing the proof.

Part (1) is the statement that prescribing an algebra homomorphism out of $C(q)$ costs exactly one k-linear map $V \to A$ whose values square to the scalars $q(v)$; the relations impose nothing further. Part (2) is what makes $C(q)$ an invariant of the isometry class rather than of a chosen presentation, so it may be computed from any convenient diagonalization $\langle a_1, \ldots, a_n\rangle$ produced by theorem 22.2.1. The smallest case already shows that the answer depends on the field.

Example 22.7.4: Clifford Algebras of a Line

Let k be a field with $\operatorname{char} k \neq 2$, let $\dim_k V = 1$, say $V = kv$, and let $a := q(v)$, so that $q \cong \langle a\rangle$. Then

$$C(q) \cong k[x]/(x^2 - a)$$

To see this, write $A := k[x]/(x^2 - a)$ and let $\bar{x}$ be the class of x. The map $f\colon V \to A$ sending $\lambda v \mapsto \lambda\bar{x}$ is k-linear, and $f(\lambda v)^2 = \lambda^2\bar{x}^2 = \lambda^2 a = q(\lambda v)$ by definition 22.1.1(1), so proposition 22.7.3(1) gives an algebra homomorphism $F\colon C(q) \to A$ with $F(\iota(v)) = \bar{x}$. In the other direction, evaluation at $\iota(v)$ is an algebra homomorphism $k[x] \to C(q)$, since the powers of $\iota(v)$ commute with one another and k is central in $C(q)$; it kills $x^2 - a$ because $\iota(v)^2 = q(v) = a$, so it factors through a homomorphism $G\colon A \to C(q)$ with $G(\bar{x}) = \iota(v)$. Now $F \circ G$ is an algebra endomorphism of A fixing $\bar{x}$, and the elements it fixes form a subalgebra, which therefore contains the algebra generated by $\bar{x}$, namely all of A; so $F \circ G = \mathrm{id}_A$. Likewise $G \circ F$ is an algebra endomorphism of $C(q)$ fixing $\iota(V)$ pointwise, hence equals $\mathrm{id}_{C(q)}$ by the uniqueness clause of proposition 22.7.3(1). So F is an isomorphism.

Three cases occur, according to the square class of a (definition 22.1.6).

1. $a = 0$. Then $C(q) \cong k[x]/(x^2)$, the algebra of dual numbers, in agreement with example 22.7.2, which identifies it with $\bigwedge(V)$.
2. $a = c^2$ with $c \in k^\times$. Then $x^2 - a = (x-c)(x+c)$ is a product of two distinct monic irreducibles, distinct because $c \neq -c$ when $\operatorname{char} k \neq 2$, so proposition 11.1.6 gives
$$C(q) \cong k[x]/(x-c) \times k[x]/(x+c) \cong k \times k$$
an algebra with zero divisors: $(\bar{x} - c)(\bar{x} + c) = 0$.
3. $a \in k^\times$ is not a square. Then $C(q)$ is a field, the quadratic extension of k obtained by adjoining a square root of a. Indeed $(u + w\bar{x})(u - w\bar{x}) = u^2 - w^2 a$ for $u, w \in k$, and this is nonzero unless $u = w = 0$: if $w \neq 0$ then $u^2 = w^2 a$ would make $a = (u/w)^2$ a square, and if $w = 0$ then $u \neq 0$ gives $u^2 \neq 0$. So every nonzero element of $C(q)$ has the inverse $(u - w\bar{x})/(u^2 - w^2 a)$.

Over $k = \mathbb{R}$ the three cases are $\mathbb{R}[x]/(x^2)$, $\mathbb{R} \times \mathbb{R}$ and $\mathbb{C}$, the last being $C(\langle -1 \rangle)$.

All three algebras above have dimension $2 = 2^1$, and the same count holds in every dimension. Proving it is the one place where the construction has to be shown not to collapse: nothing said so far rules out $C(q) = 0$, since a quotient by a large ideal can kill everything. The proof below produces an explicit $C(q)$-module of dimension 2^n, and the independence of the monomials of $C(q)$ follows from the independence of that module's basis.

Theorem 22.7.5: The Monomial Basis of $C(q)$

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of dimension n, let q be a quadratic form on V, and let $e_1, \ldots, e_n$ be an orthogonal basis of V for the associated symmetric bilinear form φ, with $a_i := q(e_i)$. For a subset $S = \{i_1 < i_2 < \cdots < i_r\}$ of $\{1, \ldots, n\}$ write
$$e_S := \iota(e_{i_1})\iota(e_{i_2}) \cdots \iota(e_{i_r}) \in C(q), \qquad e_\emptyset := 1$$
Then the 2^n elements e_S form a k-basis of $C(q)$. In particular $\dim_k C(q) = 2^n$ and $\iota \colon V \to C(q)$ is injective.

***Proof*:**
An orthogonal basis exists by theorem 22.2.1, whose hypotheses are the standing ones here. Throughout, ι is suppressed from the notation for products, so e_i denotes $\iota(e_i) \in C(q)$.

Step 1: the relations. By (22.3) applied to $v = w = e_i$ and to $v = e_i$, $w = e_j$ with $i \neq j$,
$$e_i^2 = q(e_i) = a_i, \qquad e_i e_j = -e_j e_i \quad (i \neq j)$$
the second because $\varphi(e_i, e_j) = 0$ for $i \neq j$ by the choice of basis. Also, for $v = \sum_i c_i e_i$, expanding

φ by bilinearity and using orthogonality gives

$$q(v) = \varphi(v,v) = \sum_i c_i^2 a_i \tag{22.4}$$

Step 2: the e_S span $C(q)$. As established in the uniqueness half of the proof of proposition 22.7.3, $C(q)$ is spanned as a k-vector space by 1 together with the products $\iota(v_1)\cdots\iota(v_m)$ with $v_t \in V$. Expanding each v_t in the basis $e_1,\dots,e_n$ and multiplying out, which is legitimate because multiplication in $C(q)$ is k-bilinear, shows that the products $e_{j_1}e_{j_2}\cdots e_{j_m}$ of basis vectors already span.

It remains to reduce such a product to a scalar multiple of some e_S, which is done by induction on m. For $m = 0$ the product is $1 = e_\emptyset$. If the indices $j_1,\dots,j_m$ are pairwise distinct, then finitely many interchanges of adjacent factors sort them into increasing order, each interchange multiplying the product by -1 by Step 1, so the product equals $\pm e_S$ with $S = \{j_1,\dots,j_m\}$. If two indices coincide, choose a pair $j_s = j_t = i$ with $s < t$ and $t - s$ minimal; then $j_u \neq i$ for $s < u < t$, so moving the factor e_{j_t} leftwards past those $t - s - 1$ factors changes the product only by a sign and makes the two occurrences of e_i adjacent. Replacing e_ie_i by the scalar a_i produces a scalar multiple of a product of length $m - 2$, which the inductive hypothesis handles. Hence the e_S span, and $\dim_k C(q) \leq 2^n$.

Step 3: a module of dimension 2^n. Let M be the k-vector space with basis $\{u_S \mid S \subseteq \{1,\dots,n\}\}$, so $\dim_k M = 2^n$. For an index i and a subset S write $S\triangle i$ for the symmetric difference, that is $S \cup \{i\}$ if $i \notin S$ and $S \setminus \{i\}$ if $i \in S$, and set

$$d(i,S) := \#\{l \in S \mid l < i\}, \qquad \sigma(i,S) := (-1)^{d(i,S)}, \qquad c_i(S) := \begin{cases} a_i & i \in S \\ 1 & i \notin S \end{cases}$$

Define $E_i \in \mathrm{End}_k(M)$ on the basis by

$$E_i(u_S) := c_i(S)\,\sigma(i,S)\,u_{S\triangle i}$$

Three facts about these coefficients are what the computation uses. First, $\sigma(i, S\triangle j) = \sigma(i,S)$ whenever $j \geq i$, since $d(i,-)$ counts only elements smaller than i and is therefore unchanged by inserting or deleting j. Second, $\sigma(j, S\triangle i) = -\sigma(j,S)$ whenever $i < j$, since $d(j,-)$ then changes by exactly one. Third, $c_i(S\triangle j) = c_i(S)$ for $j \neq i$, since membership of i is unchanged, while $c_i(S)c_i(S\triangle i) = a_i$, one of the two factors being a_i and the other 1.

Applying E_i twice and using the first and third facts,

$$E_i^2(u_S) = c_i(S)\sigma(i,S)\,c_i(S\triangle i)\sigma(i,S\triangle i)\,u_S = a_i\,\sigma(i,S)^2\,u_S = a_i\,u_S$$

so $E_i^2 = a_i\,\mathrm{id}_M$. For $i < j$, the same facts give

$$E_iE_j(u_S) = c_j(S)\sigma(j,S)\,c_i(S\triangle j)\sigma(i,S\triangle j)\,u_{S\triangle\{i,j\}} = c_i(S)c_j(S)\sigma(i,S)\sigma(j,S)\,u_{S\triangle\{i,j\}}$$

whereas, the second fact now contributing a sign,

$$E_jE_i(u_S) = c_i(S)\sigma(i,S)\,c_j(S\triangle i)\sigma(j,S\triangle i)\,u_{S\triangle\{i,j\}} = -\,c_i(S)c_j(S)\sigma(i,S)\sigma(j,S)\,u_{S\triangle\{i,j\}}$$

so $E_iE_j = -E_jE_i$ for $i \neq j$.

Step 4: the e_S are independent. Let $\rho\colon V \to \mathrm{End}_k(M)$ be the k-linear map with $\rho(e_i) = E_i$. For $v = \sum_i c_i e_i$, expanding the square and grouping the cross terms in pairs,

$$\rho(v)^2 = \sum_i c_i^2 E_i^2 + \sum_{i<j} c_i c_j \big(E_i E_j + E_j E_i\big) = \Big(\sum_i c_i^2 a_i\Big)\mathrm{id}_M = q(v)\,\mathrm{id}_M$$

by Step 3 and (22.4). So proposition 22.7.3(1), applied to the k-algebra $A = \mathrm{End}_k(M)$, yields an algebra homomorphism $\Phi\colon C(q) \to \mathrm{End}_k(M)$ with $\Phi(e_i) = E_i$.

Evaluate $\Phi(e_S)$ at $u_\emptyset$ for $S = \{i_1 < \cdots < i_r\}$, by descending induction on t on the claim that $E_{i_{t+1}} \cdots E_{i_r}(u_\emptyset) = u_T$ with $T = \{i_{t+1}, \ldots, i_r\}$. For $t = r$ the product is empty and the claim reads $u_\emptyset = u_\emptyset$. Given it for t, every element of T exceeds i_t, so $d(i_t, T) = 0$ and $\sigma(i_t, T) = 1$, while $i_t \notin T$ gives $c_{i_t}(T) = 1$; hence $E_{i_t}(u_T) = u_{T\cup\{i_t\}}$, which is the claim for $t-1$. At $t = 0$ it reads $\Phi(e_S)(u_\emptyset) = u_S$. Now if $\sum_S \lambda_S e_S = 0$ in $C(q)$, applying Φ and evaluating at $u_\emptyset$ gives $\sum_S \lambda_S u_S = 0$ in M, and the u_S are a basis, so every $\lambda_S = 0$.

The e_S therefore span and are independent, so they are a basis and $\dim_k C(q) = 2^n$. The images $\iota(e_1), \ldots, \iota(e_n)$ are the basis elements e_S with $\#S = 1$, hence linearly independent, so the linear map ι carries a basis of V to an independent set and is injective, completing the proof.

Two consequences are recorded before any further computation. First, ι being injective, V is identified with its image from here on and the symbol ι is dropped: a vector $v \in V$ is treated as an element of $C(q)$ with $v^2 = q(v)$. Second, $C(q)$ and $\bigwedge(V)$ have the same dimension for every q, by theorem 22.7.5 and example 22.7.2, and the same monomials index a basis of each; the multiplications differ, since squares of vectors are 0 in one and $q(v)$ in the other. For $n \geq 2$ neither algebra is commutative: $e_1e_2 = -e_2e_1$ in both, and $e_1e_2 \neq 0$ because it is a basis element (theorem 22.7.5 and theorem 16.4.3), so $e_1e_2 = e_2e_1$ would give $2e_1e_2 = 0$ against $\operatorname{char} k \neq 2$.

Example 22.7.6: The Quaternions as a Clifford Algebra

Let $V = \mathbb{R}^2$ and $q = \langle -1, -1\rangle$, with orthogonal basis e_1, e_2 and $q(e_1) = q(e_2) = -1$. By theorem 22.7.5 the algebra $C(q)$ has basis $1, e_1, e_2, e_1e_2$. Writing $\omega := e_1e_2$ and using $e_2e_1 = -e_1e_2$ to move factors past one another,

$$e_1^2 = e_2^2 = -1, \qquad \omega^2 = e_1e_2e_1e_2 = -e_1e_1e_2e_2 = -(-1)(-1) = -1$$

$$e_1e_2 = \omega, \qquad e_2\omega = e_2e_1e_2 = -e_1e_2e_2 = e_1, \qquad \omega e_1 = e_1e_2e_1 = -e_1e_1e_2 = e_2$$

These are the relations $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$ satisfied by the quaternion units, so $C(q)$ and the quaternions $\mathbb{H}$ are the same algebra; $\mathbb{H}$ appears in theorem 20.2.3 as one of the three associative real division algebras, though that theorem gives no basis or multiplication rule, so both are recalled here. To see it as an isomorphism rather than a coincidence of tables, recall that $\mathbb{H}$ is the 4-dimensional $\mathbb{R}$-algebra with basis $1, i, j, k$ and $ij = k = -ji$. Those two relations already force the rest of the table: putting $k = ij$ and using associativity,

$$jk = j(ij) = (ji)j = -ij^2 = i, \qquad ki = (ij)i = i(ji) = -i^2j = j$$

and in the same way $ik = i(ij) = i^2j = -j$ and $kj = (ij)j = ij^2 = -i$, so the products computed in $C(q)$ above are exactly the quaternion products, and they agree with the relations displayed for Q_8 in section 1.2.4. Now let $f\colon V \to \mathbb{H}$ be the $\mathbb{R}$-linear map with $f(e_1) = i$ and $f(e_2) = j$. For $v = c_1e_1 + c_2e_2$,

$$f(v)^2 = c_1^2i^2 + c_1c_2(ij + ji) + c_2^2j^2 = -(c_1^2 + c_2^2) = q(v)$$

so proposition 22.7.3(1) gives an algebra homomorphism $F\colon C(q) \to \mathbb{H}$ with $F(e_1) = i$ and $F(e_2) = j$. Its image is a subalgebra containing i, j and hence $ij = k$, so it contains the spanning set $1, i, j, k$ and F is surjective; both algebras have dimension 4 over $\mathbb{R}$, so

$$C(\langle -1, -1\rangle) \cong \mathbb{H}$$

Compare example 22.7.4, where $C(\langle -1\rangle) \cong \mathbb{C}$: the first two entries of a negative definite real form produce the complex numbers and then the quaternions.

The $\mathbb{Z}$-grading of $\mathcal{T}(V)$ was lost when the inhomogeneous relations were imposed, but the parity of a monomial was unchanged by the reduction in Step 2 of theorem 22.7.5: cancelling a repeated index removes two factors at a time. The following proposition makes that observation into a decomposition of the algebra, obtained not from the monomials but from the universal property, which is what makes it independent of the chosen orthogonal basis.

Proposition 22.7.7: The Parity Grading and the Even Subalgebra

Let k be a field with $\operatorname{char} k \neq 2$, let V be a k-vector space of dimension n, and let q be a quadratic form on V.

1. There is a unique k-algebra automorphism $\alpha\colon C(q) \to C(q)$ with $\alpha(v) = -v$ for all $v \in V$, and $\alpha^2 = \mathrm{id}$.
2. Setting $C^0(q) := \{x \in C(q) \mid \alpha(x) = x\}$ and $C^1(q) := \{x \in C(q) \mid \alpha(x) = -x\}$,
$$C(q) = C^0(q) \oplus C^1(q), \qquad C^\epsilon(q)\, C^\delta(q) \subseteq C^{\epsilon+\delta}(q)$$
the superscripts being read modulo 2. In particular $C^0(q)$ is a subalgebra, the *even part* of $C(q)$, and $V \subseteq C^1(q)$.
3. With the notation of theorem 22.7.5, $C^0(q)$ has basis the e_S with $\#S$ even and $C^1(q)$ has basis the e_S with $\#S$ odd. For $n \geq 1$ both have dimension 2^{n-1}.

***Proof*:**

1. The map $f\colon V \to C(q)$ with $f(v) = -v$ is k-linear and satisfies $f(v)^2 = (-v)^2 = v^2 = q(v)$, so proposition 22.7.3(1) with $A = C(q)$ gives a unique algebra homomorphism α with $\alpha(v) = -v$ on V. Then $\alpha \circ \alpha$ is an algebra homomorphism $C(q) \to C(q)$ restricting on V to $v \mapsto -(-v) = v$, and $\mathrm{id}_{C(q)}$ restricts to that same map; the uniqueness clause of the same proposition, applied to $f = \iota$, forces $\alpha^2 = \mathrm{id}$. Hence α is bijective, so it is an automorphism.
2. For $x \in C(q)$ write
$$x = \tfrac{1}{2}\big(x + \alpha(x)\big) + \tfrac{1}{2}\big(x - \alpha(x)\big)$$
which is legitimate since $\operatorname{char} k \neq 2$. Applying α and using $\alpha^2 = \mathrm{id}$ shows the first summand is fixed by α and the second is negated, so $C(q) = C^0(q) + C^1(q)$. If x lies in both then $x = -x$, so $2x = 0$ and $x = 0$; the sum is direct. For the product rule, let $x \in C^\epsilon(q)$ and $y \in C^\delta(q)$. Then $\alpha(xy) = \alpha(x)\alpha(y) = (-1)^\epsilon(-1)^\delta xy = (-1)^{\epsilon+\delta}xy$, which is the stated containment. Taking $\epsilon = \delta = 0$ shows $C^0(q)$ is closed under multiplication, and $\alpha(1) = 1$ puts 1 in it, so it is a subalgebra. Finally $\alpha(v) = -v$ for $v \in V$ is the statement $V \subseteq C^1(q)$.

3. Since α is an algebra homomorphism, $\alpha(e_S) = \alpha(e_{i_1}) \cdots \alpha(e_{i_r}) = (-1)^r e_S$ for S of size r. Writing an arbitrary $x = \sum_S \lambda_S e_S$ in the basis of theorem 22.7.5,

$$\alpha(x) = \sum_S (-1)^{\# S} \lambda_S e_S$$

and comparing coefficients, $\alpha(x) = x$ holds if and only if $\lambda_S = 0$ for every S of odd size, while $\alpha(x) = -x$ holds if and only if $\lambda_S = 0$ for every S of even size. So the even monomials are a basis of $C^0(q)$ and the odd ones a basis of $C^1(q)$. Their numbers are equal for $n \geq 1$, since

$$\sum_{r \text{ even}} \binom{n}{r} - \sum_{r \text{ odd}} \binom{n}{r} = (1-1)^n = 0$$

by the binomial theorem, and they sum to 2^n, so each is 2^{n-1}, as we sought to show.

The decomposition is a $\mathbb{Z}/2\mathbb{Z}$-grading, and $C^0(q)$ is a subalgebra of half the dimension of $C(q)$. In example 22.7.6 it has basis $1, e_1e_2$ with $(e_1e_2)^2 = -1$, so evaluation at e_1e_2 is a surjective algebra homomorphism $\mathbb{R}[x] \to C^0$ killing x^2+1, and comparing dimensions gives

$$C^0(\langle -1, -1\rangle) \cong \mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$$

sitting inside $\mathbb{H}$ as the subalgebra spanned by 1 and one imaginary unit. The general rank-two computation, which recovers $\operatorname{disc}(q)$ of definition 22.1.7 from the even part, is the third exercise below.

Exercise 22.7.1

1. Let $\operatorname{char} k \neq 2$, let q be a quadratic form on V and regard $V \subseteq C(q)$ as in theorem 22.7.5. Show that a vector $v \in V$ is invertible in $C(q)$ if and only if $q(v) \neq 0$, and that its inverse is then $q(v)^{-1}v$. Deduce that if $q(v) = 0$ for some nonzero $v \in V$, then $C(q)$ is not a division algebra.

2. Let $\operatorname{char} k \neq 2$ and let $q = \langle 1, -1\rangle$ on k^2. Show that $C(q) \cong M_2(k)$. (Send e_1 and e_2 to two explicit matrices, check the hypothesis of proposition 22.7.3(1) for every vector rather than only for e_1 and e_2, and finish with theorem 22.7.5.)

3. Let $\operatorname{char} k \neq 2$ and let $q \cong \langle a, b\rangle$ be nondegenerate of rank 2. Show that $C^0(q) \cong k[x]/(x^2+ab)$, and that this algebra is a field exactly when $-ab$ is not a square in k. Using corollary 22.2.3, express the condition in terms of $\operatorname{disc}(q)$, and check the answer against example 22.7.6.

4. Let $\operatorname{char} k \neq 2$ and let $q \cong \langle a_1, \ldots, a_n\rangle$ be nondegenerate, so every $a_i \neq 0$, and set $\omega := e_1e_2 \cdots e_n$ for an orthogonal basis as in theorem 22.7.5. Show that $\omega e_i = (-1)^{n-1} e_i \omega$ for every i, and deduce that ω lies in the centre of $C(q)$ if and only if n is odd.

5. Let $\operatorname{char} k \neq 2$ and let $C(g)$ be as in proposition 22.7.3(2). Show that $C(\mathrm{id}_V) = \mathrm{id}_{C(q)}$ and that $C(h \circ g) = C(h) \circ C(g)$ for isometries g and h. Deduce that $g \mapsto C(g)$ is a group homomorphism from the orthogonal group $O(\varphi)$ of definition 22.1.10 to the group of k-algebra automorphisms of $C(q)$, and that every automorphism in its image commutes with the α of proposition 22.7.7 and therefore preserves $C^0(q)$ and $C^1(q)$.

6. Let $\operatorname{char} k \neq 2$ and let $C(q)^{\mathrm{op}}$ denote the k-algebra with the same underlying vector space as $C(q)$ and multiplication $x * y := yx$. Show that there is a unique k-algebra homomorphism

$t\colon C(q) \to C(q)^{\mathrm{op}}$ with $t(v) = v$ for all $v \in V$, that the resulting map on $C(q)$ satisfies $t(xy) = t(y)t(x)$ and $t^2 = \mathrm{id}$, and that $t(e_{i_1}e_{i_2}\cdots e_{i_r}) = (-1)^{r(r-1)/2}e_{i_1}e_{i_2}\cdots e_{i_r}$ for $i_1 < \cdots < i_r$.

7. Let $\operatorname{char} k \neq 2$ and let q be a quadratic form on V. Show that $C^0(q)$ is spanned as a k-vector space by the products $v_1v_2\cdots v_{2m}$ of an even number of vectors of V, and hence that it is generated as a k-algebra by the products vw with $v, w \in V$.